yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

LEVEL-1

1. (a)	2. (b)	3. (b)	4. (b)	5. (c)	6. (d)	7. (c)	8. (b)	9. (a)	10. (c)
11. (c)	12. (a)	13. (b)	14. (a)	15. (a)	16. (c)	17. (a)	18. (c)	19. (a)	20. (d)
21. (b)	22. (a)	23. (d)	24. (c)	25. (c)	26. (a)	27. (b)	28. (c)	29. (c)	30. (b)
31. (d)	32. (a)	33. (c)	34. (c)	35. (c)	36. (a)	37. (a)	38. (a)	39. (b)	40. (a)
41. (b)	42. (d)	43. (d)	44. (c)	45. (a)	46. (c)	47. (a)	48. (c)	49. (b)	50. (b)
51. (c)	52. (c)	53. (b)	54. (c)	55. (d)					

LEVEL-2

1. (c)	2. (b)	3. (c)	4. (b)	5. (a)	6. (b)	7. (c)	8. (a)	9. (d)	10. (b)
11. (a)	12. (d)	13. (c)	14. (d)	15. (b)	16. (b)	17. (a)	18. (b)	19. (b)	20. (a)
21. (a)	22. (c)	23. (c)	24. (d)	25. (b)	26. (a)	27. (a)	28. (b)	29. (c)	30. (d)
31. (c)	32. (a)	33. (b)	34. (c)	35. (d)	36. (b)	37. (a)	38. (a)	39. (b)	40. (a)
41. (a)	42. (d)	43. (b)	44. (a)	45. (c)	46. (c)	47. (a)	48. (b)	49. (a)	50. (b)
51. (a)	52. (a)	53. (c)	54. (b)	55. (b)	56. (a)	57. (b)	58. (c)	59. (a)	60. (a)
61. (b)	62. (c)	63. (b)	64. (c)	65. (a)	66. (a)	67. (c)	68. (b)		

TEST OF YOUR LEARNING

1. (c)	2. (b)	3. (a)	4. (c)	5. (c)	6. (a)	7. (b)	8. (c)	9. (d)	10. (a)
11. (b)	12. (a)	13. (c)	14. (b)	15. (c)	16. (c)	17. (b)	18. (a)		

www.LearnEngineering.in

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Hints & Solutions

INTRODUCTORY EXERCISE 10.1

1. $2(16x + 9x) = 750$; $l = 16x$ and $b = 9x$

2. $$\text{Area} = \frac{110}{0.5} = 220 \text{ sq. m}$$
   and $$11x \times 5x = 220$$

3. Length : breadth = 3 : 2
   Also, area of floor = area of roof

   and 30 is almost 50% of 62. So you need not to solve just look out for the appropriate option thus 1411.2 is almost 50% of 2916.48 and rest of the options are not satisfactory.

4. 1 hectares = 10000 m^2, find length and breadth, then perimeter
   Perimeter = 1000 m = 1 km
   $$\therefore \quad \text{Time} = \frac{\text{distance}}{\text{speed}} = \frac{1}{3} \text{ hrs} = 20 \text{ min}$$

5. 5 hectares and 76 ares = 57600 m^2, find perimeter then multiply it by 6.

6. $l \times 0.7 = 19 \times 3.5 \Rightarrow l = 95$ m (70 cm = 0.7 m)
   Also $(95)^2 = 9025$
   So $95 \times 0.95 = 90.25$

7. $$\text{Number of stones} = \frac{\text{Area of courtyard}}{\text{Area of one stone}} = 14400$$
   Cost = Rate × number of stones
   $= (0.5 \times 14400)$

8. Area $= (63 + 54 - 6) \times 6$ [$\because$ Area $= (l + b - 2w)\, 2w$]
   Area $= (111 \times 6)$
   and $$\text{Rate} = \frac{37}{2} = \frac{37 \times 3}{2 \times 3} = \frac{111}{6}$$
   $$\therefore \quad \text{Cost} = (111 \times 6) \times \frac{111}{6}$$
   $$= 111 \times 111 = 12321$$

9. Original area $= l \times b$
   New area $= 2l \times \frac{b}{2} = l \times b$
   Hence, no change.

10. $(116 + 68 - 2w)\, 2w = 720$
    Solve through quadratic equation.
    Alternatively: Go through options and put $w = 2$ from option (c) you will find that both sides are equal. So the presumed choice is correct.

11. Net area = Total area of 4 walls – $8 m^2$
    and Area of 4 walls $= 2(l + b) \times h$

12. $$d^2 = 2 \times \text{area} \qquad \left(\because \text{Area} = \frac{d^2}{2}\right)$$
    $$d^2 = 2 \times 2 \Rightarrow d = 2 \text{ km}$$

13. $100 \times 100 = 10000 \text{ m}^2 = 1$ hectare
    $111 \times 111 = 12321 \text{ m}^2$
    $\therefore$ Difference in area = 2321 m^2

14. $$\text{Rate} = \frac{759 - 561}{6} = \text{Rs. } 33 \text{ per metre}$$

15. $$l \times b = \frac{1}{25} \times 10000 = 1000 \times b$$
    $$\Rightarrow \quad b = 0.4 \text{ m}$$

16. Side $= \sqrt{(289)^2 - (240)^2}$

17. $$\frac{1600 \times 900}{40 \times 20} = 1800$$

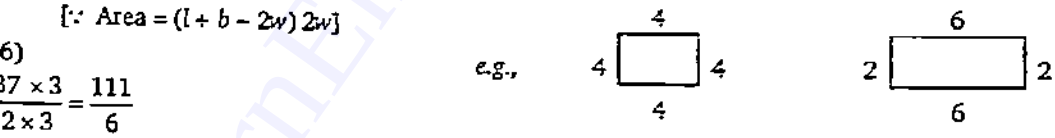
18. $$\text{Area} = \frac{d^2}{2}$$

19. $$\frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$$
    $\therefore \quad 90 \times \frac{1}{9} = 10 g$ (It depends upon area)

20. $l : b = 2 : 1$
    $2x^2 = 24200$, find diagonal.
    $\sqrt{(110)^2 + (220)^2} = 110\sqrt{1^2 + 2^2}$
    $= 110 \times \sqrt{5}$
    $= 110 \times 2.236 = 246$ m

21. $2(2x + x) \times 11 = 2640$, find $2x \times x = ?$

22. Consider some appropriate values.

    e.g., 

    $4a = 16$ cm $\qquad 2(l + b) = 16$ cm
    but $a^2 = 16 \text{ cm}^2$ and $l \times b = 12 \text{ cm}^2$
    $\Rightarrow \quad a^2 > l \times b$

    Remember, when $a + b$ is constant, then the maximum value of $a \times b$ is at when $a = b$.
    So, area of square (i. e., $l \times b = a \times a$) is always greater than area of rectangle with same perimeter.
    Alternatively: Let each side of a square be a and length and breadth of the rectangle be l and b, then
    $$4a = 2(l + b) \Rightarrow a = \left(\frac{l + b}{2}\right)$$
    $\therefore$ Area of the rectangle $= l \times b$
    and Area of the square $= a^2 = \frac{1}{4}(l + b)^2$
    But since we know that for any (different) certain values
    AM > GM
    $$\therefore \quad \frac{l + b}{2} > \sqrt{l \times b}$$
    $$\Rightarrow \quad \left(\frac{l + b}{2}\right)^2 > l \times b$$
    $\Rightarrow$ Area of square > Area of rectangle

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

23. Best way is to go through options and verify the result.

Alternatively: $2(l+b)=4a=80$

$\Rightarrow \quad l+b=40$ and $a=20 \Rightarrow a^2=400$

Also $a^2-lb=100$

$\Rightarrow \quad 400-lb=100$

$\Rightarrow \quad lb=300$

Now $(l-b)^2=(l+b)^2-4lb$

$(l-b)^2=1600-1200$

$\Rightarrow \quad l-b=20$

(Solving $l+b=40$ and $l-b=20$)

$\therefore \quad l=30$ and $b=10$

24. $\frac{100\times100}{25\times25}=16 \quad (1\text{ m}=100\text{ cm})$

25. $(120+80-24)\times24=176\times24=4224$

26. $(90+40+10)\times10=1400$

27. $2[(x+2)+x]=48 \Rightarrow x=11$ cm

$\therefore \quad (x+2)=13$ cm

28. $\frac{52000}{325}=160$

3.25 cm means 325 m, as per scale.

29. Area $=\frac{d^2}{2}$; $d \rightarrow$ diagonal

30. $1.2\times1.2=1.44 \Rightarrow$ 44% increase

31. $a^2:(a\sqrt{2})^2=1:2$

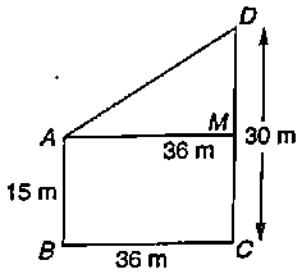
32. $1.6\times1.4=2.24 \Rightarrow$ increase $=1.24 \Rightarrow 124\%$

INTRODUCTORY EXERCISE 10.2

1. **Use Hero's formula:**

Area of scalene triangle $=\sqrt{s(s-a)(s-b)(s-c)}$

2.



$AM=BC=36$ m

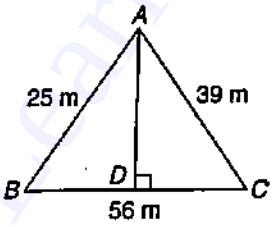
$MD=CD-MC=30-15 \quad (\because MC=AB=15\text{ m})$

$\therefore \quad MD=15$ m

Now using Pythagorus theorem

$AD^2=AM^2+MD^2$, find AD

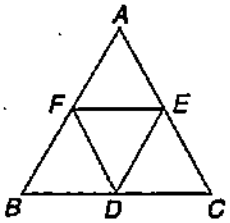
3. Find the area, using Hero's formula, then



$\text{Area}=\frac{1}{2}\times b\times h$

$=\frac{1}{2}\times56\times AD$

4. All the triangles are similar to each other.



$\therefore \quad \frac{\text{Area of }\Delta ABC}{\text{Area of }\Delta DEF}=\frac{4}{1}$

(Since in ΔABC there are 4 similar triangles having same area as ΔDEF.)

5. $60=\frac{b}{4}\sqrt{4a^2-b^2}$

$\Rightarrow \quad 60=\frac{b}{4}\sqrt{676-b^2}$

$\Rightarrow \quad 57600=b^2(676-b^2)$

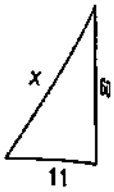
By solving the above equation we can get the value of b.

Alternatively: You can go through option.

6. Use pythagorus theorem

$x^2=(60)^2+(11)^2$

[Pythagorus theorem:
(Hypotenuse)^2 = (base)^2 + (height)^2



7. Area of field $=\frac{495.72}{36.72}$

$=13.5$ hectare $=135000$ m^2

Now, if $h=x, \quad b=3x$

$\therefore \quad \frac{1}{2}\times b\times h=135000$

8. Area of triangle $=\frac{1}{2}\times b\times h$

Let initially area of triangle $=1\times1=1$ unit

Now, the area of triangle $=2\times2=4$ unit

Increase in area $=\frac{4-1}{1}\times100=300\%$

(For your convenience assume any value of b and h.)

9. Height of an equilateral triangle $=\frac{\sqrt{3}}{2}\times$ side

$\therefore \quad 2\sqrt{3}=\frac{\sqrt{3}}{2}\times$ side

$\Rightarrow \quad$ Side $=4$ cm

$\therefore$ Area of an equilateral triangle $=\frac{\sqrt{3}}{4}\times(\text{side})^2$

$=\frac{\sqrt{3}}{4}\times4\times4=4\sqrt{3}$ cm^2

10. Let the each side of a square be $3x$ and each side of an equilateral triangle be $4x$

then perimeter of square $=4\times3x=12x$

and perimeter of equilateral triangle $=3\times4x=12x$

Now area of square $=(3x)^2=9x^2$

Area of equilateral triangle $=\frac{\sqrt{3}}{4}\times(4x)^2=4\sqrt{3}x^2$

Since $9x^2>4\sqrt{3}x^2 \Rightarrow 9>4\sqrt{3}$

Hence area of square is greater than area of equilateral triangle, when the perimeter of each is same.

Alternatively: Consider any suitable values and verify.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

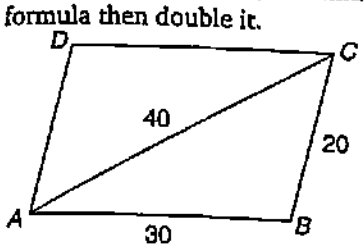
yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

INTRODUCTORY EXERCISE 10.3

1. Area $= 6 \times 8 \sin 30°$

$= 6 \times 8 \times \frac{1}{2} = 24 \text{ cm}^2$ $\left(\sin 30° = \frac{1}{2}\right)$

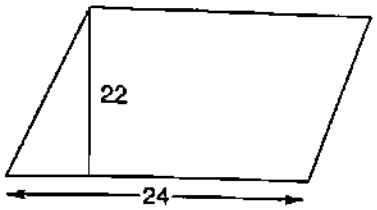
2. To find the area of parallelogram, first find the area of ΔABC by Hero's formula then double it.



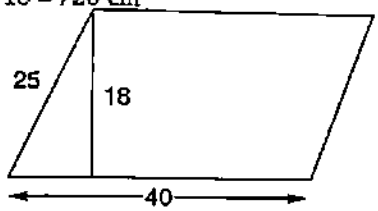
Since, area of $ABCD = 2 \times$ Area of ΔABC

$= 2 \times$ Area of ΔACD

3. Area $= 22 \times 24 = 528 \text{ cm}^2$



4. Area $= 40 \times 18 = 720 \text{ cm}^2$



5. $\frac{\text{Area of parallelogram } ABCD}{\text{Area of triangle } ABN} = \frac{BC \times AN}{\frac{1}{2} \times BN \times AN} = \frac{2 \times 4x}{x} = \frac{8}{1}$

6. Area of rhombus $= \frac{1}{2} \times$ product of diagonals

$= \frac{1}{2} \times a \times b = \frac{ab}{2}$

7. Area of rhombus $= \frac{1}{2} \times 2x \times 5x = \frac{10x^2}{2} = 5x^2$

and square of the shorter diagonal $= (2x)^2 = 4x^2$

$\therefore \quad \frac{5x^2}{4x^2} = \frac{5}{4}$

8. $\frac{1}{2} \times x \times 2x = 256$

$\Rightarrow \quad x^2 = 256$

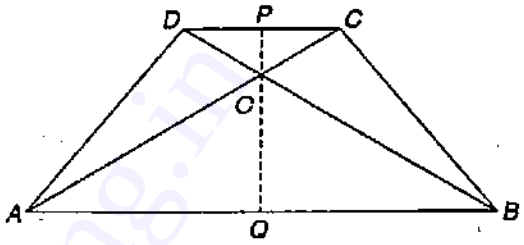
$\Rightarrow \quad x = 16$

$\therefore \quad 2x = 32$

Hence, $x + 2x = 16 + 32 = 48$

9. Area $= \frac{1}{2} \times (30 + 50) \times 16 = 640 \text{ cm}^2$

10. $\frac{\text{Area of } \Delta AOB}{\text{Area of } \Delta COD} = \frac{\frac{1}{2} \times AB \times OQ}{\frac{1}{2} \times CD \times PO}$



$= \frac{AB \times OQ}{CD \times PO}$

$= \frac{2CD \times 2OP}{CD \times OP} = \frac{4}{1}$ $\left(\because \begin{matrix} AB = 2CD \\ \text{and } OQ = 2PO \end{matrix}\right)$

This is due to the similarity of triangles AOB and COD.

11. Area of trapezium $= 441$

$\frac{1}{2}(5x + 9x) \times 21 = 441$

$\Rightarrow \quad 14x = 42 \Rightarrow x = 3$

$\therefore \quad 9x = 27 \text{ cm}$

12. $\frac{1}{2} \times (12 + 8) \times h = 360$

$\Rightarrow \quad h = 36 \text{ m}$

13. $6 \times \frac{\sqrt{3}}{4} \times (\text{Side})^2 = \frac{3\sqrt{3}}{2} (\text{Side})^2$

$= \frac{3\sqrt{3}}{2} \times 4 \times 4 = 24\sqrt{3} \text{ m}^2$

14. Area of quadrilateral $= \frac{1}{2} \times 19 \times (5 + 7) = 114 \text{ cm}^2$

15. $AB = BC = CD = AD = 4 \text{ cm}$ $\left(\because \text{Side} = \frac{\text{Diagonal}}{\sqrt{2}}\right)$

and $EF = 1.5 \text{ cm}$ (By Pythagorus theorem)

$\therefore$ Area of $ABCDE$ = Area of $ABCD$ + Area of AED

$= (4)^2 + \frac{1}{2} \times 4 \times 1.5 = 19 \text{ cm}^2$

INTRODUCTORY EXERCISE 10.4

1. $2\pi r = 704 \Rightarrow r = 112$

$\therefore \quad \pi r^2 = \frac{22}{7} \times 112 \times 112 = 39424 \text{ cm}^2$

2. $2\pi r = 4.4 \Rightarrow r = \frac{7}{10} \text{ m}$

$\therefore \quad \pi r^2 = 0.49\pi \text{ m}^2$

3. $2\pi r = 2 \times \frac{22}{7} \times 4.2$

$= 26.4 \text{ metre}$

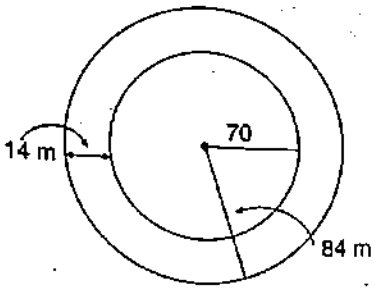
$26.4 = 2(6x + 5x)$

$\Rightarrow \quad 6x = 7.2 \text{ m}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

4. $2\pi r = 440 \Rightarrow r = 70$ m



$\therefore \quad R = 70 + 14 = 84$ m

5. $r + 2\pi r = 51 \Rightarrow r\left(1 + \frac{44}{7}\right) = 51 \Rightarrow r = 7,$

Find area.

6. $(2\pi r - 2r) = 15 \Rightarrow 2r\left(\frac{22}{7} - 1\right) = 15 \Rightarrow r = \frac{7}{2}$

$\therefore \quad \pi r^2 = \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} = 38.5 \text{ m}^2$

7. $\frac{\text{New area}}{\text{Original area}} = \frac{\pi \times 7 \times 7}{\pi \times 5 \times 5} = \frac{49}{25}$

Change in area $= \frac{24}{25} \times 100 = 96\%$

8. $\frac{\text{New circumference}}{\text{Original circumference}} = \frac{6}{5} = \frac{\text{New radius}}{\text{Original radius}}$

$\therefore \quad \frac{\text{New area}}{\text{Original area}} = \left(\frac{6}{5}\right)^2 = \frac{36}{25}$

$\therefore \quad$ Change in area $= \frac{11}{25} \times 100 = 44\%$

Alternatively: Change $= (1.2 \times 1.2) - (1 \times 1) = 0.44$

$\therefore \quad$ Change $= 44\%$

9. $\pi r^2 = 124.74$ hectare

$\pi r^2 = 1247400 \text{ m}^2$

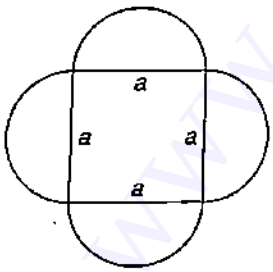
$\Rightarrow \quad r = 630$ m

$\therefore \quad 2\pi r = 3960$

$\therefore \quad$ Cost $= 3960 \times 0.8 = 3168$

10. $\pi r^2 = \frac{158400}{1400} \Rightarrow r^2 = 36 \Rightarrow r = 6$ m

11.



Area of square $= a^2$

Area of circular parts $= 4 \times \frac{1}{2} \pi \left(\frac{a}{2}\right)^2 = \frac{\pi a^2}{2}$

$\therefore \quad$ Total area $= a^2 + \frac{\pi a^2}{2} = a^2\left(1 + \frac{\pi}{2}\right)$

12. $\pi r^2 = 2464 \Rightarrow r = 28$ m

13. Basically there are 12 equilateral triangles each of side a.

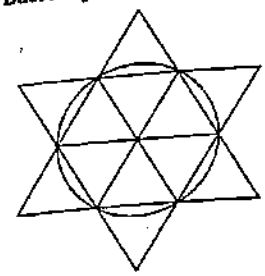


Fig. (i)

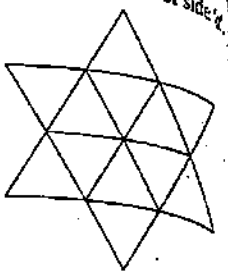


Fig. (ii)

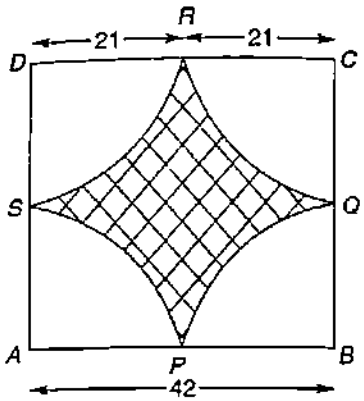
$\therefore \quad 12 \times \frac{\sqrt{3}}{4} \times (a)^2 = 3\sqrt{3}a^2$

14. $\frac{1}{2} \times \pi \times (7)^2 + 2\left[\frac{1}{2} \times \pi \times \left(\frac{7}{2}\right)^2\right]$

Area of larger semicircle + 2 (area of smaller semicircle)

15. $2\pi \times R \times \frac{75}{360} = 25 \Rightarrow R = \frac{60}{\pi}$

16. Ungrazed area = Area of square − 4 (area of quadrants)



$= (42)^2 - 4 \times \frac{1}{4} \times \pi (21)^2 = (21)^2 [4 - \pi] = 378$

17. $20 + \frac{1}{2} \times [2\pi \times 10] = 20 + 10\pi$

18. $\frac{40}{360} = \frac{1}{9} \quad \therefore \quad \frac{(360 - 40)}{360} = \frac{8}{9}$

$\therefore$ Area of major sector $= 8 \times 8.25 = 66 \text{ cm}^2$

19. Area of a sector $= \frac{1}{2} \times$ arc $\times$ radius $= \frac{1}{2} \times 8 \times 5.6 = 22.4 \text{ cm}^2$

20. $2\pi r = 2 \times \frac{22}{7} \times 700 = 4400 \text{ m} = 4.4 \text{ km}$

Time $= \frac{\text{Distance}}{\text{Speed}} = \frac{4.4}{13.2} = \frac{1}{3}$ h $= 20$ minute

21. $\pi R^2 = \pi [r_1^2 + r_2^2 + r_3^2]$, find R.

22. $\pi [23^2 - 12^2] = \frac{22}{7} \times [529 - 144] = 1210$

23. $4a = 4 \times 66 = 264 = 2\pi r \Rightarrow r = 42$

$\therefore \quad d = 2r = 84$ m

24. $r = 42 \quad \therefore 2\pi r = 264 = 4a \Rightarrow a = 66$

$\therefore \quad d = a\sqrt{2} = 66\sqrt{2}$

25. Circumference $= \frac{1100}{560} = \frac{110}{56} = 2\pi r$

$\therefore \quad 2r = \frac{110}{56} \times \frac{7}{22} = \frac{5}{8}$ m $= 62.5$ cm

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

INTRODUCTORY EXERCISE 10.5

1. Volume of original cube = $(4)^3 = 64$ cm^3
and its weight = 400 kg
Since weight of the larger cube is 8 times the weight of smaller cube. Hence, the volume of new cube will be 8 times the volume of smaller cube.
Hence volume of required cube = $8 \times 64 = (8)^3$
$\therefore$ Edge of this cube = 8 cm

2. Volume of the tank = 3 m^3
$\therefore$ Base area $\times$ height = 3 m^3
$\Rightarrow$ $\text{Base area} = \dfrac{3}{2.56} = 1.171875 \text{ m}^2$
[$\because$ Volume of cuboid = $(l \times b) \times h$ = (base area) $\times$ height]

3. Volume of cube = Volume of cuboid
$a^3 = lbh$
$\Rightarrow$ $a^3 = 36 \times 75 \times 80 = 216000$
$\Rightarrow$ $a = 60$ cm

4. Base area $\times$ height = Volume
$10 \times 4 \times 1 = 40$ m^3
But 1 m^3 = 1000 litre = 1 kilolitre
$\therefore$ 40 m^3 = 40,000 litre = 40 kilolitre

5. Volume of smaller (required) cube = 8 (dm)^3 = 0.008 m^3
$\therefore$ Number of required cubes $= \dfrac{\text{Volume of larger cube}}{\text{Volume of each smaller cube}}$
$= \dfrac{1}{0.008} = 125$

6. Volume of new cube = $(3)^3 + (4)^3 + (5)^3 = 216$ cm^3
$\therefore$ Each edge of the new cube = 6 cm
and hence, surface area = $6 \times (a)^2 = 216$ cm^2

7. Total length of tape = $2(l + b) + 3.75$
$= 2(39.5 + 9.35) + 3.75$
$= 101.45$ cm

8. Area of surface to be cemented = $2 \times (l + b) \times h + (l \times b)$
i.e., area of four walls + area of floor
$= 2 \times (21) \times 4 + (106.25)$
$= 274.25$ m^2
$\therefore$ Cost of cementing = 24×274.25 = Rs. 6582

9. Total volume of water displaced by 250 men
$= 250 \times 4 = 1000$ m^3
$\therefore$ Rise in water level $(h) = \dfrac{\text{Volume}}{\text{Base area}}$
$= \dfrac{1000}{80 \times 50} = 25$ cm

10. Let each edge of smaller cube = 1 m
$\therefore$ Each edge of larger cube = 2 m
and Surface area of smaller cube = $6 \times (1)^2 = 6$ m^2
$\therefore$ Surface area of larger cube = $6 \times (2)^2 = 24$ m^2
$\therefore$ % increase in surface area $= \dfrac{24 - 6}{6} \times 100 = 300\%$

NOTE It can be determined by using variables *e.g., x* (edge of cube) instead of solving by assuming some numerals.

Alternatively : $\dfrac{S_2}{S_1} = \left(\dfrac{e_2}{e_1}\right)^2 \Rightarrow \dfrac{S_2}{S_1} = \dfrac{4}{1}$
$\therefore$ Percentage increase in surface area $= \dfrac{4 - 1}{1} \times 100 = 300\%$
where S = surface area, e = edge of cube.

11. $\dfrac{V_2}{V_1} = \left(\dfrac{e_2}{e_1}\right)^3$ $\{\because V = (e)^3\}$ {e = edge of cube}
$\therefore$ $\dfrac{V_2}{V_1} = \left(\dfrac{2}{1}\right)^3 = \dfrac{8}{1}$
$\therefore$ $V_2 = 8V_1$

12. External volume of the box = $24 \times 16 \times 10 = 3840$ cm^3
Thickness of the wood = 5 mm = 0.5 cm
$\therefore$ Internal length of box = $24 - 2 \times 0.5 = 23$ cm
Internal breadth of box = $16 - 2 \times 0.5 = 15$ cm
Internal height of box = $10 - 2 \times 0.5 = 9$ cm
$\therefore$ Internal volume of the box = $23 \times 15 \times 9 = 3105$ cm^3
$\therefore$ Volume of the wood = $3840 - 3105 = 735$ cm^3
Now, total weight of wood = Volume $\times$ weight of 1 cm^3 wood
$7350 = 735 \times$ weight of 1 cm^3 wood
$\therefore$ Weight of 1 cm^3 wood = 10 gm

13. Length of tank = 120 cm
But since, $\dfrac{120}{7} = 17\dfrac{1}{7}$, hence 17 cubes can be placed along length and breadth of tank = 80 cm
But since, $\dfrac{80}{7} = 11\dfrac{3}{7}$, hence 11 cubes can be placed along breadth and height of tank = 50 cm.
But since, $\dfrac{50}{7} = 7\dfrac{1}{7}$, hence only 7 cubes can be placed along height of the tank.
$\therefore$ Total volume occupied by these cubes = $(7)^3 \times 17 \times 11 \times 7$
$= 448987$ cm^3
Total volume of the tank = $120 \times 80 \times 50 = 480000$ cm^3
$\therefore$ Area of unoccuppied space = $480000 - 448987$
$= 31013$ cm^3 = 31.013 dm^3

14. Surface area of the cuboid = $2(lb + bh + hl) = 11.6$ m^3
$\therefore$ Cost of canvas = 11.6×25 = Rs. 290

15. Required surface area = $2(9 \times 3 + 3 \times 3 + 3 \times 9) = 126$ cm^2

16. Area of 4 walls = $2(36 + 12) \times 10 = 960$ m^2
Total area of (windows + door + chimney) = 120 m^2
$\therefore$ Net area for papering = $960 - 120 = 840$ m^2
$\therefore$ Length of required paper $= \dfrac{840}{1.2} = 700$ m
Hence, cost of papering = 700×0.7 = Rs. 490

17. Number of children $\times$ Space required for one child
= Volume of room
$\therefore$ Number of children $= \dfrac{30 \times 12 \times 6}{8} = 270$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

18. $(x+2)^3 - x^3 = 1016$

$\Rightarrow$ $x = 12$ cm

and $x^3 - (x-2)^3 = (12)^2 - (10)^2 = 728$

19. You can see the figure shown in the question number 15. Now let us consider that surface area of each face of the cube 1 cm^3.

$\therefore$ Total surface area of the cuboid = 14 cm^2

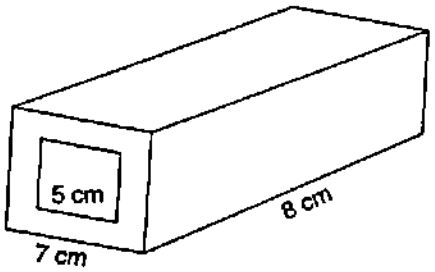
and Total surface area of the 3 cubes = 18 cm^2

Hence, required ratio = 14 : 18 = 7 : 9

NOTE In the arrangement of cuboid there are only 14 faces visible to us.

20. Iron used in the tube

= Difference in external and internal volumes of the tube



$\therefore$ $192 = 8x^2 - 8(5)^2$

$\Rightarrow$ $x = 7$ cm

Hence, the thickness of the tube $= \frac{7-5}{2} = 1$ cm

21. Base area of vessel $\times$ rise in water level = Volume of cube

$15 \times 12 \times h = 11 \times 11 \times 11$

$\Rightarrow$ $h = 7.39$ cm

22. (Initial volume of water + required volume of water + volume of cube)

= Base area of vessel $\times$ 10

$\therefore$ $25 \times 20 \times 5$ + required volume of water + 1000 $= 25 \times 20 \times 10$

$\Rightarrow$ Required volume of water = 1500 cm^3 = 1.5 litre

23. Net volume = $(10 \times 8 \times 2) - (2 \times 2 \times 2) = 152$ cm^3

Net surface area = $2(10 \times 8 + 8 \times 2 + 2 \times 10) + 4(2 \times 2) - 2(2 \times 2)$

= 240 cm^2

24. Available area for spreading the earth

$= (600 \times 200) - (24 \times 12)$

$= 119712$ m^2

Volume of the earth = 119712 $\times$ rise in level

$24 \times 12 \times 8 = 119712 \times h$

$\Rightarrow$ $h = \frac{2304}{119712}$

$= 0.01924$ m $= 1.92$ cm

25. Net volume of the wall

= Total volume – Volume taken away due to doors

$= (30 \times 0.3 \times 5) - 2(4 \times 2.5 \times 0.3) = 39$ m^3

Volume of the bricks $= 39 \times \frac{8}{9}$

(Since $\frac{1}{9}$ part is lime in the wall)

$\therefore$ Number of bricks $= \frac{39 \times 8}{9 \times 0.2 \times 0.16 \times 0.08}$

$= 13541.66 \approx 13600$

26. Volume of water which flow in 25 minutes

$= 25 \times 60 \times 0.05 \times 0.03 \times 16 = 36$ m^3

$\therefore$ Rise in water level $= \frac{36}{15 \times 12} = \frac{1}{5}$ m $= 0.2$ m

27. A tetrahedron is a pyramid with triangular base hence it has total $(3+1) = 4$ vertices.

(See the figure shown after exercise 10.6)

28. $a\sqrt{3} = 6\sqrt{3}$ $\Rightarrow$ $a = 6$ cm

29. $\sqrt{l^2 + b^2} = 12$ $\Rightarrow$ $l^2 + b^2 = 144$

and $\sqrt{l^2 + b^2 + h^2} = 15$

$\Rightarrow$ $l^2 + b^2 + h^2 = 225$

$\Rightarrow$ $h^2 = 81$ $\Rightarrow$ $h = 9$ m

30. $l + b + h = 12$ cm, $\sqrt{l^2 + b^2 + h^2} = 5\sqrt{2}$

$\Rightarrow$ $l^2 + b^2 + h^2 = 50$

Now, $(l+b+h)^2 = l^2 + b^2 + h^2 + 2(lb + bh + hl)$

$\Rightarrow$ $144 = 50 + 2(lb + bh + hl)$

$\Rightarrow$ $2(lb + bh + hl) = 94$ cm^2

31. $h : b = 3 : 1$ and $l : h = 8 : 1$

$\Rightarrow$ $l : h : b = 24 : 3 : 1$

$\therefore$ $24x \times 3x \times x = 36.864$

$\Rightarrow$ $x^3 = 0.512$ $\Rightarrow$ $x = 0.8$

$\therefore$ $h = 3x = 2.4$ m

32. $lb = x$, $bh = y$, $hl = z$

$\therefore$ $lb \times bh \times hl = xyz$

$\Rightarrow$ $(lbh)^2 = xyz$ $\Rightarrow$ $lbh = \sqrt{xyz}$

INTRODUCTORY EXERCISE 10.6

1. Required volume of water $= \frac{22}{7} \times (0.04)^2 \times 3500$

$= 17.6$ m^3

2. $h = \frac{10 \times 10 \times 10}{\pi \times 10 \times 10} = 3\frac{2}{11}$ cm

3. $h = \frac{511}{36.5} = 14$ m

4. $2\pi rh = 1056$ cm^2

$\Rightarrow$ $r = \frac{33}{\pi}$ cm

$\therefore$ $\pi r^2 h = \pi \times \left(\frac{33}{\pi}\right)^2 \times 16$

$\Rightarrow$ Volume = 5544 cm^3

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

5. Diameter of the cylinder = edge of the cube = 2 cm
Height of the cylinder = edge of the cube
$\therefore$ Required volume $= (2)^3 - \pi \times (1)^2 \times 2$
$= 8 - \left(\frac{44}{7}\right) = \frac{12}{7}$ cm^3

6. $h = \frac{2\pi rh}{2\pi r} = \frac{17.6}{8.8} = 2$ m
and $2\pi r = 8.8 \Rightarrow r = 1.4$ m
$\therefore$ $\pi r^2 h = \frac{22}{7} \times (1.4)^2 \times 2 = 12.32$ m^3

7. $2r = h \Rightarrow r = \frac{h}{2}$
$\therefore$ $2\pi r(h + r) = 2\pi \times \frac{h}{2}\left(h + \frac{h}{2}\right) = \frac{3}{2}\pi h^2$

8. $2\pi rh = 2 \times \pi \times 2 \times 10 = 40\pi$ m^2

9. $\frac{2\pi rh}{2\pi r(h + r)} = \frac{2}{3} \Rightarrow \frac{h}{h + r} = \frac{2}{3} \Rightarrow \frac{h}{r} = \frac{2}{1}$
$\therefore$ $2\pi r(h + r) = 924$
$\therefore$ $2\pi rh = 924 \times \frac{2}{3} = 616$ cm^2
$\Rightarrow$ $2 \times \pi \times x \times 2x = 616$
$\Rightarrow$ $x = 7$ $\therefore r = 7$ cm and $h = 14$ cm
$\therefore$ $\pi r^2 h = \frac{22}{7} \times (7)^2 \times 14 = 2156$ cm^3

10. $\frac{\pi \times r \times r \times h}{2 \times \pi \times r \times h} = \frac{269.5}{154}$
$\Rightarrow$ $r = 3.5$ cm
Now, $2\pi \times 3.5 \times h = 154$
$\Rightarrow$ $h = 7$ cm

11. $2\pi rh = 1320 \Rightarrow h = 10$ cm
$\therefore$ $2\pi r(h + r) = 2 \times \frac{22}{7} \times 21 \times 31 = 4092$ cm^2

12. $\frac{r}{h} = \frac{3x}{4x}$, $\pi r^2 h = 4851 \Rightarrow x = 3.5$
$\therefore$ $r = 10.5$ m and $h = 14$ m
$\therefore$ $2\pi rh = 2 \times \frac{22}{7} \times 10.5 \times 14 = 924$ m^2

13. $\frac{2\pi r(h + r)}{2\pi rh} = \frac{4}{1} \Rightarrow \frac{h + r}{h} = \frac{4}{1}$
$\Rightarrow$ $\frac{r}{h} = \frac{3}{1}$

Alternatively : Go through options.

14. $2\pi r = 33 \Rightarrow r = \frac{21}{4}$
$\therefore$ Volume $= \pi r^2 h = \frac{22}{7} \times \left(\frac{21}{4}\right)^2 \times 330$
$= 28586.25$ cm^3

15. $\frac{r_1}{r_2} = \frac{4x}{x}$, but $V_1 = V_2$
$\therefore$ $\pi (4x)^2 \times h_1 = \pi (x)^2 h_2$
$\Rightarrow$ $h_2 = 16h_1$

16. $2\pi r(h + r) = 2640$
$\Rightarrow$ $2\pi r(30) = 2640$
$\Rightarrow$ $r = 14$ m
$\Rightarrow$ $h = 16$ m $(\because r + h = 30$ m$)$
$\therefore$ $h : r = 8 : 7$

17. $\frac{V_1}{V_2} = \frac{\pi (r_1)^2 h_1}{\pi (r_2)^2 h_2} = \frac{3 \times 3 \times 4}{5 \times 5 \times 3} = \frac{12}{25}$

18. $\frac{V_1}{V_2} = \frac{1}{1}$
$\left(\frac{r_1}{r_2}\right)^2 \times \frac{h_1}{h_2} = \frac{1}{1} \Rightarrow \left(\frac{r_1}{r_2}\right)^2 \times \frac{3}{1} = \frac{1}{1}$
$\Rightarrow$ $\left(\frac{r_1}{r_2}\right)^2 = \frac{1}{3} \Rightarrow \frac{r_1}{r_2} = \frac{1}{\sqrt{3}}$

19. $\frac{2\pi r_1 h_1}{2\pi r_2 h_2} = \frac{3}{2} \times \frac{6}{7} = \frac{9}{7}$

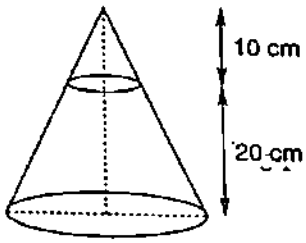
20. $h = 42$ cm, $2\pi R = 132$ cm
$\Rightarrow$ $R = 21$ cm
$\therefore$ $r = 21 - 3 = 18$ cm
$\therefore$ Required volume $= \pi (R^2 - r^2) h$
$= \frac{22}{7}[(21)^2 - (18)^2] \times 42$
$= 15444$ cm^3

21. Since radius and height of the cylinder are same as that of cone. Therefore cylinder can contain $15 \times 3 = 45$ litre of milk.

HINT Volume of cone $= \frac{1}{3}\pi r^2 h$ and volume of cylinder $= \pi r^2 h$.

22. Best way is to go through option.

Alternatively : $\frac{V_1}{V_2} = \frac{1}{27}$



$\frac{(r_1)^2 h_1}{(r_2)^2 h_2} = \frac{1}{27}$
$\Rightarrow$ $\frac{x_1^3}{x_2^3} = \frac{(1)^3}{(3)^3} \Rightarrow \frac{x_1}{x_2} = \frac{1}{3}$
$\Rightarrow$ $\frac{h_1}{h_2} = \frac{1}{3}$
$\therefore$ $h_1 = \frac{h_2}{3} = \frac{30}{3} = 10$ cm

Hence, cone is cut off at $(30 - 10 =)$ 20 cm above the base.

HINT For more clarification of the concept seek help from geometry. (Similarity of triangles)

23. Volume of conical tent $= \frac{1}{3} \times \frac{22}{7} \times \frac{13}{2} \times \frac{13}{2} \times 10.5$
$\therefore$ Average space per man $= \frac{\text{Volume of tent}}{8} = 58\frac{3}{32}$

24. $\frac{1}{3} \times \pi \times (5x)^2 \times (12x) = 314\frac{2}{7} = \frac{2200}{7}$
$\Rightarrow$ $x = 1$
$\therefore$ $r = 5$ and $h = 12$
$\therefore$ $l = 13$ m

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

25. $\frac{1}{3} \times \frac{22}{7} \times r^2 \times 24 = 1232$

$\Rightarrow \quad r = 7$ cm

$\therefore \quad \pi r l = \frac{22}{7} \times 7 \times 25 = 550$ cm^2 $\quad (\because l = \sqrt{r^2 + h^2})$

26. $2\pi r = 220 \Rightarrow r = 35$ cm

$\therefore \quad l = \sqrt{r^2 + h^2} = 91$ cm

$\therefore \quad \pi r l = \frac{22}{7} \times 35 \times 91 = 10010$ cm^2

27. Area of cloth $= \pi r l = \frac{22}{7} \times 14 \times 50 = 2200$ m^2

$\therefore \quad$ Length of cloth $= \frac{2200}{10} = 220$ m

28. $\pi r l = 23.10$

$\Rightarrow \quad r l = 2.1 \times 3.5$

Now, we know that $h = 2.8$. So we can assume the value of r from the given option.

at $\quad r = 2.1, \quad l = 3.5 \quad (\because l = \sqrt{r^2 + h^2})$

Alternatively : $r^2 (r^2 + h^2) = (2.1 \times 3.5)^2$

$\Rightarrow \quad r^4 + (2.8)^2 r^2 = (7.35)^2$

$\Rightarrow \quad r^4 + 7.84 r^2 = 54.0225$

$\Rightarrow \quad k^2 + 7.84k - 54.0225, \quad k = r^2$

Now, solve the above quadratic equation, if you can else substitute the value of r from the given choices.

29. $\frac{\pi r_1 l_1}{\pi r_2 l_2} = \frac{3}{2} \quad (\because r_1 = r_2)$

$\therefore \quad \frac{A_1}{A_2} = \frac{3}{2} \Rightarrow \frac{A_1}{300} = \frac{3}{2}$

$\therefore \quad A_1 = 450$ cm^2

30. $\frac{\text{Volume of cylinder}}{\text{Volume of cone}} = \frac{\pi r^2 h}{\left(\frac{1}{3}\right) \pi r^2 h} = \frac{1}{\left(\frac{1}{3}\right)} = \frac{3}{1}$

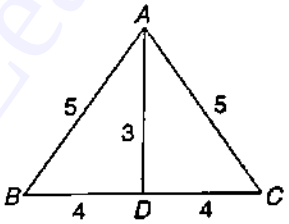
31. By Pythagorus theorem

$AD = 3$ cm

$\therefore$ Area of axial section

$= \frac{1}{2} \times 8 \times 3$

$= 12$ cm^2



32. $\frac{V_2}{V_1} = \frac{(1.4)^3}{(1)^3} = \frac{2.744}{1}$

$\therefore$ % increase in volume $= \left[\frac{2.744 - 1}{1}\right] \times 100 = 174.4\%$

33. Area of circular sheet $= 625\pi$

Since length of arc and area of sector are directly proportional to the central angle.

Therefore, length of remaining arc $= \frac{96}{100} \times 2 \times \pi \times 25 = 48\pi$

But the remaining arc is equal to the circumference of the base of circular cone.

$\therefore \quad 2\pi R = 48\pi \Rightarrow R = 24$ cm

Now, since the slant height of cone is equal to the radius of the original circular sheet.

Hence, $\quad l = 25$ cm

$h = 7$ cm $\quad (\because l = \sqrt{r^2 + h^2})$

$\therefore \quad \frac{\text{Radius}}{\text{Height}} = \frac{24}{7}$

34. $\frac{V_2}{V_1} = \frac{(r_2)^2 h_2}{(r_1)^2 h_1} = \frac{(2r)^2}{(r)^2} \quad (\because h_1 = h_2)$

$\Rightarrow \quad \frac{V_2}{V_1} = \frac{4}{1}$

35. Each edge of cube $= 10$ cm

Radius of base of cone $= 5$ cm

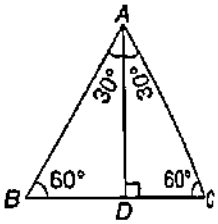
Height of cone $= 10$ cm

$\therefore$ Volume of cone $= \frac{1}{3} \times \frac{22}{7} \times 25 \times 10 = 261.9$ cm^2

36. $\frac{V_2}{V_1} = \frac{r_1^2 \times h_1}{r_2^2 \times h_2} = \frac{2 \times 2 \times 2}{1 \times 1 \times 1} = \frac{8}{1}$

37. $\frac{BD}{AB} = \cos 60°$

$\frac{BD}{AB} = \frac{1}{2}$



$BD = CD$, are the radii of the base and $AB = AC$ are the slant heights of the cone. A is the vertex and BC is the base.

38. Volume of cone $= \frac{1}{3} \pi \times 144 \times 35$

Volume of water flowing per second $= \pi \times (0.8)^2 \times \frac{500}{60}$

$\therefore \quad$ Required time $= \frac{\left(\frac{\pi}{3}\right) \times 144 \times 35}{\pi \times 0.64 \times \left(\frac{500}{60}\right)}$

$= 315$ seconds

39. Use the given formula :

$$\text{Volume} = \frac{\pi}{3} h (r^2 + Rr + R^2)$$

where $\pi = \frac{22}{7}, \quad h = 6, \quad R = 4 \quad$ and $\quad r = 2$

40. $\frac{1}{3} \pi \times (10)^2 \times 72 = \pi \times (30)^2 \times h$

$\Rightarrow \quad h = \frac{8}{3} = 2\frac{2}{3}$ cm

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

INTRODUCTORY EXERCISE 10.7

1. $\frac{4}{3}\pi(r_1^3 + r_2^3 + r_3^3) = \frac{4}{3}\pi(6)^3$

$\Rightarrow \quad 27 + 64 + r_3^3 = 216$

$\Rightarrow \quad r_3^3 = 125$

$\Rightarrow \quad r_3 = 5\text{ cm}$

2. $\pi r^2 \times 144 = \frac{4}{3}\pi \times (12)^3$ (∵ Volume remains constant)

$\Rightarrow \quad r = 4\text{ cm}$

3. Number of bottles × Volume of each bottle = Volume of hemisphere

$n \times \pi \times (3)^2 \times 1 = \frac{2}{3}\pi \times (6)^3$

$\Rightarrow \quad n = 16$

4. $\frac{1}{3}\pi \times (r)^2 \times 14 = \frac{2}{3}\pi \times (7)^3$

5. Volume of the spherical shell $= \frac{4}{3}\pi(R^3 - r^3)$

$= \frac{4}{3}\pi(7^3 - 5^3)$

$= \frac{872}{3}\pi$

6. Since volume is constant

$\therefore \quad n \times \frac{4}{3}\pi(1)^3 = \frac{4}{3}\pi \times (8)^3$

$\Rightarrow \quad n = 512$

7. Volume of hemisphere $= \frac{2}{3}\pi r^3$

$= \frac{2}{3} \times \frac{22}{7} \times (21)^3 = 19404\text{ m}^3$

8. Change in height (or level) of water $= \frac{\text{Volume of sphere}}{\text{Base area of cylinder}}$

$= \frac{\frac{4}{3}\pi \times (9)^3}{\pi \times (12)^2} = \frac{27}{4}\text{ cm}$

9. Volume of cone = Volume of sphere

$\frac{1}{3}\pi r^2 \times \frac{7}{2} = \frac{4}{3}\pi(7)^3$

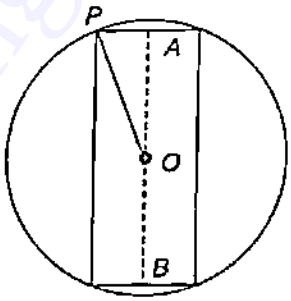
$\Rightarrow \quad r = 14\sqrt{2}\text{ cm}$

10. **Note:** The exact value of the required surface area cannot be found using the general formula. For the exact value of total surface area we have to solve it with the help of calculus. In the following solution we are calculating lumpsum value of total surface area.

Now, $OA = \sqrt{(OP)^2 - (AP)^2}$

$= \sqrt{(15)^2 - (9)^2} = 12\text{ cm}$

$\therefore \quad AB = 2 \times OA = 24\text{ cm}$



$\therefore$ Height of cylinder = 24 cm

$\therefore$ Required area = Surface area of sphere − 2 (area of each end of cylinder) + curved surface area of cylinder

$= 900\pi - 2(81\pi) + 432\pi = 1170\pi\text{ cm}^2$

11. Volume of pyramid $= \frac{1}{3} \times$ base area × height

$= \frac{1}{3} \times 25 \times 12 = 100\text{ cm}^3$

12. Volume of prism = Base area × height

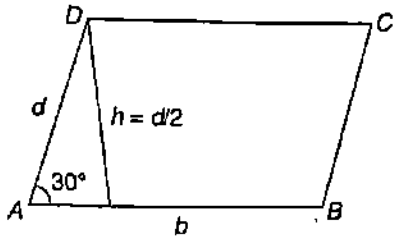
Since base area is constant and height is being halved therefore volume will also be halved. Hence, its volume will be reduced by 50%.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

LEVEL (1)

1. With the given perimeter the area of parallelogram will be maximum when it will be a rhombus.
Hence (a) is correct.
Alternatively :



$$\text{Perimeter} = 2d + 2b = 4h + 2b = 64 \text{ cm}$$

$$\Rightarrow \quad b = \frac{64 - 4h}{2}$$

$$\therefore \quad \text{Area} = b \times h = \frac{64h - 4h^2}{2} = f(A)$$

$$\therefore f'(A) = \frac{1}{2}(64 - 8h) \text{ (Refer the differentiation)}$$

For the maximum area $= f'(A) = 0$

$$\therefore \quad \frac{1}{2}(64 - 8h) = 0$$

$$\Rightarrow \quad h = 8 \text{ cm} \quad \text{and} \quad d = 16 \text{ cm}$$

$$\therefore \quad b = 16 \text{ cm}$$

2. Volume of water displaced = volume of sphere

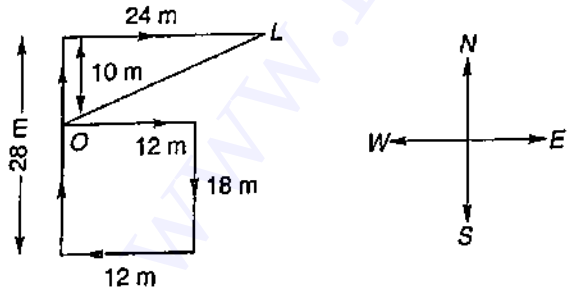
$$\pi \times (40)^2 \times h = \frac{4}{3}\pi \times (30)^3$$

$$\Rightarrow \quad h = \frac{90}{4} = 22.5 \text{ cm}$$

Thus, the level of water rises by 22.5 cm.

NOTE The volume of water will be calculated by considering it in the cylindrical shape since the water takes the shape of vessel in which it is filled.

3. O is the starting point and L is the end point.

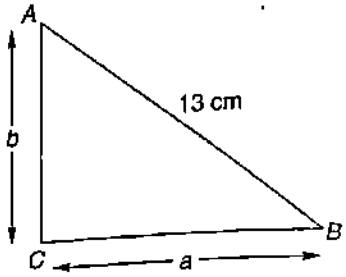


$$(OL)^2 = (24)^2 + (10)^2$$

$$OL^2 = 676$$

$$\Rightarrow \quad OL = 26 \text{ m}$$

4.



$$a^2 + b^2 = 13^2 = 169$$

and $$\frac{a \times b}{2} = 30$$

Now $$(a + b)^2 = a^2 + b^2 + 2ab$$

$$(a + b)^2 = 169 + 120$$

$$\Rightarrow \quad a + b = 17 \quad \ldots(i)$$

Again $$(a - b)^2 = (a + b)^2 - 4ab$$

$$(a - b)^2 = 289 - 240$$

$$\Rightarrow \quad (a - b) = 7 \quad \ldots(ii)$$

$\therefore$ From Eq. (i) and (ii), we get

$$a = 12 \text{ cm} \quad \text{and} \quad b = 5 \text{ cm}$$

Now, $$\text{inradius} = \frac{(\text{Altitude} + \text{Base} - \text{Hypotenuse})}{2}$$

$$= \frac{(12 + 5 - 13)}{2} = 2 \text{ cm}$$

Alternatively: $$\text{Inradius } (r) = \frac{\text{Area}}{\text{Semiperimeter}}$$

$$= \frac{30}{15} = 2 \text{ cm}$$

$$\left(\text{Semiperimeter} = \frac{(12 + 5 + 13)}{2} = 15\right)$$

NOTE If you are well aware with Pythagorian triplets then it is very easy to find the sides (*i.e.*, altitude and base) of the right angle triangle.

5. General phenomenon: A convex polygon in which there is maximum number of sides, it has the greatest enclosed area when the perimeter of the polygon is constant. Remember that in a circle there are infinite sides of minimum possible length. So, the area of circle will be maximum.
Alternatively: Perimeters of hexagon and circumference of circle are same *i. e.*,

$$6a = 2\pi r \quad \Rightarrow \quad 21a = 22r$$

where a is the side of hexagon and r is the radius of circle.

then $$\text{Area of circle} = \pi r^2$$

and $$\text{Area of hexagon} = 6 \times \frac{\sqrt{3}}{4} \times a^2$$

$$= \frac{3\sqrt{3}}{4} \times \left(\frac{22r}{21}\right)^2$$

$$= \frac{3\sqrt{3}}{4} \times \frac{22}{7 \times 3} \times \frac{22}{7 \times 3} \times r^2$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$$= \frac{\sqrt{3}}{4} \times \frac{22}{21} \pi r^2 = 0.45\pi r^2$$

Therefore, area of circle πr^2 is greater than area of hexagon.
Similarly you can compare other figures.

Alternatively: Consider some suitable values. Let us assume perimeter of square and hexagon is 24 cm, then each side of square = 6 cm.

and Area = 36 cm^2

Similarly, each side of hexagon = 4 cm

and area of hexagon $= \frac{3\sqrt{3}}{2} \times (4)^2 = 24\sqrt{3}$ cm^2

Thus, we can say that for a particular (or constant) value of perimeter, area of hexagon is greater than that of square.

Similarly each side of octagon = 3 cm $\left(= \frac{24}{8}\right)$

$\therefore$ Area of octagon $= 2a^2(1 + \sqrt{2})$
$= 2 \times 9 (1 + \sqrt{2})$
≈ 43.5 cm^2

Thus, the area of octagon is greater then the area of hexagon.

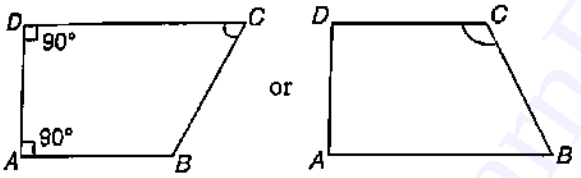
Similarly, $2\pi r = 24 \Rightarrow r = \frac{12}{\pi}$ cm

$\therefore$ Area of circle $= \pi \times r^2$

$$= \pi \times \frac{12}{\pi} \times \frac{12}{\pi} = 45.81 \text{ cm}^2$$

Thus, we can say that ultimately the area of the circle is greatest.

6.



Since sum of all the angles of a quadrilateral is 360°.

Therefore, $\angle A + \angle B + \angle C + \angle D = 360°$

$\Rightarrow$ $\angle C + \angle D = 180°$

$\Rightarrow$ $\angle D = 180° - \angle C$

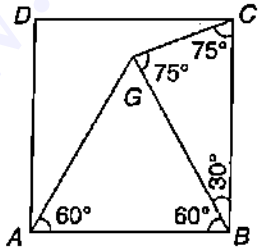
7. $\angle ABC = 60°$

$\therefore$ $\angle GBC = 90° - 60° = 30°$

Again $\angle BGC + \angle BCG = 180° - 30°$
$= 150°$

Now, since $\angle BGC = \angle BCG$

$\therefore$ $\angle BGC = 75°$



NOTE In a triangle when two sides are equal, then the two angles opposite to these sides are also equal.

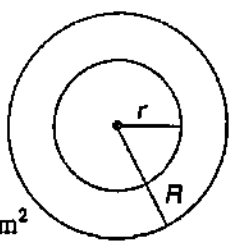
8. $\frac{r}{R} = \frac{3x}{4x}$

$\because$ $4x - 3x = 2$

$\Rightarrow$ $x = 2$

$\therefore$ Outer radius = 8 cm

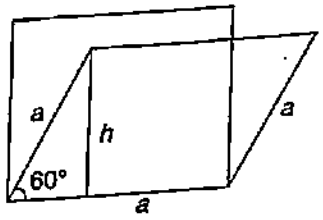
$\therefore$ Area of outer circle $= \pi \times (8)^2 = 64\pi$ cm^2



9. $\frac{h}{a} = \sin 60°$

$\frac{h}{a} = \frac{\sqrt{3}}{2}$

$\Rightarrow$ $h = \frac{a\sqrt{3}}{2}$

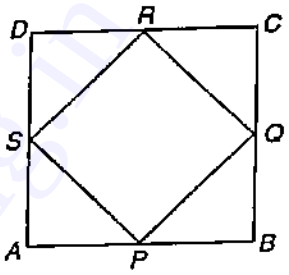


$\therefore$ Area of rhombus $= a \times h = \frac{a \times \sqrt{3}a}{2}$

and Area of square $= a^2$

$\therefore$ Required ratio $= \frac{a^2}{\sqrt{3}a^2} \times 2 = \frac{2}{\sqrt{3}} \times \left(\frac{\sqrt{3}}{\sqrt{3}}\right) = \frac{2\sqrt{3}}{3}$

10.



$ABCD$ is a square,

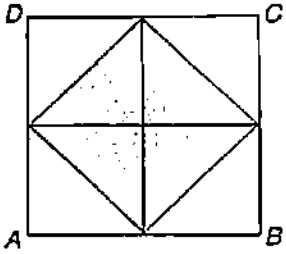
$AB = BC = CD = AD = 20$ cm

and P, Q, R and S are the mid-points.

$\therefore$ $PS = SR = RQ = PQ = 10\sqrt{2}$ cm ($\because AP = BP$ etc $= 10$ cm)

$\therefore$ Area of square $PQRS = (10\sqrt{2})^2 = 200$ cm^2

Alternatively: We can see that there are total 8 equal parts (or triangles) out of 8 we have taken only 4 parts *i.e.*, 50% area has been cut out.
So, the remaining area = 50% of 400 = 200 cm^2.



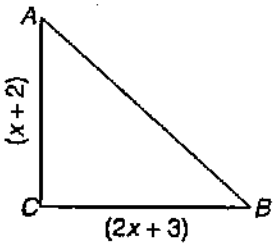
11. Area of right angle triangle

$$= \frac{(x + 2) \times (2x + 3)}{2} = 60$$

$\Rightarrow$ $2x^2 + 7x + 6 = 120$

$\Rightarrow$ $2x^2 + 7x - 114 = 0$

Solving the above quadratic equation, we get



$x = 6$

$\therefore$ $x + 2 = 8$ cm

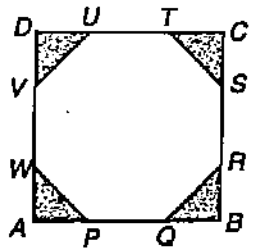
and $2x + 3 = 15$ cm

$\therefore$ Hypotenuse $AB = \sqrt{(8)^2 + (15)^2} = 17$ cm

(Using Pythagorus theorem)

12. Let us assume $PQRSTUVW$ is a regular octagon with each side a unit. Again, if we produce all the sides of octagon in both sides we get a square. Since each of the side of octagon is 'a' then

$UV = WP = QR = ST = a$



For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$\therefore \quad DU = DV \text{ (etc.)} = \frac{a}{\sqrt{2}}$

$\therefore \quad DU + UT + TC = \frac{a}{\sqrt{2}} + a + \frac{a}{\sqrt{2}}$

$= a(1 + \sqrt{2})$

$\therefore \quad$ Area of square $= a^2 (1 + \sqrt{2})^2$

$= a^2 (3 + 2\sqrt{2})$ sq unit

Again, area of each shaded portion (i.e., an isosceles right angle triangle)

$= \frac{1}{2} \times \frac{a}{\sqrt{2}} \times \frac{a}{\sqrt{2}} = \frac{a^2}{4}$

$\therefore$ Total area of all the shaded region $= 4 \times \frac{a^2}{4} = a^2$

$\therefore$ Area of octagon = Area of square $ABCD$ – Area of total shaded region

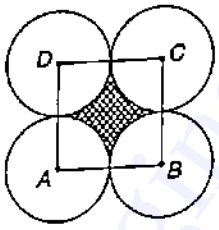
$= a^2 (3 + 2\sqrt{2}) - a^2$

$= 2a^2 (1 + \sqrt{2})$ sq. unit

13. Let the each side of the square be 2 cm, then

area of square = 4 cm^2 and

area of 4 quadrants of the four circles (i.e., unshaded part inside the square)



$= 4 \times \frac{1}{4} \pi \times (1)^2 = \pi$ cm^2

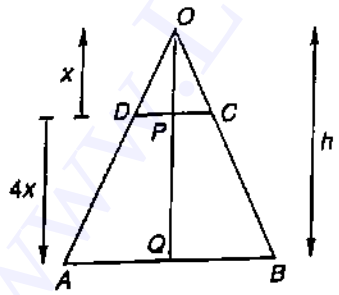
$\therefore \quad$ Area of shaded region $= (4 - \pi)$ cm^2

Therefore, the required ratio $= \frac{4 - \pi}{\pi} = \frac{6/7}{22/7} = \frac{3}{11}$

14. From the concept of mid point theorem, it is the average of the length of the parallel sides.

15. From the concept of similarity of triangles

$\frac{OP}{OQ} = \frac{1}{5} = \frac{PC}{BQ}$



Since, the ratio in radii of the two cones is 1 : 5. Therefore, the radius of smaller cone ODC is $\frac{21}{5} = 4.2$ cm.

Alternatively: Solve it in the proper way, which is actually a very tedious process. Still you have to apply the concept of similarity of triangles, which you will study in Geometry.

16. $l \times b = 2(l + b)$

$\Rightarrow \quad l = \frac{2b}{(b-2)} = \frac{2(b-2)+4}{b-2} = 2 + \frac{4}{(b-2)}$

Since, l is an integer, so 4 must be divisible by $(b - 2)$. Thus, l can be 4 or 6 or 3.

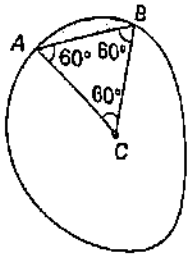
Therefore, if $b = 4$, $l = 4$, it will be a square. If $b = 6$, $l = 3$ and if $b = 3$, $l = 6$.

Hence, $l = 6$ and $b = 3$

$\therefore \quad l - b = 3$

17. Area of $\Delta ABC = \frac{\sqrt{3}}{4} \times (14)^2$

$= 49\sqrt{3}$ cm^2



Area of sector $ABC = \pi \times (14)^2 \times \frac{60}{360}$

$= 102\frac{2}{3}$ cm^2

$\therefore$ Area of the shaded region $= \left(102\frac{2}{3} - 49\sqrt{3}\right)$ cm^2

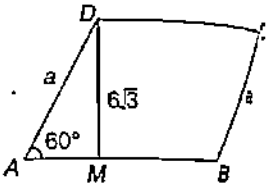
18. $l_1 \times b_1 = l_2 \times b_2$

New area of first rectangle $= \frac{5}{4} l_1 \times \frac{3}{4} b_1 = \frac{15}{16} l_1 b_1$

New area of the second rectangle $= \frac{3}{4} l_2 \times \frac{5}{4} b_2 = \frac{15}{16} l_2 b_2$

Hence, areas of both the new rectangles are same.

19. $\frac{DM}{AD} = \sin 60°$



$\frac{6\sqrt{3}}{AD} = \frac{\sqrt{3}}{2}$

$\Rightarrow \quad AD = 12$ cm

20. $\frac{4}{3} \pi r^3 = \frac{1}{3} \pi r^2 h$

(Since radii of sphere and cone are same)

$\Rightarrow \quad 4r = h$

$\therefore \quad \frac{h}{r} = \frac{4}{1} \Rightarrow h : r = 4 : 1$

21. Area of the shaded region $= \left[\frac{1}{2} \pi (22)^2 - 2\left(\frac{1}{2} \pi \times (11)^2\right)\right]$

$= \frac{1}{2} \pi \times (11)^2 [4 - 2] = 121\pi$ cm^2

22. Area of path $= (l + b - w) w$

$= (60 + 40 - 5) 5 = 475$ m^2

Cost = Area × rate

$= 475 \times 0.8 =$ Rs. 380

23. Go through options. Let the length and breadth of the second rectangle is l and b respectively, then the area of second field $= l \times b$.

Now consider option (d) i.e., $x = 80$, then

area of first field $= \frac{1}{5} l \times 5b = lb$

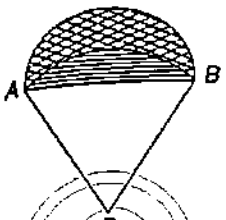
Hence, area of first field is same as that of second field, hence the presumed option is correct.

24. Area of quadrant $= \frac{1}{4} \pi a^2$

Area of triangle $ACB = \frac{a^2}{2}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in



→ Area of required region

→ Area of segment

$\therefore$ Area of segment $= \frac{\pi a^2}{4} - \frac{a^2}{2} = \frac{a^2}{4}[\pi - 2]$

Area of semi-circle $= \frac{1}{2}\pi\left(\frac{a\sqrt{2}}{2}\right)^2 = \frac{\pi a^2}{4}$

$\therefore$ Area of required region $= \frac{\pi a^2}{4} - \frac{a^2}{4}[\pi - 2]$

$= \frac{a^2}{2}$ sq. unit

25. Wrongly calculated area $= 1.05 \times 0.92 = 0.966 = 96.6\%$

$\therefore$ % error $= 100 - 96.6 = 3.4\%$

Alternatively: Actual area $= l \times b$

Wrongly calculated area $= 1.05l \times 0.92b = 0.966lb$

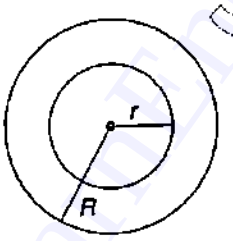
Deficit in area $= lb - 0.966lb = 0.034lb$

% error in area $= \frac{0.034lb}{lb} \times 100 = 3.4\%$

26. Volume of water = Area of cross-section × Length of pool

$= \frac{(10+6)}{2} \times 6 \times 120 = 5760 \text{ m}^3$

27. Area of path $= \frac{22176}{1} = 22176 \text{ m}^2$



$\pi(R^2 - r^2) = 22176$

$(R^2 - r^2) = \frac{22176}{22} \times 7$

$R^2 - (112)^2 = 7056$

$\Rightarrow \quad R^2 = 19600 \Rightarrow R = 140 \text{ m}$

$\therefore$ Width of the path $= 140 - 112 = 28$ m

28. From the figure itself it is clear that there are total 9 equilateral (congruent) triangles. Out of 9 triangles, 3 triangles are cut out.

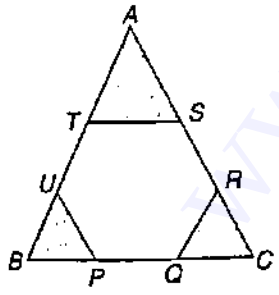


Fig. (i)

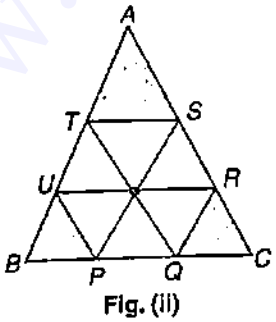
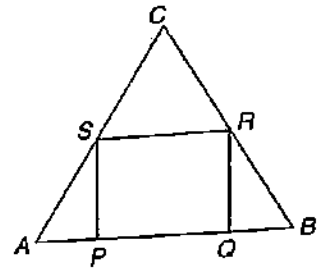


Fig. (ii)

It means $\frac{1}{3}$ (*i.e.*, 33.33%) area has been removed.

29.



$\frac{SP}{AS} = \sin 60° = \frac{\sqrt{3}}{2}$

$\Rightarrow \quad SP = \frac{\sqrt{3}}{2} AS$

Also, $SP = SR = CS = CR$

(since, $SR \parallel AB \therefore \angle CSR = \angle SRC = \angle RCS = 60°$)

$\therefore \quad CS = \frac{\sqrt{3}}{2} AS$

$\therefore \quad \frac{CS}{AS} = \frac{\sqrt{3}}{2} = \frac{CR}{RB}$

30. Let the each side of a square be a then its area will be a^2. Therefore, area of circle will also be a^2.

Again since the perimeter of square and equilateral triangle is same then, the each side of equilateral triangle is $\frac{4a}{3}$.

$\therefore$ Area of equilateral triangle $= \frac{\sqrt{3}}{4}\left(\frac{4a}{3}\right)^2$

$= \frac{4\sqrt{3}}{9} a^2$

Required ratio of area $= \frac{a^2 \times 9}{4\sqrt{3} \times a^2} = 9 : 4\sqrt{3}$

31. *ABCD* is a square *O* is the point of intersection of diagonals. *P*, *Q*, *R* and *S* are the mid-points on the sides *AB*, *BC*, *CD*, *DA* respectively.

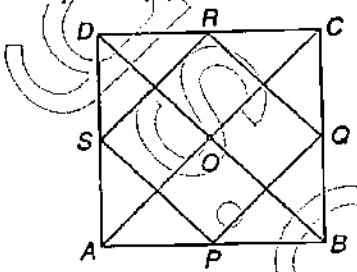


Fig. (i)

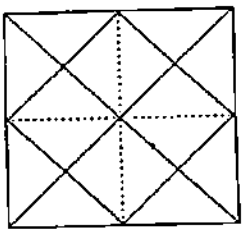


Fig. (ii)

In the above figure you can see that there are total 16 congruent isosceles right angled triangles.

In figure (iii) you can see that *OMPN* is a square of maximum possible area which is made up of 2 isosceles right angled triangles *OMP* and *ONP*. Thus, there are 4 smaller squares around *O* and thus the total area of these 4 squares is half of the larger square. Hence, the required area $= 800 \text{ cm}^2$.

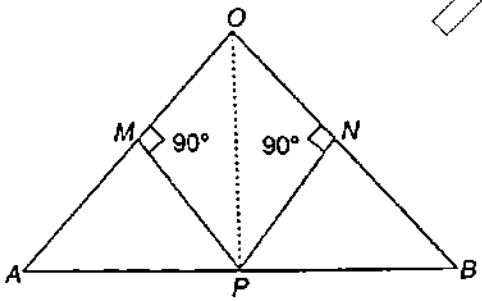


Fig. (iii)

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

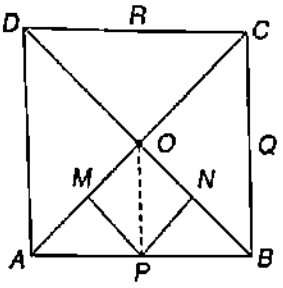
For More Visit : www.LearnEngineering.in

Alternatively: $ABCD$ is a square

$AB = 40$ cm

$OA = OB = 20\sqrt{2}$ cm

and $OP = 20$ cm



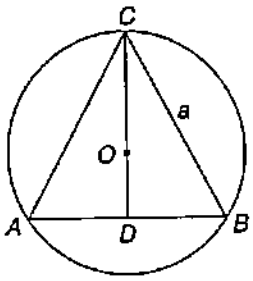
Square $OMPN$ has maximum area inside the triangle and OP is the diagonal of square $OMPN$.

$$\therefore \quad \text{Area of } OMPN = \frac{(20)^2}{2} = 200 \text{ cm}^2$$

Since, there are total 4 similar squares in each of the 4 triangles AOB, BOC, COD, DOA inside the larger square. So, the total required area of 4 smaller squares

$$= 4 \times 200 = 800 \text{ cm}^2$$

32. Let the each side of equilateral triangle be 'a' then circumradius of the circle $= \frac{a}{\sqrt{3}}$



$\therefore$ Area of circum circle

$$= \pi r^2 = \pi \times \left(\frac{a}{\sqrt{3}}\right)^2 = \frac{\pi}{3}a^2$$

Area of square whose side is equal to that of equilateral triangle $= a^2$.

$$\therefore \quad \text{Required ratio} = \frac{\frac{\pi}{3}a^2}{a^2} = \frac{\pi}{3}$$

NOTE A circumcircle always passes through the vertices of the inscribed figure (say triangle).

33. Best way is to go through option. Given that height of room $= 10$ m.

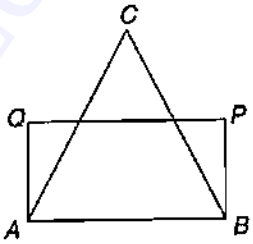
Volume of room $= 25 \times 400 = 10000 \text{ m}^3$

and Surface area of walls $= 2h(l + b) = 1300 \text{ m}^2$

Now, consider option (c) and verify it.

where $l = 40$ and $b = 25$

34. Let the each side of equilateral triangle be a then its perimeter $= 3a$



Again, $2(l + b) = 3a$

$$\Rightarrow \quad l + b = \frac{3}{2}a$$

$$\Rightarrow \quad a + b = \frac{3a}{2}$$

$AB = l = a$ (for the rectangle)

$$\Rightarrow \quad b = \frac{a}{2}$$

$$\therefore \quad \text{Area of rectangle} = a \times \frac{a}{2} = \frac{a^2}{2}$$

and $$\text{Area of triangle} = \frac{\sqrt{3}}{4} \times a^2$$

$$\therefore \quad \text{Required ratio} = \frac{a^2/2}{\sqrt{3}a^2/4} = \frac{2}{\sqrt{3}}$$

35. Since, b, l and $2(l + b)$ are in GP, therefore

$$\frac{l}{b} = \frac{2(l + b)}{l}$$

Suppose $\frac{l}{b} = x$,

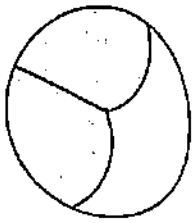
then $$x = 2\left(1 + \frac{1}{x}\right) \Rightarrow x^2 - 2x - 2 = 0$$

$$\Rightarrow \quad x = \sqrt{3} + 1 = \frac{l}{b}$$

36. Non-polished area $= 4\pi(r)^2$

Polished area $= 4 \times \left(2 \times \frac{\pi r^2}{2}\right)$

$= 4\pi r^2$



In the adjoining one of the four parts of the sphere is shown (To understand it properly, take an apple and cut it in the four parts one across horizontal and another cut make vertical to it then you will notice that in a piece there are 2 semicircles.) Therefore, required ratio $= 1 : 1$.

37. Number of cubes $= \frac{85 \times 17 \times 5.1}{1.7 \times 1.7 \times 1.7}$

$= 1500$ (1.7 is the HCF of 85, 17 and 5.1)

Area of each cube $= 6 \times (1.7)^2$

$\therefore$ Area of all the 1500 cubes $= 1500 \times 6 \times (1.7)^2$

$= 26010 \text{ cm}^2$

38. Area of large cube $= 6 \times (5)^2 = 150 \text{ (unit)}^2$

Area of cuboid $= 2(1 \times 1 + 1 \times 125 + 125 \times 1)$

$= 502 \text{ (unit)}^2$

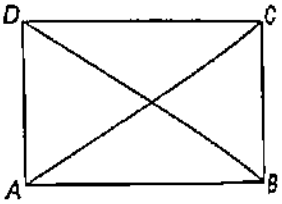
Therefore, increase in surface area $= \frac{(502 - 150)}{150} \times 100$

$= 234\frac{2}{3}\%$

39. $AO = BO = 13$ cm

$\Rightarrow$ $AC = BD = 26$ cm ($\because O$ is the point of bisector)

Now, since the diagonals are equal, it means the given figure is actually a rectangle.



$\therefore$ $BC^2 = AC^2 - AB^2$

$BC^2 = (26)^2 - (24)^2$

$BC^2 = 100$

$\Rightarrow$ $BC = 10$ cm

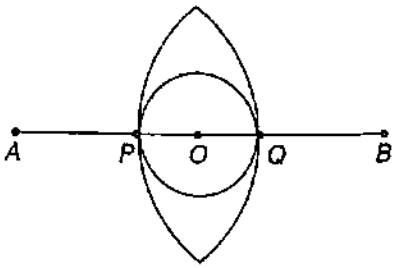
40. $AB = 16$ cm

$AQ = 10$ cm

and $AO = \frac{AB}{2} = 8$ cm

$\therefore$ $OQ = AQ - AO$

$= 10 - 8 = 2$ cm



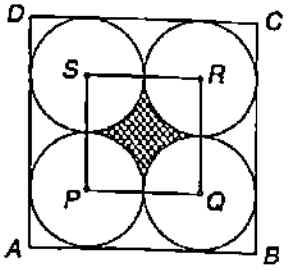
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

OQ ($= OP$) is the radius of smaller enclosed circle between two arcs.

$\therefore$ Area of circle with centre O is $\pi \times (2)^2 = 4\pi$ cm^2.

41. Area of the shaded region

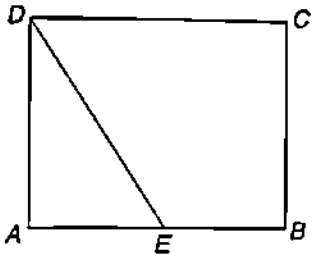


= (Area of square $PQRS$ – 4 × area of each quadrant of circles)

$$= \left[(2)^2 - 4 \times \frac{1}{4}\pi \times (1)^2\right]$$

$$= (4 - \pi) \text{ cm}^2$$

42. Let each side of the square be a, then



$$AE = AB - BE = (a - 17)$$

$\therefore$ Area of triangle $= \frac{1}{2} \times AD \times AE$

$$= \frac{1}{2} a \times (a - 17) = 84$$

$\Rightarrow$ $a^2 - 17a - 168 = 0$

$\Rightarrow$ $a = 24$ cm

$\therefore$ Area of square $= (24)^2 = 576$ cm^2

43. The solid with the least number of sides will have maximum surface area. So, tetrahedron will have maximum surface area. Notice that in a sphere there are infinite number of sides with least possible length. So, the surface area of sphere will be least.

44. $$\frac{l}{b} = \frac{(l + b)}{l}$$

$\Rightarrow$ $l^2 = b(l + b) = lb + b^2$

$\Rightarrow$ $l^2 - b^2 = lb$ ...(A)

$\Rightarrow$ $(l + b)(l - b) = lb$ ...(B)

and $$\frac{(l + b)}{l} = \frac{b}{(l - b)}$$ ...(C)

Therefore, statements (ii) and (iii) are true from equation (A)

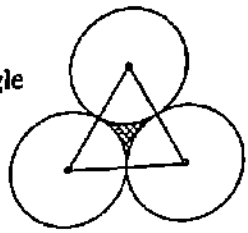
$$\frac{l^2}{b^2} = \frac{bl + b^2}{b^2}$$

$\Rightarrow$ $$\frac{l^2}{b^2} = 1 + \frac{l}{b}$$

$\Rightarrow$ $$\frac{l}{b} = \frac{l^2}{b^2} - 1$$

Hence, statement (i) is wrong.

45. Area of triangle $= \frac{\sqrt{3}}{4} \times (2)^2 = \sqrt{3}$ cm^2



Area of 3 circles enclosed by the triangle

$$= 3 \times \pi \times (1)^2 \times \frac{60}{360} = \frac{\pi}{2} \text{ cm}^2$$

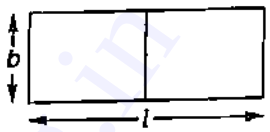
$\therefore$ Area of shaded region

$$= \sqrt{3} - \frac{\pi}{2} = \frac{(2\sqrt{3} - \pi)}{2} \text{ cm}^2$$

46. Diameter of cube = 3 cm

$\therefore$ Number of cubes $= \frac{9 \times 9 \times 9}{3 \times 3 \times 3} = 27$ cubes

47.



$l \times b = 20000$

$\Rightarrow$ $\frac{l}{2} \times b = 10000$, Area of Kaurav's land

For the given area, a square gives the minimum perimeter.

$\therefore$ $\frac{l}{2} = b$

$\Rightarrow$ $b \times b = 10000$

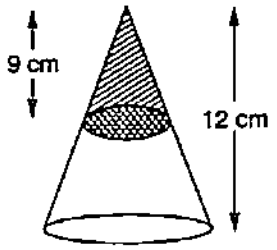
$\Rightarrow$ $b = 100$ m

$\therefore$ Perimeter of Kaurav's land $= 4 \times 100 = 400$ m

$\therefore$ Expenses $= 2 \times 400 =$ Rs. 800

48. Ratio of height

$$\frac{h_1}{h_2} = \frac{9}{12} = \frac{3}{4}$$



$\therefore$ Ratio of volumes $= \frac{(3)^3}{(4)^3} = \frac{27}{64}$

Hence, the volume of smaller cone

$= 27x$

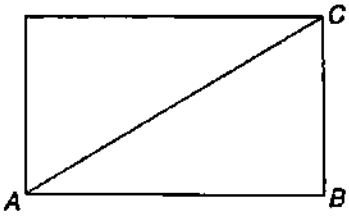
and the volume of larger (actual) cone $= 64x$

$\therefore$ the volume of frustum $= 64x - 27x = 37x$

$\therefore$ Required ratio $= \frac{64x}{37x} = 64 : 37$

49. $AB = 45$ km, $AC = 53$ km

(Since distance = speed × time)



$\therefore$ $BC = \sqrt{AC^2 - AB^2}$

$BC = \sqrt{(53)^3 - (45)^2}$

or $BC = 28$ km

$\therefore$ Area of field $= AB \times BC$

$= 45 \times 28 = 1260$ km^2

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

50.

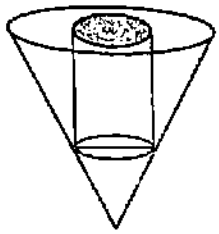


Fig. (I)

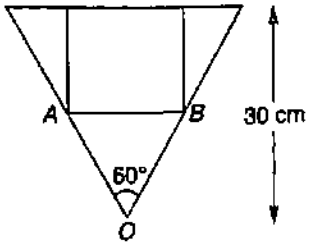


Fig. (II)

$\angle AOB = 60°$

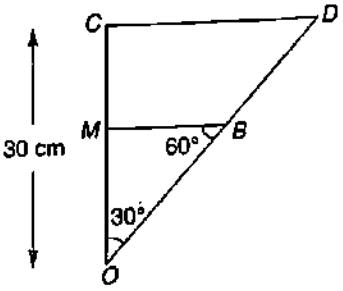
$AB = 8\sqrt{3}$ cm

$OC = 30$ cm

$\frac{OM}{MB} = \tan 60° = \sqrt{3}$

$OM = \sqrt{3}MB$

$OM = 12$ cm



Now, $\frac{OM}{MB} = \frac{OC}{CD}$

(Triangles OMB and OCD are similar)

$\therefore$ $CD = 10\sqrt{3}$ cm (which is the radius of cone)

$\therefore$ Volume of cone $= \frac{1}{3}\pi r^2 h$

$= \frac{1}{3}\pi \times (10\sqrt{3})^2 \times 30$

$= 3000\pi$ cm^3

51. Surface area of one cube $= 6a^2$

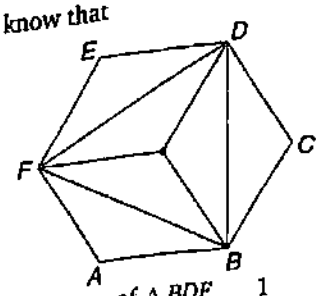
when 6 cubes are fixed on the 6 faces of a cube then only 5 faces of a cube are visible of each cube. Since, the central cube is completely covered. So, the only 6 cubes are visible each with 5 faces. Hence, the total surface area of this solid $= 5a^2 \times 6$.

$\therefore$ Required ratio $= \frac{5a^2 \times 6}{6a^2} = \frac{5}{1} \Rightarrow 5:1$

52. Ratio of areas = (Ratio of diagonals)^2

$= \left(\frac{3}{2}\right)^2 = \frac{9}{4}$

53. You should know that



$\frac{\text{Area of } \Delta BDF}{\text{Area of hexagon}} = \frac{1}{2}$

Actually there is a perfect symmetricity.

$\therefore$ Area of hexagon $= \frac{3\sqrt{3}}{2} \times (6)^2 = 54\sqrt{3}$ cm^2

$\therefore$ Area of $\Delta BDF = 27\sqrt{3}$ cm^2

Alternatively: Subtract the areas of three triangles DEF, BAF, BCD from the area of hexagon.

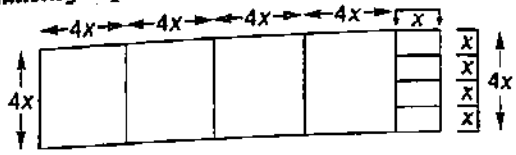
Area of $\Delta DEF = \frac{1}{2} \times 6 \times 6 \sin 120° = 9\sqrt{3}$ cm^2

$\therefore$ Area of all the three triangles $= 27\sqrt{3}$ cm

Therefore, required area of $\Delta BDF = 54\sqrt{3} - 27\sqrt{3}$

$= 27\sqrt{3}$ cm^2

Solutions for question number 54 and 55:



Total length of plot $= 17x$

and total breadth of the plot $= 4x$

$\therefore$ *Area of plot* $= 17x \times 4x = 9792$

$\Rightarrow$ $x = 12$

$\therefore$ $l = 17 \times 12 = 204\ m$

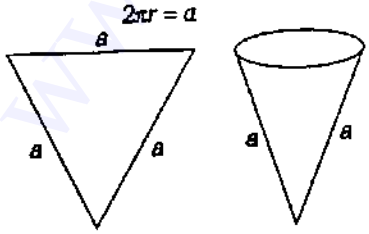
and $b = 4 \times 12 = 48\ m$

54. $\frac{4x \times 4x}{x \times x} = \frac{16}{1} \Rightarrow 16:1$

55. $l = 204$ m and $b = 48$ m

LEVEL 2

1.



Also, slant height $(l) = a$ $\therefore$ $r = \frac{a}{2\pi}$

2. $\because$ $l^2 = h^2 + r^2$

$\Rightarrow$ $h^2 = l^2 - r^2 = a^2 - \left(\frac{a}{2\pi}\right)^2$

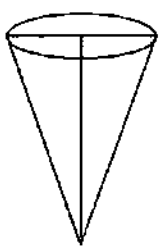
$h^2 = a^2\left[\frac{4\pi^2 - 1}{4\pi^2}\right]$

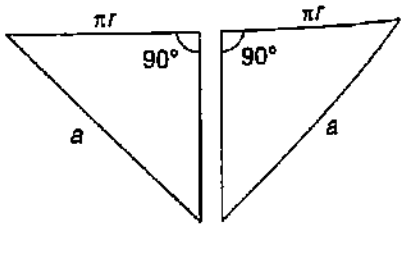
$\therefore$ $h = \frac{a}{2\pi}\sqrt{4\pi^2 - 1}$

$\therefore$ Volume $= \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \times \frac{a^2}{4\pi^2} \times \frac{a}{2\pi}\sqrt{4\pi^2 - 1}$

$= \frac{a^3}{24\pi^2}\sqrt{4\pi^2 - 1}$

3. It will be in the form of a right angled triangle.





For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

4. $2\pi r(r+h) = 1540 \text{ cm}^2$

and $(r+h) = 35 \text{ cm}$

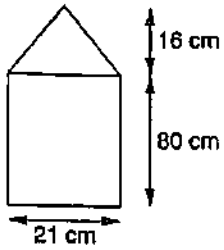
$\therefore \quad 2\pi r = \dfrac{1540}{35} = 44 \text{ cm}$

5. Total volume $= \pi r^2 h_1 + \dfrac{1}{3}\pi r^2 h_2$

$= \pi r^2 \left[h_1 + \dfrac{h_2}{3}\right]$

$= \dfrac{22}{7} \times (21)^2 \left[80 + \dfrac{16}{3}\right]$

$= \dfrac{22}{7} \times 441 \times \dfrac{256}{3}$



Weight $= \dfrac{22}{7} \times 441 \times \dfrac{256}{3} \times \dfrac{8.45}{1000} = 999.39 \text{ kg}$

6. *ABCD* is a square, each side of square is '*a*'.

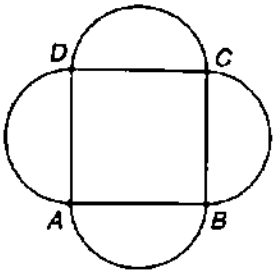


Fig. (i)

In figure (ii),

$\angle DOC = 120°$

and $\angle ODC = \angle OCD = 30°$

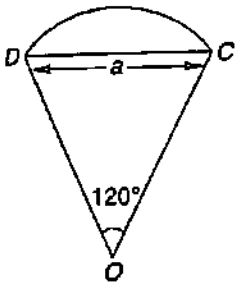


Fig. (ii)

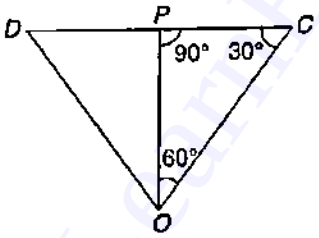


Fig. (iii)

In figure (iii),

$\dfrac{PC}{OC} = \sin 60°$

$\dfrac{a/2}{OC} = \dfrac{\sqrt{3}}{2}$

$\Rightarrow \quad OC = \dfrac{a}{\sqrt{3}} \Rightarrow$ radius of the arc '*CD*'.

$\therefore$ Area of triangle $OCD = \dfrac{1}{2} \times CD \times OP$

$= \dfrac{1}{2} \times a \times \dfrac{a}{2\sqrt{3}} = \dfrac{a^2}{4\sqrt{3}}$

$\left(\because \dfrac{OP}{PC} = \tan 30° \text{ and } \tan 30° = \dfrac{1}{\sqrt{3}}\right)$

and area of sector *COD* (figure ii)

$= \pi r^2 \dfrac{120}{360}$

$= \pi \times \left(\dfrac{a}{\sqrt{3}}\right)^2 \times \dfrac{1}{3} = \dfrac{\pi a^2}{9}$

$\therefore$ Area of segment = (Area of sector – Area of triangle)

$= \dfrac{\pi a^2}{9} - \dfrac{a^2}{4\sqrt{3}}$

Total area of all the four segments $= 4\left(\dfrac{\pi a^2}{9} - \dfrac{a^2}{4\sqrt{3}}\right)$

and the total area of whole figure $= a^2 + 4\left(\dfrac{\pi a^2}{9} - \dfrac{a^2}{4\sqrt{3}}\right)$

7. $2(l+b) = 26 \Rightarrow l + b = 13$

$12 + 1 = 13$

$11 + 2 = 13$

$10 + 3 = 13$

$9 + 4 = 13$

$8 + 5 = 13$

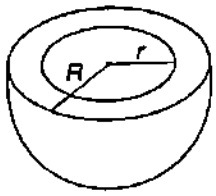
$7 + 6 = 13$

Since, $l > b$, therefore, there are only 6 integral values of the length *viz.*, 7, 8, 9, 10, 11 and 12.

8. Total surface area $= 2\pi R^2 + 2\pi r^2 + (\pi R^2 - \pi r^2)$

$= 3\pi R^2 + \pi r^2$

$= \pi(3R^2 + r^2)$



$\Rightarrow \quad 1436\dfrac{2}{7} = \pi[3 \times (12)^2 + r^2]$

$\Rightarrow \quad \dfrac{10054}{7} \times \dfrac{1}{\pi} = 432 + r^2$

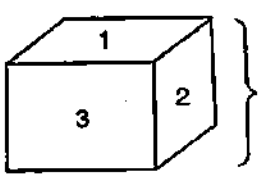
$\Rightarrow \quad r = 5 \text{ cm}$

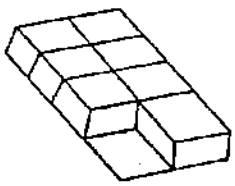
$\therefore$ Internal volume of hemisphere $= \dfrac{2}{3}\pi(R^3 - r^3)$

$= \dfrac{2}{3}\pi[(12)^3 - (5)^3]$

$= 3358\dfrac{2}{3} \text{ cm}^3$

9. Since, there are 3 faces which are visible in a corner cube. When the cube of a corner is removed then the 3 faces of other cubes will be visible from outside. So, there will not be any change in the surface area of this solid figure.





10. Number of spheres $= \dfrac{\frac{4}{3}\pi\left(\frac{15}{2}\right)^3}{\frac{4}{3}\pi\left(\frac{3}{2}\right)^3} = 125 \text{ spheres}$

Surface area of a large sphere $= 4\pi \times \left(\dfrac{15}{2}\right)^2$

and surface area of a small sphere $= 4\pi\left(\dfrac{3}{2}\right)^2$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

and total surface area of all the smaller spheres

$$= 125 \times 4\pi \left(\frac{3}{2}\right)^2$$

$$\% \text{ change in area} = \left[\frac{500\pi\left(\frac{3}{2}\right)^2 - 4\pi\left(\frac{15}{2}\right)^2}{4\pi\left(\frac{15}{2}\right)^2} \times 100\right] = 400\%$$

Alternatively: Surface area of larger sphere $= 25x$

Surface area of smaller sphere $= x$

$\because$ the ratio of radii is 5 : 1.

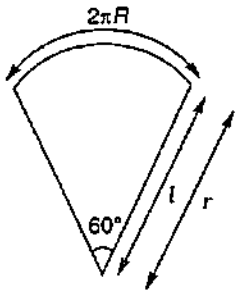
Therefore, ratio in surface areas = 25 : 1

Now, since there are 125 smaller spheres

$\therefore$ Total surface area of smaller spheres $= 125x$

$\therefore$ $\%$ change in area $= \frac{125x - 25x}{25x} \times 100 = 400\%$

11. Let the radius of cone be R and radius of sector $= r$,



$\Rightarrow l = r$

then the slant height of cone $(l) = r$

and $2\pi R = 2\pi r \times \frac{60}{360}$

$\Rightarrow$ $R = \frac{r}{6} = \frac{14}{6} = \frac{7}{3}$ cm

$\therefore$ Total surface area $= \pi r (l + r)$

$$= \frac{22}{7} \times \frac{7}{3}\left(14 + \frac{7}{3}\right)$$

$$= 119.78 \text{ cm}^2$$

12. Between 26 poles, total length is $(26 - 1) \times 4 = 100$ m

It means the length of each side of a square field is 100 m.

$\therefore$ Area of field $= (100)^2 = 10{,}000 \text{ m}^2 = 1$ hectare

13. It is clear that length of the lawn is 2 m more than the breadth of lawn.

To solve this problem quickly, go through options. Let us take option (c).

$$l = 10 \text{ m} \Rightarrow b = 8 \text{ m}$$

Area of path $= (l + b + 2w)\, 2w$

$$= (10 + 8 + 4)\, 4 = 88 \text{ m}^2$$

and area of lawn $= 10 \times 8 = 80 \text{ m}^2$

Reduced area of lawn $= 8 \times 8 = 64 \text{ m}^2$

$\therefore$ New area of path $= 88 + (80 - 64) = 104 \text{ m}^2$

$\therefore$ Ratio of areas of path $= \frac{104}{88} = \frac{13}{11}$

Hence, option (c) is correct.

Alternatively: Let the breadth of the lawn be 'b' then the length will be $(b + 2)$

$\therefore$ Area of path $= (l + 4)(b + 4) - lb$

$$= (b + 6)(b + 4) - (b + 2)b$$

$$= 8b + 24$$

and the new area of path $= (l + 4)(b + 4) - b \times b$

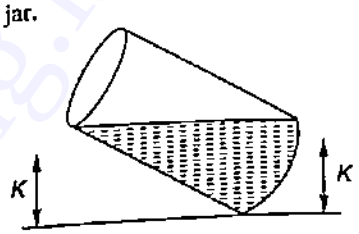
$$= (b + 6)(b + 4) - b^2$$

$$= 10b + 24$$

$\because$ $\frac{(10b + 24)}{(8b + 24)} = \frac{13}{11}$

$\Rightarrow$ $b = 8$ m $\therefore l = 10$ m

14. From the figure you can see that just half of the liquid has been flown off and half the liquid is remained in the cylindrical jar.



Thus it is clear that the capacity (or volume) of the cylinder

$$= 2 \times 2.1 = 4.2 \text{ L}$$

15. When the height and base of the cone are same as that of cylinder, then the voume of cone is $\frac{1}{3}$ that of the cylinder.

Thus the capacity of cone $= \frac{1}{3} \times 4.2 = 1.4\, l$

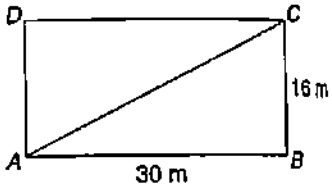
Thus the remaining volume $= 2.1 - 1.4 = 0.7\, l$

$\therefore$ the required ratio $= \frac{0.7}{4.2} = \frac{1}{6}$

16. $AC = \sqrt{(30)^2 + (16)^2}$

$AC = 34$ m

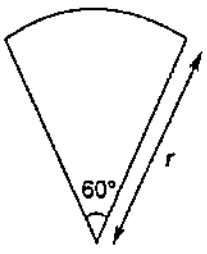
but since elephant is itself 4 m long. So he has to travel only

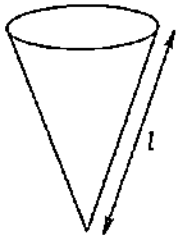


$$(34 - 4) = 30 \text{ m}.$$

$\therefore$ the speed of elephant $= \frac{30}{15} = 2$ m/s

17. Arc of sector $= \frac{2\pi r\, 60}{360} = \frac{2\pi r}{6}$





This arc of sector will be equal to the perimeter of cone. Let the radius of cone be R,

then $2\pi R = \frac{2\pi r}{6} \Rightarrow R = \frac{r}{6}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Further the radius of sector will be equal to the slant height of cone

$\therefore \quad l = r$

Now since $l^2 = h^2 + R^2$

$\Rightarrow \quad h = \sqrt{l^2 - R^2}$

$$h = \sqrt{r^2 - \left(\frac{r}{6}\right)^2}$$

$$h = \frac{\sqrt{35}}{6} r$$

18. The diagonal of cube will be equal to the diameter of sphere.

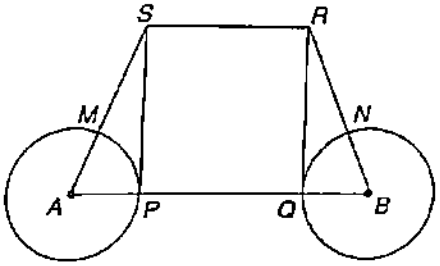
$\therefore$ Volume of sphere $= \frac{4}{3}\pi\left(\frac{d}{2}\right)^3 = \frac{\pi d^3}{6}$

and each side of cube $= a = \frac{d}{\sqrt{3}}$

$\therefore$ Volume of cube $= a^3 = \frac{d^3}{3\sqrt{3}}$

$\therefore$ Remaining volume $= \frac{\pi d^3}{6} - \frac{d^3}{3\sqrt{3}} = \frac{d^3}{3}\left(\frac{\pi}{2} - \frac{1}{\sqrt{3}}\right)$

19. Let $AP = x$ then $AM = x$ and $MS = x$

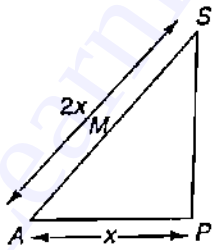


$\therefore \quad AS = AM + MS$

$AS = 2x$

$\therefore \quad PS = \sqrt{AS^2 - AP^2}$

$PS = \sqrt{3}x$



$\therefore$ Area of square $PQRS = (\sqrt{3}x)^2 = 3x^2$

Area of circle $= \pi r^2 = \pi \times x^2 = \pi x^2$

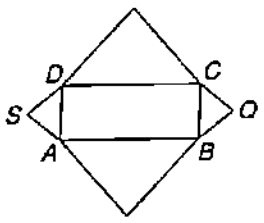
Area of both the circles $= 2\pi x^2$

$\therefore$ Required ratio $= \frac{2\pi x^2}{3x^2} = \frac{2\pi}{3}$

20. Let the length of rectangle be 'l' and breadth be 'b', then

$2(l + b) = 12$

$\Rightarrow \quad l + b = 6$ cm



and area of larger equilateral triangle $= \frac{\sqrt{3}}{4} l^2$

Similarly area of smaller equilateral triangle $= \frac{\sqrt{3}}{4} b^2$

$\therefore$ Total area of all the 4 triangles $= 2 \times \frac{\sqrt{3}}{4}(l^2 + b^2) = 10\sqrt{3}$

$\Rightarrow \quad l^2 + b^2 = 20$

$\therefore \quad (l + b)^2 = l^2 + b^2 + 2lb$

$\Rightarrow \quad 36 = 20 + 2lb$

$\Rightarrow \quad lb = 8$

$\therefore \quad (l - b)^2 = l^2 + b^2 - 2lb = 20 - 16$

$(l - b)^2 = 4$

$\Rightarrow \quad l - b = 2$

$\because \quad l + b = 6$ and $l - b = 2$

$\therefore \quad l = 4$ and $b = 2$

$\therefore$ area of rectangle $= 4 \times 2 = 8$ cm^2

$\therefore$ Total area of the figure $= 8 + 10\sqrt{3}$

$= 2(4 + 5\sqrt{3})$ cm^2

21. Area of each square $= 16$ cm^2

Area of quadrant $ADMB = \frac{1}{4}\pi \times (4)^2 = 4\pi$

and radius of smaller quadrant

$CPMQ = CM = AC - MA$

$= 4\sqrt{2} - 4 = 4(\sqrt{2} - 1)$

$\therefore$ Area of smaller quadrant $= \frac{1}{4}\pi[4(\sqrt{2} - 1)]^2$

$= 4\pi(3 - 2\sqrt{2})$

Area of shaded region inside the square $ABCD$

$= 16 - [4\pi + 4\pi(3 - 2\sqrt{2})]$

$= 16 - [4\pi(1 + 3 - 2\sqrt{2})]$

$= 16 - [4\pi(4 - 2\sqrt{2})]$

$= 8[2 - 2\pi + \sqrt{2}\pi]$

Now, area of quadrants $= AEG + EFG = 2AEG$

$= 2 \times \frac{1}{4}\pi(4)^2 = 8\pi$

$\therefore$ Area of shaded region inside the square $EAGF$

$= 8\pi - 16 = 8(\pi - 2)$

$\therefore$ Required ratio $= \frac{8(2 - 2\pi + \sqrt{2}\pi)}{8(\pi - 2)}$

$= \frac{[2 + \pi(\sqrt{2} - 2)]}{(\pi - 2)}$

22. Given that $\frac{AB}{BC} = \frac{AD}{DF}$

Also $BE = BC$

Let $AD = 1$ and $AE = x$

$\therefore \quad \frac{AE}{EF} = \frac{AE}{AD} = \frac{AE}{BC} = x$

$\therefore \quad \frac{AE}{EF} = \frac{AD}{AB}$ $\left(\because \begin{matrix} AD = BC = BE \\ \text{and } AB = AE - BE \end{matrix}\right)$

$\frac{x}{1} = \frac{1}{x - 1}$

$\Rightarrow \quad x^2 - x - 1 = 0$

$\Rightarrow \quad x = \frac{(1 \pm \sqrt{5})}{2}$

$\therefore \quad x = \frac{(1 + \sqrt{5})}{2}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

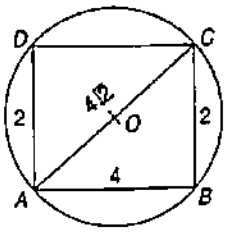
Since ratio of two sides can never be negative.

Alternatively: Since ratio of two side can never be negative therefore only option (c) is correct.

Solutions for question number 23 to 25:

$AB = 4$

$\therefore \quad AO = AC = \dfrac{4\sqrt{2}}{2} = 2\sqrt{2}$



Area of circle

$ABCD = \pi \times (2 \times \sqrt{2})^2 = 8\pi$

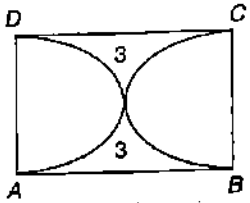
Area of region 2 (only left part)

$$= \frac{\text{Area of circle} - \text{Area of square}}{4}$$

$$= \frac{8\pi - 16}{4} = (2\pi - 4)$$

Area of region 3 = Area of square

− 2 (Area of semicircle)



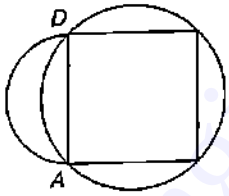
$$= 16 - 2\left(\frac{1}{2} \times \pi \times 4\right)$$

$= 16 - 4\pi = 4\,(4 - \pi)\ cm^2$

Area of region 1

= Area of semicircle AD

− Area of region 2



$$= \frac{1}{2}\pi \times (2)^2 - (2\pi - 4) = 4\ cm^2$$

23. Total area of region 1 $= 2 \times 4 = 8\ cm^2$

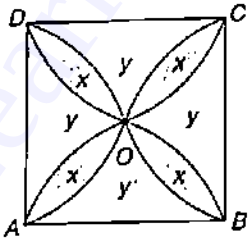
24. Total area of region 2 $= 2 \times (2\pi - 4) = 4\,(\pi - 2)\ cm^2$

25. Total area of region 3 $= 4\,(4 - \pi)\ cm^2$

26. Total area of square $= 64\ cm^2$

$\therefore \quad 4\,(x + y) = 64$

$\Rightarrow \quad x + y = 16 \quad \ldots(i)$



Again in a semicircle

$$AOB = x + y + x = \frac{1}{2}\pi \times (4)^2$$

$2x + y = 8\pi \quad \ldots(ii)$

From Eq. (i) and (ii), we get

$x = 8\pi - 16$

Total area of shaded region $= 4\,(8\pi - 16)$

$= 32\,(\pi - 2)\ cm^2$

Alternatively: Area of square − 2 (Area of semicircles) = $2y$

$(64 - 16\pi) = 2y$

$\therefore \quad 4y = (128 - 32\pi)$

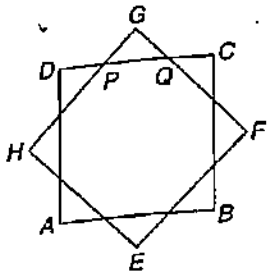
$\therefore$ Required area of shaded region $(4x)$ = (Area of square − $4y$)

$= 64 - (128 - 32\pi)$

$= 32\pi - 64$

$= 32\,(\pi - 2)\ cm^2$

27. You can see in the figure that the sides of one square is parallel to the diagonals of the other square.



Let $DP = a$, then

$DC = DP + PQ + QC$

$= a + a\sqrt{2} + a$

$DC = a\,(2 + \sqrt{2})$

$\left.\begin{array}{l} \because \quad DP = QC \\ \text{and } DP = PG = GQ = QC \\ \angle G = 90° \\ \therefore \quad PQ = \sqrt{2}a \end{array}\right\}$

$\therefore$ Area of $\Delta PGQ = \dfrac{1}{2} \times a \times a = \dfrac{a^2}{2}$

$\therefore$ Area of all the triangles outside the square $ABCD$

$$= 4 \times \frac{a^2}{2} = 2a^2$$

But $\quad DC = a\,(2 + \sqrt{2}) = 4\ cm$

$$\Rightarrow \quad a = \frac{4}{(2 + \sqrt{2})}$$

$$\therefore \quad 2\,(a)^2 = 2 \times \left(\frac{4}{2 + \sqrt{2}}\right)^2$$

$$= \frac{16}{(3 + 2\sqrt{2})} \times \frac{(3 - 2\sqrt{2})}{(3 - 2\sqrt{2})}$$

$$= 16\,(3 - 2\sqrt{2})$$

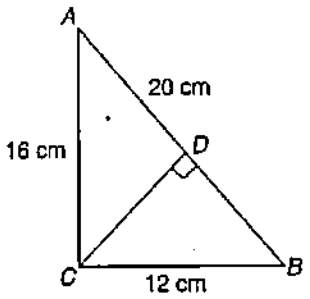
and Area of square $= 16\ cm$

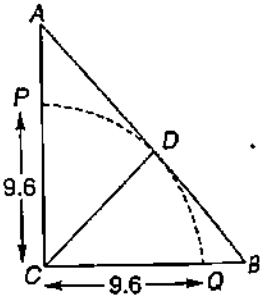
$\therefore$ Total area of the figure $= 16 + 16\,(3 - 2\sqrt{2})$

$= 16\,(4 - 2\sqrt{2})$

$= 32\,(2 - \sqrt{2})\ cm^2$

28. When $l = CD$, then the volume of cone will be maximum, where l is the slant height of the cone and the largest possible angle at the vertex of cone is 90°.





$$CD = \frac{12 \times 16}{20} = 9.6\ cm,$$

which is the radius of the sector.

Therefore, arc of the sector $= 2\pi \times 9.6 \times \dfrac{90}{360} = 4.8\pi$

Let the radius of the cone be r, then

$2\pi r =$ arc of the sector

$2\pi r = 4.8\pi$

$r = 2.4$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$\therefore$ Height of the cone $(h) = \sqrt{l^2 - r^2}$

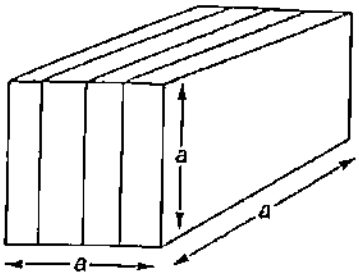
$= \sqrt{(9.6)^2 - (2.4)^2} = 2.4\sqrt{15}$ cm

$\therefore$ Volume of the cone $= \frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \frac{22}{7} \times (2.4)^2 \times 2.4\sqrt{15}$

$= 56.1 \text{ cm}^3$

29. To increase the value (or price of diamond) they should cut (divide) the diamond in such a way that the surface area will be maximum.



Thus, when four parts are parallel to each other.

In this way total surface area

$= 6a^2 + 2a^2 + 2a^2 + 2a^2 = 12a^2$

Actual surface area of cubical diamond $= 6a^2$

Therefore, percentage increase in area

$= \frac{12a^2 - 6a^2}{6a^2} \times 100 = 100\%$

Remember that for the given volume, minimum surface area is possessed by a cube. So to maximize the area we have to increase the maximum possible difference between the edges of cuboid.

30. Side of square I $= a$

Side of square II $= \frac{a}{\sqrt{2}}$

Side of square III $= \frac{a}{2}$

Side of square IV $= \frac{a}{2\sqrt{2}}$

Side of square V $= \frac{a}{4}$

Therefore, sum of perimeters of all the squares

$= 4\left(a + \frac{a}{\sqrt{2}} + \frac{a}{2} + \frac{a}{2\sqrt{2}} + \frac{a}{4}\right)$

$= 4a\left(1 + \frac{1}{\sqrt{2}} + \frac{1}{2} + \frac{1}{2\sqrt{2}} + \frac{1}{4}\right)$

$= 4a\left(\frac{4 + 2\sqrt{2} + 2 + \sqrt{2} + 1}{4}\right)$

$= a(7 + 3\sqrt{2})$

31. Total area of the five squares

$= a^2 + \left(\frac{a}{\sqrt{2}}\right)^2 + \left(\frac{a}{2}\right)^2 + \left(\frac{a}{2\sqrt{2}}\right)^2 + \left(\frac{a}{4}\right)^2$

$= a^2\left[1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16}\right]$

$= a^2\left[\frac{16 + 8 + 4 + 2 + 1}{16}\right]$

$= a^2 \times \frac{31}{16} = \frac{31a^2}{16}$

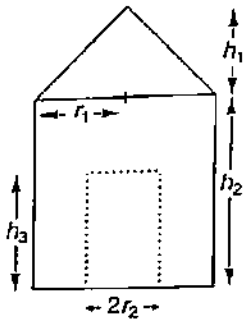
32. $(n-2)^3$

33. $6(n-2)^2$

34. $12(n-2)$

35. These are the 8 cubes at the corners, which is always fix.

36. Volume of the whole body



$V_1 = \frac{1}{3}\pi r_1^2 h_1 + \pi r_1^2 h_2$

but $\frac{h_1}{h_2} = \frac{2}{3}$

$\therefore \quad V_1 = \pi r_1^2 \frac{11h_1}{6}$

and $h_3 = \frac{2}{3}(h_1 + h_2) = \frac{5h_1}{3}$

Hence, volume of the hole $(V_2) = \pi r_2^2 h_3$

$= \frac{5}{3}\pi r_2^2 h_1$

But it is given that $V_2 = \frac{V_1 - V_2}{3}$

$\therefore \quad V_1 = 4V_2$

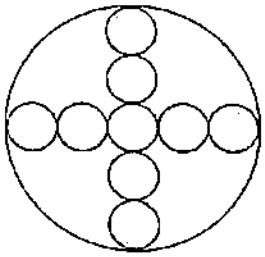
$\Rightarrow \quad 4 \times \frac{5}{3}\pi r_2^2 h_1 = \pi r_1^2 \times \frac{11}{6} h_1$

$\Rightarrow \quad r_2 = \sqrt{\frac{55}{8}}$ cm

37. $19 \times 19 = 361$

Thus, we make equal 19 measurements each of 19°, then we get $(361 - 360) = 1°$ angle at the centre. Thus, moving continuously in the similar fashion, we can get all the 360° angle *i. e.*, 360 equal sectors of 1°.

38. When we open the paper after cutting it, we will find it as shown in the following figure.



Radius of the larger circle = 5 cm

$\therefore$ Area of larger circle = 25π

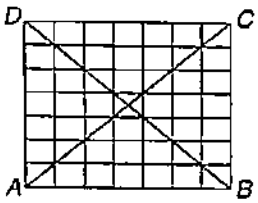
and the radius of each smaller circle is 1 cm.

Therefore, total area of all the 9 circles $= 9 \times \pi \times (1)^2 = 9\pi$

$\therefore$ Remaining area $= (25 - 9)\pi = 16\pi$

Hence, the required ratio = 25 : 16

39. In the above layer we can see that total 13 cubes get a cut. So, in 7 layers total $13 \times 7 = 91$ cubes will get a cut and the remaining $(7^3 - 91) = 252$ cubes are without any cut.



Total number of pieces which are not a cube

$= 12 \times 2 \times 7 + 4 \times 7 = 196$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

(Since 84 cubes are diagonally cut into two parts and 7 cubes which are in the centre are divided into 4 parts.)
Thus, total 196 children will get one-one piece and 252 adults get one-one piece.
Thus total 252 + 196 = 448 people can get a piece of cake.

NOTE It is clear that everyone get equal number of pieces but not according to the volume of pieces.

Solutions for questions number 40 to 42: Diameter (2R) of the outermost circle is equal to the diagonal of larger square.

Hence, the side of square $= \frac{2R}{\sqrt{2}} = \sqrt{2}R$

Again the side of larger square is equal to the diameter of middle most circle.

Hence, the radius of mid-circle is $\frac{R}{\sqrt{2}}$.

Once again the diameter of the mid-circle is equal to the diagonal of smaller square. Hence, side of the smaller square = R.
Similarly the diameter of innermost circle is equal to the side of the smaller square. Hence, radius of the innermost circle $= \frac{R}{2}$.

40. $\frac{R}{2}$

41. Area of larger square $= (\sqrt{2}R)^2 = 2R^2$
and area of smaller square $= R^2$
$\therefore$ Total area of both squares $= 3R^2$

42. Sum of all the circumferences $= 2\pi\left(R + \frac{R}{\sqrt{2}} + \frac{R}{2}\right)$

$= 2\pi R\left(\frac{2+\sqrt{2}+1}{2}\right)$

$= (3+\sqrt{2})\pi R$

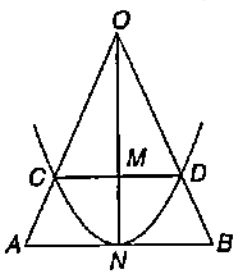
Sum of perimeters of all the squares $= 4(\sqrt{2}R + R)$

$= 4R(\sqrt{2}+1)$

$\therefore$ Required ratio $= \frac{(3+\sqrt{2})\pi R}{(\sqrt{2}+1)4R} = \frac{(3+\sqrt{2})\pi}{(\sqrt{2}+1)4}$

Alternatively: Since circumference and perimeter both has R. So R will be cancelled in the ratio of circumference to the perimeter. Thus, neither of the choices *a*, *b* and *c* are admissible. Hence, the only correct choice is (d).

Solutions for questions number 43 and 44: Each side of outer (larger) hexagon is equal to the radius of circle which is R.



Now, $OC = ON = OD$

radii of the inner (smaller) circle

But $\frac{ON}{OA} = \sin 60° = \frac{\sqrt{3}}{2}$

$\Rightarrow$ $ON = \frac{\sqrt{3}}{2}OA = \frac{\sqrt{3}}{2}R$, *radius of the inner circle*

and this is also equal to the side of the inner hexagon.

43. Sum of perimeters of both the hexagons $= 6R + 6 \times \frac{\sqrt{3}}{2}R$

$= 6R\left(1 + \frac{\sqrt{3}}{2}\right) = 3(2+\sqrt{3})R$

44. $\frac{\text{Area of inner circle}}{\text{Area of outer circle}} = \frac{\pi}{\pi}\frac{\left[\left(\frac{\sqrt{3}}{2}\right)R\right]^2}{(R)^2} = \frac{3}{4}$

45. Radius of the first hexagon $= R$

Radius of the second hexagon $= \frac{\sqrt{3}}{2}R$

Radius of the third hexagon $= \frac{3}{4}R$

Radius of the fourth hexagon $= \frac{3\sqrt{3}}{8}R$

$\therefore$ Required ratio $= \frac{R}{\frac{3\sqrt{3}}{8}R} = \frac{8}{3\sqrt{3}}$

46. From the concept of similarity of triangles. All the five quadrilaterals viz., AOA', BOB', COC', DOD' and EOE' are similar.
From the figure (ii)

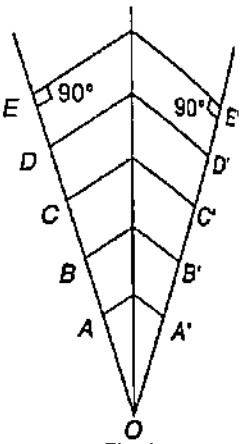


Fig. (i)

$\frac{r_2 - r_1}{r_2 + r_1} = \frac{r_3 - r_2}{r_3 + r_2}$

$= \frac{r_4 - r_3}{r_4 + r_3} = \frac{r_5 - r_4}{r_5 + r_4} = K$

$\Rightarrow$ $\frac{r_2}{r_1} = \frac{r_3}{r_2} = \frac{r_4}{r_3} = \frac{r_5}{r_4} = K$

(By componendo and dividendo)

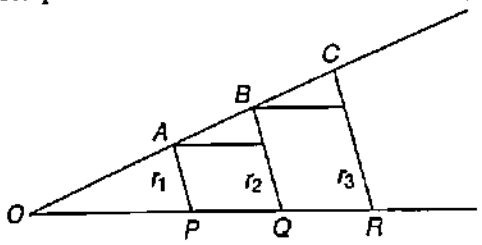


Fig. (ii)

It means all the radii are in GP

Therefore, $\frac{r_5}{r_1} = (K)^4 = \frac{81}{16} = \left(\frac{3}{2}\right)^4$

$\Rightarrow$ $K = \frac{3}{2}$ $\therefore$ $r_3 = r_1(K)^2$

46. $r_3 = r_1 \times \frac{9}{4} = \frac{9r_1}{4} = \frac{9}{4} \times 16 = 36$ cm

47. $\because$ $r_1 = 16$, $r_2 = 24$, $r_3 = 36$, ... etc.

$\therefore$ $\frac{OP}{AP} = \frac{OQ}{BQ}$

$\frac{h + r_1}{r_1} = \frac{h + 2r_1 + r_2}{r_2}$

$\frac{h+16}{16} = \frac{h+56}{24}$ $\Rightarrow$ $h = 64$ cm

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

48. $60 = 1 \times 1 \times 60$
$= 1 \times 2 \times 30$
$= 1 \times 3 \times 20$
$= 1 \times 4 \times 15$
$= 1 \times 5 \times 12$
$= 1 \times 6 \times 10$
$= 2 \times 2 \times 15$
$= 2 \times 3 \times 10$
... ...
... ...
$= 3 \times 4 \times 5$

Out of the given different combinations the first combination $(1 \times 1 \times 60)$ gives maximum length of diagonal of cuboid, but in this case two of the edges are same. So, the second combination gives the proper value *i.e.*, which gives the maximum length of diagonal whose all sides are different. Hence, the length of such a pencil is equal to the diagonal of cuboid

$$= \sqrt{1^2 + 2^2 + 30^2} = \sqrt{905}$$

49.

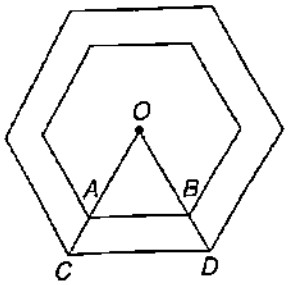


Fig. (i)

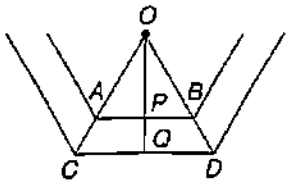


Fig. (ii)

In figure (ii)

$$OP = \frac{\sqrt{3}}{2} OA = 4\sqrt{3} \text{ cm}$$

Again $\quad \frac{OP}{OQ} = \frac{OA}{OC}$

$$\frac{4\sqrt{3}}{6\sqrt{3}} = \frac{8}{OC} \qquad (OQ = OP + PQ = 4\sqrt{3} + 2\sqrt{3})$$

$$\Rightarrow \quad OC = 12 \text{ cm}$$

$\therefore$ Each side of the outer hexagon is 12 cm.

$\therefore$ Required area = (Area of outer hexagon – Area of inner hexagon)

$$= \frac{3\sqrt{3}}{2}[12^2 - 8^2] = 120\sqrt{3} \text{ cm}^2$$

50. Area of region x = Area of square – Area of inscribed circle $= (4 - \pi)$

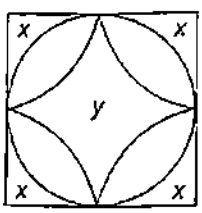


Fig. (i)

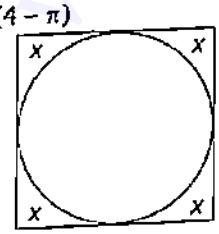


Fig. (ii)

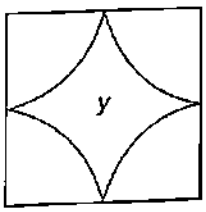


Fig. (iii)

Area of region y = Area of square – 4 (area of quadrant)

$$= 4 - 4\left(\frac{1}{4}\pi \times (1)^2\right) = (4 - \pi)$$

$\therefore$ Required area (of shaded region)
= Area of square – [Area of region x + Area of region y]
$= 4 - [4 - \pi + 4 - \pi] = 2\pi - 4$

51. Let the volume of solid block be V and radius of the spheres formed from the first block be r_1, then the volume of each sphere be V_1.
Similarly, let the radius of each sphere obtained from second block be r_2 $(= 2r_1)$, then the volume of each sphere be

$$V_2 = (8V_1)$$

$\therefore \quad V = kV_1 + 14 \qquad ...(i)$

and $\quad V = lV_2 + 36$

or $\quad V = 8lV_1 + 36 \qquad ...(ii)$

From equation (i) and (ii)

$$kV_1 + 14 = 8lV_1 + 36$$

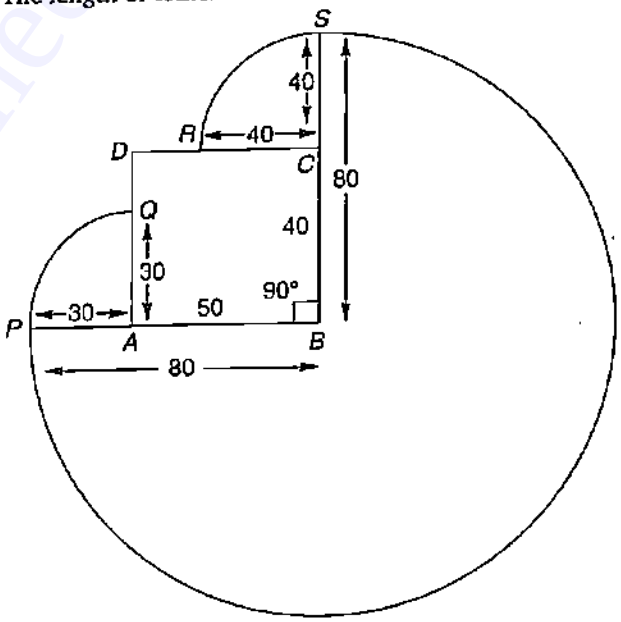
$$\Rightarrow \quad V_1(k - 8l) = 22$$

The possible value of $V_1 = 22, 11, 2$ or 1

But V_1 can never be equal to or less than 14 (since remainder is always less than divisior) So, the only possible value of $V_1 = 22$

$$\therefore \quad V_2 = 8 \times V_1 = 176 \text{ cm}^3$$

52. The length of tether of the horse is 80 m.



Area grazed by horse

$$= \left[\pi \times (80)^2 \times \frac{270}{360} + \pi \times (30)^2 \times \frac{90}{360} + \pi \times (40)^2 \times \frac{90}{360}\right]$$

$$= \pi\left(6400 \times \frac{3}{4} + 900 \times \frac{1}{4} + 1600 \times \frac{1}{4}\right) = \pi\left[\frac{21700}{4}\right]$$

$$= 5425\pi \text{ m}^2$$

HINT When horse is tethered at B, then he can move freely 270° *i.e.*, from P to S with the full length of his tether as the radius of the arc PS. Again when horse reaches P and tries to move further in the right direction *i.e.*, towards D, his tether gets fixed at A and now only 30 m tether is free which works as a radius of the quadrant PAQ. Similarly when horse reaches S and tries to move left further in the direction of D, due to wall BC his tether gets fixed at C and now only 40 m tether is free to move further. At this moment point C behaves like a fixed point of tether (as centre of quadrant RSC).

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

NOTE You can tether horse at any of the four corners A, B, C and D area grazed by horse remains same.

53. Here each side is broken up into 6 parts *i.e.*, $n = 6$

Now, $N_0 = (n-2)^3 = (4)^3 = 64$

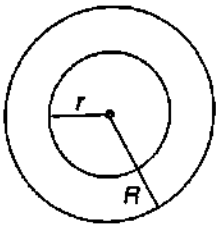
$N_1 = 6(n-2)^2 = 6 \times (4)^2 = 96$

$N_2 = 12(n-2) = 12(4) = 48$

$\therefore \quad N_0 : N_1 : N_2 = 64 : 96 : 98$

$= 4 : 6 : 3$

54. Let the radius of seed be r and radius of the whole fruit (pulp + seed) be R, then thickness of the pulp $= (R - r)$



Volume of mango fruit $= \frac{4}{3}\pi R^3$

and Volume of pulp $= \frac{4}{3}\pi (R^3 - r^3)$

but $= \frac{4}{3}\pi\left[R^3 - \left(\frac{2}{7}R\right)^3\right]$

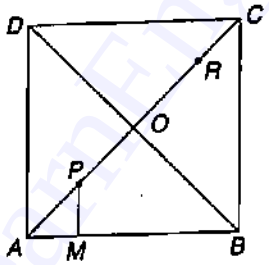
$\left[\because \frac{r}{R-r} = \frac{2}{5} \Rightarrow r = \frac{2}{7}R\right]$

$\therefore$ Percentage of volume of pulp to the total volume of fruit

$$= \frac{\frac{4}{3}\pi R^3\left[1 - \frac{8}{343}\right]}{\frac{4}{3}\pi R^3}$$

$$= \frac{335}{343} \times 100 = 97.66\%$$

55. Let the radius of each smaller circle is r and radius of the larger circle is R, then



$\pi R^2 = 4\pi r^2 \Rightarrow R = 2r$

$OR = OP = R + r = 3r$

Also $PM = r$

(PM is the perpendicular on AB)

$\therefore \quad AP = \sqrt{2}r$

$\therefore \quad AO = AP + PO$

$= r\sqrt{2} + 3r = r(3 + \sqrt{2})$

$\therefore AC = 2AO = 2r(3 + \sqrt{2})$, which is the diagonal of square

$\therefore$ Required ratio $= \frac{2r(3+\sqrt{2})}{\sqrt{2}r} = (2 + 3\sqrt{2})$

56. Initial radius = 14 cm

Radius at a time when the balloon explodes = 35 cm

Change in volume $= \frac{4}{3}\pi[(35)^3 - (14)^3]$

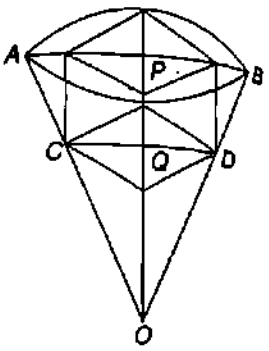
$= \frac{4}{3}\pi(7)^3[125 - 8]$

$= \frac{4}{3}\pi \times 343 \times 117$

Required time to explode $= \frac{\frac{4}{3}\pi \times 343 \times 117}{156}$

$= 1078\ s$

57. Let the each side of cube be a, then



$CD = \sqrt{2}a$

$\therefore \quad CQ = \frac{a}{\sqrt{2}}$

Let the radius of cone be r and height be h, then

$r = h\sqrt{2}$

$\therefore$ In ΔAPO and ΔCQO (Similar triangles)

$$\frac{AP}{PO} = \frac{CQ}{OQ} = \frac{r}{h} = \frac{\frac{a}{\sqrt{2}}}{(h-a)}$$

$\Rightarrow \quad \frac{\frac{a}{\sqrt{2}}}{(h-a)} = \sqrt{2}$

$\Rightarrow \quad a = 2(h - a)$

$\Rightarrow \quad h = \frac{3a}{2}$

$\therefore \quad r = \frac{3a}{2} \times \sqrt{2}$

and $h = \frac{3a}{2}$

$\therefore$ Volume of cone $= \frac{1}{3}\pi \times \left(\frac{3a\sqrt{2}}{2}\right)^2 \times \frac{3a}{2}$

$= \frac{9}{4}a^3\pi$

and Volume of cube $= a^3$

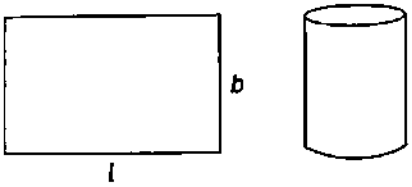
$\therefore$ Required ratio $= \frac{\frac{9}{4}\pi a^3}{a^3} = \frac{9}{4}\pi = 2.25\pi$

58. For the given volume, cube has minimum possible length of diagonal.

Therefore each side of cube = 4 cm

and its diagonal $= 4\sqrt{3}$ cm.

59. $l = 2\pi r \Rightarrow r = \frac{l}{2\pi}$



where r is the base radius of cylinder and l is the length of paper and $h = b$, where h is the height of cylinder and b is the breadth of the paper.

$\therefore$ Volume of cylinder $= \pi r^2 h = \pi \times \left(\frac{l}{2\pi}\right)^2 \times b$

$\Rightarrow \quad \frac{\pi \times l^2 b}{4\pi^2} = 48.125 = \frac{385}{8}$

$\Rightarrow \quad l^2 b = 11 \times 11 \times 5$

$\Rightarrow \quad l = 11$ and $b = 5$ $(\because l > b)$

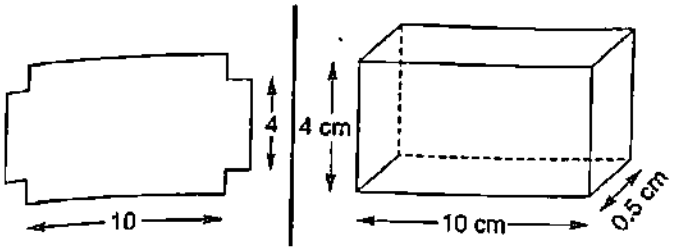
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmanboob.wordpress.com

For More Visit : www.LearnEngineering.in

$\therefore$ Volume of the box $= l \times b \times 4$

$= 10 \times 4 \times 0.5 = 20 \text{ cm}^3$



60. Vertical spacing between any two turns $= \dfrac{\text{Height of cone}}{\text{Number of turns}}$

$= \dfrac{h}{n}$

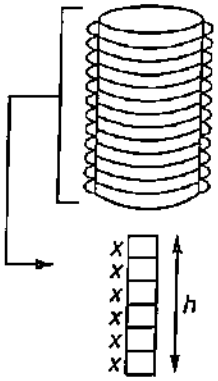
61. Number of turns $= \dfrac{h}{x}$

Length of string in each turn

$= 2\pi r = 2\pi \times \dfrac{4}{\pi} = 8 \text{ cm}$

$\therefore$ Total length of string in all the n turns

$= \dfrac{h}{x} \times 8 = \dfrac{8h}{x} \text{ cm}$

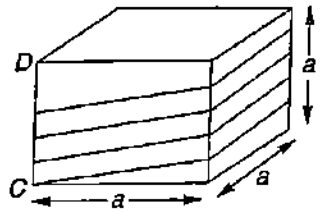


62. Total length of string $= 8n$ cm

Since, total length of string

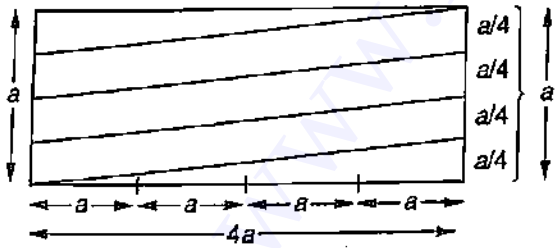
= number of turns × perimeter of cylinder

$= 8 \times n = 8n$ cm



Length of string required for 1 turn (or round) $= \dfrac{8n}{4} = 2n$

but $2n = \sqrt{\left(\dfrac{a}{4}\right)^2 + (4a)^2}$



$\Rightarrow \quad a = \dfrac{8n}{\sqrt{257}}$

where a is the side of cube.

63. From the sheet of 10 ft long, maximum $\dfrac{10}{2} = 5$ circular discs can be cut along the length of the iron sheet.

$CM = \sqrt{AC^2 - AM^2} = \sqrt{4x^2 - x^2}$

$CM = x\sqrt{3} = \sqrt{3}$ ft (Since $x = 1$ ft)

$\therefore$ Width of the sheet $= AK + MC + CT$

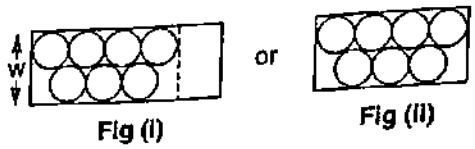


Fig (i) Fig (ii)

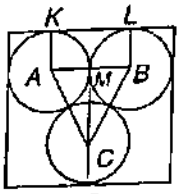


Fig. (iii)

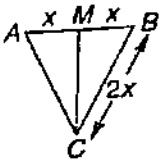


Fig. (iv)

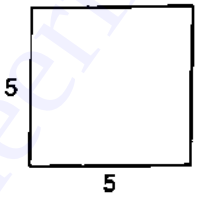
$= 1 + \sqrt{3} + 1$

$= (2 + \sqrt{3})$ ft

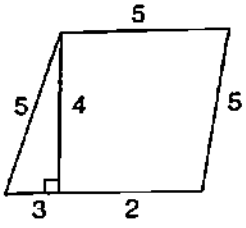
64. Recall that for given perimeter the polygon of minimum number of sides has minimum area and the polygon of maximum number of sides has maximum area. So, the correct relation is $h > s > r$.

Thus, hexagon (6 sides) has maximum area.

Now, between square and rhombus, square has greater area than rhombus. For easier understanding consider some values.



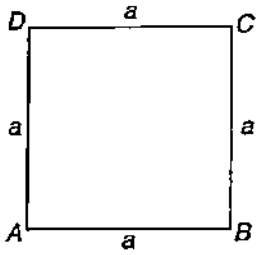
Area $= 25 \text{ cm}^2$



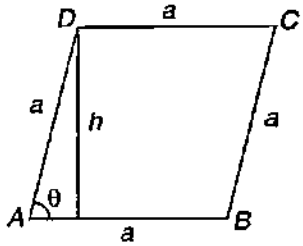
Area = base × height

$= 5 \times 4 = 20 \text{ cm}^2$

Alternatively:



Square



Rhombus

In rhombus *ABCD*: Area of rhombus $= a \times h$

$= a \times a \sin\theta \qquad \left(\dfrac{h}{a} = \sin\theta\right)$

$= a^2 \sin\theta$

As you know the maximum value of $\sin\theta$ is 1 at $\theta = 90°$ but at $\theta = 90°$ rhombus will become a square. So except $\theta = 90°$ for all the rest values the area of rhombus will be less than the area of square.

65.

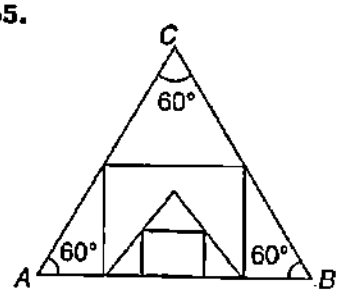


Fig. (i)

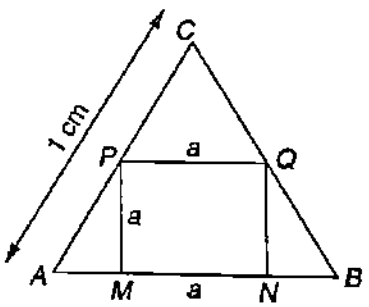


Fig. (ii)

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmartbook.wordpress.com
 For More Visit : www.LearnEngineering.in

PCQ is also an equilateral triangle

$\therefore \quad PC = PQ = PM = a$

$\therefore \quad \frac{a}{PA} = \frac{\sqrt{3}}{2}$

$\therefore \quad PA = \frac{2a}{\sqrt{3}}$

$\therefore \quad AC = AP + PC = \frac{2a}{\sqrt{3}} + a = 1 \text{ cm}$

$\Rightarrow \quad a = \frac{\sqrt{3}}{(2+\sqrt{3})} = \sqrt{3}(2-\sqrt{3})$

Now, in figure (iii)

$PM = MT = a$

Let the each side of square $RSYX$ be K, then $RT = K$ also (since RTS is an equilateral triangle)

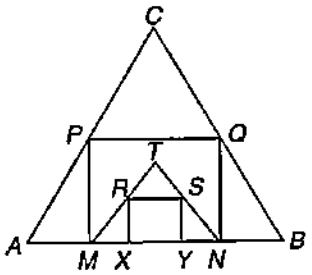


Fig. (iii)

$\therefore \quad \frac{K}{RM} = \frac{\sqrt{3}}{2}$

$\therefore \quad RM = \frac{2K}{\sqrt{3}}$

$\therefore \quad MT = RT + RM = K + \frac{2K}{\sqrt{3}}$

$MT = \frac{(\sqrt{3}+2)}{\sqrt{3}} K$

but $MT = a$

$\therefore \quad a = \left(\frac{\sqrt{3}+2}{\sqrt{3}}\right) K$

$\therefore \quad K = \frac{\sqrt{3}a}{(\sqrt{3}+2)}$

But $a = \sqrt{3}(2-\sqrt{3})$

$\therefore \quad K = \frac{\sqrt{3}}{(\sqrt{3}+2)}[\sqrt{3}(2-\sqrt{3})]$

$K = \frac{3(2-\sqrt{3})}{(2+\sqrt{3})} \times \frac{(2-\sqrt{3})}{(2-\sqrt{3})}$

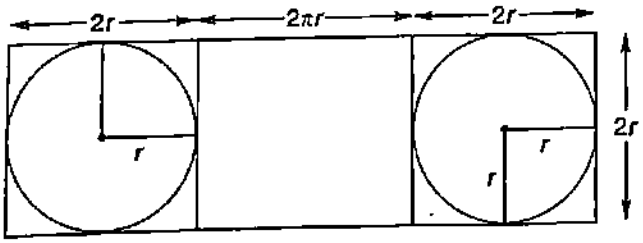
$\Rightarrow \quad K = \frac{3(2-\sqrt{3})^2}{1} = 3(7-4\sqrt{3})$

$\therefore$ Area of square $RSYX = K^2 = [3(7-4\sqrt{3})]^2$

$K^2 = 9(49 + 48 - 56\sqrt{3})$

$K^2 = (873 - 504\sqrt{3}) \text{ cm}^2$

66. For the minimum wastage of sheet he has to cut the sheet in the given manner.



Total area of sheet required

$(2\pi r + 4r) \times 2r = 4r^2(\pi + 2)$

Area of sheet utilised $= (2\pi r \times 2r) + 2(\pi r^2) = 6\pi r^2$

Area of wastage sheet $= 4r^2(\pi + 2) - 6\pi r^2$.

$= 8r^2 - 2\pi r^2$

$\therefore \quad$ Required ratio $= \frac{8r^2 - 2\pi r^2}{6\pi r^2}$

$= \frac{2r^2(4-\pi)}{6r^2\pi} = \frac{1}{11}$

67. Short cut: Very quickly check the options. If all the options have values.

Alternatively: The required capacity of box is 864 m^3. Let the length of the base be l and height of the box is 'h', then

$864 = l^2 h \quad \Rightarrow \quad h = \frac{864}{l^2}$

Now, surface area of the box $A = l^2 + 4lh$

$= l^2 + \frac{4l \times 864}{l^2}$

$= l^2 + \frac{3456}{l}$

Differentiating w.r. to l, we get

$\frac{dA}{dl} = 2l - \frac{3456}{l^2}$

For the minimum area $= \frac{dA}{dl} = 0$

$\therefore \quad 2l = \frac{3456}{l^2}$

$\Rightarrow \quad l^3 = 1728$

$\Rightarrow \quad l = 12$

$\therefore \quad$ Base area $= (12)^2 = 144$

and Height $= \frac{864}{l^2} = 6 \text{ m}$

68. Let the initial radius be r and volume be V, then

$V = \pi r^2 \times 4$

Ist case: $V_1 = \pi(r+12)^2 \times 4$

IInd case: $V_2 = \pi r^2 \times (4+12)$

But $V_1 = V + K$

and $V_2 = V + K$

$\therefore \quad V_1 = V_2$

$\Rightarrow \quad \pi(r+12)^2 \times 4 = \pi r^2 (16)$

$\Rightarrow \quad r = 12 \text{ ft}$

$\therefore$ Increased volume $= V_1 = V_2$

$= \pi \times (24)^2 \times 4$

$= 2304\pi \text{ cubic ft}$

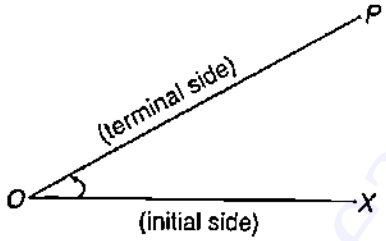
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

11 TRIGONOMETRY

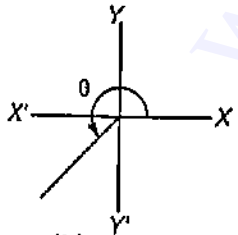
This chapter is basically written for those students who were either not so good in the secondary classes maths or who are not in touch with the following concepts from 4-5 years. Hardly a question is asked in CAT, directly from this chapter, but almost every year there are one or two questions in which we can find the application of the concepts of trigonometry *e.g.*, sometimes in geometry or in height and distance problems. So just refresh the concepts of trigonometry. Therefore CAT aspirants are not required to waste so much of precious time on this chapter. But this chapter is useful for IIFT, NIFT, FMS etc.

Angle : An angle is thought of as traced out by the rotation of a line (called the revolving line) from the initial position OX to a terminal position OP.

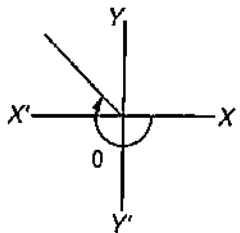


Then O is the vertex, OX is called the initial side and OP is called the terminal side of the angle.

Convention of Sign of Angles : It is customary to regard the angle traced out by a **counter-clockwise** rotation as positive and one traced out by a **clockwise** rotation as negative. The angle is indicated by using an arrow, starting from the initial line and ending at the terminal line.

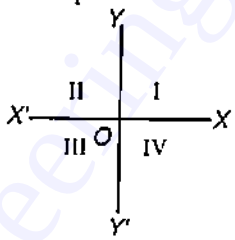


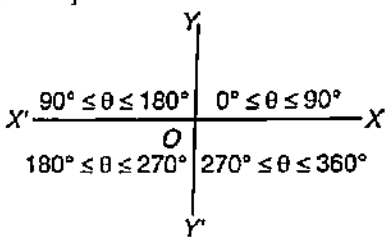
(θ is positive)



(θ is negative)

Quadrants : Let XOX' and YOY' be two mutually perpendicular lines in any plane. These lines divide the plane into four regions called quadrants, numbered counter-clockwise. Thus the quadrants XOY, YOX', $X'OY'$ and $Y'OX$ are respectively called the first quadrant, the second quadrant, the third quadrant and the fourth quadrant.





NOTE θ will be more than 360°, if after making one complete revolution, the revolving line does not stop at the initial position OX but proceeds on further on.

Measures of Angles : There are three different systems of units for measurement in trigonometry, *viz.* (i) Sexagesimal system (ii) Centesimal system (iii) Circular system.

Sexagesimal System : In this system, 1 right angle is divided into 60 equal parts and each part (or division) is called a degree (*i.e.*, 1°, read as 1 degree).

$1° = 60'$ (60 sexagesimal minutes)

$1' = 60''$ (60 sexagesimal seconds)

This system is called the Common or the English system.

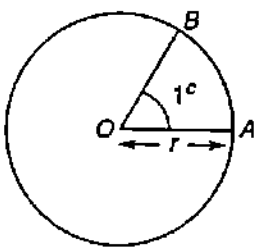
Centesimal System : In this system, 1 right angle is divided into 100 equal parts and each part (or division) is called a grade (or 1^g).

$90° = 100^g$

$1^g = 100'$ ($K' \to K$ centesimal minute)

$1' = 100''$ ($K'' \to K$ centesimal second)

Circular Measure : The unit of measurement of angles in this system is a radian (or 1^c)



A **radian** is defined as the angle subtended at the centre of a circle by an arc whose length is equal to the radius of the circle and it is denoted by 1^c.

In the figure $OA = OB = r$, (radius of the circle) and $\angle AOB = 1$ radian or 1^c.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

IMPORTANT

1. In all circles, the ratio of the circumference to its diameter is constant.
   (Circumference = $\pi \times$ diameter)
   π is an incommensurable number, *i.e.*, it cannot be expressed exactly either by a whole number or by a fraction. Its approximate value is 22/7 or 3.1416, correct to four places of decimal only.
2. A radian is a constant angle.
   1 radian (1^c) = a constant angle
3. π radian $= 180°$ (or $1° = \frac{\pi}{180}$ radian)
4. If an arc of length 's' subtends an angle θ radian at the centre of a circle of radius r, then $s = r\theta$. $\left(\theta = \frac{\text{arc}}{\text{radius}}\right)$
5. (i) Area of the sector $AOB = \frac{1}{2}r^2\theta$
   (ii) Area of the sector $AOB = \frac{1}{2}rs$
   ($s \rightarrow$ length of arc AB)

EXAMPLE 1 *Express the following angles in radian measure and centesimal measure :*
(i) 45° (ii) 20° 35′ (iii) 50° 38′ 40″

SOLUTION (i) $\because$ $180° = \pi$ rad

$\therefore$ $1° = \frac{\pi}{180}$ rad

$\therefore$ $45° = 45 \times \frac{\pi}{180} = \frac{\pi}{4}$ rad

Now, $90° = 100$ grade

$\Rightarrow$ $45° = 50$ grade

(ii) $20° 35' = 20\frac{35}{60}$ degree $= \frac{247}{12}$ degree

Now, 180 degree = π rad

$\therefore$ $\frac{247}{12}$ degree $= \frac{247}{12} \times \frac{\pi}{180} = \frac{247}{2160}\pi$ rad

and $\because$ $90° = 100$ grade

$\therefore$ $\left(\frac{247}{12}\right)^\circ = \frac{247}{12} \times \frac{100}{90} = \frac{2470}{108}$ grade $= \frac{1235}{54}$ grade

(iii) $50° 38' 40'' = 50° 38\frac{40}{60}$ min $= 50°\frac{116}{3}$ min

$= 50° \frac{116}{3 \times 60}$ degree

$= \frac{2279}{45}$ degree

Now $180° = \pi$ rad

$\because$ $\left(\frac{2279}{45}\right)^\circ = \frac{2279}{45} \times \frac{\pi}{180} = \frac{2279}{8100}\pi$ rad

and $90° = 100^g$

$\Rightarrow$ $\left(\frac{2279}{8100}\right)^\circ = \frac{2279}{8100} \times \frac{100}{90} = \frac{2279}{7290}$ grade.

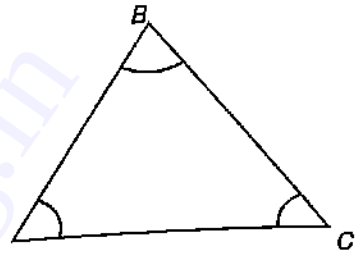
EXAMPLE 2 *One angle of a triangle is 54° and another angle is $\frac{\pi}{4}$ radian. Find the third angle in centesimal unit.*

SOLUTION Let $\angle A = 54°$ and $\angle B = \frac{\pi}{4}$ rad

$\therefore$ $B = 45°$

Thus, $\angle A + \angle B = 99°$

$\therefore$ $\angle C = 180 - 99 = 81°$ ($\because \angle A + \angle B + \angle C = 180°$)



Now, since $90° = 100$ grade

$\Rightarrow$ $81° = 81 \times \frac{100}{90} = 90$ grade

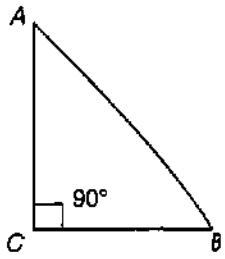
EXAMPLE 3 *The difference between two acute angles of a right-angled triangle is $\frac{\pi}{6}$ radians. Find the angle in degrees.*

SOLUTION $\angle A + \angle B = 90°$

and $\angle A - \angle B = \frac{\pi}{6} = 30°$

$\therefore$ $\angle A = 60°$ and $\angle B = 30°$

Hence the two acute angles are 60° and 30°.



EXAMPLE 4 *In a circle of diameter 60 m, the length of a chord is 30 m. Find the length of the minor arc on one side of the chord. What is the length of the major arc?*

SOLUTION $2r = 60$ m $\Rightarrow$ $r = 30$ m

$\therefore$ $OA = OB = AB = 30$ m

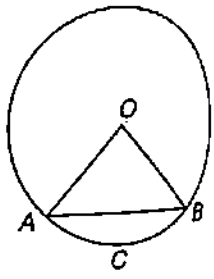
$\therefore$ ΔOAB is an equilateral triangle.

Thus, $\angle AOB = 60° = \frac{\pi}{3}$ rad

Now, since $s = \theta$ $r = \frac{\pi}{3} \times 30$

$= 10 \times 3.142$

$= 31.42$ m (approx.)



Thus the length of minor arc $ACB = 31.42$ m and the length of major arc

$= (2\pi r -$ minor arc)

$= 2\pi \times 30 - 31.42 = 157.1$ m (approx.)

EXAMPLE 5 *An arc AB of a circle subtends an angle x radians at the centre O of the circle. Given that the area of the sector AOB is equal to the square of the length of the arc AB, find the value of x.*

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

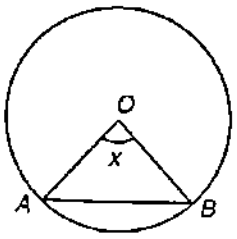
SOLUTION Let arc $AB = s$

Given that area of the sector

$$AOB = (\text{arc } AB)^2$$

$$\therefore \quad \frac{1}{2}r^2x = s^2 \quad \text{and} \quad s = rx \quad (\because \theta = x)$$

$$\therefore \quad \frac{1}{2}r^2x = r^2x^2 \quad \text{or} \quad x = \frac{1}{2} \text{ radian}$$



EXAMPLE 6 *Two arcs of two different circles are of equal lengths. If these arcs subtends angles of 45° and 60° at the centres of the circles. Find the ratio of the radii of the two circles.*

SOLUTION $45° = \frac{\pi}{4}$ rad and $60° = \frac{\pi}{3}$ rad

Let r_1 and r_2 be the radii of the two circles and s be the length of each arc.

$$\therefore \quad s = r_1\frac{\pi}{4} = r_2\frac{\pi}{3} \qquad (\because s = r\theta)$$

$$\Rightarrow \quad \frac{r_1}{r_2} = \frac{4}{3}$$

Hence the required ratio of radii = 4 : 3

EXAMPLE 7 *The wheel of a railway carriage is 4 ft in diameter and makes 6 revolution per second, how fast is the train going ? ($\pi = 3.14$)*

SOLUTION Radius (r) of the wheel = 2 ft

$\therefore$ Circumference of wheel = $2 \times 3.14 \times 2 = 12.56$ ft

Since the wheel makes 6 revolutions per second,

$\therefore$ Distance traversed in 1 second = $6 \times 12.56 = 75.36$ ft.

Hence the velocity of the train = 75.36 ft/s

$$= \frac{75.36 \times 60 \times 60}{3 \times 1760}$$

$$= 51.38 \text{ mile/h}$$

EXAMPLE 8 *A horse trots uniformly along a circular track of radius 27 m. The angle subtended at the centre of the track by the arc passed over by the horse in 3 seconds is 70°. What distance will the horse pass over in $\frac{1}{2}$ minute.*

SOLUTION $r = 27$ m

$$\theta = 70° = \frac{7\pi}{18} \text{ radian}$$

$$\therefore \quad s = r\theta = 27 \times \frac{7\pi}{18} = 33 \text{ m (approx.)}$$

$\therefore$ Distance passed over by the horse in $\frac{1}{2}$ minute

$$= \frac{33}{3} \times 30 = 330 \text{ m (approx.)}$$

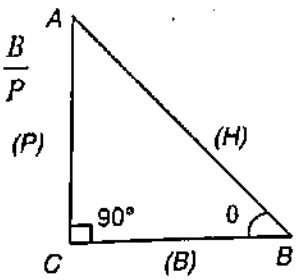
INTRODUCTORY EXERCISE-11.1

1. The perimeter of a certain sector of a circle is equal to the length of the arc of the semicircle having the same radius. The angle of the sector (approx.) is :
   (a) 65° 27′ 16″ (b) 68° 18′ 19″
   (c) 56° 52′ 18″ (d) none of these
2. The length of a pendulum is 8 m while the pendulum swings through 1.5 rad, find the length of the arc through which the tip of the pendulum passes :
   (a) 8 m (b) 9 m
   (c) 12 m (c) none of these
3. The angles of a triangle are in AP and the greatest angle is 75°. Find all the three angles in degrees :
   (a) 55°, 55°, 70° (b) 45°, 60°, 75°
   (c) 40°, 65°, 75° (d) none of these
4. A circular wire of radius 2.5 cm is cut and bent so as to lie along the circumference of a hoop whose radius is 1.29 m. Find (in degrees) the angle which is subtended at the centre of the hoop :
   (a) 9.67° (b) 7.69°
   (c) 6.97° (d) none of these
5. A circle is drawn on AB as diameter. The centre of the circle is O and the length $AB = 13$ cm. P is a point on the circumference of the circle such that the chord $AP = 12$ cm. Calculate the value of the angles PAB and POB in radians :
   (a) 0.395, 0.789 (b) 0.786, 0.735
   (c) 0.398, 0.689 (d) 0.786, 0.753

TRIGONOMETRICAL RATIOS

$$\sin\theta = \frac{P}{H}, \cos\theta = \frac{B}{H}, \tan\theta = \frac{P}{B}$$

$$\text{cosec}\,\theta = \frac{H}{P}, \sec\theta = \frac{H}{B}, \cot\theta = \frac{B}{P}$$



Remember :

sin θ	cos θ	tan θ	
P	B	P	⇒ Pandit Badri Prasad
H	H	B	⇒ Hari Hari Bol
cosec θ	sec θ	cot θ	

$P \rightarrow$ Perpendicular (AC)
$B \rightarrow$ Base (BC)
$H \rightarrow$ Hypotenuse (AB)

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Thus, $\sin\theta \operatorname{cosec}\theta = 1$, $\cos\theta \sec\theta = 1$, $\tan\theta \cot\theta = 1$ and $\tan\theta = \frac{\sin\theta}{\cos\theta}$, $\cot\theta = \frac{\cos\theta}{\sin\theta}$

NOTE $(\sin\theta)^{-1}$ is not written as $\sin^{-1}\theta$. Thus $\sin^{-1}\theta \neq (\sin\theta)^{-1}$ etc but $(\sin\theta)^2 = \sin^2\theta$ and $(\sin\theta)^3 = \sin^3\theta$

$\sin^2\theta + \cos^2\theta = 1$

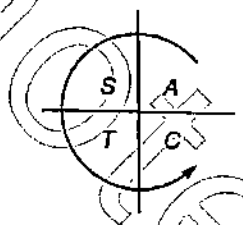
$\sec^2\theta - \tan^2\theta = 1$

$\operatorname{cosec}^2\theta - \cot^2\theta = 1$

SIGN OF TRIGONOMETRICAL FUNCTIONS

sin and cosec +ve	All +ve
tan and cot +ve	cos and sec +ve

+ve → (Positive)



A → All

S → sin/cosec

T → tan/cot

C → cos/sec

Remember : "Add Sugar To Coffee"

Values of T-Ratios :

Angle	$\theta = 0°$	$\theta = 30°$	$\theta = 45°$	$\theta = 60°$	$\theta = 90°$	$\theta = 180°$
sin	0	1/2	$1/\sqrt{2}$	$\sqrt{3}/2$	1	0
cos	1	$\sqrt{3}/2$	$1/\sqrt{2}$	1/2	0	−1
tan	0	$1/\sqrt{3}$	1	$\sqrt{3}$	∞	0
cosec	∞	2	$\sqrt{2}$	$2/\sqrt{3}$	1	∞
sec	1	$2/\sqrt{3}$	$\sqrt{2}$	2	∞	−1
cot	∞	$\sqrt{3}$	1	$1/\sqrt{3}$	0	∞

Remember :

$\operatorname{cosec}\theta = \frac{1}{\sin\theta}$, $\sec\theta = \frac{1}{\cos\theta}$, $\cot\theta = \frac{1}{\tan\theta}$

and $\tan\theta = \frac{\sin\theta}{\cos\theta}$, $\cot\theta = \frac{\cos\theta}{\sin\theta}$

and $\tan\theta = \frac{1}{\cot\theta}$ or $\cot\theta = \frac{1}{\tan\theta}$

Range of Trigonometric Ratios :

(i) $-1 \le \sin\theta \le 1 \Rightarrow |\sin\theta| \le 1$

(ii) $-1 \le \cos\theta \le 1 \Rightarrow |\cos\theta| \le 1$

(iii) $\operatorname{cosec}\theta \le -1$ and $\operatorname{cosec}\theta \ge 1 \Rightarrow |\operatorname{cosec}\theta| \ge 1$

(iv) $\sec\theta \le -1$ and $\sec\theta \ge 1 \Rightarrow |\sec\theta| \ge 1$

(v) $-\infty < \tan\theta < \infty$ *i.e.*, $\tan\theta$ may assume any value.

INCREASING AND DECREASING FUNCTIONS OF T-RATIOS

	1st quadrant	2nd quadrant	3rd quadrant	4th quadrant
sin θ	increases from 0 to 1	decreases from 1 to 0	decreases from 0 to −1	increases from −1 to 0
cos θ	decreases from 1 to 0	decreases from 0 to −1	increases from −1 to 0	increases from 0 to 1
tan θ	increases from 0 to ∞	increases from −∞ to 0	increases from 0 to ∞	increases from −∞ to 0

INTRODUCTORY EXERCISE-11.2

Directions for question number 1 to 7: Find the values of the following expressions.

1. $\frac{\sec^2\theta - 1}{\tan^2\theta} = ?$

(a) 1 (b) 2
(c) 3 (d) 4

2. $\sin^4\theta + \sin^2\theta\cos^2\theta = ?$

(a) 0 (b) 1
(c) 2 (d) $\sin^2\theta$

3. $\frac{\sin\theta \operatorname{cosec}\theta \tan\theta \cot\theta}{\sin^2\theta + \cos^2\theta} = ?$

(a) 0 (b) 1
(c) −1 (d) 2

4. $\sin^2 A \cot^2 A + \cos^2 A \tan^2 A = ?$

(a) −1 (b) 0
(c) 3 (d) 1

5. $\tan\theta + \cot\theta$ is :

(a) 1 (b) $\tan\theta$
(c) $\operatorname{cosec}\theta\cot\theta$ (d) $\sec\theta\operatorname{cosec}\theta$

6. $\frac{3 - 4\sin^2\theta}{\cos^2\theta} + \tan^2\theta$ is :

(a) 1 (b) 2
(c) 3 (d) none of these

7. $\frac{\cot A + \tan B}{\cot B + \tan A}$ is :

(a) $\tan A \cot B$ (b) $\cot A \tan B$
(c) 1 (d) none of these

8. If $\sin\theta = \frac{21}{29}$, find the value of $\sec\theta + \tan\theta$, if θ lies between 0 and $\pi/2$:

(a) 1 (b) $\pi/2$
(c) 5/2 (d) none of these

9. If A is in the fourth quadrant and $\cos A = \frac{5}{13}$, find the value of $\frac{13\sin A + 5\sec A}{5\tan A + 6\operatorname{cosec} A}$:

(a) −2/37 (b) −3/27
(c) 2/37 (d) can't be determined

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

10. Find the value of,

$\frac{4}{3}\cot^2 30° + 3\sin^2 60° - 2\,\text{cosec}^2 60° - \frac{3}{4}\tan^2 30°$:

(a) 10 / 3 (b) 11 / 3
(c) 4 (d) none of these

11. Find the perimeter of a regular octagon inscribed in a circle of radius 100 cm :

(a) 532.87 cm (b) 612.32 cm
(c) 378.32 cm (d) 875 cm

12. Find the altitude and base of an isosceles triangle whose vertical angle is 65° and whose equal sides are 415 cm :

(a) 350 cm, 646 cm (b) 350 cm, 446 cm
(c) 630 cm, 445 cm (d) none of these

13. If $5\sin^2\theta + 3\cos^2\theta = 4$, find the value of $\sin\theta$ and $\cos\theta$:

(a) $\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}}$ (b) $\pm\frac{\sqrt{3}}{2}, \pm\sqrt{2}$
(c) $\frac{\sqrt{3}}{2}, \frac{1}{\sqrt{2}}$ (d) none of these

14. Find the value of $\sqrt{\frac{1+\cos\theta}{1-\cos\theta}} + \sqrt{\frac{1-\cos\theta}{1+\cos\theta}}$

(a) $2\sec\theta$ (b) $\sec\theta$
(c) $2\,\text{cosec}\,\theta$ (d) none of these

15. If $\tan\theta = \frac{a}{b}$, find the value of $\frac{a\sin\theta - b\cos\theta}{a\sin\theta + b\cos\theta}$

(a) $\frac{a^2 - b^2}{a^2 + b^2}$ (b) $\frac{b^2 - a^2}{b^2 + a^2}$
(c) $\frac{a^2 + b^2}{a^2 - b^2}$ (d) none of these

Direction for question number 16-18 : For the every $0, 0 < \theta \le 90$, find the values of the following angles.

16. $\tan\theta + \cot\theta = 2$:

(a) 30° (b) 45°
(c) 60° (d) none of these

17. $2\sin^2\theta + 4\cos^2\theta = 3$:

(a) 30° (b) 60°
(c) 45° (d) none of these

18. $2\sin^2\theta - 3\sin\theta = -1$:

(a) 30° (b) 45°
(c) 90° (d) both (a) and (c)

TRIGONOMETRICAL RATIOS OF NEGATIVE AND ASSOCIATED ANGLES

Angle	$-\theta$	$(90-\theta)$	$(90+\theta)$	$(180-\theta)$	$(180+\theta)$	$(360-\theta)$	$(360+\theta)$
$\sin\theta$	$-\sin\theta$	$\cos\theta$	$\cos\theta$	$\sin\theta$	$-\sin\theta$	$-\sin\theta$	$\sin\theta$
$\cos\theta$	$\cos\theta$	$\sin\theta$	$-\sin\theta$	$-\cos\theta$	$-\cos\theta$	$\cos\theta$	$\cos\theta$
$\tan\theta$	$-\tan\theta$	$\cot\theta$	$-\cot\theta$	$-\tan\theta$	$\tan\theta$	$-\tan\theta$	$\tan\theta$

SUM, DIFFERENCE AND PRODUCT FORMULAE

1. $\sin(A+B) = \sin A\cos B + \cos A\sin B$
2. $\sin(A-B) = \sin A\cos B - \cos A\sin B$
3. $\cos(A+B) = \cos A\cos B - \sin A\sin B$
4. $\cos(A-B) = \cos A\cos B + \sin A\sin B$
5. $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A\tan B}$
6. $\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A\tan B}$
7. $\sin(A+B)\sin(A-B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$
8. $\cos(A+B)\cos(A-B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$
9. $2\sin A\cos B = \sin(A+B) + \sin(A-B)$
10. $2\cos A\sin B = \sin(A+B) - \sin(A-B)$
11. $2\cos A\cos B = \cos(A+B) + \cos(A-B)$
12. $2\sin A\sin B = \cos(A-B) - \cos(A+B)$
13. $\cot(A+B) = \frac{\cot A\cot B - 1}{\cot A + \cot B}$
14. $\cot(A-B) = \frac{\cot A\cot B + 1}{\cot B - \cot A}$
15. $\sin C + \sin D = 2\sin\frac{C+D}{2}\cos\frac{C-D}{2}$
16. $\sin C - \sin D = 2\cos\frac{C+D}{2}\sin\frac{C-D}{2}$
17. $\cos C + \cos D = 2\cos\frac{C+D}{2}\cos\frac{C-D}{2}$
18. $\cos C - \cos D = 2\sin\frac{C+D}{2}\sin\frac{D-C}{2}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

INTRODUCTORY EXERCISE-11.3

1. If $A + B = 45°$, find the value of $\tan A + \tan B + \tan A \tan B$:
   (a) -1 (b) $\frac{1}{2}$
   (c) $\sqrt{3}$ (d) 1

2. Find the value of $\tan 75°$:
   (a) $\frac{\sqrt{3}+1}{2\sqrt{2}}$ (b) $\frac{\sqrt{3}}{2+\sqrt{2}}$
   (c) $\frac{\sqrt{3}-1}{2\sqrt{2}}$ (d) $\frac{2+\sqrt{2}}{\sqrt{3}}$

3. Find the value of $\cos 28° \cdot \cos 32° - \sin 28° \cdot \sin 32°$:
   (a) 1 (b) 1/2
   (c) 1/3 (d) can't be determined

4. If $\cos A = \frac{3}{5}$ and $\sin B = \frac{5}{13}$, find the value of $\frac{\tan A + \tan B}{1 - \tan A \tan B}$:
   (a) $\frac{63}{16}$ (b) $\frac{36}{16}$
   (c) $\frac{61}{36}$ (d) none of these

5. Find the value of
   $\sin^2(120° - A) + \sin^2 A + \sin^2(120° + A)$:
   (a) $\frac{2}{3}$ (b) $\frac{3}{2}$
   (c) $\frac{5}{2}$ (d) $\frac{\sqrt{3}}{2}$

6. If $\cos(\alpha + \beta) = \frac{4}{5}$ and $\sin(\alpha - \beta) = \frac{5}{13}$ and α, β lie between 0° and 45°, find the value of $\tan 2\alpha$:
   (a) $\frac{56}{63}$ (b) $\frac{56}{33}$
   (c) $\frac{25}{33}$ (d) $\frac{65}{63}$

7. Find the maximum and minimum values of $7\cos\theta + 24\sin\theta$:
   (a) 25 and -25 (b) 16 and -9
   (c) 25 and 0 (d) 36 and 25

8. Find the value of $2\cos 45° \cdot \cos 15°$:
   (a) $\frac{(\sqrt{3}+1)}{2\sqrt{2}}$ (b) $\frac{(\sqrt{3}-1)}{2}$
   (c) $\frac{\sqrt{3}+1}{2}$ (d) none of these

9. Find the value of $\frac{\sin 75° + \sin 15°}{\cos 75° + \cos 15°}$:
   (a) 1 (b) 2
   (c) $\frac{\sqrt{3}}{2}$ (d) can't be determined

10. Find the value of $\cos 20° + \cos 100° + \cos 140°$:
    (a) -1 (b) 0
    (c) 1 (d) none of these

11. $\cos 20° \cdot \cos 40° \cdot \cos 60° \cdot \cos 80°$ is :
    (a) $\frac{1}{16}$ (b) $\frac{1}{50}$
    (c) $\frac{1}{24}$ (d) $\frac{1}{25}$

12. What is the value of;
    $\frac{\sin(90° - \theta)\sec(180° - \theta)\sin(-\theta)}{\sin(180° + \theta)\cot(360° - \theta)\operatorname{cosec}(90° + \theta)}$:
    (a) $\sin\theta$ (b) $\cos\theta$
    (c) 1 (d) $\frac{1}{\sqrt{2}}$

13. If $\sin(\alpha + \beta) = \frac{4}{5}$, $\sin(\alpha - \beta) = \frac{5}{13}$, find the value of $\tan 2\alpha$:
    (a) $\frac{16}{36}$ (b) $\frac{63}{16}$
    (c) $\frac{36}{16}$ (d) none of these

14. Find the value of $\frac{\cos 2B - \cos 2A}{\sin 2A + \sin 2B}$:
    (a) $\tan(A - B)$ (b) $\cos(A - B)$
    (c) $\cot(A - B)$ (d) $\tan(A + B)$

15. $\frac{\sin A + \sin 3A + \sin 5A + \sin 7A}{\cos A + \cos 3A + \cos 5A + \cos 7A}$ is :
    (a) $\tan 2A$ (b) $\tan 3A$
    (c) $\tan 4A$ (d) $\tan 8A$

TRIGONOMETRICAL RATIOS OF MULTIPLE AND SUB-MULTIPLE ANGLES

1. $\sin 2A = 2\sin A\cos A = \frac{2\tan A}{1+\tan^2 A}$

2. $\cos 2A = \cos^2 A - \sin^2 A = 1 - 2\sin^2 A$
   $= 2\cos^2 A - 1 = \frac{1-\tan^2 A}{1+\tan^2 A}$

3. $\tan 2A = \frac{2\tan A}{1-\tan^2 A}$

4. $\sin 3A = 3\sin A - 4\sin^3 A$

5. $\cos 3A = 4\cos^3 A - 3\cos A$

6. $\tan 3A = \frac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}$

7. $\sin A = 2\sin\left(\frac{A}{2}\right)\cos\left(\frac{A}{2}\right) = \frac{2\tan(A/2)}{1+\tan^2(A/2)}$

8. $\cos A = \cos^2\left(\frac{A}{2}\right) - \sin^2\left(\frac{A}{2}\right)$ etc.

9. $\tan A = \frac{2\tan(A/2)}{1-\tan^2(A/2)}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

INTRODUCTORY EXERCISE-11.4

1. If $\sin\theta = \frac{4}{5}$, find the value of $\sin 2\theta$:
   (a) 24/25 (b) 16/25
   (c) 9/20 (d) none of these

2. $\frac{\sin 2A}{1+\cos 2A}$ is :
   (a) $\tan 2A$ (b) $\cos 2A$
   (c) $\tan A$ (d) none of these

3. $\cos^4\theta - \sin^4\theta$ is :
   (a) $\cos 2\theta$ (b) $\sin 2\theta$
   (c) $\tan 2\theta$ (d) none of these

4. If $\cot x = \frac{a}{b}$, find the value of $a\cos 2x + b\sin 2x$
   (a) a (b) b
   (c) $\sin a$ (d) a/b

5. Find the value of $\cot 3\theta$:
   (a) $\frac{\cot^3\theta - 3\cot\theta}{3\cot^2\theta}$ (b) $\frac{\cot^3\theta - 3\cot\theta}{3\cot^2\theta - 1}$
   (c) $\frac{3\cot\theta - \cot^3\theta}{3\cot^2\theta - 1}$ (d) none of these

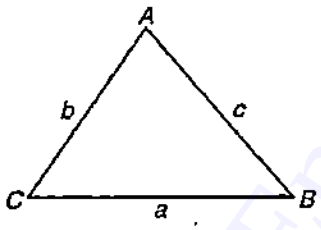
6. Find the value of $\sin\left(22\frac{1}{2}\right)^\circ$:
   (a) $\frac{\sqrt{(2-\sqrt{2})}}{2}$ (b) $\frac{\sqrt{(2-\sqrt{3})}}{3}$
   (c) $\frac{\sqrt{2-\sqrt{2}}}{4}$ (d) none of these

7. If $2\cos\theta = x + \frac{1}{x}$, find the value of $2\cos 3\theta$
   (a) $x^3 + \frac{1}{x^3}$ (b) $x^2 + \frac{1}{x^2}$
   (c) $x^3 - \frac{1}{x^3}$ (d) $\sqrt{x + \frac{1}{x}}$

8. Find the value of $\tan A + \tan B + \tan C$, if $A + B + C = \pi$:
   (a) $\tan A \tan B \tan C$ (b) 1
   (c) 0 (d) $\cot A \cot B \cot C$

PROPERTIES OF TRIANGLES

A triangle ABC has three angles and three sides. The three angles are denoted by A, B and C while the three sides are denoted by a, b and c respectively. The area of the triangle is denoted by Δ and its perimeter is denoted by $2s$, so that $a+b+c=2s$. Thus 's' is the semiperimeter of the triangle and $s=\frac{a+b+c}{2}$. Also we denote the circumradius by R and inradius by r.



1. **The law of sines :** In any triangle the sides are proportional to the sines of the opposite angles *i.e.*,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

2. **The law of cosines :** The square on any side of a triangle is equal to the sum of the squares on the other two sides, minus twice the product of those two sides and the cosine of the included angle *i.e.*,

$$a^2 = b^2 + c^2 - 2bc\cos A$$
$$b^2 = c^2 + a^2 - 2ca\cos B$$
$$c^2 = a^2 + b^2 - 2ab\cos C$$

3. **Area of a triangle :**

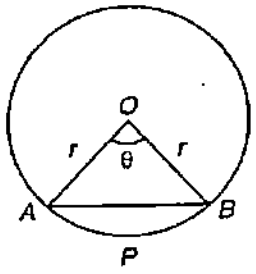
(i) $\Delta = \frac{1}{2}bc\sin A = \frac{1}{2}ca\sin B = \frac{1}{2}ab\sin C$

(ii) $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$ (Hero's formula)

4. **The area of a segment of a circle :**

Area of the segment APB
= Area of the sector AOB
− Area of ΔAOB

$$= \frac{1}{2}r^2\theta - \frac{1}{2}r^2\sin\theta$$
$$= \frac{1}{2}r^2(\theta - \sin\theta)$$



5. **Circumradius of a circle (R) :**

(i) $R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C}$

(ii) $R = \frac{abc}{4\Delta}$

6. **Inradius of a circle (r)**

$$r = \frac{\Delta}{s} \qquad \left(s = \frac{a+b+c}{2}\right)$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

INTRODUCTORY EXERCISE-11.5

1. If $\cos B = \frac{\sin A}{2\sin C}$, find the nature of triangle :
   (a) isosceles (b) scalene
   (c) can't be determined (d) none of these

2. If the angles of a triangle are in the ratio 1 : 2 : 3 and its circumradius is 10 cm, find the sides of the triangle (in cm) :
   (a) 8, 16, 24 (b) 18, 12, 6
   (c) 10, 20, $20\sqrt{3}$ (d) 10, 20, $10\sqrt{3}$

3. If $a = 1$, $b = \sqrt{3}$ and $A = 30°$, find the value of the angle B:
   (a) 60° (b) 30°
   (c) 120° (d) either (a) or (c)

4. Find B and C of a triangle ABC, if $b = 2$ cm, $c = 1$ cm and $A = 60°$:
   (a) 90° and 45° (b) 90° and 30°
   (c) 60° and 30° (d) none of these

5. In a triangle ABC, $a = 1$ cm, $b = \sqrt{3}$ cm and $C = \frac{\pi}{6}$, find the third side of the triangle :
   (a) 1 cm (b) 1.5 cm
   (c) 2 cm (d) $3\sqrt{3}$ cm

6. If the angles of a triangle are in the ratio 1 : 2 : 3, find the ratio between corresponding sides :
   (a) 3 : 2 : 1 (b) $1 : \sqrt{3} : 2$
   (c) 6 : 3 : 2 (d) $1 : \sqrt{2} : \sqrt{3}$

7. If $a = 2b$ and $A = 3B$, find the angles A, B, C of ΔABC :
   (a) 90°, 30°, 60° (b) 30°, 60°, 90°
   (c) 45°, 45°, 90° (d) data insufficient

8. If in any triangle $a = 13$ cm, $b = 14$ cm and $c = 15$ cm, find the inradius of the circle :
   (a) 2 cm (b) 6.5 cm
   (c) 4 cm (d) 7 cm

9. In a right angled triangle $a^2 + b^2 + c^2$ is :
   (a) R^2 (b) $4R^2$
   (c) $8R^2$ (d) none of these

10. Abhinav and Brijesh started walking simultaneously @ 3 km/h and 4 km/h from the same point 'C' on two different paths which diverge from each other at an angle of 120°. They walked for 6 hours and they stopped at A and B respectively. What is the least possible distance between Abhinav and Brijesh when they are at A and B respectively.
    (a) $36\sqrt{37}$ (b) $6\sqrt{37}$ km
    (c) $12\sqrt{37}$ (d) none of these

HEIGHT AND DISTANCE

Angle of Elevation : $\angle OPM$ is called as the angle of elevation.

Suppose a person P is looking an object O which is at a higher level than P, then the angle θ ($\angle OPM$) is the angle of elevation for that person.

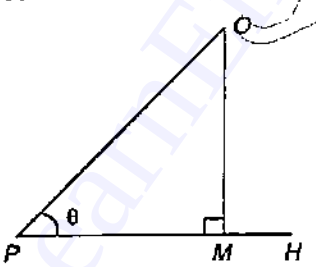


Fig. (i)

Angle of Depression : $\angle OPM$ in fig (ii) is called as angle of depression.

Suppose a person P is looking downward an object O, which is at a lower level than P, then the angle θ ($\angle OPM$) is called as the angle of depression.

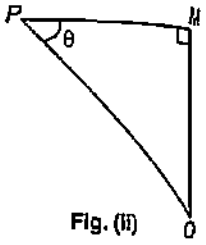
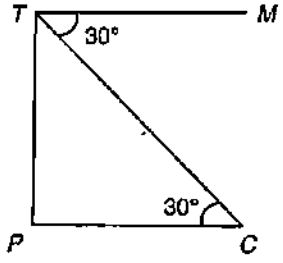


Fig. (ii)

NOTE It should be noted that the angle of elevation of one position as seen from the other is equal to the angle of depression of the latter as seen from the former.

EXAMPLE 1 *From the top of a tower 180 m high, it was observed that the angle of depression of the bottom of a cat sitting on the ground was 30°. Find the distance of the cat from the foot of the tower.*

SOLUTION Let PT be the tower and cat is sitting at C



then $\angle MTC = \angle TCP$

Now in ΔCPT, $\angle TCP = 30°$

$\therefore \quad \tan 30° = \frac{TP}{PC}$

$\frac{1}{\sqrt{3}} = \frac{180}{PC}$

$\Rightarrow \quad PC = 180\sqrt{3}$ m

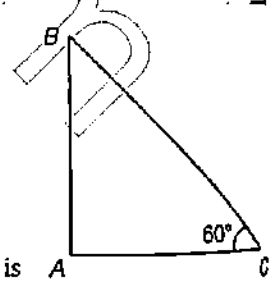
$= 311.76$ m

Hence cat is 311.76 m away from the foot of the tower.

NOTE For a particular case, the angle of depression is equal to the angle of elevation.

EXAMPLE 2 *The angle of elevation of the top of a tower at a distance of 100 m from its foot on a horizontal plane is found to be 60°. Find the height of the tower.*

SOLUTION Let AB be the tower



then $\tan 60° = \frac{AB}{AC}$

$\sqrt{3} = \frac{AB}{100}$

$AB = 100\sqrt{3}$ m $= 173.2$ m

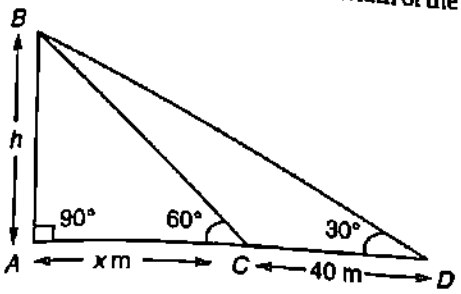
Thus the height of the tower is 173.2 m.

EXAMPLE 3 *A person standing on the bank of a river observes that the angle subtended by a tree on the opposite bank is 60°, when he retreats 40 m from the bank he finds the angle to be 30°. Find the height of the tree and the breadth of the river.*

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

SOLUTION Let height of the tree be AB and width of the river be AC.



In ΔABC, $\tan 60° = \frac{h}{x}$

$$\sqrt{3} = \frac{h}{x} \Rightarrow h = x\sqrt{3} \quad \ldots(1)$$

and in ΔABD, $\frac{AB}{AD} = \tan 30°$

$$\frac{h}{(x+40)} = \frac{1}{\sqrt{3}} \quad \ldots(2)$$

From equation (1) and (2), we get

$$\frac{x\sqrt{3}}{(x+40)} = \frac{1}{\sqrt{3}}$$

$$x = 20 \text{ m}$$

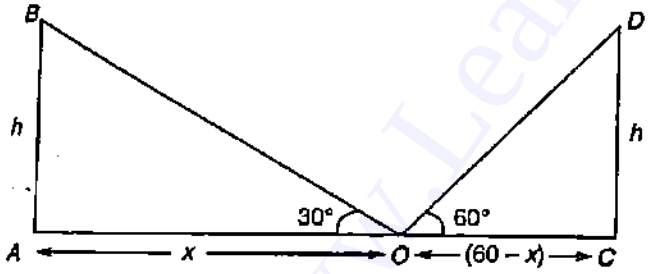
$\therefore$ $h = x\sqrt{3} = 20\sqrt{3} = 34.64$ m (approx.)

Thus the height of the tree is 34.46 m and breadth of the river is 20 m.

EXAMPLE 4 *Two towers of the same height stand on either side of a road 60 m wide. At a point on the road between the towers, the elevations of the towers are 60° and 30°. Find the height of the towers and the position of the point.*

SOLUTION In ΔAOB

$$\frac{AB}{AO} = \frac{h}{x} = \frac{1}{\sqrt{3}}$$



and in ΔCOD, $\frac{CD}{OC} = \frac{h}{(60-x)} = \sqrt{3}$ ...(1)

from equation (1) and (2)

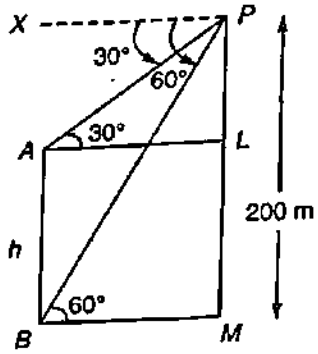
$$\frac{x}{\sqrt{3}(60-x)} = \sqrt{3} \Rightarrow x = 45 \text{ m}$$

$\therefore$ $h = \frac{45}{\sqrt{3}} = 15\sqrt{3} = 15 \times 1.732 = 25.98$ m

Hence, the height of the towers is 25.98m and the distance of the point 'O' from A is 45 m and C is 15 m.

EXAMPLE 5 *From the top of a cliff, 200 m high, the angle of depression of the top and bottom of a tower are observed to be 30° and 60°, find the height of the tower.*

SOLUTION Let the height of the tower be h m, then



In ΔPBM, $\tan 60° = \frac{PM}{BM}$

$$\sqrt{3} = \frac{200}{BM}$$

$BM = \frac{200}{\sqrt{3}}$ m $= AL$ (also)

Now, in ΔALP,

$$\tan 30° = \frac{PL}{AL}$$

$$\frac{1}{\sqrt{3}} = \frac{PL}{(200/\sqrt{3})}$$

$\Rightarrow$ $PL = \frac{200}{3}$

$\therefore$ $LM = PM - PL = 200 - \left(\frac{200}{3}\right)$

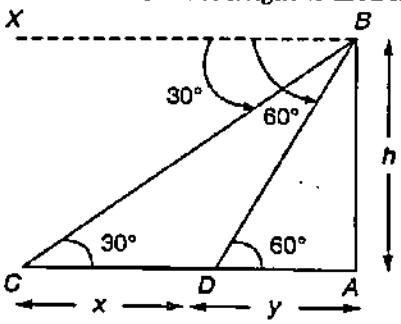
$$LM = \frac{400}{3}$$

But $LM = AB = h$

$\therefore$ Height of the tower $(h) = \frac{400}{3}$ m

EXAMPLE 6 *A man on the top of a rock rising on a seashore observes a boat coming towards it. If it takes 10 minutes for the angle of depression to change form 30° to 60°, how soon the boat reach the shore ?*

SOLUTION Let AB be the rock of height 'h' metres.



Let C and D be the two positions of boat such that

$$\angle ACB = \angle XBC = 30°$$

and $$\angle ADB = \angle XBD = 60°$$

Let $CD = x$ m and $AD = y$ m

Now, in ΔABD, $\tan 60° = \frac{h}{y}$

$$\sqrt{3} = \frac{h}{y} \quad \ldots(1)$$

and in ΔABC, $\tan 30° = \frac{h}{x+y} = \frac{1}{\sqrt{3}}$ ...(2)

from equation (1) and (2), we get

$$\frac{\sqrt{3}y}{x+y} = \frac{1}{\sqrt{3}} \Rightarrow x = 2y \Rightarrow y = \frac{x}{2}$$

Since the boat takes 10 minutes to cover x m, hence it will take $\frac{10}{2} = 5$ minutes to cover $y = \frac{x}{2}$ metres. Thus the required time

$= 5$ minutes.

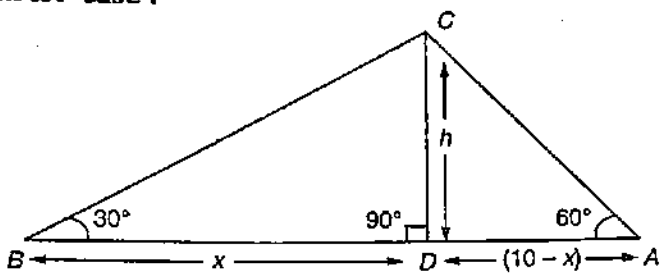
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

EXAMPLE 7 *The angles of elevation of an aeroplane from two places 10 km apart are found to be 60° and 30° respectively. Find the height of the aeroplane.*

SOLUTION Let A and B be the two places such that $AB = 10$ kms and let C be the position of the aeroplane at a height of h metres above AB. Let CD be perpendicular to AB.

In Ist case :



In ΔBCD, $\frac{h}{x} = \frac{1}{\sqrt{3}}$ ...(1)

and $\frac{h}{10 - x} = \sqrt{3}$ ...(2)

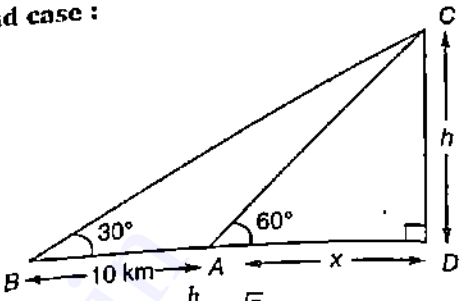
from equation (1) and (2)

$$\frac{x}{\sqrt{3}(10 - x)} = \sqrt{3} \Rightarrow x = 7.5 \text{ km}$$

$$\therefore \quad h = \frac{7.5}{\sqrt{3}} = \frac{7.5}{3} \times \sqrt{3} = 2.5 \times \sqrt{3}$$

$$= 4.33 \text{ km (approx.)}$$

In IInd case :



In ΔACD, $\frac{h}{x} = \sqrt{3}$ ...(1)

In ΔBCD, $\frac{h}{10 + x} = \frac{1}{\sqrt{3}}$ ...(2)

Therefore from equation (1) and (2)

$$h = 5\sqrt{3} \text{ km} \Rightarrow h = 8.66 \text{ km}$$

INTRODUCTORY EXERCISE-11.6

1. Two towers of the same height stand on opposite sides of a road 100 m wide. At a point on the road between the towers, the elevations of the towers are 30° and 45°. Find the height of the towers and the position of the point from the nearest tower :
   (a) 36.6 m and 63.4 m (b) 63.3 m and 63.4 m
   (c) 66.3 m and 63.4 m (d) 36.6 m and 86.4 m
2. The elevation of a tower at a station A due north of it is 45° and at a station B due west of A is 30°. If $AB = 40$ m, find the height of the tower :
   (a) 26.26 m (b) 28.28 m
   (c) 38.5 m (d) none of these
3. If the shadow of a tower is 30 m when the sun's altitude is 30° what is the length of the shadow when the sun's altitude is 60° ?
   (a) $10\sqrt{3}$ m (b) 20 m
   (c) 10 m (d) 12 m
4. A helicopter is in a stationary position at a certain height over the lake. At a point 200 m above the surface of the lake, the angle of elevation of the helicopter is 45°. At the same time, the angle of depression of its reflection in the lake is 75°. Calculate the height of the helicopter from the surface of the lake :
   (a) $\frac{200}{\sqrt{3}}$ m (b) $200\sqrt{3}$ m
   (c) 200 m (d) 150 m
5. A vertical tower stands on a horizontal plane and surmounted by a flagstaff of height 'h'. At a point on the plane, the angle of elevation of bottom of the flagstaff is α and that of the top of the flagstaff is β. Find the height of the tower :
   (a) $h \tan \alpha$ (b) $\frac{h \tan \alpha}{\tan \alpha - \tan \beta}$
   (c) $\frac{h \tan \alpha}{\tan \beta - \tan \alpha}$ (d) $\frac{\tan \alpha + \tan \beta}{h \tan \alpha}$
6. A round balloon of radius 'r' subtends an angle α at the eye of the observer, while the angle of elevation of its centre is β. Find the height of the centre of balloon.
   (a) $r \operatorname{cosec}\left(\frac{\alpha}{2}\right) \sin \beta$ (b) $r \sin \alpha \operatorname{cosec} \beta$
   (c) $\frac{r}{2} \sin \beta$ (d) $r \sec\left(\frac{\alpha}{2}\right)$
7. At the foot of a mountain the elevation of its summit is 45°, after ascending 1000 m towards the mountain up a stop of 30° inclination, the elevation is found to be 60°. Find the height of the mountain :
   (a) 1.3 km (b) 1.366 km
   (c) 2.72 km (d) none of these
8. An aeroplane when 3000 m high passes vertically above another aeroplane at an instant when their angles of elevation at the same observing point are 60° and 45° respectively. How many metres higher is the one than the other?
   (a) 1248 m (b) 1188 m
   (c) 1752 m (d) 1268 m
9. The angle of elevation of a cloud from a height h above the level of water in a lake is α and the angle of depression of its image in the lake is β. Find the height of the cloud above the surface of the lake :
   (a) $\frac{h \sin(\beta - \alpha)}{\sin(\alpha + \beta)}$ (b) $h \sin \alpha$
   (c) $\frac{h \sin(\alpha + \beta)}{\sin(\beta - \alpha)}$ (d) none of these
10. A palm tree 90 ft high, is broken by the wind and its upper part meet the ground at an angle of 30°. Find the distance of the point where the top of the tree meets the ground , from its root :
    (a) 43.69 ft (b) 51.96 ft
    (c) 60 ft (d) 30 ft

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

EXERCISE

LEVEL 1

1. If $0 < \theta < 90°$, then $(\sin\theta + \cos\theta)$ is :
   (a) less than 1 (b) equal to 1
   (c) greater than 1 (d) greater than 2

2. The value of x satisfying the equation $\sin x + \frac{1}{\sin x} = \frac{7}{2\sqrt{3}}$ is :
   (a) 10° (b) 30°
   (c) 45° (d) 60°

3. If $\sin\theta - \cos\theta = 0$ and $0 < \theta \le \pi/2$, then θ is equal to :
   (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$
   (c) $\frac{\pi}{6}$ (d) 0

4. Given that θ is acute and then $\sin\theta = \frac{3}{5}$. Let x, y be positive real numbers such that $3(x - y) = 1$, then one set of solutions for x and y expressed in terms of θ is given by :
   (a) $x = \sec\theta, y = \text{cosec}\,\theta$ (b) $x = \cot\theta, y = \tan\theta$
   (c) $x = \text{csoec}\,\theta, y = \cot\theta$ (d) $x = \sec\theta, y = \tan\theta$

5. Which one of the following pairs is correctly matched ?

If	then
(a) $x = \frac{1 + \sin 60° - \cos 60°}{1 + \sin 60° + \cos 60°}$	$x = \tan 60°$
(b) $x = \frac{1 + \sin 90° - \cos 90°}{1 + \sin 90° - \cos 90°}$	$x = \tan 30°$
(c) $x = \frac{2\tan 30°}{1 - \tan^2 30°}$	$x = \tan 60°$
(d) $x = \frac{1 - \tan^2 30°}{1 + \tan^2 30°}$	$x = \cos 60°$

6. If $x \tan 45° \cdot \cos 60° = \sin 60° \cot 60°$, then x is equal to
   (a) 1 (b) 1/2
   (c) $\sqrt{3}$ (d) $1/\sqrt{2}$

7. If θ lies in the second quadrant, then $\sqrt{\frac{1 - \sin\theta}{1 + \sin\theta}} + \sqrt{\frac{1 + \sin\theta}{1 - \sin\theta}}$ is equal to :
   (a) $-2\sec\theta$ (b) $2\sec\theta$
   (c) $2\,\text{cosec}\,\theta$ (d) $2\tan\theta$

8. $\sin^6 A + \cos^6 A$ is equal to :
   (a) $1 - 3\sin^2 A\cos^2 A$ (b) $1 - 3\sin A\cos A$
   (c) $1 + 3\sin^2 A\cos^2 A$ (d) 1

9. If $\sec x = P$, $\text{cosec}\,x = Q$, then :
   (a) $P^2 + Q^2 = PQ$ (b) $P^2 + Q^2 = P^2Q^2$
   (c) $P^2 - Q^2 = P^2Q^2$ (d) $P^2 + Q^2 = -P^2Q^2$

10. $\sin^2 A\cos^2 B - \cos^2 A\sin^2 B$ simplifies to :
    (a) $\sin^2 A + \sin^2 B$ (b) $\cos^2 A + \cos^2 B$
    (c) $\sin^2 A - \sin^2 B$ (d) $\sin^2 A - \cos^2 B$

11. If $\sin 2x = n\sin 2y$, then the value of $\frac{\tan(x + y)}{\tan(x - y)}$ is :
    (a) $\frac{n + 1}{n - 1}$ (b) $\frac{n - 1}{n + 1}$
    (c) $\frac{1 - n}{n + 1}$ (d) $\frac{1 + n}{1 - n}$

12. The least value of $2\sin^2\theta + 3\cos^2\theta$ is :
    (a) 1 (b) 2
    (c) 3 (d) 5

13. The value of $\tan(180 + \theta)\cdot\tan(90 - \theta)$ is :
    (a) 1 (b) −1
    (c) 0 (d) none of these

14. $\log\tan 1° + \log\tan 2° + \ldots + \log\tan 89°$ is :
    (a) 1 (b) $1/\sqrt{2}$
    (c) 0 (d) −1

15. If we convert $\sin(-566°)$ to same trigonometrical ratio of a positive angle lying between 0° and 45°, then we get :
    (a) $\cos 26°$ (b) $-\cos 26°$
    (c) $\sin 26°$ (d) $-\sin 26°$

16. From the mast head of ship, the angle of depression of a boat is 60°. If the mast head is 150 m, then the distance of the boat from the ship is :
    (a) 86.6 m (b) 68.6 m
    (c) 66.8 m (d) none of these

17. A portion of a 30 m long tree is broken by tornado and the top struck up the ground making an angle 30° with ground level. The height of the point where the tree is broken is equal to :
    (a) $30/\sqrt{3}$ m (b) 10 m
    (c) $30\sqrt{3}$ m (d) 60 m

18. Two posts are 25 m and 15 m high and the line joining their tips makes an angle of 45° with horizontal. The distance between these posts is :
    (a) 5 m (b) $10/\sqrt{2}$ m
    (c) 10 m (d) $10\sqrt{2}$ m

19. The angle of elevation of the top of a tower at a point G on the ground is 30°. On walking 20 m towards the tower the angle of elevation becomes 60°. The height of the tower is equal to :
    (a) $10/\sqrt{3}$ m (b) $20\sqrt{3}$ m
    (c) $20/\sqrt{3}$ m (d) $10\sqrt{3}$ m

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

20. If $x = \sec\theta + \tan\theta$, $y = \sec\theta - \tan\theta$, then the relation between x and y is :
(a) $x^2 + y^2 = 0$ (b) $x^2 = y^2$
(c) $x^2 = y$ (d) $xy = 1$

21. The value of θ for which $\sqrt{3}\cos\theta + \sin\theta = 1$ is :
(a) 0 (b) $\pi/3$
(c) $\pi/6$ (d) $\pi/2$

22. If $\tan\theta = \frac{4}{3}$, then the value of $\sqrt{\frac{1+\cos\theta}{1-\cos\theta}}$ is :
(a) 1 (b) 2
(c) 3 (d) 4

23. If the arcs of the same length in two circles subtend angles of 60° and 90° at the centre, then the ratio of their radii is :
(a) $\frac{1}{3}$ (b) $\frac{1}{2}$
(c) $\frac{3}{2}$ (d) 2

24. In the third quadrant, the values of $\sin\theta$ and $\cos\theta$ are :
(a) positive and negative respectively
(b) negative and positive respectively
(c) both positive
(d) both negative

25. The value of $\frac{\cot 40°}{\tan 50°} - \frac{1}{2}\frac{\cos 35°}{\sin 55°}$ is :
(a) 1 (b) -1
(c) $\frac{1}{2}$ (d) $-\frac{1}{2}$

26. The value of $\theta (0 \le \theta \le \pi/2)$ satisfying the equation $\sin^2\theta - 2\cos\theta + \frac{1}{4} = 0$ is :
(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$
(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$

27. If $\cos\theta = \frac{4}{5}$ and $0 < \theta < 90°$, then the value of $\frac{3\cos\theta + 2\,\text{cosec}\,\theta}{4\sin\theta - \cot\theta}$ is :
(a) $-\frac{43}{2}$ (b) $-\frac{41}{2}$
(c) $\frac{43}{8}$ (d) $-\frac{43}{6}$

28. Maximum value of $(\cos\theta - \sin\theta)$ is :
(a) $\sqrt{2}$ (b) 1
(c) $\frac{1}{2}$ (d) $\frac{1}{\sqrt{2}}$

29. The value of sin 105° is :
(a) $\frac{\sqrt{3}-1}{2\sqrt{2}}$ (b) $\frac{\sqrt{3}-1}{\sqrt{2}}$
(c) $\frac{\sqrt{3}+1}{2\sqrt{2}}$ (d) $\frac{\sqrt{3}+1}{\sqrt{2}}$

30. If $\tan\theta = t$, then $\sin 2\theta$ is equal to :
(a) $\frac{1}{1+t^2}$ (b) $\frac{2t}{1+t^2}$
(c) $\frac{t^2}{1+t}$ (d) $\frac{1+t^2}{1+t}$

31. If $\tan\theta = \sqrt{2}$, then the value of θ is :
(a) less than $\frac{\pi}{4}$ (b) equal to $\frac{\pi}{4}$
(c) between $\frac{\pi}{4}$ and $\frac{\pi}{3}$ (d) greater than $\frac{\pi}{3}$

32. If $\tan\theta = 2 - \sqrt{3}$, then $\tan(90 - \theta)$ is equal to :
(a) $2 + \sqrt{3}$ (b) $2 - \sqrt{3}$
(c) $3 + \sqrt{2}$ (d) $3 - \sqrt{2}$

33. If from point 100 m above the ground the angles of depression of two objects due south on the ground are 60° and 45°, then the distance between the objects is :
(a) $\frac{50(3-\sqrt{3})}{3}$ m (b) $\frac{50(3+\sqrt{3})}{3}$ m
(c) $\frac{100(3+\sqrt{3})}{3}$ m (d) $\frac{100(3-\sqrt{3})}{3}$ m

34. If the length of shadow of a vertical pole on the horizontal ground is $\sqrt{3}$ times of its height, then the angle of elevation of sun is :
(a) 15° (b) 30°
(c) 45° (d) 60°

35. A kite is flown with a thread of 250 m length. If the thread is assumed to be stretched and makes an angle of 60° with the horizontal, then the height of the kite above the ground is (approx.) :
(a) 216.25 m (b) 215. 25 m
(c) 212.25 m (d) 210.25 m

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Answers

INTRODUCTORY EXERCISE- 11.1

1. (a)	2. (c)	3. (b)	4. (c)	5. (a)					

INTRODUCTORY EXERCISE- 11.2

1. (a)	2. (d)	3. (b)	4. (d)	5. (d)	6. (c)	7. (b)	8. (c)	9. (a)	10. (a)
11. (b)	12. (b)	13. (a)	14. (c)	15. (a)	16. (b)	17. (c)	18. (d)		

INTRODUCTORY EXERCISE- 11.3

1. (d)	2. (a)	3. (b)	4. (a)	5. (b)	6. (b)	7. (a)	8. (c)	9. (a)	10. (b)
11. (a)	12. (a)	13. (b)	14. (a)	15. (c)					

INTRODUCTORY EXERCISE- 11.4

1. (a)	2. (c)	3. (a)	4. (a)	5. (b)	6. (a)	7. (a)	8. (a)		

INTRODUCTORY EXERCISE- 11.5

1. (a)	2. (d)	3. (d)	4. (b)	5. (a)	6. (b)	7. (a)	8. (c)	9. (c)	10. (b)

INTRODUCTORY EXERCISE- 11.6

1. (a)	2. (b)	3. (c)	4. (b)	5. (c)	6. (a)	7. (b)	8. (d)	9. (c)	10. (b)

LEVEL-1

1. (c)	2. (d)	3. (b)	4. (c)	5. (c)	6. (a)	7. (b)	8. (a)	9. (b)	10. (c)
11. (a)	12. (b)	13. (b)	14. (c)	15. (c)	16. (a)	17. (b)	18. (c)	19. (d)	20. (d)
21. (d)	22. (b)	23. (c)	24. (d)	25. (c)	26. (b)	27. (c)	28. (b)	29. (c)	30. (b)
31. (c)	32. (a)	33. (d)	34. (b)	35. (a)					

www.LearnEngineering.in
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Hints & Solutions

LEVEL 1

1. Let $z = \sin\theta + \cos\theta$

$\Rightarrow z^2 = 1 + \sin 2\theta$

$\because 0 < \theta < 90°$ so $\sin 2\theta < 1$, so that $z^2 < 2$,

Thus $z < \sqrt{2}$ i.e., z is greater than 1.

2. Go through option.

3. $\sin\theta - \cos\theta = 0$

$\Rightarrow \sin\theta = \cos\theta$

$\Rightarrow \tan\theta = 1 \Rightarrow \theta = \frac{\pi}{4}$

4. Go through option.

HINT $\sin\theta = \frac{3}{5}$ and $\operatorname{cosec}\theta = \frac{5}{3}$

7. $\sqrt{\frac{1-\sin\theta}{1+\sin\theta}} + \sqrt{\frac{1+\sin\theta}{1-\sin\theta}} = \frac{(1-\sin\theta)+(1+\sin\theta)}{\cos\theta}$

$= 2\sec\theta$

8. $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$

Let $a = \sin^2\theta$, $b = \cos^2\theta$, so that

$a + b = \sin^2\theta + \cos^2\theta = 1$

$\Rightarrow \sin^6\theta + \cos^6\theta = 1 - 3\sin^2\theta\cos^2\theta$

9. $\cos x = \frac{1}{P}$ and $\sin x = \frac{1}{Q}$

$1 = \cos^2 x + \sin^2 x = \frac{1}{P^2} + \frac{1}{Q^2}$

$\Rightarrow P^2 + Q^2 = P^2Q^2$

10. $\sin^2 A(1 - \sin^2 B) - (1 - \sin^2 A)\sin^2 B = \sin^2 A - \sin^2 B$

11. $\frac{n}{1} = \frac{\sin 2x}{\sin 2y}$

$\Rightarrow \frac{n+1}{n-1} = \frac{\sin 2x + \sin 2y}{\sin 2x - \sin 2y} = \frac{2\sin(x+y)\cos(x-y)}{2\cos(x+y)\sin(x-y)}$

$= \frac{\tan(x+y)}{\tan(x-y)}$

12. The value is least when $\theta = 90°$.

14. $\log\tan 1° + \log\tan 89° = \log\tan 1° + \log\tan(90-1)$

$= \log\tan 1° + \log\cot 1°$

$= \log\tan 1°.\cot 1° = \log 1 = 0$

Similarly, $\log\tan 2° + \log\tan 88° = 0$

Also, $\log\tan 45° = \log 1 = 0$

Thus, the value of the expression is 0.

15. $\sin(-566°) = -\sin(566°)$

$= -\sin(90 \times 6 + 26) = \sin 26°$.

www.LearnEngineering.in
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

12 GEOMETRY

This chapter tests nothing but your imagination. I think it is the drawing of MATHS. You need not to be an artist or painter, but at least you must be capable to visualize the geometrical figures. So before going for the tedious and complex geometrical problems it is prerequisite to be familiar with the fundamentals of the geometry. Basically it consists of a very large proportion of problems in CAT. Sometimes 8-10 problems related to this chapter in addition with mensuration are asked in CAT. Therefore it is advised that you must learn this chapter religiously for the sake of your score in CAT. Apart from CAT other entrance tests for MBA ask a plethora of questions as well.

Remember if you don't take seriously any theorem, axiom or concept you cannot perform well since the complexity of ideas, figures and concepts makes it a daunting task for you.

For the sake of convenience this chapter is divided into 5 parts.

(i) Lines, angles and planes
(ii) Triangles
(iii) Quadrilaterals
(iv) Polygons
(v) Circles and Loci

"Well begun is half done"–Anonymous

(1) LINE AND ANGLES

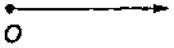
Point : The figure of which length, breadth and height cannot be measured is called a point. It is infinitesimal.

Line : A line is made up of a infinite number of points and it has only length *i.e.*, it does not has any thickness (or width). A line is endless *i.e.*, it can be extended in both directions.

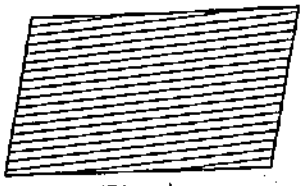
Line segment : A line segments has two end points, but generally speaking line segment is called a line.

(line segment $\overrightarrow{AB}$)

Ray : A ray extends indefinitely in one direction from any given point. This is exhibited by an arrow. The starting point is called as the initial point.



Plane : It is a flat surface, having length and breadth both, but no thickness. It is a two dimensional figure.



(Plane)

Type of Lines	Definition	Diagram
Parallel lines	Two lines, lying in a plane and having no common intersecting point are called parallel lines. The distance between two parallel lines is constant.	Symbol : $\parallel$
Perpendicular lines	Two lines, which lie in a single plane and intersect eachother at right angle, are called perpendicular lines.	Symbol : $\perp$
Concurrent lines	More than two straight lines intersecting at the same point.	

POINTS TO REMEMBER

1. A line contains infinitely many points.
2. The intersection of two different lines is a point.
3. Through a given point, there pass an infinite number of lines and these lines are called **concurrent lines.**
4. Only one line can pass through any two particular points.
5. When more than two points lie on a line, they are called as **collinear points** else they are called as **non-collinear** points.
6. Two lines can intersect maximum at one point. This point is called as **point of intersection** and these lines are called as **intersecting lines.**
7. There are an infinite number of planes which pass through a single (particular) point.
8. When more than three points lie in the same plane, they are called as **coplanar** else they are called as **non-coplanar.**
9. When more than one line lie in the same plane, then these lines are called as coplanar else they are called as non-coplanar.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

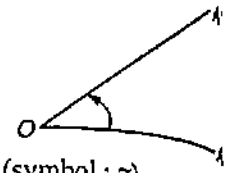
10. When two plane intersect each other, they form a line *i.e.*, intersecting region is a line.
11. Two different lines which are perpendicular to the same (a third line) line, necessarily parallel to each other, lying on the same plane.
12. When two or more parallel lines are intercepted by some other intercepting lines, then the ratio of corresponding intercepts are equal. An intercepting line is generally called as a transversal.

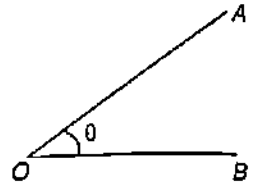
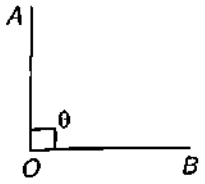
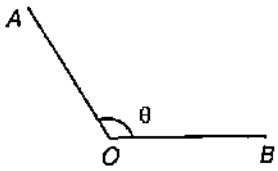
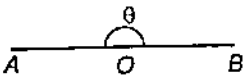
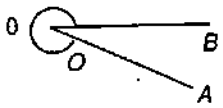
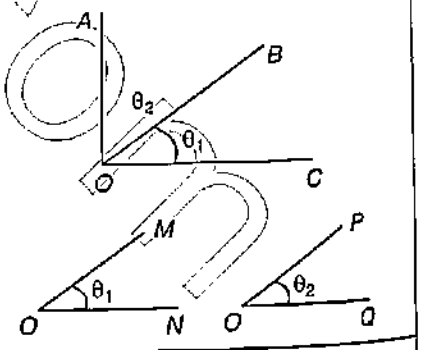
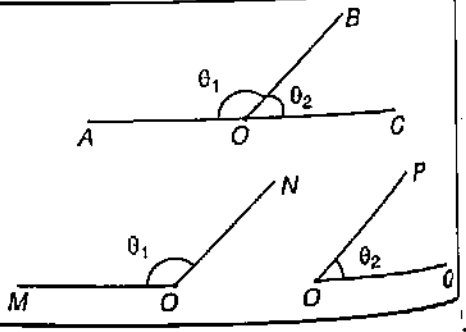


i.e., $\frac{AB}{BC} = \frac{DE}{EF}$

Angles : The amount of rotation about O, the vertex of the angle AOA', is called the magnitude of the angle.

$m\angle AOA'$ denotes the measure of $\angle AOA'$. Angles are said to be congruent when their measure is same. (symbol : $\cong$)

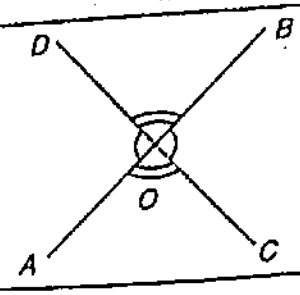
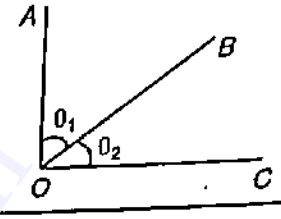
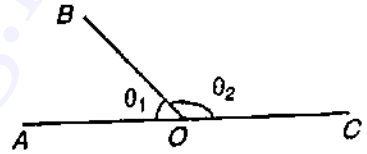
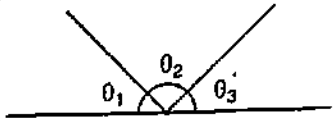
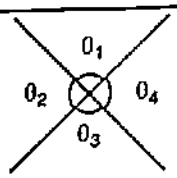
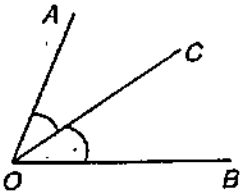
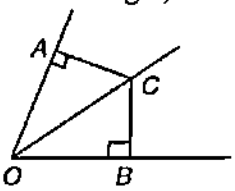
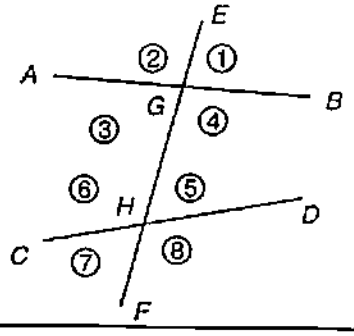


Types of Angles	Property	Diagram
Acute	$0° < \theta < 90°$ ($\angle AOB$ is an acute angle)	
Right	$\theta = 90°$ ($\angle AOB$ is a right angle)	
Obtuse	$90° < \theta < 180°$ ($\angle AOB$ is an obtuse angle)	
Straight	$\theta = 180°$ ($\angle AOB$ is a straight angle)	
Reflex	$180° < \theta < 360°$ ($\angle AOB$ is a reflex angle)	
Complementary	$\theta_1 + \theta_2 = 90°$ Two angles whose sum is 90°, are complementary to each other	
Supplementary	$\theta_1 + \theta_2 = 180°$ Two angles whose sum is 180°, are supplementary to each other.	

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

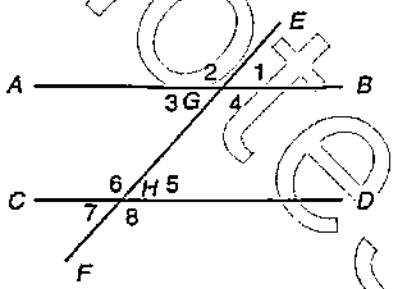
Types of Angles	Property	Diagram
Vertically opposite	$\angle DOA = \angle BOC$ and $\angle DOB = \angle AOC$	
Adjacent angles	$\angle AOB$ and $\angle BOC$ are adjacent angles Adjacent angles must have a common side. (*e.g.*, OB)	
Linear pair	$\angle AOB$ and $\angle BOC$ are linear pair angles. One side must be common (*e.g.*, OB) and these two angles must be supplementary.	
Angles on the one side of a line	$\theta_1 + \theta_2 + \theta_3 = 180°$	
Angles round the point	$\theta_1 + \theta_2 + \theta_3 + \theta_4 = 360°$	
Angle bisector	OC is the angle bisector of $\angle AOB$. *i.e.*, $\angle AOC = \angle BOC = \frac{1}{2}(\angle AOB)$ When a line segment divides an angle equally into two parts, then it is said to be the angle bisector (OC)	 (Angle bisector is equidistant from the two sides of the angle) *i.e.*,  $AC = BC$
Corresponding angles	When two lines are intersected by a transversal, then they form four pairs of corresponding angles (a) $\angle AGE, \angle CHG \Rightarrow (\angle 2, \angle 6)$ (b) $\angle AGH, \angle CHF \Rightarrow (\angle 3, \angle 7)$ (c) $\angle EGB, \angle GHD \Rightarrow (\angle 1, \angle 5)$ (d) $\angle BGH, \angle DHF \Rightarrow (\angle 4, \angle 8)$	
Exterior angles	These are following four angles (i) $\angle AGE \Rightarrow \angle 2$ (ii) $\angle CHF \Rightarrow \angle 7$ (iii) $\angle EGB \Rightarrow \angle 1$ (iv) $\angle DHF \Rightarrow \angle 8$	

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Types of Angles	Property	Diagram
Interior angles	These are following four angles (i) $\angle AGH \Rightarrow \angle 3$ (ii) $\angle GHC \Rightarrow \angle 6$ (iii) $\angle BGH \Rightarrow \angle 4$ (iv) $\angle DHG \Rightarrow \angle 5$	
Alternate angles	These are two pairs of angles as following : (i) $\angle AGH, \angle GHD$ $(\angle 3, \angle 5)$ (ii) $\angle GHC, \angle BGH$ $(\angle 6, \angle 4)$	

When two parallel lines are intersected by a transversal then



1. The pairs of corresponding angles so formed are congruent.

i.e., $\angle 2 = \angle 6$

$\angle 3 = \angle 7$

$\angle 1 = \angle 5$

$\angle 4 = \angle 8$

NOTE If one pair of corresponding angles is congruent, then the rest pairs of corresponding angles are also congruent.

2. The pairs of alternate angles so formed are congruent

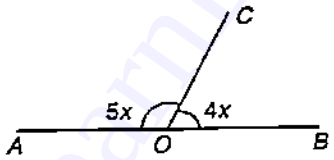
i.e., $\angle 3 = \angle 5$ and $\angle 4 = \angle 6$

3. The pair of interior angles (*i.e.*, the interior angles on the same side of a transversal) are supplementary.

NOTE The converse of all the 3 rules is also true *i.e.*, if the pair of corresponding angles, or alternate angles or interior angles is equal (*i.e.*, congruent) then the two line are parallel which are intersected by a transversal.

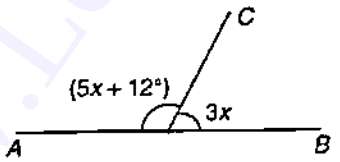
INTRODUCTORY EXERCISE-12.1

1. What is the value of x in the adjoining figure?



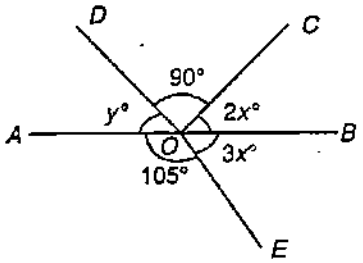
(a) 80
(b) 40
(c) 20
(d) 25

2. What is the value of x in the adjoining figure ?



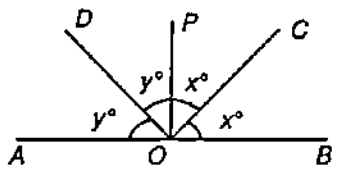
(a) 18°
(b) 20°
(c) 21°
(d) 24°

3. In the following figure AB is a straight line. Find $(x + y)$:



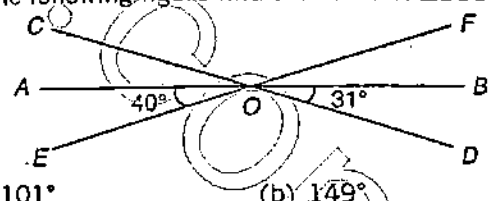
(a) 55°
(b) 65°
(c) 75°
(d) 80°

4. In the following figure $\angle BOP = 2x°$, $\angle AOP = 2y°$, OC and OD are angle bisectors of $\angle BOP$ and $\angle AOP$ respectively. Find the value of $\angle COD$:



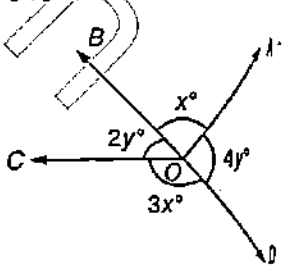
(a) 75°
(b) 90°
(c) 100°
(d) 120°

5. In the following figure find the value of $\angle BOC$:



(a) 101°
(b) 149°
(c) 71°
(d) 140°

6. Find y, if $x° = 36°$, as per the given diagram :



(a) 36°
(b) 16°
(c) 12°
(d) 42°

7. If $(2x + 17)°$, $(x + 4)°$ are complementary, find x:
(a) 63°
(b) 53°
(c) 35°
(d) 23°

8. If $(5y + 62)°$, $(22° \div y)$ are supplementary, find y:
(a) 16°
(b) 32°
(c) 8°
(d) 1°

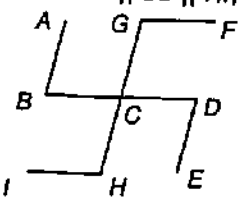
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

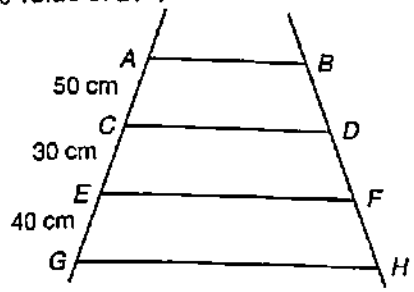
9. If two supplementary angles are in the ratio 13 : 5 find the greater angle :
(a) 130° (b) 65°
(c) 230° (d) 28°

10. An angle is 30° more than one half of its complement. Find the angle in degrees :
(a) 60° (b) 50°
(c) 45° (d) 80°

11. In the given diagram $AB \parallel GH \parallel DE$ and $GF \parallel BD \parallel HI$, $\angle FGC = 80°$. Find the value of $\angle CHI$:
(a) 80°
(b) 120°
(c) 100°
(d) 160°



12. In the given figure $AB \parallel CD \parallel EF \parallel GH$ and $BH = 100$ cm. Find the value of DF :



(a) 26 cm (b) 40 cm
(c) 25 cm (d) 24 cm

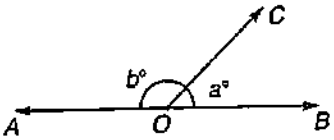
13. The complement of 65°50′ is :
(a) 24° 50′ (b) 24°10′
(c) 14° 50′ (d) 34° 10′

14. The supplement of 123° 45′ is :
(a) 56° 55′ (b) 56° 15′
(c) 55° 56′ (d) none of these

15. If two angles are complementary of each other, then each angle is :
(a) a right angle
(b) a supplementary angle
(c) an obtuse angle
(d) an acute angle

16. How many degrees are there in an angle which equals one-fifth of its supplement?
(a) 15° (b) 30°
(c) 75° (d) 150°

17. In the given figure, $\angle a$ is greater than one sixth of right angle, then :
(a) $b > 165°$
(b) $b < 165°$
(c) $b \leq 165°$
(d) $b \geq 165°$

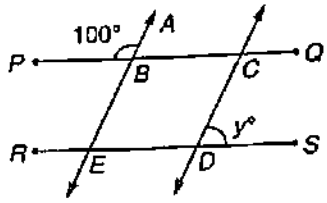


18. Let D be the mid-point of a straight line AB and let C be a point different from D such that $AC = BC$, then :
(a) $AC \perp AB$ (b) $\angle BDC = 90°$
(c) $\angle BDC$ is acute (d) $\angle BDC > 90°$

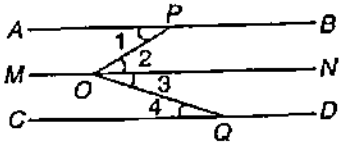
19. Answer on the basis of the following statements :
When two straight lines intersect, then :
1. adjacent angles are complementary
2. adjacent angles are supplementary
3. opposite angles are equal
4. opposite angles are supplementary
(a) 2 and 3 are correct (b) 1 and 4 are correct
(c) 1 and 3 are correct (d) 2 and 4 are correct

20. AB is a straight line and O is a point on AB, if a line OC is drawn not coinciding with OA or OB, then $\angle AOC$ and $\angle BOC$ are :
(a) equal (b) complementary
(c) supplementary (d) together equal to 100

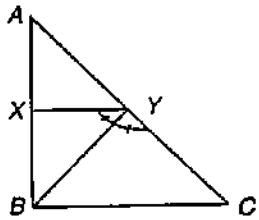
21. In the adjoining figure $AE \parallel CD$ and $BC \parallel ED$, then find Y :
(a) 60°
(b) 80°
(c) 90°
(d) 75°



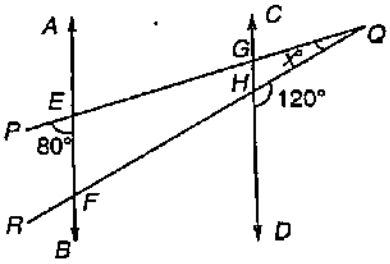
22. In the adjoining figure $\angle APO = 42°$ and $\angle CQO = 38°$. Find the value of $\angle POQ$:
(a) 68°
(b) 72°
(c) 80°
(d) 126°



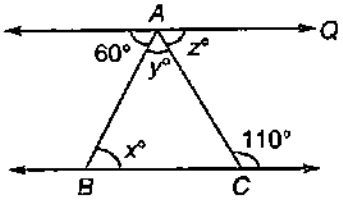
23. In a ΔABC, a line XY parallel to BC intersects AB at X and AC at Y :
If BY bisects $\angle XYC$, then $m\angle CBY : m\angle CYB$ is :
(a) 5 : 4
(b) 4 : 5
(c) 1 : 1
(d) 6 : 5



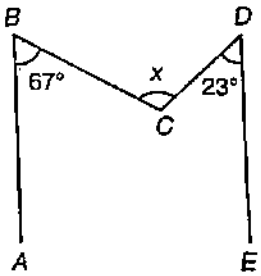
24. In the adjoining figure $AB \parallel CD$ and PQ, QR intersects AB and CD both at E, F and G, H respectively. Given that $m\angle PEB = 80°$, $m\angle QHD = 120°$ and $m\angle PQR = x°$, find the value of x :
(a) 40° (b) 20°
(c) 100° (d) 30°



25. In the following figure, find the value of y :
(a) 70°
(b) 60°
(c) 50°
(d) 80°



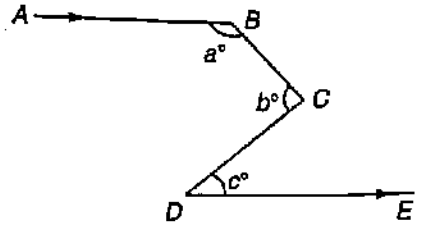
26. In the adjoining figure $AB \parallel DE$, $\angle ABC = 67°$ and $\angle EDC = 23°$. Find $\angle BCD$:
(a) 90°
(b) 44°
(c) 46°
(d) none of the above



For More Visit : www.LearnEngineering.in
Scanned by CamScanner

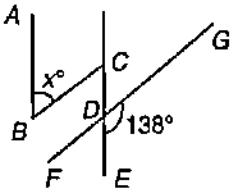
yourshahbood.wordpress.com
For More Visit : www.LearnEngineering.in

27. In the given figure $AB \parallel DE$. Find $a° + b° - c°$:



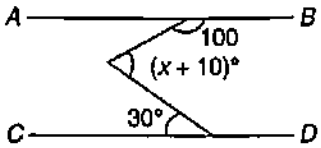
(a) 160°
(b) 120°
(c) 180°
(d) 210°

28. In the given figure $AB \parallel CE$ and $BC \parallel FG$. Find the value of $x°$:



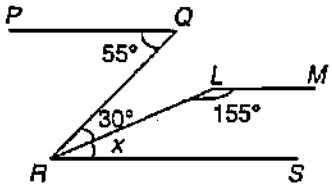
(a) 52°
(b) 32°
(c) 42°
(d) 36°

29. $AB \parallel CD$, shown in the figure. Find the value of x :



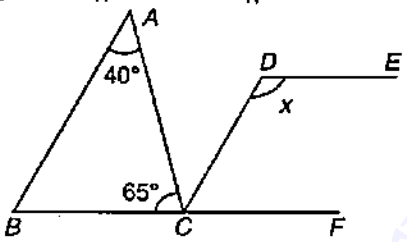
(a) 100°
(b) 90°
(c) 110°
(d) 140°

30. In the figure $PQ \parallel LM \parallel RS$. Find the value of $\angle LRS$:



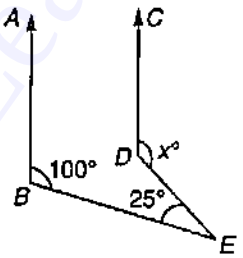
(a) 30°
(b) 25°
(c) 35°
(d) 40°

31. In the figure $AB \parallel DC$ and $DE \parallel BF$. Find the value of x :



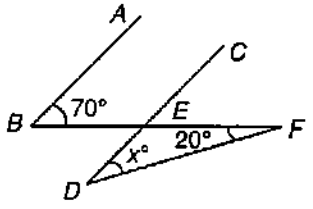
(a) 140°
(b) 155°
(c) 105°
(d) 115°

32. In the figure $AB \parallel CD$, $\angle ABE = 100$. Find $m\angle CDE$:



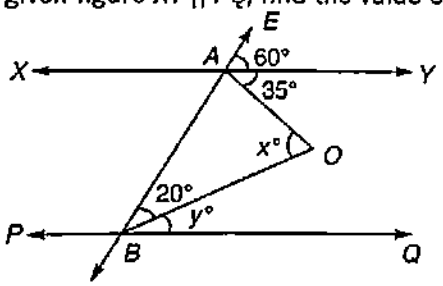
(a) 125°
(b) 55°
(c) 65°
(d) 75°

33. In the figure $AB \parallel CD$, find $x°$ (*i.e.*, $\angle CDF$) :



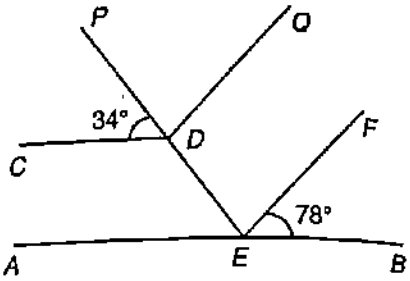
(a) 50°
(b) 90°
(c) 30°
(d) 70°

34. In the given figure $XY \parallel PQ$, find the value of x :



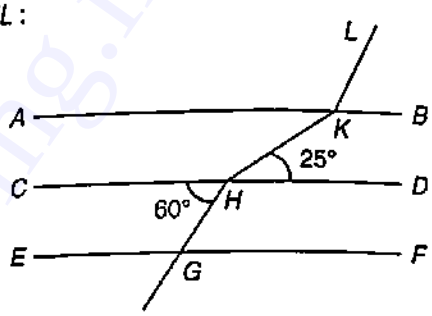
(a) 70°
(b) 40°
(c) 75°
(d) 15°

35. In the given figure $AB \parallel CD$ and $EF \parallel DQ$. Find the value of $\angle DEF$:



(a) 68°
(b) 78°
(c) 34°
(d) 39°

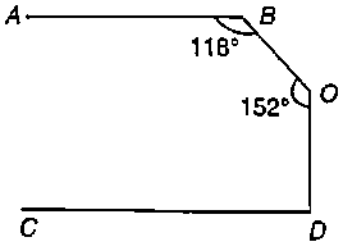
36. In the given figure $AB \parallel CD \parallel EF$ and $GH \parallel KL$. Find $m\angle HKL$:



(a) 85°
(b) 145°
(c) 120°
(d) 95°

37. Which one of the following statements is false?
(a) Two straight lines can intersect at only one point.
(b) Through a given point, only one straight line can be drawn.
(c) A line segment can be produced to any desired length.
(d) Through two given points, it is possible to draw one and only one straight line.

38. $AB \parallel CD$. $\angle ABO = 118°$, $\angle BOD = 152°$, find $\angle ODC$:



(a) 70°
(b) 80°
(c) 90°
(d) 34°

39. Two parallel lines AB and CD are intersected by a transversal EF at M and N respectively. The lines MP and NP are the bisectors of interior angles $\angle BMN$ and $\angle DNM$ on the same side of the transversal. Then $\angle MPN$ is equal to :
(a) 60°
(b) 90°
(c) 45°
(d) 120°

40. If the arms of one angle are respectively parallel to the arms of another angle, then the two angles are :
(a) either equal or supplementary
(b) neither equal not supplementary
(c) equal but not supplementary
(d) not equal but supplementary

www.LearnEngineering.in
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

41. AB and CD are two parallel lines. PQ cuts AB and CD at E and F respectively. EL is the bisector of $\angle BEF$. If $\angle LEB = 35°$ then $\angle CFQ$ will be :
(a) 70° (b) 55°
(c) 110° (d) 125°

42. A plane figure is bounded by straight lines only. If n is the number of these lines, then the least value of n is :
(a) 1 (b) 2
(c) 3 (d) 4

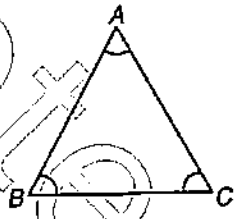
43. How many planes can pass through any three arbitrary points?
(a) 1 (b) 2
(c) 3 (d) 0

44. Out of the four arbitrary non-collinear points three points are taken at a time, then the number of planes that can be drawn through the three points is :
(a) 3 (b) 4
(c) 6 (d) 12

45. Maximum number of points of intersection of four lines on a plane is :
(a) 4 (b) 6
(c) 8 (d) 5

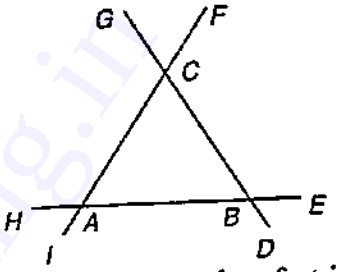
(2) TRIANGLES

Triangle : A three sided closed plane figure, which is formed by joining the three non-collinear points, is called as a triangle. It is denoted by the symbol Δ.



In the above Δ (triangle) ABC, A, B and C are three vertices, line segments AB, BC and AC are the three sides of the triangle. $\angle A$, $\angle B$ and $\angle C$ are the three **interior angles** of a triangle ABC.

In the adjoining figure $\angle FCB$, $\angle CBE$, $\angle ABD$, $\angle IAB$, $\angle HAC$, $\angle GCA$ are the **exterior angles** of the ΔABC.



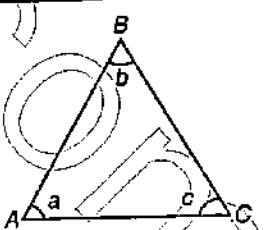
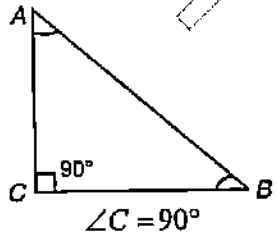
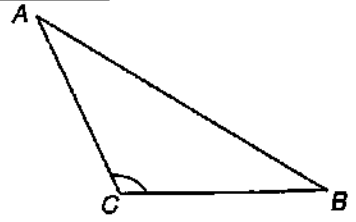
- Sum of the three interior angles of a triangle is always 180°.
- Exterior angle = Sum of two interior opposite angles
  e.g., $\angle CBE = \angle CAB + \angle BCA$

Perimeter of triangle is equal to sum of all the three sides
i.e., $a + b + c$

Semiperimeter of a triangle is half of the perimeter
i.e., $s = \frac{a+b+c}{2}$, a, b, c are the length of three sides of a triangle.

Types of Triangles

(A) According to interior angles

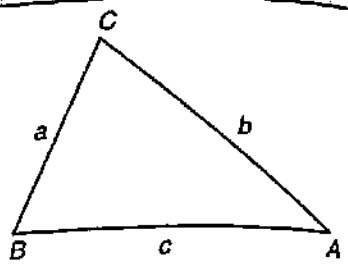
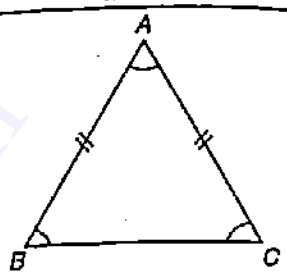
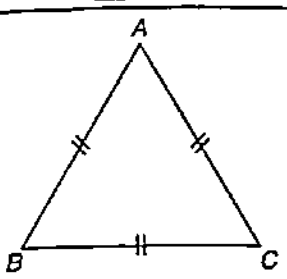
Types of Triangles	Property/Definition	Diagram
Acute angle triangle	Each of the angle of a triangle is less than 90° *i.e.*, $a < 90°$, $b < 90°$, $c < 90°$	 $\{\angle a, \angle b, \angle c\} < 90°$
Right angled triangle	One of the angle is equal to 90°, then it is called as right angled triangle. Rest two angles are complementary to each other.	 $\angle C = 90°$
Obtuse angle triangle	One of the angle is obtuse (*i.e.*, greater than 90°), then it is called as obtuse angle triangle.	 $\angle C > 90°$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

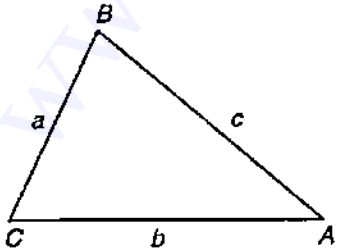
For More Visit : www.LearnEngineering.in

(B) According to the length of sides.

Types of triangles	Property/Definition	Diagram
Scalene triangle	A triangle in which none of the three sides is equal is called a scalene triangle (all the three angles are also different).	 $a \neq b \neq c$
Isosceles triangle	A triangles in which at least two sides are equal is called as isosceles triangle. In this triangle, the angles opposite to the congruent sides are also equal.	 $AB = AC$ $\angle B = \angle C$
Equilateral triangle	A triangle in which all the three sides are equal is called as, an equilateral triangle. In this triangle each angle is congruent and equal to 60°.	 $AB = BC = AC$ $\angle A = \angle B = \angle C = 60°$

FUNDAMENTAL PROPERTIES OF TRIANGLES

1. Sum of any two sides is always greater than the third side.
2. The difference of any two sides is always less than the third side.
3. Greater angle has a greater side opposite to it and smaller angle has a smaller side opposite to it *i.e.*, if two sides of a triangle are not congruent then the angle opposite to the greater side is greater.
4. Let a, b and c be the three sides of a ΔABC and c is the largest side. Then



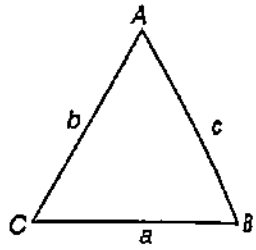
(i) if $c^2 < a^2 + b^2$, the triangle is **acute angle triangle**

(ii) if $c^2 = a^2 + b^2$, the triangle is **right angled triangle**

(iii) if $c^2 > a^2 + b^2$, the triangle is **obtuse angle triangle**

5. **Sine rule :** In a ΔABC, if a, b, c be the three sides opposite to the angles A, B, C respectively, then

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$



6. **Cosine rule :** In a ΔABC, if a, b, c be the sides opposite to angle A, B and C respectively, then

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc},$$

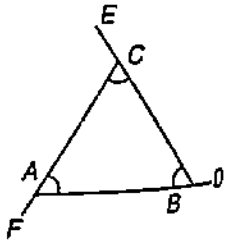
$$\cos B = \frac{c^2 + a^2 - b^2}{2ca}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

(These rules have been discussed already in trigonometry.)

7. The sum of all the three interior angles is always 180°

i.e., $\angle CAB + \angle ABC + \angle BCA = 180°$



For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

8. The sum of three (ordered) exterior angles of a triangle is 360°

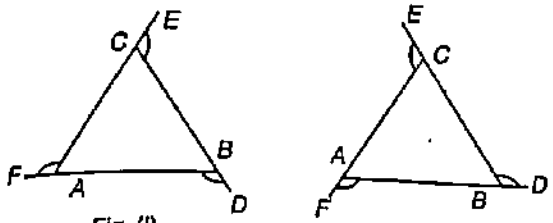
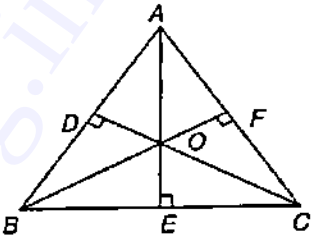
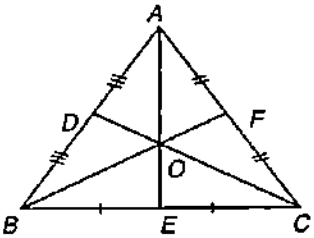
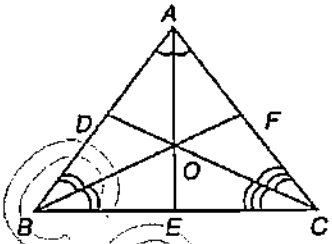
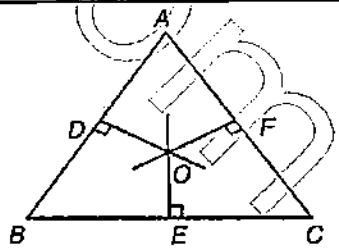
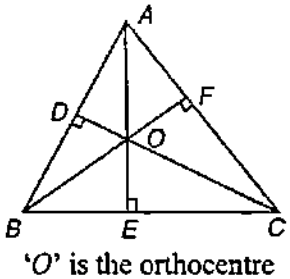


Fig. (i) Fig. (ii)

In fig. (i) : $(\angle FAC + \angle ECB + \angle DBA) = 360°$
In fig. (ii) : $(\angle FAB + \angle DBC + \angle ECA) = 360°$

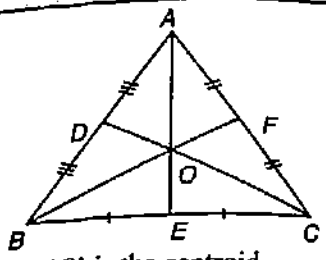
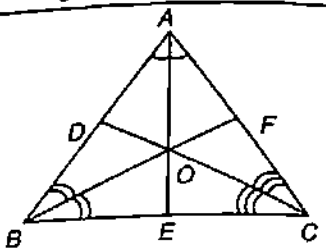
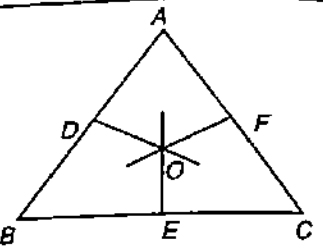
9. The sum of an interior angle and its adjacent exterior angle is 180°.
10. A triangle must has at least two acute angles.
11. In a triangle, the measure of an exterior angle equals the sum of the measures of the interior opposite angles.
12. The measure of an exterior angle of a triangle is greater than the measure of each of the opposite interior angles.

Some Important Definitions

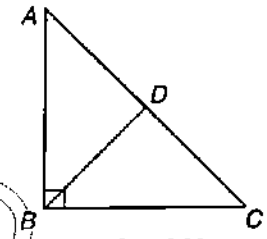
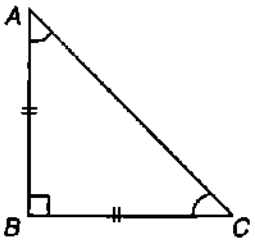
Nomenclature	Property/Definition	Diagram
Altitude (or height)	The perpendicular drawn from the opposite vertex of a side in a triangle is called an altitude of the triangle. ➢ There are three altitudes in a triangle.	 AE, CD and BF are the altitudes
Median	The line segment joining the mid-point of a side to the vertex opposite to the side is called a median. ➢ There are three medians in a triangle. ➢ A median bisects the area of the triangle *i.e.*, $A(ABE) = A(AEC) = \frac{1}{2} A(\Delta ABC)$ etc.	 AE, CD and BF are the medians $(BE = CE, AD = BD, AF = CF)$
Angle bisector	A line segment which originates from a vertex and bisects the same angle is called an angle bisector. $\left(\angle BAE = \angle CAE = \frac{1}{2} \angle BAC\right)$ etc.	 AE, CD and BF are the angle bisectors.
Perpendicular bisector	A line segment which bisects a side perpendicularly (i.e, at right angle) is called a perpendicular bisector of a side of triangle. ➢ All points on the perpendicular bisector of a line are equidistant from the ends of the line.	 DO, EO and FO are the perpendicular bisectors.
Orthocentre	The point of intersection of the three altitudes of the triangle is called as the orthocentre. $\angle BOC = 180 - \angle A$ $\angle COA = 180 - \angle B$ $\angle AOB = 180 - \angle C$	 'O' is the orthocentre

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Types of Triangles	Property/Definition	Diagram
Centroid	The point of intersection of the three medians of a triangle is called the centroid. A centroid divides each median in the ratio 2 : 1 (vertex : base) $\frac{AO}{OE}=\frac{CO}{OD}=\frac{BO}{OF}=\frac{2}{1}$	 'O' is the centroid.
Incentre	The point of intersection of the angle bisectors of a triangle is called the incentre. Incentre O is the always equidistant from all three sides *i.e.*, the perpendicular distance between the sides and incentre is always same for all the three sides.	 'O' is the incentre.
Circumcentre	The point of intersection of the perpendicular bisectors of the sides of a triangle is called the circumcentre. $OA = OB = OC$ = (circum radius) Circumcentre O is always equidistant from all the three vertices A, B and C.	 'O' is the circumcentre.

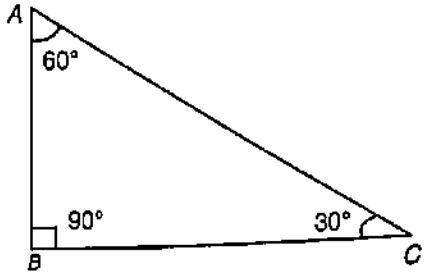
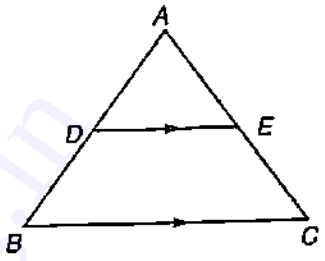
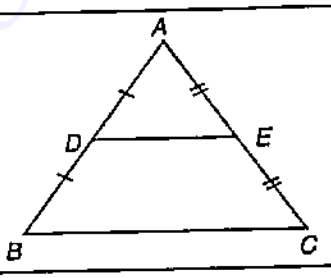
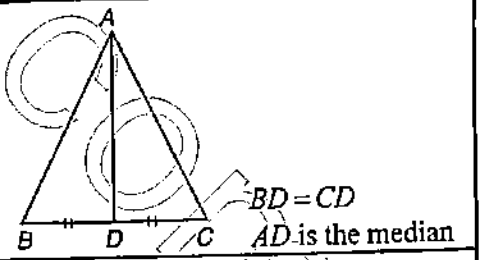
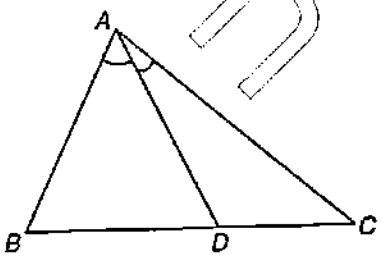
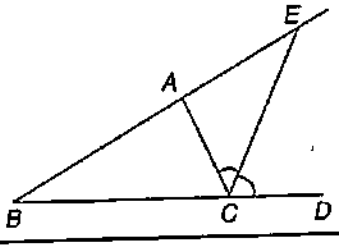
Important Theorems on Triangles

Theorem	Statement/Explanation	Diagram
Pythagoras theorem	The square of the hypotenuse of a right angled triangle is equal to the sum of the squares of the other two sides. *i.e.*, $(AC)^2 = (AB)^2 + (BC)^2$ ➢ The converse of this theorem is also true. ➢ The numbers which satisfy this relation, are called Pythagorean triplets. *e.g.*, (3, 4, 5), (5, 12, 13), (7, 24, 25), (8, 15, 17), (9, 40, 41), (11, 60, 61), (12, 35, 37), (16, 63, 65), (20, 21, 29), (28, 45, 53), (33, 56, 65) **Note :** All the multiples (or submultiples) of Pythagorean triplets also satisfy the relation. *e.g.*, (6, 8,10), (15,36, 39), (1.5, 2, 2.5) etc	 $\angle B = 90°$ $AC \rightarrow$ Hypotenuse $AD = CD = BD$ (D is the mid-point of AC)
$45° - 45° - 90°$ triangle theorem	If the angles of a triangle are 45°, 45° and 90°, then the hypotenuse (*i.e.*, longest side) is $\sqrt{2}$ times of any smaller side. ➢ Excluding hypotenuse rest two sides are equal. *i.e.*, $AB = BC$ and $AC = \sqrt{2}AB = \sqrt{2}BC$	 $\angle A = 45°$ $\angle B = 90°$ $\angle C = 45°$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

Theorem	Statement/Explanation	Diagram
$30°-60°-90°$ triangle theorem	If the angles of a triangle are 30°, 60° and 90°, then the sides opposite to 30° angle is half of the hypotenuse and the side opposite to 60° is $\frac{\sqrt{3}}{2}$ times the hypotenuse. *e.g.*, $AB = \frac{AC}{2}$ and $BC = \frac{\sqrt{3}}{2} AC$ $\therefore AB : BC : AC = 1 : \sqrt{3} : 2$	
Basic proportionality theorem (BPT) or Thales theorem	Any line parallel to one side of a triangle divides the other two sides proportionally. So if DE is drawn parallel to BC, it would divide sides AB and AC proportionally *i.e.*, $\frac{AD}{DB} = \frac{AE}{EC}$ or $\frac{AD}{AB} = \frac{AE}{AC}$ $\frac{AD}{DE} = \frac{AB}{BC} = \frac{AE}{DE} = \frac{AC}{BC}$	
Mid-point theorem	If the mid-points of two adjacent sides of a triangle are joined by a line segment, then this segment is parallel to the third side. *i.e.*, if $AD = BD$ and $AE = CE$ then $DE \parallel BC$	
Apollonius theorem	In a triangle, the sum of the squares of any two sides of a triangle is equal to twice the sum of the square of the median to the third side and square of half the third side. *i.e.*, $AB^2 + AC^2 = 2(AD^2 + BD^2)$	
Interior angle bisector theorem	In a triangle the angle bisector of an angle divides the opposite side to the angle in the ratio of the remaining two sides. *i.e.*, $\frac{BD}{CD} = \frac{AB}{AC}$ and $BD \times AC - CD \times AB = AD^2$	
Exterior angle bisector theorem	In a triangle the angle bisector of any exterior angle of a triangle divides the side opposite to the external angle in the ratio of the remaining two sides *i.e.*, $\frac{BE}{AE} = \frac{BC}{AC}$	

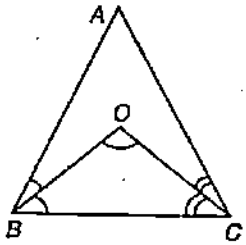
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

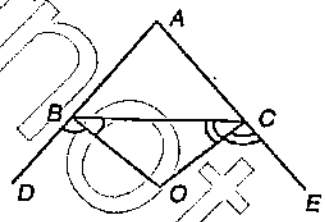
SOME USEFUL RESULTS

1. In a ΔABC, if the bisectors of $\angle B$ and $\angle C$ meet at O then

$$\angle BOC = 90° + \frac{1}{2}\angle A$$

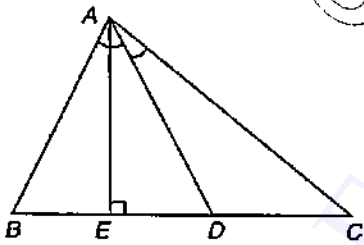


2. In a ΔABC, if sides AB and AC are produced to D and E respectively and the bisectors of $\angle DBC$ and $\angle ECB$ interseet at O, then

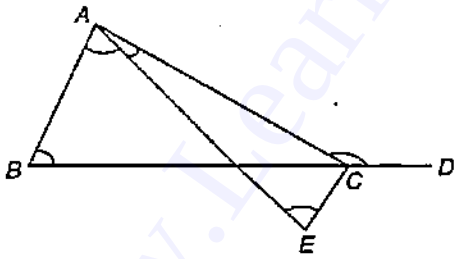


$$\angle BOC = 90° - \frac{1}{2}\angle A$$

3. In a ΔABC, if AD is the angle bisector of $\angle BAC$ and $AE \perp BC$, then $\angle DAE = \frac{1}{2}(\angle ABC - \angle ACB)$

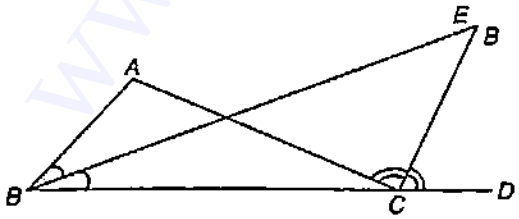


4. In a ΔABC, if BC is produced to D and AE is the angle bisector of $\angle A$, then



$$\angle ABC + \angle ACD = 2\angle AEC.$$

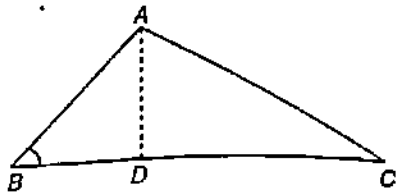
5. In a ΔABC, if side BC is produced to D and bisectors



of $\angle ABC$ and $\angle ACD$ meet at E, then

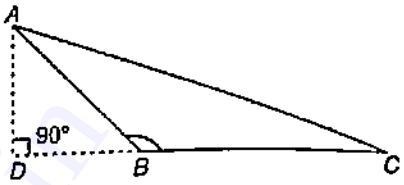
$$\angle BEC = \frac{1}{2}\angle BAC$$

6. In an acute angle ΔABC, AD is a perpendicular dropped on the opposite side of $\angle A$, then



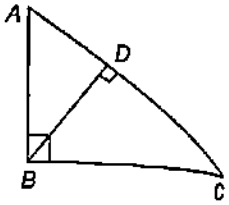
$$AC^2 = AB^2 + BC^2 - 2BD \cdot BC \quad (\angle B < 90°)$$

7. In an obtuse angle ΔABC, AD is perpendicular dropped on BC. BC is produced to D to meet AD, then



$$AC^2 = AB^2 + BC^2 + 2BD \cdot BC \quad (\angle B > 90°)$$

8. In a right angle ΔABC, $\angle B = 90°$ and AC is hypotenuse. The perpendicular BD is dropped on hypotenuse AC from right angle vertex B, then

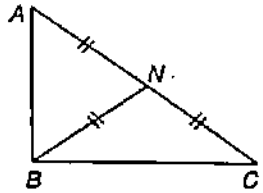


(i) $BD = \frac{AB \times BC}{AC}$ (ii) $AD = \frac{AB^2}{AC}$

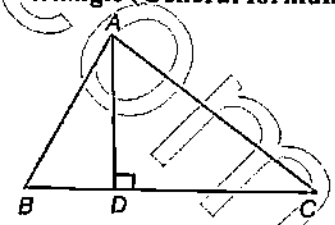
(iii) $CD = \frac{BC^2}{AC}$ (iv) $\frac{1}{BD^2} = \frac{1}{AB^2} + \frac{1}{BC^2}$

- In a right angled triangle, the median to the hypotenuse $= \frac{1}{2} \times$ hypotenuse

i.e., $BN = \frac{AC}{2}$ (as per the fig.)



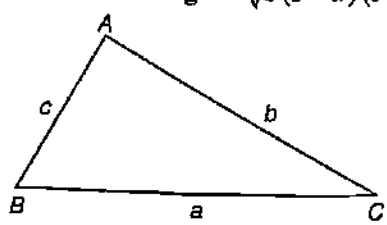
9. **Area of a triangle (General formula)**



$$A(\Delta) = \frac{1}{2} \times \text{base} \times \text{height}$$

$$A(\Delta) = \frac{1}{2} \times BC \times AD \quad \text{(as per the figure.)}$$

10. **Area of scalene triangle** $= \sqrt{s(s-a)(s-b)(s-c)}$



For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

Also, $A(\Delta) = r \times s = \frac{abc}{4R}$

where a, b and c are the sides of the triangle.

$s \rightarrow$ semiperimeter $= \frac{a+b+c}{2}$

$r \rightarrow$ inradius

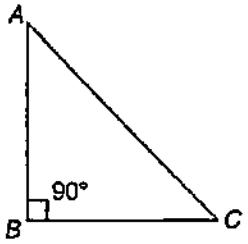
$R \rightarrow$ circumradius

11. Area of right angled triangle

$= \frac{1}{2} \times$ base $\times$ height

$= \frac{1}{2} \times BC \times AB$

(as per the figure)



12. Area of an isosceles triangle

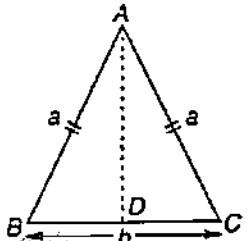
$= \frac{b}{4}\sqrt{4a^2 - b^2}$

$AB = AC$

and $\angle B = \angle C$

$\Delta ABD \cong \Delta ACD$

($AD \rightarrow$ Angle bisector, median, altitude and perpendicular bisector)

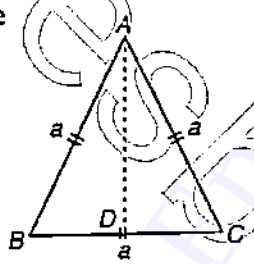


13. Area of an equilateral triangle

$= \frac{\sqrt{3}}{4} a^2$

$\left[A(\Delta) = \frac{1}{2} BC \times AD \right.$

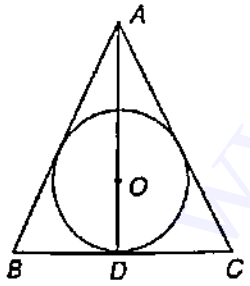
$\left. = \frac{1}{2} \times a \times \frac{\sqrt{3}}{2} a = \frac{\sqrt{3}}{4} a^2 \right]$

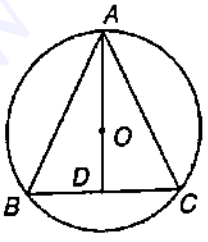


($a \rightarrow$ each side of the triangle)

$AD \rightarrow$ Altitude, median, angle bisector and perpendicular bisector also.

Inradius : $\frac{1}{3} \times$ height $= \frac{\text{side}}{2\sqrt{3}}$, $OD \rightarrow$ Inradius





Circumradius $= \frac{2}{3} \times$ height $= \frac{\text{side}}{\sqrt{3}}$

$OA \rightarrow$ Circumradius

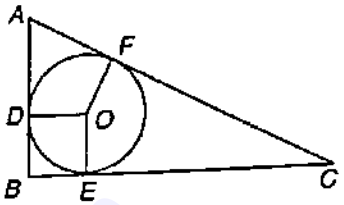
NOTE In equilateral triangle orthocentre centroid, incentre and circumcentre coincide at the same point.

- Circumradius $= 2 \times$ inradius
- For the given perimeter of a triangle, the area of equilateral triangle is maximum.
- For the given area of a triangle, the perimeter of equilateral triangle is minimum.

14. In a right angled triangle

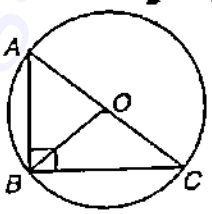
(i) Inradius $(r) = \frac{AB + BC - AC}{2}$

(ii) Inradius $(r) = \frac{\text{Area}}{\text{Semiperimeter}}$



$DO = EO = FO\ (r)$

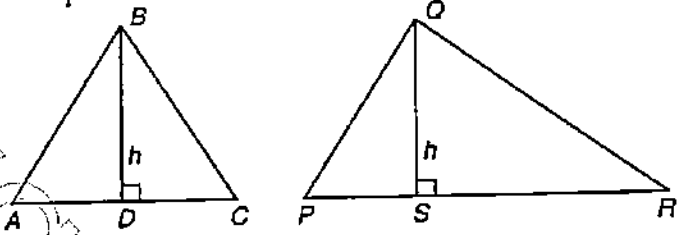
(iii) Circumradius $(R) = \frac{AC}{2} = \left(\frac{\text{hypotenuse}}{2}\right.$



$AO = CO = BO = (R)$

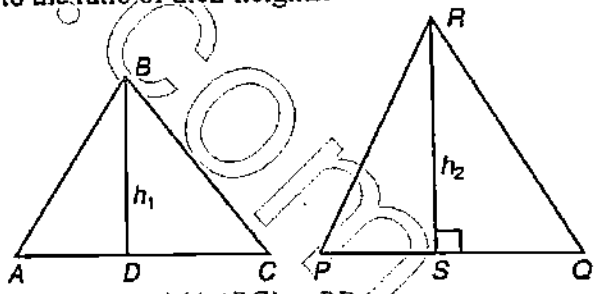
AC is the diameter.

15. The ratio of areas of two triangles of equal heights is equal to the ratio of their corresponding bases. *i.e.*,



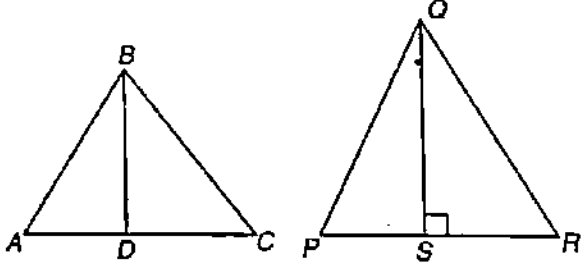
$\frac{A(\Delta ABC)}{A(\Delta PQR)} = \frac{AC}{PR}$

16. The ratio of areas of triangles of equal bases is equal to the ratio of their heights.



i.e., $\frac{A(\Delta ABC)}{A(\Delta PQR)} = \frac{BD}{RS}$

17. The ratio of the areas of two triangles is equal to the



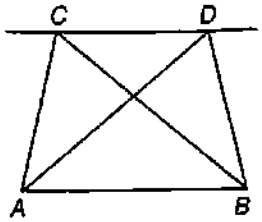
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

ratio of the products of base and its corresponding height *i.e.*,

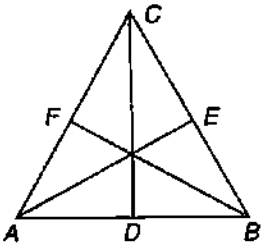
$$\frac{A(\Delta ABC)}{A(\Delta PQR)} = \frac{AC \times BD}{PR \times QS}$$

18. If the two triangles have the same base and lie between the same parallel lines (as shown in figure), then the area of two triangles will be equal.



i.e., $A(\Delta ABC) = A(\Delta ADB)$

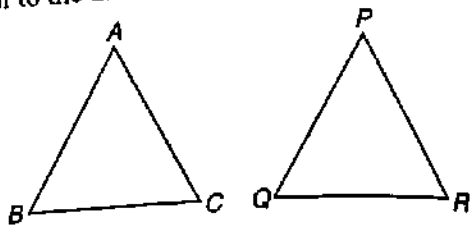
19. In a triangle AE, CD and BF are the medians then



$$3(AB^2 + BC^2 + AC^2) = 4(CD^2 + BF^2 + AE^2)$$

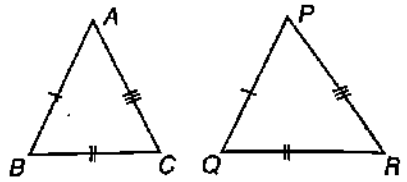
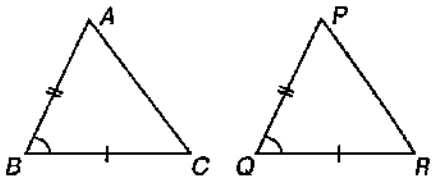
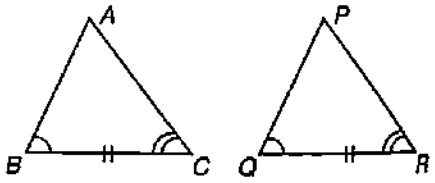
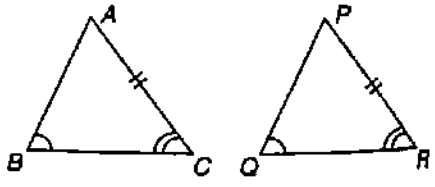
Congruency of triangles : Two triangles are said to be congruent if they are equal in all respects. *i.e.*,

1. Each of the three sides of one triangle must be equal to the three respective sides of the other.
2. Each of the three angles of the one triangle must be equal to the three respective angles of the other.



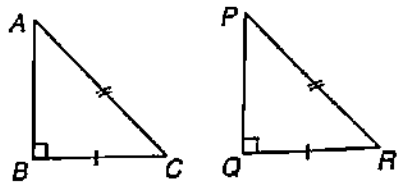
i.e., $\left.\begin{matrix} AB = PQ \\ AC = PR \\ BC = QR \end{matrix}\right\}$ and $\left.\begin{matrix} \angle A = \angle P \\ \angle B = \angle Q \\ \angle C = \angle R \end{matrix}\right\}$

Tests of congruency : With the help of the following given tests, we can deduce without having detailed information about triangles that whether the given two triangles are congruent or not.

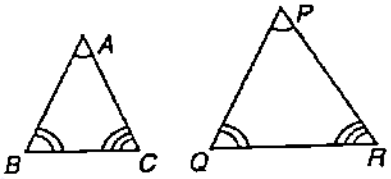
Test	Property	Diagram
$S-S-S$	(Side–Side–Side) If the three sides of one triangle are equal to the corresponding three sides of the other triangle, then the two triangles are congruent. $AB \cong PQ, AC \cong PR, BC \cong QR$ $\therefore$ $\Delta ABC \cong \Delta PQR$	
$S-A-S$	(Side–Angle–Side) If two sides and the angle included between them are congruent to the corresponding sides and the angle included between them, of the other triangle then the two triangles are congruent. $AB \cong PQ, \angle ABC \cong \angle PQR, BC \cong QR$ $\therefore$ $\Delta ABC \cong \Delta PQR$	
$A-S-A$	(Angle–Side–Angle) If two angles and the included side of a triangle are congruent to the corresponding angles and the included side of the other triangle, then the two triangles are congruent. $\angle ABC \cong \angle PQR, BC \cong QR, \angle ACB \cong \angle PRQ$ $\therefore$ $\Delta ABC \cong \Delta PQR$	
$A-A-S$	(Angle–Angle–Side) If two angles and a side other than the included side of a one triangle are congruent to the corresponding angles and a corresponding side other than the included side of the other triangle, then the two triangles are congruent. $\angle ABC \cong \angle PQR, \angle ACB \cong \angle PRQ$ and $AC \cong PR$ (or $AB \cong PQ$)	

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Test	Property	Diagram
R—H—S	(Right angle–Hypotenuse–Side) If the hypotenuse and one side of the right angled triangle are congruent to the hypotenuse and a corresponding side of the other right angled triangle, then the two given triangles are congruent. $AC \cong PR$, $\angle B = \angle Q$ and $BC \cong QR$ $\therefore \Delta ABC \cong \Delta PQR$	

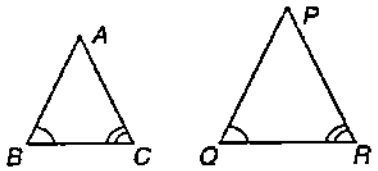
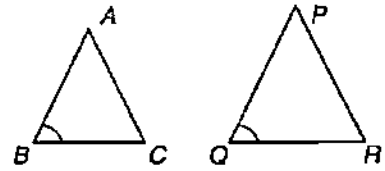
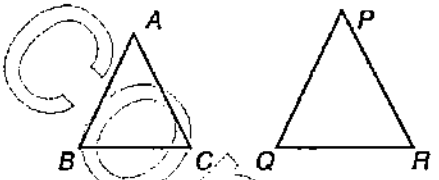
Similarity of triangles : Two triangles are said to be similar if the corresponding angles are congruent and their corresponding sides are in proportion. The symbol for similarity is '~'.



If $\Delta ABC \sim \Delta PQR$ then

$\angle ABC \cong \angle PQR$, $\angle BCA \cong \angle QRP$, $\angle BAC \cong \angle QPR$

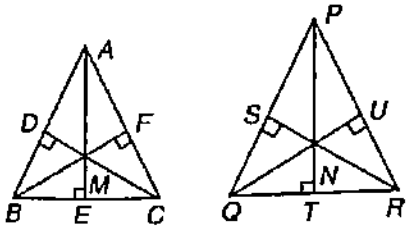
Tests for Similarity : Through the tests for similarity we can deduce the similarity of triangles with minimum required information.

Test	Property/Definition	Diagram
A – A	Angle-Angle If the two angles of one triangle are congruent to the corresponding two angles of the other triangle, then the two triangles are said to be similar. $\angle ABC \cong \angle PQR$ $\angle ACB \cong \angle PRQ$ $\therefore \quad \Delta ABC \sim \Delta PQR$	
S-A-S	Side-Angle-Side If the two sides of one triangle are proportional to the corresponding two sides of the other triangle and the angle included by them are congruent, then the two triangles are similar. *i.e.,* $\frac{AB}{PQ} = \frac{BC}{QR}$ and $\angle ABC = \angle PQR$ $\therefore \quad \Delta ABC \sim \Delta PQR.$	 $\frac{AB}{PQ} = \frac{BC}{QR} = K$ (K is any constant)
S-S-S	Side-Side-Side If the three sides of one triangle are proportional to the corresponding three sides of the other triangle, then the two triangles are similar. *i.e.,* $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR}$ $\therefore \quad \Delta ABC \sim \Delta PQR$	 $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR} = K$

NOTE When the corresponding sides are in proportion , then the corresponding angles are in proportion.

PROPERTIES OF SIMILAR TRIANGLES

If the two triangles are similar, then for the proportional/corresponding sides we have the following results.



1. Ratio of sides = Ratio of heights (altitudes)
   = Ratio of medians
   = Ratio of angle bisectors
   = Ratio of inradii
   = Ratio of circumradii
2. Ratio of areas = Ratio of squares of corresponding sides.

i.e., if $\Delta ABC \sim \Delta PQR$, then

$$\frac{A(\Delta ABC)}{A(\Delta PQR)} = \frac{(AB)^2}{(PQ)^2} = \frac{(BC)^2}{(QR)^2} = \frac{(AC)^2}{(PR)^2}$$

NOTE Rule 1 can also apply with rule 2.

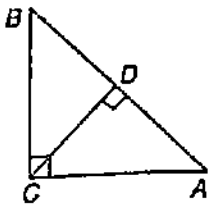
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

3. In a right angled triangle, the triangles on each side of the altitude drawn from the vertex of the right angle to the hypotenuse are similar to the original triangle and to each other too.

i.e., $\Delta BCA \sim \Delta BDC \sim \Delta CDA$.

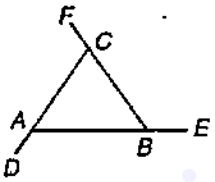


INTRODUCTORY EXERCISE-12.2

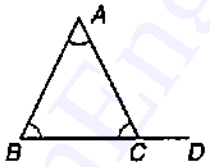
1. In a triangle ABC, if AB, BC and AC are the three sides of the triangle, then which of the statements is necessarily true?
(a) $AB + BC < AC$ (b) $AB + BC > AC$
(c) $AB + BC = AC$ (d) $AB^2 + BC^2 = AC^2$

2. The sides of a triangle are 12 cm, 8 cm and 6 cm respectively, the triangle is :
(a) acute (b) obtuse
(c) right (d) can't be determined

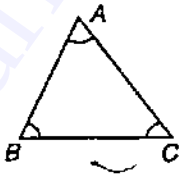
3. If the sides of a triangle are produced then the sum of the exterior angles *i.e.*, $\angle DAB + \angle EBC + \angle FCA$ is equal to :
(a) 180°
(b) 270°
(c) 360°
(d) 240°



4. In the given figure BC is produced to D and $\angle BAC = 40°$ and $\angle ABC = 70°$. Find the value of $\angle ACD$:
(a) 30°
(b) 40°
(c) 70°
(d) 110°

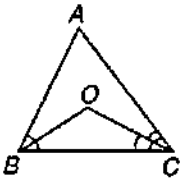


5. In a ΔABC, $\angle BAC > 90°$, then $\angle ABC$ and $\angle ACB$ must be :
(a) acute
(b) obtuse
(c) one acute and one obtuse
(d) can't be determined

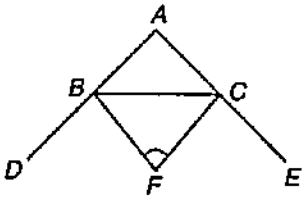


6. If the angles of a triangle are in the ratio 1 : 4 : 7, then the value of the largest angle is :
(a) 135° (b) 84°
(c) 105° (d) none of these

7. In the adjoining figure $\angle B = 70°$ and $\angle C = 30°$. BO and CO are the angle bisectors of $\angle ABC$ and $\angle ACB$. Find the value of $\angle BOC$:
(a) 30° (b) 40°
(c) 120° (d) 130°

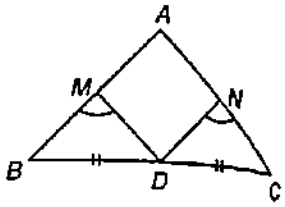


8. In the given diagram of ΔABC, $\angle B = 80°$, $\angle C = 30°$. BF and CF are the angle bisectors of $\angle CBD$ and $\angle BCE$ respectively. Find the value of $\angle BFC$:
(a) 110° (b) 50°
(c) 125° (d) 55°

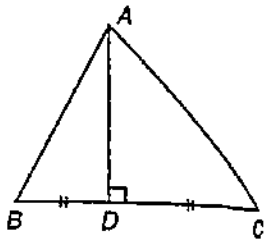


9. In an equilateral triangle, the incentre, circumcentre, orthocentre and centroid are :
(a) concylic (b) coincident
(c) collinear (d) none of these

10. In the adjoining figure D is the midpoint of BC of a ΔABC, DM and DN are the perpendiculars on AB and AC respectively and $DM = DN$, then the ΔABC is :
(a) right angled (b) isosceles
(c) equilateral (d) scalene



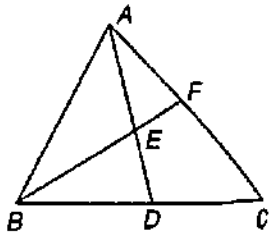
11. In the adjoining figure of ΔABC, AD is the perpendicular bisector of side BC. The triangle ABC is :
(a) right angled
(b) isosceles
(c) scalene
(d) equilateral



12. Triangle ABC is such that $AB = 9$ cm, $BC = 6$ cm, $AC = 7.5$ cm. Triangle DEF is similar to ΔABC. If $EF = 12$ cm then DE is :
(a) 6 cm (b) 16 cm
(c) 18 cm (d) 15 cm

13. In ΔABC, $AB = 5$ cm, $AC = 7$ cm. If AD is the angle bisector of $\angle A$. Then $BD : CD$ is :
(a) 25 : 49 (b) 49 : 25
(c) 6 : 1 (d) 5 : 7

14. In a ΔABC, D is the mid-point of BC and E is mid-point of AD, BF passes through E. What is the ratio of $AF : FC$?
(a) 1 : 1
(b) 1 : 2
(c) 1 : 3
(d) 2 : 3



15. In a ΔABC, $AB = AC$ and $AD \perp BC$, then :
(a) $AB < AD$ (b) $AB > AD$
(c) $AB = AD$ (d) $AB \leq AD$

16. The difference between altitude and base of a right angled triangle is 17 cm and its hypotenuse is 25 cm. What is the sum of the base and altitude of the triangle is :

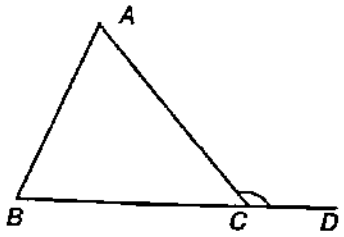
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

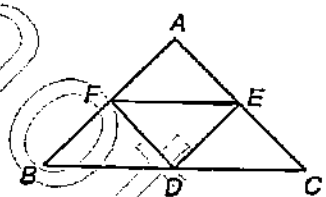
For More Visit : www.LearnEngineering.in

(a) 24 cm (b) 31 cm
(c) 34 cm (d) can't be determined

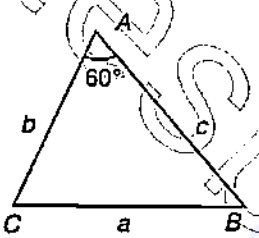
17. If AB, BC and AC be the three sides of a triangle ABC, then which one of the following is true?
(a) $AB - BC = AC$ (b) $(AB - BC) > AC$
(c) $(AB - BA) < AC$ (d) $AB^2 - BC^2 = AC^2$

18. In the triangle ABC, side BC is produced to D. $\angle ACD = 100°$ if $BC = AC$, then $\angle ABC$ is :


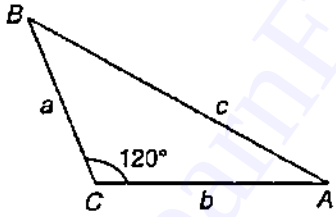
(a) 40°
(b) 50°
(c) 80°
(d) can't be determined

19. In the adjoining figure D, E and F are the mid-points of the sides BC, AC and AB respectively. ΔDEF is congruent to triangle :


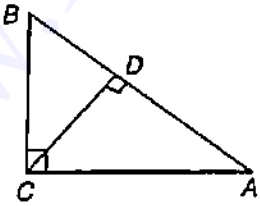
(a) ABC
(b) AEF
(c) CDE, BFD
(d) AFE, BFD and CDE

20. In the adjoining figure $\angle BAC = 60°$ and $BC = a$, $AC = b$ and $AB = c$, then :


(a) $a^2 = b^2 + c^2$
(b) $a^2 = b^2 + c^2 - bc$
(c) $a^2 = b^2 + c^2 + bc$
(d) $a^2 = b^2 + 2bc$

21. In the adjoining figure of ΔABC, $\angle BCA = 120°$ and $AB = c$, $BC = a$, $AC = b$ then:


(a) $c^2 = a^2 + b^2 + ba$
(b) $c^2 = a^2 + b^2 - ba$
(c) $c^2 = a^2 + b^2 - 2ba$
(d) $c^2 = a^2 + b^2 + 2ab$

22. In a right angled ΔABC, $\angle C = 90°$ and CD is the perpendicular on the hypotenuse AB, $AB = c$, $BC = a$, $AC = b$ and $CD = p$, then :


(a) $\frac{p}{a} = \frac{p}{b}$
(b) $\frac{1}{p^2} + \frac{1}{b^2} = \frac{1}{a^2}$
(c) $p^2 = b^2 + c^2$
(d) $\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}$

23. If the medians of a triangle are equal, then the triangle is :
(a) right angled (b) isosceles
(c) equilateral (d) scalene

24. The incentre of a triangle is determined by the :
(a) medians
(b) angle bisector
(c) perpendicular bisectors
(d) altitudes

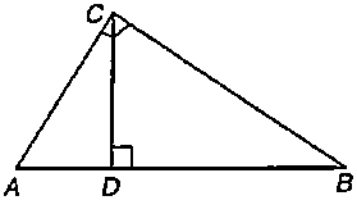
25. The circumcentre of a triangle is determined by the :
(a) altitudes
(b) median
(c) perpendicular bisector
(d) angle bisectors

26. The point of intersection of the angle bisectors of a triangle is :
(a) orthocentre (b) centroid
(c) incentre (d) circumcentre

27. A triangle PQR is formed by joining the mid-points of the sides of a triangle ABC. 'O' is the circumcentre of ΔABC, then for ΔPQR, the point 'O' is :
(a) incentre (b) circumcentre
(c) orthocentre (d) centroid

28. If in a ΔABC, 'S' is the circumcentre then :
(a) S is equidistant from all the vertices of a triangle
(b) S is equidistant from all the sides of a triangle
(c) AS, BS and CS are the angular bisectors
(d) AS, BS and CS produced are the altitudes on the opposite sides.

29. If AD, BE, CF are the altitudes of ΔABC whose orthocentre is H, then C is the orthocentre of :
(a) ΔABH (b) ΔBDH
(c) ΔABD (d) ΔBEA

30. In a right angled ΔABC, $\angle C = 90°$ and CD is the perpendicular on hypotenuse AB if $BC = 15$ cm and $AC = 20$ cm then CD is equal to :


(a) 18 cm (b) 12 cm
(c) 17.5 cm (d) can't be determined

31. In an equilateral ΔABC, if a, b and c denote the lengths of perpendiculars from A, B and C respectively on the opposite sides, then :
(a) $a > b > c$ (b) $a > b < c$
(c) $a = b = c$ (d) $a = c \neq b$

32. What is the ratio of side and height of an equilateral triangle?
(a) 2 : 1 (b) 1 : 1
(c) $2 : \sqrt{3}$ (d) $\sqrt{3} : 2$

33. The triangle is formed by joining the mid-points of the sides AB, BC and CA of ΔABC and the area of ΔPQR is 6 cm^2, then the area of ΔABC is :
(a) 36 cm^2 (b) 12 cm^2
(c) 18 cm^2 (d) 24 cm^2

34. One side other than the hypotenuse of right angle isosceles triangle is 6 cm. The length of the perpendicular on the hypotenuse from the opposite vertex is :
(a) 6 cm (b) $6\sqrt{2}$ cm
(c) 4 cm (d) $3\sqrt{2}$ cm

35. Any two of the four triangles formed by joining the mid-points of the sides of a given triangle are :
(a) congruent

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

(b) equal in area but not congruent
(c) unequal in area and not congruent
(d) none of the above

36. The internal bisectors of $\angle B$ and $\angle C$ of ΔABC meet at O. If $\angle A = 80°$ then $\angle BOC$ is :
(a) 50° (b) 160°
(c) 100° (d) 130°

37. The point in the plane of a triangle which is at equal perpendicular distance from the sides of the triangle is :
(a) centroid (b) incentre
(c) circumcenre (d) orthocentre

38. Incentre of a triangle lies in the interior of :
(a) an isosceles triangle only
(b) a right angled triangle only
(c) any equilateral triangle only
(d) any triangle

39. In a triangle PQR, $PQ = 20$ cm and $PR = 6$ cm, the side QR is :
(a) equal to 14 cm (b) less than 14 cm
(c) greater than 14 cm (d) none of these

40. The four triangles formed by joining the pairs of mid-points of the sides of a given triangle are congruent if the given triangle is :
(a) an isosceles triangle
(b) an equilateral triangle
(c) a right angled triangle
(d) of any shape

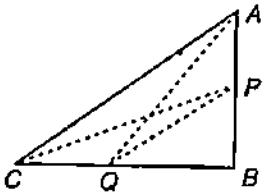
41. O is orthocentre of a triangle PQR, which is formed by joining the mid-points of the sides of a ΔABC, O is :
(a) orthocentre (b) incentre
(c) circumcentre (d) centroid

42. In a ΔABC, a line PQ parallel to BC cuts AB at P and AC at Q. If BQ bisects $\angle PQC$, then which one of the following relation is always true :
(a) $BC = CQ$ (b) $BC = BQ$
(c) $BC \neq CQ$ (d) $BC \neq BQ$

43. Which of the following is true, in the given figure, where AD is the altitude to the hypotenuse of a right angled ΔABC?
(i) ΔCAD and ΔABD are similar
(ii) ΔCDA and ΔADB are congruent
(iii) ΔADB and ΔCAB are similar
Select the correct answer using the codes given below :
(a) 1 and 2 (b) 2 and 3
(c) 1 and 3 (d) 1, 2 and 3

44. If D is such a point on the side, BC of ΔABC that $\frac{AB}{AC} = \frac{BD}{CD}$, then AD must be a/an :
(a) altitude of ΔABC
(b) median of ΔABC
(c) angle bisector of ΔABC
(d) perpendicular bisector of ΔABC

45. In right angled ΔABC, $\angle B = 90°$, if P and Q are points on the sides AB and BC respectively, then :
(a) $AQ^2 + CP^2 = 2(AC^2 + PQ^2)$
(b) $AQ^2 + CP^2 = AC^2 + PQ^2$
(c) $(AQ^2 + CP^2) = \frac{1}{2}(AC^2 + PQ^2)$
(d) $(AQ + CP) = \frac{1}{2}(AC + PQ)$



46. If ABC is a right angled triangle at B and M, N are the mid-points of AB and BC, then $4(AN^2 + CM^2)$ is equal to :
(a) $4AC^2$ (b) $6AC^2$
(c) $5AC^2$ (d) $\frac{5}{4}AC^2$

47. If ΔABC and ΔDEF are so related that $\frac{AB}{FD} = \frac{BC}{DE} = \frac{CA}{EF}$, then which of the following is true ?
(a) $\angle A = \angle F$ and $\angle B = \angle D$
(b) $\angle C = \angle F$ and $\angle A = \angle D$
(c) $\angle B = \angle F$ and $\angle C = \angle D$
(d) $\angle A = \angle E$ and $\angle B = \angle D$

48. ABC is a right angle triangle at A and AD is perpendicular to the hypotenuse. Then $\frac{BD}{CD}$ is equal to :
(a) $\left(\frac{AB}{AC}\right)^2$ (b) $\left(\frac{AB}{AD}\right)^2$
(c) $\frac{AB}{AC}$ (d) $\frac{AB}{AD}$

49. Let ABC be an equilateral triangle. Let $BE \perp CA$ meeting CA at E, then $(AB^2 + BC^2 + CA^2)$ is equal to :
(a) $2BE^2$ (b) $3BE^2$
(c) $4BE^2$ (d) $6BE^2$

50. If D, E and F are respectively the mid-points of sides BC, CA and AB of a ΔABC. If $EF = 3$ cm, $FD = 4$ cm and $AB = 10$ cm, then DE, BC and CA respectively will be equal to :
(a) 6, 8 and 20 cm (b) 4, 6 and 8 cm
(c) 5, 6 and 8 cm (d) $\frac{10}{3}$, 9 and 12 cm

51. In the right angle triangle $\angle C = 90°$. AE and BD are two medians of a triangle ABC meeting at F. The ratio of the area of ΔABF and the quadrilateral $FDCE$ is :
(a) 1 : 1 (b) 1 : 2
(c) 2 : 1 (d) 2 : 3

52. ABC is a triangle and DE is drawn parallel to BC cutting the other sides at D and E. If $AB = 3.6$ cm, $AC = 2.4$ cm and $AD = 2.1$ cm, then AE is equal to :
(a) 1.4 cm (b) 1.8 cm
(c) 1.2 cm (d) 1.05 m

53. Consider the following statements :
(1) If three sides of a triangle are equal to three sides of another triangle, then the triangles are congruent.
(2) If three angles of a triangle are equal to three angles of another triangle respectively, then the two triangles are congruent. Of these statements :
(a) 1 is correct and 2 is false
(b) both 1and 2 are false
(c) both 1 and 2 are correct
(d) 1 is false and 2 is correct

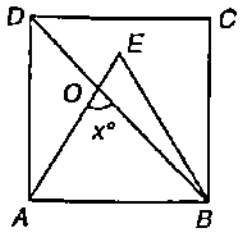
www.LearnEngineering.in

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

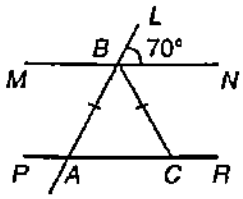
For More Visit : www.LearnEngineering.in

54. In the figure ΔABE is an equilateral triangle in a square $ABCD$. Find the value of angle x in degrees :



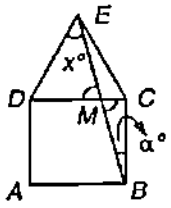
(a) 60°
(b) 45°
(c) 75°
(d) 90°

55. In the given diagram $MN \parallel PR$ and $m\angle LBN = 70°$, $AB = BC$. Find $m\angle ABC$:



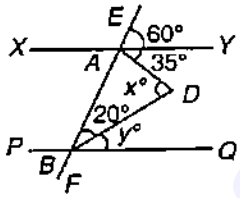
(a) 40°
(b) 30°
(c) 35°
(d) 55°

56. In the given diagram, equilateral triangle EDC surmounts square $ABCD$. Find the $m\angle BED$ represented by x. Where $m\angle EBC = \alpha°$:



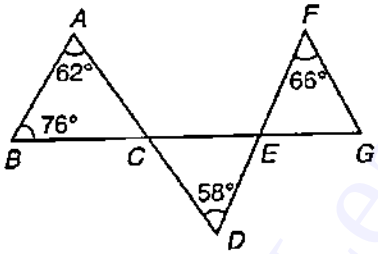
(a) 45°
(b) 60°
(c) 30°
(d) none of the above

57. In the given diagram $XY \parallel PQ$. Find $m\angle x°$ and $m\angle y°$:



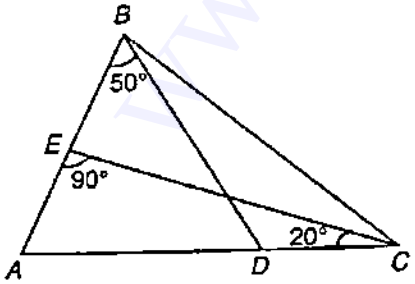
(a) 75° and 40°
(b) 45°, 60°
(c) 75°, 45°
(d) 60° and 45°

58. In the adjoining figure $m\angle CAB = 62°$, $m\angle CBA = 76°$ $m\angle ADE = 58°$ and $\angle DFG = 66°$,



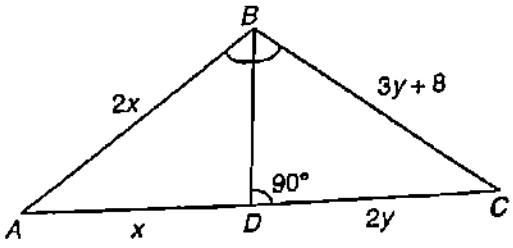
Find $m\angle FGE$:
(a) 44°
(b) 34°
(c) 36°
(d) none of these

59. In the given figure $CE \perp AB$, $m\angle ACE = 20°$ and $m\angle ABD = 50°$. Find $m\angle BDA$:



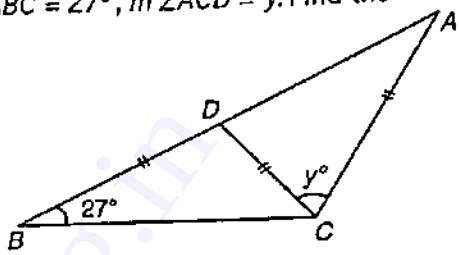
(a) 50°
(b) 60°
(c) 70°
(d) 80°

60. In the ΔABC, BD bisects $\angle B$, and is perpendicular to AC. If the lengths of the sides of the triangle are expressed in terms of x and y as shown, find the value of x and y :



(a) 6, 12
(b) 10, 12
(c) 16, 8
(d) 8, 15

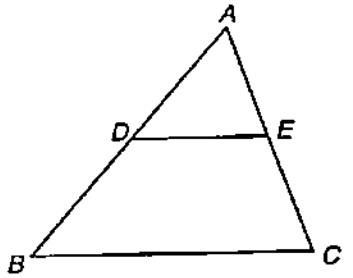
61. In the following figure $ADBC$. $BD = CD = AC$, $m\angle ABC = 27°$, $m\angle ACD = y$. Find the value of y :



(a) 27°
(b) 54°
(c) 72°
(d) 58°

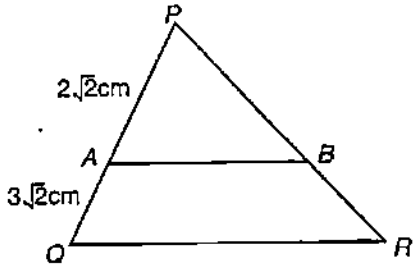
62. ABC is an isosceles triangle with $AB = AC$. Side BA is produced to D such that $AB = AD$. Find $m\angle BCD$:
(a) 60°
(b) 90°
(c) 120°
(d) can't be determined

63. In ΔABC, $AC = 5$ cm. Calculate the length of AE where $DE \parallel BC$. Given that $AD = 3$ cm and $BD = 7$ cm :



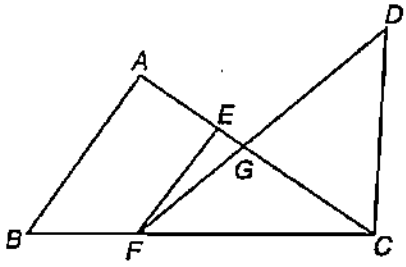
(a) 2 cm
(b) 1 cm
(c) 1.5 cm
(d) 2.5 cm

64. In ΔPQR, $AP = 2\sqrt{2}$ cm, $AQ = 3\sqrt{2}$ cm and $PR = 10$ cm, $AB \parallel QR$. Find the length of BR :



(a) $6\sqrt{2}$ cm
(b) 6 cm
(c) $5\sqrt{2}$ cm
(d) none of these

65. In the adjoining figure (not drawn to scale) AB, EF and CD are parallel lines. Given that $EG = 5$ cm, $GC = 10$ cm and $DC = 18$ cm. Calculate AC, if $AB = 15$ cm :

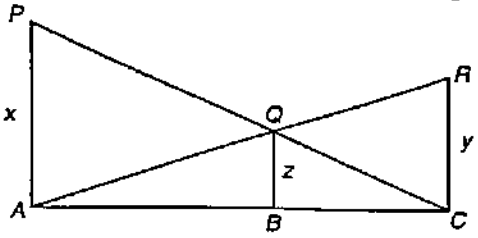


(a) 21 cm
(b) 25 cm
(c) 18 cm
(d) 28 cm

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

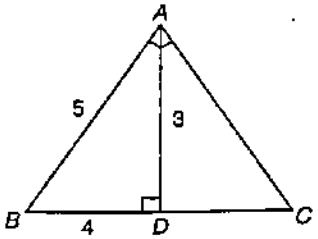
yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

66. In the adjoining figure PA, QB and RC are each perpendicular to AC. Which one of the following is true :



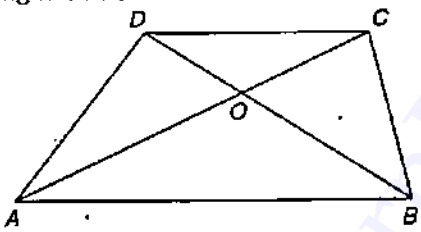
(a) $x + y = z$ (b) $xy = 2z$
(c) $\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$ (d) $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 0$

67. In the adjoining figure the angle BAC and $\angle ADB$ are right angles.
$BA = 5$ cm, $AD = 3$ cm and $BD = 4$ cm, what is the length of DC ?
(a) 2.5
(b) 3
(c) 2.25
(d) 2



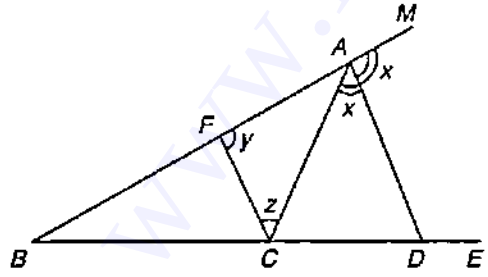
68. The areas of the similar triangles are in the ratio of 25 : 36. What is the ratio of their respective heights?
(a) 5 : 6 (b) 6 : 5
(c) 1 : 11 (d) 2 : 3

69. In the given diagram $AB \parallel CD$, then which one of the following is true ?



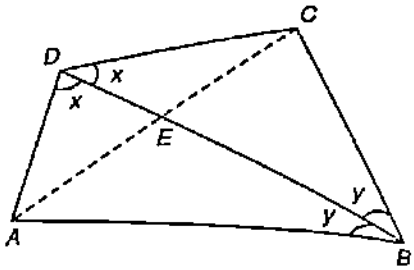
(a) $\frac{AB}{CD} = \frac{AO}{OC}$ (b) $\frac{AB}{CD} = \frac{BO}{OD}$
(c) $\Delta AOB \sim \Delta COD$ (d) all of these

70. The bisector of the exterior $\angle A$ of ΔABC intersects the side BC produced to D. Here CF is parallel to AD.



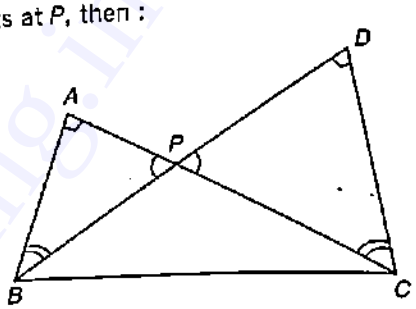
(a) $\frac{AB}{AC} = \frac{BD}{CD}$ (b) $\frac{AB}{AC} = \frac{CD}{BD}$
(c) $\frac{AB}{AC} = \frac{BC}{CD}$ (d) none of these

71. The diagonal BD of a quadrilateral $ABCD$ bisects $\angle B$ and $\angle D$, then :



(a) $\frac{AB}{CD} = \frac{AD}{BC}$ (b) $\frac{AB}{BC} = \frac{AD}{CD}$
(c) $AB = AD \times BC$ (d) none of these

72. Two right triangles ABC and DBC are drawn on the same hypotenuse BC on the same side of BC. If AC and DB intersects at P, then :

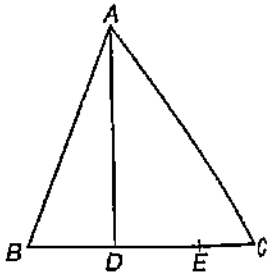


(a) $\frac{AP}{PC} = \frac{BP}{DP}$ (b) $AP \times DP = PC \times BP$
(c) $AP \times PC = BP \times DP$ (d) $AP \times BP = PC \times PD$

73. A man goes 150 m due east and then 200 m due north. How far is he from the starting point?
(a) 200 m (b) 350 m
(c) 250 m (d) 175 m

74. From a point O in the interior of a ΔABC perpendiculars OD, OE and OF are drawn to the sides BC, CA and AB respectively, then which one of the following is true ?
(a) $AF^2 + BD^2 + CE^2 = AE^2 + CD^2 + BF^2$
(b) $AB^2 + BC^2 = AC^2$
(c) $AF^2 + BD^2 + CE^2 = OA^2 + OB^2 + OC^2$
(d) $AF^2 + BD^2 + CE^2 = OD^2 + OE^2 + OF^2$

75. In an equilateral triangle ABC, the side BC is trisected at D. Find the value of AD^2 :
(a) $\frac{9}{7} AB^2$
(b) $\frac{7}{9} AB^2$
(c) $\frac{3}{4} AB^2$
(d) $\frac{4}{5} AB^2$



76. ABC is a triangle in which $\angle A = 90°$, $AN \perp BC$, $AC = 12$ cm and $AB = 5$ cm. Find the ratio of the areas of ΔANC and ΔANB :
(a) 125 : 44 (b) 25 : 144
(c) 144 : 25 (d) 12 : 5

77. A vertical stick 15 cm long casts it shadow 10 cm long on the ground. At the same time a flag pole casts a shadow 60 cm long. Find the height of the flag pole :

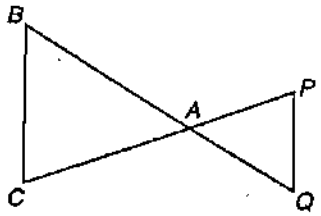
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

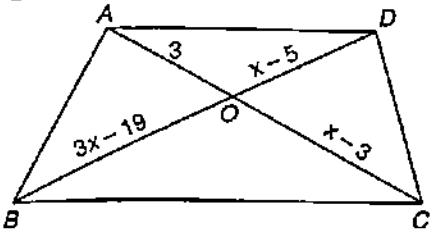
(a) 40 cm (b) 45 cm
(c) 90 cm (d) none of these

78. Vertical angles of two isoceles triangles are equal. Then corresponding altitudes are in the ratio 4 : 9. Find the ratio of their areas :
(a) 16 : 49 (b) 16 : 81
(c) 16 : 65 (d) none of these

79. In the figure $\Delta ACB \sim \Delta APQ$. If $BC = 8$ cm, $PQ = 4$ cm, $AP = 2.8$ cm, find CA:
(a) 8 cm
(b) 6.5 cm
(c) 5.6 cm
(d) none of these



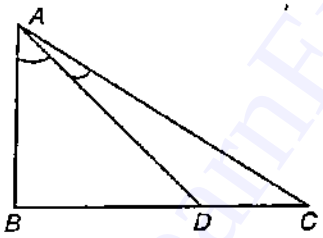
80. In the figure $BC \| AD$. Find the value of x :



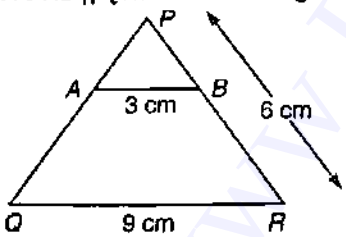
(a) 9, 10 (b) 7, 8
(c) 10, 12 (d) 8, 9

81. In an equilateral triangle of side $2a$, calculate the length of its altitude :
(a) $2a\sqrt{3}$ (b) $a\sqrt{3}$
(c) $a\frac{\sqrt{3}}{2}$ (d) none of these

82. In figure AD is the bisector of $\angle BAC$. If $BD = 2$ cm, $CD = 3$ cm and $AB = 5$ cm. Find AC :
(a) 6 cm
(b) 7.5 cm
(c) 10 cm
(d) 15 cm

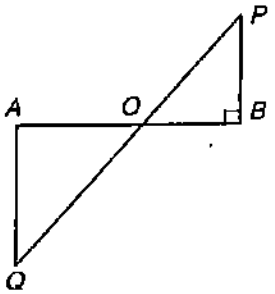


83. In the figure $AB \| QR$. Find the length of PB :

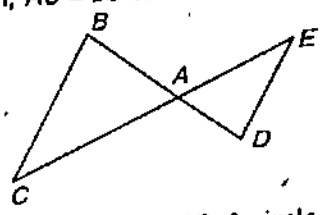


(a) 2 cm (b) 3 cm
(c) 2.5 cm (d) 4 cm

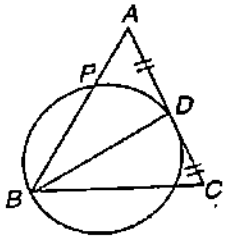
84. In the figure QA and PB are perpendicular to AB. If $AO = 10$ cm, $BO = 6$ cm and $PB = 9$ cm. Find AQ:
(a) 8 cm
(b) 9 cm
(c) 15 cm
(d) 12 cm



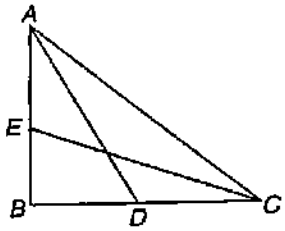
85. In the given figure $AB = 12$ cm, $AC = 15$ cm and $AD = 6$ cm. $BC \| DE$, find the length of AE :
(a) 6 cm
(b) 7.5 cm
(c) 9 cm
(d) 10 cm



86. In the figure, ABC is a triangle in which $AB = AC$. A circle through B touches AC at D and intersects AB at P. If D is the mid-point of AC, Find the value of AB :
(a) $2AP$
(b) $3AP$
(c) $4AP$
(d) none of the above



87. In figure, ABC is a right triangle, right angled at B. AD and CE are the two medians drawn from A and C respectively. If $AC = 5$ cm and $AD = \frac{3\sqrt{5}}{2}$ cm, find the length of CE :
(a) $2\sqrt{5}$ cm
(b) 2.5 cm
(c) 5 cm
(d) $4\sqrt{2}$ cm



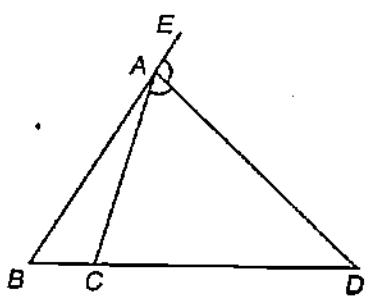
88. In a ΔABC, $AB = 10$ cm, $BC = 12$ cm and $AC = 14$ cm. Find the length of median AD. If G is the centroid , find length of GA :
(a) $\frac{5}{3}\sqrt{7}, \frac{5}{9}\sqrt{7}$ (b) $5\sqrt{7}, 4\sqrt{7}$
(c) $\frac{10}{\sqrt{3}}, \frac{8}{3}\sqrt{7}$ (d) $4\sqrt{7}, \frac{8}{3}\sqrt{7}$

89. ΔABC is a right angled at A and AD is the altitude to BC. If $AB = 7$ cm and $AC = 24$ cm. Find the ratio of AD is to AM if M is the mid–point of BC :
(a) 25 : 41 (b) 32 : 41
(c) $\frac{336}{625}$ (d) $\frac{625}{336}$

90. Area of $\Delta ABC = 30$ cm^2. D and E are the mid-points of BC and AB respectively. Find $A(\Delta BDE)$:
(a) 10 cm (b) 7.5 cm
(c) 15 cm (d) none of these

91. The three sides of a triangles are given which one of the following is not a right angle :
(a) 20, 21, 29 (b) 16, 63, 65
(c) 56, 90, 106 (d) 36, 35, 74

92. In the figure AD is the external bisector of $\angle EAC$, intersects BC produced to D. If $AB = 12$ cm, $AC = 8$ cm and $BC = 4$ cm, find CD :
(a) 10 cm
(b) 6 cm
(c) 8 cm
(d) 9 cm

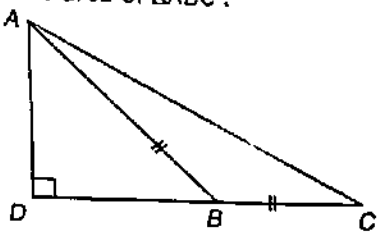


For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

93. In ΔABC, $AB^2 + AC^2 = 2500$ cm^2 and median $AD = 25$ cm. Find BC :
(a) 25 cm (b) 40 cm
(c) 50 cm (d) 48 cm

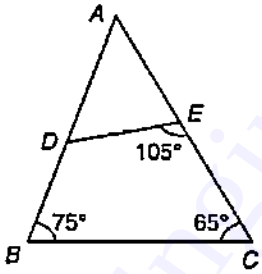
94. In the given figure, $AB = BC$ and $\angle BAC = 15°$, $AB = 10$ cm. Find the area of ΔABC :



(a) 50 cm^2 (b) 40 cm^2
(c) 25 cm^2 (d) 32 cm^2

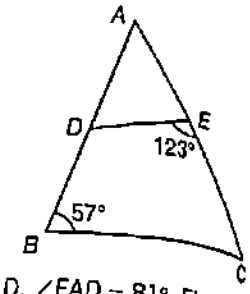
95. In ABC, G is the centroid, $AB = 15$ cm, $BC = 18$ cm and $AC = 25$ cm. Find GD, where D is the mid-point of BC :
(a) $\frac{1}{3}\sqrt{86}$ cm (b) $\frac{2}{3}\sqrt{86}$ cm
(c) $\frac{8}{3}\sqrt{15}$ cm (d) none of these

96. In the given figure , if $\frac{DE}{BC} = \frac{2}{3}$ and if $AE = 10$ cm. Find AB :
(a) 16 cm
(b) 12 cm
(c) 15 cm
(d) 18 cm

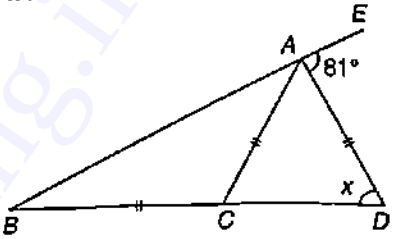


97. Find the maximum area that can be enclosed in a triangle of perimeter 24 cm :
(a) 32 cm^2 (b) $16\sqrt{3}$ cm^2
(c) $16\sqrt{2}$ cm^2 (d) 27 cm^2

98. In the figure $AD = 12$ cm, $AB = 20$ cm and $AE = 10$ cm. Find EC :
(a) 14 cm
(b) 10 cm
(c) 8 cm
(d) 15 cm



99. In the given figure, $BC = AC = AD$, $\angle EAD = 81°$. Find the value of x :



(a) 45° (b) 54°
(c) 63° (d) 36°

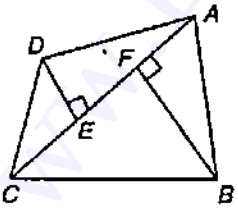
100. What is the ratio of inradius to the circumradius of a right angled triangle?
(a) 1 : 2 (b) 1 : $\sqrt{2}$
(c) 2 : 5 (d) can't be determined

.3) QUADRILATERALS

A four sided closed figure is called a quadrilateral. It is denoted by symbol '▭'.

Properties

1. Sum of four interior angles is 360°.
2. The figure formed by joining the mid-points of a quadrilateral is a parallelogram.



3. The sum of opposite sides of a quadrilateral circumscribed about a circle, is always equal.
4. Area of quadrilateral $= \frac{1}{2} \times$ one of the diagonals $\times$ sum of the perpendiculars drawn to the diagonals from the opposite vertices.

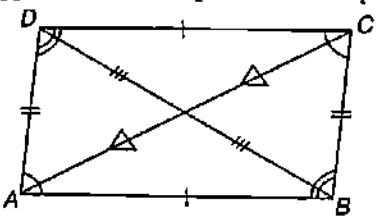
i.e., $A(\square ABCD) = \frac{1}{2} \times AC \times (DE + BF)$

DIFFERENT TYPES OF QUADRILATERALS ARE GIVEN BELOW

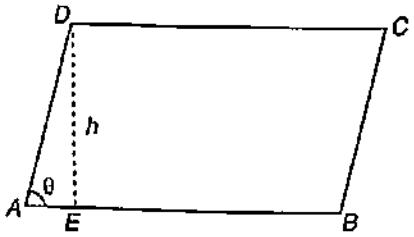
Parallelogram : A quadrilateral whose opposite sides are parallel.

Properties

1. The opposite sides are parallel and equal.



2. Opposite angles are equal.
3. Sum of any two adjacent angles is 180°.
4. Diagonals bisect each other.



For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

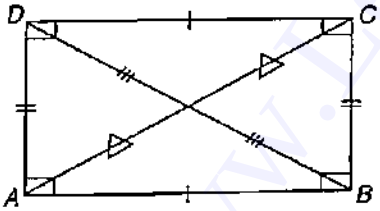
For More Visit : www.LearnEngineering.in

5. Diagonals need not be equal in length.
6. Diagonals need not bisect at right angle.
7. Diagonals need not bisect angles at the vertices.
8. Each diagonal divides a parallelogram into two congruent triangles.
9. Lines joining the mid-points of the adjacent sides of a **quadrilateral** form a **parallelogram.**
10. Lines joining the mid-points of of the adjacent sides of a **parallelogram** is a **parallelogram.**
11. The parallelogram that is inscribed in a circle is a rectangle.
12. The parallelogram that is circumscribed about a circle is a rhombus.
13. (a) Area of a parallelogram = base × height
    (b) Area of parallelogram
    = product of any two adjacent sides × sine of the included angles.
    $= AB \times AD \times \sin\theta$
14. Perimeter of a parallelogram = 2 (sum of any two adjacent sides)
15. $(AC)^2 + (BD)^2 = (AB)^2 + (BC)^2 + (CD)^2 + (AD)^2$
    $= 2(AB^2 + BC^2)$
16. Parallelogram that lie on the same base and between the same parallel lines are equal in area.
17. Area of a triangle is half of the area of a parallelogram which lie on the same base and between the same parallel lines.
18. A parallelogram is a rectangle if its diagonals are equal.

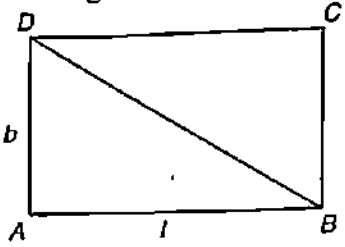
Rectangle : A parallelogram in which all the four angles at vertices are right (*i.e.*, 90°), is called a rectangle.

Properties

1. Opposite sides are parallel and equal.
2. Opposite angles are equal and of 90°.



3. Diagonals are equal and bisect each other, but not necessarily at right angles.
4. When a rectangle is inscribed in a circle, the diameter of the circle is equal to the diagonal of the rectangle.
5. For the given perimeter of rectangles, a square has maximum area.
6. The figure formed by joining the mid-points of the adjacent sides of a rectangle is a rhombus.
7. The quadrilateral formed by joining the mid-points of intersection of the angle bisectors of a parallelogram is a rectangle.
8. Every rectangle is a parallelogram.
9. Area of a rectangle = length × breadth $(= l \times b)$

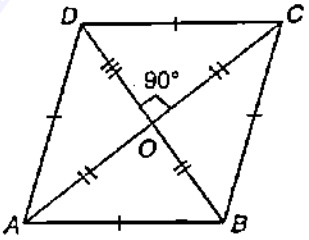


10. Diagonals of a rectangle $= \sqrt{l^2 + b^2}$
11. Perimeter of a rectangle $= 2(l + b)$
    $l \rightarrow$ length and $b \rightarrow$ breadth

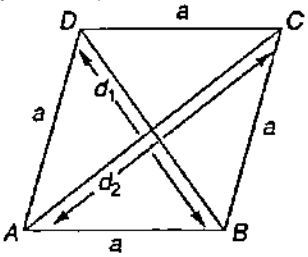
Rhombus : A parallelogram in which all sides are equal, is called a rhombus.

Properties

1. Opposite sides are parallel and equal.
2. Opposite angles are equal.
3. Diagonals bisect each other at right angle, but they are not necessarily equal.



4. Diagonals bisect the vertex angles.
5. Sum of any two adjacent angles is 180°.



6. Figure formed by joining the mid-points of the adjacent sides of a rhombus is a rectangle.
7. A parallelogram is a rhombus if its diagonals are perpendicular to each other.
8. (a) Area of a rhombus $= \frac{1}{2} \times$ product of diagonals
   $= \frac{1}{2} \times d_1 \times d_2$
   (b) Area of a rhombus = Product of adjacent sides × sine of the included angle.

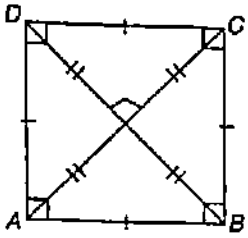
Square : A rectangle whose all sides are equal or a rhombus whose all angles are equal is called a square. Thus each rhombus is a parallelogram, a rectangle and a rhombus.

Properties

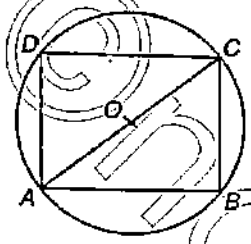
1. All side are equal and parallel.
2. All angles are right angles.
3. Diagonals are equal and bisect each other at right angle.

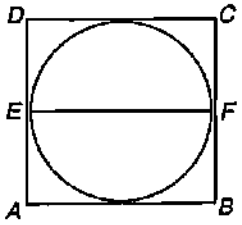
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in



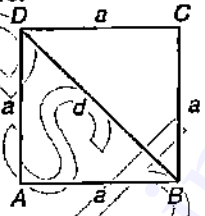
4. Diagonal of an inscribed square is equal to diameter of the inscribing circle.
5. Side of a circumscribed square is equal to the diameter of the inscribed circle.





6. The figure formed by joining the mid-points of the adjacent sides of a square is a square.
7. Area $= (\text{side})^2$

$$= a^2 = \frac{(\text{diagonal})^2}{2} = \frac{d^2}{2}$$

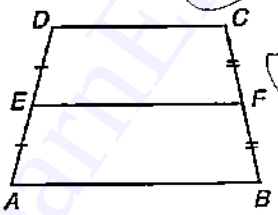


8. Diagonal $=$ side $\sqrt{2} = a\sqrt{2}$
9. Perimeter $= 4 \times$ side $= 4a$

Trapezium : A quadrilateral whose only one pair of sides is parallel and other two sides are not parallel.

Properties

1. The line joining the mid-points of the oblique (non-parallel) sides is half the sum of the parallel sides and is called the median.



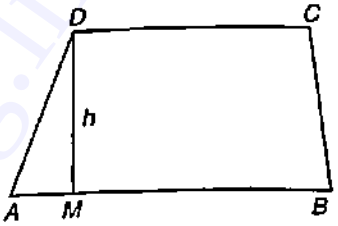
(*i.e.*, Median $= \frac{1}{2} \times$ sum of parallel sides

$$= \frac{1}{2} \times (AB + DC) = EF)$$

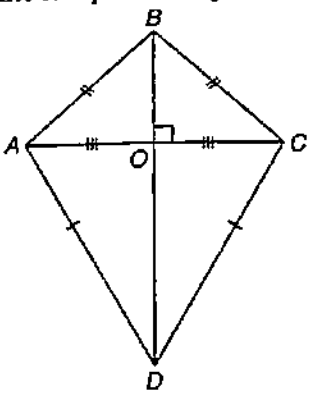
2. If the non-parallel sides are equal then the diagonals will also be equal to each other.
3. Diagonals intersect each other proportionally in the ratio of lengths of parallel sides.
4. By joining the mid-points of adjacent sides of a trapezium four similar triangles are obtained.
5. If a trapezium is inscribed in a circle, then it is an isosceles trapezium with equal oblique sides.
6. Area of trapezium $= \frac{1}{2} \times$ (sum of parallel sides $\times$ height)

$$= \frac{1}{2} \times (AB + CD) \times h$$

7. $AC^2 + BD^2 = BC^2 + AD^2 + 2AB \cdot CD$



Kite : In a kite two pairs of adjacent sides are equal.



Properties

1. $AB = BC$ and $AD = CD$
2. Diagonals intersect at right angles.
3. Shorter diagonal is bisected by the longer diagonal.
4. Area $= \frac{1}{2} \times$ product of diagonals

S. No.	Property	Parallelogram	Rectangle	Rhombus	Square
1.	Opposite sides are equal	✔	✔	✔	✔
2.	All sides are equal	✗	✗	✔	✔
3.	Opposite sides are parallel	✔	✔	✔	✔
4.	Opposite angles are equal	✔	✔	✔	✔
5.	All angles are equal and right angle	✗	✔	✗	✔
6.	Diagonals bisect each other	✔	✔	✔	✔
7.	Diagonals bisect each other at right angles	✗	✗	✔	✔
8.	Diagonals bisect vertex angles	✗	✗	✔	✔
9.	Diagonals are equal	✗	✔	✗	✔
10.	Diagonals form four triangles of equal area	✔	✔	✔	✔
11.	Diagonals form four congruent triangles	✗	✗	✔	✔

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

INTRODUCTORY EXERCISE-12.3

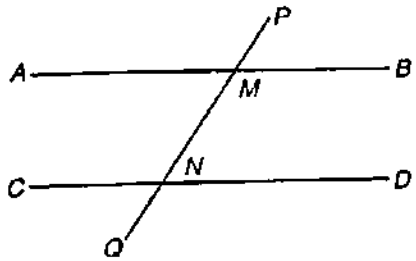
1. The measures of the angles of a quadrilateral taken in order are proportional to 1 : 2 : 3 : 4, then the quadrilateral is :
(a) parallelogram (b) trapezium
(c) rectangle (d) rhombus

2. Find the measure of largest angle of a quadrilateral if the measures of its interior angles are in the ratio of 3 : 4 : 5 : 6 :
(a) 60° (b) 120°
(c) 90° (d) can't be determined

3. $ABCD$ is a parallelogram, P and Q are the points on the diagonal AC such that $AP = QC$, then quadrilateral $BPDQ$ is a :
(a) trapezium (b) parallelogram
(c) square (d) none of these

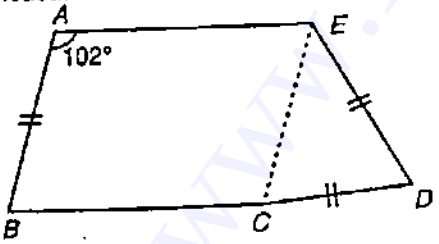
4. In a parallelogram $ABCD$, bisectors of consecutive angles A and B intersect at P. Find the measure of $\angle APB$:
(a) 90° (b) 60°
(c) 120° (d) data insufficient

5. AB and CD are two parallel lines and a transversal PQ intersects AB and CD at M and N respectively. The bisector of the interior angles form a quadrilateral :



(a) rectangle (b) square
(c) parallelogram (d) none of these

6. In the given figure $AE = BC$ and $AE \parallel BC$ and the three sides AB, CD and ED are equal in length. If $m\angle A = 102°$, find measures of $\angle BCD$:



(a) 138° (b) 162°
(c) 88° (d) none of these

7. $ABCD$ is a square, A is joined to a point P on BC and D is joined to a point Q on AB. If $AP = DQ$ and AP intersects DQ at R then $\angle DRP$ is :
(a) 60° (b) 120°
(c) 90° (d) can't be determined

8. A point X inside a rectangle $PQRS$ is joined to the vertices then, which of the following is true :
(a) $A(\Delta PSX) = A(\Delta RXQ)$
(b) $A(\Delta PSX) + A(\Delta PXQ) = A(\Delta RSX) + A(\Delta RQX)$
(c) $A(\Delta PXS) + A(\Delta RXQ) = A(\Delta SRX) + A(\Delta PXQ)$
(d) none of the above

9. $\square ABCD$ is a parallelogram, $m\angle DAB = 30°$, $BC = 20$ cm and $AB = 20$ cm. Find the area of parallelogram :
(a) 150 cm^2 (b) 200 cm^2
(c) 400 cm^2 (d) 260 cm^2

10. The length of a side of a rhombus is 10 m and one of its diagonal is 12 m. The length of the other diagonal is :
(a) 15 m (b) 18 m
(c) 16 m (d) can't be determined

11. $ABCD$ is a parallelogram and BD is a diagonal. $\angle BAD = 65°$ and $\angle DBC = 45°$, then $m\angle BDC$ is :
(a) 65° (b) 70°
(c) 20° (d) none of these

12. If $ABCD$ is a parallelogram in which P and Q are the centroids of ΔABD and ΔBCD, then, PQ equals :
(a) AQ (b) AP
(c) BP (d) DQ

13. Two parallelograms stand on equal bases and between the same parallels. The ratio of their areas is :
(a) 1 : 1 (b) $\sqrt{2}$: 1
(c) 1 : 3 (d) 1 : 2

14. If a rectangle and a parallelogram are equal in area and have the same base and are situated on the same side, then the ratio of perimeter of rectangle and that of parallelogram is k, then k is :
(a) $k > 1$ (b) $k < 1$
(c) $k = 1$ (d) can't be determined

15. If area of a parallelogram with sides l and b is A and that of a rectangle with sides l and b is S, then :
(a) $A < B$ (b) $A = B$
(c) $A > B$ (d) none of these

16. $ABCD$ is a parallelogram and M is the mid-point of BC. AB and DM are produced to meet at N, then :
(a) $AN = \sqrt{3}AB$ (b) $AN = 2AB$
(c) $AN^2 = \frac{3}{2}AB^2$ (d) $AN^2 = 2AB^2$

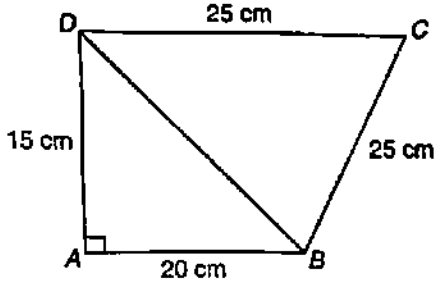
17. In a rectangle $ABCD$, P, Q are the mid-points of BC and AD respectively and R is any point on PQ, then ΔARB equals :
(a) $\frac{1}{2}(\square ABCD)$ (b) $\frac{1}{3}(\square ABCD)$
(c) $\frac{1}{4}(\square ABCD)$ (d) none of these

18. Diagonals of a parallelogram are 8 m and 6 m respectively. If one of side is 5m, then the area of parallelogram is :
(a) 18 m^2 (b) 30 m^2
(c) 24 m^2 (d) 48 m^2

19. In the given figure $AD = 15$ cm, $AB = 20$ cm and $BC = CD = 25$ cm. Find the area of $\square ABCD$:

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

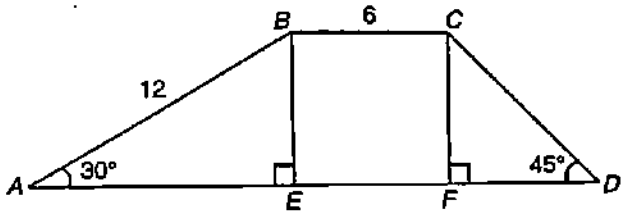
yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in



(a) $\frac{25}{4}(24+25\sqrt{3})$ cm^2 (b) $24(25+24\sqrt{3})$ cm^2

(c) $\frac{25}{2}(24+25\sqrt{3})$ cm^2 (d) none of these

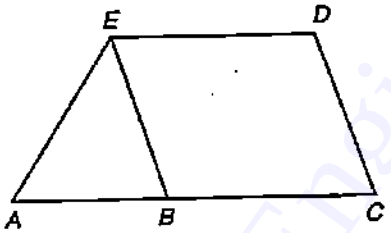
20. In the trapezium $ABCD$, $\angle BAE = 30°$, $\angle CDF = 45°$. $BC = 6$ cm and $AB = 12$ cm. Find the area of trapezium :



(a) $18(3+\sqrt{3})$ cm^2 (b) $36\sqrt{3}$ cm^2

(c) $12(3+2\sqrt{3})$ cm^2 (d) none of these

21. Area of quadrilateral $ACDE$ is 36 cm^2, B is the mid-point of AC. Find the area of ΔABE if $AC \parallel DE$ and $BE \parallel DC$:



(a) 10 cm^2

(b) 9 cm^2

(c) 12 cm^2

(d) can't be determined

22. A square and a rhombus have the same base and the rhombus is inclined at 30°. What is the ratio of area of the square to the area of the rhombus :

(a) $\sqrt{2}:1$ (b) $2:1$

(c) $1:1$ (d) $2:\sqrt{3}$

23. Find the area of a quadrilateral with sides 17, 25, 30 and 28 cm and one of its diagonal is 26 cm :

(a) 450 cm^2 (b) 360 cm^2

(c) 540 cm^2 (d) 720 cm^2

24. $PQRS$ is a parallelogram in which $\angle P = 70°$, $\angle Q = 90°$ and $\angle R = 100°$. How many points in the plane of the quadrilateral are there such that a point is equidistant from its vertices ?

(a) 0 (b) 1

(c) 2 (d) 3

25. $ABCD$ is a parallelogram. The diagonals AC and BD intersect at a point O. If E, F, G and H are the mid-points of AO, DO, CO and BO respectively, then the ratio of $(EF + FG + GH + HE)$ to $(AD + DC + CB + BA)$ is :

(a) $1:1$ (b) $1:2$

(c) $1:3$ (d) $1:4$

26. If $ABCD$ is a rhombus, then :

(a) $AC^2 + BD^2 = 4AB^2$ (b) $AC^2 + BD^2 = AB^2$

(c) $AC^2 + BD^2 = 2AB^2$ (d) $2(AC^2 + BD^2) = 3AB^2$

27. If P is a point within a rectangle $ABCD$, then :

(a) $AP^2 + PC^2 = BP^2 + PD^2$

(b) $AP^2 + BP^2 = PC^2 + PD^2$

(c) $AP + PC = BP + PD$

(d) $AP \times PC = BP \times PD$

28. $\square ABCD$ is a parallelogram, $AB = 14$ cm, $BC = 18$ cm and $AC = 16$ cm. Find the length of the other diagonal:

(a) 24 cm (b) 28 cm

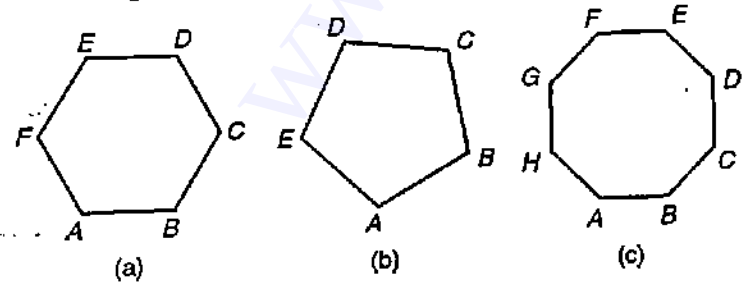
(c) 36 cm (d) 32 cm

(4) POLYGONS

Polygon : It is a closed plane figure bounded by three or more than three straight lines.

e.g., triangle, quadrilateral, pentagon, hexagon, heptagon, octagon, nonagon and decagon.

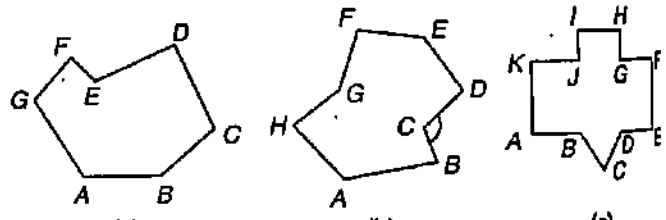
Convex Polygon: A polygon in which none of its interior angle is more than 180°, is known as a convex polygon.



(a) (b) (c)

Convex Polygons

Concave Polygon: A polygon in which atleast one interior angle is more than 180°, then it is said to be concave.



(a) (b) (c)

Concave Polygons

Regular Polygon : A polygon in which all the sides are equal and also all the interior angles are equal, is called a regular polygon.

Formulae : For regular polygons

1. Sum of all interior angles $= (n-2)\times 180$
$= (2n-4)\times 90°$
2. Each interior angle $= 180 -$ exterior angle
3. Each exterior angle $= \left(\frac{360}{\text{Number of sides}}\right)$ (in degrees)

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

4. Sum of all exterior angle = 360° (always constant)

5. Number of sides in a polygon $= \frac{360}{\text{Exterior angle}}$

$= 2(x+1)$ $\left(x = \frac{\text{Interior angle}}{\text{Exterior angle}}\right)$

6. Number of diagonals $= {}^{n}C_2 - n = \frac{n(n-1)}{2} - n$

$= \frac{n^2-3n}{2} = \frac{n(n-3)}{2}$

where $n \rightarrow$ number of sides of a polygon.

7. Star : Sum of angles of a n point star $= (n-4)\pi$

8. Area of a polygon $= \frac{na^2}{4} \times \cot\left(\frac{180}{n}\right)$;

where $n \rightarrow$ number of sides
$a \rightarrow$ length of sides

No. of sides	Polygon	Sum of all the angles	Each interior angle	Each exterior angle	No. of diagonals
3	Triangle	180°	60°	120°	0
4	Quadrilateral	360°	90°	90°	2
5	Pentagon	540°	108°	72°	5
6	Hexagon	720°	120°	60°	9
7	Heptagon	900°	$\left(128\frac{4}{7}\right)^\circ$	$\left(51\frac{3}{7}\right)^\circ$	14
8	Octagon	1080°	135°	45°	20
9	Nonagon	1260°	140°	40°	27
10	Decagon	1440°	144°	36°	35

INTRODUCTORY EXERCISE-12.4

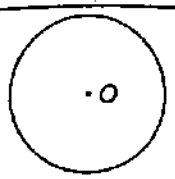
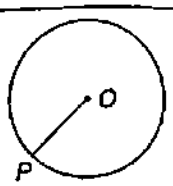
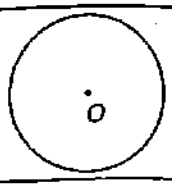
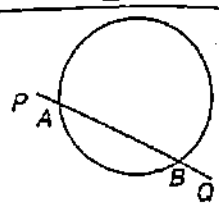
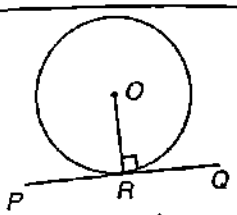
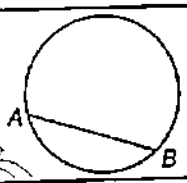
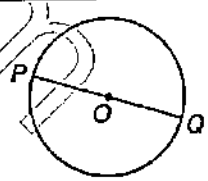
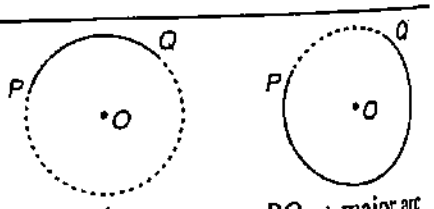
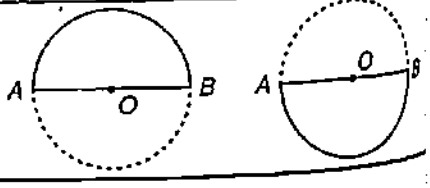
1. Each interior angle of a regular polygon is 140°. The number of sides is :
   (a) 10 (b) 8
   (c) 6 (d) 9
2. Each angle of a regular hexagon is :
   (a) 60° (b) 120°
   (c) 90° (d) none of these
3. If one of the interior angles of a regular polygon is equal to 5/6 times of one of the interior angles of a regular pentagon, then the number of sides of the polygon is :
   (a) 3 (b) 4
   (c) 6 (d) 8
4. The sum of the interior angles of a polygon is 1260°. The number of sides of the polygon is :
   (a) 6 (b) 7
   (c) 8 (d) 9
5. If each interior angle of a regular polygon is 3 times its exterior angle, the number of sides of the polygon is :
   (a) 4 (b) 5
   (c) 6 (d) 8
6. Difference between the interior and exterior angles of regular polygon is 60°. The number of sides in the polygon is :
   (a) 5 (b) 6
   (c) 8 (d) 9
7. A polygon has 54 diagonals. The number of sides in the polygon is :
   (a) 7 (b) 9
   (c) 12 (d) none of these
8. The ratio between the number of sides of two regular polygon 1 : 2 and the ratio between their interior angle is 3 : 4. The number of sides of these polygons are respectively :
   (a) 3, 6 (b) 4, 8
   (c) 6, 9 (d) 5, 10
9. The sum of all the interior angles of a regular polygon is four times the sum of its exterior angles. The polygon is :
   (a) hexagon (b) triangle
   (c) decagon (d) nonagon
10. The ratio of the measure of an angle of a regular nonagon to the measure of its exterior angle is :
   (a) 3 : 5 (b) 5 : 2
   (c) 7 : 2 (d) 4 : 5

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

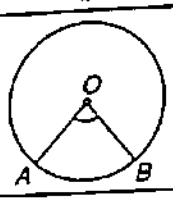
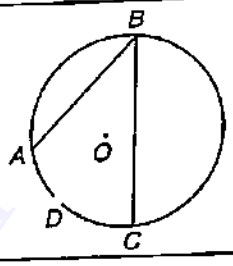
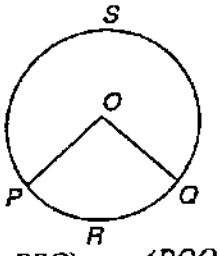
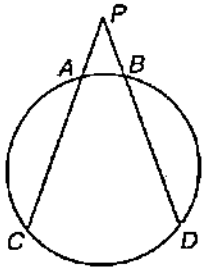
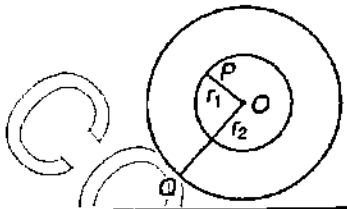
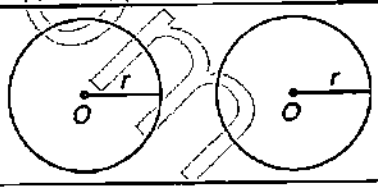
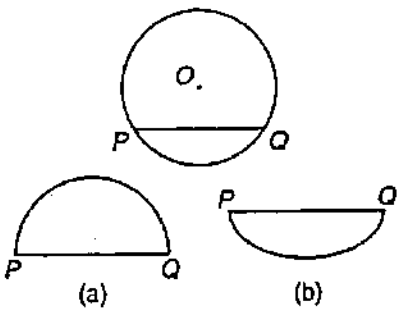
(5) CIRCLE

Circle : A circle is a set of points on a plane which lie at a fixed distance from a fixed point.

Nomenclature	Definition	Diagram
Centre	The fixed point is called the centre. In the given diagram 'O' is the centre of the circle.	
Radius	The fixed distance is called a radius. In the given diagram OP is the radius of the circle. (Point P lies on the circumference)	
Circumference	The circumference of a circle is the distance around a circle, which is equal to $2\pi r$. ($r \rightarrow$ radius of the circle)	
Secant	A line segment which intersects the circle in two distinct points, is called as secant. In the given diagram secant PQ intersects circle at two points at A and B.	
Tangent	A line segment which has one common point with the circumference of a circle *i.e.*, it touches only at only one point is called as tangent of circle. The common point is called as point of contact. In the given diagram PQ is a tangent which touches the circle at a point R.	 (R is the point of contact) **Note:** Radius is always perpendicular to tangent.
Chord	A line segment whose end points lie on the circle. In the given diagram AB is a chord.	
Diameter	A chord which passes through the centre of the circle is called the diameter of the circle. The length of the diameter is twice the length of the radius. In the given diagram PQ is the diameter of the circle. ($O \rightarrow$ is the centre of the circle)	
Arc	Any two points on the circle divides the circle into two parts the smaller part is called as minor arc and the larger part is called as major arc. It is denoted as '$\frown$'. In the given diagram $\overset{\frown}{PQ}$ is arc.	 $PQ \rightarrow$ minor arc $PQ \rightarrow$ major arc
Semicircle	A diameter of the circle divides the circle into two equal parts. Each part is called as semicircle.	

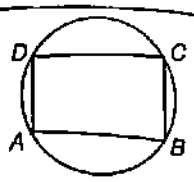
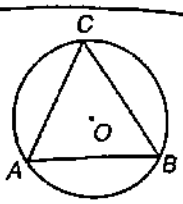
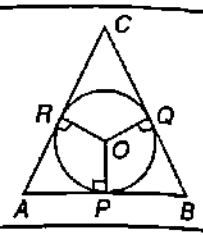
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmabhoob.wordpress.com
For More Visit : www.LearnEngineering.in

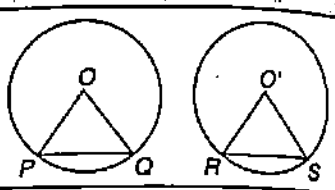
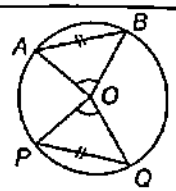
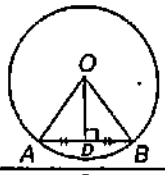
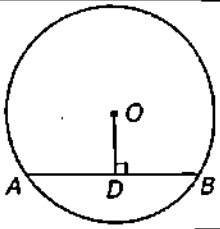
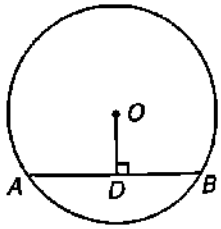
Nomenclature	Definition	Diagram
Central angle	An angle formed at the centre of the circle, is called the central angle. In the given diagram $\angle AOB$ is the central angle.	
Inscribed angle	When two chords have one common end point, then the angle included between these two chords at the common point is called the inscribed angle. $\angle ABC$ is the inscribed angle by the arc $\overset{\frown}{ADC}$.	
Measure of an arc	Basically it is the central angle formed by an arc, *e.g.*, (a) measure of a circle = 360° (b) measure of a semicircle = 180° (c) measure of a minor arc $\overset{\frown}{PQ} = \angle POQ$ (d) measure of a major arc $\overset{\frown}{PQ} = 360 - \angle POQ$	 $m(\text{arc } PRQ) = m\angle POQ$ $m(\text{arc } PSQ) = 360° - m(\text{arc } PRQ)$
Intercepted arc	In the given diagram $\overset{\frown}{AB}$ and $\overset{\frown}{CD}$ are the two intercepted arcs, intercepted by $\angle CPD$. The end points of the arc must touch the arms of $\angle CPD$ *i.e.*, CP and DP.	
Concentric circles	Circles having the same centre at a plane are called the concentric circles. In the given diagram there are two circles with radii r_1 and r_2 having the common (or same) centre. These are called as concentric circles.	
Congruent circles	Circles with equal radii are called as congruent circles.	
Segment of a circle	A chord divides a circle into two regions. These two regions are called the segments of a circle. (a) major segment (b) minor segment	

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

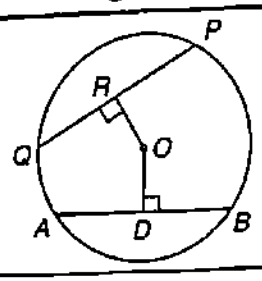
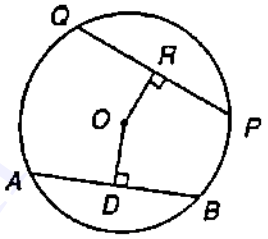
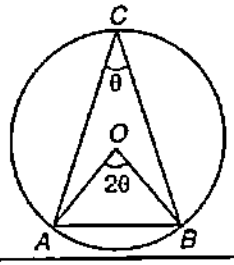
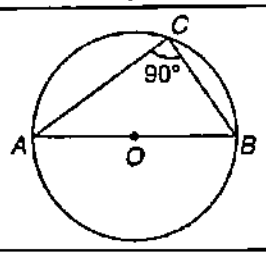
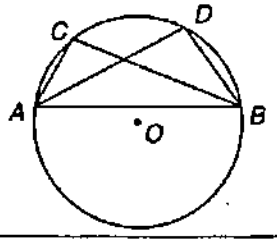
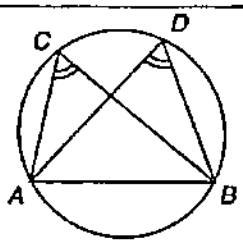
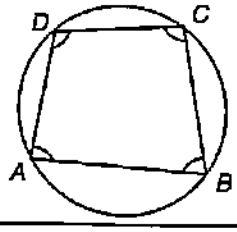
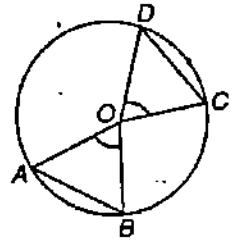
Nomenclature	Definition	Diagram
Cyclic quadrilateral	A quadrilateral whose all the four vertices lie on the circle.	
Circum-circle	A circle which passes through all the three vertices of a triangle. Thus the circumcentre is always equidistant from the vertices of the triangle. $OA = OB = OC$ (circumradius)	
Incircle	A circle which touches all the three sides of a triangle *i.e.*, all the three sides of a triangle are tangents to the circle is called an incircle. Incircle is always equidistant from the sides of a triangle. $OP = OQ = OR$ (inradius of the circle)	

- Two arcs of a circle (or of congruent circles) are congruent if their degree measures are equal.
- There is one and only one circle passes through three non-collinear points.

S. No.	Theorem	Diagram
1.	In a circle (or in congruent circles) equal chords are made by equal arcs. $\{OP = OQ\} = \{O'R = O'S\}$ and $\overset{\frown}{PQ} = \overset{\frown}{RS}$ $\therefore$ $PQ = RS$	
2.	Equal arcs (or chords) subtend equal angles at the centre $\overset{\frown}{PQ} = \overset{\frown}{AB}$ (or $PQ = AB$) $\therefore$ $\angle POQ = \angle AOB$	
3.	The perpendicular from the centre of a circle to a chord bisects the chord *i.e.*, if $OD \perp AB$ $\therefore$ $AB = 2AD = 2BD$	
4.	The line joining the centre of a circle to the mid-point of a chord is perpendicular to the chord. $\because$ $AD = DB$ $\therefore$ $OD \perp AB$	
5.	Perpendicular bisector of a chord passes through the centre. *i.e.*, if $OD \perp AB$ and $AD = DB$ $\therefore$ O is the centre of the circle.	

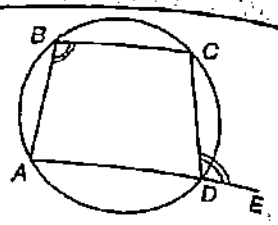
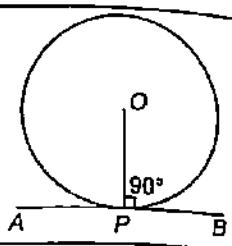
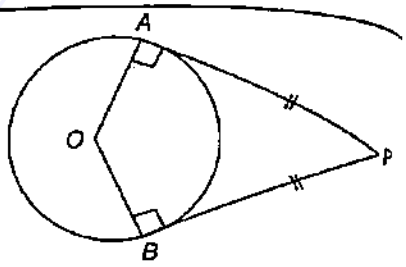
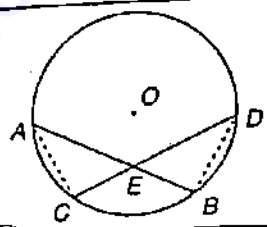
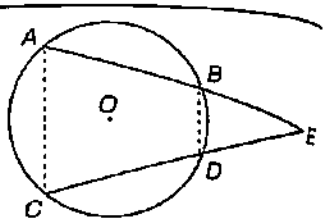
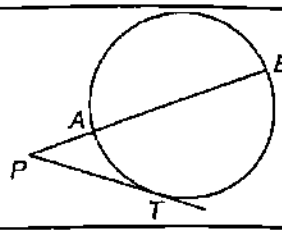
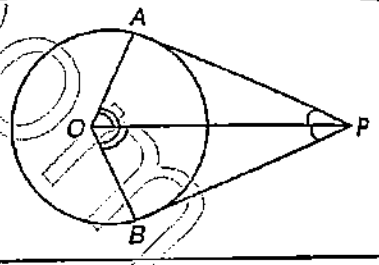
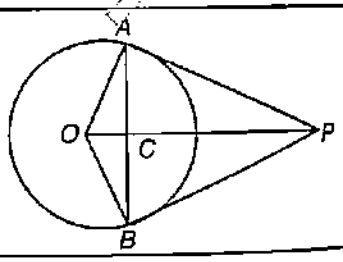
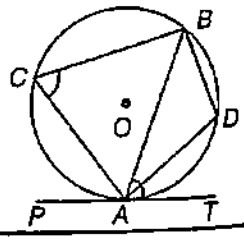
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
 For More Visit : www.LearnEngineering.in

S. No.	Theorem	Diagram
6.	Equal chords of a circle (or of congruent circles) are equidistant from the centre. $\because$ $AB = PQ$ $\therefore$ $OD = OR$	
7.	Chords which are equidistant from the centre in a circle (or in congruent, circles) are equal. $\because$ $OD = OR$ $\therefore$ $AB = PQ$	
8.	The angle subtended by an arc (the degree measure of the arc) at the centre of a circle is twice the angle subtended by the arc at any point on the remaining part of the circle. $m\angle AOB = 2m\angle ACB$.	
9.	Angle in a semicircle is a right angle.	
10.	Angles in the same segment of a circle are equal *i.e.*, $\angle ACB = \angle ADB$	
11.	If a line segment joining two points subtends equal angle at two other points lying on the same side of the line containing the segment, then the four points lie on the same circle. $\angle ACB = \angle ADB$ $\therefore$ Points A, C, D, B are concyclic *i.e.*, lie on the circle.	
12.	The sum of pair of opposite angles of a cyclic quadrilateral is 180°. $\angle DAB + \angle BCD = 180°$ and $\angle ABC + \angle CDA = 180°$ (Inverse of this theorem is also true)	
13	Equal chords (or equal arcs) of a circle (or congruent circles) subtended equal angles at the centre. $AB = CD$ (or $\overset{\frown}{AB} = \overset{\frown}{CD}$) $\therefore$ $\angle AOB = \angle COD$ (Inverse of this theorem is also true)	

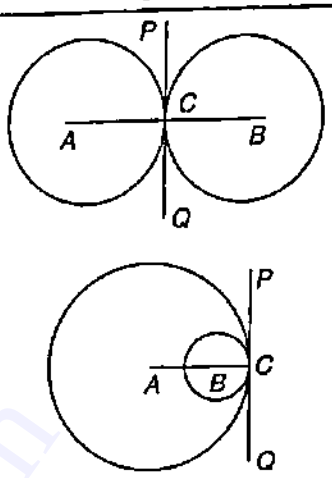
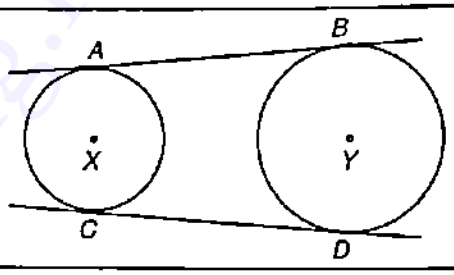
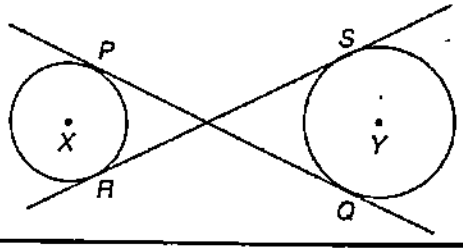
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

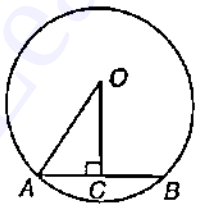
S. No.	Theorem	Diagram
14.	If a side of a cyclic quadrilateral is produced, then the exterior angle is equal to the interior opposite angle. $m\angle CDE = m\angle ABC$	
15.	A tangent at any point of a circle is perpendicular to the radius through the point of contact. (Inverse of this theorem is also true)	
16.	The lengths of two tangents drawn from an external point to a circle are equal. *i.e.*, $AP = BP$	
17.	If two chords AB and CD of a circle, intersect inside a circle (outside the circle when produced at a point E), then $AE \times BE = CE \times DE$	 
18.	If PB be a secant which intersects the circle at A and B and PT be a tangent at T then $PA \cdot PB = (PT)^2$	
19.	From an external point from which the tangents are drawn to the circle with centre O, then (a) they subtend equal angles at the centre. (b) they are equally inclined to the line segment joining the centre of that point. $\angle AOP = \angle BOP$ and $\angle APO = \angle BPO$	
20.	If P is an external point from which the tangents to the circle with centre O touch it at A and B then OP is the perpendicular bisector of AB. $OP \perp AB$ and $AC = BC$	
21.	**Alternate segment theorem :** If from the point of contact of a tangent, a chord is drawn then the angles which the chord makes with the tangent line are equal respectively to the angles formed in the corresponding alternate segments. In the adjoining diagram. $\angle BAT = \angle BCA$ and $\angle BAP = \angle BDA$	

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

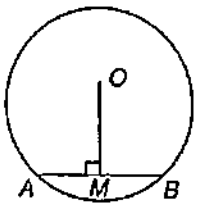
yoursmahboob.wordpress.com

S. No.	Theorem	Diagram
22.	The point of contact of two tangents lies on the straight line joining the two centres. (a) When two circles touch externally then the distance between their centres is equal to sum of their radii, *i.e.*, $AB = AC + BC$ (b) When two circles touch internally the distance between their centres is equal to the difference between their radii *i.e.*, $AB = AC - BC$	
23.	For the two circles with centre X and Y and radii r_1 and r_2. AB and CD are two Direct Common Tangents (DCT), then the length of DCT $= \sqrt{(\text{distance between centres})^2 - (r_1 - r_2)^2}$	
24.	For the two circles with centre X and Y and radii r_1 and r_2. PQ and RS are two transverse common tangent, then length of TCT $= \sqrt{(\text{distance between centres})^2 - (r_1 + r_2)^2}$	

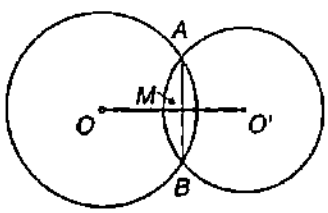
INTRODUCTORY EXERCISE-12.5

1. In the given figure, O is the centre of the circle. Radius of the circle is 17 cm. If $OC = 8$ cm, then the length of the chord AB is :


(a) 35 cm
(b) 30 cm
(c) 15 cm
(d) 18 cm

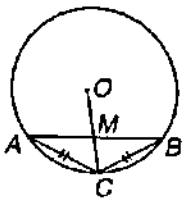
2. In the given figure $OM \perp AB$, radius of the circle is 5 cm and length of the chord $AB = 8$ cm. Find the measure of OM :


(a) 3 cm
(b) 2.5 cm
(c) 2 cm
(d) 6 cm

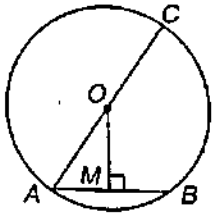
3. In the given figure, two circles with their respective centres intersect each other at A and B and AB intersects OO' at M, then $m\angle OMA$ is :


(a) 60°
(b) 80°
(c) 90°
(d) can't be determined

4. In the above question (no. 3) what is the ratio of $AM : BM$?
(a) 5 : 6 (b) 3 : 2
(c) 1 ; 1 (d) can't be determined

5. In the given figure the two chords AC and BC are equal. The radius OC intersect AB at M, then $AM : BM$ is :


(a) 1 : 1
(b) $\sqrt{2} : 3$
(c) $3 : \sqrt{2}$
(d) none of the above

6. In the adjoining figure, O is the centre of circle and diameter $AC = 26$ cm. If chord $AB = 10$ cm, then the distance between chord AB and centre O of the circle is :


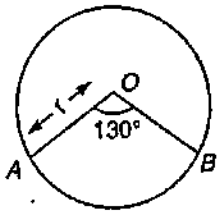
(a) 24 cm
(b) 16 cm
(c) 12 cm
(d) none of the above

7. In the adjoining circle $C(0, r)$ the degree measure of minor arc $AB = 130°$. Find the degree measure of major arc :

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

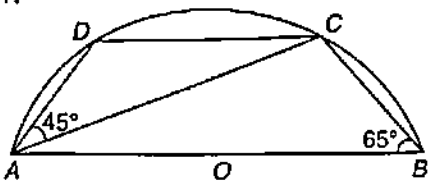
yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

(a) 230°
(b) 260°
(c) 310°
(d) none of the above



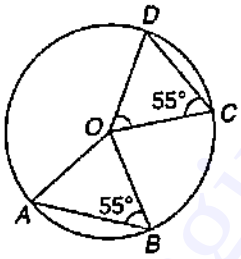
8. If the diagonals of a cyclic quadrilateral are equal, then the quadrilateral is :
(a) rhombus (b) square
(c) rectangle (d) none of these

9. In the given figure, AB is diameter of the circle. C and D lie on the semicircle. $\angle ABC = 65°$ and $\angle CAD = 45°$. Find $m\angle DCA$:



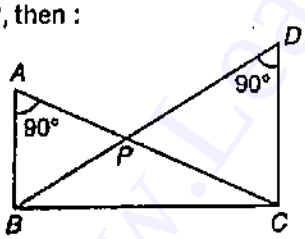
(a) 45° (b) 25°
(c) 20° (d) none of these

10. In the given figure, chords AB and CD are equal. If $\angle OBA = 55°$, then $m\angle COD$ is :
(a) 65°
(b) 55°
(c) 70°
(d) 50°



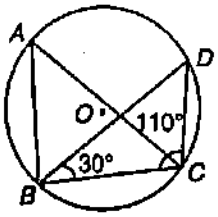
11. The quadrilateral formed by angle bisectors of a cyclic quadrilateral is a :
(a) rectangle (b) square
(c) parallelogram (d) cyclic quadrilateral

12. ΔABC and ΔDBC have a common base and drawn towards one sides. $\angle BAC = \angle BDC = 90°$. If AC and DB intersect at P, then :

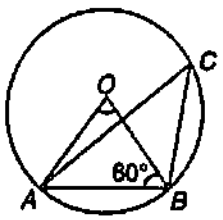


(a) $AP \times PC = BP \times PD$ (b) $AP \times BP = PC \times PD$
(c) $AP \times PD = PC \times BP$ (d) none of these

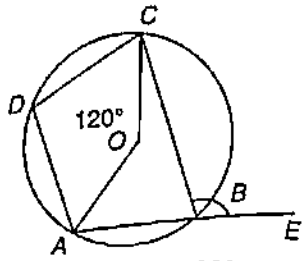
13. In the given figure, $\angle BAC$ and $\angle BDC$ are the angles of same segments. $\angle DBC = 30°$ and $\angle BCD = 110°$. Find $m\angle BAC$ is :
(a) 35°
(b) 40°
(c) 55°
(d) 60°



14. In the given figure, O is the centre of the circle. $\angle ABO = 60°$. Find the value of $\angle ACB$:
(a) 40°
(b) 60°
(c) 50°
(d) 30°

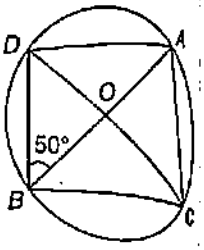


15. In the given figure, $\angle AOC = 120°$. Find $m\angle CBE$, where O is the centre :

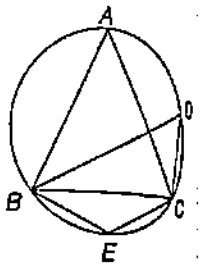


(a) 60° (b) 100°
(c) 120° (d) 150°

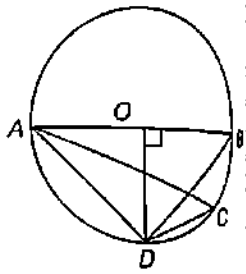
16. In the adjoining figure, O is the centre of the circle and $\angle OBD = 50°$. Find the $m\angle BAD$:
(a) 60°
(b) 40°
(c) 80°
(d) 45°



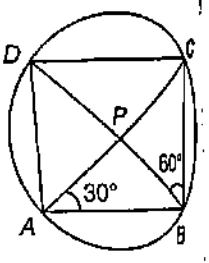
17. In the given figure, ΔABC is an equilateral triangle. Find $m\angle BEC$:
(a) 120°
(b) 60°
(c) 80°
(d) none of the above



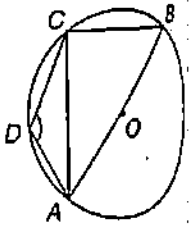
18. In the given figure, AB is the diameter of the circle. Find the value of $\angle ACD$:
(a) 30°
(b) 60°
(c) 45°
(d) 25°



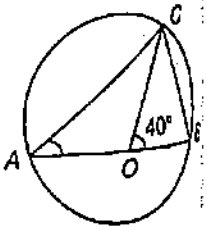
19. In the given figure, $ABCD$ is a cyclic quadrilateral and diagonals bisect each other at P. If $\angle DBC = 60°$ and $\angle BAC = 30°$, then $\angle BCD$ is :
(a) 90°
(b) 60°
(c) 80°
(d) none of the above



20. In the given figure, $ABCD$ is a cyclic quadrilateral and AB is the diameter. $\angle ADC = 140°$, then find $m\angle BAC$:
(a) 45°
(b) 40°
(c) 50°
(d) none of the above



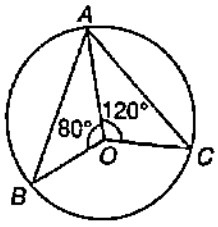
21. In the given figure, $\angle COB = 40°$, AB is the diameter of the circle. Find $m\angle CAB$:
(a) 40°
(b) 20°
(c) 30°
(d) none of the above



For More Visit : www.LearnEngineering.in
Scanned by CamScanner

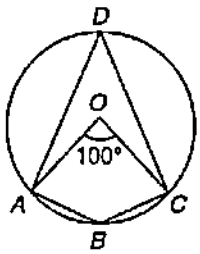
yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

22. In the given figure, O is the centre of circle. $\angle AOB = 80°$ and $\angle AOC = 120°$. Find $m\ \angle BAC$:



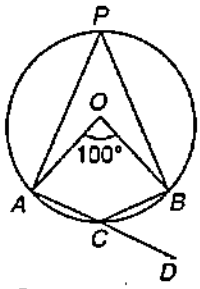
(a) 120°
(b) 80°
(c) 100°
(d) none of the above

23. In the given figure, O is the centre of the circle and $\angle AOC = 100°$. Find the ratio of $m\ \angle ADC : m\ \angle ABC$:



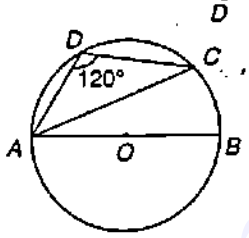
(a) 5 : 6
(b) 1 : 2
(c) 5 : 13
(d) none of the above

24. In the given figure, O is the centre of circle, $\angle AOB = 100°$. Find $m\ \angle BCD$:



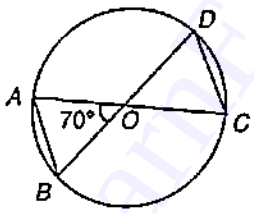
(a) 80°
(b) 60°
(c) 50°
(d) 40°

25. In the given figure, AB is the diameter of the circle. $\angle ADC = 120°$. Find $m\ \angle CAB$:



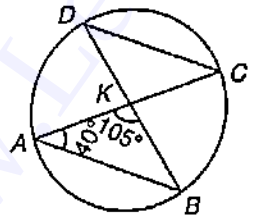
(a) 20°
(b) 30°
(c) 40°
(d) can't be determined

26. In the given figure, O is the centre of the circle. $\angle AOB = 70°$, find $m\ \angle OCD$:



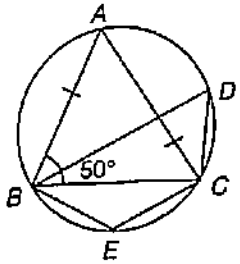
(a) 70°
(b) 55°
(c) 65°
(d) 110°

27. In the given figure, $\angle CAB = 40°$ and $\angle AKB = 105°$. Find $\angle KCD$:



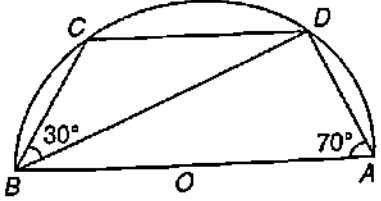
(a) 65°
(b) 35°
(c) 40°
(d) 72°

28. In the given figure, ABC is an isosceles triangle in which $AB = AC$ and $m\ \angle ABC = 50°$, $m\ \angle BDC$:



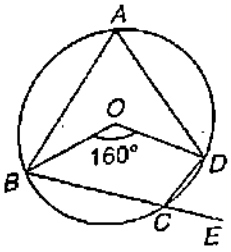
(a) 80°
(b) 60°
(c) 65°
(d) 100°

29. In the given figure, AB is the diameter. $m\ \angle BAD = 70°$ and $m\ \angle DBC = 30°$. Find $m\ \angle BDC$:



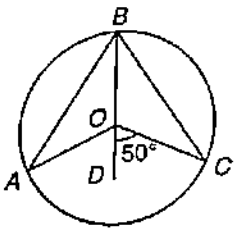
(a) 25°
(b) 30°
(c) 40°
(d) 60°

30. Find the value of $\angle DCE$:



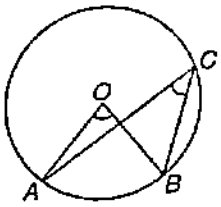
(a) 100°
(b) 80°
(c) 90°
(d) 75°

31. 'O' is the centre of the circle, line segment BOD is the angle bisector of $\angle AOC$, $m\ \angle COD = 50°$. Find $m\ \angle ABC$:



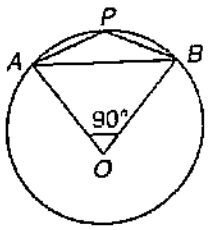
(a) 25°
(b) 50°
(c) 100°
(d) 120°

32. In the given figure, O is the centre of the circle and $\angle ACB = 25°$. Find $\angle AOB$:



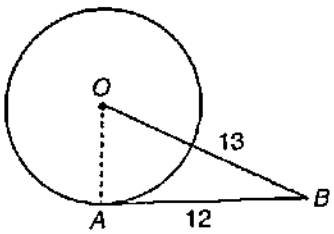
(a) 25°
(b) 50°
(c) 75°
(d) 60°

33. In the given figure, O is the centre of the circle. $\angle AOB = 90°$. Find $m\ \angle APB$:



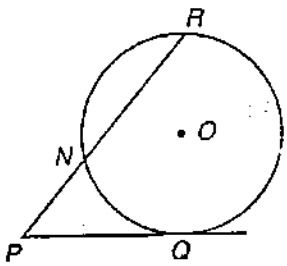
(a) 130°
(b) 150°
(c) 135°
(d) can't be determined

34. In the given figure, O is the centre of the circle. AB is tangent. $AB = 12$ cm and $OB = 13$ cm. Find OA :



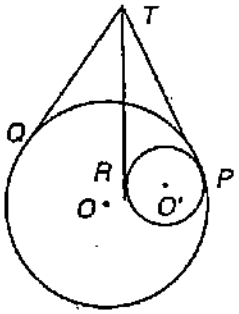
(a) 6.5 cm
(b) 6 cm
(c) 5 cm
(d) none of these

35. In the given figure, PQ is the tangent of the circle. Line segment PR intersects the circle at N and R. $PQ = 15$ cm, $PR = 25$ cm, find PN :



(a) 15 cm
(b) 10 cm
(c) 9 cm
(d) 6 cm

36. In the given figure, there are two circles with the centres O and O' touching each other internally at P. Tangents TQ and TP are drawn to the larger circle and tangents TP and TR are drawn to the smaller circle. Find $TQ : TR$:



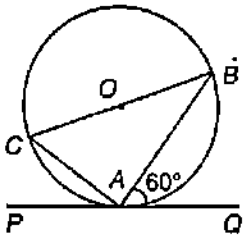
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

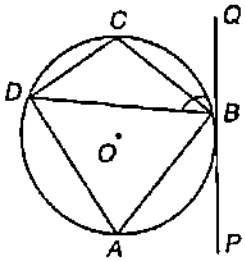
(a) 8 : 7 (b) 7 : 8
(c) 5 : 4 (d) 1 : 1

37. In the given figure, PAQ is the tangent. BC is the diameter of the circle. $m\angle BAQ = 60°$, find $m\angle ABC$:



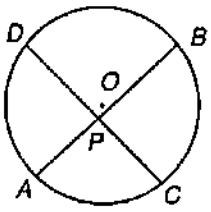
(a) 25°
(b) 30°
(c) 45°
(d) 60°

38. $ABCD$ is a cyclic quadrilateral PQ is a tangent at B. If $\angle DBQ = 65°$, then $\angle BCD$ is :



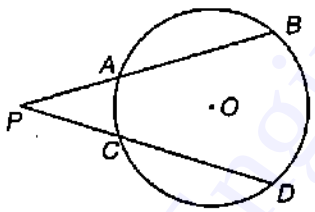
(a) 35°
(b) 85°
(c) 115°
(d) 90°

39. In the given figure, $AP = 2$ cm, $BP = 6$ cm and $CP = 3$ cm. Find DP :



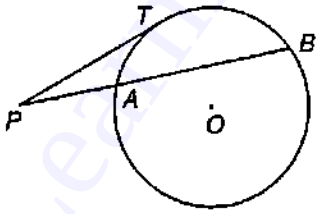
(a) 6 cm
(b) 4 cm
(c) 2 cm
(d) 3 cm

40. In the given figure, $AP = 3$ cm, $BA = 5$ cm and $CP = 2$ cm. Find CD :



(a) 12 cm
(b) 10 cm
(c) 9 cm
(d) 6 cm

41. In the given figure, tangent $PT = 5$ cm, $PA = 4$ cm, find AB :

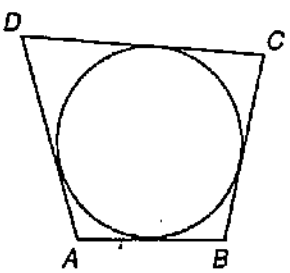


(a) $\frac{7}{4}$ cm
(b) $\frac{11}{4}$ cm
(c) $\frac{9}{4}$ cm
(d) can't be determined

42. Two circles of radii 13 cm and 5 cm touch internally each other. Find the distance between their centres :
(a) 18 cm (b) 12 cm
(c) 9 cm (d) 8 cm

43. Three circles touch each other externally. The distance between their centre is 5 cm, 6 cm and 7 cm. Find the radii of the circles :
(a) 2 cm, 3 cm, 4 cm (b) 3 cm, 4 cm, 1 cm
(c) 1cm, 2.5 cm, 3.5 cm (d) 1 cm, 2 cm, 4 cm

44. A circle touches a quadrilateral $ABCD$. Find the true statement :

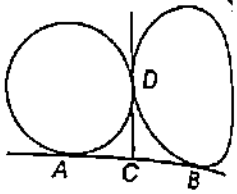


(a) $AB + BC = CD + AD$
(b) $AB + CD = BC + AD$
(c) $BD = AC$
(d) none of the above

45. O and O' are the centres of two circles which touch each other externally at P. AB is a common tangent. Find $\angle APO$:
(a) 90° (b) 120°
(c) 60° (d) data insufficient

46. If AB is a chord of a circle, P and Q are two points on the circle different from A and B, then :
(a) the angle subtended by AB at P and Q are either equal or supplementary.
(b) the sum of the angles subtended by AB at P and Q always equal two right angles.
(c) the angles subtended at P and Q by AB are always equal.
(d) the sum of the angles subtended at P and Q is equal to four right angles.

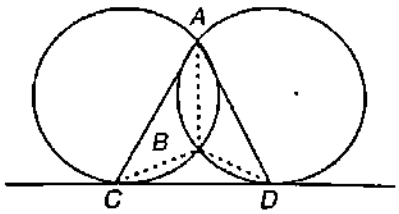
47. In the given figure, AB and CD are two common tangents to the two touching circle. If $CD = 6$ cm, then AB is equal to :



(a) 9 cm
(b) 15 cm
(c) 12 cm
(d) none of the above

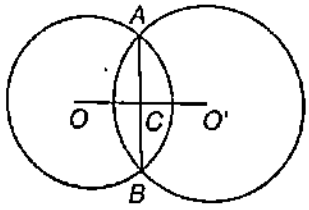
48. In the given figure, CD is a direct common tangent to two circles intersecting each other at A and B, then :

$\angle CAD + \angle CBD = ?$



(a) 120° (b) 90°
(c) 360° (d) 180°

49. O and O' are the centres of circle of radii 20 cm and 15 cm. $AB = 24$ cm. What is the distance OO' ?



(a) 51 cm (b) 45 cm
(c) 35 cm (d) 48 cm

50. In a circle of radius 5 cm, AB and AC are the two chords such that $AB = AC = 6$ cm. Find the length of the chord BC
(a) 4.8 cm (b) 10.8 cm
(c) 9.6 cm (d) none of these

51. In a circle of radius 17 cm, two parallel chords are drawn on opposite sides of a diameter. The distance between the chords is 23 cm. If the length of one chord is 16 cm then the length of the other is :
(a) 23 cm (b) 30 cm
(c) 15 cm (d) none of these

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

52. If two circles are such that the centre of one lies on the circumference of the other, then the ratio of the common chord of two circles to the radius of any of the circles is :
(a) $\sqrt{3}:2$ (b) $\sqrt{3}:1$
(c) $\sqrt{5}:1$ (d) none of these

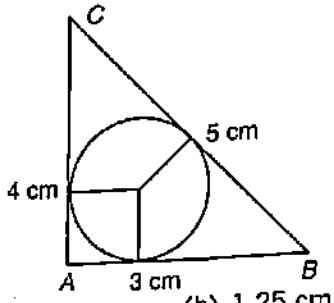
53. Two circles touch each other internally. Their radii are 2 cm and 3 cm. The biggest chord of the other circle which is outside the inner circle, is of length :
(a) $2\sqrt{2}$ cm (b) $3\sqrt{2}$ cm
(c) $2\sqrt{3}$ cm (d) $4\sqrt{2}$ cm

54. Through any given set of four points P, Q, R, S it is possible to draw :
(a) atmost one circle
(b) exactly one circle
(c) exactly two circles
(d) exactly three circles

55. The distance between the centres of equal circles each of radius 3 cm is 10 cm. The length of a transverse tangent is :
(a) 4 cm (b) 6 cm
(c) 8 cm (d) 10 cm

56. The number of common tangents that can be drawn to two given circles is at the most :
(a) 1 (b) 2
(c) 3 (d) 4

57. ABC is a right angled triangle $AB = 3$ cm, $BC = 5$ cm and $AC = 4$ cm, then the inradius of the circle is :



(a) 1 cm (b) 1.25 cm
(c) 1.5 cm (d) none of these

58. A circle has two parallel chords of lengths 6 cm and 8 cm. If the chords are 1 cm apart and the centre is on the same side of the chords, then a diameter of the circle is of length :
(a) 5 cm (b) 6 cm
(c) 8 cm (d) 10 cm

59. Three equal circles of unit radius touch each other. Then, the area of the circle circumscribing the three circles is :
(a) $6\pi(2+\sqrt{3})^2$ (b) $\frac{\pi}{6}(2+\sqrt{3})^2$
(c) $\frac{\pi}{3}(2+\sqrt{3})^2$ (d) $3\pi(2+\sqrt{3})^2$

60. The radius of a circle is 20 cm. The radii (in cm) of three concentric circles drawn in such a manner that the whole area is divided into four equal parts, are :
(a) $20\sqrt{2}, 20\sqrt{3}, 20$ (b) $\frac{10\sqrt{3}}{3}, \frac{10\sqrt{2}}{3}, \frac{10}{3}$
(c) $10\sqrt{3}, 10\sqrt{2}, 10$ (d) 17, 14, 9

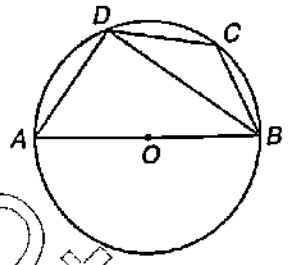
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

EXERCISE

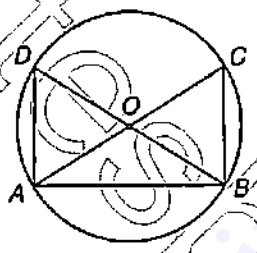
LEVEL 1

1. In the adjoining figure AB is a diameter of the circle and $\angle BCD = 130°$. What is the value of $\angle ABD$?



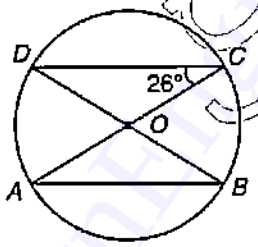
(a) 30°
(b) 50°
(c) 40°
(d) none of the above

2. In the given figure O is the centre of the circle and $\angle BAC = 25°$, then the value of $\angle ADB$ is :



(a) 40°
(b) 55°
(c) 50°
(d) 65°

3. In the given figure O is the centre of the circle and $\angle OCD = 26°$, find $\angle AOD$:

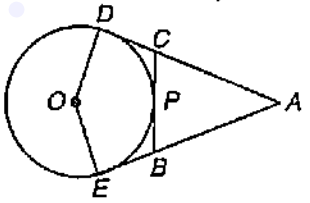


(a) 52°
(b) 154°
(c) 128°
(d) data insufficient

4. Three sides of a triangle ABC are a, b, c. $a = 4700$ cm, $b = 4935$ cm and $c = 6815$ cm. The internal bisector of $\angle A$ meets BC at P, and the bisector passes through incentre $'O'$. What is ratio of $PO : OA$?

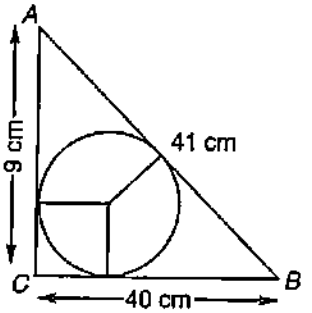
(a) 3 : 2
(b) 2 : 3
(c) 2 : 5
(d) can't be determined

5. In the given circle O is the centre of the circle and AD, AE are the two tangents. BC is also a tangent, then :



(a) $AC + AB = BC$
(b) $3AE = AB + BC + AC$
(c) $AB + BC + AC = 4AE$
(d) $2AE = AB + BC + AC$

6. What is the inradius of the incircle shown in the figure?

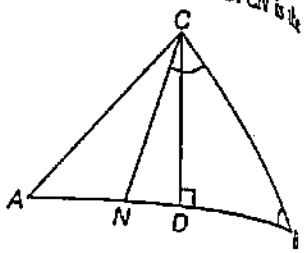


(a) 9 cm
(b) 4 cm
(c) can't be determined
(d) none of the above

7. One of the diagonal of a parallelogram is 18 cm, whose adjacent sides are 16 cm and 20 cm respectively. What is the length of other diagonal?

(a) $2\sqrt{247}$ cm
(b) 13 cm
(c) 15.2 cm
(d) 28.5 cm

8. In the adjoining figure CD is perpendicular on AB. CN is the angle bisector of $\angle ACB$. $\angle ACN = 50°$, then the angle NCD is :



(a) $\frac{1}{3}\angle ACB$
(b) $\angle CAB - \angle CBA$
(c) $\frac{1}{2}(\angle CBA - \angle BAC)$
(d) can't be determined

9. In a ΔPQR, points M and N are on the sides PQ and PR respectively such that $PM = 0.6.PQ$ and and $NR = 0.4PR$. What percentage of the area of the triangle PQR does that of triangle PMN form?

(a) 60%
(b) 50%
(c) 36%
(d) 55%

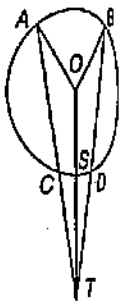
10. In a trapezium $ABCD$, the diagonals AC and BD intersect each other at O such that $OB : OD = 3 : 1$ then the ratio of areas of $\Delta AOB : \Delta COD$ is :

(a) 3 : 1
(b) 1 : 4
(c) 9 : 1
(d) can't be determined

11. Three cities Fatehpur, Barabanki and Lucknow are very famous. Fatehpur is 42 km, away from Barakanki and the distance between Fatehpur and Lucknow is 66 km. Which of the following cannot be the distance between Barabanki and Lucknow?

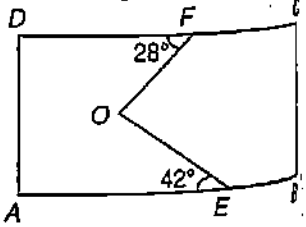
(a) 28 km
(b) 98 km
(c) 32 km
(d) 23 km

12. In the adjoining figure 'O' is the the centre of circle AC and BD are the two chords of circle which meets at T outside the circle. OT bisects CD, $OA = OB = 8$ cm and $OT = 17$ cm. What is the ratio of distance of AC and BD from the centre of the circle?



(a) 15 : 17
(b) 8 : 15
(c) 8 : 9
(d) none of the above

13. In the adjoining figure $ABCD$ is a rectangle and $DF = CF$ also $AE = 3BE$. what is the value of $\angle EOF$, if $\angle DFO = 28°$ and $\angle AEO = 42°$?

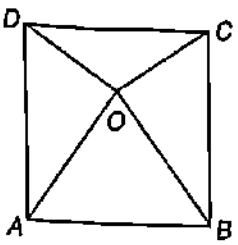


(a) 14°
(b) 42°
(c) 70°
(d) 90°

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

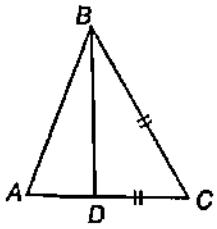
yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

14. $ABCD$ is a square and AOB is an equilateral triangle. What is the value of $\angle DOC$?



(a) 120°
(b) 150°
(c) 125°
(d) can't be determined

15. In the triangle ABC, $BC = CD$ and $(\angle ABC - \angle BAC) = 30°$. The measure of $\angle ABD$ is :

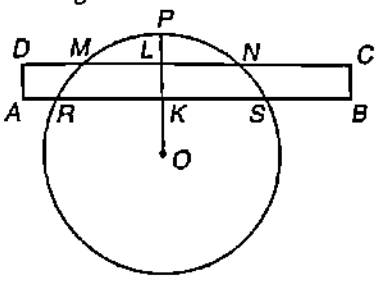


(a) 30°
(b) 45°
(c) 15°
(d) can't be determined

16. In a trapezium $ABCD$, AB is a parallel to CD. BD is perpendicular to AD. AC is perpendicular to BC. If $AD = BC = 15$ cm and $AB = 25$ cm , then the area of the trapezium is :

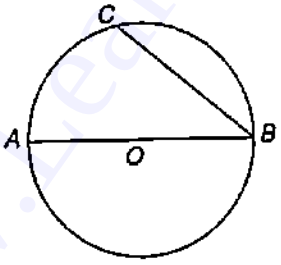
(a) 192 cm^2 (b) 232 cm^2
(c) 172 cm^2 (d) none of these

17. In the adjoining figure O is the centre of the circle. The radius OP bisects a rectangle $ABCD$, at right angle. $DM = NC = 2$ cm and $AR = SB = 1$ cm and $KS = 4$ cm and $OP = 5$ cm. What is the area of the rectangle?



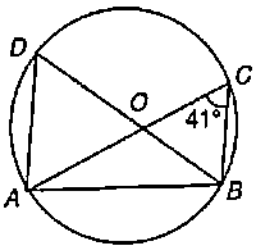
(a) 8 cm^2 (b) 10 cm^2
(c) 12 cm^2 (d) None of these

18. In the given figure of circle AB is the diameter with length 20 cm and BC is 16 cm, then find the length of CO when CO is perpendicular on AB :



(a) 9.6 cm
(b) 8.4 cm
(c) 10 cm
(d) data insufficient

19. In the adjoining figure BD is the diameter of the circle and $\angle BCA = 41°$. Find $\angle ABD$:



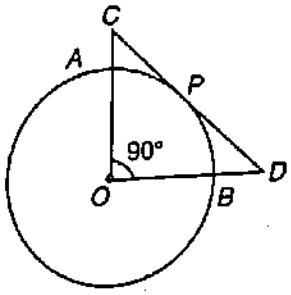
(a) 41°
(b) 49°
(c) 22.5°
(d) 20.5°

20. Each interior angle of a regular polygon exceeds its exterior angle by 132°. How many sides does the polygon have ?

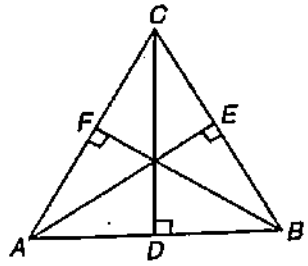
(a) 9 (b) 15
(c) 12 (d) none of these

21. In a circle O is the centre and $\angle COD$ is right angle. $AC = BD$ and CD is the tangent at P. What is the value of $AC + CP$, if the radius of the circle is 1 metre?

(a) 105 cm
(b) 141.4 cm
(c) 138.6 cm
(d) can't be determined

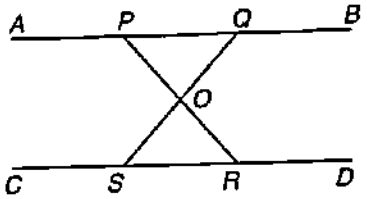


22. In the given triangle ABC, CD, BF and AE are the altitudes. If the ratio of $CD : AE : BF = 2 : 3 : 4$, then the ratio of $AB : BC : CA$ is :



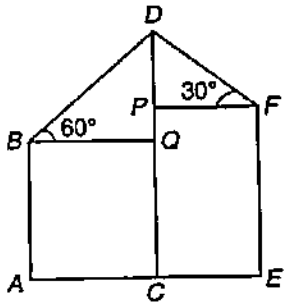
(a) 4 : 3 : 2
(b) 2 : 3 : 4
(c) 4 : 9 : 16
(d) 6 : 4 : 3

23. In the adjoining figure $AB \parallel CD$ and $PQ = SR$, then :



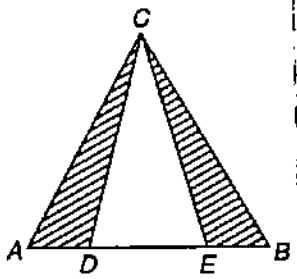
(a) $PQ = PS$
(b) $SR = RP$
(c) $PS = QR$
(d) $AP = RD$

24. In the given figure AB, CD and EF are three towers. The angle of elevation of the top of the tower CD from the top of the tower AB is 60° and that from EF is 30°. $BD = 2\sqrt{3}$ m, $CD : EF = 5 : 4$ and $DF = 4$ m. What is the height of the tower AB ?



(a) 6 m
(b) 12 m
(c) 7 m
(d) none of the above

25. In the equilateral triangle ABC, $AD = DE = BE$, D and E lies on the AB. If each side of the triangle (i.e., AB, BC and AC) be 6 cm, then the area of the shaded region is :

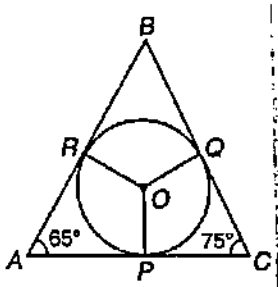


(a) 9 cm^2
(b) $6\sqrt{3}$ cm^2
(c) $5\sqrt{3}$ cm^2
(d) none of the above

26. In the above question (number 25) what is the perimeter of triangle CDE ?

(a) $(3 + 2\sqrt{5})$ cm (b) $2(1 + 2\sqrt{7})$ cm
(c) $2(1 + 3\sqrt{7})$ cm (d) none of these

27. In a triangle ABC, O is the centre of incircle PQR, $\angle BAC = 65°$, $\angle BCA = 75°$, find $\angle ROQ$:



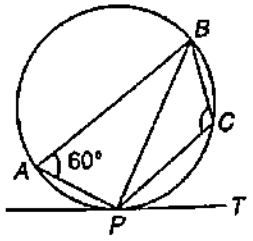
(a) 80°
(b) 120°
(c) 140°
(d) can't be determined

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmehboob.wordpress.com

For More Visit : www.LearnEngineering.in

28. In the given figure, PT is a tangent at P and $ABCP$ is a quadrilateral. $\angle BAP$ is 60°, then the value of $\angle PCB$ is :

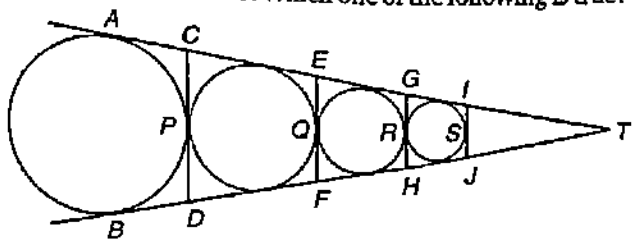


(a) 60°
(b) 90°
(c) 120°
(d) data insufficient

29. In the above question (number 28), what is the value of $\angle TPC$?

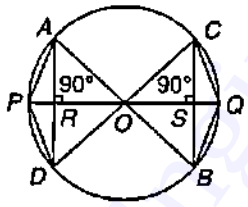
(a) 30° (b) 60°
(c) 90° (d) can't be determined

30. In the adjoining figure AT and BT are the to two tangents at A and B respectively CD is also a tangent at P.
There are some more circles touching each other and the tangents AT and BT also. Which one of the following is true?



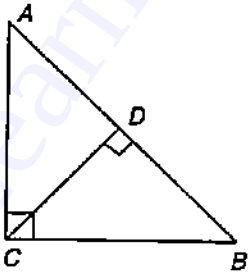
(a) $PC + CT = PD + DT$ (b) $RG + GT = RH + HT$
(c) $PC + QE = CE$ (d) all of these

31. In the adjoining figure O is the centre of the circle. $\angle AOD = 120°$. If the radius of the circle be 'r', then find the sum of the areas of quadrilaterals $AODP$ and $OBQC$:



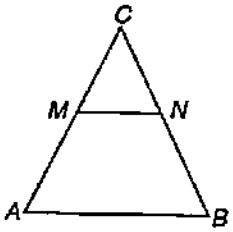
(a) $\frac{\sqrt{3}}{2}r^2$ (b) $3\sqrt{3}r^2$
(c) $\sqrt{3}r^2$ (d) none of these

32. In a right angled triangle ABC, CD is the perpendicular on the hypotenuse AB. Which of the following is correct?



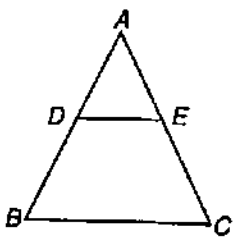
(a) $CD = \frac{AC \times BC}{AB}$
(b) $AD = \frac{AC \times AC}{AB}$
(c) $BD = \frac{BC \times BC}{AB}$
(d) All of the the above

33. In the triangle ABC, MN is parallel to AB. Area of trapezium $ABNM$ is twice the area of triangle CMN. What is ratio of $CM : AM$?



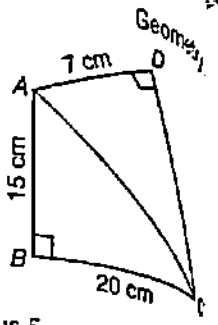
(a) $\frac{1}{\sqrt{3}+1}$ (b) $\frac{\sqrt{3}-1}{2}$
(c) $\frac{\sqrt{3}+1}{2}$ (d) none of these

34. In the given figure $AD = AE$, $\angle BDE = 100°$, then what is the value of $\angle DBC + \angle BCE$?



(a) 200°
(b) 160°
(c) 80°
(d) can't be determined

35. In the given quadrilateral $ABCD$, $AB = 15$ cm, $BC = 20$ cm and $AD = 7$ cm, $\angle ABC = \angle ADC = 90°$. Find the length of side CD :



(a) 12 cm
(b) 18 cm
(c) 24 cm
(c) none of the above

36. There are two circles each with radius 5 cm. Tangent AB is 26 cm. The length of tangent CD is :

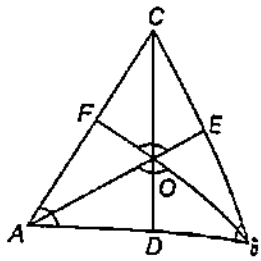


(a) 15 cm (b) 21 cm
(c) 24 cm (d) can't be determined

37. One of diagonal of a parallelogram is 10 cm and an angle of the parallelogram is $\pi/4$. If its height be 8 cm then find the area of the parallelogram :

(a) 112 cm^2 (b) 88 cm^2
(c) 92 cm^2 (d) 104 cm^2

38. ABC is a triangle in which $\angle CAB = 80°$ and $\angle ABC = 50°$, AE, BF and CD are the altitudes and O is the orthocentre. What is the value of $\angle AOB$?

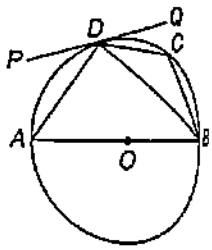


(a) 65°
(b) 70°
(c) 50°
(d) 130°

39. ABC is a triangle in which 35 times the smallest angle is equal to the 26 times largest angle. What is the measure of the second largest angle?

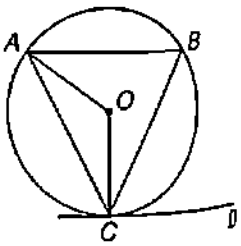
(a) 63° (b) 58°
(c) 70° (d) 42°

40. In the adjoining figure O is the centre of the circle and AB is the diameter. Tangent PQ touches the circle at D. $\angle BDQ = 48°$. Find the ratio of $\angle DBA : \angle DCB$:



(a) $\frac{22}{7}$
(b) $\frac{7}{22}$
(c) $\frac{7}{12}$
(d) can't be determined

41. In the given diagram O is the centre of the circle and CD is a tangent. $\angle CAB$ and $\angle ACD$ are supplementary to each other $\angle OAC = 30°$. Find the value of $\angle OCB$:

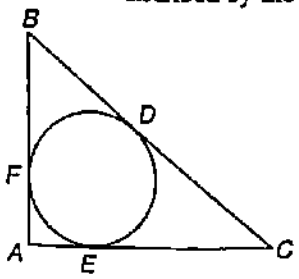


(a) 30°
(b) 20°
(c) 60°
(d) none of the above

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

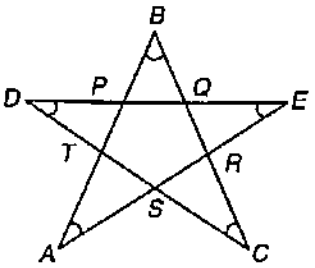
yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

42. In the given diagram an incircle DEF is circumscribed by the right angled triangle in which $AF = 6$ cm and $EC = 15$ cm. Find the difference between CD and BD :



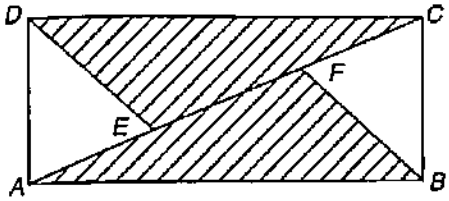
(a) 1 cm
(b) 3 cm
(c) 4 cm
(d) can't be determined

43. In the adjoining figure, a star is shown. What is the sum of the angles A, B, C, D and E?



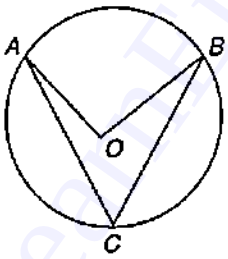
(a) 120°
(b) 180°
(c) 240°
(d) can't be determined

44. $ABCD$ is a rectangle of dimensions 6 cm × 8 cm. DE and BF are the perpendiculars drawn on the diagonal of the rectangle. What is the ratio of the shaded to that of unshaded region?



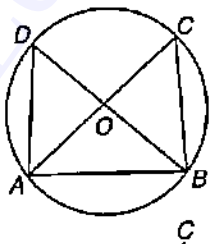
(a) 7 : 3
(b) 16 : 9
(c) $4 : 3\sqrt{2}$
(d) data insufficient

45. In the adjoining figure 'O' is the centre of circle. $\angle CAO = 25°$ and $\angle CBO = 35°$. What is the value of $\angle AOB$?



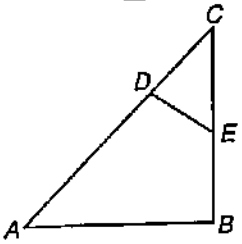
(a) 55°
(b) 110°
(c) 120°
(d) data insufficient

46. In the given diagram $ABCD$ is a cyclic quadrilateral $\angle OCB = 50°$ and $\angle BOC = 110°$. Find the value of $\angle DAO$:



(a) 20°
(b) 30°
(c) 50°
(d) can't be determined

47. ABC and CDE are right angled triangle. $\angle ABC = \angle CDE = 90°$, D lies on AC and E lies on BC. $AB = 24$ cm, $BC = 60$ cm. If $DE = 10$ cm, then CD is :

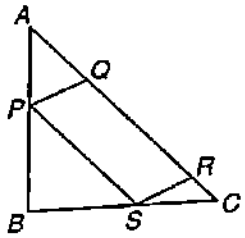


(a) 28 cm
(b) 35 cm
(c) 25 cm
(d) can't be determined

48. In the above question (number 48) what is the ratio of $CE : BE$?

(a) $\sqrt{29} : (12 - \sqrt{29})$
(b) $12 : \sqrt{29}$
(c) $7 : \sqrt{21}$
(d) none of these

49. In the given diagram ΔABC is an isosceles right angled triangle, in which a rectangle is inscribed in such a way that the length of the rectangle in twice of breadth. Q and R lie on the hypotenuse and P, S lie on the two different smaller sides of the triangle. What is the ratio of the areas of the rectangle and that of triangle :



(a) $\sqrt{2} : 1$
(b) $1 : \sqrt{2}$
(c) 1 : 2
(d) $\sqrt{3} : 2$

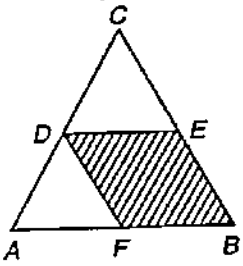
50. An n sided polygon has 'n' diagonals, then the value of n is :

(a) 4
(b) 6
(c) 7
(d) 5

51. How many distinct equilateral triangles can be formed in a regular nonagon having at two of their vertices as the vertices of nonagon?

(a) 72
(b) 36
(c) 66
(d) none of these

52. ABC is a triangle in which D, E and F are the mid-points of the sides AC, BC and AB respectively. What is the ratio of the area of the shaded to the unshaded region in the triangle?

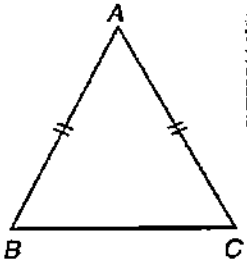


(a) 1 : 1
(b) 3 : 4
(c) 4 : 5
(d) none of the above

53. Three circles each of units radius intersects each other at P, Q and R. P, Q and R are the centres of the three circles. What is the sum of length of the arcs PQ, QR and PR ?

(a) $\sqrt{3}\pi$
(b) $2\sqrt{2}\pi$
(c) π
(d) none of these

54. ABC is an isosceles triangle in which $AB = AC$ and $(\angle A) = 2(\angle B)$. $AB = 4$ cm. What is the ratio of inradius to the circumradius?



(a) 1 : 2
(b) $(\sqrt{2} - 1) : 1$
(b) $1 : (2\sqrt{2} - 1)$
(c) none of the above

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yourshahbood.wordpress.com
 For More Visit : www.LearnEngineering.in

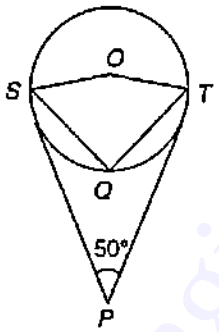
LEVEL (2)

1. The semiperimeter of a right angled triangle is 126 cm and the shortest median is 53 cm. What is the area of a triangle which has the largest median as its longest side?
(a) 1560 cm^2 (b) 1260 cm^2
(c) 1060 cm^2 (d) none of these

2. In an isosceles right angled triangle ABC, $\angle B$ is right angle. Angle bisector of $\angle BAC$ is AN cut at M to the median BO. Point 'O' lies on the hypotenuse. OM is 20 cm, then the value of AB is :
(a) 38.96 cm (b) 24.18 cm
(c) 34.134 cm (d) none of these

3. In a right angled triangle $\angle B$ and $\angle A$ are acute angles. If $\angle A - \angle B = k$, where A and B are integers, then how many integer values can 'k' take?
(a) 80 (b) 88
(c) 45 (d) 89

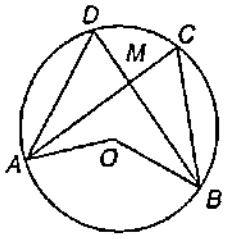
4. In the given figure 'O' is the centre of the circle SP and TP are the two tangents at S and T respectively. $\angle SPT$ is 50°, the value of $\angle SQT$ is :
(a) 125°
(b) 65°
(c) 115°
(d) none of the above



5. A ladder 6.5 m long is standing against a wall and the difference between the base of the ladder and wall is 5.2 m. If the top of the ladder now slips by 1.4 m, then by how much will the foot of the ladder slip?
(a) 1.2 m (b) 0.8 m
(c) 0.75 (d) can't be determined

6. In a triangle all the three angles A, B, C are in integers, then the number of values that A, B and C can take :
(a) 89 (b) 90
(c) 178 (d) 180

7. In the given figure of circle, 'O' is the centre of the circle $\angle AOB = 130°$. What is the value of $\angle DMC$?
(a) 65°
(b) 125°
(c) 85°
(d) can't be determined



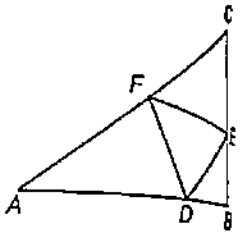
8. In a right angled triangle ABC, $\angle B$ is right angle, side AB is half of the hypotenuse. AE is parallel to median BD and CE is parallel to BA. What is the ratio of length of BC to that of EC?
(a) $\sqrt{2} : 1$ (b) $\sqrt{3} : 2$
(c) $\sqrt{5} : \sqrt{3}$ (d) can't be determined

9. In an equilateral triangle ABC, AO, BO and CO are the angle bisectors meet at the incentre 'O'. D, E and F are the mid-points of AO, BO and CO respectively. A circle with centre 'O' passes through D, E and F. Area of the circle is 3π cm^2. What is the perimeter of triangles (Δ) ABC?
(a) $12\sqrt{3}$ cm (b) 18 cm
(c) $6\sqrt{3}$ cm (d) none of these

10. Two trains Punjab mail and Lucknow mail start simultaneously from Patiyala and Lakhimpur respectively towards each other with the speed of 40 km/h and 60 km/h respectively on the same track Lakhimpur is 500 km due east of Patiyala. A plane starts flying at 200 km/h at the same time from Patiyala to Jalandhar. Jalandhar is 100 km due north of Patiyala. After travelling sometime two trains Punjab mail and Lucknow mail collides with each other. The plane moves continuously to and fro between Patiyala to Jalandhar till the collision of the trains. How far would the plane have travelled?
(a) 100 km (b) 1000 km
(c) 2000 km (d) can't be determined

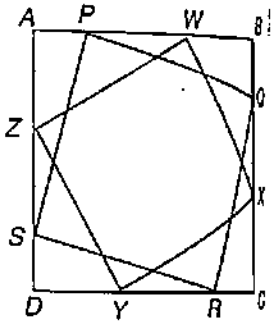
11. In the above question (number 10) what is the distance between the place of accident and the plane at the moment of accident of two trains?
(a) 200 km (b) 250 km
(c) 400 km (d) data insufficient

12. In the given figure $\angle B$ is right angle. $AD : BD = 3 : 2$ and $CE : BE = 5 : 2$ and $AF : FC = 1 : 1$. What is the area of ΔABC, if the area of ΔBDE is 20 cm^2?
(a) 40 cm^2
(b) 35 cm^2
(c) 52.5 cm^2
(d) none of the above

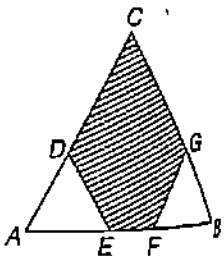


13. In a triangle ABC with side $AB = AC$ and $\angle BAC = 20°$, D is a point on side AC and $BC = AD$. Find $\angle DBC$:
(a) 50° (b) 45°
(c) 65° (d) 70°

14. In the adjoining figure $ABCD$, $PQRS$ and $WXYZ$ are three squares. Find number of triangles and quadrilaterals in the figure :
(a) 24 and 16
(b) 28 and 15
(c) 27 and 16
(d) none of the above



15. In the given figure ABC is a triangle in which $CDEFG$ is a pentagon. Triangles ADE and BFG are equilateral triangles each with side 2 cm and $EF = 2$ cm. Find the area of the pentagon :
(a) $8\sqrt{3}$ cm
(b) $7\sqrt{3}$ cm^2
(c) $15\sqrt{3}$ cm^2
(d) 11.28 cm^2



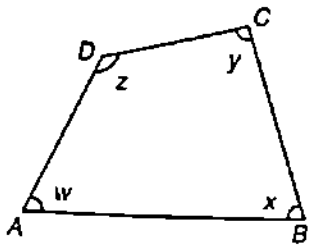
16. $PQRS$ is a quadrilateral which is formed by joining the mid-points of a quadrilateral $ABCD$, such that $\angle A = 75°$, $\angle B = 95°$, $\angle C = 110°$.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

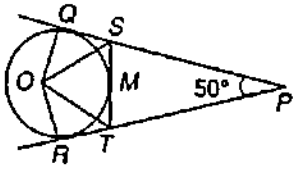
If $\angle PQR = 90°$, what is the value of $\angle PSR$?
(a) 90° (b) 110°
(c) 60° (d) 75°

17. $ABCD$ is a quadrilateral in which $\frac{z}{y} = \frac{y}{x} = \frac{x}{w} = k$ and k is an integer. Also $\{w, x\} < 90$ and $\{y, z\} > 90$, then the difference between the greatest angle and smallest angle (*i.e.*, $z - w$) is :



(a) 168° (b) 172°
(c) 128° (d) 212°

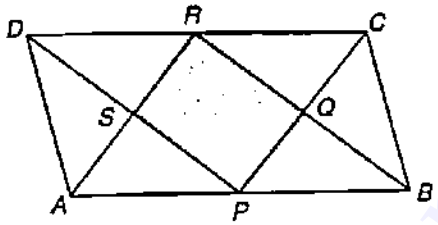
18. In the adjoining figure 'O' is the centre of the circle and PQ, PR and ST are the three tangents. $\angle QPR = 50°$, then the value of $\angle SOT$ is :



(a) 30° (b) 75°
(c) 65° (d) can't be determined

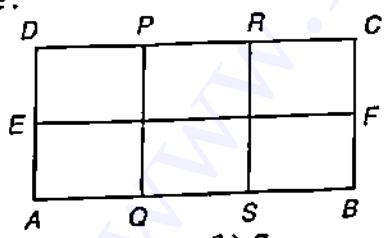
19. The number of points of intersection of the diagonals of a regular hexagon is :
(a) 10 (b) 15
(c) 18 (d) 19

20. In the adjoining figure $ABCD$, P and R are the mid-points of the sides AB and CD. $ABCD$ is a parallelogram. What is the ratio of the shaded to the unshaded region?



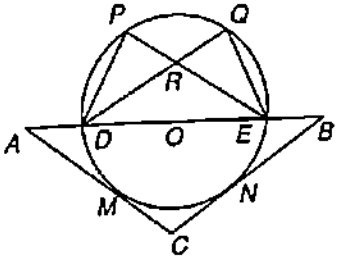
(a) 1/2 (b) 1/3
(c) 1/4 (d) none of these

21. In the adjoining figure $ABCD$ is a rectangle. Find the maximum number of rectangles including the largest possible rectangle :



(a) 16 (b) 7
(c) 18 (d) 24

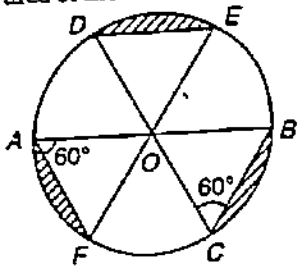
22. ABC is an isosceles triangle and AC, BC are the tangents at M and N respectively. DE is the diameter of the circle. $\angle ADP = \angle BEQ = 100°$. What is value of $\angle PRD$?



(a) 60°
(b) 50°
(c) 20° (d) can't be determined

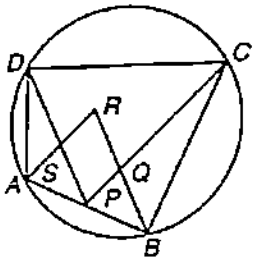
23. In the above question if OC is the half of the AB, then the value of $\angle ACB$ is :
(a) 60° (b) 90°
(c) 80° (d) can't be determined

24. In the adjoining figure O is the centre of the circle with radius r. AB, CD and EF are the diameters of the circle. $\angle OAF = \angle OCB = 60°$. What is the area of the shaded region?



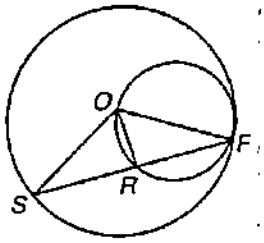
(a) $\frac{r^2}{2}\left(\pi - \frac{3\sqrt{3}}{2}\right)$
(b) $\frac{r^2}{2}\left(\pi - \frac{3\sqrt{3}}{4}\right)$
(c) $\frac{r^2}{3}\left(\pi - \frac{2\sqrt{3}}{3}\right)$
(d) data insufficient

25. $ABCD$ is a cyclic quadrilateral. The angle bisector of $\angle A$, $\angle B$, $\angle C$ and $\angle D$ intersect at P, Q, R and S as shown in the figure. These four points form a quadrilateral $PQRS$. Quadrilateral $PQRS$ is a :



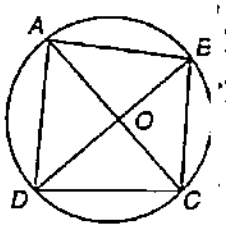
(a) square
(b) rhombus
(c) rectangle
(d) cyclic quadrilateral

26. In the adjoining figure the diameter of the larger circle is 10 cm and the smaller circle touches internally the lager circle at P and passes through O, the centre of the larger circle. Chord SP cuts the smaller circle at R and OR is equal to 4 cm. What is the length of the chord SP ?



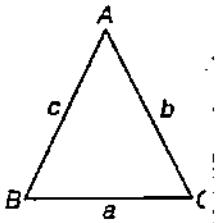
(a) 9 cm (b) 12 cm
(c) 6 cm (d) $8\sqrt{2}$ cm

27. In the given figure $ABCD$ is a cyclic quadrilateral $DO = 8$ cm and $CO = 4$ cm. AC is the angle bisector of $\angle BAD$. The length of AD is equal to the length of AB. DB intersects diagonal AC at O, then what is the length of the diagonal AC ?



(a) 20 cm (b) 24 cm
(c) 16 cm (d) none of these

28. In the given triangle ABC, the length of sides AB and AC is same (*i.e.*, $b = c$) and $60° < A < 90°$, then the possible length of BC is :

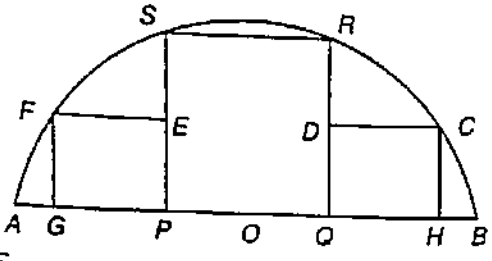


(a) $b < a < 2b$
(b) $\frac{c}{3} < a < 3a$
(c) $b < a < b\sqrt{3}$
(d) $c < a < c\sqrt{2}$

29. In the following figure there is semicircle with centre 'O' and diameter AB ($= 2r$). $PQRS$ is a square of maximum possible area. P and Q lie on the diameter AB and R, S lie on the arc of the semicircle, there are two more squares of maximum possible area $EFGP$ and $CDQH$. What is the sum of lengths of RC and FS?

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

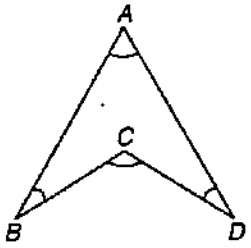
yourSmanboob.wordpress.com
For More Visit : www.LearnEngineering.in



(a) $\sqrt{\frac{2}{5}}(2r)$ (b) $\frac{2\sqrt{2}r}{5}$

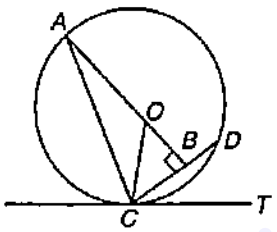
(c) $\frac{\sqrt{2}}{5}r$ (d) none of these

30. In the adjoining figure $\angle BAD = a$, $\angle ABC = b$ and $\angle BCD = c$ and $\angle ADC = d$, find the value of $\angle ABC$ in terms of a, c and d :



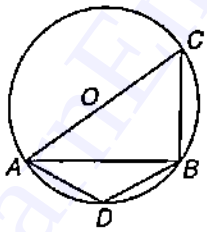
(a) $c-(a+d)$
(b) $a-(c+d)$
(c) $b-(c+d)$
(d) none of the above

31. In the given diagram CT is tangent at C, making an angle of $\frac{\pi}{4}$ with CD. O is the centre of the circle. $CD = 10$ cm. What is the perimeter of the shaded region (ΔAOC) approximately?



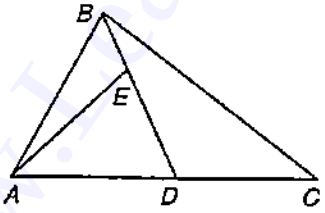
(a) 27 cm (b) 30 cm
(c) 25 cm (d) 31 cm

32. In the given diagram, 'O' is the centre of the circle and AC is the diameter. $\angle ADB$ is 120°. Radius of the circle is 6 cm, what is the area of the triangle ABC?



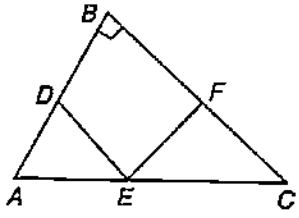
(a) $18\sqrt{3}$ cm^2
(b) $24\sqrt{3}$ cm^2
(c) 27 cm^2
(d) data insufficient

33. In the given figure ABC is a triangle in which $3AD = CD$ and E lies on BD, $DE = 2BE$. What is the ratio of area of ΔABE and area of ΔABC ?



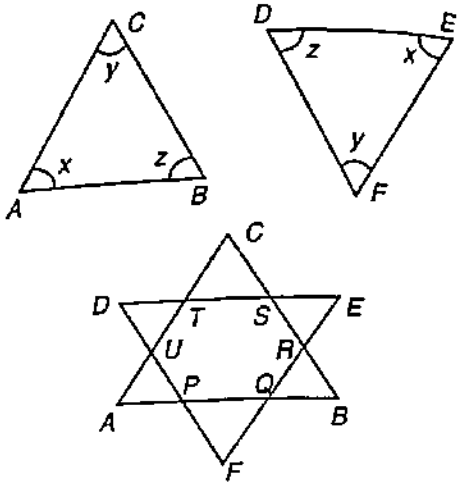
(a) 1/12 (b) 1/3
(c) 1/2 (d) none of these

34. In the adjoining figure ABC is a right angled triangle, $BDEF$ is a square, $AE = 7.5$ cm and $AC = 18$ cm. What is the area of triangle ABC ?



(a) 76.621 cm^2
(b) 70.054 cm^2
(c) 83.25 cm^2
(d) 90.90 cm^2

35. There are two congruent triangles each with area 198 cm^2. Triangle DEF is placed over triangle ABC in such a way that the centroid of both the triangles coincides with each other and $AB \parallel DE$, as shown in figure, thus forming a star . What is the area of the common region ($PQRSTU$)?



(a) 99 cm^2 (b) 132 cm^2
(c) 148 cm^2 (d) can't be determined

36. What is the sum of all the angles of a 9 pointed star (i.e., $\angle 1 + \angle 2 + \angle 3 + \ldots \angle 8 + \angle 9$) :

(a) 909° (b) 900°
(c) 720° (d) 540°

37. A circle is circumscribed by the rhombus which in turn is made up by joining the mid-points of a rectangle whose sides, 12 cm and16 cm respectively. What is the area of the circle?

(a) $\frac{625}{26}\pi$ (b) $\frac{676}{25}\pi$

(c) $\frac{576}{25}\pi$ (d) can't be determined

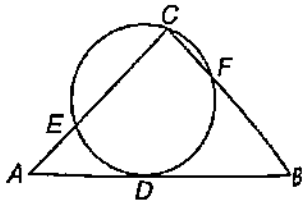
38. There are n rectangles each with area 200 cm^2. If the dimensions of each n rectangles are in integers then the value of n is :

(a) 4 (b) 6
(c) 12 (d) none of these

39. There are 8 points on a plane, out of which 4 points lies on the circumference of the same circle and rest 4 points do not lie on a single circumference of a circle and also they are non-collinear. Maximum how many circles can be drawn such that each contains at least three of the given points?

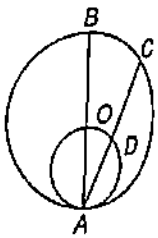
(a) 53 (b) 32
(c) 35 (d) 56

40. ABC is an isosceles triangle a circle is such that it passes through vertex C and AB acts as a tangent at D for the same circle. AC and BC intersects the circle at E and F respectively $AC = BC = 4$ cm and $AB = 6$ cm. Also D is the mid-point of AB. What is the ratio of $EC : (AE + AD)$?



(a) 1 : 2 (b) 1 : 3
(c) 2 : 5 (d) none of these

41. A smaller circle touches internally to a larger circle at A and passes through the centre of the larger circle. O is the centre of the larger circle and BA, OA are of the diameters of the larger and smaller circles respectively. Chord AC intersects the smaller circle at a point D. If $AC = 12$ cm, then AD is :



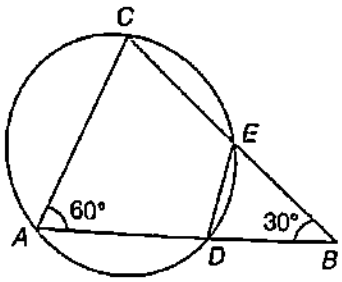
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

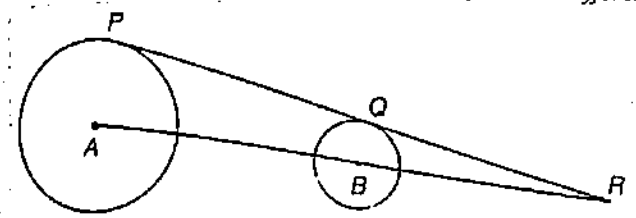
(a) 4 cm (b) 6 cm
(c) 5.6 cm (d) data insufficient

42. In the given figure $ADEC$ is a cyclic quadrilateral, CE and AD are extended to meet at B, $\angle CAD = 60°$ and $\angle CBA = 30°$. $BD = 6$ cm and $CE = 5\sqrt{3}$ cm,
What is the ratio of $AC : AD$?



(a) $\frac{3}{4}$
(b) $\frac{4}{5}$
(c) $\frac{2\sqrt{3}}{5}$
(d) cant' be determined

Directions for the question number 43 and 44 : *In the following diagram A and B are the centres of the two different*

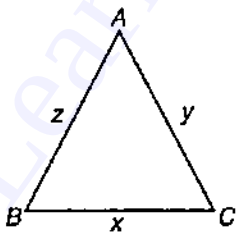


circles. PQR is a common tangent. Points A, B, and R lie on the straight line.
Distance between A and B is 25 cm and the distance between P and Q is 24 cm. Diameter of the larger circle is 24 cm.

43. What is the ratio of $AB : BR$?
(a) 7 : 5 (b) 7 : 6
(c) 7 : 10 (d) data insufficient

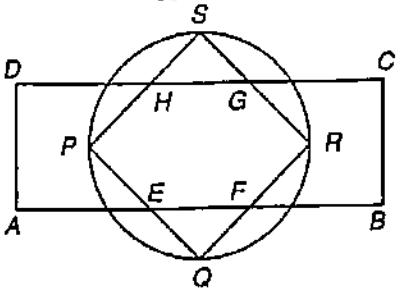
44. What is the ratio of area of ΔAPR and ΔBQR ?
(a) 169 : 36 (b) 144 : 25
(c) 625 : 144 (d) can't be determined

45. If $x^2 + y^2 + z^2 = xy + yz + zx$, then the triangle is :



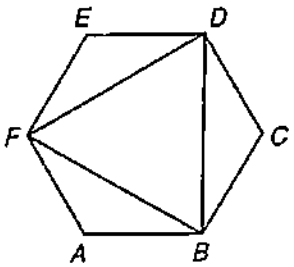
(a) isosceles
(b) right angled
(c) equilateral
(d) scalene

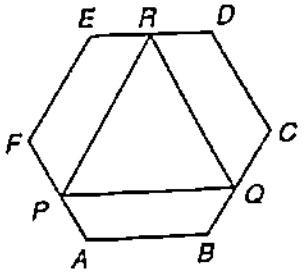
46. In the adjoining figure $ABCD$ is a rectangle in which length is twice of breadth. H and G divide the line CD into three equal parts. Similarly points E and F trisect the line AB. A circle $PQRS$ is circumscribed by a square $PQRS$ which passes through the points E, F, G and H. What is the ratio of areas of circle to that of rectangle?



(a) $3\pi : 7$ (b) $3 : 4$
(c) $25\pi : 72$ (d) $32\pi : 115$

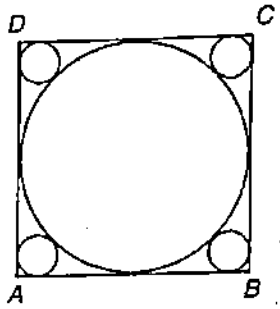
47. In the adjoining figure there are two congruent regular hexagons each with side 6 cm.





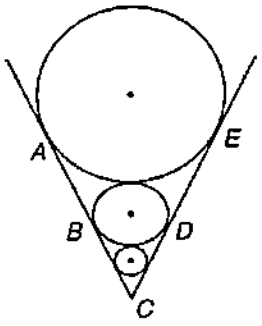
What is the ratio of area of ΔBDF and ΔPQR, if P, Q and R are the mid-points of side AF, BC and DE ?
(a) 6 : 5 (b) 7 : 6
(c) 4 : 3 (d) 1 : 1

48. $ABCD$ is a square, in which a circle is inscribed touching all the sides of square. In the four corners of square 4 smaller circles of equal radii are drawn, containing maximum possible area.
What is the ratio of the area of larger circle to that of sum of the areas of four smaller circles ?



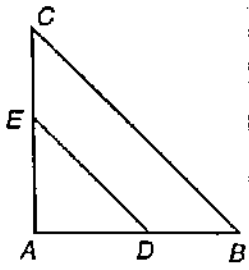
(a) $1 : (68 - 48\sqrt{2})$ (b) $1 : 17\sqrt{2}$
(c) $3 : (34 - 12\sqrt{2})$ (d) none of these

49. In the adjoining figure $\angle ACE$ is a right angle there are three circles which just touch each other and AC and EC are the tangents to all the three circles. What is the ratio of radii of the largest circle to that of the smallest circle?



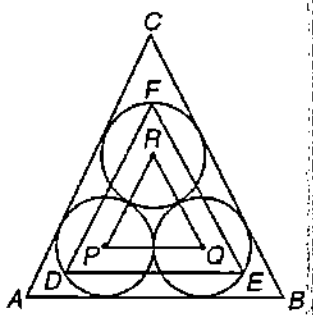
(a) $17 : 12\sqrt{2}$
(b) $1 : (17 - 12\sqrt{2})$
(c) $12 : 17\sqrt{2}$
(d) none of the above

50. In a right angle triangle ABC, $\angle A$ is right angle DE is parallel to the hypotenuse BC and the length of DE is 65% the length of BC, what is the area of ΔADE, if the area of ΔABC is 68 cm^2 ?



(a) 27.83 cm^2
(b) 41.6 cm^2
(c) 28.73 cm^2
(d) none of the above

51. In the adjoining figure three congruent circles are touching each other. Triangle ABC circumscribes all the three circles. Triangle PQR is formed by joining the centres of the circle. There is a third triangle DEF. Points A, D, P and B, E, Q and C, F, R lie in the same straight lines respectively.



For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmanboob.wordpress.com

For More Visit : www.LearnEngineering.in

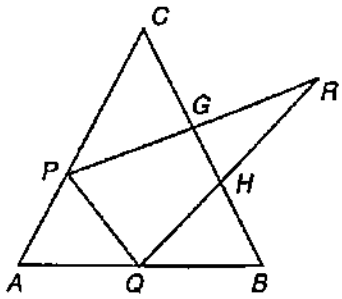
What is the ratio of perimeters of $\Delta ABC : \Delta DEF$ ΔPQR

(a) $3\sqrt{2} : 2\sqrt{2} : 1$ (b) $2(4+\sqrt{3}) : (2+\sqrt{3}) : \sqrt{3}$

(c) $2(1+\sqrt{3}) : (2+\sqrt{3}) : 2$ (d) $2(1+\sqrt{3}) : 2\sqrt{3} : \sqrt{3}$

52. In the given figure, P and Q are the mid-points of AC and AB. Also, $PG = GR$ and $HQ = HR$.

What is the ratio of area of ΔPQR : area of ΔABC?



(a) $\frac{1}{2}$

(b) $\frac{2}{3}$

(c) $\frac{3}{5}$

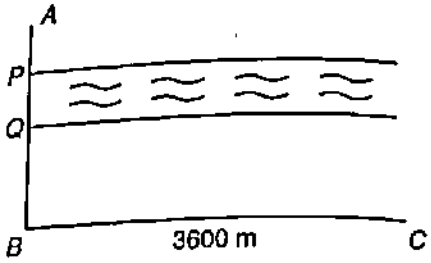
(d) none of the above

53. A trapezium $PQRS$ inscribes a circle which touches the circle at M, A, N, B, Radius of circle is 10 cm. The length of each non-parallel side is 21 cm. What is the perimeter of trapezium?

(a) 82 cm (b) 84 cm

(c) 85.5 cm (d) can't be determined

54. In the given diagram, river PQ is just perpendicular to the national highway AB. At a point B highway just turns at right angle and reaches to C. $PA = 500$ m and $BQ = 700$ m and width of the uniformly wide river (*i.e.*, PQ) is 300m. Also $BC = 3600$ m. A bridge has to be constructed across the river

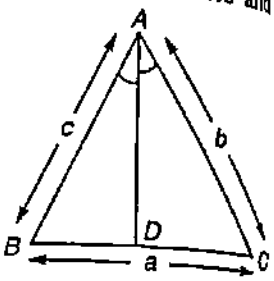


perpendicular to its stream in such a way that a person can reach from A to C via bridge covering least possible distance. PQ is the widthness of the river, then what is the minimum possible required distance from A to C including the length of bridge?

(a) 4100 m (b) 3900 m

(c) $3000\sqrt{2}$ m (d) none of these

55. In a triangle ABC, AD is the angle bisector of $\angle BAC$ and $\angle BAD = 60°$. What is the length of AD ?



(a) $\frac{b+c}{bc}$

(b) $\frac{bc}{b+c}$

(c) $\sqrt{b^2+c^2}$

(d) $\frac{(b+c)^2}{bc}$

www.LearnEngineering.in

For More Visit : www.LearnEngineering.in

Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

Answers

INTRODUCTORY EXERCISE-12.1

1. (c)	2. (c)	3. (b)	4. (b)	5. (b)	6. (a)	7. (d)	8. (a)	9. (a)	10. (b)
11. (a)	12. (c)	13. (b)	14. (b)	15. (d)	16. (b)	17. (b)	18. (b)	19. (a)	20. (c)
21. (b)	22. (c)	23. (c)	24. (b)	25. (c)	26. (a)	27. (c)	28. (c)	29. (a)	30. (b)
31. (c)	32. (a)	33. (a)	34. (c)	35. (a)	36. (b)	37. (b)	38. (c)	39. (b)	40. (a)
41. (c)	42. (c)	43. (a)	44. (b)	45. (b)					

INTRODUCTORY EXERCISE-12.2

1. (b)	2. (b)	3. (c)	4. (d)	5. (a)	6. (c)	7. (d)	8. (d)	9. (b)	10. (b)
11. (b)	12. (c)	13. (d)	14. (b)	15. (b)	16. (b)	17. (c)	18. (b)	19. (d)	20. (b)
21. (a)	22. (d)	23. (c)	24. (b)	25. (c)	26. (c)	27. (c)	28. (a)	29. (a)	30. (b)
31. (c)	32. (c)	33. (d)	34. (d)	35. (a)	36. (d)	37. (b)	38. (d)	39. (c)	40. (d)
41. (c)	42. (a)	43. (c)	44. (c)	45. (b)	46. (c)	47. (a)	48. (a)	49. (c)	50. (c)
51. (a)	52. (a)	53. (a)	54. (c)	55. (a)	56. (a)	57. (a)	58. (b)	59. (b)	60. (c)
61. (c)	62. (b)	63. (c)	64. (b)	65. (b)	66. (c)	67. (c)	68. (a)	69. (d)	70. (a)
71. (b)	72. (c)	73. (c)	74. (a)	75. (b)	76. (c)	77. (c)	78. (b)	79. (c)	80. (d)
81. (b)	82. (b)	83. (a)	84. (c)	85. (b)	86. (c)	87. (a)	88. (d)	89. (c)	90. (b)
91. (d)	92. (c)	93. (c)	94. (c)	95. (b)	96. (c)	97. (b)	98. (a)	99. (b)	100. (d)

INTRODUCTORY EXERCISE-12.3

1. (b)	2. (b)	3. (b)	4. (a)	5. (a)	6. (b)	7. (c)	8. (c)	9. (b)	10. (c)
11. (b)	12. (b)	13. (a)	14. (b)	15. (a)	16. (b)	17. (c)	18. (c)	19. (a)	20. (a)
21. (c)	22. (b)	23. (c)	24. (a)	25. (b)	26. (a)	27. (a)	28. (b)		

INTRODUCTORY EXERCISE-12.4

1. (d)	2. (b)	3. (b)	4. (d)	5. (d)	6. (b)	7. (c)	8. (d)	9. (c)	10. (c)

INTRODUCTORY EXERCISE-12.5

1. (b)	2. (a)	3. (c)	4. (c)	5. (a)	6. (c)	7. (a)	8. (c)	9. (c)	10. (c)
11. (d)	12. (a)	13. (b)	14. (d)	15. (c)	16. (b)	17. (a)	18. (c)	19. (a)	20. (c)
21. (b)	22. (b)	23. (c)	24. (c)	25. (b)	26. (b)	27. (b)	28. (a)	29. (c)	30. (b)
31. (b)	32. (b)	33. (c)	34. (c)	35. (c)	36. (d)	37. (b)	38. (c)	39. (b)	40. (b)
41. (c)	42. (d)	43. (a)	44. (b)	45. (a)	46. (a)	47. (c)	48. (d)	49. (a)	50. (c)
51. (b)	52. (b)	53. (d)	54. (a)	55. (c)	56. (b)	57. (a)	58. (d)	59. (c)	60. (c)

LEVEL-1

1. (c)	2. (d)	3. (a)	4. (c)	5. (d)	6. (b)	7. (a)	8. (c)	9. (c)	10. (c)
11. (d)	12. (d)	13. (c)	14. (b)	15. (c)	16. (a)	17. (b)	18. (a)	19. (b)	20. (b)
21. (b)	22. (d)	23. (c)	24. (c)	25. (b)	26. (b)	27. (c)	28. (c)	29. (d)	30. (d)
31. (c)	32. (d)	33. (c)	34. (b)	35. (c)	36. (c)	37. (a)	38. (d)	39. (b)	40. (b)
41. (a)	42. (a)	43. (b)	44. (b)	45. (c)	46. (a)	47. (c)	48. (a)	49. (c)	50. (d)
51. (c)	52. (a)	53. (c)	54. (b)						

LEVEL-2

1. (b)	2. (d)	3. (d)	4. (c)	5. (b)	6. (c)	7. (d)	8. (b)	9. (b)	10. (b)
11. (a)	12. (d)	13. (d)	14. (b)	15. (b)	16. (a)	17. (a)	18. (c)	19. (d)	20. (b)
21. (c)	22. (c)	23. (b)	24. (a)	25. (d)	26. (c)	27. (a)	28. (d)	29. (a)	30. (a)
31. (a)	32. (a)	33. (a)	34. (a)	35. (b)	36. (b)	37. (c)	38. (b)	39. (a)	40. (b)
41. (b)	42. (a)	43. (a)	44. (b)	45. (c)	46. (c)	47. (c)	48. (a)	49. (b)	50. (c)
51. (c)	52. (a)	53. (b)	54. (a)	55. (b)					

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Hints & Solutions

INTRODUCTORY EXERCISE 12.1

1. $4x + 5x = 180° \Rightarrow x = 20$
($4x$ and $5x$ are linear pair angles)

2. $(5x + 12) + 3x = 180° \Rightarrow x = 21°$

3. $105 + 3x = 180 \Rightarrow x = 25$
and $y + 90 + 2x = 180°$
$\Rightarrow y = 40°$
$\therefore x + y = 65°$

4. $2x + 2y = 180$
$\Rightarrow x + y = 90°$
$\therefore \angle COD = 90°$

5. $\angle DOE = 180 - (40 + 31) = 109°$
and $\angle BOF = 40°$ ($\because \angle AOE = \angle BOF$)
$\therefore \angle BOC = \angle BOF + \angle FOC$
$= 40 + 109$ ($\because \angle FOC = \angle DOE$)
$= 149°$

6. $2y + 3x = 180 \Rightarrow y = 36°$ ($\because x = 36°$)
or $x + 4y = 180° \Rightarrow y = 36°$ ($\because x = 36°$)

7. $(2x + 17) + (x + 4) = 90°$
$\Rightarrow 3x + 21 = 90°$
$\Rightarrow x = 23°$

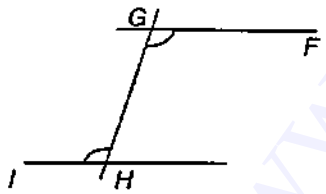
8. $(5y + 62)° + (22° + y) = 180$
$\Rightarrow 6y + 84 = 180$
$\Rightarrow y = 16°$

9. $13x + 5x = 180° \Rightarrow x = 10°$
$\therefore 13x = 130°$

10. Let the angle be x, then its complementary angle be $(90 - x)$
$\therefore x - \frac{(90 - x)}{2} = 30 \Rightarrow x = 50$

Alternatively: The best way is to solve through options.

11.



(Pair of alternate angles)

12. $\frac{AG}{BH} = \frac{CE}{DF}$

$\frac{120}{100} = \frac{30}{DF} \Rightarrow DF = 25$ cm

13. $65°50' + x = 90° = 89° 60'$ ($1° = 60'$)
$\Rightarrow x = 89°60' - 65°50' = 24° 10'$

14. $123°45' + x = 180° = 179°60'$ ($1° = 60'$)
$\Rightarrow x = 179° 60' - 123° 45'$
$\Rightarrow x = 56° 15'$

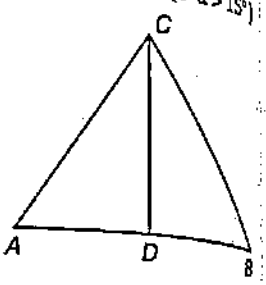
15. $x + y = 90° \Rightarrow x < 90°$
If $y \neq 0$ i.e., $y > 0$
and $y < 90°$ if $x \neq 0$ i.e., $x > 0$

16. Let he angle be x, then is supplement be $(180° - x)$
$\therefore x = \frac{1}{5}(180° - x) \Rightarrow x = 30°$

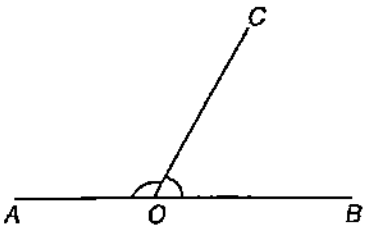
Alternatively: Go through options.
$30 = \frac{1}{5}(180° - 30°)$

17. $a > \frac{90°}{6} \Rightarrow a > 15°$
$\because a + b = 180°$
$\therefore b < 165°$ ($\because a > 15°$)

18. $\because AC = BC$
and $AD = BD$
and CD is common
$\therefore \Delta CDA$ is similar to ΔCDB and AB is a straight line
$\therefore \angle CDA = \angle CDB = 90°$



20.



21. $\angle ABP = \angle CDR = 100°$
$\therefore y = 80°$ ($\because \angle CDR + \angle CDS = 180°$)

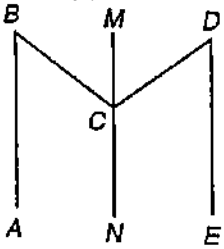
22. $\angle APO = 42°$ and $\angle CQO = 38°$
$\angle POQ = \angle PON + \angle NOQ$
$= \angle APO + \angle OQC = 42° + 38° = 80°$

23. $\angle XYB = \angle YBC$ and $\angle XYB = \angle BYC$
$\Rightarrow \angle YBC = \angle BYC$

24. $\angle PGH = 80° \Rightarrow \angle QGH = 100°$
$\angle QHD = 120° \Rightarrow \angle CHQ = 60°$
$\therefore \angle x + 100 + 60 = 180°$
$\Rightarrow x = 20°$

25. $\angle z = 180° - 110 = 70°$ ($\because \angle QAC = \angle BCA$)
and $60° + y° + z° = 180°$
$\therefore y = 180° - (70° + 60°)$
$y = 50°$

26. $\angle BCD = \angle BCM + \angle DCM$
$= \angle ABC + \angle EDC$
$= 67° + 23° = 90°$



(Draw a line parallel to AB or DE through C.)

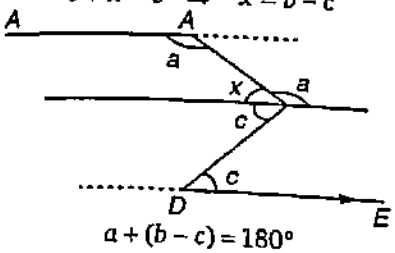
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

27. $a + x = 180°$

$c + x = b \Rightarrow x = b - c$



$\therefore \quad a + (b - c) = 180°$

$\Rightarrow \quad a + b - c = 180°$

28. $\angle x = \angle ABC = \angle CDG = 180° - 138° = 42°$

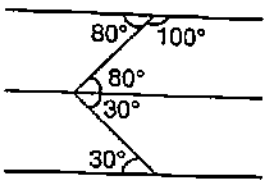
29. Use the same method as in question number 27.

i.e., $(a + b - c) = 180°$

$\Rightarrow \quad 100° + (x + 10°) - 30° = 180°$

$\Rightarrow \quad x = 100°$

Alternatively :



$\therefore \quad x + 10° = 80° + 30°$

$\Rightarrow \quad x = 100°$

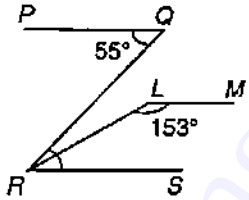
30. $\angle PQR = \angle QRS$

$55° = \angle QRS$

$55° = \angle QRL + \angle LRS$

$55° = 30° + \angle LRS$

$\Rightarrow \quad \angle LRS = 25°$



31. $\angle ABC = 180° - (40° + 65°)$

$\angle ABC = 75°$

and $\angle DCF = \angle ABC = 75°$

$\therefore \quad \angle EDC = 180° - \angle DCF = 105°$

32. Extend CD to M, then

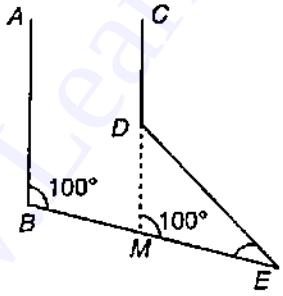
$\angle DME = \angle ABE = 100°$

$\angle MED = 25°$

$\therefore \angle MDE = 180° - (100° + 25°)$

$= 55°$

$\therefore \angle CDE = 180° - 55° = 125°$



33. $\angle CEF = \angle ABF = 70°$

$\therefore \quad \angle FED = 180° - 70° = 110°$

$\angle EDF + \angle DFE + \angle FED = 180°$

$\therefore \quad \angle FDE = 180° - (110° + 20°) = 50°$

34. $\angle BAO = 180° - (60° + 35°) = 85°$

$\angle AOB = 180° - (85° + 20°) = 75°$

35. $\angle AEP = \angle CDP = 34°$

$\angle AEP + \angle BEF + \angle DEF = 180°$

$34° + 78° + \angle DEF = 180°$

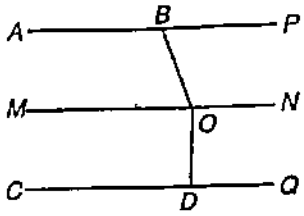
$\Rightarrow \quad \angle DEF = 68°$

36. $\angle AKH = \angle KHD = 25°$

$\angle EGH = \angle AKL = 180° - 60° = 120°$

$\therefore \quad \angle HKL = 120° + 25° = 145°$

38.



$\angle ABO = 118°$

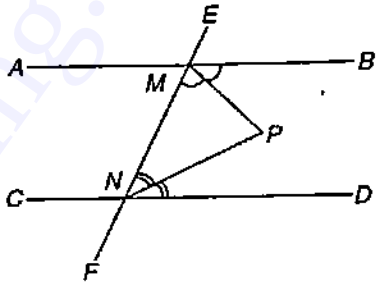
$\therefore \quad MOB = 180° - 118° = 62°$

$\therefore \quad \angle MOD = 152° - 62° = 90°$

$\therefore \quad \angle ODC = 180° - 90° = 90°$

39. Let $\angle BMN = 2x$ and $\angle DNM = 2y$

$\therefore \quad 2x + 2y = 180°$

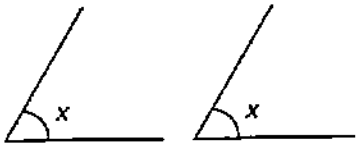


$\Rightarrow \quad x + y = 90°$

$\therefore \quad \angle MPN = 180° - 90° = 90°$

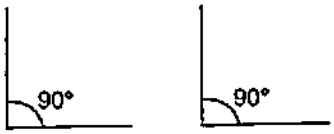
40.

Equal



$x \neq 90°$

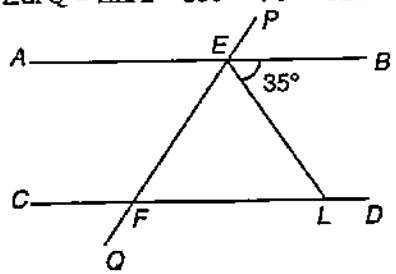
Equal and Supplementary



$x = 90°$

41. $\angle BEF = 2\angle BEL = 70°$

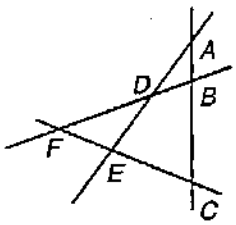
$\therefore \quad \angle CFQ = \angle EFL = 180° - 70° = 110°$



44. $^4C_3 = \dfrac{4!}{3!1!} = \dfrac{4 \cdot 3 \cdot 2 \cdot 1}{3 \cdot 2 \cdot 1 \cdot 1} = 4$

45. $^4C_2 = \dfrac{4!}{2!2!}$

$= \dfrac{4 \cdot 3 \cdot 2 \cdot 1}{(2 \cdot 1) \cdot (2 \cdot 1)} = 6$



For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yourshahboob.wordpress.com
For More Visit : www.LearnEngineering.in

INTRODUCTORY EXERCISE 12.2

1. The sum of the smaller sides cannot be equal to or less than the largest side.

2. $12^2 > 8^2 + 6^2 \Rightarrow$ obtuse triangle

3. Sum of exterior angles = 360°

4. $\angle ACD = \angle CBA + \angle BAC = 70° + 40° = 110°$

5. $\angle ABC + \angle ACB < 90°$

6. $x + 4x + 7x = 180°$

$\Rightarrow \quad x = 15°$

$\therefore \quad 7x = 105°$

7. $\angle OBC + \angle OCB = \frac{70}{2} + \frac{30}{2} = 50°$

$\therefore \quad \angle BOC = 180° - 50° = 130°$

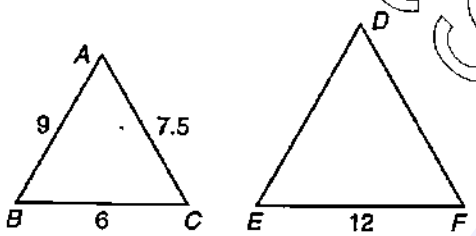
8. $\angle BFC = 90° - \frac{1}{2}(\angle A)$

$= 90° - \frac{1}{2}[180° - (80° + 30°)] = 55°$

10. Prove with the help of similarity.

11. Prove with the help of the similarity.

12. $\Delta ABC \sim \Delta DEF$



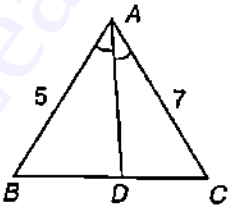
$\therefore \quad \frac{AB}{BC} = \frac{DE}{EF}$

$\frac{9}{6} = \frac{DE}{12}$

$\Rightarrow \quad DE = 18$ cm

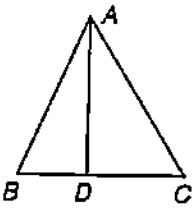
13. $\frac{AB}{AC} = \frac{BD}{CD} = \frac{5}{7}$

(By angle bisector theorem)



15. AB is hypotenuse and AD is perpendicular in ΔABD,

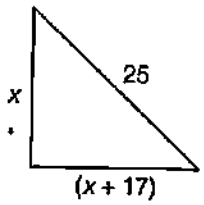
$\therefore \quad AB > AD$



16. $x^2 + (x - 17)^2 = 25^2$ (Using Pythagorus theorem)

$\therefore \quad x = 24$ cm

$\therefore$ Altitude + Base = 24 + 7 = 31 cm



17. Go back to the basics.

18. $\angle ACB = 180° - 100° = 80°$

and $\angle BAC + \angle ABC = 180° - 80° = 100°$

$\therefore \quad \angle ABC = 50°$ ($\because \angle BAC = \angle ABC$)

20. $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$ (cosine rule)

$\frac{1}{2} = \frac{b^2 + c^2 - a^2}{2bc}$ ($\because \cos 60° = 1/2$)

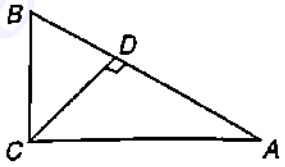
$\therefore \quad a^2 = b^2 + c^2 - bc$

21. $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$

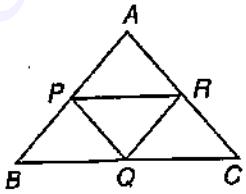
$-\frac{1}{2} = \frac{a^2 + b^2 - c^2}{2ab}$

$\Rightarrow \quad c^2 = a^2 + b^2 + ab$

22. $\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}$ (Apply similarity of triangles)



27.



30. $CD = \frac{AC \times BC}{AB}$

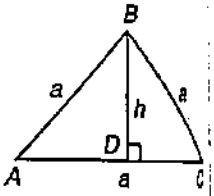
or $\frac{1}{(CD)^2} = \frac{1}{(AC)^2} + \frac{1}{(BC)^2}$

32. $AB^2 = BD^2 + AD^2$ (Pythagorus theorem)

$a^2 = h^2 + \left(\frac{a}{2}\right)^2$

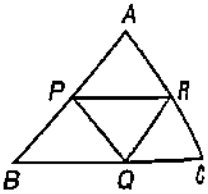
$\Rightarrow \quad h^2 = \frac{3}{4}a^2 \Rightarrow h = \frac{\sqrt{3}}{2}a$

$\Rightarrow \quad \frac{a}{h} = \frac{2}{\sqrt{3}}$



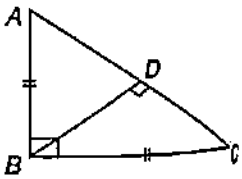
33. There are 4 congruent triangles.

Hence, $A(\Delta ABC) = 6 \times 4 = 24$ cm^2



34. $BD = \frac{AB \times BC}{AC}$

$= \frac{6 \times 6}{6\sqrt{2}} = 3\sqrt{2}$ cm



36. $\angle BOC = 90° + \frac{1}{2}\angle A$

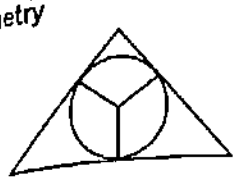
$= 90° + 40° = 130°$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

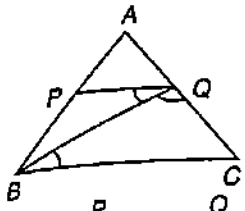
yoursmahboob.wordpress.com

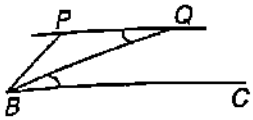
For More Visit : www.LearnEngineering.in

37.



42.



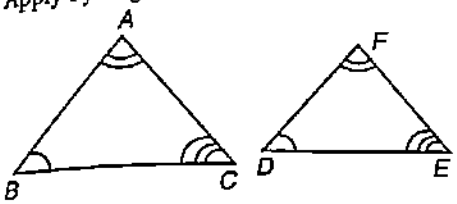
i.e., 

BQ is a transversal of PQ and BC.

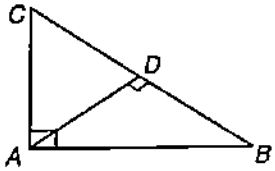
45. Apply Pythagorus theorem.

46. Apply Pythagorus theorem.

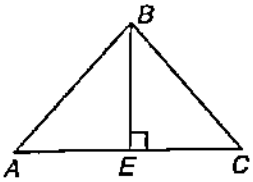
47.



48. $\frac{BD}{CD} = \frac{(AB)^2}{(AC)^2}$



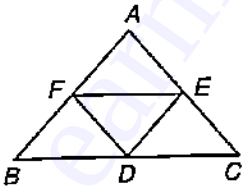
49. Apply Pythagorus theorem



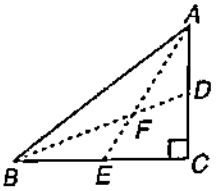
50. $EF = 3$ cm $\quad \therefore BC = 6$ cm

$FD = 4$ cm $\quad \therefore AC = 8$ cm

$AB = 10$ cm $\quad \therefore DE = 5$ cm



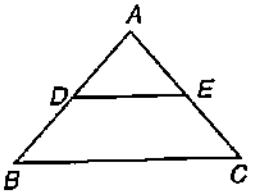
51.



52. $\frac{AB}{AC} = \frac{AD}{AE}$

$\frac{3.6}{2.4} = \frac{2.1}{AE}$

$\Rightarrow \quad AE = 1.4$ cm



54. $\angle EAB = \angle OAB = 60°$

$\angle ABD = 45°$

$\therefore \quad \angle AOB = 180° - (60° + 45°) = 75°$

55. $\angle LBN = \angle BAC = \angle BCA = 70°$

$\therefore \quad \angle ABC = 180° - (70° + 70°) = 40°$

56. $\angle ECB = 60° + 90° = 150°$

$\therefore \quad \angle CEB + \angle EBC = 180° - 150° = 30°$

$\therefore \quad \angle BEC = 15°$ $\quad (\because \angle BEC = \angle EBC)$

$\therefore \quad \angle DEC = 60°$

$\therefore \quad \angle DEM = 60° - 15° = 45°$

58. $\angle ACB = 42° = \angle ECD$

$\therefore \quad \angle CED = 180° - (58° + 42°) = 80° = \angle FEG$

$\therefore \quad \angle FGE = 180° - (66° + 80°) = 34°$

59. $\angle BAD = 70°$

$\therefore \quad \angle BDA = 180° - (70° + 50°) = 60°$

60. $\frac{2x}{x} = \frac{3y + 8}{2y} \Rightarrow y = 8$

$\therefore \quad x = 2y = 16$

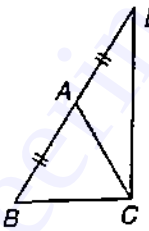
61. $\angle BCD = 27°$

$\angle BDC = 180° - (27° + 27°) = 126°$

$\therefore \quad \angle ADC = 54° = \angle DAC$

$\therefore \quad \angle ACD = 180° - (54° + 54°) = 72°$

62.



63. By mid-point theorem

64. $\frac{PB}{BR} = \frac{AP}{AQ}$

$\frac{PB}{BR} = \frac{2\sqrt{2}}{3\sqrt{2}} = \frac{2}{3}$

$\because \quad \frac{PB}{BR} = \frac{2x}{3x}$

$\therefore \quad BR = 6$ cm

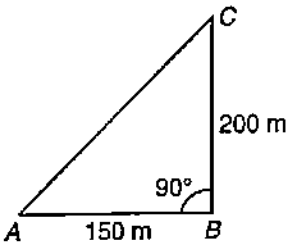
66. Apply similarity of triangles

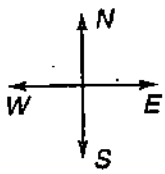
67. Apply similarity of triangles

$\Delta ADB \sim \Delta CDA$

71. $\Delta ABD \cong \Delta CBD$

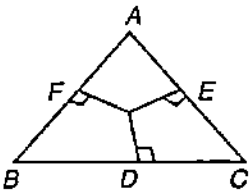
73.





$AC = \sqrt{AB^2 + BC^2}$

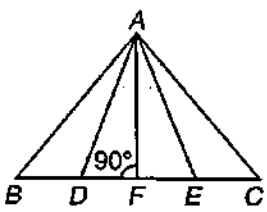
74.



75. $BD = 2DF$

$AB^2 = BF^2 + AF^2$

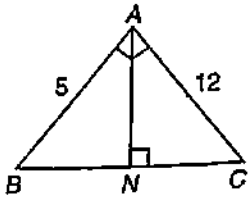
and $\quad AD^2 = DF^2 + AF^2$



For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmanboob.wordpress.com
For More Visit : www.LearnEngineering.in

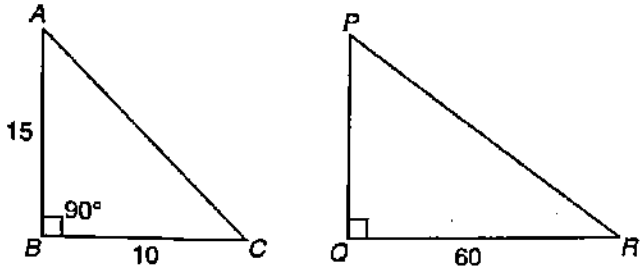
76. $\dfrac{A(\Delta ANC)}{A(\Delta ANB)} = \dfrac{\frac{AN \times NC}{2}}{\frac{AN \times BN}{2}}$



$= \dfrac{NC}{NB} = \dfrac{\frac{(12 \times 12)}{13}}{\frac{5 \times 5}{13}} = \dfrac{144}{25}$

77. $\dfrac{AB}{BC} = \dfrac{PQ}{QR}$

$\dfrac{15}{10} = \dfrac{PQ}{60}$



$\Rightarrow$ $PQ = 90$ cm

78. 16 : 81

79. $\dfrac{BC}{PQ} = \dfrac{AC}{AP} = \dfrac{AB}{AQ}$

80. $\dfrac{3}{x-3} = \dfrac{x-5}{3x-19}$

$\Rightarrow$ $x = 8$ or 9

81. $h = \dfrac{\sqrt{3}}{2}\text{ side} = \dfrac{\sqrt{3}}{2} \times 2a = a\sqrt{3}$

82. $\dfrac{AB}{AC} = \dfrac{BD}{CD}$

$\Rightarrow$ $\dfrac{5}{AC} = \dfrac{2}{3}$

$\Rightarrow$ $AC = 7.5$ cm

83. $\dfrac{AB}{QR} = \dfrac{PB}{PR}$

$\dfrac{3}{9} = \dfrac{PB}{6} \Rightarrow PB = 2$ cm

84. $\dfrac{AO}{BO} = \dfrac{AQ}{BP}$

$\dfrac{10}{6} = \dfrac{AQ}{9} \Rightarrow AQ = 15$ cm

85. $\dfrac{AB}{AC} = \dfrac{AD}{AE}$

$\dfrac{12}{15} = \dfrac{6}{AE}$

$\therefore$ $AE = 7.5$ cm

86. $AB = AC$ and $AD = CD$

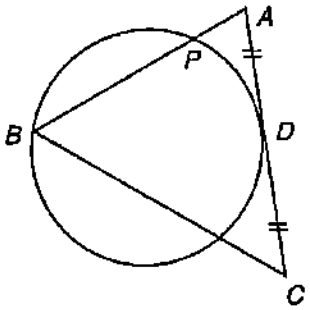
$\therefore$ $AB = 2AD$

Now, since AD is a tangent

$\therefore$ $AD^2 = AP \times AB$

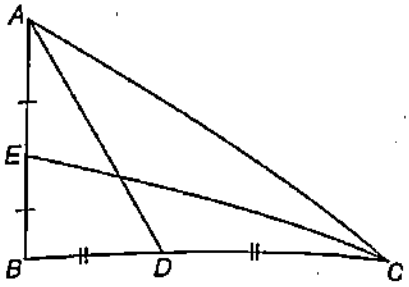
$\left(\dfrac{AB}{2}\right)^2 = AP \times AB$

$\Rightarrow$ $AB = 4AP$



87. $AC = 5$ cm

$AD = \dfrac{3\sqrt{5}}{2}$ cm



$AE = BE$ and $BD = CD$

$AB^2 = AC^2 - BC^2 = 25 - BC^2$

and $AB^2 = AD^2 - BD^2 = \left(\dfrac{3\sqrt{5}}{2}\right)^2 - BD^2$

$= \dfrac{45}{4} - \dfrac{BC^2}{4}$

From eq. (1) and (2)

$BC^2 = \dfrac{55}{3}$

Now, from eq. (1)

$AB^2 = 25 - \dfrac{55}{3} = \dfrac{20}{3}$

Also $CE^2 = BE^2 + BC^2$

$= \left(\dfrac{1}{2}AB\right)^2 + BC^2$

$= \dfrac{AB^2}{4} + BC^2 = CE^2$

$= \dfrac{5}{3} + \dfrac{55}{3} = \dfrac{60}{3} = 20$

$\therefore$ $CE = 2\sqrt{5}$ cm

88. By Apollonius theorem

$AB^2 + AC^2 = 2(AD^2 + BD^2)$

$100 + 196 = 2(AD^2 + 36)$

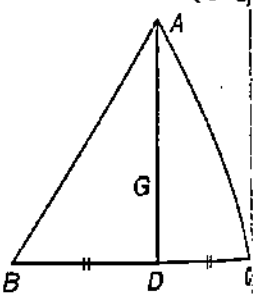
$\Rightarrow$ $AD^2 = 112$

$\therefore$ $AD = 4\sqrt{7}$

Now, since $\dfrac{AG}{GD} = \dfrac{2}{1}$

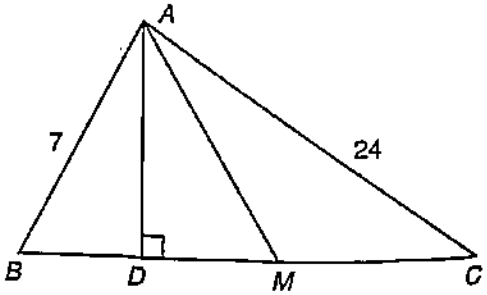
$\Rightarrow$ $AG = \dfrac{2}{3}AD$

$\therefore$ $AG = \dfrac{2}{3} \times 4\sqrt{7} = \dfrac{8}{3}\sqrt{7}$ cm



89. $\because$ AM is the median of a right angled triangle.

$\therefore$ $AM = \dfrac{BC}{2} = \dfrac{25}{2}$



For More Visit : www.LearnEngineering.in
Scanned by CamScanner

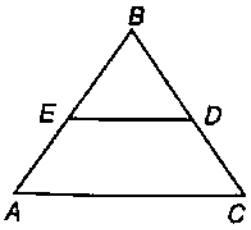
yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

and $AD = \frac{AB \times AC}{BC} = \frac{7 \times 24}{25}$

$\therefore \quad \frac{AD}{AM} = \frac{7 \times 24 \times 2}{25 \times 25} = \frac{336}{625}$

90. $\Delta BED \sim \Delta BAC$



$\therefore \quad \frac{BE}{BA} = \frac{ED}{AC} = \frac{BD}{BC} = \frac{1}{2}$

$\therefore \quad \frac{A(\Delta BED)}{A(\Delta BAC)} = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$

$\therefore \quad A(\Delta BDE) = 7.5\,cm^2$

91. Apply Pythagorus theorem.

NOTE The sides given in option (d) cannot form any type of triangle.

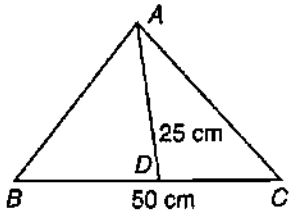
92. $\frac{AB}{BD} = \frac{AC}{CD}$

$\frac{12}{BC + CD} = \frac{8}{CD}$

$\frac{12}{4 + CD} = \frac{8}{CD} \Rightarrow CD = 8\,cm$

93. $AB^2 + AC^2 = 2\left[AD^2 + \left(\frac{BC}{2}\right)^2\right]$

(Appollonius theorem)



$2500 = 2\left[625 + \left(\frac{BC}{2}\right)^2\right]$

$\Rightarrow \quad BC = 50\,cm$

94. $\angle BAC = 15°$

$\therefore \quad \angle BCA = 15° \qquad (\because AB = BC)$

$\therefore \quad \angle ABC = 180° - (15° + 15°) = 150°$

$\therefore \quad \angle ABD = 30° \qquad (180° - 150°)$

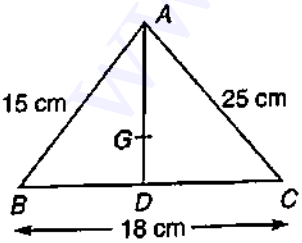
$\therefore \quad \sin 30° = \frac{AD}{AB}$

$\frac{1}{2} = \frac{AD}{AB} \Rightarrow AD = 5\,cm \qquad (\because AB = 10\,cm)$

$\therefore \quad$ Area of $\Delta ABC = \frac{1}{2} \times BC \times AD$

$= \frac{1}{2} \times 10 \times 5 = 25\,cm^2$

95. $AB^2 + AC^2 = 2(AD^2 + BD^2)$



$\therefore \quad 225 + 625 = 2(AD^2 + 81)$

$\Rightarrow \quad AD^2 = 344$

$\Rightarrow \quad AD = 2\sqrt{86}$ and $GD = \frac{1}{3}AD$

$\therefore \quad GD = \frac{2}{3}\sqrt{86}\,cm$

96. $\angle BDE = 115°$

$\therefore \quad \angle ADE = 65°$

and $\angle AED = 75°$

$\therefore \quad \Delta AED \sim \Delta ABC$

$\therefore \quad \frac{DE}{BC} = \frac{AE}{AB} = \frac{AD}{AC}$

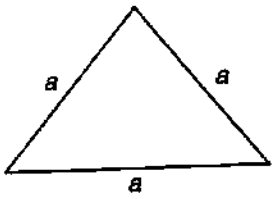
$\therefore \quad \frac{2}{3} = \frac{10}{AB}$

$\Rightarrow \quad AB = 15\,cm$

97. For the given perimeter of a triangle the maximum area is enclosed by an equilateral triangle.

$\therefore \quad 3a = 24\,cm$

$\Rightarrow \quad a = 8\,cm$



$\therefore \quad$ Area $= \frac{\sqrt{3}}{4}a^2$

$= \frac{\sqrt{3}}{4} \times (8)^2 = 16\sqrt{3}\ cm^2$

98. $\Delta ADE \sim \Delta ACB$ (A-A-A property)

$\therefore \quad \frac{AE}{AB} = \frac{AD}{AC}$

$\frac{10}{20} = \frac{12}{AC} \Rightarrow AC = 24\,cm$

$EC = AC - AE = 24 - 10 = 14\,cm$

99. $\angle ACD = \angle ADC = x$

$\therefore \quad \angle CAD = (180° - 2x)$

$\angle ABC = \angle BAC = \frac{x}{2}$

$(\because \angle ABC + \angle BAC = \angle ACD = x)$

$\therefore \quad \angle BAC + \angle CAD + 81° = 180°$

$\therefore \quad \frac{x}{2} + (180° - 2x) + 81° = 180°$

$\therefore \quad \frac{3}{2}x = 81° \Rightarrow x = 54°$

Alternatively: Applying the same process go through options.

100. It is value (side) dependent *i.e.*, it is different for different triangles unlike in equilateral triangle.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmanboob.wordpress.com

For More Visit : www.LearnEngineering.in

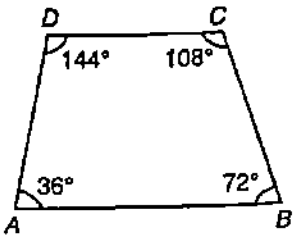
INTRODUCTORY EXERCISE 12.3

1. $x + 2x + 3x + 4x = 360$

$\Rightarrow \quad x = 36$

$\therefore$ The angles of quadrilateral (in order) are

$36°, 72°, 108°, 144°$



Since opposite angles are supplementary, therefor $AB \| CD$. Hence it is a trapezium.

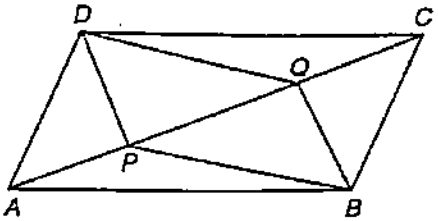
2. $3x + 4x + 5x + 6x = 360°$

$\Rightarrow \quad x = 20$

$\therefore \quad 6x = 120°$

Hence, the largest angle of a quadrilateral is 120°.

3.



4. $\angle DAB = 2x$

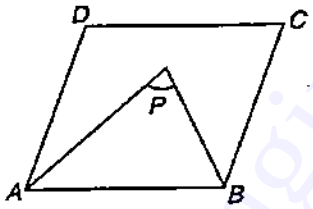
$\angle ABC = 2y$

$2x + 2y = 180$

$x + y = 90°$

i.e., $\angle PAB + \angle PBA = 90°$

$\therefore \quad \angle APB = 90°$



5. Prove that opposite sides are parallel, and diagonals are equal. Also all the interior angles are right angle.

6. $\angle BCE = 102°$ and $\angle ECD = 60°$

$\therefore \quad \angle BCD = 102° + 60° = 162°$

7. 90°

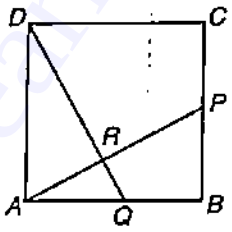
$\because \quad \Delta ADQ \cong \Delta BAP$

$\angle DCR = 90°$

$\angle CDR$ and $\angle CRP$ are supplementary

$\therefore \quad \angle DCR + \angle DRP = 180°$

$\therefore \quad \angle DRP = 90°$

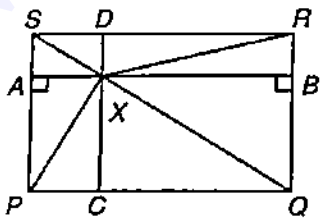


8. Area of $\Delta SXP = \dfrac{(SP) \times (AX)}{2}$

$A(\Delta RXQ) = \dfrac{(RQ) \times (BX)}{2}$

$A(\Delta SXR) = \dfrac{(SR) \times (DX)}{2}$

$A(\Delta PXQ) = \dfrac{(PQ) \times (CX)}{2}$



Area of $\Delta SXP + RXQ = \dfrac{SP \times AX}{2} + \dfrac{RQ \times BX}{2}$

$= \dfrac{1}{2}(l \times b)$ ...(1)

where $SP = RQ = b$ and $PQ = SR = l$

and $AX + BX = AB = PQ = SR = l$

and Area of $\Delta SXR + PXQ = \dfrac{(SR) \times (DX)}{2} + \dfrac{(PQ \times CX)}{2}$

$= \dfrac{(SR) \times (DX)}{2} + \dfrac{(SR) \times (CX)}{2}$

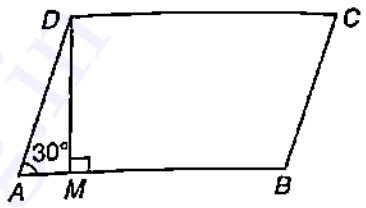
$= \dfrac{SR}{2}(DX + CX)$

$= \dfrac{l}{2} \times b = \dfrac{lb}{2}$

From eq. (1) and (2), we have option (c) is correct.

9. $AD = BC = 20$ cm

$AB = 20$



In ΔADM, $\dfrac{DM}{AD} = \sin 30° = \dfrac{1}{2}$

$\therefore \quad DM = \dfrac{1}{2} AD = 10$ cm

$\therefore$ Area of parallelogram = base × height

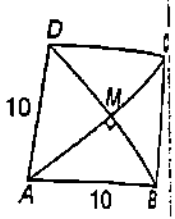
$= 20 \times 10 = 200$ cm^2

10. Let $DB = 12$ m

$\therefore \quad BM = 6$ m

$\therefore \quad AM = \sqrt{10^2 - 6^2}$

$AM = 8$ m



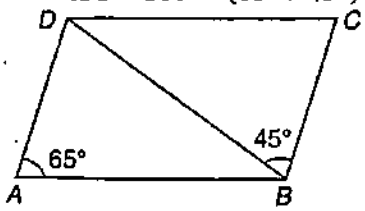
(Remember: Diagonals bisects each other at right angles)

(Bisect: Divide equally into two parts)

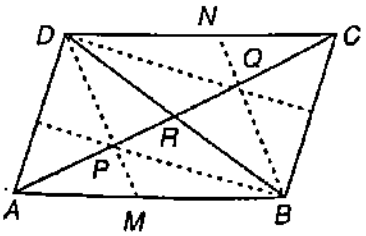
$\therefore \quad AC = 8 \times 2 = 16$ m

11. $\angle BCD = \angle DAB = 65°$

$\therefore \quad CDB = 180° - (65° + 45°) = 70°$



12. Let AC and BD bisects each other at R, then $AP = 2PR$ and $QC = 2RQ$



($\because AR$ and CR are the medians of ΔADB and ΔBDC)

and $AP = QC$ and $PR = RQ$ ($\because \Delta DAB \cong \Delta BDC$)

$\therefore \quad AP = PQ$

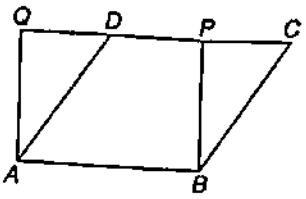
For More Visit : www.LearnEngineering.in

Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

13. Since their base and perpendicular heights are same.

14. $ABCD$ is a parallelogram and $ABPQ$ is a rectangle having same base and equal perpendicular height. Therefore the area of both is same.



Now, since $AD > AQ$

and $BC > BP$

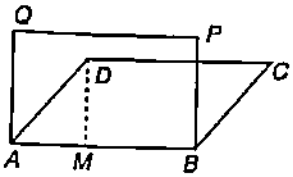
$\therefore \quad 2(AB + BC) > 2(AB + BP)$

$\therefore$ Perimeter of parallelogram > perimeter of rectangle

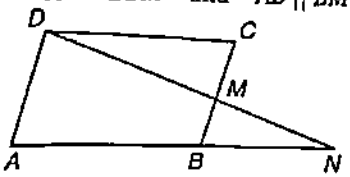
15. $AB = CD = PQ = l$

and $AQ = BP = AD = CB = b$

Bases of rectangle $ABPQ$ and parallelogram $ABCD$ is same, but perpendicular heights of parallelogram is less than that of rectangle. Therefore area of parallelogram is less than that of rectangle.



16. $AD = 2BM$ and $AD \parallel BM$

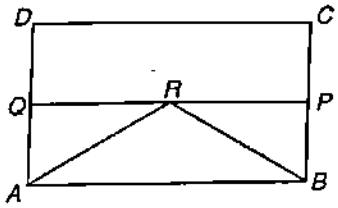


$\therefore$ From mid-point theorem

$\therefore \quad \frac{NM}{ND} = \frac{NB}{NA} = \frac{BM}{AD} = \frac{1}{2}$

$\therefore \quad AN = 2AB$

17. $A(\square ABPQ) = \frac{1}{2} A(\square ABCD)$

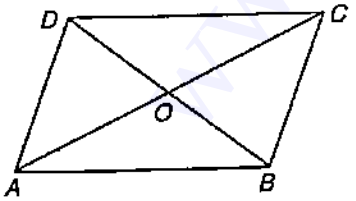


and $A(\Delta ARB) = \frac{1}{2} A(\square ABPQ)$

$\therefore \quad A(\Delta ARB) = \frac{1}{4} A(\square ABCD) \quad \left[A(\Delta ARB) = \frac{1}{2} \times AB \times BP\right]$

18. Let $BD = 6$ cm and $AC = 8$ cm

$\therefore \quad AO = 4$ cm and $BO = 3$ cm



Let $AB = 5$ cm

$\therefore \quad \angle AOB = 90°$

$\Rightarrow \quad \angle BOC = \angle AOD = \angle DOC = 90°$

$\therefore BC$ is also 5 cm.

Hence, $ABCD$ is a rhombus

$\therefore$ Area of rhombus $ABCD = \frac{AC \times BD}{2}$

$= \frac{6 \times 8}{2} = 24 \text{ cm}^2$

19. $BD = \sqrt{15^2 + 20^2} = 25$ cm (ΔABD is right angled triangle)

$\therefore \Delta BCD$ is equilateral triangle.

$\therefore$ Area of $\square ABCD$ = Area of ΔABD + Area of ΔBCD

$= \frac{1}{2} \times 15 \times 20 + \frac{\sqrt{3}}{4} \times (25)^2$

$= \frac{25}{4}(24 + 25\sqrt{3}) \text{ cm}^2$

20. $\frac{BE}{AB} = \sin 30° = \frac{1}{2}$

$\therefore \quad BE = \frac{1}{2} \times AB = 6 \text{ cm} = CF$

and $\frac{CF}{DF} = \tan 45° = 1$

$\therefore \quad DF = CF = 6$ cm

and $\frac{AE}{AB} = \cos 30° = \frac{\sqrt{3}}{2}$

$\therefore \quad AE = 6\sqrt{3}$ cm

$\therefore \quad AD = AE + EF + FD = 6\sqrt{3} + 6 + 6$

$= 6(2 + \sqrt{3}) \text{ cm}^2$

Area of trapezium $ADCB = \frac{1}{2} \times (AD + BC) \times BE$

$= \frac{1}{2} \times [6(2 + \sqrt{3}) + 6] \times 6$

$= \frac{6}{2}(2 + \sqrt{3} + 1) \times 6 = 18 \times (3 + \sqrt{3})$

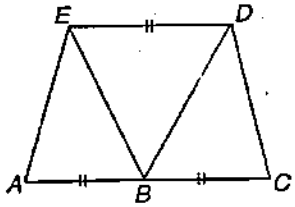
$= 18(3 + \sqrt{3}) \text{ cm}^2$

21. $A(\Delta ABE) = A(\Delta BDE)$

$= A(\Delta BCE)$

$A(\Delta ABC) = \frac{1}{3} A(\square ACDE)$

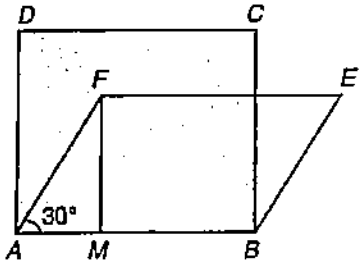
$= \frac{1}{3} \times 36 = 12 \text{ cm}^2$



22. $ABCD$ is a square and $ABEF$ is a rhombus.

$\frac{FM}{AF} = \sin 30° = \frac{1}{2}$

$\therefore \quad FM = \frac{AF}{2}, \quad AF = AB$



Area of square $= a^2$ $\quad (AB = AD = a)$

Area of rhombus $= \frac{a \times a}{2}$ $\quad \left(FM = \frac{a}{2}\right)$

(Area of rhombus = base × height)

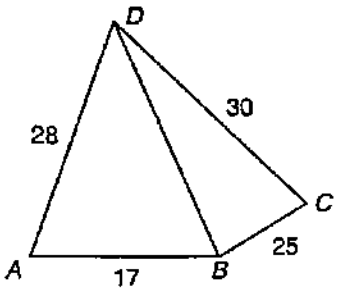
$\therefore \quad \frac{\text{Area of square}}{\text{Area of rhombus}} = \frac{2}{1}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yourmathhub.wordpress.com
For More Visit : www.LearnEngineering.in

23. Area of quadrilateral = Area of ΔABD + Area of ΔBCD

where $BD = 26$ cm



HINT Apply Hero's formula :

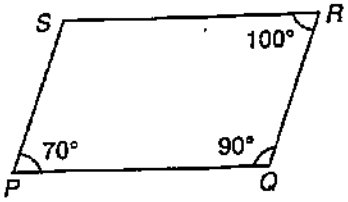
$$\text{Area of } \Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

where $s = \dfrac{a+b+c}{2}$

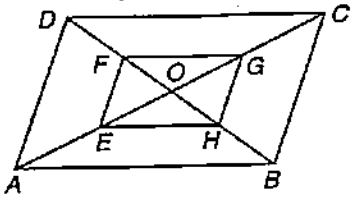
24. It is not possible

$\because \angle P + \angle R = 170° \neq 180°$

$\therefore$ The given quadrilateral is not cyclic so there is no any point in the given plane which is equidistant from its vertices.



25. By mid-point theorem,

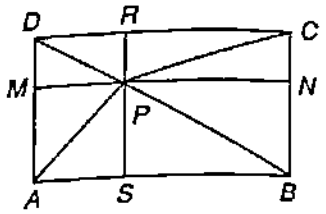


$$\frac{EF}{AD} = \frac{FG}{DC} = \frac{GH}{BC} = \frac{EH}{AB} = \frac{1}{2}$$

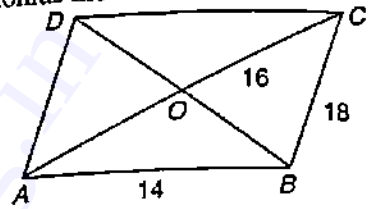
26. Since diagonals bisect each other at right angles and all sides are equal.

Apply Pythagorus theorem or Apollonius theorem.

27. Use Pythagorus theorem



28. Apply Apollonius theorem.



$$AB^2 + BC^2 = 2(BO^2 + AO^2)$$

$$AB^2 + BC^2 = 2(BO^2 + 64)$$

$$196 + 324 = 2(BO^2 + 64)$$

$\therefore \quad BO^2 = 196$

$\therefore \quad BO = 14$ cm

$\therefore \quad BD = 2BO = 28$ cm

INTRODUCTORY EXERCISE 12.4

1. Each interior angle $= 140°$

$\therefore$ Exterior angle $= 180° - 140° = 40°$

$\therefore$ Number of sides $= \dfrac{360°}{\text{Exterior angle}} = \dfrac{360°}{40°} = 9$

Alternatively : It can be checked easily through options.

2. Exterior angle of a regular hexagon $= \dfrac{360°}{6} = 60°$

$\therefore$ Interior angle $= 180° - 60° = 120°$

3. $\therefore$ Interior angle of pentagon $= 180° - \dfrac{360°}{5} = 108°$

$\therefore$ Interior angle of required polygon $= \dfrac{5}{6} \times 108° = 90°$

$\therefore$ Each exterior angle of the requird polygon

$= 180° - 90° = 90°$

$\therefore$ Number of sides $= \dfrac{360°}{\text{Exterior angle}} = \dfrac{360°}{90°} = 4$

Alternatively : Best way is to go through options.

4. Sum of all the interior angles $= (n-2)\,180°$

$1260° = (n-2)\,180°$

$\Rightarrow \quad n = 9 \qquad (n \rightarrow$ number of sides)

5. Interior angle $= 3 \times$ exterior angle

$$180° - \frac{360°}{n} = 3 \times \left(\frac{360}{n}\right) \Rightarrow n = 8$$

Alternatively : Best way is to go through options.

6. Go through option

Exterior angle $= \dfrac{360°}{6} = 60°$

Interior angle $= 180° - 60° = 120°$

Difference between interior angle and exterior angle

$= 120° - 60° = 60°$

7. $\dfrac{n\,(n-3)}{2} = 54° \Rightarrow n^2 - 3n = 108° \Rightarrow n = 12$

Alternatively : Put the value of n from the choices given below.

8. Go through options. Let us consider the correct option (d)

Number of sides	5	:	10
Exterior angle =	72°		36°
Interior angle =	180° − 72° (= 108°)		180° − 36° (= 144°)
	3	:	4

Hence, option (d) is correct.

9. Go through options

Alternatively : Sum of al exterior angles $= 360°$

$\therefore$ Sum of interior angles $= 4 \times 360° = 1440°$

$\therefore \quad 1440° = (n-2)\,180°$

$\Rightarrow \quad n = 10$ (no. of sides)

10. Exterior angle $= \dfrac{360°}{9} = 40°$

Interior angle $= 180° - 40° = 140°$

$\therefore$ Interior angle : exterior angle $= 140° : 40 = 7 : 2$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

INTRODUCTORY EXERCISE 12.5

1. $AC = \sqrt{OA^2 - OC^2} = \sqrt{(17)^2 - (8^2)} = 15$ cm

$\therefore$ $AB = 2AC = 30$ cm $\quad (\because AC = BC)$

2. $AB = 8$ cm

$\therefore$ $AM = 4$ cm

and $OA = 5$ cm

$\therefore$ $OM = \sqrt{(OA)^2 - (AM)^2}$

$= \sqrt{25 - 16} = 3$ cm

3. $\because AB$ is the perpendicular bisector of OO'

$\therefore$ $\angle OMA = 90°$

4. $AM : BM : 1 : 1$

5. $\angle AOC = \angle BOC \quad (\because AC = BC)$

$\therefore OC$ is the perpendicular bisector of AB.

$\therefore$ $AM = BM$

6. $OA = 13$ cm

$AM = 5$ cm

$\therefore$ $OM = \sqrt{(13)^2 - (5)^2} = 12$ cm

7. Degree measure of major arc $= 360° - 130° = 230°$

8. It is necessarily a rectangle.

9. $\angle ADC = 180° - 65° = 115°$

$\therefore$ $\angle DCA = 180° - (115° + 45°) = 20°$

($\square ADCB$ is a cyclic quadrilateral)

10. $\angle OBA = \angle OAB = 55°$

$\therefore$ $\angle AOB = 180° - (55° + 55°) = 70°$

and $\angle AOB = \angle COD = 70°$

12. $\Delta BAP \sim \Delta CDP \quad (\because \angle A = \angle D$ and $\angle APB = \angle DPC)$

$\therefore$ $\frac{AB}{CD} = \frac{AP}{DP} = \frac{BP}{CP}$

$\therefore$ $AP \times PC = BP \times DP$

13. $\angle BDC = 180° - (110° + 30°) = 40°$

and $\angle BAC = \angle BDC = 40°$ (angles of the same segment)

14. $\angle ABO = 60° = \angle BAO$ ($\because AO = BO$, radii of the circle)

$\therefore$ $\angle AOB = 180° - (60° + 60°) = 60°$

$\therefore$ $\angle ACB = \frac{1}{2}\angle AOB = 30°$

15. $\angle CBA = \frac{1}{2}\angle COA = 60°$

$\therefore$ $\angle CBE = 180° - \angle CBA$

$180° - 60° = 120°$

16. $\angle BDA = 90°$

$\angle DBA = 50°$

$\therefore$ $\angle DAB = 180° - (90° + 50°) = 40°$

17. $\angle BAC = 60°$

$\therefore$ $\angle BEC = 180° - 60° = 120°$

18. $\angle ABD = 45°$

$\therefore$ $\angle ACD = 45°$

19. $\angle BDC = \angle BAC = 30°$

$\therefore$ $\angle BCD + \angle BDC + \angle DBC = 180°$

$\therefore$ $\angle BCD = 180° - (30° + 60°)$

$\angle BCD = 90°$

20. $\angle ADC = 140°$

$\therefore$ $\angle ABC = 40°$

$\angle ACB = 90°$

$\therefore$ $\angle BAC = 180° - (90° + 40°) = 50°$

21. $\angle CAB = \frac{1}{2}\angle BOC = \frac{1}{2} \times 40° = 20°$

22. $\angle OBA = \angle OAB = \left(\frac{180° - 80°}{2}\right) = 50°$

and $\angle OCA = \angle OAC = \left(\frac{180° - 120°}{2}\right) = 30°$

$\therefore$ $\angle BAC = \angle BAO + \angle OAC = 50° + 30° = 80°$

23. $\angle ADC = \frac{100°}{2} = 50°$

$\therefore$ $\angle ABC = 180° - 50° = 130°$

$\therefore$ $\frac{m\angle ADC}{m\angle ABC} = \frac{50°}{130°} = \frac{5}{13}$

24. $\angle APB = 50° \quad \left(\because m\angle APB = \frac{1}{2} m\angle AOB\right)$

$\therefore$ $\angle ACB = 180° - 50° = 130°$

$\therefore$ $\angle BCD = 180° - 130° = 50°$

25. $\angle ABC = 180° - 120° = 60°$

$\angle ACB = 90°$

$\therefore$ $\angle CAB = 180° - (90° + 60°) = 30°$

26. $\angle COD = \angle AOB = 70°$

$\therefore$ $\angle OCD = \frac{180° - 70°}{2} = 55°$

27. $\angle ABK = 180° - (105° + 40°) = 35°$

$\therefore$ $\angle ACD = \angle ABD = 35°$

28. $\angle BAC = 80° \quad (\because \angle ABC = \angle ACB = 50°)$

$\therefore$ $\angle BDC = \angle BAC = 80°$

29. $\angle ADB = 90°$

$\therefore$ $\angle DBA = 20° \quad [180° - (90° + 70°)]$

$\therefore$ $\angle CBA = 30° + 20° = 50°$

$\therefore$ $\angle CDA = 180° - 50° = 130°$

$\therefore$ $\angle CDB = 130° - 90° = 40°$

30. $\angle BAD = 80°$

$\therefore$ $\angle DCE = 80°$

31. $\angle AOC = 2 \times 50 = 100°$

$\therefore$ $\angle ABC = \frac{100}{2} = 50°$

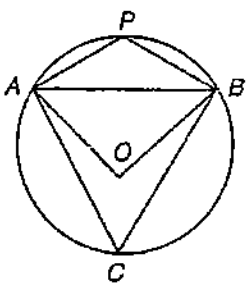
NOTE There are some more ways to solve it. Find them.

32. $\angle AOB = 2\angle ACB = 2 \times 25 = 50°$

33. $\angle APB = 180° - \frac{1}{2} \times 90° = 135°$

$\left[\because \angle ACB = 45° = \left(\frac{1}{2} \times 90°\right)\right.$

$\left.\text{and } \angle APB + \angle ACB = 180°\right]$



For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yourSmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

34. $OA = \sqrt{(OB)^2 - (AB)^2}$

$\therefore \quad OA = \sqrt{169 - 144} = 5$

35. $PN \times PR = PQ^2$

$PN \times 25 = (15)^2$

$\therefore \quad PN = 9 \text{ cm}$

36. $TQ = TP \quad \text{and} \quad TP = TR$

$\therefore \quad TQ = TP = TR$

37. $\angle BAC = 90°$

$\angle BCA = 60° \qquad (\because \angle BCA = BAQ)$

$\therefore \quad \angle ABC = 180° - (90° + 60°)$

$\angle ABC = 30°$

38. $\angle DBQ = 65°$

$\therefore \quad \angle DAB = 65°$

$\therefore \quad \angle DCB = 180° - 65° = 115°$

39. $AP \times BP = DP \times CP$

$2 \times 6 = DP \times 3$

$\Rightarrow \quad DP = 4 \text{ cm}$

40. $PA \times PB = PC \times PD$

$3 \times (3 + 5) = 2 \times PD$

$\therefore \quad PD = 12 \text{ cm}$

$\therefore \quad CD = PD - PC$

$= 12 - 2 = 10 \text{ cm}$

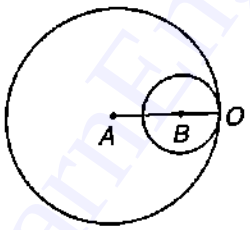
41. $PT^2 = PA \times PB$

$5 \times 5 = 4 \times (4 + x)$

$\Rightarrow \quad x = \frac{9}{4} \text{ cm}, \qquad$ where $AB = x$

42. Let they touch eachother at O and A, B be the centres of the two circles, then $OA = 13$ cm and $OB = 5$ cm

$\therefore \quad AB = OA - OB = 8 \text{ cm}$



43. $AB = 5 \text{ cm} = x + y$

$BC = 6 \text{ cm} = y + z$

$AC = 7 \text{ cm} = z + x$

$\therefore \quad 2(x + y + z) = 5 + 6 + 7 = 18 \text{ cm}$

$\therefore \quad x + y + z = 9 \text{ cm}$

$\therefore \quad x = (x + y + z) - (y + z) = 9 - 6 = 3 \text{ cm}$

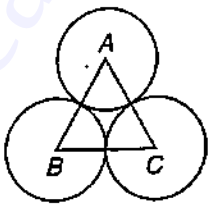
$\therefore \quad x + y = 5 \text{ cm}$

$\therefore \quad y = 5 - x = 5 - 3 = 2 \text{ cm}$

$\therefore \quad z = 6 - y = 6 - 2 = 4 \text{ cm}$

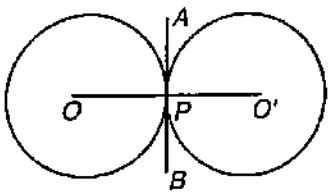
$\therefore \quad x = 3 \text{ cm}$

$y = 2 \text{ cm} \quad \text{and} \quad z = 4 \text{ cm}$



44. Prove with the help of property of tangents : "Tangents drawn from the same point on a circle are always equal."

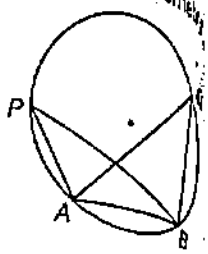
45. Tangent is always perpendicular to the radius.



46. $\angle APB = \angle AQB$

when $\angle APB = \angle AQB = 90°$

then they are supplementary. Also they are supplementary when they are in different segments.



$CD = 6 \text{ cm}$

47. $AC = 6 \text{ cm} \quad \text{and} \quad BC = 6 \text{ cm}$

$\therefore$ ($\because AC = CD$ and $BC = CD$. Two tangents from the same point are always equal.)

$AB = 12 \text{ cm}$

48. $\therefore \quad \angle CAB = \angle BCD$

and $\quad \angle DAB = \angle BDC$

$\therefore \quad \angle CAD = \angle CAB + \angle DAB = \angle BCD + \angle BDC$

$\Rightarrow \quad \angle CAD = \angle BCD + \angle BDC$

$\therefore \quad \angle CAD + \angle CBD = \angle BCD + \angle BDC + \angle CBD = 180°$

49. $AB = 24 \text{ cm}$

$\therefore \quad AC = BC = 12 \text{ cm}$

$OA = 20 \text{ cm}$

$\therefore \quad OC = \sqrt{(OA)^2 - (AC)^2}$

$= \sqrt{400 - 144} = 16 \text{ cm}$

and $\quad O'A = 37 \text{ cm}$

$\therefore \quad O'C = \sqrt{(O'A)^2 - (AC)^2}$

$= \sqrt{(37)^2 - (12)^2} = 35 \text{ cm}$

$\therefore \quad OO' = OC + CO' = 16 + 35 = 51 \text{ cm}$

50. $AB = AC, OB = OC$

and BC is common and OA is common

also $\quad \angle BOA = \angle COA$

$\therefore \quad \Delta BOA \cong \Delta COA$

$\therefore BC$ and OA are perpendicular bisector to each other

[Area of $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$]

$s = \frac{a+b+c}{2} = \frac{5+5+6}{2} = 8 \text{ cm}$

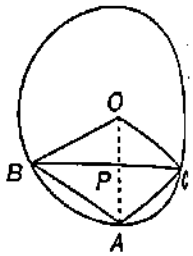
Area of $\Delta BOA = \sqrt{8 \times 3 \times 3 \times 2} = 12 \text{ cm}^2$

Again area of $\Delta BOA = \frac{1}{2} OA \times BP = \frac{1}{2} \times 5 \times BP$

$12 = \frac{1}{2} \times 5 \times BP$

$\Rightarrow \quad BP = 4.8 \text{ cm}$

$\therefore \quad BC = 2BP = 2 \times 4.8 = 9.6 \text{ cm}$



51. $OA = OC = 17 \text{ cm}$

$AP = \frac{1}{2} AB = \frac{1}{2} \times 16 = 8 \text{ cm}$

$\therefore \quad OP = \sqrt{OA^2 - AP^2} = 15 \text{ cm}$

Now, $\quad PQ = OP + OQ = 23 \text{ cm}$

$\therefore \quad 15 + OQ = 23 \text{ cm}$

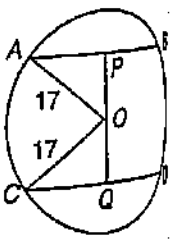
$\therefore \quad OQ = 8 \text{ cm}$

Again, $\quad CQ = \sqrt{(OC)^2 - (OQ)^2}$

$CQ = \sqrt{(17)^2 - (8)^2}$

$\therefore \quad CQ = 15 \text{ cm}$

$\therefore \quad CD = 2CQ = 30 \text{ cm}$



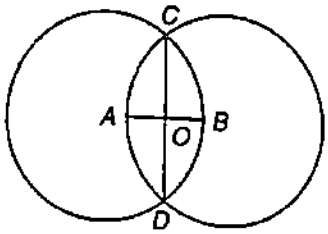
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

52. $AB = r$ (say)

then $AC = BC = r$, also

$\therefore$ $OA = OB = \frac{r}{2}$ (CD is a common chord)

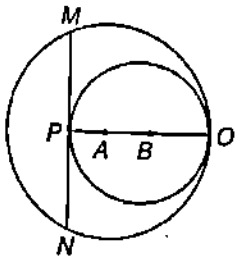


$$\therefore OC = \sqrt{(AC)^2 - (OA)^2} = \sqrt{r^2 - \left(\frac{r}{2}\right)^2} = \sqrt{\frac{3r^2}{4}} = \frac{\sqrt{3}}{2}r$$

$$\therefore \quad CD = 2CO = 2 \times \frac{\sqrt{3}}{2}r = \sqrt{3}r$$

$$\therefore \quad \frac{l(CD)}{l(AC)} = \frac{\sqrt{3}r}{r} = \frac{\sqrt{3}}{1}$$

53. Let A be the centre of larger circle and B be the centre of the smaller circle.



$OA = 3$ cm and $BO = 2$ cm

and $AB = (OA - OB) = 1$ cm

also $BP = 2$ cm

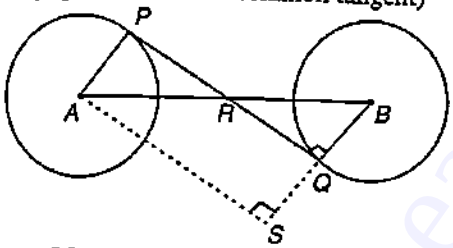
$\therefore$ $AP = 1$ cm

P is the point of contact of tangent MN

$\therefore$ $MP = \sqrt{AM^2 - AP^2} = \sqrt{3^2 - 1^2} = 2\sqrt{2}$ cm

$\therefore$ $MN = 2MP = 4\sqrt{2}$ cm

55. $PQ = AS$ (PQ is a transverse common tangent)



and $BS = BQ + SQ = BQ + AP$

$BS = 2BQ$ ($BQ = AP$ = radius)

$BS = 6$ cm

$\therefore$ $AS = \sqrt{(AB)^2 - (BS)^2} = \sqrt{(10)^2 - (6)^2} = 8$ cm

$\therefore$ $AS = 8$ cm

$\therefore$ $PQ = 8$ cm

Alternatively : The length of transverse common tangent

$$= \sqrt{(\text{distance beween the centres})^2 - (r_1 + r_2)^2}$$

$$= \sqrt{(10)^2 - (3+3)^2} = 8 \text{ cm}$$

57. Inradius of an incircle of right angled triangle

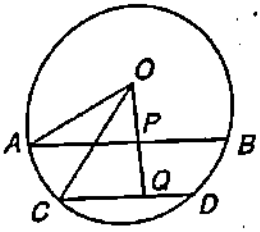
$$= \frac{\text{Base} + \text{Altitude} - \text{Hypotenuse}}{2}$$

$$= \frac{4+3-5}{2} = 1 \text{ cm}$$

Alternatively : Inradius $= \dfrac{\text{Area of triangle}}{\text{Simiperimeter of triangle}}$

$$= \frac{(3 \times 4)/2}{(3+4+5)/2} = \frac{6}{6} = 1 \text{ cm}$$

58. Let AB and CD be the two chords, then $AB = 8$ cm and $CD = 6$ cm.



$\therefore$ $AP = 4$ cm and $CQ = 3$ cm

and $PQ = 1$ cm (given)

Let $OP = x$

$\therefore$ $AO^2 = AP^2 + OP^2$

$\therefore$ $AO^2 = 4^2 + x^2 = 16 + x^2$ ...(1)

and $OC^2 = CQ^2 + OQ^2$

$= CQ^2 + (OP + PQ)^2$

$= (3)^2 + (x+1)^2$

$OC^2 = 9 + x^2 + 1 + 2x$ ...(2)

But $OA = OC$, are the radii of the same circle.

$\therefore$ $16 + x^2 = 9 + x^2 + 1 + 2x$

$\Rightarrow$ $x = 3$ cm

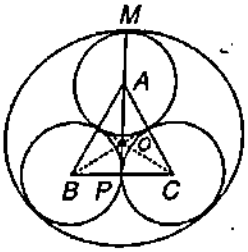
$\therefore$ $OA^2 = 16 + x^2 = 16 + 9 = 25$

$\therefore$ $OA = 5$ cm

$\therefore$ Diameter of the circle $= 2(OA) = 10$ cm

59. $AB = BC = AC = 2$ cm

($\because$ radius of each circle = 1 cm)



$\therefore$ $AP = \frac{\sqrt{3}}{2} \times 2$

$AP = \sqrt{3}$ cm

Let O be the centroid, then

$$OA = \frac{2}{3} \times \sqrt{3} = \frac{2}{\sqrt{3}} \text{ cm}$$

and $AM = 1$ cm

$\therefore$ $OM = \frac{2}{\sqrt{3}} + 1 = \frac{2+\sqrt{3}}{\sqrt{3}}$ cm

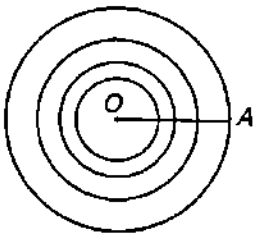
OM is the radius of the larger circle.

$\therefore$ Area of the circumscribing circle $= \pi R^2$

$$= \pi \times \left(\frac{2+\sqrt{3}}{\sqrt{3}}\right)^2$$

$$= \frac{\pi}{3}(2+\sqrt{3})^2$$

60. OA is the radius of the largest circle and O is the centroid of the all concentric circles.



Let r_1, r_2, r_3 and r_4 be the radii of the concentric circles in increasing order then

Total area of given circle

$= \pi \times (20)^2 = 400\pi$

Area of each region $= \frac{1}{4} \times 400\,\pi = 100\pi$

$\therefore$ Area of central region $= 100\pi = \pi r_1^2$

$\Rightarrow$ $r_1 = 10$ cm

Similarly, area of second region $= 100\pi = \pi (r_2^2 - r_1^2)$

$100\pi = \pi (r_2^2 - 100)$

$\Rightarrow$ $r_2 = 10\sqrt{2}$ cm

Again, area of third region $= 100\pi = \pi (r_3^2 - r_2^2)$

$100\pi = \pi (r_3^2 - 200)$

$\Rightarrow$ $r_3^2 = 10\sqrt{3}$ cm

$\therefore$ The required radii are 10, $10\sqrt{2}$ and $10\sqrt{3}$.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmanboob.wordpress.com

For More Visit : www.LearnEngineering.in

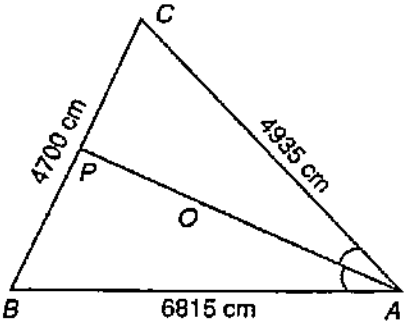
LEVEL 1

1. $\angle BAD = 180° - 130° = 50°$ ($\because$ □ $ABCD$ is cyclic)
and $\angle ADB = 90°$ (angle of semicircle)
$\therefore$ $\angle ABD = 180 - (90 + 50) = 40°$

2. $\angle ACB = 180 - (90 + 25) = 65°$ ($\because \angle ABC = 90°$)
But since $\angle ADB = \angle ACB$
$\therefore$ $\angle ADB = 65°$

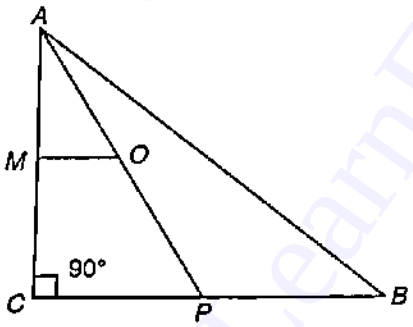
3. $\angle AOD = \angle ODC + \angle OCD$ ($\because \angle AOD$ is an external angle)
$= 52°$ ($\angle ODC = \angle OCD = 26°$)
($\because OD = OC$)

4. Note that $4700 : 4935 : 6815 = 20 : 21 : 29$



$\therefore$ ΔACB is right angled triangle and $\angle ACB$ is a right angle.
$\therefore$ The inradius of the right angled triangle

$$= \frac{20 + 21 - 29}{2} = 6 \text{ cm}$$



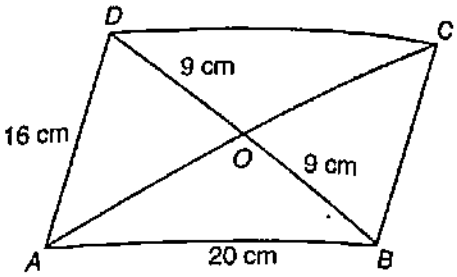
$\therefore$ $MO = MC = 6$ cm
$\therefore$ $AM = 21 - 6 = 15$ cm
$\therefore$ $\frac{AM}{MC} = \frac{AO}{OP} = \frac{15}{6} = \frac{5}{2}$
$\Rightarrow$ $\frac{PO}{AO} = \frac{2}{5}$

5. $CD = CP$ and $BE = BP$, since tangent from the same point are always same.
$\therefore$ $AE = AD$
Now, $AD = AC + CD = AC + CP$
and $AE = AB + BE = AB + BP$
$\therefore$ $AE + AD = AC + CP + AB + BP$
$2AE = AC + AB + (CP + BP)$
$2AE = AB + AC + BC$, Hence (d)

6. Inradius of a right angled triangle $= \frac{AC + BC - AB}{2}$

$$= \frac{9 + 40 - 41}{2} = 4 \text{ cm}$$

7. Let $AD = 16$ cm



then $AB = 20$ cm and $BD = 18$ cm
$\therefore$ $DO = BO = 9$ cm

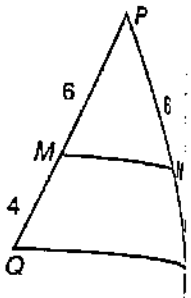
NOTE In a parallelogram the two diagonal bisect each other, therefore by Appollonious theorem

$\therefore$ $16^2 + 20^2 = 2(9^2 + AO^2)$
$\Rightarrow$ $AO = \sqrt{247}$
$\therefore$ $AC = 2\sqrt{247}$ cm

8. Go back to the basics

9. It is clear that

$$\frac{PM}{MQ} = \frac{PN}{NR} = \frac{6}{4} = \frac{3}{2}$$

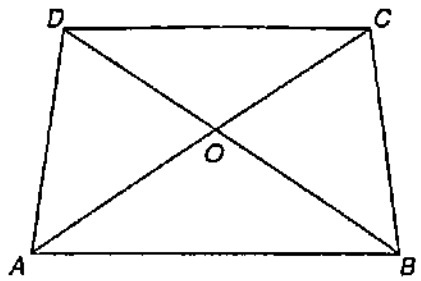


Therefore ΔPMN and ΔPQR are similar
Now Ratio of areas of $\Delta PMN : \Delta PQR$

$= 3 \times 3 : 5 \times 5 = 9 : 25$

$\therefore$ Percentage of area of ΔPMN over

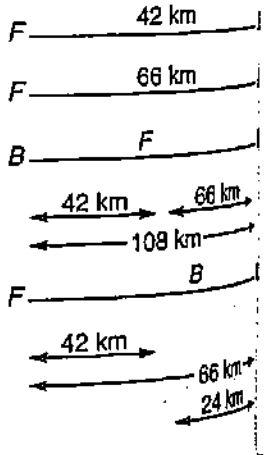
$$\Delta PQR = \frac{9}{25} \times 100 = 36\%$$



10. Since, ΔAOB is similar to ΔCOD.

$\therefore$ Ratio of areas of $\frac{\Delta AOB}{\Delta COD} = \left(\frac{3}{1}\right)^2 = \frac{9}{1}$

11. The maximum possible distance between Barabanki and Lucknow is 108 km and minimum possible distance between Barabanki and Lucknow is 24 km. Hence (d)



For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

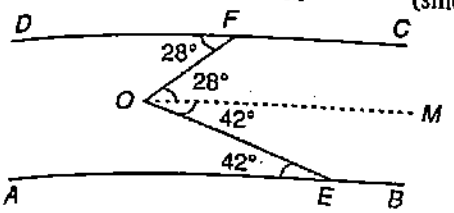
12. Since $CS = SD$

$\therefore$ The two chords must be equidistant from the centre O.

Thus, the required ratio is 1 : 1

13. $\angle DFO = \angle FOM$

and $\angle AEO = \angle EOM$ (since $CD \parallel AB$)



$\therefore$ $\angle FOE = (28° + 42°) = 70°$

14. $\angle AOB = 60°$

and $\angle AOD = \angle ADO = \dfrac{(180° - 30°)}{2} = 75°$

Also $\angle BOC = 75°$

$\therefore$ $\angle COD = 360° - (75° + 60° + 75°) = 150°$

15. $\angle ABD = \angle ABC - \angle DBC$

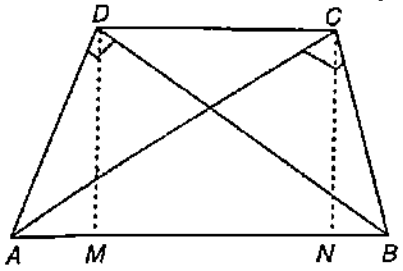
$= \angle ABC - \angle BDC$

$= \angle ABC - (\angle ABD + \angle BAD)$

$\therefore$ $2(\angle ABD) = \angle ABC - \angle BAD = 30°$

$\therefore$ $\angle ABD = 15°$ ($\because \angle BAD = \angle BAC$)

16. $BD = AC = 20$ cm (by Pythagorus theorem)



$\therefore$ $DM = \dfrac{AD \times BD}{AB}$

$\therefore$ $DM = \dfrac{15 \times 20}{25} = 12$ cm, also $CN = DM$

$\therefore$ $AM = \dfrac{AD^2}{AB}$

$\therefore$ $AM = \dfrac{15 \times 15}{25} = 9$ cm, also $AM = BN$

$\therefore$ $MN = AB - (AM + BN)$

$= 25 - (18) = 7$ cm

but $MN = CD = 7$ cm

$\therefore$ Area of trapezium $= \dfrac{1}{2} \times (CD + AB) \times DM$

$= \dfrac{1}{2} \times 32 \times 12 = 192\ \text{cm}^2$

17. $(OS)^2 = (OK)^2 + (KS)^2$

$25 = OK^2 + 16 \Rightarrow OK = 3$

and $(OS)^2 = (OL)^2 + (LN)^2$

$25 = (OL)^2 + 9$

$\Rightarrow$ $OL = 4$ cm

$\therefore$ $KL = OL - OK = 1$ cm

$\therefore$ Area of rectangle $= 1 \times 10 = 10\ \text{cm}^2$

18. $\angle ACB = 90°$

$\therefore$ $AC = 12$ cm

Now, since CO is perpendicular on AB

$\therefore$ $CO = \dfrac{AC \times BC}{AB} = \dfrac{16 \times 12}{20} = 9.6$ cm

19. $\angle ADB = \angle ACB = 41°$

(angles of the same segment)

$\therefore$ $\angle ABD = 180 - (\angle ADB + 90) = 49°$ ($\angle DAB = 90°$)

20. Go through options for quicker answer

Exterior angle $= \dfrac{360}{15} = 24°$ (for $n = 15$)

$\therefore$ Interior angle $= 180 - 24° = 156°$

$\therefore$ Interior – Exterior $= 156 - 24 = 132°$

Hence, option (b) is correct.

21. $OC = OD$ and $OA = OP = OB$

$OP = 1$ m

$\therefore$ $PC = 1$ m

$OC = \sqrt{2}$ m

$\therefore$ $AC = OC - OA$

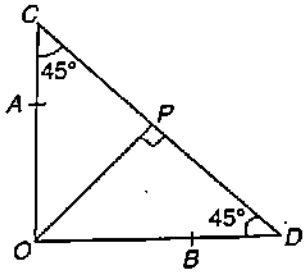
$= (\sqrt{2} - 1)$ m

and $AC + CP = (\sqrt{2} - 1) + 1$

$= \sqrt{2}$ m

$= 1.414$ m

$= 141.4$ cm



22. $AB : BC : CA = \dfrac{1}{CD} : \dfrac{1}{AE} : \dfrac{1}{BF} = \dfrac{1}{2} : \dfrac{1}{3} : \dfrac{1}{4} = 6 : 4 : 3$

23. ΔPOQ and ΔROS are similar

$\therefore$ $PS = QR$

Option a, b and d are irrelevent.

24. $\dfrac{DQ}{BD} = \sin 60° \Rightarrow DQ = BD \times \dfrac{\sqrt{3}}{2} = 3$ m

and $\dfrac{DP}{DF} = \sin 30° = \dfrac{1}{2}$

$\therefore$ $DP = 2$ m ($\because DF = 4$ m)

$\therefore$ $PQ = 3 - 2 = 1$ m

Let $AB = CQ = x$ m

$\therefore$ $\dfrac{CD}{EF} = \dfrac{5}{4}$

$\dfrac{x + 3}{x + 1} = \dfrac{5}{4} \Rightarrow x = 7$ m.

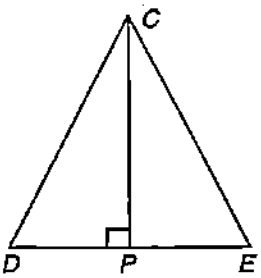
25. Area of equilateral triangle

$ABC = \dfrac{\sqrt{3}}{4} \times (6)^2 = 9\sqrt{3}\ \text{cm}^2$

Area of $\Delta ADE = \dfrac{1}{2} \times DE \times CP$

$= \dfrac{1}{2} \times 2 \times \dfrac{\sqrt{3}}{2} \times 6$

$= 3\sqrt{3}\ \text{cm}^2$



$\therefore$ Area of shaded region $= 9\sqrt{3} - 3\sqrt{3} = 6\sqrt{3}\ \text{cm}^2$

26. $CD = CP^2 + DP^2 = (3\sqrt{3})^2 + (1)^2 = 2\sqrt{7}$ cm

and $CE = CD = 2\sqrt{7}$ and $DE = 2$ cm

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmanboob.wordpress.com
 For More Visit : www.LearnEngineering.in

$\therefore$ Perimeter of $\Delta CDE = 2(2\sqrt{7}) + 2$
$= 2(2\sqrt{7} + 1)$ cm

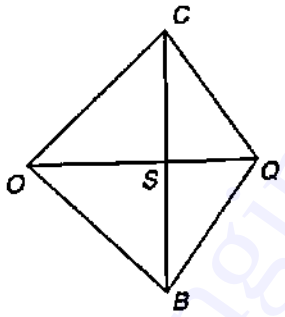
27. $\angle ABC = 180 - (65 + 75) = 40°$
$\angle ORB = \angle OQB = 90°$
$\therefore \quad \angle ROQ = 360 - (90 + 90 - 40)$
$\therefore \quad \angle ROQ = 140°$

28. $\angle BAP + \angle BCP = 180°$
($\square ABCP$ is a cyclic quadrilateral)
$\therefore \quad \angle BCP = 180 - 60° = 120°$

29. Since, we don't have requisite information.

30. $\because \quad AT = BT,$
(∵ Tangents on the same circle from a fixed point is equal)
$AC = PC \quad \text{and} \quad BD = PD$
$\therefore \quad AT = BT$
$\Rightarrow \quad AC + CT = BD + DT$
$PC + CT = PD + DT$
Similarly, all the relations can be verified.

31. $OQ = OB = OC = r$ (say)
$\angle AOD = \angle BOC = 120°$
$\therefore \quad \angle BOQ = \angle COQ = 60°$
$\therefore \quad \frac{SB}{OB} = \sin 60° = \frac{\sqrt{3}}{2}$
$\Rightarrow \quad SB = \frac{r\sqrt{3}}{2}$
$\therefore \quad BC = 2SB = r\sqrt{3}$



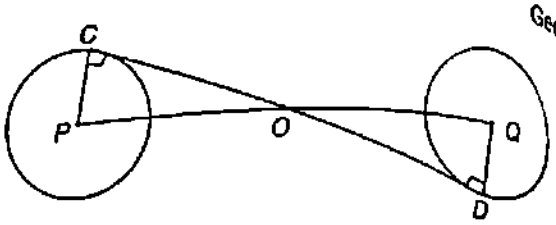
$\therefore$ Area of quadrilateral $BQCO$
$= \frac{1}{2} \times BC \times OQ$
$= \frac{1}{2} \times r\sqrt{3} \times r = \frac{r^2\sqrt{3}}{2}$ cm^2
$\therefore$ Area of both the quadrilateral $= 2\left(\frac{r^2\sqrt{3}}{2}\right) = r^2\sqrt{3}$ cm^2

32. Go back to the basics.

33. $\frac{(A)\,\Delta CMN}{(A)\,\square ABNM} = \frac{1}{2}$
$\therefore \quad \frac{(A)\,\Delta CMN}{(A)\,\Delta CAB} = \frac{1}{3}$
$\Rightarrow \quad \frac{MN}{AB} = \frac{CM}{CA} = \frac{1}{\sqrt{3}}$
$\Rightarrow \quad \frac{CM}{MA} = \frac{1}{\sqrt{3}-1} = \frac{\sqrt{3}+1}{2} \qquad MA = (CA - CM)$

34. $\angle ADE = \angle AED = 80°$
$\therefore \quad \angle BDE = \angle CED = 100°$
$\therefore \quad \angle DBC + \angle BCE = 360° - (100° + 100°) = 160°$

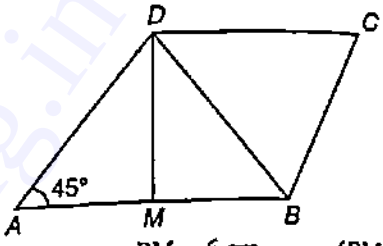
35. $AC = \sqrt{(15)^2 + (20)^2} = 25$ cm
$CD = \sqrt{(25)^2 - (7)^2} = 24$ cm

36.


$AB = PQ = 26$ cm
and $PO = OQ = 13$ cm
$CO = \sqrt{(PO)^2 - (PC)^2}$
$CO = \sqrt{(13)^2 - (5)^2}$
$CO = 12$ cm
$\therefore \quad CD = 2CO = 24$ cm

Alternatively: Solve by using the formula of tangent

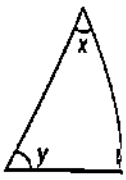
37. $\angle DAM = 45°$
$DM = 8$ cm
and
and $BD = 10$ cm



$\therefore \quad BM = 6$ cm $\qquad (BM = \sqrt{BD^2 - DM^2})$
and $\frac{AM}{DM} = \tan 45° = 1$
$\therefore \quad AM = 8$ cm
and $AB = AM + MB = 8 + 6 = 14$ cm
$\therefore$ Area of parallelogram $ABCD = AB \times MD$
$= 14 \times 8 = 112$ cm^2

38. $\angle ACB = 50° \qquad (\because \angle A + \angle B + \angle C = 18$
$\angle CFO = \angle CEO = 90°$
$\therefore \quad \angle FOE = 360° - (90° + 90° + 50°) = 130°$
but $\angle AOB = \angle FOE = 130°$

39. $35x = 26y \Rightarrow \frac{x}{y} = \frac{26k}{35k}$



$\therefore \quad x + y = 61k$
if $x = 1$, then $z = 119°$
But $\quad x < z < y$
again consider $k = 2$, then
$x = 52° \quad \text{and} \quad y = 70°$
$\therefore \quad z = 180° - (70° + 52°) = 58°$

40. $\angle DAB = \angle BDQ = 48°$
$\therefore \quad \angle DBA = 180° - (90° + 48°) = 42°$ (∵ $\angle ADB =$
$\angle DCB = 180° - \angle DAB = 132°$
$\therefore \quad \frac{\angle DBA}{\angle DCB} = \frac{42}{132} = \frac{7}{22}$

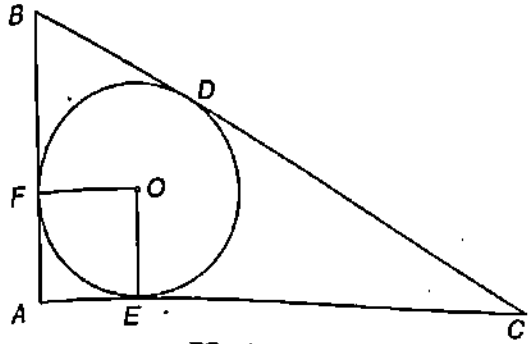
41. $\angle OCD = 90°$
$\angle OAC = \angle OCA = 30°$
$\angle ACD = \angle ACO + \angle OCD = 30° + 90° = 120°$
$\therefore \quad \angle BAC = 180° - 120° = 60°$
$\Rightarrow \quad \angle BCD = 60° \qquad (\angle BCD =$
$\Rightarrow \quad \angle OCB = \angle OCD - \angle BCD$
$= 90° - 60° = 30°$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
 For More Visit : www.LearnEngineering.in

42. $OE = OF$ (inradius)

$BF = BD$



and $EC = DC = 15$ cm

Let $BF = x$ cm

then $AB = (6 + x)$ cm

and $AC = 6 + 15 = 21$ cm

$\therefore \quad (BC)^2 = (AB)^2 + (BC)^2$

$(x + 15)^2 = (6 + x)^2 + (21)^2 \quad (\because BC = BD + CD)$

$\Rightarrow \quad x = 14$ cm

$\Rightarrow \quad BD = 14$ cm

$\therefore \quad CD - BD = 15 - 14 = 1$ cm

43. $(\angle BPQ + \angle BQP) + (\angle EQR + \angle ERQ)$

$+ (\angle CRS + \angle CSR) + (\angle TSA + \angle ATS)$

$+ (\angle DTP + \angle DPT) = 2 \times 360$

$\angle A + \angle D + \angle B + \angle E + \angle C = 5 \times 180° - 2 \times 360° = 180°$

44. $\dfrac{\text{Area of } \Delta DAE}{\text{Area of } \Delta DEC} = \dfrac{\frac{1}{2} \times DE \times AE}{\frac{1}{2} \times DE \times CE}$

$= \dfrac{AE}{CE} = \dfrac{(AD)^2}{(DC)^2} = \left(\dfrac{6}{8}\right)^2 = \dfrac{9}{16}$

Similarly, in ΔABC,

$\dfrac{\text{Area of } \Delta BCF}{\text{Area of } \Delta BFA} = \dfrac{9}{16}$

$\therefore$ The area of shaded to unshaded region $= \dfrac{16}{9}$

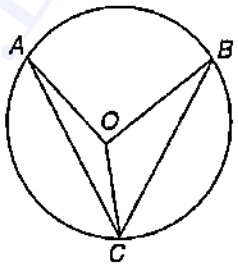
45. $\angle OCA = \angle OAC = 25°$

and $\angle OCB = \angle OBC = 35°$

$\therefore \quad \angle ACB = \angle ACO + \angle BCO$

$= 25° + 35° = 60°$

$\therefore \quad \angle AOB = 2\angle ACB = 120°$



46. $\angle DOA = 110°$

and $\angle ADB = 50°$

$\therefore \quad \angle DAO = 180 - (110° + 50°) = 20°$

47. ΔABC is similar to ΔEDC

$\therefore \quad \dfrac{AB}{ED} = \dfrac{BC}{DC} = \dfrac{AC}{EC}$

$\therefore \quad \dfrac{AB}{DE} = \dfrac{BC}{DC} \Rightarrow \dfrac{24}{10} = \dfrac{60}{DC}$

$\Rightarrow \quad DC = 25$ cm

48. $AC = \sqrt{(24)^2 + (60)^2} \quad (\because AC^2 = AB^2 + BC^2)$

$AC = 12\sqrt{29}$

$\therefore \quad \dfrac{BC}{DC} = \dfrac{AC}{EC}$

$\Rightarrow \quad \dfrac{60}{25} = \dfrac{12\sqrt{29}}{EC}$

$\Rightarrow \quad EC = 5\sqrt{29}$

$\therefore \quad BE = BC - EC = 60 - 5\sqrt{29}$

$\Rightarrow \quad BE = 5(12 - \sqrt{29})$

$\therefore \quad \dfrac{CE}{BE} = \dfrac{5\sqrt{29}}{5(12 - \sqrt{29})} = \dfrac{\sqrt{29}}{(12 - \sqrt{29})}$

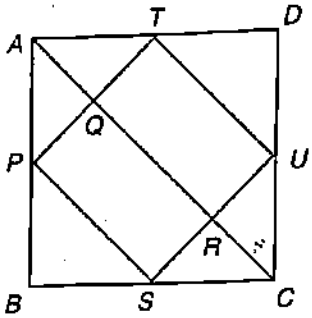
49. $PTUS$ is a square inscribed by a square $ABCD$.

Let each side of the square $ABCD$ be 'a', then

Area of square $ABCD = a^2$

also $PU = ST = a$

$\therefore$ Area of square $PTUS = \dfrac{a^2}{2}$



$\therefore \quad \dfrac{\text{Area of } PTUS}{\text{Area of } ABCD} = \dfrac{a^2/2}{a^2} = \dfrac{1}{2}$

$\therefore \quad \dfrac{\text{Area of } \square PQRS}{\text{Area of } \Delta ABC} = \dfrac{1}{2}$

50. Go through options. It is pentagon

Alternatively : ${}^nC_2 - n = n$

$\Rightarrow \quad {}^nC_2 = 2n$

$\Rightarrow \quad \dfrac{n(n-1)}{2} = 2n$

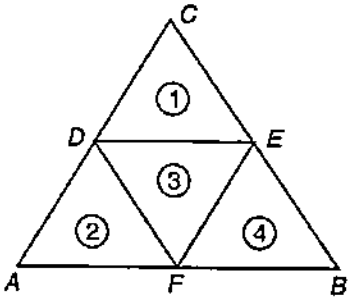
$\Rightarrow \quad n = 5$

51. Let $P_1, P_2, P_3, P_4, \ldots, P_9$ be the vertices of the regular nonagon then, we can select two vertices out of 9 vertices in ${}^9C_2$ ways and ${}^9C_2 = 36$ ways and number of equilateral triangles formed $= 36 \times 2 = 72$

But the three triangles $\{P_1, P_4, P_7\}$, $\{P_2, P_5, P_8\}$ and $\{P_3, P_6, P_9\}$ are each counted 3 times *i.e.*, counted as 9 triangles instead by 3 triangles.

So, net number of equilateral triangles $= 72 - 6 = 66$.

52. There are 4 triangles (smaller) are congruent.



So, out of these 4 triangles 2 triangles are taken thus the ratio of the shaded to the unshaded region is 1 : 1 (since two triangles are shaded and 2 are unshaded).

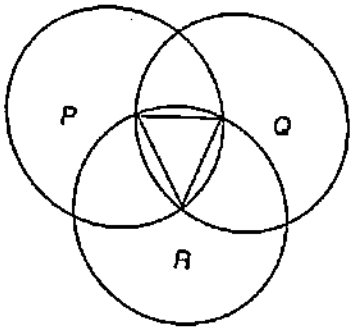
53. ΔPQR is an equilateral triangle *i.e.*, all the three angles are 60° each.

$\therefore \quad PQ = QR = PR$

For More Visit : www.LearnEngineering.in

Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in



Now, $\text{arc } PQ = 2\pi \times (1) \times \frac{60}{360} = \frac{\pi}{3}$

$\therefore$ Sum of all three arcs $= 3 \times \frac{\pi}{3} = \pi$

LEVEL 2

1. $BD = 53$ cm

$\therefore$ $AD = CD = BD = 53$ cm

$\therefore$ $AC = 2 \times 53 = 106$ cm

$AB + BC + AC = 2 \times 126$ cm

$= 252$ cm

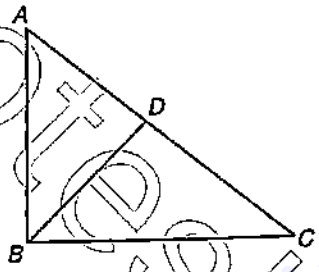
$\therefore$ $AB + BC = 146$ cm

Let $AB = x$ cm

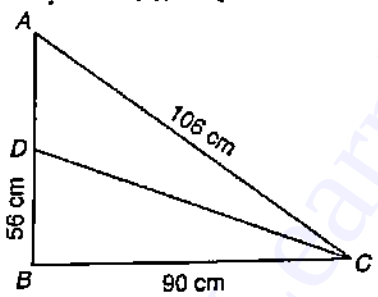
$\therefore$ $BC = (146 - x)$ cm

$\therefore$ $AB^2 + BC^2 = AC^2$

$x^2 + (146 - x)^2 = (106)^2$ ...(1)



Solving the equation (1), we get $x = 56$ and $x = 90$



Consider $AB = 56$ cm

then $BC = 90$ cm

Longest median will fall on the shortest side.

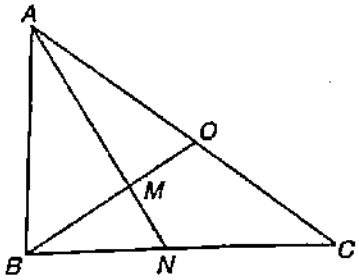
Now, the area of $\Delta BCD = \frac{1}{2} \times BD \times BC$

$= \frac{1}{2} \times 28 \times 90 = 1260 \text{ cm}^2$

2. Let $AB = BC = a$

then $AC = \sqrt{2}a$

$\therefore$ $AO = OC = BO = \frac{\sqrt{2}a}{2} = \frac{a}{\sqrt{2}}$



54. $\angle A = 90°$ and $\angle B = \angle C = 45°$

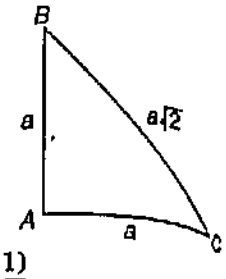
Inradius $\frac{(a + a) - a\sqrt{2}}{2} = \frac{a(2 - \sqrt{2})}{2}$

and Circumradius $= \frac{a\sqrt{2}}{2}$

$\therefore$ $\frac{\text{Inradius}}{\text{Circumradius}} = \frac{a(2 - \sqrt{2})/2}{(a\sqrt{2})/2}$

$= \frac{2 - \sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}(\sqrt{2} - 1)}{\sqrt{2}}$

$= \frac{(\sqrt{2} - 1)}{1}$



Now, by angle bisector theorem

$\frac{AB}{AO} = \frac{BM}{MO} \Rightarrow \frac{BM}{MO} = \frac{a}{a/\sqrt{2}} = \frac{\sqrt{2}}{1}$

$\therefore$ $MO = 20$ cm

$\therefore$ $BM = 20\sqrt{2}$ cm

$\therefore$ $BO = 20 + 20\sqrt{2} = 20(1 + \sqrt{2})$ cm

Now, since $BO = \frac{a}{\sqrt{2}} = \frac{AB}{\sqrt{2}}$

$\therefore$ $AB = \sqrt{2}(BO) = 1.414\ [20(1 + 1.414)]$

$= 68.2679 = 68.27$ cm

3. $\angle A + \angle B = 90°$

$\angle A - \angle B$

$89 - 1 = 88$

$88 - 2 = 86$

$87 - 3 = 84$

... ...

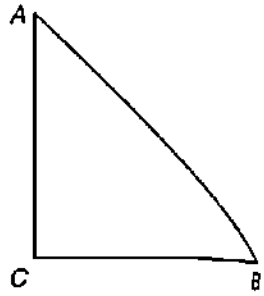
$45 - 45 = 0$

$44 - 46 = -2$

... ...

$1 - 89 = -88$

Thus k can assume total $44 + 1 + 44 = 89$ values

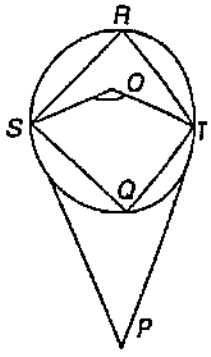


4. $\angle SPT$ and $\angle SOT$ are supplementary, then

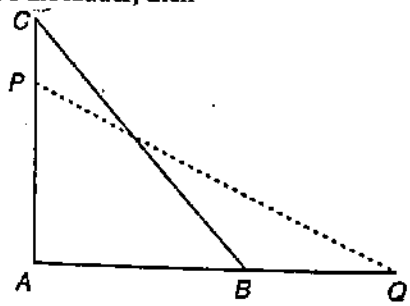
$\angle SOT = 180° - 50° = 130°$

$\therefore$ $\angle SRT = \frac{1}{2}(\angle SOT) = 65°$

$\therefore$ $\angle SQT = 180 - 65° = 115°$



5. Let BC be the ladder, then



For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$BC = 6.5$ m and $AB = 5.2$ m

$AC = \sqrt{(BC)^2 - (AB)^2}$

$\therefore \quad AC = 3.9$ m

$PQ^2 = PA^2 + AQ^2$

Now, $(6.5)^2 = (2.5)^2 + AQ^2$

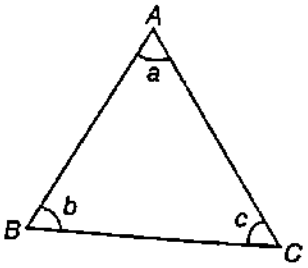
$AQ = 6$ m

$\Rightarrow \quad BQ = AQ - AB = 6 - 5.2 = 0.8$ m

$\therefore$ The foot of the ladder will slip by 0.8 m

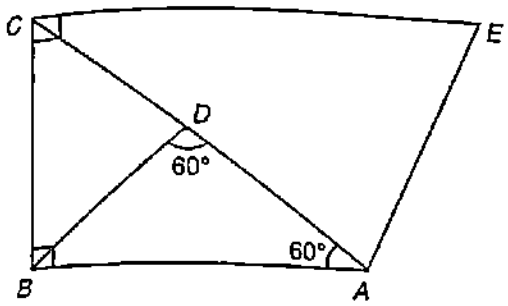
6. $\angle A + \angle B + \angle C = 180$

Any one of angle can posses the values from 1 to 178.



7. Cannot be determined.

8. ΔABC is a right angled



and $\angle ABC = 90°$

Let $AB = x$

then $AB = AD = CD = BD = x$

$\therefore \Delta ABD$ is equilateral triangle

$\therefore \quad \angle CAE = 60°$

$\therefore \quad \angle BCA = 30°$

$\Rightarrow \quad \angle ACE = 60°$

$\therefore \quad \angle CEA = 60°$, also

Hence, ΔACE is an equilateral triangle

Thus, $AC = AE = CE = 2x$

and $\dfrac{BC}{AB} = \tan 60° = \sqrt{3}$

$\Rightarrow \quad BC = AB\sqrt{3} = x\sqrt{3}$

$\therefore \quad \dfrac{BC}{AE} = \dfrac{x\sqrt{3}}{2x} = \dfrac{\sqrt{3}}{2}$

9. $\pi r^2 = 3\pi \Rightarrow r = \sqrt{3}$

$\therefore \quad DE = 2r^2 - 2r^2 \cos 120°$

$DE = r^2 \quad (\because \sqrt{3} = r)$

But $AB = 2DE$

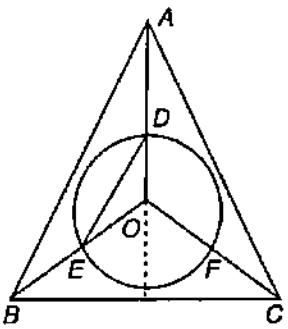
(By similarly of triangles)

$\therefore \quad AB = 2r^2 = 2 \times (\sqrt{3})^2 = 6$

($\because D$ and E are the mid-points of OA and OB)

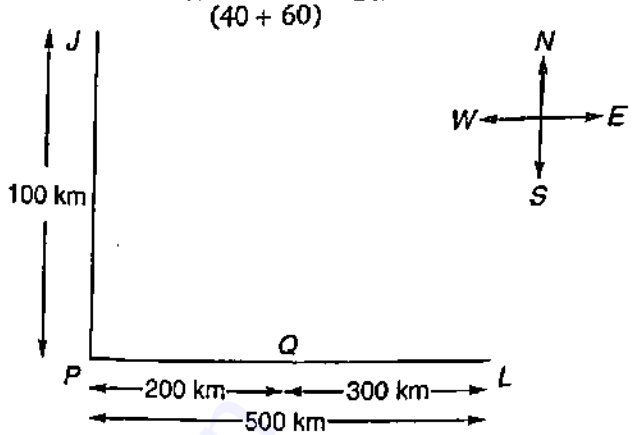
$\therefore$ Perimeter of triangle

$ABC = 3 \times 6 = 18$ unit



10. Time taken in the collision the two trains

$= \dfrac{500}{(40 + 60)} = 5$ h



In 5 hours, plane will cover $5 \times 200 = 1000$ km distance

11. Two trains meet with accident at a place $200 (= 40 \times 5)$ km away from Patiyala.

$\therefore$ The required distance = 200 km

12. Area of $\Delta BDE = \dfrac{1}{2} \times \dfrac{2}{5} AB \times \dfrac{2}{7} BC$

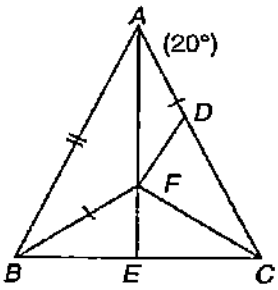
$= \dfrac{4}{35} \times \dfrac{1}{2} (AB \times BC)$

$= \dfrac{4}{35}$ area of ΔABC

$\therefore \quad$ Area of $\Delta ABC = \dfrac{35}{4} \times 20 = 175$ cm^2

13. Let E be on BC and $BE = EC$

Let F be on AE so that triangle FBC is equilateral.



$\angle DAB = \angle ABF = 20°$

and $DA = BF$

$\therefore$ Trapezoid $ADFB$ is isosceles.

$\angle FAD = \angle DBF = 10°$

$\therefore \quad \angle DBC = 10 + 60 = 70°$

14. Calculate them physically (or manually)

15. $AB = 6$ cm, $\angle C = 60°$

and $\angle A = \angle B = 60°$

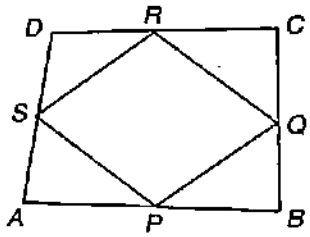
$\therefore \Delta ABC$ is an equilateral triangle

$\therefore \quad$ Area of $\Delta ABC = \dfrac{\sqrt{3}}{4} \times (6)^2 = 9\sqrt{3}$

Area of $(\Delta ADE + \Delta BFG) = 2 \times \dfrac{\sqrt{3}}{4} \times (2)^2 = 2\sqrt{3}$

$\therefore \quad$ Area of pentagon $= 9\sqrt{3} - 2\sqrt{3} = 7\sqrt{3}$ cm^2

16. Since, $PQRS$ is a parallelogram



$\therefore \quad \angle PSR = 90° \quad (\angle PSR + \angle PQR = 180°)$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

17. Best way in to go through option

$r > 1$ and $r \in I$

Let us assume $r = 2$ $(r \to k)$

$w = a$

$x = ar$

$y = ar^2$

and $z = ar^3$

$\therefore \quad ar^3 - a = a(r^3 - 1) = 168$

$\Rightarrow \quad a(2^3 - 1) = 168$

$\Rightarrow \quad a = 24$

Note only option (a) gives a value (168) which is divisible by 7. Now,

$a = 24, \quad ar = 48, \quad ar^2 = 96$ and $ar^3 = 192$

This value satisfies all the required conditions hence it is correct.

NOTE For other values of r and other options are not being matched.

18. $\angle ROQ = 180 - 50 = 130°$

$(\because \angle OQP + \angle ORP + \angle QOR + \angle QPR = 360°$ and $\angle OQP = \angle ORP = 90°)$

Now, since $RT = TM$ and $QS = SM$

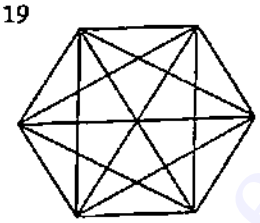
$\therefore$ also $OR = OM = OQ$

$\therefore \quad \angle ROT = \angle TOM$ and $\angle MOS = \angle SOQ$

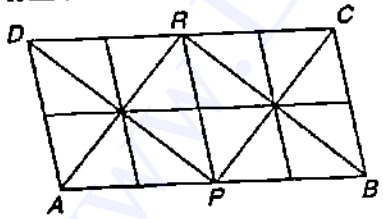
$\therefore \quad \angle SOT = \frac{1}{2}\angle ROQ$

$\therefore \quad \angle SOT = \frac{130}{2} = 65°$

19. $12 + 6 + 1 = 19$



20. There are total 16 similar triangles each with equal area.



Here, 4 out of 16 triangles are taken. So the number of shaded triangles = 4 and number of unshaded triangles = 12

$\therefore \quad$ Required ratio $= \frac{1}{3}$

21. Number of total rectangles $= {}^4C_2 \times {}^3C_2 = 6 \times 3 = 18$

22. $\angle PDB = \angle QEA = 80°$

$\therefore \quad \angle PED = \angle QDE = 10°$ $(\because \angle DPE = \angle DQE = 90°)$

$\therefore \quad \angle DRE = 180 - (10 + 10) = 160°$

$\therefore \quad \angle PRD = 180 - \angle DRE = 20°.$

23. $OC = \frac{AB}{2}$

$AO = OC = OB$

$\therefore \quad \angle OAC = \angle OCA$

$\therefore \quad = \angle BCO = \angle OBC = 45°$

$\angle ACB = 90°$

$\therefore$ Notice that all the given triangles are equilateral

24. Area of shaded region $= 3\left[\pi r^2 \frac{60}{360} - \frac{\sqrt{3}}{4} \times r^2\right]$

$= \frac{r^2}{2}\left[\pi - \frac{3\sqrt{3}}{2}\right]$

26. Notice $\angle ORP = 90°$ ($\because$ OP is a diameter of smaller circle)

$OS = 5$ cm and $OR = 4$ cm

$\therefore \quad SR = \sqrt{(5)^2 - (4)^2} = 3$ cm

$\therefore \quad SP = 2(SR) = 6$ cm

(Since, OR passes through centre O and perpendicular to SP, therefore OR bisects SP.)

27. $\frac{AD}{AB} = \frac{DO}{BO} = 1$

$\Rightarrow \quad OB = OD = 8$ cm

$\because$ $ABCD$ is a cyclic quadrilateral

$\therefore \quad DO \times BO = CO \times AO$

$8 \times 8 = 4 \times AO$

$\Rightarrow \quad AO = 16$ cm

$\therefore \quad AC = 16 + 4 = 20$ cm

28. At $\angle A = 60°$, $BC = b = c$

and at $\angle A = 90°$, $BC = \sqrt{2}\, b = \sqrt{2}c$

$\therefore \quad 60° < A < 90°, \quad BC = c < a < c\sqrt{2}$

29. $OR^2 = (OQ)^2 + (RQ)^2$ $RQ = 2(OQ)$

$OR^2 = 5OQ^2$

$\Rightarrow \quad$ radius $(r) = OQ\sqrt{5}$

$\therefore \quad OQ = \frac{r}{\sqrt{5}}$

Again, $OC^2 = OH^2 + HC^2$

$r^2 = (OQ + QH)^2 + (QH)^2$ $(\because HQ = HC)$

$r^2 = \left(\frac{r}{\sqrt{5}} + QH\right)^2 + (QH)^2$

$\Rightarrow \quad QH = \frac{r}{\sqrt{5}}$

$\therefore \quad HC = \frac{r}{\sqrt{5}} = \frac{RQ}{2}$

$RC = \sqrt{(RD)^2 + (DC)^2}$

$= \sqrt{\left(\frac{r}{\sqrt{5}}\right)^2 + \left(\frac{r}{\sqrt{5}}\right)^2} = r\sqrt{\frac{2}{5}}$

$\therefore \quad RC + FS = 2r\sqrt{\frac{2}{5}}$

30. Best way is to consider some values and verify the results.

31. $\angle OCT = 90°, \quad \angle DCT = 45°$

$\therefore \quad \angle OCB = 45°$

$\therefore \quad \angle COB = 45°$ (ΔBOC is a right angled triangle)

$\therefore \quad \angle AOC = 180° - 45° = 135°$

Now, $\because \quad CD = 10 \Rightarrow BC = 5$ cm $= OB$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

$OC = 5\sqrt{2}$ cm $= OA$

$\Rightarrow$ Again, $AC^2 = OA^2 + OC^2 - 2OA \cdot OC \cos 135°$

$= 2(OA)^2 - 2(OA)^2 . \cos 135°$

$= 2(5\sqrt{2})^2 - 2(5\sqrt{2})^2 \times \left(-\frac{1}{\sqrt{2}}\right)$

$= 100 + \frac{100}{\sqrt{2}}$

$AC^2 \approx 170.70$

$AC \approx 13$ cm

$\Rightarrow$ $\therefore$ Perimeter of $\Delta OAC = OA + OC + AC$

$= 5\sqrt{2} + 5\sqrt{2} + 13 = 27$ cm

32. $\angle ACB = 60°$ ($\because \angle ACB + \angle ADB = 180°$)

and $\angle CAB = 30°$ ($\because \angle ACB + \angle CAB = 90°$)

$AC = 2 \times 6 = 12\,cm^2$

$\frac{BC}{AC} = \sin 30° = \frac{1}{2}$

$BC = 6$ cm

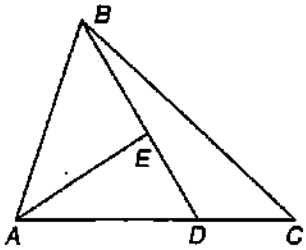
$\Rightarrow$ and $\frac{BC}{AB} = \tan 30° = \frac{1}{\sqrt{3}}$

$AB = 6\sqrt{3}$ cm

$\Rightarrow$ $\therefore$ Area of $\Delta ABC = \frac{1}{2} \times 6 \times 6\sqrt{3} = 18\sqrt{3}$ cm

33. Area of $\Delta BAE = \frac{1}{4} AC \times \frac{1}{3} BD$

$= \frac{1}{12}$ Area of ΔABC



34. $\frac{AE}{EC} = \frac{AB}{BC} = \frac{7.5}{10.5} = \frac{5}{7}$

Now, $AB^2 + BC^2 = AC^2$

$(5k)^2 + (7k)^2 = (18)^2$

$74k^2 = 324 \Rightarrow k^2 = \frac{324}{74}$

Area of $\Delta ABC = \frac{1}{2} \times AB \times BC = \frac{1}{2} \times 5k \times 7k$

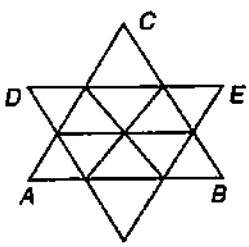
$= \frac{35}{2} k^2 = \frac{35}{2} \times \frac{324}{74}$

$= 76.621$ cm

35. There are total 12 similar triangles each with equal area. But a larger triangle ABC (or DEF) has only 9 smaller triangles. Out of 9 triangles only 6 triangles are common.

$\therefore$ Area of common region

$= \frac{6}{9} \times 198 = 132\ cm^2$



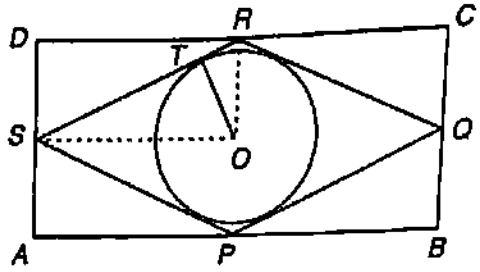
36. $9 \times 180 - 2 \times 360$

$= 180 \times 5 = 900°$

$\because \begin{Bmatrix} n \times 180 - 2 \times 360 \\ = 180\,(n-4) \end{Bmatrix}$

37. $DS = \frac{AD}{2} = 6$ cm

and $DR = \frac{DC}{2} = 8$ cm $= OS$



$\therefore SR = 10$ cm and $OR = 6$ cm

$\therefore$ Area of $\Delta ORS = \frac{OS \times OR}{2} = \frac{SR \times OT}{2}$

$\frac{8 \times 6}{2} = \frac{10 \times OT}{2}$

$\Rightarrow$ $OT = \frac{48}{10}$ cm

$\therefore$ Area of circle $= \pi r^2 = \pi \times \left(\frac{48}{10}\right)^2 = \frac{576}{25}\pi$

38. $200 = 2^3 \times 5^2$

Number of total factors $= (3+1) \times (2+1) = 12$

$\therefore$ Total number of required rectangles $= \frac{12}{2} = 6$

Area $= b \times l$

$200 = 1 \times 200 = 2 \times 100$

$= 4 \times 50 = 5 \times 40$

$= 8 \times 25 = 10 \times 20$

39. Maximum probable number of circles $= {}^8C_3$

(Since, a circle can pass through any three non-collinear points)

But since 4 points are lie on the same circle so which reduces the formation of some circles.

$\therefore$ Actual number of circles $= {}^8C_3 - {}^4C_3 + 1$

$= 56 - 4 + 1 = 53$

NOTE Since when we subtract ${}^4C_3$, then actually we reduce one actually possible circle, so we have to add up 1.

40. Here, AC and BC are the secants of the circle and AB is tangent at D

$\therefore$ $AE \times AC = AD^2$

$AE \times 4 = (3)^2 \Rightarrow AE = \frac{9}{4}$

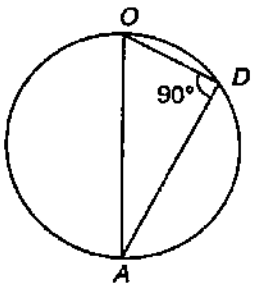
$\therefore$ $CE = 4 - \frac{9}{4} = \frac{7}{4}$

$\therefore$ $CE : (AE + AD) = \frac{7}{4} : \left(\frac{9}{4} + 3\right) = \frac{7}{4} : \frac{21}{4} = 1 : 3$

41. $\angle ADO$ is a right angle (angle of semicircle)

Again when OD is perpendicular on the chord AC and OD passes through the centre of circle ABC, then it must bisect the chord AC at D.

$\therefore$ $AD = CD = 6$ cm



For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

42. $\angle CED = 120^\circ$ ($\square ACED$ is cyclic)

$\angle BED = 60^\circ$

$\therefore \quad \angle EDB = 90^\circ$

$\therefore \quad \frac{BD}{BE} = \cos 30^\circ$

$\frac{6}{BE} = \frac{\sqrt{3}}{2}$

$\Rightarrow \quad BE = 4\sqrt{3}$ cm

$\therefore \quad BC = BE + CE$

$= 4\sqrt{3} + 5\sqrt{3} = 9\sqrt{3}$ cm

Now, since AB and CB are the secants of the circle.

$\therefore \quad BD \times BA = BE \times BC$

$6 \times BA = 4\sqrt{3} \times 9\sqrt{3}$

$\Rightarrow \quad BA = 18$ cm

Again, ΔACB is a right angled triangle ($\because \angle C = 90^\circ$)

$\therefore \quad AC = AB \sin 30^\circ$ ($\sin 30^\circ = 1/2$)

$AC = 9$ cm (Alternatively apply Pythagorus theorem).

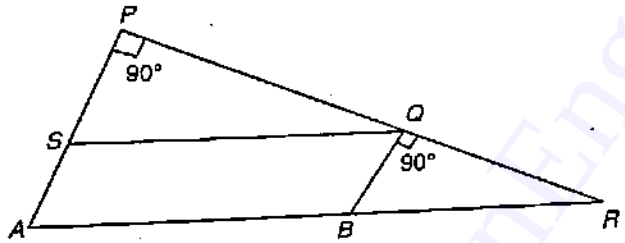
and $AD = AB - BD = 12$ cm

$\therefore \quad \frac{AC}{AD} = \frac{9}{12} = \frac{3}{4}$

43. $AB = SQ = 25$ cm

$PQ = 24$ cm

$\therefore \quad SP = 7$ cm



$AP = 12$ cm and $SP = 7$ cm

(ΔPSQ, ΔQBR and ΔPAR are similar)

$\therefore \quad AS (= BQ) = 12 - 7 = 5$ cm

$\therefore \quad \frac{PS}{SQ} = \frac{PA}{AR}$

$\frac{7}{25} = \frac{12}{AR} \Rightarrow AR = \frac{300}{7}$

$\therefore \quad BR = AR - AB = \frac{300}{7} - 25$

$BR = \frac{125}{7}$

$\therefore \quad \frac{AB}{BR} = \frac{25 \times 7}{125} = \frac{7}{5}$

$\frac{SP}{PQ} = \frac{BQ}{QR}$

44. $\frac{7}{24} = \frac{5}{QR} \Rightarrow QR = \frac{120}{7}$

$\therefore \quad \frac{\text{Area of } \Delta APR}{\text{Area of } \Delta BQR} = \frac{\frac{1}{2} \times AP \times PR}{\frac{1}{2} \times BQ \times QR} = \frac{12 \times \frac{288}{7}}{5 \times \frac{120}{7}} = \frac{144}{25}$

($PR = PQ + QR$)

Alternatively:

$\frac{\text{Area of } \Delta APR}{\text{Area of } \Delta BQR} = \left(\frac{AP}{BQ}\right)^2 = \left(\frac{12}{5}\right)^2 = \frac{144}{25}$

45. $x^2 + y^2 + z^2 = xy + yz + zx$

$x^2 + y^2 + z^2 - xy - yz - zx = 0$

$\Rightarrow \quad 2(x^2 + y^2 + z^2 - xy - yz - zx) = 0$

$\Rightarrow \quad (x^2 + y^2) + (y^2 + z^2) + (z^2 + x^2) - 2xy - 2yz - 2zx = 0$

$\Rightarrow \quad (x - y)^2 + (y - z)^2 + (z - x)^2 = 0$

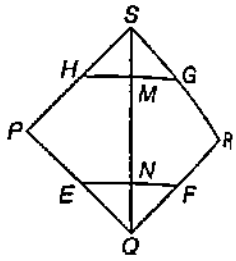
$\Rightarrow \quad x = y = z$

$\therefore$ The given triangle is an equilateral triangle.

46. Let $AD = 3a$ and $DC = 6a$

$\therefore \quad DH = HG = GC = \frac{6a}{3} = 2a$

$HM = MG = \frac{2a}{2} = a = SM$



(also)

$\therefore \quad NQ = a$

$\therefore \quad SQ = SM + MN + NQ$

$= a + 3a + a = 5a$

Since diagonal of square $SQ = 5a$

But, diameter of circle SQ = diagonal of square SQ

$\therefore \quad$ Radius of the circle $= \frac{5a}{2}$

$\Rightarrow \quad$ Area of the circle $= \pi \times \left(\frac{5a}{2}\right)^2$

Here $\frac{\text{Area of circle}}{\text{Area of rectangle}} = \frac{\frac{25}{4}(a^2\pi)}{3a \times 6a} = \frac{25\pi}{72}$

47. Area of hexagon $ABCDEF = 6 \times \frac{\sqrt{3}}{4} \times (6)^2$

$= 54\sqrt{3}$ cm^2

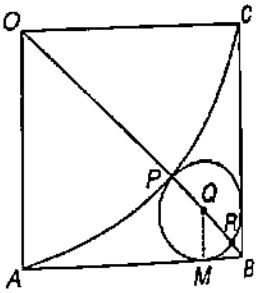
$\therefore \quad$ Area of $BDF = \frac{1}{2}$(Area of hexagon)

$= 27\sqrt{3}$ cm^2

48. $OA = AB = BC = OC = OP$

Let $OA = R$ (radius of the larger circle) then $OB = R\sqrt{2}$

Similarly $PQ = MQ = QR = r$ (radius of the smaller circle)



then $BQ = r\sqrt{2}$

$\therefore \quad BP = r + r\sqrt{2}$

and $BP = OB - OP = R\sqrt{2} - R$

$R\sqrt{2} - R = r + r\sqrt{2}$

$R(\sqrt{2} - 1) = r(\sqrt{2} + 1)$

$\Rightarrow \quad r = R(\sqrt{2} - 1)^2$

$r = R(3 - 2\sqrt{2})$

$\therefore \quad \frac{\text{Area of larger circle}}{\text{Area of smaller circle}} = \frac{\pi R^2}{4\pi r^2}$

$= \frac{R^2}{4(3 - 2\sqrt{2})^2 R^2} = \frac{1}{4(17 - 12\sqrt{2})}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

49. Apply the same logic as in the previous problem.

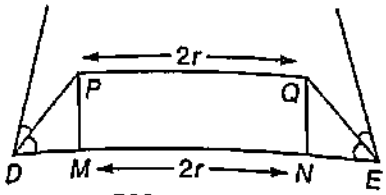
50. $\frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta AED} = \left(\frac{BC}{0.65\,BC}\right)^2 = \frac{1}{0.4225}$

$\therefore$ Area of $\Delta AED = 0.4225 \times 68 = 28.73 \text{cm}^2$

51. Let the radius of each circle be r unit then

$PQ = QR = PR = 2r$

$\angle PDM = \angle QEN = 30°$



$\therefore \quad \frac{DM}{DP} = \cos 30°$

$DM = DP \times \frac{\sqrt{3}}{2} \qquad [DP = QE = (r)]$

$DM = \frac{r\sqrt{3}}{2}$

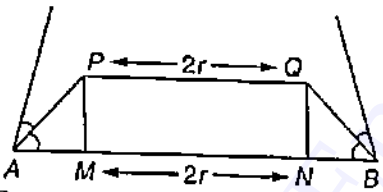
$\therefore \quad DE = DM + MN + NE$

$= \frac{r\sqrt{3}}{2} + 2r + \frac{r\sqrt{3}}{2} = (2+\sqrt{3})r$

$\therefore \quad DE = DF = EF = (2+\sqrt{3})r$

Again, $\angle PAM = \angle QBN = 30°$

$\therefore \quad \frac{PM}{AM} = \tan 30° = \frac{1}{\sqrt{3}}$



$\frac{r}{AM} = \frac{1}{\sqrt{3}}$

$\Rightarrow \quad AM = r\sqrt{3} = BN$

$\therefore \quad AB = AM + MN + NB$

$= r\sqrt{3} + 2r + r\sqrt{3} = 2r(1+\sqrt{3})$

$\therefore \quad AB = BC = AC = 2r(1+\sqrt{3})$

Ratio of perimeter of equilateral triangle = ratio of their sides

$\therefore$ Ratio of perimeter of $\Delta ABC : \Delta DEF : \Delta PQR$

$= 2(1+\sqrt{3}) : (2+\sqrt{3}) : 2$

52. $\Delta APQ \sim \Delta ACB, \quad BC = 2PQ$

and $\quad BC \parallel PQ$

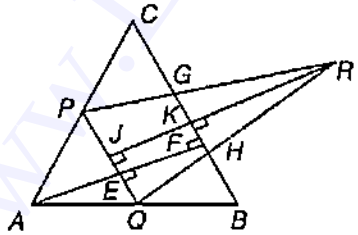
$\Rightarrow \quad AE = 2AF$

$\Rightarrow \quad AE = EF$

Again $\Delta RGH \sim \Delta RPQ$

and $\quad PQ = 2GH$

(By mid-point theorem)



$\Rightarrow \quad RJ = 2RK$

$\Rightarrow \quad RK = JK$

But since $\quad EF = JK$

$\therefore \quad AE = EF = JK = RK$

$\therefore \quad RJ = RK + JK \quad$ and $\quad AF = AE + EF$

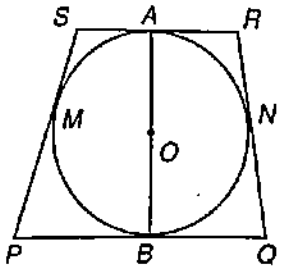
and $\quad RJ = AF = h$ (say),

then $\quad \frac{\text{Area of } \Delta PQR}{\text{Area of } \Delta ABC} = \frac{\frac{1}{2} \times PQ \times h}{\frac{1}{2} \times BC \times h} = \frac{PQ}{BC} = \frac{1}{2}$

53. It can be solved using the property of tangents.

(Tangents on the circle drawn from the same points are same in length)

Points M, A, N and B are the points of tangent.



$\therefore \quad PS + QR = PQ + SR = 2(21) = 42 \text{cm}$

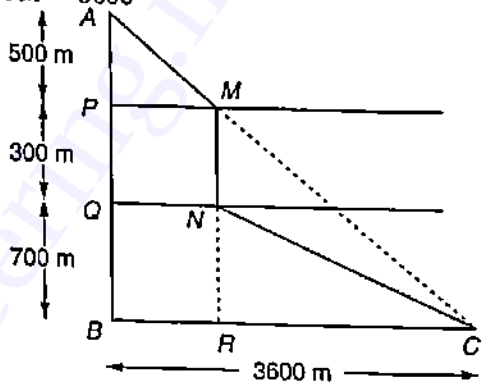
$\therefore \quad$ Perimeter of trapezium $= 2(42) = 84$ cm

54. Let MN be the bridge.

$\Delta APM \sim \Delta ABC$

$\therefore \quad \frac{AP}{PM} = \frac{AB}{BC}$

$\frac{500}{PM} = \frac{1500}{3600} \Rightarrow PM = 1200 = QN = BR$



$\therefore \quad RC = BC - BR = 2400 \text{ m}$

and $\quad NR = BQ = 700 \text{ m}$

$\therefore \quad NC = \sqrt{NR^2 + RC^2}$

$NC = 2500 \text{ m}$

Also $\quad AM = \sqrt{AP^2 + PM^2}$

$AM = 1300 \text{ m}$

$\therefore$ Total distance to be travelled $= AM + MN + NC$

$= 1300 + 300 + 2500$

$= 4100 \text{ m}$

55. Let $AD = h$ (say)

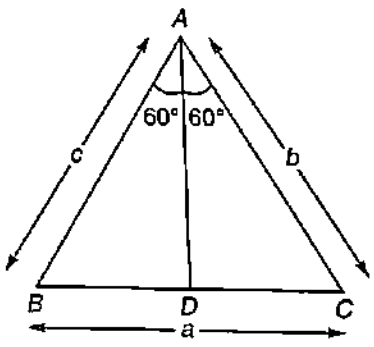
then Area of ΔABC

$= \frac{1}{2} bc \sin 120° = \frac{\sqrt{3}}{4} bc$

Area of ΔBAD

$= \frac{1}{2} ch \sin 60°$

$= \frac{\sqrt{3}}{4} ch$



and area of $\Delta CAD = \frac{1}{2} bh \sin 60° = \frac{\sqrt{3}}{4} bh$

Now, $\quad A(\Delta ABC) = A(\Delta BAD) + A(\Delta CAD)$

$\frac{\sqrt{3}}{4} bc = \frac{\sqrt{3}}{4} ch + \frac{\sqrt{3}}{4} bh$

$\Rightarrow \quad bc = h(b + c)$

$\Rightarrow \quad h = \frac{bc}{b+c}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

13 ELEMENTS OF ALGEBRA

Algebra plays a very crucial role in CAT since on an average 35–40% problems of 'quantitative aptitude' section are related to algebra and the question asked from this section are not very calculative and can be solved quickly through options. According to the past years question papers, the questions of this section are very logical even being algebraic. Their level can be of High School Standard or IIT-JEE level. But you need not to go for an IIT coaching or for higher studies in maths. For this, first you get the command over the fundamentals, then solve the plenty of problems so that you must be aware with variety and diversity of problems. This section is written in this book as per the needs of those students who are really very poor in algebra and for the same, I have started from the zero level and created the standard of problems higher than CAT level. So in this journey of learning of algebra you have to learn maximum from the problems given in the chapter, because theory is always not sufficient in itself.

FUNDAMENTAL CONCEPT OF ALGEBRA

1. Literal Numbers (or literals) : Algebra is a branch of mathematics in which all the arithmetical operations are performed by using letters (like, $a, b, c, \ldots, p, q, r, \ldots x, y, z$) alongwith numerals. The letters used in algebra represent generalised numbers (or unknown) numbers. They are called literals.

2. Operations on Literals : Since literals represent numbers, so they obey all the rules of operations of addition, subtraction, multiplication and division of numbers.

Addition of Literals and Numbers

(i) $x + y = y + x$

(ii) $x + 3 = 3 + x$

(iii) $(x + y) + z = x + (y + z)$

(iv) $(x + 0) = x = (0 + x)$

(v) $x + (-x) = 0 = (-x) + x$

Subtraction of Literals and Numbers

(i) $4 - x$

(ii) $x - 4$

(iii) $x - y = -(y - x)$

(iv) $y - x = -(x - y)$

(v) $x - 0 = x$

Multiplication of Literals and Numbers

(i) $x \times y = xy$ $\quad (xy = x + x + x + \ldots\ y \text{ times})$

(Ii) $4 \times x = 4x$ (not as $x4$) $\quad (4x = x + x + x + x)$

(iii) $1 \times x = x$ (not as $1x$)

(iv) $xy = yx$

(v) $(xy)z = x(yz)$

(vi) $1 \cdot x = x = x \cdot 1$

(vii) $x(y + z) = xy + xz$

(viii) $0 \cdot x = 0 = x \cdot 0$

Division of Literals and Numbers

(i) $4 \div x = \frac{4}{x}$ $\quad$ (ii) $x \div 4 = \frac{x}{4}$

(iii) $x \div y = \frac{x}{y}$ $\quad$ (iv) $0 \div x = 0$

NOTE $x \neq 0$, *i.e.*, $\frac{x}{0}$ is not defined.

Powers of a Literal

(i) $x \times x = x^2$ or $x \times x \times x = x^3$ etc.

i.e., $x \times x \times \ldots\ n \text{ times} = x^n$

(ii) $x^1 = x$

(iii) $x^0 = 1$

NOTE x^n is read as x raised to the power n or x to the power n or x power n.
$x \rightarrow$ base
$n \rightarrow$ exponent or index
$x^n \rightarrow$ exponential expression

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

LANGUAGE OF ALGEBRA

Statement	Meaning
$a=b$	a is equal to b
$a\neq b$	a is not equal to b
$a<b$	a is less than b
$a>b$	a is greater than b
$a\leq b$	a is less than or equal to b
$a\geq b$	a is greater than or equal to b
$a\nless b$	a is not less than b
$a\ngtr b$	a is not greater than b.

CONSTANTS AND VARIABLES

Constants : A symbol which has a fixed value is called a constant.

e.g., $4, 5, 0, -3, \frac{2}{5}, 3\frac{2}{7}$, etc.

Variable : A symbol which can assume various numerical values is called a variable.

e.g., Area of rectangle = length $(l)\times$ breadth $(b)=l\times b$

here length (l) and breadth (b) can assume any possible numerical values.

Perimeter of a square $=4a$ ($a\rightarrow$ side of a square)

here a can assume any possible numerical value.

Price of 10 balls $=10x=10\times 2=20$ or $10\times 5=50$, etc.

here x can assume any possible numerical value.

NOTE In $5x$, 5 is constant and x is variable but together $5x$ is variable.

Similarly, $5+x, x-5, 5-x, \frac{x}{5}$ and $\frac{5}{x}$ are variables.

Hence a combination of a constant and a variable is a variable.

ALGEBRAIC EXPRESSION

A collection of constants and literals connected by one or more of the operations of addition, subtraction, multiplication and division is called an **"algebraic expression".**

The various parts of an algebraic expression separated by + or − sign are called **"terms"** of the algebraic expression.,

e.g.,

	Algebraic expression	Number of terms	Terms
(i)	$4x$	1	$4x$
(ii)	$8x+3y$	2	$8x, 3y$
(iii)	$7+3x-2y^2$	3	$7, 3x, -2y^2$
(iv)	$\frac{7}{8}x^3+3x^2+4x-9$	4	$\frac{7}{8}x^3, 3x^2, 4x, -9$

NOTE Multiplication and division do not separate the terms of an algebraic expression, *e.g.*, $7x$ is a single term, similarly $\frac{x}{7}$ is also a single term.

TYPES OF ALGEBRAIC EXPRESSIONS

1. Monomial : An algebraic expression having only one term is called a monomial.

e.g., $3x, -5y, 12xy^2, \frac{3x}{y}, \frac{7x^2}{3y}$, etc.

2. Binomial : An algebraic expression having two terms is called a binomial, *e.g.*, $3x+4y, 7x^2+2x, 9yz^2-4x^2, 3x-\frac{2}{3}y$, etc.

NOTE A trinomial has three terms.

3. Polynomial (or multinomial) : An algebraic expression having two or more than two terms is called a multinomial or polynomial, *e.g.*,

(i) $7x^2-3y$ is a polynomial having 2 terms

(ii) $7x^4-3x^2+5x$ is a polynomial having 3 terms

(iii) $3x^4-7x^3+3xy+8$ is a polynomial having 4 terms, etc.

NOTE
Mono → single or one
Bi → two
Tri → three
Poly/Multi → many
and nomial → terms

Product : When two or more constants or literals (or both) are multiplied, then the result so obtained is called the **'product'**, *e.g.*, $7xy$ is the product of 7, x and y.

Factors : Each of the quantity (constant or literal) multiplied together to form a product is called a **'factor'** of the product.

A constant factor is called a **numerical factor** and any factor containing only literals is called a **literal factor.** *e.g.*,

(i) In $7xy$, the numerical factor is 7 and literal factors are x, y and xy.

(ii) In $-7x^2y$, the numerical factor is -7 and the literal factors are, x^2, y, x, xy and x^2y.

Constant term : The term of an algebraic expression having no literal factors is called its **constant term.** *e.g.*

In the algebraic expression $3x^2+7x+8$, the constant term is 8. (Since 8 is not associated with any literals)

Coefficients : In an algebraic expression any factor of a term is called the **coefficient** of the remaining factor of the term, *e.g.*,

$$4xy+5x^2y-17$$

In the first term $4xy$,

(i) numerical coefficient $=4$

(ii) literal coefficient $=xy$

(iii) coefficient of $4=xy$

(iv) coefficient of $x=4y$

(v) coefficient of $y=4x$

(vi) coefficient of $4x=y$

(vii) coefficient of $4y=x$

Similarly we can find the coefficient of the other two terms

Like and unlike terms : The terms having same literal coefficients are called like **terms** else the terms are called **unlike terms,**

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

e.g.,

(i) $7xy, 4xy, 9xy, -\frac{3}{2}xy$ are like terms

(ii) $3x^2y, 5x^2y, 8x^2y, \ldots -7x^2y$ are like terms

(iii) $3x, 5x^2y, 9xy^2z$ are unlike terms, etc.

Polynomials in one variable : An algebraic expression of the form $a + bx + cx^2 + dx^3 + \ldots$ where $a, b, c, d, \ldots$ are constants and x is a variable, is called a polynomial in the variable x.

The greatest power of x present in the polynomial is called the **degree of the polynomial.**

NOTE No term in a polynomial can be of the form $\frac{1}{x}, \frac{1}{x^2}$, etc.

e.g.,

(i) $7 + 3x$ is a polynomial in x of degree 1.

(ii) $7y^2 + 3y - 10$ is a polynomial in y of degree 2.

(iii) $8k^3 + 3k^2 - 15$ is a polynomial in k of degree 3.

(iv) $7x^5 + 3x^2 + 8$ is a polynomial in x of degree 5.

(v) $3x^4 + 5x^3 + 7x - \frac{8}{x}$ is not a polynomial.

(vi) 8 can be written as $8x^0$, so it is a polynomial in x of degree 0, *i.e.*, in general a single constant term is always of degree 0 (zero).

Polynomials in two or more variables : Any algebraic expression in two or more variables is called a polynomial if every variable present in it has only non-negative integral powers.

In case of a polynomial in two or more variables, the sum of the powers of the variables in each term is taken and the greatest sum is the **degree of the polynomial.**

NOTE No term in a polynomial can be of the form $\frac{1}{x}, \frac{1}{x^2}, \frac{x}{y}$, etc.

Examples :

(i) $8x^2 - 5xy + 3x - 7y + 2$ is a polynomial in two variables x and y.
The degree of its terms are 2, (1 + 1), 1, 1, 0, so the degree of the polynomial is 2.

(ii) $8a^3 + 9b^2a^3 + 12b^3 - \frac{5}{7}a + 15$ is a polynomial in two variables a and b.
The degree of its terms are 3, (2 + 3), 3, 1, 0, so the degree of the polynomial is 5.

NOTE The degree of a polynomial is equal to the maximum value of the sum of the powers of the variables in any term. Degree is always a whole number. It is not defined for polynomials having fractional and negative exponent.

Substitution : The process of finding the value of an algebraic expression by replacing the variables (literals) by their particular values is called substitution.

EXAMPLE 1 *If $x = 3$ and $y = 4$, find the value of*
(i) $3x + 2y$ (ii) $4x - 3y$ (iii) $18xy$ (iv) $\frac{15x}{12y}$

SOLUTION (i) $3x + 2y = 3 \times 3 + 2 \times 4$ (Substituting $x = 3$ and $y = 4$)
$= 9 + 8 = 17$

(ii) $4x - 3y = 4 \times 3 - 3 \times 4$ (Substituting $x = 3$ and $y = 4$)
$= 12 - 12 = 0$

(iii) $18xy = 18 \times 3 \times 4$ (Substituting $x = 3$ and $y = 4$)
$= 216$

(iv) $\frac{15x}{12y} = \frac{15 \times 3}{12 \times 4} = \frac{15}{16}$

EXAMPLE 2 *If $x = -3$, find the value of $4x^2 - 5x + 3$*

SOLUTION $4x^2 - 5x + 3 = 4 \times (-3)^2 - 5 \times (-3) + 3$
$= 36 + 15 + 3 = 54$

EXAMPLE 3 *If $a = 4$, $b = -3$, $c = 2$, find the value of $4a^2 + 5b - 12c$.*

SOLUTION $4a^2 + 5b - 12c = 4 \times (4)^2 + 5(-3) - 12(2)$
$= 64 - 15 - 24 = 25$

OPERATIONS ON ALGEBRAIC EXPRESSION

Addition of like terms : In any algebraic expression only like terms can be added or subtracted.

Thus, the sum of two or more like terms is a like term whose numerical coefficient is the sum of coefficients of given like terms. *e.g.,*

$2x = x + x$, $3x = x + x + x$, $4x = x + x + x + x$.

$\therefore 2x + 3x + 4x = (x + x) + (x + x + x) + (x + x + x + x)$
$= 9x$

Similarly, $2x^2 + 3x^2 = (2 + 3)x^2 = 5x^2$

and $5xy + (-3xy) = [5 + (-3)]xy = 2xy$

Thus, we cannot further simplify $7x + 5x^2$ into single term since both are unlike terms.

ADDITION OF ALGEBRAIC EXPRESSIONS

EXAMPLE 1 *Solve the following expression*
$(4a + 2b - 3c) + (3a - 2b + 4c)$

SOLUTION $(4a + 2b - 3c) + (3a - 2b + 4c)$
$= (4a + 3a) + (2b - 2b) + (-3c + 4c)$
$= (4 + 3)a + (2 - 2)b + (-3 + 4)c$
$= 7a + c$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

EXAMPLE 2 *Add* $2x^2 + 5xy - 2y^2$, $3x^2 - 2xy + 5y^2$ *and* $-2x^2 + 8xy + 3y^2$.

SOLUTION $(2x^2 + 5xy - 2y^2) + (3x^2 - 2xy + 5y^2) + (-2x^2 + 8xy + 3y^2)$

$= (2x^2 + 3x^2 - 2x^2) + (5xy - 2xy + 8xy) + (-2y^2 + 5y^2 + 3y^2)$

$= 3x^2 + 11xy + 6y^2$

SUBTRACTION OF ALGEBRAIC EXPRESSIONS

EXAMPLE 1 *Subtract* $3x + 2y - z$ *from* $4x - 5y + 3z$

SOLUTION $(4x - 5y + 3z) - (3x + 2y - z)$

$= (4x - 3x) + (-5y - 2y) + [3z - (-z)]$

$= x - 7y + 4z$

EXAMPLE 2 *Subtract* $7x^3 - 3x^2 + 8x + 16$ *from* $9x^3 + 6x^2 - 6x + 20$

SOLUTION $(9x^3 + 6x^2 - 6x + 20) - (7x^3 - 3x^2 + 8x + 16)$

$= (9x^3 - 7x^3) + [6x^2 - (-3x^2)] + [(-6x) - (8x)] + (20 - 16)$

$= 2x^3 + 9x^2 - 14x + 4$

MULTIPLICATION OF ALGEBRAIC EXPRESSIONS

It follows the distributive property. In this operation all the numerals are multiplied with each other individually and all the literals (or variables) multiplied with each other as per the rule.

EXAMPLE 1 *Find the product of* $6xy$ *and* $7x^2$.

SOLUTION $6xy \times 7x^2 = (6 \times 7) \times (xy \times x^2)$

$= (6 \times 7) \times (x \times y \times x \times x) = 42 \times x^3y$

EXAMPLE 2 *Find the product of* $3ab^2$ *and* $(4ab + 3b^2)$

SOLUTION $3ab^2 \times (4ab + 3b^2)$

$= 3ab^2 \times 4ab + 3ab^2 \times 3b^2$

$= 3 \times a \times b \times b \times 4 \times a \times b + 3 \times a \times b \times b \times 3 \times b \times b$

$= 12a^2b^3 + 9ab^4 = 3ab^3(4a + 3b)$

EXAMPLE 3 *Multiply* $(a + b)$ *by* $(c + d)$

SOLUTION $(a + b)(c + d) = a(c + d) + b(c + d)$

$= ac + ad + bc + bd$

EXAMPLE 4 *Multiply* $(5x - 4)$ *by* $(3x + 5)$

SOLUTION $(5x - 4) \times (3x + 5) = 5x \times (3x + 5) - 4(3x + 5)$

$= 15x^2 + 25x - 12x - 20$

$= 15x^2 + 13x - 20$

DIVISION OF ALGEBRAIC EXPRESSIONS

EXAMPLE 1 *Solve* $x^6 \div x^4$

SOLUTION $\frac{x^6}{x^4} = \frac{x \times x \times x \times x \times x \times x}{x \times x \times x \times x}$

$= x \times x = x^2$

EXAMPLE 2 *Solve* $35x^3y^2 \div 7x^2y^2$

SOLUTION $\frac{35x^3y^2}{7x^2y^2} = \frac{7 \times 5 \times x \times x \times x \times y \times y}{7 \times x \times x \times y \times y}$

$= 5x$

Thus, we can say $7x^2y^2$ and $5x$ are the two factors of $35x^3y^2$.

EXAMPLE 3 *Divide* $14x^3y^2 + 8x^2y^3 - 32x^2y^5$ *by* $-2xy^2$.

SOLUTION $\frac{14x^3y^2 + 8x^2y^3 - 32x^2y^5}{-2xy^2}$

$= \frac{(14x^3y^2)}{(-2xy^2)} + \frac{(8x^2y^3)}{(-2xy^2)} + \frac{(-32x^2y^5)}{(-2xy^2)}$

$= -7x^2 - 4xy + 16xy^3$

EXAMPLE 4 *Divide* $(4x^2 + 3x + 2)$ *by* $(x + 2)$

SOLUTION

$$
\begin{array}{r|l}
(x+2)\overline{)\,4x^2+3x+2\,} & 4x-5 \\
\cline{2-2}
\underline{4x^2+8x} & \\
-5x+2 & \\
\underline{-5x-10} & \\
12 &
\end{array}
$$

Here $(x + 2)$ is divisor
and $(4x - 5)$ is quotient
and 12 is remainder
and $(4x^2 + 3x + 2)$ is the dividend.

NOTE (i) If there is a '+' sign before a bracket, then the signs of the terms inside the bracket do not change.
(ii) If there is a '−' sign before a bracket, then the signs of all the terms inside the bracket change.

GENERAL FORM OF A POLYNOMIAL

The expression of the form $a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \ldots a_n$ in which $a_0, a_1, a_2, \ldots a_n$ are constants ($a_0 \neq 0$) and n is a positive integer is called a polynomial (or rational integral function) in x of degree n. We may denote it by $f(x)$, *i.e.*,

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \ldots a_n.$$

Algebraic Expression as a Function : A function is an operation which forms a relation between two variables. *e.g.*, $y = x^2$, $y = 3x$, $y = x^2 + 3$, etc.

This operation is represented by $f(x)$, *i.e.*,

$y = f(x) \Rightarrow f(x) = x^2$ (f is called operand)

$f(x) = 3x \Rightarrow f(x) = x^2 + 3$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

EXAMPLE 1 *If $f(x) = x^3 + 2x$, then find the value of $f(3)$.*

SOLUTION $\because \quad f(x) = x^3 + 2x$

$\therefore \quad f(3) = (3)^3 + 2(3) = 27 + 6 = 33$

EXAMPLE 2 *If $f(x) = 2x^3 + 3x^2 + 5$, then find the value of $f(2)$.*

SOLUTION $\because \quad f(x) = 2x^3 + 3x^2 + 5$

$\therefore \quad f(2) = 2(2)^3 + 3(2)^2 + 5$

$= 16 + 12 + 5 = 33$

GRAPHICAL REPRESENTATION OF ALGEBRAIC EXPRESSION

All the polynomial expressions that can be written as $y = f(x)$ can be represented in two dimensional co-ordinate system.

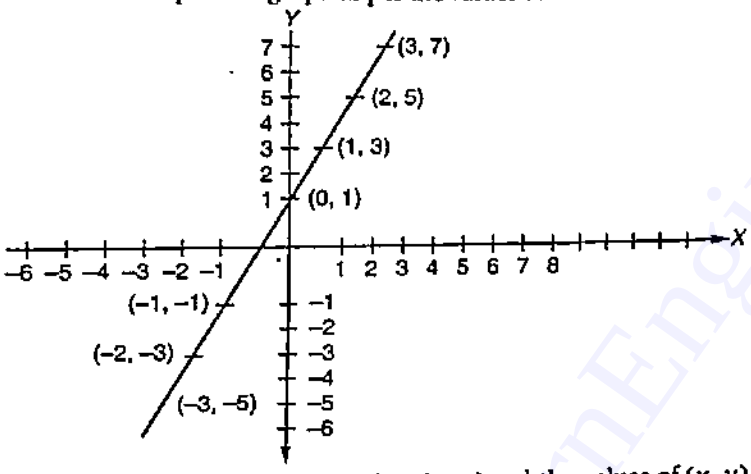
EXAMPLE 1 $y = f(x) = 2x + 1$

SOLUTION

x	−3	−2	−1	0	1	2	3
y	−5	−3	−1	1	3	5	7

HINT Find the corresponding values of y for each value of x. Generally we consider some +ve and some −ve values of x.

Now we plot the graph as per the values obtained in the table.

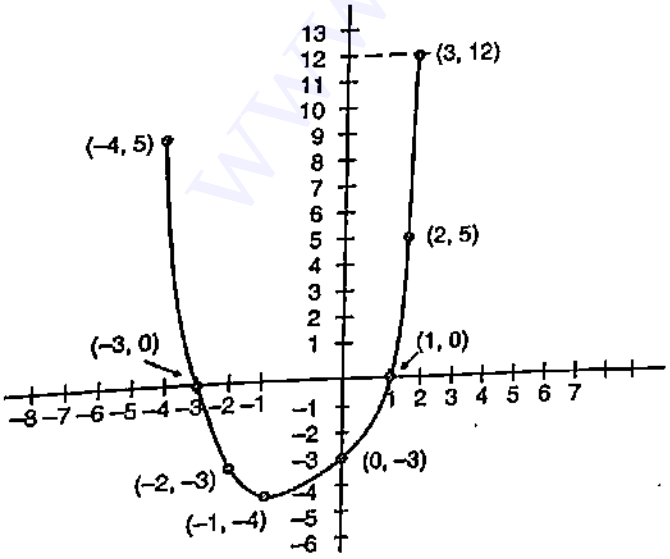


The points are represented as (x, y) and the values of (x, y) represent the points on the plane.

NOTE $y = 2x + 1$ is a linear equation, therefore the graph of the given equation shows a straight line.

EXAMPLE 2 $f(x) = x^2 + 2x - 3$

SOLUTION $y = f(x) = x^2 + 2x - 3$



x	−4	−3	−2	−1	0	1	2	3
y	5	0	−3	−4	−3	0	5	12

Since $y = x^2 + 2x - 3$ is a quadratic equation, therefore the graph is a parabola.

Remainder theorem : If a polynomial (*i.e.*, a rational integral function) $f(x)$ is divided by $(x - a)$, then the remainder is obtained by substituting a for x in $f(x)$, *i.e.*, the remainder is $f(a)$.

EXAMPLE 1 *Find the remainder when $x^2 - 6x + 5$ is divided by $(x + 2)$.*

SOLUTION Let $f(x) = x^2 - 6x + 5$. Thus by remainder theorem, the remainder when $f(x)$ is divided by $(x + 2)$, *i.e.*, $[x - (-2)]$ is $f(-2)$

$\therefore \quad f(-2) = (-2)^2 - 6(-2) + 5$

$= 4 + 12 + 5 = 21.$

$\therefore$ Thus the remainder is 21.

NOTE
1. If a polynomial $f(x)$ is divided by $(x + a)$, then remainder is $f(-a)$
2. If a polynomial $f(x)$ is divided by $(x - a)$, then remainder is $f(a)$
3. If a polynomial $f(x)$ is divided by $(ax + b)$, then the remainder is $f(-b/a)$

HINT To get the value of remainder first of all write the divisor equal to zero, *i.e.*, if divisor is $(x + a)$, then write $x + a = 0$
Then find the value of x, *i.e.*, $x + a = 0 \Rightarrow x = -a$
Then put the obtained value of x in the polynomial, *i.e.*, $f(x) = 3x^2 + 4x - 5 \Rightarrow f(-a) = 3(-a)^2 + 4(-a) - 5$ then solve the function (or expression) and get the simplified value of remainder.

EXAMPLE 2 *If $2x^3 + 3x^2 + 7x + 6$ be divided by $2x + 1$, find the remainder.*

SOLUTION Let $f(x) = 2x^3 + 3x^2 + 7x + 6$.

Then the remainder when $f(x)$ is divided by $(2x + 1)$

$\therefore \quad 2x + 1 = 0$

$\Rightarrow \quad x = -\frac{1}{2}$

$$\therefore \quad f\left(-\frac{1}{2}\right) = 2\left(-\frac{1}{2}\right)^3 + 3\left(-\frac{1}{2}\right)^2 + 7\left(-\frac{1}{2}\right) + 6$$

$$= -\frac{1}{4} + \frac{3}{4} - \frac{7}{2} + 6$$

$$= \frac{-1 + 3 - 14 + 24}{4} = \frac{12}{4} = 3$$

EXAMPLE 3 *Find the remainder when the expression $3x^3 + 8x^2 - 6x + 1$ is divided by $(x + 3)$.*

SOLUTION Let $f(x) = 3x^3 + 8x^2 - 6x + 1$

$\because \quad x + 3 = 0 \Rightarrow x = -3$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$\therefore \quad f(-3) = 2(-3)^3 + 3(-3)^2 - 6(-3) + 1$

$= -81 + 72 + 18 + 1 = 10$

Hence the remainder is 10.

EXAMPLE 4 *When $x^3 + 3x^2 - kx + 4$ is divided by $x - 2$, the remainder is k. Find the value of the constant k.*

SOLUTION Let $f(x) = x^3 + 3x^2 - kx + 4$

Now, since $x - 2 = 0 \Rightarrow x = 2$

$\therefore \quad f(2) = (2)^3 + 3(2)^2 - k(2) + 4 = 24 - 2k$

$\therefore \quad$ Remainder $= 24 - 2k = k \Rightarrow k = 8$

(∵ the given remainder is k)

Factor Theorem : *If $f(x)$ be a polynomial (i.e. rational integral function) in x and $f(a) = 0$, then $(x - a)$ is a factor of $f(x)$.*

By remainder theorem if $f(x)$ be divided by $(x - a)$ the remainder is $f(a)$. But $f(a) = 0$, i.e., there is no remainder.

$\therefore$ $f(x)$ is exactly divisible by $(x - a)$

Hence $(x - a)$ is a factor of $f(x)$

NOTE 1. *If $f(x)$ be a polynomial and $f(-a) = 0$, then $(x + a)$ is a factor of $f(x)$.*

2. *If $f(x)$ be a polynomial and $f(-b/a) = 0$, then $(ax + b)$ is a factor of $f(x)$.*

EXAMPLE 1 *Show that $(x - 2)$ is a factor of the polynomial $2x^2 - 5x + 2$*

SOLUTION Let $f(x) = 2x^2 - 5x + 2$

$\because \quad (x - 2) = 0 \Rightarrow x = 2$

$\therefore \quad f(2) = 2(2)^2 - 5(2) + 2$

$f(2) = 0$

$\therefore$ By factor theorem $(x - 2)$ is a factor of $2x^2 - 5x + 2$

EXAMPLE 2 *Prove that $(2x + 5)$ is a factor of the polynomial $2x^2 + 11x + 15$.*

SOLUTION Let $f(x) = 2x^2 + 11x + 15$

$\because \quad 2x + 5 = 0 \Rightarrow x = -5/2$

$\therefore \quad f(-5/2) = 2(-5/2)^2 + 11(-5/2) + 15$

$= \frac{25}{2} - \frac{55}{2} + 15$

$f(-5/2) = 0$

Hence, $(2x - 5)$ is a factor of $f(x) = 2x^2 + 11x - 15$

EXAMPLE 3 *Prove that $(a - b)$ is a factor of $(a^5 - b^5)$*

SOLUTION Let $f(a) = a^5 - b^5$

$\because \quad (a - b) = 0 \Rightarrow a = b$

$\therefore \quad f(b) = (b)^5 - b^5$

$f(b) = 0$

$\therefore \quad f(a) = 0$

Hence $(a - b)$ is a factor of $a^5 - b^5$.

EXAMPLE 4 *If $(x - 3)$ and $(x - 4)$ are both the only factors of $f(x)$, then find $f(x)$.*

SOLUTION Since there are only two factors of $f(x)$, therefore the product of $(x - 3)$ and $(x - 4)$ is equal to $f(x)$

$\therefore \quad f(x) = (x - 3)(x - 4)$

$f(x) = x^2 - x - 12$

H.C.F. AND L.C.M. OF POLYNOMIALS

1. A polynomial $D(x)$ is a divisor of the polynomial $P(x)$ if it is a factor of $P(x)$. Where $Q(x)$ is another polynomial such that $P(x) = D(x) \cdot Q(x)$
2. **HCF/GCD (Greatest Common Divisor) :** A polynomial $h(x)$ is called the HCF or GCD of two or more given polynomials, if $h(x)$ is a polynomial of highest degree dividing each one of the given polynomials.

NOTE The coefficient of highest degree term is always taken as positive.

3. **L.C.M. (Least Common Multiple) :** A polynomial $P(x)$ is called the LCM of two or more given polynomials, if it is a polynomial of smallest degree which is divided by each one of the given polynomials.

For any two polynomials $P(x)$ and $Q(x)$. We have :

$P(x) \times Q(x) =$ [HCF of $P(x)$ and $Q(x)$]

$\times$ [LCM of $P(x)$ and $Q(x)$]

EXAMPLE 1 *Find the HCF and LCM of the following polynomials $(x + 2)^2(x - 3)^2(x + 1)^2, (x + 1)^3(x + 2)^3(x - 3)$*

SOLUTION Let $P(x) = (x + 2)^2(x - 3)^2(x + 1)^2$

and $Q(x) = (x + 1)^3(x + 2)^3(x - 3)$

HCF $= (x + 1)^2(x + 2)^2(x - 3)$

(For the HCF we take the highest power of factors common to both $P(x)$ and $Q(x)$)

and LCM $= (x + 1)^3(x + 2)^3(x - 3)^2$

[For LCM we take the highest powers of the factors either of $P(x)$ or $Q(x)$]

EXAMPLE 2 *Find the HCF and LCM of the following polynomials $22x(x + 1)^2$, $36x^2(2x^2 + 3x + 1)$*

SOLUTION Let $P(x) = 22x(x + 1)^2$

and $Q(x) = 36x^2(2x^2 + 3x + 1)$

$P(x) = 2 \times 11 \times x \times (x + 1)^2$

and $Q(x) = 2 \times 18 \times x^2 \times (x + 1)(2x + 1)$

$\therefore \quad$ HCF $= 2 \times x \times (x + 1) = 2x(x + 1)$

and LCM $= 2x(x + 1) \times 11 \times 18 \times x \times (x + 1)(2x + 1)$

$= 396x^2(x + 1)^2(2x + 1)$

(To find the LCM, first take the HCF and then take all the rest factors).

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursnahboob.wordpress.com

For More Visit : www.LearnEngineering.in

EXAMPLE 3 *Find the HCF and LCM of* $(3+13x-30x^2)$ *and* $(25x^2-30x+9)$.

SOLUTION Let $P(x) = (3+13x-30x^2)$

and $Q(x) = (25x^2-30x+9)$

or $P(x) = 3+13x-30x^2$

$= 3+18x-5x-30x^2$

$= -(6x+1)(5x-3)$

$Q(x) = 25x^2-30x+9 = (5x-3)^2$

$\therefore$ HCF of $P(x)$ and $Q(x) = (5x-3)$

and LCM of $P(x)$ and $Q(x) = -(5x-3)^2(6x+1)$.

EXAMPLE 4 *The HCF of two polynomials is* $(x-2)(x+3)$ *and their LCM is* $(x-2)^2(x+3)(x+1)$. *If one of the polynomial is* x^2-x-2, *find the other.*

SOLUTION $HCF = (x-2)(x+3)$

$LCM = (x-2)^2(x+3)(x+1)$

and $HCF \times LCM = (x-2)^3(x+3)^2(x+1)$

$\therefore$ $P(x) = x^2-x-2$

Let $= x^2-2x+x-2$

$= (x-2)(x+1)$

$\therefore$ $P(x)\cdot Q\cdot(x) = HCF \times LCM$

$\therefore$ $Q(x) = \dfrac{HCF \times LCM}{P(x)}$

$= \dfrac{(x-2)^3(x+3)^2(x+1)}{(x-2)(x+1)}$

$= (x-2)^2(x+3)^2$

$\therefore$ The other polynomial [say $Q(x)$] is $(x-2)^2(x+3)^2$.

Rational Expressions : If $P(x)$ and $Q(x)$ are two polynomials such that $Q(x) \neq 0$, then the quotient $\dfrac{P(x)}{Q(x)}$ is called a rational expression.

- Every polynomial is a rational expression but a rational expression need not be a polynomial.
- The rational expression $\dfrac{P(x)}{Q(x)}$ is said to be in its simplest form (or in lowest terms) if the G.C.D. of $P(x)$ and $Q(x)$ is 1.
- To express a rational expression in its simplest form, express the polynomials in the numerator and in the denominators as product of simplest factors (*i.e.*, no further factorization is possible), and then cancel the common factors.

EXAMPLE 1 *Which of the following algebraic expressions are polynomials?*

(a) $x^2-\sqrt{3}x+4$ (b) $x^3-3x^2+4\sqrt{x}$

(c) $\sqrt{2}x^2-\sqrt{3}x+\sqrt{5}$ (d) x^3-2x+x^{-2}

SOLUTION (a) $x^2-\sqrt{3}x+4$ is a polynomial.

and (c) $\sqrt{2}x^2-\sqrt{3}x+\sqrt{5}$ is also a polynomial

(b) $x^3-3x^2+4\sqrt{x}$ is not polynomial since $4\sqrt{x} = 4x^{1/2}$, *i.e.*, power of x is $\dfrac{1}{2}$ (*i.e.*, fractional)

(d) x^3-2x+x^{-2} is not a polynomial since x^{-2} has negative power of x.

EXAMPLE 2 *Express the following in the lowest terms.* $\dfrac{x^2+6x+9}{x^2-9}$

SOLUTION $\dfrac{x^2+6x+9}{x^2-9} = \dfrac{(x+3)(x+3)}{(x+3)(x-3)} = \dfrac{x+3}{x-3}$

IMPORTANT FORMULAE

1. $(a+b)^2 = a^2+b^2+2ab$
2. $(a-b)^2 = a^2+b^2-2ab$
3. $(a+b)^2 = (a-b)^2+4ab$
4. $(a-b)^2 = (a+b)^2-4ab$

{ By taking the difference of formula 1 and 2.

5. $(a^2+b^2) = \dfrac{1}{2}[(a+b)^2+(a-b)^2]$
6. $a^2-b^2 = (a+b)(a-b)$
7. $(a+b)^3 = a^3+b^3+3ab(a+b)$
8. $(a-b)^3 = a^3-b^3-3ab(a-b)$
9. $a^3+b^3 = (a+b)(a^2+b^2-ab)$
10. $a^3-b^3 = (a-b)(a^2+b^2+ab)$
11. $(a+b+c)^2 = (a^2+b^2+c^2)+2(ab+bc+ac)$
12. $a^3+b^3+c^3-3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ac)$

if $a+b+c=0$, then $a^3+b^3+c^3 = 3abc$

13. $a^4+b^4+a^2b^2 = (a^2+b^2+ab)(a^2+b^2-ab)$
14. $a^4-b^4 = (a-b)(a+b)(a^2+b^2)$
15. $a^8-b^8 = (a-b)(a+b)(a^2+b^2)(a^4+b^4)$

BASIC FACTS TO REMEMBER

1. To factorize the cyclic expressions :

(a) Arrange the terms in decreasing or increasing powers of one of the letters (or literals)

(b) Make one factor common by taking two or three terms together.

(c) Rewrite the terms of the other factor in decreasing or increasing power of the next letter.

(d) Repeat the process till all the factors are found out.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

2. Sigma (Σ) notation :

$\Sigma x = x + y + z$, for letters x, y, z

$\Sigma xy = xy + yz + zx$

$\Sigma x^2(y - z) = x^2(y - z) + y^2(z - x) + z^2(x - y)$

3. Some important results :

(a) $\Sigma(x - y) = 0$

(b) $\Sigma(x^2 - y^2) = 0$

(c) $\Sigma(x^3 - y^3) = 0$

(d) $\Sigma x(y - z) = 0$

(e) $\Sigma x^2(y^2 - z^2) = 0$

(f) $\Sigma x^2(y - z) = -(x - y)(y - z)(z - x)$

(g) $\Sigma x(y^2 - z^2) = (x - y)(y - z)(z - x)$

(h) $\Sigma(x - y)^3 = 3(x - y)(y - z)(z - x)$

EXAMPLE 1 *Factorize* $x(y^2 + z^2) + y(z^2 + x^2) + z(x^2 + y^2) + 2xyz$

SOLUTION $x(y^2 + z^2) + y(z^2 + x^2) + z(x^2 + y^2) + 2xyz$

$= xy^2 + xz^2 + yz^2 + x^2y + x^2z + y^2z + 2xyz$

Arranging in decreasing powers of x

$= x^2y + x^2z + xy^2 + xz^2 + 2xyz + y^2z + yz^2$

$= x^2(y + z) + x(y^2 + z^2 + 2yz) + yz(y + z)$

$= x^2(y + z) + x(y + z)^2 + yz(y + z)$

$= (y + z)[x^2 + x(y + z) + yz]$

$= (y + z)(x^2 + xy + xz + yz)$

$= (y + z)[x(x + y) + z(x + y)]$

$= (y + z)(x + y)(x + z)$

$= (x + y)(y + z)(z + x)$

EXAMPLE 2 *Factorize* $\Sigma xy(x + y) + 2xyz$

SOLUTION $\Sigma xy(x + y) + 2xyz$

$= xy(x + y) + yz(y + z) + zx(z + x) + 2xyz$

$= x^2y + x^2z + y^2z + y^2x + z^2x + z^2y + 2xyz$

$= x^2(y + z) + x(y^2 + z^2 + 2yz) + yz(y + z)$

$= x^2(y + z) + x(y + z)^2 + yz(y + z)$

$= (y + z)[x^2 + x(y + z) + (yz)]$

$= (y + z)[x(x + y) + z(x + y)]$

$= (y + z)(x + y)(y + z)$

INTRODUCTORY EXERCISE-13.1

1. Sunny Gupta scored x marks in Maths, y marks in *DI/DS* and 40 marks in English. What is his total score in these three subjects?
   (a) $100 - (x + y)$ (b) $100 - (x + y + 40)$
   (c) $x + y - 40$ (d) $x + y + 40$

2. Quotient of z by 4 is multiplied by thrice the number y.
   (a) $\frac{3y}{4z}$ (b) $\frac{3z}{4y}$
   (c) $\frac{3yz}{4}$ (d) $\frac{3}{4yz}$

3. Express $6 \times p \times q \times r \times p \times p \times p$ in the exponential form :
   (a) $6(p^4 + q + r)$ (b) $6 \times 4p \times q \times r$
   (c) $6p^4qr$ (d) none of these

4. Find the number of terms in $2x^2 - \frac{3}{x} + \frac{5}{x^2} + 9$.
   (a) 1 (b) 2
   (c) 3 (d) 4

5. Find the number of terms in
   $$4xyz - \frac{3}{x^2} + 6xy^2 - 5z^3 + 6x :$$
   (a) 3 (b) 4
   (c) 5 (d) 6

6. Find the number of terms in $7 \times a + b \div 3 - c + b$.
   (a) 3 (b) 5
   (c) 4 (d) 6

7. Find the coefficient of z^3 in $-7xy^2z^3$.
   (a) xy^2 (b) -7
   (c) $7xy^2$ (d) $-7xy^2$

8. In the following, which pair contains like terms?
   (a) $2x^2y, 2xy^2$ (b) $2xy, -3yx$
   (c) $6, 6x$ (d) $2xy, 3xyz$

9. Find the degree of the polynomial $4x^3 - 2x^2y^3 + 6$.
   (a) 3 (b) 4
   (c) 5 (d) 6

10. Simplify the following :
    $$2x^2 + 3y^2 - 5xy + 5x^2 - y^2 + 6xy - 3x^2$$
    (a) $7x^2 + 4y^2 - 11xy$ (b) $7x^2 + 2y^2 - 30xy$
    (c) $4x^2 + 2y^2 - 11xy$ (d) $4x^2 + xy + 2y^2$

11. If sum of two polynomials is $5x^2 + 3x - 1$. If one of them is $3x^3 - 2x + 7$. Find the other.
    (a) $3x^3 + 5x^2 + 5x - 8$ (b) $-3x^3 + 5x^2 + 5x - 8$
    (c) $2x^2 + x + 6$ (d) none of these

12. Multiply $\frac{1}{2}a^2b - \frac{2}{3}ab^2 + 5$ by $6abc$:
    (a) $3a^3b^2c - 4a^2b^3c + 30abc$
    (b) $3a^3b^3c - 4a^3b^2c + 6abc$
    (c) $6a^3b^2 + 18ab^2c + 30abc$
    (d) none of the above

13. Find the product of $\left(\frac{1}{2}x^2 - \frac{1}{3}y^2\right)$ and $\left(\frac{1}{2}x^2 + \frac{1}{3}y^2\right)$.
    (a) $\frac{1}{9}y^4 - \frac{1}{4}x^4$ (b) $\frac{1}{4}x^4 - \frac{1}{9}y^4$
    (c) $4x^2 - 9y^4$ (d) $\frac{1}{4}x^4 - \frac{1}{9}y^9$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yourshahbaz.wordpress.com
For More Visit : www.LearnEngineering.in

14. Divide $36x^2y^5 + 42xy^3 - 24x^3y^2 - 12y^5$ by $-6y^2$.
(a) $6x^2y^3 - 7xy + 4x^3 + 2y^3$
(b) $-6x^2y^3 - 7xy^2 + 4x^2 + 2y^3$
(c) $-6x^2y^3 - 7xy + 4x^3 + 2y^3$
(d) none of the above

15. Find the remainder when the expression $3x^2 + 8x^2 - 6x + 1$ is divided by $x + 3$.
(a) 1 (b) 10
(c) 6 (d) 0

16. Find the value of a if the division of $ax^3 + 9x^2 + 4x - 10$ by $(x + 3)$ leaves a remainder 5.
(a) 1 (b) 2
(c) 3 (d) 4

17. If $(x + 1)$ and $(x - 2)$ are factors of $x^3 + ax^2 - bx - 6$, then find the values of a and b respectively.
(a) 2, 3 (b) 3, 5
(c) 5, 3 (d) 2, 5

18. If $(5x^2 + 14x + 2)^2 - (4x^2 - 5x + 7)^2$ is divided by $(x^2 + x + 1)$ then, quotient 'q' and remainder 'r' are given by :
(a) $q = (x^2 + 19x - 5), r = 0$ (b) $q = 9(x^2 + 19x - 5), r = 0$
(c) $q = (x^2 + 19x - 5), r = 1$ (d) $q = 9(x^2 + 19x - 5), r = 1$

19. On dividing $(x^3 - 6x + 7)$ by $(x + 1)$, then the remainder is :
(a) 2 (b) 0
(c) 12 (d) 6

20. When $(x^4 - 3x^3 + 2x^2 - 5x + 7)$ is divided by $(x - 2)$, then the remainder is :
(a) 0 (b) –3
(c) 2 (d) 3

21. If $x^3 + 5x^2 + 10k$ leaves remainder $-2x$ when divided by $x^2 + 2$, then the value of k is :
(a) –2 (b) –1
(c) 1 (d) 2

22. If $(x^{11} + 1)$ is divided by $(x + 1)$, then the remainder is :
(a) 0 (b) 2
(c) 1 (d) 10

23. If $5x^3 + 5x^2 - 6x + 9$ is divided by $(x + 3)$, then the remainder is :
(a) 0 (b) –32
(c) 23 (d) –63

24. If $f(x)$ is divided by $(2x + 3)$, then the remainder is :
(a) $f(2/3)$ (b) $f(-2/3)$
(c) $f(3/2)$ (d) $f(-3/2)$

25. When $(x^3 - 2x^2 + px - q)$ is divided by $(x^2 - 2x - 3)$, the remainder is $(x - 6)$. The values of p and q are :
(a) $p = -2, q = -6$ (b) $p = 2, q = -6$
(c) $p = -2, q = 6$ (d) $p = 2, q = 6$

26. If $(x - 2)$ is a factor of $(x^2 + 3qx - 2q)$, then the value of q is :
(a) –2 (b) –1
(c) 1 (d) 2

27. Find the value of k, if $(x + 2)$ exactly divides $x^3 + 6x^2 + 4x + k$.
(a) 4 (b) 6
(c) –8 (d) –10

28. Which one of the following is a factor of $x^4 - 5x^3 + 5x^2 - 10x + 24$?
(a) $(x + 2)$ (b) $(x + 4)$
(c) $(x - 2)$ (d) none of these

29. If $(x + k)$ is a common factor of $(x^2 + px + q)$ and $(x^2 + lx + m)$, then the value of k is :
(a) $l + p$ (b) $m - q$
(c) $\frac{l - p}{m - q}$ (d) $\frac{m - q}{l - p}$

30. If $(x - a)$ is a factor of $(x^3 - 3x^2a + 2a^2x + b)$, then the value of b is :
(a) 0 (b) 1
(c) 2 (d) 3

31. $(x^{29} - x^{25} + x^{13} - 1)$ is divisible by :
(a) $(x + 1)$ but not by $(x - 1)$
(b) $(x - 1)$ but not by $(x + 1)$
(b) both $(x + 1)$ and $(x - 1)$
(d) neither $(x - 1)$ nor $(x + 1)$

32. One of the factors of $3x^3 + x^2 - 12x - 4$ is :
(a) $3x - 2$ (b) $3x - 1$
(c) $3x + 1$ (d) $3x + 2$

33. $x^{100} + 2x^{99} + k$ is divisible by $(x + 1)$, then the value of k is :
(a) –3 (b) –2
(c) 1 (d) 2

34. If $(x - 1)$ is a factor of $(x^3 - m)$, then the value of m is :
(a) –6 (b) –5
(c) 1 (d) 3

35. If the polynomial $f(x)$ is such that $f(-3) = 0$, then a factor of $f(x)$ is :
(a) –3 (b) 3
(c) $\frac{1}{3}$ (d) $x + 3$

36. If $\left(x + \frac{1}{x}\right) = 2$, then $\left(x^2 + \frac{1}{x^2}\right)$ is equal, to :
(a) 2 (b) 4
(c) 5 (d) 6

37. If $\left(x - \frac{1}{x}\right) = 4$, then the value of $\left(x^2 + \frac{1}{x^2}\right)$ is :
(a) 16 (b) 18
(c) 14 (d) none of these

38. If $\left(x + \frac{1}{x}\right) = 2\sqrt{3}$, then the value of $\left(x^3 + \frac{1}{x^3}\right)$ is :
(a) $12\sqrt{3}$ (b) 18
(c) $18\sqrt{3}$ (d) none of these

39. If $\left(x + \frac{1}{x}\right) = 3$, then the value of $\left(x^3 + \frac{1}{x^3}\right)$ is equal to :
(a) $18\sqrt{3}$ (b) 18
(c) $9\sqrt{3}$ (d) 12

40. If $\left(x + \frac{1}{x}\right) = 2$, then the value of $\left(x^6 + \frac{1}{x^6}\right)$ is :
(a) 2 (b) 4
(c) 8 (d) none of these

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

41. If $\left(x^2+\frac{1}{x^2}\right)=6$, then the value of $\left(x-\frac{1}{x}\right)$ is :
(a) 2 (b) 3
(c) -2 (d) both (a) and (c)

42. If $\left(x^3-\frac{1}{x^3}\right)=36$, then the value of $\left(x-\frac{1}{x}\right)$ is :
(a) 1 (b) 2
(c) 3 (d) 6

43. If $\left(x^3+\frac{1}{x^3}\right)=2$, then the value of $\left(x+\frac{1}{x}\right)$ is :
(a) 1 (b) 2
(c) 3 (d) 4

44. If $\left(x^4+\frac{1}{x^4}\right)=34$, then the value of $\left(x-\frac{1}{x}\right)$ is :
(a) 1 (b) 2
(c) 3 (d) 4

45. If $\left(x^4+\frac{1}{x^4}\right)=119$, then the value of $\left(x^3-\frac{1}{x^3}\right)$ is :
(a) 49 (b) 27
(c) 36 (d) 63

46. If $\left(3x-\frac{2}{x}\right)=5$, then the value of $\left(9x^2-\frac{4}{x^2}\right)$ is :
(a) 25 (b) 35
(c) 30 (d) none of these

47. If $m^2-4m+1=0$, then the value of $\left(m^3+\frac{1}{m^3}\right)$ is :
(a) 48 (b) 52
(c) 64 (d) none of these

48. If $x+y=13$ and $xy=40$, then the value of (x^2+y^2) is :
(a) 69 (b) 80
(c) 89 (d) none of these

49. If $(x+y)=13$ and $xy=36$, then the value of (x^3+y^3) is :
(a) 369 (b) 936
(c) 693 (d) none of these

50. If $(x+y)=8$ and $xy=15$, then the value of (x^3-y^3) is :
(a) 120 (b) 60
(c) 98 (d) 82

51. If $(x+y+z)=6$ and $(xy+yz+zx)=11$, then the value of $(x^3+y^3+z^3-3xyz)$ is :
(a) 81 (b) 54
(c) 18 (d) none of these

52. If $(x+y+z)=0$, then the value of $(x^3+y^3+z^3)$ is :
(a) $3x^3y^3z^3$ (b) $3xyz$
(c) xyz (d) $3(x+y+z)$

53. If $a^{1/3}+b^{1/3}+c^{1/3}=0$, then :
(a) $a+b+c=0$ (b) $a+b+c=3abc$
(c) $a^3+b^3+c^3=0$ (d) $(a+b+c)^3=27abc$

54. If $2^x-2^{x-1}=8$, then the value of x^3 is :
(a) 46 (b) 27
(c) 64 (d) 8

55. If $3^x+3^{x+1}=36$, then the value of x^x is :
(a) 64 (b) 3125
(c) 81 (d) 4

56. If a and b are non zero rational unequal numbers, then $\frac{(a+b)^2-(a-b)^2}{a^2b-ab^2}$ is equal to :
(a) $\frac{1}{a-b}$ (b) $\frac{2}{a-b}$
(c) $\frac{4}{a-b}$ (d) $\frac{1}{ab}$

57. If $\frac{x}{(b-c)(b+c-2a)}=\frac{y}{(c-a)(c+a-2b)}=\frac{z}{(a-b)(a+b-2c)}$, then the value of $(x+y+z)$ is :
(a) $a+b+c$ (b) 0
(c) $a^2+b^2+c^2$ (d) can't be determined

58. $(a^4+5a^3+6a^2)$ is equal to :
(a) $a(a^2+3)(a+2)$ (b) $a(a+3)(a^2+2)$
(c) $a^2(a+3)(a+2)$ (d) $a^2(a-2)(a-3)$

59. The factors of (x^4+16) are :
(a) $(x^2+4)(x^2-4)$ (b) $(x^2+4)(x^2+4)$
(c) $x(x^2+2)(x^4+4)$ (d) do not exist

60. If $(a^{3/2}-ab^{1/2}+a^{1/2}b-b^{3/2})$ is divided by $(a^{1/2}-b^{1/2})$, then the quotient is :
(a) $a+b$ (b) $a-b$
(c) $a^{1/2}+b^{1/2}$ (d) a^2-b^2

61. The factors of $x(y^2-z^2)+y(z^2-x^2)+z(x^2-y^2)$ are :
(a) $(x-y)(y-z)(z-x)$ (b) $(x+y)(y+z)(z+x)$
(c) $(y-x)(z-y)(x-z)$ (d) $(x+y)(z-y)(x-z)$

62. $(a+b)^3-(a-b)^3$ can be factorized as :
(a) $2a(a^2+3b^2)$ (b) $2a(3a^2+b^2)$
(c) $2b(3a^2+b^2)$ (d) $2a(a^2+3b^2)$

63. The factors of (x^4+4) are :
(a) $(x^2+2)^2$ (b) $(x^2+2x+2)(x^2-2x+2)$
(c) $(x^2+2)(x^2-2)$ (d) none of these

64. The factors of x^3-7x+6 are :
(a) $(x-1)(x+3)(x-2)$ (b) $(x+1)(x+2)(x-3)$
(c) $x(x-6)(x-1)$ (d) $(x^2-6)(x-1)$

65. The factors of $(64x^3-216y^3)$ are :
(a) $(4x-6y)(16x^2-36y^2)$
(b) $(4x-6y)[16x^2+36y^2+24xy]$
(c) $(4x+6y)(4x-6y)$
(d) none of the above

66. The factors of $(a^2+4b^2+4b-4ab-2a-8)$ are :
(a) $(a-2b-4)(a-2b+2)$
(b) $(a-b+2)(a-4b-4)$
(c) $(a+2b-4)(a+2b+2)$
(d) none of the above

67. The factors of $(a-b)^3+(b-c)^3+(c-a)^3$ are :
(a) $(a+b+c)(a^2+b^2+c^2+ab+bc+ac)$
(b) $(a+b+c)(a^2+b^2+c^2-ab-bc-ac)$
(c) $3(a-b)(b-c)(c-a)$
(d) $(a-b)(b-c)(c-a)$

68. If $a+b+c=0$, then $a^2+b^2+c^2$ is :
(a) 0 (b) $2(a^2-bc)$
(c) $-4(ab+bc+ca)$ (d) $-2(ab+bc+ca)$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

69. Which of the following statements are true ?
1. LCM of $10a^2bc, 15abc, 20ab^2c^2$ is $120a^2b^2c^2$
2. HCF of $(x^2 - 6x + 9)$ and $(x^3 - 27)$ is $(x - 3)$
3. HCF of $(6x^2 - 7x - 3)$ and $(2x^2 + 11x - 21)$ is $2x - 3$.

(a) 1, 2, 3 (b) 1, 2
(c) 1, 3 (d) 2, 3

70. The LCM of $(2x^2 - 3x - 2)$ and $(x^3 - 4x^2 + 4x)$ is :
(a) $x(2x^2 + 1)(x^2 + 2)$ (b) $x(2x + 1)(x - 2)^2$
(c) $x(2x^2 + 1)(x - 1)^2$ (d) $x(2x + 1)(x^2 - 1)$

71. The G.C.D. of $2(x^2 - y^2)$ and $5(x^3 - y^3)$ is :
(a) $2(x^2 - y^2)$ (b) $x + y$
(c) $x - y$ (d) $10(x^2 - y^2)(x^2 + y^2 + xy)$

72. The GCD of $(x^4 - 4x^2 + 3)$ and $(x^4 - x^2 - 6)$ is :
(a) $(x - 1)$ (b) $(x + 1)$
(c) $(x^2 - 1)$ (d) $(x^2 - 3)$

73. The HCF of $(x^2 - 4)$, $(x^2 - 5x - 6)$ and $(x^2 + x - 6)$ is :
(a) 1 (b) $x + 2$
(c) $x - 2$ (d) $(x^2 + x - 6)$

74. The GCD of $[x^2 - ax - (a + 1)]$ and $[ax^2 - x - (a + 1)]$ is :
(a) $(x - 1)$ (b) $(x + 1)$
(c) $(x + a + 1)$ (d) $(x - a - 1)$

75. LCM of the polynomials P and Q, where
$P = (x - 2)(x + 1)^2(x + 3)^2$
$Q = (x + 1)^2(x + 3)(x + 4)$ is given by :
(a) $(x - 2)(x + 1)(x + 3)^2(x + 4)$
(b) $(x - 2)(x + 4)(x + 3)^2(x + 1)^2$
(c) $(x - 2)(x + 1)^2(x + 3)(x + 4)$
(d) $(x + 1)(x - 2)(x + 3)(x + 4)$

76. The LCM of $(a^3 + b^3)$ and $(a^4 - b^4)$ is :
(a) $(a^3 + b^3)(a^2 + b^2)(a - b)$
(b) $(a^3 + b^3)(a^2 + b^2)(a + b)$
(c) $(a^3 + b^3)(a^2 + b^2 + ab)(a + b)$
(d) $(a^3 + b^3)(a^2 - b^2)(a - b)$

77. The HCF of $(x^4 - 1)$ and $(x^3 + x^2 + x + 1)$ is :
(a) $(x^2 - 1)(x^2 + 1)$ (b) $(x^2 + 1)(x + 1)$
(c) $(x + 1)(x^2 - 1)$ (d) $(x + 1)(x^2 + 1)(x^3 + 1)$

78. The GCD of $(2x^2 - 4x)$, $(3x^4 - 12x^2)$ and $(2x^5 - 2x^4 - 4x^3)$ is :
(a) $x(x - 2)$ (b) $2x(x - 2)$
(c) $2x(2 - x)$ (d) $2x(2 + x)$

79. The HCF of two expressions P and Q is 1. Their LCM is :
(a) $(P + Q)$ (b) $(P - Q)$
(c) PQ (d) $(P)^Q$

80. The LCM of $(x + 2)^2(x - 2)$ and $(x^2 - 4x - 12)$ is :
(a) $(x + 2)(x - 2)^2$ (b) $(x - 2)(x + 2)(x - 6)$
(c) $(x - 2)(x + 2)$ (d) $(x - 2)(x - 6)(x + 2)^2$

81. The HCF of $a^2 - ab - 2b^2$ and $2a^2 - ab - b^2$ is :
(a) $(a + b)$ (b) $(a - b)$
(c) $(2a - 3b)$ (d) none of these

82. HCF and LCM of $a^2b^3c^4$ and $a^5b^4c^3$ are :
(a) $a^2b^3c^4$, $a^5b^4c^3$ (b) $a^2b^3c^3$, $a^2b^3c^4$
(c) $a^2b^3c^3$, $a^5b^4c^4$ (d) $a^5b^3c^3$, $a^5b^4c^3$

Directions for question number 83–86 : Express each of the following as a rational expressions.

83. $\frac{(x + 3)}{(x - 2)} - \frac{(x + 1)}{(x - 3)}$:
(a) $\frac{x - 7}{(x^2 + 5x - 6)}$ (b) $\frac{7 - x}{(x^2 - 5x + 6)}$
(c) $\frac{x - 7}{2x^2 - 5x + 6}$ (d) $\frac{x - 7}{x^2 - 5x + 6}$

84. $\frac{x + 1}{x - 1} + \frac{x - 1}{x + 1}$:
(a) $\frac{1}{2}\frac{(x^2 + 1)}{(x^2 - 1)}$ (b) $\frac{(x^2 + 1)}{(x^2 - 1)}$
(c) $\frac{2(x^2 + 1)}{(x^2 - 1)}$ (d) $\frac{x^2 - 1}{x^2 + 1}$

85. $\frac{x^2 - 5x + 6}{x^2 - 9x + 20} + \frac{x^2 - 3x + 2}{x^2 - 5x + 4}$:
(a) $\frac{(x - 4)}{(x - 3)}$ (b) $\frac{(x - 3)}{(x - 5)}$
(c) $\frac{(x - 2)}{(x - 5)}$ (d) $\frac{(x - 5)}{(x - 3)}$

86. Sum of $\frac{2x^2 + 3}{x - 3}$ and its reciprocal is :
(a) $\frac{4x^4 + 13x^2 - 6x + 18}{2x^3 - 6x^2 + 3x - 9}$ (b) $\frac{4x^4 - 13x^2 + 6x - 18}{3x^2 - 6x^3 + 3x + 9}$
(c) $\frac{4x^4 - 13x^2 - 6x + 18}{4x^3 - 6x^2 + 3x - 9}$ (d) none of these

Directions for question number 87-89 : Express the following in the lowest terms.

87. $\frac{(x - 3)(x^2 - 5x + 4)}{(x - 4)(x^2 - 2x - 3)}$:
(a) $\frac{x - 1}{x + 1}$ (b) $\frac{x + 1}{x - 1}$
(c) $\frac{x - 1}{x - 2}$ (d) $\frac{x + 1}{x + 3}$

88. $\left(\frac{2x^2 + 1}{x - 1} + \frac{x - 1}{x + 1}\right) \times \left(\frac{x^2 - 1}{2x}\right)$:
(a) $\frac{2x^2 + 3x}{2}$ (b) $\frac{2x^3 + 3x^2 - x + 2}{2x}$
(c) $\frac{2x^3 - 3x^2 + x - 2}{2x}$ (d) none of these

89. $\sqrt{\frac{(x^2 + 3x + 2)(x^2 + 5x + 6)}{x^2(x^2 + 4x + 3)}}$:
(a) $\frac{x + 2}{2}$ (b) $\frac{x + 2}{x}$
(c) $\frac{x - 2}{x}$ (d) $\frac{x + 4}{x + 3}$

90. Simplify $\frac{1}{(a - b)(a - c)} + \frac{1}{(b - c)(b - a)} + \frac{1}{(c - a)(c - b)}$:
(a) 0 (b) 1
(c) 3 (d) none of these

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

91. Express $\frac{1}{(1-x)}+\frac{1}{(1+x)}+\frac{2}{(1+x^2)}+\frac{4}{(1+x^4)}$ as a rational expression :

(a) 1 (b) $\frac{1-x^4}{1+x^4}$

(c) $\frac{8}{1-x^8}$ (d) none of these

LINEAR EQUATIONS

Equation : A statement of equality which involves literal number(s) is called an equation.

e.g., $4x=12$, $4+x=10$, $7-2x=5$ etc.

NOTE The literal numbers involved in each equation are called its variables (unknowns). Usually the variables are denoted by letters towards the end of English alphabet, *e.g.*, x, y, z, u, v, w, etc.

Linear Equation : An equation in which the highest power of the variables involved is one, is called a linear equation. *e.g.*, $x+y=10$, $7x=21$, $\frac{x}{3}=8$.

Linear Equation in one Variable : An equation containing only one variable (literal) with highest power 1 is called a linear equation in one variable. *e.g.*, $17x=51$, $23+x=30$, $5y=30$, etc.

Linear Equation in two Variables : An equation of the form $ax+by=c$, where a, b, c are real numbers is called a linear equation in two variables x and y.

The graph of a linear equation $ax+by=c$ is a straight line.

$3x+2y=18$, is an example of a linear equation in two variables.

- The value of the variables that satisfy the equation is called the solution (or solution set) of the equation.

An equation can be solved by using the following rules :

(i) The same number may be added to both sides of the equation.

(ii) The same number may be subtracted from both sides of the equation.

(iii) Both sides of the equation may be multiplied by same non zero number.

(iv) Both sides of the equation may be divided by same non-zero number.

(v) A term may be transposed from one side of the equation to the other side, but its sign will change. This is called rule of transposition.

Important points about linear equation in two variables:

(i) The graph of an equation of the type $x=k$ (where k is a constant) is a straight line parallel to the y-axis at a distance of k units from the y-axis.

(ii) The graph of an equation of the type $y=k$ (where k is a constant) is a straight line parallel to the x-axis at a distance of k units from the x-axis.

(iii) The point of intersection of the two lines gives the solution of the two equations.

(iv) A single linear equation in two variables has infinite no. of solutions.

Simultaneous Linear Equations : Two or more linear equations in two variables form a system of linear simultaneous equations. *e.g.*, $a_1x+b_1y=c_1$ and $a_2x+b_2y=c_2$

Consistent System : A system consisting of two simultaneous linear equations is said to be consistent, if it has at least one solution.

Inconsistent System : A system consisting of two simultaneous linear equations is said to be inconsistent, if it has no solution.

If there are two simultaneous equations :

$a_1x+b_1y=c_1$ and $a_2x+b_2y=c_2$ then

Unique solution	Infinite solution	No solution
$\frac{a_1}{a_2}\neq\frac{b_1}{b_2}$	$\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$	$\frac{a_1}{a_2}=\frac{b_1}{b_2}\neq\frac{c_1}{c_2}$
(Independent equation)	(Dependent equation)	(Inconsistent equation)

- The homogeneous system has a non-zero solution only when $\frac{a_1}{a_2}=\frac{b_1}{b_2}$ and in this case, the system has an infinite number of solution.
- A system of equations has unique solution, when only one variable satisfies the equation.
- For a system of equations a unique solution is possible only when the number of variables is equal to or less than number of independent and consistent equations.

e.g., $2x+3y=5$ and $7x+5y=20$

and $2x+3y=5$, $7x+5y=12$, $5x+8y=13$

- The equation of the type $ax+by=c$ and $kax+kby=kc$ are known as dependent equations.

e.g., $2x+y=11$ ...(i)

$6x+3y=33$ ...(ii)

Here eq. (ii) is the multiple of eq. (i). Thus, there are basically both the equations same, *i.e.*, eq. (ii) is dependent on eq. (i). In this case there are infinite number of solutions.

- The equation of the type $ax+by=c$ and $kax+kby=lc$ are known as inconsistent equations.

e.g., $3x+2y=8$ ...(i)

and $6x+4y=6$ ...(ii)

Properties of graphs of $a_1x+b_1y=c_1$ and $a_2x+b_2y=c_2$

(i) intersecting if the system has a unique solution.

(ii) coincident if the system has infinite number of solution

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

(iii) parallel if the system has no solution.

Graphical representation of simultaneous equations :

(i) intersecting lines (unique solution)

e.g., $x + y = 4$ ...(i)

and $3x + 2y = 11$ ...(ii)

$\therefore \quad 3(x + y) = 3 \times 4 \Rightarrow 3x + 3y = 12$

$\therefore \quad 3x + 3y = 12$

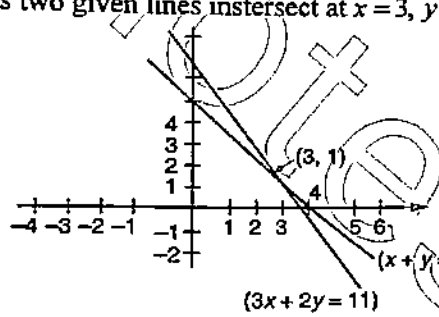
$3x + 2y = 11$

$y = 1$

$\therefore \quad x + y = 4 \Rightarrow x + 1 = 4 \Rightarrow x = 3$

$\therefore \quad x = 3$ and $y = 1$

Thus two given lines instersect at $x = 3$, $y = 1$.

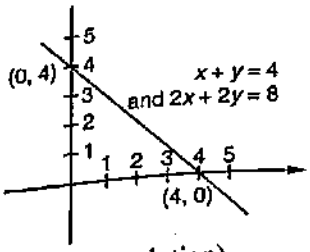


(ii) Coincident lines (infinite number of solutions)

e.g., $x + y = 4$ ...(i)

$2x + 2y = 8$ ...(ii)

The two equations are dependent, therefore the graph of two lines will coincide.

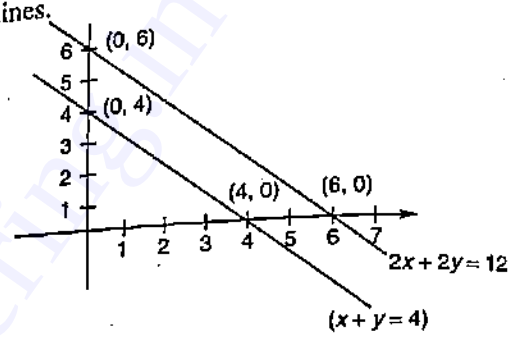


(iii) Parallel lines (no solution)

e.g., $x + y = 4$ ...(i)

$2x + 2y = 12$ ...(ii)

Since the system of equations is inconsistent therefore there is no any solution and in this case we obtain two different parallel lines.



Algebraic Methods of Solving Simultaneous Equations in Two Variables.

- Elimination method
  - Substitution method
  - Comparison method
  - Elimination method or Addition or Subtraction of Equations
- Cross multiplication method (Cramer's rule)

EXAMPLE 1 *Solve the following equations* $3x + 5y = 9$ *and* $2x + 3y = 4$

(a) by substitution method

(b) by elimination method

(c) by comparison method

SOLUTION (a) $3x + 5y = 9$ ...(i)

$2x + 3y = 4$ ...(ii)

Express y in terms of x from Eq. (i)

$$y = \frac{9 - 3x}{5} \quad \text{...(iii)}$$

Substitute this value of y in Eq. (ii)

$$2x + 3\left(\frac{9 - 3x}{5}\right) = 4$$

$\Rightarrow \quad 10x + 27 - 9x = 20 \Rightarrow x = -7$

Substitute $x = -7$ in Eq. (iii)

$\therefore \quad x = -7, \quad y = 6.$

NOTE We can express x in terms of y for one equation and substitute this value in other.

(b) $3x + 5y = 9$ ...(i)

$2x + 3y = 4$ ...(ii)

Multiply Eq. (i) by 3 and (2) by 5.

$9x + 15y = 27$ ...(iii)

$10x + 15y = 20$ ...(iv)

$-x + 0 = 7$

$x = -7$

$\Rightarrow$ Substitute $x = -7$ in Eq. (i)

$3(-7) + 5y = 9$

$\Rightarrow \quad -21 + 5y = 9$

$\Rightarrow \quad y = 6$

$\therefore \quad x = -7$ and $y = 6$

NOTE We can eliminate x by multiplying (i) by 2 and (ii) by 3 and then subtract the resulting equations.

(c) $3x + 5y = 9$ ...(i)

$2x + 3y = 4$ ...(ii)

From Eq. (i) $x = \frac{9 - 5y}{3}$ ...(iii)

and from Eq. (ii) $x = \frac{4 - 3y}{2}$ ...(iv)

Comparing Eq. (iii) and (iv)

$$\frac{9 - 5y}{3} = \frac{4 - 3y}{2}$$

$\Rightarrow \quad 2(9 - 5y) = 3(4 - 3y)$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$$18 - 10y = 12 - 9y$$

$$\Rightarrow \quad y = 6 \qquad \ldots(v)$$

$\Rightarrow$ Substitute the value of y in Eq. (iii)

$$x = \frac{9 - 5(6)}{3} = \frac{-21}{3} = -7$$

$\therefore \quad x = -7$ and $y = 6$

NOTE We can compare the value of y in terms of x.

EXAMPLE 2 *Solve the equations* $3x + 5y = 9$ *and* $2x + 3y = 4$ *by cross multiplication method.*

SOLUTION If there are two simultaneous equations

$a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$, where $c_2 \neq 0$

then, $$\frac{x}{b_1c_2 - b_2c_1} = \frac{y}{c_1a_2 - c_2a_1} = \frac{-1}{a_1b_2 - a_2b_1}$$

$$\left[\frac{x}{b_1 \quad c_1} = \frac{y}{c_1 \quad a_1} = \frac{-1}{a_1 \quad b_1}\right]$$
$$\quad b_2 \quad c_2 \quad\quad c_2 \quad a_2 \quad\quad a_2 \quad b_2$$

Here $a_1 = 3 \quad b_1 = 5 \quad c_1 = 9$

$a_2 = 2 \quad b_2 = 3 \quad c_2 = 4$

$$\therefore \quad \frac{x}{(20 - 27)} = \frac{y}{(18 - 12)} = \frac{-1}{(9 - 10)}$$

$$\Rightarrow \quad \frac{x}{-7} = \frac{y}{6} = 1$$

$$\Rightarrow \quad x = -7 \quad \text{and} \quad y = 6.$$

EXAMPLE 3 *Find the correct choice.*

(A) $5x + 2y = 16$, $7x - 4y = 2$ — (P) *no solution*

(B) $5x + 2y = 16$, $3x + \frac{6}{5}y = 2$ — (Q) *unique solution*

(C) $5x + 2y = 16$, $\frac{15}{2}x + 3y = 24$ — (R) *infinite number of solution*

(a) $A - P, B - Q, C - R$ (b) $A - R, B - P, C - Q$

(c) $A - Q, B - P, C - R$ (d) $A - Q, B - R, C - P$

SOLUTION (A) $\frac{5}{7} \neq \frac{2}{-4}$.

$\therefore$ The system of equations has unique solution.

(B) $$\frac{5}{3} = \frac{2}{6/5} \neq \frac{16}{2}.$$

$$\Rightarrow \quad \frac{5}{3} = \frac{5}{3} \neq \frac{8}{1}$$

$\therefore$ The system of equations has no solution.

(C) $$\frac{5}{15/2} = \frac{2}{3} = \frac{16}{24}$$

$$\Rightarrow \quad \frac{2}{3} = \frac{2}{3} = \frac{2}{3}$$

$\therefore$ The system of equations has infinite number of solution.

EXAMPLE 4 *Solve for* x *and* y

$$4(x - 3) = 3(y + 3)$$
$$17 - 2x = 8 - (2y + 1)$$

SOLUTION $4(x - 3) = 3(y + 3)$

$$\Rightarrow \quad 4x - 3y = 21 \qquad \ldots(1)$$

and $17 - 2x = 8 - (2y + 1)$

$$\Rightarrow \quad x - y = 5 \qquad \ldots(2)$$

Multiply Eq. (2) by 3 and subtract from (1)

$$4x - 3y = 21$$
$$3x - 3y = 15$$
$$- \quad + \quad -$$
$$x = 6$$

Substitute $x = 6$ in Eq. (2), we get $y = 1$

$\therefore \quad x = 6$ and $y = 1$.

EXAMPLE 5 *Solve for* x *and* y

$$0.04x + 0.02y = 5$$
$$0.5(x - 2) - 0.4y = 29$$

SOLUTION $0.04x + 0.02y = 5$

$$\Rightarrow \quad 4x + 2y = 500 \quad \text{(multiply both sides by 100)}$$

$$\Rightarrow \quad 2x + y = 250 \qquad \ldots(1)$$

and $0.5(x - 2) - 0.4y = 29$

$$\Rightarrow \quad 5(x - 2) - 4y = 290 \quad \text{(multiply both sides by 10)}$$

$$\Rightarrow \quad 5x - 4y = 300 \qquad \ldots(2)$$

Multiply Eq. (1) by 4 and add to Eq. (2)

$$8x + 4y = 1000$$
$$5x - 4y = 300$$
$$13x = 1300$$

$$\Rightarrow \quad x = 100$$

Substitute $x = 100$, in Eq. (1), we get

$$y = 50$$

$\therefore \quad x = 100$ and $y = 50$.

EXAMPLE 6 *Solve for* x *and* y

$$\frac{3x - y}{5} = 2y - 1, \quad \frac{3x}{8} - \frac{y}{4} = \frac{1}{2}$$

SOLUTION $3x - y = 5(2y - 1)$

$$\Rightarrow \quad 3x - 11y = -5 \qquad \ldots(1)$$

and $$\frac{3x}{8} - \frac{y}{4} = \frac{1}{2}$$

$$\Rightarrow \quad 3x - 2y = 4 \qquad \ldots(2)$$

Subtract Eq. (2) from (1)

$$3x - 11y = -5$$
$$3x - 2y = 4$$
$$- \quad + \quad -$$
$$-9y = -9 \Rightarrow y = 1$$

Substitute $y = 1$ in Eq. (2), we get

$$x = 2$$

$\therefore \quad x = 2$ and $y = 1$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

EXAMPLE 7 *Solve for x and y*

$$\frac{2}{x}+\frac{3}{y}=2,\ \frac{8}{x}-\frac{6}{y}=2$$

SOLUTION

$$\frac{2}{x}+\frac{3}{y}=2 \quad ...(1)$$

$$\frac{8}{x}-\frac{6}{y}=2 \quad ...(2)$$

Let $\frac{1}{x}=A$ and $\frac{1}{y}=B$

$\therefore$ $2A+3B=2$ ...(3)

and $8A-6B=2$ ...(4)

Multiply Eq. (3) by 2

$$4A+6B=4$$
$$8A-6B=2$$
$$12A=6 \Rightarrow A=1/2$$

Substitute $A=\frac{1}{2}$ in Eq. (1), we get

$$B=\frac{1}{3}$$

$\therefore$ $x=2$ and $y=3$.

EXAMPLE 8 *Solve for x and y*

$$\frac{2}{3x+2y}+\frac{3}{3x-2y}=\frac{17}{5}$$

$$\frac{5}{3x+2y}+\frac{1}{3x-2y}=2$$

SOLUTION Let $\frac{1}{3x+2y}=A$ and $\frac{1}{3x-2y}=B$

$\therefore$ $2A+3B=\frac{17}{5} \Rightarrow 10A+15B=17$ ...(1)

and $5A+B=2$ ...(2)

Multiply Eq. (2) by 2 and subtract from (1)

$$10A+15B=17$$
$$10A+2B=4$$
$$13B=13 \Rightarrow B=1$$

Substitute $B=1$ in Eq. (2), we get $A=\frac{1}{5}$

Now, $A=\frac{1}{3x+2y}=\frac{1}{5} \Rightarrow 3x+2y=5$ ...(3)

and $B=\frac{1}{3x-2y}=1 \Rightarrow 3x-2y=1$ ...(4)

$\therefore$ Add Eq. (3) to Eq. (4)

$$3x+2y=5$$
$$3x-2y=1$$
$$6x=6 \Rightarrow x=1$$

Substitute $x=1$ in Eq. (3), we get $y=1$

$\therefore$ $x=1$ and $y=1$.

INTRODUCTORY EXERCISE-13.2

Directions for question number 1–4 : Solve the following system of equations :

1. $x-y=0.9, \frac{11}{2(x+y)}=1$:
   (a) 3.2, 3.5 (b) 3.2, 2.3
   (c) 2.3, 2.5 (d) 4, 3.1

2. $\frac{x+y}{xy}=2, \frac{x-y}{xy}=6$:
   (a) $-\frac{1}{2},\frac{1}{4}$ (b) $\frac{1}{2},-\frac{1}{4}$
   (c) $\frac{1}{2},-\frac{3}{2}$ (d) $-\frac{1}{4},-\frac{3}{4}$

3. $\frac{1}{2x}-\frac{1}{y}=-1, \frac{1}{x}+\frac{1}{2y}=8$:
   (a) $\frac{2}{3},\frac{2}{5}$ (b) $\frac{1}{4},\frac{1}{5}$
   (c) $\frac{1}{6},\frac{1}{4}$ (d) $-\frac{1}{6},-\frac{1}{4}$

4. $\frac{2x}{a}+\frac{y}{b}=2, \frac{x}{a}-\frac{y}{b}=4$:
   (a) $\frac{2}{a},\frac{2}{b}$ (b) $2a,-2b$
   (c) $-2a, 2b$ (d) $\frac{a}{2},-\frac{b}{2}$

5. For what value of k, the following equations have no solutions?
   $$9x+4y=9$$
   $$7x+ky=5$$
   (a) 3 (b) 4.7
   (c) 28/9 (d) 9/28

6. For what value of k, the system of equations $kx+2y=2$ and $3x+y=1$ will be coincident?
   (a) 2 (b) 3
   (c) 5 (d) 6

7. For what value of k, will the equations $2x+3y-5=0$, $6x+ky-15=0$ have an infinite number of solutions?
   (a) 0 (b) 9
   (c) 2 (d) 3

8. The total cost of 8 buckets and 5 mugs is Rs. 92 and the total cost of 5 buckets and 8 mugs is Rs. 77. Find the cost of 2 mugs and 3 buckets.
   (a) Rs. 35 (b) Rs. 70
   (c) Rs. 30 (d) Rs. 38

9. The sum of two numbers is 8. If their sum is four times their difference, find the numbers.
   (a) 6, 2 (b) 7, 1
   (c) 5, 3 (d) 6, 3

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

10. The numerator of a fraction is 4 less than its denominator. If the numerator is decreased by 2 and the denominator is increased by 1, then the denominator is eight times the numerator. Find the fraction.
(a) 3/7 (b) 4/8
(c) 2/7 (d) 3/8

11. Points A and B are 90 km apart from each other on a highway. A car starts from A and another from B at the same time. If they go in the same direction, they meet in 9 hours and if they go in opposite directions, they meet in 9/7 hours. Find their speed (in km/h).
(a) 25, 45 (b) 20, 50
(c) 30, 40 (d) 35, 65

12. Priyanka has only 25 paise coins and 50 paise coins in her purse. If in all she has 40 coins totalling Rs. 12.75, how many coins of 25 paise does she have?
(a) 30 (b) 29
(c) 32 (d) 26

13. Astha has pens and pencils which together are 40 in number. If she has 5 more pencils and 5 less pens, the number of pencils would have become 4 times the number of pens. Find the original number of pens with Astha.
(a) 11 (b) 12
(c) 13 (d) 14

14. A father is three times as old as his son. After twelve years, his age will be twice as that of the age of his son then, find their present ages.
(a) 10 y, 30 y (b) 12 y, 36 y
(c) 6 y, 36 y (d) 124 y, 24 y

15. Five years ago, A was three times as old as B and ten years later, A shell be twice as old as B. What are the present ages of A and B (in years)?
(a) 45, 15 (b) 30, 40
(c) 50, 30 (d) 50, 20

16. Astha and Saumya each have certain number of oranges. Astha says to Saumya, "If you give me 10 of your oranges, I will have twice the number of oranges left with you. "Saumya replies, "If you give me 10 of your oranges, I will have the same number of oranges as left with you." Find the number of oranges with Astha and Saumya respectively.
(a) 60, 40 (b) 70, 50
(c) 60, 80 (d) 70, 90

17. Shubham travels 760 km to his home partly by train and partly by car. He takes 8 hours if he travels 160 km by train and the rest by car. He takes 12 minutes more if he travels 240 km by train and the rest by car. Find the speed of the train and the car respectively (in km/hr.).
(a) 40, 80 (b) 60, 120
(c) 80, 100 (d) 100, 120

18. A person invested some amount at the rate of 12% simple interest and some other amount at the rate of 10% simple interest. He received yearly interest of Rs. 130. But if he had interchanged the amounts invested he would have received Rs. 4 more as interest. How much amount did he invest at 10% simple interest?
(a) Rs. 700 (b) Rs. 500
(c) Rs. 800 (d) Rs. 400

19. The solution of the equations
$\frac{3x-y+1}{3} = \frac{2x+y+2}{5} = \frac{3x+2y+1}{6}$ given by :
(a) $x = 2, y = 1$ (b) $x = 1, y = 1$
(c) $x = -1, y = -1$ (d) $x = 1, y = 2$

20. The course of an enemy submarine as plotted on a set of rectangular axes gives the equation $2x + 3y = 5$. On the same axes, the course of destroyer is indicated by $x - y = 10$. The point (x, y) at which the submarine can be destroyed is :
(a) $(-3, 7)$ (b) $(7, -3)$
(c) $(-7, 3)$ (d) $(3, -7)$

21. Let a, b, c be the positive numbers. The following system of equations in x, y and z.
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$
$$-\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1, \text{ has}$$
(a) no solution
(b) unique solution
(c) infinitely many solutions
(d) finitely many solutions

22. If the system of equations, $x - ky - z = 0, kx - y - z = 0$, $x + y + z = 0$, has a non zero solution, then the possible values of k are :
(a) $-1, 2$ (b) $1, 2$
(c) $0, 1$ (d) $-1, 1$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

EXERCISE

LEVEL 1

1. If $a+b+c=13$, what is the maximum value of $(a-3)(b-2)(c+1)$?
(a) 26 (b) 27
(c) 30 (d) 19

2. If $abcd = 81$, what is the minimum value of $a+b+c+d$?
(a) 18 (b) 9
(c) 12 (d) 36

3. Which one of the following is correct?
(a) $x+\frac{1}{x} \le -2$ (b) $x+\frac{1}{x}=0$
(c) $x+\frac{1}{x} \ge 2$ (d) both (a) and (c)

4. If a, b, c are all distinct positive numbers, then $(a+b)(b+c)(c+a)$:
(a) less than 4 (b) less than $4abc$
(c) greater than $8abc$ (d) greater than $(abc)^3$

5. If a, b, c are the distinct positive numbers then $(a+b+c)(ab+bc+ac)$ is :
(a) greater than $9abc$ (b) less than $8abc$
(c) equal to $10abc$ (d) greater than $25abc$

6. If $x+y=25$ and $x^2y^3+y^2x^3=25$, what is the value of xy?
(a) 0 (b) ± 1
(c) 5 (d) 4

7. If x and y are both positive, then the minimum value of $(x+y)\left(\frac{1}{x}+\frac{1}{y}\right)$ is :
(a) 0 (b) 1
(c) 2 (d) 4

8. If a, b, c and d are four positive numbers such that $a+b+c+d=4$, then what is the maximum value of $(a+1)(b+1)(c+1)(d+1)$?
(a) 32 (b) 8
(c) 16 (d) 81

9. If x, y, z are three distinct positive real numbers such that $x+y+z=1$, then the value of $\left(\frac{1}{x}-1\right)\left(\frac{1}{y}-1\right)\left(\frac{1}{z}-1\right)$ is :
(a) 16 (b) 8
(c) 4 (d) 1

10. If x, y, z are real numbers such that $x+y+z=4$ and $x^2+y^2+z^2=6$, then x, y, z lie in :
(a) $\left[\frac{3}{2}, 2\right]$ (b) $\left[\frac{2}{3}, 2\right]$
(c) $\left[0, \frac{2}{3}\right]$ (d) none of these

11. If $x=7+4\sqrt{3}$ and $xy=1$, then the value of $\left(\frac{1}{x^2}+\frac{1}{y^2}\right)$ is :
(a) 194 (b) 57
(c) $85\sqrt{3}$ (d) $\frac{7+4\sqrt{3}}{7-4\sqrt{3}}$

12. If a, b, c are positive real numbers, the least value of $(a+b+c)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$ is :
(a) 1 (b) 9
(c) 12 (d) none of these

13. If $a>0$, $b>0$ and $ab=1$, then the least value of the expression $(1+a)(1+b)$ is :
(a) 2 (b) 4
(c) 1 (d) 1/4

14. If a, b, c are all positive and not equal, then the value of $\frac{(a+b+c)(ab+bc+ca)}{abc}$ is :
(a) less than 9 (b) greater than 9
(c) less than or equal to 9 (d) can't be determined

15. If a, b, c are all positive, then the minimum value of the expression $\frac{(a^2+a+1)(b^2+b+1)(c^2+c+1)}{abc}$ is :
(a) 3 (b) 9
(c) 27 (d) 1

16. If $1 \le x \le 3$ and $2 \le y \le 4$, what is the maximum value of $\left(\frac{x}{y}\right)$?
(a) $\frac{2}{3}$ (b) 4
(c) $\frac{3}{2}$ (d) 2

17. If $a+b+c=3$, $a^2+b^2+c^2=6$ and $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=1$ where a, b, c are all non-zero, then abc is :
(a) $\frac{1}{3}$ (b) $\frac{2}{3}$
(c) $\frac{3}{2}$ (d) 1

18. $2^x=4^y=8^z$ and $xyz=288$, then value of $\frac{1}{2x}+\frac{1}{4y}+\frac{1}{8z}$ is :
(a) $\frac{11}{12}$ (b) $\frac{11}{96}$
(c) $\frac{29}{96}$ (d) none of these

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

19. If $a^x = b, b^y = c, c^z = a$, then the value of xyz is :
(a) 3 (b) 0
(c) 1 (d) can't be determined

20. Let $x, y \in N$ and $7x + 12y = 220$. The number of solutions is :
(a) 1 (b) 2
(c) 3 (d) infinitely many

21. If $a^x = (x + y + z)^y$, $a^y = (x + y + z)^z$, $a^z = (x + y + z)^x$, then :
(a) $3(x + y + z) = a$ (b) $2a = x + y + z$
(c) $x + y + z = 0$ (d) $x = y = z = \frac{a}{3}$

22. If $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, then $xy + yz + zx$ is equal to :
(a) $\frac{(a+b+c)^2}{x^2+y^2+z^2}$
(b) $\frac{x^2(a+b+c)^2 - a^2(x^2+y^2+z^2)}{2a^2}$
(c) $\frac{ax+by+cz}{(a+b+c)^2}$
(d) none of the above

23. In the farm-house of Bhuwan Kisan, there were total 14 deer and ducks. One day a thief entered secretely in the farm-house and after killing all the deer and ducks he took away all the stuff except their legs. When Bhuwan noticed it he has found there 38 legs of deer and ducks. The number of deer in the farm house was :
(a) 5 (b) 8
(c) 9 (d) 10

24. In an autofair at Pragati Maidan, New Delhi, there are 600 vehicles consisting of only two wheelers and four wheelers. A world renowned tyre agency visits the fair to survey the trye market which finds that there are total 2000 tyres are fixed in all the 600 vehicles, but without any stepney (or spare tyre). The number of two wheelers in the autofair is :
(a) 300 (b) 400
(c) 180 (d) 200

25. In the battle of Mahabharat Veer Arjun destroyed all the 125 raths (horse-carts). Out of which few were driven by 3 horses each and rest were driven by 5 horses each. Thus, he captured 125 damaged raths along with 575 horses. The raths which were driven by 5 horses belonged to the Kaurav and the raths which were driven by 3 horses belonged to the allied army of Kaurav. It is known that no any horse was injured in the battle. The number of raths originally belonged to the Kaurav (not to their allied army) is :
(a) 60 (b) 25
(c) 75 (d) 100

26. In an electric appliances shop there were some fans and coolers containing 3 blades and 4 blades each respectively along with a motor each. The cost of a fan is Rs. 1200 and that of cooler is Rs. 3000. One day due to short circuit the shop caught the fire. After extinguishing the fire, shopkeeper has estimated that 90 sets of motor bodies and 320 blades were burned and then nothing was left useful in his shop. What is the loss incurred from the coolers alone?
(a) Rs. 1,50,000 (b) Rs. 12,000
(c) Rs. 4800 (d) can't be determined

27. In the air talent show in the eve of independence day 90 aircraft (only Sukhoi and Mig-21) participated. It is known that for every one hour flight a Sukhoi and a Mig-21 consume 2 litres and 3 litres of fuel respectively. The price of fuel for Sukhoi is $ 10 per litre and price for the fuel of Mig-21 is $ 15 per litre. Thus all the aircrafts accounted $ 3050 for the fuel in one hour show. The number of Sukhoi planes is :
(a) 20 (b) 30
(c) 40 (d) 50

Directions for question number 28 to 30: *In the fruit market adjacent to IIM Lucknow, a fruit vendor has total 20 kg of fruits containing only mangoes and apples. The vendor told to Ritika, an IIM student, that one kg apple contains only 7 apples while 1 kg of mangoes contains 10 mangoes and it is known that all the mangoes and apples are uniform in size individually. The vendor has initially 176, pieces of apples and mangoes, in all. Ritika purchased 55 fruits to distribute equally among the group of 11 students, irrespective of the kind of fruit i.e., mango or apple, and then the vendor is left with only 13 kg of fruits.*

28. The number of apples remaining with the vendor when Ritika has purchased 7 kg of fruits, is :
(a) 21 (b) 35
(c) 80 (d) none of these

29. If the selling price of mangoes and apples be Rs. 35 per kg and Rs. 40 per kg respectively, then how many rupees Ritika has spent for the 7 kg of fruits?
(a) Rs. 262.5 (b) Rs. 255
(c) Rs. 270 (d) Rs. 155

30. If the vendor has received 16.66% profit in selling of mangoes and 33.33% profit in the selling of apples to Ritika, but for the rest of the selling he exchanges the profit per cent between mangoes and apples. What is his overall profit per cent for the rest of the fruits? (Data can be used from the previous question if necessary)
(a) 29.48% (b) 5.92%
(c) no any profit (d) can't be determined

31. One day my assistant Rajesh went to a stationary shop to purchase marker pens. He purchased the pens for Rs. 52 and he gave a Rs. 100 note to the shopkeeper. The shopkeeper offered Rajesh to take back Rs. 48 in the denominations of only Re. 1, Rs. 2 and Rs. 5.
So, Rajesh received total 26 coins from the shopkeeper. Minimum number of Re. 1 coins which Rajesh has with him if he has at least one coin of each denomination offered by the shopkeeper :
(a) 4 (b) 5
(c) 7 (d) 10

32. What is the solutions of the following simultaneous equations?
$x + y + z = 6, \quad x + 2y + 3z = 14, \quad x + 3y + z = 10$
(a) $x = 1, y = 2, z = 4$ (b) $x = 1, y = 2, z = 3$
(c) $x = 2, y = 1, z = 3$ (d) none of these

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Answers

INTRODUCTORY EXERCISE-13.1

1 (d)	2. (c)	3. (c)	4. (d)	5. (c)	6. (c)	7. (d)	8. (b)	9. (c)	10. (d)
11. (b)	12. (a)	13. (b)	14. (c)	15. (b)	16. (b)	17. (d)	18. (b)	19. (c)	20. (b)
21. (c)	22. (a)	23. (d)	24. (d)	25. (c)	26. (b)	27. (c)	28. (c)	29. (d)	30. (a)
31. (b)	32. (c)	33. (c)	34. (c)	35. (d)	36. (a)	37. (b)	38. (c)	39. (b)	40. (a)
41. (d)	42. (c)	43. (b)	44. (b)	45. (c)	46. (b)	47. (b)	48. (c)	49. (c)	50. (c)
51. (c)	52. (b)	53. (d)	54. (c)	55. (d)	56. (c)	57. (b)	58. (c)	59. (d)	60. (a)
61. (a)	62. (c)	63. (b)	64. (a)	65. (b)	66. (a)	67. (c)	68. (d)	69. (d)	70. (b)
71. (c)	72. (d)	73. (a)	74. (b)	75. (b)	76. (a)	77. (b)	78. (a)	79. (c)	80. (d)
81. (d)	82. (c)	83. (d)	84. (c)	85. (b)	86. (a)	87. (a)	88. (b)	89. (b)	90. (a)
91. (c)									

INTRODUCTORY EXERCISE-13.2

1 (b)	2. (a)	3. (c)	4. (b)	5. (c)	6. (d)	7. (b)	8. (a)	9. (c)	10. (a)
11. (c)	12. (b)	13. (c)	14. (b)	15. (d)	16. (b)	17. (c)	18. (a)	19. (b)	20. (b)
21. (d)	22. (d)								

LEVEL-1

1 (b)	2. (c)	3. (d)	4. (c)	5. (a)	6. (b)	7. (d)	8. (c)	9. (b)	10. (b)
11. (a)	12. (b)	13. (b)	14. (b)	15. (c)	16. (c)	17. (c)	18. (b)	19. (c)	20. (c)
21. (d)	22. (b)	23. (a)	24. (d)	25. (d)	26. (a)	27. (c)	28. (a)	29. (c)	30. (a)
31. (c)	32. (b)								

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Hints & Solutions

INTRODUCTORY EXERCISE 13.1

1. Total score $= x + y + 40$

2. Quotient $= \frac{z}{4}$

$\therefore \quad 3y \times \frac{z}{4} = \frac{3yz}{4}$

3. $6 \times p \times q \times r \times p \times p \times p = 6 \times p \times p \times p \times p \times q \times r$
$= 6 \times p^4 \times q \times r = 6p^4qr$

4. (i) $2x^2$ (ii) $-\frac{3}{x}$ (iii) $\frac{5}{x^2}$ (iv) 9

5. (i) $4xyz$ (ii) $-\frac{3}{x^2}$ (iii) $6xy^2$ (iv) $-5z^3$ (v) $6x$

6. $7 \times a + b \div 3 - c + b = 7a + \frac{b}{3} - c + b$

$\therefore$ (i) $7a$ (ii) $\frac{b}{3}$ (iii) $-c$ (iv) b

7. $\frac{-7xy^2z^3}{z^3} = -7xy^2$

9. $4x^3 \rightarrow$ degree 3
$2x^2y^3 \rightarrow$ degree $(2 + 3) = 5$
$6x^0y^0 \rightarrow$ degree 0
$\therefore$ the degree of whole expression is 5

10. $2x^2 + 3y^2 - 5xy + 5x^2 - y^2 + 6xy - 3x^2$
$= 2x^2 - 3x^2 + 5x^2 + 3y^2 - y^2 - 5xy + 6xy$
$= 4x^2 + 2y^2 + xy$

11. $(5x^2 + 3x - 1) - (3x^3 - 2x + 7)$
or
$$\begin{array}{r} 5x^2 + 3x - 1 \\ 3x^3 - 2x + 7 \\ - \quad + \quad - \\ \hline -3x^3 + 5x^2 + 5x - 8 \end{array}$$
(In subtraction we change the sign)

12. $\left(\frac{1}{2}a^2b - \frac{2}{3}ab^2 + 5\right) \times (6abc)$
$= \frac{1}{2}a^2b \times 6abc - \frac{2}{3}ab^2 \times 6abc + 5 \times 6abc$
$= 3a^3b^2c - 4a^2b^3c + 30abc$

13. $\left(\frac{1}{2}x^2 - \frac{1}{3}y^2\right) \times \left(\frac{1}{2}x^2 + \frac{1}{3}y^2\right)$
$= \frac{1}{2}x^2\left(\frac{1}{2}x^2 + \frac{1}{3}y^2\right) - \frac{1}{3}y^2\left(\frac{1}{2}x^2 + \frac{1}{3}y^2\right)$
$= \left(\frac{1}{2}x^2 \times \frac{1}{2}x^2 + \frac{1}{2}x^2 \times \frac{1}{3}y^2\right) - \frac{1}{3}y^2 \times \frac{1}{2}x^2 - \frac{1}{3}y^2 \times \frac{1}{3}y^2$
$= \frac{1}{4}x^4 + \frac{1}{6}x^2y^2 - \frac{1}{6}x^2y^2 - \frac{1}{9}y^4$
$= \frac{1}{4}x^4 - \frac{1}{9}y^4$

14. $\frac{36x^2y^5 + 42xy^3 - 24x^3y^2 - 12y^5}{-6y^2}$
$= \frac{36x^2y^5}{-6y^2} + \frac{42xy^3}{-6y^2} - \frac{24x^3y^2}{-6y^2} - \frac{12y^5}{-6y^2}$
$= -6x^2y^3 - 7xy + 4x^3 + 2y^3$

15. Since divisor is $x + 3$
$\therefore \quad x + 3 = 0 \rightarrow x = -3$
$\therefore$ put $x = -3$ in the expression
$3x^3 + 8x^2 - 6x + 1 = 3(-3)^3 + 8(-3)^2 - 6(-3) + 1$
$= -81 + 72 + 18 + 1 = 10$

16. Put $x = -3$ in the given expression
$ax^3 + 9x^2 + 4x - 10$
$\because \quad a(-3)^3 + 9(-3)^2 + 4(-3) - 10 = 5$
$\therefore \quad -27a + 81 - 12 - 10 = 5$
$\Rightarrow \quad -27a = -54 \Rightarrow a = 2$

17. Put $x = -1$, then
$(-1)^3 + a(-1)^2 - b(-1) - 6 = 0$
or $\quad -1 + a + b - 6 = 0$
or $\quad a + b - 7 = 0 \quad \ldots(1)$
and put $x = 2$, then
$(2)^3 + a(2)^2 - b(2) - 6 = 0$
$\Rightarrow \quad 8 + 4a - 2b - 6 = 0$
$\Rightarrow \quad 4a - 2b + 2 = 0 \quad \ldots(2)$
By solving Eq. (1) and (2)
$a + b - 7 = 0 \Rightarrow a + b = 7$
and $\quad 4a - 2b + 2 = 0 \Rightarrow 2a - b = -1$
$\therefore \quad 3a = 6 \Rightarrow a = 2$
Substituting $a = 2$ in Eq. (1) we get $b = 5$.
$\therefore \quad a = 2$ and $b = 5$

18. $\because A^2 - B^2 = (A + B) \cdot (A - B)$
$\therefore (5x^2 + 14x + 2)^2 - (4x^2 - 5x + 7)^2$
$= (5x^2 + 14x + 2 + 4x^2 - 5x + 7)$
$\times (5x^2 + 14x + 2 - 4x^2 + 5x - 7)$
$= 9(x^2 + 19x - 5)(x^2 + x + 1)$
$\therefore \frac{(5x^2 + 14x + 2)^2 - (4x^2 - 5x + 7)^2}{(x^2 + x + 1)} = 9(x^2 + 19x - 5)$
$\therefore \quad q = 9(x^2 + 19x - 5)$ and $r = 0$.

19. Putting $x = -1$ in the given polynomial, we get
$(-1)^3 - 6(-1) + 7 = -1 + 6 + 7 = 12$ $(\because x + 1 = 0)$

20. Putting $x = 2$, we get $(\because x - 2 = 0)$
$(2)^4 - 3(2)^3 + 2(2)^2 - 5(2) + 7$
$= 16 - 24 + 8 - 10 + 7 = -3$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yourshahbaab.wordpress.com
For More Visit : www.LearnEngineering.in

21. Since divisor is $x^2 + 2$

$\therefore$ Substitute $x^2 = -2$, in the given expression

$x^3 + 5x^2 + 10k$ gives the remainder $-2x$

$$(-2)x + 5(-2) + 10k = -2x$$

$$\Rightarrow \quad k = 1$$

22. Putting $x = -1$, we get

$$(-1)^{11} + 1 = -1 + 1 = 0$$

23. Putting $x = -3$, we get

$$5(-3)^3 + 5(-3)^2 - 6(-3) + 9 = -135 + 45 + 18 + 9 = -63$$

25. Best way is that to consider some appropriate value of x (say $x = 1$), then check through options :

Let $x = 1$, then

$x^3 - 2x^2 + px - q = 1 - 2 + p - q \rightarrow$ dividend

and $\quad x - 6 = 1 - 6 = -5 \rightarrow$ remainder

and $\quad x^2 - 2x - 3 = 1 - 2 - 3 = -4 \rightarrow$ divisor

$\therefore$ now assume choice (c) as $p = -2$ and $q = 6$ then

$$\frac{-1 + p - q}{-4} = \frac{-1 + (-2) - 6}{-4} = \frac{-9}{-4} \rightarrow \text{Remainder} - 5$$

NOTE To avoid negative remainder you can assume $x = 7, 8, \ldots$, etc.)

Alternatively : Divide the given expression as conventionally and then compare the actual remainder with obtained remainder.

$$x^2 - 2x - 3 \overline{)x^3 - 2x^2 + px - q} (\times\times\times\times$$
$$\underline{\times \quad \times \quad \times \quad \times}$$

$$\text{Remainder} = (p + 3)x - q$$

$$\therefore \quad (p + 3)x - q = x - 6$$

$$\therefore \quad p + 3 = 1 \quad \text{and} \quad q = 6$$

$$\Rightarrow \quad p = -2 \quad \text{and} \quad q = 6$$

26. Put $x = 2$ in the given expression as :

$$x^2 + 3qx - 2q = 0$$

$$(2)^2 + 3q(2) - 2q = 0$$

$$\Rightarrow \quad 4 + 6q - 2q = 0 \quad \Rightarrow \quad 4 + 4q = 0$$

$$\Rightarrow \quad q = -1$$

27. Put $x = -2$ in the given polynomial.

$$(-2)^3 + 6(-2)^2 + 4(-2) + k = 0$$

$$\Rightarrow \quad k = -8$$

28. The best way is check by options. Let us consider choice (c). $(x - 2)$ is a factor of the given expression then at $x = 2$ the given expression must be zero.

$$\therefore \quad (2)^4 - 5(2)^3 + 5(2)^2 - 10(2) + 24 = 0$$

Hence, $(x - 2)$ is the factor of $x^4 - 5x^3 + 5x^2 - 10x + 24$.

29. Since $(x + k)$ is a factor of each one of the given expression, $x = -k$ will make each zero.

$$\therefore \quad k^2 - pk + q = k^2 - lk + m = 0$$

$$\Rightarrow \quad k = \frac{m - q}{l - p}$$

30. Substitute $x = a$ in the given expression since $(x - a)$ is a factor, which will make it equal to zero.

$$\therefore \quad (a)^3 - 3a^3 + 2a^3 + b = 0 \quad \Rightarrow \quad b = 0$$

31. Go through options

32. Go through options

33. Put $x = -1$ in the given expression

34. Put $x = 1$ in the given expression

36.
$$\left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2x \cdot \frac{1}{x}$$

$$(2)^2 = x^2 + \frac{1}{x^2} + 2 \quad \left(\text{Putting } x + \frac{1}{x} = 2\right)$$

$$\Rightarrow \quad x^2 + \frac{1}{x^2} = 2$$

37.
$$\left(x - \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} - 2x \cdot \frac{1}{x}$$

$$(4)^2 = x^2 + \frac{1}{x^2} - 2$$

$$\Rightarrow \quad x^2 + \frac{1}{x^2} = 18$$

38.
$$\left(x + \frac{1}{x}\right)^3 = x^3 + \frac{1}{x^3} + 3x \cdot \frac{1}{x}\left(x + \frac{1}{x}\right)$$

$$\left(x + \frac{1}{x}\right)^3 = x^3 + \frac{1}{x^3} + 3\left(x + \frac{1}{x}\right)$$

$$\Rightarrow \quad (2\sqrt{3})^3 = x^3 + \frac{1}{x^3} + 3(2\sqrt{3})$$

$$\Rightarrow \quad x^3 + \frac{1}{x^3} = 18\sqrt{3}$$

39.
$$\left(x + \frac{1}{x}\right)^3 = x^3 + \frac{1}{x^3} + 3x \cdot \frac{1}{x}\left(x + \frac{1}{x}\right)$$

$$(3)^3 = x^3 + \frac{1}{x^3} + 3(3)$$

$$\Rightarrow \quad x^3 + \frac{1}{x^3} = 18$$

40.
$$\left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2 \cdot x \cdot \frac{1}{x}$$

$$(2)^2 = x^2 + \frac{1}{x^2} + 2$$

$$x^2 + \frac{1}{x^2} = 2$$

$$\text{Again} \left(x^2 + \frac{1}{x^2}\right)^3 = x^6 + \frac{1}{x^6} + 3x^2 \cdot \frac{1}{x^2}\left(x^2 + \frac{1}{x^2}\right)$$

$$(2)^3 = x^6 + \frac{1}{x^6} + 3(2)$$

$$\Rightarrow \quad x^6 + \frac{1}{x^6} = 2$$

41.
$$\left(x - \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} - 2x \cdot \frac{1}{x}$$

$$\left(x - \frac{1}{x}\right)^2 = 6 - 2$$

$$\left(x - \frac{1}{x}\right)^2 = 4$$

$$\Rightarrow \quad \left(x - \frac{1}{x}\right) = \pm 2$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

42. $\left(x-\frac{1}{x}\right)^3 = x^3 - \frac{1}{x^3} - 3x\cdot\frac{1}{x}\left(x-\frac{1}{x}\right)$

$\left(x-\frac{1}{x}\right)^3 = 36 - 3\left(x-\frac{1}{x}\right)$

Let $\left(x-\frac{1}{x}\right) = P$

then $P^3 = 36 - 3P$

or $P^3 + 3P = 36 \Rightarrow P^3 + 3P - 36 = 0$

Clearly $P = 3$ satisfies the equation.

$\therefore$ $x - \frac{1}{x} = 3$

Alternatively: Go through options

43. $\left(x+\frac{1}{x}\right)^3 = x^3 + \frac{1}{x^3} + 3x\cdot\frac{1}{x}\left(x+\frac{1}{x}\right)$

$\Rightarrow \left(x+\frac{1}{x}\right)^3 = x^3 + \frac{1}{x^3} + 3\left(x+\frac{1}{x}\right)$

Now go through the given choices.

44. $x^4 + \frac{1}{x^4} = \left(x^2+\frac{1}{x^2}\right)^2 - 2$

$\therefore$ $34 = \left(x^2+\frac{1}{x^2}\right)^2 - 2$

$\Rightarrow \left(x^2+\frac{1}{x^2}\right)^2 = 36 \Rightarrow x^2 + \frac{1}{x^2} = 6$

Again $\left(x-\frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} - 2$

$\Rightarrow \left(x-\frac{1}{x}\right)^2 = 6 - 2 = 4$

$\Rightarrow x - \frac{1}{x} = 2$

Alternatively : Go through options.

45. $\left(x^2+\frac{1}{x^2}\right)^2 = x^4 + \frac{1}{x^4} + 2$

$\Rightarrow \left(x^2+\frac{1}{x^2}\right)^2 = 121$

$\Rightarrow x^2 + \frac{1}{x^2} = 11$

Again $\left(x-\frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} - 2$

$\therefore$ $\left(x-\frac{1}{x}\right)^2 = 9 \Rightarrow x - \frac{1}{x} = 3$

$\therefore$ $\left(x-\frac{1}{x}\right)^3 = x^3 - \frac{1}{x^3} - 3x\cdot\frac{1}{x}\left(x-\frac{1}{x}\right)$

$(3)^3 = x^3 - \frac{1}{x^3} - 3(3)$

$\Rightarrow x^3 - \frac{1}{x^3} = 36$

46. $\left(3x-\frac{2}{x}\right)^2 = 25$

$9x^2 + \frac{4}{x^2} - 12 = 25 \Rightarrow 9x^2 + \frac{4}{x^2} = 37$

Again $9x^2 + \frac{4}{x^2} + 12 = 37 + 12 = 49$

$\Rightarrow \left(3x+\frac{2}{x}\right)^2 = (7)^2 \Rightarrow 3x + \frac{2}{x} = 7$

$\therefore \left(9x^2 - \frac{4}{x^2}\right) = \left(3x-\frac{2}{x}\right)\left(3x+\frac{2}{x}\right) = 5\times 7 = 35$

47. $m^2 - 4m + 1 = 0 \Rightarrow m^2 + 1 = 4m \Rightarrow \frac{m^2+1}{m} = 4$

$\Rightarrow m + \frac{1}{m} = 4$

$\therefore \left(m+\frac{1}{m}\right)^3 = m^3 + \frac{1}{m^3} + 3\left(m+\frac{1}{m}\right) = 64$

$\Rightarrow m^3 + \frac{1}{m^3} = 52$

48. $(x+y)^2 = x^2 + y^2 + 2xy$

$169 = x^2 + y^2 + 2\times 40$

$\Rightarrow x^2 + y^2 = 89$

49. $(x+y)^3 = x^3 + y^3 + 3xy(x+y)$

$2097 = x^3 + \frac{1}{x^3} + 3\times 36(13)$

$\Rightarrow x^3 + \frac{1}{x^3} = 2097 - 1404$

$x^3 + \frac{1}{x^3} = 693$

50. $(x-y)^2 = (x+y)^2 - 4xy = 64 - 60 = 4$

$\Rightarrow (x-y) = 2$

$\therefore x^3 - y^3 = (x-y)^3 + 3xy(x-y)$

$= 8 + 45(2)$

$x^3 - y^3 = 98$

51. $(x^3 + y^3 + z^3 - 3xyz)$

$= (x+y+z)(x^2+y^2+z^2-xy-yz-zx)$

$= (x+y+z)[(x+y+z)^2 - 3(xy+yz+zx)]$

$= 6\times[36 - 3\times 11] = 6\times 3 = 18$

52. $x + y + z = 0 \Rightarrow x^3 + y^3 + z^3 = 3xyz.$

53. $\because x + y + z = 0 \Rightarrow x^3 + y^3 + z^3 = 3xyz$

$\therefore a^{1/3} + b^{1/3} + c^{1/3} = 0$

$\Rightarrow a + b + c = 3a^{1/3}b^{1/3}c^{1/3}$

$\therefore (a+b+c)^3 = 27abc$

54. $2^x - \frac{2^x}{2} = 8 \Rightarrow k - \frac{k}{2} = 8 \Rightarrow k = 16$

$\therefore 2^x = 16 \Rightarrow$ (say $2^x = k$)

$\Rightarrow 2^x = 2^4 \Rightarrow x = 4.$

$\therefore x^3 = (4)^3 = 64$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

55. $3^x + 3^{x+1} = 36 \Rightarrow 3^x + 3.3^x = 36$

Let $3^x = k$

$\therefore \quad k + 3k = 36 \Rightarrow k = 9$

$\therefore \quad 3^x = 9 \Rightarrow x = 2$

$\therefore \quad x^x = 2^2 = 4.$

56. $\dfrac{(a+b)-(a-b)^2}{a^2b - ab^2} = \dfrac{(a^2+b^2+2ab)-(a^2+b^2-2ab)}{ab(a-b)}$

$= \dfrac{4ab}{ab(a-b)} = \dfrac{4}{(a-b)}$

57. Let $\dfrac{x}{(b-c)(b+c-2a)} = \dfrac{y}{(c-a)(c+a-2b)}$

$= \dfrac{z}{(a-b)(a+b-2c)} = k$, then

$x = k(b-c)(b+c-2a),\ y = k(c-a)(c+a-2b),$

$z = k(a-b)(a+b-2c)$

$\therefore \quad (x+y+z) = k[(b^2-c^2+c^2-a^2+a^2-b^2)$

$-2\{a(b-c)+b(c-a)+c(a-b)\}]$

$= 0$

58. $a^4 + 5a^3 + 6a^2 = a^2(a^2+5a+6) = a^2(a+3)(a+2)$

60. $a^{3/2} - ab^{1/2} + a^{1/2}b - b^{3/2}$

$= a(a^{1/2} - b^{1/2}) + b(a^{1/2} - b^{1/2})$

$= (a+b)(a^{1/2} - b^{1/2})$

$\therefore \quad \dfrac{(a+b)(a^{1/2}-b^{1/2})}{(a^{1/2}-b^{1/2})} = (a+b)$

61. $x(y^2-z^2) + y(z^2-x^2) + z(x^2-y^2)$

$= x^2(z-y) - x(z^2-y^2) + yz(z-y)$

(Writing in descending power of x)

$= (z-y)[x^2 - x(z+y) + yz]$

$= (z-y)(x-z)(x-y)$

Alternatively: Consider $x = 1$, $y = 2$ and $z = 3$ and then verify the correct option.

62. Since $x^3 - y^3 = (x-y)(x^2+y^2+xy)$

$\therefore (a+b)^3 - (a-b)^3$

$= [(a+b)-(a-b)][(a+b)^2 + (a-b)^2 + (a+b)(a-b)]$

$= 2b[2(a^2+b^2) + (a^2-b^2)] = 2b(3a^2+b^2)$

63. $(x^2+4) = (x^2+2^2) = (x)^2 + (2)^2 + 4x^2 - 4x^2$

$= (x^2+2)^2 - (2x)^2$

$= (x^2+2+2x)(x^2+2-2x)$

64. Best way is to go through options.

$(x-1)(x+3)(x-2) = (x^2+2x-3)(x-2)$

$= x^3 + 2x^2 - 3x - 2x^2 - 4x + 6$

$= x^3 - 7x + 6$

Hence option (a) is correct.

65. $(64x^3 - 216y^3) = (4x)^3 - (6y)^3$

$= (4x-6y)[(4x)^2 + (6y)^2 + 24xy]$

66. $a^2 + 4b^2 + 4b - 4ab - 2a - 8$

$= a^2 + 4b^2 - 4ab + 4b - 2a - 8$

$= (a-2b)^2 - 2(a-2b) - 8$

$= k^2 - 2k - 8,$ where $k = a-2b$

$= k^2 - 4k + 2k - 8$

$= k(k-4) + 2(k-4)$

$= (k-4)(k+2)$

$= (a-2b-4)(a-2b+2)$

67. Let $(a-b) = x$, $(b-c) = y$ and $(c-a) = z$

Then $x + y + z = 0$

$\therefore \quad x^3 + y^3 + z^3 = 3xyz$

Hence $(a-b)^3 + (b-c)^3 + (c-a)^3 = 3(a-b)(b-c)(c-a)$

68. $(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab+bc+ac)$

$0 = a^2 + b^2 + c^2 + 2(ab+bc+ac)$

$\therefore \quad a^2 + b^2 + c^2 = -2(ab+bc+ac)$

69. First of all factorize the expression then find the HCF or LCM as per the requirement.

(1) LCM of $10a^2b^2$, $15abc$, $20ab^2c^2$ is $3 \times 2^2 \times 5a^2b^2c^2 = 60a^2b^2c^2$, Hence (a) is wrong.

(2) $(x^2 - 6x + 9) = (x-3)^2$

and $(x^3 - 27) = (x-3)(x^2+3x+9)$

$\therefore \quad$ HCF $= (x-3)$

(3) $6x^2 - 7x - 3 = (2x-3)(3x+1)$

$(2x^2 + 11x - 21) = (x+7)(2x-3)$

$\therefore \quad$ HCF $= (2x-3)$.

Hence 2 and 3 are correct.

70. $2x^2 - 3x - 2 = 2x^2 - 4x + x - 2$

$= (x-2)(2x+1)$

and $(x^3 - 4x^2 + 4x) = x(x^2 - 4x + 4) = x(x-2)^2$

$\therefore \quad$ LCM $= x(x-2)^2(2x+1)$

71. $2(x^2 - y^2) = 2(x-y)(x+y)$

and $5(x^3 - y^3) = 5(x-y)(x^2+y^2+xy)$

$\therefore \quad$ HCF $= (x-y)$

72. $(x^4 - 4x^2 + 3) = x^4 - 3x^2 - x^2 + 3$

$= x^2(x^2-3) - 1(x^2-3)$

$= (x^2-3)(x^2-1)$

$= (x^2-3)(x-1)(x+1)$

and $x^4 - x^2 - 6 = x^4 - 3x^2 + 2x^2 - 6$

$= x^2(x^2-3) + 2(x^2-3)$

$= (x^2-3)(x^2+2)$

$\therefore \quad$ HCF $= (x^2-3)$

73. $x^2 - 4 = (x+2)(x-2)$

$(x^2 - 5x - 6) = (x+1)(x-6)$

$(x^2 + x - 6) = (x-2)(x+3)$

Since there is no any common factor. So HCF is 1.

74. $x^2 - ax - (a+1) = x^2 - (a+1)x + x - (a+1)$

$= (x+1)(x-a+1)$

and $ax^2 - x - (a+1) = ax^2 - a - x - 1$

$= a(x^2-1) - 1(x+1)$

$= (x+1)(ax-a-1)$

$\therefore \quad$ HCF $= (x+1)$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

75. $P = (x-2)(x+1)^2(x+3)^2$

$Q = (x+1)^2(x+3)(x+4)$

and

$\therefore$ LCM of P and $Q = (x-2)(x+1)^2(x+3)^2(x+4)$

76. $a^3 + b^3 = (a+b)(a^2 + b^2 - ab)$

$(a^4 - b^4) = (a-b)(a+b)(a^2+b^2)$

$\text{LCM} = (a-b)(a+b)(a^2+b^2)(a^2+b^2-ab)$

$\therefore \quad = (a-b)(a^2+b^2)(a+b)(a^2+b^2-ab)$

$= (a-b)(a^2+b^2)(a^3+b^3)$

77. $(x^4-1) = (x-1)(x+1)(x^2+1)$

$(x^3+x^2+x+1) = x^2(x+1)+1(x+1) = (x^2+1)(x+1)$

$\text{HCF} = (x+1)(x^2+1)$

$\therefore$

78. $2x^2 - 4x = 2x(x-2)$

$3x^4 - 12x^2 = 3x^2(x^2-4)$

$= 3x^2(x+2)(x-2)$

$2x^5 - 2x^4 - 4x^3 = 2x^3(x^2 - x - 2)$

$= 2x^3(x-2)(x+1)$

$\text{HCF} = x(x-2)$

$\therefore$

79. Since all the factors are distinct

80. $(x+2)^2(x-2)$ and $x^2 - 4x - 12 = (x-6)(x+2)$

$\therefore \quad \text{LCM} = (x-6)(x+2)^2(x-2)$

81. $a^2 - ab - 2b^2 = (a^2 - b^2) - (ab + b^2)$

$= (a+b)(a-b-b) = (a+b)(a-2b)$

and $2a^2 - ab - b^2 = (a^2 - ab) + (a^2 - b^2)$

$= (a-b)a + (a^2 - b^2) = (a-b)(2a+b)$

Since there is no common factor, hence HCF = 1.

82. $\text{HCF} = a^2b^3c^3$

$\text{LCM} = a^5b^4c^4$

83. $\dfrac{(x+3)}{(x-2)} - \dfrac{(x+1)}{(x-3)} = \dfrac{(x+3)(x-3) - (x-2)(x+1)}{(x-2)(x-3)}$

$= \dfrac{(x^2-9) - (x^2-x-2)}{(x^2-5x+6)} = \dfrac{x-7}{(x^2-5x+6)}$

84. $\dfrac{x+1}{x-1} + \dfrac{x-1}{x+1} = \dfrac{(x+1)^2 + (x-1)^2}{(x-1)(x+1)} = \dfrac{2(x^2+1)}{(x^2-1)}$

85. $\dfrac{x^2-5x+6}{x^2-9x+20} \div \dfrac{x^2-3x+2}{x^2-5x+4}$

$= \dfrac{x^2-5x+6}{x^2-9x+20} \times \dfrac{x^2-5x+4}{x^2-3x+2}$

$= \dfrac{(x-2)(x-3)}{(x-4)(x-5)} \times \dfrac{(x-1)(x-4)}{(x-1)(x-2)}$

$= \left(\dfrac{x-3}{x-5}\right)$

86. $\dfrac{2x^2+3}{x-3} + \dfrac{x-3}{2x^2+3}$

$= \dfrac{(2x^2+3)^2 + (x-3)^2}{(x-3)(2x^2+3)} = \dfrac{4x^4 + 9 + 12x^2 + x^2 + 9 - 6x}{2x^3 - 6x^2 + 3x - 9}$

$= \dfrac{4x^4 + 13x^2 - 6x + 18}{2x^3 - 6x^2 + 3x - 9}$

87. $\dfrac{(x-3)(x^2-5x+4)}{(x-4)(x^2-2x-3)} = \dfrac{(x-3)(x-1)(x-4)}{(x-4)(x-3)(x+1)} = \dfrac{(x-1)}{(x+1)}$

88. $\dfrac{(2x^2+1)(x+1)+(x-1)^2}{(x^2-1)} \times \dfrac{(x^2-1)}{2x}$

$= \dfrac{2x^3 + 2x^2 + x + 1 + (x-1)^2}{(2x)}$

$= \dfrac{2x^3 + 3x^2 - x + 2}{2x}$

89. $\sqrt{\dfrac{(x^2+3x+2)(x^2+5x+6)}{x^2(x^2+4x+3)}}$

$= \sqrt{\dfrac{(x+1)(x+2)(x+2)(x+3)}{x^2(x+1)(x+3)}} = \sqrt{\dfrac{(x+2)^2}{x^2}} = \dfrac{x+2}{x}$

90. $\dfrac{1}{(a-b)(a-c)} + \dfrac{1}{(b-c)(b-a)} + \dfrac{1}{(c-a)(c-b)}$

$\dfrac{(b-c)(b-a)(c-a)(c-b) + (a-b)(a-c)(c-a)(c-b) + (a-b)(a-c)(b-c)(b-a)}{(a-b)(a-c)(b-c)(b-a)(c-a)(c-b)} = 0$

91. $\dfrac{1}{(1-x)} + \dfrac{1}{(1+x)} + \dfrac{2}{(1+x^2)} + \dfrac{4}{(1+x^4)}$

$= \dfrac{(1+x)+(1-x)}{(1-x^2)} + \dfrac{2}{(1+x^2)} + \dfrac{4}{(1+x^4)}$

$= \dfrac{2}{(1-x^2)} + \dfrac{2}{(1+x^2)} + \dfrac{4}{(1+x^4)}$

$= \dfrac{4}{(1-x^4)} + \dfrac{4}{(1+x^4)} = \dfrac{8}{1-x^8}$

INTRODUCTORY EXERCISE 13.2

1. $x - y = \dfrac{9}{10}$ and $(x+y) = \dfrac{11}{2}$

$\Rightarrow \quad x - y = \dfrac{9}{10}$

and $\quad x + y = \dfrac{11}{2}$

By addition

$2x = \dfrac{64}{10}$

$\Rightarrow \quad x = \dfrac{32}{10} = 3.2$

$\Rightarrow \therefore \quad y = \dfrac{23}{10} = 2.3$

2. $\dfrac{x+y}{xy} = 2 \Rightarrow \dfrac{1}{x} + \dfrac{1}{y} = 2 \quad \ldots(1)$

and $\dfrac{x-y}{xy} = 6 \Rightarrow \dfrac{1}{y} - \dfrac{1}{x} = 6 \quad \ldots(2)$

By adding Eq. (1) and (2), we get

$\dfrac{2}{y} = 8 \Rightarrow y = \dfrac{1}{4}$

$\therefore \quad x = -\dfrac{1}{2}$

$\therefore \quad x = -\dfrac{1}{2}$ and $y = \dfrac{1}{4}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Alternatively:

$$x + y = 2xy \quad \text{and} \quad x - y = 6xy \qquad \ldots(1)$$

$$\Rightarrow \quad 3(x + y) = 6xy \quad \text{and} \quad x - y = 6xy$$

$$\Rightarrow \quad 3x + 3y = x - y$$

$$\Rightarrow \quad 2x = -4y$$

$$\Rightarrow \quad x = -2y$$

Substituting $x = -2y$ in the Eq. (1), we get

$$-3y = 6xy \quad \Rightarrow \quad x = -\frac{1}{2}$$

and $\therefore \quad y = \frac{1}{4}$

3. $\frac{1}{2x} - \frac{1}{y} = -1$ and $\frac{1}{x} + \frac{1}{2y} = 8$

$\Rightarrow \quad \frac{1}{2x} - \frac{1}{y} = -1$ and $\frac{2}{x} + \frac{1}{y} = 16$

By adding these two equations, we get

$$\frac{5}{2x} = 15 \quad \Rightarrow \quad x = \frac{1}{6}$$

$\therefore \quad y = \frac{1}{4}$

4. $\frac{2x}{a} + \frac{y}{b} = 2 \qquad \ldots(i)$

$\frac{x}{a} - \frac{y}{b} = 4 \qquad \ldots(ii)$

Adding Eq. (i) and (ii), we get

$$\frac{3x}{a} = 6 \quad \Rightarrow \quad \frac{x}{a} = 2 \quad \Rightarrow \quad x = 2a$$

Now put $x = 2a$ in Eq. (ii)

$$\frac{y}{b} = -2$$

$\therefore \quad x = 2a$ and $y = -2b$.

5. For no solution

$$\frac{9}{7} = \frac{4}{k} \neq \frac{9}{5} \quad \Rightarrow \quad k = \frac{28}{9}$$

6. Two linear equations will coincide if there are infinite number of solutions,

i.e., $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

for $a_1x + b_1y = c_1$

and $a_2x + b_2y = c_2$

$\therefore \quad \frac{k}{3} = \frac{2}{1} = \frac{2}{1} \quad \Rightarrow \quad k = 6$

7. For an infinite number of solutions

$$\frac{2}{6} = \frac{3}{k} = \frac{-5}{-15} \quad \Rightarrow \quad k = 9$$

8. $8B + 5M = 92 \qquad \ldots(1)$

$5B + 8M = 77 \qquad \ldots(2)$

Multiply Eq. (1) by 8 Eq. (2) by 5, we get

$$\left.\begin{aligned} 64B + 40M &= 736 \\ 25B + 40M &= 385 \end{aligned}\right\} \quad \text{[Subtracting]}$$

$$- \quad - \quad -$$

$$39B = 351$$

$$B = 9$$

$$M = 4$$

$\therefore \quad 2M + 3B = 35$

$\therefore$ The required cost of 2 mugs and 3 buckets is Rs. 35.

(Here $M \rightarrow$ Price of a mug and $B \rightarrow$ Price of a bucket)

9. Let the two numbers be x and y.

$$x + y = 8$$

then $\quad (x + y) = 4(x - y)$

and $\quad x - y = 2$

$\Rightarrow \quad x + y = 8$ and $x - y = 2$

$\therefore \quad 2x = 10 \quad \Rightarrow \quad x = 5$

$y = 3$

$\therefore$

10. Let numerator be x and denominator be y.

$$x = y - 4 \qquad \ldots(1)$$

then $\quad 8(x - 2) = (y + 1) \qquad \ldots(2)$

The above equations can be written as

$$\left.\begin{aligned} y - x &= 4 \\ y - 8x &= -17 \end{aligned}\right\} \qquad \ldots(3)$$

and

$$- \quad + \quad +$$

$$7x = 21$$

$$x = 3 \quad \therefore \quad y = 7$$

$\Rightarrow$

$\therefore$ the required fraction $= \frac{x}{y} = \frac{3}{7}$

11. Distance = 90 km

Let the speeds of car be x km/h and y km/h

$\therefore \quad$ Distance = Speed × Time

$\therefore \quad 90 = (x + y) \times \frac{9}{7} \quad \Rightarrow \quad x + y = 70$

and $\quad 90 = (x - y) \times 9 \quad \Rightarrow \quad x - y = 10$

$\therefore \quad x = 40$ and $y = 30$

$\therefore$ the required speeds of cars = 40 km/h and 30 km/h

12. Let there be x coins of 25 paise denomination and y coins of 50 paise denomination, then

$$x + y = 40 \quad \Rightarrow \quad 25x + 25y = 1000 \qquad \ldots(1)$$

and $\quad 25x + 50y = 1275 \qquad \ldots(2)$

$\therefore$ Subtracting Eq. (1) from (2) we get

$$25y = 275 \quad \Rightarrow \quad y = 11$$

$\therefore \quad x = 29.$

13. Let there be x pencils and y pens, then

$\therefore \quad x + y = 40 \quad \Rightarrow \quad 4x + 4y = 160$

$$(x + 5) = 4(y - 5)$$

$\Rightarrow \quad x - 4y = -25$

$\therefore \quad 4x + 4y = 160$

$$x - 4y = -25$$

$$5x = 135$$

$\Rightarrow \quad x = 27 \quad \therefore \quad y = 13$

14. Let the present age of the son be 'x' years and the present age of his father be 'y' years.

then $\quad y = 3x \qquad \ldots(1)$

Again $\quad (y + 12) = 2(x + 12) \qquad \ldots(2)$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Substituting $y = 3x$ in Eq. (2), we get

$$x = 12$$

$$y = 36$$

$\therefore$

15. Let the present age of A is x and the present age of B is y (in years).

then, $(x-5) = 3(y-5) \Rightarrow x - 3y = -10$ ...(1)

$(x+10) = 2(y+10) \Rightarrow x - 2y = 10$ ...(2)

Subtracting Eq. (2) from Eq. (1), we get

$$-y = -20 \Rightarrow y = 20$$

$$x = 50$$

$\therefore$

16. Let Astha will have x number of oranges and Saumya will have y number of oranges. Then,

$(x+10) = 2(y-10) \Rightarrow x - 2y = -30$ ...(1)

and $(x-10) = (y+10) \Rightarrow x - y = 20$ ...(2)

Subtracting Eq. (2) from Eq. (1), we get

$$-y = -50 \Rightarrow y = 50$$

$$x = 70$$

$\therefore$

17. Total distance = 760 km.

Let the speed of car be x km/h and the speed of train be y km/h

$\therefore$ Distance = Speed × Time

$\Rightarrow$ $T = \frac{D}{S}$

$\therefore$ $\frac{600}{x} + \frac{160}{y} = 8$ ...(1)

and $\frac{520}{x} + \frac{240}{y} = 8\frac{1}{5}$ ...(2)

$\therefore$ Eq. (1) can be written as

$$600A + 160B = 8 \quad ...(3)$$

and Eq. (2) can be written as

$$520A + 240B = \frac{41}{5}. \quad ...(4)$$

where, $\frac{1}{x} = A$ and $\frac{1}{y} = B$

$\therefore$ Solving Eq. (3) and (4) we get

$$1800A + 480B = 24$$

$$1040A + 480B = \frac{82}{5}$$

$$760A = \frac{38}{5}$$

$\Rightarrow$ $A = \frac{1}{100} \Rightarrow x = 100$ km/h

$\therefore$ $B = \frac{1}{80} \Rightarrow y = 80$ km/h

18. Let he has invested Rs. x at 12% S.I. and Rs. y at 10% SI

$\therefore$ $\left(\because SI = \frac{P \times r \times t}{100}\right)$

$$\frac{x \times 12 \times 1}{100} + \frac{y \times 10 \times 1}{100} = 130$$

$\Rightarrow$ $12x + 10y = 13000$ ...(1)

$\Rightarrow$ $10x + 12y = 134000$ ...(2)

$\therefore$ $x = 500$ and $y = 700$

19. $\frac{3x - y + 1}{3} = \frac{2x + y + 2}{5}$

$\Leftrightarrow$ $5(3x - y + 1) = 3(2x + y + 2)$

$\Leftrightarrow$ $9x - 8y = 1$ ...(1)

$\Leftrightarrow$ $\frac{2x + y + 2}{5} = \frac{3x + 2y + 1}{6}$

$\Leftrightarrow$ $6(2x + y + 2) = 5(3x + 2y + 1)$

$\Leftrightarrow$ $3x + 4y = 7$ ...(2)

Multiplying (2) by 2 and adding to (1), we get

$$15x = 15 \text{ or } x = 1$$

Putting $x = 1$ in (1), we get

$$9 \times 1 - 8y = 1 \Rightarrow y = 1.$$

20. Submarino can be destroyed when it meets the destroyer. Thus

$$2x + 3y = 5 \quad ...(1)$$

$$x - y = 10 \quad ...(2)$$

$\therefore$ $x = 7$ and $y = -3$

So the required point is $(7, -3)$

21. $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ ...(1)

$\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ ...(2)

$-\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ ...(3)

Adding Eq. (1), (2) and (3), we have

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 3 \quad ...(4)$$

$\therefore$ $\frac{x^2}{a^2} = 1 \Rightarrow x = \pm a$

$\frac{y^2}{b^2} = 1 \Rightarrow y = \pm b$

$\frac{z^2}{c^2} = 1 \Rightarrow z = \pm c$

Hence (d) is correct.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmashbook.wordpress.com

LEVEL 1

1. If $x + y + z$ is constant, the product xyz takes maximum value when each of x, y, z takes equal value.

$\because \quad a + b + c = 13$

$\therefore \quad (a-3) + (b-2) + (c+1) = 13 - 3 - 2 + 1 = 9$

For the maximum value of $(a-3)(b-2)(c+1)$

$$= (a-3) = (b-2) = (c+1) = \frac{9}{3} = 3$$

So, $\quad (a-3)(b-2)(c+1) = 3 \times 3 \times 3 = 27$

2. If xyz is constant, then the sum of x, y, z (*i.e.*, $x + y + z$) takes minimum value when each of x, y, z takes equal value.

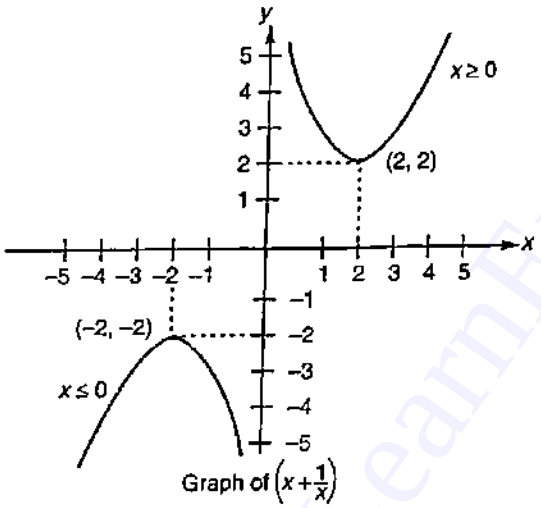
$\therefore$ Minimum value of $a + b + c + d$ for given constant product $abcd$ will be when $a = b = c = d$

$\therefore \quad a = b = c = d = 3$

$\therefore$ Minimum value of $a + b + c + d = 3 + 3 + 3 + 3 = 12$

3. For $x \geq 0$, $\quad x + \frac{1}{x} \geq 2$

and for $x \leq 0$, $\quad x + \frac{1}{x} \leq 2$



Graph of $\left(x + \frac{1}{x}\right)$

4. This is the standard inequality formula.

5. This is the standard inequality formula.

NOTE The result can be proved by considering suitable values.

6.
$$x^2y^3 + y^2x^3 = 25$$
$$\Rightarrow \quad x^2y^2(x + y) = 25$$
$$\Rightarrow \quad (xy)^2(x + y) = 25$$
$$\Rightarrow \quad (xy)^2 = 1 \qquad (\because x + y = 25)$$
$$\Rightarrow \quad xy = \pm 1$$

7. $x > 0$ and $y > 0$

$$\therefore \quad (x + y)\left(\frac{1}{x} + \frac{1}{y}\right) = 2 + \frac{x}{y} + \frac{y}{x}$$
$$= 2 + \left(k + \frac{1}{k}\right), \quad \text{where } k = \frac{x}{y}$$

Since the minimum value of the expression $\left(k + \frac{1}{k}\right)$ is 2.

$\therefore$ Minimum value of the given expression is 4.

8. If $a + b + c + d$ is constant, then the product $abcd$ is maximum when $a = b = c = d$.

$\therefore \quad (a+1) = (b+1) = (c+1) = (d+1)$

Given that $(a+1) + (b+1) + (c+1) + (d+1) = 8$

$$4(a+1) = 8$$
$$\therefore \quad (a+1) = 2$$
$$\Rightarrow \quad x + y + z = 1$$

$\therefore$ Maximum value $= 2 \times 2 \times 2 \times 2 = 16$

9. As
$$\left(\frac{1}{x} - 1\right)\left(\frac{1}{y} - 1\right)\left(\frac{1}{z} - 1\right) = \frac{y+z}{x} \cdot \frac{z+x}{y} \cdot \frac{x+y}{z}$$

But $\quad \frac{y+z}{2} \geq yz$ etc.

Hence, $\quad \text{LHS} \geq \frac{8xyz}{xyz} = 8$

10. $x + y + z = 4$ and $x^2 + y^2 + z^2 = 6$

$$\therefore \quad y + z = 4 - x$$

and
$$y^2 + z^2 = 6 - x^2$$
$$yz = \frac{1}{2}\{(y+z)^2 - (y^2 + z^2)\}$$
$$= \frac{1}{2}\{(4-x)^2 - (6 - x^2)\}$$
$$\Rightarrow \quad yz = x^2 - 4x + 5$$

Hence, y and z are the roots of

$$t^2 - (4 - x)t + (x^2 - 4x + 5) = 0$$

Since the roots y and z are real

$$(4-x)^2 - 4(x^2 - 4x + 5) \geq 0$$
$$\Rightarrow \quad 3x^2 - 8x + 4 \leq 0$$
$$\Rightarrow \quad (3x - 2)(x - 2) \leq 0$$
$$\Rightarrow \quad x \in \left[\frac{2}{3}, 2\right]$$

By symmetry y, z also $\in \left[\frac{2}{3}, 2\right]$.

11. $\frac{1}{x^2} + \frac{1}{y^2} = \frac{x^2 + y^2}{(xy)^2}$

$$= \frac{(7 + 4\sqrt{3})^2 + (7 - 4\sqrt{3})^2}{[(7 + 4\sqrt{3})(7 - 4\sqrt{3})]^2} = \frac{2(49 + 48)}{(1)^2} = 194$$

12. $\text{AM} \geq \text{GM} \Rightarrow \frac{a+b+c}{3} \geq (abc)^{1/3}$

(AM → Arithmetic mean, GM → Geometric mean)

and
$$\frac{1}{3}\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \geq \left(\frac{1}{abc}\right)^{1/3}$$
$$\Rightarrow \quad \frac{1}{3}(a+b+c)\frac{1}{3}\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \geq (abc)^{1/3}\left(\frac{1}{abc}\right)^{1/3} = 1$$
$$\Rightarrow \quad (a+b+c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \geq 9$$

Putting $a = b = c = 1$, expression takes the value 9, which is therefore, its least value.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

13. If ab is constant, then $(a+b)$ takes minimum value when
$a = b$
$a = b = 1$
$\therefore (1+a)(1+b) = (1+1)(1+1) = 4$

14. $$\frac{(a+b+c)(ab+bc+ac)}{abc} = (a+b+c)\left(\frac{ab}{abc}+\frac{bc}{abc}+\frac{ac}{abc}\right)$$
$$= (a+b+c)\left(\frac{1}{c}+\frac{1}{a}+\frac{1}{b}\right)$$
$$= (a+b+c)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) > 9$$

(See the problem number 12 in this exercise.)

15. The expression will have minimum value of the expression when $a = b = c$.
Therefore, the required minimum value
$$= \frac{(1+1+1)}{1} \times \frac{(1+1+1)}{1} \times \frac{(1+1+1)}{1} = 27$$

16. $\text{Max}\left(\frac{x}{y}\right) = \frac{\text{Max}(x)}{\text{Min}(y)} = \frac{3}{2}$

17. $$\frac{1}{a}+\frac{1}{b}+\frac{1}{c} = 1$$
$$\Rightarrow \quad \frac{bc+ac+ab}{abc} = 1$$
$$\Rightarrow \quad bc+ac+ab = abc \qquad \ldots(1)$$
Again, $(a+b+c)^2 = a^2+b^2+c^2+2(ab+bc+ac)$
$(a+b+c)^2 = a^2+b^2+c^2+2abc$
$(3)^2 = 6+2abc \Rightarrow abc = \frac{3}{2}$

18. $2^x = 4^y = 8^z \Rightarrow 2^x = 2^{2y} = 2^{3z}$
$\Rightarrow x = 2y = 3z = k$ (say)
Then $xyz = \frac{k^3}{6} = 288 \Rightarrow k = 12$
$\therefore x = 12, \quad y = 6, \quad z = 4$
$\Rightarrow \frac{1}{2x}+\frac{1}{4y}+\frac{1}{8z} = \frac{11}{96}$

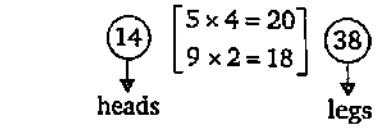
19. $a = c^z$
$\Rightarrow a = (b^y)^z \qquad (\because c = b^y)$
$\Rightarrow a = b^{yz}$
$\Rightarrow a = (a^x)^{yz} \qquad (\because b = a^x)$
$\Rightarrow a = a^{xyz}$
$\Rightarrow a^1 = a^{xyz}$
$\Rightarrow xyz = 1$

20. $7x + 12y = 220$
$\Rightarrow 7x = 220 - 12y$
$x = \frac{220-12y}{7} = \frac{4(55-3y)}{7}$
It means $55 - 3y$ must be divisible by 7. Since 4 is not divisible by 7.
$\therefore$ at $y = 2, 9, 16$
we get $x = 28, 16, 4$
Thus we have 3 solutions of x, y
$(x, y) = (28, 2), (16, 9)$ and $(4, 16)$

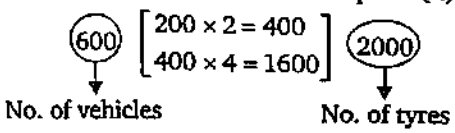
21. $a^x \cdot a^y \cdot a^z = (x+y+z)^{x+y+z}$
$\Rightarrow a^{x+y+z} = (x+y+z)^{(x+y+z)}$
$\Rightarrow a = x+y+z \qquad \ldots(1)$
$\therefore (x+y+z)^y = a^x = (x+y+z)^x$
$\Rightarrow x = y$
Similarly $y = z$ and $z = x$
$\therefore x = y = z = \frac{a}{3}$ [using (1)]

22. Let $\frac{x}{a} = \frac{y}{b} = \frac{z}{c} = k$
$\therefore x = ak, \quad y = bk, \quad z = ck$
$\therefore (x+y+z) = k(a+b+c)$
$\therefore (x+y+z)^2 = k^2(a+b+c)^2$
$\therefore x^2+y^2+z^2+2(xy+yz+zx) = k^2(a+b+c)^2$
$\therefore 2(xy+yz+zx) = k^2(a+b+c)^2 - (x^2+y^2+z^2)$
$\therefore xy+yz+zx = \frac{k^2}{2}(a+b+c)^2 - \frac{1}{2}(x^2+y^2+z^2)$
$= \frac{x^2(a+b+c)^2 - a^2(x^2+y^2+z^2)}{2a^2} \qquad \left(\because k = \frac{x}{a}\right)$

23. Let x and y be the number of deer and ducks respectively.
$\therefore x + y = 14 \qquad \ldots(1)$
and $4x + 2y = 38 \qquad \ldots(2)$
($\because$ A deer has 4 legs and a duck has 2 legs)
By solving the above two Eq. (1) and (2), we get
$x = 5$ and $y = 9$
Thus, the number of deer is 5.
Alternatively: Assume that there are only ducks (because a duck has less number of legs than that of deer). Hence, there must be $9 \times 2 = 2 = 18$ legs, but in this way there are 10 legs $(= 28 - 18)$ more.
So, there must be some deer which fulfill this gap. Now this difference of 10 legs is compensated by 5 deer because a deer has 2 extra legs (*i.e.*, 2 more legs than a duck has).
Hence, there are 5 deer and $14 - 5 = 9$ ducks.
In short: Step 1. $2 \times 14 = 28$
Step 2. $38 - 28 = 10$
Step 3. $10 \div 2 = 5$
Step 4. $14 - 5 = 9$
Hence, there were 5 deer and 9 ducks.
Alternatively: Going through options. First of all option (d) is ruled out. Now we have to check just 3 options. Again we can leave option (b) and (c) also.
So, we can check the option (a).



24. Going through options, we find that option (d) is correct *i.e.*,



For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Alternatively:

Step 1. $2 \times 600 = 1200$

($2 \rightarrow$ number of tyres in two wheelers)

Step 2. $2000 - 1200 = 800$

Step 3. $800 \div 2 = 400 \rightarrow$ number of 4 wheelers

Step 4. $600 - 400 = 200 \rightarrow$ number of 2 wheelers

25. Go through options. Let us consider option (d).

(125) [$100 \times 5 = 500$, $25 \times 3 = 75$] (575)

(125) → No. of raths; (575) → No. of horses

Alternatively:

Step 1. $3 \times 125 = 375$ → Number of horses in a rath carried by 3 horses

Step 2. $575 - 375 = 200$

Step 3. $200 \div 2 = 100$ → Number of extra horses in a rath carried by 5 horses.

Step 4. $125 - 100 = 25$

$\therefore$ Number of raths of Kaurav = 100

Number of raths of allied army of Kaurav = 25

26. Let us assume option (a) is correct

then the number of coolers $= \frac{150000}{3000} = 50$

$\therefore$ the number of coolers = 50

and the number of fans = 40 (= 90 − 50)

Now (90) [$50 \times 4 = 200$, $40 \times 3 = 120$] (320)

Thus, the assumed option (a) is correct.

Alternatively: Step 1. $3 \times 90 = 270$

Step 2. $320 - 270 = 50$

Step 3. $50 \div 1 = 50$

Step 4. $90 - 50 = 40$

Hence, number of fans = 40

and number of coolers = 50

$\therefore$ Cost of coolers = $50 \times 3000 = 150000$

27. Let us assume option (c), then

(90) [$40 \times 2 = 80 \rightarrow 80 \times 10$, $50 \times 3 = 150 \rightarrow 150 \times 15$] 800, 2250 (3050)

Hence, assumed option (c) is correct.

Alternatively: Per hour expenditure in fuel for Sukhoi

$= 2 \times 10 = \$20$

and per hour expenditure in fuel for Mig-21 $= 3 \times 15 = \$45$

Now, Step 1. $90 \times 20 = 1800$

Step 2. $3050 - 1800 = 1250$

Step 3. $1250 \div 25 = 50$ $(25 = 45 - 20)$

Step 4. $90 - 50 = 40$

Hence, the number of Sukhoi is 40.

28. Step 1. $20 \times 7 = 140$

Step 2. $176 - 140 = 36$

Step 3. $36 \div 3 = 12 \rightarrow$ weight of mangoes, initially

Step 4. $20 - 12 = 8 \rightarrow$ weight of apples, initially

$\therefore$ Number of mangoes $= 12 \times 10 = 120$

and Number of apples $= 8 \times 7 = 56$

Again, since she is left with 13 kg of mangoes and apples containing 121 fruits $(176 - 55 = 121)$

$\therefore$ Step 1. $13 \times 7 = 91$

Step 2. $121 - 91 = 30$

Step 3. $30 \div 3 = 10 \rightarrow$ weight of mangoes

Step 4. $13 - 10 = 3 \rightarrow$ weight of apples

Thus, the number of apples left with vendor $= 3 \times 7 = 21$.

Alternatively: In the first instance option (c) is ruled out since,

$\frac{80}{7} + \frac{41}{10} \neq 13$ $\quad 176 - 55 = 121$ and $121 - 80 = 41$

Again $\frac{35}{7} + \frac{86}{10} \neq 13$ kg, option (b) is wrong.

Once again $\frac{21}{7} + \frac{100}{10} = 13$ kg

Hence, option (a) is correct.

29. Since we know that Ritika has purchased 2 kg mangoes and 5 kg apples. Thus, she has spent $(2 \times 35 + 5 \times 40) =$ Rs. 270.

30.

	Mangoes	Apples
SP	35	40
CP	30	30
New SP	40	35

Therefore, CP $= 30 \times 10 + 30 \times 3 = 390$

and New SP $= 40 \times 10 + 35 \times 3 = 505$

$\therefore$ Profit $= 505 - 390 = 115$

and Profit percentage $= \frac{115}{390} \times 100 = 29.48\%$

31. Solving with the help of options: In this type of questions we start with least valued options and tend towards higher valued option.

Option (a): If we take 4 coins of Re 1, then

$(5 \times 1) + (2 \times 21) = 47 \neq 44$ $\quad (48 - 4 = 44)$

$5 \times 2 + 2 \times 20 = 50 \neq 44$ $\quad$ and $(26 - 4 = 22)$

Thus, we need to move further since the value of money is exceeding.

Option (b): If we take 5 coins of Re 1, then

$5 \times 1 + 2 \times 20 = 45 \neq 43$ $\quad (43 = 48 - 5)$

$5 \times 2 + 2 \times 19 = 48 \neq 43$ $\quad$ and $(21 = 26 - 5)$

Option (c): If we take 7 coins of Re 1, then

$5 \times 1 + 2 \times 18 = 41 = 41$ $\quad \{(48 - 7 = 41)$ and $(26 - 7 = 19)\}$

Thus, the option (c) is correct.

Alternatively:

$5a + 2b + c = 48$ $\quad a \rightarrow$ Rs. 5 coin

$a + b + c = 26$ $\quad b \rightarrow$ Rs. 2 coin

$4a + b = 22$ $\quad c \rightarrow$ Re. 1 coin

$(a, b) \rightarrow (c)$

$(1, 18) \rightarrow 7$

$(2, 14) \rightarrow 10$

$(3, 10) \rightarrow 13$

$(4, 6) \rightarrow 16$

$(5, 2) \rightarrow 19$

32. Go through options.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

14 THEORY OF EQUATIONS

As per the past years CAT papers, on an average two-three questions are being asked. The problems of this chapter are not so much difficult to solve. Since this chapter is concerned with secondary level syllabus therefore it is important in JMET, XAT and SNAP etc.

In this chapter we study quadratic equation, inequation and higher degree polynomial expressions.

QUADRATIC EQUATIONS

Quadratic Equation : An equation of the form

$$ax^2 + bx + c = 0 \quad \ldots(i)$$

where $a, b, c \in C$ and $a \neq 0$ is called a quadratic equation.

The numbers a, b, c are called the coefficients of this equation.

A root of the quadratic Eq (i) is a complex number α such that $a\alpha^2 + b\alpha + c = 0$.

Discriminant $D = b^2 - 4ac$

The roots of Eq (i) are given by the formula

$$x = \frac{-b \pm \sqrt{D}}{2a}$$

or

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Properties of Quadratic Equations

1. A quadratic equation has two and only two roots.
2. A quadratic equation cannot have more than two different roots.
3. If α be a root of the quadratic equation $ax^2 + bx + c = 0$, then $(x - \alpha)$ is a factor of $ax^2 + bx + c = 0$.

NOTE The possible values of x which satisfy the quadratic equation are called the roots of the quadratic equation.

Sum and Product of the roots of a Quadratic Equation

Let α, β be the roots of a quadratic equation $ax^2 + bx + c = 0; a \neq 0$, then

$$\alpha + \beta = \frac{-b}{a} = -\left(\frac{\text{coefficient of } x}{\text{coefficient of } x^2}\right)$$

and

$$\alpha \cdot \beta = \frac{c}{a} = \left(\frac{\text{constant term}}{\text{coefficient of } x^2}\right)$$

Therefore,

- If the two roots α and β be reciprocal to each other, then $a = c$.
- If the two roots α and β be equal in magnitude and opposite in sign, then $b = 0$.

Sign of the Roots

1. The roots α, β are both negative, if $(\alpha + \beta)$ and $\alpha\beta$ are both positive (*i.e.*, if $\frac{-b}{a}$ and $\frac{c}{a}$ are both positive).
2. The roots α, β are both positive, if $\alpha + \beta$ is negative and $\alpha\beta$ is positive (*i.e.*, if $\frac{b}{a}$ is negative and $\frac{c}{a}$ is positive).
3. The roots α, β are of opposite signs, if $\alpha\beta$ is negative (*i.e.*, $\frac{c}{a}$ is negative).

Sign of $(\alpha + \beta)$	Sign of $(\alpha\beta)$	Sign of the α, β
+ve	+ve	α and β are positive
–ve	+ve	α and β are negative
+ve	–ve	α is positive and β is negative if $\alpha > \beta$ (numerically)
–ve	–ve	α is negative and β is positive if $\alpha > \beta$ (numerically)

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

Nature of Roots

For a quadratic equation $ax^2 + bx + c = 0$ where $a, b, c \in R$ and $a \neq 0$ and $D = b^2 - 4ac$.

(i) If $D < 0$, roots are imaginary (ii) If $D \geq 0$ roots are real.

- $D < 0$ (roots are complex with non-zero imaginary part)
- $D = 0$ (roots are rational and equal)
- $D > 0$
  - D is a perfect square → roots are rational and unequal
  - D is not a perfect square → roots are irrational and conjugate pairs

- If $a, b, c \in R$ and $p + iq$ is one root of quadratic equation (where $q \neq 0$) then the other root must be conjugate $p - iq$ and vice-versa. ($p, q \in R$ and $i = \sqrt{-1}$)
- If $a, b, c \in Q$ and $p + \sqrt{q}$ is one root of the quadratic equation, then the other root must be the conjugate $p - \sqrt{q}$ and vice-versa (where p is a rational and $\sqrt{q}$ is a surd).
- If $a = 1$ and $b, c \in I$ and the roots of quadratic equation are rational numbers, then these roots must be integers.
- If the quadratic equation has more than 2 roots (complex numbers), then the equation becomes an identity *i.e.*, $a = b = c = 0$.

Condition for Common Roots

Consider two quadratic equations

$$ax^2 + bx + c = 0 \quad \ldots(i) \quad a \neq 0$$

and

$$a'x^2 + b'x + c' = 0 \quad \ldots(ii) \quad a' \neq 0$$

(A) If one root is common then,

$$(ab' - a'b)(bc' - b'c) = (ca' - c'a)^2$$

(B) If two roots are common then,

$$\frac{a}{a'} = \frac{b}{b'} = \frac{c}{c'}$$

Condition that $ax^2 + bx + c = 0$, is Factorizable into two Linear Factors

When $D \geq 0$, then the equation $ax^2 + bx + c = 0$ is factorizable into two linear factors.

i.e., $ax^2 + bx + c = 0 \Rightarrow (x - \alpha)(x - \beta) = 0$, where α and β are the roots of quadratic equation.

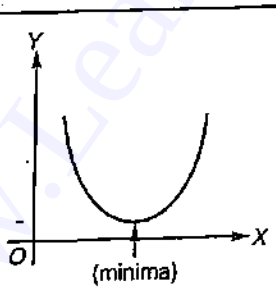
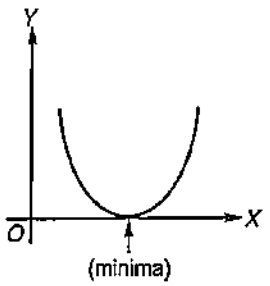
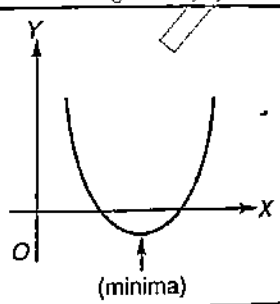
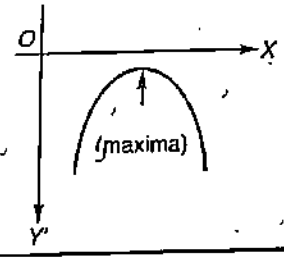
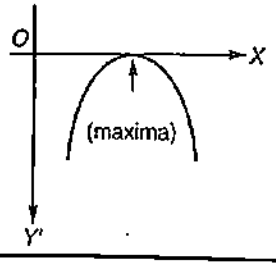
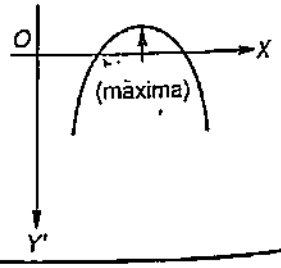
Formation of a Quadratic Equation : Let α, β be the two roots then we can form a quadratic equation as follows:

$x^2 - (\text{sum of roots})\, x + (\text{product of roots}) = 0$

i.e., $x^2 - (\alpha + \beta)x + (\alpha\beta) = 0$

or $(x - \alpha)(x - \beta) = 0$

Graph of a Quadratic Function $ax^2 + bx + c = 0, a \neq 0$

Characteristics of the Function	$b^2 - 4ac < 0$	$b^2 - 4ac = 0$	$b^2 - 4ac > 0$
When a is positive *i.e.*, $a > 0$			
When a is negative *i.e.*, $a < 0$			

- The graph of a quadratic function (expression) is called a parabola. The point at which its direction changes is called its turning point, commonly called the vertex of the parabola.
- The graph of the function is concave upwards when $a > 0$ and concave downwards when $a < 0$.
- If the graph has no points in common with the x-axis, the roots of the equation are imaginary and cannot be determined from the graph.
- If the graph is tangent to the x-axis, the roots are real and equal.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

- If the graph cuts the x-axis, the roots of the equation will be real and unequal. Their values will be given by the abscissae of the points of intersection of the graph and the x-axis.

QUADRATIC INEQUATIONS

The following properties of inequations will be useful in solving the problems on inequations :

1. Square of any real quantity ≥ 0
2. If a, b, c are three real numbers such that $a \geq b$ then, $a \pm c \geq b \pm c$
3. (i) If a, b, c are three real numbers such that $a \geq b$ and $c > 0$, then $ac \geq bc$.
   (ii) If $a \leq b$ and $c > 0$, then $ac \leq bc$
4. (i) If a, b, c are three real numbers such that $a \geq b$ and $c < 0$, then $ac \leq bc$
   (ii) If $a \leq b$ and $c < 0$, then $ac \geq bc$
5. (i) If $ab > 0$, then a and b are either both positive or both negative.
   (ii) If $ab < 0$, then a and b are of opposite signs, *i.e.*, either $a > 0$ and $b < 0$ or $a < 0$ and $b > 0$.

Formation of a New Quadratic Equation by Changing the Roots of a given Quadratic Equation

Let α, β be the roots of a quadratic equation $ax^2 + bx + c = 0$; then we can form a new quadratic equation as per the following rules.

(a) A quadratic equation whose roots are p more than the roots of the equation $ax^2 + bx + c = 0$ (*i.e.*, the roots are $\alpha + p$ and $\beta + p$).
The required equation is
$$a(x - p)^2 + b(x - p) + c = 0$$

(b) A quadratic equation whose roots are less by p than the roots of the equation $ax^2 + bx + c = 0$, (*i.e.*, the roots are $\alpha - p$ and $\beta - p$)
The required equation is
$$a(x + p)^2 + b(x + p) + c = 0$$

(c) A quadratic equation whose roots are p times the roots of the equation $ax^2 + bx + c = 0$ (*i.e.*, the roots are $p\alpha$ and $p\beta$)
The required equation is $a\left(\frac{x}{p}\right)^2 + b\left(\frac{x}{p}\right) + c = 0$

(d) A quadratic equation whose roots are $1/p$ times the roots of the equation $ax^2 + bx + c = 0$ (*i.e.*, the roots are α / p and β / p)
The required equation is $a(px)^2 + b(px) + c = 0$

(e) A quadratic equation whose roots are the reciprocals of the roots of the equation $ax^2 + bx + c = 0$, (*i.e.*, the roots are $1/\alpha$ and $1/\beta$).
The required equation is $a\left(\frac{1}{x}\right)^2 + b\left(\frac{1}{x}\right) + c = 0$
$$\Rightarrow \quad cx^2 + bx + a = 0$$

(f) A quadratic equation whose roots are the negative of the roots of the equation $ax^2 + bx + c = 0$ (*i.e.*, the roots are $-\alpha$ and $-\beta$)
The required equation is $a(-x)^2 + b(-x) + c = 0$
$$\Rightarrow \quad ax^2 - bx + c = 0$$

(g) A quadratic equation whose roots are the squares of the roots of the equation $ax^2 + bx + c = 0$ (*i.e.*, the roots are α^2 and β^2)
The required equation is $a(\sqrt{x})^2 + b(\sqrt{x}) + c = 0$
$$\Rightarrow \quad ax + b\sqrt{x} + c = 0$$

(h) A quadratic equation whose roots are the cubes of the roots of the equation $ax^2 + bx + c = 0$ (*i.e.*, the roots are α^3 and β^3)
The required equation is $a(x^{1/3})^2 + b(x^{1/3}) + c = 0$
$$\Rightarrow \quad ax^{2/3} + bx^{1/3} + c = 0$$

Maximum or Minimum value of a Quadratic Equation

At $x = -b/2a$ we get the maximum or minimum value of the quadratic expression.

(i) When $a > 0$ (in the equation $ax^2 + bx + c$) the expression gives minimum value $y = \frac{4ac - b^2}{4a}$

(ii) When $a < 0$ (in the equation $ax^2 + bx + c$) the expression gives maximum value $y = \frac{4ac - b^2}{4a}$

QUADRATIC INEQUALITY

By a quadratic inequality we mean an inequality of the following kinds :

(a) $ax^2 + bx + c \geq 0$ (b) $ax^2 + bx + c < 0$

Quadratic inequations can be solved by the following methods

(i) Algebraic method (ii) Method of intervals

(i) Algebraic method : This will be explained by the application of examples.

(ii) Method of intervals :

Step 1. Express the quadratic inequation $ax^2 + bx + c$ whose roots are α and β in the form $(x - \alpha)(x - \beta)$, where $\alpha < \beta$ and $a > 0$.

Step 2. On the number line, the numbers α and β must be arranged in order of increasing magnitude then place the plus sign in the interval to the right of the largest number (*i.e.*, β). In the next interval (from right to left) place the minus sign, then in the next interval plus sign, etc.

The solution of the inequality, *i.e.*, $ax^2 + bx + c \geq 0$ will then consists of the intervals having the plus sign.

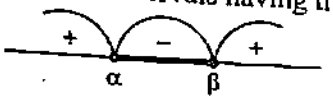


$(\alpha < \beta)$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

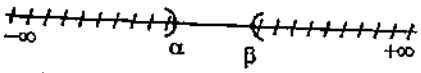
yoursmahboob.wordpress.com
 For More Visit : www.LearnEngineering.in

The solution of the inequality, *i.e.*, $ax^2 + bx + c < 0$ will then consists of the intervals having the minus sign.

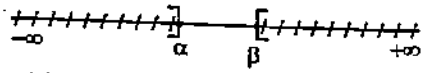


$(\alpha < \beta)$

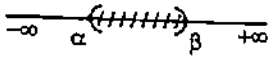
Thus, if $(x-\alpha)(x-\beta) > 0$, then x lies outside the interval $[\alpha, \beta]$, *i.e.*, $x \in (-\infty, \alpha) \cup (\beta, \infty)$



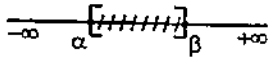
If $(x-\alpha)(x-\beta) \geq 0$, then x lies on and outside the interval $(-\infty, \alpha] \cup [\beta, \infty)$



If $(x-\alpha)(x-\beta) < 0$, then x lies inside the interval (α, β), *i.e.*, $x \in (\alpha, \beta)$



If $(x-\alpha)(x-\beta) \leq 0$, then x lies in the interval $[\alpha, \beta]$, *i.e.*, $x \in [\alpha, \beta]$



Quadratic fraction (or Quadratic Rational Expression) :

An expression of the form $\frac{a_1x^2 + b_1x + c}{a_2x^2 + b_2x + c}$ is known as Quadratic fraction (or rational expression)

VALUES OF A RATIONAL EXPRESSION

Step 1. Equate the rational expression to y.

Step 2. Obtain a quadratic equation in x by simplifying the expression in step 1.

Step 3. Obtain the discriminant (D) of the quadratic equation in step 2.

Step 4. Put $D \geq 0$ and solve the inequation for y. The values of y so obtained determined the set of values attained by the given rational expression.

Solution of Rational Inequalities

Let $P(x) = a_1x^2 + b_1x + c$ and $Q(x) = a_2x^2 + b_2x + c$, then

$$\frac{P(x)}{Q(x)} > 0, \quad \frac{P(x)}{Q(x)} \geq 0, \quad \frac{P(x)}{Q(x)} < 0 \quad \text{and} \quad \frac{P(x)}{Q(x)} \leq 0$$

are known as rational algebraic inequations.

To obtain the range of values of solution set we apply the following algorithm.

Step 1. Factorize $P(x)$ and $Q(x)$ into linear factors, *i.e.*, in the form of $\frac{(x-\alpha_1)(x-\beta_1)}{(x-\alpha_2)(x-\beta_2)}$, where the coefficient of x must be positive in all factors.

Step 2. On the number line represent $\alpha_1, \beta_1, \alpha_2$ and β_2 in the increasing order which divide the number line into $(4+1) = 5$ regions.

[If there are $\alpha_1, \alpha_2, ..., \alpha_m$, and $\beta_1, \beta_2, \beta_3, ..., \beta_n$ roots then the number line is being divided into $(m+n+1)$ regions]

Step 3. In the right region the expression $\frac{P(x)}{Q(x)}$ bears positive sign and in other regions the expression bears alternate positive and negative signs.

(Note the greatest value of root will be written in the rightmost)

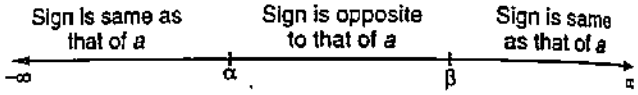
Sign of Quadratic Expression $ax^2 + bx + c$

As we know that if α, β are the roots of the corresponding quadratic equation, then for $x = \alpha$ and $x = \beta$, the value of the expression is equal to zero. *i.e.*, $f(x) = ax^2 + bx + c = 0$.

But for other real values of x (*i.e.*, except α and β) the expression is either less than zero or greater than zero, *i.e.*, $f(x) < 0$ or $f(x) > 0$.

Thus the sign of $ax^2 + bx + c, x \in R$, is determined by the following rules :

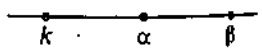
1. If $D < 0$ *i.e.*, α and β are imaginary, then
   $ax^2 + bx + c > 0$, if $a > 0$
   and $ax^2 + bx + c < 0$, if $a < 0$
2. If $D = 0$ *i.e.*, α and β are real and equal, then
   $ax^2 + bx + c \geq 0$, if $a > 0$
   and $ax^2 + bx + c \leq 0$, if $a < 0$
3. If $D > 0$ *i.e.*, α and β are real unequal $(\alpha < \beta)$, then the sign of the expression $ax^2 + bx + c$, $x \in R$ is determined as follows :



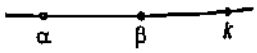
POSITION OF ROOTS OF A QUADRATIC EQUATION WITH RESPECT TO ONE OR TWO REAL NUMBERS

Let $f(x) = ax^2 + bx + c$, where $a, b, c \in R, a \neq 0$. Let α, β, $(\alpha < \beta)$ be the roots of the corresponding quadratic equation. Let $k, k_1, k_2 \in R$ and $k_1 < k_2$.

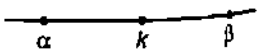
1. **Position of roots with respect to one real number:**
   (a) If both the roots of $f(x) = 0$ are greater than k, then
   (i) $D \geq 0$
   (ii) $af(k) > 0$
   (iii) $k < -\frac{b}{2a}$



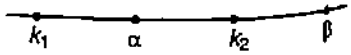
   (b) If both the roots of $f(x) = 0$ are less than k, then
   (i) $D \geq 0$
   (ii) $af(k) > 0$
   (iii) $k > -\frac{b}{2a}$



   (c) If one root is less than k and other is greater than k, then
   (i) $D > 0$
   (ii) $af(k) < 0$



2. **Position of roots with respect to two real numbers:**
   (a) If exactly one root of $f(x) = 0$ lies in the interval (k_1, k_2), then
   (i) $D > 0$
   (ii) $f(k_1) \cdot f(k_2) < 0$

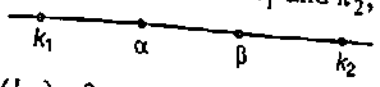


For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

(b) If both roots of $f(x)=0$ lie between k_1 and k_2, then

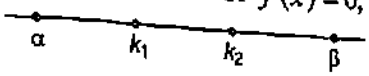


(i) $D \geq 0$

(ii) $af(k_1)>0, af(k_2)>0$

(iii) $k_1 < -\frac{b}{2a} < k_2$

(c) If k_1 and k_2 lie between the roots of $f(x)=0$, then



(i) $D \geq 0$

(ii) $af(k_1)<0, af(k_2)<0$

POLYNOMIAL OF HIGHER DEGREE

Let $a_0, a_1, a_2, ..., a_n$ be real numbers and x is a real variable. Then $f(x)=a_0+a_1x+a_2x^2+...a_nx^n$ is called a real polynomial of degree n.

e.g., $ax^2+bx+c=0$ is a quadratic polynomial of degree 2.

and $ax^3+bx^2+cx+d=0$ is a cubic polynomial of degree 3.

and $ax^4+bx^3+cx^2+dx+e=0$ is a biquadratic polynomial of degree 4.

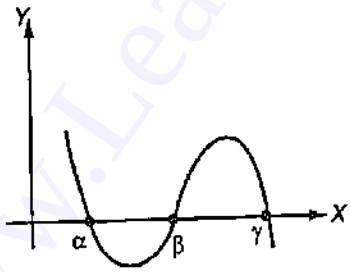
NOTE If $a_0, a_1, a_2, ..., a_n$ be complex numbers and x is a varying complex number, then $f(x)=a_0+a_1x+a_2x^2+...a_nx^n$ is called a complex polynomial.

Roots of a polynomial (expression) : The values of the variable satisfying the given equation are called its roots *i.e.*, $x=\alpha$ is a root of the equation $f(x)=0$

if $f(\alpha)=0$.

NOTE The real roots of an equation $f(x) = 0$ are the x-coordinates of the points through which the curve $y = f(x)$ passes the x-axis.

In the given graph α, β and γ are the roots of a particular cubic polynomial. Simply when $y=0$, then the corresponding values of x are called the roots (or zeros) of the polynomial expression.



- An equation of degree n has n roots (real or imaginary)
- An odd degree equation has at least one real root having sign opposite to that of its last term, provided that the coefficient of highest degree term is positive.
- An even degree equation whose constant term is negative and the coefficient of highest degree term is positive, has at least two real roots, one positive and one negative.
- Surd and imaginary roots always occur in conjugate pairs.

i.e., if $a+\sqrt{b}$ is a root of an equation, then $a-\sqrt{b}$ is also a root of the equation

Similarly if $a+ib$ is a root of an equation, then $a-ib$ is also a root of the given equation.

Sign Rules

1. The maximum number of positive real roots of a polynomial equation $f(x)=0$ is the number of changes of signs from positive to negative and negative to positive in $f(x)$. *e.g.*,

Let an equation be

$$ax^3+bx^2-cx+d=0$$

$\therefore$ The signs of the various terms are

$$+ \; + \; - \; +$$

Clearly there are two changes of sign in the given expression. So, the given equation has at most two positive real roots.

2. The maximum number of negative real roots of the polynomial equation $f(x)=0$ is the number of changes from positive to negative and negative to positive in $f(-x)$.

e.g., Let $f(x)=ax^4-bx^3+cx^2+dx-e=0$

$\therefore \quad f(-x)=ax^4+bx^3+cx^2-dx-e=0$

The signs of various terms in $f(-x)$ are :

$$+ \; + \; + \; - \; -$$

Clearly there is only one change of sign. So, the given equation has at most one negative real solution (or root)

RELATION BETWEEN ROOTS AND COEFFICIENTS

If $f(x)=a_0x^n+a_1x^{n-1}+a_2x^{n-2}+...a_{n-1}x+a_n=0$ and $\alpha_1, \alpha_2, \alpha_3, ..., \alpha_n$ be the roots of the polynomial, then

$$S_1=\alpha_1+\alpha_2+\alpha_3+...\alpha_n=\Sigma\alpha_i=-\frac{a_1}{a_0}$$

$$S_2=\alpha_1\alpha_2+\alpha_1\alpha_3+...=\sum_{i\neq j}\alpha_i\alpha_j=(-1)^2\frac{a_2}{a_0}$$

$$S_3=\alpha_1\alpha_2\alpha_3+\alpha_2\alpha_3\alpha_4+...=\sum_{i\neq j\neq k}\alpha_i\alpha_j\alpha_k=(-1)^3\frac{a_3}{a_0}$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots$$

Similarly

$$S_n=\alpha_1\alpha_2\alpha_3...\alpha_n=(-1)^n\frac{a_n}{a_0}=(-1)^n\frac{a_n}{a_0}$$

$a_n \rightarrow$ constant term

Hence for **quadratic equation** $ax^2+bx+c=0$, having the roots α and β, then

$$\alpha+\beta=\frac{-b}{a} \quad \text{and} \quad \alpha\beta=\frac{c}{a}$$

and for **cubic equation** $ax^3+bx^2+cx+d=0$, having roots α, β and γ, then

$$\alpha+\beta+\gamma=\frac{-b}{a},$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yousmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$$\alpha\beta+\beta\gamma+\gamma\alpha=(-1)^2\frac{c}{a}=\frac{c}{a}$$

and $$\alpha\beta\gamma=(-1)^3\frac{d}{a}=\frac{-d}{a}$$

Similarly for **biquadratic equation**

$$ax^4+bx^3+cx^2+dx+e=0$$

having roots $\alpha, \beta, \gamma, \delta$, then

$$S_1=\alpha+\beta+\gamma+\delta=\frac{-b}{a}$$

$$S_2=\alpha\beta+\beta\gamma+\gamma\delta+\alpha\gamma+\alpha\delta+\beta\delta$$
$$=(-1)^2\frac{c}{a}=\frac{c}{a}$$

or $$S_3=\alpha\beta\gamma+\beta\gamma\delta+\gamma\delta\alpha+\alpha\beta\delta=(-1)^3\frac{d}{a}=-\frac{d}{a}$$

and $$S_4=\alpha\beta\gamma\delta=(-1)^4\frac{e}{a}=\frac{e}{a}$$

Formation of a Polynomial Equation from given Roots

If $\alpha_1, \alpha_2, \ldots, \alpha_n$ are the roots of an nth degree equation, then the equation is

$$x^n-S_1x^{n-1}+S_2x^{n-2}-S_3x^{n-3}+\ldots+(-1)^nS_n=0$$

where S_k $(k=1,2,3,\ldots n)$ denotes the sum of the products of roots taken k at a time.

Formation of a Quadratic Equation

$$x^2-S_1x+S_2=0$$

$$\Rightarrow \quad x^2-(\alpha+\beta)x+(\alpha\beta)=0$$

Formation of a Cubic Equation

$$x^3-S_1x^2+S_2x-S_3=0$$

$$i.e.,\quad x^3-(\alpha+\beta+\gamma)x^2+(\alpha\beta+\beta\gamma+\gamma\alpha)x-\alpha\beta\gamma=0$$

Formation of a Biquadratic Equation

$$x^4-S_1x^3+S_2x^2-S_3x+S_4=0$$

$$i.e.,\ x^4-(\alpha+\beta+\gamma+\delta)x^3+(\alpha\beta+\beta\gamma+\gamma\delta+\alpha\delta$$
$$+\beta\delta+\alpha\gamma)x^2-(\alpha\beta\gamma+\alpha\beta\delta+\beta\gamma\delta+\gamma\delta\alpha)x+\alpha\beta\gamma\delta=0$$

METHODS OF SOLVING THE QUADRATIC EQUATIONS

(1) By factorization method
(2) By Sridharacharya's method

(1) By factorization method : If ax^2+bx+c can be factorized as $(x-\alpha)(x-\beta)$, then $ax^2+bx+c=0$ is equivalent to $(x-\alpha)(x-\beta)=0$

Thus, $$(x-\alpha)(x-\beta)=0$$

$$\Rightarrow \quad (x-\alpha)=0 \text{ or } (x-\beta)=0$$

$$i.e., \quad x=\alpha \text{ or } x=\beta$$

Here α and β are called the roots of the equation $ax^2+bx+c=0$.

The following steps are required to solve the quadratic equation by factorization :

(1) Clear fractions and brackets, if necessary.
(2) Transfer all the terms to LHS and combine like terms.
(3) Write the equation in the standard form, *i.e.*, $ax^2+bx+c=0$
(4) Factorize the LHS.
(5) Put each factor equal to zero and solve.

(2) By Sridharacharya's method : Write the equation in the standard form $ax^2+bx+c=0$, then the roots are

$$\alpha=\frac{-b+\sqrt{b^2-4ac}}{2a}$$

and $$\beta=\frac{-b-\sqrt{b^2-4ac}}{2a}; \qquad b^2-4ac=D$$

EXAMPLE 1 *Which of the following are quadratic equations?*

(a) $3x^2-2=0$ (b) $3x^2-2x+4=\frac{1}{3}(9x^2-x+3)$

(c) $x^3-6x=x^2-3$ (d) $x+\frac{1}{x}=4$

SOLUTION (a) and (d) are only two quadratic equations. On simplifying (b) gives a linear equation and (c) is a cubic equation.

EXAMPLE 2 *Determine whether the given values are solutions (i.e., roots) of the equation or not.*

$$3x^2+5x+2=0, \quad x=3, x=-1$$

SOLUTION On substituting $x=3$, we get

$$3(3)^2+5(3)+2=44\neq 0$$

Therefore $x=3$ is not a solution of the given equation.

Again, substitute $x=-1$, we get

$$3(-1)^2+5(-1)+2=0$$

Therefore $x=-1$ is the solution (*i.e.*, root) of the given equation.

EXAMPLE 3 *Solve the following equation by factorization.*

$$6x^2-x-2=0$$

SOLUTION

$$6x^2-x-2=0$$

$$\Rightarrow \quad 6x^2-4x+3x-2=0$$

$$\Rightarrow \quad 2x(3x-2)+1(3x-2)=0$$

$$\Rightarrow \quad (3x-2)(2x+1)=0$$

Either $(2x+1)=0$ or $(3x-2)=0$

$$\Rightarrow \quad x=-\frac{1}{2} \text{ or } x=\frac{2}{3}$$

$$x=-\frac{1}{2},\frac{2}{3}$$

EXAMPLE 4 *Find the value of K for which $x=2$ is a root of the equation $3x^2-2Kx+5=0$.*

SOLUTION If $x=2$ is a root of the given equation, then $x=2$ must satisfy it.

$$\therefore \quad 3(2)^2-2K(2)+5=0$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$$12 - 4K + 5 = 0$$

$$\Rightarrow \quad -4K = -17 \quad \Rightarrow \quad K = \frac{17}{4}$$

$$\Rightarrow$$

EXAMPLE 5 *Solve the following equation by using the formula :*

$$8x - 2x^2 = 5$$

SOLUTION $8x - 2x^2 = 5 \Rightarrow 2x^2 - 8x + 5 = 0$

Here $a = 2, \quad b = -8 \quad \text{and} \quad c = 5$

Substituting the values of a, b and c in the formula

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \quad \text{and} \quad \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

$$\alpha = \frac{8 + \sqrt{64 - 40}}{4} \quad \text{and} \quad \beta = \frac{8 - \sqrt{64 - 40}}{4}$$

$$\alpha = \frac{8 + 2\sqrt{6}}{4} \quad \text{and} \quad \beta = \frac{8 - 2\sqrt{6}}{4}$$

$$\alpha = \frac{4 + \sqrt{6}}{2} \quad \text{and} \quad \beta = \frac{4 - \sqrt{6}}{2}$$

NOTE The values of α and β can be interchanged.

EXAMPLE 6 *Write the discriminant of the following quadratic equations and mention the nature of the root :*

(a) $9x^2 - 6x + 1 = 0$ *(b)* $x^2 + x - 12 = 0$

(c) $4\sqrt{3}x^2 + 5x + 2\sqrt{3} = 0$

SOLUTION (a) $D = b^2 - 4ac = (-6)^2 - 4(9)(1) = 0$

Since $D = 0$, therefore the roots are real and equal.

(b) $D = b^2 - 4ac = (1)^2 - 4 \times 1 \times (-12) = 49$

Since $D > 0$ and D is a perfect square, therefore the roots are real, unequal and rational.

(c) $D = b^2 - 4ac = (5)^2 - 4(4\sqrt{3}) \times (2\sqrt{3}) = -71$

Since $D < 0$, therefore the roots are complex (*i. e.*, no real roots).

SOLUTIONS OF EQUATIONS REDUCIBLE TO QUADRATIC FORM

Equation which are not quadratic at a glance but can be reduced to quadratic equations by suitable transformations. Some of the common types are :

Type 1. $ax^4 + bx^2 + c = 0.$

This can be reduced to a quadratic equation by substituting $x^2 = y$. *i.e.*, $ay^2 + by + c = 0$

Tpye 2. $px + \frac{q}{x} = r,$

$$p(x - a) + \frac{q}{(x - b)} = r$$

Multiply both sides by the LCD of LHS to get a quadratic equation :

$$px + \frac{q}{x} = r \quad \Rightarrow \quad px^2 - rx + q = 0$$

and $$p(x - a) + \frac{q}{(x - b)} = r$$

$$\Rightarrow \quad p(x - a)(x - b) - r(x - b) + q = 0$$

Type 3. $\sqrt{a - x^2} = bx + c$

Square both sides and simplify.

i.e., $a - x^2 = b^2x^2 + 2bcx + c^2$

$$\Rightarrow \quad (1 + b^2)x^2 + 2bcx + (c^2 - a) = 0$$

Type 4. $\sqrt{ax + b} + \sqrt{cx + d} = e$

Transform one of the radicals to RHS and square $\sqrt{ax + b} = e - \sqrt{cx + d}$, such equations may require squaring and your solution must satisfy $ax + b \geq 0$ and $cx + d \geq 0$

Type 5. $a\left(x^2 + \frac{1}{x^2}\right) + b\left(x + \frac{1}{x}\right) + c = 0$

For this type of equations we use the following identity

$$x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2$$

Thus $a\left[\left(x + \frac{1}{x}\right)^2 - 2\right] + b\left(x + \frac{1}{x}\right) + c = 0$

Now put $x + \frac{1}{x} = y$ to get a quadratic equation *i.e.*, $ay^2 + by + (c - 2a) = 0$

Type 6. $a\left(x^2 + \frac{1}{x^2}\right) + b\left(x - \frac{1}{x}\right) + c = 0$

Use the following identity.

$$x^2 + \frac{1}{x^2} = \left(x - \frac{1}{x}\right)^2 + 2$$

Thus, $a\left[\left(x - \frac{1}{x}\right)^2 + 2\right] + b\left(x - \frac{1}{x}\right) + c = 0$, put $x - \frac{1}{x} = y$ to get a quadratic equation in y.

i.e., $ay^2 + by + (c - 2a) = 0$

Type 7. (i) $x^{2a} + x^a + b = 0$ (ii) $x^a + x^{-a} = b$

Put $x^a = y$ to get a quadratic equation in y,

i.e., (i) $y^2 + y + b = 0$ (ii) $y + \frac{1}{y} = b$

NOTE In all the equations involving radical signs, the answer must be checked by substituting in the original equation.

Type 8. $(x + a)(x + b)(x + c)(x + d) + k = 0.$

When sum of the quantities a, b, c, d is equal to the sum of the other two, can be solved as shown in example 16.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

EXAMPLE 7 *Solve for* $y : 9y^4 - 29y^2 + 20 = 0$.

SOLUTION $9y^4 - 29y^2 + 20 = 0$

Put $y^2 = x$.

$$9x^2 - 29x + 20 = 0$$
$$\Rightarrow \quad 9x^2 - 20x - 9x + 20 = 0$$
$$\Rightarrow \quad (x-1)(9x-20) = 0$$
$$\Rightarrow \quad x = 1 \quad \text{or} \quad x = \frac{20}{9}$$
$$\Rightarrow \quad y^2 = 1 \quad \text{or} \quad y^2 = \frac{20}{9}$$
$$\Rightarrow \quad y = \pm 1 \quad \text{and} \quad y = \pm \frac{2\sqrt{5}}{3}$$

EXAMPLE 8 *Solve for* $x : x^6 - 26x^3 - 27 = 0$.

SOLUTION $x^6 - 26x^3 - 27 = 0$

Let $x^3 = y$

then, $y^2 - 26y - 27 = 0$

$\Rightarrow \quad y^2 - 27y + y - 27 = 0$

$\Rightarrow \quad (y+1)(y-27) = 0$

$\Rightarrow \quad y = -1 \quad \text{or} \quad y = 27$

$\Rightarrow \quad x^3 = -1 \quad \text{or} \quad x^3 = 27$

$\Rightarrow \quad x = -1 \quad \text{or} \quad x = 3$

EXAMPLE 9. *Solve :* $2x - \frac{3}{x} = 5$

SOLUTION $2x - \frac{3}{x} = 5$

$\Rightarrow \quad 2x^2 - 3 = 5x$

$\Rightarrow \quad (2x+1)(x-3) = 0$

$\Rightarrow \quad x = -\frac{1}{2} \quad \text{or} \quad x = 3$

EXAMPLE 10 *Solve* $\sqrt{2x+9} + x = 13$

SOLUTION $\sqrt{2x+9} + x = 13$

$\Rightarrow \quad \sqrt{2x+9} = 13 - x$

Squaring both sides

$$2x + 9 = (13-x)^2$$
$$\Rightarrow \quad x^2 - 28x + 160 = 0$$
$$\Rightarrow \quad (x-8)(x-20) = 0$$
$$\Rightarrow \quad x = 8, \quad x = 20$$

NOTE Here we must have $2x + 9 \geq 0$ Thus for $x = 8, \sqrt{2x+9} > 0$
Also for $x = 20, \sqrt{2x+9} > 0$.
$\therefore$ Both the values satisfy the condition.

EXAMPLE 11 *Solve* $\sqrt{2x+9} - \sqrt{x-4} = 3$

SOLUTION $\sqrt{2x+9} - \sqrt{x-4} = 3$

$\Rightarrow \quad \sqrt{2x+9} = 3 + \sqrt{x-4}$

Squaring both sides and simplifying, we get

$$x + 4 = 6\sqrt{x-4}$$

Again squaring both sides,

$$(x+4)^2 = 36(x-4)$$
$$\Rightarrow \quad x^2 - 28x + 160 = 0$$
$$\Rightarrow \quad (x-8)(x-20) = 0$$
$$\Rightarrow \quad x = 8, \quad x = 20$$

Verification: $2x + 9 \geq 0$ and $x - 4 \geq 0$

$$\Rightarrow \quad x \geq \frac{-9}{2} \quad \text{and} \quad x \geq 4$$

Since the values $x = 8$ and 20 satisfy both these conditions

$$\therefore \quad x = 8, \quad x = 20$$

EXAMPLE 12 *Solve for* $x : 2\left(x^2 + \frac{1}{x^2}\right) - 9\left(x + \frac{1}{x}\right) + 14 = 0$

SOLUTION Put $x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2$

$$\therefore \quad 2\left(x^2 + \frac{1}{x^2}\right) - 9\left(x + \frac{1}{x}\right) + 14 = 0$$
$$\Rightarrow \quad 2\left[\left(x + \frac{1}{x}\right)^2 - 2\right] - 9\left(x + \frac{1}{x}\right) + 14 = 0$$

Substitute $x + \frac{1}{x} = y$

$$\therefore \quad 2(y^2 - 2) - 9y + 14 = 0$$
$$\Rightarrow \quad 2y^2 - 4 - 9y + 14 = 0$$
$$\Rightarrow \quad (y-2)(2y-5) = 0$$
$$\Rightarrow \quad y = 2 \quad \text{or} \quad y = \frac{5}{2}$$

Since $x + \frac{1}{x} = 2 \Rightarrow x^2 - 2x + 1 = 0$

$$\Rightarrow \quad (x-1)^2 = 0 \Rightarrow x = 1$$

Also, $y = \frac{5}{2} \Rightarrow x + \frac{1}{x} = \frac{5}{2}$

$$\Rightarrow \quad 2x^2 - 5x + 2 = 0$$
$$\Rightarrow \quad x = 2 \quad \text{or} \quad x = \frac{1}{2}$$
$$\therefore \quad x = \frac{1}{2}, 1, 2$$

EXAMPLE 13 *Solve* $6\left(x^2 + \frac{1}{x^2}\right) - 25\left(x - \frac{1}{x}\right) + 12 = 0$

SOLUTION Put $x^2 + \frac{1}{x^2} = \left(x - \frac{1}{x}\right)^2 + 2$

$$\therefore \quad 6\left[\left(x - \frac{1}{x}\right)^2 + 2\right] - 25\left(x - \frac{1}{x}\right) + 12 = 0$$

Let $x - \frac{1}{x} = y$

$$\Rightarrow \quad 6(y^2 + 2) - 25y + 12 = 0$$
$$\Rightarrow \quad y = \frac{3}{2}, \quad y = \frac{8}{3}$$

Since $y = x - \frac{1}{x}$ and $x - \frac{1}{x} = \frac{8}{3}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

$\therefore \quad x - \frac{1}{x} = \frac{3}{2}$ and $x = -\frac{1}{3}$, $x = 3$

$\Rightarrow \quad x = -\frac{1}{2}$ and $x = 2$

$\therefore \quad x = -\frac{1}{3}, -\frac{1}{2}, 2, 3$

EXAMPLE 14 *Solve for*

$x : \sqrt{x^2 + x - 6} - x + 2 = \sqrt{x^2 - 7x + 10}, \quad x \in R$

SOLUTION $\sqrt{x^2 + x - 6} - x + 2 = \sqrt{x^2 - 7x + 10}$

$\sqrt{(x+3)(x-2)} - (x-2) = \sqrt{(x-5)(x-2)}$

$\Rightarrow \sqrt{(x-2)}[\sqrt{(x+3)} - \sqrt{(x-2)} - \sqrt{(x-5)}] = 0$

Either $\sqrt{(x-2)} = 0 \Rightarrow x = 2$

or $\sqrt{(x+3)} - \sqrt{(x-2)} - \sqrt{(x-5)} = 0$

$\Rightarrow \sqrt{(x+3)} - \sqrt{(x-2)} = \sqrt{x-5}$

Squaring both sides

$x^2 + 12x + 36 = 4(x^2 + x - 6)$

$\Rightarrow \quad x = 6, \quad x = -\frac{10}{3}$

Since the equation involves radical therefore substituting $x = 2, 6$ and $-\frac{10}{3}$ in the original equation, we find that $x = -\frac{10}{3}$ does not satisfy the equation.

$\therefore \quad x = 2, 6$

EXAMPLE 15 *Solve for* $x : 3^{x+2} + 3^{-x} = 10$

SOLUTION $3^{x+2} + 3^{-x} - 10 = 0$

$\Rightarrow \quad 3^x \cdot 3^2 + \frac{1}{3^x} - 10 = 0$

Let $3^x = y \Rightarrow 9y + \frac{1}{y} - 10 = 0$

$\Rightarrow \quad 9y^2 - 10y + 1 = 0$

$\Rightarrow \quad (9y-1)(y-1) = 0 \Rightarrow y = \frac{1}{9}$ or $y = 1$

When $y = \frac{1}{9} \Rightarrow 3^x = \frac{1}{3^2}$

$\Rightarrow \quad x = -2$

when $y = 1 \Rightarrow 3^x = 1$

$\Rightarrow \quad x = 0$

$\therefore \quad x = -2, 0$

EXAMPLE 16 *Solve for* $x : (x+1)(x+2)(x+3)(x+4) = 24$

SOLUTION $(x+1)(x+2)(x+3)(x+4) = 24$

$\Rightarrow \quad [(x+1)(x+4)][(x+2)(x+3)] = 24$

$\Rightarrow \quad (x^2 + 5x + 4)(x^2 + 5x + 6) = 24$

Let $x^2 + 5x = y$

$\therefore \quad (y+4)(y+6) = 24$

$\Rightarrow \quad y^2 + 10y = 0$

$\Rightarrow \quad y = 0$ and $y = -10$

Now, $y = x^2 + 5x$

When $y = 0$, $x^2 + 5x = 0 \Rightarrow x = 0, x = -5$

Again when $y = -10$, $x^2 + 5x = -10$

$\Rightarrow \quad x^2 + 5x + 10 = 0$

Since LHS expression cannot be factorized, therefore we should use the formula for finding the value of x.

Here $D = b^2 - 4ac$

$= 25 - 4 \times 1 \times 10 = -15$

Since $D < 0$, the equation $x^2 + 5x + 10 = 0$ has no real solution

$\therefore \quad x = 0, -5$

EXAMPLE 17 *Find the sum and product of the roots of the quadratic equation :*

$3x^2 + 2x - 1 = 0$

SOLUTION Let α, β be the two roots of the equation then,

$\alpha + \beta = -\frac{b}{a} = -\frac{2}{3}$

and $\alpha\beta = \frac{c}{a} = -\frac{1}{3}$

EXAMPLE 18 *Find the quadratic equation whose roots are, –3, 5*

SOLUTION **Method (I):**

$\alpha + \beta = -3 + 5 = 2$

and $\alpha\beta = -3 \times 5 = -15$

$\therefore$ The required quadratic equation

$x^2 - (\alpha + \beta)x + (\alpha\beta) = 0$

$\Rightarrow \quad x^2 - (2)x + (-15) = 0$

$\Rightarrow \quad x^2 - 2x - 15 = 0$

Method (II): Let $\alpha = -3$ and $\beta = 5$, then the required equation

$(x - \alpha)(x - \beta) = 0$

$\Rightarrow \quad (x+3)(x-5) = 0$

$\Rightarrow \quad x^2 - 2x - 15 = 0$

EXAMPLE 19 *If α and β are the roots of the equation $ax^2 + bx + c = 0$, find the values of the following expressions in terms of a, b and c*

(a) $\alpha^2 + \beta^2$ *(b)* $\frac{1}{\alpha} + \frac{1}{\beta}$ *(c)* $\alpha^4 - \beta^4$

SOLUTION (a) Let $ax^2 + bx + c = 0$ and α, β be the roots of the equation, then

$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}$

$\therefore \quad \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$= \left(-\frac{b}{a}\right)^2 - \frac{2c}{a} = \frac{b^2}{a^2} - \frac{2c}{a}$

$= \frac{b^2 - 2ac}{a^2}$.

(b) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{-b/a}{c/a} = -\frac{b}{c}$

(c) $\alpha^4 - \beta^4 = (\alpha^2 + \beta^2)(\alpha + \beta)(\alpha - \beta)$

$= [(\alpha + \beta)^2 - 2\alpha\beta](\alpha + \beta)\sqrt{(\alpha + \beta)^2 - 4\alpha\beta}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

$$= \left(\frac{b^2 - 2ac}{a^2}\right)\left(-\frac{b}{a}\right)\sqrt{\frac{b^2}{a^2} - 4\frac{c}{a}}$$

$$= -\frac{b}{a^4}(b^2 - 2ac)\sqrt{b^2 - 4ac}$$

EXAMPLE 20 *If α, β are the roots of the equation $2x^2 - 3x + 2 = 0$, form the equation whose roots are α^2, β^2.*

SOLUTION $2x^2 - 3x + 2 = 0$, and α, β are the roots

$$\therefore \quad \alpha + \beta = \frac{3}{2}, \quad \alpha\beta = 1$$

For the new equation, roots are α^2 and β^2

∴ Sum of the roots

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \left(\frac{3}{2}\right)^2 - 2(1)$$

$$= \frac{9}{4} - 2 = \frac{1}{4}$$

and product of the roots $= \alpha^2\beta^2 = (\alpha\beta)^2 = (1)^2 = 1$

∴ the required equation is

$x^2 -$ (sum of roots) $x +$ product of roots $= 0$

$$\Rightarrow \quad x^2 - \frac{1}{4}x + 1 = 0$$

$$\Rightarrow \quad 4x^2 - x + 4 = 0$$

EXAMPLE 21 *Determine k for which the roots of the equation $9x^2 + 2kx + 4 = 0$ are equal.*

SOLUTION Let the roots be α and α.

$$\therefore \quad \text{Sum of roots} = \alpha + \alpha = 2\alpha = -\frac{2k}{9}$$

$$\Rightarrow \quad \alpha = -\frac{k}{9}$$

and product of the roots $= \alpha^2 = \frac{4}{9}$

$$\therefore \quad \left(-\frac{k}{9}\right)^2 = \frac{4}{9} \Rightarrow \frac{k^2}{81} = \frac{4}{9}$$

$$\Rightarrow \quad k^2 = 36 \Rightarrow k = \pm 6.$$

Alternatively: In order that roots of a quadratic equation are equal, its discriminant must be zero.

i.e., $b^2 - 4ac = 0$

$$\therefore \quad (2k)^2 - 4 \times 9 \times 4 = 0$$

$$\Rightarrow \quad k = \pm 6.$$

EXAMPLE 22 *Which of the following quadratic expression can be expressed as a product of real linear factors?*

(a) $x^2 - 2x + 3$ *(b)* $3x^2 - \sqrt{2}x - \sqrt{3}$

(c) $2x^2 + 3x - 4$ *(d)* $\sqrt{2}x^2 - \sqrt{5}x + 3$

SOLUTION (a) $D = b^2 - 4ac < 0$ cannot be expressed

(b) $D = b^2 - 4ac > 0$ can be expressed

(c) $D = b^2 - 4ac > 0$ can be expressed

(d) $D = b^2 - 4ac < 0$, can't be expressed.

NOTE If $D < 0$, then the quadratic equation cannot be expressed into two linear factors.

EXAMPLE 23 *Find the set of values of p for which the quadratic equation has real linear factors.*

$$9x^2 - px + 4$$

SOLUTION For any quadratic polynomial to have real linear factors, we must have $D \geq 0$

$$\therefore \quad b^2 - 4ac \geq 0$$

$$\therefore \quad p^2 - 4 \times 9 \times 4 \geq 0$$

$$p^2 - 144 \geq 0$$

$$\Rightarrow \quad p^2 \geq 144$$

$$\Rightarrow \quad p \geq \pm 12$$

$$\Rightarrow \quad \text{Either } p \leq -12 \text{ or } p \geq 12$$

EXAMPLE 24 *The length of a rectangle is 7 cm more than its width. The area is 60 cm^2. Find the dimensions of the rectangle.*

SOLUTION Let width of the rectangle $= x$

$$\therefore \quad \text{Length of rectangle} = (x + 7) \text{ cm.}$$

$$\therefore \quad \text{Area of rectangle} = (x + 7) \times x$$

$$\Rightarrow \quad (x + 7)(x) = 60$$

$$\Rightarrow \quad x^2 + 7x - 60 = 0$$

$$\Rightarrow \quad (x + 12)(x - 5) = 0$$

$$\Rightarrow \quad x = -12 \text{ and } x = 5$$

Since width can never be negative, therefore $x = 5$ cm and $x + 7 = 12$ cm are the required values.

i.e., Length $= 12$ cm and width $= 5$ cm

EXAMPLE 25 *A passenger train takes 2 hours less for a journey of 300 km if its speed is increased by 5 km/hr from its usual speed. Find the usual speed of the train.*

SOLUTION Let the usual speed of the train be x km/hr

$$\therefore \quad \text{Usual time} = \frac{\text{Distance}}{\text{speed}} = \frac{300}{x} \text{ hour}$$

If the speed is increased by 5 km/hr *i.e.,* $(x + 5)$ km/hr,

the time $= \frac{300}{x + 5}$ hour

$$\therefore \quad \frac{300}{x} - \frac{300}{x + 5} = 2$$

$$\Rightarrow \quad 2x(x + 5) = 1500$$

$$\Rightarrow \quad x^2 + 5x - 750 = 0$$

$$\Rightarrow \quad x = 25 \text{ or } x = -30$$

$$\therefore \quad \text{Speed} = 25 \text{ km/hr } (x = -30 \text{ is inadmissible})$$

EXAMPLE 26 *A two digit number is such that the product of its digits is 12. When 36 is added to the number, its digits are reversed. Find the number.*

SOLUTION Let the number be $10x + y$

$$\therefore \quad xy = 12 \Rightarrow y = \frac{12}{x}$$

$$\therefore \text{ the number is } 10x + \frac{12}{x}$$

and the number with digits reversed is

$$10y + x = 10\left(\frac{12}{x}\right) + x = \frac{120}{x} + x$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$$\left(10x + \frac{12}{x}\right) + 36 = \frac{120}{x} + x$$

$$\therefore \quad 10x - x + \frac{12}{x} - \frac{120}{x} + 36 = 0$$

$$\Rightarrow \quad 9x + \frac{1}{x}(12 - 120) + 36 = 0$$

$$\Rightarrow \quad 9x - \frac{108}{x} + 36 = 0$$

$$\Rightarrow \quad 9x^2 + 36x - 108 = 0$$

$$\Rightarrow \quad x = 2 \text{ or } x = -6$$

$$\therefore \quad y = \frac{12}{x} = \frac{12}{2} = 6 \quad (x = -6 \text{ is inadmissible})$$

$$\therefore$$

Hence, the number is $10x + y = 10 \times 2 + 6 = 20 + 6 = 26$.

EXAMPLE 27 *Two candidates attempt to solve a quadratic equation of the form $x^2 + px + q = 0$. One starts with a wrong value of p and finds the roots to be 2 and 6. The other starts with a wrong value of q and finds the roots to be 2 and –9. Find the correct roots and the equation.*

SOLUTION When p is wrong i.e., $\frac{-b}{a}(= \alpha + \beta)$ is wrong but $\frac{c}{a}(= \alpha \cdot \beta)$ is correct.

Hence $\alpha\beta = \frac{c}{a} = 2 \times 6 = 12$

Again when q is wrong i.e., $\frac{c}{a} = (\alpha\beta)$ is wrong but $\frac{-b}{a} = \alpha + \beta$ is correct.

$$\therefore \quad \frac{-b}{a} = \alpha + \beta = 2 + (-9) = -7$$

$\therefore$ the required correct quadratic equation is

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

$$x^2 - (-7)x + 12 = 0$$

$$\Rightarrow \quad x^2 + 7x + 12 = 0$$

and the correct roots of this equation are $-3, -4$.

EXAMPLE 28 *For which value of k will the equations $x^2 - kx - 21 = 0$ and $x^2 - 3kx + 35 = 0$ have one common root.*

SOLUTION Let the common root be α, then,

$$\alpha^2 - k\alpha - 21 = 0 \quad \ldots(i)$$

$$\alpha^2 - 3k\alpha + 35 = 0 \quad \ldots(ii)$$

Solving by the rules of cross multiplication,

$$\frac{\alpha^2}{-35k - 63k} = \frac{\alpha}{-21 - 35} = \frac{1}{-3k + k}$$

$$\therefore \quad \alpha = \frac{-98k}{-56} = \frac{7k}{4}$$

and $\therefore \frac{7k}{4} = \frac{28}{k} \Rightarrow 7k^2 = 28 \times 4 \Rightarrow k = \pm 4$

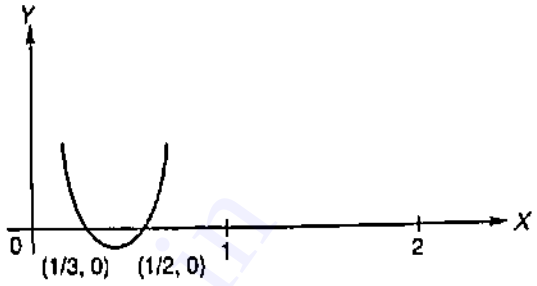
EXAMPLE 29 *Find the sign of $6x^2 - 5x + 1$ for all real values of x.*

SOLUTION $\because \quad D = b^2 - 4ac$

$$D = 25 - 4 \times 6 \times 1 = 1$$

$\because$ The roots are real and different and the roots are $\frac{5 \pm \sqrt{25 - 24}}{12}$ i.e., $\frac{1}{2}$ and $\frac{1}{3}$

$\therefore$ The given expression has the same sign as the coefficient of x^2, i.e., positive for all real values of x except for those which lie between $\frac{1}{3}$ and $\frac{1}{2}$



EXAMPLE 30 *Find the ranges of the values of x for which $x^2 - 4x + 2$ lies between +1 and –1.*

SOLUTION $x^2 - 4x + 2 > -1$ or $x^2 - 4x + 3 > 0$

Since the coefficient of x^2 is positive and the roots of $x^2 - 4x + 3 = 0$ are 1 and 3, therefore

$$x^2 - 4x + 3 > 0 \text{ for } x < 1 \text{ and } x > 3 \quad \ldots(i)$$

Again, $x^2 - 4x + 2 < 1$ or $x^2 - 4x + 1 < 0$

Since the coefficient of x^2 is positive and the roots of $x^2 - 4x + 1 = 0$ are $2 - \sqrt{3}$ and $2 + \sqrt{3}$, therefore

$$x^2 - 4x + 1 < 0 \text{ for } 2 - \sqrt{3} < x < 2 + \sqrt{3} \quad \ldots(ii)$$

Combining (i) and (ii), we get $2 - \sqrt{3} < x < 1$ or $3 < x < 2 + \sqrt{3}$

Which are the required ranges of the values of x.

EXAMPLE 31 *If x be real, find the maximum value of $7 + 10x - 5x^2$.*

SOLUTION

NOTE When $a < 0$, we get maxima otherwise if $a > 0$ we get minima.

As we know, at $x = \frac{-b}{2a}$, we get the maxima,

$$\therefore \quad y = \frac{4ac - b^2}{4a} = \frac{4 \times (-5) \times 7 - (10)^2}{4 \times (-5)}$$

$$= \frac{-140 - 100}{-20} = \frac{-240}{-20} = 12$$

Thus the maximum value of the given quadratic equation is 12.

EXAMPLE 32 *If x is real, prove that the value of the expression $\frac{(x-1)(x+3)}{(x-2)(x+4)}$ cannot be between $\frac{4}{9}$ and 1.*

SOLUTION Let $\frac{(x-1)(x+3)}{(x-2)(x+4)} = y$, then

$$x^2 + 2x - 3 = y(x^2 + 2x - 8)$$

or $\quad x^2(1 - y) + 2x(1 - y) + (8y - 3) = 0$

For x to be real, its discriminant $D \geq 0$

or $\quad [2(1 - y)]^2 - 4(1 - y)(8y - 3) \geq 0$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

or $(1-y)(4-9y) \geq 0$

or $(1-y)\left(\frac{4}{9}-y\right) \geq 0$

It shows that $(1-y)$ and $\left(\frac{4}{9}-y\right)$ should be of the same sign.

$\therefore$ we get $y = \frac{4}{9}, 1$

If $y < \frac{4}{9}$, $(1-y)\left(\frac{4}{9}-y\right) > 0$

If $\frac{4}{9} < y < 1$, $(1-y)\left(\frac{4}{9}-y\right) < 0$

If $y > 1$ $(1-y)\left(\frac{4}{9}-y\right) > 0$

Hence for x to be real, y, (i.e., the value of the given expression) cannot lie between $\frac{4}{9}$ and 1.

EXAMPLE 33 *If x be real, find the maximum value of* $\frac{(x+2)}{(2x^2+3x+6)}$.

SOLUTION Let $y = \frac{x+2}{2x^2+3x+6}$

then $2x^2y + (3y-1)x + 6y - 2 = 0$

For x to be real, $(3y-1)^2 - 8y(6y-2) \geq 0$

or $(1+13y)(1-3y) \geq 0$

or $(13y+1)(3y-1) \leq 0$

Putting each factor equal to zero, we get

$$y = -\frac{1}{13}, \frac{1}{3}$$

If $y < -\frac{1}{13}$, $(1+13y)(1-3y) < 0$

If $-\frac{1}{13} < y < \frac{1}{3}$, $(1+13y)(1-3y) > 0$

If $y > \frac{1}{3}$, $(1+13y)(1-3y) < 0$

Thus, y will lie between $-\frac{1}{13}$ and $\frac{1}{3}$.

Hence the maximum value of y is $\frac{1}{3}$ and minimum value is $-\frac{1}{13}$.

EXAMPLE 34 *Solve $x^2 - 5x + 4 > 0$.*

SOLUTION $x^2 - 5x + 4 = (x-1)(x-4)$



Since $f(x) = x^2 - 5x + 4 > 0$

therefore the required range is $x < 1$ and $x > 4$.

EXAMPLE 35 *Solve $-x^2 + 6x - 8 > 0$*

SOLUTION $-x^2 + 6x - 8 > 0$

$\Rightarrow$ $x^2 - 6x + 8 < 0$

$\Rightarrow$ Now $x^2 - 6x + 8 = (x-2)(x-4)$



Since $f(x) = x^2 - 6x + 8 < 0$

therefore the required range is $2 < x < 4$.

EXAMPLE 36 *Solve the inequality* $\frac{x-2}{x+2} > \frac{2x-3}{4x+1}$.

SOLUTION The range of the variable x in this inequality consists of all values of x except $x = -2$ and $x = 1/4$. Hence we cannot cross multiply. So we adopted another method

$$\frac{x-2}{x+2} - \frac{2x-3}{4x-1} > 0 \quad \ldots(i)$$

or $$\frac{(x-2)(4x-1) - (x+2)(2x-3)}{(x+2)(4x-1)} > 0$$

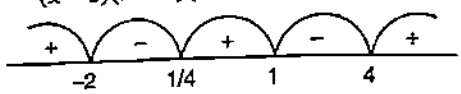
or $$\frac{2(x^2-5x+4)}{(x+2)(4x-1)} > 0$$

or $$\frac{(x-1)(x-4)}{(x+2)(x-1/4)} > 0 \quad \ldots(ii)$$

Now multiply both sides of (ii) by the expression $(x+2)^2\left(x-\frac{1}{4}\right)^2$, which is positive for the x under consideration.

$\therefore$ $$\frac{(x-1)(x-4)(x+2)^2\left(x-\frac{1}{4}\right)^2}{(x+2)\left(x-\frac{1}{4}\right)} > 0$$

or $$(x-1)(x-4)(x+2)(x-1/4) > 0 \quad \ldots(iii)$$



Thus, the range is $x < -2$ or $\frac{1}{4} < x < 1$ or $x > 4$

EXAMPLE 37 *Determine the range of values of x for which* $\frac{x^2-2x+5}{3x^2-2x-5} > \frac{1}{2}$.

SOLUTION $$\frac{x^2-2x+5}{3x^2-2x-5} > \frac{1}{2},$$

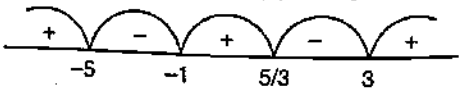
or $$\frac{x^2-2x+5}{3x^2-2x-5} - \frac{1}{2} > 0$$

or $$\frac{x^2+2x-15}{3x^2-2x-5} < 0$$

or $$\frac{(x+5)(x-3)}{(x+1)(3x-5)} < 0$$

or $$\frac{(x+5)(x-3)(x+1)^2(3x-5)^2}{(x+1)(3x-5)} < 0$$

or $$(x+5)(x-3)(x+1)(3x-5) < 0$$



$\therefore$ $-5 < x < -1$ or $\frac{5}{3} < x < 3$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

EXAMPLE 38 *Find the range of real values of x for which* $\frac{x-1}{(4x+5)} < \frac{x-3}{4x-3}$.

SOLUTION $\frac{x-1}{4x+5} < \frac{x-3}{4x-3}$ or $\frac{x-1}{4x+5} - \frac{x-3}{4x-3} < 0$

or $\frac{(x-1)}{(4x+5)} \times \frac{(4x-3)}{(4x-3)} - \frac{(x-3)}{(4x-3)} \times \frac{(4x+5)}{(4x+5)} < 0$

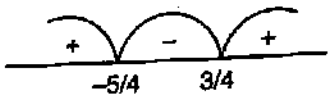
or $\frac{4x^2-7x+3}{(4x+5)(4x-3)} - \frac{(4x^2-7x-15)}{(4x+5)(4x-3)} < 0$

or $\frac{18}{(4x+5)(4x-3)} < 0$

Since 18 is positive

$\therefore (4x+5)(4x-3) < 0$

or $16\left(x+\frac{5}{4}\right)\left(x-\frac{3}{4}\right) < 0$



$\therefore \frac{-5}{4} < x < \frac{3}{4}$

INTRODUCTORY EXERCISE-14.1

Directions for question number 1–12: Find the roots of the following equations.

1. $15x^2 - 7x - 36 = 0$:
(a) $\frac{5}{9}, -\frac{4}{3}$ (b) $\frac{9}{5}, -\frac{4}{3}$
(c) $\frac{9}{5}, -\frac{3}{4}$ (d) none of these

2. $\sqrt{7}y^2 - 6y - 13\sqrt{7} = 0$:
(a) $\sqrt{7}, 2\sqrt{7}$ (b) $3, \frac{2}{\sqrt{7}}$
(c) $\frac{13}{\sqrt{7}}, -\sqrt{7}$ (d) none of these

3. $6x^2 + 40 = 31x$:
(a) $\frac{3}{8}, \frac{2}{5}$ (b) $\frac{3}{8}, \frac{3}{2}$
(c) $0, \frac{8}{3}$ (d) $\frac{8}{3}, \frac{5}{2}$

4. $6\left(\frac{y-3}{2y+1}\right) + 1 = 5\sqrt{\frac{y-3}{2y+1}}, y \neq \frac{1}{2}$:
(a) $4, \frac{13}{2}$ (b) $4, \frac{11}{2}$
(c) $\frac{13}{2}, \frac{9}{2}$ (d) none of these

5. $(x+2)(x-5)(x-6)(x+1) = 144$:
(a) $-1, -2, -3$ (b) $7, -3, 2$
(c) $2, -3, 5$ (d) none of these

6. $\sqrt{x+5} + \sqrt{x+21} = \sqrt{6x+40}$:
(a) 1 (b) 2
(c) 4 (d) 6

7. $2\left(x^2 + \frac{1}{x^2}\right) - 3\left(x + \frac{1}{x}\right) - 1 = 0$:
(a) $2, \frac{1}{2}$ (b) $-2, 4$
(c) $2, \frac{3}{2}$ (d) none of these

8. $\sqrt{\frac{x}{1-x}} + \sqrt{\frac{1-x}{x}} = \frac{13}{6}$:
(a) $\frac{13}{4}, 5$ (b) $\frac{4}{13}, \frac{9}{13}$
(c) $\frac{4}{13}, 6$ (d) none of these

9. $\sqrt{2x^2 - 2x + 1} - 2x + 3 = 0$:
(a) 2 (b) 3
(c) 4 (d) 5

10. $\sqrt{x^2 - 9x + 20} - \sqrt{x^2 - 12x + 32} = \sqrt{2x^2 - 25x + 68}$:
(a) 4, 9 (b) 3, 9
(c) 2, 4 (d) 2, 6

11. $\left(x - \frac{1}{x}\right)^2 + 8\left(x + \frac{1}{x}\right) = 29, x \neq 0$:
(a) $-5, 4$ (b) 3, 2
(c) 1, 3 (d) $\frac{3+\sqrt{5}}{2}, \frac{-11 \pm 3\sqrt{13}}{2}$

12. $\left(\frac{1}{x+1} + \frac{1}{x+5}\right) = \left(\frac{1}{x+2} + \frac{1}{x+4}\right)$:
(a) 7 (b) -3
(c) -5 (d) 4

13. Determine k such that the quadratic equation $x^2 + 7(3 + 2k) - 2x(1 + 3k) = 0$ has equal roots :
(a) 2, 7 (b) 7, 5
(c) $2, \frac{-10}{9}$ (d) none of these

14. Discriminant of the equation $-3x^2 + 2x - 8 = 0$ is :
(a) -92 (b) -29
(c) 39 (d) 49

15. The nature of the roots of the equation $x^2 - 5x + 7 = 0$ is :
(a) no real roots (b) 1 real root
(c) can't be determined (d) none of these

16. The roots of $a^2x^2 + abx = b^2, a \neq 0$ are :
(a) equal (b) non-real
(c) unequal (d) none of these

17. The equation $x^2 - px + q = 0, p, q \in R$ has no real roots if:
(a) $p^2 > 4q$ (b) $p^2 < 4q$
(c) $p^2 = 4q$ (d) none of these

18. Determine the value of k for which the quadratic equation $4x^2 - 3kx + 1 = 0$ has equal roots :
(a) $\pm\left(\frac{2}{3}\right)$ (b) $\pm\left(\frac{4}{3}\right)$
(c) ± 4 (d) ± 6

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

19. Find the value of k such that the equation $x^2 - (k+6)x + 2(2k-1) = 0$ has sum of the roots equal to half of their product :
(a) 3 (b) 4
(c) 7 (d) 10

20. Find the value of k so that the sum of the roots of the quadratic equation is equal to the product of the roots :
$$(k+1)x^2 + 2kx + 4 = 0$$
(a) −2 (b) −4
(c) 6 (d) 8

21. If −4 is a root of the quadratic equation $x^2 - px - 4 = 0$ and the quadratic equation $x^2 - px + k = 0$ has equal roots, find the value of k :
(a) 9/4 (b) 1
(c) 2.5 (d) 3

22. Find the value of k such that the sum of the squares of the roots of the quadratic equation $x^2 - 8x + k = 0$ is 40 :
(a) 12 (b) 2
(c) 5 (d) 8

23. Find the value of p for which the quadratic equation $x^2 + p(4x + p - 1) + 2 = 0$ has equal roots :
(a) $-1, \frac{2}{3}$ (b) 3, 5
(c) $1, -\frac{4}{3}$ (d) $\frac{3}{4}, 2$

24. If α and β are the roots of the equation $3x^2 + 7x + 3 = 0$ Find the value of $\alpha\beta$:
(a) 1 (b) 0
(c) 2 (d) 3

25. If α and β are the roots of the equation $x^2 - 3x + 2 = 0$. Find the quadratic equation whose roots are $-\alpha$ and $-\beta$:
(a) $x^2 - 3x + 2 = 0$ (b) $x^2 + 3x + 2 = 0$
(c) $x^2 + 3x - 2 = 0$ (d) none of these

26. Find the quadratic equation whose roots are $\sqrt{3}$ and $2\sqrt{3}$:
(a) $x^2 + 3\sqrt{3}x - 6 = 0$ (b) $x^2 - 3\sqrt{3}x + 6 = 0$
(c) $x^2 + 3\sqrt{3}x + 5 = 0$ (d) none of these

27. If α and β are the roots of equation $6x^2 + x - 2 = 0$, find the value of $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$:
(a) $-\frac{25}{12}$ (b) $-\frac{12}{75}$
(c) $-\frac{4}{9}$ (d) none of these

28. If a and c are such that the quadratic equation $ax^2 - 5x + c = 0$ has 10 as the sum of the roots and also as the product of the roots, find a and c respectively :
(a) $\frac{1}{2}, 6$ (b) $\frac{1}{2}, 5$
(c) $\frac{1}{2}, 8$ (d) none of these

29. If α and β are the roots of the equation $x^2 - x - 4 = 0$, find the value of $\frac{1}{\alpha} + \frac{1}{\beta} - \alpha\beta$:
(a) $\frac{16}{7}$ (b) $\frac{8}{5}$
(c) $\frac{15}{4}$ (d) none of these

30. If α and β are the roots of the equation $x^2 - 2x - 1 = 0$, find the value of $\alpha^2\beta + \alpha\beta^2$:
(a) −8 (b) −5
(c) 6 (d) −2

31. If α and β are the roots of $x^2 - x - 2 = 0$, find the quadratic equation in x whose roots are $(2\alpha + 1)$ and $(2\beta + 1)$:
(a) $x^2 + 5x - 4 = 0$ (b) $x^2 - 4x - 5 = 0$
(c) $x^2 + 4x - 5 = 0$ (d) none of these

32. If α, β be the roots of the quadratic equation $3x^2 - 6x + 4 = 0$, find the value of
$$\left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right) + 2\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) + 3\alpha\beta :$$
(a) 6 (b) 8
(c) 7 (d) 5

33. If α, β be the roots of the quadratic equation $x^2 - 5x + k = 0$, find the value of k such that $\alpha - \beta = 1$:
(a) 2 (b) 4
(c) 6 (d) 8

34. If one root of the quadratic equation $ax^2 + bx + c = 0$ is double the other then which one of the following is correct?
(a) $b^2 = 3ac$ (b) $2b^2 = 5ac$
(c) $2b^2 = 9ac$ (d) $2b^2 > 9ac$

35. The length of a hypotenuse of a right triangle exceeds the length of its base by 2 cm and exceeds twice the length of the altitude by 1 cm. Find the length of each side of the triangle (in cm) :
(a) 6, 8, 10 (b) 7, 24, 25
(c) 8, 15, 17 (d) 7, 40, 41

36. A two digit number is such that the product of its digits is 12. When 9 is addded to the number, the digits interchange their places, find the number :
(a) 62 (b) 34
(c) 26 (d) 43

37. A plane left 40 minutes late due to bad weather and in order to reach its destination, 1600 km away in time, it had to increase its speed by 400 km/h from its usual speed. Find the usual speed of the plane :
(a) 600 km/h (b) 750 km/h
(c) 800 km/h (d) none of these

38. The sum of the squares of two consecutive positive odd numbers is 290. Find the sum of the numbers :
(a) 14 (b) 24
(c) 28 (d) none of these

39. A shopkeeper buys a number of books for Rs. 80. If he had bought 4 more for the same amount, each book would have cost Re. 1 less. How many books did he buy?
(a) 8 (b) 16
(c) 24 (d) 28

40. Two square have sides x cm and $(x + 4)$ cm. The sum of their areas is 656 cm^2. Find the sides of the square :
(a) 8 cm, 12 cm (b) 12 cm, 15 cm
(c) 6 cm, 10 cm (d) 16 cm, 20 cm

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

EXERCISE

LEVEL 1

1. If p and q ($\neq 0$) are the roots of the equation $x^2 + px + q = 0$, then the least value of $x^2 + px + q$ ($x \in R$) is :
(a) $-\frac{1}{4}$ (b) $\frac{1}{4}$
(c) $\frac{-9}{4}$ (d) $\frac{9}{4}$

2. The value of p for which the sum of the squares of the roots of the equation $x^2 - (p-2)x - p - 1 = 0$ assume the least value is :
(a) -1 (b) 1
(c) 0 (d) 2

3. If the equation $x^2 + 2(p+1)x + 9p - 5 = 0$ has only negative roots, then :
(a) $p \leq 0$ (b) $p \leq -6$
(c) $p \geq 6$ (d) $p \geq 36$

4. The ratio of the roots of the equation $ax^2 + bx + c = 0$ is same as the ratio of the roots of the equation $px^2 + qx + r = 0$. If D_1 and D_2 are the discriminants of $ax^2 + bx + c = 0$ and $px^2 + qx + r = 0$ respectively then $D_1 : D_2 =$
(a) $\frac{a^2}{q^2}$ (b) $\frac{a^2}{p^2}$
(c) $\frac{b^2}{q^2}$ (d) none of these

5. If every pair from among the equations $x^2 + px + qr = 0$ $x^2 + qx + rp = 0$ and $x^2 + rx + pq = 0$ has a common root, then the sum of the three common roots is :
(a) $2(p+q+r)$ (b) $p+q+r$
(c) $-(p+q+r)$ (d) pqr

6. If the roots of the equation, $ax^2 + bx + c = 0$, are of the form $\frac{\alpha}{\alpha - 1}$ and $\frac{\alpha + 1}{\alpha}$ then the value of $(a+b+c)^2$ is :
(a) $b^2 - 2ac$ (b) $b^2 - 4ac$
(c) $2b^2 - ac$ (d) $4b^2 - 2ac$

7. Equation $ax^2 + 2x + 1$ has one double root if :
(a) $a = 0$ (b) $a = -1$
(c) $a = 1$ (d) $a = 2$

8. The number of quadratic equations which remain unchanged by squaring their roots, is :
(a) 0 (b) 2
(c) 4 (d) infinitely many

9. The value of $\sqrt{6+\sqrt{6+\sqrt{6+\sqrt{6+\ldots\infty}}}}$ is :
(a) 2 (b) 3
(c) 4 (d) 5

10. If α, β, γ are the roots of the equation $x^3 + a_0x^2 + a_1x + a_2 = 0$, then $(1-\alpha^2)(1-\beta^2)(1-\gamma^2)$ is equal to :
(a) $(1-a_1)^2 + (a_0 - a_2)^2$ (b) $(1+a_1)^2 - (a_0 + a_2)^2$
(c) $(1+a_1)^2 + (a_0 + a_2)^2$ (d) none of these

11. If $x \in R$, and $K = \frac{(x^2 - x + 1)}{(x^2 + x + 1)}$, then :
(a) $x \leq 0$ (b) $\frac{1}{3} \leq k \leq 3$
(c) $k \geq 5$ (d) none of these

12. If $x = \sqrt{2+\sqrt{2+\sqrt{2+\sqrt{2+\ldots\infty}}}}$ then x is :
(a) 1 (b) 2
(c) 3 (d) none of these

13. If α, β, γ are the roots of the equation $x^3 - 3x + 11 = 0$ then the equation whose roots are $(\alpha + \beta)$, $(\beta + \gamma)$ and $(\gamma + \alpha)$ is :
(a) $x^3 + 3x + 11 + 0$ (b) $x^3 - 3x + 11 = 0$
(c) $x^3 + 3x - 11 = 0$ (d) $x^3 - 3x - 11 = 0$

14. If $x^2 - ax - 21 = 0$ and $x^2 - 3ax + 35 = 0$; $a > 0$ have a common root, then a is equal to :
(a) 1 (b) 2
(c) 4 (d) 5

15. If α, β, γ are such that $\alpha + \beta + \gamma = 2$, $\alpha^2 + \beta^2 + \gamma^2 = 6$, $\alpha^3 + \beta^3 + \gamma^3 = 8$, then $\alpha^4 + \beta^4 + \gamma^4$ is equal to :
(a) 10 (b) 12
(c) 18 (d) none of these

16. The real values of a for which the quadratic equation $2x^2 - (a^3 + 8a - 1)x + a^2 - 4a = 0$ possesses roots of opposite signs are given by :
(a) $a > 6$ (b) $a > 9$
(c) $0 < a < 4$ (d) $a < 0$

17. If α, β are the roots of $ax^2 + bx + c = 0$ and $\alpha + k, \beta + k$ are the roots of $px^2 + qx + r = 0$, then $k =$
(a) $-\frac{1}{2}\left(\frac{a}{b} - \frac{p}{q}\right)$ (b) $\left(\frac{a}{b} - \frac{p}{q}\right)$
(c) $\frac{1}{2}\left(\frac{b}{a} - \frac{q}{p}\right)$ (d) $(ab - pq)$

18. If $\alpha, \beta \in R$, are the roots of the equation $ax^2 + bx + c = 0$, $k \in R$ lies between α and β, if :
(a) $a^2k^2 + bk + c < 0$ (b) $a^2k^2 + abk + ac < 0$
(c) $ak^2 + abk + ac > 0$ (d) none of these

19. If the equation $(3x)^2 + (27 \times 3^{1/k} - 15)x + 4 = 0$ has equal roots, then $k =$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

(a) -2 (b) $-\frac{1}{2}$
(c) $\frac{1}{2}$ (d) 0

20. The least possible integral value of a for which the equation $x^2 - 2(a-1)x + (2a+1) = 0$ has both the roots positive, is :
(a) 1 (b) 3
(c) 4 (d) 6

21. The number of real solutions of the equation $2^{3x^2-7x+4} = 1$ is :
(a) 0 (b) 4
(c) 2 (d) infinitely many

22. The number of real solutions of the equation $2|x|^2 - 5|x| + 2 = 0$ is :
(a) 0 (b) 4
(c) 2 (d) none of these

23. The number of real roots of the equation $(x-1)^2 + (x-2)^2 + (x-3)^2 = 0$:
(a) 0 (b) 2
(c) 3 (d) 6

24. The equation $\sqrt{x+1} - \sqrt{x-1} = \sqrt{4x-1}$ has :
(a) no solution
(b) one solution
(c) two solutions
(d) more than two solutions

25. If α, β are the roots of the equation $8x^2 - 3x + 27 = 0$, then the value of $[(\alpha^2/\beta)^{1/3} + (\beta^2/\alpha)^{1/3}]$ is :
(a) 1/3 (b) 1/4
(c) 1/5 (d) 1/6

26. The quadratic equation $3x^2 + 2(a^2+1)x + a^2 - 3a + 2 = 0$ possesses roots of opposite sign then a lies in :
(a) $(-\infty, 0)$ (b) $(-\infty, 1)$
(c) (1, 2) (d) (4, 9)

27. The expression $\frac{(x-a)(x-b)}{(x-c)}$ will assume all real values for every $x \in R$, if :
(a) $a < b < c$ (b) $a > b > c$
(c) c lies between a and b (d) none of these

28. If α and β are the roots of the equation $ax^2 + bx + c$, where $(a, b, c) > 0$, then α and β are :
(a) rational numbers (b) real and negative
(c) negative real parts (d) none of these

29. The number of solutions of the equation $3^{x-1} + 5^{x-1} = 34$:
(a) 0 (b) 1
(c) 2 (d) none of these

30. If $(2+\sqrt{3})^{x^2-2x+1} + (2-\sqrt{3})^{x^2-2x-1} = \frac{2}{2-\sqrt{3}}$, then x is equal to :
(a) 0 (b) 1
(c) 2 (d) both (a) and (c)

31. The values of a for which the quadratic equations $(1-2a)x^2 - 6ax - 1 = 0$ and $ax^2 - x + 1 = 0$ have at least one root in common are :
(a) $\frac{1}{2}, \frac{2}{9}$ (b) $0, \frac{1}{2}$
(c) $\frac{2}{9}$ (d) $0, \frac{1}{2}, \frac{2}{9}$

32. The integer k for which the inequality $x^2 - 2(4k-1)x + 15k^2 - 2k - 7 > 0$ is valid for any x is :
(a) 2 (b) 3
(c) 4 (d) 6

33. The values of a for which $2x^2 - 2(2a+1)x + a(a+1) = 0$ may have one root less than a and other root greater than a are given by :
(a) $-1 < a < 0$ (b) $0 < a < 1$
(c) $a \geq 0$ (d) $a < -1$ or $a > 0$

34. The condition that $x^3 - ax^2 + bx - c = 0$ may have two of its roots equal to each other but of opposite signs is :
(a) $ab = c$ (b) $\frac{2}{3}a = bc$
(c) $a^2b = c$ (d) none of these

35. If the roots of $ax^2 + bx + c = 0$, $a > 0$, be each greater than unity, then :
(a) $a + b + c = 0$ (b) $a + b + c > 0$
(c) $a + b + c < 0$ (d) none of these

36. The number of real solutions of $x - \frac{1}{x^2-4} = 2 - \frac{1}{x^2-4}$ is :
(a) 0 (b) 1
(c) 2 (d) infinite

37. The number of real solutions of the equation $2^{x/2} + (\sqrt{2}+1)^x = (5+2\sqrt{2})^{x/2}$ is :
(a) one (b) four
(c) six (d) infinite

38. If $xy = 2(x+y)$, $x \leq y$ and $x, y \in N$, the number of solutions of the equation :
(a) 2 (b) 3
(c) 0 (d) infinitely many

39. If $a, b, c \in R$ and equality $ax^2 + bx + c = 0$ has complex roots which are reciprocal to each other, then one has :
(a) $a = c$ (b) $|b| \leq |c|$
(c) $|b| \leq |a|$ (d) all of these

40. The equation $|x+1||x-1| = a^2 - 2a - 3$ can have real solutions for x if a belongs to :
(a) $(-\infty, -1] \cup [3, \infty)$
(b) $[1-\sqrt{5}, 1+\sqrt{5}]$
(c) $[1-\sqrt{5}, -1] \cup [3, 1+\sqrt{5}]$
(d) none of the above

41. $p, q, r \in R$ and k is a prime number such that $p + qk^{1/3} + rk^{2/3} = 0$, then :
(a) $p = \frac{q}{r}$ (b) $p = q^{1/3} \neq r^{2/3}$
(c) $p = q = r = 0$ (d) none of these

42. The expression $x^4 + 2px^3 + x^2 + 2px + 1 = 0$ has at least two distinct negative roots if :
(a) $p < 1$ (b) $p = 0$
(c) $p > \frac{3}{4}$ (d) $p \in \left(-\frac{1}{2}, \frac{1}{2}\right)$

43. The roots of the expression $x^2 - 2a|x-a| - 3a^2 = 0; a < 0$:
(a) $(1+\sqrt{2})a, (-1-\sqrt{6})a$
(b) $(1-\sqrt{2})a, (\sqrt{6}-1)a$
(c) $(1 \pm \sqrt{2})a$
(d) none of the above

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

44. The possible values of x for the equation $2^{x+2} - |2^{x+1} - 1| - 2^{x+1} = 1$ is :
(a) $x = -3$ (b) $x \geq -1$
(c) $x = 0$ (d) $(x \geq -1) \cup (x = -3)$

45. The values of x which satisfy the expression :
$$(5 + 2\sqrt{6})^{x^2-3} + (5 - 2\sqrt{6})^{x^2-3} = 10$$
(a) $\pm 2, \pm\sqrt{3}$ (b) $\pm\sqrt{2}, \pm 4$
(c) $\pm 2, \pm\sqrt{2}$ (d) $2, \sqrt{2}, \sqrt{3}$

46. If $a, b \in \{1, 2, 3, 4\}$, then the number of quadratic equations of the form $ax^2 + bx + 1 = 0$, having real roots is :
(a) 6 (b) 7
(c) 8 (d) none of these

47. The minimum value of the expression $a + \frac{1}{a}; a > 0$ is :
(a) 0 (b) 1
(c) 2 (d) 4

48. If $a^2 + b^2 + c^2 = 1$, then $ab + bc + ac$ lies in the interval :
(a) $[1, 2/3]$ (b) $[-1/2, 1]$
(c) $[-1, 1/2]$ (d) $[2, -4]$

49. The integral values of x satisfying the equation $2^{x+1} - 2^x = |2^x - 1| + 1$ is :
(a) $(-\infty, 0)$ (b) $[1, \infty)$
(c) $[0, \infty)$ (d) $2n : n \in I$

50. Number of values of x satisfying the equation $(15 + 4\sqrt{14})^t + (15 - 4\sqrt{14})^t = 30$, where $t = x^2 - 2|x|$:
(a) 0 (b) 2
(c) 4 (d) 6

51. The inequality $|2x - 3| < 1$ is valid when x lies in the interval :
(a) $(-4, 3)$ (b) $(-1, 2)$
(c) $(1, 2)$ (d) $(3, 4)$

52. Let α and β be the roots of the equation $x^2 + x + 1 = 0$. The equation whose roots are α^{19}, β^7 is :
(a) $x^2 - x - 1 = 0$ (b) $x^2 - x + 1 = 0$
(c) $x^2 + x - 1 = 0$ (d) $x^2 + x + 1 = 0$

53. The value of p for which the difference between the roots of the equation $x^2 + px + 8 = 0$ is 2 are :
(a) ± 2 (b) ± 4
(c) ± 6 (d) ± 8

54. If $x > 0$, $xy = 1$, minimum value of $x + y$ is :
(a) 1 (b) 0
(c) 2 (d) $\sqrt{2}$

55. If $x = 7 + 4\sqrt{3}$ and $xy = 1$, then the value of $\frac{1}{x^2} + \frac{1}{y^2}$ is :
(a) 64 (b) 128
(c) 184 (d) 194

56. If $x \in R$, and $\alpha = \frac{x^2}{(1 + x^4)}$, then :
(a) $0 \leq \alpha \leq 2$ (b) $0 \leq \alpha \leq 1$
(c) $0 \leq \alpha \leq \frac{1}{4}$ (d) $0 \leq \alpha \leq \frac{1}{2}$

57. The set of values for which $x^3 + 1 \geq x^2 + x$ is :
(a) $x \geq 0$ (b) $x \leq 0$
(c) $x \geq -1$ (d) $-1 \leq x \leq 1$

LEVEL 2

1. The value of x satisfying the equation
$$|x - 1|^{\log_3 x^2 - 2\log_x 9} = (x - 1)^7 :$$
(a) $\sqrt{3}$ (b) 3^4
(c) 3 (d) $\log_4 3$

2. If the roots of $10x^3 - cx^2 - 54x - 27 = 0$ are in HP, then find the value of c :
(a) 2 (b) 6
(c) 9 (d) none of these

3. Find the number of pairs for (x, y) from the following equations :
$$\log_{100} |x + y| = \frac{1}{2}$$
$$\log_{10} y - \log_{10} |x| = \log_{100} 4$$
(a) 0 (b) 1
(c) 2 (d) none of these

4. The number of solutions for real x, which satisfy the equation $2\log_2 \log_2 x + \log_{1/2} \log_2 (2\sqrt{2}x) = 1$:
(a) 1 (b) 2
(c) 4 (d) none of these

5. The solution set for which the equation satisfies $|x^2 + 4x + 3| + 2x + 5 = 0$:
(a) $(-1 + \sqrt{3}, -1 - \sqrt{3})$ (b) $[-2, (-1 + \sqrt{3})]$
(c) $[-4, (-1 - \sqrt{3})]$ (d) none of these

6. The real numbers x_1, x_2, x_3 satisfying the equation $x^3 - x^2 + \beta x + \gamma = 0$ are in A.P. Find the intervals in which β and γ lie, respectively :
(a) $(-\infty, 1/3]\,[-1/27, \infty)$
(b) $(-\infty, 3)$
(c) $(-1/3, 1/3)(-1/27, 1/27)$
(d) none of these

7. For all $x \in (0, 1)$:
(a) $e^x < 1 + x$ (b) $\log_e (1 + x) < x$
(c) $\sin x > x$ (d) $\log_e x > x$

8. Let $p \geq 3$ be an integer and α, β be the roots of $x^2 - (p + 1)x + 1 = 0$, then the value of $\alpha^n + \beta^n$, where $n \in N$
(a) is divisible by 'p'
(b) is an integer
(c) is a rational number
(d) both (b) and (c)

9. Let a, b, c be real, if $ax^2 + bx + c = 0$ has two real roots α, β, where $\alpha < -1$ and $\beta > 1$, then the value of $1 + \frac{c}{a} + \left|\frac{b}{a}\right|$ is :
(a) less than zero (b) greater than zero
(c) equal to zero (d) equal to $b^2 - 4ac$

10. If p and q are the roots of $x^2 + px + q = 0$, then :
(a) $p = 1$ (b) $p = 1$ or 0
(c) $p = -2$ (d) $p = -2$ or 0

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

11. If p, q, r are positive and are in AP, the roots of quadratic equation $px^2 + qx + r = 0$ are real for :

(a) $\left|\frac{r}{p} - 7\right| \geq 4\sqrt{3}$ (b) $\left|\frac{p}{r} - 7\right| \geq 4\sqrt{3}$

(c) all p and r (d) no p and r

12. The sum of all the real roots of the equation $|x-2|^2 + |x-2| - 2 = 0$ is :

(a) 2 (b) 3

(c) 4 (d) none of these

13. Let p and q be the roots of the equation $x^2 - 2x + A = 0$ and let r and s be the roots of the equation $x^2 - 18x + B = 0$. If $p < q < r < s$ are in arithmetic progression, then A, B respectively equal to :

(a) 8, 17 (b) 3, 7

(c) –3, 11 (d) none of these

14. If the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are real and less than 3, then :

(a) $a < 2$ (b) $2 \leq a \leq 3$

(c) $3 < a \leq 4$ (d) $a > 4$

15. If α and β $(\alpha < \beta)$, are the roots of the equation $x^2 + bx + c = 0$, where $c < 0 < b$, then :

(a) $0 < \alpha < \beta$

(b) $\alpha < 0 < \beta < |\alpha|$

(c) $\alpha < \beta < 0$

(d) $\alpha < 0 < |\alpha| < \beta$

16. If $b > a$, then the equation $(x-a)(x-b) - 1 = 0$ has :

(a) both roots in $[a, b]$

(b) both roots in $(-\infty, 0)$

(c) both roots in $(b, +\infty)$

(d) one root in $(-\infty, a)$ and other root in $(b, +\infty)$

17. Let α, β be the roots of $x^2 - x + p = 0$ and γ, δ be the roots of $x^2 - 4x + q = 0$. If $\alpha, \beta, \gamma, \delta$ are in GP, then the integral values of p and q respectively are :

(a) –2, –32 (b) –2, 3

(c) –6, 3 (d) –6, –32

18. If α, β are the roots of the equation $x^2 - px + q = 0$, then find the quadratic equation, the roots of which are $(\alpha^2 - \beta^2)(\alpha^3 - \beta^3)$ and $(\alpha^3\beta^2 + \alpha^2\beta^3)$:

(a) $px^2 - (5p + 7q)x - (p^6q^6 + 4p^2q^6) = 0$

(b) $x^2 - (p^5 - 5p^3q + 5pq^2)x + (p^6q^2 - 5p^4q^3 + 4p^2q^4) = 0$

(c) $x^2 - (p^3q - 5p^5 + p^4q) - (p^6q^2 - 5p^2q^6) = 0$

(d) all of the above

19. Given that α, γ are the roots of the equation $Ax^2 - 4x + 1 = 0$ and β, δ are the roots of the equation $Bx^2 - 6x + 1 = 0$, then the values of A and B respectively such that α, β, γ and δ are in HP :

(a) –5, 9 (b) 3/2, 5

(c) 3, 8 (d) none of these

20. Let α, β be the roots of the equation $(x-a)(x-b) = c, c \neq 0$, then the roots of the equation $(x-\alpha)(x-\beta) + c = 0$ are :

(a) a, c (b) b, c

(c) a, b (d) $a + c, b + c$

www.LearnEngineering.in

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Answers

INTRODUCTORY EXERCISE-14.1

1 (b)	2. (c)	3. (d)	4. (a)	5. (b)	6. (c)	7. (a)	8. (b)	9. (c)	10. (a)
11. (d)	12. (b)	13. (c)	14. (a)	15. (a)	16. (c)	17. (b)	18. (b)	19. (c)	20. (a)
21. (a)	22. (a)	23. (a)	24. (a)	25. (b)	26. (b)	27. (a)	28. (b)	29. (c)	30. (d)
31. (b)	32. (b)	33. (c)	34. (c)	35. (c)	36. (b)	37. (c)	38. (b)	39. (b)	40. (d)

LEVEL-1

1 (c)	2. (b)	3. (c)	4. (c)	5. (b)	6. (b)	7. (c)	8. (c)	9. (b)	10. (b)
11. (b)	12. (b)	13. (d)	14. (c)	15. (c)	16. (c)	17. (c)	18. (b)	19. (b)	20. (c)
21. (c)	22. (b)	23. (a)	24. (a)	25. (b)	26. (c)	27. (c)	28. (c)	29. (b)	30. (d)
31. (c)	32. (b)	33. (d)	34. (a)	35. (b)	36. (a)	37. (a)	38. (a)	39. (d)	40. (a)
41. (c)	42. (c)	43. (b)	44. (d)	45. (c)	46. (b)	47. (c)	48. (b)	49. (d)	50. (c)
51. (c)	52. (d)	53. (c)	54. (c)	55. (d)	56. (d)	57. (c)			

LEVEL-2

1 (b)	2. (c)	3. (c)	4. (a)	5. (c)	6. (a)	7. (b)	8. (d)	9. (a)	10. (b)
11. (b)	12. (c)	13. (d)	14. (a)	15. (b)	16. (d)	17. (a)	18. (b)	19. (c)	20. (c)

www.LearnEngineering.in
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Hints & Solutions

INTRODUCTORY EXERCISE 14.1

1. $\because \quad x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$\therefore \quad x = \dfrac{7 \pm \sqrt{49 - 4 \times 15 \times (-36)}}{2 \times 15}$

$\Rightarrow \quad x = \dfrac{7 \pm 47}{30}$

$\therefore \quad x = \dfrac{54}{30}$ or $x = \dfrac{-40}{30}$

$x = \dfrac{9}{5}$ or $x = \dfrac{-4}{3}$

NOTE (i) The above problem can also be solved by factorization.
(ii) The same problem can also be solved through options.

2. $\because \quad x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$x = \dfrac{6 \pm \sqrt{36 - 4\sqrt{7} \times (-13\sqrt{7})}}{2 \times \sqrt{7}}$

$\Rightarrow \quad x = \dfrac{6 \pm \sqrt{36 + 364}}{2\sqrt{7}}$

$\Rightarrow \quad x = \dfrac{6 \pm 20}{2\sqrt{7}}$

$\therefore \quad x = \dfrac{13}{\sqrt{7}}$ or $x = \dfrac{-7}{\sqrt{7}}$

$\Rightarrow \quad x = \dfrac{13}{\sqrt{7}}$ or $x = -\sqrt{7}$

3. $6x^2 - 31x + 40 = 0$

$\Rightarrow \quad \left(x - \dfrac{8}{3}\right)\left(x - \dfrac{5}{2}\right) = 0$

$\Rightarrow \quad x = \dfrac{8}{3}, \dfrac{5}{2}$

4. Consider $\sqrt{\dfrac{y-3}{2y+1}} = t$

then, $6t^2 + 1 = 5t$

$\Rightarrow \quad 6t^2 - 5t + 1 = 0$ ...(i)

get the values of t by solving the eq. (i) and then substitute the value of t in the relation

$t = \sqrt{\dfrac{y-3}{2y+1}}$ ...(ii)

Solve the eq. (i) and then get the values of t and subsequently y.

5. $(x+2)(x-5)(x-6)(x+1) = 144$

$\Rightarrow \quad [(x+2)(x-6)][(x-5)(x+1)] = 144$

$\Rightarrow \quad (x^2 - 4x - 12)(x^2 - 4x - 5) = 144$

$\Rightarrow \quad \{(x^2 - 4x - 5) - 7\}\{x^2 - 4x - 5\} = 144$

Let $t = x^2 - 4x - 5$ ...(i)

$\therefore \quad (t-7)(t) = 144$

$\Rightarrow \quad t^2 - 7t - 144 = 0$

$\Rightarrow \quad (t-16)(t+9) = 0$

$\Rightarrow \quad t = 16$ or $t = -9$

Substituting the values of t in eq. (i), we get

$x = 7, -3, 2$

6. Check through options
Alternatively, square both the sides twice.

7. $2\left(x^2 + \dfrac{1}{x^2}\right) - 3\left(x + \dfrac{1}{x}\right) - 1 = 0$

$\Rightarrow \quad 2\left[x^2 + \dfrac{1}{x^2} + 2 - 2\right] - 3\left(x + \dfrac{1}{x}\right) - 1 = 0$

$\Rightarrow \quad 2\left[\left(x + \dfrac{1}{x}\right)^2 - 2\right] - 3\left(x + \dfrac{1}{x}\right) - 1 = 0$

$\Rightarrow \quad 2\left(x + \dfrac{1}{x}\right)^2 - 3\left(x + \dfrac{1}{x}\right) - 5 = 0$

$\Rightarrow \quad 2t^2 - 3t - 5 = 0$ (Substituting $x + \dfrac{1}{x} = t$)

Now solve it and you will get

$t = -1$ and $t = \dfrac{5}{2}$

Now if $t = -1$, then $x + \dfrac{1}{x} = -1$.

$\Rightarrow \quad x^2 + 1 + x = 0$

$\Rightarrow \quad x^2 + x + 1 = 0$

$\therefore \quad x = \dfrac{-1 \pm \sqrt{1-4}}{2} = \dfrac{-1 \pm \sqrt{-3}}{2}$

and if $t = \dfrac{5}{2}$ then $x + \dfrac{1}{x} = \dfrac{5}{2}$

$\Rightarrow \quad 2x^2 - 5x + 2 = 0$

$\Rightarrow \quad x = \dfrac{1}{2}, 2$

8. Assume $\sqrt{\dfrac{x}{1-x}} = t$

$\therefore \quad \sqrt{\dfrac{1-x}{x}} = \dfrac{1}{t}$

$\therefore \quad t + \dfrac{1}{t} = \dfrac{13}{6}$

Now, solve it as in the previous problem.

9. $\sqrt{2x^2 - 2x + 1} = 2x - 3$

Square on both sides and simplify
(Hint: Type -3 problem)

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

10. Hint: Type-3 problem. It requires double squaring.

Alternatively : Go through options.

11. Hint: Type 6 problem.

12. $$\frac{1}{x+1}+\frac{1}{x+5}=\frac{1}{x+2}+\frac{1}{x+4}$$

$$\Rightarrow \left(\frac{1}{x+1}-\frac{1}{x+4}\right)+\left(\frac{1}{x+5}-\frac{1}{x+2}\right)=0$$

$$\Rightarrow \frac{3}{(x+1)(x+4)}-\frac{3}{(x+5)(x+2)}=0$$

$$\Rightarrow \frac{3[(x^2+7x+10)-(x^2+5x+4)]}{(x+1)(x+4)(x+5)(x+2)}=0$$

$$\Rightarrow (x^2+7x+10)-(x^2+5x+4)=0$$

$$\Rightarrow 2x+6=0 \Rightarrow x=-3$$

13. For equal roots $D=0$

i.e., $b^2-4ac=0$

$$\Rightarrow [-2(1+3k)]^2-4\times1\times7\times(3+2k)=0$$

Solve it and get the value of k.

14. $D=b^2-4ac$

$=4-4\times(-3)\times(-8)=-92$

15. $D=b^2-4ac=25-4\times1\times7=-3$

Since $D<0$, therefore roots are not real, *i.e.*, roots will be imaginary.

16. $$a^2x^2+abx-b^2=0$$

$$D=b^2-4ac=(ab)^2-4\times a^2\times(-b^2)$$

$$=(ab)^2+4a^2b^2$$

$$=(ab)^2[1+4]=(ab)^2(5)$$

$\therefore$ The roots are real and unequal

17. For non-real roots $D<0$

$$\therefore b^2-4ac<0$$

$$\therefore (-p)^2-4\times1\times q<0$$

$$\Rightarrow p^2-4q<0$$

$$\Rightarrow p^2<4q$$

18. $$4x^2-3kx+1=0$$

$$D=b^2-4ac=0$$

$$\therefore 9k^2-4\times1\times4=0$$

$$\Rightarrow k^2=\frac{16}{9} \Rightarrow k=\pm\frac{4}{3}$$

19. Let α, β be the roots of the equation, then

$$\alpha+\beta=\frac{1}{2}\alpha\beta$$

$$-\frac{b}{a}=\frac{1}{2}\cdot\frac{c}{a}$$

$$\Rightarrow -b=\frac{c}{2}$$

$$\Rightarrow (k+6)=\frac{2(2k-1)}{2}$$

$$\Rightarrow k=7$$

20. $$\alpha+\beta=\alpha\beta$$

$$\Rightarrow \frac{-b}{a}=\frac{c}{a} \Rightarrow -b=c$$

$$\therefore -2k=4$$

$$\Rightarrow k=-2$$

21. Since -4 is a root of $x^2-px-4=0$

$$\therefore (-4)^2-p(-4)-4=0$$

$$\Rightarrow p=-3$$

$\therefore$ The equation becomes $x^2+3x+k=0$

Since both the roots of this equation are equal.

$\therefore$ Sum of the roots $(\alpha+\alpha)=2\alpha=-3 \Rightarrow \alpha=-\frac{3}{2}$

$\therefore$ Product of the roots $(\alpha\cdot\alpha)=\alpha^2=(-3/2)^2=\frac{9}{4}=k$

$$\therefore k=\frac{9}{4}$$

22. $$\alpha^2+\beta^2=40$$

$$\Rightarrow (\alpha+\beta)^2-2\alpha\beta=40$$

$$\Rightarrow (8)^2-2k=40$$

$$\Rightarrow k=12$$

23. For equal roots, $D=0$

i.e., $b^2-4ac=0$

But now $x^2+p(4x+p-1)+2=0$

$$\Rightarrow x^2+4px+p^2-p+2=0$$

$$\Rightarrow x^2+4px+(p^2-p+2)=0$$

$$\therefore (4p)^2-4\times1\times(p^2-p+2)=0$$

$$\Rightarrow 16p^2-4p^2+4p-8=0$$

$$\Rightarrow 12p^2+4p-8=0$$

$$\Rightarrow 12p^2+12p-8p-8=0$$

$$\Rightarrow 12p(p+1)-8(p+1)=0$$

$$\Rightarrow (p+1)(12p-8)=0$$

$$\Rightarrow p=-1 \text{ or } p=\frac{8}{12}=\frac{2}{3}$$

24. $\alpha\beta=\frac{c}{a}=\frac{3}{3}=1$

25. $$x^2-3x+2=0$$

$$\Rightarrow x^2-x-2x+2=0$$

$$\Rightarrow (x-1)(x-2)=0$$

$$\Rightarrow \alpha=1 \text{ and } \beta=2$$

$$\therefore -\alpha=-1 \text{ and } -\beta=-2$$

$\therefore$ The required equation is

$$[x-(-\alpha)][x-(-\beta)]=0$$

$$\Rightarrow (x+1)(x+2)=0$$

$$\Rightarrow x^2+3x+2=0$$

26. Method I. If α and β be the roots of a quadratic equation, then the equation will be

$$(x-\alpha)(x-\beta)=0$$

$$\therefore (x-\sqrt{3})(x-2\sqrt{3})=0$$

$$\Rightarrow x^2-3\sqrt{3}x+6=0$$

www.LearnEngineering.in

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yourshahbaab.wordpress.com
For More Visit : www.LearnEngineering.in

Method II. The required equation

$$x^2 - (\alpha + \beta)x + (\alpha \cdot \beta) = 0$$

$$\Rightarrow \quad x^2 - (\sqrt{3} + 2\sqrt{3})x + (\sqrt{3} \cdot 2\sqrt{3}) = 0$$

$$\Rightarrow \quad x^2 - 3\sqrt{3}x + 6 = 0$$

27.

$$\alpha + \beta = \frac{-b}{a} = \frac{-1}{6}$$

$$\alpha\beta = \frac{c}{a} = -\frac{2}{6} = -\frac{1}{3}$$

$$\therefore \quad \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta}$$

$$= \frac{(-1/6)^2 - 2(-1/3)}{-1/3}$$

$$= \frac{\frac{1}{36} + \frac{2}{3}}{-\frac{1}{3}} = \frac{\frac{25}{36}}{-\frac{1}{3}} = -\frac{25}{12}$$

Alternatively : $6x^2 + x - 2 = 0$

$$\Rightarrow \quad 6x^2 - 3x + 4x - 2 = 0$$

$$\Rightarrow \quad 3x(2x - 1) + 2(2x - 1) = 0$$

$$\Rightarrow \quad x = \frac{1}{2} \text{ or } x = -\frac{2}{3}$$

$$\therefore \quad \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{1/2}{-2/3} + \frac{-2/3}{1/2}$$

$$= -\frac{3}{4} - \frac{4}{3} = -\frac{25}{12}$$

28. $\alpha + \beta = 10$ and $\alpha \cdot \beta = 10$

$$\Rightarrow \quad -\frac{b}{a} = 10 \text{ and } \frac{c}{a} = 10$$

$$\Rightarrow \quad \frac{5}{a} = 10$$

$$\Rightarrow \quad a = \frac{1}{2} \quad \therefore \frac{c}{1/2} = 10$$

$$\Rightarrow \quad c = 5$$

$$\therefore \quad a = \frac{1}{2} \text{ and } c = 5$$

29. $\alpha + \beta = -\frac{b}{a} = 1$ and $\alpha\beta = \frac{c}{a} = -4$

$$\therefore \quad \frac{1}{\alpha} + \frac{1}{\beta} - \alpha\beta = \frac{\alpha + \beta}{\alpha\beta} - \alpha\beta$$

$$= \frac{1}{-4} - (-4) = 4 - \frac{1}{4} = \frac{15}{4}$$

30. $\alpha + \beta = 2$ and $\alpha\beta = -1$

$$\therefore \quad \alpha^2\beta + \alpha\beta^2 = \alpha\beta(\alpha + \beta)$$

$$= 2(-1) = -2$$

31. Sum of roots $= (2\alpha + 1) + (2\beta + 1)$

$$= 2\alpha + 2\beta + 2$$

$$= 2(\alpha + \beta) + 2$$

and product of roots $= (2\alpha + 1)(2\beta + 1)$

$$= 4\alpha\beta + 2\alpha + 2\beta + 1$$

$$= 4\alpha\beta + 2(\alpha + \beta) + 1$$

Now since $\alpha + \beta = 1$

and $\alpha\beta = -2$

$\therefore$ Sum of roots $= 2(\alpha + \beta) + 2 = 2(1) + 2 = 4$

and product of roots $= 4\alpha\beta + 2(\alpha + \beta) + 1$

$$= 4 \times (-2) + 2(1) + 1$$

$$= -8 + 2 + 1 = -5$$

$\therefore$ Required equation is $x^2 - 4x - 5 = 0$.

32. $\alpha + \beta = \frac{6}{3} = 2$ and $\alpha\beta = \frac{4}{3}$

$$\therefore \quad \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right) + 2\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) + 3\alpha\beta$$

$$= \frac{\alpha^2 + \beta^2}{\alpha\beta} + 2\left(\frac{\alpha + \beta}{\alpha\beta}\right) + 3\alpha\beta$$

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} + 2\left(\frac{\alpha + \beta}{\alpha\beta}\right) + 3\alpha\beta$$

$$= \frac{4 - 8/3}{4/3} + 2\left(\frac{2}{4/3}\right) + 3 \times \frac{4}{3}$$

$$= \frac{4/3}{4/3} + 3 + 4 = 8$$

33.

$$\alpha + \beta = 5$$

$$\alpha - \beta = 1$$

$$\therefore \quad \alpha = 3 \text{ and } \beta = 2$$

$$\therefore \quad \alpha\beta = \frac{c}{a} = \frac{k}{1} = 3 \times 2 = 6$$

$$\Rightarrow \quad k = 6$$

34. Let the one root be α and other root be β, then $\beta = 2\alpha$.

$$\therefore \quad \alpha + \beta = 3\alpha = \frac{-b}{a} \quad \Rightarrow \quad \alpha = \frac{-b}{3a}$$

and $\alpha\beta = 2\alpha^2 = \frac{c}{a}$

$$\therefore \quad 2\alpha^2 = \frac{c}{a}$$

$$\Rightarrow \quad 2 \times \left(\frac{-b}{3a}\right)^2 = \frac{c}{a}$$

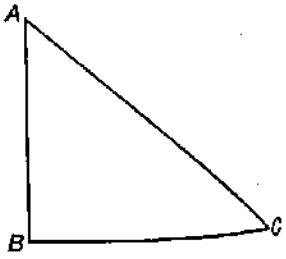
$$\Rightarrow \quad \frac{2b^2}{9a^2} = \frac{c}{a}$$

$$\Rightarrow \quad 2b^2 = 9ac$$

35. Let the base of the triangle be x cm, then $BC = x$ cm and hypotenuse $AC = (x + 2)$ cm and the perpendicular

$$AB = \frac{(x + 2) - 1}{2} = \frac{(x + 1)}{2}$$

[Let the length of AB be k, then $AC = 2k + 1 \Rightarrow k = \frac{AC - 1}{2}$]



$\therefore$ By Pythagorus theorem

$AB^2 + BC^2 = AC^2$

(see the figure)

$$\left(\frac{x + 1}{2}\right)^2 + x^2 = (x + 2)^2$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

$$\frac{x^2+1+2x}{4}+x^2=x^2+4+4x$$

$\Rightarrow$ $x^2+1+2x+4x^2=4x^2+16+16x$

$\Rightarrow$ $x^2-14x-15=0$

$\Rightarrow$ $(x-15)(x+1)=0$

$\Rightarrow$ $x=15$ cm ($\because x=-1$ cm is inadmissible)

$\Rightarrow$ base = 15 cm

$\therefore$ perpendicular = 8 cm

and hypotenuse = 17 cm

and

36. Let the tens digit be x and unit digit be y. Then the two digit number $= 10x + y$

but $x \times y = 12 \Rightarrow y = \frac{12}{x}$

$\therefore$ the number is $\left(10x+\frac{12}{x}\right)$

Again $10x+\frac{12}{x}+9=10\times\left(\frac{12}{x}\right)+x$

$\Rightarrow$ $10x^2+12+9x=120+x^2$

$\Rightarrow$ $9x^2+9x-108=0$

$\Rightarrow$ $x^2+x-12=0$

$\Rightarrow$ $(x+4)(x-3)=0$

$\Rightarrow$ $x=-4, 3$

but x cannot be negative

$\therefore$ $x=3$ only.

$\therefore$ $y=\frac{12}{x}=\frac{12}{3}=4$

$\therefore$ the number $= 10x+y=10\times3+4=34$

37. Let the usual speed of plane be x km/hr

$\therefore$ Usual time taken $=\frac{1600}{x}$ hr

New speed $= (x+400)$ km/hr

and the new time $=\frac{1600}{x}-\frac{40}{60}=\left(\frac{1600}{x}-\frac{2}{3}\right)$ hr

$\therefore$ Distance = speed $\times$ time

$$1600=(x+400)\left(\frac{1600}{x}-\frac{2}{3}\right)$$

$$1600=1600-\frac{2x}{3}+\frac{640000}{x}-\frac{800}{3}$$

$\Rightarrow$ $4800x=4800x-2x^2+1920000-800x$

$\Rightarrow$ $2x^2-1920000+800x=0$

$\Rightarrow$ $x^2-960000+400x=0$

$\Rightarrow$ $x^2+400x-960000=0$

$\Rightarrow$ $x^2+1200x-800x-960000=0$

$\Rightarrow$ $x(x+1200)-800(x+1200)=0$

$(x+1200)(x-800)=0$

$\Rightarrow$ $x=800$ km/hr ($\because x=-1200$ is inadmissible)

38. Let the two consecutive odd numbers be x and $x+2$

$\therefore$ $x^2+(x+2)^2=290$

$\Rightarrow$ $x^2+x^2+4+4x=290$

$\Rightarrow$ $2x^2+4x+4=290$

$\Rightarrow$ $x^2+2x+2=145$

$\Rightarrow$ $x^2+2x-143=0$

$\Rightarrow$ $(x+13)(x-11)=0$

$\Rightarrow$ $x=11$ or $x=-13$, but x is positive

$\therefore$ $x+2=13$

$\therefore$ Sum of the numbers $= 11+13=24$

39. Let the shopkeeper buys x books

$\therefore$ Rate of the book $=\frac{80}{x}$

the new number of books $= (x+4)$

$\therefore$ the new price $=\frac{80}{(x+4)}$

$\therefore$ $\frac{80}{(x+4)}+1=\frac{80}{x}$

$\Rightarrow$ $\left(\frac{1}{x}-\frac{1}{(x+4)}\right)=\frac{1}{80}$

$\Rightarrow$ $\frac{4}{x^2+4x}=\frac{1}{80}$

$\Rightarrow$ $x^2+4x-320=0$

$\Rightarrow$ $(x+20)(x-16)=0$

$\Rightarrow$ $x=16$, ($x=-20$, is inadmissible)

$\therefore$ He bought 16 books.

40. $x^2+(x+4)^2=656$

$\Rightarrow$ $x^2+x^2+16+8x=656$

$\Rightarrow$ $x^2+4x-320=0$

$\Rightarrow$ $(x+20)(x-16)=0$

$\Rightarrow$ $x=16$, ($x=-20$ is inadmissible)

$\therefore$ Sides of the square are $x=16$ cm

and $(x+4)=20$ cm.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

LEVEL 1

1. $y_{\min.} = \dfrac{-b^2 + 4ac}{4a}$ at $x = -\dfrac{b}{2a}$

$\therefore \quad y_{\min.} = \dfrac{-p^2 + 4\times1\times q}{4\times1} = \dfrac{4q - p^2}{4}$

Now, since p and q are the roots of the equation $x^2 + px + q = 0$.

$\therefore \quad p + q = -p \Rightarrow q = -2p$

and $\quad pq = q \Rightarrow p = 1 \quad (\because q \neq 0)$

$\therefore \quad q = -2$

$\therefore \quad y_{\min} = \dfrac{4q - p^2}{4} = \dfrac{4\times(-2) - (1)^2}{4} = -\dfrac{9}{4}.$

2. Let α, β be the roots of the given equation, then, $\alpha + \beta = p - 2$ and $\alpha\beta = -(p + 1)$.

Now, $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = (p - 2)^2 + 2(p + 1)$

$= p^2 - 2p + 6 = (p - 1)^2 + 5$

Clearly $\alpha^2 + \beta^2 \geq 5$. So minimum value of $\alpha^2 + \beta^2$ is 5, which attains at $p = 1$.

3. Let $f(x) = x^2 + 2(p + 1)x + 9p - 5$. Let α, β be the roots of $f(x) = 0$. The equation $f(x) = 0$ will have both negative roots, if

(i) $D \geq 0$ (ii) $\alpha < 0, \beta < 0$, *i.e.*, $(\alpha + \beta) < 0$

(iii) $f(0) > 0$

Now, $D \geq 0 \Rightarrow 4(p + 1)^2 - 36p + 20 \geq 0$

$\Rightarrow \quad p^2 - 7p + 6 \geq 0$

$\Rightarrow \quad (p - 6)(p - 1) \geq 0$

$\Rightarrow \quad p \leq 1 \text{ or } p \geq 6 \quad \ldots(i)$

$\alpha + \beta < 0 \Rightarrow -2(p + 1) < 0$

$\Rightarrow \quad p + 1 > 0 \Rightarrow p > -1 \quad \ldots(ii)$

and $\quad f(0) > 0 \Rightarrow 9p - 5 > 0$

$\Rightarrow \quad p > \dfrac{5}{9} \quad \ldots(iii)$

from (i), (ii) and (iii) we get $p \geq 6$.

4. Let α_1, β_1 be the roots of $ax^2 + bx + c = 0$ and α_2, β_2 be the roots of $px^2 + qx + r = 0$, then,

$$\frac{\alpha_1}{\beta_1} = \frac{\alpha_2}{\beta_2} \Rightarrow \frac{\alpha_1 + \beta_1}{\alpha_1 - \beta_1} = \frac{\alpha_2 + \beta_2}{\alpha_2 - \beta_2}$$

$$\Rightarrow \frac{(\alpha_1 + \beta_1)^2}{(\alpha_1 - \beta_1)^2} = \frac{(\alpha_2 + \beta_2)^2}{(\alpha_2 - \beta_2)^2}$$

$$\Rightarrow \frac{(\alpha_1 + \beta_1)^2}{(\alpha_1 + \beta_1)^2 - 4\alpha_1\beta_1} = \frac{(\alpha_2 + \beta_2)^2}{(\alpha_2 + \beta_2)^2 - 4\alpha_2\beta_2} = \frac{b^2/a^2}{\dfrac{b^2 - 4ac}{a^2}}$$

$$= \frac{q^2/p^2}{\dfrac{q^2 - 4rp}{p^2}} = \frac{b^2}{D_1} = \frac{q^2}{D_2}$$

$$\Rightarrow \quad \frac{D_1}{D_2} = \frac{b^2}{q^2}$$

Alternatively : Let 2, 3 be the roots of the equation $ax^2 + bx + c = 0$

$\therefore$ the equation is $x^2 - 5x + 6 = 0$

$\therefore \quad a = 1, \quad b = -5 \quad \text{and} \quad c = 6$

$\therefore \quad D_1 = b^2 - 4ac = 25 - 4\times1\times6 = 1$

Again, if 4, 6 be the roots of the equation $px^2 + qx + r = 0$ then the equation will be $x^2 - 10x + 24 = 0$

$\Rightarrow \quad p = 1, \quad q = -10 \quad \text{and} \quad r = 24$

$\therefore \quad D_2 = q^2 - 4pr = 100 - 4\times1\times24 = 4$

$\therefore \quad \dfrac{D_1}{D} = \dfrac{1}{4}.$

Now, check through option, only option (c) is correct as

$$\frac{b^2}{q^2} = \frac{(-5)^2}{(-10)^2} = \frac{1}{4} = \frac{D_1}{D_2}.$$

Caution : $\dfrac{\alpha_1}{\beta_1} \neq \dfrac{\alpha_1}{\alpha_2}$, when solving through numerical values.

5. The given equations are :

$$x^2 + px + qr = 0 \quad \ldots(i)$$

$$x^2 + qx + rp = 0 \quad \ldots(ii)$$

and $\quad x^2 + rx + pq = 0 \quad \ldots(iii)$

Let α, β be the roots of (i), β, γ be roots of (ii) and γ, α be the roots of (iii). Since β is a common root of (i) and (ii)

$\therefore \quad \beta^2 + p\beta + qr = 0$

and $\quad \beta^2 + q\beta + rp = 0$

$\Rightarrow \quad (p - q)\beta + r(q - p) = 0 \Rightarrow \beta = r$

Now, $\quad \alpha\beta = qr \Rightarrow \alpha r = qr \Rightarrow \alpha = q$

Since β and γ are roots of (ii). Therefore

$$\beta r = rp \Rightarrow \gamma r = rp \Rightarrow \gamma = p$$

$\therefore \quad \alpha + \beta + \gamma = q + r + p = p + q + r$

NOTE $\alpha + \beta + \gamma$ can also be equal to 0 and $-\dfrac{1}{2}(p + q + r)$.

6. By hypothesis,

$$\frac{\alpha}{\alpha - 1} + \frac{\alpha + 1}{\alpha} = -\frac{b}{a} \quad \text{and} \quad \frac{\alpha}{\alpha - 1} \cdot \frac{\alpha + 1}{\alpha} = \frac{c}{a}$$

$$\Rightarrow \quad \frac{2\alpha^2 - 1}{\alpha^2 - \alpha} = -\frac{b}{a} \text{ and } \alpha = \frac{c + a}{c - a}$$

$$\Rightarrow \quad (c + a)^2 + 4ac = -2b(c + a)$$

$$\Rightarrow \quad (c + a)^2 + 2b(a + c) + b^2 = b^2 - 4ac$$

$$\Rightarrow \quad (a + b + c)^2 = b^2 - 4ac$$

7. One double root means a single root appears two times i.e., two roots are equal.

$\therefore \quad D = 0 \Rightarrow (2)^2 - 4a\cdot1 = 0$

$\Rightarrow \quad a = 1.$

8. Let α, β be the roots of a quadratic equation and α^2, β^2 be the roots of another quadratic, Since the quadratic remains the same, we have

$$\alpha + \beta = \alpha^2 + \beta^2 \quad \ldots(i)$$

and $\quad \alpha\beta = \alpha^2\beta^2 \quad \ldots(ii)$

Now, $\quad \alpha\beta = \alpha^2\beta^2 \Rightarrow \alpha\beta(\alpha\beta - 1) = 0$

Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$\alpha = 0$ or $\beta = 0$ or $\alpha\beta = 1$.

$\Rightarrow$ If $\alpha = 0$, then $\beta = \beta^2$ [putting $\alpha = 0$ in (i)]

$\Rightarrow$ $\beta(1-\beta) = 0 \Rightarrow \beta = 0, \beta = 1$.

Thus, we get two sets of values of α and β viz. $\alpha = 0, \beta = 0$ and $\alpha = 0, \beta = 1$. Now, if $\alpha\beta = 1$,

then $\alpha + \frac{1}{\alpha} = \alpha^2 + \frac{1}{\alpha^2}$ [Putting $\beta = \frac{1}{\alpha}$ in (i)]

$\Rightarrow$ $\alpha + \frac{1}{\alpha} = \left(\alpha + \frac{1}{\alpha}\right)^2 - 2$

$\Rightarrow$ $\left(\alpha + \frac{1}{\alpha}\right)^2 - \left(\alpha + \frac{1}{\alpha}\right) - 2 = 0$

$\Rightarrow$ $\alpha + \frac{1}{\alpha} = 2$ or $\alpha + \frac{1}{\alpha} = -1$

$\Rightarrow$ $\alpha = 1$ or $\alpha = \omega, \omega^2$

Putting $\alpha = 1$, in $\alpha\beta = 1$, we get $\beta = 1$, and putting $\alpha = \omega$ in $\alpha\beta = 1$, we get $\beta = \omega^2$

Putting $\alpha = \omega^2$ in $\alpha\beta = 1$, we get $\beta = \omega$. Thus, we get four sets of values of α, β viz., $\alpha = 0, \beta = 0$, $\alpha = 0, \beta = 1$, $\alpha = \omega, \beta = \omega^2$, $\alpha = 1, \beta = 1$.

Thus, there are four quadratics which remain unchanged by squaring their roots.

9. Let $x = \sqrt{6 + \sqrt{6 + \ldots \infty}} \Rightarrow x^2 = x + 6 \Rightarrow x^2 - x - 6 = 0 \Rightarrow x = 3, -2$, but $x > 0$.

10. The best way is to consider some values of the roots α, β and γ.

Let $\alpha = 2$, $\beta = 3$ and $\gamma = 4$, then

$(x-\alpha)(x-\beta)(x-\gamma) = 0$

$\Rightarrow$ $(x-2)(x-3)(x-4) = 0$

$\Rightarrow$ $x^3 - 9x^2 + 26x - 24 = 0$

$\Rightarrow$ $a_0 = -9$

$a_1 = 26$

$a_2 = -24$

Now, $(1-\alpha^2)(1-\beta^2)(1-\gamma^2) = (1-4)(1-9)(1-16)$

$= -3 \times -8 \times -15 = -360$

Now, go through options and verify.

Consider option (b)

$(1 + a_1)^2 - (a_0 + a_2)^2 = (1 + 26)^2 - (-9 - 24)^2$

$= (27)^2 - (33)^2$

$= (27 + 33)(27 - 33)$

$= 60 \times (-6) = -360$

Hence option (b) is correct.

11. $k = \frac{x^2 - x + 1}{x^2 + x + 1}$

$\Rightarrow$ $kx^2 + kx + k = x^2 - x + 1$

$\Rightarrow$ $(k-1)x^2 + (k+1)x + k - 1 = 0$

Since x is real, the discriminant

$D = (k+1)^2 - 4(k-1)^2 \geq 0$

$\Rightarrow$ $(3k-1)(-k+3) \geq 0$

$\Rightarrow$ $\left(k - \frac{1}{3}\right)(k-3) \leq 0$

$\Rightarrow$ $\frac{1}{3} \leq k \leq 3$

12. Let $x = \sqrt{2 + \sqrt{2 + \sqrt{2 + \sqrt{2 + \ldots \infty}}}}$

$\therefore$ $x = \sqrt{2 + x}$

$\Rightarrow$ $x^2 = 2 + x$

$\Rightarrow$ $x^2 - x - 2 = 0$

$\Rightarrow$ $x = \frac{1 \pm \sqrt{9}}{2} = \frac{1 \pm 3}{2} = 2, -1$

$\because$ $x > 0$, $\therefore x = 2$

13. We have, $\alpha + \beta + \gamma = 0$

$\therefore$ $\beta + \gamma = -\alpha$, $\gamma + \alpha = -\beta$ and $\alpha + \beta = -\gamma$.

$\therefore$ The equation whose roots are $-\alpha, -\beta, -\gamma$ can be obtained by replacing x by $-x$ in the given equation.

Therefore the required equation is :

$(-x)^3 - 3(-x) + 11 = 0$

or $x^3 - 3x - 11 = 0$

14. Let α be a common root of the two given equations, then $\alpha^2 - 3a\alpha + 35 = 0$ and $\alpha^2 - a\alpha - 21 = 0$. On subtracting we get $-2a\alpha + 56 = 0$ or $\alpha = \frac{28}{a}$.

As α is a root of $x^2 - ax - 21 = 0$,

$\therefore$ $\left(\frac{28}{a}\right)^2 - a\left(\frac{28}{a}\right) - 21 = 0$

or $a^2 = 4^2$ or $a = \pm 4$

$\therefore a > 0$, we get $a = 4$

15. $(\alpha + \beta + \gamma)^2 = \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \beta\gamma + \gamma\alpha)$

$4 = 6 + 2(\alpha\beta + \beta\gamma + \gamma\alpha)$

$\Rightarrow$ $\alpha\beta + \beta\gamma + \gamma\alpha = -1$

Also, $\alpha^3 + \beta^3 + \gamma^3 - 3\alpha\beta\gamma$

$= (\alpha + \beta + \gamma)(\alpha^2 + \beta^2 + \gamma^2 - \alpha\beta - \beta\gamma - \gamma\alpha)$

$8 - (3\alpha\beta\gamma) = 2(6 + 1)$

$\Rightarrow$ $\alpha\beta\gamma = -2$

Now, $(\alpha^2 + \beta^2 + \gamma^2)^2 = \Sigma\alpha^4 + 2\Sigma\beta^2\gamma^2$

$= \Sigma\alpha^4 + 2[(\Sigma\beta\gamma)^2 - 2\alpha\beta\gamma(\Sigma\gamma)]$

$\Rightarrow$ $\Sigma\alpha^4 = 36 - 2[(-1)^2 - 2(-2)(2)] = 18$

16. The roots of the given equation will be of opposite signs if they are real and their product is negative, i.e.,

$D \geq 0$ and product of roots < 0

$\Rightarrow$ $(a^3 + 8a - 1)^2 - 8(a^2 - 4a) \geq 0$

and $\frac{a^2 - 4a}{2} < 0$

$\Rightarrow$ $a^2 - 4a < 0$

[$\because a^2 - 4a < 0 \Rightarrow (a^3 + 8a - 1)^2 - 8(a^2 - 4a) \geq 0$]

$\Rightarrow$ $0 < a < 4$.

17. We have $\alpha + \beta = -\frac{b}{a}$, $\alpha\beta = \frac{c}{a}$

and $\alpha + k + \beta + k = -\frac{q}{p}$

and $(\alpha + k)(\beta + k) = \frac{r}{p}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yourSmartBook.wordpress.com
For More Visit : www.LearnEngineering.in

$$\Rightarrow \quad \alpha+\beta+2k=-\frac{q}{p}$$

$$\Rightarrow \quad -\frac{b}{a}+2k=-\frac{q}{p} \qquad \left(\because\ \alpha+\beta=-\frac{b}{a}\right)$$

$$\therefore \quad k=\frac{1}{2}\left(\frac{b}{a}-\frac{q}{p}\right)$$

NOTE The problem can also be solved by assuming some appropriate values.

18. Let $f(x)=ax^2+bx+c$. It is given that α, β are real roots of $f(x)=0$. So, k lies between α and β, if

$$af(k)<0 \Rightarrow a(ak^2+bk+c)<0$$

$$\Rightarrow \quad a^2k^2+abk+ac<0$$

19. The given equation will have equal roots iff $D=0$.

$$\Rightarrow \quad (27\times 3^{1/k}-15)^2-144=0$$

$$\Rightarrow \quad 27\times 3^{1/k}-15=\pm 12$$

$$\Rightarrow \quad 27\times 3^{1/k}=27 \text{ or } 3$$

$$\Rightarrow \quad 3^{1/k}=1 \text{ or } 3^{1/k}=\frac{1}{9}$$

$$\Rightarrow \quad \frac{1}{k}=0 \text{ or } -2$$

But $\frac{1}{k}$ cannot be zero. So $k=-\frac{1}{2}$.

20. Let $f(x)=x^2-2(a-1)x+(2a+1)=0$.
Then, $f(x)=0$ will have both roots positive, if
(i) $D>0$ (ii) Sum of the roots >0 and (iii) $f(0)>0$
Now, $D\geq 0 \Rightarrow 4(a-1)^2-4(2a+1)\geq 0$

$$\Rightarrow \quad a^2-4a\geq 0 \Rightarrow a\leq 0 \text{ or } a\geq 4 \qquad \ldots(i)$$

Sum of the roots >0

$$\Rightarrow \quad 2(a-1)>0 \Rightarrow a>1 \qquad \ldots(ii)$$

and $f(0)>0$

$$\Rightarrow \quad (2a+1) \Rightarrow a>-\frac{1}{2} \qquad \ldots(iii)$$

From (i), (ii) and (iii) we get $a\geq 4$.
Hence, the least integral value of a is 4.

21.

$$2^{3x^2-7x+4}=1=2^0$$

$$\Rightarrow \quad 3x^2-7x+4=0$$

$$\Rightarrow \quad 3x^2-3x-4x+4=0$$

$$\Rightarrow \quad 3x(x-1)-4(x-1)=0$$

$$\Rightarrow \quad x=1 \text{ or } x=\frac{4}{3}$$

$$\therefore \quad x=1, \frac{4}{3}$$

22.

$$2|x|^2-5|x|+2=0$$

$$\Rightarrow \quad (2|x|-1)(|x|-2)=0$$

$$\therefore \quad |x|=\frac{1}{2}, 2$$

$$\therefore \quad x=\pm\frac{1}{2}, \pm 2$$

23.

$$(x-1)^2+(x-2)^2+(x-3)^2=0 \qquad \ldots(i)$$

It is possible only when $x-1=0,\ x-2=0,\ x-3=0$

$$x=1, \quad x=2, \quad x=3$$

but $x=1, 2, 3$ do not satisfy eq. (i).

24.

$$\sqrt{x+1}-\sqrt{x-1}=\sqrt{4x-1}$$

$$\therefore \quad x+1\geq 0, \quad x-1\geq 0, \quad 4x-1\geq 0$$

All these inequalities are satisfied when $x\geq 1$. Squaring both the sides of the equation we get

$$(x+1)+(x-1)-2\sqrt{x^2-1}=4x-1$$

$$\Rightarrow \quad 2\sqrt{x^2-1}=1-2x$$

Squaring again, we get

$$4(x^2-1)=1-4x+4x^2$$

$$\Rightarrow \quad x=\frac{5}{4}$$

For $x=\frac{5}{4}$, $\quad \text{LHS}=\sqrt{\frac{5}{4}+1}-\sqrt{\frac{5}{4}-1}=1$

and $\quad \text{RHS}=\sqrt{4\left(\frac{5}{4}\right)-1}=\sqrt{4}=2$

but $\quad \text{LHS}\neq\text{RHS}$

$\therefore\ x=\frac{5}{4}$ is not a root of the equation.

25.

$$\alpha+\beta=\frac{3}{8}, \quad \alpha\beta=\frac{27}{8}$$

$$\therefore \quad \left(\frac{\alpha^2}{\beta}\right)^{1/3}+\left(\frac{\beta^2}{\alpha}\right)^{1/3}=\left(\frac{\alpha^3}{\alpha\beta}\right)^{1/3}+\left(\frac{\beta^3}{\alpha\beta}\right)^{1/3}$$

$$=\frac{\alpha+\beta}{(\alpha\beta)^{1/3}}=\frac{3/8}{(27/8)^{1/3}}$$

$$=\frac{3/8}{3/2}=\frac{1}{4}$$

26. The quadratic equation $3x^2+2(a^2+1)x+(a^2-3a+2)=0$ will have two roots of opposite sign if it has real roots and the product of the roots is negative, i.e., if

$$D\geq 0 \quad \text{and} \quad \alpha\cdot\beta<0$$

$$\Rightarrow \quad 4(a^2+1)^2-12(a^2-3a+2)\geq 0$$

and $\quad \frac{a^2-3a+2}{3}<0$

$$\therefore \quad a^2-3a+2<0,$$

i.e., if $\quad (a-1)(a-2)<0 \quad \text{or} \quad 1<a<2$

27. Let $\quad y=\frac{(x-a)(x-b)}{(x-c)}$

$$\Rightarrow \quad x^2-(a+b+y)x+ab+cy=0$$

Since x is real, then $D\geq 0$

$$\Rightarrow \quad (a+b+y)^2-4(ab+cy)\geq 0$$

$$\Rightarrow \quad y^2+2(a+b-2c)y+(a+b)^2-4ab\geq 0$$

$$\Rightarrow \quad y^2+2(a+b-2c)y+(a-b)^2\geq 0 \qquad \ldots(i)$$

Since y takes all real values (i) is possible iff

$$4(a+b-2c)^2-4(a-b)^2<0$$

$$\Rightarrow \quad (a+b-2c+a-b)(a+b-2c-a+b)<0$$

$$\Rightarrow \quad 4(a-c)(b-c)<0$$

$$\Rightarrow \quad (a-c)(b-c)<0$$

This is possible if c lies between a and b, that is if $a<c<b$ or $a>c>b$.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$$D = b^2 - 4ac$$

28. ∵ and, if $D \geq 0$, then the roots of the equation are given by

$$x = \frac{-b \pm \sqrt{D}}{2a}$$

As $D = b^2 - 4ac < b^2$ (∵ $a > 0$ and $c > 0$)

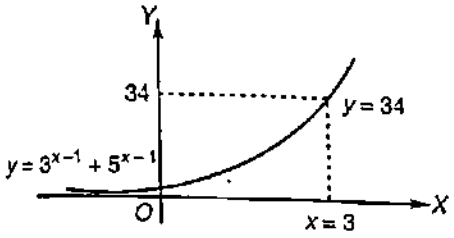
It follows that the roots of the quadratic equation are negative.

In case $D < 0$, then the roots of the equation are given by

$$x = \frac{-b \pm i\sqrt{-D}}{2a}$$

which have negative real parts.

29. It is very obvious that at $x = 3$ the given expression satisfies. Now, $y = 3^{x-1}$ and $y = 5^{x-1}$ are both increasing function of x (exponential functions with base greater than 1). Therefore their sum $y = 3^{x-1} + 5^{x-1}$ is also an increasing function of x. It means for $x < 3$, $y = 3^{x-1} + 5^{x-1} < 34$ and for $x > 3$, $y = 3^{x-1} + 5^{x-1} > 34$.



Thus, the equation has no other solution.

30. ∵ $$2 - \sqrt{3} = \frac{1}{2 + \sqrt{3}}$$

∴the given equation can be expressed as

$$(2+\sqrt{3})^{x^2-2x+1} + \frac{1}{(2+\sqrt{3})^{x^2-2x-1}} = 2(2+\sqrt{3}) \quad \ldots(i)$$

Let us put $y = (2+\sqrt{3})^{x^2-2x+1}$, then the eq. (i) can be expressed as

$$y + \frac{(2+\sqrt{3})^2}{y} = 2(2+\sqrt{3})$$

$$\Rightarrow \quad y^2 - 2(2+\sqrt{3})y + (2+\sqrt{3})^2 = 0$$

$$\Rightarrow \quad [y - (2+\sqrt{3})]^2 = 0 \quad \Rightarrow \quad y = 2+\sqrt{3}$$

$$\therefore \quad (2+\sqrt{3})^{x^2-2x+1} = (2+\sqrt{3})$$

$$\Rightarrow \quad x^2 - 2x + 1 = 1 \quad \Rightarrow \quad x(x-2) = 0$$

$$\Rightarrow \quad x = 0 \text{ or } 2.$$

31. For $a = 0$ or $a = \frac{1}{2}$ one of the quadratic equation becomes linear, So $a \neq 0$, $a \neq \frac{1}{2}$.

Hence the only answer is $a = \frac{2}{9}$

32. Let $f(x) = x^2 - 2(4k-1)x + 15k^2 - 2k - 7$. Then $f(x) > 0 \Rightarrow D < 0$ [∵ Coefficient of $x^2 > 0$]

$$\Rightarrow \quad 4(4k-1)^2 - 4(15k^2 - 2k - 7) < 0$$

$$\Rightarrow \quad k^2 - 6k + 8 < 0 \quad \Rightarrow \quad 2 < k < 4$$

33. According to the given condition a lies between the roots.

Let $f(x) = 2x^2 - 2(2a+1)x + a(a+1)$.

For a to lie between the roots, we must have $D \geq 0$ and $f(a) < 0$

Now, $D \geq 0 \Rightarrow 4(2a+1)^2 - 8a(a+1) \geq 0$

$$\Rightarrow \quad 8\left[a^2 + a + \frac{1}{2}\right] \geq 0, \quad \text{which is always true}$$

$$\therefore \quad f(a) < 0 \Rightarrow 2a^2 - 2a(2a+1) + a(a+1) < 0$$

$$\Rightarrow \quad -a^2 - a < 0 \Rightarrow a^2 + a > 0$$

$$\Rightarrow \quad a > 0 \text{ or } a < -1$$

34. Let α, β and γ be the roots of the given equation such that $\alpha = -\beta$. Then

$$\alpha + \beta + \gamma = a \Rightarrow \gamma = a$$

Since γ is a root of the given equation, so it satisfies the equation, i.e.,

$$\gamma^3 - a\gamma^2 + b\gamma - c = 0$$

$$\Rightarrow \quad a^3 - a^3 + ab - c = 0 \Rightarrow ab = c$$

35. Let $f(x) = ax^2 + bx + c$. Since 1 lies outside the roots of $f(x) = 0$. So,

$$af(1) > 0 \Rightarrow f(1) > 0 \quad (\because a > 0)$$

$$\Rightarrow \quad a + b + c > 0$$

36. $$x - \frac{1}{x^2-4} = 2 - \frac{1}{x^2-4}$$

Here $x^2 - 4 \neq 0$

$$\Rightarrow \quad x \neq \pm 2 \Rightarrow (x-2) = 0$$

$\Rightarrow x = 2$, which is inadmissible.

37. $$2^{x/2} + (\sqrt{2}+1)^x = (5+2\sqrt{2})^{x/2}$$

$$\Rightarrow \quad \left(\frac{\sqrt{2}}{\sqrt{5+2\sqrt{2}}}\right)^x + \left(\frac{\sqrt{2}+1}{\sqrt{5+2\sqrt{2}}}\right)^x = 1$$

which is of the form

$$\cos^x \alpha + \sin^x \alpha = 1$$

$$\therefore \quad x = 2.$$

38. $$xy = 2(x+y) \Rightarrow y(x-2) = 2x$$

$\therefore y = \frac{2x}{x-2}$ but $x, y \in N$ by trial, we get $x = 3, 4, 6$

$$\therefore \quad y = 6, 4, 3 \qquad \text{but } x \leq y$$

$$\therefore \quad x = 3, 4 \text{ and } y = 6, 4$$

Thus two solutions are possible

39. Let the roots be α and $\frac{1}{\alpha}$

$$\therefore \quad \alpha \cdot \frac{1}{\alpha} = \frac{c}{a} \Rightarrow c = a$$

$$\because \quad (|a| - |b|)^2 \geq 0$$

$$\text{or} \quad |a| \geq |b|$$

$$\text{but} \quad c = a \quad \therefore |c| \geq |b|$$

40. $$|x+1||x-1| = a^2 - 2a - 3$$

$$\Rightarrow \quad |x^2 - 1| = a^2 - 2a - 3$$

$$\Rightarrow \quad x^2 - 1 = \pm(a^2 - 2a - 3)$$

$$\text{but} \quad a^2 - 2a - 3 \geq 0$$

$$\therefore \quad x^2 - 1 = a^2 - 2a - 3$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yousmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$\Rightarrow \quad x^2 = a^2 - 2a - 2, \quad -a^2 + 2a + 4 \geq 0$

for real solutions,

$a^2 - 2a - 2 \geq 0$ and $-a^2 + 2a + 4 \geq 0$,

Thus we have $a \in (-\infty, -1] \cup [3, \infty)$

41. Since k is a prime number

$\therefore \quad k \neq 0$

Again $p = 0, \; qk^{1/3} = 0$ and $rk^{2/3} = 0$

$\Rightarrow \quad p = 0$ and $q = 0$ and $r = 0$

$\therefore \quad p = q = r = 0$

42. The equation $x^4 + 2px^3 + x^2 + 2px + 1 = 0$ cannot have 0 as a root. Dividing by x^2, we can write as

$$x^2 + \frac{1}{x^2} + 2p\left(x + \frac{1}{x}\right) + 1 = 0$$

$$\Rightarrow \quad \left(x + \frac{1}{x}\right)^2 + 2p\left(x + \frac{1}{x}\right) - 1 = 0 \qquad \ldots(1)$$

Put $x + \frac{1}{x} = y$. Then we can write eq. (1) as

$$y^2 - 2py - 1 = 0 \qquad \ldots(2)$$

Since $D > 0$, *i.e.*, $4p^2 + 4 > 0$, hence eq. (2) has two distinct real roots. These roots are

$$y = \frac{-2p \pm \sqrt{4p^2 + 4}}{2} = -p \pm \sqrt{p^2 + 1}$$

Since $-p + \sqrt{p^2 + 1} > 0$, the equation

$$x + \frac{1}{x} = -p + \sqrt{p^2 + 1} \qquad \ldots(3)$$

has either positive roots or non-real complex roots.

Since $-p - \sqrt{p^2 + 1} < 0$, both the roots of the equation $x + \frac{1}{x} = -p - \sqrt{p^2 + 1}$ are negative.

We can write eq. (3) as

$$\left(\sqrt{-x} - \frac{1}{\sqrt{-x}}\right)^2 = p - 2 + \sqrt{p^2 + 1}$$

As $\left(\sqrt{-x} - \frac{1}{\sqrt{-x}}\right)^2 > 0$, we must have,

$p - 2 + \sqrt{p^2 + 1} > 0$, *i.e.*, $\sqrt{p^2 + 1} > |2 - p|$.

Since $p^2 + 1 = (p - 2)^2 + 4a - 3$,

$\sqrt{p^2 + 1} > |2 - p|$ iff $4p - 3 > 0$.

i.e., iff $p > \frac{3}{4}$.

43. **Case 1.** Suppose $x \geq a$. Thus the given equation becomes

$$x^2 - 2a(x - a) - 3a^2 = 0$$

$$\Rightarrow \quad x^2 - 2ax - a^2 = 0$$

$$\Rightarrow \quad x = \frac{2a \pm 2\sqrt{2}a}{2} = (1 \pm \sqrt{2})a$$

As $a < 0$ and $1 + \sqrt{2} > 1$, $(1 + \sqrt{2})a < a$

Therefore, $x \neq 1(1 + \sqrt{2})a$

Again as $1 - \sqrt{2} < 1$, $(1 - \sqrt{2})a > a$,

$\therefore \quad x = (1 - \sqrt{2})a$

Case 2. Suppose $x < a$, then the given equation becomes

$$x^2 - 2a(a - x) - 3a^2 = 0$$

$$x^2 + 2ax - 5a^2 = 0$$

$$\Rightarrow \quad x = \frac{-2a \pm \sqrt{24}a}{2} = (-1 \pm \sqrt{6})a$$

As $a < 0, -1 - \sqrt{6} < 0 < 1$, so $(-1 - \sqrt{6})a > a$,

$\therefore \quad x \neq (-1 - \sqrt{6})a$ $(\because x < a)$

Again, as $a < 0, -1 + \sqrt{6} > 1$ and $(-1 + \sqrt{6})a < a$

$\therefore \quad x = (-1 + \sqrt{6})a = (\sqrt{6} - 1)a$

44. **Case 1.** Let $x + 2 \geq 0$.

$\therefore |x + 2| = x + 2$ and the equation becomes

$$2^{x+2} - 2^{x+1} - 1 - |2^{x+1} - 1| = 0$$

$$\Rightarrow \quad 2^{x+1}(2 - 1) - 1 - |2^{x+1} - 1| = 0$$

$$\Rightarrow \quad 2^{x+1} - 1 = |2^{x+1} - 1|$$

$$\Rightarrow \quad 2^{x+1} - 1 \geq 0 \text{ or } x + 1 \geq 0 \text{ or } x \geq -1$$

Case 2. Let $x + 2 < 0$.

$\therefore \quad |x + 2| = -(x + 2)$ and the equation becomes

$$2^{-(x+2)} - |2^{x+1} - 1| = 2^{x+1} + 1$$

Put $2^{x+1} = y$. Then eq. (i) becomes

$$\frac{1}{2y} - |y - 1| = y + 1$$

or $\quad 2y^2 + 2y + |y - 1|\,2y = 1$

If $y \geq 1$, then LHS of eq. (ii) becomes

$$2y^2 + 2y - 2y(y - 1) - 1 = 0$$

$$\Rightarrow \quad 4y = 1 \text{ or } y = \frac{1}{4}$$

Also $\quad 2^{x+1} = 2^{-2}$ or $x = -3$

45. Let $y = 5 + 2\sqrt{6}$, then $\frac{1}{y} = 5 - 2\sqrt{6}$. Thus the given equation becomes

$$y^{x^2-3} + \left(\frac{1}{y}\right)^{x^2-3} = 10$$

Again let $y^{x^2-3} = t$. Then eq. (i) becomes

$$t + \frac{1}{t} = 10 \quad \Rightarrow \quad t^2 - 10t + 1 = 0$$

Thus, $\quad t = 5 \pm 2\sqrt{6}$

$\therefore \quad (5 + 2\sqrt{6})^{x^2-3} = 5 \pm 2\sqrt{6} = (5 + 2\sqrt{6})^{\pm 1}$

$\Rightarrow \quad x^2 - 3 = 1$ or $x^2 - 3 = -1$

$\Rightarrow \quad x^2 = 4$ or $x^2 = 2$

$\Rightarrow \quad x = \pm 2$ or $x = \pm\sqrt{2}$

46. For the real roots we must have $D \geq 0$. *i.e.*, $b^2 - 4ac \geq 0$

$b^2 \geq 4a \; (\because c = 1)$

b	b^2	a	$4a$	
1	1	×	×	
2	4	1	4	→ 1
3	9	1, 2	4, 8	→ 2 (7)
4	16	1, 2, 3, 4	4, 8, 12, 16	→ 4

Thus, there are 7 possible equations of the given form.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

47. Let $y = a + \frac{1}{a}$ ...(i)

Differentiating above eq. (i), we get

$$\frac{dy}{da} = 1 - \frac{1}{a^2} \quad ...(ii)$$

Equating eq. (ii) with zero, we get

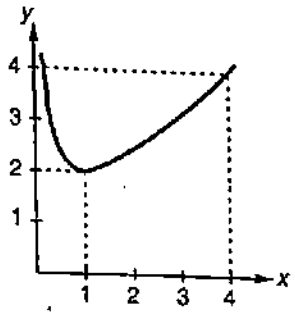
$$1 - \frac{1}{a^2} = 0 \Rightarrow a^2 = 1 \Rightarrow a = \pm 1$$

But since $a > 0$, hence only $a = 1$ is acceptable. Now substituting $a = 1$ in the original eq. (i) we get the required minimum value of the expression.

Hence, minimum of

$$\therefore \quad y = 1 + \frac{1}{1} = 2$$

Alternatively: See the following graph of $y = x + \frac{1}{x}$.



Here, $x = a$

$$y = x + \frac{1}{x} = a + \frac{1}{a}$$

a	$\frac{1}{a}$	$a + \frac{1}{a}$
1	1	2
2	1/2	2.5
3	1/3	3.33
4	1/4	4.25
5	1/5	5.20
1/2	2	2.5
1/3	3	3.33
1/4	4	4.25

48. $(a+b+c)^2 \geq 0$

$\therefore (a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca) \geq 0$ ...(i)

$\therefore \quad 1 + 2(ab + bc + ca) \geq 0 \quad (a^2 + b^2 + c^2 = 1)$

$\Rightarrow \quad (ab + bc + ca) \geq -\frac{1}{2}$ ...(ii)

Also, $a^2 + b^2 \geq 2ab$, $b^2 + c^2 \geq 2bc$, $c^2 + a^2 \geq 2ac$

$(a^2 + b^2 + b^2 + c^2 + c^2 + a^2) \geq 2ab + 2bc + 2ac$

$\Rightarrow \quad a^2 + b^2 + c^2 \geq ab + bc + ac \quad (\because a^2 + b^2 + c^2 = 1)$

$\Rightarrow \quad ab + bc + ac \leq 1$

$\therefore \quad -\frac{1}{2} \leq ab + bc + ac \leq 1.$

49. Best way is to go through options. It satisfies for every even integer.

50. Let $(15 + 4\sqrt{14})^t = k$

$\therefore \quad (15 - 4\sqrt{14}) = \frac{1}{k}$

$\therefore$ The given equation can be written as

$$k + \frac{1}{k} = 30$$

$\Rightarrow \quad k^2 - 30k + 1 = 0$

$\Rightarrow \quad k = 15 \pm 4\sqrt{14}$

Now, the equation will hold only at $t = \pm 1$.

$\therefore \quad x^2 - 2|x| = \pm 1$

Again if $x^2 - 2|x| = 1$

$\Rightarrow \quad x = \pm (1 + \sqrt{2})$

and if $x^2 - 2|x| = -1$

$\Rightarrow \quad x = \pm 1$

Hence $x = \pm 1, \ \pm(1 + \sqrt{2})$

51. We have $|2x - 3| < 1 \Leftrightarrow -1 < 2x - 3 < 1$

$\Leftrightarrow \quad 1 < x < 2 \Leftrightarrow x \in (1, 2)$

52. The roots of the equation are ω and ω^2.

Let $\alpha = \omega$ and $\beta = \omega^2$. Then

$$\alpha^{19} = \omega^{19} = \omega$$

and $\beta^7 = \omega^{14} = \omega^2$

Hence α^{19} and β^7 are roots of the same equation.

53. Let α, β be the roots of the equation $x^2 + px + 8 = 0$. Then $\alpha + \beta = -p$ and $\alpha\beta = 8$

But $\alpha - \beta = 2 \Rightarrow (\alpha + \beta)^2 - 4\alpha\beta = (2)^2$

$\Rightarrow \quad p^2 - 32 = 4 \Rightarrow p = \pm 6.$

54. $(x + y)^2 = x^2 + y^2 + 2xy$

$(x + y)^2 = x^2 + y^2 + 2 \quad (\because xy = 1)$

For the minimum value of the expression

$x = y \quad \therefore x = 1$ and $y = 1$

Hence, $(x + y)_{min} = \sqrt{x^2 + y^2 + 2} = \sqrt{4} = 2$

$(x + y)_m = 2$ (Since $x > 0$ and $xy > 0 \therefore y > 0$)

Alternatively :

xy	x	y	$\sqrt{x^2 + y^2 + 2} = (x + y)$
1	1/10	10	101/10
1	1/5	5	26/5
1	1/2	2	5/2
1	1	1	2
1	2	1/2	5/2
1	4	1/4	17/4
1	10	1/10	101/10

55. $x = 7 + 4\sqrt{3}$

$\therefore \quad y = \frac{1}{7 + 4\sqrt{3}} = 7 - 4\sqrt{3}$

$\therefore \quad \frac{1}{x^2} + \frac{1}{y^2} = \frac{x^2 + y^2}{(xy)^2}$

$$= \frac{(7 + 4\sqrt{3})^2 + (7 - 4\sqrt{3})^2}{[(7 + 4\sqrt{3})(7 - 4\sqrt{3})]^2}$$

$$= \frac{2(49 + 48)}{1} = 194$$

56. For real x, we have $x^2/(1 + x^4) \geq 0$ Also

$$1 + x^4 - 2x^2 = (1 - x^2)^2 \geq 0$$

$\Rightarrow \quad 1 + x^4 \geq 2x^2$

$\therefore \quad \left(\frac{x^2}{1 + x^4}\right) \leq \frac{1}{2}$

So $0 \leq \frac{x^2}{1 + x^4} \leq \frac{1}{2}$

57. $x^3 + 1 \geq x^2 + x \Rightarrow x^3 - x^2 - x + 1 \geq 0$

$\Rightarrow \quad (x - 1)^2(x + 1) \geq 0 \Rightarrow x \geq -1$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmartboob.wordpress.com
For More Visit : www.LearnEngineering.in

LEVEL (2)

1. Best way is to go through options.

Consider option (b)

$$|3^4-1|^{\log_3(3^4)^2-2\log_{81}9}=(3^4-1)^7$$

$$|80|^{\log_3 3^8-\log_{81}81}=(80)^7$$

$\Rightarrow \quad \log_3 3^8-\log_{81}81=7$

$\Rightarrow \quad 8-1=7$

$7=7$

Hence option (b) is correct.

2. Putting $x=\frac{1}{y}$, we get

$$27y^3+54y^2+cy-10=0 \qquad \ldots(i)$$

This above eq. (i) must be in AP.

Let the roots of equation in y be

$$\alpha-\beta,\quad \alpha,\quad \alpha+\beta \qquad (\because \text{roots are in AP})$$

$\therefore \quad \Sigma\alpha=\alpha-\beta+\alpha+\alpha+\beta=3\alpha$

$\Rightarrow \quad 3\alpha=\frac{-54}{27} \Rightarrow \alpha=\frac{-2}{3}$

Now $\alpha=\frac{-2}{3}$ will satisfy the eq. (i) we get

$$27\times\frac{-8}{27}+54\times\frac{4}{9}-\frac{2c}{3}-10=0$$

$\Rightarrow \quad c=9$

3. $\log_{100}|x+y|=\frac{1}{2} \Rightarrow (100)^{1/2}=|x+y|$

$\Rightarrow \quad |x+y|=10 \qquad \ldots(1)$

Again, $\log_{10}y-\log_{10}|x|=\log_{100}4$

$\log_{10}y-\log_{10}|x|=\log_{10}2$

$\Rightarrow \quad \log_{10}\frac{y}{|x|}=\log_{10}2$

$\Rightarrow \quad y=2|x| \qquad \ldots(2)$

From eq. (2) we can conclude that y is always positive.

Now, when $x>0$ and $y>0$ (always)

$$|x+y|=10 \Rightarrow |x+2|x||=10$$

$\Rightarrow \quad x+2|x|=10 \qquad (\because x>0)$

$\Rightarrow \quad x+2x=10$

$\Rightarrow \quad x=\frac{10}{3}$

$\therefore \quad y=\frac{20}{3}$

Again, $x<0$ and $y>0$ (always positive)

$$|-x+2|-x||=10$$

$\Rightarrow \quad |-x+2x|=10$

$\Rightarrow \quad |x|=10$

$\Rightarrow \quad x=-10 \qquad (\because x<0)$

$\therefore \quad y=20$

Hence, $x=-10$, $y=20$ and $x=\frac{10}{3}$ and $y=\frac{20}{3}$.

4. $2\log_2\log_2 x+\log_{1/2}\log_2(2\sqrt{2}x)=1$

$\Rightarrow \quad 2\log_2\log_2 x-\log_2\log_2(2\sqrt{2}x)=\log_2 2$

$\Rightarrow \quad \log_2(\log_2 x)^2-\log_2[\log_2 2\sqrt{2}x]=\log_2 2$

$\Rightarrow \quad \log_2\frac{(\log_2 x)^2}{[\log_2 2\sqrt{2}x]}=\log_2 2$

$\Rightarrow \quad \frac{(\log_2 x)^2}{\log_2(2\sqrt{2}x)}=2$

$\Rightarrow \quad (\log_2 x)^2=2\log_2(2\sqrt{2}x)=2\log_2(2^{3/2}x)$

$\Rightarrow \quad (\log_2 x)^2=2[3/2\log_2 x+\log_2 x]=3+2\log_2 x$

$\Rightarrow \quad (\log_2 x)^2-2\log_2 x-3=0$

$\Rightarrow \quad \log_2 x=-1 \quad \text{or} \quad \log_2 x=3$

$\Rightarrow \quad x=\frac{1}{2} \quad \text{or} \quad x=8$

But for $x=\frac{1}{2}$, $\log_2\log_2(1/2)$ is undefined

$\therefore$ Only possible value of $x=8$.

5. Consider $x^2+4x+3\geq 0$ ...(i)

then $(x^2+4x+3)+2x+5=0$

$\Rightarrow \quad x=-2 \quad \text{and} \quad x=-4$

but $x=-2$ does not satisfy eq (i)

Now, if $x^2+4x+3<0$ ...(ii)

then $-(x^2+4x+3)+2x+5=0$

$\Rightarrow \quad x=-1-\sqrt{3} \quad \text{or} \quad x=-1+\sqrt{3}$

but only $x=-1-\sqrt{3}$ satisfies the eq. (ii).

Hence the solution set of x is $(-4,-1-\sqrt{3})$.

Alternatively, check the options by substituting the value from the options given in the question.

6. $\because x_1, x_2, x_3$ are in AP.

$\Rightarrow \quad x_1=a-d,\quad x_2=a,\quad x_3=a+d$

where d is the common difference

Now, since x_1, x_2, x_3 are the roots of the given equation

$$x^3-x^2+\beta x+\gamma=c$$

So, $\Sigma\alpha=x_1+x_2+x_3=1$

$\Rightarrow \quad (a-d)+(a)+(a+d)=1$ ...(i)

$\Sigma\alpha\beta=x_1x_2+x_2x_3+x_1x_3$

$\Rightarrow \quad \beta=(a-d)a+a(a+d)+(a-d)(a+d)$ ...(ii)

and $\Sigma=\alpha\beta\gamma=x_1x_2x_3=-\gamma=(a-d)(a)(a+d)$ ...(iii)

hence from (i) we get $a=\frac{1}{3}$

and from eq. (ii) we get

$$\beta=3a^2-d^2$$

$\Rightarrow \quad \beta=\frac{1}{3}-d^2 \qquad \left(\because a=\frac{1}{3}\right)$

Thus $\beta=\frac{1}{3}-d^2\leq\frac{1}{3} \qquad [\because d^2\geq 0]$

$\Rightarrow \quad \beta\leq\frac{1}{3}$

$\therefore \quad \beta\in\left(-\infty,\frac{1}{3}\right]$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

your smahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Again from eq. (iii)

$$\alpha(\alpha^2 - d^2) = -\gamma$$

$$\Rightarrow \quad \left(\frac{1}{27}\right) + \left(-\frac{d^2}{3}\right) = -\gamma$$

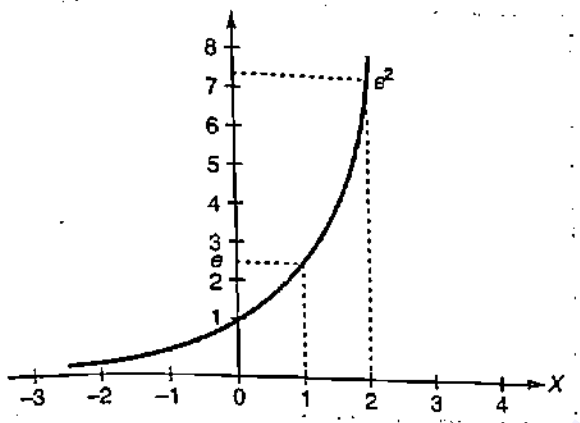
$$\Rightarrow \quad \gamma = \frac{d^3}{3} - \frac{1}{27}$$

$$\Rightarrow \quad \gamma \geq \frac{-1}{27} \qquad (\because\ d^2 \geq 0)$$

Hence option (a) is correct.

7. (a) $\rightarrow e^x < 1 + x$
(b) $\rightarrow e^x > (1 + x) \Leftrightarrow \log_e (1 + x) < x$
(c) $\rightarrow \sin x > x$
(d) $\rightarrow e^x < x \Leftrightarrow \log_e x > x$

NOTE $e \approx 2.714$ and $e^2 \approx 7.366$



from the graph it is clear that when $0 < x < 1, e^x > (1 + x)$

Option (c) is clearly wrong.

8. Let us consider some value of $p = 3$ (say), then

$$x^2 - 4x + 1$$

and $\quad (\alpha, \beta) = 2 \pm \sqrt{3}$ $\quad$ (α, β are the roots)

Now, $\quad \alpha^n + \beta^n = (2 + \sqrt{3})^n + (2 - \sqrt{3})^n$ ...(i) $(n \in N)$

then $\alpha^n + \beta^n$ will always be an integer, for the validity of statement you can put $n = 1, 2, 3, \ldots$ etc. in Eq. (i).
Similarly for $p = 4, 5, 6, \ldots$ etc. we can conclude the same results.

NOTE In this question, the discriminant D is always positive, *i.e.*, $b^2 - 4ac > 0$. So the roots will always be real, unequal and irrational.
But the fact is that $\alpha^n + \beta^n$, for $n \in N$, always yields integral value. This can be easily proved by mathematical induction method. So if the answer is an integer, then it must be a rational number, hence (d) is correct.

9. Just assume some values of α, β conforming the basic constraints of the problem.
e.g., $\alpha = -2, \beta = 8$, then the equation becomes $x^2 - 6x - 16$

$$\Rightarrow \quad b = -6 \quad \text{and} \quad c = -16$$

$$\therefore \quad 1 + \frac{c}{a} + \left|\frac{b}{a}\right| = 1 - 16 + 6 = -9$$

$\therefore$ The value of the expression is negative, hence choice (a) is correct.

NOTE Since $\quad \alpha < 1$ and $\beta > 1$

$$\therefore \quad \alpha\beta < 1 \Rightarrow \frac{c}{a} < 1$$

Further the product of any two numbers $(n_1, n_2 \neq 0)$ is less than the sum of the number if any one of them is negative.

So, $\quad \alpha\beta < \alpha + \beta$ $\quad$ ($\because$ Here $\alpha\beta$ is negative)

$$\therefore \quad \frac{c}{a} < \left|\frac{b}{a}\right|;$$

but $\frac{c}{a}$ is numerically greater than $\left|\frac{b}{a}\right|$.

$$\Rightarrow \quad 1 + \frac{c}{a} + \left|\frac{b}{a}\right| < 0.$$

10. Since p and q are the roots of given equation $x^2 + px + q = 0$
then $p + q = -p \Rightarrow q = -2p$
and $\quad pq = q \Rightarrow p = 1$
So, when $p = 1$, then $q = -2$
Again, when $q = 0$, then $p = 0$ hence,
$p = 1, 0$ and $q = -2, 0$
Thus, option (b) is most appropriate.

11. $\because p, q, r$ are in AP.

$$\therefore \quad q = \frac{p + r}{2} \qquad [\because\ p + r = 2q]$$

For the real roots $\quad q^2 - 4pr \geq 0$

$$\Rightarrow \quad \left(\frac{p + r}{2}\right)^2 - 4pr \geq 0$$

$$\Rightarrow \quad p^2 + r^2 - 14pr \geq 0$$

$$\Rightarrow \quad \left(\frac{p}{r}\right)^2 - 14\left(\frac{p}{r}\right) + 1 \geq 0$$

$$\Rightarrow \quad \left(\frac{p}{r} - 7\right)^2 \geq 48$$

$$\Rightarrow \quad \left|\frac{p}{r} - 7\right| \geq 4\sqrt{3}$$

12. The given equation is $|x - 2|^2 + |x - 2| - 2 = 0$.
Let us assume $|x - 2| = m$
then $\quad m^2 + m - 2 = 0$
$(m - 1)(m + 2) = 0$
Only admissible value is
$m = 1$ $\quad$ ($\because\ m \neq -2$ as $m \geq 0$)

$$\therefore \quad |x - 2| = 1$$

$$\Rightarrow \quad x - 2 = 1 \Rightarrow x = 3$$

or $\quad -(x - 2) = 1 \Rightarrow x = 1$

Hence $\quad x = 1, 3$

$\therefore$ Sum of the roots of equation $= 1 + 3 = 4$.

13. Just consider an option, and then substitute the values of A and B from assumed option, if the roots p, q, r, s are in A.P., then the presumed option is correct, else not.
thus we get options a, b and c are incorrect, hence (d) is the right choice.
Alternatively : Let us consider $p = a - 3d$, $q = a - d$, $r = a + d$, and $s = a + 3d$

$$\therefore \quad 2a - 4d = 2 \qquad \ldots(i)$$

and $\quad 2a + 4d = 18 \qquad \ldots(ii)$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Solving equations (i) and (ii), we get

$$a = 5, d = 2$$

$\therefore$ $p = -1, \quad q = 3, \quad r = 7, \quad s = 11$

Hence $A = pq = -3$

and $B = rs = 77$

Thus (d) is the right choice

14. Let $f(x) = x^2 - 2ax + a^2 + a - 3$

Since $f(x)$ has real roots both less than 3.

Therefore, $D > 0$ and $f(3) > 0$

$\Rightarrow$ $a^2 - (a^2 + a - 3) > 0$

and $a^2 - 5a + 6 > 0$

$\Rightarrow$ $a < 3$ and $(a - 2)(a - 3) > 0$

$\Rightarrow$ $a < 3$ and $a < 2$ or $a > 3$

$\Rightarrow$ $a < 2$

15. Considering the given constraints in the problem.

Let us consider $\alpha, \beta = (-3, 2)$

then the given equation becomes

$$x^2 + x - 6 = 0$$

$\Rightarrow$ $b = 1$ and $c = -6$

Now, we check for the given choices, which satisfy the aforesaid conditions

(a) It is clearly wrong

(b) It is correct

(c) It is also wrong

(d) It is also wrong

Hence option (b) is appropriate.

NOTE Students don't be worried about this technique. It must be clear to them that while assuming the values of α, β, they have to strictly follow the basic given constraints, *i.e.*, α must be less than β and after forming the similar equation they should have $c < 0 < b$. So I think it is even very intelligent method to solve the problems rather than conventional methods.

16. Let us assume $a = 3$, $b = 4$, given that $a < b$ then the given equation becomes

$$(x - 3)(x - 4) - 1 = 0$$

$$x^2 - 7x + 11 = 0$$

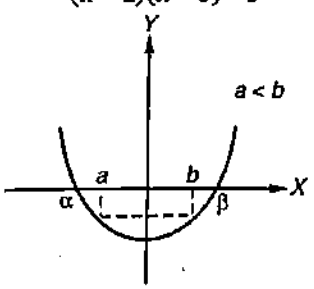
$\therefore$ $x = \dfrac{7 \pm \sqrt{49 - 44}}{2} \Rightarrow x = \dfrac{7 \pm \sqrt{5}}{2}$

$\Rightarrow$ $x = \dfrac{7 + \sqrt{5}}{2} > 4$ and $\dfrac{7 - \sqrt{5}}{3} < 3$

Hence only option (d) is satisfied, hence correct.

Alternatively: Since $f(x) = (x - a)(x - b) - 1 = 0$

$\Rightarrow$ $(x - a)(x - b) = 1$



$\alpha < a$ and $\beta > b$

It means x-axis is shifted to -1 unit below the original position. So it is clear from the graph that $\alpha < a$ and $\beta > b$

$$\alpha\beta = p \quad \text{and} \quad \gamma\delta = q$$

17. Now since, $\alpha, \beta, \gamma, \delta$ are in GP and integral values. So the options (b) and (c) are ruled out as they have no required prop integral factors. Now let us look for option (a). We see that

$$\alpha\beta = -2 = -1 \times 2$$

$$\gamma\delta = -32 = -4 \times 8$$

So, $-1, 2, -4, 8$ are in GP satisfying the above conditions. Again in option (d) the two values don't have the factors wit common ratio, hence its wrong. Finally option (a) is correct.

Alternatively : The sum of roots and product of roots is a follows :

$$\alpha + \beta = 1, \qquad \alpha\beta = p$$

$$\gamma + \delta = 4, \qquad \gamma\delta = q$$

Let r be the common ratio of the GP $\alpha, \beta, \gamma, \delta$.

Then, $\alpha + \beta = 1$ and $\gamma + \delta = 4$

$\alpha + \alpha r = 1$ and $\alpha r^2 + \alpha r^3 = 4$

$\Rightarrow$ $\alpha(1 + r) = 1$ and $\alpha r^2(1 + r) = 4$

So $1 + r = \dfrac{1}{2}$ and $\alpha r^2 \left(\dfrac{1}{\alpha}\right) = 4$

$\Rightarrow$ $r = \pm 2$

When $r = 2$, then $\alpha = \dfrac{1}{3}$ which is inadmissible.

Hence, $r = -2$ is to be considered, then $\alpha = -1$

Thus $\alpha\beta = p = -2$

and $r\delta = q = -32$

18. When this problem will be solved by algebraic methods, it will take too much time to solve beyond the normal required time. So, the best way to get the correct and quick answer is to assume some simple roots (*i.e.*, α and β) then go through options.

Let us take two arbitrary values $\alpha = -1$, $\beta = 2$, then the equation will be $x^2 - x - 2 = 0$

Comparing with the equation $x^2 - px + q = 0$

$\Rightarrow$ $p = 1, \quad q = -2$

Now, the sum of the roots of the required equation

$$= [(\alpha^2 - \beta^2)(\alpha^3 - \beta^3)] + [\alpha^3\beta^2 + \alpha^2\beta^3]$$

$$= 27 + 4 = 31$$

and product of roots $[(\alpha^2 - \beta^2)(\alpha^3 - \beta^3)[\alpha^3\beta^2 + \alpha^2\beta^3]$

$= 27 \times 4 = 108$

Hence equation is $x^2 - 31x + 108 = 0$

Now putting the values of p and q in the equation options a, b and c we get option (b) is correct. as :

$x^2 - [1 - \{5 \times 1 (-2)\} + 5 \times 1 \times 4] + [1 \times 4 - \{5 \times 1 \times (-8)\}$

$+ 4 \times 1 \times 16] = 0$

$$= x^2 - 31x + 108 = 0$$

19. Let us consider choice (a). When we put the values of A and B respectively, we get the values of α, β, γ and δ as -1, $1/3$, $1/5$, $1/3$, which are not in HP. So this option is not correct.

Now for our convenience we consider choice (c), then by substituting the values of A and B, we get the values of α, β, γ and δ as 1, 1/2, 1/3 and 1/4 which are in HP. Hence this could be the correct choice.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Alternatively :

$$Ax^2 - 4x + 1 = 0 \quad \ldots(i)$$

$$\alpha + \gamma = \frac{4}{A} \quad \ldots(1)$$

$$\alpha\gamma = \frac{1}{A} \quad \ldots(2)$$

and

$$Bx^2 - 6x + 1 = 0 \quad \ldots(ii)$$

$$\beta + \delta = \frac{6}{B} \quad \ldots(3)$$

$$\beta\delta = \frac{1}{B} \quad \ldots(4)$$

Since it is given that $\alpha, \beta, \gamma, \delta$ are in HP.

$$\therefore \quad \beta = \frac{2\alpha\gamma}{\alpha + \gamma} = \frac{1}{2}$$

and

$$\gamma = \frac{2\beta\delta}{\beta + \delta} = \frac{1}{3}$$

Again since β and γ are the roots of the given equation hence they must satisfy the equation. So

$$B\beta^2 - 6\beta + 1 = 0$$

and

$$A\gamma^2 - 4\gamma + 1 = 0$$

$$\Rightarrow \quad B = 8 \quad \text{and} \quad A = 3$$

Hence option (c) is correct.

20. Assume some convenient and appropriate values of a, b, c as

$$a = 3, \quad b = 4, \quad c = 6.$$

then $(x-3)(x-4) - 6 = 0 \quad [\because (x-a)(x-b) = c, c \neq 0]$

$$\Rightarrow \quad x^2 - 7x + 6 = 0$$

$$\Rightarrow \quad \alpha = 6, \quad \beta = 1$$

Again $(x-6)(x-1) + 6 \; (\because (x-\alpha)(x-\beta) + c = 0)$

$$x^2 - 7x + 6 + 6 = 0$$

$$x^2 - 7x + 12 = 0$$

$\therefore$ the roots $k_1 = 3$ and $k_2 = 4$

which are same as a and b

Hence, option (c) is correct.

Alternatively : $x^2 - (a+b)x + (ab - c) = 0$

$$\therefore \quad \alpha + \beta = a + b$$

and

$$\alpha\beta = ab - c$$

Again if k_1 and k_2 be the roots of the other equation, then

$$(x-\alpha)(x-\beta) + c = 0$$

i.e., $x^2 - (\alpha + \beta)x + (\alpha\beta + c) = 0$

$$\therefore \quad k_1 + k_2 = \alpha + \beta = a + b \quad \ldots(i)$$

and $k_1 \cdot k_2 = \alpha\beta + c = (ab - c) + c = ab \quad \ldots(ii)$

Thus, from eq. (i) and (ii) it is clear that the roots are a and b.
Hence correct choice is c.



www.LearnEngineering.in
For More Visit :.www.LearnEngineering.in
Scanned by CamScanner

yoursmanboob.wordpress.com
For More Visit : www.LearnEngineering.in

15 SET THEORY

It is one of the easiest and most interesting chapter. Almost every year 2-3 questions appear in CAT from this chapter, either in QA or in DI/LR section. The important feature of this chapter is Venn diagrams which play a vital role in the logical reasoning. That's why this chapter is interesting logical and important in CAT, XLRI, FMS, CET etc. Since we have to solve most of the problems through Venn diagrams, therefore the whole emphasis is given on the theory and concepts of Venn diagrams. Formulae and algebraic theory is not important particularly in CAT, still we have to know some fundamentals of sets and their applications.

SET

A well defined collection of objects is called a set.

The objects in a set are called its members or elements.

If a is an element of a set A, then we write $a \in A$ (and say a belongs to A) *i.e.*, 'a' is a member of 'A'.

If a does not belong to A, then we write $a \notin A$

By convention the sets are denoted by capital alphabets *e.g.*, $A, B, C, \ldots X, Y, Z$ etc and elements of a set are denoted by $a, b, c, \ldots, x, y, z$ etc.

Some Important Notations of Standard Sets

N : the set of natural numbers
Z : the set of integers
Z^+ : the set of positive integers
Q : the set of all rational numbers
Q^+ : the set of all positive rational numbers
R : the set of real numbers
R^+ : the set of all positive real numbers
C : the set of all complex numbers.

METHODS OF REPRESENTATION OF SETS

1. Roster Method : In this method a set is described by listing all the elements, separated by commas, within braces { }, *e.g.*,

(1) The set of perfect squares upto 100 may be described as {0, 1, 4, 9, 16, 25, 36, 49, 64, 81, 100}

(2) The set of all the days of a week may be described as {Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday}.

NOTE The order of elements (*i.e.*, position of elements) is not important. Hence any element can be written anywhere within braces { }.

2. Set Builder Method : In this method, a set is described by listing its property in the given manner.

The set $P(x)$ is described as

$$P = \{x : P(x) \text{ holds}\} \quad \text{or} \quad \{x \mid P(x) \text{ holds}\}$$

or $P = \{x : x$ has the property $abc \ldots\}$

which is read as 'the set of all x such that $P(x)$ holds'. The symbol '|' or ':' is read as 'such that', *e.g.*,

1. The set of perfect squares upto 100 can be written as

$$P = \{x : x \text{ is a perfect square and } 0 \le x \le 100, x \in N\}$$

or $P = \{x : x \in N,\ x = n^2,\ 0 \le x < 100, n \in Z\}$

or $P = \{x^2 : x \in Z\}$

2. The set of all the odd natural numbers can be expressed as

$$A = \{x : x \in N,\ x = 2n - 1,\ n \in N\}$$

TYPES OF SETS

Singleton Set : A set consisting of a single element is called a singleton set.

e.g., 1. {cat}, {7} etc.

2. $\{x : x + 5 = 7\} = \{2\}$

3. {0} is a singleton set.

Empty Set : A set containing no element at all is called an empty set or a null set or a void set and is denoted by ϕ.

In set builder form it is expressed as $\phi = \{\}$

e.g., 1. $\{x : x \in N,\ 9 < x < 10\} = \phi$

2. $\{x : x \in R,\ x^2 = -8\} = \phi$

NOTE A set containing atleast one element is called a non-empty or non-void set.

Finite Set : If a set has finite number of elements that can be counted or listed *i.e.*, the counting of elements surely comes to an end, is called a finite set, *e.g.*,

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

1. Set of all the management institutes in the world.
2. Set of all the recognised countries in the world.

Cardinal Number : The number of distinct elements in a finite set is called the 'cardinal number' of the set *e.g.*, $n(A) \to n$ is the number of distinct elements in the set A.

Infinite Set : If the number of elements of a set is infinite *i.e.*, which cann't be counted by natural numbers, is called infinite set in other words if a set is not finite then it must be an infinite set, *e.g.*

1. Set of all points on the arc of a circle
2. Set of all concentric circles with a given centre
3. Number of all lines in a particular plane.
4. $\{x : x \in R,\ 1 < x < 2\}$ and $\{x : x \in I,\ x < 10\}$

Equivalent Sets : Two finite sets A and B are equivalent if their cardinal numbers are same

i.e., $n(A) = n(B)$

e.g., $A = \{a, e, i, o, u\}$ and $B = \{1, 3, 5, 7, 9\}$

$\Rightarrow$ $n(A) = n(B)$

Equal Sets : Two sets A and B are said to be equal, if every element of A is in B and every element of B is in A and it is denoted as $A = B$

When the elements of A and B are distinct, then $A \neq B$ *i.e.*, A and B are not equal *e.g.*,

1. $A = \{a, e, i, o, u\}$ and $B = \{e, o, i, a, u\}$

$\Rightarrow$ $A = B$

2. $A = \{p, q, r\}$, $B = \{q, p, r\}$ and $C = \{r, q, p\}$

$\Rightarrow$ $A = B = C$

3. $A = \{C, A, T\}$
   $B = \{C, A, T, A\}$
   $C = \{A, C, A, A, T\}$

$\Rightarrow$ $A = B = C$

Remark

1. Order of elements in the list of elements is not important *i.e.*, meaningless.
2. The repetition of elements in a set is meaningless.
3. $\phi \neq \{0\} \neq 0$, since ϕ is a set containing no element at all, $\{0\}$ is a singleton set and 0 is not a set.
   Remember that equivalent sets are not always equal but equal sets are always equivalent.
   Thus $\{1, 2, 3, 4\}$ and $\{W, X, Y, Z\}$ are equivalent but not equal.

Subsets : Let A and B be two sets such that every element of A is the element of B (*i.e.*, every element of A is in B), then A is called a subset of B and is denoted as-$A \subseteq B$ (Read it as-A is a subset of B)

Superset : If $A \subseteq B$, we say that B is a superset of A and we write $B \supseteq A$.

Read it as-B is the superset of A.

e.g., $B = \{2, 4, 6, 8, 10\}$, $A = \{4, 8\} \Rightarrow B \supseteq A$

Proper Subset : If $A \subseteq B$ and $A \neq B$, then A is called a proper subset of B and denoted as $A \subset B$ *e.g.*,

1. $\{5, 15, 25\} \subset \{5, 10, 15, 20, 25\}$
2. $N \subset W \subset I \subset Q \subset R$

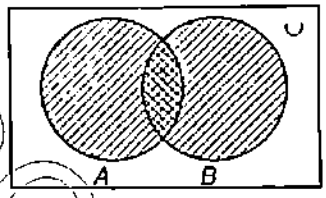
Comparable Sets : Two sets A and B are said to be comparable if either $A \subseteq B$ or $B \subseteq A$.

Results on Subsets

1. The empty set ϕ is a subset of every set.
2. Every set is a subset of itself.
3. The number of all possible subsets of a set containing n elements is 2^n.
4. The number of all proper subsets of a set containing n elements is $(2^n - 1)$.
5. The set of all subsets of a given set A is called the power set of A, denoted by $P(A)$. Thus, if A has n elements $P(A)$ has 2^n elements.

Universal Set : A set that contains all sets in a given context is called the universal set *e.g.*,

1. In a two dimensional coordinate geometry, the set of all points in x-y plane is the universal set.
2. Sets containing natural numbers have a universal set N.
3. If $A = \{1, 2, 4\}$, $B = \{2, 2, 4, 6\}$, $C = \{1, 2, 5, 6, 7\}$, then $U = \{1, 2, 3, 4, 5, 6, 7\}$



Power Set : The collection or family of all the subsets of a set A is called the power set of A and is denoted by $P(A)$.

NOTE The power set is always non-empty, because the empty set and the given set itself are the members or elements of the power set.
$A = \{1, 2, 3\}$, then the number of subsets of the $P(A)$ will be $2^3 = 8$
$P(A) = \{\}, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\}$

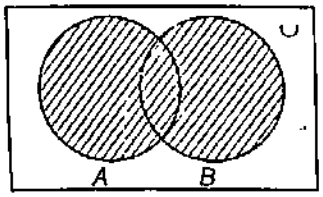
VENN DIAGRAMS

First of all Swiss mathematician Euler elucidated the idea of representing the sets by points in a closed curve. Later on British mathematician Venn brought this into practice.That's why this whole concept is known as Euler-Venn diagram or simply Venn diagrams.

In Venn diagrams the universal set U is represented by points within a rectangle and its subsets are represented by points in closed curves (usually circles) within a rectangle.

OPERATIONS ON SETS

1. Union of Sets : If A and B be two sets, then the union of A and B is the set of the all those elements which belong either to A or to B or to both A and B. It is denoted as $A \cup B$ (read as 'A union B') *e.g.*,



[The shaded region is representing the union of sets]

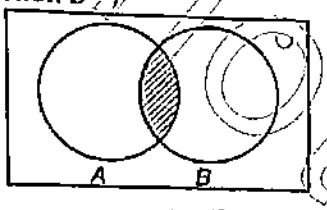
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$A = \{2, 3, 5\}$, $B = \{1, 2, 4, 5\}$

$A \cup B = \{1, 2, 3, 4, 5\}$

2. Intersection of Sets : If A and B be two sets then the intersection of A and B is the set of all those elements that belong to both A and B. It is denoted by $A \cap B$ (read as "A intersection B").



[The shaded region is representing the intersection of sets]

e.g., $A = \{3, 6, 9, 12, 15, 18\}$,

$B = \{4, 6, 8, 10, 12, 14, 16, 18\}$

$\therefore \quad A \cap B = \{6, 12, 18\}$

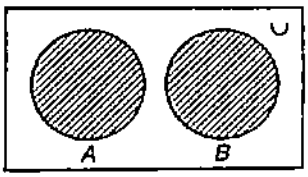
Disjoint Sets : Two sets A and B are said to be disjoint if $A \cap B = \phi$. If $A \cap B \neq \phi$ then A and B are said to be intersecting or overlapping sets.

e.g., $A = \{1, 3, 5, 7, 9\}$ $B = \{2, 4, 6, 8, 10\}$,

$C = \{4, 8, 12\}$

Here $A \cap B = \phi$ $\therefore A$ and B are disjoint sets

and $B \cap C \neq \phi$ $\therefore B$ and C are intersecting sets.



Sets A and B have no elements common.

Difference of Sets : The difference of set A and set B (written as $A - B$) is the set of all those elements of A which do not belong to B.

Thus, $A - B = \{x : x \in A \text{ and } x \notin B\}$

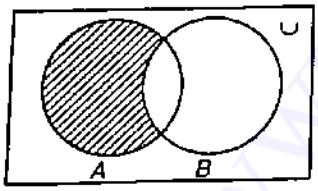
or $A - B = \{x \in A : x \notin B\}$

Clearly $x \in A - B \Leftrightarrow x \in A$ and $x \notin B$

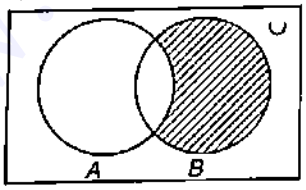
e.g., If $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{2, 4, 6, 8, 10\}$

$A - B = \{1, 3, 5\}$

and $B - A = \{8, 10\}$

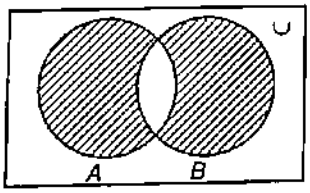


$(A - B)$ is represented by the shaded region



$(B - A)$ is represented by the shaded region

Symmetric difference of two Sets : The symmetric difference of sets A and B is the set $(A - B) \cup (B - A)$ and is denoted by $A \Delta B$



Thus

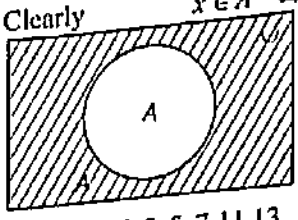
$A \Delta B = (A - B) \cup (B - A)$

$= \{x : x \notin A \cap B\}$

Complement of a Set : If U be the universal set and a set A is such that $A \subset U$ then, the complement of A with respect to U is denoted by A' or A^c or $U - A$ and is defined by the set of all those elements of U which are not in A.

Thus $A' = \{x \in U : x \notin A\}$

Clearly $x \in A' \Leftrightarrow x \notin A$



A' is shown by the shaded region

e.g., $U = \{2, 3, 5, 7, 11, 13 \ldots\}$

$A = \{2\}$

$A' = U - A = \{3, 5, 7, 11, 13, \ldots\}$

ALGEBRAIC LAWS OF SETS

1. **Idempotent Laws :** For any set A, we have
   (i) $A \cup A = A$ (ii) $A \cap A = A$
2. **Identity Laws :** For any set A, we have
   (i) $A \cup \phi = A$ (ii) $A \cap U = A$
3. **Commutative Laws :** For any two sets A and B, we have
   (i) $A \cup B = B \cup A$ (ii) $A \cap B = B \cap A$
4. **Associative Laws :** If A, B and C are any three sets, then
   (i) $(A \cup B) \cup C = A \cup (B \cup C)$
   (ii) $A \cap (B \cap C) = (A \cap B) \cap C$
5. **Distributive Laws :** If A, B and C are any three sets, then
   (i) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
   (ii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
6. **De-Morgan's Laws :** If A and B are any two sets, then
   (i) $(A \cup B)' = A' \cap B'$ (ii) $(A \cap B)' = A' \cup B'$

SOME IMPORTANT RESULTS ON OPERATIONS ON NUMBERS

1. $A - B = A \cap B'$
2. $B - A = B \cap A'$
3. $A - B = A \Leftrightarrow A \cap B = \phi$
4. $(A - B) \cup B = A \cup B$
5. $(A - B) \cap B = \phi$
6. $A \subseteq B \Leftrightarrow B' \subseteq A'$
7. $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$

SOME IMPORTANT RESULTS ON NUMBERS OF ELEMENTS IN SETS

If A, B and C are finite sets, and U be the finite universal set then

(i) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

(ii) $n(A \cup B) = n(A) + n(B) \Leftrightarrow A, B$ are disjoint non-void sets.

(iii) $n(A - B) = n(A) - n(A \cap B)$ *i.e.*,

$n(A - B) + n(A \cap B) = n(A)$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

(iv) $n(A \Delta B)$ = Number of elements which belong to exactly one of A or B.

$$= n(A) + n(B) - 2n(A \cap B)$$

(v) $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$

(vi) number of elements in exactly two of the sets A, B, C

$$= n(A \cap B) + n(B \cap C) + n(C \cap A) - 3n(A \cap B \cap C)$$

(vii) number of elements in exactly one of the sets A, B, C

$$= n(A) + n(B) + n(C) - 2n(A \cap B) - 2n(B \cap C) - 2n(A \cap C) + 3n(A \cap B \cap C)$$

(viii) $n(A' \cup B') = n[(A \cap B)'] = n(U) - n(A \cap B)$

(ix) $n(A' \cap B') = n[(A \cup B)'] = n(U) - n(A \cup B)$

INTRODUCTORY EXERCISE-15.1

1. If $A = \{S\}$, which of the following is a correct statement ?
(a) $S \in A$ (b) $S \subset A$
(c) $\{S\} \in A$ (d) $A = S$

2. Consider the following statements :
1. Any set A is comparable with itself
2. $\{0\}$ is a singleton set
3. $\{\phi\}$ is an empty set.
Of these statements, the correct ones are :
(a) 1 and 2 (b) 2 and 3
(c) 1 and 3 (d) 1, 2 and 3

3. The total number of elements in the power set of a set A containing n elements is :
(a) n^2 (b) 2^n
(c) $2^n - 1$ (d) none of these

4. Let $A = \{3, \pi, \sqrt{2}, -5, 3 + \sqrt{7}, 2/7\}$. The subset of A containing all the elements from it which are irrational numbers is :
(a) $\{\pi, \sqrt{2}, 3 + \sqrt{7}\}$
(b) $\{3, \pi, 2/7, -5, 3 + \sqrt{7}\}$
(c) $\{3, 2/7, -5\}$
(d) $\{3, -5\}$

5. The set $\{x : (x - 3)(x - 5) > 0\}$ is equal to :
(a) $\{x : 3 < x < 5\}$ (b) $\{x : x < 3\} \cup \{x : x < 5\}$
(c) $\{x : x < 3\} \cup (x : x > 5\}$ (d) none of these

6. If A is the set of all integral multiples of 3 and B is the set of all integral multiples of 5, then $A \cap B$ is the set of all integral multiples of :
(a) 3 + 5 (b) 5 - 3
(c) GCD (3, 5) (d) LCM (3, 5)

7. If $X = \{4^n - 3n - 1 : n \in N\}$ and $Y = \{9(n - 1) : n \in N$, then precisely :
(a) $X \subset Y$ (b) $X \subseteq Y$
(c) $X = Y$ (d) $X \supseteq Y$

8. The number of elements of the set $\{x : x \in N, x^2 = 1\}$, where N is the set of all natural numbers is :
(a) 2 (b) 1
(c) 0 (d) 3

9. If $A = \{1, 2, 3, 4\}$ and $B = \{5, 6, 7\}$, then $A \cap B$ is :
(a) $\{1, 2, 3\}$ (b) $\{5, 6, 7\}$
(c) $\{4\}$ (d) ϕ

10. If $A = \{1, 2, 3, ... 9\}$, $B = \{2, 4, 6, 7, 8\}$ and $C = \{3, 4, 5, 8, 9, 10\}$, then $(A - B) \cup C$ is :
(a) $\{1, 3, 4, 5, 8, 9, 10\}$ (b) $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$
(c) $\{2, 4, 6, 7, 8\}$ (d) $\{1, 3, 4, 5, 8, 9\}$

11. $A \cup B = A$, then :
(a) $A \subset B$ (b) $B \subset A$
(c) $A \not\subset B$ and $B \not\subset A$ (d) none of these

12. If A has 3 elements and B has 6 elements, then the minimum number of elements in $A \cup B$ is :
(a) 3 (b) 6
(c) 9 (d) 18

13. Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ be the universal set and $A = \{2, 4, 6\}$, $B = \{1, 3, 7\}$, then $A^c \cap B^c$ is equal to :
(a) $\{2, 4, 5, 6, 8, 9, 10\}$ (b) $\{1, 3, 5, 7, 8, 9, 10\}$
(c) X (d) $\{5, 8, 9, 10\}$

14. If A, B, C are any three sets, then :
(a) $A - (B \cup C) = (A - B) \cup (A - C)$
(b) $A - (B \cup C) = (A - B) \cap (A - C)$
(c) $A - (B \cap C) = (A - B) \cup (A - C)$
(d) $A - (B \cup C) = (A \cup B) - (A \cup C)$

15. Consider the following statements :
For any two sets A and B
(1) $(A - B) \cup B = A$ (2) $(A - B) \cup A = A$
(3) $(A - B) \cap B = \phi$ (4) $A \subseteq B \Rightarrow A \cup B = B$
Of these statements :
(a) 1, 2, 3 are correct (b) 2, 3, 4 are correct
(c) 1, 3, 4 are correct (d) 1, 2, 4 are correct

16. If X and Y are two sets and X' denotes the complement of X, then $X \cap (X \cup Y)'$ equals :
(a) X (b) Y
(c) ϕ (d) none of these

17. If $B \subseteq A$, which of the following statements is correct?
(a) $A \cup B = B \Rightarrow A = B$ (b) $A \cap B = B \Rightarrow A \supseteq B$
(c) $A - B = A \Rightarrow A = B$ (d) $B - A = B \Rightarrow A = B$

18. If A, B and C be any three sets, then $A \cup (B \cap C)$ is the same as :
(a) $(A \cap C) \cup (B \cap C)$ (b) $(A \cup B) \cap (A \cup C)$
(c) $(A \cup B) \cap (B \cup C)$ (d) $(A \cup C) \cap (B \cup C)$

19. $A - (B \cup C)$ equals :
(a) $(A - B) \cup (A - C)$ (b) $(A - B) \cap (A \cap C)$
(c) $(A - B) \cup C$ (d) $(A - C) \cup B$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

your.shahbood.wordpress.com
For More Visit : www.LearnEngineering.in

PROBLEMS BASED ON VENN DIAGRAMS

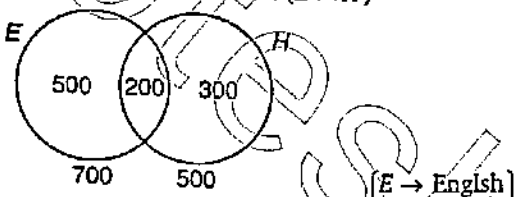
EXAMPLE 1 *In a group of 1000 people 700 can speak English and 500 can speak Hindi. If all the people speak atleast one of the two languages, find :*

(a) how many can speak both the languages?

(b) how many can speak exactly one language?

SOLUTION (a) $n(E \cup H) = n(E) + n(H) - n(E \cap H)$

$1000 = 700 + 500 - n(E \cap H)$



$E \rightarrow$ English
$H \rightarrow$ Hindi

$\Rightarrow \quad n(E \cap H) = 200$

(b) People speaking exactly one language

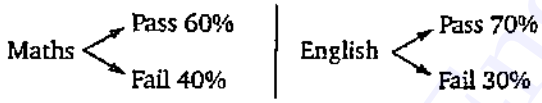
$n(E \cup H) - n(E \cap H) = 1000 - 200 = 800$

Alternatively: People speaking exactly one language

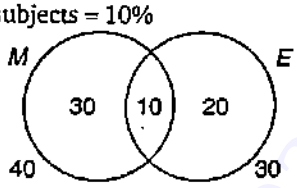
$= 500 + 300 = 800$

EXAMPLE 2 *In an exam 60% of the candidates passed in Maths and 70% candidates passed in English and 10% candidates failed in both the subjects. If 300 candidates passed in both the subjects. The total number of candidates appeared in the exam, if they took test in only two subjects viz-Maths and English.*

SOLUTION



Fail in both subjects = 10%



Failed number of candidates

$\therefore$ Total number of candidates failed in atleast one subject

$= 30 + 10 + 20 = 60\%$

$\therefore$ Total number of candidates passed in both the subjects

$= 100 - 60 = 40\%$

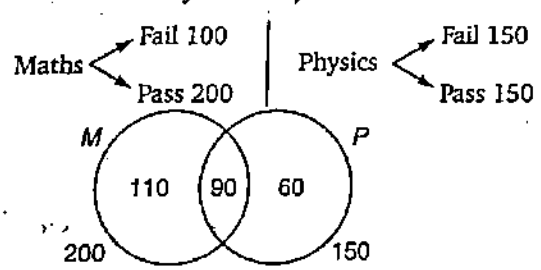
Let total number of candidates appeared in exam $= x$

$\therefore \quad \frac{40x}{100} = 300 \Rightarrow x = 750$

Hence total 750 candidates appeared in exam.

EXAMPLE 3 *A class took two tests, one in Physics and other in Maths. $\left(\frac{1}{3}\right)^{rd}$ class failed in Maths and $\frac{1}{2}$ class failed in Physics. 60% of those passed in Physics also passed in Maths. If 300 students took both the tests then how many students failed in both?*

SOLUTION



Total number of students passed in atleast one subject

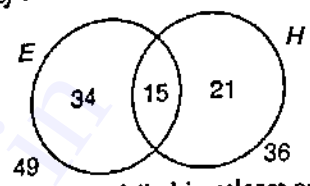
$= 110 + 90 + 60 = 260$

Therefore number of students failed in both the subjects

$= 300 - 260 = 40$

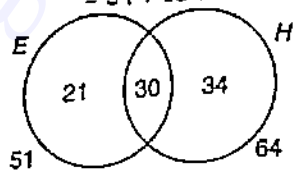
EXAMPLE 4 *In an exam 49% candidates failed in English and 36% failed in Hindi and 15% failed in both subjects. If the total number of candidates who passed in English alone is 630. What is the total number of candidates appeared in exam?*

SOLUTION



Number of candidates failed in atleast one subject

$= 34 + 15 + 21 = 70\%$



$\therefore$ Number of candidates passed in both the subjects

$= 100 - 70 = 30\%$

Number of candidates passed in English only $= 51 - 30 = 21\%$

Let the total number of candidates be x, then

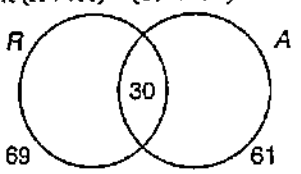
$\frac{21x}{100} = 630 \Rightarrow x = 3000$

Alternatively: Since 21% candidates failed in Hindi only it means they are pass in only English.

Hence number of candidates who are passed only in English $= 21\%$

EXAMPLE 5 *In a survey it was found that 69% of Indians like romantic movies and 61% like action movies. What percentage of Indians like both type of movies?*

SOLUTION $n(R \cap A) = (69 + 61) - 100$



NOTE In general we cannot determine the solution but this can be done if we assume that all the Indians like atleast one of these two types of movies.

EXAMPLE 6 *In a certain city only two cellular (mobile) phone services are available. One is Radiance and other is DSML. It is known that 25% of the city population uses the services of Radiance and 20% uses the services of DSML, while 8% uses both Radiance and DSML. It is also known that 30% of those who uses Radiance services but not services of DSML have prepaid cards and 40% of those who uses DSML but not Radiance have prepaid cards while 50% of those who uses the services of both Radiance and DSML have prepaid cards. What percentage of the population have prepaid cards?*

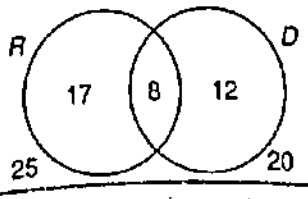
(a) 12·5% (b) 10% (c) 13·9% (d) 20%

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Set Theory

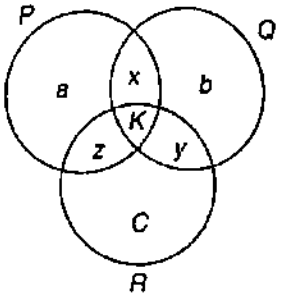
SOLUTION



$\therefore$ Percentage of people have prepaid cards
$= (30\% \text{ of } 17) + (40\% \text{ of } 12) + (50\% \text{ of } 8)$
$= 13.9\%$

VENN DIAGRAMS CONTAINING THREE CIRCLES

Explanation of the concept using an example



Let P, Q, R be three books, then
$a \rightarrow$ number of people reading only P
$b \rightarrow$ number of people reading only Q
$c \rightarrow$ number of people reading only R
$x \rightarrow$ number of people reading P and Q both but not R
$y \rightarrow$ number of people reading Q and R both but not P
$z \rightarrow$ number of people reading P and R both but not Q
$k \rightarrow$ number of people reading all the three books
$x + k \rightarrow$ number of people reading P and Q both
$y + k \rightarrow$ number of people reading Q and R both
$z + k \rightarrow$ number of people reading P and R both
$a + x + k + z \rightarrow$ number of people reading P
$b + x + k + y \rightarrow$ number of people reading Q
$c + y + k + z \rightarrow$ number of people reading R
$a + x + b \rightarrow$ number of people reading P or Q but not R
$b + y + c \rightarrow$ number of people reading Q or R but not P
$c + z + a \rightarrow$ number of people reading P or R but not Q.
$a + b + c \rightarrow$ number of people reading only one book
$x + y + z \rightarrow$ number of people reading only two books
$k \rightarrow$ number of people reading all of three books
$(a + b + c) + (x + y + z) + (k) \rightarrow$ number of people reading atleast one book
$(x + y + z) + (k) \rightarrow$ number of people reading atleast two books
$(a + b + c) + (x + y + z) + (k) \rightarrow$ total number of readers.

FINDING THE VALUES OF COMPONENTS

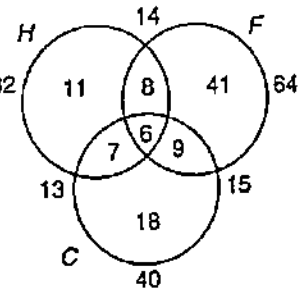
$$P + Q + R = (a + x + k + z) + (b + x + k + y) + (c + y + k + z)$$
$$P + Q + R = (a + b + c) + 2(x + y + z) + 3k$$
$$P + Q + R = \alpha + 2\beta + 3\gamma$$
where $\alpha = a + b + c, \quad \beta = x + y + z, \quad \gamma = k$
Again $(x + k) + (y + k) + (z + k) = (x + y + z) + 3k$
$= \beta + 3\gamma$
$\therefore \quad (\alpha + 2\beta + 3\gamma) - (\beta + 3\gamma) = \alpha + \beta$
and $\quad (\alpha + \beta + \gamma) - (\alpha + \beta) = \gamma$

NOTE 'And' means intersection, 'or' means union

EXAMPLE 1 *In a group of 100 children, 32% play hockey, 64% play football and 40% play cricket. 14% play hockey and football, 15% play football and cricket, 13% play hockey and cricket. Only 6% play all the three games.*

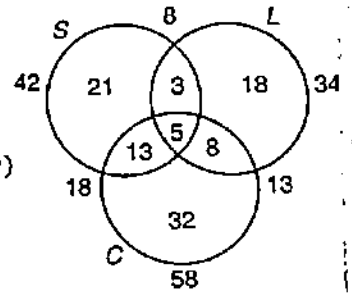
(a) How many children play only one game?
(b) How many children play atleast two game?
(c) How many children play hockey or football but not cricket?

SOLUTION (a) $11 + 41 + 18 = 70$
(b) $(8 + 7 + 9) + 6 = 30$
(c) $(11 + 41) + 8 = 60$



EXAMPLE 2 *In a hostel of 100 students capacity, 42 are science graduate, 34 are law graduate and 58 are commerce graduate. 8 students are science and law graduate, 13 students are law and commerce graduate, 18 students are commerce and science graduate. How many students are only science graduate?*

SOLUTION $\alpha + \beta + \gamma = 100$
$\alpha + 2\beta + 3\gamma = 42 + 34 + 58$
$= 134$
$\beta + 3\gamma = 8 + 13 + 18 = 39$
$\alpha + \beta = (\alpha + 2\beta + 3\gamma) - (\beta + 3\gamma)$
$= 134 - 39 = 95$
$\therefore \quad \gamma = (\alpha + \beta + \gamma) - (\alpha + \beta)$
$\Rightarrow \quad \gamma = 100 - 95 = 5$



$\therefore$ Number of students who are only science graduate
$= 42 - [(18 + 8) - 5]$
$= 42 - (13 + 3 + 5) = 21$

EXAMPLE 3 *In a premier B-school having 100 students in the MBA final year, 15 students specializes in Marketing and HR. 17 students specializes in HR and Systems, 16 students specializes in Marketing and Systems. The number of students specializes in only one field is equal and is 22 in each field of specialization. How many students specializes in all three fields viz- Marketing, HR and Systems?*

SOLUTION $\alpha = a + b + c = 22 + 22 + 22 = 66$
$\therefore \quad \beta + \gamma = 100 - \alpha = 100 - 66 = 34$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yourshekhoob.wordpress.com

For More Visit : www.LearnEngineering.in

Again $\beta + 3\gamma = x + y + z + 3k$

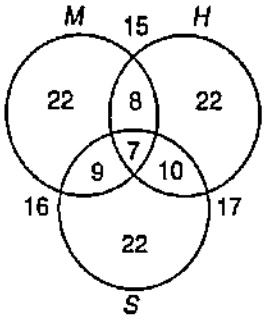
$= 15 + 16 + 17 = 48$

$\therefore \quad (\beta + 3\gamma) - (\beta + \gamma) = 2\gamma$

$= 48 - 34 = 14$

$\Rightarrow \quad \gamma = 7$

Hence, 7 students specializes in all the three fields.

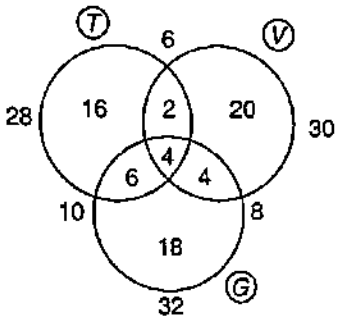


EXAMPLE 4 *In a music school 28 students learn trumpet, 30 students learn violin and 32 students learn guitar. 6 students learn trumpet and violin, 8 students learn violin and guitar, 10 students learn guitar and trumpet. The number of students who learn only one instrument is 54. Also 20 students learn only violin. Every student learn atleast one instrument out of the three instruments.*

(a) Find the number of students in the music school.

(b) Find the number of students who learn trumpet and guitar but not violin?

SOLUTION



$x + b + k + y = 30$

$b = 20$

$\Rightarrow \quad x + k + y = 10$

But, $(x + k) + (y + k) = 14$

$\Rightarrow \quad (x + y + k) + k = 14$

$\Rightarrow \quad 10 + k = 14$

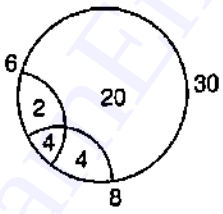
$\Rightarrow \quad k = 4$

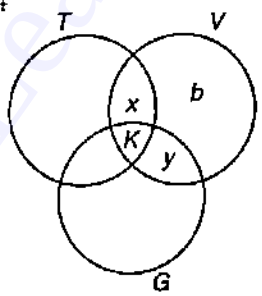
(a) Number of students in music school

$= (16 + 20 + 18) + (2 + 4 + 6) + 4$

$= 70$

(b) Number of students who learn trumpet and guitar but not violin = 6





EXAMPLE 5 *Out of 180 students in a class 7 students failed in all the three subjects viz., Physics, Chemistry, Maths. Only 144 students pass in only one subject and 21 students pass in only two subjects.*

(a) How many students passed in all the three subjects?

(b) How many students passed in atleast two subjects?

(c) How many students passed in at most two subjects but atleast in one subject?

SOLUTION

$\alpha = a + b + c = 144$

$\beta = x + y + z = 21$

$\alpha + \beta + \gamma = 180 - 7 = 173$

$\therefore \quad \gamma = (\alpha + \beta + \gamma) - (\alpha + \beta)$

$\Rightarrow \quad \gamma = (173) - (144 + 21)$

$\Rightarrow \quad \gamma = 8 \qquad (\gamma = k)$

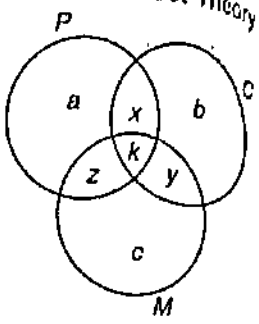
(a) 8

(b) Atleast two subjects

$= \beta + \gamma = 21 + 8 = 29$

(c) Atleast one subject but atmost two subjects

$= \alpha + \beta = 144 + 21 = 165$



EXAMPLE 6 *In a marriage party total 278 guests were present. 20 guests took Pepsi and Dew, 23 guests took Dew and Sprite and 21 guests took Pepsi and Sprite and 9 guests took all the three cold drinks viz. Pepsi, Sprite and Dew. It is also known that there were equal number of bottles of each of three kinds viz. Pepsi, Dew and Sprite*

(a) How many guests took only one bottle colddrink?

(b) How many guests enjoyed Dew or Sprite but not Pepsi?

(c) How many guests enjoyed Pepsi and Dew but not Sprite.

SOLUTION Number of guests

$= (a + b + c) + (11 + 12 + 14) + (9)$

$= 278$

Number of bottles

$= (a + b + c) + 2(11 + 12 + 14) + 3(9)$

$= [(a + b + c) + (11 + 12 + 14) + (9)] + (11 + 12 + 14) + 2(9)$

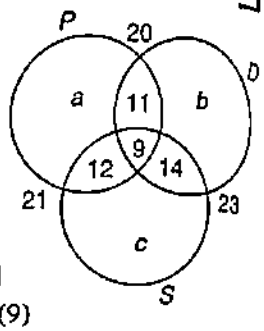
$= 278 + 55 = 333$

$\therefore \quad$ Each kind of bottles $= \dfrac{333}{3} = 111$

$\because \quad a = 111 - (11 + 9 + 12) = 79$

$b = 111 - (11 + 9 + 14) = 77$

$c = 111 - (12 + 9 + 14) = 76$



(a) Number of guests who took only one bottle $= a + b + c$

$= (79 + 77 + 76) = 232$

(b) Dew or Sprite but not Pepsi $= b + 14 + c$

$= 77 + 14 + 76 = 167$

(c) Dew and Pepsi but not Sprite $= 20 - 9 = 11$

EXAMPLE 7 *250 students appeared in MOCK CAT consisting of three sections viz. Maths, Data Interpretation (DI) and English. 20 students did not qualify any of the three sections and 66 students qualified all the three sections, 24 students did not qualify DI only. 12 did not qualify DI and Maths only. Those who qualified Maths only exceeded those who did not qualify any of the section by 16. Those who did not qualify Maths only exceeded those who did not qualify only DI and Maths by 8. Those who did not qualify English only were twice the number of students who did not qualify Maths only. Find the number of students who has qualified atleast two sections.*

SOLUTION Total students = 250

Number of students who did not qualify any of the three sections = 20

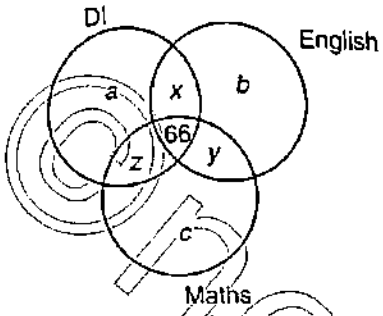
$\therefore$ Number of students who qualified atleast one section

$= 250 - 20 = 230$

For More Visit : www.LearnEngineering.in

Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in



EXTRA $y = 24$ (It means 24 students must have qualified English and Maths)

$b = 12$ (It means they have qualified neither DI nor Maths but English only)

$c = 36 = (20 + 16)$

$x = 20 = (12 + 8)$

$z = 40 = (20 \times 2)$

$\therefore \quad (a + b + c) + (x + y + z) + 66 = 230$

$(a + 12 + 36) + (20 + 24 + 40) + 66 = 230$

$\Rightarrow \quad a = 32$

but the required answer $= (x + y + z) + 66$

$= 84 + 66 = 150$

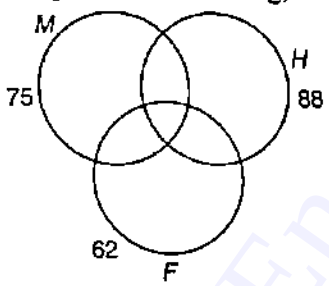
CONCEPT OF MAXIMUM OR MINIMUM

Type 1. In a Business School there are 3 electives and atleast one elective is compulsory to opt. 75% students opted for Marketing, 62% students opted for Finance and 88% students opted for HR. A student can have dual or triple specialization.

(a) What is the minimum number of students that are specialize in all three streams?

(b) What is the maximum number of students that can specialize in all three streams?

SOLUTION (a) 75% students opted for Marketing, it means atleast 25% students opted for HR or Finance or HR and Finance both.



Similarly atleast 12% students opted for Marketing or Finance or Marketing and Finance both.

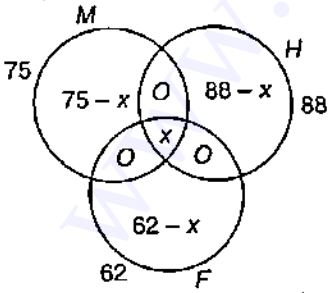
Again atleast 38% students opted for Marketing or HR or Marketing and HR both.

$\therefore \quad 25 + 12 + 38 = 75\%$

It means that if there is no intersection in these three courses, 75% would be maximum number of students in M, H, F alone or $(M \cap H)$, $(M \cap F)$, $(H \cap F)$.

Thus it gives atleast $25\% = (100 - 75)$ students in all 3 departments.

(b) $(75 - x) + (88 - x) + (62 - x) + x = 100 \Rightarrow x = 62.5\%$



Type 2. In a Business school there are 3 electives offered to the students, where students have a choice of not choosing any electives.

70% students opted for Marketing, 60% students opted for HR and 50% students opted for Finance. 30% students opted for Marketing and HR both, 40% opted for HR and Finance both, 20% opted for Marketing and Finance both.

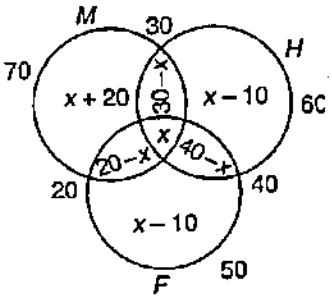
(a) What is the minimum number of students opted for all 3 electives?

(b) What is the maximum number of students opted for all 3 electives?

SOLUTION Total number of students

$= 70 + (x - 10) + (40 - x) + (x - 10)$

$= 90 + x$



(a) The minimum value of x so that none of the categories becomes negative $= 10$

(b) The maximum value of x so that none of the categories becomes negative $= 20$

Type 3. In a hostel there are 250 students. 120 watch news channels, 80 watch sports channels and 90 watch music channels. Each channel telecasts the content as per their nature 50 students watch both news and sports channel 60 students watch both sports and music channel and 65 students watch both news and music channels.

(a) What is the minimum number of students who watch atleast one of the given channel?

(b) What is the maximum number of students who watch atleast one of the given channels?

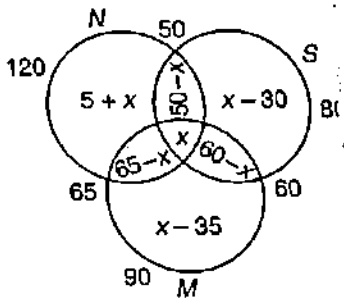
(c) What is the minimum number of students who watch all the three types of channels?

(d) What is the maximum number of students who watch all the three types of channels?

SOLUTION Since the number of students cannot be negative

$\therefore \quad \left.\begin{matrix} x - 30 \geq 0 \\ x - 35 \geq 0 \end{matrix}\right\}$

Also $\left.\begin{matrix} 50 - x \geq 0 \\ 60 - x \geq 0 \\ 65 - x \geq 0 \end{matrix}\right\}$



Total number of students playing atleast one game

$= 120 + (x - 30) + (60 - x) + (x - 35) = 115 + x$

(a) Minimum number of students watching atleast one channel $= 115 + 35 = 150$

(b) Maximum number of students watching atleast one channel $= 115 + 50 = 165$

(c) Minimum number of students watching all the 3 channels $= 35$

(d) Maximum number of students watching all the 3 channels $= 50$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

EXERCISE

LEVEL 1

1. In a city 45% of the people read English and Hindi newspapers. 15% read only Urdu newspaper and 75% read Hindi newspaper. If nobody reads all the three newspapers and everybody read atleast one newspaper out of English, Hindi and Urdu newspapers, how many read Urdu newspaper?
(a) Max 45% (b) Max. 55%
(c) Min 25% (d) Min 30%

✍ ***Directions for question number 2-5 :*** *In our coaching institute there are total 170 students and they use different vehicles for transportation viz. bike, car and taxi.*
The ratio of students using all 3 vehicles to students using atleast 2 vehicles is 2 : 9. The ratio of students using only one vehicle to students using atleast 2 vehicles is 8 : 9.
Number of students using car only exceeds number of students using bike only by 14.
Number of students using taxi only exceeds number of students using bike only by 12. Number of students using taxi, bike, car is 90, 93, 97 respectively.

2. Number of students using all three vehicles is :
(a) 18 (b) 12
(c) 20 (d) none of these

3. Number of students using no more than one vehicles is :
(a) 76 (b) 80
(c) 60 (d) can't be determined

4. Number of students using exactly two vehicles is :
(a) 38 (b) 55
(c) 70 (d) none of these

5. The number of students who are using both bike and car but not taxi is :
(a) 23 (b) 40
(c) 36 (d) data insufficient

6. In a survey among *B*-school students, 68% of those surveyed were in favour of atleast one of the three magazines- *A*, *B* and *C*. 38% of those surveyed favoured magazine *A*, 26% favoured magazine *B* and 36% favoured magazine *C*. If 11% of those surveyed favoured all three magazines. What per cent of those surveyed favoured more than one of the three magazines?
(a) 25% (b) 33%
(c) 21% (d) 26%

✍ ***Directions of question number 7 and 8:*** *In a class of 80 students 25 passed in QA and DI, 25 passed in DI and English, 20 passed in QA and English. 10 student passed in all the three subjects.*

7. How many students passed only in Q.A.?
(a) 20 (b) 15
(c) 12 (d) can't be determined

8. If no student failed in all three subjects, what is the total number of students who passed in QA only, DI only and English only?
(a) 50 (b) 30
(c) 25 (d) can't be determined

✍ ***Directions for question number 9 and 10:*** *A survey shows that 41%, 35% and 60% of the people watch "Maine Pyaar Kiya" "Maine Pyaar Kyun Kiya" and "Pyaar to Hona Hi Tha" respectively. 27% people watch exactly two of the three movies and 3% watch none.*

9. What percentage of people watch all the three movies ?
(a) 40% (b) 6%
(c) 9% (d) 12%

10. If another survey indicates that 16% of the people watch Maine Pyaar Kiya and Pyaar to Hona Hi Tha, and 14% watch Maine Pyaar Kyun Kiya and Pyaar to Hona Hi Tha, then what percentage of the people watch only Maine Pyaar Kyun Kiya? (Use the data from the previous question, if necessary)
(a) 10% (b) 8%
(c) 12% (d) 15%

11. Find the number of positive integers up to 100 which are not divisible by any 2, 3 and 5 ?
(a) 24 (b) 25
(c) 26 (d) 27

12. A survey was conducted at a coaching institution and it was found that there were 34 students who appeared in MAT. There were 37 students who appeared in CAT of which 17 students appeared in MAT. 30 students appeared in XAT of which 13 students appeared in MAT. Of the XAT applicants (*i.e.*, appeared students) 14 appeared in CAT and these 6 appeared in MAT. How many students appeared in CAT but not in MAT or XAT ?
(a) 9 (b) 10
(c) 12 (d) none of these

13. A survey among 151 persons is conduced regarding their favourite channel of radio - Radio city, Radio mirchi and Radio life. It was found that every listener of Radio mirchi also listen either Radio city or Radio life. The number of persons listening all the radio channels is the same as the number of persons who listen none of the channels. 55 persons listen exactly two channels and 70 persons listen only

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

one channel. The number of people who listen all the three channels ?
(a) 16 (b) 13
(c) 9 (d) data insufficient

14. In a group of 132 people 50, 60, 70 people like three different sweets-Barfi, Jalebi, Rasgulla, respectively. The number of people who like all the three sweets is half the number of people who like exactly two sweets. The number of people who like exactly any two out of the three sweets is the same as those who like exactly any other two of the three sweets. The number of people who like the three sweets :
(a) 12 (b) 6
(c) 8 (d) none of these

15. In the previous question (number 14) find the number of people who like Rasgulla or Jalebi but not Barfi.
(a) 82 (b) 42
(c) 48 (d) 38

16. In a group of 80 employees, the number of employees who are engineers is twice that of the employees who are MBAs. The number of employees who are not engineers is 32 and that of those who are not MBAs is 56. The number of employees who are both engineers and MBAs is twice that of the employees who are only MBAs. How many employees are neither engineer (B. Tech) nor MBAs ?
(a) 24 (b) 38
(c) 36 (d) can't be determined

Direction for question number 17-19: There are 60 workers who work for M/s. Nottan Dibbawala Pvt. Ltd. Mumbai, out of which 25 are women. Also :
(i) 28 workers are married
(ii) 26 workers are graduate
(iii) 20 married workers are graduate of which 9 are men
(iv) 15 men are graduate
(v) 15 men are married.

17. How many unmarried women are graduate ?
(a) 20 (b) 8
(c) 0 (d) can't be determined

18. How many unmarried women work in the company ?
(a) 11 (b) 12
(c) 9 (d) none of these

19. How many graduate men are married ?
(a) 9 (b) 15
(c) 13 (d) none of these

Directions for question number20-23: In the year 222 B.C. 100 CAT aspirants appeared in CAT- (Common Admission Test for MBA in some prestigious institutions). They had to show their competency in all the following four areas viz- Maths, Data Interretation (DI), Logical Reasoning (LR) and English.
Number of aspirants who had qualified Maths is 55.
Number of aspirants who had qualified LR is 38
Number of aspirants who had qualified Maths and English is 30.
Number of aspirants who had qualified LR and English is 15.
Number of aspirants who had qualified Maths and LR is 20.
Number of aspirants who had qualified Maths, LR and English is 5.
Number of aspirants who had qualified DI is 22.
Number of aspirants who had qualified DI and LR is 5.
Number of aspirants who had qualified DI and Maths is 5.
Number of aspirants who had qualified DI, Maths and LR is 5.
Number of aspirants who had qualified English is 50.
Number of aspirants who had qualified English could not qualify DI section.

20. Number of aspirants who had qualified only in DI section :
(a) 17 (b) 22
(c) 27 (d) none of these

21. Number of aspirants who had qualified in only Maths and LR sections :
(a) 20 (b) 10
(c) 28 (d) 38

22. Find the number of aspirants who had qualified in atleast two sections out of 4 sections.
(a) 45 (b) 35
(c) 55 (d) can't be determined

23. Find the number of aspirants who had qualified none of the sections.
(a) 12 (b) 18
(c) 7 (d) 0

24. In a survey among 80 people, 50 people like arrange marriage and 70 people like love marriage. What is the minimum and maximum number of people like both the marriages respectively?
(a) 40, 45 (b) 40,50
(c) 30, 40 (d) can't be determined

25. In a car agency one day 120 cars were decorated with three different accessories viz., power window, AC and music system. 80 cars were decorated with power windows, 65 cars were decorated with AC and 80 cars were decorated with music systems. What is the minimum and maximum number of cars which were decorated with all of three accessories ?
(a) 10 ,61 (b) 10, 45
(c) 25, 35 (d) none of these

26. In our coaching there were 200 students enrolled for DI, 150 for English and 150 for Maths. Of these 80 students enrolled for both DI and English. 60 students enrolled for Maths and English, while 70 students enrolled for DI and Maths. Some of these students enrolled for all the three subjects. Diwakar teaches those students who are enrolled for DI classes only. Priyanka teaches those students who are enrolled for English only and Varun teaches those students who are enrolled for Maths only. Sarvesh is a senior most faculty therefore, he can teach all the three subjects. Students always prefer a specialist for their respective subjects. If Diwakar teaches 80 students then the other three faculty can be arranged in terms of the number of students taught as :
(a) Sarvesh > Varun > Priyanka
(b) Sarvesh > Priyanka > Varun
(c) Varun > Sarvesh > Priyanka
(d) none of the above

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Directions : ***Answer the question no. 27 to 29 on the basis of the information given below :***

Sanskaram Karoti (SK) is a spiritual organisation involved in performing spiritual rites. Currently it has 37 volunteers. They are involved in three jobs : Body Massage (BM), Yoga and Pooja. Each volunteer working with Sanskaram Karoti (SK) has to be involved in atleast one of the three jobs mentioned above.

- *A maximum number of volunteers are involved in Yoga. Among them, the number of volunteers involved in Yoga alone is equal to the volunteers having additional involvement in the Pooja.*
- *The number of volunteers involved in Pooja alone is double the number of volunteers involved in all the three jobs.*
- *17 volunteers are involved in Body Massage (BM).*
- *The number of volunteers involved in Body Massage alone is one less than the number of volunteers involved in Pooja alone.*
- *Ten volunteers involved in the Body Massage are also involved in atleast one more job.*

27. Based on the information given above, the minimum number of volunteers involved in both Yoga and Body Massage, but not in the Pooja is :

(a) 1 (b) 3
(c) 4 (d) 5

28. Which of the following additional information would enable to find the exact number of volunteers involved in various jobs?

(a) Twenty volunteers are involved in Yoga
(b) Four volunteers are involved in all the three jobs
(c) Twenty three volunteers are involved in exactly one task
(d) No need for any additional information

29. After some time, the volunteers who were involved in all the three tasks were asked to withdraw from one task. As a result, one of the volunteers opted out of the Body Massage and one opted out of the Pooja, while the remaining ones involved in all the three tasks opted out of the Yoga. Which of the following statements, then, necessarily follows?

(a) The least number of volunteers is now performing Body Massage
(b) More volunteers are now associated with Yoga as compared to Pooja
(c) More volunteers are now performing Body Massage as compared to Pooja
(d) None of the above

Answers

INTRODUCTORY EXERCISE-15.1

1. (a)	2. (a)	3. (b)	4. (a)	5. (c)	6. (d)	7. (a)	8. (b)	9. (d)	10. (a)
11. (b)	12. (b)	13. (d)	14. (b)	15. (b)	16. (c)	17. (b)	18. (b)	19. (b)	

LEVEL-1

1. (b)	2. (c)	3. (b)	4. (c)	5. (a)	6. (c)	7. (d)	8. (b)	9. (b)	10. (c)
11. (c)	12. (c)	13. (b)	14. (a)	15. (a)	16. (a)	17. (c)	18. (b)	19. (a)	20. (a)
21. (b)	22. (c)	23. (d)	24. (b)	25. (d)	26. (a)	27. (c)	28. (a)	29. (b)	

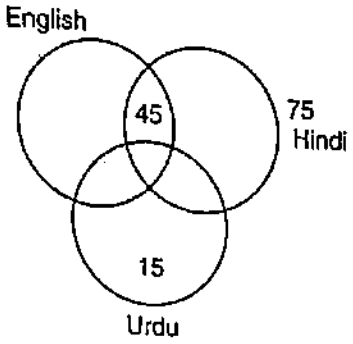
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

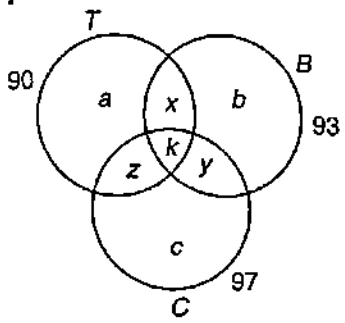
Hints & Solutions

LEVEL 1

1. It is clear that 45% people cannot read another third (*i.e.*, Urdu) newspaper. Besides them all of the rest people can read Urdu newspaper.
Hence maximum 55% (100 – 45) people can read Urdu newspaper.



Solution for question number 2-5:



$a+b+c=\alpha,\quad x+y+z=\beta,\quad k=\gamma$

$\alpha+\beta+\gamma=170$

$\alpha+2\beta+3\gamma=90+93+97=280$

$\gamma:(\beta+\gamma)=2:9$

$\Rightarrow \quad \gamma:\beta=2:7$

and $\quad \alpha:(\beta+\gamma)=8:9$

$\Rightarrow \quad \alpha:\beta:\gamma=8:7:2$

$\therefore \quad \alpha=80, \beta=70$ and $\gamma=20$

$\Rightarrow \quad a+b+c=80,\quad x+y+z=70 \qquad \ldots(1)$

and $\quad k=20$

Again $\quad c-b=14$ and $a-b=12 \qquad \ldots(2)$

On solving eq. (1) and (2) we get $a=30,\ b=18,\ c=32$

Again $(a+x+k+z)-(a+k)=(x+z)$

$=90-(30+20)=40$

and $\quad (x+y+z)-(x+z)=y=70-40=30$

similarly $\quad x=25$ and $z=15$

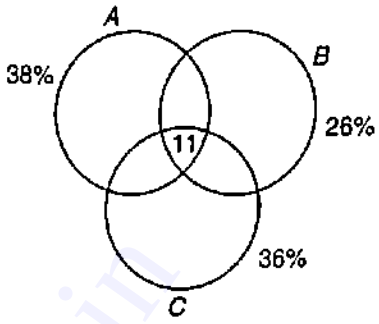
2. 20

3. 80

4. 70

5. 14

6.



$\alpha+\beta+\gamma=68$

$\alpha+2\beta+3\gamma=(38+26+36)=100$

and $\quad \gamma=11$

$\therefore (\alpha+2\beta+3\gamma)-[(\alpha+\beta+\gamma)+\gamma)]=\beta+\gamma$

$=100-[68+11]$

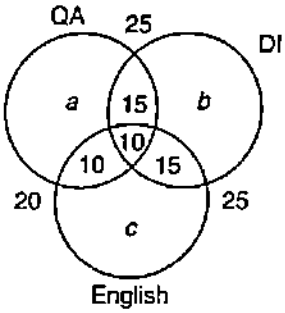
$=21$

Hence 21% favoured more than one magazine.

($\alpha \rightarrow$ favoured only one magazine, $\beta \rightarrow$ favoured only two magazines, $\gamma \rightarrow$ favoured all three magazines)

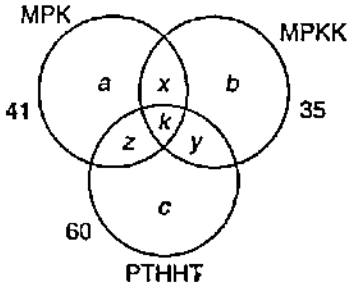
Solution for question number 7-8 :

7. Since we don't know how many students failed in all the three subjects, the questions cannot be answered. Hence (d).



8. $(a+b+c)=80-[(15+15+10)+(10)]=30$

9. $\alpha+\beta+\gamma=97\%$



$\alpha+2\beta+3\gamma=41+35+60=136\%$

But $\quad \beta=(x+y+z)=27\%$

$\therefore \quad (\alpha+2\beta+3\gamma)-(\alpha+\beta+\gamma)=\beta+2\gamma=39\%$

$\therefore \quad (\beta+2\gamma)-\beta=2\gamma=39-27=12\%$

$\therefore \quad \gamma=6\%=(k)$

$\therefore$ 6% people watch all the three movies

10. $z+k=16 \Rightarrow z=10 \qquad \{\because \gamma=k=6\}$

$y+k=14 \Rightarrow y=8$

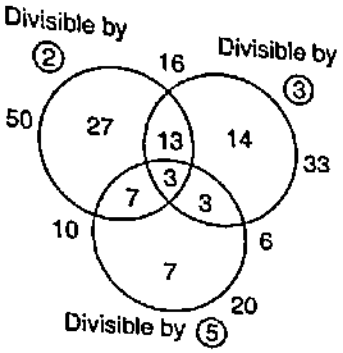
$\therefore \quad x=9 \qquad (\because x+y+z=27)$

$\therefore \quad b=35-(x+k+y)=35-(9+6+8)=12\%$

11. Total numbers divisible by 2 upto 100 = 50
Total numbers divisible by 3 upto 100 = 33
Total numbers divisible by 5 upto 100 = 20
Total numbers divisible by 2 and 3 *i.e.*, 6 upto 100 = 16

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in



Total numbers divisible by 3 and 5 *i.e.*, 15 upto 100 = 6

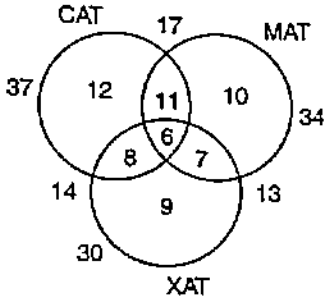
Total numbers divisible by 2 and 5 *i.e.*, 10 upto 100 = 10

Total numbers divisible by 2, 3 and 5 *i.e.*, 30 upto 100 = 3

∴Total number of numbers upto 100 which are divisible by at least one of 2, 3 and 5 = 74

∴Total number of numbers upto 100 which are not divisible by any 2, 3 or 5 = 100 − 74 = 26

12.



Hence there are 12 students who appeared in CAT but not in MAT or XAT

13. $\beta = (x + y + z) = 55$

$\alpha = (a + b + c) = 70$

$\gamma = k$

Let m people listen none of the three channels, then $m = \gamma = k$

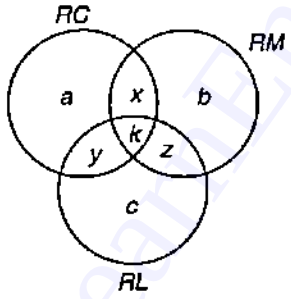
$$(\alpha + \beta + \gamma) + m = 151$$

$$\Rightarrow \quad \alpha + \beta + \gamma + \gamma = 151$$

$$\Rightarrow \quad (55 + 70) + 2\gamma = 151$$

$$\Rightarrow \quad \gamma = 13$$

Hence, there are 13 people listen all three channels.



14. $\alpha = a + b + c$

$\beta = x + y + z$

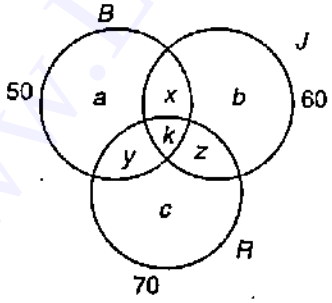
$\gamma = k$

Here $\gamma = \frac{1}{2}\beta \Rightarrow 2\gamma = \beta$

Again $x = y = z = p$

$\therefore \quad \beta = 3p$

$\Rightarrow \quad \gamma = \frac{3}{2}p$



Now, $\alpha + 2\beta + 3\gamma = 50 + 60 + 70 = 180$

$\Rightarrow \quad \alpha + 4\gamma + 3\gamma = 180$

$\Rightarrow \quad \alpha + 7\gamma = 180 \quad ...(1)$

Again $\alpha + \beta + \gamma = 132$

$\alpha + 2\gamma + \gamma = 132$

$\Rightarrow \quad \alpha + 3\gamma = 132 \quad ...(2)$

From eq. (1) and (2), we get

$\gamma = 12$

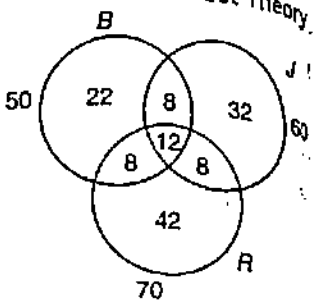
Hence 12 people like all 3 sweets.

15. $\gamma = \frac{3}{2}p \Rightarrow 12 = \frac{3}{2}p$

$\Rightarrow \quad p = 8$

Hence the number of persons who like Rasgulla or Jalebi but not Barfi

$= 32 + 8 + 42 = 82$

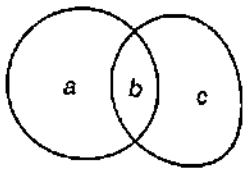


16. Let a be the number of engineers only

c be the number of MBAs only

b be the number of employees who are both engineers and MBAs and

d be the number of employees who are neither engineer nor MBA



$$\therefore \quad a + b + c + d = 80 \quad ...(1)$$

$$(a + b) = 2(b + c) \Rightarrow (a - b) = 2c \quad ...(2)$$

and $c + d = 32 \quad ...(3)$

and $a + d = 56 \quad ...(4)$

and $b = 2c \quad ...(5)$

From eq. (2) and (5), we get

$a = 2b \quad ...(6)$

From eq. (1) and (3), we get

$a + b = 48$

From eq. (6), we get

$b = 16$

$\therefore \quad a = 32$ (from eq. 6)

and $c = 8$ (from eq. 5)

and $d = 24$

Hence 24 employees are neither engineer nor MBAs.

Solutions for question number 17-19:

Total number of employees = 60

Women = 25

Men = 35

Married workers = 28

Graduate workers = 26

a → unmarried men who are not graduate

b → married women who are not graduate

c → unmarried women who are graduate

x → married men who are not graduate

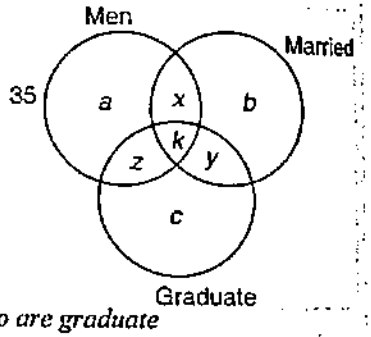
y → married women who are graduate

z → unmarried men who are graduate

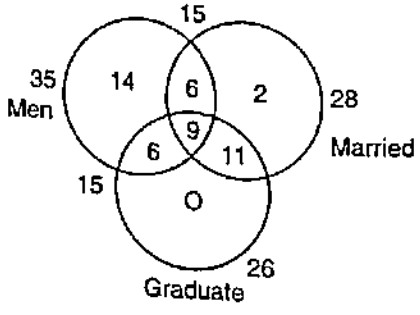
k → married men who are graduate

p → unmarried women who are not graduate

According to the given information the Venn diagram can be completed as given below.



17. No one unmarried woman is graduate. Hence (c)



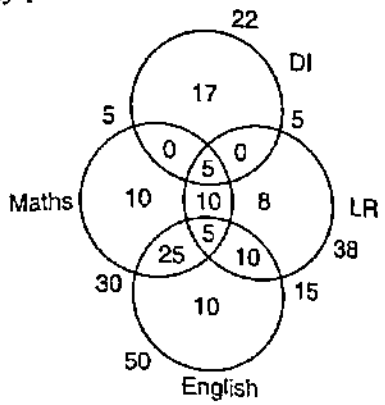
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

18. Number of unmarried women
$= 60 - [14 + 2 + 6 + 6 + 11 + 9] = 12$

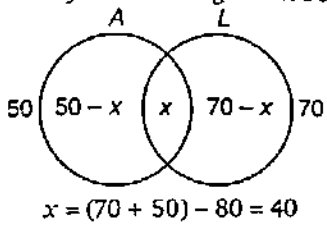
19. There are 9 graduate men who are married

Solutions of question number 20-23:



20. 17
21. 10
22. 55
23. 0

24. For the minium value of x people who like only arrange marriage and only love marriage must be greater

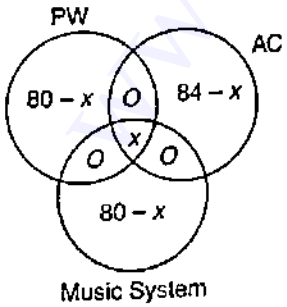


$\therefore \quad x = (70 + 50) - 80 = 40$

For the maximum value of x: $(50 - x)$ and $(70 - x)$ must not be negative, therefore max. possible value of x is 50.

25. 80 cars were decorated with power windows it means at least 40 cars were decorated with AC or music system or both.
84 cars were decorated with ACs, it means atleast 36 cars were decorated with power windows or music systems.
80 cars were decorated with music system means at least 40 cars were decorated with power windows or ACs.
It means if there is no intersection in these three, then at most $40 + 36 + 40 = 116$ cars had been decorated with one or two accessories.
Hence at least 4 cars would have been decorated with all three accessories.

For maximum value of x :



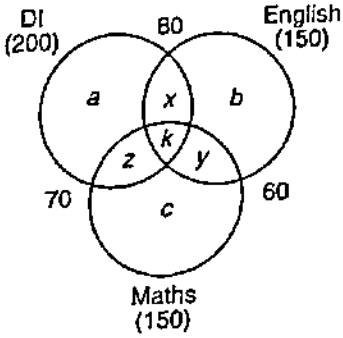
Total number of cars $= (80 - x) + (84 - x) + (80 - x) + x$

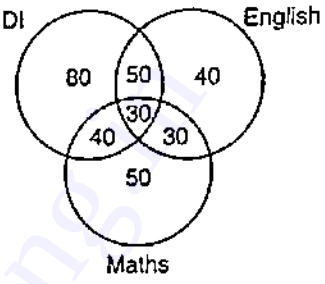
$120 = 244 - 2x$

$\Rightarrow \quad 2x = 124 \Rightarrow x = 62$

minimum $\rightarrow$ 4 cars and maximum $\rightarrow$ 62 cars

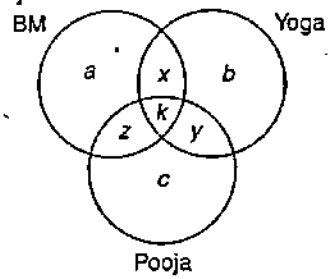
26. $a + x + k + z = 200$ ...(i)
$b + x + k + y = 150$ ...(ii)
$c + y + k + z = 150$ ...(iii)
But since Diwakar teaches only 80 students of DI.
Therefore, $a = 80$
Hence, $x + k + z = 120$
but $(x + k) + (k + z) = 150$
$\therefore \quad k = 30$
Hence, $x = 50, \quad z = 40, \quad y = 30, \quad b = 40$





$\therefore$ No. of students taught by Diwakar $= a = 80$
No. of students taught by Priyanka $= b = 40$
No. of students taught by Varun $= c = 50$
No. of students taught by Sarvesh $= x + y + z + k = 150$
Hence choice (a) is correct.

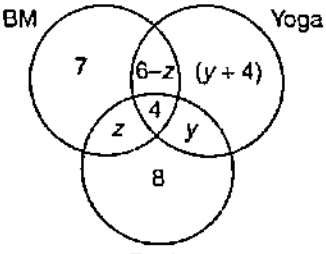
Solutions for question no. 27-29 :



Remember : *Maximum number of volunteers are involved in Yoga.*

Now,
$b = k + y$ ...(1)
$c = 2k$ ...(2)
$a + x + k + z = 17$ ...(3)
$a = c - 1$ ...(4)
$x + k + z = 10$ ...(5)

From eq. (3) *and* (5), *we obtain*
$a = 7$ ...(6)
and from eq. (4) *and* (6), *we obtain*
$c = 8 \Rightarrow k = 4$
$\therefore \quad (z + x) = 6 \Rightarrow x = (6 - z)$
$\therefore \quad b = (y + 4)$ *from eq.* (1).

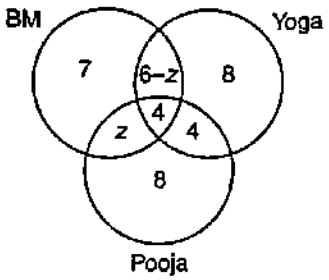


Again $7 + (6 - z) + 4 + z + (y + 4) + y + 8 = 37$
$\Rightarrow \quad 29 + 2y = 37 \Rightarrow y = 4$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

27. Since no. of volunteers involved in Yoga are maximum so we can compare it from the no. of volunteers involved in Pooja and that of Body Massage.



Since

$6 - z > z \forall\ z \in (0, 1, 2, 3, \ldots)$

$\therefore \quad z = 0, 1, 2 \Rightarrow (6 - z) = 4, 5, 6$

Hence the minimum possible value of $(6 - z) = 4$.

28. See the venn diagram shown in solution no. 27, then you will notice that you are required to know the value of y.

Thus from the data provided by choice (a) enable us to calculate all the required details.

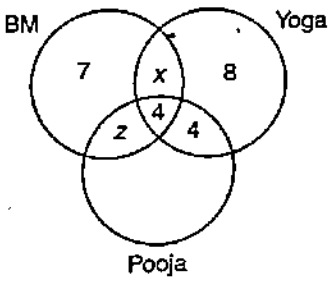
$\because \quad \{(6 - z) + 4 + 4 + 8\}_{\text{Yoga}} = 20$

$\Rightarrow \quad z = 4$

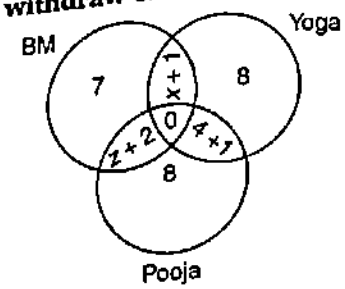
Hence, we can find the exact number of volunteers involved in various projects.

NOTE Coincidentally we obtain $z = 4$ in both the questions 27 and 28 but actually these two answers have no any relation between them.

29. Initially



After the withdraw of volunteers :



The volunteer who is opted out of the BM will be involved in the Yoga and Pooja.

Similarly the volunteer who is opted out of Pooja will be involved in the BM and Yoga.

and the remaining two volunteers who are opted out of Yoga will be involved in BM and Pooja.

NOTE Since initially 4 volunteers are involved in all the three jobs, but when a volunteer leave one job, he or she still continue to perform other two jobs.

Total no. of volunteers in BM $= 7 + (x + 1) + 0 + (z + 2)$

$= 16 \qquad (\because\ x + z = 6)$

Since we know that $x = 4, 5, 6$

Therefore corresponding values of $z = 2, 1, 0 \quad (\because\ x + z = 6)$

$\therefore$ No. of volunteers involved in Yoga = 18, 19 or 20

and No.of volunteers involved in Pooja = 17, 16 or 15

Hence it is clear that choice (b) is correct.

www.LearnEngineering.in

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

16 LOGARITHM

Atleast one question from this chapter is necessarily being asked in CAT. Whenever a question appears related with logarithm, it is of applied nature and the logic behind the question is the application of logarithm.

One more thing I would like to mention that many a student are scared of this chapter. But In my opinion they should not bother about the word 'logarithm' instead they should look the problems of logarithm as problems of exponents and most of the problems can be easily solved by plotting the graph. Notice that this chapter is important for SNAP, MAT, IIFT, JMET and XAT etc.

EXPONENTIAL FUNCTION

For every $x \in R$, $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots + \frac{x^n}{n!} + \ldots$

or $$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

Here e^x is called as exponential function and it is a finite number for every $x \in R$.

Properties : For every $x \in R$, e^x is defined, then

(i) $e^x > 0$ for all $x \in R$ and $e^0 = 1$

(ii) $e^a > e^b$ if $a > b$ and $a, b \in R$

(iii) $e^a \times e^b = e^{(a+b)}$ for all $a, b \in R$

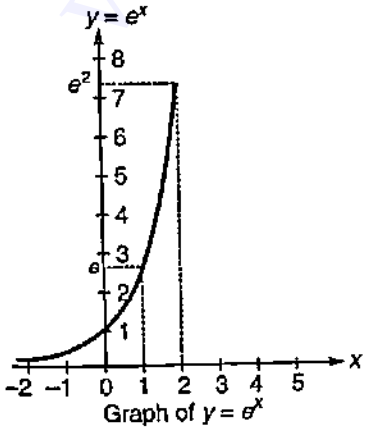
(iv) $e^a \div e^b = e^{(a-b)}$ for all $a, b \in R$

(v) $(e^a)^b = e^{ab}$ for all $a, b \in R$

(vi) For each positive real number x there exists one and only one real number y such that $e^y = x$

(vii) e^x is one-one function

(viii) $e \approx 2.714$



Graph of $y = e^x$

Logarithm : Let a, b be positive real numbers then $a^x = b$ can be written as

$$\log_a b = x; \quad a \neq 1, a > 0, b > 0$$

e.g., $2^5 = 32 \Leftrightarrow \log_2 32 = 5$

$10^3 = 1000 \Leftrightarrow \log_{10} 1000 = 3$

$3^{-4} = \frac{1}{81} \Leftrightarrow \log_3\left(\frac{1}{81}\right) = -4$ etc.

(i) Natural Logarithm : $\log_e N$ is called Natural logarithm or Naperian Logarithm, denoted by $\ln N$ *i.e.*, when the base is 'e' then it is called as Natural logarithm.

e.g., $\log_e 5, \log_e\left(\frac{1}{81}\right), \log_e 72 \ldots$ etc.

(ii) Common Logarithm : $\log_{10} N$ is called common logarithm or Brigg's Logarithm *i.e.*, when base of log is 10, then it is called as common logarithm

e.g., $\log_{10}(100), \log_{10} 25, \log_{10} 248$ etc.

PROPERTIES OF LOGARITHM

1. $\log_a 1 = 0, \quad a > 0, a \neq 1$
2. $\log_a a = 1, \quad a > 0, a \neq 1$
3. $\log_a a^x = x \; \forall \; x \in R, x > 0$
4. $a^{\log_a x} = x \; \forall \; x \in R, x > 0$

NOTE $\log_a a^x$ is the inverse function of a^x.

5. $\log_a (m.n) = \log_a m + \log_a n \; \forall \; m, n > 0, a > 0, a \neq 1$
6. $\log_a (m/n) = \log_a m - \log_a n \; \forall \; m, n > 0, a > 0, a \neq 1$
7. $\log_a (m^n) = n \log_a m \; \forall \; m > 0, a > 0, a \neq 1$
8. $\log_a\left(\frac{1}{m}\right) = -\log_a m \; \forall \; m > 0, a > 0, a \neq 1$
9. $\log_a b = \frac{1}{\log_b a} = \frac{\log_c b}{\log_c a}$

   $\forall \; a, b, c > 0$ and $a \neq 1, b \neq 1, c \neq 1$
10. If $\log_a b = x \; \forall \; a > 0, a \neq 1, b > 0$ and $x \in R$

    (i) $\log_{(1/a)} b = -x$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

(ii) $\log_a\left(\frac{1}{b}\right) = -x$

(iii) $\log_{1/a}\left(\frac{1}{b}\right) = x$

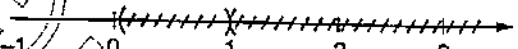
11. $\log_{a^m}(b) = \frac{1}{m}\log_a b$

12. $\log_a x$ is a decreasing function, if $0 < a < 1$

13. $\log_a x$ is an increasing function, if $a > 1$

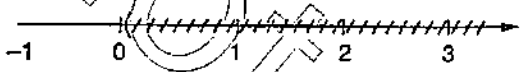
NOTE The base of logarithm can never be equal to 1.
i.e., $\log_1 x$ is undefined $\forall x$

14. On the real number line the base of logarithm can assume any positive real value except 1 (*i.e.*, unity)
Hence at $a = 1$, $\log_a x$ is not defined; $x > 0$



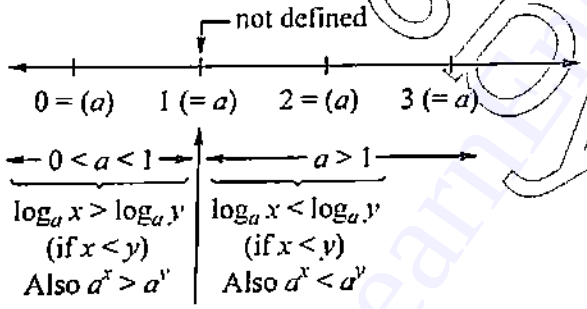
Thus at $a > 0$ except $a = 1$, $\log_a x$ is defined; $x > 0$

15. On the real number line x can assume any positive real value (number) for $\log_a x$; $a > 0, a \neq 1$



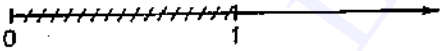
Thus the logarithm of only positive values is defined
i.e., $\log_a x$ is defined for every $x > 0, a > 0, a \neq 1$
Hence for $f(x) = \log_a x$
Domain $\to R^+$ and Range $\to R$

16.



at $a = 1$, $\log_a x$ is not defined.

17.



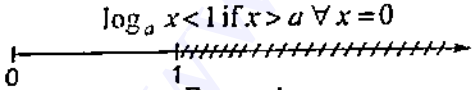
For $0 < a < 1$

$\log_a x > 1$ if $x < a$ $\forall x > 0$

$\log_a x = 1$ if $x = a$ $\forall x > 0$

$\log_a x < 1$ if $x > a$ $\forall x = 0$

18.



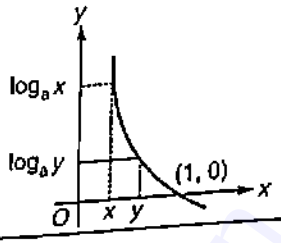
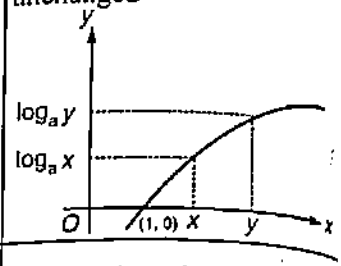
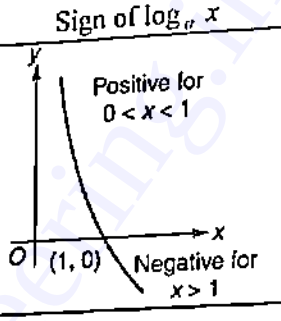
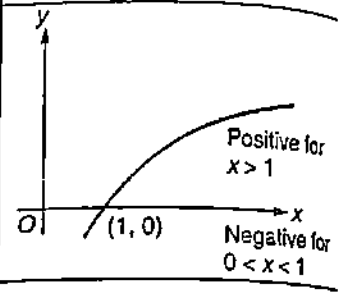
For $a > 1$

$\log_a x < 1$ if $x < a$ $\forall x > 0$

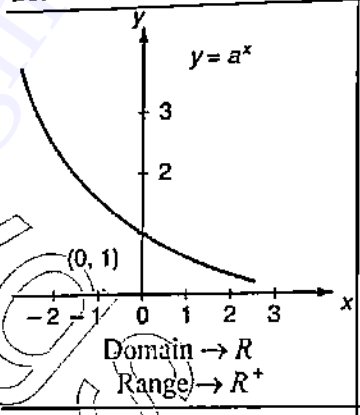
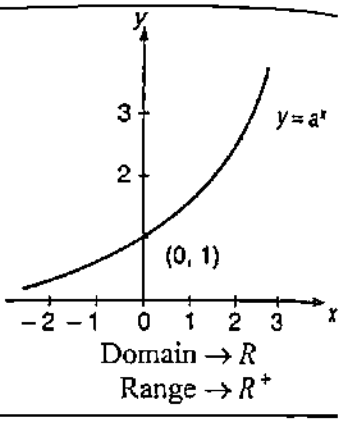
$\log_a x = 1$ if $x = a$ $\forall x > 0$

$\log_a x > 1$ if $x > a$ $\forall x > 0$

19.

$0 < a < 1$	$a > 1$
$\log_a x$ is a decreasing function	$\log_a x$ is an increasing function
$\log_a x > \log_a y$ $(x < y)$ *i.e.*, inequality gets reversed	$\log_a x < \log_a y$ $(x < y)$ *i.e.*, the inequality remains unchanged
	
Sign of $\log_a x$	Sign of $\log_a x$
	

20.

$0 < a < 1$	$a > 1$
 Domain $\to R$ Range $\to R^+$	 Domain $\to R$ Range $\to R^+$
Decreasing function	Increasing function

21. For $|x| < 1$

(i) $\log(1+x) = \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots\right)$

(ii) $\log(1-x) = -\left(x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \ldots\right)$

(iii) $\log\left(\frac{1+x}{1-x}\right) = 2\left[x + \frac{x^3}{3} + \frac{x^5}{5} + \ldots\right]$

(iv) $\log 2 = \left(1 - \frac{1}{2} + \frac{1}{3} - \ldots\right)$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

(v) $\log_{10} 2 = 0.3010$

(vi) $\log_{10} 3 = 0.4771$

22. Logarithmic Inequalities

$0 < a < 1$	$a > 1$
$\log_a b \geq \log_a c \Leftrightarrow b \leq c$	$\log_a b \geq \log_a c \Leftrightarrow b \geq c > 0$
$\log_a b \geq c \Leftrightarrow b \leq a^c$	$\log_a b \geq c \Leftrightarrow b \geq a^c$

CHARACTERISTICS AND MANTISSA

Characteristic : The integral part of logarithm is known as characteristic.

Mantissa : The decimal part is known as mantissa and is always positive.

e.g., In $\log_a x$, the integral part of x is called the characteristic and the decimal part of x is called the mantissa.

For example : In $\log 3274 = 3.5150$, the integral part is 3 *i.e.*, characteristic is 3 and the decimal part is .5150 *i.e.*, mantissa is .5150.

To find the characteristic of common logarithm $\log_{10} x$:

(a) When the number is greater than 1 *i.e.*, $x > 1$
In this case , the characteristic is one less than the number of digits in the left of the decimal point in the given number.

(b) When the number is less than 1 *i.e.*, $0 < x < 1$
In this case the characteristic is one more than the number of zeros between the decimal point and the first significant digit of the number and is negative.

NOTE The characteristic may be positive or negative. If the characteristic is negative, it is always represented by putting a bar on the number *i.e.*,

$-1 \to \bar{1}$
$-2 \to \bar{2}$
$-3 \to \bar{3}$

EXAMPLE

Number	Characteristic	Number	Characteristic
8.3145	0	0.457	$\bar{1}$
74.8120	1	0.0546	$\bar{2}$
568.31	2	0.001324	$\bar{3}$

Remark : (1) Students must note that $\bar{2}.784$ is not the same as -2.784

i.e., $\bar{2}.784 = (-2) + (.784)$

$-2.784 = (-2) - (.784)$

(2) If the logarithm of a number is negative, then we have to make the **mantissa** positive.
For this we subtract 1 from the integral part and add up 1 to the decimal part.

e.g., $-2.5436 = (-2) + (-.5436)$ Step 1.
$\Rightarrow \quad = (-2-1) + (-.5436+1)$ Step 2.
$\Rightarrow \quad = (-3) + (.4564)$ Step 3.
$\Rightarrow \quad = \bar{3}.4564$ Step 4.

POINTS TO REMEMBER ABOUT CHARACTERISTICS

1. The characteristic of common logarithm of positive number less than unity (*i.e.*, 1) is negative.
2. The characteristic of common logarithm of a positive number greater than 1 is positive.
3. If the logarithm to any base a gives the characteristic 'n', then the number of possible integral values is given by $a^{n+1} - a^n$.
   For example $\log_{10} x = n.abcd$, then the number of integral values that x can have given by $10^{n+1} - 10^n$
   e.g., $\log_{10} x = 2.abcd$ gives $10^{2+1} - 10^2 = 900$ integral values for x *i.e.*, $x = 100, 101, 102, 103, \ldots 998, 999$. For all these 900 integral numbers (100 – 999) all the characteristic is 2.
   Similarly $\log_{10} x = 3.abcd$ give $10^{3+1} - 10^3 = 9000$ integral values for x *i.e.*, $x = 1000, 1001, 1002, \ldots, 9999$. For all these 9000 integral numbers (1000 – 9999) the characteristic is 3.
4. If the characteristic of $\log_{10} x$ is negative (*i.e.*, $-n$), then the number of zeros between the decimal and the first significant number after the decimal is $(n-1)$
   e.g., $\log_{10} x = -3.abcd$ has $a = 0$ and $b = 0$.
   i.e., first two places after the decimal point are filled with zeros since characteristic is -3.

Hence $\log_{10} x = -3.abcd = -3.00cd$

IMPORTANT CONVERSIONS

(i) For $a > 1$, $a \neq 1$, $c > 0$

$$a^b > c \Leftrightarrow \log_a c < b$$

(ii) For $0 < a < 1$, $a \neq 1$, $c > 0$

$$a^b > c \Leftrightarrow \log_a c > b$$

EXAMPLE 1 *The value of log 6 is equal to :*
(a) $\log 1 + \log 2 + \log 3$ (b) $\log (1 + 2 + 3)$
(c) $\log (1 \times 2 \times 3)$ (d) *all of the above*

SOLUTION $\log 6 = \log (1 \times 2 \times 3) = \log 1 + \log 2 + \log 3$
Also $\log 6 = \log (1 + 2 + 3)$, Hence (d)

EXAMPLE 2 *Find the value of* $\log_8 128$.

SOLUTION Let $\log_8 128 = x$
$\Rightarrow \quad 8^x = 128$
$\Rightarrow \quad 2^{3x} = 2^7$
$\Rightarrow \quad 3x = 7$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$\Rightarrow \quad x = \frac{7}{3}$

$\therefore \quad \log_8 128 = \frac{7}{3}$

EXAMPLE 3 *Find the value of* $\log_{5\sqrt{5}}(125)$

SOLUTION Let $\log_{5\sqrt{5}}(125) = x$

$\Rightarrow \quad (5\sqrt{5})^x = 125$

$\Rightarrow \quad 5^{(3/2)x} = 5^3$

$\Rightarrow \quad \frac{3}{2}x = 3 \Rightarrow x = 2$

$\therefore \quad \log_{5\sqrt{5}}(125) = 2$

EXAMPLE 4 *Find the value of* $\log_5 \log_5(3125)$

SOLUTION $\log_5 \log_5(3125) = \log_5 \log_5 5^5 = \log_5 5 \log_5 5$

$= \log_5 5.1 \qquad (\because \log_5 5 = 1)$

$= \log_5 5 = 1$

$\therefore \quad \log_5 \log_5(3125) = 1$

EXAMPLE 5 *If* $\log_{10} x - \log_{10} \sqrt{x} = \frac{2}{\log_{10} x}$, *find the value of* x.

SOLUTION $\log_{10} x - \log_{10} \sqrt{x} = \frac{2}{\log_{10} x}$

$\Rightarrow \quad \log_{10}\left(\frac{x}{\sqrt{x}}\right) = \frac{2}{\log_{10} x}$

$\Rightarrow \quad \log_{10}(x)^{1/2} = \frac{2}{\log_{10} x}$

$\Rightarrow \quad \frac{1}{2}\log_{10} x = \frac{2}{\log_{10} x}$

$\Rightarrow \quad \frac{1}{2}(\log_{10} x)^2 = 2$

$\Rightarrow \quad (\log_{10} x)^2 = 4$

$\Rightarrow \quad \log_{10} x = \pm 2$

If $\log_{10} x = 2 \Rightarrow x = 10^2 \Rightarrow x = 100$

If $\log_{10} x = -2 \Rightarrow x = 10^{-2} \Rightarrow x = \frac{1}{100}$

$\therefore \quad x = 100, \frac{1}{100}$

EXAMPLE 6 *Find the value of* $\log_y x \times \log_z y \times \log_x z$.

(a) 0 (b) 1 (c) $\log x$ (d) xyz

SOLUTION $\log_y x \times \log_z y \times \log_x z$

$= \frac{\log_a x}{\log_a y} \times \frac{\log_a y}{\log_a z} \times \frac{\log_a z}{\log_a x} = 1$

EXAMPLE 7 *If* $\frac{\log a}{b-c} = \frac{\log b}{c-a} = \frac{\log c}{a-b}$, *then find the value of* $a^a \cdot b^b \cdot c^c$.

SOLUTION We have,

$\frac{\log a}{b-c} = \frac{\log b}{c-a} = \frac{\log c}{a-b} = k$ (say)

Then, $\log a = k(b-c)$

$\log b = k(c-a)$

$\log c = k(a-b)$

Let $A = a^a \cdot b^b \cdot c^c$

Then $\log A = a \log a + b \log b + c \log c$

$= ak(b-c) + bk(c-a) + ck(a-b)$

$= abk - ack + bck - abk + cak - bck = 0$

Thus $\log A = 0 = \log 1$

$\therefore \quad A = 1$

i.e., $a^a \cdot b^b \cdot c^c = 1$

EXAMPLE 8 *Find the value of* $\log_8 25$, *given that* $\log_{10} 2 = 0.3010$

SOLUTION $\log_8 25 = \frac{\log_{10} 25}{\log_{10} 8} = \frac{2}{3}\frac{\log 5}{\log 2} = \frac{2\log_{10}(10/2)}{3\log_{10} 2}$

$= \frac{2(\log_{10} 10 - \log_{10} 2)}{3 \times 0.3010}$

$= \frac{2(1 - 0.3010)}{0.9030} = 1.5482$

EXAMPLE 9 *Find the value of* $\frac{\log\sqrt{27} + \log 8 + \log\sqrt{1000}}{\log 120}$.

SOLUTION $\frac{\log(27)^{1/2} + \log 2^3 + \log(10^3)^{1/2}}{\log(3 \times 2^2 \times 10)}$

$= \frac{\frac{1}{2}\log 3^3 + 3\log 2 + \frac{1}{2}\log(10)^3}{\log 3 + \log 2^2 + \log 10}$

$= \frac{\frac{3}{2}\log 3 + \frac{3}{2} \cdot 2\log 2 + \frac{3}{2}\log 10}{\log 3 + 2\log 2 + \log 10}$

$= \frac{\frac{3}{2}(\log 3 + 2\log 2 + \log 10)}{\log 3 + 2\log 2 + \log 10} = \frac{3}{2}$

EXAMPLE 10 *Solve for* x, *if* $\log_x(8x-3) - \log_x 4 = 2$

SOLUTION $\log_x(8x-3) - \log_x 4 = 2$

$\Rightarrow \quad \log_x\left(\frac{8x-3}{4}\right) = 2$

$\Rightarrow \quad x^2 = \frac{8x-3}{4}$

$\Rightarrow \quad 4x^2 - 8x + 3 = 0$

$\Rightarrow \quad x = \frac{3}{2}$ or $\frac{1}{2}$

EXAMPLE 11 *Find the value of*

$x^{\log y - \log z} \times y^{\log z - \log x} \times z^{\log x - \log y}$

SOLUTION Let $P = x^{\log y - \log z} \times y^{\log z - \log x} \times z^{\log x - \log y}$

Then $\log P = \log x^{\log y - \log z} + \log y^{\log z - \log x} + \log z^{\log x - \log y}$

$= (\log y - \log z)\log x + (\log z - \log x)\log y$

$+ (\log x - \log y)\log z$

$= \log x \log y - \log z \log x + \log y \log z - \log x \log y$

$+ \log z \log x - \log y \log z = 0$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

$\log P = 0 = \log 1$

i.e., $P = 1$

∴

EXAMPLE 12 *If* $\frac{xy\log(xy)}{x+y} = \frac{yz\log(yz)}{y+z} = \frac{zx\log(zx)}{z+x}$, *then show that* $x^x = y^y = z^z$.

SOLUTION $\frac{xy\log(xy)}{x+y} = \frac{yz\log(yz)}{y+z} = \frac{zx\log(zx)}{z+x} = k$ (say)

Then $xy\log xy = k(x+y)$

or $\log xy = k\left(\frac{1}{x}+\frac{1}{y}\right)$

or $\log x + \log y = k\left(\frac{1}{x}+\frac{1}{y}\right)$ ...(1)

Similarly, $\log y + \log z = k\left(\frac{1}{y}+\frac{1}{z}\right)$ ...(2)

and $\log z + \log x = k\left(\frac{1}{z}+\frac{1}{x}\right)$ ...(3)

Adding (1), (2) and (3), we get

$$2(\log x + \log y + \log z) = 2k\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)$$

or $\log x + \log y + \log z = k\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)$ ...(4)

Subtracting (1), (2) and (3) successively from (4), we get

$$\log z = k\cdot\frac{1}{z},\quad \log x = k\cdot\frac{1}{x},\quad \log y = \frac{k}{y}$$

or $z\log z = k,\quad x\log x = k$ and $y\log y = k$

or $x\log x = y\log y = z\log z$

or $\log(x^x) = \log(y^y) = \log(z^z)$

Hence $x^x = y^y = z^z$

EXAMPLE 13 *If* $\log_a bc = x$, $\log_b ca = y$, $\log_c ab = z$, *prove that*

$$\frac{1}{x+1}+\frac{1}{y+1}+\frac{1}{z+1} = 1$$

SOLUTION $\frac{1}{x+1}+\frac{1}{y+1}+\frac{1}{z+1}$

$$= \frac{1}{(\log_a bc)+1}+\frac{1}{(\log_b ca)+1}+\frac{1}{(\log_c ab)+1}$$

$$= \frac{1}{\log_a bc + \log_a a}+\frac{1}{\log_b ca + \log_b b}+\frac{1}{\log_c ab + \log_c c}$$

$$= \frac{1}{\log_a abc}+\frac{1}{\log_b abc}+\frac{1}{\log_c abc}$$

$$= \log_{abc} a + \log_{abc} b + \log_{abc} c \qquad \left(\because \log_a b = \frac{1}{\log_b a}\right)$$

$$= \log_{abc}(abc) = 1$$

EXAMPLE 14 *Given* $\log 2 = 0.3010$ *and* $\log 3 = 0.4771$, *find the value of* $\log 45$.

SOLUTION $\log 45 = \log 3^2 \cdot 5 = \log 3^2 + \log 5 = 2\log 3 + \log 5$

$$= 2\times 0.4771 + \log\frac{10}{2}$$

$$= 0.9542 + \log 10 - \log 2$$

$$= 0.9542 + 1 - 0.3010$$

$$= 1.6532$$

EXAMPLE 15 *Given* $\log 2 = 0.3010$ *and* $\log 3 = 0.4771$. *Find the value of* $\log 0.0075$

SOLUTION $\log 0.0075 = \log\frac{75}{10000} = \log 75 - \log 10000$

$$= \log 3\times 5^2 - \log 10^4$$

$$= \log 3 + 2\log 5 - 4\log 10$$

$$= \log 3 + 2[\log 10 - \log 2] - 4\log 10$$

$$= 0.4771 + 2[1 - 0.3010] - 4$$

$$= -2.1249$$

EXAMPLE 16 *What is the number of digits in* 3^{3^3}? *Given that* $\log 3 = 0.47712$?

SOLUTION Let $x = 3^{3^3} = (3)^{3^3}$

Then $\log x = 3^3 \log 3$

$$= 27\times 0.47712 = 12.88224$$

Since the characteristic in the resultant value of $\log x$ is 12

∴ The number of digits in x is $(12+1) = 13$

Hence the required number of digits in 3^{3^3} is 13.

EXAMPLE 17 *If* $\log 3 = 0.4771$, *find the number of digits in* 3^{43}.

SOLUTION Let $x = 3^{43}$

∴ $\log x = \log 3^{43} = 43\log 3 = 43\times 0.4771$

$$= 20.5153$$

Since the characteristic of $\log 3^{43}$ is 20, then number of digits in 3^{43} is $(20+1) = 21$.

EXAMPLE 18 *How many zeros are there between the decimal point and the first significant figure in* $(0.5)^{100}$? *Given that* $\log\frac{1}{2} = \bar{1}.6990$

SOLUTION Let $x = (0.5)^{100}$

∴ $\log x = 100\log(0.5) = 100\times\bar{1}.6990$

$$= 100(-1 + 0.6990)$$

$$= -100 + 69.90$$

$$= -31 + 0.90 = \bar{31}.90$$

Since the characteristics of $\log(0.5)^{100}$ is $\bar{31}$, the number of zeros between the decimal point and the first significant figure is $31 - 1 = 30$.

EXAMPLE 19 *If* $\frac{x(y+z-x)}{\log x} = \frac{y(z+x-y)}{\log y} = \frac{z(x+y-z)}{\log z}$, *then :*

(a) $xy = yz = zx$ (b) $x^y y^x = z^y y^z = x^z z^x$

(c) $xyz = 1$ (d) *none of these*

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yourshaikhsab.wordpress.com
For More Visit : www.LearnEngineering.in

SOLUTION Let $\frac{x(y+z-x)}{\log x} = \frac{y(z+x-y)}{\log y}$

$$= \frac{x(x+y-z)}{\log z} = \frac{1}{k} \quad \text{(say)}$$

$\Rightarrow$ $\log x = kx(y+z-x)$

$\log y = ky(z+x-y)$

$\log z = kz(x+y-z)$

Hence $y \log x + x \log y = 2kxyz$

$y \log z + z \log y = 2kxyz$

$z \log x + x \log z = 2kxyz$

Thus $y \log x + x \log y = y \log z + z \log y$

$= z \log x + x \log z$

$\Rightarrow$ $\log(x^y y^x) = \log(z^y y^z) = \log(x^z z^x)$

$\Rightarrow$ $x^y y^x = z^y y^z = x^z z^x$

INTRODUCTORY EXERCISE-16.1

1. If $x^y = k$, then :
(a) $\log_y k = x$ (b) $\log_x y = k$
(c) $\log_k x = y$ (d) $\log_x k = y$

2. If $\log_m n = p$, then :
(a) $m^n = p$ (b) $p^n = m$
(c) $m^p = n$ (d) $n^p = m$

3. $\log_a(mn)$ is equal to :
(a) $n \log_a(m)$ (b) $\log_a m \times \log_a n$
(c) $\log_a m + \log_a n$ (d) $\log_a m - \log_a n$

4. $\log_a\left(\frac{m}{n}\right)$ is equal to :
(a) $\log_a(m-n)$ (b) $\log_a m - \log_a n$
(c) $\frac{(\log_a m)}{n}$ (d) $\log_a m \div \log_a n$

5. The value of $\log_{81} 27$ is :
(a) 3^7 (b) $\frac{4}{3}$
(c) $\frac{3}{4}$ (d) $\frac{1}{3}$

6. The value of $\log_{36} \frac{1}{216}$ is :
(a) $\frac{2}{3}$ (b) $-\frac{3}{2}$
(c) $\frac{1}{6}$ (d) none of these

7. The value of $\log_{\sqrt{6}} 216$ is :
(a) $\frac{1}{6}$ (b) $\sqrt{6}$
(c) 3 (d) 6

8. The value of $\log_{3\sqrt{2}} 5832$ is :
(a) 12 (b) 9
(c) 6 (d) none of these

9. The value of $\log_{(-1/5)} 625$ is :
(a) $\frac{1}{4}$ (b) -5
(c) -4 (d) -3

10. The value of $\log_{10}(0.0001)$ is :
(a) $\frac{1}{1000}$ (b) -3
(c) -4 (d) none of these

11. The value of $\log_{(0.01)}(10000)$ is :
(a) 1/2 (b) -2
(c) 1/4 (d) -4

12. The value of $(\log_4 32 + \log_9 243)$:
(a) 5 (b) 4
(c) 3 (d) none of these

13. The value of $(\log_2 8 + \log_3 9 + \log_5 25)$ is :
(a) 5 (b) 6
(c) 7 (d) none of these

14. The value of $\log_2(\log_3 81)$ is :
(a) 0 (b) 1
(c) 2 (d) 6

15. The value of $\log_2 \log_2 \log_3 \log_5(125)^3$ is :
(a) 0 (b) 1
(c) 2 (d) 3

16. The value of $\log_2 \log_2 \log_2 \log_2(65536)$ is :
(a) 4 (b) 2
(c) 1 (d) 0

17. The value of
$\log_{10} 1 + \log_{10} 10 + \log_{10} 100 + \ldots \log_{10} 10000000000$
(a) 10 (b) 11
(c) 11111111111 (d) 55

18. The value of $\log_5 5 + \log_5 5^2 + \log_5 5^3 + \ldots + \log_5 5^n$:
(a) $n!$ (b) $n^2 - 1$
(c) $\frac{(n+1)n}{2}$ (d) none of these

19. The value of $27^{\log_3 81}$ is :
(a) 27381 (b) 531441
(c) 65631 (d) none of these

20. The value of ${}_{216}\log_6 49$ is :
(a) 117694 (b) 117649
(c) 216649 (d) none of these

21. The value of $25^{(-1/4\log_5 25)}$ is :
(a) $\frac{1}{5}$ (b) $-\frac{1}{25}$
(c) -25 (d) none of these

22. $(\log_{10} 500000 - \log_{10} 5)$ is equal to :
(a) 5 (b) 50
(c) 49995 (d) 100000

23. If $\log_5[\log_3(\log_2 x)] = 1$, then x is :
(a) 2^{234} (b) 243
(c) 2^{243} (d) none of these

24. If $\log_2 \log_{10} 4x = 1$, then the value of x is :
(a) 1/25 (b) 25
(c) 5 (d) none of these

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

25. If $\log_{27} x = \frac{4}{3}$, then the value of x is :
(a) 9 (b) 27
(c) 81 (d) none of these

26. If $\log_x\left(\frac{4}{9}\right) = -\frac{1}{2}$, then the value of x is :
(a) $\frac{16}{81}$ (b) $-\frac{2}{3}$
(c) $\frac{81}{16}$ (d) none of these

27. If $\log_{100} x = -4$, then the value of x is :
(a) $\frac{1}{(10)^{-4}}$ (b) $(10)^{-4}$
(c) $\frac{1}{(10)^8}$ (d) none of these

28. If $\log_{100} x = -\frac{1}{2}$, then the value of x is :
(a) 0.001 (b) $\frac{1}{10}$
(c) 0.00001 (d) 10^{-2}

29. If $\log_x(0.2) = -\frac{1}{4}$, then the value of x is :
(a) 4^{-5} (b) 0.016
(c) 5^4 (d) none of these

30. The value of x satisfying $\log_{243} x = 0.8$ is :
(a) 81 (b) 1.8
(c) 2.43 (d) none of these

31. If $\log_e(x-1) + \log_e x + \log_e(x+1) = 0$, then
(a) $x^2 + e^{-1}$ (b) $x^3 - x - 1 = 0$
(c) $x^2 + e - 1 = 0$ (d) $x^3 - x - e = 0$

32. The value of $\log_3 5 \times \log_{25} 9$ is :
(a) 0 (b) 1
(c) 2 (d) 4

33. The value of $\left(\log\frac{14}{3} + \log\frac{11}{5} - \log\frac{22}{15}\right)$ is :
(a) log 77 (b) log 11
(c) log 7 (d) none of these

34. The value of $\left[3\log\left(\frac{81}{80}\right) + 5\log\left(\frac{25}{24}\right) + 7\log\left(\frac{16}{15}\right)\right]$ is :
(a) log 3 (b) log 5
(c) log 7 (d) log 2

35. $\left(\log\frac{a^3}{bc} + \log\frac{b^3}{ac} + \log\frac{c^3}{ab}\right)$ is equal to :
(a) 1 (b) log abc
(c) abc (d) none of these

36. $\frac{1}{(\log_a bc)+1} + \frac{1}{(\log_b ac)+1} + \frac{1}{(\log_c ab)+1}$ is equal to :
(a) 1 (b) 2
(c) 0 (d) abc

37. $\frac{1}{\log_{ab} abc} + \frac{1}{\log_{bc} abc} + \frac{1}{\log_{ca} abc}$ is equal to :
(a) 0 (b) 1
(c) 2 (d) 3

38. The value of $\left(\frac{1}{\log_5 210} + \frac{1}{\log_6 210} + \frac{1}{\log_7 210}\right)$ is :
(a) 0 (b) 1
(c) 18 (d) 21

39. The value of $\left[\frac{1}{\log_{a/b} x} + \frac{1}{\log_{b/c} x} + \frac{1}{\log_{c/a} x}\right]$ is :
(a) 0 (b) 1
(c) abc (d) x^3

40. If $\log_{10} 2 = 0.3010$, then $\log_2 10$ is :
(a) 3.010 (b) 1.5050
(c) 3.3222 (d) none of these

41. If $2^x \cdot 3^{2x} = 100$, then the value of x is (log 2 = 0.3010, log 3 = 0.4771)
(a) 2.3 (b) 1.59
(c) 1.8 (d) 1.41

42. If $\log_{10} 2 = 0.3010$ and $5^x = 400$, then the value of x is :
(a) 3.72 (b) 2.52
(c) 0 (d) none of these

43. If log 2 = 0.3010, the number of digits in 5^{20} is :
(a) 16 (b) 14
(c) 8 (d) none of these

44. If $\log(m+n) = \log m + \log n$, then :
(a) $mn = 1$ (b) $m = -n$
(c) $\frac{m}{(m-1)} = n$ (d) none of these

45. If $\log_{10} a + \log_{10} b = c$, then the value of a is :
(a) bc (b) $\frac{c}{b}$
(c) $\frac{(10)^c}{b}$ (d) $\frac{10b}{c}$

46. The mantissa of log 3274 is 0.5150. The value of log (0.3274) is :
(a) $\bar{1}.5150$ (b) 1.5150
(c) $\bar{2}.5150$ (d) none of these

47. If $\log_{10} x = 1.9675$, then the value of $\log_{10}(100x)$ is :
(a) 196.75 (b) 3.9675
(c) 2.9675 (d) none of these

48. If $5^x = (0.5)^y = 1000$, then the value of $\left(\frac{1}{x} - \frac{1}{y}\right)$ is :
(a) $\frac{1}{4}$ (b) $\frac{1}{3}$
(c) $\frac{1}{2}$ (d) 1

49. If $\frac{1}{2}\log x + \frac{1}{2}\log y + \log 2 = \log(x+y)$, then :
(a) $x + y = 0$ (b) $x = y$
(c) $x = 2, y = 0$ (d) $x = \log y$

50. If $p = \log_3 5$ and $q = \log_{17} 25$, which one of the following is correct ?
(a) $p < q$ (b) $p = q$
(c) $p > q$ (d) can't say

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

51. If $x^2 + 4y^2 = 12xy$, then $\log(x + 2y)$ is equal to :
(a) $\frac{1}{2}(\log x + \log y + 2\log 2)$
(b) $\frac{1}{2}(\log x + \log y + 4\log 2)$
(c) $\frac{1}{2}(\log x + \log y - \log 2)$
(d) none of the above

52. The value of $(2^{\log_3 7} - 7^{\log_3 2})$ is :
(a) 0 (b) 1
(c) $\log_7 5$ (d) $\log_7 6$

53. $\frac{1}{\log_2 a} + \frac{1}{\log_4 a} + \frac{1}{\log_8 a} + \ldots$ to n terms $= \frac{n(n+1)}{k}$, then k is equal to :
(a) $\log_2 a^2$ (b) $\log_2\left(\frac{a}{2}\right)$
(c) $\log_a 2$ (d) none of these

54. The value of $(\log \tan 1° + \log \tan 2° + \ldots + \log \tan 89°)$ is :
(a) -1 (b) 0
(c) $\frac{1}{\sqrt{2}}$ (d) 1

55. The value of x for which $\log_9 x - \log_9\left(\frac{x}{10} + \frac{1}{9}\right) = 1$ is :
(a) 2 (b) 4
(c) 9 (d) 10

56. If $(x^4 - 2x^2y^2 + y^4)^{a-1} = (x - y)^{2a}(x + y)^{-2}$, then the value of a is :
(a) $(x^2 - y^2)$ (b) $\frac{x - y}{x + y}$
(c) $\frac{\log(x + y)}{\log(x - y)}$ (d) $\frac{\log(x - y)}{\log(x + y)}$

57. $\log_2 7$ is :
(a) an integer
(b) a prime number
(c) a rational number
(d) an irrational number

58. $\log_y x = ?$
(a) $x \log_e y$ (b) $\frac{x}{\log_e y}$
(c) $\frac{\log_e x}{\log_e y}$ (d) $\frac{\log_e y}{\log_e x}$

59. The value of
$\log_{10} 2 + 16\log_{10}\left(\frac{16}{15}\right) + 12\log_{10}\left(\frac{25}{24}\right) + 7\log_{10}\left(\frac{81}{80}\right)$ is :
(a) 0 (b) 1
(c) 2 (d) 3

60. If $\log_{10} x = a$, $\log_{10} y = b$ and $\log_{10} z = c$, then antilog $(pa + qb - rc) = ?$
(a) $\frac{pxqy}{rz}$ (b) $px + qy - rz$
(c) $\frac{x^p y^q}{z^r}$ (d) $x^p y^q z^r$

61. $\log_{10} a^p \cdot b^q \cdot c^r = ?$
(a) $pqr(\log_{10} abc)$
(b) $p\log_{10} a + q\log_{10} b + r\log_{10} c$
(c) $pqr(\log_{10} a + \log_{10} b + \log_{10} c)$
(d) $(\log_{10} a)^p + (\log_{10} b)^q + (\log_{10} c)^r$

62. If a, b, c are in GP then $\log_{10} a, \log_{10} b, \log_{10} c$ are in :
(a) GP (b) HP
(c) AP (d) none of these

63. If $\log_{10} x, \log_{10} y, \log_{10} z$ are in AP then x, y, z are in :
(a) AP (b) GP
(c) HP (d) none of these

64. If a, b, c are in GP then $\frac{1}{\log_a x}, \frac{1}{\log_b x}, \frac{1}{\log_c x}$ are in :
(a) AP (b) GP
(c) HP (d) none of these

65. If $\log(x - 1) + \log(x + 1) = 3\log 2$, then x is equal to :
(a) ± 1 (b) 4
(c) ± 3 (d) none of these

66. If a, b, c are three consecutive integers, then $\log(ac + 1)$ has the value :
(a) $\log b$ (b) $(\log b)^2$
(c) $2\log b$ (d) $\log 2b$

www.LearnEngineering.in

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

EXERCISE

LEVEL (1)

1. Find the value of $\log_{10}\sqrt[7]{13}$ if $\log_{10}13 = 1.1139$:
(a) 0.15913 (b) 0.5119
(c) 1.15913 (d) none of these

2. Find the logarithm of 144 to the base $2\sqrt{3}$:
(a) 8 (b) 4
(c) $2\sqrt{3}$ (d) none of these

3. If $\log_a N = (\log_b N) \times P$, then find P in terms of a and b :
(a) b^a (b) a^b
(c) $\log_a b$ (d) none of these

4. The value of $\log (ab)^2 - \log (ac) + \log (bc^4) - 3 \log (bc)$ is :
(a) 0 (b) $\log b$
(c) $\log c$ (d) $\log a$

5. If $\log_2 x + \log_4 x + \log_{64} x = 5$, find x :
(a) 8 (b) 16
(c) 7 (d) 2

6. If $\log_q (xy) = 3$ and $\log_q (x^2y^3) = 4$, find the value of $\log_q x$:
(a) 4 (b) 5
(c) 3 (d) 2

7. If $\frac{\log x}{l+m-2n} = \frac{\log y}{m+n-2l} = \frac{\log z}{n+l-2m}$, then xyz is equal to :
(a) 0 (b) 1
(c) lmn (d) 2

8. If a, b, c be the pth, q^{th}, r^{th} terms of a GP then the value of $(q-r)\log a + (r-p)\log b + (p-q)\log c$ is :
(a) 0 (b) 1
(c) -1 (d) pqr

9. If $a^{3-x} b^{5x} = a^{x+5} b^{3x}$, then the value of $x \log\left(\frac{b}{a}\right)$ is :
(a) $\log a$ (b) $\log b$
(c) $\log x$ (d) 1

10. If $u = v^2 = w^3 = z^4$, then $\log_u (uvwz)$ is equal to :
(a) $1 + \frac{1}{2} - \frac{1}{3} - \frac{1}{4}$ (b) 24
(c) $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}$ (d) $\frac{1}{24}$

11. Find the value of $\frac{\log\sqrt{27} + \log\sqrt{8} - \log\sqrt{125}}{\log 6 - \log 5}$:
(a) $\frac{2}{3}$ (b) $\frac{1}{3}$
(c) $\frac{3}{2}$ (d) none of these

12. The first term and the last term of a GP are a and k respectively. If the number of terms be n, then n is equal to ($r \rightarrow$ common ratio) :
(a) $1 - \frac{\log k - \log a}{\log r}$ (b) $1 + \frac{\log a + \log k}{\log r}$
(c) $1 + \frac{\log k - \log a}{\log r}$ (d) $\log r = \log k - \log a$

13. If a, b, c are in GP, then $\log_a d$, $\log_b d$, $\log_c d$ are in :
(a) AP (b) HP
(c) GP (d) none of these

14. Find the value of x for $\log_x 2 \cdot \log_{x/16} 2 = \log_{x/64} 2$:
(a) 4 (b) 4, 16
(c) 4,8 (d) 4, 8, 32

15. Find the value of x and y respectively for
$\log_{10}(x^2y^3) = 7$ and $\log_{10}(x/y) = 1$:
(a) $x = 10, y = 100$ (b) $x = 100, y = 10$
(c) $x = 10, y = 20$ (d) none of these

16. If $y = a^{\left(\frac{1}{1-\log_a x}\right)}$ and $z = a^{\left(\frac{1}{1-\log_a y}\right)}$, then x is equal to :
(a) $1 + \log yz$ (b) $\frac{1}{1-\log_a z}$
(c) $a^{\left(\frac{1}{1-\log_a z}\right)}$ (d) $\frac{y}{z}$

17. If $x = \log_c b + \log_b c$, $y = \log_a c + \log_c a$ and $z = \log_b a + \log_a b$, then $x^2 + y^2 + z^2$ is equal to :
(a) 1 (b) 2
(c) xyz (d) $xyz + 4$

18. Find the value of $\frac{1}{\log_3 e} + \frac{1}{\log_3 e^2} + \frac{1}{\log_3 e^4} + \ldots$ is :
(a) $\log_e 9$ (b) 0
(c) 1 (d) $\log_e 270$

19. If $\log_{10} x^2 - \log_{10}\sqrt{y} = 1$, find the value of y, when $x = 2$
(a) 2/9 (b) 4/25
(c) 25/4 (d) 4/9

20. Find the value of $(b^2)^{5\log_b x}$:
(a) 10^x (b) x^4
(c) $10x$ (d) x^{10}

21. Find the value of $(7^3)^{-2\log_7 8}$:
(a) 8^{-7} (b) 6^{-8}
(c) 8^{-6} (d) none of these

22. log 12900 is equal to :
(a) 2 log 129 (b) log 129 + log 2
(c) 4 + log 1.29 (d) none of these

23. log 0.786 is equal to :
(a) log 786 – log 3 (b) log 78.6 – log 100
(c) log 786 – 3 (d) both (b) and (c)

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

24. If $\log_5 x = y$, then 5^{5y} is equal to :
(a) $\frac{x}{5}$ (b) $5x$
(c) $\log_x 5$ (d) x^5

25. If $A = \log_{13} 189$ and $B = \log_{23} 521$, then which one of the following is correct?
(a) $A < B$ (b) $A = B$
(c) $A > B$ (d) none of these

26. If $A = \frac{(\log_3 2187)}{5}$ and $B = \log_{243} 2187$, then which one of the following is correct?
(a) $A < B$ (b) $A = B$
(c) $A > B$ (d) can't be determined

27. If $(150)^x = 7$, then x is equal to :
(a) $\frac{\log 7}{(\log 3) + (\log 5) + 1}$ (b) $\frac{\log 7}{(\log 3) + (\log 6)}$
(c) $\frac{\log 7}{(\log 3) + (\log 5) + 10}$ (d) none of these

28. The value of x satisfying the following relation
$$\log_{1/2} x = \log_2 (3x - 2)$$
(a) $\frac{1}{3}$ (b) $-\frac{1}{3}$
(c) 3 (d) none of these

29. If $\log_2 (x + y) = 3$ and $\log_2 x + \log_2 y = 2 + \log_2 3$ then the values of x and y are
(a) $x = 1, y = 8$ (b) $x = 4, y = 4$
(c) $x = 4, y = 8$ (d) $x = 2, y = 6$

30. The set of all the solutions of the equation
$\log_5 x \log_6 x \log_7 x = \log_5 x \cdot \log_6 x + \log_6 x \cdot \log_7 x + \log_7 x \cdot \log_5 x$ is :
(a) {0, 1} (b) {1, 210}
(c) {1, 5, 6, 7, 210} (d) none of these

31. The value of $\frac{\log_3 54}{\log_{486} 3} - \frac{\log_3 1458}{\log_{18} 3}$ is :
(a) 0 (b) 1
(c) 2 (d) 3

32. The number of solutions of $\log_9 (2x - 5) = \log_3 (x - 4)$ is :
(a) 0 (b) 1
(c) 2 (d) 3

33. If $\log_3 2$, $\log_3 (2^x - 5)$ and $\log_3 (2^x - 7/2)$ are in AP then x is equal to :
(a) 2 (b) 3
(c) 4 (d) 5

34. If 1, $\log_y x$, $\log_z y$, $-15 \log_x z$ are in AP, then
(a) $z^3 = x$ (b) $x = y^{-1}$
(c) $z^{-3} = y$ (d) $x = y^{-1} = z^3$

35. If $\log_x a$, $a^{x/2}$ and $\log_b x$ are in GP, then x is equal to :
(a) $\log_a (\log_b a)$ (b) $\log_a (\log_e a) - \log_a (\log_e b)$
(c) $-\log_a (\log_a b)$ (d) both (a) and (b)

36. If 1, $\log_9 (3^{1-x} + 2)$ and $\log_3 (4 \cdot 3^x - 1)$ are in AP, then x is equal to :
(a) $\log_4 3$ (b) $\log_3 4$
(c) $1 + \log_3 4$ (d) $\log_3 (3/4)$

37. The value of $\frac{1}{\log_{100} n} + \frac{1}{\log_{99} n} + \frac{1}{\log_{98} n} + \ldots + \frac{1}{\log_2 n}$ is :
(a) 1 (b) $\frac{1}{\log_{100!} n}$
(c) $\frac{1}{\log_n 99!}$ (d) $\frac{1}{\log_n 100!}$

38. If $\log_2 (1 + 3^{x-1}) + 2 = \log_2 (3^{2x-2} + 7)$, then x is :
(a) 0 (b) 1
(c) 2 (d) both (b) and (c)

39. $x^{\log_5 x} > 5$ implies :
(a) $x \in (0, \infty)$ (b) $x \in (1, \infty)$
(c) $x \in (1, 2)$ (d) $x \in (0, 1/5) \cup (5, \infty)$

40. Find x, if $\log x^3 - \log 3x = 2 \log 2 + \log 3$:
(a) 3 (b) 6
(c) 9 (d) none of these

41. If x satisfies $\log_5 (2x + 3) < \log_5 7$, then x lies in :
(a) $\left(-\frac{3}{2}, 2\right)$ (b) $\left(\frac{3}{2}, 2\right)$
(c) $(0, \infty)$ (d) $(0, 2)$

42. Find x, if $\log_{2x} \sqrt{x} + \log_{2\sqrt{x}} x = 0$:
(a) $1, 2^{-5/6}$ (b) $1, 2^{-6/5}$
(c) $4, -2$ (d) none of these

43. For a positive real x $(x > 1)$, which one of the following is correct ?
(a) $\frac{x}{1+x} < x < \log (x + 1)$ (b) $x < \frac{x}{1+x} < \log (x + 1)$
(c) $\log (x + 1) < \frac{x}{1+x} < x$ (d) $\frac{x}{1+x} < \log (x + 1) < x$

44. For $x \in N$, $x > 1$, if $P = \log_x (x + 1)$ and $Q = \log_{(x+1)} (x + 2)$, then which one of the following is correct ?
(a) $P < Q$ (b) $P = Q$
(c) $P > Q$ (d) can't be determined

45. If $a = 1 + \log_x yz$, $b = 1 + \log_y zx$ and $c = 1 + \log_z xy$, then $ab + bc + ca$ is :
(a) 1 (b) 0
(c) abc (d) none of these

46. Find the values of x and y for the given equation
$xy^2 = 4$ and $\log_3 (\log_2 x) + \log_{1/3} (\log_{1/2} y) = 1$:
(a) $x = \frac{1}{8}, y = 64$ (b) $x = \frac{1}{4}, y = 48$
(c) $x = 64, y = \frac{1}{4}$ (d) none of these

47. Find x if $\log_{1/\sqrt{2}} (1/\sqrt{8}) = \log_2 (4^x + 1) \cdot \log_2 (4^{x+1} + 4)$:
(a) 0 (b) 1
(c) 2 (d) none of these

48. The value of x satisfying
$$\log_3 4 - 2 \log_3 \sqrt{3x + 1} = 1 - \log_3 (5x - 2)$$
(a) 1 (b) 2
(c) 3 (d) none of these

49. The solution set of $|3 - 4x| > 2$ is :
(a) $\left(\frac{7}{4}, \infty\right)$ (b) $\left(-\frac{1}{4}, \frac{7}{4}\right)$
(c) $\left(-\infty, -\frac{1}{4}\right) \cup \left(\frac{7}{4}, \infty\right)$ (d) $\left(-\infty, \frac{1}{4}\right) \cup \left(\frac{5}{4}, \infty\right)$

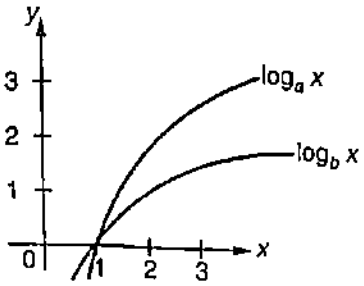
50. Solve the following equations for x and y
$\log_{100} |x + y| = \frac{1}{2}$, $\log_{10} y - \log_{10} |x| = \log_{100} 4$:
(a) $\left(\frac{8}{3}, \frac{16}{3}\right), (-8, -16)$ (b) $\left(\frac{10}{3}, \frac{20}{3}\right), (-10, 20)$
(c) $\left(-\frac{10}{3}, -\frac{20}{3}\right), (70, 20)$ (d) none of these

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

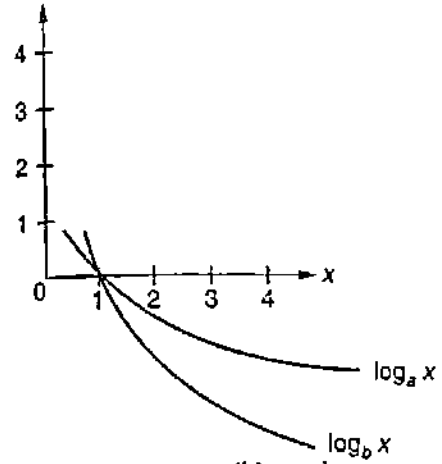
LEVEL 2

1. In the adjoining diagram there are two curvature graphs of $\log_a x$ and $\log_b x$ shown for $x > 0$. If $(a, b) > 0$ and $a \neq b \neq 1$, then :



(a) $a < b$
(b) $a > b$
(c) can't be determined
(d) none of the above

2. In the adjoining diagram graphs of $\log_a x$ and $\log_b x$ are shown for $x > 0$ and $0 < a < 1$, then :



(a) $a < b$
(b) $a > b$
(c) can't be determined
(d) none of these

3. Find the sum of 'n' terms of the series.

$$\log_2\left(\frac{x}{y}\right) + \log_4\left(\frac{x}{y}\right)^2 + \log_8\left(\frac{x}{y}\right)^3 + \log_{16}\left(\frac{x}{y}\right)^4 + \ldots :$$

(a) $\log_2\left(\frac{x}{y}\right)^{4n}$
(b) $n\left(\log_2 \frac{x}{y}\right)$
(c) $\log_2\left(\frac{x^{n-1}}{y^{n-1}}\right)$
(d) $\frac{1}{2}\log_2\left(\frac{x}{y}\right)^{n(n+1)}$

4. Find the value of $\log m + \log m^2 + \log m^3 + \ldots + \log m^n$:

(a) $\frac{n(n+1)}{2}$
(b) $\frac{mn}{2}$
(c) $\frac{n(n+1)}{2}\log m$
(d) $n(n+1)\log m^2$

5. The greatest possible value of n could be if $9^n < 10^6$, given that $\log 3 = 0.4771$ and $n \in N$:

(a) 7 (b) 8 (c) 9 (d) 10

6. The set of solutions for all x satisfying the equation

$$x^{\log_3 x^2 + (\log_3 x)^2 - 10} = \frac{1}{x^2} \text{ is :}$$

(a) $\{1, 9\}$
(b) $\{1, 4, 16\}$
(c) $\{1, 9, 81\}$
(d) $\left\{9, \frac{1}{81}, 1\right\}$

7. The set of all the solutions of the inequality $\log_{(2-x)}(x-3) \geq 1$ is :

(a) $x < 2$
(b) $x > 3$
(c) $(x < 2) \cup (x > 3)$
(d) none of these

8. If $\log_3 30 = \frac{1}{a}$ and $\log_5 30 = \frac{1}{b}$ then the value of $3\log_{30} 2$ is :

(a) $3(1 + a + b)$
(b) $2(1 - a - b)$
(c) $3(1 - a - b)$
(d) $3(1 + a - b)$

9. The set of all values of x satisfying $x^{\log_x |3-x|^2} = 4$:

(a) $(\sqrt{3}, 2)$ (b) $(1, 5)$ (c) $[1, \infty)$ (d) $\{1, 5\}$

10. The number of solutions of the expression satisfying $4^{x^2+2} - 9 \cdot 2^{x^2+2} + 8 = 0$ is :

(a) 0
(b) 1
(c) 2
(d) none of these

11. The number of solutions satisfying, for $x \in R$

$$x^{\left[(\log_3 x)^2 - \frac{9}{2}\log_3 x + 5\right]} = 3\sqrt{3}$$

(a) 0 (b) 1 (c) 2 (d) 3

12. Find the values of x satisfying

$$\log_{x^2+6x+8} \log_{2x^2+2x+3} (x^2 - 2x) = 0 \text{ is :}$$

(a) 0 (b) -1 (c) 2 (d) -3

13. For every p being real number the solution set of the inequality

$$\log_{10} p + (\log_{10} 10p)^2 + (\log_{10^{100}} p)^2 \leq \log_{10}(10)^9$$

(a) $p \geq 10^{-4}$
(b) $p > 10^{1/2}$
(c) $10^{-4} \leq p \leq 10^{1/2}$
(d) none of these

14. The number of real solutions of the equation

$$2\log_2 \log_2 x + \log_{1/2} \log_2(2\sqrt{2x}) = 1 \text{ is :}$$

(a) 1
(b) 2
(c) 3
(d) more than 3

15. The number of solutions of the equation

$$\log_{x/2} x^2 + 40 \log_{4x} \sqrt{x} - 14 \log_{16x} x^3 = 0 \text{ is :}$$

(a) 0 (b) 1 (c) 2 (d) 3

16. Find the solution set of x, for the given inequality $\log_m n > 1$, where $m = \frac{(5)^2 - (x)^2}{(4)^2}$ and $n = \frac{2(12-x) - x^2}{14}$:

(a) $(-5, 4)$
(b) $\{-3, 3\}$
(c) $(-3, 1) \cup (3, 4)$
(d) none of these

17. Find the values of x satisfying the following system of inequalities

$$\frac{\sqrt{(2-x)(x-8)}}{\log_{3/10}\frac{10}{7}\left[\log_2\left(\frac{5}{2}\right)\right]} \geq 0 \quad \text{and} \quad \frac{2^x}{8} - (2^5 - 1) > 0$$

(a) 6 (b) 8 (c) 4 (d) (2, 8)

18. The least value of expression $2\log_{10} x - \log_x(1/100)$ for $x > 1$ is :

(a) 2 (b) 3 (c) 4 (d) 5

19. The equation $x^{[(3/4)(\log_2 x)^2 + \log_2 x - (5/4)]} = \sqrt{2}$ has :

(a) at least one real solution
(b) exactly three real solutions
(c) exactly one irrational solution
(d) all of these

20. The values of x for which the given equation satisfy $\log_{(2x+3)}(6x^2 + 23x + 21) = 4 - \log_{(3x+7)}(4x^2 + 12x + 9)$ are :

(a) $(-8, 5)$ (b) $-\frac{1}{4}$ only (c) $-4, -\frac{1}{4}$ (d) $-\frac{1}{4}, -2$

21. Find all real values of x satisfying equation

$$|x-1|^{\log x^2 - 2\log_x 9} = (x-1)^7$$

(a) (2, 9)
(b) (9, 81)
(c) (2, 81)
(d) none of these

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Answers

INTRODUCTORY EXERCISE-16.1

1. (d)	2. (c)	3. (c)	4. (b)	5. (c)	6. (b)	7. (d)	8. (c)	9. (c)	10. (c)
11. (b)	12. (a)	13. (c)	14. (c)	15. (a)	16. (c)	17. (d)	18. (c)	19. (b)	20. (b)
21. (a)	22. (a)	23. (c)	24. (b)	25. (c)	26. (c)	27. (c)	28. (b)	29. (c)	30. (a)
31. (b)	32. (b)	33. (c)	34. (d)	35. (b)	36. (a)	37. (c)	38. (b)	39. (a)	40. (c)
41. (b)	42. (a)	43. (b)	44. (c)	45. (c)	46. (a)	47. (b)	48. (b)	49. (b)	50. (c)
51. (b)	52. (a)	53. (a)	54. (b)	55. (d)	56. (d)	57. (d)	58. (c)	59. (b)	60. (c)
61. (b)	62. (c)	63. (b)	64. (a)	65. (d)	66. (c)				

LEVEL-1

1. (a)	2. (b)	3. (c)	4. (d)	5. (a)	6. (b)	7. (c)	8. (a)	9. (a)	10. (c)
11. (c)	12. (c)	13. (b)	14. (c)	15. (b)	16. (c)	17. (d)	18. (a)	19. (b)	20. (d)
21. (c)	22. (c)	23. (d)	24. (d)	25. (c)	26. (b)	27. (a)	28. (d)	29. (d)	30. (b)
31. (d)	32. (b)	33. (b)	34. (d)	35. (d)	36. (d)	37. (b)	38. (d)	39. (d)	40. (b)
41. (a)	42. (b)	43. (d)	44. (c)	45. (c)	46. (c)	47. (a)	48. (a)	49. (d)	50. (b)

LEVEL-2

1. (a)	2. (a)	3. (b)	4. (c)	5. (b)	6. (d)	7. (d)	8. (c)	9. (d)	10. (c)
11. (d)	12. (b)	13. (c)	14. (a)	15. (d)	16. (b)	17. (c)	18. (c)	19. (d)	20. (b)
21. (c)									

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Hints & Solutions

INTRODUCTORY EXERCISE 16.1

1. $x^y = k \Leftrightarrow \log_x k = y$

2. $\log_m n = p \Leftrightarrow m^p = n$

3. $\log_a(mn) \Leftrightarrow \log_a m + \log_a n$

4. $\log_a\left(\frac{m}{n}\right) = \log_a m - \log_a n$

5. $\log_{27} 27 = \frac{\log_3 27}{\log_3 81} = \frac{\log_3 3^3}{\log_3 3^4} = \frac{3\log_3 3}{4\log_3 3} = \frac{3}{4}$

6. $\log_{36}\left(\frac{1}{216}\right) = \log_{36} 6^{-3} = \frac{\log 6^{-3}}{\log 36}$

$= \frac{-3\log 6}{2\log 6} = -\frac{3}{2}$

7. $\log_{\sqrt{6}} 216 = \frac{\log 216}{\log \sqrt{6}} = \frac{\log (\sqrt{6})^6}{\log \sqrt{6}} = \frac{6\log \sqrt{6}}{\log \sqrt{6}} = 6$

8. $\log_{3+\sqrt{2}} 5832 = \log_{3+\sqrt{2}}(18)^3 = \log_{3+\sqrt{2}}[(3+\sqrt{2})^2]^3$

$= \log_{3+\sqrt{2}}(3+\sqrt{2})^6 = 6 \qquad (\because \log_a a^m = m)$

9. Let $\log_{(-1/5)} 625 = x$

$\therefore \quad \left(-\frac{1}{5}\right)^x = 625$

$\Rightarrow \quad \left(-\frac{1}{5}\right)^x = (5)^4$

$\Rightarrow \quad (-5)^{-x} = (5)^4 = (-5)^4$

$\Rightarrow \quad x = -4$

10. Let $\log_{10}(0.0001) = x$

$\therefore \quad 10^x = 0.0001 = 10^{-4}$

$\Rightarrow \quad x = -4$

11. Let $\log_{0.01}(10000) = x$

$\therefore \quad (0.01)^x = 10000$

$\Rightarrow \quad (10^{-2})^x = 10000$

$\Rightarrow \quad 10^{-2x} = 10^4$

$\Rightarrow \quad x = -2$

12. $\log_4 32 + \log_9 243 = \frac{\log 32}{\log 4} + \frac{\log 243}{\log 9}$

$= \frac{5\log_2 2}{2\log_2 2} + \frac{5\log_3 3}{2\log_3 3}$

$= \frac{5}{2} + \frac{5}{2} = 5$

13. $\log_2 2^3 + \log_3 3^2 + \log_5 5^2 = 3 + 2 + 2 = 7$

14. $\log_2(\log_3 81) = \log_2(\log_3 3^4) = \log_2(4\log_3 3)$

$= \log_2 4 = \log_2 2^2 = 2$

15. $\log_2 \log_2 \log_3 \log_5(125)^3 = \log_2 \log_2 \log_3 \log_5 5^9$

$= \log_2 \log_2 \log_3 9$

$= \log_2 \log_2(2) = \log_2 1 = 0$

16. $\log_2 \log_2 \log_2 \log_2(65536) = \log_2 \log_2 \log_2 \log_2(2^{16})$

$= \log_2 \log_2 \log_2 16 = \log_2 \log_2 4$

$= \log_2 2 = 1$

17. $\log_{10} 1 + \log_{10} 10 + \log_{10} 100 + \ldots + \log_{10} 10000000000$

$= \log_{10}(1 \times 10 \times 100 \times \ldots \times 10000000000)$

$= \log_{10} 10^{(0+1+2+3+\ldots+10)}$

$= \log_{10} 10^{55} = 55$

Alternatively :

$\log_{10} 1 + \log_{10} 10 + \log_{10} 10^2 + \ldots + \log_{10} 10^{10}$

$= 0 + 1 + 2 + 3 + 4 + \ldots + 10 = 55$

18. $\log_5 5 + \log_5 5^2 + \log_5 5^3 + \ldots + \log_5 5^n = 1 + 2 + 3 + \ldots + n$

$= \frac{n(n+1)}{2}$

19. $27^{\log_3 81} = 3^{3\log_3 81}$

$= 3^{\log_3(81)^3} = (81)^3 = 531441 \qquad (\because a^{\log_a m} = m)$

20. $216^{\log_6 49} = 6^{3\log_6 49}$

$= 6^{\log_6(49)^3} = (49)^3 = 117649$

21. $25^{[(-1/4)\log_5 25]} = 5^{[2(-1/4)\log_5 25]}$

$= 5^{(-1/2)\log_5 25} = 5^{\log_5(25)^{-1/2}}$

$= 25^{-1/2} = \frac{1}{5}$

22. $\log_{10} 500000 - \log_{10} 5 = \log_{10}\left(\frac{500000}{5}\right) = \log_{10} 100000$

$= \log_{10} 10^5 = 5$

23. $\log_5[\log_3(\log_2 x)] = 1 = \log_5 5$

$\Rightarrow \quad \log_3(\log_2 x) = 5 = \log_3 3^5$

$\Rightarrow \quad \log_2 x = 3^5 = 243$

$\Rightarrow \quad 2^{243} = x$

24. $\log_2 \log_{10} 4x = 1 = \log_2 2$

$\Rightarrow \quad \log_{10} 4x = 2 = \log_{10} 10^2 = \log_{10} 100$

$\Rightarrow \quad 4x = 100 \Rightarrow x = 25$

25. If $\log_{27} x = \frac{4}{3}$

$\Rightarrow \quad (27)^{4/3} = x$

$\Rightarrow \quad (3^3)^{4/3} = x$

$\Rightarrow \quad 3^4 = x$

$\Rightarrow \quad x = 81$

26. $\log_x\left(\frac{4}{9}\right) = -\frac{1}{2}$

$\Rightarrow \quad x^{-1/2} = \frac{4}{9}$

$\Rightarrow \quad \frac{1}{x^{1/2}} = \frac{4}{9}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$\Rightarrow \left(\frac{1}{x}\right)^{1/2} = \frac{4}{9}$

$\Rightarrow \frac{1}{x} = \left(\frac{4}{9}\right)^2 \Rightarrow x = \frac{81}{16}$

27. $\log_{100} x = -4$

$\Rightarrow (100)^{-4} = x$

$\Rightarrow x = (10^2)^{-4} = 10^{-8}$

$\Rightarrow x = \frac{1}{10^8}$

28. $\log_{100} x = -\frac{1}{2} \Rightarrow (100)^{-1/2} = x$

$\Rightarrow (10^2)^{-1/2} = x$

$\Rightarrow 10^{-1} = x$

$\Rightarrow x = \frac{1}{10}$

29. $\log_x 0.2 = -\frac{1}{4}$

$\Rightarrow x^{-1/4} = 0.2 = \frac{2}{10}$

$\Rightarrow x^{-1} = \left(\frac{2}{10}\right)^4$

$\Rightarrow x = \left(\frac{10}{2}\right)^4 \Rightarrow x = 5^4$

30. $\log_{243} x = 0.8$

$\Rightarrow 243^{0.8} = x$

$\Rightarrow (3^5)^{4/5} = x$

$\Rightarrow 3^4 = x \Rightarrow x = 81$

31. $\log_e(x-1) + \log_e x + \log_e(x+1) = 0$

$\Rightarrow \log_e(x-1) \times x \times (x+1) = \log_e 1$

$\Rightarrow \log_e(x^2-1)x = \log_e 1$

$\Rightarrow (x^2-1)x = 1$

$\Rightarrow x^3 - x - 1 = 0$

32. $\log_3 5 \times \log_{25} 9 = \frac{\log 5}{\log 3} \times \frac{\log 3^2}{\log 5^2}$

$= \frac{\log 5}{\log 3} \times \frac{2\log 3}{2\log 5} = 1$

33. $\log \frac{14}{3} + \log \frac{11}{5} - \log \frac{22}{15}$

$= \log\left(\frac{14}{3} \times \frac{11}{5} \times \frac{15}{22}\right) = \log 7$

34. $3\log \frac{81}{80} + 5\log \frac{25}{24} + 7\log \frac{16}{15}$

$= \log\left[\left(\frac{81}{80}\right)^3 \times \left(\frac{25}{24}\right)^5 \times \left(\frac{16}{15}\right)^7\right]$

$= \log\left(\frac{3^{12} \times 5^{10} \times 2^{28}}{2^{12} \times 5^3 \times 2^{15} \times 3^5 \times 3^7 \times 5^7}\right)$

$= \log 2$

35. $\log \frac{a^3}{bc} + \log \frac{b^3}{ac} + \log \frac{c^3}{ab} = \log\left(\frac{a^3b^3c^3}{ab \cdot bc \cdot ac}\right)$

$= \log abc$

36. $\frac{1}{(\log_a bc)+1} + \frac{1}{(\log_b ac)+1} + \frac{1}{(\log_c ab)+1}$

$= \frac{1}{\log_a bc + \log_a a} + \frac{1}{\log_b ac + \log_b b} + \frac{1}{\log_c ab + \log_c c}$

$= \frac{1}{\log_a abc} + \frac{1}{\log_b abc} + \frac{1}{\log_c abc}$

$= \log_{abc} a + \log_{abc} b + \log_{abc} c$

$= \log_{abc} abc = 1$

37. $\frac{1}{\log_{ab} abc} + \frac{1}{\log_{bc} abc} + \frac{1}{\log_{ca} abc}$

$= \log_{abc}(ab) + \log_{abc}(bc) + \log_{abc}(ca)$

$= \log_{abc}(abc)^2 = 2$

38. $\frac{1}{\log_5 210} + \frac{1}{\log_6 210} + \frac{1}{\log_7 210}$

$= \log_{210} 5 + \log_{210} 6 + \log_{210} 7$

$= \log_{210}(5 \times 6 \times 7) = \log_{210} 210 = 1$

39. $\frac{1}{\log_{a/b} x} + \frac{1}{\log_{b/c} x} + \frac{1}{\log_{c/a} x}$

$= \log_x \frac{a}{b} + \log_x \frac{b}{c} + \log_x \frac{c}{a}$

$= \log_x\left(\frac{a}{b} \times \frac{b}{c} \times \frac{c}{a}\right) = \log_x 1 = 0$

40. $\log_2 10 = \frac{1}{\log_{10} 2} = \frac{1}{0.3010} = 3.3222$

41. $2^x \cdot 3^{2x} = 100$

$\Rightarrow x\log 2 + 2x\log 3 = \log 100$

$\Rightarrow x(0.3010 + 2 \times 0.4771) = 2$

$\Rightarrow x = \frac{1}{1.2552} = 1.59$

42. $\log_{10} 2 = 0.3010$ and $5^x = 400$

$\therefore 5^x = 400$

$\Rightarrow x\log 5 = \log 4 + \log 100$

$\Rightarrow x(\log 10 - \log 2) = 2\log 2 + 2$

$\Rightarrow x(1 - 0.3010) = 2(0.3010 + 1)$

$\Rightarrow x = 3.72$

43. $5^{20} = 20\log 5$

$= 20\log\left(\frac{10}{2}\right) = 20(\log 10 - \log 2)$

$= 20 \times [1 - 0.3010]$

$= 20 \times 0.6990 = 13.98$

Characteristic = 13

$\therefore$ Number of digits = (13 + 1) = 14

44. $\log(m+n) = \log m + \log n$

$\Rightarrow \log(m+n) = \log(mn)$

$\Rightarrow m + n = mn$

$m = n(m-1)$

$\Rightarrow n = \frac{m}{(m-1)}$

45. $\log_{10} a + \log_{10} b = c$

$\Rightarrow \log_{10}(ab) = c$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$$10^c = ab$$

$$\Rightarrow \quad a = \frac{(10)^c}{b}$$

45. Since 0.3274 gives characteristic $\bar{1}$. Therefore value of $\log(0.3274) = \bar{1}.5150$

47. $\log_{10}(100x) = \log_{10} 100 + \log_{10} x$

$$= 2 + 1.9675 = 3.9675$$

48. $$(5)^x = 1000$$

$$\Rightarrow \quad 5 = 10^{3/x} \qquad \ldots(1)$$

And $$(0.5)^y = 1000$$

$$\Rightarrow \quad \left(\frac{5}{10}\right)^y = 10^3$$

$$\Rightarrow \quad 5^y = 10^3 \times 10^y$$

$$\Rightarrow \quad 5 = 10^{\frac{(3+y)}{y}} \qquad \ldots(2)$$

From eq. (1) and (2)

$$5 = 10^{3/x} = 10^{\frac{(3+y)}{y}}$$

$$\Rightarrow \quad \frac{3}{x} = \frac{3+y}{y}$$

$$\Rightarrow \quad \frac{1}{x} - \frac{1}{y} = \frac{1}{3}$$

49. $$\frac{1}{2}\log x + \frac{1}{2}\log y + \log 2 = \log(x+y)$$

$$\Rightarrow \quad \log(x^{1/2}y^{1/2} \times 2) = \log(x+y)$$

$$\Rightarrow \quad 2\sqrt{xy} = (x+y)$$

$$\Rightarrow \quad 4xy = (x+y)^2$$

$$\Rightarrow \quad (x+y)^2 - 4xy = 0$$

$$\Rightarrow \quad (x-y)^2 = 0$$

$$\Rightarrow \quad x - y = 0 \Rightarrow x = y$$

50. $$q = \log_{17}(5)^2 = 2\log_{17} 5$$

$$\Rightarrow \quad \frac{1}{q} = \frac{1}{2}\log_5 17$$

And $$\frac{1}{p} = \log_5 3 = \frac{1}{2} \times (2\log_5 3)$$

$$= \frac{1}{2}(\log_5 9)$$

$$\therefore \quad \frac{1}{p} < \frac{1}{q} \Rightarrow p > q$$

51. $$x^2 + 4y^2 = 12xy$$

$$\Rightarrow \quad x^2 + (2y)^2 + 4xy = 12xy + 4xy$$

$$\Rightarrow \quad (x+2y)^2 = 16xy$$

$$\Rightarrow \quad (x+2y)^2 = (2^4)xy$$

$$\Rightarrow \quad 2\log(x+2y) = 4\log 2 + \log x + \log y$$

$$\Rightarrow \quad \log(x+2y) = \frac{1}{2}(\log x + \log y + 4\log 2)$$

52. $$\log_3 7 = \frac{\log_2 7}{\log_2 3} = (\log_2 7)(\log_3 2)$$

$$\log_3 7 = (\log_3 2)(\log_2 7) = \log_2 7^{\log_3 2}$$

$$\therefore \quad 2^{\log_3 7} = 2^{\log_2 7^{\log_3 2}}$$

$$\Rightarrow \quad 2^{\log_3 7} = 7^{\log_3 2}$$

$$\Rightarrow \quad 2^{\log_3 7} - 7^{\log_3 2} = 0$$

53. $$\frac{1}{\log_2 a} + \frac{1}{\log_4 a} + \frac{1}{\log_8 a} + \ldots + \frac{1}{\log_{2^n} a} = \frac{n(n+1)}{k}$$

$$= \log_a 2 + \log_a 2^2 + \log_a 2^3 + \ldots + \log_a 2^n = \frac{n(n+1)}{k}$$

$$\Rightarrow \quad (\log_a 2)(1 + 2 + 3 + \ldots + n) = \frac{n(n+1)}{k}$$

$$\Rightarrow \quad (\log_a 2)\left[\frac{n(n+1)}{2}\right] = \frac{n(n+1)}{k}$$

$$\Rightarrow \quad \frac{1}{2}\log_a 2 = \frac{1}{k}$$

$$\Rightarrow \quad \log_a 2 = \frac{2}{k}$$

$$\Rightarrow \quad \log_2 a = \frac{k}{2}$$

$$\Rightarrow \quad 2\log_2 a = k$$

$$\Rightarrow \quad \log_2 a^2 = k$$

54. $\log \tan 1° + \log \tan 2° + \ldots + \log \tan 89°$

$= (\log \tan 1° + \log \tan 89°) + (\log \tan 2° + \log \tan 88°)$
$+ \ldots + \log \tan 45°$

$= \log(\tan 1° \cdot \tan 89°) + \log(\tan 2° \cdot \tan 88°) + \ldots \log 1$

$= \log(\tan 1° \cdot \cot 1°) + \log(\tan 2° \cdot \cot 2°) + \ldots \log 1$

$= \log 1 + \log 1 + \ldots + \log 1$

$(\because \tan(90 - \theta) = \cot\theta$ and $\tan 45° = 1)$

$= 0$

55. $$\log_9 x - \log_9\left(\frac{x}{10} + \frac{1}{9}\right) = 1 = \log_9 9$$

$$\Rightarrow \quad \log_9 \frac{x}{\left(\frac{x}{10} + \frac{1}{9}\right)} = \log_9 9$$

$$\Rightarrow \quad \frac{x}{\left(\frac{x}{10} + \frac{1}{9}\right)} = 9$$

$$\Rightarrow \quad \frac{x \times 90}{9x + 10} = 9$$

$$\Rightarrow \quad 10x = 9x + 10$$

$$\Rightarrow \quad x = 10$$

Alternatively : Go through options.

56. $$[(x^2 - y^2)^2]^{a-1} = (x-y)^{2a}(x+y)^{-2}$$

$$\Rightarrow \quad (x^2 - y^2)^{2(a-1)} = (x-y)^{2a}(x+y)^{-2}$$

$$\Rightarrow (x-y)^{2(a-1)}(x+y)^{2(a-1)} = (x-y)^{2a}(x+y)^{-2}$$

$$\Rightarrow \quad \frac{(x-y)^{2a-2}}{(x-y)^{2a}} \frac{(x+y)^{2a-2}}{(x+y)^{-2}} = 1$$

$$\Rightarrow \quad (x-y)^{-2}(x+y)^{2a} = 1$$

$$\Rightarrow \quad -2\log(x-y) + 2a\log(x+y) = \log 1$$

$$\Rightarrow \quad 2a\log(x+y) = 2\log(x-y)$$

$$a = \frac{\log(x-y)}{\log(x+y)}$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmartboob.wordpress.com
For More Visit : www.LearnEngineering.in

57. Let $\log_2 7$ be a rational

$$\log_2 7 = \left(\frac{p}{q}\right) \qquad q \neq 0$$

$\Rightarrow 2^{(p/q)} = 7 \Rightarrow 2^p = 7^q$

Here p and q being the integers cannot satisfy the above relation.

Hence $\log_2 7$ is an irrational number.

(Clearly (a) and (b) are wrong)

58. $\log \frac{x}{y} = \frac{\log_e x}{\log_e y}$ (see the theory)

59. $\log_{10} 2 + \log_{10}\left(\frac{16}{15}\right)^{16} + \log_{10}\left(\frac{25}{24}\right)^{12} + \log_{10}\left(\frac{81}{80}\right)^{7}$

$= \log_{10} 2 + \log_{10}\left(\frac{2^{64}}{3^{16} \times 5^{16}}\right) + \log\left(\frac{5^{24}}{2^{36} \times 3^{12}}\right)$

$+ \log\left(\frac{3^{28}}{2^{28} \times 5^{7}}\right)$

$= \log_{10} \frac{2^{65} \times 3^{28} \times 5^{24}}{2^{64} \times 3^{28} \times 5^{23}}$

$= \log_{10}(2 \times 5) = \log_{10} 10 = 1$

60. $(pa + qb - rc) = p \log_{10} x + q \log_{10} y - r \log_{10} z$

$= \log_{10}(x^p) + \log_{10}(y^q) - \log_{10}(z^r)$

$= \log_{10}\left(\frac{x^p y^q}{z^r}\right)$

$\Rightarrow$ antilog $(pa + qb - rc) = \frac{x^p y^q}{z^r}$.

61. $\log_{10} a^p . b^q . c^r = \log_{10} a^p + \log_{10} b^q + \log_{10} c^r$

$= p \log_{10} a + q \log_{10} b + r \log_{10} c$

62. $\because$ a, b, c are in GP

$b^2 = ac$

$\Rightarrow 2 \log b = \log a + \log c$

$\Rightarrow \log_x a, \log_x b, \log_x c$ are in AP

$\Rightarrow \log_{10} a, \log_{10} b, \log_{10} c$ are in AP

63. $\log_{10} x, \log_{10} y, \log_{10} z$ are in AP

$\therefore 2 \log_{10} y = \log_{10} x + \log_{10} z$

$\Rightarrow \log_{10} y^2 = \log_{10}(xz)$

$\Rightarrow y^2 = xz$

$\Rightarrow x, y, z$ are in GP

64. $\because$ a, b, c are in GP

$\therefore \log_x a, \log_x b, \log_x c$ are in AP

$\Rightarrow \frac{1}{\log_a x}, \frac{1}{\log_b x}, \frac{1}{\log_c x}$ are in AP

65. $\log(x-1) + \log(x+1) = 3 \log 2 = \log 2^3$

$\Rightarrow \log[(x-1)(x+1)] = \log 8$

$\Rightarrow \log(x^2 - 1) = \log 8$

$\Rightarrow x^2 - 1 = 8 \Rightarrow x^2 = 9 \Rightarrow x = \pm 3$

But at $x = -3$ logarithm equation is not defined.

Hence, $x = 3$

66. a, b, c are consecutive integers

$\therefore b = a + 1$ and $c = a + 2$

$\therefore \log(ac + 1) = \log[a(a+2) + 1]$

$= \log[(b-1)(b-1+2) + 1]$ $(\because a = b-1)$

$= \log b^2 = 2 \log b$

LEVEL 1

1. $\log_{10} \sqrt[7]{13} \Rightarrow \log_{10}(13)^{1/7}$

$\Rightarrow \frac{1}{7} \log_{10} 13 = \frac{1}{7} \times 1.1139 = 0.15913$

2. $\log_{2\sqrt{3}} 144 = \log_{2\sqrt{3}} (2\sqrt{3})^4 = 4$

3. $\log_a N = \log_b N \times P$

$\Rightarrow \frac{\log_a N}{\log_b N} = P$

$\Rightarrow \frac{1}{\frac{\log_N a}{\log_N b}} = P$

$\Rightarrow \frac{\log_N b}{\log_N a} = P \Rightarrow \log_a b = P$

4. $\log(ab)^2 - \log(ac) + \log(bc^4) - 3\log(bc)$

$= \log \frac{(ab)^2}{ac} + \log \frac{(bc^4)}{(bc)^3}$

$= \log\left[\frac{(ab)^2}{ac} \times \frac{(bc^4)}{(bc)^3}\right] = \log a$

5. $\log_2 x + \log_4 x + \log_{64} x = 5$

$\Rightarrow \frac{1}{\log_x(2)} + \frac{1}{2\log_x(2)} + \frac{1}{6\log_x(2)} = 5$

$\Rightarrow \frac{10}{6\log_x(2)} = 5$

$\Rightarrow \log_x(2) = \frac{1}{3}$

$\Rightarrow \log_2 x = 3$

$\Rightarrow 2^3 = x \quad \therefore x = 8$

Alternatively: Go through options.

6. $\log_q(xy) = 3$ and $\log_q(x^2y^3) = 4$

$\therefore \log_q(x^2y^3) = 4 \Rightarrow \log_q[(xy)^2 \cdot y]$

$\Rightarrow \log_q(xy)^2 + \log_q y = 4$

$\Rightarrow 2\log_q(xy) + \log_q y = 4$

$\Rightarrow 2 \times 3 + \log_q y = 4$

$\Rightarrow \log_q y = -2$

Again $\log_q(xy) = \log_q x + \log_q y = 3$

$\Rightarrow \log_q x + (-2) = 3$

$\Rightarrow \log_q x = 5$

7. $\frac{\log x}{l+m-2n} = \frac{\log y}{m+n-2l} = \frac{\log z}{n+l-2m} = k$ (say)

$\Rightarrow \log x = k(l+m-2n)$ ...(1)

$\log y = k(m+n-2l)$ ...(2)

$\log z = k(n+l-2m)$ ...(3)

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$\therefore \log x + \log y + \log z$
$= k(l + m - 2n) + k(m + n - 2l) + k(n + l - 2m)$
$\log(xyz) = 0$
$\log xyz = \log 1$
$\Rightarrow \quad xyz = 1$
$\Rightarrow$

8. Let m be the first term and k be the common ratio of GP, then
$a = mk^{p-1}$
$b = mk^{q-1}$
$c = mk^{r-1}$

$\therefore (q-r)\log a + (r-p)\log b + (p-q)\log c$
$= \log[mk^{(p-1)}]^{q-r} + \log b[mk^{(q-1)}]^{r-p}$
$+ \log[mk^{(r-1)}]^{p-q}$
$= \log(m^{q-r+r-p+p-q})(k^{p-1})^{q-r}(k^{q-1})^{r-p}(k^{r-1})^{p-q}$
$= \log m^0 k^0 = \log 1 = 0$

9. $a^{3-x} \cdot b^{5x} = a^{x+5} \cdot b^{3x}$
$\Rightarrow \quad \frac{a^{3-x}}{a^{x+5}} = \frac{b^{3x}}{b^{5x}}$
$\Rightarrow \quad a^{-2(x+1)} = b^{-2x}$
$\Rightarrow \quad a^{-2x} \cdot a^{-2} = b^{-2x}$
$\Rightarrow \quad \left(\frac{b}{a}\right)^{-2x} = a^{-2}$
$\Rightarrow \quad \left(\frac{b}{a}\right)^{x} = a$
$\Rightarrow \quad \log\left(\frac{b}{a}\right)^{x} = \log a$
$\Rightarrow \quad x\log\left(\frac{b}{a}\right) = \log a$

10. $v = u^{1/2}, \quad w = u^{1/3}, \quad z = u^{1/4}$
$\therefore \quad uvwz = u^{\left(1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}\right)}$
$\therefore \quad \log_u uvwz = \log_u u^{\left(1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}\right)}$
$= \left(1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}\right)\log_u u = \left(1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}\right)$

11. $\frac{\log\sqrt{27} + \log\sqrt{8} - \log\sqrt{125}}{\log 6 - \log 5} = \frac{\log(3^{3/2} \times 2^{3/2}) - \log(5)^{3/2}}{\log 6 - \log 5}$
$= \frac{\log 6^{3/2} - \log 5^{3/2}}{\log 6 - \log 5}$
$= \frac{\frac{3}{2}\log\left(\frac{6}{5}\right)}{\log\left(\frac{6}{5}\right)} = \frac{3}{2}$

12. Last term $k = a \cdot r^{n-1}$, $r \rightarrow$ common ratio
$\therefore \quad \log k = \log(a \cdot r^{n-1})$
$\Rightarrow \quad \log k = \log a + (n-1)\log r$
$\Rightarrow \quad \frac{\log k - \log a}{\log r} = n - 1$
$\Rightarrow \quad n = 1 + \left(\frac{\log k - \log a}{\log r}\right)$

13. $\because a, b, c$ are in GP
$\therefore \quad b^2 = ac$
$\Rightarrow \quad 2\log b = \log a + \log c$
$\therefore \log a, \log b, \log c$ are in AP
i. e., $\log_d a, \log_d b, \log_d c$ are in AP
$\therefore \log_a d, \log_b d, \log_c d$ are in HP

14. Best way is to go through options

15. Best way is to go through options
Alternatively: $\log_{10} x^2y^3 = 7$
$\Rightarrow \quad x^2y^3 = 10^7 \quad \ldots(1)$
and $\log_{10}\left(\frac{x}{y}\right) = 1$
$\Rightarrow \quad \frac{x}{y} = 10 \quad \ldots(2)$
$\therefore \quad \frac{x^2y^3}{(x/y)^2} = \frac{10^7}{(10)^2} \Rightarrow y^5 = 10^5$
$\Rightarrow \quad y = 10 \quad \therefore x = 100$

16. $y = a^{\left(\frac{1}{1-\log_a x}\right)}$
$\Rightarrow \quad \log_a y = \frac{1}{1-\log_a x}$ (taking log of both sides)
$\Rightarrow \quad \log_a x = 1 - \frac{1}{\log_a y} \quad \ldots(1)$
Again $z = a^{\left(\frac{1}{1-\log_a y}\right)}$
$\Rightarrow \quad \log_a z = \frac{1}{1-\log_a y}$
$\Rightarrow \quad \log_a y = 1 - \frac{1}{\log_a z} \quad \ldots(2)$
$\therefore$ From eq. (1) and (2), we get
$\log_a x = 1 - \frac{1}{\left(1 - \frac{1}{\log_a z}\right)}$
$\Rightarrow \quad \log_a x = \frac{1}{(1-\log_a z)}$
$\Rightarrow \quad x = a^{\frac{1}{1-\log_a z}}$

Alternatively: Consider some numerical values suppose $a = 10$ and $x = \sqrt{10}$
$\therefore \quad y = a^{\left(\frac{1}{1-\log_a x}\right)}$
$y = 10^{\left(\frac{1}{1-\log_{10}\sqrt{10}}\right)}$
$\Rightarrow \quad y = 100$
and $z = a^{\left(\frac{1}{1-\log_a y}\right)}$
$\therefore \quad z = 10^{\left(\frac{1}{1-\log_{10}100}\right)}$
$\Rightarrow \quad z = \frac{1}{10}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmabhoob.wordpress.com

For More Visit : www.LearnEngineering.in

Now, go through options. Consider option (c)

$$x = a^{\frac{1}{1-\log_a z}} = 10^{\left(\frac{1}{1-\log_{10}(1/10)}\right)}$$

$$x = 10^{1/2} = \sqrt{10}$$

Hence option (c) is correct.

NOTE You can check other options but they do not satisfy.

17. Let $a = 10$, $b = 100$ and $c = 1000$, then

$$x = \log_{1000}(100) + \log_{100}(1000) = \frac{13}{6}$$

$$y = \log_{10} 1000 + \log_{1000} 10 = \frac{10}{3}$$

$$z = \log_{100} 10 + \log_{10} 100 = \frac{5}{2}$$

$$\therefore \quad x^2 + y^2 + z^2 = \left(\frac{13}{6}\right)^2 + \left(\frac{10}{3}\right)^2 + \left(\frac{5}{2}\right)^2 = \frac{794}{36}$$

and $$xyz = \frac{13}{6} \times \frac{10}{3} \times \frac{5}{2} = \frac{650}{36}$$

$$\therefore \quad xyz + 4 = \left(\frac{650}{36}\right) + 4 = \frac{794}{36}$$

Hence choice (d) is correct.

18. $$\frac{1}{\log_3 e} + \frac{1}{\log_3 e^2} + \frac{1}{\log_3 e^4} + \ldots$$

$$= \frac{1}{\log_3 e} + \frac{1}{2\log_3 e} + \frac{1}{4\log_3 e} + \ldots$$

$$= \frac{1}{\log_3 e}\left(1 + \frac{1}{2} + \frac{1}{4} + \ldots\right)$$

$$= \log_e 3\left(\frac{1}{1-1/2}\right)$$

(Using sum of infinite GP) $\left(s_\infty = \frac{a}{1-r}; |r| < 1\right)$

$$= 2\log_e 3 = \log_e 3^2 = \log_e 9$$

19. $$\log_{10} x^2 - \log_{10} \sqrt{y} = 1$$

$$\Rightarrow \quad \log_{10}\left(\frac{x^2}{\sqrt{y}}\right) = \log_{10} 10$$

$$\Rightarrow \quad \frac{x^2}{\sqrt{y}} = 10 \quad \Rightarrow \quad \frac{4}{\sqrt{y}} = 10$$

$$\Rightarrow \quad \sqrt{y} = \frac{4}{10} = \frac{2}{5}$$

$$\Rightarrow \quad y = \frac{4}{25}$$

20. $(b^2)^{5\log_b x} = b^{10\log_b x} = b^{\log_b (x)^{10}} = x^{10}$

21. $(7^3)^{-2\log_7 8} = 7^{-6\log_7 8} = 7^{(\log_7 8^{-6})}$

$$= 8^{-6} = \frac{1}{8^6}$$

22. $\log 12900 = \log (1.29 \times 10000)$

$= \log 1.29 + \log 10000$

$= \log 1.29 + 4$

23. $\log 0.786 = \log\left(\frac{786}{1000}\right) = \log 786 - \log 1000$

$= \log 786 - 3$

Also $\log 0.786 = \log \frac{78.6}{100} = \log 78.6 - \log 100$

24. $\log_5 x = y \quad \Rightarrow \quad 5^y = x$

$\therefore \quad (5^y)^5 = x^5 \quad \Rightarrow \quad 5^{5y} = x^5$

25. $A = \log_{13} 189 = \log_{13} (13^2 + 20)$

and $B = \log_{23} 521 = \log_{23}(23^2 - 8)$

$\therefore \quad A > B$ $\quad (\because A > 2 \text{ and } B < 2)$

26. $A = \frac{\log_3 2187}{5} = \frac{1}{5}\log_3 2187$

$= \frac{1}{5}\log_3(3^7) = \frac{7}{5}\log_3 3 = \frac{7}{5}$

and $B = \log_{243} 2187 = \frac{\log_3 2187}{\log_3 243} = \frac{\log_3 3^7}{\log_3 3^5} = \frac{7}{5}$

$\therefore \quad A = B$

27. $(150)^x = 7$

$\Rightarrow \quad \log_{150} 7 = x$

$\Rightarrow \quad \frac{\log 7}{\log 150} = x$

$\Rightarrow \quad x = \frac{\log 7}{\log (3 \times 5 \times 10)} = \frac{\log 7}{\log 3 + \log 5 + \log 10}$

$= \frac{\log 7}{\log 3 + \log 5 + 1}$

28. $\log_{1/2} x = \log_2 (3x - 2)$

$\Rightarrow \quad -\log_2 x = \log_2 (3x - 2)$

$\Rightarrow \quad \log_2 (x^{-1}) = \log_2 (3x - 2)$

$\Rightarrow \quad x^{-1} = (3x - 2)$

$\Rightarrow \quad 3x^2 - 2x - 1 = 0$

$\Rightarrow \quad 3x^2 - 3x + x - 1 = 0$

$\Rightarrow \quad 3x(x-1) + 1(x-1) = 0$

$\Rightarrow \quad x = 1, -1/3$

but at $x = -1/3$, $\log x$ is not defined

The only admissible value of x is 1.

Alternatively : Go through options.

29. $\log_2 (x + y) = 3$

and $\log_2 x + \log_2 y = 2 + \log_2 3$

$\because \quad \log_2 (x + y) = 3 = \log_2 2^3 = \log_2 8$

$\Rightarrow \quad x + y = 8,$

Hence option (c) is wrong.

Again, $\log_2 x + \log_2 y = 2 + \log_2 3 = \log_2 4 + \log_2 3$

$\Rightarrow \quad \log_2 (xy) = \log_2 12$

$\Rightarrow \quad xy = 12$

Hence option (d) is correct.

30. Go through option.

at $x = 0$, $\log x$ is not defined.

at $x = 1$, both sides are equal.

Again at $x \neq 1$

$\log_5 x \log_6 x \log_7 x = \log_5 x \log_6 x + \log_6 x . \log_7 x + \log_7 x . \log_5 x$

$\Rightarrow \quad 1 = \frac{1}{\log_7 x} + \frac{1}{\log_5 x} + \frac{1}{\log_6 x}$

(By dividing both sides by $\log_5 x \log_6 x \log_7 x$)

$\therefore \quad 1 = \log_x 7 + \log_x 5 + \log_x 6$

For More Visit : www.LearnEngineering.in

Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$$1 = \log_x(7 \cdot 5 \cdot 6) = \log_x 210$$
$$\log_x 210 = 1$$
$$\Rightarrow \quad x = 210$$
$$\Rightarrow \quad x = \{1, 210\}$$

Hence

31. Let $\log_3 18 = a$

$$\therefore \frac{\log_3 54}{\log_{486} 3} - \frac{\log_3 1458}{\log_{18} 3}$$

$$= \log_3(18 \times 3)\log_3(18 \times 27) - \log_3(18 \times 81)\log_3 18$$
$$= (\log_3 18 + \log_3 3)(\log_3 18 + 3\log_3 3) - (\log_3 18 + 4\log_3 3)\log_3 18$$
$$= (a+1)(a+3) - (a+4)a$$
$$= a^2 + 3 + 4a - a^2 - 4a = 3$$

32.
$$\log_9(2x-5) = \log_3(x-4)$$
$$\Rightarrow \quad \frac{1}{2}\log_3(2x-5) = \log_3(x-4)$$
$$\Rightarrow \quad \log_3(2x-5) = \log_3(x-4)^2$$
$$\Rightarrow \quad (2x-5) = (x-4)^2$$
$$\Rightarrow \quad (2x-5) = x^2 + 16 - 8x$$
$$\Rightarrow \quad x^2 - 10x + 21 = 0$$
$$\Rightarrow \quad (x-3)(x-7) = 0$$
$$\Rightarrow \quad x = 3 \quad \text{or} \quad x = 7$$

but at $x = 3$, $\log_3(x-4)$ is not defined since $(x-4)$ becomes negative

Hence $x = 7$ is the only possible solution.

33. Since we know that when a, b, c are in GP, then $\log a$, $\log b$ and $\log c$ are in AP.

Therefore 2, $(2^x - 5)$ and $(2^x - 7/2)$ must be in GP

Now, going through options, we get

at $x = 3$ the three terms 2, $(2^x - 5)$ and $(2^x - 7/2)$ are in GP

Hence $\log 2$, $\log(2^x - 5)$ and $\log(2^x - 7/2)$ are in AP.

Alternatively :

We have $2\log_3(2^x - 5) = \log_3 2 + \log_3(2^x - 7/2)$

$$\Rightarrow \quad \log_3(2^x - 5)^2 = \log_3 2(2^x - 7/2)$$
$$\Rightarrow \quad (2^x - 5)^2 = 2(2^x - 7/2)$$
$$\Rightarrow \quad (2^x)^2 - 10 \cdot 2^x + 25 = 2 \cdot 2^x - 7$$

$[10 \cdot 2^x + 2 \cdot 2^x = 12\ 2^x]$ $\quad [\because (a-b)^2 = a^2 - 2ab + b^2]$

$$\Rightarrow \quad (2^x)^2 - 12 \cdot 2^x + 32 = 0$$
$$\Rightarrow \quad (2^x - 8)(2^x - 4) = 0$$
$$\Rightarrow \quad x = 3, \quad x = 2$$

but at $x = 2$, $\log(2^x - 5)$ is undefined

Hence $x = 3$

34. Go through options.

35. As $\log_x a$, $a^{x/2}$, $\log_b x$ are in GP.

$$\therefore \quad (a^{x/2})^2 = \log_x a \cdot \log_b x$$
$$\Rightarrow \quad a^x = \frac{\log a}{\log x} \cdot \frac{\log x}{\log b} = \frac{\log a}{\log b} = \log_b a$$
$$\Rightarrow \quad a^x = \log_b a$$
$$\Rightarrow \quad x = \log_a(\log_b a)$$
$$= \log_a\left(\frac{\log_e a}{\log_e b}\right)$$
$$= \log_a(\log_e a) - \log_a(\log_e b)$$

36. $1, \log_3(3^{1-x} + 2), \log_3(4 \cdot 3^x - 1)$ are in AP

$1, \log_2(3^{1-x} + 2)^{1/2}, \log_2(4 \cdot 3^x - 1)$ are in AP

$\therefore \log_3 3, \log_3(3^{1-x} + 2)^{1/2}, \log_3(4 \cdot 3^x - 1)$ are in AP

$\therefore 3, (3^{1-x} + 2)^{1/2}, (4 \cdot 3^x - 1)$ are in GP

$$\therefore \quad [(3^{1-x} + 2)^{1/2}]^2 = 3 \cdot (4 \cdot 3^x - 1)$$
$$\Rightarrow \quad 3^{1-x} + 2 = 4 \cdot 3^{x+1} - 3$$
$$\Rightarrow \quad 4 \cdot 3^{x+1} - 3^{1-x} = 5$$

Solving we get $\quad x = \log_3\left(\frac{3}{4}\right)$

Alternatively: Go through options. Consider option (d)

$$x = \log_3 \frac{3}{4} \Rightarrow 3^x = \frac{3}{4}$$
$$\Rightarrow 1, \log_9\left(\frac{3}{3^x} - 2\right), \log_3\left(4 \cdot \frac{3}{4} - 1\right)$$
$$\Rightarrow 1, \log_9 6, \log_3 2$$
$$\Rightarrow \log_9 9, \log_9 6, \log_9 4 \text{ are in AP}$$

Since 9, 6, 4 are in GP

37. $\frac{1}{\log_{100} n} + \frac{1}{\log_{99} n} + \ldots + \frac{1}{\log_2 n}$

$$= \log_n 100 - \log_n 99 + \ldots + \log_n 2$$
$$= \log_n(100 \times 99 \times 98 \times \ldots \times 2)$$
$$= \log_n 100! = \frac{1}{\log_{100!} n}$$

38. Best way is to go through options.

At $x = 1$,

$$\log_2(1 + 3^0) + 2 = \log_2(3^0 + 7)$$
$$\Rightarrow \quad \log_2 2 + 2 = \log_2 8 \qquad (\because x^0 = 1)$$
$$\Rightarrow \quad 1 + 2 = \log_2 2^3$$
$$\Rightarrow \quad 3 = 3$$

Similarly, at $\quad x = 2$

$$\log_2(1 + 3) + 2 = \log_2(9 + 7)$$
$$\Rightarrow \quad (\log_2 4) + 2 = \log_2(16)$$
$$\Rightarrow \quad 4 = 4$$

Hence (d) is the correct answer

39. $x^{\log_5 x} > 5$

Taking logarithm with base 5 we have

$$(\log_5 x)(\log_5 x) > 1$$
$$\Rightarrow \quad (\log_5 x)^2 > 1$$
$$\Rightarrow \quad (\log_5 x)^2 - 1 > 0$$
$$\Rightarrow \quad (\log_5 x - 1)(\log_5 x + 1) > 0$$
$$\Rightarrow \quad \log_5 x > 1 \quad \text{Or} \quad \log_5 x < -1$$
$$\Rightarrow \quad x > 5 \quad \text{or} \quad x < \frac{1}{5}$$

Also, x must be greater than zero

$$\therefore \quad x \in (0, 1/5) \cup (5, \infty)$$

Alternatively: Go through options

Let us assume $x = 25$

then $\quad 25^{\log_5 25} = 625 > 5,$ $\quad$ which is true

Hence option (d) is correct.

Again consider $x = 5$, then

$5^{\log_5 5} = 5$, which is not greater than 5.

Hence choices (a) and (b) are wrong.

Thus option (d) is correct.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yourshahbooks.wordpress.com
For More Visit : www.LearnEngineering.in

40. $\log x^3 - \log 3x = 2\log 2 + \log 3$

$$\log \frac{x^2}{3} = \log 12$$

$$\Rightarrow \quad \frac{x^2}{3} = 12 \Rightarrow x = \pm 6$$

But at $x = -6$, $\log 3x$ and $\log x^3$ are not defined.

Hence $x = 6$ is the only correct answer.

41. $\log_5(2x+3) < \log_5 7$

$$\Rightarrow \quad (2x+3) < 7$$

$$\Rightarrow \quad 2x < 4$$

$$\Rightarrow \quad x < 2$$

But $(2x+3) > 0 \Rightarrow x > -\frac{3}{2}$

$$\therefore \quad x \in \left(-\frac{3}{2}, 2\right)$$

42. Best way is to go through options.

Alternatively: Suppose $\log_2 x = t$, then

$$\log_{2\sqrt{x}} x = \frac{\log_2 x}{\log_2 2\sqrt{x}} = \frac{t}{1+\frac{t}{2}}$$

and $$\log_{2x}\sqrt{x} = \frac{\log_2 \sqrt{x}}{\log_2 2x} = \frac{\frac{1}{2}\log_2 x}{1+\log_2 x} = \frac{t/2}{1+t}$$

$$\log_{2\sqrt{x}} x + \log_{2x}\sqrt{x} = \frac{2t}{2+t} + \frac{t}{2+2t} = 0$$

$$2t(2+2t) + t(2+t) = 0$$

$$\Rightarrow \quad 4t + 4t^2 + 2t + t^2 = 0$$

$$\Rightarrow \quad 5t^2 + 6t = 0$$

$$\Rightarrow \quad t(5t+6) = 0$$

$$\Rightarrow \quad t = 0 \text{ or } t = -\frac{6}{5}$$

$$\Rightarrow \quad \log_2 x = 0 \Rightarrow x = 2^0 = 1$$

and $$\log_2 x = -\frac{6}{5} \Rightarrow x = 2^{-6/5}$$

$$\therefore \quad x = 1 \text{ and } x = 2^{-6/5}$$

43. Clearly option (b) is wrong

$$\therefore \quad \frac{x}{1+x} < x \qquad \text{for every } x > 1$$

Again $$\frac{x}{1+x} < \log(x+1); \; x > 1$$

Hence option (c) is also wrong

and $$\log(x+1) < x$$

$\therefore$ Option (d) is correct.

44. $$\frac{k}{l} > \frac{k+1}{l+1} \quad \text{for } (k, l) > 0 \text{ and } k > l$$

Let $k = x+1$ and $l = x$

$$\therefore \quad \frac{x+1}{x} > \frac{(x+1)+1}{(x)+1}$$

$\because \quad (x+1) > x$

$$\therefore \quad \frac{\log(x+1)}{\log x} > \frac{\log(x+2)}{\log(x+1)}$$

$$\Rightarrow \quad \log_x(x+1) > \log_{x+1}(x+2)$$

45. $a = 1 + \log_x yz = \log_x x + \log_x yz = \log_x xyz$

Similarly $b = \log_y xyz$

and $c = \log_z xyz$

Now, $ab + bc + ca = abc\left[\frac{1}{c} + \frac{1}{a} + \frac{1}{b}\right]$

$$= abc\left[\frac{1}{\log_x xyz} + \frac{1}{\log_y xyz} + \frac{1}{\log_z xyz}\right]$$

$$= abc\left[\log_{xyz} x + \log_{xyz} y + \log_{xyz} z\right]$$

$$= abc\left[\log_{xyz} xyz\right] = abc$$

46. $\log_3(\log_2 x) + \log_{1/3}(\log_{1/2} y) = 1$

$$\Rightarrow \quad \log_3(\log_2 x) - \log_3(\log_{1/2} y) = 1$$

$$\Rightarrow \quad \log_3\left(\frac{\log_2 x}{\log_{1/2} y}\right) = 1 = \log_3 3$$

$$\Rightarrow \quad \frac{\log_2 x}{\log_{1/2} y} = 3$$

$$\Rightarrow \quad \log_2 x = 3\log_{1/2} y = -3\log_2 y$$

$$\Rightarrow \quad x = y^{-3}$$

$$\Rightarrow \quad xy^3 = 1 \qquad \text{...(1)}$$

Again dividing eq. (1) by $xy^2 = 4$, we get $y = \frac{1}{4}$

$$\therefore \quad x = 64, \qquad \text{[From eq. (1)]}$$

Alternatively : Go through options.

47. Best way is to check the options.

Alternatively :

$$\log_{(1/\sqrt{2})}(1/\sqrt{8}) = \log_2(4^x + 1) \cdot \log_2(4^{x-1} + 4)$$

$$\Rightarrow \quad \frac{\log_2(1/\sqrt{8})}{\log_2(1/\sqrt{2})} = \log_2(4^x + 1) \cdot \log_2(4^{x-1} + 4)$$

$$\Rightarrow \quad 3 = [\log_2 4 + \log_2(4^x + 1)][\log_2(4^x + 1)]$$

$$\Rightarrow \quad (2+t)t = 3, \qquad \text{where } t = \log_2(4^x + 1)$$

$$\Rightarrow \quad t = -3, 1$$

If $\log_2(4^x + 1) = -3$, then $4^x = -7/8$ which is not possible

If $\log_2(4^x + 1) = 1$, then $4^x + 1 \Rightarrow 2 \Rightarrow 4^x = 1$

$$\Rightarrow \quad x = 0$$

48. $\log_3 4 - \log_3(3x+1) = \log_3 \frac{3}{(5x-2)}$

$$\log_3 \frac{4}{3x+1} = \log_3 \frac{3}{(5x-2)}$$

$$\Rightarrow \quad \frac{4}{(3x+1)} = \frac{3}{(5x-2)}$$

$$\Rightarrow \quad 20x - 8 = 9x + 3$$

$$\Rightarrow \quad 11x = 11$$

$$\Rightarrow \quad x = 1$$

Alternatively : Go through options.

49. $|3 - 4x| > 2$

$$\Rightarrow \quad (3-4x) > 2 \text{ or } -(3-4x) > 2$$

$$\Rightarrow \quad x < \frac{1}{4} \text{ or } x > \frac{5}{4}$$

$$\therefore \quad x \in \left(-\infty, \frac{1}{4}\right) \cup \left(\frac{5}{4}, \infty\right)$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

50. $\log_{100} |x+y| = \frac{1}{2}$

$\Rightarrow |x+y| = 10$ (– 10 is inadmissible)

and $\log_{10} y - \log_{10} |x| = \log_{100} 4$

$\Rightarrow \log_{10}\left(\frac{y}{|x|}\right) = \log_{10} 2$

$\Rightarrow \frac{y}{|x|} = 2 \Rightarrow y = 2|x|$

Hence we have $|x+2|x||=10$

Now, if $x > 0$, then we have

$3x = 10 \Rightarrow x = \frac{10}{3}$

and if $x < 0$, then

$|-x| = 10$

$\Rightarrow x = -10$

Thus, $x = \frac{10}{3}, \quad y = \frac{20}{3}$

and $x = -10, \quad y = 20$

LEVEL 2

1. See the theory of this chapter

3. $\log_2\left(\frac{x}{y}\right) + \log_4\left(\frac{x}{y}\right)^2 + \log_8\left(\frac{x}{y}\right)^3 + \ldots$

$= \log_2\left(\frac{x}{y}\right) + \log_2\left(\frac{x}{y}\right) + \log_2\left(\frac{x}{y}\right) + \ldots$

$= \log_2\left(\frac{x}{y} \times \frac{x}{y} \times \frac{x}{y} \times \ldots n \text{ times}\right)$

$= \log_2\left(\frac{x}{y}\right)^n = n \log_2\left(\frac{x}{y}\right)$

4. $\log m + \log m^2 + \log m^3 + \ldots + \log m^n$

$= \log (m \cdot m^2 \cdot m^3 \ldots m^n)$

$= \log m^{(1+2+3+\ldots n)}$

$= \log (m)^{\frac{n(n+1)}{2}}$

$= \frac{n(n+1)}{2} \log m$

5. Given that $9^n < 10^8$

Taking log to both sides

$\log 9^n < \log 10^8$

$\Rightarrow 2n \log 3 < 8 \log 10$

$\Rightarrow 2n \times 0.4771 < 8$

$\Rightarrow n \times 0.9542 < 8$

$\Rightarrow n < \frac{8}{0.9542}$

$n < 8.3839$

$\therefore n = 8 \quad (\because n \in N)$

6. Taking log of both sides with base 3 we have,

$[\log_3 x^2 + (\log_3 x)^2 - 10] \log_3 x = -2 \log_3 x$

$\Rightarrow \log_3 x = 0$

or $(\log_3 x)^2 + 2 \log_3 x - 8 = 0$

$\Rightarrow x = 1, \quad (\log_3 x + 4)(\log_3 x - 2) = 0$

$\Rightarrow x = 1, \quad \log_3 x = -4$ and $\log_3 x = 2$

$\Rightarrow x = 1, \quad x = \frac{1}{81}$ and $x = 9$

$\therefore x = \left\{1, \frac{1}{81}, 9\right\}$

7. $(x-3) > 0 \Rightarrow x > 3$

and $(2-x) > 0$ and $2 - x \neq 1$

$\therefore x < 2$ and $x \neq 1$

Clearly there is no single value for which these inequalities are satisfied. Thus the set of its solutions is empty.

8. $\log_3 30 = \frac{1}{a} \Rightarrow a = \log_{30} 3$ and $\log_5 30 = \frac{1}{b}$

$\Rightarrow b = \log_{30} 5$

$3 \log_{30} 2 = 3[\log_{30}(30/15)]$

$= 3[\log_{30} 30 - \log_{30} 15]$

$= 3[\log_{30} 30 - (\log_{30} 3 + \log_{30} 5)]$

$= 3[1 - a - b]$

9. $x^{\log_x |3-x|^2} = 4$

$\Rightarrow |3-x|^2 = 4$

$\Rightarrow |3-x| = 2$ (– 2 is inadmissible)

$\Rightarrow (3-x) = 2$ or $-(3-x) = 2$

$\Rightarrow x = 1$ or $x = 5$

10. Let $2^{x^2+2} = t$, then

$4^{x^2+2} - 9 \cdot 2^{x^2+2} + 8 = 0$ becomes

$t^2 - 9t + 8 = 0$

$\Rightarrow t = 1, 8$

$\Rightarrow 2^{x^2+2} = 1$

$\Rightarrow x^2 + 2 = 0$ but this has no solution

If $2^{x^2+2} = 8 \Rightarrow x^2 + 2 = 3$

$\Rightarrow x^2 = 1 \Rightarrow x = \pm 1$

11. Taking log of both sides, we get

$\left[(\log_3 x)^2 - \frac{9}{2} \log_3 x + 5\right] \log_3 x = \frac{3}{2}$

$\Rightarrow 2(\log_3 x)^3 - 9(\log_3 x)^2 + 10(\log_3 x) - 3 = 0$

$\Rightarrow (\log_3 x - 1)(\log_3 x - 3)(2 \log_3 x - 1) = 0$

$\Rightarrow \log_3 x = 1, \log_3 x = 3, 2 \log_3 x = 1$

$\Rightarrow x = 3, x = 27, x = \sqrt{3}$

i.e., $x = (\sqrt{3}, 3, 27)$

12. $x^2 + 6x + 8 > 0$ and $2x^2 + 2x + 3 > 0$

$\Rightarrow (x+4)(x+2) > 0$ and $\left(x + \frac{1}{2}\right)^2 + \frac{5}{4} > 0$

$\Rightarrow x \in (-\infty, -4) \cup (-2, \infty)$ ...(1)

The given equation can be written as

$\log_{(2x^2+2x+3)}(x^2 - 2x) = 1$

$\Rightarrow x^2 - 2x = 2x^2 + 2x + 3$

$\Rightarrow x^2 + 4x + 3 = 0$

$\Rightarrow x = -1, -3$

but at $x = -3$, $\log_{(x^2+6x+8)}$ is not defined

Hence, $x = -1$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

13. Let $u = \log_{10} p$, then the given inequality reduces to

$$(2+u)^2 + (1+u)^2 + u \le 9$$
$$2u^2 + 7u + 5 \le 9$$
$$2u^2 + 7u - 4 \le 0$$
$$\Rightarrow \quad 2u^2 + 8u - u - 4 \le 0$$
$$\Rightarrow \quad 2u(u+4) - 1(u+4) \le 0$$
$$\Rightarrow \quad (u+4)(2u-1) \le 0$$
$$\Rightarrow \quad -4 \le u \le \frac{1}{2}$$
$$\Rightarrow \quad -4 \le \log_{10} p \le \frac{1}{2}$$
$$\Rightarrow \quad 10^{-4} \le p \le 10^{1/2}$$

14. Let $u = \log_2 x$, then

$$2\log_2 \log_2 x + \log_{1/2} \log_2 (2\sqrt{2}x) = 1$$
$$\Rightarrow \quad 2\log_2 u + \log_{1/2} (\log_2 2^{3/2} + u) = 1$$
$$\Rightarrow \quad \log_2 u^2 - \log_2 (3/2 + u) = 1$$
$$\Rightarrow \quad \log_2 \left(\frac{u^2}{3/2+u}\right) = 1$$
$$\Rightarrow \quad u^2 = 2\left(\frac{3}{2} + u\right)$$
$$\Rightarrow \quad u^2 - 2u - 3 = 0$$
$$\Rightarrow \quad u = -1, 3$$
$$\Rightarrow \quad x = \frac{1}{2}, 8$$

but at $x = \frac{1}{2}$, $2\log_2 \log_2 x$ is undefined

Hence, $x = 8$

15. By changing the base to 2 the given equation becomes

$$\frac{\log_2 x^2}{\log_2 x/2} + \frac{40 \log_2 \sqrt{x}}{\log_2 4x} - 14\frac{\log_2 x^3}{\log_2 16x} = 0$$
$$\Rightarrow \quad \frac{2\log_2 x}{\log_2 x - 1} + 20\frac{\log_2 x}{2 + \log_2 x} - 42\frac{\log_2 x}{4 + \log_2 x} = 0$$

Let $t = \log_2 x$, then we have

$$2t(4+t)(2+t) - 42t(t-1)(t+2) + 20t(t-1)(t+4) = 0$$
$$\Rightarrow \quad 2t[t^2 + 6t + 8 - 21t^2 - 21t + 42 + 10t^2 + 30t - 40] = 0$$
$$\Rightarrow \quad t[2t^2 - 3t - 2] = 0$$
$$\Rightarrow \quad t = 0, \quad t = 2, \quad t = -\frac{1}{2}$$
$$\Rightarrow \quad x = 1, \quad x = 4, \quad x = \frac{1}{\sqrt{2}}$$

16. $m > 0$ and $n > 0$ and $m \ne 1$

i.e., $25 - x^2 > 0$ and $x \ne \pm 3$

and $24 - 2x - x^2 > 0$

$\Rightarrow \quad -5 < x < 5, \; x \ne \pm 3$

and $x^2 + 2x - 24 < 0$

$-5 < x < 5, \quad x \ne \pm 3$

and $-6 < x < 4$

$x \in (-5, 4) - \{-3, 3\}$ ...(1)

Case 1. $0 < m < 1 \Rightarrow 9 < x^2 < 25$

$\Rightarrow \quad x \in (-5, -3) \cup (3, 5)$ ...(2)

$\therefore$ The given inequality can be written as

$$\frac{24 - 2x - x^2}{14} < \frac{25 - x^2}{16}$$
$$\Rightarrow \quad x^2 + 16x - 17 > 0$$
$$\Rightarrow \quad (x+17)(x-1) > 0$$
$$\Rightarrow \quad x < -17 \text{ or } x > 1$$

From (1) and (2), we have

$$x \in (3, 4)$$

Case 2. If $m > 1$, *i.e.*, $\frac{25 - x^2}{16} > 1$

$$\Rightarrow \quad x \in (-3, 3)$$ ...(3)

The given inequality reduces to

$$\frac{24 - 2x - x^2}{14} > \frac{25 - x^2}{16}$$
$$\Rightarrow \quad x^2 + 16x - 17 < 0$$
$$\Rightarrow \quad -17 < x < 1$$

Thus combining with (3), we get

$$x \in (-3, 1)$$

but $x \in \{-5, 4) - \{-3, 3\}$ by (1)

thus $x \in (-3, 1)$

Hence the required value of x should lie in $(-3, 1) \cup (3, 4)$

17. $$\log_2 \left(\frac{5}{2}\right) = (\log_2 5) - 1$$

but $\because \quad (\log_2 5) - 1 > (\log_2 4) - 1$

$$\therefore \quad \log_{3/10}\left[\frac{10}{7}(\log_2 5 - 1)\right] < \log_{3/10}\frac{10}{7} < \log_{3/10} 1 \; (= 0)$$

($\because \log_a x < \log_a y$ if $x > y$ for $0 < a < 1$)

Since $\log_{3/10}\frac{10}{7}(\log_2 5 - 1) < 0$

Hence, the first inequality is true only if

$$\sqrt{(x-8)(2-x)} = 0$$
$$\Rightarrow \quad x = 8 \text{ or } x = 2$$

If $x = 8$, then $\frac{2^x}{8} - (2^5 - 1) = 1 > 0$

If $x = 2$ then $\frac{2^x}{8} - (2^5 - 1) = \frac{1}{2} - (2^5 - 1) < 0$

Hence $x = 8$ is the required value.

18. $$2\log_{10} x - \log_x \frac{1}{100} = 2\log_{10} x - \frac{\log_{10} 10^{-2}}{\log_{10} x}$$
$$= 2\log_{10} x + \frac{2}{\log_{10} x}$$
$$= 2\left(\log_{10} x + \frac{1}{\log_{10} x}\right)$$

Since $x > 1 \Rightarrow \log_{10} x > 0$

But since AM $\ge$ GM

$$\therefore \quad \frac{\left(\log_{10} x + \frac{1}{\log_{10} x}\right)}{2} \ge \sqrt{\log_{10} x \times \frac{1}{\log_{10} x}}$$
$$\Rightarrow \quad \log_{10} x + \frac{1}{\log_{10} x} \ge 2$$
$$\Rightarrow \quad 2\left(\log_{10} x + \frac{1}{\log_{10} x}\right) \ge 4$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

For $x = 10$, $\quad 2\left(\log_{10} x + \dfrac{1}{\log_{10} x}\right) \geq 4$

Hence the least value of $\left(\log_{10} x - \log_x \dfrac{1}{100}\right)$ is 4.

19. We have, $x^{[(3/4)(\log_2 x)^2 + \log_2 x - (5/4)]} = \sqrt{2}$

$$\Rightarrow \quad \log_x \sqrt{2} = \frac{3}{4}(\log_2 x)^2 + \log_2 x - \left(\frac{5}{4}\right)$$

$$\Rightarrow \quad \frac{\log \sqrt{2}}{\log x} = \frac{3}{4}(\log_2 x)^2 + \log_2 x - \left(\frac{5}{4}\right)$$

$$\Rightarrow \quad \log \sqrt{2} = \log x\left[\frac{3}{4}(\log_2 x)^2 + \log_2 x - \left(\frac{5}{4}\right)\right]$$

$$\Rightarrow \quad \log_2 \sqrt{2} = \log_2 x\left[\frac{3}{4}(\log_2 x)^2 + \log_2 x - \left(\frac{5}{4}\right)\right]$$

$$\Rightarrow \quad \frac{1}{2} = \alpha\left[\frac{3}{4}\alpha^2 + \alpha - \frac{5}{4}\right] \qquad (\text{say } \alpha = \log_2 x)$$

$$\Rightarrow \quad 2 = 3\alpha^3 + 4\alpha^2 - 5\alpha$$

$$\Rightarrow \quad 3\alpha^3 + 4\alpha^2 - 5\alpha - 2 = 0$$

$$\Rightarrow \quad (\alpha - 1)(3\alpha^2 + 7\alpha + 2) = 0$$

$$\Rightarrow \quad \alpha = 1 \Rightarrow \log_2 x = 1 \Rightarrow x = 2$$

Again $\quad 3\alpha^2 + 7\alpha + 2 = 0$

$$\Rightarrow \quad \alpha = -2, -\frac{1}{3}$$

$$\Rightarrow \quad \log_2 x = -2 \quad \text{and} \quad \log_2 x = \frac{-1}{3}$$

$$\Rightarrow \quad x = \frac{1}{4} \Rightarrow x = 2^{-1/3}$$

Hence $\quad x = 2, \dfrac{1}{4}, 2^{-1/3}$

Thus option (d) is most appropriate.

20. $\quad 2x + 3 > 0 \quad \text{and} \quad 2x + 3 \neq 1$

$$\Rightarrow \quad x > -\frac{3}{2} \quad \text{and} \quad x \neq -1$$

And $\quad 3x + 7 > 0 \quad \text{and} \quad 3x + 7 \neq 1$

$$\Rightarrow \quad x > \frac{-7}{3} \Rightarrow x \neq -2$$

Now, $\log_{(2x+3)}(6x^2 + 23x + 21)$

$$= 4 - \log_{(3x+7)}(4x^2 + 12x + 9)$$

$$\Rightarrow \log_{(2x+3)}(2x + 3)(3x + 7) = 4 - \log_{(3x+7)}(2x + 3)^2$$

$$\Rightarrow \log_{(2x+3)}(2x + 3) + \log_{(2x+3)}(3x + 7) = 4 - 2\log_{(3x+7)}(2x + 3)$$

$$\Rightarrow 1 + \log_{(2x+3)}(3x + 7) + 2\log_{(3x+7)}(2x + 3) - 4 = 0$$

$$\Rightarrow \quad \frac{2\log(2x + 3)}{\log(3x + 7)} + \frac{\log(3x + 7)}{\log(2x + 3)} - 3 = 0 \qquad \ldots(1)$$

Putting $\dfrac{\log(2x + 3)}{\log(3x + 7)} = y$ in eq. (1), we get

$$2y + \frac{1}{y} - 3 = 0 \Rightarrow 2y^2 - 3y + 1 = 0$$

$$(2y - 1)(y - 1) = 0$$

$$\Rightarrow \quad y = \frac{1}{2} \quad \text{and} \quad y = 1$$

Now, when $y = \dfrac{1}{2}$

$$\therefore \quad \frac{\log(2x + 3)}{\log(3x + 7)} = \frac{1}{2}$$

$$\Rightarrow \quad (2x + 3)^2 = (3x + 7)$$

$$4x^2 + 9x + 2 = 0$$

$$(4x + 1)(x + 2) = 0$$

$$x = -\frac{1}{4}, -2$$

Again if $y = 1$, then

$$\frac{\log(2x + 3)}{\log(3x + 1)} = 1$$

$$\Rightarrow \quad 2x + 3 = 3x + 7$$

$$x = -4$$

Since we know that $x > -\dfrac{3}{2}$ and $x > -\dfrac{7}{3}$

therefore $x = -2$ and $x = -4$ are not admissible values

Again since $x \neq -1$ and $x \neq -2$

Hence $x = -2$ is also inadmissible value

Thus, $x = -\dfrac{1}{4}$ is only possible value.

∴ Option (b) is correct.

Alternatively: Go through options

21. $x > 0, x \neq 1$

Since exponential function assumes positive value, so we must have $(x - 1)^7 > 0$ *i.e.*, $x > 1$.

Taking logarithm on both sides, we get

$$(\log_3 x^2 - 2\log_x 9)\log(x - 1) = 7\log(x - 1)$$

Either $\log(x - 1) = 0$ *i.e.*, $x = 2$

or $\quad \log_3 x^2 - 2\log_x 9 = 7$

$$\Rightarrow \quad 2(\log_3 x) - 4\log_x 3 = 7$$

$$\Rightarrow \quad 2t - \frac{4}{t} = 7, \qquad (\because t = \log_3 x)$$

$$\Rightarrow \quad 2t^2 - 7t - 4 = 0$$

$$\Rightarrow \quad t = 4, -\frac{1}{2}$$

$$\log_3 x = 4 \Rightarrow x = 81$$

If $\log_3 x = -\dfrac{1}{2}$, then $x = 3^{-1/2} < 1$, which is not the case

Hence, $\quad x = 2, 81$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

17 FUNCTIONS AND GRAPH

In the introduction of this chapter, I would like to mention that in the recent past this chapter contributed 3-12 questions in some years. Basically on an average 4-5 questions are being asked from this chapter. Besides most of the questions are based on the user defined (or self defined) functions which are rather easier in comparison to the problems based on the algebraic functions of pure mathematics. Still you are required to know some basic characteristic and applications of standard algebraic functions since they are not uncommon in CAT. Remember that a little bit of practice on this chapter will give you an edge over your competitor since the problems related to this chapter are very easy provided that these problems are being solved patiently. Moreover, you need not to remember any formula at all.

The structure of the chapter is as follows :

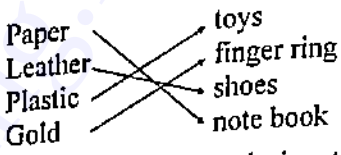
1. Function (An introduction)
2. Domain, range and graphs of different types of functions and operations on real valued functions.
3. Transformation of graphs.
4. Composite functions
5. Even and odd functions
6. Surjective and injective functions
7. Inverse functions
8. Binary operations
9. User defined (or self defined) operations (Also important for XAT)
10. Maxima and minima.

FUNCTIONS

Suppose we have to produce some required products from the raw materials, then the raw material is known as input and the desired products are known as output. For this we have to look into two points.

1. Every input must give some output (to avoid wastage of inputs)
2. No input should give more than one output. (To avoid ambiguity or variation from the desired output)

Thus, if the above conditions are satisfied by any input output relation, then this relation is called function, *e.g.*,



Here you can see that each input gives an output necessarily and no input give more than one output.

The collection of raw materials (*i.e.*, input) is called the "Domain" and the collection of products (or output) is called the "Range".

Definitions

1. **Domain :** Set of inputs (also called pre-images) is called domain.
2. **Codomain :** Set of possible outputs is called codomain.
3. **Range :** Set of actual outputs (also called images) is known as range.

Here, Range $\subseteq$ Codomain.

Mathematical Definition of Function

Let A and B be two non-empty sets, then a function from 'A' to 'B' is a rule that assigns to each element in a set A, one and only one element of a set B.

In general, the sets A and B need not be sets of real numbers. However, we consider only those functions for which A and B are both subsets of the real numbers.

If the number of elements in A be n and the number of elements in B be m, then the total number of functions that can be defined from A to B is n^m.

$$A \xrightarrow{f} B \quad \text{or} \quad f : A \longrightarrow B$$

If a pre-image is denoted by x and an image is denoted by y then we can write.

$$y = f(x)$$

Here, $f(x)$ is called the value of the function f at the point x.

Remember : A function cannot be multiple-valued. For example, the expression $\pm\sqrt{x}$ does not define a function of x, since it assigns two values to each positive x.

i.e., $y = f(x)$

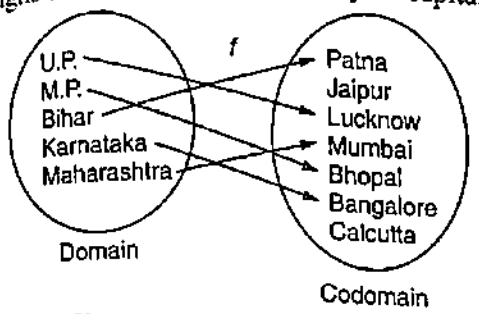
or $y = \pm\sqrt{x}$ is not a function since y assumes two values *i.e.*, $y = +\sqrt{x}$ and $y = -\sqrt{x}$, which is impossible.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

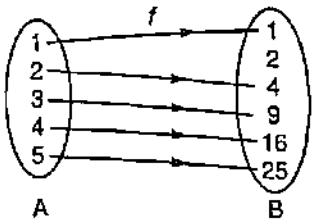
yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Methods of Representation of a Function

1. Verbal description : For example, if f is a function which assigns to each state in the country its capital city.



2. Arrow diagram :



Domain = {1, 2, 3, 4, 5}

Codomain = {1, 2, 4, 9, 16, 25}

Range = {1, 4, 9, 16, 25}

3. Tabular form :

e.g.,

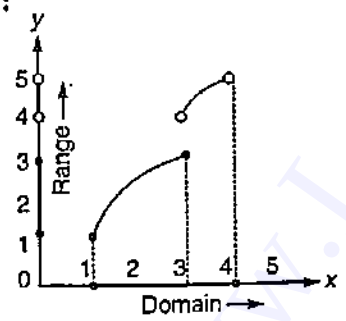
x	-1	-2	-3	0	1	2
y	-1	-8	-27	0	1	8

$D = \{-1, -2, -3, 0, 1, 2\}$

$R = \{-1, -8, -27, 0, 1, 8\}$

$f(2) = 8, \quad f(-2) = -8$ etc.

4. Graph :

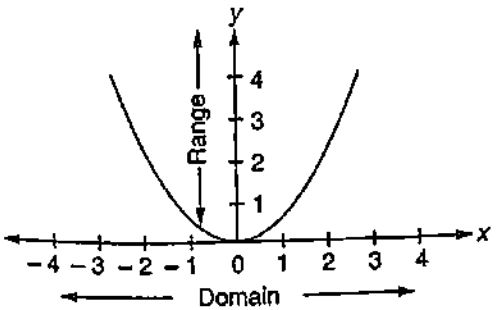


$D = [1, 4]$

$R = [1, 3] \cup (4, 5)$

5. Equation (or formula) :

$f : R \to R \quad f(x) = x^2 \ \forall x \in R$



$D = R$ (all real numbers),

Codomain = R (All real numbers)

(Range) $R = R^+ \cup \{0\}$

($R^+ \to$ set of positive real numbers)

DOMAIN & GRAPHS OF DIFFERENT TYPES OF FUNCTIONS

A function $f : A \to B$ is called a real valued if the image of every element of A, under f is a real number *i.e.*, if $f(x) \in R \ \forall x \in A$

$$y = f(x)$$

$x \to$ independent variable

$y \to$ dependent variable

Domain : The values of x must be such that for every x, y must be real is called the domain.

In other words the value of x for which the values of $f(x)$ is not a real number cannot be included in the domain of the function $y = f(x)$. There are five different common situations where the domain of $f(x)$ for some particular value(s) is not defined.

(a) When base of $(1/2n)^{th}$ power is negative for any $n \in I$:

If $x^{1/2} = k \Rightarrow x = k^2$ since k^2 is always positive even when x is negative, therefore this equality is not satisfied.

Hence the domain of definition of $y = \sqrt{x}$ is defined for only $x \geq 0$ *i.e.*, not defined for $x < 0$

EXAMPLES

Function	Domain defined for	Domain not defined for
$(x-3)^{1/2}$	$(x-3) \geq 0 \Rightarrow x \geq 3$	$(x-3) < 0 \Rightarrow x < 3$
$(x+4)^{1/6}$	$(x+4) \geq 0 \Rightarrow x \geq -4$	$(x+4) < 0 \Rightarrow x < -4$
$(\lvert x \rvert - 3)^{1/10}$	$\lvert x \rvert - 3 \geq 0 \Rightarrow \lvert x \rvert \geq 3$ $\Rightarrow x \leq -3$ and $x \geq 3$	$\lvert x \rvert - 3 < 0 \Rightarrow \lvert x \rvert < 3$ $\Rightarrow -3 < x < 3$

(b) When denominator of a function is zero : If $\frac{x^2+1}{2x}$ is such that $x = 0$, then $\frac{x^2+1}{2x}$ is not defined *i.e.*, the function $y = \frac{1}{x}$ is defined only for $x \neq 0$ (*i.e.*, non zero real numbers) therefore the domain of function $f(x) = \frac{1}{x}$ is $(-\infty, 0) \cup (0, \infty)$ or $(-\infty, \infty) - \{0\}$ or $-\infty < x < 0 \cup 0 < x < \infty$.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

EXAMPLES

Function	Domain defined for	Domain not defined for
$\frac{1}{(x-3)}$	$x \in (-\infty, \infty) - \{3\}$	$x = 3$
$\frac{19}{(x^2-16)}$	$x \in (-\infty, \infty) - \{\pm 4\}$	$x = \pm 4$
$\frac{(x^2+4)}{\lvert x\rvert - 5}$	$x \in (-\infty, \infty) - \{\pm 5\}$	$x = \pm 5$
$\frac{1}{(\log x) - 1}$	$x \in (0, \infty) - \{e\}$	$x = e$

(c) **When function becomes 0° :**
Example $y = (\lvert x\rvert - 3)^{(x-3)}$ is undefined for $x = 3$

(d) **When the value of x in $\log_a x$ becomes non-positive** (*i.e*, negative or zero)
$\therefore \log_{10}(-3)$ is not defined.

(e) **When the value of a in $\log_a x$ becomes 1 or 0 or negative.**
$\therefore \log_1 5$ is not defined.
Similarly, $\log_{(-3)} 10$ is not defined.

EXAMPLES

Function	Domain defined for	Domain not defined for
$\log_{10}(x-5)$	$(x-5) > 0 \Rightarrow x > 5$	$x - 5 \le 0 \Rightarrow x \le 5$
$\log_{15}(7-x)$	$(7-x) > 0 \Rightarrow x < 7$	$(7-x) \le 0 \Rightarrow x \ge 7$
$\log_{(x^2-4)} 12$	$(x^2-4) > 0$ and $x^2 - 4 \ne 1$ $\Rightarrow x < -2$ and $x^2 \ne 5$ and $x > 2$ and $x \ne \pm\sqrt{5}$ $\therefore x \in (-\infty, -2) \cup (2, \infty)$ excluding $\pm\sqrt{5}$	$x^2 - 4 \le 0$ and $x^2 - 4 = 1$ $\Rightarrow -2 \le x \le 2$ and $x^2 = 5$ and $x = \pm\sqrt{5}$ $\therefore x \in [-2, 2] \cup \{\pm\sqrt{5}\}$

Range : The set of all images of all the elements of domain. It can be obtained by one of the following ways.

(i) Obtaining x in terms of y and then finding the values of y for which we get real x.

e.g., $$f(x) = \frac{x+2}{x+3} = y$$

$$\Rightarrow \quad yx + 3y = x + 2$$
$$\Rightarrow \quad x(y-1) = (2 - 3y)$$
$$\Rightarrow \quad x = \frac{2-3y}{y-1}$$

for $y = 1$, there is no real value of x for the above equation is satisfied hence.

Range $= R - \{1\}$ $\quad R \to$ Set of real numbers

(ii) If the given function is continuous, then Range $= [l, m]$, where l is the least value of the function in its domain and m is the maximum value in its domain.

(iii) By sketching the graphs of the function.

Equal Function : Two functions f and g are said to be equal iff (*i.e.*, if and only if)

(i) the domain of f = domain of g
(ii) the co-domain of f = co-domain of g, and
(iii) $f(x) = g(x)$ for every x belong to their common domain.

If two functions f and g are equal then we write $f = g$, e.g.,

1. Let $A = \{1, 2\}$, $B = \{3, 6\}$ and $f : A \to B$ given by $f(x) = x^2 + 2$ and $g : A \to B$ given by $g(x) = 3x$. Then we observe that f and g have the same domain and co-domain also.

$$f(1) = 3 = g(1)$$
$$f(2) = 6 = g(2)$$

Hence $f = g$

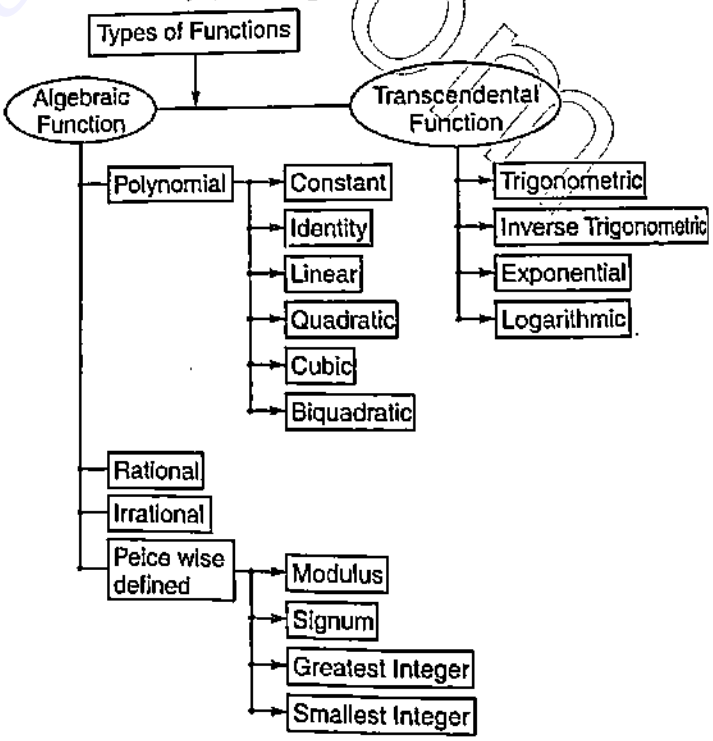
2. Let $f(x) = \sqrt{\frac{x-3}{x-5}}$ and $g(x) = \frac{\sqrt{x-3}}{\sqrt{x-5}}$

$f(x)$ is defined when $x - 3$ and $x - 5$ both are negative since $\frac{x-3}{x-5} > 0$

But when $x - 3$ and $x - 5$ are negative then both $\sqrt{x-3}$ and $\sqrt{x-5}$ are undefined. Hence $g(x)$ is undefined

$\therefore$ Domain of $f \ne$ Domain of g

Hence $f(x)$ and $g(x)$ are different.



ALGEBRAIC FUNCTIONS

A function which contains finite number of terms having different powers of independent variable (x) and the operations $+, -, \times, \div$ is called algebraic function *e.g.*,

$$4x^2 - 3x^{1/2} + 7, \quad 5x^2 + 13x + 8, \quad \frac{x^2+1}{x^2-2} \text{ etc.}$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

(A–1) Polynomial Functions : A function $f(x)$ of the following form is known as polynomial function.

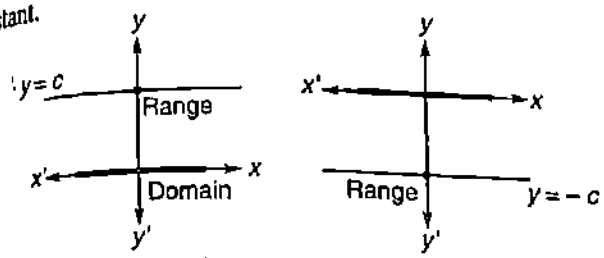
$$f(x) = a_0 + a_1x + a_2x^2 + \ldots + a_nx^n$$

where $n \in N$ and $a_1, a_2, a_3 \ldots a_n \in R$

Domain : It is the set of real numbers

Range : It is the set of real numbers

(A–1.1) Constant function : $f(x) = c$, $c \in R$, c is constant.

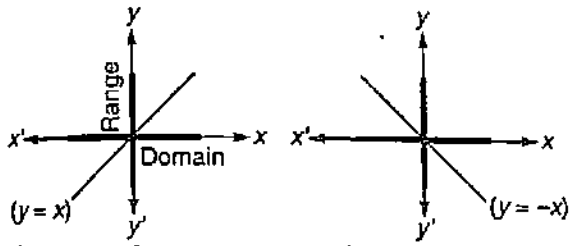


Domain : It is the set of real numbers

Range : It is a particular real number $\{c\}$

(A–1.2) Identity function

$y = x$ *i.e.*, $f(x) = x$



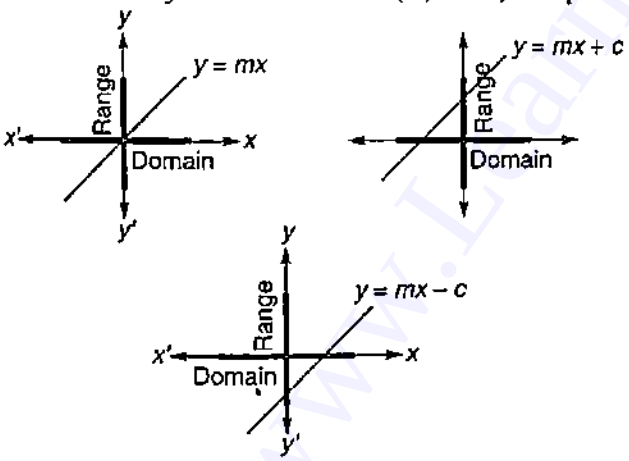
(*i.e.* input equals to output or pre image equals to image)

Domain → It is the set of real numbers.

Range → It is also the set of real numbers.

(A–1.3) Linear function

$y = mx + c$ $(m, c \in R, m$ is positive)

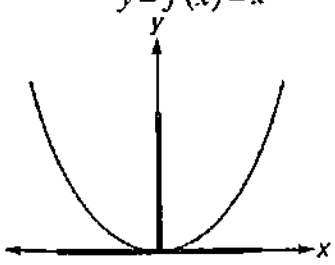


Domain → It is the set of real numbers.

Range → It is also the set of real numbers.

(A–1.4.) Quadratic function

$y = f(x) = x^2$



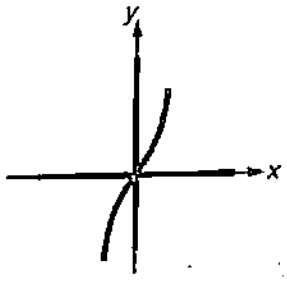
$\because \quad f(x) = f(-x)$

$\therefore$ Graph of $f(x)$ is symmetric about y-axis.

Domain → It is the set of real values.

Range → $R^+ \cup \{0\}$ *i.e.*, non-negative real numbers

(A–1.5) Cubic equation



$$y = f(x) = x^3$$

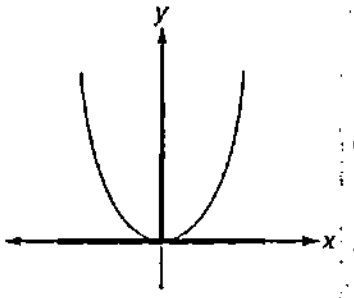
$\because \quad f(-x) = -f(x)$

$\therefore$ Graph of $f(x)$ is symmetric about origin

Domain → R (set of real numbers)

Range → R (set of real numbers)

(A–1.6) Biquadratic function



$$y = f(x) = x^4$$

$\because \quad f(x) = f(-x)$

$\therefore$ Graph of $f(x)$ is symmetric about y-axis.

Domain → R (set of real numbers)

Range → $R^+ \cup \{0\}$ *i.e.*, set of non-negative real numbers.

NOTE Graph of $f(x) = x^n$ is symmetric about
(a) y-axis if n is even
(b) origin if n is odd

Relation between the values of $x, x^2, x^3 \ldots$etc.

For	Order of the values	
$x \in (0, 1)$	$x > x^2 > x^3 > x^4 \ldots$	
$x \in (1, \infty)$	$x < x^2 < x^3 < x^4 \ldots$	
$x \in (-1, 0)$	$x < x^3 < x^5 \ldots$	(–ve values)
	$x^2 > x^4 > x^6 \ldots$	(+ve values)
$x \in (-\infty, -1)$	$x > x^3 > x^5 \ldots$	(–ve values)
	$x^2 < x^4 < x^6 \ldots$	(+ve values)

(A–2) Rational Function

$$f : A \to R; \quad f(x) = \frac{P(x)}{Q(x)}$$

Here $P(x)$ and $Q(x)$ are polynomial functions and $A = \{x : x \in R$ such that $Q(x) \neq 0\}$

EXAMPLES

	Function	Domain
1.	$\dfrac{a_0 + a_1x + a_2x^2 + \ldots + a_nx^n}{P_0 + P_1x + P_2x^2 + \ldots + P_nx^n}$	Set of all the real numbers except the real roots of the equation $P_0 + P_1x + P_2x^2 + \ldots + P_nx^n$
2.	$\dfrac{x^2 + 5x + 8}{x^2 - 5x + 6}$	$R - \{2, 3\}$
3.	$\dfrac{1}{x^n}; n \in N$	$R - \{0\}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

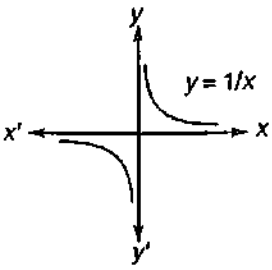
Graphs of some Rational Functions

1. $y = f(x) = \frac{1}{x}$

$\Rightarrow \quad f(x) = -f(-x)$

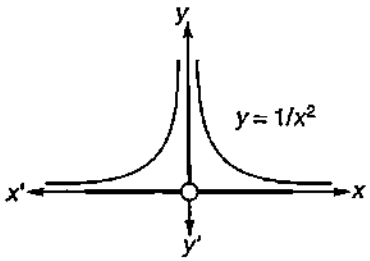
Domain $\rightarrow R - \{0\}$; set of non-zero rational numbers

Range $\rightarrow R - \{0\}$; set of non-zero rational numbers



2. $y = f(x) = \frac{1}{x^2}$

$\Rightarrow \quad f(x) = f(-x)$



Domain $\rightarrow R - \{0\}$ *i.e.*, set of non-zero rational numbers

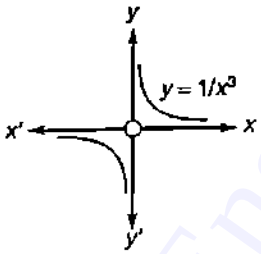
Range $\rightarrow R^+$ *i.e.*, set of positive rational numbers

3. $y = f(x) = \frac{1}{x^3}$

$f(-x) = -f(x)$

Domain $\rightarrow R - \{0\}$ *i.e.*, set of non-zero rational numbers

Range $\rightarrow R - \{0\}$ *i.e.*, set of non-zero rational numbers.



NOTE Graph of $f(x) = \frac{1}{x^n} (n \in N)$ is symmetric about

(a) y axis if n is even (b) origin if n is odd.

Relation between the values of $\frac{1}{x}, \frac{1}{x^2}, \frac{1}{x^3}, \frac{1}{x^4} \ldots$

For	The order of the values
$x \in (1, \infty)$	$\frac{1}{x} > \frac{1}{x^2} > \frac{1}{x^3} > \frac{1}{x^4} \ldots$
$x \in (0, 1)$	$\frac{1}{x} < \frac{1}{x^2} < \frac{1}{x^3} < \frac{1}{x^4} \ldots$
$x \in (-1, 0)$	$\frac{1}{x^2} < \frac{1}{x^4} < \frac{1}{x^6} \ldots$ (positive values) $\frac{1}{x} > \frac{1}{x^3} > \frac{1}{x^5} \ldots$ (negative values)
$x \in (-\infty, -1)$	$\frac{1}{x^2} > \frac{1}{x^4} > \frac{1}{x^6} \ldots$ (positive values) $\frac{1}{x} < \frac{1}{x^3} < \frac{1}{x^5} \ldots$ (negative values)

(A–3) Irrational Functions : The algebraic functions containing one or more terms having non-integral rational powers of x are called irrational functions. *e.g.*,

$y = 4x^2 + \sqrt{2x}$ then y is undefined if $x < 0$

$y = \sqrt{x}$ then y is undefined if $x < 0$

$y = \frac{5x^3 + 7x^2 + 3\sqrt{x}}{\sqrt{x+4}}$ then y is undefined if $x < 0$ and $\sqrt{x+4} \le 0$

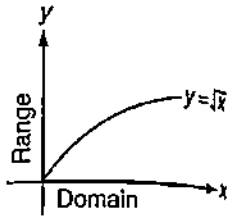
These functions are not defined for $f(x) < 0$. They are also not-defined if the denominator is zero.

Graphs of some Simple Irrational Functions

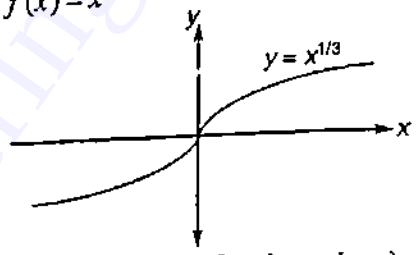
1. $y = f(x) = x^{1/2}$ (*i.e.*, $\sqrt{x}$)

Domain $\rightarrow R^+ \cup \{0\}$ *i.e.*, set of non-negative real numbers.

Range $\rightarrow R^+ \cup \{0\}$ *i.e.*, set of non-negative real numbers



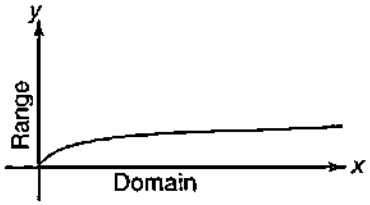
2. $y = f(x) = x^{1/3}$



Domain $\rightarrow$ (*i.e.*, set of real numbers)

Range $\rightarrow R$ (*i.e.*, set of real numbers)

3. $y = f(x) = x^{1/4}$



Domain $\rightarrow R^+ \cup \{0\}$ *i.e.*, set of non-negative real numbers

Range $\rightarrow R^+ \cup \{0\}$ *i.e.*, set of non-negative real numbers

Relation between the values of $x^{1/2}, x^{1/3}, x^{1/4} \ldots$

For	Order of the values
$x \in (1, \infty)$	$x^{1/2} > x^{1/3} > x^{1/4} > x^{1/5} > \ldots$
$x \in (0, 1)$	$x^{1/2} < x^{1/3} < x^{1/4} < x^{1/5} < \ldots$
$x \in (-1, 0)$	$x^{1/3} > x^{1/5} > x^{1/7} \ldots$
$x \in (-\infty, -1)$	$x^{1/3} < x^{1/5} < x^{1/7} \ldots$

(A–4) Piecewise Defined Functions

(A–4.1) Modulus function (or absolute-value function)

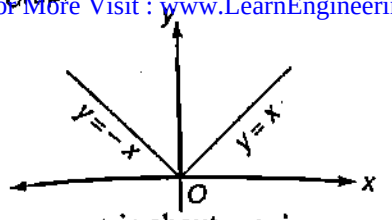
$$y = f(x) = |x| = \begin{cases} x \text{ if } x \ge 0 \\ -x \text{ if } x < 0 \end{cases}$$

$\because \quad f(-x) = f(x)$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

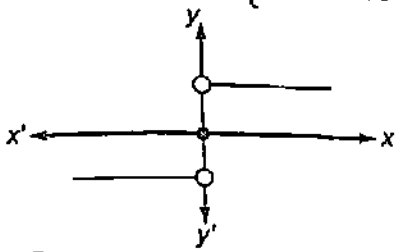
For More Visit : www.LearnEngineering.in



$\therefore f(x)$ is symmetric about y-axis.
Domain $\to R$ (*i.e.*, set of real numbers)
Range $\to R \cup \{0\}$ (*i.e.*, set of non-negative real numbers)

(A–4.2) Signum function [Sgn (x)]

$$y = f(x) = \text{Sgn}(x) = \begin{cases} 1 \text{ if } x > 0 \\ 0 \text{ if } x = 0 \\ -1 \text{ if } x < 0 \end{cases}$$



Domain $\to R$
Range $\to \{-1, 0, 1\}$
o indicates that the points (0, 1) and (0, – 1) are not included in the graphs.
• indicates that the point (0, 0) is included in the graphs

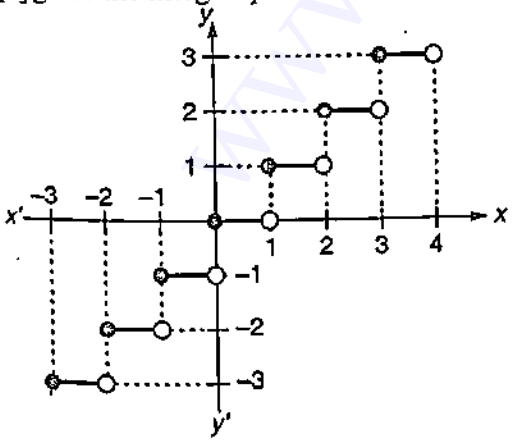
(a) $\text{Sgn}(x) = \begin{cases} \dfrac{x}{|x|} \text{ if } x \neq 0 \\ 0 \text{ if } x = 0 \end{cases}$ (b) $\text{Sgn}(x) = \begin{cases} \dfrac{|x|}{x} \text{ if } x \neq 0 \\ 0 \text{ if } x = 0 \end{cases}$

(A–4.3) Greatest integer function

$y = [x]$ denotes the greatest integer less than or equal to x.

e.g., $[2.3] = 2$
$[4.7] = 4$
$[5.2] = 5$
$[-8.3] = -9$
$[-13.2] = -14$

i.e., $[n] = n$, if n is an integer
and $[x] = n$, if $n \leq x < n+1$
Any real number x can be expressed as
Integral part + Fractional part
i.e., $I + f$ where $0 \leq f < 1$
then $[x]$ gives the integral part of x.



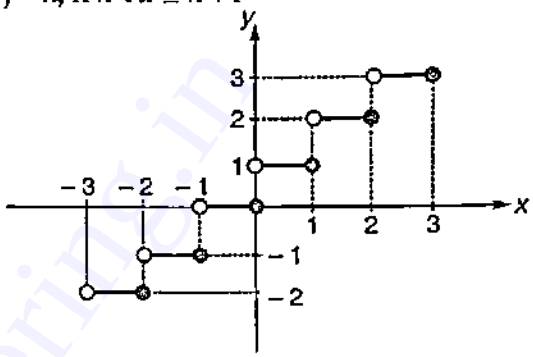
Domain $\to R$ (Set of real numbers)
Range $\to I$ (Integral values) $\{\ldots -3, -2, -1, 0, 1, 2, 3, \ldots \text{etc.}\}$
(It is also called the step functions)

(A–4.4) Smallest (or least) integer function

The function whose value at any number x is the smallest integer greater than or equal to x is called the least integer function *e.g.*, if $y = f(x) = \{x\}$ = Smallest integer, greater than or equal to x.

then, $\{2.8\} = 3$
$\{7.5\} = 8$
$\{-3.6\} = -3$
$\{-7.4\} = -7$

i.e., $\{n\} = n$, if n is an integer
and $\{x\} = n$, if $n < x \leq n+1$



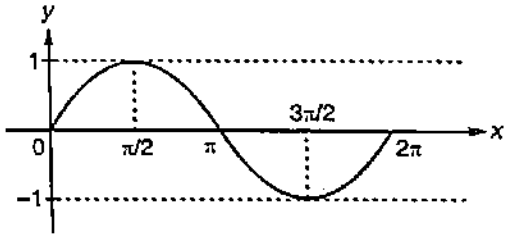
Domain $\to R$ (Set of real numbers)
Range $\to I$ (Integral values) e.g., $\{-3, -2, -1, 0, 1, 2, 3$ etc.$\}$

TRANSCENDENTAL FUNCTION

A function which is not an algebraic function is called a transcendental function.
e.g., trigonometry, logarithmic and exponential function.

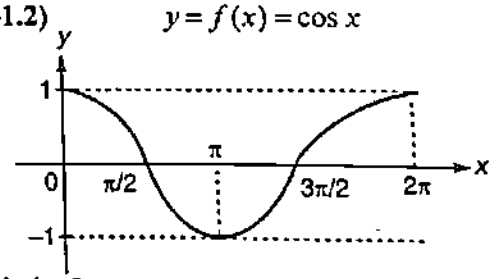
(T–1) Trigonometric Function :

(T–1.1) $y = f(x) = \sin x$



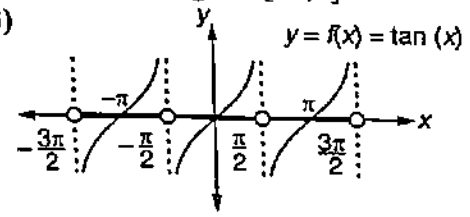
It is a periodic function with period 2π
Domain $\to R$ (Set of real numbers)
Range $\to [-1, 1]$ *i.e.*, $\{x \in R : -1 \leq x \leq 1\}$

(T–1.2) $y = f(x) = \cos x$



Period $= 2\pi$
Domain $\to R$ Range $\to [-1, 1]$

(T–1.3)



For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

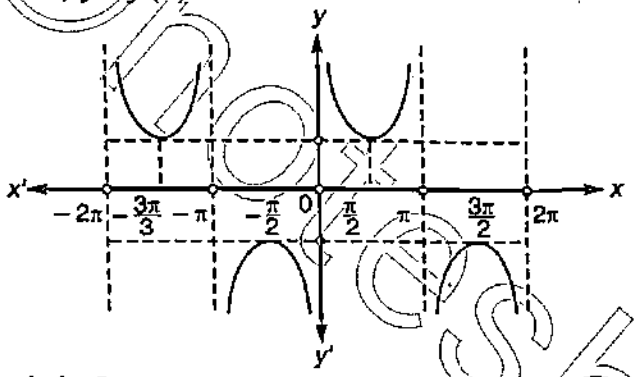
For More Visit : www.LearnEngineering.in

Period $= \pi$

Domain $\rightarrow R - \left\{(2n+1)\frac{\pi}{2},\ n \in I\right\}$

Range $\rightarrow R$ (Set of real numbers)

(T–1.4) $y = f(x) = \text{cosec } x$

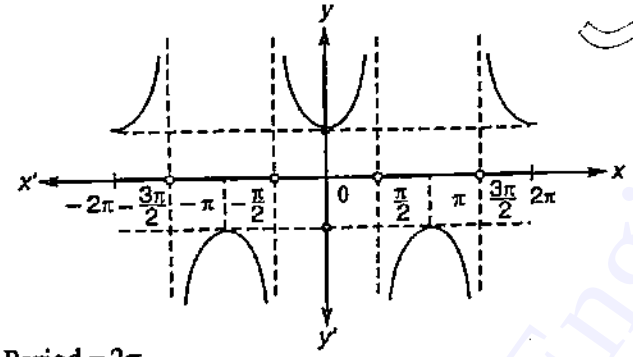


Period $= 2\pi$

Domain $\rightarrow R - \{n\pi : n \in I\}$

Range $(-\infty, -1] \cup [1, \infty)$

(T–1.5) $y = f(x) = \sec x$

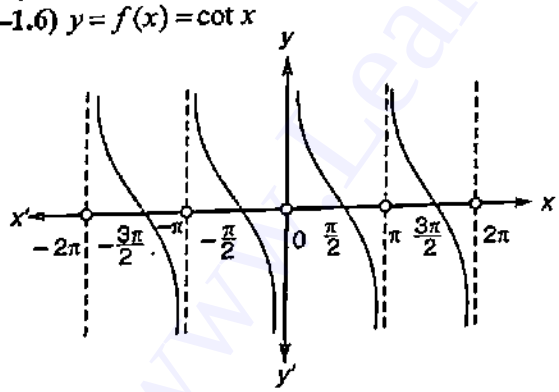


Period $= 2\pi$

Domain $\rightarrow R - \left\{(2n+1)\frac{\pi}{2} : n \in I\right\}$

Range $\rightarrow (-\infty, -1] \cup [1, \infty)$

(T–1.6) $y = f(x) = \cot x$



Period $= \pi$

Domain $\rightarrow \{n\pi : n \in I\}$

Range $\rightarrow R$ (set of real numbers)

(T–2.1) $y = \sin^{-1} x \Rightarrow x = \sin y \in [-1, 1]$

$\therefore$ Domain $= [-1, 1]$

(T–2.2) $y = \cos^{-1} x \Rightarrow x = \cos y \in [-1, 1]$

$\therefore$ Domain $= [-1, 1]$

(T–2.3) $y = \tan^{-1} x \Rightarrow x = \tan y \in R$

$\therefore$ Domain $= R$

(T–2.4) $y = \cot^{-1} x \Rightarrow x = \cot y \in R$

$\therefore$ Domain $= R$

(T–2.5) $y = \sec^{-1} x \Rightarrow x = \sec y \in (-\infty, -1] \cup [1, \infty)$

$\therefore$ Domain $= (-\infty, -1] \cup [1, \infty)$

(T–2.6) $y = \text{cosec}^{-1} x \Rightarrow x = \text{cosec } y \in (-\infty, -1] \cup [1, \infty)$

$\therefore$ Domain $= (-\infty, -1] \cup [1, \infty)$

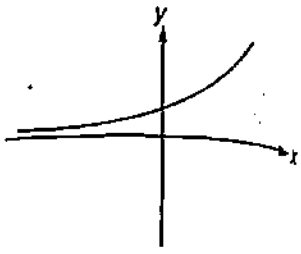
(T–3) Exponential Function

$y = a^x$

(i) $y = a^x; a > 1$

Domain $\rightarrow R$

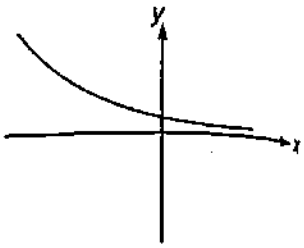
Range $\rightarrow R^+$



(ii) $y = a^x; 0 < a < 1$

Domain $\rightarrow R$

Range $\rightarrow R^+$



NOTE		
$0 < a < 1$	function is decreasing	i.e., $a^x > a^y$ if $x < y$
$a > 1$	function is increasing	i.e., $a^x < a^y$ if $x < y$

(iii) (a) $y = e^x$ and (b) $y = e^{|x|}$

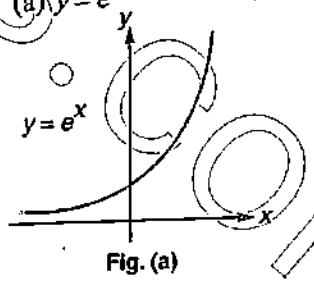


Fig. (a)

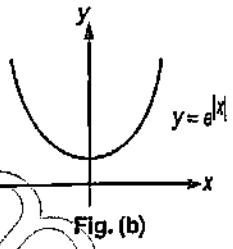
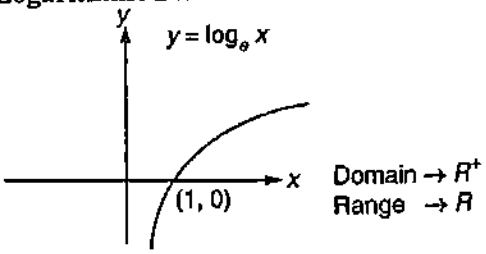


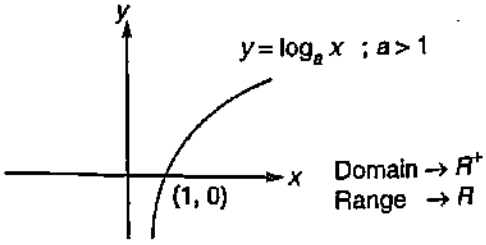
Fig. (b)

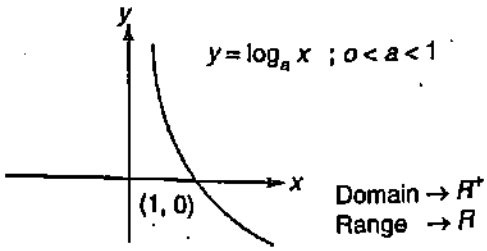
(a) Domain $\rightarrow R$
Range $\rightarrow R^+$

(b) Domain $\rightarrow R$
Range $\rightarrow R^+$

(T–4) Logarithmic Function







For More Visit : www.LearnEngineering.in

Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

ALGEBRAIC OPERATIONS ON REAL VALUED FUNCTIONS

Addition : If f and g are two functions, then the sum of the functions $(f+g)(x) = f(x)+g(x)$ is defined only for those values of x for which both f and g are defined *i.e.*,

$x \in \text{Dom } f \cap \text{Dom } g$

$\therefore$ $\text{Dom}(f+g) = \text{Dom } f \cap \text{Dom } g$

Subtraction : If f and g are two functions, then the difference of the functions $(f-g)x = f(x)-g(x)$

Where $x \in \text{Dom } f \cap \text{Dom } g$

$\therefore$ $\text{Dom of } (f-g) = \text{Dom } f \cap \text{Dom } g$

Multiplication : If k is any real number and f is a function, then kf is defined for all $x \in \text{Dom } f$ by

$$(kf)x = k\,f(x)$$

Hence,

if f and g are two functions, then the pointwise product fg is defined for all $x \in \text{Dom } f \cap \text{Dom } g$ by $(fg)x = f(x)\,g(x)$

$\therefore$ $\text{Dom of } (fg)x = \text{Dom } f \cap \text{Dom } g$

Division : If f and g are two functions, then f/g is defined for all $x \in \text{Dom } f \cap \text{Dom } g \cap \{x : g(x) \neq 0\}$ by

$$(f/g)x = \frac{f(x)}{g(x)}$$

$\therefore \text{Dom}(f/g)x = \text{Dom } f \cap \text{Dom } g - \{x : x \in \text{Dom } g \text{ and } g(x) = 0\}$

Composition : Let $f : A \to B$ and $g : B \to C$ be functions,

then $gof : A \to C$ defined by

$$(gof)x = g(f(x))$$

Alternatively : $(gof(x)) = g\,f(x)$

i.e., gof is defined when

(a) input of f (*i.e.*, x) $\in$ Domain of f and

(b) input of g (i.e., $f(x) \in$ Domain of g

Note that gof is defined, then fog may not be defined

INTRODUCTORY EXERCISE-17.1

1. The domain of the function $f(x) = x$ is :
   (a) R (b) R^+
   (c) $R - \{0\}$ (d) $R^+ \cup \{0\}$
2. The domain of the function $f(x) = \sqrt{x}$ is :
   (a) R (b) R^+
   (c) $R^+ \cup \{0\}$ (d) $R - \{0\}$
3. The domain of the function $f(x) = \frac{1}{x}$ is :
   (a) R (b) R^+
   (c) $R - \{0\}$ (d) $R^+ \cup \{0\}$
4. The domain of the function $f(x) = |x|$ is :
   (a) R (b) R^+
   (c) $R - \{0\}$ (d) $R^+ \cup \{0\}$
5. The domain of the function $f(x) = \sqrt{|x|}$ is :
   (a) $x \in (-\infty, \infty)$ (b) $x \in (0, \infty)$
   (c) $x \in (-\infty, 0) \cup (0, \infty)$ (d) $x \in [0, \infty)$
6. The domain of the function $f(x) = |\sqrt{x}|$ is :
   (a) $x \in (-\infty, +\infty)$ (b) $x \in (0, \infty)$
   (c) $x \in (9, 16)$ (d) $x \in [0, \infty)$
7. The domain of the function $f(x) = \frac{1}{\sqrt{x}}$ is :
   (a) $-\infty < x < \infty$ (b) $0 < x < \infty$
   (c) $R - \{0\}$ (d) $0 \le x < \infty$
8. The domain of the function $f(x) = \sqrt{x^2}$ is :
   (a) $-\infty < x < \infty$ (b) $0 < x < \infty$
   (c) $R - \{0\}$ (d) $R^+ \cup \{0\}$
9. The domain of the function $f(x) = (\sqrt{x})^2$ is :
   (a) $-\infty < x < \infty$ (b) $0 < x < \infty$
   (c) $0 \le x < \infty$ (d) none of these
10. The domain of the function $f(x) = \log_{10} x$ is :
    (a) $-\infty < x < \infty$ (b) $0 < x < \infty$
    (c) $0 \le x < \infty$ (d) none of these
11. The domain of the function $f(x) = \log x^2$ is :
    (a) R (b) R^+
    (c) $R - \{0\}$ (d) $R^+ \cup \{0\}$
12. The domain of the function $f(x) = \frac{1}{\log_{10} x}$ is :
    (a) R (b) $R^+ - \{1\}$
    (c) $R^+ \cup \{0\}$ (d) $R - \{0\}$
13. The domain of the function $f(x) = \frac{1}{\sqrt{\log_{10} x}}$ is
    (a) R (b) $R^+ - (0, 1]$
    (c) $R - \{0\}$ (d) none of these
14. The domain of the function $f(x) = \sqrt{3x-4}$ is :
    (a) $[0, \infty)$ (b) $\left[\frac{3}{4}, \infty\right)$
    (c) $\left[\frac{4}{3}, \infty\right)$ (d) $(4, \infty)$
15. The domain of the function $f(x) = \sqrt{4x-3} + \sqrt{2x-6}$ is :
    (a) $[0, \infty)$ (b) $\left[\frac{3}{4}, \infty\right)$
    (c) $\left[\frac{4}{3}, \infty\right)$ (d) $[3, \infty)$
16. The domain of $y = \frac{1}{\sqrt{|x| - x}}$ is :
    (a) $[0, \infty)$ (b) $(0, \infty)$
    (c) $(-\infty, 0)$ (d) $(-\infty, 0]$
17. The domain of $y = \frac{x}{\sqrt{x^2 - 3x + 2}}$
    (a) $(-\infty, \infty)$ (b) $[0, \infty)$
    (c) $(-\infty, 1] \cup [2, \infty)$ (d) none of these
18. The domain of $y = \log_{10}(\sqrt{6-x} + \sqrt{x-4})$ is :
    (a) R (b) $R - \{0\}$
    (c) $[6, \infty)$ (d) $(4, 6)$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yourshabbab.wordpress.com
For More Visit : www.LearnEngineering.in

19. The domain of definition of the function $y = \frac{1}{\log_{10}(1-x)} + \sqrt{x+2}$ is :

(a) $[-2, 1)$ (b) $[-2, 0)$
(c) $[-2, 0) \cup (0, 1)$ (d) $(-2, 1]$

20. The domain of definition of $f(x) = \frac{\log_2(x+3)}{x^2+3x+2}$ is :

(a) $(-3, -2)$ (b) $(-2, -1]$
(c) $(-3, \infty) \sim \{-1, -2\}$ (d) $[0, \infty)$

21. The domain of $f(x) = 2\sin x + 3\cos x + 4$ is :

(a) R (b) R^+
(c) $R - \{0\}$ (d) $R - \{n\pi : n \in I^+\}$

22. The domain of the function $y = \sqrt{\log \frac{1}{|\sin x|}}$ is :

(a) R (b) $R \sim \left(\frac{n}{2}\pi; n \in I\right)$
(c) $[0, \infty)$ (d) $R \sim \{n\pi, n \in I\}$

23. The domain of $f(x) = (x^2 + x + 1)^{-3/2}$ is :

(a) $(-\infty, \infty)$ (b) $(0, \infty)$
(c) $\left(\frac{3}{2}, \infty\right)$ (d) none of these

24. The domain of $f(x) = \log(5x - 6 - x^2)$ is :

(a) $(5, 6)$ (b) $(2, 3)$
(c) $[2, \infty)$ (d) none of these

25. The domain of $y = \log_x 5$ is :

(a) $x > 5$ (b) $(0, \infty)$
(c) $(0, \infty) \sim \{1\}$ (d) $[5, \infty)$

26. The domain of $y = 9 - \sqrt{9 - x^2}$ is :

(a) $x < -3$ (b) $-3 < x < 3$
(c) $x \in [-3, 3]$ (d) none of these

27. The domain of $y = \frac{x}{\sqrt{x^2 - 5x + 6}}$ is :

(a) $R \sim \{2, 3\}$ (b) $R \sim (2, 3)$
(c) $R - [2, 3]$ (d) none of these

28. The range of the function $y = \frac{x^2}{1+x^4}$ is :

(a) $[0, \infty)$ (b) $[0, 2]$
(c) $\left(0, \frac{1}{2}\right)$ (d) $\left[0, \frac{1}{2}\right]$

29. Find the range of $f(x) = \frac{x^2 - 2}{x^2 - 3}$:

(a) $(1, \infty)$ (b) $(-\infty, 2/3]$
(c) $(-\infty, 2/3)$ (d) $(-\infty, 2/3] \cup (1, \infty)$

30. If $f(x) = \frac{x^2 + 2x + 3}{x}$, then the range of the function is :

(a) $[2\sqrt{3}, 3\sqrt{2}]$ (b) $R - (2 - 2\sqrt{3}, 2 + 2\sqrt{3})$
(c) $R - (2, 2\sqrt{3})$ (d) none of these

31. Which of the following pairs are identical :

(a) $f(x) = \sqrt{x^2}$, $g(x) = (\sqrt{x})^2$

(b) $f(x) = \frac{1}{\sqrt{x^2}}$, $g(x) = \frac{x}{x^2}$

(c) $f(x) = \log(x-1) + \log(x-2)$, $g(x) = \log(x-1)(x-2)$

(d) none of the above

32. If $f(x) = \frac{1}{\sqrt{|x| - x}}$ and $g(x) = \frac{1}{\sqrt{x - |x|}}$, then :

(a) dom $f \neq \phi$ and dom $g = \phi$
(b) dom $f = \phi$ and dom $g \neq \phi$
(c) dom $f = \phi$ and dom $g = \phi$
(d) f and g have the same domain

33. If $f(x) = x$, $g(x) = \sqrt{x^2}$ and $h(x) = \frac{x^2}{x}$, then the set of values for which the given functions $f(x)$, $g(x)$ and $h(x)$ are identical is :

(a) $R - \{0\}$ (b) R
(c) R^+ (d) $R^+ \cup \{0\}$

34. If $f(x) = \log x^4$ and $g(x) = 4\log x$ then the domain for which $f(x)$ and $g(x)$ are identical ?

(a) $(-\infty, \infty)$ (b) $[0, \infty)$
(c) $(0, \infty)$ (d) none of these

35. If $f(x) = \text{Sgn } x$ and $g(x) = 1 + x - [x]$. Then for all x, $f(g(x))$ is equal to :

(a) 0 (b) 1
(c) x (d) $f(x)$

36. If $f(x) = 64x^3 + \frac{1}{x^3}$ and α, β are the roots of $4x + \frac{1}{x} = 2$ then :

(a) $f(\alpha) = -24$ (b) $f(\alpha) = -16$
(c) $f(\beta) = -16$ (d) both (b) and (c)

37. If $f(x) = \frac{x}{\sqrt{1+x^2}}$, then $f(f(f(x)))$ is :

(a) $\frac{3x}{\sqrt{1+x^2}}$ (b) $\frac{x}{\sqrt{1+3x^2}}$
(c) $\frac{x^3}{\sqrt{1+x^2}}$ (d) $\frac{x}{\sqrt{3+3x^2}}$

38. If $f(x) = \frac{4^x}{4^x + 2}$, then the value of $f(x) + f(1-x)$ is :

(a) 0 (b) -1
(c) 1 (d) can't be determined

39. If $f(x) = 1 - f(1-x)$, then the value of

$$f\left(\frac{1}{999}\right) + f\left(\frac{2}{999}\right) + \ldots + f\left(\frac{998}{999}\right) \text{ is :}$$

(a) 998 (b) 1
(c) 499 (d) none of these

40. Let $f(x) = x^2$ and $g(x) = \sqrt{x}$, then :

(a) gof $(3) = 9$ (b) gof $(-3) = 9$
(c) gof $(3) = 3$ (d) gof $(-9) = 3$

41. If $g(x) = x^2 - x + 7$ and $\left(\frac{1}{9}\right)$ gof$(x) = x^2 + x + 1$ then $f(x)$ is :

(a) $2x - 3$ (b) $3x + 2$
(c) $3x - 2$ (d) none of these

42. Let $f(x) = \frac{x}{x+3}$, then $f(x+1) = ?$

(a) $\frac{3x+2}{x+2}$ (b) $\frac{x+1}{x+4}$
(c) $\frac{x+1}{x+3}$ (d) $\frac{2x+3}{x+3}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmanboob.wordpress.com

For More Visit : www.LearnEngineering.in

43. Let $f(x)=\frac{x}{x+3}$, then $\frac{1}{f(x+1)}-f\left(\frac{1}{x+1}\right)=?$

(a) $\frac{3x^2+7x+4}{3(x^2+5x+5)}$ (b) $\frac{3(x^2+5x+5)}{3x^2+7x+4}$

(c) $\frac{-x(2x+3)}{(x+3)^2}$ (d) $\frac{-(x+3)^2}{x(2x+3)}$

44. If $f(x)=\frac{x+1}{x-1}, x\neq 1$, find $f(f(f(f(f(2)))))$

(a) 2 (b) 3

(c) 4 (d) 6

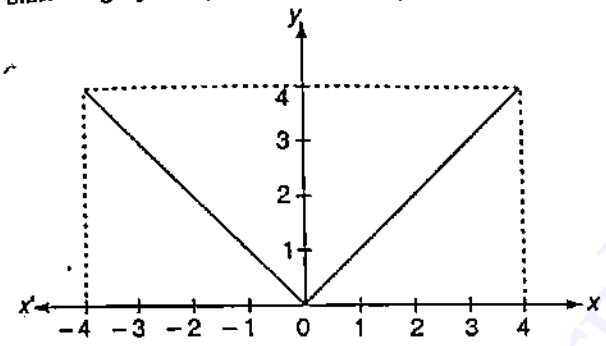
TRANSFORMATION OF GRAPHS

Here we study some techniques for the following transformation.

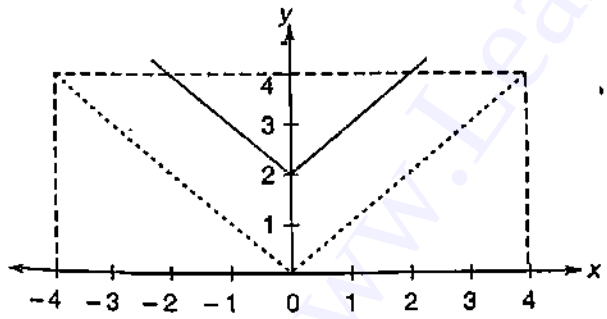
1. $f(x)\to f(x)+k$ 2. $f(x)\to f(x)-k$
3. $f(x)\to k\,f(x)$ 4. $f(x)\to \frac{1}{k}f(x)$
5. $f(x)\to -f(x)$ 6. $f(x)\to |f(x)|$
7. $f(x)\to f(x+k)$ 8. $f(x)\to f(x-k)$
9. $f(x)\to f(kx)$ 10. $f(x)\to f\left(\frac{x}{k}\right)$
11. $f(x)\to f(-x)$ 12. $f(x)\to f|x|$

1. $f(x)\to f(x)+k$, k is a positive constant.

Shift the graph of $f(x)$ upwards by 'k'.

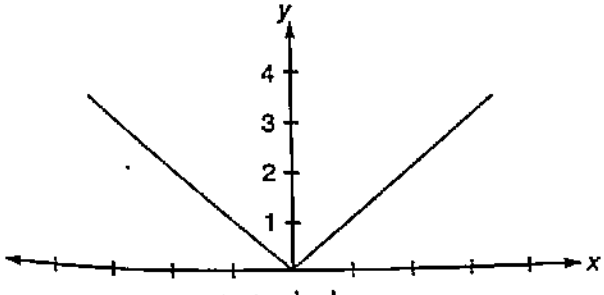


$f(x)=|x|$

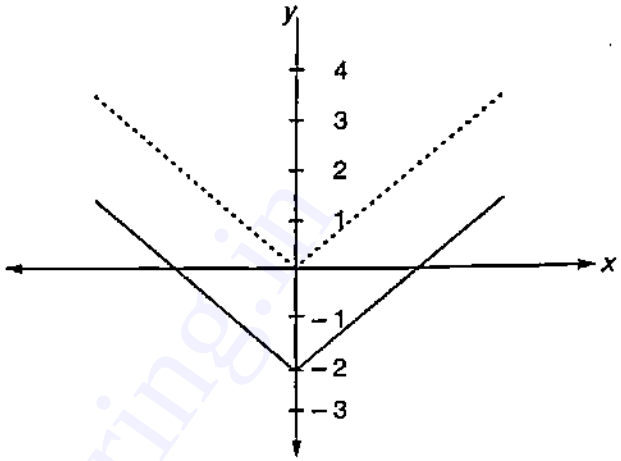


$f(x)=|x|+2$

2. $f(x)\to f(x)-k$; k is a positive constant



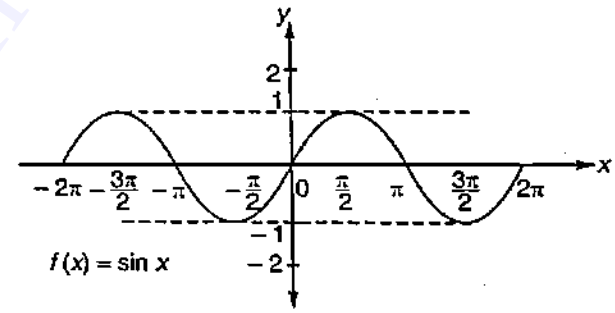
$f(x)=|x|$

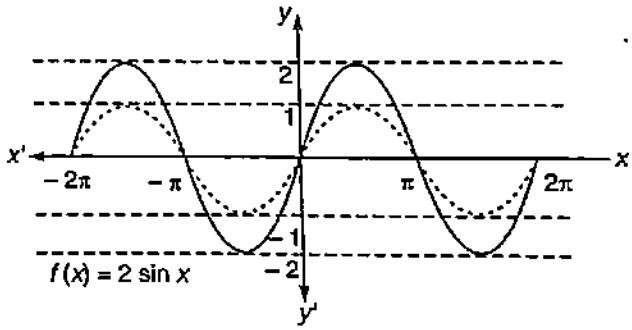


$f(x)=|x|-2$

3. $f(x)\to kf(x)$; $(k>1)$

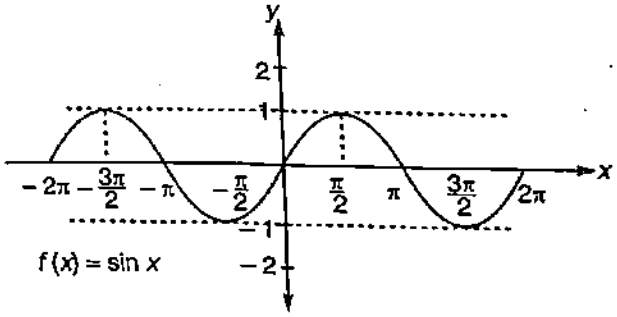
stretch the graph of $f(x)$, 'k' times along y-axis.





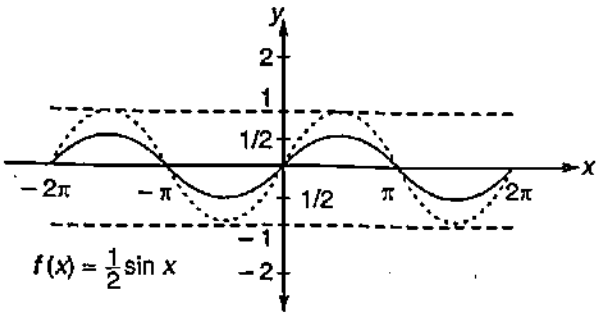
4. $f(x)\to \frac{1}{k}f(x)$; $(k>1)$

shrink the graph of $f(x)$ 'k' times along y-axis.



For More Visit : www.LearnEngineering.in
Scanned by CamScanner

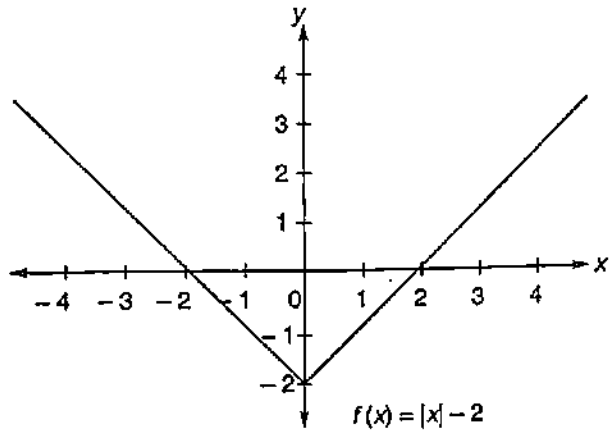
yourshabbaab.wordpress.com

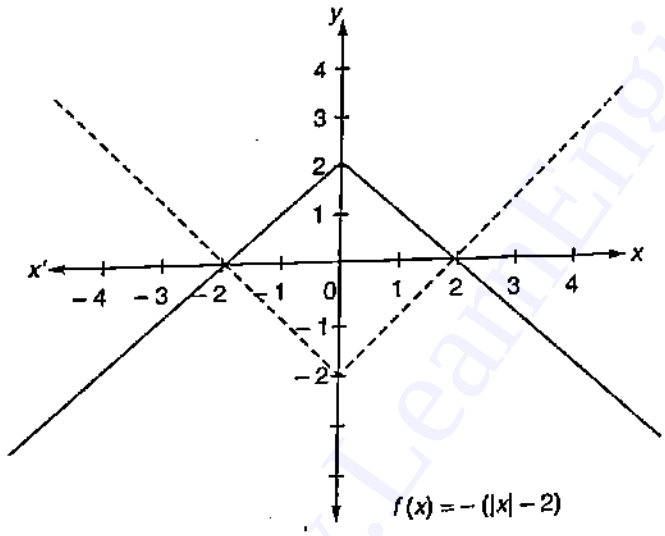


5. $f(x) \to -f(x)$

Turn the graph of $f(x)$ by 180° about x-axis.

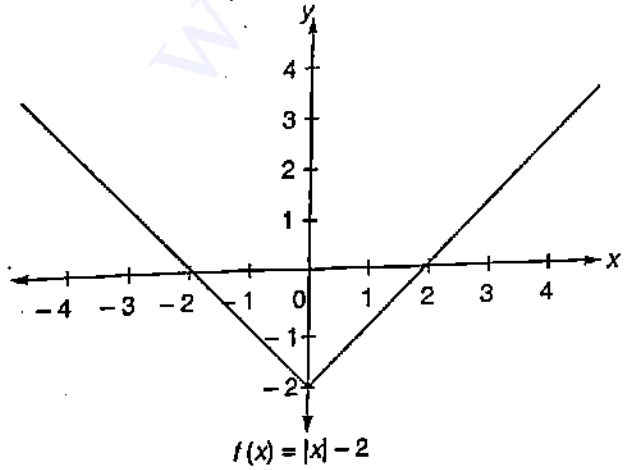
or Take the mirror image of $f(x)$ in x-axis.





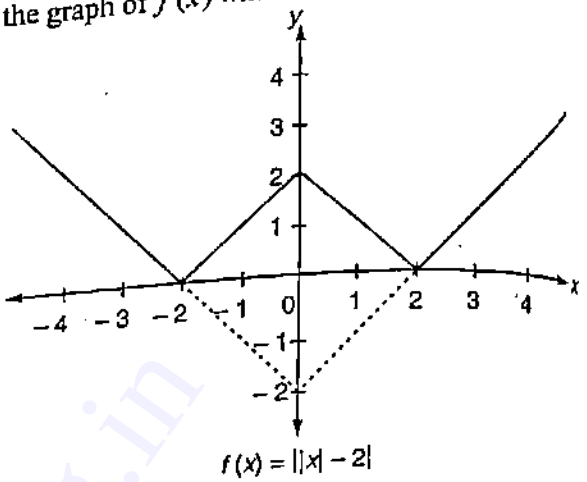
6. $f(x) = |f(x)|$,

where $|f(x)| = \begin{cases} f(x) \text{ if } f(x) \geq 0 \\ -f(x) \text{ if } f(x) < 0 \end{cases}$



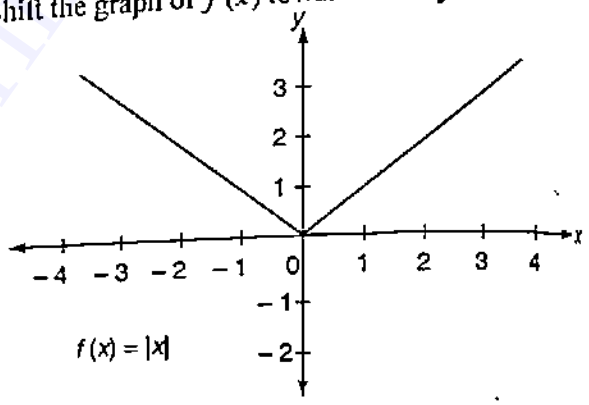
Turn the portion of the graph of $f(x)$ lying below x-axis by 180° about x-axis.

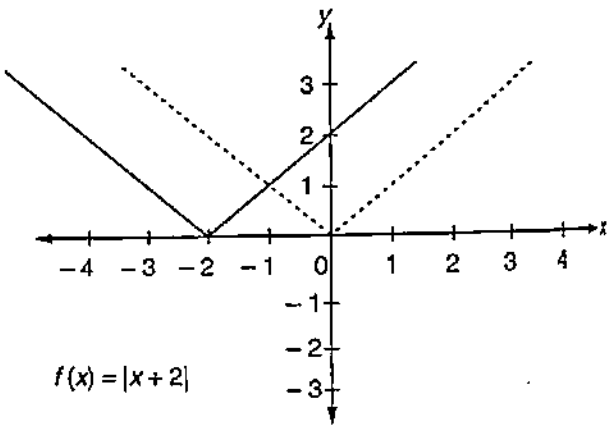
or, Take the mirror image (in x-axis) of the portion of the graph of $f(x)$ which lies below x-axis.



7. $f(x) \to f(x+k)$

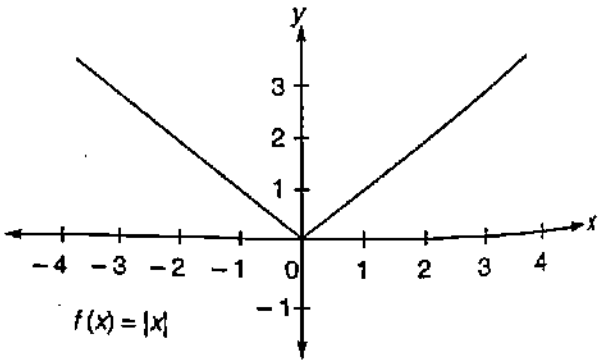
Shift the graph of $f(x)$ towards left by 'k'.





8. $f(x) \to f(x-k)$

Shift the graph of $f(x)$ towards right by 'k'

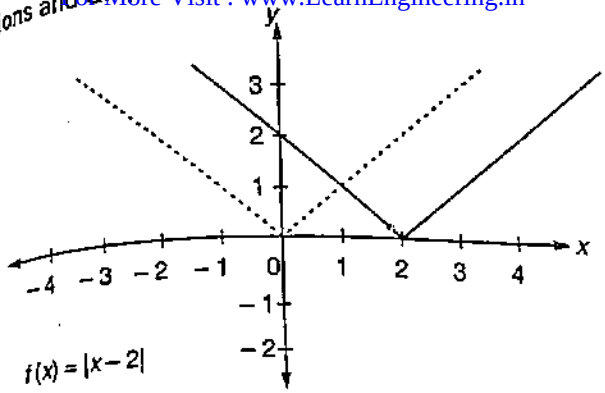


www.LearnEngineering.in

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

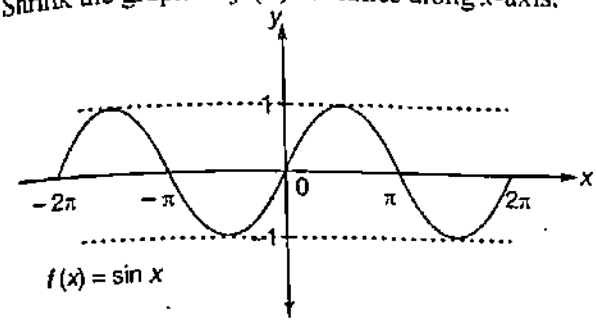
For More Visit : www.LearnEngineering.in



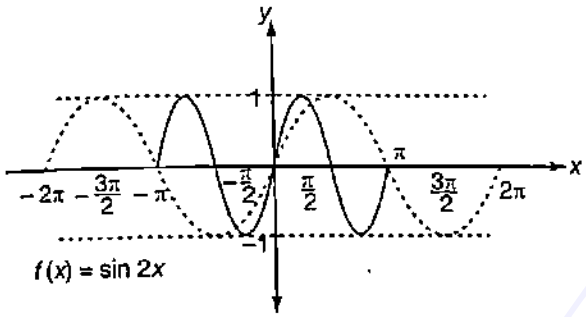
$f(x) = |x-2|$

9. $f(x) \to f(kx)\,;(k>1)$

Shrink the graph of $f(x)$ 'k' times along x-axis.



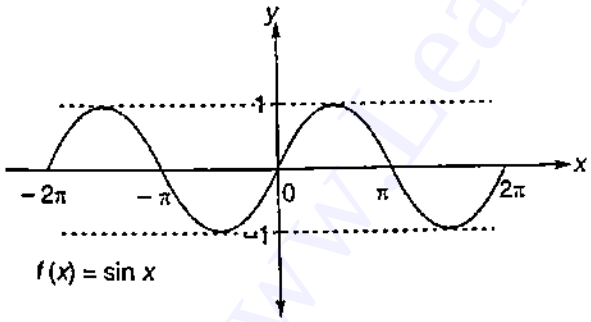
$f(x) = \sin x$



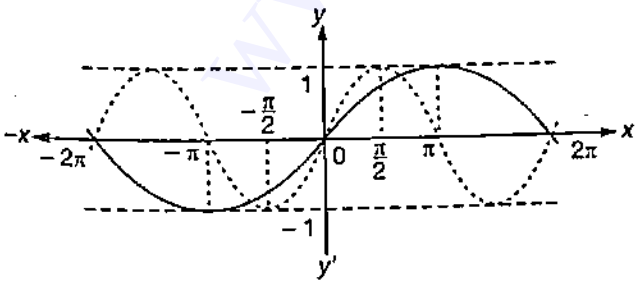
$f(x) = \sin 2x$

10. $f(x) \to f\left(\frac{x}{k}\right)$

Stretch the graph of $f(x)$ 'k' times along x-axis.



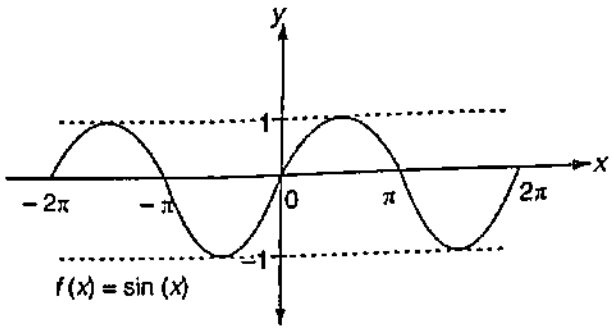
$f(x) = \sin x$



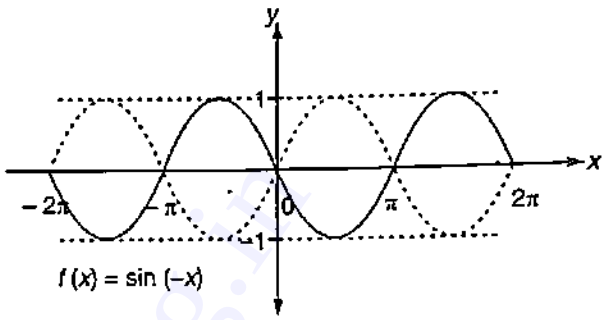
11. $f(x) \to f(-x)$

Turn the graph of $f(x)$ by 180° about y-axis.

or Take the mirror image of the graph of $f(x)$ in y-axis.



$f(x) = \sin(x)$

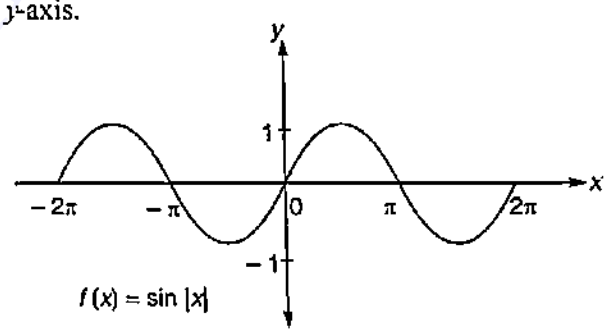


$f(x) = \sin(-x)$

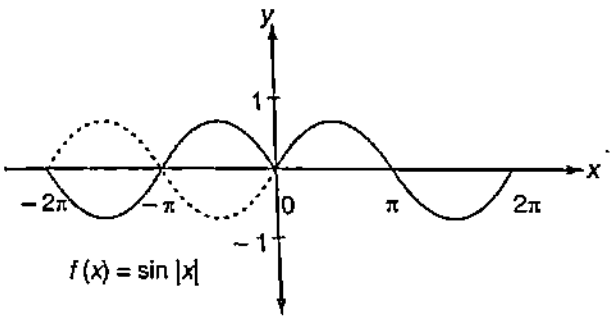
12. $f(x) \to f|x|$

$\because \quad f|x| = \begin{cases} f(x) \text{ if } x \geq 0 \\ f(-x) \text{ if } x < 0 \end{cases}$

$\therefore$ On the right of y-axis, plot the graph of $f(x)$ as such and on the left of y-axis, plot the mirror image of the graph of $f(x)$ of the portion lying on the right of y-axis.

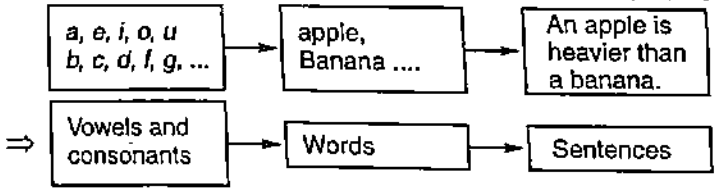


$f(x) = \sin|x|$



$f(x) = \sin|x|$

COMPOSITE FUNCTIONS



When the output of the first operation is the input for the second operation, then there is a composition of functions (operations).

If $f : A \to B$ and $g : B \to C$, then

$(gof) : A \to C$

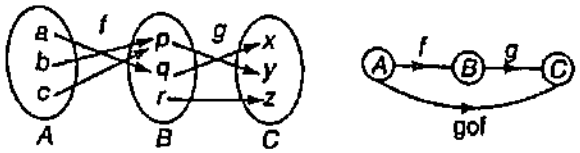
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

$\Rightarrow$ $(gof)(a) = g\{f(a)\}$

Example 1.



$(gof)(a) = g\{f(a)\} = g(q) = x$

$(gof)(b) = g\{f(b)\} = g(p) = y$

$(gof)(c) = g\{f(c)\} = g(p) = y$

Properties

1. **Associative :** If $f_1 : A \to B, f_2 : B \to C, f_3 : C \to D$ then $f_1 o(f_2 o f_3) = (f_1 o f_2) o f_3$
2. **Not commutative :** Let $f(x) = \cos x$ and $g(x) = x^3$ then, $gof(x) = g(f(x)) = g(\cos x) = \cos^3 x$ and $fog(x) = f(x^3) = \cos x^3$ Hence $gof \neq fog$
3. Range of (gof) may not be the same as the range of g



$(gof)(a) = x$

$(gof)(b) = y$

$(gof)(c) = y$

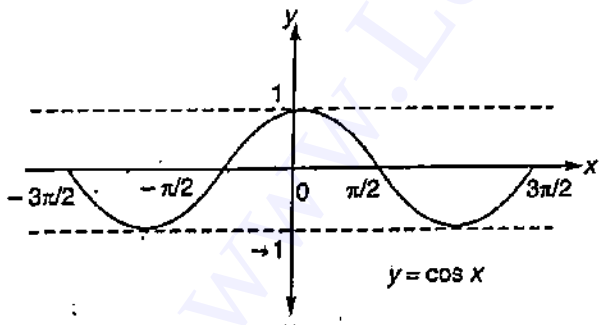
$\therefore$ Range of $(gof) = \{x, y\}$

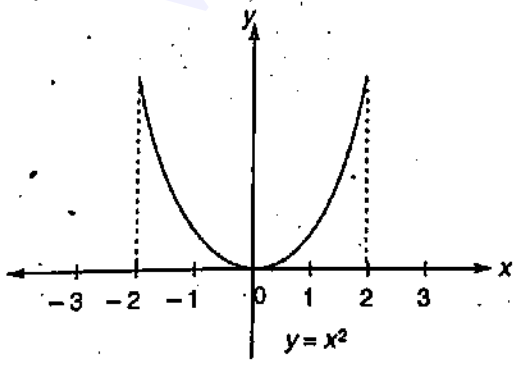
and Range of $g = \{x, y, z\}$

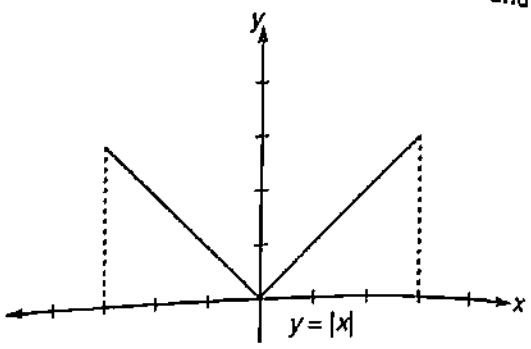
EVEN AND ODD FUNCTIONS

1. **Even function :** A function $y = f(x)$ is said to be an even function if $f(-x) = f(x)\ \forall\ x \in \text{Dom}(f)$

   e.g., $y = \cos x,\ y = x^{2n}\ \forall\ n \in N,\ y = |x|$



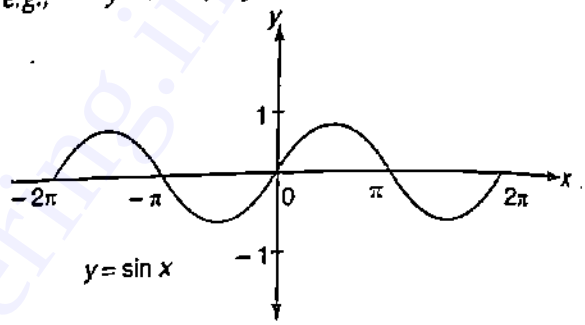


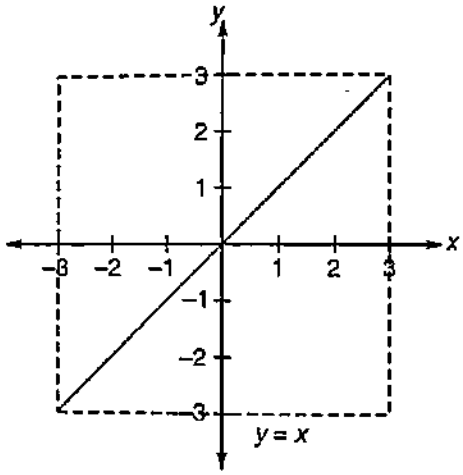


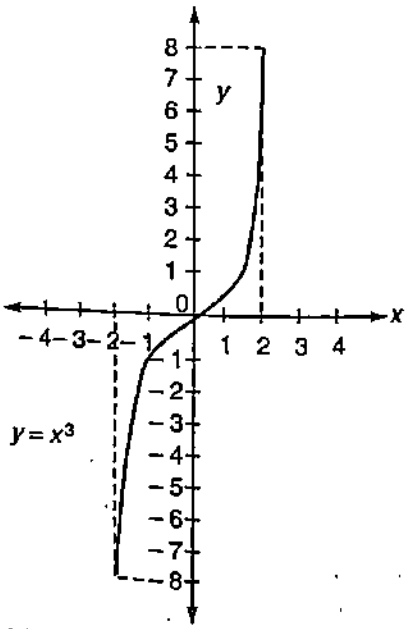
NOTE Graph of an even function is symmetric about y-axis.

2. **Odd function :** A function $y = f(x)$ is said to be an odd function if $f(-x) = -f(x), \quad \forall\ x \in \text{Dom}(f)$

   e.g., $y = \sin x,\ y = x^{2n-1}\ \forall\ n \in N$







NOTE Graph of an odd function is symmetric about origin.

For More Visit : www.LearnEngineering.in

Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

Properties of even and odd functions

1. Sum : Even + Even = Even
Odd + Odd = Odd
Even + Odd = (neither even nor odd)
Odd + Even = (neither even nor odd)

2. Difference : Even − Even = Even
Odd − Odd = Odd
Even − Odd = (neither even nor odd)
Odd − Even = (neither even nor odd)

Remember : Any function $y = f(x)$ can be expressed uniquely as the sum of an even and an odd function as follows :

$$f(x) = \frac{1}{2}[f(x) + f(-x)] + \frac{1}{2}[f(x) - f(-x)]$$

3. Product: Even × Even = Even
Odd × Odd = Even
Even × Odd = Odd
Odd × Even = Odd

NOTE If the number of odd functions in the product is odd → odd function.
If the number of odd functions in the product is even → even function

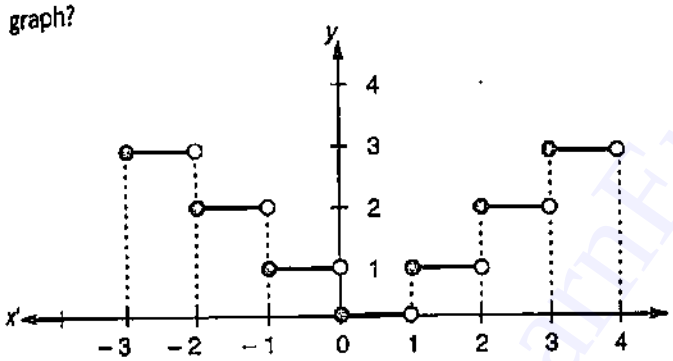
4. Division : Even ÷ Even = Even
Odd ÷ Odd = Even
Even ÷ Odd = Odd
Odd ÷ Even = Odd

5. Composition : Even (Even) = Even
Even (Odd) = Even
Odd (Even) = Even
Odd (Odd) = Odd

NOTE If all the functions in the composition are odd → Odd function
If at least one of the functions in the composition is even → even function.

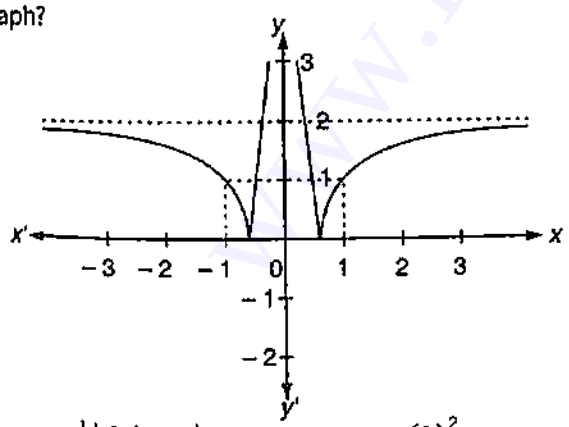
INTRODUCTORY EXERCISE-1.2

1. Which one of the following is correct for the following graph?



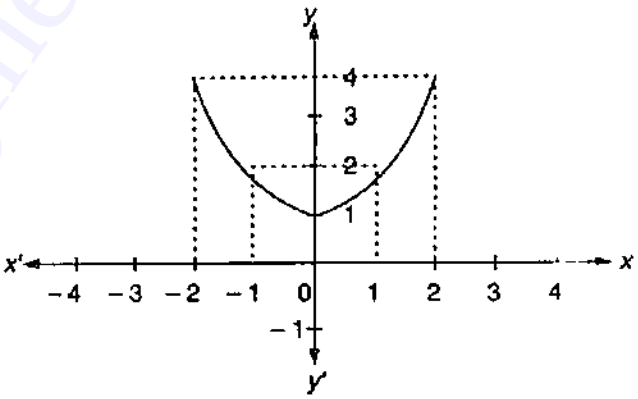
(a) $f(x) = \{x\}$ (b) $f(x) = [x]$
(c) $f(x) = |[x]|$ (d) none of these

2. Which one of the following is correct for the following graph?



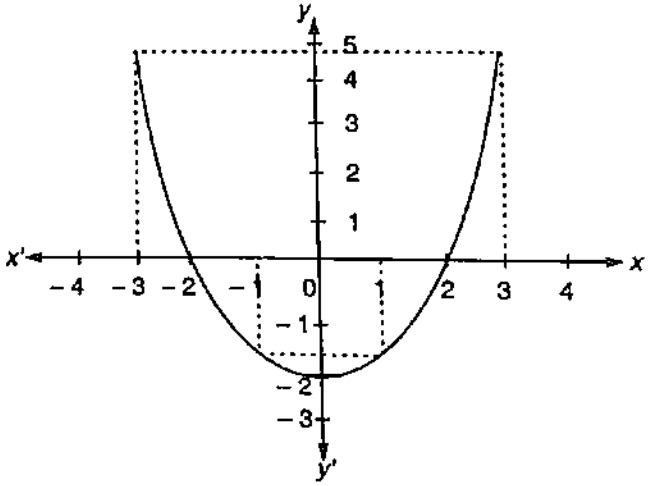
(a) $f(x) = \left|\left|\frac{1}{x}\right| - 2\right|$ (b) $f(x) = \left(\frac{1}{x}\right)^2$
(c) $f(x) = \left|\frac{1}{x^3}\right|$ (d) $f(x) = \log|x|$

3. Which one of the following is correct for the following graph?



(a) $f(x) = e^x$ (b) $f(x) = \log|x|$
(c) $f(x) = 2^{|x|}$ (d) $f(x) = x^2$

4. Which one of the following is correct for the given graph?



(a) $f(x) = |x^3 - 2|$ (b) $f(x) = x^2 - 2$
(c) $f(x) = \left|\frac{x^3}{4}\right| - 2$ (d) $f(x) = \log|x - 1|$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

5. The equation $\frac{x^2}{1-|x-2|} = 1$ has n real solutions, then :
(a) $n = 0$ (b) $n = 1$
(c) $n = 2$ (d) $n = 3$

6. The number of real solutions of the equation $\log_{0.5}|x| = 2|x|$ is :
(a) 0 (b) 1
(c) 2 (d) none of these

7. The graph of the function $f : R \to R$ defined by $f(x) = \frac{|x|^3 + |x|}{1+x^2}$ may lie in :
(a) I[st] and II[nd] quadrant
(b) II[nd] and III[rd] quadrant
(c) III[rd] and IV[rd] quadrant
(d) none of these

8. The two roots of the equation $f(x) = f\left(\frac{x+8}{x-1}\right)$ are :
(a) 2, – 2 (b) 4, – 2
(c) – 4, – 2 (d) 2, 4

9. If the functions f, g, h are defined from 'R' to 'R' by $f(x) = x^2 - 1$, $g(x) = \sqrt{x^2+1}$, $h(x) = \begin{cases} 0, & \text{if } x \le 0 \\ x, & \text{if } x \ge 0 \end{cases}$ then $hofog$ is equal to :
(a) x^2 (b) $\sqrt{x^4-1}$
(c) 0 (d) none of these

10. If $f(x) = (a - x^n)^{1/n}$, where $a > 0$, then $f(f(x))$ is :
(a) x^n (b) x^2
(c) $2x$ (d) none of these

11. If $f(x) = \left(\frac{x-1}{x+1}\right)$, then $f(f(ax))$ in terms of $f(x)$ is equal to :
(a) $\frac{f(x)+1}{a(f(x)+1)}$ (b) $\frac{f(x)+1}{a(f(x)-1)}$
(c) $\frac{f(x)-1}{a(f(x)+1)}$ (d) $\frac{f(x)+1}{a(f(x)-1)}$

12. If $f(x) = \frac{x}{\sqrt{1+x^2}}$, then $fofof(x)$ is equal to :
(a) $\frac{x}{\sqrt{(1+3x^2)}}$ (b) $\frac{3x}{\sqrt{(1-x^2)}}$
(c) $\frac{3x}{\sqrt{(1+x^2)}}$ (d) none of these

13. If $f(x) = \begin{cases} 1+|x|; x < -1 \\ [x] \quad ; x \ge -1 \end{cases}$ Also, [.] is greatest integer function then $f(f(-2.4))$ is equal to :
(a) – 3 (b) 2
(c) 3 (d) 4

14. If $f(x) = x^n$, $n \in N$ and $(gof)(x) = ng(x)$, then $g(x)$ can be :
(a) $n|x|$ (b) $\sqrt[3]{x}$
(c) e^x (d) $\log x$

15. Let $f : N \to R$; $f(x) = 2x - 1$
$g : z \to R$; $g(x) = \frac{x^2}{2}$, then $(gof)(0)$ is :
(a) $-\frac{1}{2}$ (b) $\frac{1}{2}$
(c) 1 (d) none of these

16. Let $f\left(x + \frac{1}{x}\right) = x^2 + \frac{1}{x^2}$; $(x \ne 0)$, then $f(x)$ is equal to :
(a) $x^2 - 1$ (b) $x^2 - 2$
(c) x^2 (d) none of these

17. Let $f(x) = \sqrt{x^5}$, then $f(5x)$ is equal to :
(a) $5x\sqrt{5x}$ (b) $25x^2\sqrt{5x}$
(c) $5x$ (d) $\sqrt{5x^5}$

18. If $f(x) = 4x - 5$, $g(x) = x^2$ and $h(x) = \frac{1}{x}$, then $f(g(h(x)))$ is :
(a) $\frac{4}{(x-5)}$ (b) $\frac{1}{(4x-5)^2}$
(c) $\left(\frac{x}{4} - 5\right)$ (d) $\left(\frac{4}{x^2} - 5\right)$

19. If $[x]$ is greatest integer function (GIF) and $\{x\}$ is smallest integer function (SIF), and $x \in R \sim I$, then $[x] - \{x\}$ is equal to :
(a) – 1 (b) 0
(c) 1 (d) 2

20. If the graph of the function $f(x) = \frac{a^x - 1}{x^n(a^x + 1)}$ is symmetric about y-axis, then n is equal to :
(a) $-\frac{1}{3}$ (b) $\frac{1}{4}$
(c) 2/3 (d) 2

21. $f(x) = \ln(x + \sqrt{x^2+1})$ is :
(a) an even function (b) an odd function
(c) neither even nor odd (d) none of these

22. If a function f satisfies the conditions $f(x + y) = f(x) + f(y) \forall x, y \in R$, then f is :
(a) an even function (b) an odd function
(c) neither even nor odd (d) none of these

23. Which of the following function is an even function ?
(a) $f(x) = \frac{a^x + 1}{a^x - 1}$ (b) $f(x) = \frac{a^x - a^{-x}}{a^x + a^{-x}}$
(c) $f(x) = x\frac{a^x + 1}{a^x - 1}$ (d) $f(x) = \sin x$

24. Which of the following function is odd ?
(a) $\ln\left(\frac{1+x^3}{1-x^3}\right)$ (b) $x \cdot \frac{e^{ax} - 1}{e^{ax} + 1}$
(c) $\frac{(1+2^x)^2}{2^x}$ (d) none of these

25. Which of the following function is even function ?
(a) $f(x) = \log\frac{1-x}{1+x}$
(b) $f(x) = x \cdot \frac{a^x + 1}{a^x - 1}$
(c) $f(x) = \log\left[x + \sqrt{(1+x^2)}\right]$
(d) $f(x) = \sqrt{(1+x+x^2)} - \sqrt{(1-x+x^2)}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmanboob.wordpress.com
For More Visit : www.LearnEngineering.in

26. If $f(x) = \sqrt[3]{(1-x^2)} + \sqrt[3]{(1+x^2)}$, then $f(x)$ is:
(a) an even function
(b) an odd function
(c) constant function
(d) none of these

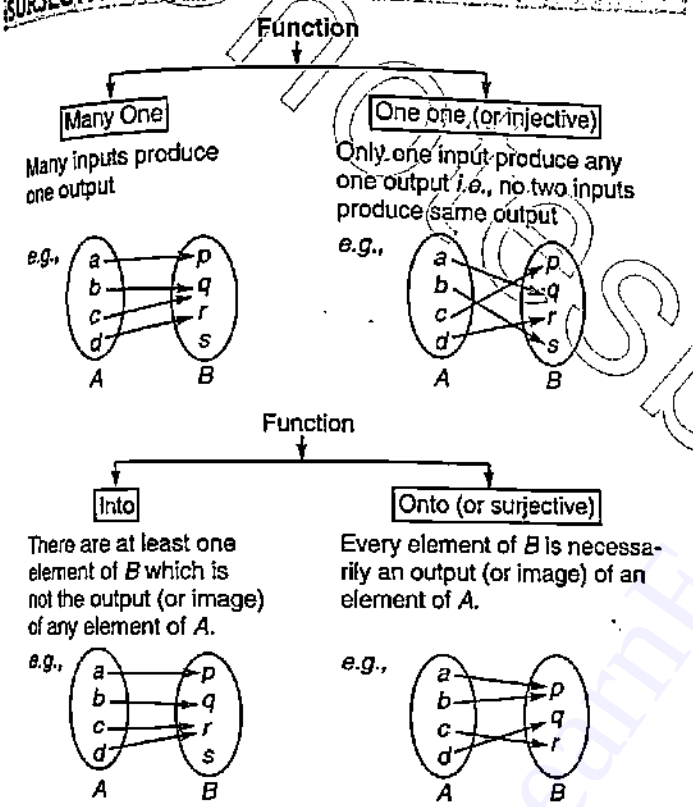
27. Which of the following function is an odd function ?
(a) $f(x) = 2^{-x^2}$ (b) $f(x) = 2^{x-x^4}$
(c) $f(x) = \cos x$ (d) none of these

28. The function $f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 1$ is :
(a) even (b) odd
(c) constant (d) neither even nor odd

29. The graph of the function $y = f(x)$ is symmetrical about the line $x = 2$, then :
(a) $f(x+2) = f(x-2)$ (b) $f(2+x) = f(2-x)$
(c) $f(x) = f(-x)$ (d) none of these

30. Given $f(x) = \frac{1}{(1-x)}$, $g(x) = f(f(x))$ and $h(x) = f(f(f(x)))$, then the value of $f(x) \cdot g(x) \cdot h(x)$ is :
(a) 0 (b) -1
(c) 1 (d) 2

SURJECTIVE AND INJECTIVE FUNCTIONS



One to One (or Injective) Function

A function f is said to be one-to-one if it does not take the same values at two distinct points in its domain. For example, $f(x) = x^3$ is one to one whereas $f(x) = x^2$ is not, as $f(1) = 1$ and $f(-1) = 1$

In order to show that a function $f : A \to B$ is one-one we may take any $x, y \to A$ such that $f(x) = f(y)$ and try to show that $x = y$.

Graphically, if every horizontal line intersects the graph of a given function at not more than one point, *i.e.*, the function is strictly increasing (or decreasing) then the function is one-one.

NOTE 1. Functions which are not one to one are called many-one function.
2. One many function does not exist.

Onto (or Surjective Function)

If a function $f : A \to B$ is such that each element in B is the f-image of at least one element in A, then we say that f is function of A 'onto' B. Equivalently a function f is an onto function if codomain of f = Range of f.

In order to show that a function $f : A \to B$ is onto we start with any $y \in B$ and try to find $x \in A$ such that $f(x) = y$.

NOTE If there exists at least one element in B which is not the f-image of any element of A, then f is called an into function.

Bijective Function : If a function f is both one-to-one and onto then f is said to be a bijective function *e.g.*, an identity function is a bijective function.

INVERSE FUNCTION

Inverse of a function : If $f : A \to B$ is a bijection, we can define a new function from B to A which associates each element $y \in B$ to its pre-image $f^{-1}(y) \in A$. Such a function is known as the inverse of function f and is denoted by f^{-1}.

- Inverse of a function is defined iff the function is one-to-one and onto.
- In the inverse function inputs and outpts interchange their roles.
- Inverse of a function is not defined if the given function is a many-one function or into function or both.
- If the graph of a function $f(x)$ is symmetric about the line $y = x$, then $f(x)$ and $f^{-1}(x)$ are identical function.
- The graph of $f(x)$ and $f^{-1}(x)$ is symmetric about $y = x$.

BINARY OPERATIONS

Binary Operation : Let S be a non-empty set. A function f from $S \times S$ to S is called a binary operation on S *i.e.*, $f : S \times S \to S$ is a binary operation on S.

Generally binary operations are represented by the symbols $*$, $\oplus$ etc. instead of letters f, g etc.

Types of Binary Operation :

1. **Commutative binary operation :** A binary operation $*$ on a set S is said to be commutative if $a * b = b * a$ for all $a, b, \in S$.
2. **Associative binary operation :** A binary operation $*$ on a set S is said to be associative if
$(a * b) * c = a * (b * c)$ for all $a, b, c \in S$.
3. **Distributive binary operation :** Let $*$ and $\oplus$ be two binary operations on a set S. Then $*$ is said to be :

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yousmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

(i) Left distributive over $\oplus$ if
$a * (b \oplus c) = (a * b) \oplus (a * c)$ for all $a, b, c \in S$,

(ii) Right distributive over $\oplus$ if
$(b \oplus c) * a = (b * a) \oplus (c * a)$ for all $a, b, c \in S$.

If $*$ is both left and right distributive over $\oplus$, then $*$ is said to be distributive over $\oplus$.

Composition table : Le $S = \{a_1, a_2, a_3, \ldots a_n\}$ be finite set and $*$ be a binary operation on S. Then the composition table is constructed in the manner given below :

$*$	a_1	a_2	...	a_i	...	a_j	...	a_n
a_1	$a_1 * a_1$	$a_1 * a_2$	...	$a_1 * a_i$	...	$a_1 * a_j$	...	$a_1 * a_n$
a_2	$a_2 * a_1$	$a_2 * a_2$	...	$a_2 * a_i$	...	$a_2 * a_j$	...	$a_2 * a_n$
...								
a_i	$a_i * a_1$	$a_i * a_2$	...	$a_i * a_i$	...	$a_i * a_j$	...	$a_i * a_n$
...			...					
a_j	$a_j * a_1$	$a_j * a_2$	...	$a_j * a_i$	...	$a_j * a_j$	...	$a_j * a_n$
a_n	$a_n * a_1$	$a_n * a_2$	...	$a_n * a_i$	...	$a_n * a_j$	...	$a_n * a_n$

Example : Consider the set $S = \{1, 2, 3, 4\}$. Define a binary operation $*$ on S as follows :

For any $a, b \in S$ define $a * b = r$, where 'r' is the least non-negative remainder when ab is divided by 5.

Solution We have

$1*1=1,\ 1*2=2,\ 1*3=3,\ 1*4=4$

$2*1=2,\ 2*2=4,\ 2*3=1,\ 2*4=3$

$3*1=3,\ 3*2=1,\ 3*3=4,\ 3*4=2$

$4*1=4,\ 4*2=3,\ 4*3=2,\ 4*4=1$

Thus the following is the composition table for the binary operation $*$ on S.

$*$	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

USER DEFINED FUNCTIONS

These functions are defined as per the requirement of user. Basically these functions are not standard function. There are some examples illustrated below :

EXAMPLE 1 *If* $f(A, B) = A \# B = A^4 - B^3 + A^2 - B + AB$, *then find the value of* $f(3, 4)$.

SOLUTION $f(3, 4) = 3 \# 4 = 3^4 - 4^3 + 3^2 - 4 + (3 \times 4)$
$= 81 - 64 + 9 - 4 + 12 = 34$

EXAMPLE 2 *If* $A \# B = A + B + AB$. *If for any* A *there is a number* C *such that* $A \# C = A$, *then* $C = ?$

SOLUTION $A \# C = A$, we get

$A + C + AC = A$

$\Rightarrow \quad C(1 + A) = 0$

$\Rightarrow \quad C = 0 \quad \text{or} \quad A = -1$

EXAMPLE 3 *If*

$$f(a, b) = \frac{a+b}{2}$$

$g(a, b) = a^2 + b^2$

$h(a, b) = \max(a, b)$

Find the value of $f(g(3, 9), h(-1, 1))$.

SOLUTION $f(g(3, 9), h(-1, 1)) = f((3^2 + 9^2), 1) = f(90, 1) = 45.5$

EXAMPLE 4

$$f_n = \begin{cases} f_{n-1} & \text{if } n \text{ is even} \\ 2f_{n-1} & \text{if } n \text{ is odd} \end{cases}$$

and $f_0 = 1$, *find the value of* $f_4 + f_5$

SOLUTION $f_4 + f_5 = f_3 + 2f_4 = 2f_2 + 2f_3$
$= 2(f_2 + f_3)$
$= 2(f_1 + 2f_2)$
$= 2(2f_0 + 2f_1)$
$= 4(f_0 + f_1)$
$= 4(1 + 2f_0)$
$= 4(1 + 2) = 12$

EXAMPLE 5

$$f_n = \begin{cases} f_{n-1} & \text{if } n \text{ is even} \\ 2f_{n-1} & \text{if } n \text{ is odd} \end{cases}$$

and $f_0 = 1$, *find the value of* f_{16}.

SOLUTION $f_{16} = f_{15} = 2f_{14} = 2f_{13} = 2^2 f_{12}$
$= 2^2 f_{11} = 2^3 f_{10} = \ldots\ldots$
$f_{16} = 2^8 f_0 = 2^8 = 256$

EXAMPLE 6 *If* $f(n)$ *defines the number of pages read by a person in* n*th hour, where* $f(n) = \{\sqrt{n}\}$; $\{x\} \rightarrow$ *least integer greater than or equal to* x *and* n *is the* n^{th} *hour. If a person reads the book for 10 hours, then find the total number of pages read by him.*

SOLUTION Total number of pages read by him
$= f_1 + f_2 + f_3 + \ldots + f_{10}$
$f_1 = \{\sqrt{1}\} = \{1\} = 1$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$f_2 = \{\sqrt{2}\} = \{1.414\} = 2$
$f_3 = \{\sqrt{3}\} = \{1.732\} = 2$
$f_4 = \{\sqrt{4}\} = 2$
$f_5 = \{\sqrt{5}\} = 3$
$f_6 = \{\sqrt{6}\} = 3$
$f_7 = \{\sqrt{7}\} = 3$
$f_8 = \{\sqrt{8}\} = 3$
$f_9 = \{\sqrt{9}\} = 3$
$f_{10} = \{\sqrt{10}\} = 4$

$\therefore$ Total number of pages read by a person
$= 1 + 2 + 2 + 2 + 3 + 3 + 3 + 3 + 3 + 4$
$= 26$ pages

MAXIMA AND MINIMA

In this topic we study the maximum and minimum value of various functions. In advance mathematics, we find the maxima and minima through differentiation of the function (As in calculus). But for those students who are not well-versed with calculus, they can solve these problems with some unconventional methods.

Recall that we have been already studied the maxima and minima in the chapter "Theory of Equations."

EXAMPLE 1 *What is the minimum value of the expression* $x^2 + 8x + 10$?

SOLUTION **Method 1.** $x^2 + 8x + 10 = x^2 + 8x + 16 - 6$

$\Rightarrow \quad = (x + 4)^2 - 6$

Now the smallest possible value of the above expression can be -6

Since $(x + 4)^2$ can be minimum 0, when $x = -4$

Method 2. Compare the given expression with the standard quadratic equation $ax^2 + bx + c = 0$

$\therefore \quad a = 1, b = 8, c = 10$

Now, $\text{minima} = \dfrac{-(b^2 - 4ac)}{4a}$ $\qquad (D = b^2 - 4ac$

$D \rightarrow$ Discriminant)

$= \dfrac{-(64 - 40)}{4} = -6$

Method 3. By differentiating the given expression twice w.r. to the given variable, we get minima or maxima.

Let $\quad y = x^2 + 8x + 10$

$\therefore \quad \dfrac{dy}{dx} = y' = 2x + 8$

Again $\quad \dfrac{d^2y}{dx^2} = y'' = 2$

Since $\dfrac{d^2y}{dx^2} = 2$ (+ve value), hence the given expression gives minima.

Now since $\quad y' = 2x + 8$

Let $\quad y' = 0$

$\Rightarrow \quad 2x + 8 = 0 \Rightarrow x = -4$

$\therefore \quad y = x^2 + 8x + 10 = (-4)^2 + 8(-4) + 10 = -6$

NOTE If $\dfrac{d^2y}{dx^2}$ is a –ve value then we get maxima.

Hence the minimum value of the expression is -6.

EXAMPLE 2 *What is the maximum value of the expression* $5 - 6x - x^2$

SOLUTION **Method 1.** $5 - 6x - x^2 = 14 - (3 + x)^2$

Thus the expression can have maximum value 14, when $x = -3$

Since the minimum value of $(3 + x)^2$ is zero.

Method 2. $\quad -x^2 - 6x + 5 = 0 \qquad \ldots(1)$

Comparing the above expression with standard quadratic equation $ax^2 + bx + c = 0$, we get

$a = -1, \quad b = -6, \quad c = 5$

Since a is negative, therefore the expression gives maxima.

Now, $\quad \text{Maxima} = \dfrac{-D}{4a} = \dfrac{-(b^2 - 4ac)}{4a}$

$= \dfrac{-(36 + 20)}{-4} = 14$

Method 3. Let $y = -x^2 - 6x + 5 \qquad \ldots(1)$

$\therefore \quad \dfrac{dy}{dx} = -2x - 6$

$\therefore \quad \dfrac{d^2y}{dx^2} = -2$

Since $\dfrac{d^2y}{dx^2} = -2$ (– ve value), therefore we get maxima

Now, let $\quad \dfrac{dy}{dx} = 0$

$\Rightarrow \quad -2x - 6 = 0$

$\Rightarrow \quad x = -3$

Substituting $x = -3$ in equation (1), we get the

$\text{maxima} = -(-3)^2 - 6(-3) + 5$

$= -9 + 18 + 5 = 14$

EXAMPLE 3 *What is the maximum value of the function*

$$y = \frac{2x^2 + 3x + 4}{x^2 + x + 3}$$

SOLUTION $\quad y = \dfrac{2x^2 + 3x + 4}{x^2 + x + 3}$

$\Rightarrow \quad (y - 2)x^2 + (y - 3)x + (3y - 4) = 0$

$\therefore \quad (y - 3)^2 - 4(y - 2)(3y - 4) \geq 0 \quad \{\because \ x \in R, \ \because b^2 - 4ac \geq 0\}$

$\Rightarrow \quad 11y^2 - 34y + 23 \leq 0$

$\therefore \quad 1 \leq y \leq \dfrac{23}{11}$

$\therefore$ The maximum value of the expression is $\dfrac{23}{11}$

(and minimum value is 1)

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmanboob.wordpress.com
For More Visit : www.LearnEngineering.in

Points to remember :

(i) When $a_1 + a_2 + a_3 + \ldots a_n = k$, then the maximum value of $a_1 \cdot a_2 \cdot a_3 \ldots a_n$ is obtained when $a_1 = a_2 = a_3 = \ldots = a_n$.

(ii) When $a_1 \cdot a_2 \cdot a_3 \ldots a_n = k$, then the minimum value of $a_1 + a_2 + a_3 + \ldots a_n$ is obtained when $a_1 = a_2 = a_3 = \ldots a_n$.

EXAMPLE 4 *If $a + b = 20$, then the maximum value of ab is :*
(a) 19 (b) 75 (c) 100 (d) 120

SOLUTION Maximum of $(ab) = 10 \times 10$ $(\because a + b = 20)$
$= 100$ $(a = b = 10)$

EXAMPLE 5 *If $a + b + c = 24$, then the maximum value of abc is :*
(a) 215 (b) 512 (c) 125 (d) 576

SOLUTION abc can be maximum only when $a = b = c$ for a given sum $a + b + c = 24$

$\therefore$ $(abc)_{max} = (8 \times 8 \times 8)$ $(\because a = b = c)$
$= 512$

EXAMPLE 6 *If $ab = 25$, then the minimum value of $a + b$ is :*
(a) 10 (b) 8 (c) 24 (d) 16

SOLUTION Minimum value of $(a + b)$ can be obtained when $a = b$
then $a + b = 5 + 5$ $(\because ab = 25$ and $a = b)$
$= 10$

EXAMPLE 7 *If $f(x) = \min(4x + 3, x + 4)$ for $x \in [0, 2]$, what is the maximum value of $f(x)$?*

SOLUTION $f(x) = \min(4x + 3, x + 4)$ is an increasing function for every $x \in [0, 2]$

It means the value of $f(x)$ increases as x increases.

$\therefore$ at $x = 2$, $f(x)$ will be maximum

$f(x)_{max} = \min(11, 6)$ at $x = 2$

$f(x)_{max} = 6$

Hence the maximum value of the function is 6.

EXAMPLE 8 *If $f(x) = \max(4x + 3, 3x + 6)$ for $x \in [-6, 10]$, Find the maximum value of $f(x)$.*

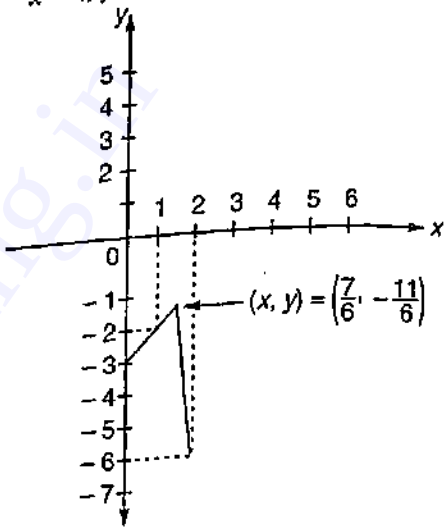
SOLUTION Since the function $f(x) = \max(4x + 3, 3x + 6)$ is an increasing function. Therefore at $x = 10$ we get the maximum value

$f(x)_{max} = \max(43, 36)$

$f(x)_{max} = 43$

EXAMPLE 9 *What is the maximum value $f(x) = \min(4 - 5x, x - 3)$ for every $x \in (0, 4)$?*

SOLUTION At $x = 0$ $f(x) = \min(4, -3) = -3$
$x = 1$, $f(x) = \min(-1, -2) = -2$
$x = 2$, $f(x) = \min(-6, -1) = -6$
$x = 3$, $f(x) = \min(-11, 0) = -11$
$x = 4$, $f(x) = \min(-16, 1) = -16$ etc.



It is clear from the graph that $f(x)_{max}$ will lie between $x = 1$ and $x = 2$ *i.e.*, when $x \in (1, 2)$ and the $f(x)_{max}$ will be obtained when

$4 - 5x = x - 3$

$\Rightarrow$ $6x = 7$

$\Rightarrow$ $x = \frac{7}{6}$

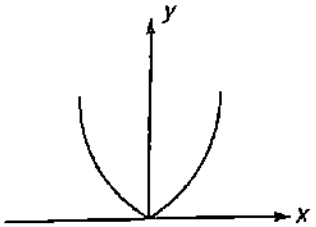
$\therefore$ $f(x)_{max} = \min(4 - 5x, x - 3)$
$= \min\left(-\frac{11}{6}, -\frac{11}{6}\right) = -\frac{11}{6}$

NOTE In case of $f(x)_{max} = \min(x, y)$
$f(x)_{max}$ will be obtained when $x = y$
Similarly in case of $f(x)_{min} = \max(x, y)$
$f(x)_{min}$ will be obtained when $x = y$

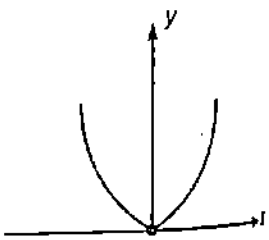
INTRODUCTORY EXERCISE-17.3

Directions for question number 1-4 :
Mark (a) if function is one-one and into
Mark (b) if function is one-one and onto
Mark (c) if function is many one and into
Mark (d) if function is many one and onto

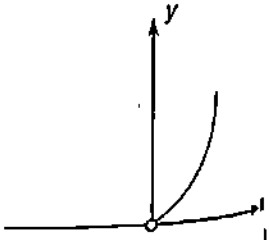
1. $f(x) = x^2$
$f : R \to R$



2. $f(x) = x^2$
$f : R \to R^+ \cup \{0\}$



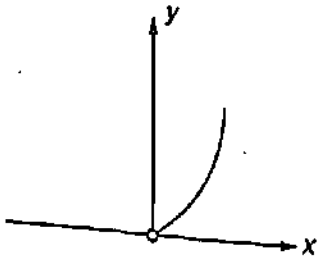
3. $f(x) = x^2$
$f : R^+ \to R$



For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

4. $f(x) = x^2$
$f : R^+ \to R^+$



5. Let $f : R \to R$ be a function defined by $f(x) = \frac{e^{|x|} - e^{-x}}{e^x + e^{-x}}$, then :
(a) f is both one–one and onto
(b) f is one–one but not onto
(c) f is onto but not one–one
(d) f is neither one–one nor onto

6. Let $n(A) = 3$ and $n(B) = 5$, then the number of one–one functions from A to B is :
(a) 15 (b) 60
(c) 125 (d) 10

7. Let $A = R - \{3\}$, $B = R - \{1\}$. Let $f : A \to B$ be defined by $f(x) = \left(\frac{x-2}{x-3}\right)$, then
(a) f is bijective
(b) f is one–one only
(c) f is onto only
(d) f is one–one but not onto

8. The function $f : N \to N$ (N is the set of natural numbers defined by $f(n) = 2n + 3$ is :
(a) surjective (b) injective
(c) bijective (d) none of these

9. The function $f : R \to R$ defined by $f(x) = (x-1)(x-2)(x-3)$ is :
(a) one–one but not onto
(b) onto but not one–one
(c) both one–one and onto
(d) neither one-one nor onto

10. If the function $f : R \to R$ be such that $f(x) = x - [x]$, where $[k]$ denotes the greatest integer less than or equal to k, then $f^{-1}(x)$ is :
(a) $[x] - x$ (b) $\frac{1}{x - [x]}$
(c) not defined (d) none of these

11. If $f(x) = x^2 + 2$ then $f^{-1}(x)$ is :
(a) $\sqrt{x-2}$ (b) $\sqrt{x+2}$
(c) $\sqrt{x} + 2$ (d) $x + 4$

12. If $f(x) = 5^x$, then $f^{-1}(x)$ is :
(a) x^5 (b) 5^{-x}
(c) $\log_5 x$ (d) none of these

13. If $f(x) = \frac{x}{x+1}$, then $f^{-1}(x)$ is :
(a) $\frac{x+1}{x}$ (b) $\frac{x}{1-x}$
(c) $\frac{x-1}{x}$ (d) none of these

14. If $f(x) = \frac{x-1}{x+1}$, then $f^{-1}(x)$ is :
(a) $\frac{1+x}{1-x}$ (b) $\frac{x+1}{x-1}$
(c) $\frac{1}{x+1}$ (d) none of these

15. Let $f : (-\infty, 1] \to (-\infty, 1]$ such that $f(x) = x(2-x)$, then $f^{-1}(x)$ is :
(a) $1 + \sqrt{1-x}$ (b) $1 - \sqrt{1-x}$
(c) $\sqrt{1-x}$ (d) none of these

16. If $f(x)$ is a one to one function, where $f(x) = x^2 - x + 1$, then find the inverse of the $f(x)$:
(a) $\left(x - \frac{1}{2}\right)$ (b) $\sqrt{x - \frac{3}{4}} + \frac{1}{2}$
(c) $\sqrt{x - \frac{3}{4}} - \frac{1}{2}$ (d) none of these

17. In the above question find the value of $f\left(\frac{3}{4}\right) + f^{-1}\left(\frac{3}{4}\right)$
(a) $\frac{21}{16}$ (b) $\frac{4}{5}$
(c) $\frac{16}{5}$ (d) none of these

www.LearnEngineering.in
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

EXERCISE

LEVEL (1)

Directions for questions number 1 and 2 :
$P(A, B) = A \times B$, $S(A, B) = A - B$, $D(A, B) = A + B$

1. $D(P(3, 5), S(5, 3))$ is equal to :
(a) 15 (b) 17
(c) 0 (d) 30

2. $P(S(D(7, 3), P(2, 3)), P(6, 5))$ is equal to :
(a) 120 (b) 60
(c) 30 (d) none of these

Directions for question number 3-5 : *For every $a, b \in N$*
$a @ b = a^2$ *when* $(a + b)$ *is even*
$= a^2 - b^2$ *when* $(a + b)$ *is odd*
$a \# b = b^2$ *when* $a \times b$ *is odd*
$= a^2 - b^2$ *when* $a \times b$ *is even*

3. Find the value of $20 \# (21 @ 22)$:
(a) 0 (b) 2249
(c) -1849 (d) can't be determined

4. What is the value of $(a \# a) @ (a @ a)$?
(a) 0 (b) a^4
(c) either (a) or (b) (d) neither (a) nor (b)

5. Find the value of $(p \# q) - (q @ p)$; p is even and q is odd :
(a) 0 (b) $2(p^2 + q^2)$
(c) $2(p^2 - q^2)$ (d) none of these

6. If $M(a, b) = a^2 + b^2 + ab$; $N(a, b) = a^2 + b^2 - ab$. Find the value of $M(7, N(9, 4))$.
(a) 5197 (b) 3197
(c) 4197 (d) none of these

7. If $f(x) = x^3 - x^2$, then $f(x + 1)$ is :
(a) $x^3 - x^2 + 1$ (b) $x(x + 1)^2$
(c) $x^2(x - 1)$ (d) none of these

Directions for question number 8-10 :

$$a \cup b = \begin{cases} a + b, \text{ if } a < b \\ a \vee b, \text{otherwise} \end{cases}$$

$$a \cup\cup b = \begin{cases} a \times b, \text{ if } a < b \\ a \vee\vee b, \text{otherwise} \end{cases}$$

$$a \cup\cup\cup b = \begin{cases} a^b, \text{ if } a < b \\ a \vee\vee\vee b, \text{otherwise} \end{cases}$$

$$a \vee b = \begin{cases} a - b, \text{ if } a \geq b \\ a \cup b, \text{otherwise} \end{cases}$$

$$a \vee\vee b = \begin{cases} \frac{a}{b}, \text{ if } a \geq b \\ a \cup\cup b, \text{otherwise} \end{cases}$$

$$a \vee\vee\vee b = \begin{cases} \frac{1}{a^b}, \text{ if } a \geq b \\ a \cup\cup\cup b, \text{otherwise} \end{cases}$$

8. Find the value of $[(5 \cup 4) \vee\vee (2 \cup\cup\cup 3)] \cup (8 \vee 3)$:
(a) 1 (b) 2
(c) 3 (d) 4

9. Which of the following is true?
(a) $2 \cup\cup 8 = 2 \cup\cup\cup 4$ (b) $8 \cup\cup 2 = 4 \cup\cup\cup 2$
(c) $8 \cup\cup\cup 2 = 2 \vee\vee\vee 5$ (d) none of these

10. Which one of the following is/are true?
(a) $a \cup b = a \vee b$ for all real values of a and b
(b) $a \vee\vee b = a \cup\cup b$ is true for integer values of a and b
(c) $a \vee\vee\vee b = a \cup\cup\cup b$ is true only for natural number a, b
(d) none of the above

Directions for question number 11-14 :
$x @ y = (x + y)^2$
$x \# y = (x - y)^2$

11. Find the value of $(20 @ 4) \# 35$:
(a) 280081 (b) 292681
(c) 262781 (d) 301756

12. Find the value of $12 \# (14 @ (12 \# 13))$:
(a) 4539 (b) 42336
(c) 45369 (d) 89719

13. Find the value of $(9 \# 7) @ (21 \# 5)$:
(a) 67600 (b) 62500
(c) 57600 (d) none of these

14. $(2 @ 2) + (3 \# 3) - (4 @ 4) + (5 \# 5)$ is equal to :
(a) 64 (b) -48
(c) 28 (d) none of these

Directions for question number 15 and 16 :

$$CAT(a, b) = \begin{cases} a^b, \text{ if } a > 0 \text{ and } b \in I \\ 0, \text{otherwise} \end{cases}$$

$$MAT(a, b) = \begin{cases} b - a - 1, \text{ if } b > 0 \\ 1, \text{otherwise} \end{cases}$$

$$XAT(a, b) = \begin{cases} \text{remainder } (b / a), \text{ if } a > 0 \\ 2, \text{otherwise} \end{cases}$$

15. $\dfrac{XAT[MAT\{2, CAT(3, XAT[3, 2])\}, 5]}{CAT[4, XAT\{MAT(2, -5), 4\}]}$:
(a) 0 (b) 2
(c) 5 (d) none of these

16. $\dfrac{XAT(r, s)}{MAT(r, s)} = 2$:
(i) $r = 3, s = 5$ (ii) $r = -3, s = -5$
(a) (i) is true (b) (ii) is true
(c) both are true (d) none is true

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Directions for question number 17-19 : a and b are any numbers and x and y are any non negative integers.
l(a) = the greatest integer less than or equal to a
g(a) = the smallest integer greater than or equal to a
r(x, y) = the remainder when x is divided by y.

17. $r(16, 7) - 2$ is equal to :
(a) 0 (b) 1
(c) 2 (d) 3

18. $g(4.7) - l(2)$ is equal to :
(a) 1 (b) 2
(c) 3 (d) 4

19. $r(x, y)$ with both x and y not equal to zero is equal to :
(a) $l\left(\frac{x}{y}\right)$ (b) $x - \left[yl\left(\frac{x}{y}\right)\right]$
(c) $y - \left[xl\left(\frac{x}{y}\right)\right]$ (d) $\left[l\left(\frac{x}{y}\right) - y\right]$

Directions for question number 20-22 :

$$x \# y = \begin{cases} x + 2y; \text{ if } |x + y| \text{ is even} \\ 2x @ 2y; \text{ if } |x + y| \text{ is odd} \end{cases}$$

$$x @ y = \begin{cases} 3x - y; \text{ if } |x + y| \text{ is even} \\ 2x \# 2y; \text{ if } |x + y| \text{ is odd} \end{cases}$$

20. Find the value of $2 \# ((7 @ (4 \# 5)) @ 3)$:
(a) 204 (b) 306
(c) 408 (d) 102

21. Which of the following is true?
(i) If $q = 2p$ and $p + q$ is even, then $p \# q = q \# p$
(ii) If $q = 3p$, $p @ q$ will always be zero
(iii) If $q = 2p$ and p is odd, then $p @ q$ will be even
(a) (ii) only (b) (iii) only
(c) (i) and (iii) only (d) (ii) and (iii) only

22. Which of the following is true?
(i) $6 @ 2 = 12 \# 2$
(ii) $6 \# 3 = (14 @ 12) + (2 @ 6)$
(iii) $(6 @ 2) + (6 \# 3) = (12 \# 2) + (14 @ 12) - (2 @ 6)$
(a) (i) and (ii) only (b) (i) and (iii) only
(c) (ii) and (iii) only (d) (i), (ii) and (iii)

Directions for question number 23-25 :

$$a \# b = a^2 + b - 1$$
$$a \$ b = a^2 + b^2 + 1$$
$$\text{Max}(a, b) = |a + b|$$
$$\text{Min}(a, b) = |a - b|$$

23. Max [(2 #3) $ (3 #4), (3 #4) $ (4 $ 4)] is equal to :
(a) 1234 (b) 1236
(c) 1335 (d) none of these

24. Max[Min ((1 $ 2), (2 $ 3)), Max ((3 # 4), (4 # 5))] is equal to :
(a) 24 (b) 40
(c) 42 (d) none of these

25. Find the value of
Min [(1 # 2) $ 3, 4#(5 $ 6)] $ Max (6 # 7), (7 # 8)] :
(a) 13574 (b) 14573
(c) 13575 (d) 14674

26. If $a \Delta b = a^b + ab + \frac{a}{b}$, then $(9 \Delta 3)$ is equal to :
(a) 729 (b) 927
(c) 759 (d) none of these

27. $a * b \# c = (ab)^c$:
$a \# b * c = a^b \cdot c$, then find the value of $3 \# 4 * 5 + 3 * 4 \# 5$
(a) 35287 (b) 249237
(c) 12345 (d) none of these

Directions for question number 28 and 29 :

$$a \# b = (-1)^{ab}(a^b + b^a)$$
$$f(x) = x^2 - 2x \text{ if } x \geq 0$$
$$= 0 \text{ if } x < 0$$
$$g(x) = 2x \text{ ; if } x \geq 0$$
$$= 1 \text{ ; if } x < 0$$

28. Find the value of $f(2 \# 3) \# g(3 \# 4)$:
(a) 2045 (b) $(17)^{18}$
(c) $(180)^{91}$ (d) none of these

29. Find the value of $f(g(2 \# 3)) + g(f(1 \# 2))$:
(a) 9401 (b) 1049
(c) 1094 (d) none of these

Directions for question number 30, 31 :

$$a \oplus b = (a + b)^2 - ab$$
$$a * b = (ab)^2 - ab$$
$$a @ b = (a - b)^2 - ab$$

30. Find the value of $(5 \oplus 6) - (6 @ 3) + (2 * 4)$
(a) 156 (b) 256
(c) 1056 (d) none of these

31. Find the vlaue of $(((1 \oplus 2) * 3) @ 20)$:
(a) 151600 (b) 141500
(c) 161500 (d) none of these

Directions for question number 32 and 33 :
If $m < n$ then $m\,Dn = m\,Cn$ else $mDn = nCm$
If $m = n$ then $mEn = mn$ else $mEn = mCn$ where $mCn = \frac{m}{n}$

32. Find the value of (3 D 4) D (12 D 5) :
(a) $\frac{9}{5}$ (b) $\frac{5}{9}$
(c) $\frac{2}{3}$ (d) $\frac{5}{3}$

33. Find the value of (12 D 60) E (2 C 10) :
(a) 1 (b) $\frac{1}{5}$
(c) 1/25 (d) 0

Directions for question number 34-38 : The operands in the Devic Mathematics are as follows :
+ → multiplication
− → division
× → addition
/ → subtraction
^ → power (or index) for exponential expression.
But Devic Mathematics must follow the same conventional BODMAS rule of mathematics. Solve the following questions in Devic Mathematics.

34. $30 + 20 - 4 \times 6$ is equal to :
(a) 156 (b) 276
(c) 330 (d) none of these

35. $50 / 40 - 5 + 4 \times 54$ is equal to :
(a) 720 (b) 72
(c) 144 (d) none of these

36. $(5 \wedge 5) \times (4 \wedge 4) / (3 \wedge 3) \times (2 \wedge 2) / (1 \wedge 1)$ is equal to
(a) 2987 (b) 4013
(c) 3357 (d) none of these

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

37. The value of the following expression in Devic mathematics is : $[(x^2 \times y^2) - (x + y)^2] + \left[(x+y)^3 - \left\{\left(\frac{x}{y}\right)^2 \times 2 + x + y\right\}\right]$

(a) $\frac{x}{y}$ (b) $x + y$

(c) $x \times y$ (d) $x \wedge y$

38. The value of the following expression in Devic mathematics is $a + a + a - a + a + a / a + a - a$:

(a) 1 (b) 0

(c) $1 - a$ (d) $\frac{1}{a}$

✍ ***Directions for question number 39 and 40 :***

$P \Delta Q = P + Q$, if $PQ < 0$

$P \Delta Q = P - Q$, if $PQ > 0$

39. $[(2\Delta 3) \Delta (-2\Delta 3)]$ is equal to :

(a) 0 (b) 1

(c) 2 (d) 3

40. If $(a \Delta b) \Delta (a\Delta - b) = 0$ and $a > 0$ then which of the following is true?

(a) $a < b$ and $b > 0$ (b) $|a| > |b|$ and $b < 0$

(c) $a > b$ and $b > 0$ (d) all of these

41. If $f(x) = x^3 + 3x^2 + 3x$, then $f(x + 1)$ is equal to

(a) $x^3 + 6x^2 + 12x + 7$ (b) $x^3 + 4x^2 + 5$

(c) $3x^3 + 12x^2 + 4x + 1$ (d) $2x^3 + 6x^2 + 3$

42. $x @ y = xy - 1$, if $x > y$

$= 0$; otherwise

and $x \# y = \frac{x + y}{x - y}$ if $x \geq 0, y \geq 0$

$= 1$; otherwise.

The value of $\frac{(4 \# 2) @ (9 \# (-1))}{(2 @ 3) \# (5 @ 2)}$:

(a) 1 (b) -2

(c) 0 (d) none of these

✍ ***Directions for question number 43 and 44 :*** *The following functions are defined for any two distinct, non zero integers a and b*

$$f_1(a, b) = |a| \times b^2$$
$$f_2(a, b) = a^2 \times |b|$$
$$f_3(a, b) = \frac{(a^2 + b^2)}{2}$$
$$f_4(a, b) = \left(\frac{a}{2} + \frac{b}{2}\right)$$

43. Which of the following is equal to $(a + b)^2$?

(a) $f_3(a, b) \div f_4(a, b)$ (b) $2 \times f_3(a, b) \times f_4(a \times b)$

(c) $4 \times f_3(a, b) \times f_4(a, b)$ (d) none of these

44. Which one of the following is necessarily greater than zero?

(a) $f_1(a, b) \times f_2(a, b) \times f_3(a, b)$

(b) $f_1(a, b) \times f_2(a, b) \times f_4(a, b)$

(c) $f_1(a, b) \times f_3(a, b) \times f_4(a, b)$

(d) $f_2(a, b) \times f_3(a, b) \times f_4(a, b)$

✍ ***Directions for question number 45-47 :***

lt (p) → the least possible integer greater than or equal to p.

gt (p) → the greatest possible integer less than or equal to p.

$\Delta(p, q) = \text{Min}\ (\text{lt}\ (p), \text{gt}\ (q))$

45. What is the value of lt (8.3) − gt (5.3) + lt (6.8) + gt (9.4)

(a) 15 (b) 20

(c) 19 (d) none of these

46. Find the value of

$\Delta(3, 2) - \Delta(2, 3) + \text{gt}\ (4.8) - \text{lt}\ (2.7) + \Delta(4, 5)$

(a) 4 (b) 5

(c) 6 (d) none of these

47. Which of the following statements is always true about $(a - b)$ where $a = \frac{p + q}{2}$ and $b = \frac{\text{lt}(p) + \text{gt}(q)}{2}$

(a) $-\frac{1}{4} \leq (a - b) \leq \frac{1}{4}$, always

(b) $-\frac{1}{2} \leq (a - b) \leq \frac{1}{2}$

(c) $-1 \leq (a - b) \leq 1$

(d) $-1 \leq (a - b) \leq 1$, except zero

✍ ***Directions for question number 48–49 :***

$$a @ b = \frac{a}{a + b}$$

and $$b \# a = \frac{a + b}{b}$$

48. Which of the following is true ?

(a) $\frac{a @ b}{b \# a} = \frac{a}{b}$ (b) $(a @ b)(b \# a) = a^2 b^2$

(c) $(a @ b)(b \# a) = \frac{a}{b}$ (d) $\frac{a @ b}{b \# a} = \frac{a + b}{b + a}$

49. Which of the following is true?

(a) $a @ b + b \# a = \frac{a^2 + b^2 + 3ab}{a(b + a)}$

(b) $a @ b - b \# a = \frac{a^2 + b^2 + ab}{b(b + a)}$

(c) $b \# a - a @ b = \frac{a^2 + b^2 + ab}{b(a + b)}$

(d) $a @ b + b \# a = 1$

✍ ***Directions for question number 50 and 51 :*** *The following operations are defined for any two real number a and b.*

$mn\ (p, q) = \min\ (p, q)$

$mx\ (p, q) = \max\ (p, q)$

$md\ (p) = |p|$

50. Find the value of $mn\ (md\ (p), mx\ (q, mn\ (p, md(q))))$ if $p = -2, q = 5$:

(a) -2 (b) 2

(c) 3 (d) -7

51. Find the value of $md\ (mn\ (P, mx\ (md\ (q), mn\ (P, r))))$ if $p = -2, q = 5, r = 7$.

(a) -1 (b) 0

(c) 2 (d) none of these

✍ ***Directions for question number 52 and 53 :***

$a @ b = (a + b)(ab + b)$

$a \# b = (a - b)(ab - b)$

$a \Delta b = (a @ b) - (a \# b)$

52. Find the value of $((1 @ (2\Delta 3)) \# 4)$:

(a) 35500 (b) 482420

(c) 123400 (d) none of these

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

53. Find the value of $(1 @ 1) \Delta (2 \# 2)$
(a) 5 (b) 22
(c) 17 (d) none of these

Directions for question number 54 and 55 :

$$a @ b = |a - b|$$
$$a * b = ab$$
$$a \# b = |a^2 - b^2|$$
$$a \Delta b = a^2 + b^2$$

where $a, b \in R$ and $a \neq b$ and the real algebraic operations are unchanged.

54. Find the value of $(a \Delta b) + (a \# b)$:
(a) $2a^2$ (b) $2b^2$
(c) $2(a^2 + b^2)$ (d) can't be determined

55. The value of the expression $[(a \# b) \div (a @ b)]^2 - 2(a * b)$:
(a) $a \# b$ (b) $a \Delta b$
(c) $(a \# b)(a @ b)$ (d) can't be determined

Directions for question number 56-58 : *A function $f(x) = \log(g(x))$, where $g(x)$ is any function of x.*

56. For what value of $g(x)$, $g(x) = g(f(x))$?
(a) e (b) e^x
(c) $\log x$ (d) none of these

57. For what value of $g(x)$ is the function of $f(x) = g(x)$?
(a) $g(x) = e$ (b) $g(x) = e^x$
(c) $g(x) = \log x$ (d) none of these

58. If $g(x) = e^x$, then $f(f(x))$ is equal to :
(a) e (b) e^x
(c) $\log x$ (d) none of these

59. If $f(x) = 2x + 3$ and $g(x) = 9x + 6$, then find $g[f(x)] - f[g(x)]$
(a) 18 (b) 22
(c) 20 (d) none of these

60. If $f(x) = x^2 + 2x + 2;\ x \geq -1$
$= -x^3 + 3x + 1;\ x < -1$
then find the value of $f(f(-2))$:
(a) 15 (b) 17
(c) 22 (d) none of these

Directions for question number 61-63 :
$[m]$ is defined as the greatest integer less than m.
$\{m\}$ is defined as the least integer greater than m.

$$f(x, y) = \{x\} + [y]$$
$$g(x, y) = [x] - \{y\}$$
$$F(f(x, y)) = \{f(x, y)\} - [g(x, y)]$$
$$G(g(x, y)) = [f(x, y)] - \{g(x, y)\}$$

61. Given that $x^2 = 4$ and $y^2 = 9$ and $|F(f(x, y))| = |G(g(x, y))|$ then find the vlaue of $x + y$:
(a) -1 (b) 0
(c) 1 (d) 5

62. If x and y are consecutive integers, then find $g(x, y)$
(a) -1 (b) -3
(c) -1 or -3 (d) none of these

63. Which of the following is false?
(a) $f(x, y) + g(x, y) = F(f(x, y)) + G(g(x, y))$
(b) $F(f(x, y)) - G(g(x, y)) = -2[g(x, y)]$
(c) $F(f(x, y) + G(g(x, y)) = 2\{f(x, y)\} + 2$
(d) none of the above

Directions for question number 64-66 : *A decimal number 'm' (say) can be expressed as*
$m = I + D$
where $I \rightarrow$ integral part
and $D \rightarrow$ decimal part, $0 \leq D < 1$
and $[m]$ = greatest integer less than or equal to m if $0 < D < 0.5$
also $\{m\}$ = least integer greater than or equal to m if $0.5 \leq D < 1$

$$\text{Hence, } f(x, y) = \begin{cases} [x + y] \text{ if } 0 < D < 0.5 \\ \{x + y\} \text{ if } 0.5 \leq D < 1 \\ (x + y) \text{ otherwise} \end{cases}$$

Also, $F(f(x, y)) = -f(x, y)$
and $G(f(x, y)) = -F(f(x, y))$

64. What is the value of the following
$f(G(f(1.2, 1.3))), f(F(f(0,1))), G(f(0.8, 0.4)))$?
(a) 1 (b) 2
(c) 3 (d) 4

65. Which of the following statements is true if x and y are positive ?
(a) $F(f(x, y)) . G(f(x, y)) = -F(f(x, y)) . G(f(x, y))$
(b) $F(f(x, y)) . G(f(x, y)) > -F(f(x, y)) . G(f(x, y))$
(c) $F(f(x, y)) . G(f(x, y)) \neq G(f(x, y)) . F(f(x, y))$
(d) $F(f(x, y)) + G(f(x, y)) + F(x, y) = F(x, y)$

66. Which of the following expressions will yield a number which has a positive integral square root if x, y are positive
(a) $G(f(1.2, 1.3)) \times f(F(f(2, 1), G(f(4, 1))$
(b) $F(f(2, 2)) + G(f(2, 2))$
(c) $f(1.1, 1.2) \times G(f(1.1, 1.2))$
(d) $G(f(0, 1)) \times G(f(0, 1))$

67. $f(x) = \frac{1}{x}; g(x) = \left(x + \frac{1}{x}\right)$, then which of the following is true?
(a) $f(g(x)) . g(f(x)) = 1$ (b) $\frac{f(g(x))}{g(f(x))} = 1$
(c) $f(g(x)) + g(f(x)) = 1$ (d) none of these

68. If $f(x) = 2x^2 + 7x - 9$ and $g(x) = 2x + 3$, then find the value of $g(f(x))$ at $x = 2$
(a) 92 (b) 39
(c) 29 (d) none of these

69. Find $f(f(f(f(f(2)))))$ if $f(x) = \frac{x + 1}{x - 1};\ x \neq 1$
(a) 1 (b) 2
(c) 3 (d) 5

Directions for question number 70 and 71 :
$f(x) = 1 - h(x), g(x) = 1 - k(x), h(x) = f(x) + 1$
$j(x) = g(x) + 1, k(x) = j(x) + 1$

70. Find the value of $\frac{j(f(x)) + k(h(x))}{h(k(x)) + f(j(x))}$:
(a) 0 (b) 1
(c) 2 (d) none of these

71. $\frac{h(f(k(x))) + j(g(h(x))) + k(j(f(x)))}{[f(x) + j(x) + k(x)][f(x) . j(x) k(x)]}$ is equal to :
(a) 0 (b) 3
(c) 3/2 (d) none of these

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

72. If $f(x)$ and $g(x)$ are odd functions of x, then which of the following is true?
(a) $f+g$ is odd, fg is odd
(b) $f+g$ is odd, fg is even
(c) $f+g$ is even, fg is odd
(d) $f+g$ is even, fg is even

73. If $f(x)$ is an even function of x and $g(x)$ is an odd function then which of the following is true?
(a) $f[g(x)]$ is odd , $g[f(x)]$ is even
(b) $f[g(x)]$ is odd, $g[f(x)]$ is odd
(c) $f[g(x)]$ is even, $g[f(x)]$ is even
(d) $f[g(x)]$ is even, $g[f(x)]$ is odd

74. If $y = \min(x^2 + 2, 6 - 3x)$, then the greatest value of y for $x > 0$
(a) 1 (b) 2
(c) 3 (d) none of these

75. If $p^2 + q^2 + r^2 = 1$, then the maximum value of pqr is :
(a) 1 (b) $\frac{1}{3\sqrt{3}}$
(c) 3 (d) none of these

Directions for question number 76-78 : *Following functions are defined for three distinct positive numbers, such that $x < y < z$.*

$f(x, y, z) = \min(\max(x+y, y+z), \max(z+x, x+y))$
$g(x, y, z) = \max(\min(x+y, y+z), \min(z+x, x+y))$
$h(x, y, z) = \max(\max(x+y, y+z), \max(z+x, x+y))$
$j(x, y, z) = \min(\min(x+y, y+z), \min(z+x, x+y))$
$k(x, y, z) = \text{int.}((x+y+z)/3)$
such that $k(x, y, z) \le (x+y+z)/3$
$l(x, y, z) = \min(x, y, z)$
$m(x, y, z) = \max(x, y, z)$

76. Which of the following expression is necessarily equal to 1?
(a) $\frac{(f(x,y,z) - m(x,y,z))}{(m(x,y,z) - l(x,y,z))}$
(b) $\frac{g(x,y,z)}{j(x,y,z)}$
(c) $\frac{(m(x,y,z) - f(x,y,z))}{k(x,y,z) + j(x,y,z)}$
(d) $\frac{(h(x,y,z) - f(x,y,z))}{f(x,y,z)}$

77. Which of the following expression gives negative value?
(a) $\frac{(m(x,y,z) - f(x,y,z))}{h(x,y,z) \times k(x,y,z) \times l(x,y,z)}$
(b) $\frac{(m(x,y,z) - l(x,y,z) + j(x,y,z))}{k(x,y,z)}$
(c) $\frac{l(x,y,z) - m(x,y,z)}{g(x,y,z) - j(x,y,z)}$
(d) $\frac{h(x,y,z) - g(x,y,z)}{k(x,y,z)}$

78. Which of the following expressions is necessarily greater than 1 ?
(a) $\frac{[h(x,y,z) - j(x,y,z)]}{k(x,y,z)}$
(b) $[g(x,y,z) - j(x,y,z)] \times m(x,y,z)$
(c) $\frac{g(x,y,z)}{j(x,y,z)}$
(d) $\frac{f(x,y,z) + h(x,y,z) - g(x,y,z)}{j(x,y,z)}$

Directions for question number 79-83 :
$A(x, y, z) = \min(x+y, y+z, z+x)$
$B(x, y, z) = \max(x-y, y-z, z-x)$
$C(x, y, z) = \max(A(x, y, z), B(x, y, z))$
$D(x, y, z) = \min(A(x, y, z), B(x, y, z))$

79. The value of $C(1, 2, 3)$ is :
(a) 1 (b) 2
(c) 3 (d) 4

80. The value of $D(0, 1, 2)$ is :
(a) -1 (b) 1
(c) 0 (d) none of these

81. What is the value of $A(2, 3, 4) + B(2, 3, 4)$?
(a) 13 (b) 4
(c) 7 (d) none of these

82. What is the value of $C(0, 1, D(1, 2, 3))$?
(a) 1 (b) 2
(c) -2 (d) none of these

83. What is the value of $D(1, 2, C(0, 1, 2))$?
(a) 1 (b) 2
(c) 3 (d) 4

84. What is the maximum possible value of xy, where $|x+5| = 1$ and $y = 9 - |x-4|$
(a) 401 (b) 124
(c) 104 (d) none of these

LEVEL 2

1. Let a function is defined as $f(x) = \frac{a^x + a^{-x}}{2}$; $a > 0$, what is the value of $f(x+y) + f(x-y)$?
(a) $f(x) + f(y)$ (b) $f(x) f(y)$
(c) $2f(x) f(y)$ (d) $4f(x) f(y)$

2. If $\left|\frac{30-x}{5}\right| < 4$, then find x :
(a) $-6 < x < 6$ (b) $10 < x < 50$
(c) $x > 10$ (d) none of these

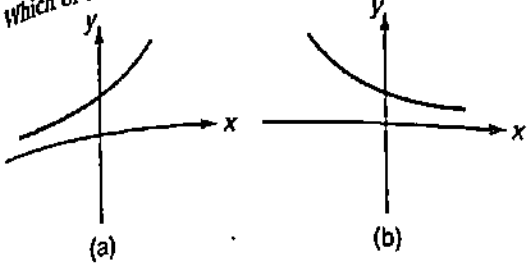
3. Let $f(x, y) = |x+y|$ and $g(x, y) = |x-y|$, then how many ordered pairs of the form (x, y) would satisfy $f(x, y) = g(x, y)$?
(a) 1 (b) 2
(c) 4 (d) infinitely many

4. What is the value of x if $|x-1| + |x| = 2$:
(a) $-\frac{1}{2}$ (b) $\frac{3}{2}$
(c) both (a) and (b) (d) none of these

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

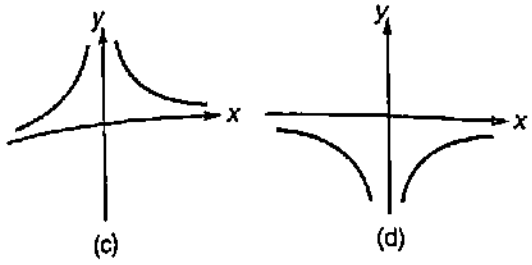
yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

5. Which of the following graph represents e^{-x} ?



(a)

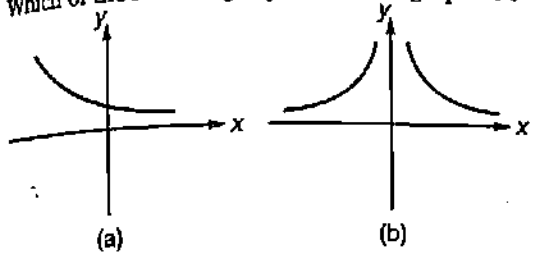
(b)



(c)

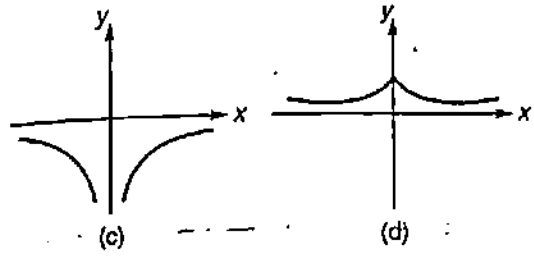
(d)

6. Which of the following represents the graph of e^{-x^2} :



(a)

(b)

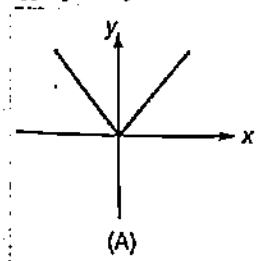


(c)

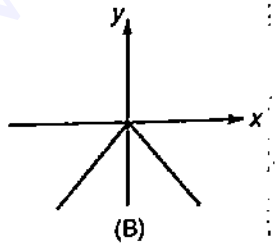
(d)

7. A function is defined as $f(x) = a^x \; \forall \; x \in R$ and $0 < a < 1$, then which of the following is true?
(i) It is an increasing function
(ii) It is a decreasing function
(iii) All such functions must meet at (0, 1)
(a) only (i) is true (b) only (ii) is true
(c) (i) and (iii) are true (d) (ii) and (iii) are true

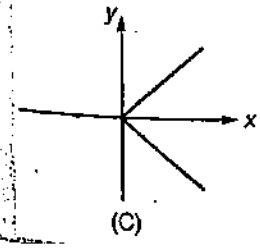
Directions for question number 8-13 : The following figure represents various functions. Match the figure with the appropriate function.



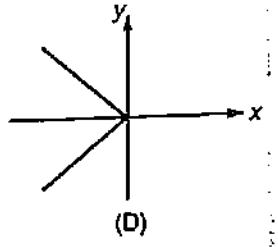
(A)



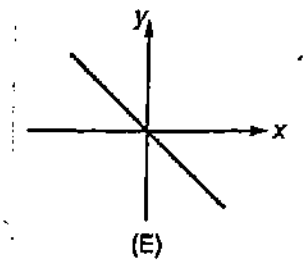
(B)



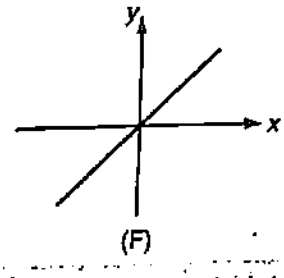
(C)



(D)



(E)



(F)

8. $f(x) = -x$:
(a) B (b) C
(c) E (d) D

9. $f(x) = |x|$:
(a) A (b) B
(c) C (d) D

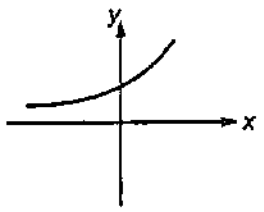
10. $f(x) = -|x|$:
(a) E (b) B
(c) A (d) F

11. $f(x) = x$:
(a) E (b) F
(c) A (d) B

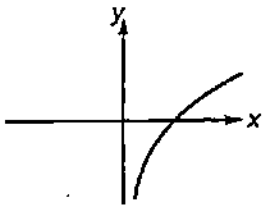
12. $f(x) = |x| - x$:
(a) A (b) C
(c) B (d) none of these

13. $f(x) = ||x| - x|$:
(a) B (b) D
(c) A (d) none of these

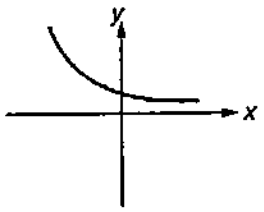
14. If $x = 10^y$ then the graph of inverse of this function is :



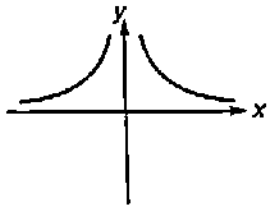
(A)



(B)



(C)



(D)

(a) A (b) B
(c) C (d) D

15. If $f(x) = \begin{cases} 1-x, & 0 \le x \le 2 \\ x-1, & 2 \le x \le 4 \\ 1, & 4 \le x \le 6 \end{cases}$

then find, $f(0) + f\left(\frac{1}{2}\right) + f(1) + f\left(\frac{45}{18}\right)$:
(a) 1 (b) 2
(c) 3 (d) none of these

16. Which one of the following is true about the given graph ?
(a) $x = \frac{|y|}{y}, y \neq 0$

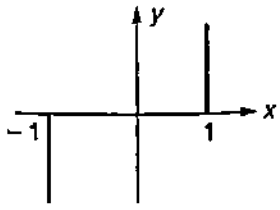
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

(b) $y = \frac{|x|}{x}, x \neq 0$

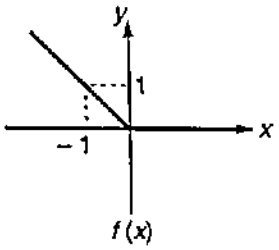
(c) $x = -\frac{|y|}{y}, y \neq 0$

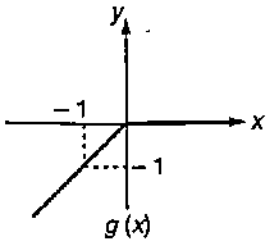
(d) $y = \frac{-|x|}{x}, x \neq 0$



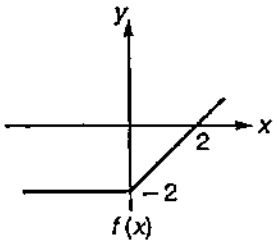
✍ ***Directions for question number 17 and 18 :*** *In each question there are two graphs $f(x)$ and $g(x)$ for the real values of x.*
Mark (a) if $g(x) = f(-x)$
Mark (b) if $g(x) = -f(x)$
Mark (c) if $g(x) = |f(x)|$
Mark (d) if none of the above is true.

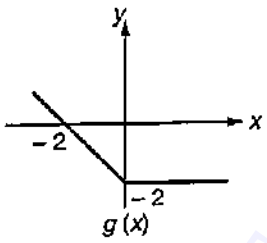
17.





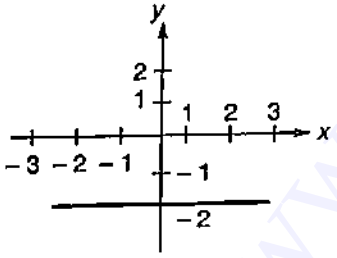
18.



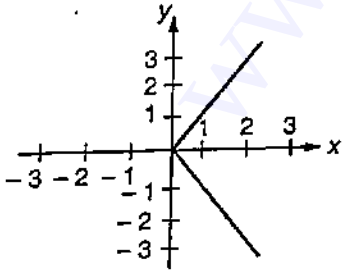


✍ ***Directions for question number19-22:*** *In the following questions a function is represented by the graph.*
Mark (a) if $f(x) = f(-x)$
Mark (b) if $f(x) = -f(-x)$
Mark (c) if $f(x)$ does not exist
Mark (d) if none of the above is true.

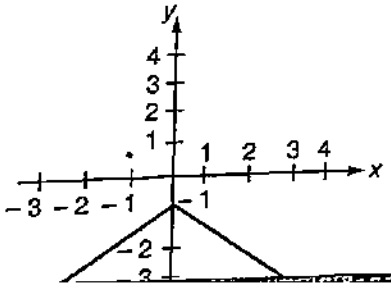
19.



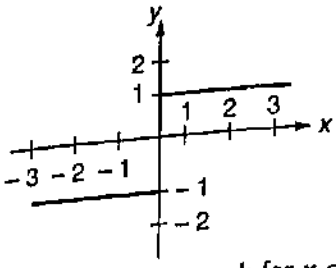
20.



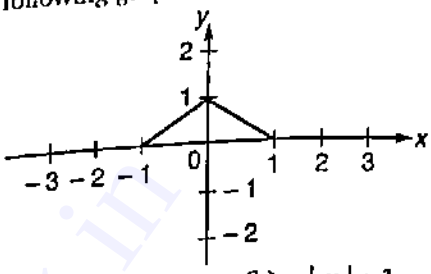
21.



22.



23. In the following graph for $x \in [-1, 1]$, $f(x)$ is defined by:

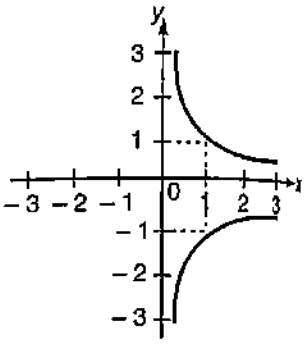


(a) $|x| + 1$ (b) $-|x| + 1$
(c) $-|x + 1|$ (d) $-|x| - 1$

24. How many times does the graph of $y = x^3 - 3x^2 - x + 3$ intersects the x-axis :
(a) 1 (b) 2
(c) 3 (d) none of these

25. Which one of the following is true for $x \in R$:
(a) $f(x) = \log|x|$ (b) $f(x) = \frac{|x|}{\log|x|}$
(c) $f(x) = \log(x^2 + 2)$ (d) $f(x) = \log_x 100$

26. Which one of the following is correct about the given graph of $f(x)$:
(a) $|xy| = 1$
(b) $|x|y = 1$
(c) $x|y| = 1$
(d) $|x||y| = 1$



27. A function is defined as $f(x) = \frac{25}{(x+5)^2}$, then the value of $f(x)$:
(a) decreases as the value of x increases, only if x is negative
(b) increases as the value of x increases, only if x is negative
(c) decreases as the value of x increases, only if x is positive
(d) both (a) and (c) are true

✍ ***Directions for question number 28 and 29 :*** *The following functions are defined for the set of variables $x_1, x_2, x_3, \ldots, x_n$*

$$f(x_i, x_j) = \begin{cases} x_{i+j} \text{ if } i + j \leq n^2 \\ x_{i+j-n} \text{ if } i + j > n^2 \end{cases}$$

$$g(x_i, x_j) = x_m,$$

where m is the remainder when $i \times j$ is divided by n.

28. Find the value of $f(f(x_2, x_3), f(x_5, x_6))$ if $n = 3$
(a) x_5 (b) x_{10}
(c) x_{13} (d) none of these

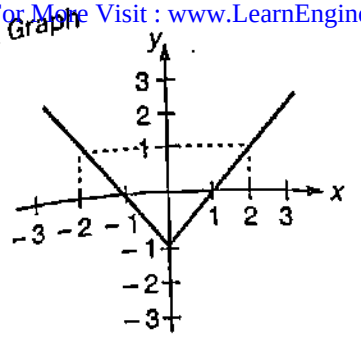
29. Find the value of $g(g(x_2, x_3), g(x_7, x_8))$ if $n = 5$
(a) x_1 (b) x_2
(c) x_5 (d) none of these

✍ ***Directions for question number 30-34 :*** *In each of the following questions a graph of a function is being shown. Select the correct equation of the function of the graph.*

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

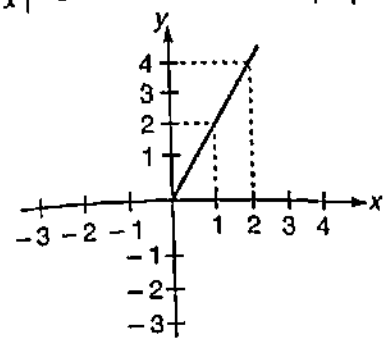
yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

30.



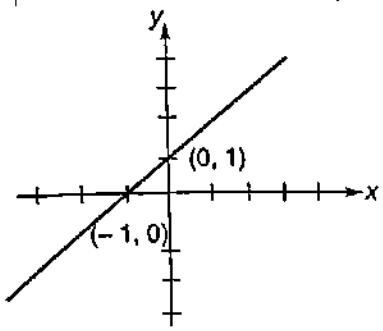
(a) $y = |x-1|$ (b) $x = |y|+1$
(c) $y = |x|-1$ (d) $y = |x|-2$

31.



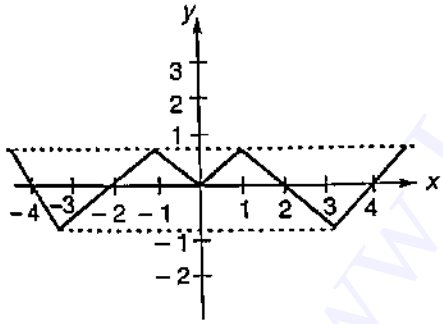
(a) $x = |y|+y$ (b) $y = |x|+x$
(c) $y = |x|$ (d) $x = |y|$

32.



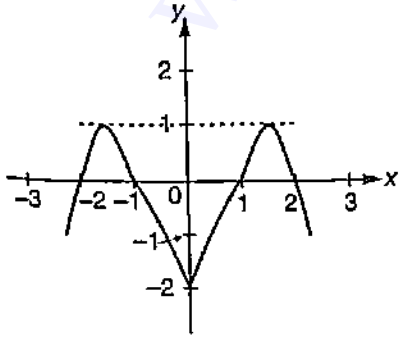
(a) $x = y+1$ (b) $y = x-1$
(c) $x = y-1$ (d) $y = 1-x$

33.



(a) $y = |x-1||x-2|$ (b) $y = |||x|-1|-2|-1$
(c) $y = |||x|-2|-1|$ (d) none of these

34.

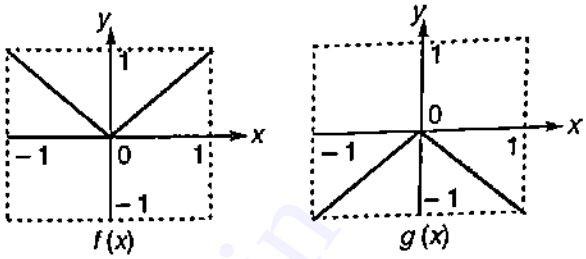


(a) $y = |x-1||x-2|$ (b) $y = -(|x|-1)(|x|-2)$
(c) $y = |||x|-1|-2|$ (d) $y = (|x|-1)(|x|-2)$

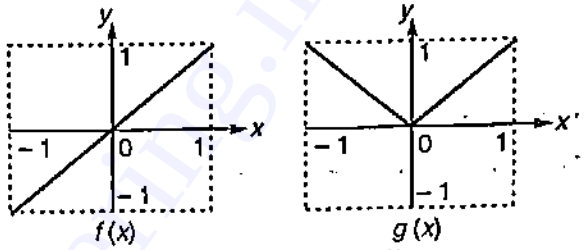
Directions for question number 35-37 : In each of the following questions a pair of graphs $f(x)$ and $g(x)$ is given, defined for every $x \in (-1, 1)$. Choose the correct answer as

(a) if $f(x) = -g(x)$ for every x
(b) if $f(x) = g(x)$ for $x < 0$ and $f(x) = -g(x)$ $x > 0$
(c) if $f(x) = -g(x)$ for $x < 0$ and $f(x) = g(x)$, $x > 0$
(d) none of the above

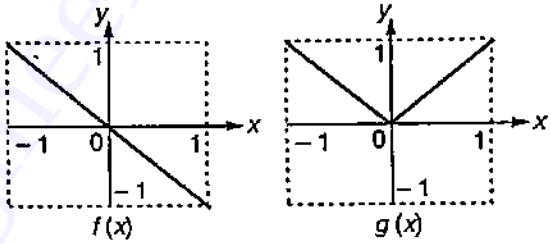
35.



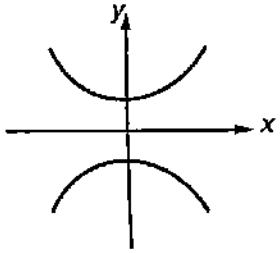
36.



37.



38. In the adjacent diagram two graphs of equal magnitude are being shown. Their maxima and minima are also same, then the correct choice for the two equation of functions is :



(a) $y = ax^2 + b$ and $y = ax^2 - b$
(b) $f(x) = ax^2$ and $f(x) = bx^2$; $a \neq b$
(c) $f(x) = ax^2 + bx$ and $f(x) = -ax^2 + bx$
(d) $y + ax^2 + c = 0$ and $y - ax^2 - c = 0$

39. If $f(x) = \frac{1}{x}$; $x \neq 0$ and $f^{n+1}(x) = f^n(f(x))$, find the product of $f^{11}(1) \cdot f^{33}(1) \cdot f^{55}(1)\ f^{77}(1) \cdot f^{99}(1)$:
(a) 110 (b) 88
(c) 275 (d) 1

40. If $f(x) = \frac{1}{x}$, $x \neq 0$ $f^n(x) = f(f^{n-1}(x))$ and $f^{12} = 11$, what is the value of $f(x)$?
(a) 1 (b) 2
(c) 1/11 (d) none of these

Directions for question number 41-43 : A program module is given below with various steps involved, which starts with the values of a, b, c as 1, 2, 3 respectively. These values remain unchanged till step 1.
The program is executed with the help of following operators.

Milan $2(P) = P \times 2$
Milaap $2(P) = P + 2$

and $b = a$ means the value of a is assigned to b.
START
Step 1. *Read (a, b, c)*

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Step 2. Milan $2(a)$
Step 3. Milaap $2(b)$
Step 4. $c = a + b$
Step 5. If, $c > 111$, then go to step 6 else go to step 2.
Step 6. STOP

41. How many times in the above program does c take a perfect square value ?
(a) 1 (b) 2
(c) 0 (d) none of these

42. In a particular cycle a, b and c get the values such that $9a = 8c$, then the value of b is :
(a) 6 (b) 8
(c) 16 (d) none of these

43. How many times is the program module repeated?
(a) 6 (b) 7
(c) 5 (d) none of these

✍ ***Directions for question number 44-46 :*** *In the following algorithm x, y and z are variables which change their values in each step such that the value of the expression on the right hand side is assigned to the variable on the left hand side of the equation. Each cycle consists of 5 steps. S_1, S_2, S_3... are the sum of each corresponding cycle 1, 2, 3....*

START
Step 0. $x = 1, y = -2, z = 3$
Step 1. $x = y - z$
Step 2. $y = z - x$
Step 3. $z = x - y$
Step 4. $S = x + y + z$
Step 5. *Go to step 1.*

44. Which one of the following is correct for each cycle?
(a) $S = 2x$ (b) $S = 3x$
(c) S is an odd number (d) none of these

45. What is the value of S_4 ?
(a) 987 (b) 754
(c) -178 (d) none of these

46. Which of the following is correct?
(a) S_{11} is an odd number
(b) S_{13} is a negative number
(c) S_{15} is an even number
(d) both (b) and (c)

✍ ***Directions for question number 47–48 :*** *There are three bottles of water K, L, M whose capacities are 9 litres, 5 litres and 4 litres respectively. For transferring water from one bottle to another and to drainout the bottles, there exists a piping system. The flow through these pipes is computer controlled. The computer that controls the flow through these pipes can be fed with three types of instructions, as explained below. Initially K is completely filled up with water and L, M are empty.*

Instruction type	*Explanation of the instruction*
Fill (X, Y)	*Fill bottle labelled X from the water in bottle labelled Y, where the remaining capacity of X is less than or equal to the amount of water in Y.*
Empty (X, Y)	*Empty out the water in bottle labelled X into bottle labelled Y, where the amount of water in X is less than or equal to remaining capacity of Y.*
Drain (X)	*Drain out all the water contained in bottle X.*

47. How much water would be there in bottle labelled L after the execution of the sequence of the following four instructions:
First instruction : Fill (L, K)
Second instruction : Fill (M, L)
Third instruction : Empty (L, K)
Fourth instruction : Empty (M, L)
(a) 0 litre (b) 4 litres
(c) 5 litres (d) none of these

48. After executing a sequence of 6 instructions, bottle K has half of the water that the bottles L and M together have and the bottle L has half of the water that the bottle M has. Find the fourth instruction if the rest instructions are given below.
Instruction 1. Fill (L, K)
Instruction 2. Fill (M, L)
Instruction 3. Empty (M, K)
Instruction 5. Fill (L, K)
Instruction 6. Fill (M, L)
(a) Fill (L, M) (b) Empty (L, M)
(c) Fill (M, K) (d) Empty (M, L)

Final Round

(The questions in this exercise given are very similar to the questions which are asked in XAT for XLRI, XIM etc.

✍ ***Directions for question number 1-10 :*** *Solve the following questions on the basis of following functions*
(a) $P_{(n+1)} = P_{(n)} - P_{(n-1)}$, P *is the term of the sequence and* $P_{(0)} = 0, P_{(1)} = 1$
(b) $Q_{(n+1)} = Q_{(n)} + Q_{(n+1)}$; Q *is the term of the sequence and* $Q_{(0)} = 0, Q_{(1)} = 1$

1. What is the 10th term of series $Q_{(n+1)}$ starting from $n = 0$?
(a) 34 (b) 89
(c) -1 (d) 55

2. What is the 12th term of the series $P_{(n+1)}$ starting from $n = 0$?
(a) -1 (b) 0
(c) 1 (d) none of these

3. The value of Q cannot be :
(a) 144 (b) 55
(c) 98 (d) 34

4. The value of P cannot be :
(a) -1 (b) 0
(c) 1 (d) 4

5. Which of the following can be a positive value of $Q_n + P_{(n+1)}$?
(a) 55 (b) 234
(c) 146 (d) none of these

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

6. The value of $Q_n + P_n$ cannot be :
(a) 2 (b) 22
(c) 55 (d) 88

7. The value of $Q_{[Q(n+1)]}$ can be :
(a) 55 (b) 13
(c) 21 (d) 89

8. The value of $Q_n \times P_n$ can be :
(a) – 34 (b) 89
(c) – 233 (d) 377

9. The value of $[Q_{(n)}]^{P_n}$ can be :
(a) – 1 (b) 0.2
(c) – 13 (d) 5.5

10. The value of $Q_{(n-1)} + Q_{(n+1)} + L_{(n-1)} + L_{(n+1)}$ cannot be :
(a) 30 (b) 75
(c) 18 (d) 122

Directions for question number 11-14 : Let S be a sequence of the form $[K(1), K(2), \ldots K(m)]$. Each term can be defined by the following four functions :

$$P[K(x)] = 3Q[K(x)] - 4$$
$$Q[K(x)] = 2R[K(x)] + R[K(2x)]$$
$$R[K(x)] = S[K(2x)] - S[K(x)]$$
$$S[K(x)] = \begin{cases} 0 \text{ if } x < 0 \\ 3x + 4 \text{ if } 0 \le x \le 6 \\ 4x + 3 \text{ if } x > 6 \end{cases}$$

Also if $Q[K(x)] = y$ then $PQ[K(x)] = P[K(y)]$

11. What is the value of $P[K(3)]$?
(a) 3 (b) 5
(c) 32 (d) 137

12. What is the value of $QRS[K(1)]$:
(a) 112 (b) 224
(c) 448 (d) none of these

13. What is the value of $R[K(5)] - S[K(10)]$:
(a) 43 (b) – 19
(c) 14 (d) none of these

14. What is the value of $\sum_{x=-\infty}^{+\infty} R[K(x)]$ if

$$\sum_{x=1}^{4} R[K(x)] = R[K(1)] \times R[K(2)] \times R[(K(3)] \times R[K(4)]?$$

(a) – 1 (b) 0
(c) 1 (d) none of these

Directions for question number 15-20 : These questions are based on the given data.
A polynomial can be represented in an equivalent sequence form.
The polynomial $P_1x^{k_1} + P_2x^{k_2} + P_3x^{k_3} + \ldots P_nx^{k_n}$ where $k_1 > k_2 > k_3 \ldots > k_n$ and $a_i \neq 0$ for $1 \le i \le n$ will be represented as the sequence $(P_1, k_1, P_2, k_2, P_3, k_3 \ldots P_n, k_n)$
Also, we add, subtract, multiply as we do for corresponding polynomials. The resulting polynomial is again represented as a sequence.

15. (3, 4, 2, 2, 5, 1) + (2, 4, 3, 3, 7, 2) equals :
(a) (5, 4, 3, 4, 9, 3, 5, 1) (b) (5, 4, 5, 5,12, 3)
(c) (5, 4, 3, 3, 9, 2,5,1) (d) none of these

16. (6, 5, 7, 4, 8,3) – (3, 5, 5,3, 7, 1) equals
(a) (3, 5, 2, 2, 1, 2) (b) (3, 5, 7, 4, 3, 3, – 7, 1)
(c) (3, 5, 7, 3, 8, 2) (d) none of these

17. (1, 1, 2, 0) × (1, 1, 2, 0) × (1, 1, 2, 0) equals :
(a) (1, 3, 6, 2, 12,1, 8, 0) (b) (1, 3, 2, 2, 3, 1, 4, 0)
(c) (1, 8, 2,4, 3, 2) (d) none of these

18. The expression $\frac{3, 3, -10, 2, 7, 1}{3, 2, -7, 1}$ equals :
(a) (1, 1, –1, 1) (b) (1, 1, – 1, 0)
(c) (2, 2, – 2, 0) (d) none of these

19. Which of the following is incorrect?
(a) (1, 1, 1, 0) × (1, 2, – 1, 1, 1, 0) = (1, 3, 1, 0)
(b) (2, 2, 1, 1) × (4, 4, 4, 4) = (16, 6, 10, 5)
(c) $\frac{(1, 2, 4, 1, 4, 0)}{(1, 1, 2, 0)} = (1, 1, 2, 0)$
(d) none of the above

20. (4, 4, 3,3) × (2, 2, 1, 1) + (2, 2, 1, 1) – (3, 5, 2, 4) equals :
(a) (8, 6, 5, 5, 7, 4, 2, 3, 2, 1)
(b) (8, 6, 7, 5, 1, 4,2, 2, 1, 1)
(c) (8,6,7,5,2, 4, 3,2, 1,1)
(d) none of the above

Directions for question number 21-25 :
If p, q, r, s be the distinct integers such that :
$f(p, q, r, s)$ = maximum of (p, q, r, s)
$g(p, q, r, s)$ = minimum of (p, q, r, s)
$h(p, q, r, s)$ = remainder when $p \times q$ is divided by $r \times s$ if $p \times q > r \times s$.
= remainder when $r \times s$ is divided by $p \times q$ if $r \times s > p \times q$
Also, the same operations are valid with two variable functions of the form $f(p, q)$
Also a function $fgh(p, q, r, s)$
$= f(p, q, r, s) \times g(p, q, r, s) \times h(p, q, r, s)$

21. What is the value of $h(3, 2, 8, 7) \div g(4, 7, 10, 8)$?
(a) 1 (b) 1/2
(c) 24 (d) none of these

22. What is the value of $h(fg(2, 5, 7, 3), 9)$?
(a) 2 (b) 3
(c) 5 (d) none of these

23. What is the value of $h(h(7, 13, 5, 9), h(4, 6, 12, 14)$?
(a) 1 (b) 1/7
(c) 7 (d) not defined

24. If $A = h(3, 7, 6, 5)$, $B = h(16, 11, 13, 3)$ $C = h(9, 4, 8, 7)$, $D = h(19, 14, 18, 7)$, then which of the following is true?
(i) $A > B > C > D$ (ii) $D > A > C$
(iii) $A < D < B$ (iv) $B = C < D$
(a) only (i) is true (b) only (iii) is true
(c) (ii) and (iii) are true (d) (iii) and (iv) are true.

25. If $a_k \times b_k$ is an integral multiple of $c_k \times d_k$ or $c_k \times d_k$ is an integral multiple of $a_k \times b_k$, which of the following is indeterminable ?
(a) $f(g(a_1, b_1, c_1, d_1), h(a_2, b_2, c_2, d_2))$
(b) $h(h(a_1, b_1, c_1, d_1), h(a_2, b_2, c_2, d_2))$
(c) $h(hh(a_1, b_1, c_1, d_1), g(a_2, b_2, c_2, d_2))$
(d) none of the above

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yousmehboob.wordpress.com

For More Visit : www.LearnEngineering.in

✍ ***Directions for question number 26-32 :*** *Following questions are based on the given information for the following functions* $f(x)$

$f(x) = 2bx + f(-x)$; if $x < 0$

$f(x) = a$ if $x = 0$

$f(x) = b + c - 2cx + f(x-1)$; if $x > 0$

26. $f(8)$ equals :
(a) $a + 8b - 32c$ (b) $a + 8(b - 8c)$
(c) $8(a + b - c)$ (d) none of these

27. $f(-19)$ equals :
(a) $a - 19b + 361c$ (b) $a + 19(b - 19c)$
(c) $a - 19(b + 19c)$ (d) none of these

28. If $a = 15$, $b = 11$, $c = -3$, then $f(7)$ equals :
(a) 239 (b) 115
(c) -147 (d) none of these

29. If $a = 10$, $b = -7$, $c = 6$, then $f(-10)$ equals :
(a) -660 (b) -520
(c) -250 (d) none of these

30. If $a = 4$, $b = -17$ and $c = -18$, then for what value of x, $f(x) = 0$?
(a) 4 or 9 (b) $\frac{1}{2}$ or $\frac{4}{9}$
(c) -1 or 18 (d) none of these

31. If $a = 12$, $b = 10$ and $c = 8$, then for what value of x, $f(x) < 0$?
(a) $\frac{3}{4}$ (b) 1
(c) -2 (d) none of these

32. If $b = -a$, $c = a$, $f(f(1))$ equals :
(a) $a + a^2 + a^3$ (b) $a + a^2 - a^3$
(c) $a - a^2 + a^3$ (d) none of these

✍ ***Directions for question number 33-37 :***

A function $f(z_1, z_2, z_3, \ldots z_n) = f(z_1, z_n) + f(z_2, z_3, \ldots z_{n-1}) + (z_1 + z_2 + \ldots z_n)$; for $n > 0$

$f(y, z) = f(z, 0) + f(0, y)$

$f(y, 0) = y + f(y - 1, 0)$

$f(0, y) = y - f(0, y - 1)$

$f(0, 0) = 1$

33. Find the value of $f(1, 2, 3, 4, \ldots n)$, where n is a perfect cube less than 50 and n is greater than 25 :
(a) 1665 (b) 1089
(c) 729 (d) not defined

34. Find the value of $f(0, 1, 0, 1)$:
(a) 2 (b) 4
(c) 8 (d) none of these

35. Find the value of $f(8, 8, 8, 2, 2, 2)$:
(a) 22 (b) 88
(c) 77 (d) 66

36. Find the value of $f(1, 1, 3, 1, 1, 3)$:
(a) 17 (b) 28
(c) 21 (d) none of these

37. $f(9, 2, k, 0, 9, 4) = 124$, then the value of k is :
(a) 0 (b) 5
(c) 9 (d) none of these

✍ ***Directions for question number 38–40 :*** *A function is defined as follows :*

$f(a_1, a_2, a_3 \ldots a_n) = a_1 2^{n-1} + a_2 2^{n-2} + a_3 2^{n-3} + \ldots a_n 2^0$

The above function is repeated until the value of function reduces to a single digit number.

38. $f(128)$ equals :
(a) 1 (b) 2
(c) 4 (d) 8

39. What is the value of $f[f(888222) + f(113113)]$:
(a) 6 (b) 7
(c) 8 (d) 9

40. $f(9235) + f(9450)$ equals :
(a) 5 (b) 3
(c) 2 (d) none of these

www.LearnEngineering.in

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Answers

INTRODUCTORY EXERCISE-17.1

1 (a)	2. (c)	3. (c)	4. (a)	5. (a)	6. (d)	7. (b)	8. (a)	9. (c)	10. (b)
11. (c)	12. (b)	13. (b)	14. (c)	15. (d)	16. (c)	17. (d)	18. (d)	19. (c)	20. (c)
21. (a)	22. (d)	23. (a)	24. (b)	25. (c)	26. (c)	27. (c)	28. (d)	29. (d)	30. (b)
31. (d)	32. (a)	33. (c)	34. (c)	35. (b)	36. (d)	37. (b)	38. (c)	39. (c)	40. (c)
41. (b)	42. (b)	43. (b)	44. (b)						

INTRODUCTORY EXERCISE-17.2

1 (c)	2. (a)	3. (c)	4. (c)	5. (a)	6. (c)	7. (a)	8. (b)	9. (a)	10. (d)
11. (c)	12. (a)	13. (c)	14. (d)	15. (d)	16. (b)	17. (b)	18. (d)	19. (a)	20. (a)
21. (b)	22. (b)	23. (c)	24. (a)	25. (b)	26. (a)	27. (d)	28. (a)	29. (b)	30. (b)

INTRODUCTORY EXERCISE-17.3

1 (c)	2. (d)	3. (a)	4. (b)	5. (d)	6. (b)	7. (a)	8. (b)	9. (b)	10. (c)
11. (a)	12. (c)	13. (b)	14. (a)	15. (b)	16. (b)	17. (a)			

LEVEL-1

1 (b)	2. (a)	3. (d)	4. (c)	5. (c)	6. (c)	7. (b)	8. (c)	9. (a)	10. (a)
11. (b)	12. (c)	13. (a)	14. (b)	15. (c)	16. (c)	17. (a)	18. (c)	19. (b)	20. (b)
21. (d)	22. (d)	23. (d)	24. (b)	25. (a)	26. (c)	27. (b)	28. (d)	29. (c)	30. (a)
31. (a)	32. (b)	33. (c)	34. (a)	35. (b)	36. (c)	37. (b)	38. (d)	39. (a)	40. (d)
41. (a)	42. (b)	43. (d)	44. (a)	45. (b)	46. (b)	47. (b)	48. (c)	49. (d)	50. (b)
51. (c)	52. (d)	53. (d)	54. (d)	55. (b)	56. (b)	57. (d)	58. (d)	59. (a)	60. (b)
61. (a)	62. (c)	63. (a)	64. (c)	65. (d)	66. (c)	67. (a)	68. (c)	69. (c)	70. (c)
71. (d)	72. (b)	73. (c)	74. (c)	75. (b)	76. (b)	77. (a)	78. (d)	79. (c)	80. (b)
81. (c)	82. (b)	83. (a)	84. (c)						

LEVEL-2

1 (c)	2. (b)	3. (d)	4. (c)	5. (b)	6. (d)	7. (d)	8. (c)	9. (a)	10. (b)
11. (b)	12. (d)	13. (d)	14. (b)	15. (c)	16. (a)	17. (b)	18. (a)	19. (a)	20. (c)
21. (a)	22. (b)	23. (b)	24. (c)	25. (c)	26. (c)	27. (c)	28. (b)	29. (a)	30. (c)
31. (b)	32. (c)	33. (b)	34. (b)	35. (a)	36. (c)	37. (b)	38. (d)	39. (d)	40. (c)
41. (d)	42. (c)	43. (a)	44. (a)	45. (b)	46. (d)	47. (b)	48. (b)		

FINAL ROUND

1 (d)	2. (b)	3. (c)	4. (d)	5. (b)	6. (a)	7. (c)	8. (d)	9. (b)	10. (b)
11. (d)	12. (c)	13. (b)	14. (b)	15. (c)	16. (b)	17. (a)	18. (b)	19. (b)	20. (b)
21. (b)	22. (c)	23. (d)	24. (b)	25. (b)	26. (b)	27. (c)	28. (a)	29. (b)	30. (b)
31. (c)	32. (b)	33. (d)	34. (b)	35. (c)	36. (b)	37. (b)	38. (d)	39. (b)	40. (a)

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yourshahbood.wordpress.com

Hints & Solutions

INTRODUCTORY EXERCISE 17.1

1. x is defined for all real numbers (R)
$\therefore$ Domain of x is R

2. $\because \sqrt{x}$ is not defined for negative values of x
hence $x \geq 0$
$\therefore \quad x \in R^+ \cup \{0\}$

3. Since a rational number is not defined if denominator is zero.
Hence $\frac{1}{x}$ is defined for only non zero numbers.
Thus $x < 0$ and $x > 0$
$\Rightarrow (-\infty < x < 0) \cup (0 < x < \infty)$
$\Rightarrow x \in (-\infty, 0) \cup (0, \infty)$
$\Rightarrow x \in R - \{0\}$

4. $\because x$ is defined for all real values
$\therefore |x|$ is also defined for all real values
Hence $|x| \in R$

5. $\because \quad |x| \geq 0 \ \forall \ x \in R$
$\therefore \sqrt{|x|}$ is defined for all real values of x.
Hence, domain of $\sqrt{|x|}$ is R.

6. Since domain of $\sqrt{x}$ is $[0, \infty)$
therefore domain of $|\sqrt{x}|$ is also $[0, \infty)$

7. $\sqrt{x}$ is defined only when $x \geq 0$
but $\frac{1}{\sqrt{x}}$ is defined only when $x > 0$
Since rational expressions are not defined for zero denominator so, $\sqrt{x} \neq 0$
$\therefore$ Domain of $\frac{1}{\sqrt{x}}$ is $(0, \infty)$ *i.e.*, R^+

8. Since x^2 is positive for all values of x
i.e., $x^2 \geq 0 \ \forall \ x \in R$
$\therefore \sqrt{x^2}$ is defined for all values of R
Hence domain of $\sqrt{x^2}$ is R.

9. Since $\sqrt{x}$ is defined for only non-negative real numbers.
Hence domain of $(\sqrt{x})^2$ is $[0, \infty)$

10. $\log_a x$ is defined only for $x > 0$
Hence the domain of $\log_{10} x$ is $(0, \infty)$

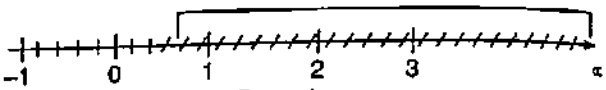
11. x^2 is positive for all values of x except at $x = 0$
Since $\log_a b$ is defined only if $b > 0$
$\therefore$ Domain of $\log_{10} x^2$ is $R - \{0\}$

12. $\log_{10} x$ is defined for only $x > 0$
also denominator must be non-zero
$\therefore$ Domain of $f(x) = \frac{1}{\log_{10} x}$ is $R^+ - \{1\}$

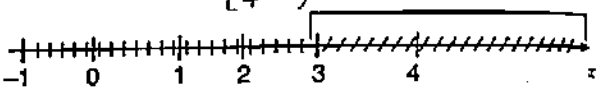
13. $\log x$ is defined for only $x > 0$
Also denominator be non-zero real number
Again square root of negative numbers is not defined.
Hence domain of $f(x) = \frac{1}{\sqrt{\log_{10} x}}$ is $1 < x < \infty$ *i.e.*, $R^+ - (0$

14. $f(x) = \sqrt{3x - 4}$ ($\because \sqrt{k}$ is not defined for $k <$
$\Rightarrow \quad 3x - 4 \geq 0$
$\Rightarrow \quad 3x \geq 4 \Rightarrow x \geq \frac{4}{3}$
$\therefore \quad \frac{4}{3} \leq x < \infty$
$\Rightarrow \quad x \in \left[\frac{4}{3}, \infty\right)$

15. $f(x) = \sqrt{4x - 3} + \sqrt{2x - 6}$
$\therefore \quad 4x - 3 \geq 0$ and $2x - 6 \geq 0$
$\Rightarrow \quad 4x \geq 3$ and $2x \geq 6$
$\Rightarrow \quad x \geq \frac{3}{4}$ and $x \geq 3$
$\therefore \quad x \in [3, \infty)$
(since if $x < 3$, $\sqrt{2x - 6}$ is not defined)
Therefore we consider only common values of domain



Domain of $\sqrt{4x - 3} = \left[\frac{3}{4}, \infty\right)$



Domain of $\sqrt{2x - 6} = [3, \infty)$

16. $|x| - x > 0 \quad (\because |x| - x \neq 0)$
$\Rightarrow |x| > x$ also $|x| - x > 0$ for the condition of square root
Since $|x|$ is positive value and $|x|$, x are numerically equ
Therefore $x < 0$
Hence, the domain of $f(x) = \frac{1}{\sqrt{|x| - x}}$ is $(-\infty, 0)$

17. $x \in R$
but $x^2 - 3x + 2 > 0$
$\Rightarrow (x - 1)(x - 2) > 0 \quad (\because x^2 - 3x + 2 \neq 0$
$\Rightarrow x < 1$ and $x > 2$



$\therefore$ Domain of $f(x) = \frac{x}{\sqrt{x^2 - 3x + 2}}$ is $(-\infty, 1) \cup (2, \infty)$

18. $(\sqrt{6 - x} + \sqrt{x - 4}) > 0$
Also $(6 - x) > 0$ and $(x - 4) > 0$
$\Rightarrow \quad 6 > x$ and $x > 4$
$\therefore \quad 4 < x < 6$
$\therefore$ domain of given function is $(4, 6)$.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

19. $(1-x)>0 \Rightarrow 1>x$

$x<1$ ...(1)

$\Rightarrow \log_{10}(1-x) \neq 0$

Also $10^{\circ} \neq (1-x)$

$\Rightarrow 1 \neq 1-x$

$\Rightarrow 0 \neq -x$

$\Rightarrow x \neq 0$ ...(2)

$\Rightarrow (x+2) \geq 0$

Again $x \geq -2$ ...(3)

$\Rightarrow x \in [-2, 0) \cup (0, 1)$

$\therefore x \in [-2, 1) - \{0\}$

or

or $-2 \leq x < 1$, excluding zero.

20. $f(x) = \dfrac{\log_2(x+3)}{x^2+3x+2}$

$(x+3)>0 \Rightarrow x>-3$ ...(1)

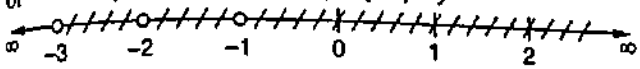
Also $x^2+3x+2 \neq 0$

$\Rightarrow (x+1)(x+2) \neq 0$

$\Rightarrow x \neq -1$ or $x \neq -2$ ...(2)

$\therefore$ Domain of $f(x) = (-3, \infty) - \{-1, -2\}$

or $(-3, -2) \cup (-2, -1) \cup (-1, \infty)$



NOTE 'o' indicates absence of domain
'###' indicates presence of domain on the real number line.

21. Domain of $\sin x$ is R
and Domain of $\cos x$ is also R
$\therefore$ Domain of given function is R

22. $\log_{10} \dfrac{1}{|\sin x|} \geq 0$

$\Rightarrow 10^0 \leq \dfrac{1}{|\sin x|} \Rightarrow 1 \leq \dfrac{1}{|\sin x|}$

$\Rightarrow |\sin x| \leq 1$

$\Rightarrow -1 \leq \sin x \leq 1$

$\Rightarrow x \in R$

but $\dfrac{1}{|\sin x|} > 0$

$\Rightarrow |\sin x| > 0$

$\Rightarrow x \in R$ ($\because |k|$ is always positive)

Again $|\sin x| \neq 0$

$\Rightarrow x \neq n\pi, \quad n \in I$

$\therefore$ Domain of $f(x)$ is $R \sim \{n\pi, n \in I\}$

23. $f(x) = (x^2+x+1)^{-3/2}$

Now $x^2+x+1 \neq 0$

$\Rightarrow \left(x+\dfrac{1}{2}\right)^2 + \dfrac{3}{4} \neq 0$

$\left(x+\dfrac{1}{2}\right)^2$ is always positive for all the values of real numbers and the whole expression is never zero and never negative for any value of R.

Hence domain of $f(x) = (x^2+x+1)^{-3/2}$ is R.

24. $5x-6-x^2>0$

$\Rightarrow x^2-5x+6<0$

$\Rightarrow (x-2)(x-3)<0$

$\Rightarrow 2<x<3$

$\Rightarrow x \in (2, 3)$



25. $y = \log_x 5 = \dfrac{\log 5}{\log x}$

$\log x \neq 0 \Rightarrow x \neq 1$

and $x>0$

$\therefore x \in (0, \infty) \sim \{1\}$

or $x \in (0, 1) \cup (1, \infty)$

26. $f(x) = 9 - \sqrt{9-x^2}$

$\therefore 9-x^2 \geq 0$

$\Rightarrow x^2-9 \leq 0$

$\Rightarrow (x-3)(x+3) \leq 0$

$\Rightarrow -3 \leq x \leq 3$

$\Rightarrow x \in [-3, 3]$



27. $x^2-5x+6 \geq 0$

but $x^2-5x+6 \neq 0$

$\therefore x^2-5x+6>0$

$\Rightarrow (x-2)(x-3)>0$

$\Rightarrow x<2$ or $x>3$

$\Rightarrow x \in (-\infty, 2) \cup (3, \infty)$

or $x \in (-\infty, \infty) - [2, 3]$



28. Clearly $\dfrac{x^2}{1+x^4} \geq 0$ for any $x \in R$

Again $(x^2-1)^2 \geq 0$

$\Rightarrow x^4+1-2x^2 \geq 0$

$\Rightarrow x^4+1 \geq 2x^2$

$\therefore \dfrac{x^2}{1+x^4} \leq \dfrac{x^2}{2x^2}$

$\Rightarrow \dfrac{x^2}{1+x^4} \leq \dfrac{1}{2}$

Hence, the range of f is $\left[0, \dfrac{1}{2}\right]$.

29. Let $y = \dfrac{x^2-2}{x^2-3}$

$\Rightarrow yx^2-3y = x^2-2$

$\Rightarrow x^2(y-1) = 3y-2$

$\Rightarrow x^2 = \dfrac{3y-2}{y-1}$

$\therefore \dfrac{3y-2}{y-1} \geq 0$

$\therefore y \in \left(-\infty, \dfrac{2}{3}\right] \cup (1, \infty)$



30. Let $y = \dfrac{x^2+2x+3}{x}$

$\Rightarrow x^2+(2-y)x+3=0$

$\because x \in R$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$\therefore \quad (2-y)^2 - 4\times1\times3 \geq 0$

$\Rightarrow \quad y^2 - 4y + 4 - 12 \geq 0$

or $\quad y^2 - 4y - 8 \geq 0$

or $\quad [y-(2-2\sqrt{3})][y-(2+2\sqrt{3})] \geq 0$

$\Rightarrow \quad y \in (-\infty, 2-2\sqrt{3}] \cup [2+2\sqrt{3}, \infty)$

i.e., $\quad y \in R - (2-2\sqrt{3}, 2+2\sqrt{3})$

31. Domain of $f(x) = \sqrt{x^2}$ is R *i.e.*, $x \in (-\infty, \infty)$

Domain of $g(x) = (\sqrt{x})^2$ is $R^+ \cup \{0\}$ *i.e.*, $x \in [0, \infty)$

$\therefore \quad \sqrt{x^2} \neq (\sqrt{x})^2$

Again co-domain of $f(x) = \dfrac{1}{\sqrt{x^2}}$ is $R^+ \sim \{0\}$

and co-domain of $g(x) = \dfrac{x}{x^2}$ is R^+ *i.e.*, $g(x) \in (0, \infty)$

i.e, $\quad g(x) \in (-\infty, 0) \cup (0, \infty)$

$\therefore \quad \dfrac{1}{\sqrt{x^2}} \neq \dfrac{x}{x^2}$

Again, Domain of $f(x) = \log(x-1) + \log(x-2)$ is $(2, \infty)$

and Domain of $g(x) = \log(x-1)(x-2)$ is $(-\infty, 1) \cup (2, \infty)$

$\therefore \quad \log(x-1) + \log(x-2) \neq \log(x-1)(x-2)$

Since in each point domain of definition of $f(x)$ is unequal, therefore the given pairs of functions are not identical.

HINT See the concept of "Equal functions".

32. Domain of $f(x) = \dfrac{1}{\sqrt{|x| - x}}$ is $(-\infty, 0)$

and domain of $g(x) = \dfrac{1}{\sqrt{x - |x|}}$ is ϕ

$\therefore \quad$ dom $f \neq \phi$ and dom $g = \phi$

HINT $\phi \rightarrow$ There is no any single value of x, which satisfy the given function.

33. Dom. of $f(x) = x$ is $(-\infty, \infty)$

Dom. of $g(x) = \sqrt{x^2}$ is $(-\infty, \infty)$, but Range $\equiv [0, \infty)$

Dom. of $h(x) = \dfrac{x^2}{x}$ is $(-\infty, \infty) \sim \{0\}$

$\therefore$ the common domain is $[-0, \infty) \sim \{0\} = R^+$

(Since for every $f(x)$, $g(x)$, $h(x) \in R^+$ the range of each $f(x)$, $g(x)$ and $h(x)$ is same)

34. Dom. $f(x)$ is $R - \{0\}$ *i.e.*, $(-\infty, 0) \cup (0, \infty)$

and Dom. $g(x)$ is R^+ *i.e.*, $(0, \infty)$

$\therefore$ Common domain of $f(x)$ and $g(x)$ is $(0, \infty)$

Hence if $x \in (0, \infty)$, then $f(x) = g(x)$

35. Since $\quad [x] \leq x \quad$ or $\quad x \geq [x]$

$\Rightarrow \quad x - [x] \geq 0$

$\therefore \quad 1 + x - [x] > 0$

$\therefore \quad f(g(x)) = f(1 + x - [x]) = 1$

HINT See the definition of Signum function.

36. $f(x) = 64x^3 + \dfrac{1}{x^3} = \left(4x + \dfrac{1}{x}\right)^3 - 3\cdot4\left(4x + \dfrac{1}{x}\right)$

So $\quad f(\alpha) = 2^3 - 3\cdot4\cdot2 = -16$

Also $\quad f(\beta) = 2^3 - 3\cdot4\cdot2 = -16$

Hence (d).

37. Let $\quad y = \dfrac{x}{\sqrt{1+x^2}}$

$\therefore \quad f(f(x)) = f(y) = \dfrac{y}{\sqrt{1+y^2}} = \dfrac{x/\sqrt{1+x^2}}{\sqrt{1+\left(\dfrac{x}{\sqrt{1+x^2}}\right)^2}}$

$= \dfrac{x}{\sqrt{1+2x^2}} = z$ (say)

$\therefore \quad f(f(f(x))) = f(z) = \dfrac{z}{\sqrt{1+z^2}} = \dfrac{x/\sqrt{1+2x^2}}{\sqrt{1+\left(\dfrac{x}{\sqrt{1+2x^2}}\right)^2}}$

$= \dfrac{x}{\sqrt{1+3x^2}}$

38. $f(x) = \dfrac{4^x}{4^x+2}$

$\therefore \quad f(1-x) = \dfrac{4^{1-x}}{4^{1-x}+2} = \dfrac{2}{2+4^x}$

$\therefore \quad f(x) + f(1-x) = \dfrac{4^x}{4^x+2} + \dfrac{2}{4^x+2} = 1$

Hence (c) is correct

39. $f(x) = 1 - f(1-x)$

$\Rightarrow \quad f(x) + f(1-x) = 1$

Now, $f\left(\dfrac{1}{999}\right) + f\left(\dfrac{2}{999}\right) + \ldots f\left(\dfrac{997}{999}\right) + f\left(\dfrac{998}{999}\right)$

$= \left[f\left(\dfrac{1}{999}\right) + f\left(\dfrac{998}{999}\right)\right] + \left[f\left(\dfrac{2}{999}\right) + f\left(\dfrac{997}{999}\right)\right] + \ldots$

$= \left[f\left(\dfrac{1}{999}\right) + f\left(1 - \dfrac{1}{999}\right)\right] + \left[f\left(\dfrac{2}{999}\right) + f\left(1 - \dfrac{2}{999}\right)\right] + \ldots$

$= 1 + 1 + 1 + \ldots 499$ times $= 499$

40. $f(3) = 9$

$\therefore \quad g(f(3)) = g(9) = 3$

Alternatively: $gof = g(f(x)) = g(x^2) = \sqrt{x^2} = x$

41. Solve through options

Let us consider $f(x) = 3x + 2$

$\therefore \quad g(f(x)) = (3x+2)^2 - (3x+2) + 7$

$\Rightarrow \quad gof(x) = 9(x^2 + x + 1)$

$\Rightarrow \quad \left(\dfrac{1}{9}\right) gof(x) = x^2 + x + 1$

Hence choice (b) is correct.

42. $f(x+1) = \dfrac{(x+1)}{(x+1)+3} = \dfrac{x+1}{x+4}$

43. $\dfrac{1}{f(x+1)} - f\left(\dfrac{1}{x+1}\right) = \dfrac{x+4}{x+1} - \dfrac{1/(x+1)}{1/(x+1)+3}$

$= \dfrac{x+4}{x+1} - \dfrac{1/(x+1)}{(3x+4)/(x+1)}$

$= \dfrac{x+4}{x+1} - \dfrac{1}{(3x+4)} = \dfrac{3(x^2+5x+5)}{3x^2+7x+4}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

44. $f(2)=\frac{2+1}{2-1}=3$

$f(f(2))=f(3)=\frac{3+1}{3-1}=2$

$f(f(f(2)))=f(f(3))=f(2)=\frac{2+1}{2-1}=3$

$f(f(f(f(2))))=f(3)=\frac{3+1}{3-1}=2$

$f(f(f(f(f(2)))))=f(2)=\frac{2+1}{2-1}=3$

INTRODUCTORY EXERCISE 17.2

1. Choice (a) is ruled out since the given graph is the graph of GIF (greatest integer function)
Again the graph of $[x]$ is parallel to the graph of $y = x$. But since in the left part (i.e., left of y-axis) all the values of y are positive which are transformed by the modulus.
Alternatively: Go through options and put the values (e.g., $-3, -2, -1, 0, 1, 2, 3$, etc) and then verify the correct option.

2. Best way is to check the option by substituting the values of x eg., $x = -3, -2, -1, 0, 1, 2, 3, \ldots$.

choice (b) is wrong since at $x = 2$, $f(x)=\frac{1}{4}$

choice (d) is wrong since at $x = 1$, $f(x) = 0$

choice (c) is wrong since at $x = 2$, $f(x)=\frac{1}{8}$

$\therefore$ choice (a) is correct, since at $x = 1$, $f(x) = 1$

at $x = 2$, $f(x)=\frac{3}{2}$

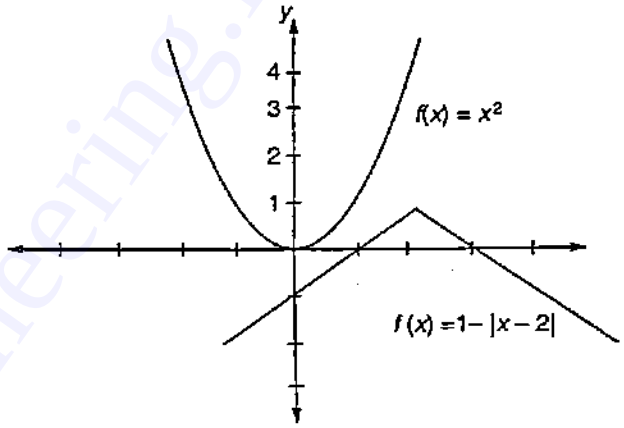
at $x=\frac{1}{2}$, $f(x) = 0$, etc.

3. Choice (a) is wrong, since graph is even
also at $x = 1$, $e^x = f(x) = 2.714$
choice (d) is wrong since at $x = 0$, $f(x) = 0$
choice (b) is wrong since at $x = 1$, $f(x) = 0$
also at $x = 0$, $f(x)$ is not defined
choice (c) is correct, since at $x = 0$, $f(x) = 1$
at $x = 1$, $f(x) = 2$
at $x = 2$, $f(x) = 4$
at $x = -1$, $f(x) = 2$
at $x = -2$, $f(x) = 4$

4. Choice (a) is wrong since at $x = 1$, $f(x) = 1$
at $x = 2$, $f(x) = 6$
choice (b) is wrong since at $x = 2$, $f(x) = 2$
choice (d) is wrong since at $x = 0$ $f(x) = 0$
choice (c) is correct since at $x = 0$, $f(x) = -2$
at $x = 2$, $f(x) = 0$
at $x = 3$, $f(x) = 4.25$
at $x = -2$, $f(x) = 0$
at $x = -3$, $f(x) = 4.25$

5. **Algebraic solution**
x^2 is always positive $\forall\ x \in R$
$1-|x-2|$ is positive, if $1 < x < 3 (x - 2 \neq 1)$
but when $1 < x < 3$, $x^2 \in (1, 9)$
and $(1-|x-2|) \in (0, 1)$
therefore there is no real solution of the given expression.
Graphical solution :

$$\frac{x^2}{1-|x-2|}=1 \Rightarrow x^2 = 1-|x-2|$$



Hence it is clear from the graph that there is no real solution, since the two graphs do not cut each other at any point.

6. **Algebraic solution :**

$$\log_{0.5}|x| = 2|x|$$

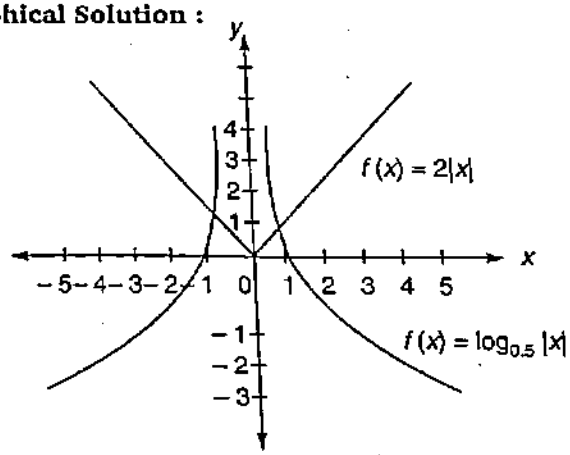
$$\Rightarrow (0.5)^{2|x|} = |x|$$

$$\Rightarrow \left(\frac{1}{2}\right)^{2|x|} = |x|$$

$$\Rightarrow \left(\frac{1}{2}\right)^{|x|} = |x|^{1/2}$$

$$\Rightarrow x = -\frac{1}{2}, \frac{1}{2}$$

Graphical Solution :

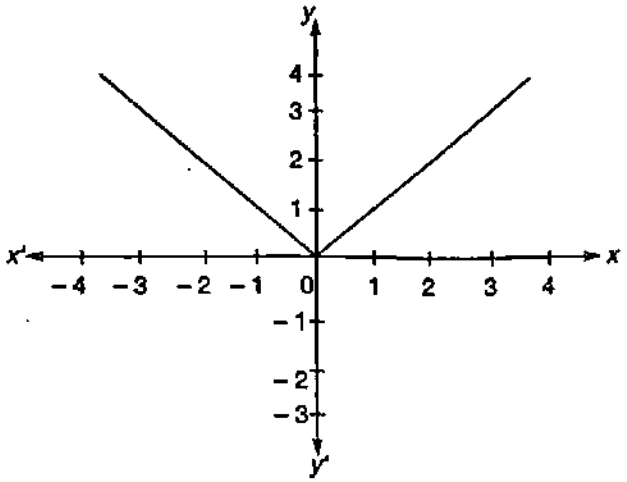


Clearly at $x = -\frac{1}{2}$ and $x = \frac{1}{2}$ the graphs of $\log_{(0.5)}|x|$ and $2|x|$ intersect each other.
Thus there are two solutions.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

7.



$$f(x)=\frac{|x|^3+|x|}{1+x^2}$$

8. $$f(x)=f\left(\frac{x+8}{x-1}\right)$$

$$\Rightarrow \quad x=\frac{x+8}{x-1}$$

$$\Rightarrow \quad x^2-2x-8=0$$

$$\Rightarrow \quad x=4,-2$$

Alternatively : Go through options

9. $$g(x)=\sqrt{x^2+1}$$

$$f(g(x))=(\sqrt{x^2+1})^2-1$$

$$\Rightarrow \quad f(g(x))=x^2$$

$$\therefore \quad h(f(g(x)))=h(x^2)=x^2 \qquad (\because h(x)=x,\ x\geq 0)$$

10. $$f(x)=(a-x^n)^{1/n}$$

$$\therefore \quad f(f(x))=(a-(f(x))^n)^{1/n}$$

$$=(a-((a-x^n)^{1/n})^n)^{1/n}$$

$$=(a-(a-x^n))^{1/n}=(x^n)^{1/n}=x$$

11. At $$y=f(x)=\frac{x-1}{x+1}$$

$$\Rightarrow \quad y(x+1)=x-1$$

$$\Rightarrow \quad x=\frac{y+1}{1-y} \qquad \ldots(1)$$

Now, $$f(f(x))=f\left(\frac{x-1}{x+1}\right)=\frac{\frac{x-1}{x+1}-1}{\frac{x-1}{x+1}+1}=-\frac{1}{x}$$

$$\therefore \quad f(f(ax))=-\frac{1}{ax}=-\frac{1}{a\left(\frac{y+1}{1-y}\right)}$$

$$=\frac{y-1}{a(y+1)}=\frac{f(x)-1}{a(f(x)+1)}$$

12. $$f(x)=\frac{x}{\sqrt{1+x^2}}$$

$$\therefore \quad f(f(x))=\frac{\frac{x}{\sqrt{1+x^2}}}{\sqrt{1+\left(\frac{x}{\sqrt{1+x^2}}\right)^2}}=\frac{x}{\sqrt{1+2x^2}}$$

$$\therefore \quad f(f(f(x)))=\frac{\frac{x}{\sqrt{1+2x^2}}}{\sqrt{1+\left(\frac{x}{\sqrt{1+2x^2}}\right)^2}}=\frac{x}{\sqrt{1+3x^2}}$$

Alternatively : Consider some value for x, then mark the correct option

$$f(2)=\frac{2}{\sqrt{1+4}}=\frac{2}{\sqrt{5}}$$

$$\therefore \quad f(f(2))=f\left(\frac{2}{\sqrt{5}}\right)=\frac{\frac{2}{\sqrt{5}}}{\sqrt{1+\left(\frac{2}{\sqrt{5}}\right)^2}}=\frac{2}{3}$$

$$\therefore \quad f(f(f(2)))=f\left(\frac{2}{3}\right)=\frac{\frac{2}{3}}{\sqrt{1+\left(\frac{2}{3}\right)^2}}=\frac{2}{\sqrt{13}}$$

Now, verify the options.

Consider option (a)

$$f(f(f(2)))=\frac{2}{\sqrt{1+3(2)^2}}=\frac{2}{\sqrt{13}}$$

Hence option (a) is correct.

13. $$f(-2.4)=1+|-2.4| \qquad (\because -2.4<-1)$$

$$f(-2.4)=3.4$$

$$\therefore \quad f(f(-2.4)=f(3.4)=[3.4] \qquad (\because 3.4>-1)$$

$$=3$$

14. Go through options

Let us consider $g(x)=\log x$,

$$\therefore \quad f(x)=x^n$$

and $$g(f(x))=g(x^n)=\log x^n$$

$$=n\log x=ng(x)$$

Hence, choice (d) is correct.

15. Since $f: N\to R$

but $0\notin N$

$\therefore f(0)$ is not defined

Hence (d)

16. Go through options.

Let us consider $f(x)=x^2-2$

$$\therefore \quad f\left(x+\frac{1}{x}\right)=\left(x+\frac{1}{x}\right)^2-2$$

$$=x^2+\frac{1}{x^2}+2-2=x^2+\frac{1}{x^2}$$

Hence choice (b) is correct.

17. $$f(x)=\sqrt{x^5}$$

$$\therefore \quad f(5x)=\sqrt{(5x)^5}=25x^2\sqrt{5x}$$

18. $$h(x)=\frac{1}{x}$$

$$g(h(x))=g\left(\frac{1}{x}\right)=\left(\frac{1}{x}\right)^2$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$$f(g(h(x))) = f\left(\left(\frac{1}{x}\right)^2\right) = \left(4\cdot\frac{1}{x^2}\right) - 5$$

$$= \frac{4}{x^2} - 5$$

19. Let $x = 2.6$, then $[x] - \{x\} = 2 - 3 = -1$
and $x = 3.8$, then $[x] - \{x\} = 3 - 4 = -1$
and $x = -3.6$, then $[x] - \{x\} = (-4) - (-3) = -1$
and $x = -7.3$, then $[x] - \{x\} = (-8) - (-7) = -1$
Hence (a) is the correct choice

20. $f(x) = \dfrac{a^x - 1}{x^n(a^x + 1)} = \underbrace{\dfrac{1}{x^n}}_{g(x)} \cdot \underbrace{\dfrac{a^x - 1}{a^x + 1}}_{h(x)}$ (By breaking the function)

$f(x)$ is even since graph of $f(x)$ is symmetric about y-axis.

Now, $h(-x) = \dfrac{a^{-x} - 1}{a^{-x} + 1} = \dfrac{1 - a^x}{1 + a^x} = -h(x)$

i.e., $h(x)$ is an odd function.
Hence, $g(x) \times$ odd function = even function
$\therefore g(x)$ is an odd function
($\because$ odd $\times$ odd = Even but even $\times$ odd = odd)
$\therefore g(x) = \dfrac{1}{x^n}$ is an odd function, which is positive only when

$n = -\dfrac{1}{3}$

21. $f(x) = \ln(x + \sqrt{x^2 + 1})$

$\therefore \quad f(-x) = \ln(-x + \sqrt{(-x)^2 + 1})$

$= \ln(-x + \sqrt{x^2 + 1})$

Here it is not obvious whether f is an odd or even function.
So in such case we check for

$f(x) + f(-x) = 0 \Rightarrow f$ is an odd function

$f(x) - f(-x) = 0 \Rightarrow f$ is an even function

Now, $f(x) + f(-x) = \ln(x + \sqrt{x^2 + 1}) + \ln(-x + \sqrt{x^2 + 1})$

$= \ln((\sqrt{x^2 + 1} + x)(\sqrt{x^2 + 1} - x))$

$= \ln(x^2 + 1 - x^2)$

($\because \ln x + \ln y = \ln x \cdot y$)

$= \ln 1 = 0$ ($\because \ln 1 = 0$)

Hence, f is an odd function.

23. $f(x) = \dfrac{a^x + 1}{a^x - 1}$

$\therefore \quad f(-x) = \dfrac{a^{-x} + 1}{a^{-x} - 1} = \dfrac{1 + a^x}{1 - a^x}$

$\Rightarrow f(x) \neq f(-x)$, hence $f(x)$ is not an even function

Again $f(x) = \dfrac{a^x - a^{-x}}{a^x + a^{-x}}$

$\therefore \quad f(-x) = \dfrac{a^{-x} - a^x}{a^{-x} + a^x}$

$\therefore f(x) \neq f(-x)$, hence $f(x)$ is not an even function

Again $f(x) = x \cdot \dfrac{a^x + 1}{a^x - 1}$

$\therefore \quad f(-x) = (-x)\dfrac{a^{-x} + 1}{a^{-x} - 1} = (-x)\dfrac{\frac{(1 + a^x)}{a^x}}{\frac{1 - a^x}{a^x}}$

$= (-x)\dfrac{1 + a^x}{1 - a^x} = (-x)\dfrac{1 + a^x}{-(a^x - 1)}$

$= x\dfrac{1 + a^x}{a^x - 1} = x \cdot \dfrac{a^x + 1}{a^x - 1}$

$\therefore \quad f(x) = f(-x)$

Hence, $f(x)$ is an even function.

24. $f(x) = \ln\left(\dfrac{1 + x^3}{1 - x^3}\right)$ (ln → natural logarithm)

$\therefore \quad f(-x) = \ln\left(\dfrac{1 + (-x)^3}{1 - (-x)^3}\right) = \ln\left(\dfrac{1 - x^3}{1 + x^3}\right)$

$\therefore \quad -f(-x) = -\ln\left(\dfrac{1 - x^3}{1 + x^3}\right)$

$= \ln\left(\dfrac{1 - x^3}{1 + x^3}\right)^{-1} = \ln\left(\dfrac{1 + x^3}{1 - x^3}\right)$

$\Rightarrow \quad f(x) = -f(-x)$

Hence $f(x) = \ln\left(\dfrac{1 + x^3}{1 - x^3}\right)$ is an odd function.

25. $f(x) = \log\dfrac{(1 - x)}{(1 + x)}$ is an odd function.

$f(x) = x \cdot \dfrac{a^x + 1}{a^x - 1}$ is an even function.

$f(x) = \log[x + \sqrt{(1 + x^2)}]$ is an odd function.

$f(x) = \sqrt{(1 + x + x^2)} - \sqrt{(1 - x + x^2)}$ is an odd function

26. $f(x) = \sqrt[3]{(1 - x^2)} + \sqrt[3]{(1 + x^2)}$

$\therefore \quad f(-x) = \sqrt[3]{(1 - (-x)^2)} + \sqrt[3]{1 + (-x)^2}$

or $\quad f(-x) = \sqrt[3]{1 - x^2} + \sqrt[3]{1 + x^2}$

$\Rightarrow \quad f(x) = f(-x)$

Hence $f(x)$ is even function

27. $f(x) = 2^{-x^2}$

$\therefore \quad f(-x) = 2^{-(-x)^2} = 2^{-x^2}$

$\because f(x) = f(-x)$, hence $f(x)$ is an even function

Again $f(x) = 2^{x - x^4} = \dfrac{2^x}{2^{x^4}}$

$\therefore \quad f(-x) = 2^{-x-(-x)^4} = 2^{-x-x^4} = \dfrac{1}{2^{x + x^4}}$

$\because f(x) = -f(-x)$, Hence $f(x)$ is not an odd function.

Again $f(x) = \cos x$ is an even function
Hence (d).

28. $f(x) = \dfrac{x}{e^x - 1} + \dfrac{x}{2} + 1$

$\therefore \quad f(-x) = \dfrac{-x}{e^{-x} - 1} + \dfrac{(-x)}{2} + 1$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$$= \frac{-xe^x}{1-e^x} - \frac{x}{2} + 1$$

$$= \frac{xe^x}{e^x - 1} - \frac{x}{2} + 1$$

If $f(x) - f(-x) = 0$, then $f(x)$ is an even function.

$$\therefore \quad f(x) - f(-x) = \left(\frac{x}{e^x - 1} + \frac{x}{2} + 1\right) - \left(\frac{xe^x}{e^x - 1} - \frac{x}{2} + 1\right)$$

$$= \frac{x(1 - e^x)}{(e^x - 1)} + \left(\frac{x}{2} + \frac{x}{2}\right) + (1 - 1)$$

$$= \frac{-x(e^x - 1)}{(e^x - 1)} + x + 0$$

$$= -x + x$$

$f(x) - f(-x) = 0$, hence the given function is an even function.

INTRODUCTORY EXERCISE 17.3

1. $f(x) = x^2$

$f(1) = 1$

and $f(-1) = 1$

$\Rightarrow f(x)$ is many-one

$\because \quad f : R \to R$

$(x) \quad (y)$

but the negative values of y are not the actual output (or images) of any element of x. Hence function is into.

2. $f : R \to R^+ \cup \{0\}$

$f(x) = x^2$

$f(1) = 1$

and $f(-1) = 1$

Hence $f(x)$ is many-one

again $f : \underbrace{R}_{(x)} \to \underbrace{R^+ \cup \{0\}}_{(y)}$

Since each element of y is the actual output of some element of x therefore function is onto.

3. $f : R^+ \to R$

$f(x) = x^2 \Rightarrow f(0) = 0, f(1) = 1, f(2) = 4$ etc.

Since non-positive values of x (*i.e.*, input) are not in the domain therefore no two inputs produce same output hence function is one-one.

Also function is into since non-positive values of y are not the images (output) of any value of x (*i.e.*, input)

4. $f : R^+ \to R^+$

$f(x) = x^2$

$f(1) = 1$

$f(2) = 4$

$f(3) = 9$ etc.

Since there are no two inputs which produce same output hence the function is one-one.

Again the function is onto since every element of y is an actual output/image of some element of x.

30. $f(x) = \frac{1}{1-x}$

$$\therefore \quad g(x) = f(f(x)) = f\left(\frac{1}{1-x}\right)$$

$$\Rightarrow \quad g(x) = \frac{1}{1 - \left(\frac{1}{1-x}\right)} = \frac{1-x}{-x} = \frac{x-1}{x}$$

$$\therefore \quad h(x) = f(f(f(x))) = f(g(x)) = f\left(\frac{x-1}{x}\right)$$

$$\Rightarrow \quad h(x) = \frac{1}{1 - \left(\frac{x-1}{x}\right)}$$

$$h(x) = x$$

$$\therefore \quad f(x) \cdot g(x) \cdot h(x) = \frac{1}{1-x} \times \frac{(1-x)}{-x} \times x = -1$$

5. f is not one-one as $f(0) = 0$ and $f(-1) = 0$

f is not onto as for $y = 1$, there is no $x \in R$ such that $f(x) = 1$

If there is such a $x \in R$ then $e^{|x|} - e^{-x} = e^x + e^{-x}$ clearly $x \neq 0$ ($\because$ at $x = 0$, $f(x) \neq 1$)

Again for $x > 0$, this equation gives $-e^{-x} = e^{-x}$ which is not possible

Again for $x < 0$, the above equation gives $-e^{-x} = e^x$, which is also not possible.

6. If $n(A) = m$ and $n(B) = n \geq m$, then the number of one-one functions from A to B is ${}^nP_m : n > m$

$${}^nP_m = \frac{n!}{(n-m)!}$$

Hence $n = 5$ and $m = 3$

$\therefore$ Number of one-one function from A to B

$$= {}^5P_3 = \frac{5!}{2!} = \frac{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{2} = 60$$

7. Let $x, y \in A$ be such that $f(x) = f(y)$

$$\Rightarrow \quad \frac{x-2}{x-3} = \frac{y-2}{y-3}$$

$$\Rightarrow \quad xy - 3x - 2y + 6 = xy - 3y - 2x + 6$$

$$\Rightarrow \quad -2y + 3y = 3x - 2x$$

$$\Rightarrow \quad x = y$$

Hence f is one-one.

Let $y \in B$, we want to got $x \in A$ such that $f(x) = y$,

i.e., $\frac{x-2}{x-3} = y$

$\Rightarrow x = \frac{3y-2}{y-1} \in A$, Hence f is onto

Thus f is bijective.

8. For any $x, y \in N$, we have

$f(x) = f(y) \Rightarrow 2x + 3 = 2y + 3 \Rightarrow x = y$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Also, f is not surjective
$\because 1 \in N$ does not have its pre image.

9. We have, $f(x) = (x-1)(x-2)(x-3)$
$f(1) = f(2) = f(3) = 0$
$\Rightarrow$
$\Rightarrow f(x)$ is not one-one
For every $y \in R$, there exists $x \in R$ such that $f(x) = y$. Therefore f is onto.
Hence $f : R \to R$ is onto but not one-one.

10. Since $x - [x] = 0$ for all integral values of x.
Therefore, the function is many-one and therefore inverse function is not defined.

11. $y = x^2 + 2 \Rightarrow x^2 = y - 2$
$\Rightarrow x = \sqrt{y-2}$
$\Rightarrow f^{-1}(x) = \sqrt{x-2}$

12. $y = 5^x \Rightarrow \log_5 y = x$
$\Rightarrow x = \log_5 y$
$\Rightarrow f^{-1}(x) = \log_5 x$

13. $y = \frac{x}{x+1} \Rightarrow yx + y - x = 0$
$\Rightarrow x(y-1) = -y$
$\Rightarrow x = -\frac{y}{y-1} = \frac{y}{1-y}$
$\Rightarrow f^{-1}(x) = \frac{x}{1-x}$

14. $y = \frac{x-1}{x+1} \Rightarrow yx + y = x - 1$
$\Rightarrow yx - x = -(y+1)$
$\Rightarrow x(y-1) = -(y+1)$
$\Rightarrow x = \frac{-(y+1)}{(y-1)} = \frac{y+1}{1-y} = \frac{1+y}{1-y}$
$\Rightarrow f^{-1}(x) = \frac{1+x}{1-x}$

15. Let $f(x) = y = x(2-x)$
$\Rightarrow x^2 - 2x - y = 0$
$\Rightarrow x = \frac{2 \pm 2\sqrt{1-y}}{2} = 1 \pm \sqrt{1-y}$
But $x < 1$
So $x = 1 - \sqrt{1-y}$
$\therefore f^{-1}(x) = 1 - \sqrt{1-x}$

16. $y = x^2 - x + 1$
$\Rightarrow y = x^2 - x + \frac{1}{4} + \frac{3}{4}$
$\Rightarrow y = \left(x - \frac{1}{2}\right)^2 + \frac{3}{4}$
$\Rightarrow y - \frac{3}{4} = \left(x - \frac{1}{2}\right)^2$
$\Rightarrow \left(x - \frac{1}{2}\right) = \sqrt{y - \frac{3}{4}}$
$\Rightarrow x = \sqrt{y - \frac{3}{4}} + \frac{1}{2}$
$\Rightarrow y = \sqrt{x - \frac{3}{4}} + \frac{1}{2}$

17. $f(x) + f^1(x) = (x^2 - x + 1) + \sqrt{x - \frac{3}{4}} + \frac{1}{2}$
$= \left(\frac{3}{4}\right)^2 - \frac{3}{4} + 1 + \sqrt{\frac{3}{4} - \frac{3}{4}} + \frac{1}{2}$
$= \frac{9}{16} + \frac{1}{4} + \frac{1}{2}$
$= \frac{9 + 4 + 8}{16} = \frac{21}{16}$

LEVEL 1

1. $P(3, 5) = 3 \times 5 = 15$
$S(5, 3) = 5 - 3 = 2$
$D(P(3, 5)), S(5, 3) = D(15, 2) = 15 + 2 = 17$

2. $P(S(10, 6), 30)) = P(4, 30) = 120$

3. $20 \# (-43)$ ($\because 21 @ 22 = -43$)
We cannot determine further the value of $20 \# (-43)$ since we don't know $a \# b$ takes what value in case of negative product as a, b is defined only for natural numbers. Hence (d)

4. $(a \# a) @ (a @ a)$
Since, it is given that a and b are natural numbers. Two cases arise.
Case I. a is even
$a \# a = a^2 - a^2 = 0$
$a @ a = a^2$
$\therefore 0 @ a^2 = 0$ ($\because 0 + a^2$ is even)
Case II. a is odd
$a \# a = a^2$, which is odd
$a @ a = a^2$
$\therefore a^2 @ a^2 = a^4$ ($\because a^2 + a^2$ will be even)
Hence (c) is correct.

5. $(p \# q) - (q @ p)$
Since p is even and q is odd
$p \# q = p^2 - q^2$
$q @ p = q^2 - p^2$
$\therefore p^2 - q^2 - (q^2 - p^2)$
$= 2p^2 - 2q^2 = 2(p^2 - q^2)$

6. $M(7, N(9, 4)) = M(7, (9^2 + 4^2 - 9.4))$
$= M(7, 61) = 7^2 + 61^2 + 7 \cdot 61 = 4197$

7. $f(x) = x^3 - x^2$
$\therefore f(x+1) = (x+1)^3 - (x+1)^2$
$= (x+1)^2 (x + 1 - 1) = (x+1)^2 (x)$

8. $[(5 \cup 4) \vee \vee (2 \cup \cup \cup 3)] \cup (8 \vee 3) = [1 \vee \vee 8] \cup (5) = 8 \cup 5 = 3$

9. $2 \cup \cup 8 = 2 \times 8 = 16$
and $2 \cup \cup \cup 4 = 2^4 = 16$
Hence (a) is true.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

10. $a \cup b = a \vee b$ if $a \geq b$

$a \vee b = a \cup b$ if $a < b$

$\therefore\ a \cup b = a \vee b$ for all real values.

11. $(20 @ 4) \# 35 = 576 \# 35$

$= (541)^2 = 292681$

12. $12 \# (14 @ (12 \# 13)) = 12 \# (14 @ 1)$

$= 12 \# 225 = (-213)^2 = 45369$

13. $(9 \# 7) @ (21 \# 5) = 4 @ 256$

$= (260)^2 = 67600$

14. $(2 @ 2) + (3 \# 3) - (4 @ 4) + (5 \# 5)$.

$= 16 + 0 - 64 + 0 = -48$

15. $$\frac{XAT\,[MAT\,\{2,\ CAT\,(3,\ XAT\,[3,\ 2])\},\ 5]}{CAT\,[4,\ XAT\,\{MAT\,(2,\ -5),\ 4\}]}$$

$$= \frac{XAT\,[MAT\,\{2,\ CAT\,(3,\ 2)\},\ 5]}{CAT\,[4,\ XAT\,\{1, 4\}]}$$

$$= \frac{XAT\,[MAT\,\{2, 9\},\ 5]}{CAT\,[4,\ 0]}$$

$$= \frac{XAT\,[6,\ 5]}{1} = 5$$

16. $$\frac{XAT\,(3, 5)}{MAT\,(3, 5)} = \frac{2}{1} = 2,$$ hence true

and $$\frac{XAT\,(-3, -5)}{MAT\,(-3, -5)} = \frac{2}{1} = 2,$$ hence true

Thus, (c) is correct.

17. $r\,(16, 7) - 2 = 2 - 2 = 0$

18. $g(4.7) - l(2) = 5 - 2 = 3$

19. $\because$ Dividend = Divisor × Quotient + Remainder

$\therefore \quad x = y \times k + r$ (k is a quotient and $k \in I$)

But $$k = I\left(\frac{x}{y}\right)$$

$\therefore \quad r = x - yk$

$\Rightarrow \quad$ $$r = x - \left[y \cdot I\left(\frac{x}{y}\right)\right]$$

20. $2 \# ((7 @ (4 \# 5)) @ 3) = 2 \# ((7 @ 14) @ 3)$

$= 2 \# (70 @ 3)$

$= 2 \# 152 = 306$

21. (i) $\begin{cases} p \# q = p \# 2p = 5p \\ q \# p = 2p \# p = 4p \end{cases}$ ($\because p + q$ is even)

(ii) $p @ q = p @ 3p = 0$ ($\because p + 3p = 4p$ is even)

(iii) $p @ q = p @ 2p$ ($p + 2p$ is odd)

$= 2p \# 4p$ ($2p + 4p = 6p$ is even)

$= 2p + 8p$

$= 10p$, which is even

Hence (ii) and (iii) are true

Thus (d) is correct.

22. $6 @ 2 = 18 - 2 = 16$

$12 \# 2 = 12 + 4 = 16$

Hence (i) is true

Again $6 \# 3 = 12 @ 6 = 36 - 6 = 30$

$(14 @ 12) + (2 @ 6) = 30 + 0 = 30$

Hence (ii) is true

Again $(6 @ 2) + (6 \# 3) = 16 + 30 = 46$

$(12 \# 2) + (14 @ 12) - (2 @ 6) = 16 + 30 - 0 = 46$

Hence (iii) is also true

Thus (d) is correct option.

23. $\text{Max}\,[(2 \# 3)\ \$\ (3 \# 4),\ (3 \# 4)\ \$\ (4\ \$\ 4)]$

$= \text{Max}\,[(6\ \$\ 12),\ (12\ \$\ 33)]$

$= \text{Max}\,[181, 1234]$

$= |181 + 1234| = 1415$

24. $\text{Max}\,[\text{Min}\,((1\ \$\ 2),\ (2\ \$\ 3)),\ \text{Max}\,((3 \# 4),\ (4 \# 5))]$

$= \text{Max}\,[\text{Min}\,(6,\ 14),\ \text{Max}\,(12,\ 20)]$

$= \text{Max}\,[8, 32] = 40$

25. $\text{Min}\,[(1 \# 2)\ \$\ 3,\ 4 \#\ (5\ \$\ 6)]\ \$\ \text{Max}\,[(6 \# 7),\ (7 \# 8)]$

$= \text{Min}\,[(2\ \$\ 3),\ (4 \# 62)]\ \$\ \text{Max}\,[42, 56]$

$= \text{Min}\,[14, 77]\ \$\ [98]$

$= 63\ \$\ 98 = 13574$

26. $$9 \Delta 3 = 9^3 + 9 \times 3 + \frac{9}{3}$$

$= 729 + 27 + 3 = 759$

27. $3 \# 4 * 5 + 3 * 4 \# 5$

$= 3^4 \times 5 + (3 \times 4)^5$

$= 81 \times 5 + (12)^5$

$= 405 + 248832 = 249237$

28. $f(2 \# 3) \# g(3 \# 4) = f(17) \# g(145)$

$= 255 \# 290$

$= 255^{290} + 290^{255}$

29. $f(g(2 \# 3)) + g(f(1 \# 2)) = f(g(17)) + g(f(3))$

$= f(34) + g(3)$

$= 1088 + 6 = 1094$

30. $(5 \oplus 6) - (6 @ 3) + (2 * 4) = 91 - (-9) + 56$

$= 91 + 9 + 56 = 156$

31. $(((1 \oplus 2) * 3) @ 20) = ((7 * 3) @ 20)$

$= (420 @ 20)$

$= 160000 - 8400$

$= 151600$

32. $$(3\,D\,4)\,D\,(12\,D\,5) = \left(\frac{3}{4}\right) D \left(\frac{5}{12}\right) = \frac{\frac{5}{12}}{\frac{3}{4}} = \frac{5}{9}$$

33. $$(12\,D\,60)\,E\,(2C\,10) = \left(\frac{1}{5}\right) E \left(\frac{1}{5}\right) = \frac{1}{25}$$

34. $30 + 20 - 4 \times 6 = 30 \times 20 \div 4 + 6$

$= 30 \times 5 + 6 = 156$

35. $50/40 - 5 + 4 \times 54$

$= 50 - 40 \div 5 \times 4 + 54 = 50 - 8 \times 4 + 54$

$= 50 - 32 + 54 = 72$

36. $(5 \wedge 5) \times (4 \wedge 4) / (3 \wedge 3) \times (2 \wedge 2) / (1 \wedge 1)$

$= (5^5) + (4^4) - (3^3) + (2^2) - (1^1)$

$= 3125 + 256 - 27 + 4 - 1 = 3357$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

37. $[(x^2 \times y^2) - (x+y)^2 + [(x+y)^3 - \{(x/y)^2 \times 2 + x + y\}]$
$\div [(x^2+y^2) \div (xy)^2] \times [(xy)^3 \div \{(x-y)^2 + 2xy\}]$

$$= \left[\frac{x^2+y^2}{(xy)^2}\right] \times \left[\frac{(xy)^3}{(x-y)^2+2xy}\right]$$

$$= \left[\frac{(x^2+y^2)}{(xy)^2} \times \frac{(xy)^3}{(x^2+y^2)}\right] = xy$$

$= x + y$ (In Devic Mathematics)

38. $a + a + a - a + a + a / a + a - a$
$= a \times a \times a \div a \times a \times a - a \times a \div a$
$= a^3 \div a^3 - a^2 \div a = 1 - a$
$= \frac{1}{a}$ (In Devic maths)

39. $[(2 \Delta 3) \Delta (-2 \Delta 3)] = [(2-3) \Delta (-2+3)]$
$[(-1) \Delta (1)] = -1 + 1 = 0$

40. $\because$ $a \Delta a = 0 \; \forall \; a \in R$
Given $a > 0$
$b > 0$
If $(a \Delta b) \Delta (a \Delta (-b)) = (a-b) \Delta (a-b) \quad (\because a \times -b < 0)$
$= 0$

Again if $b < 0$, then
$(a \Delta b) \Delta (a \Delta (-b))$ ($\because b$ is negative, $\therefore -b$ is positive)
$= (a+b) \Delta (a+b) = 0$

Hence, (d).

41. $f(x) = x^3 + 3x^2 + 3x$
$\therefore f(x+1) = (x+1)^3 + 3(x+1)^2 + 3(x+1)$
$= x^3 + 1 + 3x(x+1) + 3(x^2 + 1 + 2x) + 3x + 3$
$= x^3 + 1 + 3x^2 + 3x + 3x^2 + 3 + 6x + 3x + 3$
$= x^3 + 6x^2 + 12x + 7$

42. $\frac{(4 \# 2) @ (9 \# (-1))}{(2 @ 3) \# (5 @ 2)} = \frac{3 @ 1}{0 \# 9} = \frac{2}{-1} = -2$

43. Solve the options as per the directions.

44. Since, f_4 can assume negative values also, but f_1, f_2 and f_3 always give positive values only.
Hence choice (b), (c) and (d) are not possible.

45. $lt\,(8.3) - gt\,(5.3) + lt\,(6.8) + gt\,(9.4)$
$= 9 - 5 + 7 + 9 = 20$

46. $\Delta(3, 2) - \Delta(2, 3) + gt\,(4.8) - lt\,(2.7) + \Delta(4, 5)$
$= 2 - 2 + 4 - 3 + 4 = 5$

47. $a = \frac{p+q}{2}$, $b = \frac{lt\,(p) + qt\,(q)}{2}$

Now, $lt\,p = p + k; \quad 0 \le k \le 1$
$gt\,q = q - m; \quad 0 \le m \le 1$ where $p, q \notin I$

Also $k + m = 1$ (Actually it is just less than 1)

$\therefore \quad (a-b) = \frac{p+q}{2} - \frac{(p+k)+(q-m)}{2}$

$\Rightarrow \quad (a-b) = \frac{p+q}{2} - \frac{p+q+k-m}{2}$

$\Rightarrow \quad (a-b) = \frac{k-m}{2}$

Now, $(a-b) = \frac{k-(1-k)}{2} \quad (\because k+m=1)$
$= k - \frac{1}{2}$

and $(a-b) = \frac{(1-m)-m}{2} \quad (\because k+m=1)$
$= \frac{1}{2} - m$
$(a-b) = k - \frac{1}{2}$

$\Rightarrow \quad \frac{1}{2} \le (a-b) \le \frac{1}{2} \quad (\because 0 \le k \le 1)$

(Similarly $-\frac{1}{2} \le (a-b) \le \frac{1}{2}$, since $0 \le m \le 1$)

NOTE The correct result (in practical) is $-\frac{1}{2} < (a-b) < \frac{1}{2}$ since $0 \le k < 1$ and $0 \le m < 1$.

48. $(a @ b)(b \# a) = \frac{a}{a+b} \times \frac{a+b}{b} = \frac{a}{b}$

Hence (c) is correct.

49. $b \# a - a @ b = \frac{(a+b)}{b} - \frac{a}{(a+b)} = \frac{(a+b)^2 - ab}{b(a+b)}$
$= \frac{a^2 + b^2 + 2ab - ab}{b(a+b)}$
$= \frac{a^2 + b^2 + ab}{b(a+b)}$

50. $mn\,(md\,(-2), mx\,(5, mn\,(-2, md\,(5))))$
$= mn\,(2, mx(5, mn\,(-2, 5)))$
$= mn\,(2, mx\,(5, -2)) = mn\,(2, 5) = 2$

51. $md\,(mn\,(-2, mx\,(md(5), mn\,(-2, -7))))$
$= md(mn\,(-2, mx\,(5, -7)))$
$= md\,(mn\,(-2, 5)) = md\,(-2) = 2$

52. $((1 @ (2 \Delta 3)) \# 4) = ((1 @ ((2 @ 3) - (2 \# 3)) \# 4))$
$= ((1 @ (45 - (-3)) \# 4$
$= ((1 @ 48) \# 4)$
$= (4704 \# 4) = 88416400$

53. $(1 @ 1) \Delta (2 \# 2) = 4 \Delta 0$
$= (4 @ 0) - (4 \# 0) = 0 - 0 = 0$

54. $(a \Delta b) + (a \# b) = (a^2 + b^2) + (|a^2 - b^2|)$

Case I. When $(a^2 - b^2)$ is positive
$(a^2 + b^2) + (a^2 - b^2) = 2a^2$

Case II. When $(a^2 - b^2)$ is negative
$(a^2 + b^2) - (a^2 - b^2) = 2b^2$

Hence we cannot determine the required value.

HINT $|k| \Rightarrow \pm k$

55. $[(a \# b) \div (a @ b)]^2 - 2(a * b)$
$= [|a^2 - b^2| \div |a - b|]^2 - 2ab \quad (\because |k| \ge$
$= (|a+b|)^2 - 2ab$
$= a^2 + b^2 = a \Delta b$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

56. Consider $g(x) = e^x$ (from the given options)

$\therefore \quad e^x = g(f(x))$

$\Rightarrow \quad e^x = g \log (g(x))$

$\Rightarrow \quad e^x = g \log (e^x)$

$\Rightarrow \quad e^x = g(x)$

$\Rightarrow \quad e^x = e^x$

Hence, choice (b) is correct.

57. Going through options, we find that none of the choices (a), (b) and (c) satisfy the relation, hence (d)

58. None of the options (a), (b) and (c) satisfy the given conditions.

59. $g[f(x) - f[g(x)] = g(2x + 3) - f(9x + 6)$

$= [9(2x + 3) + 6] - [2(9x + 6) + 3]$

$= [18x + 27 + 6] - [(18x + 12) + 3] = 18$

60. $f(f(-2)) = f(8 - 6 + 1) \qquad (\because -2 < -1)$

$= f(3)$

$= (9 + 6 + 2) \qquad (\because 3 > -1)$

$= 17$

61.

$x^2 = 4 \Rightarrow x = \pm 2$

$y^2 = 9 \Rightarrow y = \pm 3$

	(2, 3)	(−2, 3)	(−2, −3)	(2, −3)
$f(x, y)$ =	5	1	−5	−1
$g(x, y)$ =	−3	−7	−1	3
$F(f(x, y))$ =	10	10	−2	−2
$G(g(x, y))$ =	2	−6	−6	2

Also, $|F(f(x, y))| = |G(g(x, y))|$, which is true for $x = 2$ and $y = -3$

$\therefore \quad x + y = -1$

62. Let $x = 8, \quad y = 9$

$\therefore \quad [x] = 7$ and $\{y\} = 10$

$\therefore \quad g(x, y) = -3$

Again if $x = 9$ and $y = 8$, then

$[x] = 8$ and $\{y\} = 9 \quad \therefore g(x, y) = -1$

Hence (c) is correct choice.

HINT There are two cases (i) when $x < y$ (ii) when $x > y$

Alternatively : Let $x = n \therefore y = n + 1$

$\therefore \quad g(x, y) = (n - 1) - (n + 1 + 1) = -3$

Also, if $x = n, \quad y = n - 1$

$g(x, y) = (n - 1) - (n - 1 + 1) = -1$

Hence (c).

63. $f(x, y) = x + 1 + y - 1 = x + y$

$g(x, y) = x - 1 - y - 1 = x - y - 2$

$F(f(x, y)) = x + y + 1 - x + y + 2 + 1 = 2y + 4$

$G(g(x, y)) = x + y - 1 + x - y - 2 + 1 = 2x - 2$

Hence, only (a) is false.

64. $F(f(x, y)) = -f(x, y)$

$G(f(x, y)) = -F(f(x, y)) = f(x, y)$

$\therefore f(G(f(1.2, 1.3)), f(F(f(0, 1)), G(f(0.8, 0.4))))$

$= f(f(1.2, 1.3)), f(-f(0, 1)), f(0.8, 0.4)))$

$= f((2.5)), f(-(1)), [1.2])))$

$= f(3, f(-1, 1)) = f(3, 0)$

$= 3$, hence (c) is correct.

65. $F(f(x, y)) + G(f(x, y)) = 0$

Hence (d) is correct.

66. $f(1.1, 1.2) \times G(f(1.1, 1.2)) = f(1.1, 1.2) \times f(1.1, 1.2)$

$= [2.3] \times [2.3] = 2 \times 2 = 4$

67. $f(g(x)) = f\left(x + \frac{1}{x}\right) = \frac{1}{x + \frac{1}{x}} = \frac{x}{x^2 + 1}$

$g(f(x)) = g\left(\frac{1}{x}\right) = \frac{1}{x} + x = \frac{x^2 + 1}{x}$

$\therefore \quad f(g(x)) \cdot g(f(x)) = \frac{x}{x^2 + 1} \times \frac{x^2 + 1}{x} = 1$

68. $g(f(x)) = g(f(2)) = g(13) = 29$

$\because \quad f(2) = 2 \cdot (2)^2 + 7(2) - 9 = 13$

and $g(13) = 2 \times 13 + 3 = 29$

69. $f(f(f(f(f(2))))) = f(f(f(f(3)))) \qquad \left(\because f(2) = \frac{2+1}{2-1} = 3\right)$

$= f(f(f(2))) \qquad \left(\because f(3) = \frac{3+1}{3-1} = 2\right)$

$= f(f(3)) = f(2) = 3$

✍ ***Solutions for question number 70 and 71 :***

$f(x) = 1 - h(x)$ ...(1)

$g(x) = 1 - k(x)$ ...(2)

$h(x) = f(x) + 1$ ...(3)

$j(x) = g(x) + 1$ ...(4)

$k(x) = j(x) + 1$ ...(5)

From (1) and (3)

$f(x) + h(x) = 1 - h(x) + f(x) + 1$

$\Rightarrow \quad h(x) = 1$ ...(6)

$\Rightarrow \quad f(x) = 0$ ...(7)

From (2) and (4)

$g(x) = 1 - k(x) = 1 - (j(x) + 1)$

$g(x) = -j(x)$ ...(8)

From (4) and (8)

$j(x) - g(x) = 1$

and $g(x) = -j(x)$

$\Rightarrow \quad j(x) = \frac{1}{2}$ ...(9)

$\therefore \quad g(x) = -\frac{1}{2}$ ...(10)

$\therefore \quad k(x) = j(x) + 1 = \frac{3}{2}$ ...(11)

Hence $f(x) = 0$

$g(x) = -\frac{1}{2}$

$h(x) = 1$

$j(x) = \frac{1}{2}$

$k(x) = \frac{3}{2}$

Thus all the functions are constant.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

70. $\dfrac{f(f(x))+k(h(x))}{h(k(x))+f(j(x))}=\dfrac{\frac{1}{2}+\frac{3}{2}}{1+0}=\dfrac{2}{1}=2$

71. $\dfrac{1+\frac{1}{2}+\frac{3}{2}}{\left(0+\frac{1}{2}+\frac{3}{2}\right)\left(0\cdot\frac{1}{2}\cdot\frac{3}{2}\right)}=\dfrac{3}{0}=\infty$

i.e., not defined

72. Since, odd function + odd function = odd function
and (odd function) (odd function) = even function.

73. Even (Odd) = Even
and Odd (Even) = Even

74. y is maximum when $x^2+2=6-3x$
i.e., $x^2+3x-4=0$
$\Rightarrow \quad x^2+4x-x-4=0$
$\Rightarrow \quad x(x+4)-1(x+4)=0$
$\Rightarrow \quad x=1 \quad \text{or} \quad x=-4$
$\therefore$ at $x=1$, $y=3 (\because x>0)$

75. $p^2\times q^2\times r^2$ is maximum when $p^2=q^2=r^2=\dfrac{1}{3}$

$\therefore \quad (pqr)^2_{max}=\dfrac{1}{27}$

$\Rightarrow \quad (pqr)_{max}=\dfrac{1}{3\sqrt{3}}$

76. $\dfrac{g(x,y,z)}{j(x,y,z)}=\dfrac{3}{3}=1$

77. $\dfrac{(m(x,y,z)-f(x,y,z))}{h(x,y,z)\times k(x,y,z)\times l(x,y,z)}=\dfrac{3-4}{5\times 2\times 1}=\dfrac{-1}{10}$

78. $\dfrac{f(x,y,z)+h(x,y,z)-g(x,y,z)}{j(x,y,z)}=\dfrac{4+5-3}{3}=2>1$

79. $C(1,2,3)=\max(A(1,2,3), B(1,2,3))$
$=\max(3,2)=3$

80. $D(0,1,2)=\min(A(0,1,2), B(0,1,2))$
$=\min(1,2)=1$

81. $A(2,3,4)+B(2,3,4)=5+2=7$

82. $C(0,1,D(1,2,3))=C(0,1,\min(A(1,2,3), B(1,2,3)))$
$=C(0,1,\min(3,2))=C(0,1,2)$
$=\max(A(0,1,2), B(0,1,2))$
$=\max(1,2)=2$

83. $D(1,2,C(0,1,2))=D(1,2,2) \qquad (\because C(0,1,2)=2)$
$=\min(A(1,2,2), B(1,2,2))$
$=\min(3,1)=1$

84. $|x+5|=8$
$\Rightarrow \quad x=3 \quad \text{and} \quad x=-13$
and $9-|x-4|=y=f(x)$
for $x=3, \quad y=f(x)=8$
and for $x=-13, y=f(x)=-8$
Max. $(x.y)=(-13)\times(-8)=104$

LEVEL (2)

1. $f(x)=\dfrac{a^x+a^{-x}}{2}, \quad f(y)=\dfrac{a^y+a^{-y}}{2}$

$\therefore \quad f(x+y)=\dfrac{a^{x+y}+a^{-(x+y)}}{2} \qquad \ldots(1)$

and $\quad f(x-y)=\dfrac{a^{x-y}+a^{-(x-y)}}{2} \qquad \ldots(2)$

adding (1) and (2), we get

$$f(x+y)+f(x-y)=\frac{a^{x+y}+a^{-x-y}}{2}+\frac{a^{x-y}+a^{-x+y}}{2}$$
$$=\frac{1}{2}[a^x(a^y+a^{-y})+a^{-x}(a^y+a^{-y})]$$
$$=\frac{1}{2}(a^x+a^{-x})(a^y+a^{-y})$$
$$=2f(x)f(y)$$

2. $\left|\dfrac{30-x}{5}\right|<4$
$\Rightarrow \quad |30-x|<20$
$\Rightarrow \quad -20<(30-x)<20$
$\Rightarrow \quad -20<30-x<20$
$\Rightarrow \quad -50<-x<-10$
$\Rightarrow \quad 50>x>10$
or $\quad 10<x<50$

3. $|x+y|=|x-y| \Rightarrow (x+y)^2=(x-y)^2 \quad \Rightarrow \quad 4xy=0$
$\Rightarrow \quad$ either $x=0 \quad$ or $\quad y=0$
If $x=0$ and y is any real number, we have infinite possible values of y as

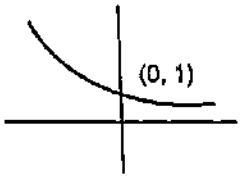
$$|0+y|=|0-y| \quad \Rightarrow \quad |y|=|-y|$$

Similarly $(x, 0)$ where x can be any real number also satisfies $|x+0|=|x-0|$
So there are infinite number of solutions.

6. $e^{-x^2}=\dfrac{1}{e^{x^2}}$

Since x^2 is an even function therefore e^{-x^2} is also an even function, hence it is symmetric about y-axis.
Also it is always positive.
and at $x=0, \quad e^{-x^2}=1$
Hence (b) is not true. Thus only choice (d) is correct.

7. $y=a^x \Rightarrow$ at $x=0$, $y=1$
Hence (iii) statement is true.



at $\quad a=\dfrac{1}{2}, f(x)=\left(\dfrac{1}{2}\right)^x$

e.g., $a=\dfrac{1}{2}$, then $f(x)=a^x=\left(\dfrac{1}{2}\right)^x$

Hence the function is decreasing.

14. Inverse of an exponential function is a logarithmic function.
$\therefore \quad x=10^y \quad \Rightarrow \quad \log_{10} x=y$
graph (A) is an exponential function
graph (B) is a logarithmic function
graph (C) is also an exponential function
graph (D) represents inverse function like $\left|\dfrac{1}{x}\right|$.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Logarithmic function is defined only for positive values hence choice (a), (c) and (d) are wrong.
Also the inverse function (i.e., $f^{-1}(x)$ of a function $f(x)$) is symmetric about $x = y$

15. $f(0) = 1 - 0 = 1, f\left(\frac{1}{2}\right) = 1 - \frac{1}{2} = \frac{1}{2} = 0.5$

$f(1) = 1 - 1 = 0, f\left(\frac{45}{18}\right) = 2.5 - 1 = 1.5$

$\therefore f(0) + f\left(\frac{1}{2}\right) + f(1) + f\left(\frac{45}{18}\right) = 1 + 0.5 + 0 + 1.5 = 3$

16. $x = \frac{|y|}{y}, y \neq 0$

if $y > 0, x = \frac{y}{y} = 1$

$\therefore$ for all values of $y > 0$, $x = 1$
and for all values of $y < 0$, $x = -1$

17. Since the reflection of $g(x)$ is about x-axis therefore $g(x) = -f(x)$

18. Since the reflection of $g(x)$ is about y-axis
$\therefore \quad g(x) = f(-x)$

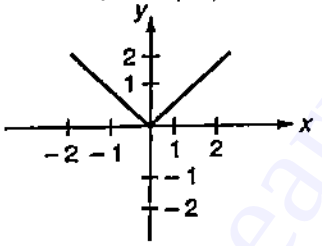
19. $f(x)$ is an even function

20. Since for each x there are more than 1 value of $f(x)$

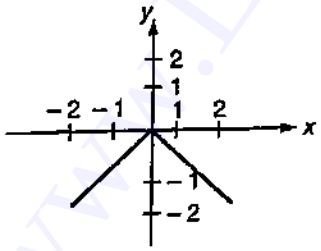
21. $f(x)$ is an even function

22. $f(x)$ is an odd function

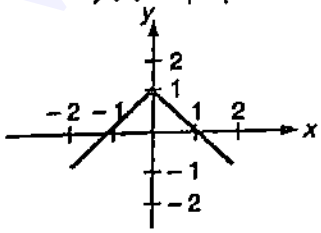
23. $f(x) = |x|$



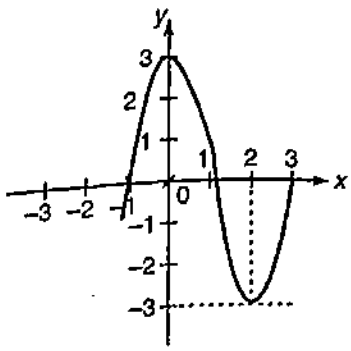
$f(x) = -|x|$



$f(x) = -|x| + 1$



24. $f(x) = x^3 - 3x^2 - x + 3$
$f(x) = (x - 1)(x + 1)(x - 3)$



$\Rightarrow \quad x = -1, 1, 3$

Thus the graph of $f(x)$ intersects the x-axis three times *viz.*, at $x = -1, 1, 3$

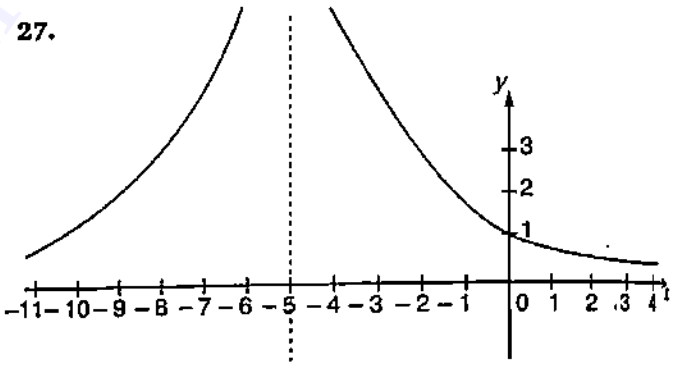
25. Choice (a) $\rightarrow$ at $x = 0$ $f(x)$ is not defined
Choice (b) $\rightarrow$ at $x = 0$ and $x = 1$, $f(x)$ is not defined
Choice (d) $\rightarrow$ x is defined for only positive real numbers except 1.

Hence choice (c) is correct as x^2 is always positive as $x^2 + 2 \neq 0$

26. Since x is always positive and y can assume both positive and negative values therefore (c) is the correct choice.

$x|y| = 1 \Rightarrow x = \frac{1}{|y|}$ for $x > 0$

27.



28. $f(x_2, x_3) = x_{2+3} = x_5 \quad (\because 2 + 3 < (3)^2)$
$f(x_5, x_6) = x_{5+6-3} = x_8 \quad (\because 5 + 6 > (3)^2)$
$\therefore \quad f(f(x_2, x_3), f(x_5, x_6)) = f(x_5, x_8) = x_{5+8-3}$
$= x_{10} \quad (\because 5 + 8 > (3)^2)$

29. $g(x_2, x_3) = x_1 \quad \left[\because \frac{2 \times 3}{5} \rightarrow m = 1\right]$

$g(x_7, x_8) = x_1 \quad \left[\because \frac{7 \times 8}{5} \rightarrow m = 1\right]$

$\therefore \quad g(g(x_2, x_3), g(x_7, x_8))$
$= g(x_1, x_1) = x_1 \quad \left[\because \frac{1 \times 1}{5} \rightarrow m = 1\right]$

30. $\because$ at $x = 1 \quad y = 0$
$x = -1 \quad y = 0$
$x = 0 \quad y = -1$
$x = 2 \quad y = 1$
$x = -2 \quad y = 1$

31. $y = |x| + x$

at		
	$x = 0$	$y = 0$
	$x = 1$	$y = 2$
	$x = 2$	$y = 4$
	$x = -1,$	$y = 0$
	$x = -2,$	$y = 0$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

32. $x = y - 1$
$y = x + 1$
$\Rightarrow$ $x = 0, \quad y = 1$
at $x = -1, \quad y = 0$
$x = 1, \quad y = 2$
$x = -2, \quad y = 1$

33. Option (a) is wrong since at $x = 1, 2, \ y = 0$
Option (c) is wrong since at $x = 0, \ y = 1$
Option (b) is correct since at $x = -2, 2; \ y = 0$
also $x = 0, \ y = 0$ etc.
Also the graph is symmetric about y-axis hence $|x|$ is must to be appeared.

34. The graph intersects x axis at four different points therefore choice (c) is wrong. Since graph is symmetric about y-axis therefore choice (a) is wrong since at $x = 1, \ y = 0$
but at $x = -1, \ y = 6$
choice (d) is wrong since at $x = 0, \ y = +2$
Hence choice (b) is correct.

35. $f(x) = \begin{cases} -x & x < 0 \\ x & x > 0 \end{cases}$

$g(x) = \begin{cases} x & x < 0 \\ -x & x > 0 \end{cases}$

$\therefore \quad f(x) = -g(x)$

36. $f(x) = \begin{cases} x \ \forall x < 0 \\ x \ \forall x > 0 \end{cases}$

$g(x) = \begin{cases} -x & x < 0 \\ x & x > 0 \end{cases}$

$\therefore \quad f(x) = -g(x)$ for $x < 0$
and $f(x) = g(x)$ for $x > 0$

37. $f(x) = \begin{cases} -x & x < 0 \\ -x & x > 0 \end{cases} \Rightarrow g(x) = \begin{cases} -x & x < 0 \\ x & x > 0 \end{cases}$

$\therefore f(x) = g(x)$ for $x < 0$ and $f(x) = -g(x)$ for $x > 0$

38. Choice (b) is wrong since at $x = 0, \ f(x) = 0$. Choices (a) and (c) are also wrong.
Now, choice (d) is correct, since
$y = -(ax^2 + c)$ and $y = (ax^2 + c)$
These two equations give equal but opposite values.

39. $f(1) = 1$
$f^2(1) = f(f(1)) = 1$
$f^3(1) = f^2(f(1)) = f^2(1) = 1$
$f^4(1) = f^3(f(1)) = f^3(1) = 1$
therefore $f^{11}(1) f^{33}(1) . f^{55}(1) f^{77}(1) f^{99}(1)$
$= 1 \times 1 \times 1 \times 1 \times 1 = 1$

40. $f^2(x) = f(f(x)) = f\left(\frac{1}{x}\right) = x$

$f^3(x) = f(f^2(x)) = f(x) = \frac{1}{x}$

$\therefore \quad f^2(x) = f^4(x) = f^6(x) = f^8(x) \ldots = x$
and $f(x) = f^3(x) = f^5(x) = f^7(x) \ldots = \frac{1}{x}$
$\therefore \quad f^{12}(x) = f^2(x) = 11$
$\therefore \quad f(f(x)) = 11$
$\Rightarrow \quad f\left(\frac{1}{x}\right) = 11 \Rightarrow f(x) = \frac{1}{11}$

✍ ***Solutions for question number 41-43 :***

Cycle	Values of		
	a	b	c
Start	1	2	3
1	2	4	6
2	4	6	10
3	8	8	16
4	16	10	26
5	32	12	44
6	64	14	78
7	128	16	144

41. 16 and 144
42. $b = 16$
43. Basically the module takes 7 cycles to reach step 6 (*i.e.*, to stop), Hence the number of repetitions $= 7 - 1 = 6$

✍ ***Solutions for question number 44-47 :***

Cycle	x	y	z	Sum
C_0	1	-2	3	$2 = S_0$
C_1	-5	8	-13	$-10 = S_1$
C_2	21	-34	55	$42 = S_2$
C_3	-89	144	-233	$-178 = S_3$
C_4	377	-610	987	$754 = S_4$

44. We can see from the above table that sum of each cycle is twice the each corresponding value of x in the same cycle.
45. $S_4 = 754$
46. $S_1, S_3, S_5, S_7 \ldots$ etc are negative integers
Hence S_{13} is negative
and each sum $S_1, S_2, S_3, S_4, S_5, S_6 \ldots$ is an even number
Hence (d)

47.

	K	L	M
initially	9	0	0
Fill (L, K)	4	5	0
Fill (M, L)	4	1	4
Empty (L, K)	5	0	4
Empty (M, L)	5	4	0

$\therefore L$ has 4 litres of water.

48. Note that we can feed the instruction of Empty (L, M) since the amount of water in L is less than the remaining capacity of M.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Final Round

Solutions for question number 1 to 10 :

P_0	P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8	P_9	P_{10}	P_{11}	P_{12}
0	1	1	0	−1	−1	0	1	1	0	−1	−1	0
Q_0	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	Q_7	Q_8	Q_9	Q_{10}	Q_{11}	Q_{12}
0	1	1	2	3	5	8	13	21	34	55	89	144

5. $Q_{13} = 233, \quad P_{14} = 1$

$\therefore \quad Q_{13} + P_{14} = 234$

6. $Q_{10} + P_{10} = 55 + (-1) = 54$

7. $Q_6 = 8 \ \therefore Q_8 = 21$

i.e. $Q_{[Q_6]} = Q_{[8]} = 21$

8. $\because$ (a), (b) and (c) are wrong

9. $(Q_5)^{P_5} = (5)^{-1} = \frac{1}{5} = 0.2$

Solutions for question number 11-14 :

11. $P[k(3)] = 3Q[k(3)] - 4$

$= 3(2R[k(3)] + R[k(6)]) - 4$

$= 3(2(S[k(6)]) - S[k(3)] + (S[k(12)] - S[k(6)])) - 4$

$= 3(2(22 - 13) + (51 - 22)) - 4$

$= 3(18 + 29) - 4 = 141 - 4 = 137$

12. $S[k(1)] = 7$

$R[k(7)] = S[k(14)] - S[k(7)]$

$= 59 - 31 = 28$

$Q[k(28)] = 2R[k(28)] + R[k(56)]$

$= 2(S[k(56)] - S[k(28)] + S[k(112)] - S(k(56))]$

$= 2(227 - 115) + (451 - 227)$

$= 2(112) + (224) = 448$

$\therefore \quad QRS[k(1)] = QR[k(7)] = Q[k(28)] = 448$

13. $R[k(5)] = S[k(10)] - S[k(5)]$

$= 43 - 19 = 24$

and $S[k(10)] = 43$

$\therefore \quad R[k(5)] - S[k(10)] = 24 - 43 = -19$

14. $\forall\, x < 0$, $R[k(x)]$ and $S[k(x)]$ are equal to zero. Therefore the whole product will be zero.

Solutions for question number 15-20 :

15. $(3x^4 + 2x^2 + 5x) + (2x^4 + 3x^3 + 7x^2)$

$= 5x^4 + 3x^3 + 9x^2 + 5x$

$= (5, 4, 3, 3, 9, 2, 5, 1)$

16. $(6, 5, 7, 4, 8, 3) - (3, 5, 5, 3, 7, 1)$

$= (6x^5 + 7x^4 + 8x^3) - (3x^5 + 5x^3 + 7x)$

$= (3x^5 + 7x^4 + 3x^3 - 7x)$

$= (3, 5, 7, 4, 3, 3, -7, 1)$

17. $(1, 1, 2, 0) \rightarrow (x + 2)$

$\therefore \quad (x + 2)^3 = x^3 + 6x^2 + 12x + 8$

$= (1, 3, 6, 2, 12, 1, 8, 0)$

18. $(3, 3, -10, 2, 7, 1) \div (3, 2, -7, 1)$

$= (3x^3 - 10x^2 + 7x) \div (3x^2 - 7x)$

$= x(3x^2 - 10x + 7) \div x(3x - 7)$

$= x(x - 1)(3x - 7) \div x(3x - 7)$

$= (x - 1)$

$= (1, 1, -1, 0)$

20. $(4x^4 + 3x^3) \times (2x^2 + x) + (2x^2 + x) - (3x^5 + 2x^4)$

$= (8x^6 + 10x^5 + 3x^4) + (2x^2 + x - 3x^5 - 2x^4)$

$= (8x^6 + 7x^5 + x^4 + 2x^2 + x)$

$= (8, 6, 7, 5, 1, 4, 2, 2, 1, 1)$

Solutions for question number 21-25 :

21. $h(3, 2, 8, 7) \div g(4, 7, 10, 8) = 2 \div 4 = \frac{1}{2}$

22. $h(fg(2, 5, 7, 3), 9) = h(f(2, 5, 7, 3) \times g(2, 5, 7, 3), 9)$

$= h((7 \times 2), 9) = h(14, 9) = 5$

23. $h(h(7, 13, 5, 9), h(4, 6, 12, 14)) = h(1, 0)$

$= \frac{1}{0}$, which is not defined

24. $A = 9, B = 20, C = 20, D = 14$

$\therefore \quad B = C > D > A$

Hence (b) is the appropriate answer

25. $h(h(a_1, b_1, c_1, d_1), h(a_2, b_2, c_2, d_2)) = h(0, 0)$

$= \frac{0}{0}$ is not defined, while $0 \times 0 = 0$ is defined

Solutions for question number 26-32 : *In case of $x > 0$ we get the following pattern.*

$f(1) = b + c - 2c + a = a + b - c$

$f(2) = b + c - 4c + a + b - c = a + 2b - 4c$

$f(3) = b + c - 6c + a + 2b - 4c = a + 3b - 9c$

$f(4) = b + c - 8c + a + 3b - 9c = a + 4b - 16c$

(i.e., $f(x) = a + bx - cx^2$)

26. Hence $f(8) = a + 8b - 64c = a + 8(b - 8c)$

Hence (b) is correct.

27. $f(-19) = 2b \times (-19) + f(-(-19))$

$= -38b + f(19)$

$= -38b + a + 19b - 361c$

$= a - 19b - 361c = a - 19(b + 19c)$

HINT: For $x < 0$ i.e., $f(-x) = a + b(-x) - c(-x)^2$

28. $f(7) = a + 7b - 49c$

$\therefore$ when $a = 15, \quad b = 11$ and $c = -3$

$f(7) = 15 + 7 \times 11 - 49(-3)$

$= 15 + 77 + 147 = 239$

29. $f(-10) = a - 10b - 100c$

$\therefore$ at $a = 10, b = -7$ and $c = 6$

$f(-10) = 10 - 10(-7) - 100 \times 6$

$= 10 + 70 - 600 = -520$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

30. $f(x) = a + b(x) - c(x)^2$ for every x

$$0 = 4 - 17x + 18x^2$$

$\therefore$ Now, for convenience go through options.

31. $$f(x) < 0$$

$$\Rightarrow \quad a + b(x) - c(x)^2 < 0$$

$$\Rightarrow \quad 12 + 10(x) - 8(x)^2 < 0 \qquad \ldots(1)$$

Now, for convenience go through options.

Alternatively : Solve the quadratic inequality (1) and then get the required set of values.

32. $$f(1) = a + b - c = -a \qquad (\because b = -a, c = a)$$

$\therefore$ $$f(f(1)) = f(-a)$$

$$= a + b(-a) - c(-a)^2$$

$$= a - ab - ca^2$$

$$= a + a^2 - a^3$$

Solutions for question number 33-37 :

$$f(y, 0) = y + f(y-1, 0)$$

$$= y + (y-1) + f(y-2, 0)$$

$$= y - (y-1) + (y-2) + \ldots + 1 + f(0, 0)$$

$$= \frac{y(y+1)}{2} + 1$$

and $$f(0, y) = y - f(0, y-1)$$

$$= y - [(y-1) - f(0, y-2)]$$

$$= y - (y-1) + f(0, y-2)$$

$$= 1 + (y-2) - f(0, y-3)$$

$$= 1 + (y-2) - (y-3) + f(0, y-4)$$

$$= 2 + f(0, y-4)$$

Thus, $$f(0, y) = \begin{cases} \frac{y-1}{2}, \text{ if } y \text{ is odd} \\ \frac{y+2}{2}; \text{ if } y \text{ is even} \end{cases}$$

33. $f(y_1, y_2, y_3, \ldots y_n)$ is not defined for every odd n.

Here $n = 27$

34. $f(0, 1, 0, 1) = f(0, 1) + f(1, 0) + (0 + 1 + 0 + 1)$

$= 0 + 2 + 2 = 4$

35. $f(8, 8, 8, 2, 2, 2) = f(8, 2) + f(8, 8, 2, 2)$

$\quad + (8 + 8 + 8 + 2 + 2 + 2)$

$= f(2, 0) + f(0, 8) + [f(8, 2) + f(8, 2) + (8 + 8 + 2 + 2)] + 30$

$= 4 + 5 + [2 f(8, 2) + 20] + 30$

$= 9 + [2(9) + 20] + 30$

$= 9 + [38] + 30 = 77$

36. $f(1, 1, 3, 1, 1, 3)$

$= f(1, 3) + f(1, 3, 1, 1) + (1 + 1 + 3 + 1 + 1 + 3)$

$= f(3, 0) + f(0, 1) + [f(1, 1) + f(3, 1) + (1 + 3 + 1 + 1)] + 10$

$= 7 + 0 + [f(1, 0) + f(0, 1) + f(1, 0) + f(0, 3) + 6] + 10$

$= 17 + [2 + 0 + 2 + 1 + 6]$

$= 17 + 11 = 28$

37. $f(9, 2, k, 0, 9, 4)$

$= f(9, 4) + f(2, k, 0, 9) + (9 + 2 + k + 0 + 9 + 4)$

$= f(4, 0) + f(0, 9) + [f(2, 9) + f(k, 0)$

$\quad + (2 + k + 0 + 9)] + 24 + k$

$= 11 + 4 + [\{f(9, 0) + f(0, 2)\} + f(k, 0)$

$\quad + (11 + k)] + 24 + k$

$= 15 + [48 + f(k, 0) + (11 + k)] + 24 + k$

$= 98 + 2k + f(k, 0)$

$$= 98 + 2k + \left[\frac{k(k+1)}{2} + 1\right] = 124$$

$$\Rightarrow \quad k^2 + 5k - 50 = 0$$

$$\Rightarrow \quad k = -10 \quad \text{or} \quad k = 5$$

Since k is a positive integer, hence $k = 5$

38. $f(128) = 1 \cdot 2^2 + 2 \cdot 2^1 + 8 \cdot 2^0$

$= 4 + 4 + 8 = 16$

$\therefore$ $f(16) = 1 \cdot 2^1 + 6 \cdot 2^0$

$= 2 + 6 = 8$

39. $f(888222) = 8 \cdot 2^5 + 8 \cdot 2^4 + 8 \cdot 2^3 + 2 \cdot 2^2 + 2 \cdot 2^1 + 2 \cdot 2^0$

$= 2^8 + 2^7 + 2^6 + 2^3 + 2^2 + 2^1$

$= 2^6(7) + 14$

$= 448 + 14 = 462$

$f(462) = 4 \cdot 2^2 + 6 \cdot 2^1 + 2 \cdot 2^0 = 30$

$f(30) = 3 \cdot 2^1 + 0 \cdot 2^0 = 6$

Again $f(113113) = 1 \cdot 2^5 + 1 \cdot 2^4 + 3 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 3 \cdot 2^0$

$= 32 + 16 + 24 + 4 + 2 + 3 = 81$

$f(81) = 8 \cdot 2^1 + 1 \cdot 2^0$

$= 16 + 1 = 17$

$f(17) = 1 \cdot 2^1 + 7 \cdot 2^0$

$= 2 + 7 = 9$

$\therefore$ $f[f(888222) + f(113113)] = f(6 + 9]$

$= f(15) = 1 \cdot 2^1 + 5 \cdot 2^0$

$= 2 + 5 = 7$

40. $f(9235) = 9 \cdot 2^3 + 2 \cdot 2^2 + 3 \cdot 2^1 + 5 \cdot 2^0$

$= 72 + 8 + 6 + 5 = 91$

$f(91) = 9 \cdot 2^1 + 1 \cdot 2^0 = 19$

$f(19) = 1 \cdot 2^1 + 9 \cdot 2^0 = 11$

$f(11) = 1 \cdot 2^1 + 1 \cdot 2^0 = 3$

$f(9450) = 9 \cdot 2^3 + 4 \cdot 2^2 + 5 \cdot 2^1 + 0 \cdot 2^0$

$= 72 + 16 + 10 = 98$

$f(98) = 9 \cdot 2^1 + 8 \cdot 2^0 = 26$

$f(26) = 2 \cdot 2^1 + 6 \cdot 2^0 = 10$

$f(10) = 1 \cdot 2^1 + 0 \cdot 2^0 = 2$

$\therefore$ $f(9235) + f(9450) = 3 + 2 = 5$



For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

18 SEQUENCE, SERIES & PROGRESSIONS

This is one of the chapters which contributes really very logical problems which can be solved easily by some innovative methods rather than conventional methods (like applying formulae etc.) Basically CAT is an exam where logic prevails over pure mathematics. Therefore the students who use their ingenuity to solve these problems score better than those who rely on the formula based approach. In the starting of the chapter I have given the proper mathematical methods, then later on I have gradually moved towards novel techniques for faster and smarter calculation. Remember that almost 1–2 questions are necessarily asked in CAT and this chapter is also important for rest of the entrance tests eg. XAT, FMS, IIFT, CET, MAT etc.

Once again I would like to advise you that maximum problems must be attempted with the help of given choices. Besides most of the problems can be solved by substituting the values of variables *e.g.*, n, x, etc. for nth term or sum of the series/progression. The above method is so smart that it reduces the time of problem solving upto 80% for some complex and typical problems.

Sequence : A set of numbers occuring in a definite order or by a rule is called a sequence. Thus any set of numbers $a_1, a_2, a_3, \ldots a_n$ such that to each positive integer n, there corresponds a number a_n, is sequence. The numbers $a_1, a_2, a_3, \ldots, a_n \ldots$ are called elements (or members) of the sequence.

e.g., $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \ldots, \frac{1}{n} \ldots$ is a sequence, since the number occur in a definite order. A sequence can be finite or infinite according to the number of elements in it. If there are finite number of elements, then it is a finite sequence and if there are infinite number of elements in it, then it is an infinite sequence and is denoted by $\{a_n\}$.

Series : Let $u_1, u_2, u_3, \ldots, u_n$ be a given sequence of n numbers, where n is a specified positive integer. Then their sum *i.e.* $u_1 + u_2 + u_3 + \ldots + u_n$ is called a finite series and the numbers $u_1, u_2, u_3 \ldots u_n$ are called terms of the series.

If $S_n = u_1 + u_2 + u_3 + \ldots + u_n$, then S_n is called the sum to n terms (or the sum of the first n terms) of the series and is denoted by

$$\sum_{r=1}^{n} u_r \text{ or } \Sigma u_r$$

e.g., $2 + 4 + 6 + \ldots + 30$ is a finite series and can be denoted as $\sum_{n=1}^{15} (2n)$ ($\Sigma \rightarrow$ Sigma, summation notation)

Progressions : Sequences of special types are called progressions. The sequences $a_1, a_2, \ldots, a_n \ldots$ or $\{a_n\}$ is known when its nth term a_n is given.

For example :

(i) If $a_n = 2n$, then the sequence is 2, 4, 6, 8, ...

(ii) If $a_n = 2n - 1$, then the sequence is 1, 3, 5, 7, ...

But to define a sequence we may not always have a formula for the nth term. For example, the sequence of the prime numbers is 2, 3, 5, 7, 11, 13, ... for which a formula for the nth term may not be given.

In this chapter we study basically Arithmetic Progression (A.P.) and Geometric Progression (G.P.) and get a overview of Harmonic Progressions (H.P.) also.

ARITHMETIC PROGRESSION (A.P.)

A sequence $a_1, a_2, a_3, \ldots, a_n$ is said to be in arithmetic progression, when $a_2 - a_1 = a_3 - a_2 = \ldots$ *i.e.*, when the quantities increase or decrease continuously by a common quantity and this constant (or common) quantity is called the common difference of the A.P. *e.g.*,

Progressions	Common Difference
1, 3, 5, 7, 9, ...	$\rightarrow 2$
12, 9, 6, 3, 0, –3, – 6, ...	$\rightarrow -3$
7, 12, 17, 22, ...	$\rightarrow 5$
– 6, – 4, – 2, 0, 2, 4, 6, ...	$\rightarrow 2$

GENERAL TERM (OR n^{th} TERM) OF AN A.P.

Let us look at the following :

1, 4, 7, 10, 13, ...

Each next term in the above sequence is being increased by 3 (which is the common difference). So you can see that from the second term onwards 3 is being added to every next term *i.e.*,

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$1 = 1$

$4 = 1 + 3$

$7 = 1 + 3 + 3$

$10 = 1 + 3 + 3 + 3$

$13 = 1 + 3 + 3 + 3 + 3$ etc.

Let if a be the first term of the progression and d be the common difference, then the

First term $\longrightarrow a$

Second term $\longrightarrow a + d$

Third term $\longrightarrow a + 2d$

Fourth term $\longrightarrow a + 3d$

Fifth term $\longrightarrow a + 4d$

And nth term $\longrightarrow a + (n-1)d$

The general term or nth term is denoted by T_n.

Hence $T_n = a + (n-1)d$

also $l = a + (n-1)d$

l is the last term of A.P.

EXAMPLE 1 *Find the 11th term of $-7, -2, 3, 8, 13, \ldots$*

SOLUTION $T_{11} = -7 + (11-1)5 = 43$

$(\because\ a = -7,\ d = 5$ and $n = 11)$

Sum of First n terms of an A.P.

Let us look at the following :

$1, 4, 7, 10, 13, \ldots$

Let the sum of 'n' terms is denoted by S_n, then

$S_1 = 1$

$S_2 = 1 + 4 = 5$

$S_3 = 1 + 4 + 7 = 12$

$S_4 = 1 + 4 + 7 + 10 = 22$

$S_5 = 1 + 4 + 7 + 10 + 13 = 35$

$\therefore\ S_n = T_1 + T_2 + T_3 + \ldots + T_n$

$= (a) + (a + d) + (a + 2d) + \ldots + (a + (n-1)d)$

$= na + \dfrac{n(n-1)d}{2}$

$S_n = \dfrac{n}{2}[2a + (n-1)d]$

Alternatively :

$S_n = \left(\dfrac{a + l}{2}\right)n$, ($l \rightarrow$ last term of the progression)

NOTE In my own experience the alternative formula of S_n is better for problem solving since in this formula we have to just take the product of average of first and last term with number of terms. Also for our convenience to use this formula we can consider any term as last term.

EXAMPLE 2 *Find the sum of 11 terms of $-7, -2, 3, 8, \ldots$*

SOLUTION $a = -7$

$l = -7 + (10)\cdot 5 = 43$

$\therefore\ S_{11} = \dfrac{-7 + 43}{2} \times 11$

$S_{11} = 198$

NOTE Personally I hardly use the formula $S_n = \dfrac{n}{2}[2a + (n-1)d]$

Since I prefer $S_n = \left(\dfrac{a + l}{2}\right)n$.

ARITHMETIC MEAN (A.M.)

When three quantities are in A.P., the middle one is called arithmetic mean (A.M.) of the other two.

Let a, m, b are in A.P., then A.M. of a and b

i.e., $m = \dfrac{a + b}{2}$

$\therefore\ \text{A.M.} = \dfrac{a + b}{2}$

In general, if $a, m_1, m_2, m_3, \ldots, m_n, b$ are in A.P. then the intermediate numbers $m_1, m_2, m_3, \ldots, m_n$ are called the n arithmetic means between a and b

also the A.M. of $a_1, a_2, a_3, \ldots, a_n$, which are in A.P. is

$$\text{A.M.} = \frac{(a_1 + a_2 + a_3 + \ldots + a_n)}{n}$$

To Insert n Arithmetic Means Between a and b.

If a and b are the first and last term respectively between which we have to insert n arithmetic means $m_1, m_2, m_3, \ldots, m_n$, then common difference $d = \dfrac{b-a}{n+1}$

Thus $m_1 = a + \left(\dfrac{b-a}{n+1}\right)$

$m_2 = a + 2\left(\dfrac{b-a}{n+1}\right)$

$m_3 = a + 3\left(\dfrac{b-a}{n+1}\right)$

... ...

... ...

$m_n = a + n\left(\dfrac{b-a}{n+1}\right)$

EXAMPLE 3 *Insert 3 arithmetic means between 5 and 21.*

SOLUTION $d = \dfrac{21-5}{3+1} = 4$

$\therefore$ First A.M. $= a + d = 5 + 4 = 9$

Second A.M. $= a + 2d = 5 + 2(4) = 13$

Third A.M. $= a + 3d = 5 + 3(4) = 17$

Hence the new sequence is 5, 9, 13, 17, 21, where all the terms are in A.P.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

PROPERTIES OF ARITHMETIC PROGRESSIONS

1. If a certain number is added or subtracted to each term of a given A.P., then the resulting sequence is also an A.P. and it has the same common difference as that of the given A.P.

   e.g., 1, 4, 7, 10, 13, 16, ... then
   3, 6, 9, 12, 15, 18, ... is also an A.P.
   and 2, 5, 8, 11, 14, 17, ... is also an A.P.
   In the second sequence each term is increased by 2 than that of the first sequence and each term of the third sequence is decreased by 1 than that of second sequence.

2. If each term of an A.P. is multiplied by a fixed constant (or divided by a non-zero fixed constant), then the resulting sequence is also an A.P.

   e.g., 6, 12, 18, 24, 30, 36, ..., then
   9, 18, 27, 36, 45, 54, ... is also an A.P.
   and 3, 6, 9, 12, 15, 18, ... is also an A.P.
   In the second sequence each term is $\frac{3}{2}$ times that of the first sequence and in the third sequence each term is half of the each term of the first sequence.

3. If $m_1, m_2, m_3, \ldots, m_k$ and $n_1, n_2, n_3, \ldots, n_k$ are in A.P., then $(m_1 + n_1), (m_2 + n_2), (m_3 + n_3), \ldots, (m_k + n_k)$ are also in A.P.

   e.g., $S_1 = 3, 7, 11, 15, 19, \ldots$
   and $S_2 = 5, 11, 17, 23, 29, \ldots$
   then $S_1 + S_2 = (3+5), (7+11), (11+17), (15+23), (19+29) \ldots$
   $\Rightarrow S_1 + S_2 = 8, 18, 28, 38, 48, \ldots$ is also an A.P.

4. If $m_1, m_2, m_3, \ldots, m_k$ and $n_1, n_2, n_3, \ldots, n_k$ are in A.P., then $(m_1 - n_1), (m_2 - n_2), (m_3 - n_3), \ldots, (m_k - n_k)$ are also in A.P.

   e.g., $S_1 = 7, 12, 17, 22, 27, \ldots$
   $S_2 = 4, 13, 22, 31, 40, \ldots$
   then $S_1 - S_2 = (7-4), (12-13), (17-22), (22-31), (27-40) \ldots$
   $\Rightarrow S_1 - S_2 = 3, -1, -5, -9, -13, \ldots$ is also an A.P.

5. If there are odd number of terms in an A.P. as $a_1, a_2, a_3, \ldots, a_n$ then the middle most term is the A.M. of the given sequence.

   e.g., 4, 10, 16, 22, 28

   then, $\frac{4+10+16+22+28}{5} = 16$

   Also $\frac{4+28}{2} = \frac{10+22}{2} = 16$ (A.M.)

   (It is the most important point for logical problems.)

6. If there are even number of terms in an A.P. as $a_1, a_2, a_3, \ldots, a_n$ then
   $a_1 + a_n = a_2 + a_{n-1} = a_3 + a_{n-2} = a_4 + a_{n-3}$ etc.

   e.g., 4, 10, 16, 22, 28, 34, 40, 46
   then, $4 + 46 = 10 + 40 = 16 + 34 = 22 + 28$
   also $\frac{4+46}{2} = \frac{10+40}{2} = \frac{16+34}{2} = \frac{22+28}{2} = 25$ (A.M.)

7. The sum of an A.P. with n terms is given by
   $S_n = n$ (A.M.)

   e.g. (1) 3, 7, 11, 15, 19, then
   $S_5 = 5 \times 11 = 55$

   (2) 5, 8, 11, 14, 17, 20
   $\therefore \quad S_6 = 6 \times \frac{25}{2} = 75$

8. If we have to take three terms in A.P. it is convenient to take them as $(m-k), m, (m+k)$ where $(m-k), m$ and $(m+k)$ are consecutive terms of an A.P. and k is the common difference.
   Similarly we can take five terms in A.P. as $(m-2k), (m-k), m, (m+k), (m+2k)$.

EXAMPLE 4 *If the sum of five terms of an A.P. is 75, then find the third term of the A.P.*

SOLUTION $(m-2k) + (m-k) + m + (m+k) + (m+2k) = 75$
$\Rightarrow \quad 5m = 75 \Rightarrow m = 15$
Hence the third term (*i.e.* middle most term) is 15.

NOTE In general to take odd number of terms, we assume m as the middlemost term and take other terms with common difference k, as per their position.

9. If we have to take 4 terms in A.P., it is convenient to take them as $(m-3k), (m-k), (m+k), (m+3k)$ where $(m-3k), (m-k), (m+k), (m+3k)$ are consecutive terms of an A.P. and '$2k$' is the common difference.
   Similarly we can take six terms in A.P. as $(m-5k), (m-3k), (m-k), (m+k), (m+3k), (m+5k)$.

EXAMPLE 5 *If the sum of 4 terms of an A.P. is 36, then find the A.M. of the A.P.*

SOLUTION $(m-3k) + (m-k) + (m+k) + (m+3k) = 36$
$\Rightarrow \quad 4m = 36$
$\therefore \quad S_4 = 4m = 36$
$\therefore \quad \text{A.M.} = \frac{S_4}{4} = \frac{4m}{4} = \frac{36}{4} = 9$

10. The sum of first n natural numbers
    $$\Sigma n = 1 + 2 + 3 + \ldots + n = \frac{n(n+1)}{2}$$

EXAMPLE 6 *Find the sum of first 10 natural numbers.*

SOLUTION $\Sigma 10 = 1 + 2 + 3 + \ldots + 10$
$= \frac{10 \times 11}{2} = 55$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yourSmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

11. The sum of squares of first 'n' natural numbers

$$\Sigma n^2 = 1^2 + 2^2 + 3^2 + \ldots + n^2$$
$$= \frac{n(n+1)(2n+1)}{6}$$

EXAMPLE 7 *Find the sum of squares of first 11 natural numbers.*

SOLUTION $\Sigma n^2 = 1^2 + 2^2 + 3^2 + \ldots + 11^2$

$$= \frac{11 \times 12 \times 23}{6} = 506$$

12. The sum of cubes of first 'n' natural numbers

$$\Sigma n^3 = 1^3 + 2^3 + 3^3 + \ldots + n^3$$
$$= \left[\frac{n(n+1)}{2}\right]^2 = [\Sigma n]^2$$

EXAMPLE 8 *Find the sum of cubes of first 10 natural numbers.*

SOLUTION $\Sigma(10)^3 = 1^3 + 2^3 + 3^3 + \ldots + 10^3$

$$= \left(\frac{10 \times 11}{2}\right)^2 = 3025$$

13. The sum of first 'n' odd numbers

i.e., $1 + 3 + 5 + 7 + \ldots (2n-1) = n^2$

EXAMPLE 9 *Find the sum of* $1 + 3 + 5 + 7 + \ldots 11$

SOLUTION $1 + 3 + 5 + 7 + \ldots 11 = (11)^2 = 121$

14. The sum of first n even numbers

i.e., $2 + 4 + 6 + \ldots 2n = n(n+1)$

EXAMPLE 10 *Find the sum of first 10 even numbers.*

SOLUTION $2 + 4 + 6 + 8 \ldots 20 = 10 \times 11 = 110$

INTRODUCTORY EXERCISE-18.1

1. Find the 41st term of the progression 3, 8, 13, 18, ... :
(a) 102 (b) 203
(c) 304 (d) none of these

2. Find the 25th term of the A.P. 10, 6, 2, –2, –6, –10, ... :
(a) –86 (b) 106
(c) 96 (d) none of these

3. What term of the A.P. 2, 5, 8, ... is 56 ?
(a) 20 (b) 21
(c) 19 (d) 15

4. What is the 10th term of the sequence 2, 4, ... ?
(a) 18 (b) 20
(c) 1024 (d) can't be determined

5. If the 3rd and 7th terms of an A.P. are 17 and 27 respectively. Find the first term of the A.P. :
(a) 9 (b) 12
(c) 14 (d) none of these

6. Find the nth term of an A.P. whose 6th and 8th terms are 12 and 22 respectively :
(a) $7n - 30$ (b) $n^2 - 24$
(c) $5n - 18$ (d) none of these

7. If 7 times the seventh term of an A.P. is equal to 11 times its eleventh term, the value of eighteenth term of the A.P. is :
(a) 0 (b) –8
(c) 18 (d) 77

8. If the pth, qth and rth terms of an A.P. are a, b, c respectively, then the value of
$a(q-r) + b(r-p) + c(p-q)$ is :
(a) 0 (b) 1
(c) abc (d) pqr

9. If the pth term of an A.P. is q and its qth term is p then its mth term is :
(a) $\frac{pq}{m}$ (b) $pq - m$
(c) $p + q - m$ (d) pqm

10. Find the sum of the A.P. 11, 13, 15, ..., 99 :
(a) 2475 (b) 2500
(c) 1122 (d) 1580

11. Find the number of terms in the A.P. 22, 28, 34, ..., 616 :
(a) 80 (b) 78
(c) 99 (d) 100

12. Find the sum of 222, 224, 226, ..., 888 :
(a) 185370 (b) 195300
(c) 183000 (d) 899000

13. If the second and seventh terms of an A.P. are 2 and 22 respectively. Find the sum of first 35 terms :
(a) 2310 (b) 3210
(c) 2130 (d) none of these

14. The 12th term of an A.P. is – 13 and the sum of the first four terms of it is 24. Find the sum of its first ten terms :
(a) – 48 (b) – 26
(c) 0 (d) 52

15. The third term of an A.P. is $\frac{1}{5}$ and the 5th term is $\frac{1}{3}$. Find the sum to 15 terms of the A.P. :
(a) 1/15 (b) 3/5
(c) 8 (d) 15

16. How many terms of the A.P. 1, 4, 7, ... are needed to give the sum 925 ?
(a) 20 (b) 22
(c) 24 (d) 25

17. How many terms of the series $20 + 16 + 12$ amounts to 48 ?
(a) 3 (b) 5
(c) 8 (d) both a and c

18. p, q, r, s, t are first five terms of an A.P. such that $p + r + t = -12$ and $p \cdot q \cdot r = 8$. Find the first term of the above A.P. :
(a) 3 (b) 2
(c) 4 (d) – 4

19. The sum of all the terms of the A.P. 7, 10, 13, ... l is 1242, where l is the last term of the A.P. Find the value of l.
(a) 67 (b) 79
(c) 85 (d) 102

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

20. Find the sum of all the integers between 55 and 5555 which are divisible by 7 :
(a) 678 (b) 786
(c) 876 (d) none of these

21. How many terms are there in A.P. whose first and fifth terms are – 14 and 2 respectively and the sum is 40 :
(a) 12 (b) 10
(c) 16 (d) 8

22. The first and last terms of an A.P. are – 7 and 233 and the sum of the A.P. is 9153. Find the number of terms in the A.P. :
(a) 83 (b) 81
(c) 49 (d) 99

23. The sum of three numbers in A.P. is 12 and sum of their cubes is 408. Find the product of the numbers :
(a) 17 (b) 35
(c) 28 (d) 36

24. The series of natural numbers is written as follows :

1
2 3 4
5 6 7 8 9
... ...
... ...

Find the sum of the numbers in the nth row :
(a) $n^3 + 1$ (b) $(2n)^2 - 1$
(c) $n^3 + (n-1)^3$ (d) none of these

25. If you save Re. 1 today, Rs. 2 the next day, Rs. 3 the succeeding day and so on, what will be your total savings in 365 days ?
(a) 66579 (b) 66795
(c) 56795 (d) none of these

26. The ratio of the 7th to the 3rd terms of an A.P. is 12 : 5, find the ratio of the 13th to 4th term :
(a) 13 : 7 (b) 4 : 3
(c) 10 : 3 (d) none of these

27. Find the sum of the first hundred even natural numbers divisible by 5 :
(a) 50500 (b) 55000
(c) 50050 (d) 50005

28. If m times the mth term of an A.P. is equal to n times its nth term, find $(m + n)$ th term of the A.P. :
(a) – 1 (b) 0
(c) mn (d) 4

29. The sum of the first fifteen terms of an A.P. is 105 and the sum of the next fifteen terms is 780. Find the common difference of A.P. :
(a) 4 (b) 3
(c) 6 (d) 5

30. If the first term of an A.P. is 2 and the sum of the first five terms is equal to one-fourth of the sum of the next five terms, then find the 20th term ?
(a) – 114 (b) – 82
(c) – 112 (d) none of these

31. The sum of the first six terms of an A.P. is 42. The ratio of the 10th term to the 30th term of A.P. is $\frac{1}{3}$. Find the 40th term of the A.P. :
(a) – 60 (b) 20
(c) 39 (d) 80

32. The sums of n terms of two arithmetic series are in the ratio of $\frac{7n+1}{4n+27}$. Find the ratio of their 11th terms :
(a) 4 : 3 (b) 5 : 4
(c) 7 : 4 (d) none of a, b, c

33. The sum of three numbes in A.P. is 15 and sum of their squares is 93. Find the greatest number :
(a) 6 (b) 10
(c) 8 (d) 7

34. If the nth term of an A.P. is $4n - 1$, find the 30th term and the sum of first 30 terms.
(a) 119 and 1830 (b) 69 and 1665
(c) 119 and 1380 (d) none of these

35. The sum of n terms of a series is $3n^2 + 5n$. Find the value of n if nth term is 152 :
(a) 15 (b) 18
(c) 20 (d) 25

36. Find the number of terms of the A.P. 98, 91, 84, ... must be taken to give a sum of zero :
(a) 14 (b) 15
(c) 31 (d) 29

37. What is the greatest possible sum of the A.P. 17, 14, 11, ... :
(a) 51 (b) 73
(c) 57 (d) 75

38. What is the least possible sum of the A.P. – 23, – 19, – 15, ... :
(a) – 3 (b) – 78
(c) – 87 (d) none of these

39. Find the sum of all odd numbers of four digits which are divisible by 9 :
(a) 2784491 (b) 2478429
(c) 2754000 (d) 2448729

40. If a, b, c be the pth, qth and rth terms of an A.P., then $p(b - c) + q(c - a) + r(a - b)$ equals to :
(a) 0 (b) 1
(c) 3 (d) $\frac{abc}{pqr}$

41. If a, b, c be respectively the sum of first p, q, r terms of an A.P. then $\frac{a}{p}(q - r) + \frac{b}{q}(r - p) + \frac{c}{r}(p - q)$ equals :
(a) pqr (b) $ap + bq + cr$
(c) 0 (d) $p + q + r$

42. Divide 20 into four parts which are in A.P. and such that the product of the first and fourth is to the product of the second and third in the ratio 2 : 3. Find the product of the first and fourth term of the A.P. :
(a) 12 (b) 16
(c) 20 (d) 25

43. The sum of the first p terms of an A.P. is q and the sum of the first q terms is p. Find the sum of the first $(p + q)$ terms :
(a) pq (b) $p - q$
(c) $-(p + q)$ (d) 0

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

GEOMETRIC PROGRESSION (G.P.)

Geometric Progression : A sequence $a_1, a_2, a_3, \ldots, a_n$ is said to be in Geometric Progression when

$$\frac{a_2}{a_1} = \frac{a_3}{a_2} = \frac{a_4}{a_3} = \ldots r \text{ (say)}$$

and $a_1, a_2, a_3, \ldots, a_n$ are all non-zero numbers and r is said to be the common ratio of the G.P.

e.g., (i) $2, 4, 8, 16, \ldots \rightarrow r = 2$

(ii) $1, 3, 9, 27, 81 \ldots \rightarrow r = 3$

(iii) $-3, 6, -12, 24, -48, 96 \ldots \rightarrow r = -2$

(iv) $\frac{1}{2}, \frac{3}{2}, \frac{9}{2}, \frac{27}{2} \ldots \rightarrow = r = 3$

(v) $k, kr, kr^2, kr^3, kr^4, \ldots \rightarrow r = r$

(vi) $64, 16, 4, 1, \frac{1}{4}, \frac{1}{16}, \ldots \rightarrow r = \frac{1}{4}$

GENERAL TERM (Or nth TERM) OF a G.P.

Let us look at the following :

$$1, 3, 9, 27, 81, 243, 729, 2187, \ldots$$

Each next term in the above sequence is becoming th (*i.e.* 3 times) which is called as common ratio.

$$1 = 1$$
$$3 = 1 \times 3$$
$$9 = 1 \times 3 \times 3$$
$$27 = 1 \times 3 \times 3 \times 3$$
$$81 = 1 \times 3 \times 3 \times 3 \times 3$$
$$243 = 1 \times 3 \times 3 \times 3 \times 3 \times 3$$
$$729 = 1 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3$$

and nth term $= a \times r \times r \times r \times \ldots \times (n-1)$ times

$$T_n = a \cdot r^{(n-1)}$$

$\therefore$ The general term or nth term is represented by T_n

Hence $T_n = a \cdot r^{(n-1)}$

also $l = a \cdot r^{(n-1)}$

where l is the last term of the G.P.

EXAMPLE 1 *Find the 7th term of the sequence 2, 10, 50, 250, ...*

SOLUTION The given sequence is a G.P. with common ratio = 5

$$\left(\because \frac{10}{2} = \frac{50}{10} = 5\right)$$

and first term $(a) = 2$

$\therefore$ $T_7 = 2 \cdot (5)^6$ $\quad (\because T_n = a \cdot r^{n-1})$

$T_7 = 31250$

Sum of First n terms of a G.P.

Let us look at the following :

$$3, 6, 12, 24, 48, 96, \ldots$$

Let the sum of n terms is denoted by S_n, then

$S_1 = 3$

$S_2 = 3 + 6 = 9$

$S_3 = 3 + 6 + 12 = 21$

$S_4 = 3 + 6 + 12 + 24 = 45$

$S_5 = 3 + 6 + 12 + 24 + 48 = 93$

$\therefore$ $S_n = T_1 + T_2 + T_3 + \ldots, T_n$

$\therefore$ $S_n = \frac{a \cdot (r^n - 1)}{(r-1)}$ if $r > 1$

and $S_n = \frac{a \cdot (1 - r^n)}{(1-r)}$ if $r < 1$

also, $S_n = \frac{lr - a}{r - 1}$

NOTE The above formulae for S_n fails when $r = 1$, as the form $\frac{0}{0}$ is undefined. However,

$S_n = a + a + a + \ldots$ to n terms $= n \cdot a$

EXAMPLE 2 *Find the sum of the first 6 terms of the G.P.*

$$3, \frac{9}{2}, \frac{27}{4}, \frac{81}{8}, \ldots$$

SOLUTION Here $a = 3$, $r = \frac{3}{2}$ and $n = 6$, then

$$S_6 = \frac{3 \cdot \left[\left(\frac{3}{2}\right)^6 - 1\right]}{\left(\frac{3}{2} - 1\right)}$$

$$= \frac{3 \cdot \left(\frac{729}{64} - 1\right)}{\frac{1}{2}} = \frac{3 \cdot \left(\frac{665}{64}\right)}{\frac{1}{2}}$$

$$= \frac{3 \times 665 \times 2}{64 \times 1} = 62.34375$$

EXAMPLE 3 *Find the sum of first five terms of the G.P. 1, 2, 4, ...*

SOLUTION $S_5 = \frac{1 \cdot (2^5 - 1)}{(2-1)} = 31$

GEOMETRIC MEAN (G.M.)

If three quantities are in G.P., the middle one is called the geometric mean (G.M.) of the other two quantities.

If a, m, b are in G.P. then m is the G.M. of a and b

and since $\frac{m}{a} = \frac{b}{m}$

$\therefore$ $m^2 = ab$

$\Rightarrow$ $m = \pm\sqrt{ab} = \pm (ab)^{1/2}$

In general, if $a_1, a_2, a_3, \ldots, a_n$ are in G.P. then

$$\text{G.M.} = (a_1 \cdot a_2 \cdot a_3 \ldots . a_n)^{1/n}$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yourmathbook.wordpress.com
For More Visit : www.LearnEngineering.in

EXAMPLE 4 *Find the G.M. of 4, 12, 36, which are in G.P.*

SOLUTION G.M. of 4, 12, 36 $= (4 \times 12 \times 36)^{1/3}$

$= (1728)^{1/3} = (12^3)^{1/3} = 12$

EXAMPLE 5 *Find the G.M. of 3, 6, 12, 24, 48.*

SOLUTION G.M. $= (3 \times 6 \times 12 \times 24 \times 48)^{1/5}$

$= (12^5)^{1/5} = 12$

To Insert n Geometric Means between a and b

If a and b are the first and last term respectively between which we have to insert n geometric means $m_1, m_2, m_3, \ldots, m_n$ then

Common ratio, $r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}}$

Thus, $m_1 = a \cdot \left(\frac{b}{a}\right)^{\frac{1}{n+1}}$

$m_2 = a \cdot \left(\frac{b}{a}\right)^{\frac{2}{n+1}}$

$m_3 = a \cdot \left(\frac{b}{a}\right)^{\frac{3}{n+1}}$

... ...

... ...

$m_n = a \cdot \left(\frac{b}{a}\right)^{\frac{n}{n+1}}$

EXAMPLE 6 *Insert 3 geometric means between 3 and 48.*

SOLUTION $r = \left(\frac{48}{3}\right)^{\frac{1}{3+1}} = (2^4)^{\frac{1}{4}} = 2$

$\therefore$ $m_1 = 3 \cdot 2 = 6$

$m_2 = 3 \cdot (2)^2 = 12$

$m_3 = 3 \cdot (2)^3 = 24$

Hence the new sequence is 3, 6, 12, 24, 48 where all the terms are in G.P.

PROPERTIES OF GEOMETRIC PROGRESSION

1. If each term of a G.P. is multiplied or divided by a certain non-zero constant, then the resulting sequence is also a G.P.
   e.g., 3, 6, 12, 24, 48, 96,... then 9, 18, 36, 72, 144, 288, ... is also a G.P. and $\frac{9}{2}$, 9, 18, 36, 72, 144, ... is also a G.P.
   In the second sequence each term is multiplied by 3 and each term of the third sequence is divided by 2 (compare with the second sequence)
2. If $a_1, a_2, a_3, \ldots$ and $b_1, b_2, b_3, \ldots$ are two geometric progressions, then the sequence $a_1b_1, a_2b_2, a_3b_3, \ldots$ is also a G.P.
   e.g., 2, 6, 18, 54, 162, ... and 1, 2, 4, 8, 16, ... then $(2 \times 1), (6 \times 2), (18 \times 4), (54 \times 8), (162 \times 16) \ldots$
   i.e. 2, 12, 72, 432, 2592 is also a G.P.
   (Similarly $\frac{2}{1}, \frac{6}{2}, \frac{18}{4}, \frac{54}{8}, \frac{162}{16}, \ldots$
   i.e., $2, 3, \frac{9}{2}, \frac{27}{4}, \frac{81}{8}, \ldots$ is also a G.P.)
3. If there are odd number of terms in a G.P. as $a_1, a_2, a_3, \ldots, a_n$ then the middle most term is the G.M. of the given sequence.
   e.g., 2, 4, 8, 16, 32
   then $(2 \times 4 \times 8 \times 16 \times 32)^{1/5} = (2^{15})^{1/5} = 2^3 = 8$
   Also $(2 \times 32)^{1/2} = (4 \times 16)^{1/2} = 8$

NOTE In this case without knowing all the terms we can find the G.M. of the G.P.

4. If there are even number of terms in a G.P. as $a_1, a_2, a_3, \ldots, a_n$, then

$\sqrt{a_1 \cdot a_n} = \sqrt{a_2 \cdot a_{n-2}} = \sqrt{a_3 \cdot a_{n-3}} = \ldots$ G.M.

5. If we have to take three terms in G.P., it is convenient to take them as

   $\frac{m}{k}, m, mk$

   where $\frac{m}{k}$, m and mk are consecutive terms of G.P. and k is the common ratio.
   Similarly we can take five terms of G.P. as $\frac{m}{k^2}, \frac{m}{k}, m, mk, mk^2$.

EXAMPLE 7 *If the product of five consecutive terms of G.P. is 1024, then find the third term of this set.*

SOLUTION $\frac{m}{k^2} \times \frac{m}{k} \times m \times mk \times mk^2 = 1024$

$\Rightarrow$ $m^5 = 1024 = 2^{10} = 4^5$

$\Rightarrow$ $m = 4$

Hence the third term (*i. e.* middle most term) is 4.

In general to take odd number of terms, we assume m as the middle most term and take other terms with common ratio k, as per their position.

6. If we have to take 4 terms in G.P. it is convenient to take them as $\frac{m}{k^3}, \frac{m}{k}, mk, mk^3$ where $\frac{m}{k^3}, \frac{m}{k}, mk$ and mk^3 are four consecutive terms of a G.P. and k^2 is the common ratio.
   Similarly we can take six terms in G.P. as

   $\frac{m}{k^5}, \frac{m}{k^3}, \frac{m}{k}, mk, mk^3, mk^5$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

EXAMPLE 8 *If the product of 4 terms of a G.P. is 729, then find the G.M. of the G.P.*

SOLUTION

$$\frac{m}{k^3} \times \frac{m}{k} \times mk \times mk^3 = 729$$

$$m^4 = 729 = 3^6$$

$$\Rightarrow \quad m = 3^{6/4} = 3^{3/2}$$

$$\Rightarrow \quad m = \sqrt{27}$$

$$\Rightarrow$$

7. If $a_1, a_2, a_3, \ldots, a_n$ are in G.P. then $a_1^m, a_2^m, a_3^m, \ldots, a_n^m$ are also in G.P.
   e.g. 1, 2, 4, 8, 16, ... then 1, 4, 16, 64, 256, ... are also in G.P.
8. If $a_1, a_2, a_3, \ldots, a_n$ are in G.P. (for every $a_i > 0$; $i \in I^+$ then $\log a_1, \log a_2, \log a_3, \ldots$ are in A.P.
   (The converse of the above is also true)

Sum of an Infinite G.P.

$$S_\infty = \frac{a}{1-r}; \qquad |r| < 1$$

where a is the first term and r is the common ratio of the G.P. of infinite terms.

EXAMPLE 9 *Find the sum of an infinite G.P. whose first term is 28 and the second term is 4.*

SOLUTION $a = 28, \quad T_2 = ar = 4$

$$\Rightarrow \quad r = \frac{1}{7}$$

$$\therefore \quad S_\infty = \frac{a}{1-r} = \frac{28}{1-\frac{1}{7}} = \frac{28}{6} \times 7 = \frac{98}{3}$$

INTRODUCTORY EXERCISE-18.2

1. Find the 22nd term of the G.P. $-2, 2, -2, \ldots$:
   (a) -22 (b) -2
   (c) 2 (d) none of these
2. Find the 7th term of the series $-\frac{1}{8} + \frac{1}{4} - \frac{1}{2} + 1 \ldots$:
   (a) 8 (b) -16
   (c) -8 (d) none of these
3. The 5th, 8th and 11th terms of a G.P. are a, b, c respectively, then which one of the following is true ?
   (a) $2b = ac$ (b) $b^2 = ac$
   (c) $a + b + c = 0$ (d) none of these
4. The 5th and 12th terms of a G.P. are 32 and 4096 respectively. Find the nth term of the G.P. :
   (a) 2^n (b) n^2
   (c) $2n^2$ (d) none of these
5. What is the least number of terms of the G.P. $5 + 10 + 20 + \ldots$ whose sum would surely exceed 10^6?
   (a) 17 (b) 18
   (c) 19 (d) 21
6. The A.M. of two positive numbers is 15 and their G.M. is 12. What is the smaller number ?
   (a) 8 (b) 12
   (c) 6 (d) 24
7. The sum of 3 numbers in G.P. is 38 and their product is 1728. Find the greatest number :
   (a) 24 (b) 18
   (c) 16 (d) 8
8. The sum of first three terms of a G.P. is to the sum of the first six terms is 125 : 152. Find the common ratio of the G.P. :
   (a) 3/5 (b) 3
   (c) 2/5 (d) 5/8
9. The sum of three numbers in G.P. is 14. If the first two terms are each increased by 1 and the third term is decreased by 1, the resulting numbers are in A.P. Find the product of these three numbers :
   (a) 125 (b) 64
   (c) 216 (d) 124
10. The third term of a G.P. is 4. The product of first five terms is :
    (a) 4^3 (b) 4^4
    (c) 4^5 (d) none of these
11. The sum of first three terms of a G.P. is 21 and sum of their squares is 189. Find the common ratio :
    (a) $\frac{1}{2}$ or 2 (b) 3 or $\frac{1}{3}$
    (c) 4 or $\frac{1}{4}$ (d) none of these
12. The sum of the first and the third term of a G.P. is 15 and that of the 5th and the 7th terms is 240. Find the 9th term :
    (a) 678 (b) 786
    (c) 867 (d) 768
13. Sum of three consecutive terms in a G.P. is 42 and their product is 512. Find the largest of these numbers :
    (a) 28 (b) 16
    (c) 32 (d) none of these
14. The sum of three numbers in G.P. is 70, if the two extremes be multiplied each by 4 and the mean by 5, the new numbers so formed are in A.P. Find the product of original numbers :
    (a) 8000 (b) 6000
    (c) 7000 (d) none of these
15. The sum of four terms in G.P. is 312. The sum of first and fourth term is 252. Find the product of second and third term :
    (a) 500 (b) 150
    (c) 60 (d) none of these
16. A bouncing tennis ball rebounds each time to a height equal to one half the height of the previous bounce. If it is dropped from a height of 16 m, find the total distance it has travelled when it hits the ground for the 10th time :

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

(a) $47\frac{15}{16}$ (b) $37\frac{5}{16}$
(c) $67\frac{11}{16}$ (d) none of these

17. Find the sum to n terms of the series
$$7 + 77 + 777 + 7777 + \ldots$$
(a) $\frac{7}{9}\{10(10^n - 1) - n\}$ (b) $\frac{7}{9}\left\{\frac{10}{9}(10^n - 1) - n\right\}$
(c) $7\left\{\frac{10}{9}(10^n - 1) - n\right\}$ (d) none of these

18. Find the sum to n terms of $0.8 + 0.88 + 0.888 + \ldots$
(a) $\frac{8}{9}\left[\frac{1}{9}\{1 - (0.1)^n\}\right]$ (b) $\frac{8}{9}\left[\frac{1}{10}\{1 - (0.1)^n\}\right]$
(c) $\frac{8}{9}\left[n - \frac{1}{9}\{1 - (0.1)^n\}\right]$ (d) none of these

19. If the pth, qth and rth terms of a G.P. be respectively a, b, and c then the value of $a^{q-r} \cdot b^{r-p} \cdot c^{p-q}$ is :
(a) 0 (b) 1
(c) 3 (d) $\frac{abc}{pqr}$

20. If a, b, c are respectively the xth, yth and zth terms of a G.P. then the value of
$(y - z)\log a + (z - x)\log b + (x - y)\log c$:
(a) 0 (b) 1
(c) 3 (d) none of these

21. There are four numbers such that the first three of them form an Arithmetic Progression and the last three form a Geometric Progression. The sum of the first and the third is 2 and that of second and fourth is 26. What is the sum of first and fourth number?
(a) $85\frac{13}{25}$ (b) $25\frac{73}{85}$
(c) 83 (d) none of these

22. Find the sum to n terms of the series $1 + 3 + 7 + 15 + \ldots$
(a) $[(2^n - 1) - n]$ (b) $[(2^n - 1) + n]$
(c) $[2(2^n - 1) - n]$ (d) none of these

23. Find the sum of $\frac{1}{2} + \frac{3}{2^2} + \frac{5}{2^3} + \ldots + \frac{2n-1}{2^n}$:
(a) $\left(\frac{2n-3}{2^n}\right)$ (b) $\left(3 - \frac{2n+3}{2^n}\right)$
(c) $\left(\frac{2n+3}{2^n}\right)$ (d) none of these

24. Find the sum to n terms of the series
$$11 + 102 + 1003 + 10004 + \ldots :$$
(a) $(10^n - 1) + \frac{n(n+1)}{2}$ (b) $\frac{10}{9}(10^n - 1) + \frac{n(n+1)}{2}$
(c) $10^n + n^2 - 1$ (d) none of these

25. Find the sum to first n groups of
$(1) + (1 + 3) + (1 + 3 + 9) + (1 + 3 + 9 + 27) + \ldots$
(a) $\frac{1}{2}(3^n - 1)$ (b) $\frac{3}{4}(3^n - 1) - \frac{n}{2}$
(c) $n^2 + 1$ (d) none of these

26. Find the sum to n terms of the following series :
$$2 + 5 + 14 + 41 + \ldots$$
(a) $\frac{3}{4}(3^{n-1}) + \frac{n}{2}$ (b) $\frac{3}{4}(3^n - 1) + \frac{n}{2}$
(c) $(3n + 1)$ (d) none of these

27. Find the sum to n terms :
$$1 + 2x + 3x^2 + 4x^3 + \ldots ; x \neq 1 :$$
(a) $1 + nx^n$ (b) $\frac{x(x+1)}{2} + x^n$
(c) $\frac{1 - x^n}{(1-x)^2}$ (d) $\left\{\frac{1 - x^n}{(1-x)^2} - \frac{nx^n}{(1-x)}\right\}$

28. Find the sum to n terms of $1 + 3x + 5x^2 + 7x^3 + \ldots$
(a) $2(1 - x^{n-1})$
(b) $\frac{2x(1 - x^{n-1})}{(1-x^2)}$
(c) $\frac{2x(1 - x^{n-1})}{(1-x)^2} + \frac{1 - (2n-1)x^n}{(1-x)}$
(d) none of these

29. Find the sum to first n terms :
$$1 + \frac{2}{3} + \frac{3}{3^2} + \frac{4}{3^3} + \ldots$$
(a) $\frac{3}{4} - \frac{3 + 2n}{4(3^{n-1})}$ (b) $\frac{9}{4} - \frac{3}{4}\left[\frac{3 + 2n}{3^n}\right]$
(c) $2n - \left(\frac{1}{n}\right)^2$ (d) none of these

30. Find the sum to n terms of
$$3 \cdot 2 + 5 \cdot 2^2 + 7 \cdot 2^3 + \ldots$$
(a) $2^{n+2} - 2^{n+1} - 2$ (b) $n \cdot 2^{n+2} - 2^{n+1} + 2$
(c) $n \cdot 2^{n+2} + 2^{n-2}$ (d) none of these

31. Find the sum of the series :
$$1 + \frac{4}{5} + \frac{7}{5^2} + \frac{10}{5^3} + \ldots \text{ to } n \text{ terms :}$$
(a) $\frac{35}{16} + \frac{7n}{5}$ (b) $\frac{12n - 7}{16(5^{n-1})}$
(c) $\frac{35}{16} - \left[\frac{12n + 7}{16(5^{n-1})}\right]$ (d) none of these

32. Find the sum of the series
$$1 \cdot 3^2 + 2 \cdot 5^2 + 3 \cdot 7^2 + \ldots \text{ to 20 terms :}$$
(a) 12896 (b) 187898
(c) 98970 (d) 188090

33. Find the sum of the series
$$\frac{1^3}{1} + \frac{1^3 + 2^3}{1 + 3} + \frac{1^3 + 2^3 + 3^3}{1 + 3 + 5} + \ldots \text{ to 16 terms :}$$
(a) 224 (b) 446
(c) 2356 (d) none of these

34. In a set of four numbers, the first three are in G.P. and the last three are in A.P. with common difference 6. If the first number is the same as the fourth, find the last number :
(a) -8 (b) 4
(c) 2 (d) can't be determined

35. The sum of an infinite G.P. is 16 and the sum of the squares of its terms is $153\frac{3}{5}$. Find the fourth term of the progression :

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

(a) $\frac{7}{16}$ (b) $\frac{5}{32}$
(c) $\frac{3}{16}$ (d) none of these

36. If $x = 1 + a + a^2 + \ldots$ to ∞ and $y = 1 + b + b^2 + \ldots$ to ∞, and $|a| < 1, |b| < 1$, then the value of :
$1 + ab + a^2b^2 + \ldots$ to ∞ is :
(a) $\frac{xy}{x+y}$ (b) $\frac{x+y+1}{xy}$
(c) $\frac{xy}{x-y}$ (d) $\frac{xy}{x+y-1}$

37. A person is entitled to receive an annual payment which for each year is less by one tenth of what it was for the year before. If the first payment is 100, then find the maximum possible payment which he can receive, however long he may live :
(a) 900 (b) 9999
(c) 1000 (d) none of these

38. Find the sum of the series $\frac{1}{4} - \frac{3}{16} + \frac{9}{64} - \frac{27}{256} + \ldots$
(a) $\frac{1}{3}$ (b) $\frac{1}{4}$
(c) $\frac{1}{7}$ (d) none of these

39. The sum of first two terms of a G.P. is $\frac{5}{3}$ and the sum to infinity of the series is 3. Find the first term :
(a) 1 (b) $\frac{2}{3}$
(c) 5 (d) both (a) and (c)

40. A ball is dropped from a height of 96 feet and it rebounds $\frac{2}{3}$ of the height it falls. If it continues to fall and rebound, find the total distance that the ball can travel before coming to rest.
(a) 240 ft (b) 360 ft
(c) 480 ft (d) none of these

HARMONIC PROGRESSION

Harmonic Progression (H.P.) : A series of quantities $a_1, a_2, a_3, a_4 \ldots, a_n$ is said to be in H.P. when their reciprocals $\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \ldots, \frac{1}{a_n}$ are in A.P.

NOTE The converse is also true.

$\therefore$ If a, b, c are in H.P., then $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P.

Simply there is no general formula for the sum of any number of quantities in H.P. All questions in H.P. are generally solved by inverting the terms and making use of the properties of the corresponding A.P.

Harmonic Mean (H.M.) : When three quantities are in H.P., the middle one is called the Harmonic Mean (H.M.) between the other two.

If a, H, b are in H.P., then H is the H.M. of a, b and $\frac{1}{a}, \frac{1}{H}, \frac{1}{b}$ are in A.P.

$\therefore \quad \frac{1}{H} - \frac{1}{a} = \frac{1}{b} - \frac{1}{H}$

or $\quad \frac{2}{H} = \frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab}$

$\therefore \quad H = \frac{2ab}{a+b}$

EXAMPLE 1 *Find the 10th term of the H.P.* $1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \frac{1}{9}, \frac{1}{11}, \ldots$

SOLUTION Since $\frac{1}{1}, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \ldots$ are in H.P.

$\therefore$ 1, 3, 5, 7, ... are in A.P.

$\therefore \quad T_{10} = 1 + 9 \times 2$

$T_{10} = 19$

Since 10th term of the A.P. is 19

therefore 10th term of the H.P. is $\frac{1}{19}$.

EXAMPLE 2 *If the 4th and the 7th terms of an H.P. are* $\frac{1}{2}$ *and* $\frac{2}{7}$ *respectively. Find the first term.*

SOLUTION 4th term in H.P. $= \frac{1}{2}$

$\therefore$ 4th term in A.P. $= 2$

7th term in H.P. $= \frac{2}{7}$

$\therefore$ 7th term in A.P. $= \frac{7}{2}$

$\therefore \quad T_4 = a + 3d = 2$

and $\quad T_7 = a + 6d = \frac{7}{2}$

$\Rightarrow \quad d = \frac{1}{2}$

$\therefore \quad a = \frac{1}{2}$ (which is the first term of the A.P.)

$\therefore$ the first term of H.P. $= 2$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yourshahbood.wordpress.com

For More Visit : www.LearnEngineering.in

ARITHMETICO-GEOMETRIC SEQUENCE

If $a_1, a_2, a_3, \ldots, a_n$ are in A.P. and $b_1, b_2, b_3 \ldots, b_n$ are in G.P., then the sequence $a_1b_1, a_2b_2, a_3b_3, \ldots, a_nb_n$ is said to be in arithmetico- geometric sequence.

An Arithmetico-Geometric sequence is of the form $ab, (a+d)br, (a+2d)\,br^2, (a+3d)br^3, \ldots$

$$\therefore \quad S_n = \frac{ab}{1-r} + \frac{dbr(1-r^{n-1})}{(1-r)^2} - \frac{[a+(n-1)d]br^n}{1-r}$$

and $$S_\infty = \frac{ab}{1-r} + \frac{dbr}{(1-r)^2}$$

EXAMPLE 3 *Find the sum to n terms of the series*

$$1 + \frac{4}{5} + \frac{7}{5^2} + \frac{10}{5^3} + \ldots$$

SOLUTION This is an A.G. series

$$\therefore \quad S_n = 1 + \frac{4}{5} + \frac{7}{5^2} + \frac{10}{5^3} + \ldots + \frac{3n-2}{5^{n-1}}$$

$$\therefore \quad \frac{S_n}{5} = \frac{1}{5} + \frac{4}{5^2} + \frac{7}{5^3} + \ldots + \frac{3n-5}{5^{n-1}} + \frac{3n-2}{5^n}$$

subtracting we get

$$\frac{4}{5}S_n = 1 + \frac{3}{5} + \frac{3}{5^2} + \frac{3}{5^3} + \ldots + \frac{3}{5^{n-1}} - \frac{3n-2}{5^n}$$

or $$\frac{4}{5}S_n = 1 + \frac{\frac{3}{5}\left(1 - \frac{1}{5^{n-1}}\right)}{1 - \frac{1}{5}} - \frac{3n-2}{5^n}$$

$$= \frac{35}{16} - \frac{12n+7}{16(5^{n-1})}$$

EXAMPLE 4 *In the above example find the sum to infinite number of terms.*

SOLUTION

$$S_\infty = \frac{ab}{1-r} + \frac{dbr}{(1-r)^2}$$

$$= \frac{1 \times 1}{1 - \frac{1}{5}} + \frac{3 \times 1 \times \frac{1}{5}}{\left(1 - \frac{1}{5}\right)^2}$$

$$= \frac{1}{\frac{4}{5}} + \frac{\frac{3}{5}}{\frac{16}{25}} = \frac{5}{4} + \frac{15}{16} = \frac{35}{16}$$

INTRODUCTORY EXERCISE-18.3

1. Which one of the following is true about A.M., G.M. and H.M. :
   (a) H.M. ≥ G.M. ≥ A.M. (b) A.M. ≥ H.M. ≥ G.M.
   (c) A.M. ≥ G.M. ≥ H.M. (d) G.M. ≥ H.M. ≥ A.M.
2. If a, b, c are in A.P. b, c, d are in G.P. and c, d, e are in H.P., then which one of the following is true :
   (a) $2c = a + e$ (b) $ae = c^2$
   (c) $c = \frac{2ae}{a+e}$ (d) $ab = de$
3. If A.M. and G.M. of two numbers is 10 and 8 respectively then find their H.M. :
   (a) 9 (b) 12
   (c) 6 (d) 6.4

www.LearnEngineering.in
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

EXERCISE

LEVEL 1

1. Find the number of terms in the G.P. whose first term is 3, sum is $\frac{4095}{1024}$ and the common ratio is $\frac{1}{4}$:
(a) 4 (b) 5
(c) 6 (d) none of these

2. Find the common ratio of the G.P. whose first and last terms are 5 and $\frac{32}{625}$ respectively and the sum of the G.P. is $\frac{5187}{625}$:
(a) $\frac{1}{5}$ (b) $\frac{2}{5}$
(c) $\frac{5}{3}$ (d) $\frac{4}{5}$

3. Find the sum of the series $2\sqrt{3}, 2\sqrt{2}, \frac{4}{\sqrt{3}}, \ldots$
(a) $6(\sqrt{3}+\sqrt{2})$ (b) $6(\sqrt{3}-\sqrt{2})$
(c) $\frac{6}{(\sqrt{3}+\sqrt{2})}$ (d) none of these

4. The number of terms in an A.P. is even, the sum of odd terms is 63 and that of even terms is 72 and the last term exceeds the first term by 16.5. Find the number of terms :
(a) 8 (b) 12
(c) 9 (d) 10

5. There are two pockets, each containing 3 coins of different denominations, which are in A.P. and the total value of coins in each pocket is Rs. 21. The common difference of the first set of coins is greater than that of the second set by 1, and the product of the first set is the product of the second set as 8 to 9. Find the value of the coin of largest denomination, among the six coins :
(a) 9 (b) 8
(c) 11 (d) 10

6. Find the sum of the three numbers in G.P. whose product is 216 and the sum of the products of them taken in pairs is 126 :
(a) 28 (b) 21
(c) 35/4 (d) none of these

7. The sum of four consecutive terms in A.P. is 36 and the ratio of product of the first and fourth is to the product of the second and third is 9 : 10. Find the largest of the numbers :
(a) 9 (b) 10
(c) 8 (d) 12

8. The sum of four integers in A.P. is 24 and their product is 945. Find the product of the smallest and greatest integers :
(a) 30 (b) 27
(c) 35 (d) 39

9. In an A.P. consisting of 23 terms, the sum of the three terms in the middle is 114 and that of the last three is 204. Find the sum of first three terms :
(a) 14 (b) 42
(c) 24 (d) 69

10. The sum of an infinite G.P. is 4 and the sum of their cubes is 192. Find the first term :
(a) 4 (b) 8
(c) 6 (d) 2

11. Vibhor joined as an area manager of Quick Corporation in the pay scale of Rs. 12,000 – 500 – 18,500. Minimum how many years he has to work in the corporation to avail the salary of Rs. 18,500 per month :
(a) 12 years (b) 10 years
(c) 13 years (d) 11 years

12. How many terms are common in two arithmetic progression 1, 4, 7, 10. ... upto 63 terms and 3, 7, 11, 15, ... upto 47 terms :
(a) 12 (b) 16
(c) 15 (d) none of these

13. The value of $3^{1/3} \cdot 9^{1/18} \cdot 27^{1/81} \ldots$:
(a) 3 (b) 9
(c) 27 (d) none of these

14. The sum to n terms of the series $1 + (1 + 3) + (1 + 3 + 5) + \ldots$ is :
(a) $\left[\frac{n(n+1)}{2}\right]^2$ (b) n^2
(c) $\frac{n(n+1)(2n+1)}{6}$ (d) none of these

15. The sum to n terms of the series
$1^2 + (1^2 + 3^2) + (1^2 + 3^2 + 5^2) + \ldots$ is :
(a) $\frac{1}{3}(n^4 + 2n^2)$ (b) $\frac{1}{3}(n^3 + 3n^2 - n)$
(c) $\frac{1}{6}n(n+1)(2n^2+2n-1)$ (d) none of these

16. If x, y, z are in G.P. and $a^x = b^y = c^z$, then :
(a) $\log_b a = \log_c b$ (b) $\log_b a = \log_a c$
(c) $\log_c b = \log_a c$ (d) none of these

17. The sum of integers from 113 to 113113 which are divisible by 7 is :
(a) 92358576 (b) 913952088
(c) 94501895 (d) 912952066

18. The sum of n terms of a progression is $3n^2 + 5$. The number of terms which equals 123 is :
(a) 12 (b) 18
(c) 21 (d) 23

19. If a, b, c are in A.P., $b - a, c - b$ and a are in G.P. then $a : b : c$ is :
(a) 4 : 3 : 2 (b) 2 : 3 : 4
(c) 1 : 2 : 3 (d) 3 : 4 : 5

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

20. The sum of first n terms of the series $\frac{1}{2}+\frac{3}{4}+\frac{7}{8}+\frac{15}{16}+\ldots$

(a) 2^n-1 (b) $1-2^{-n}$
(c) 2^n-n+1 (d) $n+2^{-n}-1$

21. The sum of all 4 digit numbers which when divided by 4 leave the remainder 1 :

(a) 12372750 (b) 12327027
(c) 12370705 (d) 109980

22. If n arithmetic means are inserted between two quantities a and b, then their sum is equal to :

(a) $n(a+b)$ (b) $\frac{n}{2}(a+b)$
(c) $2n(a+b)$ (d) $\frac{n}{2}(a-b)$

23. The product of n geometric means between two given numbers a and b is :

(a) $(ab)^n$ (b) $(ab)^{2n}$
(c) $(ab)^{n/2}$ (d) none of these

24. If $a_1, a_2, a_3, \ldots$ is an A.P. such that

$$a_1+a_5+a_{10}+a_{15}+a_{20}+a_{24}=225$$

then $a_1+a_2+a_3+\ldots+a_{23}+a_{24}$ is equal to :

(a) 999 (b) 900
(c) 1225 (d) none of these

25. The sum to n terms of the series, where n is an even number :

$$1^2-2^2+3^2-4^2+5^2-6^2+\ldots:$$

(a) $n(n+1)$ (b) $\frac{n(n+1)}{2}$
(c) $-\frac{n(n+1)}{2}$ (d) none of these

26. $\underbrace{(666\ldots6)}_{n\text{-digits}}+\underbrace{(888\ldots8)}_{n\text{-digits}}$ is equal to :

(a) $14(n^2-1)$ (b) $\frac{48}{9}(10^{2n}-1)$
(c) $\frac{14}{9}(10^n-1)$ (d) none of these

27. The cubes of the natural numbers are grouped as $1^3, (2^3, 3^3), (4^3, 5^3, 6^3), \ldots$ then the sum of the numbers in the nth group is :

(a) $\frac{n^3}{12}(n^2+2)(n^2+4)$ (b) $\frac{1}{8}n^3(n^2+1)(n^2+3)$
(c) $\frac{n^3}{6}(n^2+1)(n^2+2)$ (d) none of these

28. Three non-zero real numbers form an A.P. and the squares of these numbers taken in the same order form a G.P. Then the number of all possible common ratio of the G.P. is :

(a) 1 (b) 2
(c) 3 (d) none of these

29. If three positive real numbers a, b, c are in A.P. such that $a\cdot b\cdot c=4$, then the minimum value of b is :

(a) $2^{1/2}$ (b) $2^{1/3}$
(c) $2^{2/3}$ (d) $2^{3/2}$

30. Three numbers form an increasing G.P. If the middle number is doubled, then the new numbers are in A.P. The common ratio of the G.P. is :

(a) $2-\sqrt{3}$ (b) $2+\sqrt{3}$
(c) $\sqrt{3}-2$ (d) $3+\sqrt{2}$

31. If a, b, c $(a>0)$ are three successive terms of a G.P. with common ratio r, the value of r for which $c>4b-3a$ holds given by :

(a) $1<r<3$ (b) $-3<r<-1$
(c) $r<1$ and $r>3$ (d) none of these

32. Let a_n be the nth term of an A.P and $a_7=22$, then the value of the common difference (d) that would make $a_3\cdot a_7\cdot a_8$ greatest is :

(a) 4 (b) 2
(c) 0 (d) 7

33. If the A.M of two positive numbers a and b, $(a>b)$, is twice their G.M., then $a:b$ is :

(a) $2:\sqrt{3}$ (b) $2:7+4\sqrt{3}$
(c) $2+\sqrt{3}:2-\sqrt{3}$ (d) $7+4\sqrt{3}:7-4\sqrt{3}$

34. The number of common terms to the two sequences 17, 21, 25, ..., 417 and 16, 21, 26, ..., 466 is :

(a) 19 (b) 20
(c) 21 (d) 84

35. The sum to n terms of the series

$$\frac{1}{\sqrt{1}+\sqrt{3}}+\frac{1}{\sqrt{3}+\sqrt{5}}+\frac{1}{\sqrt{5}+\sqrt{7}}+\ldots \text{ is :}$$

(a) $\sqrt{2n+1}$ (b) $\frac{1}{2}\sqrt{2n+1}$
(c) $\sqrt{2n-1}$ (d) $\frac{1}{2}\{\sqrt{2n+1}-1\}$

36. Let T_r be the rth term of an A.P. for $r=1, 2, 3, \ldots$ If for some positive integers m, n we have $T_m=\frac{1}{n}$ and $T_n=\frac{1}{m}$ then T_{mn} equals :

(a) $\frac{1}{mn}$ (b) $\frac{1}{m}+\frac{1}{n}$
(c) 1 (d) 0

37. Find the sum to n terms of the series $3+6+10+16+\ldots$:

(a) $\frac{n(n-1)}{2}-1$ (b) $n(n+1)+2^n-1$
(c) $n(n+2)+1$ (d) $3(2n+1)-2^n$

38. In a certain colony of cancerous cells, each cell divides into two every minute. How many cells will be produced from a single cell if the rate of division continues for 12 minutes:

(a) 9180 (b) 8190
(c) 8910 (d) none of these

39. Divakar and Subhank set out to meet each other from Delhi and Saharanpur, 330 km apart. Divakar travels 30 km on the first day 28 km on the second day, 26 km on the third day and so on. Subhank travels 20 km on the first day, 24 km on the second day, 28 km on the third day and so on. In how many days they meet, if they started moving towards each other at the same time ?

(a) 4 (b) 5
(c) 6 (d) none of these

40. Find the sum to n terms of the series

$$1+(1+2)+(1+2+3)+(1+2+3+4)+\ldots:$$

(a) $\frac{n(n+1)(2n+1)}{6}$ (b) $\frac{n(n+1)(n+2)}{6}$
(c) $\frac{n(n+1)(n+2)}{12}$ (d) $\frac{n(n+1)}{2}$

41. If a, b, c are in H.P. then the value of $\left(\frac{1}{b}+\frac{1}{c}-\frac{1}{a}\right)\left(\frac{1}{c}+\frac{1}{a}-\frac{1}{b}\right)$ is :

(a) $\frac{3}{b^2}-\frac{2}{ab}$ (b) $a^2+b^2+c^2$
(c) $\frac{2}{bc}-\frac{1}{b^2}$ (d) both (a) and (c)

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

42. Find the sum to n terms of the series
$1\cdot3\cdot5 + 3\cdot5\cdot7 + 5\cdot7\cdot9 + \ldots$ is :
(a) $n(8n^3 + 11n^2 - n + 3)$ (b) $n(2n^3 + 8n^2 + 7n - 2)$
(c) $8n^3 + 12n^2 - 2n - 3$ (d) none of these

43. What is the sum of 100 terms of the series
$1 - 2 + 3 - 4 + 5 - 6 + \ldots$?
(a) 100 (b) 50
(c) 550 (d) none of these

LEVEL 2

1. An A.P. whose first term is unity and in which the sum of the first half of any even number of terms to that of the second half of the same number of terms is in constant ratio, the common difference 'd' is given by :
(a) 1 (b) 2
(c) 3 (d) 4

2. The number of divisors of 187, 637 and 1001 are in :
(a) A.P. (b) G.P.
(c) H.P. (d) none of these

3. The sum to n terms of the series
$1^2 + (1^2 + 3^2) + (1^2 + 3^2 + 5^2) + \ldots$ is :
(a) $\frac{1}{3}(n^3 + n^2 + 1)$ (b) $\frac{1}{6}n(n+1)(2n^2 + 2n - 1)$
(c) $\frac{1}{3}(2n^2 + 2n - 1)$ (d) none of these

4. If $|a| < 1$ and $|b| < 1$, then the sum of the series
$a(a+b) + a^2(a^2 + b^2) + a^3(a^3 + b^3) + \ldots$ upto ∞ is :
(a) $\frac{a^2}{b} + 1$ (b) $\frac{a^2}{1-ab}$
(c) $\frac{a^2}{1-a^2} + \frac{ab}{1-ab}$ (d) $\frac{a^2}{b^2} + \frac{1}{ab}$

5. The sum to n terms of the series
$\frac{3}{1^2\cdot2^2} + \frac{5}{2^2\cdot3^2} + \frac{7}{3^2\cdot4^2} + \ldots$ is :
(a) $\frac{n^2 - 2n}{(n-1)^2}$ (b) $\frac{n^2 + 2n}{(n+1)^2}$
(c) $\frac{2n^2 + 1}{n}$ (d) none of these

6. The natural numbers are written as follows :
1
2 3
4 5 6
7 8 9 10
... ...
... ...
The sum of numbers in the nth row is :
(a) $\frac{n(n+1)}{2}$ (b) $\frac{n^2 + 1}{2}$
(c) $\frac{n(n^2 + 1)}{2}$ (d) none of these

7. The number $\underbrace{1\,1\,1\,1\ldots1}_{119 \text{ times}}$ is a :
(a) prime number (b) composite number
(c) multiple of 3 (d) none of these

8. The interior angles of a polygon are in A.P. If the smallest angle is 120° and the common difference is 5°, then the number of sides in the polygon is :
(a) 8 (b) 12
(c) 9 (d) 6

9. If $n \in N$ and n is an odd number, then
$10^0 + 10^4 + 10^8 + 10^{12} + \ldots + 10^{4n}$ is a :
(a) prime number (b) composite number
(c) a multiple of 10 (d) none of these

10. The value of $0.1\overline{23}$ is :
(a) $\frac{61}{495}$ (b) $\frac{71}{485}$
(c) $\frac{62}{719}$ (d) $\frac{123}{999}$

11. If n is a positive integer, then $\underbrace{1\,1\,1\ldots1}_{2n \text{ times}} - \underbrace{2\,2\,2\ldots2}_{n \text{ times}}$ is :
(a) a perfect square (b) a perfect cube
(c) prime number (d) none of these

12. One side of an equilateral triangle is 24 cm. The mid-points of its sides are joined to form another triangle whose mid points are in turn joined to form still another triangle. This process continues indefinitely. Find the sum of the perimeters of all the triangles :
(a) 144 cm (b) 72 cm
(c) 536 cm (d) 676 cm

13. The odd positive integers are arranged in a triangle as follows :
1
3 5
7 9 11
13 15 17 19
21 23 25 27 29
... ...
... ...
Find the sum of the numbers in the nth row of this arrangement :
(a) $n^2 + n$ (b) $2n^2 - 2 + n$
(c) n^3 (d) none of these

14. Find the sum to n terms of the series
$\log a + \log\frac{a^3}{b} + \log\frac{a^5}{b^2} + \log\frac{a^7}{b^3} + \ldots$
(a) $\log\left(\frac{a^{2n}}{b^{n-1}}\right)^{n/2}$ (b) $\log\frac{a^{2n-1}}{b^{n-1}}$
(c) $\log\frac{a^{2n}}{b^n}$ (d) none of these

15. The value of $(0.2)^{\log_{\sqrt{5}}\left(\frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \ldots\right)}$:
(a) 2 (b) $\frac{1}{2}$
(c) 4 (d) none of these

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

16. If a, b, c are in H.P., then $\frac{b+a}{b-a} + \frac{b+c}{b-c}$ equals:
(a) 1 (b) 2
(c) $\frac{b-c}{a-b}$ (d) $\frac{ab}{c}$

17. Let a, b, c are in G.P. with common ratio r $(0 < r < 1)$. Then $a, 2b, 3c$ are in A.P., if r equals :
(a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{2}}$
(c) $\frac{1}{3}$ (d) none of these

18. The sum of the series
$\frac{5}{13} + \frac{55}{(13)^2} + \frac{555}{(13)^3} + \frac{5555}{(13)^4} + \ldots \infty$:
(a) $\frac{55}{39}$ (b) $\frac{65}{39}$
(c) $\frac{110}{169}$ (d) $\frac{65}{36}$

19. The sum of the infinite series
$\frac{1}{1\cdot4} + \frac{1}{4\cdot7} + \frac{1}{7\cdot10} + \ldots \infty$ is equal to :
(a) $\frac{1}{3}$ (b) $\frac{1}{4}$
(c) $\frac{38}{27}$ (d) none of these

20. A sequence $a_1, a_2, a_3, \ldots, a_n$ of real numbers is such that $a_1 = 0, |a_2| = |a_1 + 1|, |a_3| = |a_2 + 1|, \ldots, |a_n| = |a_{n-1} + 1|$ then the A.M. $\left(i.e.\ \frac{a_1 + a_2 + a_3 + \ldots + a_n}{n}\right)$ can not be less than :
(a) $\frac{1}{2}$ (b) $-\frac{1}{2}$
(c) 1 (d) can't be determind

21. Find the A.M. of second and third term of a G.P. of real numbers such that the sum of its first four terms is equal to 30 and the sum of the squares of the first four terms is 340.
(a) 5 (b) 6
(c) 8 (d) 10

22. A square is drawn by joining the mid points of the sides of a given square. A third square is drawn inside the second square in the same way and this process continues indefinitely. If a side of the first square is 4 cm, determine the sum of the areas of all the squares :
(a) 32 cm^2 (b) 16 cm^2
(c) 20 (d) none of these

23. An A.P. and a G.P. with positive terms have the same number of terms and their first terms as well as last terms are equal. Then what can be said about the sum of the A.P. and G.P. if A_s and G_s denote the sum of the A.P. and G. P. respectively :
(a) $A_S \geq G_S$
(b) $A_S \leq G_S$
(c) $A_S = G_S$
(d) can not be determined

24. An A.P. and a G.P. have all the terms positive. Also the first and second terms of the progressions are equal, If a_i and g_i denote the terms of A.P. and G.P. respectively for every $i \in N$ and $i > 2$, then which one of the following is true ?
(a) $a_i \geq g_i$ (b) $a_i \leq g_i$
(c) $a_i = g_i$ (d) none of these

25. $1^k + 2^k + 3^k + \ldots + n^k$ is divisible by $1 + 2 + 3 + \ldots n$ for every $n \in N$, then k is :
(a) even (b) odd
(c) only prime (d) none of these

26. If pth, qth, rth and sth terms of an A.P. are in G.P. then $p - q, q - r, r - s$ are in :
(a) A.P. (b) G.P.
(c) H.P. (d) none of these

27. In our country 'Dreamland' the currency consists of a set of notes of different denominations. If the sum of all the denominations is 63 units of currency then what is the minimum number of denominations that can be there so that all the transactions can be made (in integers).
(a) 3 (b) 5
(c) 6 (d) 9

28. A retailer has n stones by which he can measure (or weigh) all the quantities from 1 kg to 121 kg (in integers only, e.g., 1 kg, 2 kg, 3 kg, etc) keeping these stones on either side of the balance. What is the minimum value of n?
(a) 3 (b) 4
(c) 5 (d) 11

29. What will be the 33rd term of the sequence 1, 7, 25, 79, ...?
(a) $33^3 - 1$ (b) $3^{33} - 2$
(c) $3(33^3 - 2)$ (d) none of these

30. All the four angles of a quadrilateral are in G.P. with common ratio r $(r \in N)$. Any two angles are acute and other two angles are obtuse, then the measure of the smallest angle of the quadrilateral is :
(a) 18° (b) 32°
(c) 24° (d) 36°

Directions for question number 31, 32 : In the kingdom of YAMRAJ, there are certain number of souls in a particular day. Where a single soul can merge with another soul in every second and thus two souls reduced to a single soul and if any soul can not merge with anyother soul in a particular second then it has to become a ghost and suffer in the hell. Thus every soul tries to avoid to become a ghost, but after a certain time it has to go to hell after maximum possible survival.

31. In a particular day there were $(2^{43200} - 1)$ souls after how many seconds will all the souls become ghost :
(a) 43200 (b) 43199
(c) 43201 (d) 21600

32. If after n seconds there are k souls left, then what is the maximum number of souls there can be initially in a particular day ?
(a) $2^{k \cdot n}$ (b) $2^{kn} + 2k$
(c) $2^n k + 2^n - 1$ (d) $2^k n + 2^n + 1$

33. The income of HBI on the nth day is Rs. $(n^2 + 2)$ and the expenditure of HBI on the nth day is Rs. $(2n + 1)$.
Also income = expenditure + savings
In how many days his total saving will be Rs. 1240. :
(a) 10 (b) 12
(c) 15 (d) 16

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Directions for question numbers 34 and 35: In the zoological park Lucknow there are four kinds of animals viz. Elephant, monkey, lion and tiger which are in increasing G.P. and in the local zoo in Kanpur there are the same kinds of animals but in A.P. The number of elephants is least and equal in each of the places. Also the number of monkeys in each of the places is same but just greater than that of elephants. Total number of animals in zoological park is 50% more than that of local zoo. Also the common ratio of the G.P. is same as the common difference of the A.P. Number of tigers in both the places is maximum.

34. What is the number of elephants in each of the places ?
(a) 1 (b) 2
(c) 3 (d) 4

35. What is the number of lions in zoological park Lucknow :
(a) 5 (b) 6
(c) 8 (d) can not be determine

36. What is the sum of n terms of the series $-1+1^2-2+2^2-3+3^2\ldots$?
(a) $\frac{n(n^2+1)}{3}$ (b) n^2+n
(c) $\frac{n(n^2-1)}{3}$ (d) none of these

37. Find the sum of $1+\frac{2}{3}+\frac{4}{9}+\frac{6}{27}+\frac{8}{81}+\frac{10}{243}+\ldots\infty$:
(a) $\frac{2}{3}$ (b) $\frac{5}{2}$
(c) $\frac{19}{45}$ (d) $\frac{81}{17}$

38. Sum of n terms of the series
$$\log a+\log\left(\frac{a^2}{b}\right)+\log\left(\frac{a^3}{b^2}\right)+\log\left(\frac{a^4}{b^3}\right)+\ldots \text{ is :}$$
(a) $\log\left\{\frac{a^n}{b^{n-1}}\right\}^{n/2}$ (b) $\log\left\{\frac{a^{n+1}}{b^{n-1}}\right\}^{n/2}$
(c) $\log\left\{\frac{a^n}{b^n}\right\}^{n/2}$ (d) $\log\left\{\frac{a^{n+1}}{b^n}\right\}^{n/2}$

39. If $a_1, a_2, a_3, \ldots, a_n$ are in A.P. where $a_i > 0$ for all i, then the value of
$$\frac{1}{\sqrt{a_1}+\sqrt{a_2}}+\frac{1}{\sqrt{a_2}+\sqrt{a_3}}+\ldots+\frac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}}:$$
(a) $\frac{n-1}{\sqrt{a_1}+\sqrt{a_2}}$ (b) $\frac{n+1}{\sqrt{a_1}+\sqrt{a_2}}$
(c) $\frac{n}{\sqrt{a_1}-\sqrt{a_2}}$ (d) none of these

40. If $\log_5 2$, $\log_5(2^x-3)$ and $\log_5\left(\frac{17}{2}+2^{x-1}\right)$ are in A.P. then the value of x is :
(a) 2 (b) 3
(c) $\log_2 3$ (d) $\log_2 5$

41. If $1, \log_y x, \log_z y, -15\log_x z$ are in A.P., then which is correct ?
(a) $x=z^3$ (b) $x=\frac{1}{y}$
(c) $y=\frac{1}{z^3}$ (d) all of these

42. If x, y, z are in G.P. and $a^x=b^y=c^z$, then $\log_b a\cdot\log_b c$ is equal to :
(a) 0 (b) 1
(c) ac (d) none of these

43. If $\log_2(5\cdot 2^x+1)$, $\log_4(2^{1-x}+1)$ and 1 are in A.P. then x equals :
(a) $\log_2 5$ (b) $1-\log_5 2$
(c) $\log_5 2$ (d) none of these

44. If $1, \log_9(3^{1-x}+2)$ and $\log(4\cdot 3^x-1)$ are in A.P. then x is equal to :
(a) $\log_4 3$ (b) $\log_3 4$
(c) $1-\log_3 4$ (d) $\log_3\left(\frac{1}{4}\right)$

45. If $x>1, y>1, z>1$ are in G.P., then $\frac{1}{1+\ln x}, \frac{1}{1+\ln y}, \frac{1}{1+\ln z}$ are in :
(a) A.P. (b) H.P.
(c) G.P. (d) none of these

46. Let $a_1, a_2, \ldots, a_{10}$ be in A.P. and $h_1, h_2, \ldots, h_{10}$ be in H.P. If $a_1=h_1=2$ and $a_{10}=h_{10}=3$, then a_4h_7 is:
(a) 2 (b) 3
(c) 5 (d) 6

47. Let $S_1, S_2, \ldots$ be squares such that for each $n\geq 1$, the length of a side of S_n equals the length of a diagonal of S_{n+1}. If the length of a side of S_1 is 10 cm, then for which of the following values of n is the area of S_n less than 1 square cm ?
(a) 7 (b) 8
(c) 6 (d) 4

Final Round

1. A bacteria gives birth to two new bacterias in each second and the life span of each bacteria is 5 seconds. The process of the reproduction is continuous untill the death of the bacteria. Initially there is one newly born bacteria at time $t=0$, then find the total number of live bacterias just after 10 seconds :
(a) $\frac{3^{10}}{2}$ (b) $3^5(3^5-1)$
(c) $3^{10}-2^{10}$ (d) $3^{10}-2^5$

2. If S_n be defined as the sum of n terms of the series S. Such that $S\equiv 2^{66}-2^{65}-2^{64}\ldots$ Find S_{20}:
(a) 400 (b) 2^{47}
(c) 2^{20} (d) none of these

3. If U_m be defined as the sum of m terms of the series V, where $V=S_m+S_{m+1}+S_{m+2}+\ldots$ Find U_{34} :
(a) $2^{33}-1$ (b) $2^{33}+1$
(c) $2^{34}-1$ (d) none of these

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

4. Let $a_1, a_2, a_3, \ldots, a_n$ be an A.P. and S_1, S_2 and S_3 be the sum of first n, $2n$ and $3n$ terms respectively then $S_3 - S_2 - S_1$ is equal to, if a is first term and d is common difference :
(a) $3a - 2n - d$ (b) $a(n + 2d)$
(c) $3a + 2nd$ (d) $2n^2d$

5. Consider the sets
$$T_n = \{n, n + 3, n + 5, n + 7, n + 9\}$$
where $n = 1, 2, 3, \ldots, 99$. How many of these sets contain 5 or any integral multiple thereof (*i. e.* any one of the numbers 5, 10, 15, 20, 25, ...) ?
(a) 81 (b) 79
(c) 80 (d) none of these

6. If $\{x\}$ is the least integer greater than or equal to x, then find the value of the following series :
$$\{\sqrt{1}\} + \{\sqrt{2}\} + \{\sqrt{3}\} + \{\sqrt{4}\} + \ldots + \{\sqrt{99}\} + \{\sqrt{100}\}$$
(a) 715 (b) 55
(c) 157 (d) 835

7. If $[x]$ is the greatest integer less than or equal to x, then find the value of the following series
$$[\sqrt{1}] + [\sqrt{2}] + [\sqrt{3}] + [\sqrt{4}] + \ldots + [\sqrt{323}]$$
(a) 3237 (b) 2373
(c) 3723 (d) none of these

Directions for question number 8–14 : *King Dashratha of Ayodhya on his birthday decided to offer 100 coins of gold among his 4 sons and 3 wives. The denomination of each coin is Re. 1. He put all the 100 coins in 7 bags in such a way that by taking a proper combination of various bags any integral sum (i.e. Rs. 1, 2, 3, 4, ..., 100) can be obtained and it is known that the only whole sum of any bag can be taken.*

8. What is the maximum amount received by any one of the recipient ?
(a) Rs. 79 (b) Rs. 96
(c) Rs. 37 (d) none of these

9. If all the coins, of these who have odd number of coins, are combined then minimum how many people are required to combine their coins to make the same amount having even number of coins :
(a) 2 (b) 3
(c) 4 (d) can't be determined

10. If the two least amounts are combined with highest one then minimum how many persons combined their coins to get the same amount :
(a) 2 (b) 3
(c) 4 (d) 5

11. If the king wanted to distribute the amount equally among them, then how many people would have received more amount (number of coins) than that of previously retaining two coins with himself :
(a) 6 (b) 3
(c) 4 (d) none of these

12. If the king also included Hanuman another 8th candidate in this occassion and distributed 200 coins in similar fashion such that the combination of two or more person's coin could yield any integral sum from 1 to 200, then what was the maximum amount (in Rs.) that anybody had received ?
(a) 128 (b) 73
(c) 37 (d) 64

13. In the previous problem (number 12), if the people who had obtained the coins equal to odd powers of 2 (*i. e.* $2^1, 2^3, 2^5, \ldots$ etc.) can not combine their money with any other. Then maximum how many different sums can be made ?
(a) 51 (b) 28
(c) 31 (d) 158

14. In the problem number 12 if we assume Hanuman as a special guest who received the maximum amount but he distributed his sum in the same proportion as others had received previously leaving the person who had received initally the second highest amount. The number of coins that Hanuman was left with :
(a) 10 (b) 13
(c) 11 (d) 9

15. The sum of an infinite geometric progression (G.P.) is 2 and the sum of the G.P. made from the cubes of the terms of this infinite series is 24. The values a and r respectively (where a is the first term and r denotes common ratio of the series) :
(a) $-2, 3$ (b) $3, -\frac{1}{2}$
(c) $0, \frac{1}{2}$ (d) can't be determined

16. The sum of first ten terms of an A.P. is 155 and the sum of first two terms of a G.P. is 9. The first term of the A.P. is equal to the common ratio of the G.P. and the first term of the G.P. is equal to the common difference of the A.P. which can be the A.P. as per the given conditions ?
(a) 2, 4, 6, 8, 10,... (b) $\frac{25}{2}, \frac{79}{6}, \frac{83}{6}, \ldots$
(c) 2, 5, 8, 11,... (d) both (b) and (c)

17. The sum of an infinite geometric series is 162 and the sum of its first n terms is 160. The inverse of its common ratio is an integer, then how many values of the common ratio (r) are possible ?
(a) 1 (b) 2
(c) 4 (d) 5

18. In the previous question how many values of n are possible?
(a) 1 (b) 2
(c) 3 (d) 4

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Answers

INTRODUCTORY EXERCISE-18.1

1 (b)	2. (a)	3. (c)	4. (d)	5. (b)	6. (c)	7. (a)	8. (a)	9. (c)	10. (a)
11. (d)	12. (a)	13. (a)	14. (c)	15. (c)	16. (d)	17. (d)	18. (b)	19 (c)	20. (d)
21. (b)	22. (b)	23. (c)	24. (c)	25. (b)	26. (c)	27. (a)	28. (b)	29. (b)	30. (c)
31. (d)	32. (a)	33. (c)	34. (a)	35. (d)	36. (d)	37. (c)	38. (b)	39. (c)	40. (a)
41. (c)	42. (b)	43. (c)							

INTRODUCTORY EXERCISE-18.2

1 (c)	2. (c)	3. (b)	4. (a)	5. (b)	6. (c)	7. (b)	8. (a)	9. (b)	10. (c)
11. (a)	12. (d)	13. (c)	14. (a)	15. (a)	16. (a)	17. (b)	18. (c)	19 (b)	20. (a)
21. (d)	22. (c)	23. (b)	24. (b)	25. (b)	26. (b)	27. (d)	28. (c)	29. (b)	30. (b)
31. (c)	32. (d)	33. (b)	34. (c)	35. (c)	36. (d)	37. (c)	38. (c)	39. (d)	40. (c)

INTRODUCTORY EXERCISE-18.3

1 (c)	2. (b)	3. (d)							

LEVEL-1

1 (c)	2. (b)	3. (a)	4. (b)	5. (d)	6. (b)	7. (d)	8. (b)	9. (c)	10. (c)
11. (c)	12. (b)	13. (d)	14. (c)	15. (c)	16. (a)	17. (b)	18. (c)	19. (c)	20. (d)
21. (a)	22. (b)	23. (c)	24. (b)	25. (c)	26. (c)	27. (b)	28. (c)	29. (c)	30. (b)
31. (c)	32. (c)	33. (c)	34. (b)	35. (d)	36. (c)	37. (b)	38. (b)	39. (c)	40. (b)
41. (d)	42. (b)	43. (d)							

LEVEL-2

1 (b)	2. (a)	3. (b)	4. (c)	5. (b)	6. (c)	7. (b)	8. (c)	9. (b)	10. (a)
11. (a)	12. (a)	13. (c)	14. (a)	15. (c)	16. (b)	17. (c)	18. (d)	19. (a)	20. (b)
21. (b)	22. (a)	23. (a)	24. (b)	25. (b)	26. (b)	27. (c)	28. (c)	29. (b)	30. (c)
31. (a)	32. (c)	33. (d)	34. (b)	35. (c)	36. (d)	37. (b)	38. (b)	39. (a)	40. (b)
41. (d)	42. (b)	43. (b)	44. (c)	45. (b)	46. (d)	47. (b)			

FINAL ROUND

1 (b)	2. (b)	3. (a)	4. (d)	5. (b)	6. (a)	7. (c)	8. (c)	9. (b)	10. (a)
11. (c)	12. (b)	13. (c)	14. (c)	15. (b)	16. (d)	17. (d)	18. (c)		

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Hints & Solutions

INTRODUCTORY EXERCISE 18.1

1. $T_{41} = 3 + 40 \times 5 = 203$ $(\because T_n = a + (n-1)d$

2. $T_{25} = 10 + 24(-4) = -86$ $(a = 10, d = -4)$

3. $T_n = l = a + (n-1)d$

$= 2 + (n-1)3 = 56$

$\Rightarrow \quad 3(n-1) = 56 - 2$

$\Rightarrow \quad n = 19$

4. Since there are only two consecutive terms so we can not find the pattern of the sequence that whether it is a non-progression sequence or A.P. or G.P. or H.P. etc. Hence (d).

5. $T_3 = a + 2d = 17$ ...(1)

$T_7 = a + 6d = 27$ ...(2)

$\therefore$ subtracting equation (1) from equation (2), we get

$4d = 10 \Rightarrow d = 2.5$

$\therefore \quad a + 2 \times 2.5 = 17$

$\Rightarrow \quad a = 12$

6. $T_6 = a + 5d = 12$ ...(1)

$T_8 = a + 7d = 22$

$\Rightarrow \quad 2d = 10$ or $d = 5$

$\therefore \quad a + 25 = 12$ [from eq. (1)]

$\Rightarrow \quad a = -13$

$\therefore \quad T_n = -13 + (n-1)5$

$\Rightarrow \quad T_n = 5n - 18$

Alternatively : Check the options.

$5n - 18 \Rightarrow 5 \times 6 - 18 = 12$

and $5 \times 8 - 18 = 22$

Hence option (c) is correct.

7. $T_7 = a + 6d$

$T_{11} = a + 10d$

$\because \quad 7 \cdot T_7 = 11 \cdot T_{11}$

$\Rightarrow \quad 7(a + 6d) = 11(a + 10d)$

$\Rightarrow \quad 4a = -68d$

$\Rightarrow \quad a = -17d$

Now $T_{18} = a + 17d$

$\Rightarrow \quad T_{18} = -17d + 17d$

$\Rightarrow \quad T_{18} = 0$

8. $T_p = a_1 + (p-1)d$

$T_q = a_1 + (q-1)d$

$T_r = a_1 + (r-1)d$

$\therefore$ L.H.S. $= a(q-r) + b(r-p) + c(p-q)$

$= \{a_1 + (p-1)d\}(q-r) + \{a_1 + (q-1)d\}(r-p)$

$+ \{a_1 + (r-1)d\}(p-q)$

$= a_1 \cdot (q - r + r - p + p - q) + d[(p-1)(q-r)$

$+ (q-1)(r-p) + (r-1)(p-q)]$

$= 0$

Alternatively : Let us assume an A.P.

2, 5, 8, 11, 14, 17, 20, ...

Again assume $p = 2, q = 4, r = 5$

$\therefore \quad T_p = T_2 = 5 = a$

$T_q = T_4 = 11 = b$

$T_r = T_5 = 14 = c$

and $\therefore \quad a(q-r) + b(r-p) + c(p-q) = 5(-1) + 11(3) + 14(-2)$

$= -5 + 33 - 28 = 0$

Hence (a) is the correct choice.

9. $T_p = a + (p-1)d = q$ ...(1)

and $T_q = a + (q-1)d = p$ ...(2)

and $T_m = a + (m-1)d$ ...(3)

From (1) and (2)

$a = q - (p-1)d$ and $a = p - (q-1)d$

$\Rightarrow \quad q - (p-1)d = p - (q-1)d$

$\Rightarrow \quad (q-p) = -d[(q-1) - (p-1)]$

$\Rightarrow \quad d = -1$

Again, $a + (p-1)d = q$

$\Rightarrow \quad a + (p-1) \times -1 = q$

$\Rightarrow \quad a = p + q - 1$

$\therefore \quad T_m = (p + q - 1) + (m-1) \times -1$

$\Rightarrow \quad T_m = p + q - m$

10. $11 + 13 + 15 + \ldots + 99$

Here number of terms $= 50 - 5 = 45$

or Number of terms $= \left(\frac{99 - 11}{2}\right) + 1 = 45$

NOTE When both the extremes of a sequence are counted then

number of terms $= \left(\frac{l - a}{d}\right) + 1$

Now the sum of the progression $= \left(\frac{a + l}{2}\right)n$

$= \left(\frac{11 + 99}{2}\right) \times 45 = 2475$

Alternatively : $11 + 13 + \ldots + 99$

$= (1 + 3 + 5 + \ldots + 99) - (1 + 3 + \ldots + 9)$

$= (50)^2 - (5)^2 = 2475$

11. Number of terms $= \left(\frac{616 - 22}{6}\right) + 1$

$= \frac{594}{6} + 1 = 100$

Alternatively :

22, 28, 34, ..., 616 = (4, 10, 16, 22, 28, ..., 616) − (4, 10, 16)

$\Rightarrow$ (6, 12, 18, 24, 30, ..., 618) − (6, 12, 18)

$\Rightarrow$ (1, 2, 3, 4, 5, ..., 103) − (1, 2, 3)

$\Rightarrow$ 103 − 3 = 100

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Alternatively : 22, 28, 34, ..., 616
$\Rightarrow$ 24, 30, 36, ..., 618
$\Rightarrow$ 4, 5, 6, ..., 103
$\Rightarrow$ 103 – 3 = 100

12. Number of terms $= \left(\frac{888-222}{2}\right)+1 = 334$

$\therefore \quad S_{334} = \left(\frac{222+888}{2}\right)\times 334$

$= 555 \times 334 = 185370$

Alternatively : $222 + 224 + \ldots + 888$
$= (2+4+6+\ldots+888) - (2+4+6+\ldots+220)$
$= (444 \times 445) - (110 \times 111)$
$= 111(4\times 445 - 110) = 111(1780-110)$
$= 111 \times 1670 = 185370$

13. $T_2 = a + d = 2 \quad \ldots(1)$

$T_7 = a + 6d = 22 \quad \ldots(2)$

$\Rightarrow 5d = 20$ (subtracting (1) from (2))

$\Rightarrow \quad d = 4$

$\therefore a = -2$ (substituting 'd' in equation (1))

$\therefore \quad l = T_{35} = -2 + 34 \times 4 = 134$

$\therefore \quad S_{35} = \left(\frac{-2+134}{2}\right)\times 35 = 2310$

14. $T_{12} = a + 11d = -13 \quad \ldots(1)$

$S_4 = 24 = \left(\frac{a+T_4}{2}\right)\times 4 \quad \left\{\because S_n = \left(\frac{a+l}{2}\right)n\right\}$

$\Rightarrow \quad a + T_4 = 12$

$\Rightarrow \quad a + a + 3d = 12$

$\Rightarrow \quad 2a + 3d = 12 \quad \ldots(2)$

from equations (1) and (2)

$$\begin{array}{r} 2a + 22d = -26 \\ 2a + 3d = 12 \\ - \quad - \quad - \\ \hline 19d = -38 \end{array}$$

$\Rightarrow \quad d = -2$

$\therefore \quad a = 9$

Hence $S_{10} = \left(\frac{a+T_{10}}{2}\right)10$

$= \left[\frac{9+(-9)}{2}\right]\times 10 = 0 \quad \{\because T_{10} = -9\}$

15. $T_3 = a + 2d = \frac{1}{5} \quad \ldots(1)$

$T_5 = a + 4d = \frac{1}{3} \quad \ldots(2)$

$\Rightarrow \quad 2d = \frac{1}{3} - \frac{1}{5} = \frac{2}{15}$

$\Rightarrow \quad d = \frac{1}{15} \quad \therefore \quad a = \frac{1}{15}$

$\therefore \quad S_{15} = \left(\frac{a+T_{15}}{2}\right)15$

$= \left(\frac{1/15+1}{2}\right)\times 15 \quad (\because T_{15} = 1)$

$= \frac{\left(\frac{16}{15}\right)}{2}\times 15 = 8$

16. Let $n = 24$, then

$T_{24} = 1 + 23 \times 3 = 70$

$\therefore \quad S_{24} = \left(\frac{1+70}{2}\right)\times 24 = 71 \times 12 = 852$

$\therefore \quad S_{24} = 852 < 925$, hence wrong

Thus it is obvious that n must be greater than 24, which gives us $n = 25$, as per the choices given.

$\therefore \quad S_{25} = \left(\frac{a+T_{25}}{2}\right)25 \quad (\because T_{25} = T_{24} + d = 73)$

$= \left(\frac{1+73}{2}\right)\times 25 = 925$

Hence (d) is correct.

Alternatively : $S_n = 925 = \frac{n}{2}[2\times 1 + (n-1)3]$

$\Rightarrow \quad 1850 = 2n + 3n^2 - 3n$

$\Rightarrow \quad 3n^2 - n - 1850 = 0$

$\Rightarrow \quad 3n^2 - 75n + 74n - 1850 = 0$

$\Rightarrow \quad 3n(n-25) + 74(n-25) = 0$

$\Rightarrow \quad n = 25 \quad \text{or} \quad n = -\frac{74}{3}$

The only admissible value of $n = 25$
Since number of terms cannot be negative and as a fraction too.
Hence (d).

17. Best way is to go through options

$20 + 16 + 12 = 48,$

hence number of terms = 3

Again $20 + 16 + 12 + 8 + 4 + 0 + (-4) + (-8) = 48$

hence number of terms = 8

Thus choice (d) is most appropriate answer.

Alternatively :

$S_n = 48 = \frac{n}{2}[2\times 20 + (n-1)(-4)]$

$\Rightarrow \quad 96 = 40n - 4n^2 + 4n$

$\Rightarrow \quad n^2 - 11n + 24 = 0$

$\Rightarrow \quad (n-8)(n-3) = 0$

$\Rightarrow \quad n = 8 \quad \text{or} \quad n = 3$

Hence (d) is the correct answer.

18. $p = (r-2d), \quad q = (r-d), \quad r = r$

$s = (r+d), \quad t = (r+2d)$

$\therefore \quad p + r + t = 3r = -12$

$\Rightarrow \quad r = -4$

Again $p \cdot q \cdot r = 8$

$\Rightarrow \quad p \cdot q = -2 \quad (\because r = -4)$

$\Rightarrow \quad (r-2d)(r-d) = -2$

$\Rightarrow \quad (-4-2d)(-4-d) = -2$

$\Rightarrow \quad (2+d)(4+d) = -1$

$\Rightarrow \quad d^2 + 6d + 9 = 0$

$\Rightarrow \quad (d+3)^2 = 0$

$\Rightarrow \quad d = -3$

$\therefore \quad p = (r-2d) = -4 - 2\times(-3) = 2$

Alternatively : Go through options.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Let us consider choice (a)

If $p + r + t = -12$

$\Rightarrow$ $r = -4$ ($\because$ r is the A.M. of p, r, t)

Now if $p = 3$, then

p	q	r	s	t
3	$-\frac{1}{2}$	-4	$-\frac{15}{2}$	-11

$\therefore$ $p \cdot q \cdot r = 3 \times \left(-\frac{1}{2}\right) \times (-4) = 6$, which is wrong.

Again consider choice (b)

p	q	r	s	t
2	-1	-4	-7	-10

$\therefore$ $p \cdot q \cdot r = 2 \times (-1) \times (-4) = 8$, which is correct.

Hence (b) is the right choice.

19. $l = a + (n-1)d$

$l = 7 + (n-1)3$

$\Rightarrow$ $l = 3n + 4$

Let us assume option (d), then

$102 = 3n + 4$

$\Rightarrow n = \frac{98}{3}$, which is inadmissible

Since number of terms can never be a fraction.

Now, assume option (b)

$79 = 3n + 4$

$\Rightarrow$ $n = 25$

$\therefore$ $S_{25} = \left(\frac{7 + 79}{2}\right) \times 25 = 1075$

but since $S_{25} = 1075 < 1242$

Hence l must be greater than 79, therefore choice (c) is possible.

Let us consider choice (c).

$l = 85 = 3n + 4$

$\Rightarrow$ $n = 27$

$\therefore$ $S_{27} = \left(\frac{7 + 85}{2}\right) \times 27 = 1242$

Hence (c) is correct.

Alternatively : $S_n = \frac{n}{2}[2a + (n-1)d]$

$1242 = \frac{n}{2}[14 + (n-1)3]$

$\Rightarrow$ $n(3n + 11) = 2484$

$\Rightarrow$ $3n^2 + 11n - 2484 = 0$

$\Rightarrow$ $3n^2 - 81n + 92n - 2484 = 0$

$\Rightarrow$ $3n(n - 27) + 92(n - 27)$

$\Rightarrow$ $n = 27$ or $n = -\frac{92}{3}$

Hence $n = 27$ is the only admissible value.

$\therefore$ $l = a + (n-1)d = 7 + 26 \times 3 = 85$

20. The required numbers are 56, 63, 70, ..., 5551

number of terms = 786

$\therefore$ $S_{786} = 56 + 63 + \ldots + 5551$

$= \left(\frac{56 + 5551}{2}\right) \times 786 = 2203551$

21. $T_1 = a = -14$

$T_5 = a + 4d = 2$

$d = 4$

$\Rightarrow$

Now, $S_n = \frac{n}{2}[2a + (n-1)d]$

$40 = \frac{n}{2}[-28 + (n-1)4]$

$\Rightarrow$ $n^2 - 8n - 20 = 0$

$\Rightarrow$ $n = 10$ or $n = -2$

Since number of terms can not be negative, hence $n = 10$

Alternatively : Since S_n is very small, therefore we can solve it physically within seconds.

$T_1 = -14, T_5 = 2$

T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8	T_9	T_{10}
-14	-10	-6	-2	2	6	10	14	18	22
				0					40

$\Rightarrow$ $T_1 + T_2 + \ldots + T_{10} = 0 + 40 = 40$

22 $S_n = \left(\frac{a + l}{2}\right)n$

$9153 = \left(\frac{-7 + 233}{2}\right)n$

$\Rightarrow$ $9153 = 113 \times n$

$\Rightarrow$ $n = 81$

23. Let the three consecutive numbers in A.P. be p, q, r such that

$p = (q - d)$ and $r = (q + d)$

$\therefore$ $p + q + r = 12$

$\Rightarrow$ $q = 4$

and $p^3 + q^3 + r^3 = 408$

$\Rightarrow$ $p^3 + r^3 = 344$ ($\because$ $q = 4$)

and $p + q + r = 12$

$\Rightarrow$ $p + r = 8$

Now, let us assume $p = 1$, then $r = 7$

$\therefore$ $p^3 + r^3 = 344$, which is true.

Hence $p = 1$, $q = 4$ and $r = 7$

$\therefore$ $p \cdot q \cdot r = 1 \times 4 \times 7 = 28$

NOTE Another possible value of $p = 7$ and $r = 1$.

Alternatively : $p + q + r = 12$

$\Rightarrow$ $q = 4$ ($\because$ q is A.M. of p, q, r)

$\Rightarrow$ $p + r = 8$

and $p^3 + q^3 + r^3 = 408$

$\Rightarrow$ $p^3 + r^3 = 344$ ($\because$ $q^3 = 64$)

Now, $(p + r)^3 = p^3 + r^3 + 3pr(p + r)$

$(8)^3 = 344 + 3pr(8)$

$\Rightarrow$ $pr = 7$

$\therefore$ $pqr = q \cdot pr = 4 \times 7 = 28$

Alternatively : $p = (q - d)$

$q = q$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$r = (q + d)$

$(q - d) + q + (q + d) = 12$

$\therefore \quad q = 4$

$\Rightarrow \quad (q - d)^3 + q^3 + (q + d)^3 = 408$

and $\quad q^3 + 2qd^2 = 136$

$\Rightarrow \quad 8d^2 = 72 \qquad (\because\ q = 4)$

$\Rightarrow \quad d = \pm 3$

$\Rightarrow \quad p = 1 \text{ and } r = 7 \qquad$ when $d = +3$

$\therefore \quad p = 7 \text{ and } r = 1 \qquad$ when $d = -3$

and $\quad p \cdot q \cdot r = 1 \times 4 \times 7 = 7 \times 4 \times 1 = 28$

$\therefore$

24. Go through options.

Let us consider choice (c)

for $n = 1 \quad S_1 = (1)^3 + (1 - 1)^3 = 1$

for $n = 2 \quad S_2 = (2)^3 + (2 - 1)^3 = 9$

for $n = 3 \quad S_3 = (3)^3 + (3 - 1)^3 = 35$

Hence choice (c) is correct.

25. $S_{365} = \dfrac{365 \times 366}{2} \qquad \left(S_n = \dfrac{n(n+1)}{2}\right)$

or $\quad S_{365} = 66795$

26. $\dfrac{T_7}{T_3} = \dfrac{a + 6d}{a + 2d} = \dfrac{12}{5}$

$\Rightarrow \quad 5(a + 6d) = 12(a + 2d)$

$\Rightarrow \quad \dfrac{a}{d} = \dfrac{6}{7}$

Now, $\quad \dfrac{T_{13}}{T_4} = \dfrac{a + 12d}{a + 3d}$

$= \dfrac{\frac{a}{d} + 12}{\frac{a}{d} + 3} = \dfrac{\frac{6}{7} + 12}{\frac{6}{7} + 3} = \dfrac{10}{3}$

27. The numbers which are even and divisible by 5 also are 10, 20, 30 ... 1000.

$\therefore \quad S_{100} = 10 + 20 + 30 + \ldots + 1000$

$= 10(1 + 2 + 3 + \ldots + 100)$

$= 10 \times \dfrac{100 \times 101}{2} = 50500$

28. $m \cdot T_m = n \cdot T_n$

$m[a + (m - 1)d] = n[a + (n - 1)d]$

$\Rightarrow \quad m[a + md - d] = n[a + nd - d]$

$\Rightarrow \quad ma + m^2d - md = na + n^2d - nd$

$\Rightarrow \quad (m - n)a = (n^2 - m^2)d + (m - n)d$

$\Rightarrow \quad (m - n)a = -(m + n)(m - n)d + (m - n)d$

$\Rightarrow \quad (m - n)a = (m - n)d[1 - (m + n)]$

$\Rightarrow \quad a = [1 - (m + n)]d$

Now $\quad T_{(m+n)} = a + [(m + n) - 1]d$

$= [1 - (m + n)]d + [(m + n) - 1]d = 0$

29. $S_{15} = \dfrac{15}{2}[2a + 14d] = 105$

$\Rightarrow \quad 2a + 14d = 14 \qquad \ldots(1)$

and $\quad S_{30} = \dfrac{30}{2}[2a + 29d] = 105 + 780 = 885$

$\Rightarrow \quad 2a + 29d = 59 \qquad \ldots(2)$

$\therefore$ from equation (1) and (2)

$15d = 45 \quad \Rightarrow \quad d = 3$

30. $a \to 2$

and $\quad S_5 = \dfrac{1}{4}(S_{10} - S_5)$

$\Rightarrow \quad 5(S_5) = S_{10}$

$\Rightarrow \quad 5\left[\dfrac{5}{2}(4 + 4d)\right] = \dfrac{10}{2}[4 + 9d]$

$\Rightarrow \quad d = -6$

$\therefore \quad T_{20} = a + 19d$

$= 2 + 19 \times (-6) = -112$

31. $S_6 = 42$

$\Rightarrow \quad \dfrac{6}{2}[2a + 5d] = 42$

$\Rightarrow \quad 2a + 5d = 14 \qquad \ldots(1)$

Now, $\quad \dfrac{T_{10}}{T_{30}} = \dfrac{a + 9d}{a + 29d} = \dfrac{1}{3}$

$\Rightarrow \quad a = d$

$\therefore$ from equation (1)

$a = d = 2$

$\therefore \quad T_{40} = 2 + 39 \times 2 = 80$

32. $\dfrac{S_n}{S'_n} = \dfrac{7n + 1}{4n + 27}$

$\Rightarrow \quad \dfrac{\frac{n}{2}[2a + (n - 1)d]}{\frac{n}{2}[2a' + (n - 1)d']} = \dfrac{7n + 1}{4n + 27}$

$\Rightarrow \quad \dfrac{2a + (n - 1)d}{2a' + (n - 1)d'} = \dfrac{7n + 1}{4n + 27} \qquad \ldots(1)$

Now, $\quad \dfrac{T_{11}}{T'_{11}} = \dfrac{a + 10d}{a' + 10d'}$

Substituting $n = 21$ in equation (1), we get

$\dfrac{2a + 20d}{2a' + 20d'} = \dfrac{148}{111}$

$\Rightarrow \quad \dfrac{a + 10d}{a' + 10d'} = \dfrac{148}{111} = \dfrac{4}{3}$

33. $a + b + c = 15$

or $\quad (b - d) + b + (b + d) = 15$

$d \to$ common difference

$\Rightarrow \quad b = 5$

Again $\quad a^2 + b^2 + c^2 = 93$

$\Rightarrow \quad (b - d)^2 + b^2 + (b + d)^2 = 93$

$\Rightarrow \quad (b - d)^2 + (b + d)^2 = 68$

$\Rightarrow \quad (5 - d)^2 + (5 + d)^2 = 68$

$\Rightarrow \quad 2(5^2 + d^2) = 68$

$\Rightarrow \quad 25 + d^2 = 34$

$\Rightarrow \quad d = \pm 3$

$\therefore$ if $d = 3$, then $a = 2$ and $c = 8$

if $d = -3$, then $a = 8$ and $c = 2$

Therefore the greatest value is 8.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yourmathbook.wordpress.com
For More Visit : www.LearnEngineering.in

Alternatively : Go through option.
Since middle term $b = 5$
and let $c = 10$, then $a = 0$
$\therefore \quad a^2 + b^2 + c^2 = 125 \neq 93$, hence wrong
Again let $c = 8$, then $a = 2$
$\therefore \quad a^2 + b^2 + c^2 = 4 + 25 + 64 = 93$ hence correct.

34. $T_n = 4n - 1$
$\therefore \quad T_{30} = 119$
$\therefore \quad T_1 = 3, \quad T_2 = 7, \quad T_3 = 11, \ldots, 8, \quad T_{30} = 119$
$\therefore \quad S_{30} = \left(\frac{T_1 + T_{30}}{2}\right) \times 30 = \left(\frac{3 + 119}{2}\right) \times 30$
or $\quad S_{30} = 1830$

35. $S_n = 3n^2 + 5n$
$T_n = S_n - S_{(n-1)}$
$152 = [3n^2 + 5n] - [3(n-1)^2 + 5(n-1)]$
$152 = 6n + 2$
$\Rightarrow \quad n = 25$

Alternatively : Go through options.
$[3 \times (25)^2 + 5 \times 25] - [3 \times (24)^2 + 5 \times 24]$
$= (2000) - (1848) = 152$
Hence (d) is correct.

36. $S_n = \frac{n}{2}[2a + (n-1)d]$
$0 = \frac{n}{2}[196 + (n-1) \times (-7)]$
$\Rightarrow \quad 0 = n(203 - 7n)$
$\Rightarrow \quad 7n = 203 \quad \text{or} \quad n = 0$
$\Rightarrow \quad n = 29$
Since $n = 0$ is not acceptable.
The A.P. is 98, 91, 84, ... 0, ..., – 91, –.98
(Try to understand it as it is a problem of common sense, there is no need to calculate with the help of formula.)

37. It is possible only when all the terms of A.P. are non-negative.
$\therefore$ Let T_n be the least possible non-negative term of the A.P. then
$T_n \geq 0$
$\therefore \quad a + (n-1)d \geq 0$
$17 + (n-1) \times (-3) \geq 0$
$\Rightarrow \quad 20 - 3n \geq 0$
$\Rightarrow \quad n \leq \frac{20}{3}$
$\Rightarrow \quad n \leq 6$ (only integral part is considerable)
$\therefore \quad n = 6$
$\therefore \quad T_6 = 17 + 0.5 \times (-3)$
$T_6 = 2$
$\therefore \quad S_6 = \left(\frac{17 + 2}{2}\right) \times 6 = 57$
(The required A.P. is 17, 14, 11, 8, 5, 2).

38. The least possible sum can be obtained only when all the terms of A.P. be non-positive.
Let the last term be T_n which must be non-positive.
$\therefore \quad T_n \leq 0$
$a + (n-1)d \leq 0$
$-23 + (n-1)4 \leq 0$
$4n \leq 27$
$n \leq 6$
$n = 6$
$\therefore \quad T_n = T_6 = -23 + 5 \times 4 = -3$
$\therefore \quad S_6 = \left[\frac{(-23) + (-3)}{2}\right] \times 6 = -78$
(The required A.P. is – 23, – 19, – 15, – 11, – 7, – 3).

39. The required numbers are 1017, 1035, 1053, ..., 9999
Total number of terms = 500 $\quad \left[\because \ n = \left(\frac{9999 - 1017}{18}\right) + 1\right]$
$\therefore$ Sum of all the numbers $= \left(\frac{1017 + 9999}{2}\right) \times 500$
$= 2754000$

40. Let an A.P. be 3, 7, 11, 15, 19, 23, 27, ...
and Let $p = 2, \quad q = 3, \quad r = 7$
then $T_p = a = 7$
$T_q = b = 11$
$T_r = c = 27$
$\therefore \ p(b - c) + q(c - a) + r(a - b)$
$= 2(11 - 27) + 3(27 - 7) + 7(7 - 11)$
$= -32 + 60 - 28 = 0$
Hence (a) is true.

41. Let an A.P. be 2, 5, 8, 11, 14, 17, 20, 23, ...
and let $p = 3, q = 5, r = 6$
then $S_p = a = (2 + 5 + 8) = 15$
and $S_q = b = (2 + 5 + 8 + 11 + 14) = 40$
and $S_r = c = (2 + 5 + \ldots + 14 + 17) = 57$
$\therefore \frac{a}{p}(q - r) + \frac{b}{q}(r - p) + \frac{c}{r}(p - q)$
$= \frac{15}{3}(5 - 6) + \frac{40}{5}(6 - 3) + \frac{57}{6}(3 - 5)$
$= -5 + 24 - 19 = 0$
Hence (c) is correct.

42. Let p, q, r, s be the four consecutive terms in A.P. then
$p = (m - 3d)$
$q = (m - d)$
$r = (m + d)$
$s = (m + 3d)$
$\therefore \ p + q + r + s = (m - 3d) + (m - d) + (m + d) + (m + 3d) = 20$
$\Rightarrow \quad 4m = 20$
$\Rightarrow \quad m = 5 \quad \ldots(1)$
Again $(m - 3d)(m + 3d) : (m - d)(m + d) = 2 : 3$
$\Rightarrow \quad \frac{m^2 - 9d^2}{m^2 - d^2} = \frac{2}{3}$
$\Rightarrow \quad m = 5d$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

$d = 1$ (from equation 1)

$\therefore$ Hence $(m - 3d)(m + 3d) = (5 - 3)(5 + 3) = 16$

Alternatively : Since p, q, r and s are in A.P.

$p + q + r + s = 20$

and $p + s = 10$ and $q + r = 10$

$\Rightarrow$ Let $p = 1$, then $s = 9$

$\therefore$ $q = 1 + \frac{8}{3} = \frac{11}{3}$ $\left(\because\ d = \frac{8}{3}\right)$

and $r = \frac{11}{3} + \frac{8}{3} = \frac{19}{3}$

$\therefore$ $\frac{p \cdot s}{q \cdot r} = \frac{1 \times 9}{\frac{11}{3} \times \frac{19}{3}} = \frac{9 \times 9}{11 \times 19} = \frac{81}{209} \neq \frac{2}{3}$

Hence the values of p, q, r and s are wrong.

Again if $p = 2$, then $s = 8$

$\therefore$ $q = 2 + 2 = 4$ $(\because\ d = 2)$

and $r = 4 + 2 = 6$

$\therefore$ $\frac{p \cdot s}{q \cdot r} = \frac{2 \times 8}{4 \times 6} = \frac{2}{3}$, hence true.

Thus the values of p, q, r and s are correct.

Hence $p \cdot s = 2 \times 8 = 16$

43. $S_p = \frac{p}{2}[2a + (p - 1)d] = q$ ...(1)

and $S_q = \frac{q}{2}[2a + (q - 1)d] = p$ ...(2)

from (1) $2a + (p - 1)d = \frac{2q}{p}$ ...(3)

and from (2) $2a + (q - 1)d = \frac{2p}{q}$ ...(4)

subtracting equation (4) from equation (3), we get

$(p - q)d = \frac{-2(p + q)(p - q)}{pq}$

$\Rightarrow$ $d = \frac{-2(p + q)}{pq}$

Now, from equation (3)

$2a + (p - 1)\left(\frac{-2(p + q)}{pq}\right) = \frac{2p}{q}$

$\Rightarrow$ $a = \frac{p^2 + q^2 + pq - (p + q)}{pq}$

$\therefore$ $S_{p+q} = \frac{(p + q)}{2}[2a + (p + q - 1)d]$

$= \frac{(p + q)}{2}\left[\frac{2(p^2 + q^2 + pq - (p + q))}{pq} - \frac{2(p + q - 1)(p + q)}{pq}\right]$

$= \frac{(p + q)}{2}\left[\frac{2}{pq}(p^2 + q^2 + pq - p - q - p^2 - q^2 - pq + p + q - pq)\right]$

$= \frac{(p + q)}{pq}(-pq)$

$S_{(p+q)} = -(p + q)$

INTRODUCTORY EXERCISE 18.2

1. Every even term is equal to 2

common ratio of the G.P. is -1

2. $r = -2$ $\left(\because\ r = \frac{\frac{1}{4}}{-\frac{1}{8}} = \frac{\frac{-1}{2}}{\frac{1}{4}} = -2\right)$

$\therefore$ $T_7 = \frac{-1}{8} \cdot (-2)^6$

$= -\frac{1}{8} \times 64 = -8$

3. $T_5 = a_1 \cdot r^4 = a$

$T_8 = a_1 \cdot r^7 = b$

$T_{11} = a_1 \cdot r^{10} = c$

$\therefore$ Let $b^2 = ac$

then $(a_1 r^7)^2 = a_1 r^4 \cdot a_1 r^{10}$

$\Rightarrow$ $a_1^2 r^{14} = a_1^2 r^{14}$

Hence (b) is correct.

4. $T_5 = ar^4 = 32$

$T_{12} = ar^{11} = 4096$

$\therefore$ $r^7 = \frac{4096}{32} = 2^7$

$\Rightarrow$ $r = 2$

$\therefore$ $a = 2$ $(\because\ ar^4 = 32)$

$\therefore$ $T_n = ar^{n-1} = 2 \cdot 2^{n-1} = 2^{1 + n - 1} = 2^n$

5. $S_n > 10^6$, for least possible $n \in I^+$

then, $\frac{5(2^n - 1)}{(2 - 1)} > 10^6$

$\Rightarrow$ $2^n - 1 > 2 \times 10^5$

$\Rightarrow$ $2^n > 200001$

at $n = 17$, $2^n = 2^{17} = 131072$

and at $n = 18$, $2^n = 2^{18} = 262144$

Hence the least possible value of $n = 18$

6. Let a, b be the two numbers then

A.M. $= \frac{a + b}{2} = 15$

$\Rightarrow$ $a + b = 30$

and G.M. $= \sqrt{ab} = 12$

$\Rightarrow$ $ab = 144$

Now go through options

let $a = 6$, then $b = 24$

$\therefore$ $a \cdot b = 6 \times 24$

Hence choice (c) is correct.

Alternatively : $(a - b)^2 = (a + b)^2 - 4ab$

$= (30)^2 - 4 \times 144$

$= 900 - 576 = 324$

$\Rightarrow$ $a - b = \pm 18$

$\therefore$ $(a, b) = (6, 24)$ or $(24, 6)$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

7. Let a, b, c be the three numbers in G.P. then

$$a+b+c=38$$

and $$abc=1728$$

$\Rightarrow$ $$b=(abc)^{1/3}=(1728)^{1/3}=12$$

$\therefore$ $$a\times c=144$$

and $$a+c=26$$

Now, go through options

Let $c=18$, then $a=8$

$\therefore$ $$a\cdot c=8\times 18=144$$

Hence choice (b) is correct

Alternatively : Let the three numbers in G.P. be

$$\frac{a}{r}, a, ar$$

$\therefore$ $$\frac{a}{r}+a+ar=38 \quad ...(1)$$

and $$\frac{a}{r}\cdot a\cdot ar=1728 \quad ...(2)$$

$\Rightarrow$ $$a=12$$

$\therefore$ $$\frac{12}{r}+12+12r=38$$

or $$\frac{6}{r}+6+6r=19$$

$\Rightarrow$ $$6r^2-13r+6=0$$

$\Rightarrow$ $$6r^2-9r-4r+6=0$$

$\Rightarrow$ $$(2r-3)(3r-2)=0$$

$\Rightarrow$ $$r=\frac{3}{2} \text{ or } r=\frac{2}{3}$$

$\therefore$ Three number in G.P. are 8, 12, 18 or 18, 12, 8.

Hence (b) is correct.

8. $$\frac{S_3}{S_6}=\frac{125}{152}=\frac{\frac{a(r^3-1)}{(r-1)}}{\frac{a(r^6-1)}{(r-1)}}=\frac{r^3-1}{r^6-1}$$

$\Rightarrow$ $$\frac{r^3-1}{(r^3)^2-1}=\frac{125}{152}$$

$\Rightarrow$ $$\frac{(r^3-1)}{(r^3+1)(r^3-1)}=\frac{125}{152}$$

$\Rightarrow$ $$\frac{1}{r^3+1}=\frac{125}{152}$$

$\Rightarrow$ $$r^3+1=\frac{152}{125}$$

$\Rightarrow$ $$r^3=\frac{27}{125}$$

$\Rightarrow$ $$r=\frac{3}{5}$$

Alternatively : Go through options.

9. $$a+b+c=14$$

Let $$abc=125$$

$\Rightarrow$ $$b=5$$

let $a=1$, then $c=25$

$\therefore a+b+c=31$, which is wrong.

Again let $$a\cdot b\cdot c=64$$

$\Rightarrow$ $$b=4.$$

Let $a=1$ then $c=16$

$\Rightarrow a+b+c=21$, which is wrong

again if $a=2$ then $c=8$

$\therefore$ $$a+b+c=2+4+8=14$$

Hence choice (b) may be correct

$$a=2, \quad b=4, \quad c=8$$

$$a'=3, \quad b'=5, \quad c'=7$$

$\because a', b'$ and c' are in A.P.

Hence choice (b) is correct.

Alternatively : $a+b+c=14$

since $(a+1), (b+1)$ and $(c-1)$ are in A.P.

$\therefore$ $$(b+1)-(a+1)=(c-1)-(b+1)$$

$\Rightarrow$ $$2b+2=a+c$$

Now, from equation (1)

$$3b+2=14$$

$\Rightarrow$ $$b=4$$

Again $$a=\frac{b}{r} \text{ and } c=br$$

$\therefore$ $$a+b+c=14$$

$\Rightarrow$ $$\frac{b}{r}+b+br=14$$

$\Rightarrow$ $$\frac{4}{r}+4+4r=14$$

$\Rightarrow$ $$2r^2-5r+2=0$$

$\Rightarrow$ $$r=2 \text{ or } r=\frac{1}{2}$$

$\therefore$ $$a\cdot b\cdot c=2\times 4\times 8=8\times 4\times 2=64$$

Shortcut : Since $b=4$

$\therefore$ $$abc=b^3 \qquad (\because a, b, c \text{ are in G.P.})$$

$\therefore$ $$abc=64$$

10. Let a, b, c, d, e be five consecutive terms in G.P. then

$$a=\frac{c}{r^2}, \quad b=\frac{c}{r}, \quad d=cr, \quad e=cr^2$$

$\therefore$ $$a\cdot b\cdot c\cdot d\cdot e=\frac{c}{r^2}\cdot\frac{c}{r}\cdot c\cdot cr\cdot cr^2=c^5$$

$\therefore$ $$a\cdot b\cdot c\cdot d\cdot e=4^5 \qquad (\because c=4$$

11. Let a, b, c be in G.P., then

$$a+b+c=21=3(7)=3(1+2+4)$$

and $a^2+b^2+c^2=189=9(21)=9(1+4+16)$

Hence common ratio $=2$

Alternatively : $a+ar+ar^2=21$

$\Rightarrow$ $$a(1+r+r^2)=21$$

and $$a^2+(ar)^2+(ar^2)^2=189$$

$\therefore$ $$a^2(1+r^2+r^4)=189$$

$\Rightarrow$ $$a^2[(1+r^2)^2-r^2]=189$$

$\Rightarrow$ $$a^2[(1+r^2+r)(1+r^2-r)]=189$$

Dividing equation (2) by equation (1), we get

$$a(1-r+r^2)=9$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Dividing equation (1) by equation (3), we get

$$\frac{1+r^2+r}{1+r^2-r}=\frac{21}{9}=\frac{7}{3}$$

$$\Rightarrow \quad \frac{(1+r^2)+r}{(1+r^2)-r}=\frac{7}{3}$$

$$\Rightarrow \quad \frac{1+r^2}{r}=\frac{5}{2}$$

(By componendo and dividendo)

$$\Rightarrow \quad 2r^2-5r+2=0$$

$$\Rightarrow \quad r=2 \text{ or } r=\frac{1}{2}$$

12. $T_1+T_3=a+ar^2=15$ ...(1)

and $T_5+T_7=ar^4+ar^6=240$ ...(2)

From equation (1) $a(1+r^2)=15$

and equation (2) becomes $ar^4(1+r^2)=240$

$$\therefore \quad r^4=16$$

$$\Rightarrow \quad r=\pm 2$$

$\therefore a=3$ (from equation (1))

$$\therefore \quad T_9=a\cdot r^8=3\cdot(2)^8=768$$

13. $a+b+c=42$

$abc=512$

$\Rightarrow \quad b=8 \quad (\because (abc)^{1/3}=b=8)$

$\therefore \quad ac=64$

and $a+c=34$

Now, go through options (or by hit and trial)

$\therefore$ either $a=2$, then $c=32$

or $a=32$ then $c=2$

Hence, (c) is correct.

14. $a+b+c=70$

and $4a$, $5b$, $4c$ are in A.P.

$$\therefore \quad 10b=4(a+c)$$

$$\Rightarrow \quad a+c=\frac{5b}{2}$$

$$\therefore \quad a+b+c=\frac{7b}{2}=70$$

$$\Rightarrow \quad b=20$$

$$\therefore \quad a+c=50$$

By hit and trial $a=10$ and $c=40$

Now $4a=40$, $5b=100$ and $4c=160$

Hence, $a\times b\times c=10\times 20\times 40=8000$

Alternatively : $\frac{b}{r}+b+br=70$

$$\Rightarrow \quad \frac{20}{r}+20+20r=70$$

$$\Rightarrow \quad \frac{2}{r}+2+2r=7$$

$$\Rightarrow \quad 2r^2-5r+2=0$$

$$\Rightarrow \quad r=2 \text{ or } r=\frac{1}{2}$$

$\therefore a, b, c = 10, 20, 40$ or $40, 20, 10$

$$\therefore \quad abc=8000$$

15. $a+b+c+d=312$

$a+ar+ar^2+ar^3=312$

and $a+d=252$

and $b+c=312-252=60$

$\Rightarrow \quad a+ar^3=252$

and $ar+ar^2=60$

$\Rightarrow \quad a(1+r^3)=252$

and $a(r+r^2)=60$

$$\therefore \quad \frac{a(1+r^3)}{a(r+r^2)}=\frac{252}{60}=\frac{21}{5}$$

$$\Rightarrow \quad \frac{1+r^3}{r+r^2}=\frac{21}{5}$$

$$\Rightarrow \quad \frac{(1+r)(1+r^2-r)}{r(1+r)}=\frac{21}{5}$$

$(\because a^3+b^3=(a+b)(a^2+b^2-ab))$

$$\Rightarrow \quad \frac{1+r^2-r}{r}=\frac{21}{5}$$

$$\Rightarrow \quad 5r^2-26r+5=0$$

$$\Rightarrow \quad r=5 \text{ or } r=\frac{1}{5}$$

$\therefore \quad a=2$ and $ar\times ar^2=500$ (when $r=5$)

and $a=250$ and $ar\times ar^2=500$ (when $r=\frac{1}{5}$)

Alternatively : Go through options.

16. Distance travelled before the first hit = 16 m

Distance travelled before the second hit $=16+\frac{1}{2}\times 16\times 2$

$=(16+16)$ m

Distance travelled before the third hit

$=16+16+\frac{1}{2}\times 8\times 2$

$=(16+16+8)$ m

Distance travelled before the fourth hit

$=16+16+8+\frac{1}{2}\times 4\times 2$

$=(16+16+8+4)$ m

and so on.

Hence the required distance

$$=16+16+8+4+2+1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}$$

$$=16+\left(16+8+4+\ldots+\frac{1}{16}\right)$$

$$=16+\frac{16\left(1-\left(\frac{1}{2}\right)^9\right)}{1-\frac{1}{2}}=16+16\times\left(\frac{511}{512}\right)\times\frac{2}{1}$$

$$=16+\frac{511}{16}=47\frac{15}{16}$$

17. $S_n=7+77+777+\ldots$ to n terms

$=7(1+11+111+\ldots$ to n terms$)$

$=\frac{7}{9}(9+99+999+\ldots$ to n terms$)$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$$= \frac{7}{9}\{(10-1)+(100-1)+(1000-1)+\ldots \text{to } n \text{ terms}\}$$

$$= \frac{7}{9}\{(10+100+1000+\ldots \text{ to } n \text{ terms}) - (1+1+1+\ldots \text{to } n \text{ terms})\}$$

$$= \frac{7}{9}\left\{\frac{10\cdot(10^n-1)}{10-1} - n\right\}$$

$$= \frac{7}{9}\left\{\frac{10}{9}(10^n-1) - n\right\}$$

Alternatively : Best way is to go through options.

Consider $n = 2$, then $S_2 = 7 + 77 = 84$

Now, put $n = 2$ in choice (b).

NOTE This type of questions necessarily be solved by options and must take at least two values for $n = 2, 3$ etc. as a measure of safety.

18. It is very similar to the previous question.

19. Let a G.P. be 1, 3, 9, 27, 81, 243, 729, ...

Consider $p = 2, \quad q = 5, \quad r = 6$

$\therefore \quad T_p = 3 = a$

$T_q = 81 = b$

$T_r = 243 = c$

$\therefore \quad a^{q-r}\cdot b^{r-p}\cdot c^{p-q} = 3^{(5-6)}\cdot 81^{(6-2)}\cdot 243^{(2-5)}$

$= 3^{-1}\cdot 81^4\cdot 243^{-3}$

$= 3^{-1}\cdot 3^{16}\cdot 3^{-15}$

$= 3^{-16}\cdot 3^{16} = 3^0 = 1$

20. Let the G.P. be 1, 2, 4, 8, 16, 32, 64, ...

Let $x = 3, \quad y = 5$ and $z = 6$

$\therefore \quad T_x = 4 = a$

$T_y = 16 = b$

$T_z = 32 = c$

$\therefore (y-z)\log a + (z-x)\log b + (x-y)\log c$

$= -1\cdot\log 4 + 3\log 16 + (-2)\log 32$

$= \log 2^{-2} + \log 2^{12} + \log 2^{-10}$

$= \log 2^0 = \log 1 = 0$

21. Let a, b, c, d be four numbers such that a, b, c are in A.P. and b, c, d are in G.P.

$\therefore \quad a + c = 2b \quad \ldots(1)$

and $\quad bd = c^2 \quad \ldots(2)$

Also $\quad a + c = 2 \quad \ldots(3)$

and $\quad b + d = 26 \quad \ldots(4)$

$\Rightarrow \quad b = 1$

from (4), $d = 25$

$\therefore$ from (2) $c = \pm 5$

when $c = 5$, then $a = -3$

when $c = -5$, then $a = 7$

Hence the four numbers are $-3, 1, 5, 25$ or $7, 1, -5, 25$

22. $S_n = 1 + 3 + 7 + 15 + \ldots + t_n \quad \ldots(1)$

and $S_n = 1 + 3 + 7 + \ldots + t_{n-1} + t_n \quad \ldots(2)$

subtracting, we get

$0 = (1 + 2 + 4 + 8 + \ldots + \text{to } n \text{ terms}) - t_n$

$\Rightarrow \quad t_n = 1 + 2 + 4 + 8 + \ldots \text{to } n \text{ terms}$

$\Rightarrow \quad t_n = \frac{1\cdot(2^n-1)}{(2-1)} = 2^n - 1$

again $S_n = t_1 + t_2 + t_3 + \ldots + t_n$

$= (2-1) + (2^2-1) + (2^3-1) + \ldots + (2^n-1)$

$= (2 + 2^2 + 2^3 + \ldots 2^n) - (n)$

$= \frac{2(2^n-1)}{(2-1)} - n$

$= 2(2^n-1) - n$

Alternatively : Go through options.

23. $S = \frac{1}{2} + \frac{3}{2^2} + \frac{5}{2^3} + \frac{7}{2^4} + \ldots + \frac{2n-3}{2^{n-1}} + \frac{2n-1}{2^n}$

and $\frac{1}{2}S = \frac{1}{2^2} + \frac{3}{2^3} + \frac{5}{2^4} + \ldots + \frac{2n-3}{2^n} + \frac{2n-1}{2^{n+1}}$

subtracting we get,

$\frac{1}{2}S = \frac{1}{2} + \frac{2}{2^2} + \frac{2}{2^3} + \ldots + \frac{2}{2^n} - \frac{2n-1}{2^{n+1}}$

$= \frac{1}{2} + 2\left(\frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \ldots + \frac{1}{2^n}\right) - \frac{2n-1}{2^{n+1}}$

$= \frac{1}{2} + \frac{2\cdot\frac{1}{2^2}\left\{1 - \left(\frac{1}{2}\right)^{n-1}\right\}}{1-\frac{1}{2}} - \frac{2n-1}{2^{n+1}}$

$\therefore \quad S = 3 - \frac{2n+3}{2^n}$

24. Go through options.

Alternatively : $11 + 102 + 1003 + \ldots$

$= (10+1) + (100+2) + (1000+3)$

$= (10 + 100 + 1000 + \ldots) + (1 + 2 + 3 + \ldots)$

25. Go through options.

Alternatively : $T_n = 1 + 3 + 9 + \ldots$

or $\quad T_n = \frac{1}{2}(3^n - 1)$

$\therefore \quad S_n = t_1 + t_2 + t_3 + \ldots + t_n$

$S_n = \frac{1}{2}[(3^1 + 3^2 + 3^3 + \ldots + 3^n) - n]$

$= \frac{1}{2}\left[\frac{3(3^n-1)}{2} - n\right]$

$= \frac{1}{4}[3(3^n-1) - 2n]$

$= \frac{3}{4}(3^n-1) - \frac{n}{2}$

26. $S_n = 2 + 5 + 14 + 41 + \ldots \text{to } n \text{ terms} \quad \ldots(1)$

and $S_n = 2 + 5 + 14 + 41 + \ldots \text{to } n \text{ terms} \quad \ldots(2)$

subtracting we get,

$0 = 2 + 3 + 9 + 27 + \ldots - t_n$

$\Rightarrow \quad t_n = 2 + 3 + 9 + 27 + \ldots$

$\Rightarrow \quad t_n = 2 + \{3 + 9 + 27 + \ldots \text{to } (n-1) \text{ terms}\}$

$\Rightarrow \quad t_n = 2 + \frac{3(3^{n-1}-1)}{2}$

$\therefore \quad S_n = t_1 + t_2 + t_3 + \ldots + t_n$

$S_n = \Sigma t_n = t_1 + t_2 + t_3 + \ldots + t_n$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

$$= \Sigma\left\{2 + \frac{3(3^{n-1}-1)}{2}\right\}$$

$$= \Sigma\left\{2 + \frac{3^n}{2} - \frac{3}{2}\right\}$$

$$= \Sigma\left\{\frac{1}{2} + \frac{3^n}{2}\right\}$$

$$= \left\{\frac{n}{2} + \frac{1}{2}(3 + 3^2 + 3^3 + \ldots + 3^n)\right\}$$

$$= \left\{\frac{n}{2} + \frac{1}{2}\left(\frac{3\cdot(3^n-1)}{2}\right)\right\}$$

$$= \left\{\frac{n}{2} + \frac{1}{4}3\cdot(3^n-1)\right\}$$

$$= \frac{3}{4}(3^n-1) + \frac{n}{2}$$

Alternatively : You should go through options. Consider $n = 1, 2$ or 3 and verify the correct option.

As, when you put $n = 1$, in the correct formula you will get $S_1 = 2$

and at $n = 2, S_2 = 7$ $(2 + 5)$

and at $n = 3, S_3 = 21$ $(2 + 5 + 14)$

27. $S_n = 1 + 2x + 3x^2 + 4x^3 + \ldots$

$x\cdot S_n = x + 2x^2 + 3x^3 + 4x^4 + \ldots$

subtracting we get

$$(1-x)S_n = (1 + x + x^2 + x^3 + \ldots + x^{n-1}) - nx^n$$

$$(1-x)S_n = \frac{1-x^n}{(1-x)} - nx^n$$

$$\Rightarrow \quad S_n = \frac{1-x^n}{(1-x)^2} - \frac{nx^n}{(1-x)}$$

Alternatively : Go through options.

28. Solve through options.

NOTE This is very similar to the previous problem.

29. Let $S_n = 1 + \frac{2}{3} + \frac{3}{3^2} + \frac{4}{3^3} + \ldots + \frac{n}{3^{n-1}}$

$$\therefore \quad \frac{1}{3}S_n = \frac{1}{3} + \frac{2}{3^2} + \frac{3}{3^3} + \frac{4}{3^4} + \ldots + \frac{n-1}{3^{n-1}} + \frac{n}{3^n}$$

subtracting we get,

$$\frac{2}{3}S_n = \left(1 + \frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \frac{1}{3^4} + \ldots\right) - \frac{n}{3^n}$$

$$\Rightarrow \quad \frac{2}{3}S_n = \left[\frac{1\left\{1-\left(\frac{1}{3}\right)^n\right\}}{1-\frac{1}{3}}\right] - \frac{n}{3^n}$$

$$\Rightarrow \quad \frac{2}{3}S_n = \frac{3}{2}\left\{1-\left(\frac{1}{3}\right)^n\right\} - \frac{n}{3^n}$$

$$\Rightarrow \quad S_n = \frac{9}{4} - \frac{3}{4}\left(\frac{3+2n}{3^n}\right)$$

30. Let $S_n = 3\cdot 2 + 5\cdot 2^2 + 7\cdot 2^3 + \ldots + (2n-1)\cdot 2^n$

$\therefore 2S_n = 3\cdot 2^2 + 5\cdot 2^3 + \ldots + (2n-1)\cdot 2^n + (2n+1)\cdot 2^{n+1}$

subtracting, we get

$-S_n = 3\cdot 2 + 2\cdot 2^2 + 2\cdot 2^3 + \ldots + 2\cdot 2^n - (2n+1)\cdot 2^{n+1}$

$= 3\cdot 2 + 2(2^2 + 2^3 + \ldots + 2^n) - (2n+1)\cdot 2^{n+1} = \text{etc}$

Now, simplify by yourself.

31. Go through options.

32.
$$t_n = n(2n+1)^2$$
$$= 4n^3 + 4n^2 + n$$
$$\therefore \quad S_n = \Sigma t_n = 4\Sigma n^3 + 4\Sigma n^2 + \Sigma n$$

then put $n = 20$

33.
$$t_n = \frac{n^2(n+1)^2}{4n^2} = \frac{1}{4}(n^2 + 2n + 1)$$
$$\therefore \quad S_n = \Sigma t_n = \frac{1}{4}[\Sigma n^2 + 2\Sigma n + n]$$

then put $n = 16$.

34. Let a, b, c and d be the required numbers, then a, b, c are in G.P. and b, c, d are in A.P.

a	b	c	d	
↓	↓	↓	↓	
a	b	$b+6$	$b+12$	(common difference = 6)
↓	↓	↓	↓	
$b+12$	b	$b+6$	$b+12$	($\because$ $a = d$)

Now, since a, b, c are in G.P.

$$\therefore \quad \frac{c}{b} = \frac{b}{a}$$
$$\Rightarrow \quad \frac{b+6}{b} = \frac{b}{b+12}$$
$$\Rightarrow \quad (b+6)(b+12) = b^2$$
$$\Rightarrow \quad b = -4$$
$$\therefore \quad c = b + 6 = -4 + 6 = 2$$

Alternatively : Go through options.

35. Let a be the first term and r be the common ratio of the G.P. then

$$S_\infty = a + ar + ar^2 + \ldots = 16 = \frac{a}{1-r}, |r| < 1$$

$$\therefore \quad \frac{a}{1-r} = 16 \qquad \ldots(1)$$

and $\quad a^2 + (ar)^2 + (ar^2)^2 + \ldots = 153\frac{3}{5}$

$$\Rightarrow \quad a^2 + a^2r^2 + a^2r^4 + \ldots = \frac{768}{5}$$

$$\Rightarrow \quad \frac{a^2}{1-r^2} = \frac{768}{5} \qquad \ldots(2)$$

dividing equation (2) by equation (1), we get

$$\frac{\frac{a^2}{(1-r^2)}}{\frac{a}{(1-r)}} = \frac{\frac{768}{5}}{16}$$

$$\Rightarrow \quad \frac{a}{1+r} = \frac{48}{5} \qquad \ldots(3)$$

dividing equation (1) by equation (3)

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yourshahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$$\frac{\frac{a}{(1-r)}}{\frac{a}{(1+r)}} = \frac{16}{\frac{48}{5}}$$

$$\Rightarrow \quad \frac{1+r}{1-r} = \frac{5}{3}$$

$$\Rightarrow \quad \frac{1}{r} = 4 \quad \text{(By componendo-dividendo)}$$

$$\Rightarrow \quad r = \frac{1}{4}$$

$$\therefore \quad a = 16 \times \left(1 - \frac{1}{4}\right) \text{ from equation (1)}$$

$$a = 12$$

$$\therefore \quad T_4 = ar^3 = 12 \times \left(\frac{1}{4}\right)^3 = \frac{3}{16}$$

36. $x = 1 + a + a^2 + a^3 + \ldots \infty$

$$\therefore \quad x = \frac{1}{1-a} \quad \because |a| < 1$$

Similarly, $y = \frac{1}{1-b}$

Let $S = 1 + ab + a^2b^2 + \ldots \infty$

$= 1 + ab + (ab)^2 + \ldots \infty$

$= \frac{1}{1-ab}$ $(\because |ab| < 1)$ ...(1)

Now, go through option

$$\frac{xy}{x+y-1} = \frac{\frac{1}{1-a} \times \frac{1}{1-b}}{\frac{1}{1-a} + \frac{1}{1-b} - 1} = \frac{1}{1-ab}$$

Hence, $S = 1 + ab + a^2b^2$

$$= \frac{1}{1-ab} = \frac{xy}{x+y-1}$$

37. His first payment = Rs. 100

His second payment = Rs. $90 = \left(100 \times \frac{9}{10}\right)$

His third payment = Rs. 81 etc. $= \left(90 \times \frac{9}{10}\right)$

∴ The annual payments are 100, 90, 81, ... which are in G.P. with common ratio $\frac{9}{10}$ (< 1)

Therefore the sum to infinity of this G.P.

$$= 100 + 90 + 81 + \ldots \infty$$

$$= \frac{100}{1 - \frac{9}{10}} = \frac{100}{\frac{1}{10}} = 1000$$

Hence the person can receive maximum amount of Rs. 1000.

38. $S_\infty = \frac{1}{4} - \frac{3}{16} + \frac{9}{64} - \frac{27}{256} + \ldots$

$$S_\infty = \frac{\frac{1}{4}}{1 - \left(-\frac{3}{4}\right)} = \frac{\frac{1}{4}}{\frac{7}{4}}$$

$$S_\infty = \frac{1}{7}$$

$\left(\because r = \frac{1}{4}\right)$

39. $a + ar = \frac{5}{3} \Rightarrow a(1+r) = \frac{5}{3}$ ...(1)

and $\frac{a}{1-r} = 3$ ...(2)

Dividing equation (1) by equation (2), we get

$$(1+r)(1-r) = \frac{5}{9}$$

$$\Rightarrow \quad 1 - r^2 = \frac{5}{9}$$

$$\Rightarrow \quad r^2 = \frac{4}{9} \Rightarrow r = \pm\frac{2}{3}$$

If $r = \frac{2}{3}$, then $a = 1$ and if $r = -\frac{2}{3}$, then $a = 5$

40. Distance covered by ball just before it hits the ground for the first time = 96 feet.

In the first round (i. e. going up and returning to the earth) ball covers the distance $= \left(96 \times \frac{2}{3}\right) \times 2$

In the third round distance covered by ball $= 96 \times \left(\frac{2}{3}\right)^2 \times 2$

In the fourth round distance covered by ball

$$= 96 \times \left(\frac{2}{3}\right)^3 \times 2 \text{ etc.}$$

NOTE In each round ball covers equal distance while going up and coming down to the earth).

Hence the required distance

$$= 96 + 2\left[96 \times \frac{2}{3} + 96 \times \frac{4}{9} + 96 \times \frac{8}{27} + \ldots\right]$$

$$= 96 + 96 \times 2\left[\frac{2}{3} + \frac{4}{9} + \frac{8}{27} + \ldots\right]$$

$$= 96 + 192\left(\frac{\frac{2}{3}}{1 - \frac{2}{3}}\right)$$

$$= 96 + 192\left(\frac{\frac{2}{3}}{\frac{1}{3}}\right)$$

$$= 96 + 192 \times 2 = 480 \text{ ft.}$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

INTRODUCTORY EXERCISE 18.3

1. Let A, G, H be the Arithmetic mean, Geometric mean and Harmonic mean of a and b respectively.

Then, we have

$$A = \frac{a+b}{2}, \quad G = \sqrt{ab}, \quad H = \frac{2ab}{a+b}$$

$$\text{A.H.} = \frac{a+b}{2} \times \frac{2ab}{a+b} = ab = G^2$$

$\therefore$

$$\text{Again } A - G = \frac{a+b}{2} - \sqrt{ab}$$

$$= \frac{a+b-2\sqrt{ab}}{2}$$

$$= (\sqrt{a})^2 + (\sqrt{b})^2 - 2\sqrt{ab}$$

$$= \frac{(\sqrt{a}-\sqrt{b})^2}{2} \geq 0$$

$$A \geq G$$

$\therefore$

Again since $\text{A.H.} = G^2 = G \times G$

$$H \leq G \qquad (\because A \geq G)$$

$\therefore$

i.e.

$$G \geq H$$

$\therefore$

$$A \geq G \geq H$$

Alternatively : Let $a = 6$ and $b = 24$

$$\therefore \quad \text{A.M.} = \frac{6+24}{2} = 15$$

and

$$\text{G.M.} = \sqrt{6 \times 24} = 12$$

and

$$\text{H.M.} = \frac{2 \times 6 \times 24}{6+24} = 9.6$$

$\therefore$

$$\text{A.M.} > \text{G.M.} > \text{H.M.}$$

Hence (c) is the correct option

NOTE The equality depends upon values as

$$a = 2 \text{ and } b = 2$$

then A.M. = 2 and G.M. = 2 and H.M. = 2

then A.M. = G.M. = H.M.

2. Let a, b, c, d, e are taken such that a, b, c are in A.P. and b, c, d are in G.P. and c, d, e are in H.P. then

$$a, b, c, d, e$$

$$2, 4, 6, 9, 18$$

$\therefore$

$$a \cdot e = c^2$$

Hence choice (b) is correct

3.

$$\frac{a+b}{2} = 10 \Rightarrow a + b = 20$$

$$\sqrt{a \cdot b} = 8 \Rightarrow ab = 64$$

$\therefore$

$$a, b = 16, 4 \text{ or } 4, 16$$

$\therefore$

$$\text{H.M.} = \frac{2 \times 4 \times 16}{(4+16)} = 6.4$$

NOTE $\text{G.M.} = \sqrt{\text{A.M.} \times \text{H.M.}}$

HINT Choices (a) and (b) are clearly wrong since H.M. ≤ G.M. ≤ A.M. Now you are left with choices (c) and (d) only.

LEVEL 1

1.

$$S_n = \frac{a(1-r^n)}{1-r}; \qquad r < 1$$

$$S_n = \frac{4095}{1024} = \frac{3(1-r^n)}{1-\frac{1}{4}}$$

$$\Rightarrow \quad \frac{4095}{1024} = 4(1-r^n)$$

$$\Rightarrow \quad \frac{4095}{4096} = 1 - r^n$$

$$\Rightarrow \quad r^n = \frac{1}{4096}$$

$$\Rightarrow \quad r^n = \left(\frac{1}{4}\right)^6$$

$$\Rightarrow \quad \left(\frac{1}{4}\right)^n = \left(\frac{1}{4}\right)^6 \qquad \left(\because r = \frac{1}{4}\right)$$

$$\Rightarrow \quad n = 6$$

2. Since $a > l$, therefore $r < 1$

Now, since $S_n = \frac{a-lr}{1-r}$ if $r < 1$

$\therefore$

$$\frac{5187}{625} = \frac{5 - \frac{32}{625}r}{1-r}$$

$$(1-r)\frac{5187}{625} = 5 - \frac{32r}{625} \Rightarrow r = \frac{2}{5}$$

3.

$$r = \frac{2\sqrt{2}}{2\sqrt{3}} = \frac{\frac{4}{\sqrt{3}}}{2\sqrt{2}} = \frac{\sqrt{2}}{\sqrt{3}} = \sqrt{\frac{2}{3}}$$

$\therefore$

$$S_\infty = \frac{a}{1-r}; \quad |r| < 1$$

$$= \frac{2\sqrt{3}}{1-\sqrt{\frac{2}{3}}} = \frac{2\sqrt{3} \times \sqrt{3}}{\sqrt{3}-\sqrt{2}}$$

$$= \frac{6}{\sqrt{3}-\sqrt{2}}$$

$$= 6(\sqrt{3}+\sqrt{2})$$

4. (Sum of odd terms − sum of even terms) = 72 − 63

$$\Rightarrow \frac{\text{number of terms}}{2} \times \text{common difference} = 9$$

$$\Rightarrow \quad nd = 18 \qquad \ldots(1)$$

($n \rightarrow$ number of terms, $d \rightarrow$ common difference)

Again

$$l - a = 16.5$$

$$\Rightarrow \quad (n-1)d = 16.5 \qquad \ldots(2)$$

$\therefore$ from (1) and (2), we get

$$d = 1.5$$

$\therefore$

$$n = 12$$

5. Go through option

$$\{\text{First Pocket}\} \quad \{\text{Second Pocket}\}$$

$$\{4, 7, 10\} \quad \{5, 7, 9\}$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yourshahboob.wordpress.com
For More Visit : www.LearnEngineering.in

6. $a \cdot b \cdot c = a \cdot ar \cdot ar^2 = 216$

$\Rightarrow \quad b = ar = 6 \quad \ldots(1)$

and $ab + bc + ac = a \cdot ar + ar \cdot ar^2 + a \cdot ar^2 = 126$

$\Rightarrow \quad a^2r + a^2r^3 + a^2r^2 = 126$

$\Rightarrow \quad a^2r + a^2r^2 = 90 \quad (\because ar = 6)$

$\Rightarrow \quad 6a + 36r = 90$

$\Rightarrow \quad a + 6r = 15 \quad \ldots(2)$

from equation (1) and (2), we get

$$\frac{6}{r} + 6r = 15$$

$\Rightarrow \quad 6r^2 - 15r + 6 = 0$

$\Rightarrow \quad r = 2 \quad or \quad \frac{1}{2}$

$\therefore \quad a, ar, ar^2 = 3, 6, 12 \text{ or } 12, 6, 3$

$\therefore \quad a + b + c = 21$

Alternatively: Go through options

7. Let a, b, c, d be in A.P. then

$$\frac{ad}{bc} = \frac{9}{10}$$

Also $\quad a + b + c + d = 36$

$\Rightarrow \quad (a + d) = (b + c) = 18$

Now, go through options.

8. Let a, b, c, d be four integers in A.P.

then $\quad a + d = b + c = 12$

Since $a \cdot b \cdot c \cdot d = 945$, it means one of the a, b, c and d has its unit digit 5.

∴Therefore the obvious possibility is $b = 5$

So, if $b = 5$, then $c = 7$

$\therefore \quad a = 3 \quad$ and $\quad d = 9$

$\therefore \quad a \cdot b \cdot c \cdot d = 3 \times 5 \times 7 \times 9 = 945$

Hence the values of a, b, c and d are correct.

Therefore $\quad a \times d = 3 \times 9 = 27$

Alternatively : Option a is not suitable as

$30 = 1 \times 30 \quad \times$

$30 = 2 \times 15 \quad \times$

$30 = 3 \times 10 \quad \times$

$30 = 5 \times 6 \quad \times$

Since difference between a and d must be divisible by 3.

Option (c) is not suitable as

$35 = 1 \times 35 \quad \times$

$35 = 5 \times 7 \quad \times$

Since the difference between a and d is not divisible by 3.

Option (d) is also wrong as

$39 = 1 \times 39 \quad \times$

$= 3 \times 13 \quad \times$

Since the difference between a and d is not divisible by 3.

Choice (b) is correct as

$27 = 1 \times 27 \quad \times$

$27 = 3 \times 9 \quad \checkmark$

If $\quad a = 3 \quad$ and $\quad d = 9$

then $\quad b = 5 \quad$ and $\quad c = 7$

$\therefore \quad a \cdot b \cdot c \cdot d = 945$

NOTE Choices (a) and (d) are wrong since 30 and 39 are not the factors of 945.

Alternatively: Find the factors of 945 and then find the required values of b and c under the condition $a + b + c + d = 24$

9. $T_{11} + T_{12} + T_{13} = 114$

$\Rightarrow \quad T_{12} = \frac{114}{3} = 38$

$\Rightarrow \quad a + 11d = 38 \quad \ldots(1)$

and $\quad T_{21} + T_{22} + T_{23} = 204$

$\Rightarrow \quad T_{22} = 68$

$\Rightarrow \quad a + 21d = 68 \quad \ldots(2)$

from equations (1) and (2)

$10d = 30 \Rightarrow d = 3$

$a = 5$

$\therefore$

$\therefore \quad T_1 + T_2 + T_3 = 5 + 8 + 11 = 24$

Alternatively: $(T_1 + T_2 + T_3) + (T_{21} + T_{22} + T_{23})$

$= 2T_{11} + T_{12} + T_{13})$

$\Rightarrow \quad (T_1 + T_2 + T_3) + 204 = 2 \times 114$

$\Rightarrow \quad T_1 + T_2 + T_3 = 24$

10. $a + ar + ar^2 + ar^3 + \ldots \infty = 4$

$\Rightarrow \quad \frac{a}{1 - r} = 4 \quad \ldots(1)$

or $\quad \frac{a^3}{(1 - r)^3} = 64$

and $\quad a^3 + (ar)^3 + (ar^2)^3 + (ar^3)^3 + \ldots \infty = 192$

$= \frac{a^3}{1 - r^3} = 192 \quad \ldots(2)$

from equation (1) and (2)

$64(1 - r)^3 = (1 - r^3)192$

$\Rightarrow \quad 2r^3 + 3r^2 - 3r - 2 = 0$

$r = -\frac{1}{2}, 1, -2$

But $\quad |r| < 1$

$\therefore r = -\frac{1}{2}$ is the only admissible value

$\therefore \quad \frac{a}{1 - r} = 4$

$\Rightarrow \quad \frac{a}{1 - \left(-\frac{1}{2}\right)} = 4 \Rightarrow a = 6$

11. $\frac{18500 - 12000}{500} = 13$ years

He has to work minimim 13 years to reach the highest scale of 18500.

12. 1, 4, 7, 10, 13, 16, 19, 22, 25, 28, 31, 34, 37, ..., 187

3, 7, 11, 15, 19, 23, 27, 31, 35, 39, ..., 187

The Common terms are 7, 19, 31, 43, ... 187

Therefore number of such terms $= \left(\frac{187 - 7}{12}\right) + 1 = 16$

13. $3^{1/3} \cdot 9^{1/18} \cdot 27^{1/81} \ldots = 3^{1/3} \cdot 3^{1/9} \cdot 3^{1/27} \ldots$

$= 3^{1/3 + 1/9 + 1/27 + \ldots}$

$= 3^{\left(\frac{1/3}{1 - 1/3}\right)} = 3^{1/2} = \sqrt{3}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

14. $1+4+9+16+\ldots+n^2$
$=1^2+2^2+3^2+4^2+\ldots+n^2=\dfrac{n(n+1)(2n+1)}{6}$

15. Put $n=2$ and 3 and then check for the correct choice
Sum of 2 terms = 11
Sum of 3 terms = 46
and
at $n=2$, $\dfrac{1}{6}\times 2(3)(11)=11$
and at $n=3$, $\dfrac{1}{6}\times 3\times 4\times 23=46$
Hence choice (c) is correct.

16. Let x, y, z be 1, 2, 4 respectively
$$a^1=b^2=c^4$$
$$\therefore \quad 16^1=4^2=2^4$$
$$\Rightarrow \quad a=16,\quad b=4 \quad \text{and} \quad c=2$$
$$\Rightarrow \quad \log_4 16=\log_2 4$$
$$\therefore \quad 2=2$$
$$\Rightarrow$$
Hence choice (a) is correct.

17. Since 913952088 is divisible by 7
Alternatively : $a=119$
$a+d=119+7=126$
$a+2d=119+14=133$
∴The numbers which are divisible by 7 are
119, 126, 133, ..., 113113 ...
$$\therefore \text{Number of terms} = \left(\frac{113113-119}{7}\right)+1=16143$$
$$\therefore \quad S_{16143}=\left(\frac{119+113113}{2}\right)\times 16143$$
$$=913952088$$

HINT The unit digit will be 8 as $\left(\dfrac{9+3}{2}\right)\times 3$
$$\Rightarrow \quad 6\times 3 \Rightarrow 8$$
Hence only choice (b) is appropriate.

18. Going through options
$$S_{21}-S_{20}=T_{21}$$
$$\Rightarrow \quad [3(21)^2+5]-[3(20)^2+5]=3\times 441-3\times 400$$
$$=3(41)=123$$
Thus choice (c) is correct.

19. Let us consider choice (a)
then
$$b-a=3-4=-1$$
$$c-b=2-3=-1$$
$$a=4$$
Which are not in G.P., hence wrong.
Now, let us consider choice (c)
$$b-a=1$$
$$c-b=1$$
$$a=1$$
Since $b-a, c-b, a$ (i.e. 1, 1, 1) are in G.P. hence correct

20. $n=1$, $S_n=\dfrac{1}{2}$
$n=2$, $S_n=\dfrac{5}{4}=\left(\dfrac{1}{2}+\dfrac{3}{4}\right)$
$n=3$, $S_n=\dfrac{17}{8}=\left(\dfrac{1}{2}+\dfrac{3}{4}+\dfrac{7}{8}\right)$
Choice (a) is wrong
Since at $n=2$, $S_2=3\neq\dfrac{5}{4}$
Choice (b) is also wrong
Since at $n=2$, $S_2=\dfrac{3}{4}\neq\dfrac{5}{4}$
Choice (c) is also wrong
Since at $n=2$, $S_2=3\neq\dfrac{5}{4}$
Choice (d) is correct
Since at $n=1$, $S_1=\dfrac{1}{2}$
at $n=2$, $S_2=\dfrac{5}{4}$
at $n=3$, $S_3=\dfrac{17}{18}$

21. The required numbers are 1001, 1005, 1009, ..., 9997 and total number of terms in the above A.P. is 2250.
$$\therefore \quad S_{2250}=\frac{(1001+9997)}{2}\times 2250$$
$$=\frac{10998}{2}\times 2250$$
$$=5499\times 2250=12372750$$

HINT Choice (a) is the only suitable option since last two digits in the resultant value must be 50.

22. Let 3, 5, 7, 9, 11, 13, ... are in A.P.
where 5, 7, 9 and 11 are 4 arithmetic means berween 3 and 13
Hence choice (b) is correct
Since $\dfrac{4}{2}(3+13)=5+7+9+11$
$$32=32$$

23. Let 1, 2, 4, 8, 16 are in G.P.
then 2, 4, 8 are 3 geometric means between 1 and 16
Hence choice (c) is correct
Since $(1\times 16)^{3/2}=2\times 4\times 8$
$$64=64$$

24. $a_1+a_5+a_{10}+a_{15}+a_{20}+a_{24}=225$
$\Rightarrow a_1+(a_1+4d)+(a_1+9d)+(a_1+14d)$
$+(a_1+19d)+(a_1+23d)=225$
$$\Rightarrow \quad 6a_1+69d=225 \qquad \ldots(1)$$
Now, $a_1+a_2+a_3+a_4+\ldots+a_{23}+a_{24}=24a_1+276d$
$=4(6a_1+69d)$
$=4\times 225=900$

25. $1^2-2^2+3^2-4^2+5^2-6^2+7^2-8^2+\ldots$
$=(1-2)(1+2)+(3-4)(3+4)+(5-6)(5+6)$
$+(7+8)(7-8)+\ldots$
$=-(1+2)-(3+4)-(5+6)\ldots$
$=-[(1+2)+(3+4)+(5+6)+\ldots]$
$=-[1+2+3+4+5+6+\ldots]=-\left[\dfrac{n(n+1)}{2}\right]$

Alternatively : Go through options.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

26. If $n = 1$, then $6 + 8 = 14$
If $n = 2$, then $66 + 88 = 154$
If $n = 3$, then $666 + 888 = 1554$
Now, go through options.
If you put $n = 1, 2, 3$ etc. in choice (c) you will find satisfactory results.

27. If $n = 1$, then sum of 1st group $= 1$
If $n = 2$, then sum of 2nd group $= 35$
If $n = 3$, then sum of 3rd group $= 405$
Now, going through options, we get choice (b) is correct
since at $n = 1$, $Sg_1 = 1$
at $n = 2$, $Sg_2 = 35$
at $n = 3$, $Sg_3 = 405$

28. Let the numbers be $a - d, a, a + d$
then $(a-d)^2, a^2, (a+d)^2$ are in G.P.

$$\therefore \quad a^4 = (a-d)^2(a+d)^2$$
$$\Rightarrow \quad d^4 - 2a^2d^2 = 0$$
$$\Rightarrow \quad d = 0, \pm\sqrt{2}a$$

Hence d has three values.

29. Since we know that when $a \cdot b \cdot c = k$ (any constant value) then the minimum value of $a + b + c$ is obtained
when $a = b = c$

$$\therefore \quad b^3 = 4 = 2^2$$
$$\Rightarrow \quad b = 2^{2/3}$$

Alternatively : Since a, b, c are in A.P.
Therefore $a = b - d$ and $c = b + d$
where d is the common difference.
Now, $abc = 4 \Rightarrow (b-d)\cdot b \cdot (b+d) = 4$

$$\Rightarrow \quad b(b^2 - d^2) = 4$$

But $b\cdot(b^2 - d^2) < b \cdot b^2 \quad (\because\ b^2 - d^2 < b^2)$

$$\therefore \quad b(b^2 - d^2) < b^3$$
$$4 < b^3$$
$$\Rightarrow \quad 2^2 < b^3$$
$$\Rightarrow \quad 2^{2/3} < b$$

Hence the minimum value of b is $2^{2/3}$.

30. Let the three numbers be $\frac{a}{r}, a, ar$. As the numbers form an increasing G.P.
Therefore $r > 1$
Also $\frac{a}{r}, 2a, ar$ are in A.P.

$$\therefore \quad 2(2a) = \frac{a}{r} + ar$$
$$\Rightarrow \quad r^2 - 4r + 1 = 0$$
$$\Rightarrow \quad r = 2 \pm \sqrt{3}$$
$$\Rightarrow \quad r = 2 + \sqrt{3} \qquad (\because\ 2 - \sqrt{3} < 1)$$

HINT If you go through options then options (a) and (d) are inadmissible.
Since $2 - \sqrt{3} < 1$
and $\sqrt{3} - 2 < 1$
Now you have only two probable choices consider three terms in G.P. with $r = 2 + \sqrt{3}$ or $r = 3 + \sqrt{2}$ and then satisfy the condition.

31. Best approach is to go through options. Choice (c) is correct.
Let $r = 4$, then a, b, c can be $1, 4, 16$

$$\therefore \quad 16 > 4 \times 4 - 3 \times 1$$
$$\Rightarrow \quad 16 > 13$$

Let $r = -5$ then a, b, c can be $1, -5, 25$

$$\therefore \quad 25 > -5 \times 4 - 1 \times 3$$
$$\Rightarrow \quad 25 > -23$$

Let $r = 2$ then a, b, c can be $1, 2, 4$

$$\therefore \quad 4 > 4 \times 2 - 3 \times 1$$
$$\Rightarrow \quad 4 > 5, \qquad \text{which is wrong.}$$

Hence choice (a) is wrong and choice (c) is correct also the range of choice (b) is the subsect of the range of choice (c).
Alternatively : Let a, b, c be the first three consecutive terms of G.P. with common ratio r, then

$$b = ar \quad \text{and} \quad c = ar^2$$

Now, $c > 4b - 3a$

$$\Rightarrow \quad ar^2 > 4ar - 3a$$
$$\Rightarrow \quad r^2 > 4r - 3$$
$$\Rightarrow \quad r^2 - 4r + 3 > 0$$
$$\Rightarrow \quad (r-1)(r-3) > 0$$
$$\Rightarrow \quad r < 1 \quad \text{or} \quad r > 3$$

32. Let d be the common difference of the A.P.
Then $a_3 \cdot a_7 \cdot a_{11} = (22 - 4d)\cdot 22 \cdot (22 + 4d)$

$$= 88(121 - 4d^2)$$

Obviously, R.H.S. is greatest for $d = 0$

33. Given that A.M. $= 2$ G.M.

$$\Rightarrow \quad \frac{a+b}{2} = 2\sqrt{ab}$$
$$\Rightarrow \quad a + b = 4\sqrt{ab}$$

Now, let us consider choice (c)

$$\therefore \quad (2+\sqrt{3}) + (2-\sqrt{3}) = 4\sqrt{(2+\sqrt{3})(2-\sqrt{3})}$$
$$4 = 4 \cdot 1$$
$$4 = 4$$

Hence choice (c) is correct
Alternatively : $a + b = 4\sqrt{ab}$

$$\Rightarrow \quad (a+b)^2 = 16ab$$
$$\Rightarrow \quad a^2 + b^2 + 2ab = 16ab$$
$$\Rightarrow \quad a^2 + b^2 = 14ab$$
$$\Rightarrow \quad \frac{a}{b} + \frac{b}{a} = 14$$
$$\Rightarrow \quad x + \frac{1}{x} = 14 \qquad \left(\because \frac{a}{b} = x\right)$$
$$\Rightarrow \quad x^2 - 14x + 1 = 0$$
$$\Rightarrow \quad x = \frac{14 \pm \sqrt{192}}{2}$$
$$\Rightarrow \quad x = 7 \pm 4\sqrt{3}$$
$$\Rightarrow \quad \frac{a}{b} = 7 + 4\sqrt{3} = (2)^2 + (\sqrt{3})^2 + 4\sqrt{3}$$
$$\frac{a}{b} = (2+\sqrt{3})^2 = \frac{2+\sqrt{3}}{2-\sqrt{3}}$$

Now $x = 7 - 4\sqrt{3}$
$\Rightarrow x = \frac{2-\sqrt{3}}{2+\sqrt{3}} \neq \frac{a}{b}$, since a should be greater than b.
Hence, $x \neq 7 - 4\sqrt{3}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

34. The two sequences are

$17, 21, 25, 29, 33, 37, 41, \ldots, 417$

$16, 21, 26, 31, 36, 41, \ldots, 466$

The common terms are

$21, 41, 61, 81, \ldots 381, 401$

$\therefore$ number of terms (which are common) $= 20$

35. $S_n = \frac{1}{\sqrt{1}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{5}} + \ldots \frac{1}{\sqrt{2n-1}+\sqrt{2n+1}}$

$= \frac{1}{2}[(\sqrt{3}-\sqrt{1}) + (\sqrt{5}-\sqrt{3}) + (\sqrt{7}-\sqrt{5}) + \ldots$

$+ \ldots (\sqrt{2n+1}-\sqrt{2n-1})]$

$= \frac{1}{2}(\sqrt{2n+1}-1)$

Alternatively : Substitute $n = 1, 2, 3$ etc. and check the correct answer.

36. Let a be the first term and d be the common difference of the given A.P. then

$$T_m = \frac{1}{n} \Rightarrow a + (m-1)d = \frac{1}{n} \quad \ldots(1)$$

$$T_n = \frac{1}{m} \Rightarrow a + (n-1)d = \frac{1}{m} \quad \ldots(2)$$

Solving (1) and (2), we get

$$d = \frac{1}{mn} \quad \text{and} \quad a = \frac{1}{mn}$$

$\therefore$ $T_{mn} = a + (mn-1)d$

$= \frac{1}{mn} + (mn-1)\frac{1}{mn} = 1$

Alternatively : Let $m = 2$ and $n = 4$, then

$$T_2 = \frac{1}{4} \quad \text{and} \quad T_4 = \frac{1}{2}$$

$\therefore$ $a + d = \frac{1}{4}$ ...(1)

and $a + 3d = \frac{1}{2}$ ...(2)

$\Rightarrow$ $2d = \frac{1}{4} \Rightarrow d = \frac{1}{8}$

$\therefore$ $a = \frac{1}{4} - \frac{1}{8} = \frac{1}{8}$ [from (1)]

$\therefore$ $T_{mn} = T_8 = a + 7d$

$= \frac{1}{8} + 7 \times \frac{1}{8} = 1$

Hence (c) is the correct choice.

37. Go through options

Let $n = 2$, then $S_n = 3 + 6 = 9$

$\therefore$ $S_n = 2(3) + 2^2 - 1 = 9$

at $n = 3$, $S_n = 19$

$\therefore$ $S_n = 3 \times 4 + 2^3 - 1 = 19$

Hence choice (b) is correct.

Alternatively : $3 + 6 + 10 + 16 + \ldots$

$= (2 + 4 + 6 + 8 + \ldots) + (1 + 2 + 4 + 8 + \ldots)$

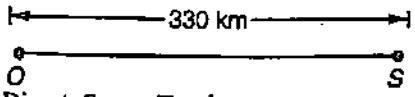
$= n(n+1) + (2^n - 1)$

38. Total number of cells $= 2 + (2^2 + 2^3 + 2^4 + \ldots + 2^{12})$

$= 2^1 + 2^2 + 2^3 + \ldots + 2^{12}$

$= \frac{2(2^{12}-1)}{2-1} = 8190$

39.



Div. + Su. → Total

Ist day $30 + 20 \rightarrow 50$

2nd day $28 + 24 \rightarrow 52$

3rd day $26 + 28 \rightarrow 54$ and so on.

$\therefore$ $50 + 52 + 54 + 56 + 58 + 60 = 330$

Hence required number of days $= 6$

40. Best way is to go through options

Let $n = 2$, then $S_n = S_2 = 1 + (1 + 2) = 4$

from choice (b)

$$S_2 = \frac{2 \times 3 \times 4}{6} = 4$$

and for $n = 3$

$S_n = S_3 = 1 + (1 + 2) + (1 + 2 + 3) = 10$

$\therefore$ from choice (b)

$$S_3 = \frac{3 \times 4 \times 5}{6} = 10$$

Hence choice (b) is correct

Alternatively : $T_n = \frac{n(n+1)}{2}$

$\therefore$ $S_n = \Sigma\left[\frac{n(n+1)}{2}\right] = \Sigma\left[\frac{n^2+n}{2}\right]$

$= \frac{1}{2}[\Sigma n^2 + \Sigma n]$

$= \frac{1}{2}\left[\frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2}\right]$

$= \frac{n(n+1)}{4}\left[\frac{2n+1}{3} + 1\right]$

$= \frac{n(n+1)}{4}\left[\frac{2n+4}{3}\right] = \frac{n(n+1)(n+2)}{6}$

41. $a = 2$ (since 3, 2, 1 are in A. P.)

$b = 3$ ($\therefore$ 2, 3, 6 are in H. P.)

and $c = 6$

$\therefore$ $\left(\frac{1}{b}+\frac{1}{c}-\frac{1}{a}\right)\left(\frac{1}{c}+\frac{1}{a}-\frac{1}{b}\right) = \left(\frac{1}{3}+\frac{1}{6}-\frac{1}{2}\right)\left(\frac{1}{6}+\frac{1}{2}-\frac{1}{3}\right) = 0$

Check the option (a)

$\frac{3}{3^2} - \frac{2}{6} = 0$, which is satisfied

again check the option (c)

$\frac{2}{bc} - \frac{1}{b^2} = \frac{2}{18} - \frac{1}{9} = 0$, which is also true.

Hence (d) is the most appropriate answer.

42. Go through options

$S_1 = 1 \cdot 3 \cdot 5 = 15$

and $S_2 = 1 \cdot 3 \cdot 5 + 3 \cdot 5 \cdot 7 = 120$ etc.

Now, put $n = 1$ in choice (b)

$1(2 + 8 + 7 - 2) = 15$

again put $n = 2$ in choice (b)

$2(16 + 32 + 14 - 2) = 120$

Hence choice (b) is correct.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

NOTE For the certainty of the result you can check for $n = 3$ also.

Alternatively : $T_n = (2n-1)(2n+1)(2n+3)$

$= 8n^3 + 12n^2 - 2n - 3$

$\therefore \quad S_n = 8\Sigma n^3 + 12\Sigma n^2 - 2\Sigma n - 3n$

$= 8\left[\frac{n(n+1)}{2}\right]^2 + \frac{12(n)(n+1)(2n+1)}{6} - \frac{2(n)(n+1)}{2} - 3n$

$= n(2n^3 + 8n^2 + 7n - 2)$

43. $(1-2) + (3-4) + (5-6) + (7-8) + \ldots + (99-100)$

$= (-1) + (-1) + (-1) + (-1) + \ldots 50 \text{ times} = -50$

LEVEL (2)

1. Let us consider $d = 1$, then the A.P. is

$1, 2, 3, 4, 5, 6, 7, 8$

$\therefore \quad 1 + 2 + 3 + 4 = 10$

and $\quad 5 + 6 + 7 + 8 = 26$

Again $1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12$

$\therefore \quad 1 + 2 + 3 + 4 + 5 + 6 = 21$

and $\quad 7 + 8 + 9 + 10 + 11 + 12 = 57$

$\because \quad \frac{10}{26} \neq \frac{21}{57}$

hence $\quad d \neq 1$

Now, consider $d = 2$, then the A.P. is

$1, 3, 5, 7, 9, 11, 13, 15$

$\therefore \quad 1 + 3 + 5 + 7 = 16$

and $\quad 9 + 11 + 13 + 15 = 48$

Again $1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23$

$\therefore \quad 1 + 3 + 5 + 7 + 9 + 11 = 36$

and $\quad 13 + 15 + 17 + 19 + 21 + 23 = 108$

$\because \quad \frac{16}{48} = \frac{36}{108} \Rightarrow \frac{1}{3} = \frac{1}{3}$

hence $d = 2$ is the correct answer.

Alternatively : Let S_n denote the sum to n terms of the A.P. then,

$$\frac{S_n}{S_{2n} - S_n} = k \qquad \text{for every } n \geq 1$$

$$\Rightarrow \quad \frac{S_1}{S_2 - S_1} = \frac{S_2}{S_4 - S_2}$$

$$\Rightarrow \quad S_1 S_4 = S_2^2$$

$$\Rightarrow \quad a \cdot [2(2a + 3d)] = (2a + d)^2$$

$$\Rightarrow \quad 2ad = d^2$$

$$\Rightarrow \quad d = 2 \qquad (\because a = 1)$$

2. Number of divisors of $187 = 4$

Number of divisors of $637 = 6$

Number of divisors of $1001 = 8$

Hence they are in A.P.

HINT To know more about how to find the number of divisors refer to chapter 1 "Fundamentals".

3. $S_1 = 1^2 = 1$

$S_2 = 1^2 + (1^2 + 3^2) = 11$

$S_3 = 1^2 + (1^2 + 3^2) + (1^2 + 3^2 + 5^2) = 46$

Now Put $n = 1, 2, 3$ in choice (b) you will get

$$S_1 = \frac{1}{6} \times 1 \times 2 \times 3 = 1$$

$$S_2 = \frac{1}{6} \times 2(3)(11) = 11$$

$$S_3 = \frac{1}{6} \times 3(4)23 = 46$$

Hence choice (b) is correct.

Alternatively : $t_r = 1^2 + 3^2 + 5^2 + \ldots + (2r-1)^2$

$$\therefore \quad = \sum_{k=1}^{r} (2k-1)^2 = \sum_{k=1}^{r} (4k^2 - 4k + 1)$$

$$= 4\left\{\frac{1}{6} r(r+1)(2r+1)\right\} - \frac{4}{2} r(r+1) + r$$

$$= \frac{1}{3}(4r^3 - r)$$

$$\therefore \quad S_n = \frac{4}{3} \sum_{r=1}^{n} r^3 - \frac{1}{3} \sum_{r=1}^{n} r$$

$$= \frac{4}{3} \cdot \frac{1}{4} n^2 (n+1)^2 - \frac{1}{3} \cdot \frac{1}{2} n(n+1)$$

$$= \frac{1}{6} n(n+1)(2n^2 + 2n - 1)$$

4. Since $|a| <$ and $|b| <$, therefore $|ab| = |a|\,|b| < 1$

Also, $a(a+b) + a^2(a^2 + b^2) + a^3(a^3 + b^3) + \ldots$

$= (a^2 + a^4 + a^6 + \ldots \infty) + (ab + a^2 b^2 + a^3 b^3 + \ldots \infty)$

$$= \frac{a^2}{1 - a^2} + \frac{ab}{1 - ab}$$

5. Best way is to go through options by substituting the values of $n = 1, 2, 3, \ldots$

Alternatively : $\frac{3}{1^2 \cdot 2^2} + \frac{5}{2^2 \cdot 3^2} + \frac{7}{3^2 \cdot 4^2} + \ldots$

$$= \left(1 - \frac{1}{2^2}\right) + \left(\frac{1}{2^2} - \frac{1}{3^2}\right) + \ldots + \left(\frac{1}{n^2} - \frac{1}{(n+1)^2}\right)$$

$$= 1 - \frac{1}{(n+1)^2} = \frac{n^2 + 2n}{(n+1)^2}$$

6. Best way is to go through options by substituting the values.

Sum of the terms of first row $= 1$

Sum of the terms of second row $= 2 + 3 = 5$

Sum of the terms of third row $= 4 + 5 + 6 = 15$ etc.

7. We have

$$\underbrace{1\,1\,1 \ldots 1}_{119 \text{ times}} = 10^{118} + 10^{117} + \ldots + 10^2 + 10^1 + 1$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

$$= \frac{(10^{119}-1)}{10-1} = \left(\frac{10^{119}-1}{10^7-1}\right)\left(\frac{10^7-1}{10-1}\right)$$

$$= (10^{112} + 10^{105} + 10^{98} + \ldots 1)(10^6 + 10^5 + \ldots 10 + 1)$$

Thus $\underbrace{1\,1\,1\ldots1}_{119 \text{ terms}}$ is not a prime number.

8. Since we know that the sum of the interior angles of a polygon of n sides is $(n-2)\pi$

$$\therefore \quad \frac{n}{2}[2(120) + (n-1)5] = (n-2)180$$

$$\Rightarrow \quad n = 9, 16$$

But if $n = 16$, then the greatest angle will be 195° and other interior angles are 190°, 185°, 180°. But no interior angle of a polygon can be equal to 180°.
Hence $n = 9$.

9. If $n = 1$, then

$$1 + 10^4 = 10001 = 73 \times 137$$

$\therefore 10^4 + 1$ is a composite number.

10. $0.1\overline{23} = \frac{1}{10} + \frac{23}{10^3} + \frac{23}{10^5} + \frac{23}{10^7} + \ldots$

$$= \frac{1}{10} + \frac{\frac{23}{10^3}}{1 - \left(\frac{1}{10^2}\right)}$$

$$= \frac{1}{10} + \frac{23}{1000 - 10} = \frac{122}{990} = \frac{61}{495}$$

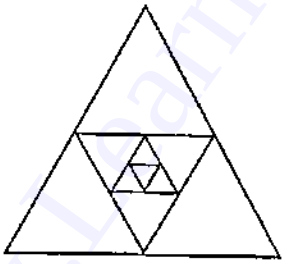
11. $11 - 2 = 9 = 3^2$

$1111 - 22 = 1089 = 33^2$

$111111 - 222 = 110889 = 333^2$

$11111111 - 2222 = 11108889 = 3333^2$ etc.

12. Perimeter of the outer most triangle is 72 cm and perimeter of the second outer most triangle is 36 cm.
Similarly perimeter of the third outermost triangle is 18 cm.
∴ Sum of perimeters of all such triangles



$$= 72 + 36 + 18 + 9 + 4.5 + \ldots \infty \qquad \left(\because S_\infty = \frac{a}{1-r}\right)$$

$$= \frac{72}{1 - \frac{1}{2}} = \frac{72}{\frac{1}{2}} = 144 \text{ cm}$$

13. Go through options.

14. $\log a + \log \frac{a^3}{b} + \log \frac{a^5}{b^2} + \log \frac{a^7}{b^3} + \ldots + \log \frac{a^{2n-1}}{b^{n-1}}$

$$= \log a \times \frac{a^3}{b} \times \frac{a^5}{b^2} \times \frac{a^7}{b^3} \times \ldots \times \frac{a^{2n-1}}{b^{n-1}}$$

$$= \log \frac{a^{(1+3+5+7+\ldots+2n-1)}}{b^{(1+2+3+4+\ldots+n-1)}}$$

$$= \log \frac{a^{n^2}}{b^{(n-1)n/2}}$$

$$= \log \left(\frac{a^{2n}}{b^{n-1}}\right)^{n/2}$$

15. $_{(0.2)}\log_{\sqrt{5}}\left(\frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \ldots\right) = {}_{(0.2)}\log_{\sqrt{5}}\left(\frac{1}{2}\right)$

$$= {}_{(5^{-1})}\log_{\sqrt{5}}\left(\frac{1}{2}\right)$$

$$= {}_5\log_{\sqrt{5}} 2 \qquad (\because m \log n = \log (n)^m)$$

$$= {}_{(\sqrt{5})^2}\log_{\sqrt{5}}(2)$$

$$= {}_{(\sqrt{5})}\log_{\sqrt{5}}(4) \qquad (\because {}_a\log_a b = b)$$

$$= 4$$

16. Since a, b, c are in H.P.
Therefore we can assume $a = 2, b = 3, c = 6$

$$\therefore \quad \frac{b+a}{b-a} + \frac{b+c}{b-c} = \frac{5}{1} + \frac{9}{-3} = 2$$

17. Let $r = \frac{1}{2}$, then $(a, b, c) = (4, 2, 1)$

$\therefore$ $(a, 2b, 3c) = (4, 4, 3)$ which are not in A.P.

Again if $r = \frac{1}{3}$ then

$$(a, b, c) = (9, 3, 1)$$

$\therefore$ $(a, 2b, 3c) = (9, 6, 3)$ which are in A.P.
Hence choice (c) is correct.

18. Let $S = \frac{5}{13} + \frac{55}{(13)^2} + \frac{555}{(13)^3} + \frac{5555}{(13)^4} + \ldots \infty$

$$\therefore \quad \frac{S}{13} = \frac{5}{(13)^2} + \frac{55}{(13)^3} + \frac{555}{(13)^4} + \frac{5555}{(13)^5} + \ldots \infty$$

$$\therefore \quad \left(S - \frac{S}{13}\right) = \frac{5}{13} + \frac{50}{(13)^2} + \frac{500}{(13)^3} + \frac{5000}{(13)^4} + \ldots \infty$$

$$S\left(1 - \frac{1}{13}\right) = \frac{5}{13} + \frac{5}{13}\left(\frac{10}{13}\right) + \frac{5}{13}\left(\frac{10}{13}\right)^2 + \frac{5}{13}\left(\frac{10}{13}\right)^3 + \ldots \infty$$

$$\frac{12}{13}S = \frac{5/13}{3/13} \qquad \left(\because S_\infty = \frac{a}{1-r}\right)$$

$$\frac{12}{13}S = \frac{5}{3} \quad \Rightarrow \quad S = \frac{65}{36}$$

19. Let $S = \frac{1}{1\cdot4} + \frac{1}{4\cdot7} + \frac{1}{7\cdot10} + \ldots \infty$

$$\Rightarrow \quad S = \frac{1}{3}\left[\frac{3}{1\cdot4} + \frac{3}{4\cdot7} + \frac{3}{7\cdot10} + \ldots \infty\right]$$

$$\Rightarrow \quad S = \frac{1}{3}\left[\left(\frac{1}{1} - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{7}\right) + \left(\frac{1}{7} - \frac{1}{10}\right) + \ldots \infty\right]$$

$$\Rightarrow \quad S = \frac{1}{3}[1] \quad \therefore \quad S = \frac{1}{3}$$

20. Let us add up one more number a_{n+1} to the given sequence such that $|a_{n+1}| = |a_n + 1|$.
Now, squaring all the numbers, we get

$$a_1^2 = 0$$
$$a_2^2 = a_1^2 + 2a_1 + 1$$
$$a_3^2 = a_2^2 + 2a_2 + 1$$
$$a_4^2 = a_3^2 + 2a_3 + 1$$
$$\ldots \ldots \ldots$$
$$\ldots \ldots \ldots$$
$$a_n^2 = a_{n-1}^2 + 2a_{n-1} + 1$$
$$a_{n+1}^2 = a_n^2 + 2a_n + 1$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yourshahoob.wordpress.com
For More Visit : www.LearnEngineering.in

$\therefore \quad a_1^2 + a_2^2 + a_3^2 + \ldots + a_{n+1}^2 = a_1^2 + a_2^2 + \ldots + a_n^2 + 2(a_1 + a_2 + \ldots + a_n) + n$

$\Rightarrow \quad 2(a_1 + a_2 + \ldots + a_n) = -n + a_{n+1}^2 \geq -n$

$\Rightarrow \quad \frac{a_1 + a_2 + \ldots + a_n}{n} \geq -\frac{1}{2}$

21. Let a, b, c, d are the first consecutive terms in G.P.

Let us consider choice (a)

$\therefore \quad \frac{b+c}{2} = 5 \Rightarrow b + c = 10$

Now if $(b, c) = (1, 9)$, then a and d can not be found so that $a + b + c + d = 30$.

Similarly $(b, c) = (2, 8), (3, 7), (4, 6), (5, 5)$ are also not suitable.

Again consider choice (b)

$\therefore \quad \frac{b+c}{2} = 6 \Rightarrow b + c = 12$

Now if $(b, c) = 4, 8$, then $a = 2$ and $d = 16$

$\therefore \quad a + b + c + d = 30$

Hence this can be the correct answer.

Further $a^2 + b^2 + c^2 + d^2 = 4 + 16 + 64 + 256 = 340$

Hence choice (b) is correct.

Alternatively : Let a, ar, ar^2, ar^3 be the 4 consecutive terms of G.P. then

$\therefore \quad a + ar + ar^2 + ar^3 = 30 \quad \ldots(1)$

and $\quad a^2 + a^2r^2 + a^2r^4 + a^2r^6 = 340 \quad \ldots(2)$

clearly $r \neq 1$

$[\because (a + ar + ar^2 + ar^3)^2 \neq 4(a^2 + a^2r^2 + a^2r^4 + a^2r^6)]$

$\therefore \quad \frac{(1 + r + r^2 + r^3)^2}{(1 + r^2 + r^4 + r^6)} = \frac{(30)^2}{340} = \frac{45}{17}$

$\Rightarrow \quad \frac{\left[\frac{(r^4 - 1)}{(r - 1)}\right]^2}{\left(\frac{r^8 - 1}{r^2 - 1}\right)} = \frac{45}{17} \quad (\because r \neq 1)$

$\Rightarrow \quad \frac{(r^4 - 1)^2}{(r - 1)^2} \cdot \frac{(r^2 - 1)}{(r^8 - 1)} = \frac{45}{17}$

$\Rightarrow \quad \frac{(r^4 - 1)(r^4 - 1)(r - 1)(r + 1)}{(r - 1)(r - 1)(r^4 - 1)(r^4 + 1)} = \frac{45}{17}$

$\Rightarrow \quad \frac{(r^4 - 1)}{(r - 1)} \cdot \frac{(r + 1)}{(r^4 + 1)} = \frac{45}{17}$

$\Rightarrow \quad (r^3 + r^2 + r - 1) \cdot \frac{(r + 1)}{(r^4 + 1)} = \frac{45}{17}$

$\Rightarrow \quad 17r^4 + 34r^3 + 34r^2 + 34r + 17 = 45r^4 + 45$

$\Rightarrow \quad 14r^4 - 17r^3 - 17r^2 - 17r + 14 = 0$

Dividing by r^2, we get

$$14\left(r^2 + \frac{1}{r^2}\right) - 17\left(r + \frac{1}{r}\right) - (17) = 0$$

$\Rightarrow \quad 14\left[\left(r + \frac{1}{r}\right)^2 - 2\right] - 17\left(r + \frac{1}{r}\right) - 17 = 0$

$\Rightarrow \quad 14y^2 - 17y - 45 = 0$

$\Rightarrow \quad y = \frac{17 \pm \sqrt{17^2 + 4 \times 14 \times 45}}{28}$

$\Rightarrow \quad y = \frac{17 \pm \sqrt{2809}}{28} = \frac{17 \pm 53}{28} = \frac{5}{2}, -\frac{9}{7}$

$\therefore \quad \left(r + \frac{1}{r}\right) = \frac{5}{2} \Rightarrow r = 2, \frac{1}{2}$

and $\left(r + \frac{1}{r}\right) = -\frac{9}{7}$ has no real roots.

$\therefore$ when $r = 2, a = 2$

and when $r = \frac{1}{2}, a = 16$

Thus the required progression is $2, 4, 8, 16, 32, 64, \ldots$ or $16, 8, 4, 2, 1, \frac{1}{2}, \ldots$

22. Side of the first square is 4 cm

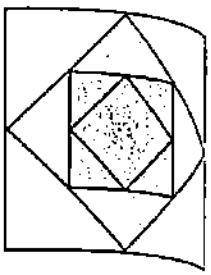
Side of the second square is $2\sqrt{2}$ cm

and side of the third square is 2 cm

and so on.

i.e., each side of first, second, third square ...

$= 4, 2\sqrt{2}, 2, \sqrt{2}, 1, \frac{1}{\sqrt{2}}, \ldots$



$\therefore$ Area of each of first, second, third square ...

$= 16, 8, 4, 2, 1, \frac{1}{2}, \ldots$

$\therefore$ Sum of the area of first, second, third square ...

$= 16 + 8 + 4 + 2 + \ldots = \frac{16}{1 - \left(\frac{1}{2}\right)} = 32 \text{ cm}^2$

23. Best way consider some suitable values to check it physically.

Let A.P. be 2, 9.5, 17, 24.5, 32

and G.P. be 2, 4, 8, 16, 32

$\therefore \quad A_s = 2 + 9.5 + 17 + 24.5 + 32 = 85$

and $\quad G_s = 2 + 4 + 8 + 16 + 32 = 62$

Since $\quad A_s > G_s$

Hence (a) is the correct option.

NOTE For the fool proof testing you can consider some more values (*i.e.*, Progressions)

24. Consider any A.P. and G.P. as per the given conditions, then

A.P. 2, 4, 6, 8, 10, 12, 14, ...

G.P. 2, 4, 8, 16, 32, 64, ...

Hence $a_i < g_i$ for $i > 2$ and $i \in N$

NOTE You can consider some more examples and then you will find that only choice (b) is correct

25. $\frac{1^1 + 2^1 + 3^1 + 4^1}{1 + 2 + 3 + 4} = 1$

$\frac{1^3 + 2^3 + 3^3 + 4^3}{1 + 2 + 3 + 4} = \frac{100}{10} = 10$ and so on.

26. Let the A.P. be 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, ...

where $\quad p = 1, \quad q = 2, \quad r = 4, \quad s = 8$

$\therefore \quad p - q = -1$ and $q - r = -2$ and $r - s = -4$

hence $(p - q), (q - r)$ and $(r - s)$ are in G.P. with common ratio 2 and first term -1.

27. $2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5 = 1 + 2 + 4 + 8 + 16 + 32 = 63$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

28. $n = 5$

The weight of each stone (in kg) is

1, 3, 9, 27, 81

1 kg = 1 kg

2 kg = 3 – 1 = 2 kg

3 kg = 3 kg

4 kg = (3 + 1) = 4 kg

5 kg = 9 – (3 + 1) = 5 kg

6 kg = (9 – 3) = 6 kg

7 kg = (9 + 1) – 3 = 7 kg

8 kg = (9 – 1) = 8 kg

9 kg = 9 kg

10 kg = (9 + 1) = 10 kg and so on.

Remember he is allowed to put the stones on either side of the balance.

29. $T_1 = 3^1 - 2$

$T_2 = 3^2 - 2$

$T_3 = 3^3 - 2$

$T_4 = 3^4 - 2$

30. Try with the options.

31. $2^1 - 1$ souls become ghost in 1 second

$2^2 - 1$ souls become ghost in 2 second

$2^3 - 1$ souls become ghost in 3 second

$2^4 - 1$ souls become ghost in 4 second

$\therefore (2^{43200} - 1)$ souls become ghost in 43200 seconds

32. Assume some values of n and k and then verify the correct answer.

Alternatively : At a particular moment there are x souls left

Just 1 second before there are

$$2^1 x + 1 = 2^1 x + 2^1 - 1 \text{ souls}$$

Just 2 seconds before there are

$$2^2 x + 2 + 1 = 2^2 x + 2^2 - 1 \text{ souls}$$

Just 3 seconds before there are

$$2^3 x + 2^2 + 2 + 1 = 2^3 x + 2^3 - 1 \text{ souls}$$

Hence n seconds before there are

$$2^n x + 2^n - 1 \text{ souls left.}$$

33. Savings = Income – Expenditure

Savings on Ist day = 3 – 3 = 0

Savings on 2nd day = 6 – 5 = 1

Savings on 3rd day = 11 – 7 = 4

Savings on 4th day = 18 – 9 = 9

$\therefore$ Total savings = 1240 = 0 + 1 + 4 + 9 + 16 + 25 + ... + 225

This will take 16 days.

34. Check through options.

Let us consider choice (a)

	E	M	L	T	
Lucknow (G.P.)	1	1	1	1	} 4
Kanpur (A.P.)	1	2	3	4	} 10

It is wrong since $M = E$

Again	E	M	L	T	
Lucknow (G.P.)	1	2	4	8	} 15
Kanpur (A.P.)	1	3	5	7	} 16

It is wrong since $15 \neq \frac{3}{2} \times 16$, also $2 \neq 3$

Again	E	M	L	T	
Lucknow (G.P.)	1	3	9	27	} 40
Kanpur (A.P.)	1	4	7	10	} 22

It is also wrong since $40 \neq \frac{3}{2} \times 22$, also $3 \neq 4$

Again	E	M	L	T	
Lucknow (G.P.)	2	4	8	16	} 30
Kanpur (A.P.)	2	4	6	8	} 20

It is correct since number of monkeys and number of elephants in both the places are equal respectively. Also total number of animals in zoological park is 50% more than that of Local zoo in Kanpur.

36. Go through options

Let $n = 4$, then

$$S_n = -1 + 1^2 - 2 + 2^2 = 2$$

form option (a) $\frac{4 \times 17}{3} = \frac{68}{3} \neq 2$

from option (b) $16 + 4 = 20 \neq 2$

from option (c) $\frac{4 \times 15}{3} = 20 \neq 2$

Hence choice (d) is correct.

37. Let $S = 1 + \frac{2}{3} + \frac{4}{9} + \frac{6}{27} + \frac{8}{81} + \frac{10}{243} + \ldots \infty$

$$\therefore \quad 3(S) = 3 + 2 + \frac{4}{3} + \frac{6}{9} + \frac{8}{27} + \frac{10}{81} + \ldots \infty$$

$$\therefore \quad 2(S) = 4 + \frac{2}{3} + \frac{2}{9} + \frac{2}{27} + \frac{2}{81} + \ldots \infty$$

$$\therefore \quad 2(S) = 4 + 2\left(\frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \frac{1}{81} + \ldots \infty\right)$$

$$\Rightarrow \quad 2(S) = 4 + 2\left[\frac{\frac{1}{3}}{1 - \left(\frac{1}{3}\right)}\right]$$

$$\Rightarrow \quad 2(S) = 4 + 1$$

$$\Rightarrow \quad 2(S) = 5$$

$$\therefore \quad S = \frac{5}{2}$$

38. $\log a + \log \frac{a^2}{b} + \log \frac{a^3}{b^2} + \log \frac{a^4}{b^3} + \ldots + \log \frac{a^n}{b^{n-1}}$

$$= \log\left(a \cdot \frac{a^2}{b} \cdot \frac{a^3}{b^2} \cdot \frac{a^4}{b^3} \cdots \frac{a^n}{b^{n-1}}\right)$$

$$= \log\left[\frac{a^{(1+2+3+4+\ldots+n)}}{b^{(1+2+3+4+\ldots+(n-1))}}\right]$$

$$= \log\left[\frac{a^{\frac{n(n+1)}{2}}}{b^{\frac{(n+1)n}{2}}}\right] = \log\left[\frac{a^{(n+1)}}{b^{(n-1)}}\right]^{n/2}$$

39. Go through options

Let 1, 2, 3, 4, 5, ... be an A.P. then

$$\frac{1}{\sqrt{a_1} + \sqrt{a_2}} + \frac{1}{\sqrt{a_2} + \sqrt{a_3}} + \frac{1}{\sqrt{a_3} + \sqrt{a_4}} + \frac{1}{\sqrt{a_4} + \sqrt{a_5}}$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$$= \frac{1}{\sqrt{1}+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} + \frac{1}{\sqrt{4}+\sqrt{5}}$$
$$= -(\sqrt{1}-\sqrt{2}) - (\sqrt{2}-\sqrt{3}) - (\sqrt{3}-\sqrt{4}) - (\sqrt{4}-\sqrt{5})$$
$$= \sqrt{5}-\sqrt{1}$$
$$= \sqrt{5}-\sqrt{1} \times \frac{\sqrt{5}+\sqrt{1}}{\sqrt{5}+\sqrt{1}} = \frac{5-1}{\sqrt{5}+\sqrt{1}}$$
$$= \frac{4}{\sqrt{1}+\sqrt{5}} = \frac{n-1}{\sqrt{a_1}+\sqrt{a_n}}$$

Alternatively : $\frac{1}{\sqrt{a_1}+\sqrt{a_2}} + \frac{1}{\sqrt{a_2}+\sqrt{a_3}} + \frac{1}{\sqrt{a_3}+\sqrt{a_4}} + \ldots + \frac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}}$

$$= \frac{\sqrt{a_2}-\sqrt{a_1}}{(\sqrt{a_2}+\sqrt{a_1})(\sqrt{a_2}-\sqrt{a_1})} + \frac{\sqrt{a_3}-\sqrt{a_2}}{(\sqrt{a_3}+\sqrt{a_2})(\sqrt{a_3}-\sqrt{a_2})} + \ldots + \frac{\sqrt{a_n}-\sqrt{a_{n-1}}}{(\sqrt{a_n}+\sqrt{a_{n-1}})(\sqrt{a_n}-\sqrt{a_{n-1}})}$$
$$= \frac{\sqrt{a_2}-\sqrt{a_1}}{a_2-a_1} + \frac{\sqrt{a_3}-\sqrt{a_2}}{a_3-a_2} + \ldots + \frac{\sqrt{a_n}-\sqrt{a_{n-1}}}{a_n-a_{n-1}}$$
$$= \frac{\sqrt{a_2}-\sqrt{a_1}}{d} + \frac{\sqrt{a_3}-\sqrt{a_2}}{d} + \ldots + \frac{\sqrt{a_n}-\sqrt{a_{n-1}}}{d}$$
$$= \frac{1}{d}[(\sqrt{a_2}-\sqrt{a_1}) + (\sqrt{a_3}-\sqrt{a_2}) + \ldots + (\sqrt{a_n}-\sqrt{a_{n-1}})]$$
$$= \frac{1}{d}[\sqrt{a_n}-\sqrt{a_1}] = \frac{a_n-a_1}{d(\sqrt{a_n}+\sqrt{a_1})}$$
$$= \frac{n-1}{\sqrt{a_n}+\sqrt{a_1}} = \frac{n-1}{\sqrt{a_1}+\sqrt{a_n}}$$

40. Go through options.
Remember If $\log a$, $\log b$, $\log c$ are in A.P., then a, b, c are in G.P.

$\therefore$ 2, $(2^x - 3)$ and $\left(\frac{17}{2} + 2^{x-1}\right)$ must be in G.P.

Let $x = 3$, then the value of $2^x - 3 = 5$

and $\frac{17}{2} + 2^{x-1} = \frac{25}{2}$

Since 2, 5, $\frac{25}{2}$ are in G.P.

Hence, the given terms of the sequence are in A.P.

41. Combining all of the three relations we get

$$x = y^{-1} = z^3$$

$\therefore 1, \log_y x, \log_z y, -15 \log_x z$

$\Rightarrow 1, -1, -3, -5$, which are in A.P.

Hence, choice (d) is most appropriate.

42. Let $x = 1, y = 2, z = 4$

$\therefore \quad a = 16, \quad b = 4, \quad c = 2 \qquad (\because a^x = b^y = c^z)$

$\therefore \quad \log_b a \cdot \log_b c = \log_4 16 \cdot \log_4 2$
$$= (2\log_4 4) \cdot \left(\frac{\log 2}{2\log 2}\right)$$
$$= (2 \cdot 1)\left(\frac{1}{2}\right) = 1$$

43. Go through options.
Let us consider choice (b)

$$x = 1 - \log_2 5 = \log_2 2 - \log_2 5 = \log_2\left(\frac{2}{5}\right)$$

$\therefore \log_2(5 \cdot 2^x + 1), \log_4(2^{1-x} + 1), 1$
$$= \log_2(5 \cdot 2^{\log_2 2/5} + 1), \log_4(2^{1-\log_2 2/5} + 1), 1$$
$$= \log_2\left(5 \times \frac{2}{5} + 1\right), \log_4\left(\frac{2}{2^{\log_2 2/5}} + 1\right), 1$$
$$= \log_2 3, \log_4\left(\frac{2}{2/5} + 1\right), 1$$
$$= \log_2 3, \log_4 6, 1 = \log_4 9, \log_4 6, \log_4 4$$

$\because$ 9, 6, 4 are in G.P.

Hence the assumed choice (b) is correct.

HINT If a, b, c are in G.P. then $\log a$, $\log b$, $\log c$ are in A.P.

44. Very similar to the previous problem.

45. Let $x = e$, $y = e^2$ and $z = e^3$, then

$$\frac{1}{1+\ln e}, \frac{1}{1+\ln e^2} + \frac{1}{1+\ln e^3} = \frac{1}{1+1}, \frac{1}{1+2}, \frac{1}{1+3}$$
$$= \frac{1}{2}, \frac{1}{3}, \frac{1}{4} \text{ which are in H.P.}$$

46. Let d be the common difference of the A.P.

then $\quad a_{10} = 3 \Rightarrow a_1 + 9d = 3$

$\Rightarrow \quad 2 + 9d = 3 \Rightarrow d = \frac{1}{9}$

$\therefore \quad a_4 = a_1 + 3d = 2 + \frac{1}{3} = \frac{7}{3}$

Let D be the common difference of $\frac{1}{h_1}, \frac{1}{h_2}, \ldots, \frac{1}{h_{10}}$

Then, $\quad h_{10} = 3 \Rightarrow \frac{1}{h_{10}} = \frac{1}{3}$

$\Rightarrow \quad \frac{1}{h_1} + 9D = \frac{1}{3} \Rightarrow \frac{1}{2} + 9D = \frac{1}{3}$

$\Rightarrow \quad D = -\frac{1}{54}$

$\therefore \quad \frac{1}{h_7} = \frac{1}{h_1} + 6D = \frac{1}{2} - \frac{1}{9} = \frac{7}{18}$

$\Rightarrow \quad h_7 = \frac{18}{7}$

$\therefore \quad a_4 h_7 = \frac{7}{3} \times \frac{18}{7} = 6$

47. $S_4 < 1 \text{ cm}^2$

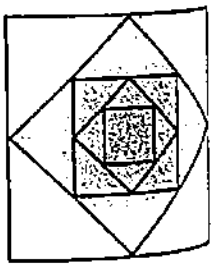
Therefore the side of square also be less than 1 cm.

$\therefore \quad l_1 = 10, \quad l_2 = \frac{10}{\sqrt{2}},$

$l_3 = \frac{10}{2}, \quad l_4 = \frac{10}{2\sqrt{2}}$

$l_5 = \frac{10}{4}, \quad l_6 = \frac{10}{4\sqrt{2}},$

$l_7 = \frac{10}{8}, \quad l_8 = \frac{10}{8\sqrt{2}}$



Hence $\quad l_8 = \frac{10}{8\sqrt{2}} < 1 \text{ cm}$

$\therefore \quad$ area of $S_8 < 1 \text{ cm}^2$

where $l_1, l_2, \ldots, l_{10}$ are the lengths of sides of different squares, from outwards to inwards.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Final Round

1. Total number of bacterias after 10 seconds
$= 3^{10} - 3^5 = 3^5(3^5 - 1)$

since just after 10 seconds all the bacterias (i.e. 3^5) are dead after living for 5-5 seconds.

2. $S_{20} = 2^{66} - 2^{65} - 2^{64} - 2^{63} - 2^{62} - \ldots - 2^{47}$
$= 2^{65}(2-1) - 2^{64} - 2^{63} - \ldots - 2^{47}$
$= 2^{65} - 2^{64} - 2^{63} - \ldots - 2^{47}$
$= 2^{64}(2-1) - 2^{63} - \ldots - 2^{47}$
$= 2^{64} - 2^{63} - 2^{62} - \ldots - 2^{47}$
$= 2^{63}(2-1) - 2^{62} - \ldots - 2^{47}$
$= 2^{63} - 2^{62} - 2^{61} - \ldots - 2^{47}$
... ...
... ...
$= 2^{47}$

3. Since $S_{67} = 2^0$ $\quad (\because S_n = 2^{67-n})$
$S_{66} = 2^1$
$S_{65} = 2^2$
$S_{64} = 2^3$
$S_{63} = 2^4$
... ...
... ...
$S_{34} = 2^{33}$ etc.

Now, $U_m = S_m + S_{m+1} + S_{m+2} + \ldots + S_{m+(m-1)}$
$\therefore \quad U_{34} = S_{34} + S_{35} + S_{36} + \ldots + S_{67}$
$\Rightarrow \quad U_{34} = 2^{33} + 2^{32} + 2^{31} + \ldots + 2^0$
$\Rightarrow \quad U_{34} = 1 + 2^1 + 2^2 + 2^3 + \ldots + 2^{32} + 2^{33}$
$\Rightarrow \quad U_{34} = (2^{33} - 1)$

4. Consider an A. P, then go through options
Let 1, 2, 3, 4, 5, 6 be an A.P. with 6 terms
i.e. $3n = 6, \quad 2n = 4 \quad$ and $\quad n = 2$
$\therefore \quad S_3 - S_2 - S_1$
$= (1 + 2 + \ldots 6) - (1 + 2 + 3 + 4) - (1 + 2)$
$= 21 - 10 - 3 = 8$

Choice (a) gives
$S_3 - S_2 - S_1 = 3a - 2n - d = 3 - 4 - 1$
$= -2$, hence wrong

Check the choice (d)
$S_3 - S_2 - S_1 = 2n^2 d = 8$
Hence it is correct.

Alternatively : Since $S_3 - S_2 - S_1$ is independent of a (i.e., first term).
Hence only choice (d) is correct.

5.

$T_n = n$	$n+3$	$n+5$	$n+7$	$n+9$
$T_1 = 1$	4	6	8	10
$T_2 = 2$	5	7	9	11
$T_3 = 3$	6	8	10	12
$T_4 = 4$	7	9	11	13
$T_5 = 5$	8	10	12	14
$T_6 = 6$	9	11	13	15
$T_7 = 7$	10	12	14	16
$T_8 = 8$	11	13	15	17
$T_n = 9$	12	14	16	18

In general when $n = 4, 9, 13, 17, \ldots, 99$ does not contain 5 or its multiple.
i.e. $n = 5k - 1$ where $k = 1, 2, 3, \ldots, 20$
Hence out of 99 sets 20 sets does not contain the 5 or its multiples.
Thus the required number of sets $= 99 - 20 = 79$

6. Let $S = \{\sqrt{1}\} + \{\sqrt{2}\} + \{\sqrt{3}\} + \{\sqrt{4}\} + \ldots + \{\sqrt{99}\} + \{\sqrt{100}\}$
$\Rightarrow \quad S = 1 + 2 + 2 + 2 + 3 + 3 + 3 + 3 + 3 + \ldots + 10$
$\Rightarrow \quad S = 1 + 3(2) + 5(3) + 7(4) + 9(5) + \ldots + 19(10)$
Since $\quad T_n = (2n-1)n = 2n^2 - n$
$\therefore \quad S_n = \Sigma(2n^2 - n) = 2\Sigma n^2 - \Sigma n$
$= \dfrac{2n(n+1)(2n+1)}{6} - \dfrac{n(n+1)}{2}$
$= \dfrac{n(n+1)}{2}\left[\dfrac{2}{3}(2n+1) - 1\right]$
$= \dfrac{n(n+1)}{2}\left[\dfrac{4n+2-3}{3}\right]$
$= \dfrac{n(n+1)(4n-1)}{6}$
$\therefore \quad S_{10} = \dfrac{10 \times 11 \times 39}{6} = 715$

7. Let $S = [\sqrt{1}] + [\sqrt{2}] + [\sqrt{3}] + \ldots + [\sqrt{323}]$
$\Rightarrow \quad S = 1 + 1 + 1 + 2 + 2 + 2 + 2 + 2 + \ldots + 17 + 17$
$\Rightarrow \quad S = 1(3) + 2(5) + 3(7) + 4(9) + \ldots + 17(35)$
$\because \quad T_n = n(2n+1)$
$\therefore \quad S_n = \Sigma\, n(2n+1) = 2\Sigma n^2 + \Sigma n$
$= \dfrac{2 \cdot n(n+1)(2n+1)}{6} + \dfrac{n(n+1)}{2}$
$= \dfrac{n(n+1)}{2}\left[\dfrac{2}{3}(2n+1) + 1\right]$
$= \dfrac{n(n+1)}{2}\left[\dfrac{4n+5}{3}\right]$
$= \dfrac{n(n+1)(4n+5)}{6}$
$\therefore \quad S = \dfrac{17 \times 18 \times 73}{6} = 3723$

8. This can be done only by giving the number of coins as $2^0, 2^1, 2^2, 2^3, 2^4, \ldots$ etc.
So, the amounts are 1, 2, 4, 8, 16, 32, 37.
Hence (c) is the correct answer.

9. $(1 + 37) = (32 + 4 + 2)$
Hence, 3 people are required.

10. $(1 + 2 + 37) = (32 + 8)$
Hence 2 people are required.

12. 1, 2, 4, 8, 16, 32, 64, 73

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

13. Since there are only 5 people left with their amount viz. 1, 4, 16, 64, 73, (excluding 2, 8, 32).
So total number of combinations are $2^5 - 1 = 31$
Hence option (c) is correct.

14.
$$1 + 2 + 4 + 8 + 16 + 32 = 63$$
$$3(1 + 2 + 4 + 8 + 16 + 32) = 189$$
$$200 - 189 = 11$$

15. First of all choice (c) is ruled out since 'a' can not be zero.
Again choice (a) is ruled out because $|r| \neq 1$
Now let us check the option (b)

$$S_\infty = 3 + \left(-\frac{3}{2}\right) + \frac{9}{4} + \left(-\frac{27}{8}\right) + \ldots \infty$$

$$\therefore \quad S_\infty = \frac{3}{1-\left(-\frac{1}{2}\right)} = \frac{3}{\frac{3}{2}} = 2$$

Again, $$27 - \frac{27}{8} + \frac{729}{64} - \ldots \infty$$

$$S_\infty = \frac{27}{1-\left(-\frac{1}{8}\right)} = \frac{27}{\frac{9}{8}} = 24$$

Hence choice (b) is the appropriate choice.

Alternatively : $S_1 = \frac{a}{1-r} = 2$

$$S_2 = \frac{a^3}{1-r^3} = 24$$

By solving these two equations, we get

$$a = 3, \quad r = -\frac{1}{2}$$

16. When you check option (a) it will be proved wrong. Again for convenience consider option (c)

2, 5, 8, 11, ...

then first term of that G.P. = 3
and the common ratio of G.P. = 2
Hence G.P. = 3, 6, 12, 24, ...
So $a_1 + a_2 = 9$, which is correct
and S_n of A.P. $= \frac{10}{2}[2 \times 2 + 9 \times 3] = 155$

which is also correct. Hence choice (c) is correct.

Similarly choice (b) is also correct.
Hence choice (d) is most appropriate.

Alternatively : $a, a + d, a + 2d, \ldots$ be an A.P.
and $A, Ar, Ar^2, Ar^3 \ldots$ be a G.P.

then, $$\frac{n}{2}[2a + (n-1)d] = 155$$

$$5[2a + 9d] = 155$$

$$2a + 9d = 31 \quad \ldots(1)$$

Again $$A + Ar = 9$$

$$d + ad = 9 \quad \ldots(2)$$

Solving equations (1) and (2) by substitution method,

$$a = 2, \frac{25}{2} \quad \text{and} \quad d = 3, \frac{2}{3}$$

Thus A.P. → 2, 5, 8, 11, ...
G.P. → 3, 6, 12, 24, ...
or A.P. → $\frac{25}{2}, \frac{79}{6}, \frac{83}{6}, \ldots$
G.P. → $\frac{2}{3}, \frac{25}{3}, \frac{625}{6}, \ldots$

Thus choice (d) is correct.

17. $$\frac{a}{1-r} = 162, \quad \frac{a(1-r^n)}{1-r} = 160$$

$$\therefore \quad 1 - r^n = \frac{80}{81} \Rightarrow r^n = \frac{1}{81}$$

Now, since $\frac{1}{r} \in Z$

$$\therefore \quad \frac{1}{r^n} = 81 \Rightarrow \frac{1}{r} = 3^{4/n}$$

$$\Rightarrow \therefore \quad n = 1, 2, 4$$

Here 1, 2, 4 are the factors of 4.

Hence

n	r	a
1	$\frac{1}{81}$	160
2	$\frac{1}{9}, -\frac{1}{9}$	144 and 180
4	$\frac{1}{3}, -\frac{1}{3}$	108 and 216

Hence choice (d) is correct.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

19 PERMUTATIONS & COMBINATIONS

In this chapter we study the selection and arrangement of things, persons, groups etc. which is the most logical phenomenon of mathematics. Since this chapter is logic based, that's why in CAT and XAT it is very important. Every year CAT asks 2-3 questions while XAT asks almost 5-10 questions from this chapter only.

I would like to advise those students who are not familiar with this topic or who are afraid of it, they blend their common sense while solving the problem by formula and beginners should look at the problem through various angles so that they become aware of all the cases inherent in the problem.

One thing is clear about the problems of this chapter is that they do not involve any algebraic equations/expressions. They just require the ability to interpret the situations with open mind and quick multiplication is must.

FUNDAMENTAL PRINCIPLE OF COUNTING

(i) Multiplication : If one operation can be performed in 'm' ways and corresponding to each way of performing the first operation a second operation can be performed in n ways, then the two operations together can be performed in $m \times n$ ways.

If after two operations are performed in any one of the $m \times n$ ways a third operation can be performed in p ways, then the three operations together can be performed in $m \times n \times p$ ways and so on.

In general if there are n jobs (works/operations) $j_1, j_2, j_3, ..., j_n$ such that job j_i can be performed independently in m_i ways. $(i = 1, 2, 3, ..., n)$. Then the total number of ways in which all the jobs can be performed is $m_1 \times m_2 \times m_3 \times ... \times m_n$.

Here the different jobs/operations are mutually inclusive. It implies that all the jobs are being done in succession. In this case we use the word **'and'** to complete the all stages of operation and the meaning of **'and'** is multiplication.

e.g., A student has to select a letter from vowels and another letter from consonants, then in how many ways can he make this selection?

SOLUTION Out of 5 vowels he can select one vowel (letter) in 5 ways and out of 21 consonants he can select one consonant (letter) in 21 ways. Thus a letter from the vowels and a letter from the consonants can be selected together in $5 \times 21 = 105$ ways.

(ii) Addition : If there are two jobs such that they can be performed independently in m and n ways respectively, then either of the two jobs can be performed in $(m + n)$ ways.

In general if there are various jobs which are mutually exclusive, then they can be performed in $m_1 + m_2 + m_3 + ... + m_n$ ways.

In this case we use the word **'or'** between various jobs and the meaning of **'or'** is addition.

e.g. A student has to select a letter either from vowels 'or' from consonants, then in how many ways can he make this selection?

SOLUTION Out of 5 vowels he can select one letter in 5 ways.

Similarly he can select one letter from 26 consonants in 21 ways.

Thus he can select one letter in $5 + 21 = 26$ ways.

EXAMPLE 1 *If a die is cast and then a coin is tossed, find the number of all possible outcomes.*

SOLUTION A die can fall in 6 different ways showing six different points 1,2 , 3, 4, 5, 6, ... and a coin can fall in 2 different ways showing head (H) or tail (T).

∴ The number of all possible outcomes from a die and a coin $= 6 \times 2 = 12$

HINT $(1, H), (1, T), (2, H), (2, T), (3, H), (3, T), (4, H), (4, T), (5, H), (5, T), (6, H), (6, T)$ *i.e.,* the total number of all possible outcomes is 12. *i.e.,* the total number of all possible outcomes is 12.

EXAMPLE 2 *There are 6 trains running between Indore and Bhopal. In how many ways can a man go from Indore to Bhopal and return by a different train?*

SOLUTION A man can go from Indore to Bhopal in 6 ways by any one of the 6 trains available. Then he can return from Bhopal to Indore in 5 ways by the remaining 5 trains, since he cannot return by the same train by which he goes to Bhopal from Indore.

Thus the required number of ways $= 6 \times 5 = 30$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

EXAMPLE 3 *In the previous question, if the person can return by any one of the 6 trains, then find the number of ways in which he can go from Indore to Bhopal and return from Bhopal to Indore.*

SOLUTION He can go from Indore to Bhopal in 6 ways and he can return from Bhopal to Indore in 6 ways also.

Hence the number of ways in which he can go from Indore to Bhopal and return from Bhopal to Indore = $6 \times 6 = 36$ ways.

EXAMPLE 4 *Four persons entered the lift cabin on the ground floor of a six floor office. Suppose each of them can leave the cabin independently at any floor beginning with the first. Find the total number of ways in which each of the four persons can leave the cabin at different floors.*

SOLUTION Suppose M_1, M_2, M_3 and M_4 are four persons. M_1 can leave at any of the 5 floors. So M_1 can leave the cabin in 5 ways. Now, M_2 can leave the cabin at any of remaining 4 floors. So M_2 can leave the cabin in 4 ways. Similarly M_3 and M_4 can leave the cabin in 3 ways and 2 ways respectively. Thus, the total number of ways in which each of the 4 persons can leave the cabin at different floors is $5 \times 4 \times 3 \times 2 = 120$.

EXAMPLE 5 *In the above question if the 4 persons can leave the cabin at any one of the 5 floors, then find the number of ways in which this can be done.*

SOLUTION Since each of the four person can leave the cabin at any one of the 5 floors. Hence each one of M_1, M_2, $M_3$and M_4 can leave the cabin in 5 ways.

Hence total number of ways in which each of the four persons can leave the cabin at any of the 5 floors is $5 \times 5 \times 5 \times 5 = 625$

EXAMPLE 6 *In a MOCK CAT there are 3 sections viz, QA, English and D I containing 4, 6 and 11 questions respectively. In how many ways can a candidate select one question from each of the three sections?*

SOLUTION The candidate can select one question from Q A in 4 ways, one question from English in 6 ways and one question from D I in 11 ways. Hence he can select one question from each of three sections in $4 \times 6 \times 11 = 264$ ways.

EXAMPLE 7 *A tricolour flag is to be formed having three adjacent strips of three different colours choosen from six different colours. How many different coloured flags can be formed with different design in which all the three strips are always in horizontal position ?*

SOLUTION First strip can be coloured in 6 ways and second strip can be coloured in 5 ways and third strip can be coloured in 4 ways.

Hence all the three strips can be coloured in $6 \times 5 \times 4 = 120$ ways.

Hence there can be 120 differently coloured flags.

EXAMPLE 8 *In how many ways can a chairman and vice-chairman be elected from a committee of 8 members who are equally eligible for the post?*

SOLUTION The chairman can be elected in 8 ways, and the vice chairman can be elected in 7 ways from the remaining 7 members.

Hence the required number of ways = $8 \times 7 = 56$ ways.

EXAMPLE 9 *In how many ways can 5 prizes be given away to 7 boys when each boy is eligible for all the prizes?*

SOLUTION First prize can be given away to any one of 7 boys in 7 ways. Again second prize can be given away to any one of the 7 boys since each boy is eligible for all the prizes. Similarly third, fourth and fifth prize can also be given away in 7 ways.

Hence the required number of ways = $7 \times 7 \times 7 \times 7 \times 7 = 7^5$

EXAMPLE 10 *A test paper consists of 10 questions and each question has 4 choices. If each question is necessarily attempted, then find the number of ways of answering the test paper.*

SOLUTION Since each question can be answered in 4 ways. So the total number of ways of answering 10 questions

$= 4 \times 4 \times 4 \times 4 \times 4 \ldots 10$ times $= 4^{10}$.

EXAMPLE 11 *A set of 6 questions contains true/false type questions. Maximum how many students can take the test if all the students answer differently from others and must attempt all the questions?*

SOLUTION A question can be answered in two ways *i.e.* either true or false. Hence all the 6 questions can be answered in $2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^6 = 64$ ways.

Thus there can be maximum 64 students.

EXAMPLE 12 *In a test paper first section contains 5 question each with 4 choices and second section contains 4 questions each with 3 choices. In how many different ways can the paper be answered if all the questions are attempted?*

SOLUTION First section can be answered in $4 \times 4 \times 4 \times 4 \times 4 = 4^5$ ways

Second section can be answered in $3 \times 3 \times 3 \times 3 = 3^4$ ways.

Hence the whole test paper can be answered in $4^5 \times 3^4$ ways.

EXAMPLE 13 *How many three letter words can be formed using only consonants but each only once?*

SOLUTION There are only 21 consonants in English alphabet. So, there are 21 ways of filling up the first place. Now, there are only 20 ways of filling up the second place and the third place can be filled up in 19 ways.

Hence, the required number of words = $21 \times 20 \times 19 = 7980$

EXAMPLE 14 *How many four digit numbers can be formed with the digits 3, 5, 7 and 9 only?*

4	4	4	4
Th	*H*	*T*	*U*

A four digit number has 4 places *viz*, thousands, hundreds tens and unit place. All the four places can be filled up in $4 \times 4 \times 4 \times 4 = 256$ ways since repetition of digits is not restricted.

EXAMPLE 15 *How many 3 digit number can be formed whose unit digit is always zero and repetition of digits is not allowed?*

SOLUTION

9	8	1

$\Rightarrow \quad 9 \times 8 \times 1 = 72$

Since zero is fixed for the unit digit place. So only 9 digits are available for the hundreds place. Now only 8 digits are available for

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

the tens place since out of 10 digits (*i.e.*, 0, 1, 2, 3, ..., 9) two digits at two places have been used. Therefore required possible numbers $= 9 \times 8 \times 1 = 72$

EXAMPLE 16 *How many numbers can be formed with digits 2, 4, 6, 8 without repetition?*

SOLUTION **Case 1.** There are 4 numbers of 1 digit.

Case 2. | 4 | 3 | $\Rightarrow 4 \times 3 = 12$

There are 12 numbers of 2 digits.

Case 3. | 4 | 3 | 2 | $\Rightarrow 4 \times 3 \times 2 = 24$

There are 24 numbers of 3 digits.

Case 4. | 4 | 3 | 2 | 1 | $\Rightarrow 4 \times 3 \times 2 \times 1 = 24$

There are 24 numbers of 4 digits.

Hence total required numbers $= 4 + 12 + 24 + 24 = 64$

NOTE No any number of 5, 6, 7, ... digits can be formed since we have only four digits and repetition of digits is not allowed.

EXAMPLE 17 *How many four digit numbers can be formed using 5 only once but 5 must be used in all such numbers?*

SOLUTION **Case 1.** | 8 | 9 | 9 | 1 | $\Rightarrow 8 \times 9 \times 9 \times 1 = 648$

When 5 is fixed at unit place, then there are only 8 digits available for thousands place (*i.e.*, except to 0 and 5).

Again hundreds and tens places can be filled by any one of digits from 0 to 9 except 5.

Case 2. | 8 | 9 | 1 | 9 | $\Rightarrow 8 \times 9 \times 1 \times 9 = 648$

When 5 is fixed at tens, place then there are only 8 digits available for thousands place and 9 digits are available for each of the hundreds and unit places.

Case 3. | 8 | 1 | 9 | 9 | $\Rightarrow 8 \times 1 \times 9 \times 9 = 648$

Here 5 is fixed at hundreds place.

Case 4. | 1 | 9 | 9 | 9 | $\Rightarrow 9 \times 9 \times 9 = 729$

Here 5 is fixed at thousands place and each of the hundreds, tens and unit place can be filled up in 9 ways each.

Therefore, the total required numbers of ways

$= 648 + 648 + 648 + 729 = 2673$

EXAMPLE 18 *How many four digit numbers can be formed with the digits 0, 2, 3, 5, 8, 9 if*

(i) repetition of digits is allowed?

(ii) repetition of digits is not allowed?

SOLUTION (i) | 5 | 6 | 6 | 6 | $\Rightarrow 5 \times 6 \times 6 \times 6 = 1080$

Since at thousands place zero can not be placed.

(ii) | 5 | 5 | 4 | 3 | $\Rightarrow 5 \times 5 \times 4 \times 3 = 300$

Since at thousands place zero cannot be placed so only 5 digits are available and there are only 5 remaining digits available for hundreds place and 4 digits for tens place and 3 digits for unit place.

PERMUTATIONS

Each of the different "arrangements" that can be made out of a given number of things by taking some or all of them at a time is called "permutation".

Thus the permutations of three letters (say) *a*, *b*, *c* taken two at a time are

ab, *ba*, *bc*, *cb*, *ac*, *ca*

Therefore the number of permutations of three different things taken two at a time is ${}^3P_2 = 6$

EXAMPLE 1 *Write down all the permutations of the set of four letters E, O, N, T, taken two at a time.*

SOLUTION *EO, OE, EN, NE, ET, TE, ON, NO, OT, TO, NT, TN*

Thus there are total 12 permutations

i.e., ${}^4P_2 = 12$

Notations : The numbers of permutations of '*n*' different things taken *r* at a time is denoted by the symbol $P(n, r)$ or ${}^nP_r$, for every $1 \le r \le n$

Thus ${}^nP_r$ = Total number of permutations of *n* distinct things taken *r* at a time.

$$ {}^nP_r = \frac{n!}{(n-r)!} $$

and $n! = 1.2.3.4...(n-1).n$

and $(n-r)! = 1.2.3.4.....(n-r-1).(n-r)$

NOTE $0! = 1 = 1!$

- PERMUTATIONS of *n* different things taken *r* at a time $= {}^nP_r = \frac{n!}{(n-r)!}$
- PERMUTATIONS of *n* different things taken all at a time $= {}^nP_n = n!$
- PERMUTATIONS of *n* different things taken *r* at a time, when a particular things always occurs $= r \cdot {}^{(n-1)}P_{(r-1)}$
- PERMUTATIONS of *n* different things taken *r* at a time when a particular thing never occurs $= {}^{(n-1)}P_r$.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

INTRODUCTORY EXERCISE-19.1

1. Find the value of ${}^{9}P_{3}$.
(a) 504 (b) 309
(c) 405 (d) 600

2. Find the value of n, if ${}^{n}P_{5} = 20\ {}^{n}P_{3}$.
(a) 5 (b) 8
(c) 6 (d) 4

3. If ${}^{56}P_{r+6} : {}^{54}P_{r+3} = 30800 : 1$, find r.
(a) 14 (b) 20
(c) 41 (d) 21

4. How many different numbers of 3 digits can be formed with the digits 1, 2, 4, 5, 7, 8 ; none of the digits being repeated in any of the numbers so formed?
(a) 120 (b) 1200
(c) 180 (d) 270

5. How many even numbers less than 10,000 can be formed with the digits 3, 5, 7,8, 9 without any repetition?
(a) 32 (b) 16
(c) 44 (d) 41

6. How many numbers with different digits each greater than 4000 can be formed from the digits 0, 2, 5, 7, 8?
(a) 160 (b) 168
(c) 320 (d) 270

7. How many numbers greater than 5000 can be formed with the digits, 3, 5, 7, 8, 9 no digit being repeated?
(a) 216 (b) 126
(c) 512 (d) 252

8. How many 6 digit telephone numbers can be constructed with the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 if each number starts with 35 and no digit appears more than once?
(a) 1600 (b) 1680
(c) 900000 (d) 9000

9. How many 6 digit numbers can be formed from the digits 1, 2, 4, 5, 6, 7 (no digit being repeated) which are divisible by 5?
(a) 555 (b) 156
(c) 120 (d) none of these

10. Find the sum of all the four digit numbers which are formed by the digits 1, 2, 5, 6.
(a) 933510 (b) 93324
(c) 65120 (d) 8400

11. How many even numbers greater than 300 can be formed with the digits 1, 2, 3, 4, 5 no digit being repeated?
(a) 111 (b) 600
(c) 900 (d) none of these

12. How many numbers each lying between 100 and 1000 can be formed with the digits 0, 2, 3, 4, 5 (no digit being repeated)?
(a) 24 (b) 48
(c) 72 (d) 96

13. In how many ways the letters of the word RAINBOW be arranged?
(a) 5040 (b) 4050
(c) 3040 (d) 8040

14. In the above question how many words begin with R?
(a) 720 (b) 360
(c) 1440 (d) none of these

15. In the question number 13 how many words begin with R and ends with W?
(a) 120 (b) 240
(c) 180 (d) 360

16. In the question number 13 how many words are there in which R and W are at the end positions ?
(a) 120 (b) 180
(c) 210 (d) 240

17. In the question number 13 how many words are there in which N and B are together ?
(a) 720 (b) 360
(c) 540 (d) 1440

18. In the question number 13 how many words are there in which I and O are never together?
(a) 1440 (b) 720
(c) 3600 (d) none of these

19. In the question number 13 how many words are there in which vowels are never together?
(a) 720 (b) 1440
(c) 360 (d) 3600

20. In the question number 13 how many words are there in which A is always before I and I is always before O?
(a) 840 (b) 420
(c) 720 (d) none of these

21. In the question number 13 how many words are there in which vowels are always before the consonants?
(a) 72 (b) 144
(c) 96 (d) none of these

22. In the question number 13 how many words are there in which first and last letters are vowels?
(a) 360 (b) 720
(c) 1440 (d) 2880

23. If all the words formed by the letters of the word RAINBOW are arranged in a dictionary form, then what is the position of the word RAINBOW in that dictionary?
(a) 3136 (b) 3631
(c) 3361 (d) 1363

24. If no two consonants are together then in how many ways can the word RAINBOW be arranged?
(a) 441 (b) 420
(c) 360 (d) 144

25. How many words can be formed of the letters in the word RAINBOW, so that the vowels may occupy only even positions?

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

(a) 567 (b) 144
(c) 576 (d) 625

25. How many different words can be formed with the letters of the word RAINBOW so that the vowels occupy odd places?
(a) 676 (b) 625
(c) 343 (d) 576

27. In how many ways can the letters of the word MOBILE be arranged so that the consonants always occupy the odd places?
(a) 18 (b) 36
(c) 54 (d) 72

28. In how many ways can the letters of the word RAINBOW be arranged so that only two vowels always remain together?
(a) 2880 (b) 1440
(c) 3200 (d) none of these

29. In how many ways can the letters of the word MOBILE be arranged so that at least two consonants remain together?
(a) 2880 (b) 3200
(c) 576 (d) 1600

30. In how many ways can the letters of the word STRANGE be arranged so that the vowels may appear in the odd places?
(a) 720 (b) 1092
(c) 1440 (d) none of these

31. Four men and three women are to be seated for a dinner such that no two women sit together and no two men sit together. Find the number of ways in which this can be arranged.
(a) 144 (b) 72
(c) 36 (d) none of these

32. 5 men and 4 women are to be seated in a row so that the women occupy the even places. How many such arrangements are possible?
(a) 2880 (b) 1440
(c) 720 (d) 2020

33. In how many ways 6 students and 4 teachers be arranged in a row so that no two teachers are together?
(a) 604800 (b) 24680
(c) 25860 (d) none of these

34. In how many ways can 5 girls and 3 boys stand in a row so that no two boys are together?
(a) 14400 (b) 7200
(c) 3600 (d) 3200

35. In how many different ways can 8 examination papers be arranged in a row, so that the best and the worst papers may never come together?
(a) 30240 (b) 2340
(c) 12340 (d) none of these

36. There are 6 books on Physics, 3 on Chemistry and 4 on Biology. In how many ways can these be placed on a shelf if the books on the same subject are to be together?
(a) 622080 (b) 888000
(c) 222000 (d) 413080

37. In how many ways three different rings can be worn in four fingers with at most one in each finger?
(a) 24 (b) 12
(c) 36 (d) 120

38. How many different signals can be given using any number of flags from 6 flags of different colours?
(a) 1240 (b) 1956
(c) 3976 (d) 1976

39. A person has 3 shirts 4 coats and 6 ties. In how many ways can he wear them?
(a) 27 (b) 36
(c) 72 (d) 130

40. Four friends have 7 shirts, 6 paints and 8 ties. In how many ways can they wear them?
(a) 336000 (b) 33600
(c) 12480 (d) 508032000

41. In how many ways can 7 soldiers stand in a queue?
(a) 5040 (b) 2880
(c) 280 (d) none of these

42. In how many ways can 10 soldiers stand in two rows having 5 soldiers in each row?
(a) 7257600 (b) 235600
(c) 923500 (d) 35400

43. How many different signals can be made by taking 3 different coloured flags at a time from 5 different coloured flags?
(a) 120 (b) 60
(c) 15 (d) 125

44. 10 students took a test and each one got different score. In how many ways can three different prizes be given?
(a) 270 (b) 360
(c) 720 (d) none of these

45. There are 10 stations on a certain railway line. How many different kinds of tickets of class IInd must be printed in order that a passenger may go from one station to any other by purchasing ticket?
(a) 90 (b) 45
(c) 135 (d) 100

46. Find the number of ways in which 12 different books can be arranged on a shelf so that two particular books shall not be together.
(a) 12! (b) $1320 \times 9!$
(c) 387892 (d) 399168000

47. Find the number of ways in which 12 different books can be arranged on a shelf so that three particular books shall not be together.
(a) $126 \times 10!$ (b) 125000
(c) 357500 (d) 123040

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

PERMUTATIONS OF n THINGS NOT ALL DIFFERENT

- Number of permutations of n things, taken all at a time, of which p are alike of one kind, q are alike of second kind, r are alike of third kind and rest are different

$$= \frac{n!}{p!\,q!\,r!}$$

- Number of permutations of n things, of which P_1 are alike of one kind, P_2 are alike of second kind, P_3 are alike of third kind, ... P_r are alike of rth kind such that $P_1 + P_2 + P_3 + ... + P_r = n$ is $\frac{n!}{P_1!\,P_2!\,P_3!...P_r!}$

INTRODUCTORY EXERCISE-19.2

1. In how many ways can the letters of the word REPEAT be arranged ?
(a) 2240 (b) 232230
(c) 360 (d) 235760

2. In how many ways can the letters of the word TAMANNA be arranged?
(a) 120 (b) 420
(c) 840 (d) none of these

3. In how many ways can the letters of the word RECUPERATE be arranged?
(a) 234000 (b) 123000
(c) 232300 (d) 302400

4. In how many ways can the letters of the word ASSASSINATION be arranged?
(a) 181800 (b) 818100
(c) 108108 (d) 10810800

5. In the above question (number 4) find the number of arrangements of four different letters of the word in a row?
(a) 360 (b) 180
(c) 540 (d) none of these

6. How many arrangements can be made out of the letters of the word COMMITTEE, taken all at a time, such that the four vowels do not come together?
(a) 216 (b) 45360
(c) 1260 (d) 43200

7. In how many ways can the letters of the word DUMDUMDIGADIGA be arranged?
(a) 180 (b) 90
(c) 270 (d) none of these

8. How many different words can be formed with the letters of the word NAINITAL such that each of the word begin with L and end with T?
(a) 90 (b) 80
(c) 88 (d) 82

9. How many words can be made from the word MATHEMATICS in which vowels are together?
(a) 12960 (b) 120960
(c) 15400 (d) none of (a), (b), (c)

10. How many words can be made from the word IMPORTANT in which both T do not come together?
(a) 141120 (b) 112244
(c) 113113 (d) 888222

11. How many words can be made from the word TING TING TRING in which vowels occupy even positions?
(a) 336000 (b) 85360
(c) 113600 (d) none of these

12. If the different permutations of the word PRODIGIOUS are listed as in a dictionary, then what is the rank of the word PRODIGIOUS?
(a) 200966 (b) 609902
(c) 12500 (d) 24800

13. If all the letters of the word SEQUESTERED be arranged as in a dictionary, what is 50th word?
(a) DEEEEQURSTS (b) RUQDESTESEE
(c) ESSTREEUQDE (d) DQUESTREEES

14. How many 7 digit numbers can be formed using the digits 1, 2, 0, 2, 4, 2, 4?
(a) 120 (b) 360
(c) 240 (d) 424

15. How many 6 digit numbers can be formed out of the digits of the number 113113?
(a) 15 (b) 180
(c) 888 (d) 222

16. How many 6 digit numbers can be formed out of the number 567724, which are even?
(a) 360 (b) 480
(c) 180 (d) 220

17. How many numbers greater than a million can be formed with the digits 5, 5, 2, 2, 1, 7, 0?
(a) 980 (b) 1080
(c) 920 (d) 1200

18. How many different words can be formed with the letters of the word REGURGITATE so that the two T's are always together?
(a) 453600 (b) 543600
(c) 183600 (d) 5040

19. The number of ways in which the letters of the word SUMPTUOS can be arranged so that the two U's do not come together is :
(a) 5040 (b) 7560
(c) 38920 (d) none of (a), (b), (c)

20. In how many ways the letters of the word AFLATOON be arranged if the consonants and vowels must occupy alternate places?
(a) 144 (b) 143
(c) 288 (d) 248

21. In how many ways can the letters of the word SOOTHSAYER be arranged so that the two S's and two O's come together?
(a) 23400 (b) 302050
(c) 12340 (d) 40320

22. There are three copies of each of 4 different books. In how many ways can they be arranged on a shelf?

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

(a) 369600 (b) 366900
(c) 396600 (d) 306609

23. There are 3 red, 4 green and 5 pink marbles in a bag. They are drawn one by one and arranged in a row. Assuming that all the 12 marbles are drawn, determine the number of different arrangements.
(a) 22770 (b) 27720
(c) 22077 (d) 27270

PERMUTATIONS WHERE REPETITIONS ARE ALLOWED

- The number of permutations of n different things taken r at a time, when each may be repeated any number of times in each arrangement is 'n^r'.

INTRODUCTORY EXERCISE-19.3

1. How many numbers of 5 digits can be formed with the digits 0, 2, 3, 4 and 5 if the digits may repeat?
(a) 2500 (b) 250
(c) 120 (d) 2400

2. How many numbers each lying between 9 and 1000 can be formed with the digits 0, 1, 2, 3, 7, 8 (numbers can be repeated) ?
(a) 30 (b) 120
(c) 210 (d) 90

3. How many five-figures numbers can be formed with the digits 1, 2, 3, 4, 5 given that an even digit may not occupy an even place if the digits may be repeated?
(a) 1125 (b) 5210
(c) 1152 (d) 2120

4. How many of the numbers from 1000 to 9999 (both inclusive) do not have four different digits?
(a) 4446 (b) 4664
(c) 4464 (d) 6444

5. Find the number of three digit numbers (repetitions allowed) such that atleast one of the digits is 9.
(a) 252 (b) 648
(c) 468 (d) 864

6. How many natural numbers not exceeding 4321 can be formed with the digits 1, 2, 3, 4 if repetition is allowed?
(a) 123 (b) 113
(c) 222 (d) 313

7. In how many ways 4 rings of different types can be worn in 3 fingers?
(a) 49 (b) 12
(c) 24 (d) 81

8. In how many ways can 5 prizes be given away to 4 boys, when each boy is eligible for all the prizes?
(a) 1024 (b) 20
(c) 625 (d) 540

9. In how many ways can n balls be randomly distributed in n cells?
(a) $n!$ (b) n^n
(c) $n(n-1)$ (d) 2^n

10. In the above question, what will be the number of ways if it is specified that each cell is to be occupied?
(a) $n!$ (b) n^n
(c) n^3 (d) 2^n

11. In how many ways can 5 letters be posted in 4 letter boxes?
(a) 512 (b) 1024
(c) 625 (d) 20

CIRCULAR PERMUTATIONS

In the previous articles we have learned linear permutations. Now we have to know about circular permutations. In circular permutation things are to be arranged in the form of a ring or a circle, *e.g.*, arrangements of persons around circular dinning table, conferencing table, garland etc.

Since in the circular permutation there is no any end point *i.e.*, we can not say which one is the beginning article or ending article in the possible arrangements.

Hence the number of circular permutations of n objects $= \frac{n!}{n} = (n-1)!$

Thus in a circular permutation one thing is kept fixed and the remaining $(n-1)$ things are arranged in $(n-1)!$ ways.

- If the clockwise and counter clockwise orders are not distinguishable, then the number of ways $= \frac{1}{2}\{(n-1)!\}$

INTRODUCTORY EXERCISE-19.4

1. In how many ways can 6 boys form a ring?
(a) 120 (b) 720
(c) 119 (d) none of (a), (b), (c)

2. In how many ways can 6 boys be arranged at a round table so that 2 particular boys may be together?
(a) 24 (b) 48
(c) 360 (d) 60

3. In how many ways can 6 beads be strung into a necklace?
(a) 60 (b) 360
(c) 720 (d) 120

4. In how many ways can 5 men and 2 ladies be arranged at a round table if two ladies are never together?
(a) 5040 (b) 480
(c) 240 (d) 720

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

5. 9 persons were invited for a business meeting by Ambani, where the host be seated at a circular table. How many different arrangements are possible if two invitees *viz.*, Sahara and Mahindra be seated on either side of the host Ambani?
(a) 10080 (b) 10800
(c) 9200 (d) 4600

6. In a G-20 meeting there were total 20 people representing their own country. All the representative sat around a circular table. Find the number of ways in which we can arrange them around a circular table so that there is exactly one person between two representatives namely Manmohan and Musharraf.
(a) $2 \times (17!)$ (b) $2 \times (18!)$
(c) $(3!) \times (18!)$ (d) $(17!)$

7. 6 men and 6 women are to be seated at a circular table such that no two women are adjacent, then the number of arrangements is :
(a) 86400 (b) 46800
(c) 12! (d) $\frac{1}{2}(12!)$

8. There are 3 boys and 4 girls, seated around a circular table so that no two boys are together. Find the number of ways in which this can be done :
(a) 36 (b) 120
(c) 144 (d) 132

9. 7 sisters dine at a round table. They dine together till each of them dine with different neighbours *i.e.*, they do not like to dine with same neighbours in any two arrangements. In a year how many days they dine together?
(a) 180 (b) 504
(c) 720 (d) 360

10. Find the number of ways in which 10 different flowers can be strung to form a garland so that 3 particular flowers are always together.
(a) 30240 (b) 30420
(c) 23400 (d) none of (a), (b), (c)

11. 3 Male and 3 female singers sat around a circular table. Alka is a female singer did not sat adjacent to a male singer and Sonu is a male singer did not sat adjacent to a female singer. In how many ways this could be done?
(a) 4 (b) 6
(c) 9 (d) 10

COMBINATIONS

Let *A, B, C* be three letters, then we can combine any two of them in the following ways :

AB, BC, AC

Similarly if *A, B, C, D* are four letters, then we can combine any two of them in the following manner :

AB, AC, AD, BC, BD, CD

Similarly we can combine any 3 of *A, B, C, D* as :

ABC, ABD, ACD, BCD

NOTE In the combinations order of the letters (or things) is not considered.
Here *AB, BA* are same and *BC, CB* are also same and so on.
Hence *ABC, ACB, BAC, BCA, CAB* and *CBA* are same and counted as a single combination of *A, B, C*.

NOTE The word combination is generally used for selection of things and the word permutation is used for arrangements of things.

Combinations of 'n' different things taken 'r' at a time

The number of all combinations of n distinct things taken r at a time $(r \le n)$ is

$${}^nC_r = \frac{n!}{(n-r)!\,r!}, \quad C \text{ denotes combination.}$$

PROPERTIES/FACTS ABOUT COMBINATIONS

1. ${}^nC_r = ({}^nP_r)/(r!)$
2. ${}^nC_0 = {}^nC_n = 1$
3. ${}^nC_r = {}^nC_{n-r}$ $(0 \le r \le n)$
4. ${}^nC_x = {}^nC_y$ if $x + y = n$ or $x = y$ $(x, y \in W)$
5. ${}^nC_{r-1} + {}^nC_r = {}^{n+1}C_r$
6. If n is even, the greatest value of ${}^nC_r = {}^nC_m$ where, $m = \frac{n}{2}$.
7. If n is odd, the greatest value of ${}^nC_r = {}^nC_m$ where, $m = \frac{(n-1)}{2}$ or $m = \frac{(n+1)}{2}$
8. ${}^rC_r + {}^{r+1}C_r + {}^{r+2}C_r + \ldots + {}^nC_r = {}^{n+1}C_{r+1}$; $r \le n$
9. ${}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + \ldots + {}^nC_n = 2^n$
10. ${}^nC_1 + {}^nC_2 + {}^nC_3 + \ldots + {}^nC_n = 2^n - 1$
11. ${}^nC_0 + {}^nC_2 + {}^nC_4 + \ldots = 2^{2n}$

INTRODUCTORY EXERCISE-19.5

1. Find the vlaue of ${}^8C_3$.
(a) 56 (b) 8!
(c) 65 (d) 3^8

2. Find the value of ${}^{10}C_5$.
(a) 525 (b) 126
(c) 252 (d) 50

3. If ${}^{13}C_x = {}^{13}C_y$ and $x \ne y$, what is the value of $x + y$?
(a) 6 (b) 7
(c) 13 (d) 26

4. ${}^{17}C_r = {}^{17}C_{r+3}$. Find the value of r.
(a) 17 (b) 6
(c) 7 (d) 13

5. ${}^7P_3 = n.{}^7C_3$, find the value of n.
(a) 4 (b) 8
(c) 6 (d) 9

6. If $C(2n, 3) : C(n, 2) = 12 : 1$, find n.
(a) 3 (b) 5
(c) 6 (d) 4

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

7. What is the least possible value of n if : ${}^{n-1}C_3 + {}^{n-1}C_4 > {}^{n}C_3$
(a) 6 (b) 8
(c) 7 (d) 12

8. If ${}^{n}C_x = 56$ and ${}^{n}P_x = 336$, find n and x.
(a) 7, 3 (b) 8, 4
(c) 8, 3 (d) 9, 6

9. What is the value of x when ${}^{11}C_x$ is maximum ?
(a) 7 (b) 6
(c) 5 (d) both (b) and (c)

10. What is the maximum value of ${}^{12}C_x$ for any natural number x ?
(a) 6 (b) 7
(c) 8 (d) 4

11. For what value of x, ${}^{14}C_x$ is maximum?
(a) 5 (b) 6
(c) 7 (d) 8

12. How many different committees of 5 members may be formed from 6 gentlemen and 4 ladies?
(a) 181 (b) 357
(c) 603 (d) 252

13. In a test paper there are total 10 questions. In how many different ways can you choose 6 questions to answer?
(a) 210 (b) 540
(c) 336 (d) none of these

14. In the above question (no. 13) if the question number 1 is compulsory in how many ways can you choose to answer 6 questions in all?
(a) 53 (b) 63
(c) 126 (d) 210

15. Droupdi has 5 friends. In how many ways can she invite one or more of them to a dinner?
(a) 31 (b) 5^5
(c) 13 (d) 2^5

16. In how many ways can a committee of 6 members be formed from 7 men and 6 ladies consisting of 4 men and 2 ladies?
(a) 252 (b) 525
(c) 625 (d) 256

17. A committee of 5 persons is to be formed from a group of 6 gentlemen and 4 ladies. In how many ways can this be done if the committee is to be included atleast one lady?
(a) 123 (b) 113
(c) 246 (d) 945

18. A committee of 7 persons is to be chosen from 13 persons of whom 6 are Americans and 7 are Indians. In how many ways can the selection be made so as to retain a majority of Indians?
(a) 945 (b) 1057
(c) 923 (d) 1056

19. In how many ways can a committee of 4 men and 3 women be appointed from 6 men and 8 women?
(a) 480 (b) 408
(c) 420 (d) 840

20. In the previous question (no. 19) what will be the number of committees in which Miss A refuses to serve if Mr. B is a member?
(a) 210 (b) 420
(c) 630 (d) none of these

21. In how many ways 7 members forming a committee out of 11 be selected so that 3 particular members must be included?
(a) 60 (b) 130
(c) 80 (d) 70

22. In the previous question (no. 21) how many committees can be formed if 3 particular members must not be included?
(a) 6 (b) 8
(c) 10 (d) 61

23. A committee of 3 experts is to be selected out of a panel of 7 persons, three of them are engineers, three of them are managers and one is both engineer and manager. In how many ways can the committee be selected if it must have at least an engineer and a manager?
(a) 33 (b) 22
(c) 11 (d) 66

24. A committee of 5 persons is to be formed out of 6 gents and 4 ladies. In how many ways this can be done, when at most two ladies are included?
(a) 186 (b) 168
(c) 136 (d) 169

25. In an election, a voter may vote for any number of candidates not greater than the number to be chosen. There are 7 candidates and 4 members are to be chosen. In how many ways can a person vote?
(a) 89 (b) 98
(c) 79 (d) 101

26. At an election there are 5 candidates and 3 members are to be elected and a voter is entitled to vote for any number to be elected. In how many ways a voter can a vote?
(a) 25 (b) 5^3
(c) 3^5 (d) 15

27. A question has two parts, Part A and Part B, each containing 8 questions. If the students has to choose 6 from part A and 5 questions from part B, in how many ways can he choose the questions?
(a) 1268 (b) 1788
(c) 1024 (d) 1568

28. An examinee is required to answer six questions out of twelve questions which are divided into two groups each containing six questions and he is not permitted to answer more than four questions from any group. In how many ways can he answer six questions?
(a) 750 (b) 850
(c) 580 (d) 570

29. An examination paper consists of 12 questions divided into two parts, part A and part B. Part A contains 7 questions and part B contains 5 questions. A candidate is required to attempt 8 questions, selecting atleast 3 from each part. In how many ways can he select the questions?
(a) 240 (b) 60
(c) 420 (d) 480

30. In MOCK CAT Quantitative Aptitude Section was divided into 3 groups of 5, 5 and 4 questions respectively. A candidate is required to answer 6 questions taking at least two questions from each of the first two groups and atleast one question from the third group. In how many ways can he make up his choice?
(a) 1200 (b) 1400
(c) 1284 (d) 1560

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yourshahbood.wordpress.com

For More Visit : www.LearnEngineering.in

31. Mr. Daruwala has 18 acquaintances of whom 13 are relatives. In how many ways he may invite 10 guests so that 8 of whom are relatives?
(a) 12870 (b) 22022
(c) 20222 (d) 12780

32. If there are 11 players in a cricket team, all of whom shake hands with each other, how many shake hands take place in the team?
(a) 36 (b) 55
(c) 66 (d) none of these

33. Amitabh has a list of 24 friends. He wishes to invite some of them in such a manner that he can enjoy maximum number of parties, but in each party the number of friends (*i.e.*, invitees) be same and each party must have different set of persons. Then how many parties can Amitabh enjoy?
(a) 2704156 (b) 357600
(c) 235763 (d) 270156

34. An ice-cream parlour offers only family packs of ice-creams with 11 different flavours. If each member of a family loves different flavours, then maximum how many such families can purchase the ice-cream if each family contains equal number of persons?
(a) 246 (b) 462
(c) 123 (d) $^{11}C_2$

35. In the previous question (no. 34) what is the maximum possible number of member in a family?
(a) 4 (b) 5
(c) 6 (d) 8

36. In a meeting everyone had shaken hands with everyone else, it was found that 66 handshakes were exchanged. How many members were present in the meeting?
(a) 10 (b) 14
(c) 12 (d) 8

37. In how many ways can a cricket team of eleven players be chosen out of a batch of 16 players if a particular player is always chosen?
(a) 2002 (b) 3003
(c) 1003 (d) 7603

38. In the previous question (no. 37) if a particular player is never chosen, then the number of ways in which a cricket team of eleven players can be chosen is :
(a) 2365 (b) 2359
(c) 1365 (d) 1056

39. A cricket team of 11 players is to be formed from 16 players including 4 bowlers and 2 wicket keepers. In how many different ways can a team be formed so that the team has atleast 3 bowlers and atleast one wicket keeper?
(a) 2472 (b) 2274
(c) 2427 (d) 1236

40. A cricket team of 11 players is to be formed from 20 players including 6 bowlers and 3 wicket keepers. In how many different ways can a team be formed so that the team contain exactly 2 wicket keepers and atleast 4 bowlers?
(a) 22725 (b) 27225
(c) 22275 (d) none of (a), (b), (c)

41. A team of 11 cricketers containing 1 wicket keeper, 2 bowlers, 3 all rounders and 5 batsmen, is to be formed from a group of 25 cricketers containing 2 wicket keepers, 8 bowlers, 5 all rounders and 10 batsmen?
(a) 141120 (b) 111240
(c) 101240 (d) none of (a), (b), (c)

42. Out of 3 books on Economics, 4 books on Corporate Strategy and 5 books on Philosophy, how many collection consists of exactly one book on each subject?
(a) 40 (b) 36
(c) 60 (d) 120

43. In the previous question (no. 42) what is the number of different collections of books if atleast one book on each subject is taken?
(a) 2365 (b) 3255
(c) 4224 (d) 1236

44. A box contains 7 red, 6 white and 4 blue balls. How many selection of three balls can be made so that all three are red balls?
(a) 35 (b) 70
(c) 42 (d) 17

45. In the previous question (no. 44) if the red coloured ball is not taken, then the number of selections is :
(a) 30 (b) 120
(c) 60 (d) none of (a), (b), (c)

46. In the question (no. 44), if there is one ball of each colour, then the number of selections is :
(a) 148 (b) 124
(c) 168 (d) 186

47. An urn contains 5 different red and 6 different green balls. In how many ways can 6 balls be selected so that there are atleast two balls of each colour?
(a) 425 (b) 245
(c) 125 (d) 625

48. Find the number of straight lines formed by joining 6 different points on a plane, no three of them being collinear.
(a) 21 (b) 15
(c) 16 (d) 24

49. How many different straight lines can be formed by joining 12 different points on a plane of which 4 are collinear and the rest are non-collinear?
(a) 16 (b) 32
(c) 61 (d) 64

50. Find the number of diagonals in a decagon.
(a) 25 (b) 35
(c) 45 (d) $^nC_2 - n$

51. Find the number of diagonals in an n-sided polygon.
(a) n^2 (b) $\frac{n(n-3)}{2}$
(c) $n!$ (d) 2^n

52. A polygon has 54 diagonals. Find the number of its sides.
(a) 6 (b) 12
(c) 27 (d) 18

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

53. Find the number of triangles that can be formed by joining the 6 non-collinear points on a plane.
(a) 18 (b) 20
(c) 24 (d) 36

54. Find the number of triangles formed by joining 12 different points on a plane, no three of them being collinear (with the exception of 4 points which are collinear).
(a) 126 (b) 216
(c) 220 (d) 222

55. Two parallel lines each have a number of distinct points marked on them. On one line there are 2 points P and Q. On the other line there are 8 points. Find the number of different triangles which could be formed having 3 of the 10 points as vertices.
(a) 216 (b) 126
(c) 64 (d) 32

56. In the above question (no. 55) how many of these triangles have P as a vertex?
(a) 63 (b) 64
(c) 36 (d) 31

57. If 20 straight lines be drawn in a plane, no two of them being parallel and no three of them concurrent, how many points of intersection will there be?
(a) 95 (b) 380
(c) 400 (d) 190

58. Find the number of different straight lines obtained by joining n points on a plane, no three of which are collinear.
(a) n^3 (b) $\frac{n(n-1)}{2}$
(c) $n!$ (d) 2^4

59. In the previous question (no. 58) what will be the number of straight lines if p points out of n points are in the same straight line?
(a) ${}^{m+n}C_2$ (b) ${}^{n-m}C_2$
(c) ${}^nC_2 - {}^pC_2 + 1$ (d) none of (a), (b), (c)

60. There are n points in a plane no three of which are in the same straight line with the exception of them m ($m < n$) points which are all in the same straight line. Find the number of triangles formed by joining them.
(a) ${}^{n-m}C_3$ (b) ${}^nC_3 - {}^mC_3$
(c) ${}^nC_m$ (d) none of (a), (b), (c)

61. If m parallel lines in plane are intersected by a family of n parallel lines. Find the number of parallelograms formed.
(a) m^n (b) $(m+1)(n+1)$
(c) $\frac{(m-n)}{n!}$ (d) $\frac{mn(m-1)(n-1)}{4}$

62. In a plane there are 37 straight lines, of which 13 pass through the point A and 11 pass through the point B. Besides, no three lines pass through one point, no line passes through both points A and B, and no two are parallel. Find the number of points of intersection of the straight lines.
(a) 535 (b) 525
(c) 235 (d) 355

63. Find the number of different words that can be formed from 15 consonants and 5 vowels by taking 2 consonants and 4 vowels in each word.
(a) 625 (b) 630
(c) 525 (d) 615

64. Find the number of ways of selecting 4 letters from the word EXAMINATION.
(a) 136 (b) 126
(c) 252 (d) 525

65. How many words can be formed by using 4 letters at a time out of the letters of the word MATHEMATICS?
(a) 2445 (b) 2454
(c) 1243 (d) 1454

66. In how many ways can 3 ladies and 3 gentlemen be seated at a round table, so that any two and only two of ladies sit together?
(a) 72 (b) 36
(c) 24 (d) 144

67. Eighteen guests have to be seated half on each side of a long table. Four particular guests desire to sit on one particular side and three others on other side. Determine the number of ways in which the sitting, arrangements can be made.
(a) 18! (b) $462 \times (9!)^2$
(c) $(9!)^2$ (d) $7! \times (9!)^2$

68. There are 4 different letters and 4 addressed envelopes. In how many ways can the letters be put in the envelopes so that atleast one letter goes to the correct address?
(a) 15 (b) 16
(c) 18 (d) 12

69. Find the number of ways in which none of the letters goes to the correct address? (as per the previous question)
(a) 6 (b) 9
(c) 10 (d) 11

SOME RESTRICTED COMBINATIONS

1. Number of combinations of n things taken 'r' at a time in which x **particular things always occur** is ${}^{n-x}C_{r-x}$.
2. Number of combinations of n things taken 'r' at a time in which x **particular things never occur** is ${}^{n-x}C_r$.
3. Number of ways of selections of **zero or more** things from a group of 'n' **distinct things** is 2^n.
i.e., ${}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + \ldots + {}^nC_n = 2^n$
4. Number of ways of selection of **one or more** things from a group of 'n' **distinct things** is $2^n - 1$.
i.e., ${}^nC_1 + {}^nC_2 + {}^nC_3 + \ldots + {}^nC_n = 2^n - 1$
($\because {}^nC_0 = 1$)

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

5. Number of ways of selection of 'r' **things** out of 'n' **identical things** is 1. $(r \le n)$
6. Number of ways of selection of **zero or more** things from a group of 'n' **identical things** is $(n+1)$.
7. Number of ways of selection of **some or all things** *i.e.*, atleast 1 out of $(p+q+r+\ldots)$ things of which p are alike of one kind, q are alike of second kind r are alike of third kind and so on, is $\{(p+1)(q+1)(r+1)\ldots\}-1$.
8. Total number of ways of selecting one or more things from 'p' identical things of one kind, 'q' identical things of second kind, 'r' identical things of third kind and n different things is $\{(p+1)(q+1)(r+1)2^n\}-1$
9. Number of selections of k consecutive things out of n things in a row is $(n-k+1)$.
10. Number of selection of k consecutive things out of n things in a circle $= \begin{cases} n & \text{when } k<n \\ 1 & \text{when } k=n \end{cases}$

INTRODUCTORY EXERCISE-19.6

1. Given 3 different red dyes, 4 different blue dyes, and 5 different green dyes, how many combinations of dyes can be made taking atleast one green and one blue dye?
(a) 2300 (b) 31
(c) 3720 (d) 3560
2. In how many ways is it possible to make a selection by taking any number of 15 fruits, namely 3 oranges, 5 apples and 7 mangoes?
(a) 131 (b) 191
(c) 68 (d) 3720
3. In how many ways can a person choose one or more beverages out of Sodawater, Mirinda, Whisky and Coffee?
(a) 16 (b) 4
(c) 15 (d) 4!
4. A bag contains 4 mangoes and 5 oranges. In how many ways can I make a selection so as to take atleast one mango and one orange?
(a) 465 (b) 365
(c) 8 (d) 19
5. In how many ways is it possible to make a selection by taking any number of all 20 fruits, namely 9 mangoes, 7 oranges and 4 apples?
(a) 199 (b) $^{20}C_3$
(c) $^{20}P_3$ (d) 399
6. A student is allowed to select at most n books from a collection of $(2n+1)$ books. If the total number of ways in which he can select atleast one book is 63, find the value of n.
(a) 5 (b) 6
(c) 3 (d) 7

DIVISION INTO GROUPS

Division of Items into Groups of Unequal Size

1. The number of ways in which $(m+n)$ different things can be divided into two groups containing m and n things respectively is $\dfrac{(m+n)!}{m!n!}$.

   Since m things can be selected out of $(m+n)$ things in $^{m+n}C_m$ ways and each time when m things are taken, n things left out to form the other group of n things in ^nC_n ways *i.e.*, in 1 way only.

   Hence the required number of ways
   $$= {}^{m+n}C_n = \frac{(m+n)!}{m!\cdot n!}$$
2. The number of ways in which $(m+n+p)$ different things can be divided into three groups containing m, n and p things respectively is $\dfrac{(m+n+p)!}{m!\cdot n!\cdot p!}$.

NOTE If $(m+n)$ things are distributed among two persons in the groups of m, n things, then the number of ways $= \dfrac{(m+n)!}{m!\cdot n!} \times 2!$

Similarly : If $(m+n+p)$ things are distributed among three persons in the groups of m, n and p things then the number of ways
$$= \frac{(m+n+p)!}{m!\cdot n!\cdot p!} \times 3!$$

3. The number of ways in which 'n' distinct things can be distributed to r different persons $= r^n$.

Division of Items into Groups of Equal Size

The number of ways in which mn different things can be divided equally into m groups, each group containing n things
$$= \left(\frac{(mn)!}{(n!)^m}\right)\frac{1}{m!}$$

The number of ways in which mn different things can be distributed equally into m groups, each group containing n things $= \dfrac{(mn)!}{(n!)^m}$.

NOTE In the distribution order is important hence the divisible things can be arranged in $m!$ ways.
Since things are divided into m groups.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

DIVISION OF IDENTICAL ITEMS INTO GROUPS

1. The number of ways of dividing n identical things among r persons (or groups), each of whom, can receive zero or more things is ${}^{n+r-1}C_{r-1}$ where $0 \le r \le n$.
2. The number of ways of dividing n identical things among r persons, each one of whom, receives atleast one item is ${}^{n-1}C_{r-1}$ where $1 \le r \le n$.
3. The number of ways in which 'n' identical things can be divided into r groups so that no group contains less than m items and more than k (where $m < k$) is coefficient of x^n in the expansion of $(x^m + x^{m+1} + \ldots + x^k)^r$.

INTRODUCTORY EXERCISE-19.7

1. Find the number of ways in which 12 students may be equally divided into three groups.
(a) 3355 (b) 5775
(c) 7755 (d) 8822

2. Find the number of ways in which 12 mangoes may be equally divided among 3 boys.
(a) 57750 (b) 34650
(c) 23450 (d) 12345

3. Five balls of different colours are to be placed in three boxes of different sizes. Each box can hold all the five balls. In how many ways can we place the balls so that no box remains empty?
(a) ${}^5C_3$ (b) 5!
(c) 150 (d) 5^3

4. In how many ways can a pack of 52 cards be divided equally among four players in order?
(a) $(52!)^4$ (b) $4 \times (13!)$
(c) $\frac{52!}{(13!)^4}$ (d) none of these

5. In how many ways can a pack of 52 cards be divided equally into four groups?
(a) $\frac{52!}{(13!)^4 \times 4!}$ (b) $4!\left[\frac{52!}{(13!)^4}\right]$
(c) $\frac{52!}{(13!)^4}$ (d) none of these

6. In how many ways can a pack of 52 cards be divided into 4 sets, three of them having 16 cards each and the fourth just 4 cards?
(a) $16! \times 52!$ (b) $\frac{(52!)}{(16!)^3}$
(c) $\frac{52!}{(3!)^{16}}$ (d) $\frac{52!}{(16!)^3 \times (3!)}$

7. In how many ways can $2n$ students be divided into n pairs?
(a) n^2 (b) $\frac{n!}{2}$
(c) $\frac{2n!}{(2!)^n \cdot (n!)}$ (d) none of these

8. Total number of ways in which 20 mangoes can be distributed among 4 persons is :
(a) ${}^{20}C_3$ (b) ${}^{23}C_3$
(c) ${}^{24}C_4$ (d) $\frac{20!}{4!}$

9. Find the number of ways of distributing 7 identical balls into three boxes so that no box is empty and each box being large enough to accomodate all balls.
(a) 15 (b) 7^3
(c) 3^7 (d) $\frac{7!}{3!}$

10. Find the number of non negative integral solutions of :
$x_1 + x_2 + x_3 + x_4 = 44$
(a) ${}^{44}C_4$ (b) ${}^{43}C_3$
(c) ${}^{47}C_3$ (d) none of these

11. How many integral solutions are there to $a + b + c = 18$ when $a \ge 1$, $b \ge 2$, $c \ge 3$?
(a) ${}^{19}C_3$ (b) ${}^{17}C_2$
(c) ${}^{15}C_3$ (d) ${}^{14}C_2$

12. Find the number of ways in which 21 balls can be distributed among 3 persons such that each person does not receive less than 5 balls.
(a) 28 (b) 14
(c) 21 (d) 7

SOME IMPORTANT RESULTS

1. Number of squares in a square of having 'n' columns and 'n' rows
$$= 1^2 + 2^2 + 3^2 + \ldots + n^2 = \Sigma n^2$$
$$= \frac{n(n+1)(2n+1)}{6}$$
2. Number of rectangles in a square having 'n' columns and 'n' rows
$$= 1^3 + 2^3 + 3^3 + \ldots + n^3 = \Sigma n^3 = \left[\frac{n(n+1)}{2}\right]^2$$
3. Number of squares in a rectangle having 'm' columns and 'n' rows
$$= m \cdot n + (m-1)(n-1) + (m-2)(n-2) + \ldots 0.$$
4. Number of rectangles in a rectangle having 'm' columns and 'n' rows
$$= (1 + 2 + 3 + \ldots + m) \cdot (1 + 2 + 3 + \ldots + n).$$
5. Number of quadrilaterals if m parallel lines intersect 'n' parallel lines $= {}^mC_2 \times {}^nC_2$.
6. Number of terms in $(a_1 + a_2 + \ldots + a_n)^m$ is ${}^{m+n-1}C_{n-1}$.
7. Number of terms in $(1 + x + x^2 + \ldots + x^n)^m$ is $m \cdot n + 1$.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

EXERCISE

LEVEL 1

1. A letter lock consists of 4 rings, each ring contains 9 non-zero digits. This lock can be opened by setting a 4 digit code with the proper combination of each of the 4 rings. Maximum how many codes can be formed to open the lock?
(a) 4^9 (b) ${}^9P_4$
(c) 9^4 (d) none of these

2. In the previous question, if the lock opens in only one arrangement of 4 digits, how many unsuccessful attempts are possible in which lock can not be opened?
(a) $4^9 - 1$ (b) $9^4 - 1$
(c) ${}^9C_4$ (d) none of these

3. A group consists of 4 couples in which each of the 4 persons have one wife each. In how many ways could they be arranged in a straight line such that the men and women occupy alternate positions?
(a) 1152 (b) 1278
(c) 1296 (d) none of these

4. In the previous question (no. 3) in how many ways can they be arranged in a straight line such that no two men were sitting together?
(a) 1242 (b) 1440
(c) 3680 (d) 2880

5. In question number 3, in how many ways can they be seated such that each husband sit with his own wife?
(a) 384 (b) 275
(c) 184 (d) none of these

6. In question no. 3, in how many ways can they be seated around a circular table such that the men and women occupy the alternate position?
(a) 288 (b) 144
(c) 72 (d) 720

7. In question (no. 3) in how many ways can they be seated on the single bench such that all the men sit together and all the women sit together?
(a) 1512 (b) 1296
(c) 1152 (d) none of these

8. How many different words can be made using the letters of the word 'HALLUCINATION' if all consonants are together?
(a) 129780 (b) 1587600
(c) 35600 (d) none of these

9. If all vowels occupy odd places, how many words can be formed from the letters of the word HALLUCINATION?
(a) 129650 (b) 1587600
(c) 78500 (d) none of these

10. In how many ways can the letters of the word SUCCESSFUL be arranged?
(a) 1215700 (b) 1251600
(c) 151200 (d) none of these

11. If all S's come together, then in how many ways the letters of the word SUCCESSFUL be arranged?
(a) 10080 (b) 40080
(c) 2378 (d) none of these

12. If all C's occur together and all U's also occur together, then how many arrangements are possible of the word SUCCESSFUL?
(a) 5745 (b) 2760
(c) 6720 (d) 5432

13. What is the sum of all the 4 digit numbers which can be formed with the digits 1, 2, 3, 4 without repetition?
(a) 15560 (b) 87660
(c) 45600 (d) 66660

14. What is the sum of all 5 digit numbers which can be formed with the digits 0, 1, 2, 3, 4 without repetition?
(a) 2599980 (b) 235500
(c) 923580 (d) 765432

15. Which one of the following polygon has as many diagonals as the number of sides in it?
(a) 4 (b) 6
(c) 5 (d) 7

16. In how many ways can the letters of the word PROPORTION be arranged by taking 4 letters at a time?
(a) 123 (b) 758
(c) 658 (d) 578

17. In the previous question (no. 16) how many words can be formed without changing the relative positions of the vowels and consonants?
(a) 217 (b) 720
(c) 920 (d) 1040

18. There are 16 executives including two brothers, Lehman and Mckinsey. In how many ways can they be arranged around the circular table if the two brothers can not be seated together?
(a) $(14!)\cdot 13$ (b) ${}^{14}P_3$
(c) $\frac{14!}{3!}$ (d) none of these

19. In the previous question how many arrangements are possible if there is exactly one executive between these two brothers?
(a) $(14!)^2$ (b) $(14!)$
(c) $2\cdot(14!)$ (d) none of these

20. In how many different ways can 6 different balls be distributed to 4 different boxes, when each box can hold any number of ball?
(a) 2048 (b) 1296
(c) $(24)^2$ (d) 4096

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

21. In how many different ways can 6 identical balls be distributed to 4 different boxes, when each box can have any number of balls?
(a) 12 (b) 18
(c) 84 (d) 88

22. If $a+b+c=21$, what is the total number of non-negative integral solutions?
(a) 123 (b) 253
(c) 321 (d) 231

23. If $a+b+c=21$ what is the total number of positive integral solutions?
(a) 109 (b) 190
(c) 901 (d) 910

24. What is the total number of ways of selecting atleast one object from 2 sets of 12 different objects, each set contains 6 objects?
(a) 4096 (b) 4095
(c) 2048 (d) none of these

25. What is the total number of ways of selecting atleast one object from 2 different sets, each set containing 6 identical objects?
(a) 24 (b) 48
(c) 76 (d) 96

26. What is the total number of ways of selecting atleast one item from each of the two sets containing 6 different items each?
(a) 2856 (b) 3969
(c) 480 (d) none of these

27. What is the total number of ways of selecting atleast one item from each of the two sets containing 6 identical items each?
(a) 16 (b) 36
(c) 32 (d) 18

28. A girl has to climb 12 steps. She climbs in either a single step or 2 steps simultaneously. In how many ways can she do it?
(a) 233 (b) 223
(c) 322 (d) 232

29. In how many ways can 8 identical apples be divided among 3 sisters?
(a) 25 (b) 65
(c) 45 (d) 24

30. In how many ways can 100 soldiers be divided into 4 squads of 10, 20, 30, 40 respectively?
(a) 1700 (b) 18!
(c) 190 (d) none of these

31. What is the total number of 4 digit numbers that can be formed using the digits 0, 1, 2, 3, 4, 5 without repetition, such that the number is divisible by 9?
(a) 36 (b) 28
(c) 15 (d) 18

32. How many 5 digit numbers contain exactly two 7's in them?
(a) 4268 (b) 6804
(c) 2340 (d) 1269

33. In how many ways can 15 billiard balls be arranged in a row if 3 are red, 4 are white and 8 are black?
(a) 12 (b) 18
(c) 96 (d) none of these

34. Seven delegates are to address a meeting. If a particular speaker is to speak before another particular speaker, find the number of ways in which this can be arranged.
(a) 1220 (b) 2520
(c) 3250 (d) 7826

35. How many 7 digit telephone numbers can be formed from the digits 0, 1, 2, ..., 9, where each telephone number begins with digit 2?
(a) 10^6 (b) 6^{10}
(c) 10! (d) ${}^{10}P_6$

36. A man has 4 sons. There are 6 proposals of marriage for his sons. In how many ways can they select a proposal for their marriages, such that none of them marry with more than one girl?
(a) 180 (b) 270
(c) 360 (d) none of these

37. In how many ways can 4 books be selected out of 16 books on different subjects?
(a) 1208 (b) 1820
(c) 1296 (d) 1860

38. In how many ways can 4 books be arranged out of 16 books on different subjects?
(a) 34650 (b) 43680
(c) 43890 (d) none of these

39. In how many ways can 16 books on different subjects be divided equally into 4 groups?
(a) $\frac{16!}{(4!)}$ (b) $4!\times(16!)$
(c) $\frac{16!}{(4!)^5}$ (d) none of these

40. In a badminton tournament each player played one game with all the other players. How many players participated in the tournament if they played 105 games in all?
(a) 35 (b) 12
(c) 15 (d) none of these

41. There are 9 subjects in eighth standard but there are only 6 periods in a day. In how many ways can the time table be formed?
(a) 60480 (b) 23460
(c) 56780 (d) 61280

42. How many 6 digit numbers can be formed using the digit 2 two times and the digit 4 four times?
(a) 16 (b) 15
(c) 18 (d) 24

43. There are 4 different monitors and 6 different mother boards. How many different arrangements can be made to purchase a monitor and a motherboard?
(a) 12 (b) 24
(c) 18 (d) 6^4

44. The number of five digit numbers having atleast one of their digits repeated is :
(a) 900 (b) 1000
(c) 62784 (d) none of these

45. Four dice are rolled. The number of possible outcomes in which atleast one die shows 4 is :
(a) 671 (b) 168
(c) 176 (d) none of these

46. The number of signals that can be generated by using 5 differently coloured flags, when any number of them may be hoisted at a time is :

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

(a) 235 (b) 253
(c) 325 (d) none of these

47. If the letters of the word VERMA are arranged in all possible ways and these words are written out as in a dictionary, then the rank of the word VERMA is :
(a) 108 (b) 117
(c) 810 (d) 180

48. How many 5 digit numbers divisible by 3 can be formed using the digits 0, 1, 2, 3, 4 and 5, without repetition?
(a) 108 (b) 216
(c) 810 (d) 180

49. In how many ways can a mixed double game can be arranged from amongst 8 married couples if no husband and wife play in the same game?
(a) 840 (b) 240
(c) 480 (d) none of these

50. If n objects are arranged in a row, then the number of ways of selecting three of these objects so that no two of them are next to each other is :
(a) ${}^nC_3$ (b) ${}^{n-2}C_3$
(c) ${}^nC_2$ (d) none of these

51. The number of times the digit 8 will be written when listing the integers from 1 to 1000 is :
(a) 100 (b) 200
(c) 300 (d) 400

52. Five balls of different colours are to be placed in three boxes of different sizes. Each box can hold all five balls. The number of ways in which we can place the balls in the boxes so that no box remains empty is :
(a) 30 (b) 150
(c) 600 (d) 900

53. The number of squares on a chessboard is :
(a) 102 (b) 108
(c) 216 (d) 204

54. A train going from Lucknow to Mumbai stops at 7 intermediate stations. Five persons enter the train during the journey with five different tickets of the same class. How many different set of tickets they could have had ?
(a) 98280 (b) 2898
(c) 7325 (d) none of these

55. In how many ways can a committee of 4 women and 5 men be chosen from 9 women and 7 men, if Mr. A refuses to serve on the committee if Ms. B is a member?
(a) 1608 (b) 1860
(c) 1680 (d) 1806

Directions for question number 56-63 : There are 5 different caps c_1, c_2, c_3, c_4 and c_5 and 5 different boxes B_1, B_2, B_3, B_4 and B_5. The capacity of each box is sufficient to accomodate all the 5 caps.

56. If any box can have any number of caps, in how many ways can all the caps be distributed?
(a) 3125 (b) 1235
(c) 2358 (d) 1248

57. In how many arrangements does B_1 have cap C_1?
(a) 5! (b) 5^4
(c) ${}^5P_4$ (d) none of these

58. If all the caps are of different colours and each box can have only one cap, in how many ways can you arrange the caps among the 5 boxes?
(a) 120 (b) 180
(c) 360 (d) none of these

59. If atleast one cap has to be distributed and the caps have to be arranged such that any box can have a maximum of one cap only, in how many ways can you arrange the caps among 5 boxes?
(a) 1 (b) 16
(c) 1545 (d) none of these

60. If all the caps are identical, in how many ways can the caps be arranged in the different boxes such that no box is empty?
(a) 1 (b) 2
(c) 6 (d) 8

61. If C_1 and C_5 are similar in all aspects, in how many ways can you arrange the caps in such a way that all the boxes have one cap?
(a) 70 (b) 60
(c) 75 (d) 80

62. If B_3 can have only C_1 or C_5, in how many ways can you arrange the caps such that all boxes have one cap?
(a) 480 (b) 420
(c) 48 (d) 88

63. If B_1 and B_5 have the caps C_1 and C_5 among themselves, in how many ways can you arrange the caps among the 5 boxes ?
(a) 400 (b) 485
(c) 500 (d) 365

LEVEL 2

1. In how many ways can the following prizes be given away to a class of 30 students, first and second in Mathematics, first and second in Physics, first in Chemistry and first in English?
(a) $\frac{30!}{4!}$ (b) $(30)^4 \times (29)^2$
(c) $(30)^3 - 1$ (d) $(30)^4 \times (29)^4$

2. If P_r stands for ${}^rP_r$, then the value of
$1 + 1.P_1 + 2.P_2 + 3.P_3 + \ldots + n.P_n$ is :
(a) $\frac{(n-1)!}{2}$ (b) $\frac{(n+1)!}{2}$
(c) $2(n-1)!$ (d) $(n+1)!$

3. In an examination hall there are four rows of chairs. Each row has 6 chairs one behind the other. There are two classes sitting for the examination with 12 students in each class. It is desired that in each row, all students belong to the same class and that no two adjacent rows are alloted to the same class. In how many ways can these 24 students be seated?

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

(a) $2(12!)^2$ (b) $3! \times (12!)^2$
(c) $(6!)^2$ (d) none of these

4. How many 3 digit even numbers are there such that if 3 is one of the digits then 9 is the next digit?
(a) 365 (b) 536
(c) 210 (d) 156

5. A number plate of a vehicle has always a fixed code UP–32 for Lucknow city followed by the number of particular vehicle which is in two parts. First part is occupied by 2 English alphabets and second part is occupied by 4 digit numbers (0001, 0002, ... 9999). If the latest registration number of vehicle is $\begin{Bmatrix} UP-32 \\ SK-0123 \end{Bmatrix}$, find the number of vehicles registered before this vehicle number in Lucknow.
(a) 2449744 (b) 4779644
(c) 4669235 (d) 9235888222

6. How many 4 digit numbers divisible by 4 can be formed without using the digits 0, 6, 7, 8, 9 if the repetition of digits is not allowed?
(a) 67 (b) 68
(c) 24 (d) 48

7. In how many ways three girls and nine boys can be seated in two vans, each having numbered seats, 3 in the front and 4 at the back?
(a) $^{14}C_{12}$ (b) 9^4
(c) $^{14}P_{12}$ (d) none of these

8. In the previous question, how many seating arrangements are possible if 3 girls sit together in a back row on adjacent seats?
(a) $(3!)^2$ (b) 6!
(c) 5! (d) none of these

9. In how many ways 5 plus (+) signs and 5 minus (–) signs be arranged in a row so that no two minus signs are together?
(a) 6 (b) 7
(c) 8 (d) 10

10. In how many ways can 11 identical books on English and 9 identical books on Maths be placed in a row on a shelf so that two books on Maths may not be together?
(a) 110 (b) 220
(c) 330 (d) 440

11. Find the total number of factors of 1680.
(a) 40 (b) 50
(c) 60 (d) 30

12. $\sum_{k=m}^{n} {}^kC_r$ equals :
(a) $^{n+1}C_{r-1}$ (b) $^{n+1}C_{r+1} - {}^mC_r$
(c) $^{n+1}C_{r+1} - {}^mC_{r+1}$ (d) $^{n+1}C_{r+1} + {}^mC_{r+1}$

13. In a certain test there are n questions. In this test 2^{n-k} students gave wrong answers to atleast k questions, where $k = 1, 2, 3, \ldots, n$. If the total number of wrong answers given is 2047, then n is equal to :
(a) 10 (b) 11
(c) 12 (d) 13

14. The number of triangles whose vertices are at the vertices of an octagon but none of the sides of such triangles are taken from the sides of the octagon.
(a) 8 (b) 15
(c) 16 (d) none of these

15. The number of ways in which we can select 5 numbers from the set of numbers {1, 2, 3, ..., 25} such that none of the selections includes four consecutive numbers is :
(a) 53109 (b) 13350
(c) 10035 (d) none of these

16. The number of integral solutions for the equation $a + b + c + d = 12$, where $(a, b, c, d) \geq -1$ is :
(a) $^{19}C_3$ (b) $^{18}C_4$
(c) $^{20}C_4$ (d) none of these

17. If $n = {}^kC_2$, the value of $^nC_2$ is :
(a) $2({}^{k+2}C_4)$ (b) $(n-2)!$
(c) $(k-2)!$ (d) $3({}^{k+3}C_4)$

18. The number of positive integral solutions of $abc = 42$ is :
(a) 17 (b) 27
(c) 21 (d) $3! \times 42$

19. There are three piles of identical red, green and blue balls and each pile contains atleast 10 balls. The number of ways of selecting 10 balls if twice as many red balls as green balls are to be selected is :
(a) 1 (b) 2
(c) 4 (d) 6

20. Between Shirdee and Khandala, there are 10 intermediate stations. The number of ways in which a train can be made to stop at 4 of these stations so that no two of these halting stations are consecutive is :
(a) 35 (b) 70
(c) 105 (d) 10

21. The number of ways in which an examiner can assign 30 marks to 8 questions, giving not less than 2 marks to any question is :
(a) 11 (b) $^{21}C_7$
(c) 18 (d) 235

22. The exponent of 3 in 33! is :
(a) 15 (b) 2
(c) 11 (d) none of these

23. The total number of natural numbers of 6 digits that can be formed with digits 1, 2, 3, 4, if all the digits are to appear in the same number atleast once, is :
(a) 740 (b) 1830
(c) 1560 (d) none of these

24. The total number of seven digit numbers, the sum of whose digits is even are :
(a) 250000 (b) 4500000
(c) 35×10^5 (d) none of these

25. The number of ways in which 4 pictures can be hung from 6 picture nails on the wall is :
(a) 4^6 (b) $^4P_6$
(c) $^6P_4$ (d) 6^4

26. The total number of ways of selecting 6 coins out of 10 one rupee coins, 6 fifty paise coins and 8 twenty paise coins is :
(a) 28 (b) 14
(c) 13 (d) 19

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

27. Nargis has 8 children and she takes 3 at a time to children's park as often as she can without taking the same 3 children together more than once. The number of times she will go to the park is :
(a) 56 (b) 14
(c) 28 (d) 76

28. In the previous problem (no. 27), the number of times each child will go the park is :
(a) 14 (b) 8
(c) 21 (d) none of these

29. The sides AB, BC, CA of a triangle ABC have 3, 4 and 5 interior points respectively on them. The total number of triangles that can be constructed by using these points as vertices is :
(a) 105 (b) 307
(c) 420 (d) 205

30. The number of all the possible selections which a student can make for answering one or more questions out of 10 given questions in a paper, when each question has an alternative is :
(a) 1345 (b) 23560
(c) 541340 (d) 59048

31. The number of ways in which a team of eleven players can be selected from 22 players including 2 of them and excluding 4 of them is :
(a) ${}^{15}C_{10}$ (b) ${}^{16}C_{10}$
(c) ${}^{16}C_{9}$ (d) none of these

32. There are 10 lamps in a hall. Each one of them can be switched on independently. The number of ways in which the hall can be illuminated is :
(a) 1022 (b) 1023
(c) 1024 (d) none of these

33. How many 10 digits numbers can be formed by using the digits 2 and 3?
(a) 2^{10} (b) 10^2
(c) 10! (d) none of these

34. How many subsets containing at most n elements from the set of $(2n + 1)$ elements can be selected?
(a) 2^n (b) 2^{n-1}
(c) 2^{n+1} (d) 2^{2n}

35. The sum of the divisors of $2^3 \cdot 3^4 \cdot 5^2$ is :
(a) 15625 (b) 1234
(c) 56265 (d) 56789

36. The number of ways of selecting 10 balls out of an unlimited number of white, red, blue and green balls is :
(a) 268 (b) 286
(c) 246 (d) 468

37. If 10 objects are arranged in a row, then the number of ways of selecting three of these objects so that no two of them are next to each other is :
(a) 56 (b) 65
(c) 28 (d) 13

38. The number of non-negative integral solutions of $x_1 + x_2 + x_3 + x_4 \le n$ (where n is a positive integer) is :
(a) ${}^{n+3}C_3$ (b) ${}^{n+2}C_3$
(c) ${}^{n+4}C_4$ (d) ${}^{n+4}C_3$

39. The total number of permutations of $(n > 1)$ different things taken nor more than r at a time, when each thing may be repeated any number of times is :
(a) $\dfrac{n^r - 1}{(n-1)}$ (b) $r^n - 1$
(c) $\dfrac{n(n^r - 1)}{(n-1)}$ (d) none of these

40. Number of rectangles on a chessboard is :
(a) 1008 (b) 1296
(c) 1124 (d) 1600

41. Ten persons, amongst whom A, B and C are to speak at a function. The number of ways in which it can be done if A wants to speak before B and B wants to speak before C is :
(a) $\dfrac{10!}{6}$ (b) $\dfrac{9!}{6}$
(c) ${}^{9}C_{6}$ (d) none of these

42. The number of natural numbers which are smaller than $2 \cdot 10^5$ and which can be written by means of the digits 1 and 2 is :
(a) 678 (b) 786
(c) 766 (d) 677

43. Tweleve persons are arranged in a row. The number of ways of selecting four persons so that no two persons sitting next to each other are selected is :
(a) 124 (b) 136
(c) 126 (d) none of these

44. The number of n-bit (digits 0 and 1) strings having exactly k zeros with no two zeros consecutive is, where $2k < n$.
(a) ${}^{n+1-k}C_k$ (b) ${}^{n+k}C_k$
(c) ${}^{n-k}C_k$ (d) ${}^{n-1}C_k$

45. A question paper has two parts–Part A and Part B. Part A contains 5 questions and part B has 4. Each question in part A has an alternative. A student has to attempt atleast one question from each part. Find the number of ways in which the student can attempt the question paper.
(a) 3360 (b) 1258
(c) 3850 (d) 3630

46. How many different vehicle licence plates can be made if the licences contain 2 letters of the English alphabet followed by a three digit number if repetitions are allowed?
(a) 54320 (b) 67534
(c) 23456 (d) 675324

47. In the previous question (no. 46), how many different plates can be formed if the repetitions are not allowed?
(a) 468000 (b) 13000
(c) 15680 (d) none of these

48. $2m$ people are arranged along two sides of a long table with m chairs each side. r men wish to sit on one particular side and s on the other. In how many ways can they be seated? $(r, s \le m)$
(a) ${}^{48}C_r$ (b) $68\ m$
(c) $\dfrac{(2m + r)!}{r!\,s!}$ (d) none of these

49. Management city has m parallel roads running East-West and n parallel roads running North-South. How many shortest possible routes are possible to go from one corner of the city to its diagonally opposite corner?

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

(a) ${}^{m+n}C_{m-1}$ (b) ${}^{m+n}C_n$
(c) ${}^{(m+n-2)}C_{(m-1)}$ (d) none of these

50. Priyanka has 11 different toys and Supriya has 8 different toys. Find the number of ways in which they can exchange their toys so that each keeps her initial number of toys.
(a) ${}^{19}C_{11}$ (b) ${}^{18}C_{10}$
(c) ${}^{20}C_{11}$ (d) ${}^{19}C_{11} - 1$

51. The sum of the numbers of the nth term of the series $(1) + (1+2) + (1+2+3) + (1+2+3+4) + \ldots (1+2+3+\ldots n)$:
(a) ${}^{n+1}C_3$ (b) ${}^{n+1}C_2$
(c) ${}^{n}C_2$ (d) ${}^{n+2}C_3$

52. Two packs of 52 playing cards are shuffled together. Find the number of ways in which a man can be dealt 26 cards so that he does not get two cards of the same suit and same denomination.
(a) ${}^{52}C_{26} \times 2^{26}$ (b) $\frac{52!}{(2!)^{26}}$
(c) 2^{26} (d) none of these

53. In how many ways we can select four cards of different suits and different values from a deck of 52 cards?
(a) 10! (b) $\frac{13!}{10!}$
(c) $\frac{13!}{9!}$ (d) none of these

54. How many 5 digit numbers can be formed having the digits 0 three times and 3 two times?
(a) 4 (b) 6
(c) 8 (d) 10

55. How many different four digit numbers can be formed with the digits 1, 2, 3, 4, 5, 6, 7, 8 and 9 such that the digit 5 appears exactly once?
(a) 1024 (b) 2048
(c) 4096 (d) none of these

56. How many different four digit numbers can be formed using digits 1, 2, 3, 4, 5, 6, 7, 8, 9 without repetition such that the digit 5 appears exactly once?
(a) 1243 (b) 1234
(c) 1344 (d) 1355

57. How many different eight digit numbers can be formed using only four digits 1, 2, 3, 4 such that the digit 2 occurs twice?
(a) 20412 (b) 12042
(c) 25065 (d) none of these

58. In how many ways can 8 distinct things be divided among three people such that any one can receive any number of things?
(a) 1556 (b) 6561
(c) ${}^{8}C_2$ (d) ${}^{7}C_2$

59. There are 6 numbered chairs placed around a circular table. 3 boys and 3 girls want to sit on them such that neither of two boys nor two girls sit adjacent to each other. How many such arrangements are possible?
(a) 36 (b) 58
(c) 72 (d) none of these

60. In how many ways one black and one white rook can be placed on a chessboard so that they are never in an attacking position?
(a) 1234 (b) 3136
(c) 9516 (d) 1024

61. There are 9 pairs of white shoes and 6 pairs of black shoes contained in a box. We are allowed to draw only one shoe at a time. Minimum how many shoes are required to be drawn out to get one pair of white shoes?
(a) 11 (b) 22
(c) 33 (d) 15

62. In the previous question, minimum how many shoes must be drawn out to get at least 1 pair of either black or white shoes?
(a) 15 (b) 18
(c) 16 (d) 22

63. There are four different coloured balls and 4 boxes of the same colours as that of the balls. Find the number of ways in which exactly one ball can be put in a different colour box that of the ball.
(a) 16 (b) 10
(c) ${}^{4}C_2$ (d) none of these

64. In the previous question (no. 63) find the number of ways in which only two balls can be put in the correct boxes *i.e.*, the colour of box and the colour of contained ball be same.
(a) 3 (b) 6
(c) 9 (d) none of these

65. Find the number of digits required to write down the number of pages in a 300 pages book.
(a) 792 (b) 279
(c) 729 (d) none of these

66. Find the number of numbers that can be formed using all the digits 1, 2, 3, 4, 3, 2, 1 only once so that the odd digits occupy odd places only.
(a) 9 (b) 16
(c) 18 (d) 27

Directions for question numbers 67–74 : There are 5 different boxes and 7 different balls. All the 7 balls are to be distributed in the 5 boxes placed in a row so that any box can receive any number of balls.

67. In how many ways can these balls be distributed into these boxes?
(a) 5^7 (b) 7^5
(c) ${}^{7}C_5$ (d) ${}^{7}P_5$

68. In how many ways can these balls be distributed so that no box is empty?
(a) 71 (b) 16800
(c) 1775 (d) none of these

69. Suppose all the balls are identical, then in how many ways can all these balls be distributed into these boxes?
(a) 110 (b) 220
(c) 330 (d) 440

70. Suppose all the balls and all the boxes are also identical. Then in how many ways can all these balls be distributed into these boxes?
(a) 12 (b) 13
(c) 16 (d) 255

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

71. In the previous question (no. 70) in how many ways can all these balls be distributed into these boxes so that no box remains empty and no two consecutive boxes have the same number of balls?
(a) 1 (b) 4
(c) 12 (d) none of these

72. In how many ways can these balls be distributed so that box 2 and box 4 contain only 1 and 2 balls respectively?
(a) 5522 (b) 8505
(c) 2305 (d) none of these

73. In how many ways can these balls be distributed into these boxes if ball 2 can be put into either box 2 or box 4?
(a) 12360 (b) 31250
(c) 13490 (d) 31526

74. In how many ways can these balls be distributed such that no box is empty and ball 2 and ball 4 cannot be put in the same box?
(a) 1200 (b) 15000
(c) 3800 (d) none of these

Directions for question numbers 75–82 : There are 3 pots and 4 coins. All these coins are to be distributed into these pots where any pot can contain any number of coins.

75. In how many ways these coins can be distributed if all the coins and all the pots are different?
(a) 3^4 (b) 4^3
(c) ${}^3P_4$ (d) ${}^4P_3$

76. In how many ways these coins can be distributed if all the coins and all the pots are identical?
(a) 4 (b) 6
(c) 8 (d) 1

77. In how many ways all these coins can be distributed if all coins are identical but all pots are different?
(a) 15 (b) 16
(c) 17 (d) 81

78. In how many ways all these coins can be distributed if all coins are different but all pots are identical?
(a) 14 (b) 21
(c) 27 (d) none of these

79. In how many ways all these coins can be distributed such that no pot is empty if all coins are different but all pots are identical?
(a) 16 (b) 6
(c) 42 (d) 21

80. In how many ways all these coins can be distributed such that no pot is empty if all coins are identical but all pots are different?
(a) 6 (b) 3
(c) 9 (d) 27

81. In how many ways all these coins can be distributed if all coins are identical and two pots are also identical?
(a) 1 (b) 10
(c) 9 (d) 11

82. In how many ways all these coins can be distributed if out of 4 coins 2 coins are identical and all pots are different?
(a) 45 (b) 27
(c) 54 (d) none of these

Final Round

1. Maximum number of points of intersection of 6 circles, is :
(a) 30 (b) 28
(c) 15 (d) none of these

2. Maximum number of points of intersection of 6 straight lines is :
(a) 30 (b) 15
(c) 28 (d) none of these

3. Maximum number of points into which 3 circles and 3 lines intersect is :
(a) 21 (b) 9
(c) 27 (d) 31

4. Eight identical coins are arranged in a row. The total number of ways in which the number of heads is equal to the number of tails, is :
(a) 35 (b) 15
(c) 140 (d) 70

5. Two straight lines intersect at a point O. Points $A_1, A_2, A_3, A_4, A_5, \ldots, A_m$ are taken on one line and points $B_1, B_2, B_3, \ldots, B_n$ on the other. If the point O is not included, the number of triangles that can be drawn using these points as vertices, is :
(a) ${}^nC_2 + {}^mC_2$ (b) ${}^{2n}C_2$
(c) ${}^{m+n}C_2$ (d) none of these

6. How many different nine digit numbers can be formed from the number 22 33 55 888 by rearranging its digits so that the odd digits occupy even positions?
(a) 60 (b) 75
(c) 88 (d) 77

7. The straight lines l_1, l_2, l_3 are parallel and lie in the same plane. A total number of m points on l_1, n points on l_2, k points on l_3 are used to produce the triangles, the maximum number of triangles formed with vertices at these points are :
(a) ${}^mC_3 \times {}^nC_3 \times {}^kC_3$
(b) ${}^{(m+n+k)}C_3 - ({}^mC_3 + {}^nC_3 + {}^kC_3)$
(c) ${}^{(m+n+k)}C_3$
(d) none of the above

8. Given that n is the odd, the number of ways in which three numbers in AP can be selected from 1, 2, 3, 4, ..., n is :
(a) $\frac{(n-1)^2}{4}$ (b) n^2
(c) n^3 (d) $(n-2)^2$

9. Eight straight lines are drawn in the plane such that no two lines are parallel and no three lines are concurrent. The number of parts into which these lines divide the plane, is :
(a) 73 (b) 37
(c) 17 (d) 72

10. Number of divisors of the form $4n + 2$ $(n \geq 0)$ of the integer 240 is :
(a) 6 (b) 4
(c) 3 (d) 12

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

11. The number of rectangles excluding squares from a rectangle of size of 12×8 is :
(a) 1234 (b) 625
(c) 2460 (d) 256

12. Two lines intersect at O. Points $A_1, A_2, \ldots, A_n$ are taken on one of them and $B_1, B_2, \ldots, B_n$ on the other the number of triangles that can be drawn with the help of these $(2n + 1)$ points is :
(a) n (b) n^2
(c) n^3 (d) n^4

13. The number of ways in which 9 identical balls can be placed in three identical boxes is :
(a) 12 (b) 6
(c) 9 (d) $\frac{9!}{3!}$

14. The number of ways in which 30 coins of one rupee each be given to six persons so that none of them receives less than 4 rupees is :
(a) 246 (b) 462
(c) $\frac{30!}{6!}$ (d) none of these

15. Find the number of circles that can be drawn out of 10 points, of which 7 points are collinear is :
(a) 35 (b) 85
(c) 70 (d) 124

16. The number of natural numbers of two or more than two digits in which digits from left to right are in increasing order is :
(a) 205 (b) 520
(c) 502 (d) none of these

17. In how many ways a cricketer can make a century with fours and sixes only?
(a) 6 (b) 9
(c) 8 (d) 10

18. The number of permutations of the letters of the word LUMINARY such that neither the pattern LURY nor MINA occurs is :
(a) 46800 (b) 24600
(c) 40086 (d) none of these

19. 10 students are to be seated in two rows equally for the MOCK CAT in a room. There are two sets of papers, code A and code B. Each of the two rows can have only one set of paper but different that from the other row. In how many ways these students can be arranged?
(a) 2775600 (b) 1200560
(c) 125600 (d) 7257600

20. Aman Verma and Mini Mathur jointly host a TV programme in which one day n guests attended the show. There each guest shakes hands with every other guest and each guest has to shake hand with each of the hosts. If there happens to be total 65 shake hands, find the number of guests attended the show.
(a) 13 (b) 14
(c) 10 (d) 9

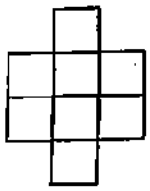
21. The number of ways in which an examiner can assign 50 marks to 10 questions giving not less than 3 marks to any question is :
(a) ${}^{29}C_9$ (b) ${}^{47}C_3$
(c) ${}^{52}C_2$ (d) ${}^{40}C_{10}$

22. For $x \in R$, let $[x]$ denote the largest integer less than or equal to x, then the value of :
$$E = \left[\frac{1}{3}\right] + \left[\frac{1}{3} + \frac{1}{100}\right] + \left[\frac{1}{3} + \frac{2}{100}\right] + \ldots + \left[\frac{1}{3} + \frac{99}{100}\right] \text{ is :}$$
(a) 22 (b) 66
(c) 33 (d) none of these

23. The number of positive integers from 1 to 10^6 (both inclusive) which are neither perfect squares, nor cubes, nor perfect fourth powers is :
(a) 918990 (b) 998911
(c) 998910 (d) none of these

24. Kumar's family consists of a grandfather, 'm' sons and daughters and '$2n$' grand children. They are to be seated in a row for dinner. The grandchildren wish to occupy the n seats at each end and the grandfather refuses to have a grandchild on either side of him. In how many ways can the Kumar's family be made to sit?
(a) $(2n!)(m-1)!$ (b) $(2n!)(m!)(m-1)$
(c) $(2m!)(n!)$ (d) none of these

25. Six Ps have to be placed in the squares of the diagram given below such that each column contains at least one P. In how many ways this can be done?



(a) 26 (b) 13
(c) 8 (d) 24

26. A box contains two red, three green and four blue balls. In how many ways can three balls be drawn from the box if atleast one green ball is to be included in the draw?
(a) 23 (b) 64
(c) 46 (d) none of these

27. Ajay has 7 relatives, 3 men and 4 women. His wife Kajol also has 7 relatives, 3 women and 4 men. In how many ways can they invite 3 men and 3 women so that 3 of them are the Ajay's relatives and 3 his wife Kajol's?
(a) 485 (b) 458
(c) 365 (d) none of these

28. Serena and Venus were only two women participiting in a chess tournament. Every participant played two games with every other participant. The number of games that men played between themselves proved to exceed by 66, compared to the number of games the men played with women. How many participants were there?
(a) 156 (b) 610
(c) 13 (d) 108

29. In the previous questions (no. 28) how many games were played?
(a) 208 (b) 156
(c) 316 (d) none of these

30. Six Burfis and six rasgullas are to be distributed among ten girls so that each girl receives at least one piece of sweets out of burfis and rasgullas. Find the number of ways in which the sweets can be distributed.
(a) 26250 (b) 22560
(c) 22330 (d) none of these

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

31. Mr. John has x children by his first wife and Ms. Bashu has $x+1$ children by her first husband. They marry and have children of their own. The whole family has 10 children. Assuming that two children of the same parents do not fight, find the maximum number of fights that can take place among children.
(a) 33 (b) 22
(c) 111 (d) none of these

32. Find the number of natural numbers which are smaller than $2 \cdot 10^8$ and are divisible by 3 which can be written by means of the digits 0, 1 and 2.
(a) 7272 (b) 4373
(c) 3437 (d) none of these

33. Find the number of whole numbers formed on the screen of a calculator which can be recognised as numbers with (unique) correct digits when they are read inverted. The greatest number that can be formed on the screen of the calculator is 999999.
(a) 98970 (b) 89912
(c) 110050 (d) 100843

34. Find the number of ways of putting five distinct rings on four fingers of the left hand. (Ignore the difference of size of rings and the fingers)
(a) 1250 (b) 6720
(c) 5260 (d) none of these

35. If n distinct things are arranged in a circle, find the number of ways of selecting three of these things so that no two of them are next to each other.
(a) $n(n-2)(n-3)$ (b) $\frac{1}{6}n(n-4)(n-5)$
(c) $\frac{1}{3}(n^2+3n-5)$ (d) none of these

36. The number of different selections of 5 letters from 1 A, 2 B's, 3 C's, 4 D's and 5 E's is :
(a) 71 (b) 41
(c) 117 (d) none of these

37. In the entrance test of Wharton Business School, the maximum marks for each of three papers is n and that for the fourth paper is $2n$. Find the number of ways in which a candidate can get $3n$ marks.
(a) $\frac{1}{6}(n+1)(n+2)$
(b) $\frac{1}{6}(n+1)(5n^2+10n+6)$
(c) $\frac{1}{2}(n^2+3n-6)$
(d) none of the above

38. An examination consists of 4 papers. Each paper has a maximum of 'n' marks. Find the number of ways in which a students can get $2n$ marks in the examination.
(a) $\frac{1}{3}(n^2-5n+4)$
(b) $\frac{1}{3}(n+1)(2n^2+4n+3)$
(c) $\frac{1}{6}(n+1)(n+4)$
(d) none of the above

www.LearnEngineering.in
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Answers

INTRODUCTORY EXERCISE-19.1

1 (a)	2. (b)	3. (c)	4. (a)	5. (d)	6. (b)	7. (a)	8. (b)	9. (c)	10. (b)
11. (a)	12. (b)	13. (a)	14. (a)	15. (a)	16. (d)	17. (d)	18. (c)	19. (b)	20. (a)
21. (b)	22. (b)	23. (b)	24. (d)	25. (b)	26. (d)	27. (b)	28. (a)	29. (c)	30. (c)
31. (a)	32. (a)	33. (a)	34. (a)	35. (a)	36. (a)	37. (a)	38. (b)	39. (c)	40. (d)
41. (a)	42. (a)	43. (b)	44. (c)	45. (a)	46. (d)	47. (a)			

INTRODUCTORY EXERCISE-19.2

1 (c)	2. (b)	3. (d)	4. (d)	5. (a)	6. (d)	7. (d)	8. (a)	9. (b)	10. (a)
11. (a)	12. (b)	13. (a)	14. (b)	15. (a)	16. (c)	17. (b)	18. (a)	19. (b)	20. (c)
21. (d)	22. (a)	23. (b)							

INTRODUCTORY EXERCISE-19.3

1 (a)	2. (c)	3. (a)	4. (c)	5. (a)	6. (d)	7. (d)	8. (a)	9. (b)	10. (a)
11. (b)									

INTRODUCTORY EXERCISE-19.4

1 (a)	2. (b)	3. (a)	4. (b)	5. (a)	6. (b)	7. (a)	8. (c)	9. (d)	10. (a)
11. (a)									

INTRODUCTORY EXERCISE-19.5

1 (a)	2. (c)	3. (c)	4. (c)	5. (c)	6. (b)	7. (b)	8. (c)	9. (d)	10. (a)
11. (c)	12. (d)	13. (a)	14. (c)	15. (a)	16. (b)	17. (c)	18. (b)	19. (d)	20. (c)
21. (d)	22. (b)	23. (a)	24. (a)	25. (b)	26. (a)	27. (d)	28. (b)	29. (c)	30. (b)
31. (a)	32. (b)	33. (a)	34. (b)	35. (c)	36. (c)	37. (b)	38. (c)	39. (a)	40. (b)
41. (a)	42. (c)	43. (b)	44. (a)	45. (b)	46. (c)	47. (a)	48. (b)	49. (c)	50. (b)
51. (b)	52. (b)	53. (b)	54. (b)	55. (c)	56. (c)	57. (d)	58. (b)	59. (c)	60. (b)
61. (d)	62. (a)	63. (c)	64. (a)	65. (b)	66. (a)	67. (b)	68. (a)	69. (b)	

INTRODUCTORY EXERCISE-19.6

1 (c)	2. (b)	3. (c)	4. (a)	5. (d)	6. (c)				

INTRODUCTORY EXERCISE-19.7

1 (b)	2. (b)	3. (c)	4. (c)	5. (a)	6. (d)	7. (c)	8. (b)	9. (a)	10. (c)
11. (d)	12. (a)								

LEVEL-1

1 (c)	2. (b)	3. (a)	4. (d)	5. (a)	6. (b)	7. (c)	8. (b)	9. (b)	10. (c)
11. (a)	12. (c)	13. (d)	14. (a)	15. (c)	16. (b)	17. (b)	18. (a)	19. (c)	20. (d)
21. (c)	22. (b)	23. (b)	24. (b)	25. (b)	26. (b)	27. (b)	28. (a)	29. (c)	30. (d)
31. (a)	32. (b)	33. (d)	34. (b)	35. (a)	36. (c)	37. (b)	38. (b)	39. (c)	40. (c)
41. (a)	42. (b)	43. (b)	44. (c)	45. (a)	46. (c)	47. (a)	48. (b)	49. (a)	50. (b)
51. (c)	52. (b)	53. (d)	54. (a)	55. (d)	56. (a)	57. (b)	58. (a)	59. (a)	60. (a)
61. (b)	62. (c)	63. (c)							

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

LEVEL-2

1 (b)	2. (d)	3. (a)	4. (a)	5. (b)	6. (c)	7. (c)	8. (d)	9. (a)	10. (b)
11. (a)	12. (c)	13. (b)	14. (c)	15. (a)	16. (a)	17. (d)	18. (b)	19. (c)	20. (a)
21. (b)	22. (a)	23. (c)	24. (b)	25. (c)	26. (a)	27. (a)	28. (c)	29. (d)	30. (d)
31. (c)	32. (b)	33. (a)	34. (d)	35. (c)	36. (b)	37. (a)	38. (c)	39. (c)	40. (b)
41. (a)	42. (c)	43. (c)	44. (a)	45. (d)	46. (d)	47. (a)	48. (d)	49. (c)	50. (d)
51. (b)	52. (a)	53. (c)	54. (a)	55. (b)	56. (c)	57. (a)	58. (b)	59. (c)	60. (b)
61. (b)	62. (c)	63. (d)	64. (b)	65. (a)	66. (c)	67. (a)	68. (b)	69. (c)	70. (b)
71. (a)	72. (b)	73. (b)	74. (b)	75. (a)	76. (a)	77. (a)	78. (a)	79. (b)	80. (b)
81. (c)	82. (c)								

FINAL ROUND

1. (a)	2. (b)	3. (c)	4. (d)	5. (d)	6. (a)	7. (b)	8. (a)	9. (b)	10. (b)
11. (c)	12. (c)	13. (a)	14. (b)	15. (b)	16. (c)	17. (b)	18. (c)	19. (d)	20. (c)
21. (a)	22. (c)	23. (c)	24. (b)	25. (a)	26. (b)	27. (a)	28. (c)	29. (b)	30. (a)
31. (a)	32. (b)	33. (d)	34. (b)	35. (b)	36. (a)	37. (b)	38. (b)		

www.LearnEngineering.in
For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Hints & Solutions

INTRODUCTORY EXERCISE 19.1

1. ${}^9P_3 = \frac{9!}{6!}$ $\left(\because {}^nP_r = \frac{n!}{(n-r)!}\right)$

$= \frac{9 \times 8 \times 7 \times 6!}{6!}$

$= 9 \times 8 \times 7 = 504$

2. ${}^nP_5 = 20 \,.\, {}^nP_3$

$\Rightarrow \frac{n!}{(n-5)!} = 20 \,.\, \frac{n!}{(n-3)!}$

$\Rightarrow \frac{(n-3)!}{(n-5)!} = 20$

$\Rightarrow \frac{(n-3)(n-4)(n-5)!}{(n-5)!} = 20$

$\Rightarrow (n-3)(n-4) = 20$

$\Rightarrow n = 8$ ($n = -1$ is inadmissible)

Alternatively : Go through options.

3. $\frac{{}^{56}P_{r+6}}{{}^{54}P_{r+3}} = \frac{30800}{1}$

$\frac{\frac{56!}{(50-r)!}}{\frac{54!}{(51-r)!}} = \frac{30800}{1}$

$\Rightarrow \frac{56 \times 55 \times (51-r)!}{(50-r)!} = \frac{30800}{1}$

$\Rightarrow \frac{(51-r)!}{(50-r)!} = \frac{10}{1}$

$\Rightarrow \frac{(51-r) \times (50-r)!}{(50-r)!} = \frac{10}{1}$

$\Rightarrow 51 - r = 10$

$\Rightarrow r = 41$

Alternatively : Go through options

4. Since there are 6 digits available and out of 6 digits we have to take only 3 digits for the arrangement. Therefore, required number of 3 digits numbers $= {}^6P_3 = \frac{6!}{3!} = 120$

Alternatively : | 6 | 5 | 4 | $\Rightarrow 6 \times 5 \times 4 = 120$

We have 6 digits to fill up hundreds place and only 5 digits for tens place and only 4 digits for unit place since repetition is not allowed.

5. **Case 1.** There is only one even number of one digit.

Case 2. | 4 | 1 | $= 4 \times 1 = 4$

There are only 4 even numbers of two digit
(Numbers are 38, 58, 78 and 98)

Case 3. | 4 | 3 | 1 | $\Rightarrow 4 \times 3 \times 1 = 12$

There are only 12 even numbers of 3 digit.

Case 4. | 4 | 3 | 2 | 1 | $\Rightarrow 4 \times 3 \times 2 \times 1 = 24$

There are 24 even numbers of 4 digits
Thus there are total $1 + 4 + 12 + 24 = 41$ even numbers.

NOTE In each case 8 is fixed at unit place, since if the unit digit is an even digit then the whole number is an even number.

6. | 3 | 4 | 3 | 2 | $\Rightarrow 3 \times 4 \times 3 \times 2 = 72$

Thousands place can assume only 3 values *viz.*, 5, 7, 8. Since required numbers are greater than 4000.

| 4 | 4 | 3 | 2 | 1 | $\Rightarrow 4 \times 4 \times 3 \times 2 \times 1 = 96$

Ten thousands place can assume all the remaining non-zero digits and thousands place can assume zero also except the digit which has been filled up at thousands place.
Therefore total required numbers
= number of 4 digit numbers + number of 5 digit numbers
$= 72 + 96 = 168$

7. | 4 | 4 | 3 | 2 | $\Rightarrow 4 \times 4 \times 3 \times 2 = 96$

Thousands place can not assume 3 since required numbers are greater than 5000.

| 5 | 4 | 3 | 2 | 1 | $\Rightarrow 5 \times 4 \times 3 \times 2 \times 1 = 120$

Total required numbers $= 96 + 120 = 216$

8. | 8 | 7 | 6 | 5 | $\Rightarrow 8 \times 7 \times 6 \times 5 = 1680$

3 and 5 have been already used, so we have 8 digits for thousands place, then 7 digits for hundreds place, 6 digits for tens digit and remaining 5 digits for unit place.

Alternatively : ${}^8P_4 = \frac{8!}{4!} = \frac{8 \times 7 \times 6 \times 5 \times 4!}{4!}$

$= 8 \times 7 \times 6 \times 5 = 1680$

9. A number is divisible by 5 only if its unit digit is either 5 or 0. So we fix 5 as unit digit and then we fill up the remaining places.

| 5 | 4 | 3 | 2 | 1 | $\Rightarrow 5 \times 4 \times 3 \times 2 \times 1 \times 1 = 120$

Hence required number = 120

10. | 4 | 3 | 2 | 1 | $\Rightarrow 4 \times 3 \times 2 \times 1 = 24$

There are total 24 numbers of 4 digit.
Since we have only 4 digits it means we use each of the digit $6 \left(= \frac{24}{4}\right)$ times in each of the unit, tens hundreds and thousands place.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Therefore the sum of the digits in the units place
$= 6(1 + 2 + 5 + 6) = 84$
Similarly, the sum of the digits in each of tens, hundreds, and thousands places = 84
Hence the required sum of all the numbers
$= 84 \times 1 + 84 \times 10 + 84 \times 100 + 84 \times 1000$
$= 84(1 + 10 + 100 + 1000)$
$= 84 \times 1111 = 93324$

11. **Case 1.** | 3 | 3 | 1 | $\Rightarrow 3 \times 3 \times 1 = 9$

When 2 is fixed at unit place we have 3, 4 and 5 *i.e.*, 3 digits for hundreds place and remaining 3 digits (out of 5) for the tens place.

| 2 | 3 | 1 | $\Rightarrow 2 \times 3 \times 1 = 6$

When 4 is fixed at unit place, we have 3 and 5 *i.e.*, only two digits for hundreds place and remaining 3 digits for tens place.

Case 2. | 4 | 3 | 2 | 2 | = 48

4 digit numbers = 48

Case 3. | 4 | 3 | 2 | 1 | 2 | = 48

5 digit numbers = 48
Thus there are total 15 + 48 + 48 = 111 even numbers greater than 300.

12. | 4 | 4 | 3 | = 48

Hundred's place can not assume 0.

13. There are 7 letters in the word RAINBOW and each letter is used only once. So all the 7 letters can be arranged in 7 !ways.
$7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 5040.$

14. If we fix *R* as the initial letter, then we have to arrange only 6 remaining letters.
Hence required number of permutations = 6!
$= 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$

15. If we fix *R* and *W* as the first and last letters then we have to arrange only 5 remaining letters which can be arranged in 5! = 120 ways.

16. When *R* and *W* are the first and last letters of all the words then we can arrange them in 5!ways. Similarly when *W* and *R* are the first and last letters of the words then the remaining letters can be arrange in 5! ways.
Thus the total number of permutations = 2 × 5!
$= 2 \times 120 = 240$

Alternatively : Except to *R* and *W* all the remaining 5 letters can be arranged in 5! ways and *R* and *W* can be arranged in 2! ways at the end positions.
Thus total number of permutations = 2! × 5! = 2 × 120 = 240

17. Assume *N* and *B* as a single letter, then we have only 6 letters for the arrangement
i.e., *A, I, R, O, W, BN*
which can be arranged in 6! ways.
Now *B* and *N* can also be mutually arranged in 2! ways
Therefore total number of arrangements = 2! × 6!
$= 2 \times 720 = 1440$

18. Number of permutations when *I* and *O* are together
$= 2! \times 6! = 1440$
(Very similar to the previous question)
Total number of permutations = 7 ! = 5040
∴Number of permutations when *I* and *O* are not together
$= 5040 - 1440 = 3600$

19. There are three vowels *viz.*,A, *I* and *O*
First of all arrange all the four consonants which can be done in 4! ways.
Now we have 5 places available for the three vowels to be filled up, which can be done in ${}^5P_3$ ways.
Thus the total number of arrangements $= 4! \times {}^5P_3 = 24 \times \frac{5!}{2!}$
$= 24 \times 60 = 1440$

20. First of all we arrange all the 4 consonants in 4! ways.
Now we have 5 places for the 3 vowels as
1. C_1 2. C_2 3. C_3 4. C_4 5.
1. When all the three vowels *AIO* are together then 5 places can be filled in 5 ways.
2. When *A* and *I* are together and *O* is separated from "*AI*" then the 5 places can be filled up in 10 ways.
3. When *I* and *O* are together and *A* is separated from "IO" then the 5 places can be filled up in 10 ways.
4. When *A*, *I* and *O* are separated from each other then the 5 places can be filled up in 10 ways.

Thus there are total 5 + 10 + 10 + 10 = 35 ways in which 5 places can be filled up such that *A* is before *I* and *I* is before O.
Hence total number of ways in which all the 7 letters of RAINBOW can be arranged in which *A* is before *I* and *I* is before *O* = 4! × 35 = 24 × 35 = 840.

Alternatively : All the 7 letters of the word RAINBOW can be arranged in 7! ways.
and 3 particular letters (A,I,O) can be arranged in 3! ways.
But the given condition is satisfied by only 1 out of 6 ways.
Hence required number of arrangements
$= \frac{7!}{3!} = 7 \times 6 \times 5 \times 4 = 840$

21. Let all the vowels be in a single packet and all the consonants be in the other packet.
Now all the vowels can be arranged in 3! ways and all the consonants can be arranged in 4! ways.
Hence total number of arrangements = 3! × 4! = 6 × 24 = 144

22. The first and last letters can be arranged in ${}^3P_2$ ways and the remaining letters can be arranged in 5! ways.
Hence required number of permutations
$= {}^3P_2 \times 5! = 6 \times 120 = 720$

23. The correct order of the letters is as follows :
A, B, I, N, O, R, W
Number of words begin with *A* = 6!
Number of words begin with *B* = 6!
Number of words begin with *I* = 6!
Number of words begin with *N* = 6!
Number of words begin with *O* = 6!
Number of words begin with *RAB* = 4!
Number of words begin with *RAIB* = 3!

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Now the next word is RAINBOW (it is the first word which begins with RAIN)
So the ranking of the word RAINBOW
$= 5 \times 6! + 4! + 3! + 1 = 3631$

24. First arrange the 3 vowels which can be done in 3! ways.
Now there are 4 places created by the 3 vowels which can be filled up by 4 consonants in ${}^4P_4$ ways.
Therefore total number of required permutations
$= 3! \times {}^4P_4 = 6 \times 24 = 144$

25. 1 2 3 4 5 6 7
In order that vowels may occupy even places arrange all the 3 vowels, then arrange all the four consonants in four places.
Thus the required number of arrangements $= 3! \times 4! = 144$

26. 1 2 3 4 5 6 7
First of all arrange any 3 consonants at even places in ${}^4P_3$ ways. Now the newly created four odd places can be filled by the remaining letters which includes 3 vowels and 1 consonants, which can be done in ${}^4P_4$ ways. Hence the required number of permutation
$= {}^4P_3 \times {}^4P_4 = 24 \times 24 = 576$

27. $\frac{}{1}\ \frac{2}{}\ \frac{}{3}\ \frac{4}{}\ \frac{}{5}\ \frac{6}{}$
First of all arrange 3 vowels at 3 even places, which can be done 3! ways. Again arrange the 3 consonants at 3 odd places which can be done in 3! ways.
Hence required number of permutations $= 3! \times 3! = 36$ ways

Alternatively :

3	3	2	2	1	1	= 36 ways

Assume vowels are even numbers say 2, 4, 6 and consonants are odd numbers say 1, 3, 5.
Now, if we have to form the numbers in which odd integers always occupy odd places then this can be done as above. (*O, E, O, E, O, E*)

28. First of all arrange all the four consonants *R, N, B, W* in 4! ways.
Then there are 5 places to be filled up by the vowels. But any two vowels are always together then we assume that there are only two vowels which can be filled in 5 places in ${}^5P_2$ ways. But we have to take any two vowels together out of 3 vowels then this can be done in ${}^3P_2$ ways.
Hence the total number of permutation $= {}^5P_2 \times {}^3P_2 \times 4!$
$= 20 \times 6 \times 24$
$= 2880$

Alternatively : Since there can be only 3 cases
(i) When *A, I* and *O* are separate from each other
(ii) When *A, I* and *O* can always together
(iii) When any two vowels out of three vowels *A, I* and *O* are together.
Now we need to calculate the value of Case (iii).
∴ Required number of permutations
= Total number of permutations – [Case (i) + Case (ii)]
∴ $7! - (1440 + 720) = 5040 - (2160)$
$= 2880$

29. At least two consonants remain together means
(i) Any two consonants are together
(ii) All the three consonants are together
which implies that all of the consonants are not separated from each other.
Hence required number of permutations
= (Total number of permutations
– number of permutations when all of the vowels are separated from each other)
$= 6! - (3! \times {}^4P_3) = 720 - 144 = 576$

30. There are 5 consonants and 2 vowels in the word STRANGE out of 7 places for the 7 letters, 4 places are odd and 3 places are even.
2 vowels can be arranged in 4 odd places in ${}^4P_2$ ways and then 5 consonants can be arranged in the remaining 5 places in ${}^5P_5$ ways.
Hence, the required number of ways $= {}^4P_2 \times {}^5P_5 = 1440$

31. Since no two men or two women sit together it means they sit on the alternate positions.
Therefore first of all we arrange 3 women in 3! ways then we arrange 4 men in newly created 4 places in ${}^4P_4$ ways.
Thus the total number of arrangements $= 3! \times {}^4P_4$
$= 6 \times 24 = 144$

32. There are total 9 places out of which 4 are even and rest 5 places are odd.
4 women can be arranged at 4 even places in 4! ways.
and 5 men can be placed in remaining 5 places in ${}^5P_5$ ways.
Hence, the required number of permutations
$= 4! \times {}^5P_5 = 24 \times 120 = 2880$

33. First of all the 6 students can be arranged in 6! ways then 4 teachers can be arranged in 7 places in ${}^7P_4$ ways.
Hence, the required number of arrangements $= 6! \times {}^7P_4$
$= 720 \times 840 = 604800$

34. First of all we arrange 5 girls in 5! ways, then we arrange 3 boys in the remaining 6 places in ${}^6P_3$ ways.
Hence, the required number of permutations $= 5! \times {}^6P_3$
$= 120 \times 120 = 14400$

35. Total number of arrangements $= 8! = 40320$
Number of arrangements when best and worst papers are together $= 7! \times 2! = 10080$
∴ Number of arrangements in which best and worst papers are not together $= 40320 - 10080 = 30240$

36. All the 6 books on Physics can be mutually arranged in 6! ways.
Similarly 3 books on Chemistry and 4 books on Biology can be arranged in 3! and 4! ways respectively.
Besides it the three sets of Physics, Chemistry and Biology also can be arranged in 3! ways
∴ Required number of permutations $= 3! \times 6! \times 3! \times 4!$
$= 622080$

37. Since we can not wear more than one ring in any finger means repetition is not allowed.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Hence, the first ring can be worn in any of the available 4 fingers in 4 ways and second ring can be worn in any of the remaining 3 fingers in 3 ways and third ring can be worn in any of the remaining 2 fingers in 2 ways.
So the required number of ways in which all the 3 rings can be worn in 4 fingers = $4 \times 3 \times 2 = 24$.
Alternatively : We have 4 places to be filled up by 3 different things (*i.e.*, rings), which can be done in ${}^4P_3$ ways.
${}^4P_3 = 24$

38. The signals can be made by using at a time one or two or three or four or five or six flags out of 6 different coloured flags.
∴ Required number of flags
$$= {}^6P_1 + {}^6P_2 + {}^6P_3 + {}^6P_4 + {}^6P_5 + {}^6P_6$$
$$= 6 + 30 + 120 + 360 + 720 + 720 = 1956$$

39. The required number of permutation = $3 \times 4 \times 6 = 72$

40. Required number of permutations = ${}^7P_4 \times {}^6P_4 \times {}^8P_4$
$$= 840 \times 360 \times 1680$$
$$= 508032000$$

41. 7 soldiers can be stand in a queue in 7! ways
∴ $7! = 5040$

42. Any 5 soldiers can stand in one row and the remaining 5 soldiers can stand in other row.
∴ Required number of arrangements
$$= {}^{10}P_5 \times {}^5P_5 \times 2! = 7257600$$

43. Required number of signals = ${}^5P_3 = 60$

44. Required number of permutations = ${}^{10}P_3 = 720$
Alternatively : First prize can be given to any of the 10 students and second prize can be given to any one of the remaining 9 students and the third prize can be given to any one of the remaining 8 students.
Therefore it can be done in
$${}^{10}P_1 \times {}^9P_1 \times {}^8P_1 = 10 \times 9 \times 8 = 720 \text{ ways.}$$

45. A passanger from any station may purchase ticket for anyone of the other 9 stations. Therefore, there must be 9 tickets in each station.
Therefore total number of different tickets = $9 \times 10 = 90$

46. Required number of ways = $12! - 11! \times 2!$
$$= 11!(12 - 2)$$
$$= 10 \times 11!$$
$$= 399168000$$

47. Total number of permutations = 12!
Number of permutations when three particulars books are together = $10! \times 3!$
∴ Number of permutations when three particular books are not together = $12! - 10! \times 3!$
$$= 10!(12 \times 11 - 6)$$
$$= 10! \times 126 = 457228800$$

INTRODUCTORY EXERCISE 19.2

1. In the given word there are 6 letters of which *E* occurs 2 times.
Hence the required number of ways = $\frac{6!}{2!}$
$$= \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{2} = 360$$

2. There are total 7 letters of which *A* occurs 3 times and *N* occurs 2 times.
Hence, the required number of ways = $\frac{7!}{3!.\,2!} = 420$

3. There are total 10 letters of which *R* occurs 2 times, *E* occurs 3 times.
Hence the required number of ways = $\frac{10!}{2!.\,3!} = 302400$

4. Required number of permutations = $\frac{13!}{3!.\,4!.\,2!.\,2!}$
$$= 10810800$$

5. There are 6 different letters in the word which are *A, S, I, N, T, O*.
The number of arrangements of 6 different letters taken 4 at a time
$$= {}^6P_4 = 360$$

6. There are total 9 letters in the word COMMITTEE in which there are 2*M*'s, 2*T*'s, 2*E*'s.
∴ The number of ways in which 9 letters can be arranged
$$= \frac{9!}{2! \times 2! \times 2!} = 45360$$
There are 4 vowels *O, I, E, E* in the given word. If the four vowels always come together, taking them as one letter we have to arrange 5 + 1 = 6 letters which include 2Ms and 2Ts and this be done in $\frac{6!}{2! \times 2!} = 180$ ways.
In which of 180 ways, the 4 vowels *O, I, E, E* remaining together can be arranged in $\frac{4!}{2!} = 12$ ways.
∴ The number of ways in which the four vowels always come together = $180 \times 12 = 2160$.
Hence, the required number of ways in which the four vowels do not come together = 45360 − 2160 = 43200

7. $\frac{14!}{4! \times 2! \times 2! \times 2! \times 2!} = 113513400$

8. When *L* and *T* are fixed as first and last letters of the word, then we have only 6 letters to be arranged.
Hence required number of permutations
$$= \frac{6!}{2! \times 2! \times 2!} = 90$$

9. $\underbrace{M, M, T, T, H, C, S}_{7}\ \underbrace{A, A, E, I}_{4}$
When all the vowels are together then $n = 7 + 1 = 8$
∴ Required number of permutations = $\frac{8! \times 4!}{2! \times 2! \times 2!} = 120960$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

HINT (*M*, *T* and *A* occur 2 times and *A A E I* can be arranged mutually in $\frac{4!}{2!}$ ways)

10. *I,M,P,O,R,T,A,N,T*

Total number of permutations $= \frac{9!}{2!} = 181440$

Number of permutations when 2*T*'s are together

$$= \frac{8! \times 2!}{2!} = 8! = 40320$$

∴Number of permutations when 2*T*'s are not together

$$= 181440 - 40320 = 141120$$

11. *T,T,T, N,N,N, G,G,G, R, I,I,I*

There are total 13 leters of which 3 are *T*'s, 3 are *N*'s, 3 are *G*'s, 3 are *I*'s.

Also there are 6 even places and 7 odd places. *3Is* (i.e., vowels) can be arranged in the 6 even places in $\frac{^6P_3}{3!} = 20$ ways.

Now, we are left with 10 places in which 10 letters can be filled up in $\frac{10!}{3! \times 3! \times 3!} = 16800$

∴The total number of permutations in which vowels occupy even places $= 20 \times 16800 = 336000$

12. Correct order of the letters in *D, G, I, I, O, O, P, R, S, U*

Word begin with	**Number of words**
D	$\frac{9!}{2!2!}$
G	$\frac{9!}{2!2!}$
I	$\frac{9!}{2!}$
O	$\frac{9!}{2!}$
PD	$\frac{8!}{2!2!}$
PG	$\frac{8!}{2!2!}$
PI	$\frac{8!}{2!}$
PO	$\frac{8!}{2!}$
PRD	$\frac{7!}{2!2!}$
PRG	$\frac{7!}{2!2!}$
PRI	$\frac{7!}{2!}$
PRODG	$\frac{5!}{2!}$

Now the next two words are

PRODIGIOSU and PRODIGIOUS

Hence the rank of the word PRODIGIOUS

$$= 2\left(\frac{9!}{2!2!}\right) + 2\left(\frac{9!}{2!}\right) + 2\left(\frac{8!}{2!2!}\right) + 2\left(\frac{8!}{2!}\right) + 2\left(\frac{7!}{2!2!}\right) + \frac{7!}{2!} + \frac{5!}{2!} + 2$$

$$= 3\left(\frac{9!}{2!}\right) + 3\left(\frac{8!}{2!}\right) + 7! + \frac{5!}{2!} + 2$$

$$= 544320 + 60480 + 5040 + 60 + 2$$

$$= 609902$$

13. The correct order of letters is D, E, E, E, E, Q, R, S, S, T, U.

Number of words begin with DEEEEQR $= \frac{4!}{2!} = 12$

Number of words begin with DEEEEQS $= 4! = 24$

Number of words begin with DEEEEQT $= \frac{4!}{2!} = 12$

Now, the next two words are

DEEEEQURSST and DEEEEQURSTS

Hence the 50th word is DEEEEQURSTS

14. There are 7 digits 1, 2, 0, 2, 4, 2, 4 in which 2 occurs 3 times, 4 occurs 2 times.

∴ Number of 7 digit numbers $= \frac{7!}{3! \times 2!} = 420$

But out of these 420 numbers there are some numbers which begin with '0' and they are not 7-digit numbers. The number of such numbers beginning with '0'.

$$= \frac{6!}{3! \times 2!} = 60$$

Hence the required number of 7 digits numbers

$$= 420 - 60 = 360$$

Alternatively :

6	6	5	4	3	2	1

$\Rightarrow 6.(6!)$

Since the left most place can assume only non-zero digit hence it can consider only 6 digits. Now the remaining 6 places can be filled up in normal ways.

But since 2 occurs 3 times, 4 occurs 2 times, therefore required number of 7 digit numbers $= \frac{6.(6!)}{3!.2!} = 360$

15. Required number of 6 digit numbers $= \frac{6!}{4!.2!} = 15$

16.

5	4	3	2	1	3

$\Rightarrow 3.(5!)$

Unit place can assume any of the three even digits *viz.*,2 ,4 ,6 and rest of the places can be filled up in 5! ways.

But since digit 7 occurs 2 times.

∴ Required number of numbers $= \frac{3.(5!)}{2!} = 180$

17.

6	6	5	4	3	2	1

Since left most place can assume only non-zero digit and rest can assume all the given digits.

Therefore the number of numbers $= 6.(6!)$

But since 5 occurs 2 times and 2 occurs 2 times

Hence the required number of numbers $= \frac{6(6!)}{2!.2!} = 1080$

18. *REGUITA*

REG T

There are total 11 letters, but since 2*T*'s are together.

Therefore number of numbers when 2*T*'s are together

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$$= \frac{10!}{2! \times 2! \times 2!} \times \frac{2!}{2!} = 453600$$

$\uparrow R \quad \uparrow E \quad \uparrow G \quad \uparrow T$

Here R is 2 times, E is 2 times, G is 2 times and T is 2 times.

19. $SUMPTO$

$S \quad U$

Number of ways in which 2U's are always together is

$$\frac{7!}{2!} \times \frac{2!}{2!} = 2520$$

Total number of permutations $= \frac{8!}{2!.\ 2!} = 10080$

$\therefore$ Number of ways in which 2U's are not together

$= 10080 - 2520 = 7560$

20. $AFLTON$

$A \quad O$

There are 4 consonants F, L, T and N and 4 vowels A, A, O, O

4 vowels can be arranged in $\frac{4!}{2!.\ 2!} = 6$ ways.

Now the 4 consonants can be arranged in 4 places in 4! = 24 ways.

Therefore required number of permutations

$= (6 \times 24) \times 2! = 288$

HINT There are two cases :
(i) $OCOCOCOC$ (ii) $COCOCOCO$
where $O \rightarrow$ vowel, $C \rightarrow$ consonant)

21. $SOTHAYER$

$S \quad O$

Since two S's and two O's together therefore we can consider there are only 8 letters, which can be permuted in $\frac{8! \times 2! \times 2!}{2! \times 2!} = 8! = 40320$ ways

22. Total number of books $= 3 \times 4 = 12$ in which each of 4 different books is repeated 3 times.

Hence the required number of permutations

$$= \frac{12!}{3! \times 3! \times 3! \times 3!} = \frac{12!}{(3!)^4} = 369600$$

23. There are total $3 + 4 + 5 = 12$ marbles of which 3 are red (alike), 4 are green (alike) and 3 are pink (alike).

$\therefore$ The required number of arrangements $= \frac{12!}{3! \times 4! \times 5!}$

$= 27720$

INTRODUCTORY EXERCISE 19.3

1. Ten thousands place can assume only non-zero digits hence ten thousands place can be filled up in 4 ways and thousands place can be filled up in 5 ways since repetition is allowed (and 0 can be filled up in this place). Similarly hundreds, tens and unit places can be filled up in 5 ways each.

$\therefore$ The required number of numbers $= 4 \times 5 \times 5 \times 5 \times 5 = 2500$

2. The numbers lying between 9 and 1000 consist of 2 or 3 digits in which repetitions are allowed.

Case 1. (For two digit numbers)

As the tens place can be filled by only non-zero digits and unit digit can be filled up in 6 ways.

$\therefore$ The number of 2 digit numbers $= 5 \times 6 = 30$

Case 2. (For three digit numbers)

Hundreds place can be filled up by only non-zero digits hence it can be done in 5 ways and the rest of the places *i.e.*, tens and unit places can be filled up in 6 ways each.

$\therefore$ The number of 3 digit numbers $= 5 \times 6 \times 6 = 180$

$\therefore$ Total number of numbers $= 30 + 180 = 210$

3. Since even digit may not occupy an even place therefore even places can be filled up in only 3 ways (with odd digits) and rest of the 3 odd places can be filled up in 5 ways with all the 5 digits.

Therefore the required number of numbers

$= \boxed{5}\boxed{3}\boxed{5}\boxed{3}\boxed{5}$

$= 5 \times 3 \times 5 \times 3 \times 5 = 1125$

4. Number of numbers of 4 digit in which repetition allowed $= 9 \times 10 \times 10 \times 10 = 9000$

Number of numbers of 4 digit in which repetitions is not allowed $= 9 \times 9 \times 8 \times 7 = 4536$

Hence the required number of numbers of 4 digit

$= 9000 - 4536 = 4464$

5. Total number of 3 digit numbers $= 9 \times 10 \times 10 = 900$

Number of 3 digit numbers in which none of the digits is 9

$= 8 \times 9 \times 9 = 648$

$\therefore$ Number of 3 digit numbers in which there is atleast one digit is $9 = 900 - 648 = 252$

6. Total one digit numbers = 4

Total 2 digit numbers $= 4 \times 4 = 16$

Total 3 digit numbers $= 4 \times 4 \times 4 = 64$

Number of 4 digit numbers beginning with $1 = 4 \times 4 \times 4 = 64$

Number of 4 digit numbers beginning with $2 = 4 \times 4 \times 4 = 64$

Number of 4 digit numbers beginning with $3 = 4 \times 4 \times 4 = 64$

Number of 4 digit numbers beginning with $41 = 4 \times 4 = 16$

Number of 4 digit numbers beginning with $42 = 4 \times 4 = 16$

Number of 4 digit numbers beginning with 431 = 4

Number of 4 digit numbers beginning with 432 = 1

Total number of 4 digit numbers

$= 64 + 64 + 64 + 16 + 16 + 4 + 1 = 229$

Hence the total number of natural numbers not exceeding 4321 is $4 + 16 + 64 + 229 = 313$

7. First ring can be worn in any of the 3 fingers similarly second, third and fourth ring can be worn in any of the three fingers.

$\therefore$ 4 rings can be worn in $3 \times 3 \times 3 \times 3 = 81$ ways,

8. First prize can be given away to 4 boys in 4 ways.

Similarly second, third, fourth and fifth prizes can also be given away to four boys in 4 ways.

Hence the required number of way in which all the 5 prizes can be given away to 4 boys

$= 4 \times 4 \times 4 \times 4 \times 4 = 1024$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

9. The first ball can be placed in any one of the n cells in n ways.
The second ball can also be placed in any one of the n cells in n ways.
$\therefore$ The first and second balls can be placed in n cells in $n \times n$ i.e., n^2 ways.
Similarly each of the rest balls can be placed in n ways.
Hence the required number of ways $= n \times n \times \ldots \times n$ times
$= n^n$

10. Since each cell is to be occupied, 1 ball is to be placed in one cell. First ball can be placed in any one of the n cells in n ways, 2nd ball can be placed in any one of the remaining $(n-1)$ cells in $(n-1)$ ways, similarly 3rd, 4th... ball can be arranged in $(n-2), (n-3)\ldots$ ways.
Hence the required number of ways
$= n(n-1)(n-2)\ldots 2.1 = n!$

11. First letter can be posted in 4 letter boxes in 4 ways.
Similarly second letter can be posted in 4 letter boxes in 4 ways and so on.
Hence all the 5 letters can be posted in
$= 4 \times 4 \times 4 \times 4 \times 4 = 1024$

INTRODUCTORY EXERCISE 19.4

1. Required number of permutations $= 5! = 120$

2. Assume there are only 5 boys since two boys always remain together and these two boys can be arranged mutually in 2 ways.
$\therefore$ Required number of permutations $= 4! \times 2 = 48$

3. Required number of permutations $= \frac{1}{2} \times 5! = 60$
(Since there is no difference in clockwise and anticlockwise arrangements)

4. Total number of arrangements $= 6! = 720$
Number of arrangements in which two ladies are together $= 2 \times 5! = 240$
$\therefore$ Number of arrangements in which two ladies are never together $= 720 - 240 = 480$
Alternatively : First of all place the five men in 4! ways, then place the two ladies in any 5 spaces in ${}^5P_2$ ways.
Hence, the required number of ways $= 4! \times {}^5P_2 = 480$

5. Assuming Sahara, Ambani and Mahindra as a single personality, then there are total $(10-3)+1 = 8$ person, which can be arranged in 7! ways.
Also Sahara and Mahindra can be mutually permuted in 2 ways.
Therefore required number of arrangements $= 2 \times 7!$
$= 10080$

6. A person can be chosen out of 18 people in 18 ways to be seated between Musharraf and Manmohan. Now consider Musharraf, Manmohan and the third person, sitting between them, as a single personality, we can arrange them in 17! ways but Musharraf and Manmohan can also be arranged in 2 ways.
$\therefore$ Required number of permutations $= 18 \times (17!) \times 2$
$= 2 \times 18!$

7. 6 Men can be arranged in 5! ways then 6 women can be arranged in 6 places in 6! ways.
Hence the required number of ways $= 5! \times 6!$
$= 120 \times 720 = 86400$

8. First of all 4 girls can be arranged in 3! ways then 3 boys can be seated in 4 places in ${}^4P_3$ ways.
$\therefore$ Required number of arrangements $= 3! \times {}^4P_3$
$= 6 \times 24 = 144$

9. Clearly 7 sisters can sit around a table in 6! ways. But in clockwise and anticlockwise arrangements each of the ladies have the same neighbours.
So, the required number of ways $= \frac{1}{2} \times (6!) = 360$
Hence they dine together only 360 days.

10. Since 3 particulars flowers are together
Hence there can be total $(10-3)+1 = 8$ flowers. These 8 flowers can be arranged 7! ways but the 3 flowers which are together can be arranged mutually in 3! ways.
Hence, the required number of ways $= 3! \times 7!$
$= 6 \times 5040$
$= 30240$

11. Sonu can have only male singer as his neighbours which can be mutually arranged in 2! ways.
Similarly Alka can have only female singers as her neighbours which can be arranged in 2! ways.
Hence the required number of arrangements $= 2! \times 2! = 4$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

INTRODUCTORY EXERCISE 19.5

1. $^{6}C_3 = \frac{8!}{3!5!} = \frac{8\times7\times6}{6} = 56$

2. $^{10}C_5 = \frac{10\times9\times8\times7\times6\times5!}{5\times4\times3\times2\times1\times(5!)} = 252$

3. $\because\ ^{13}C_x = {}^{13}C_y$

$\Rightarrow$ either $x = y$ or $x + y = 13$

But since $x \neq y$, therefore $x + y = 13$

4. Since $^{17}C_r = {}^{17}C_{r+3}$

$\Rightarrow$ either $r = r + 3$ or $r + r + 3 = 17$

But since $r \neq r + 3$

$\therefore\quad r + r + 3 = 17 \Rightarrow r = 7$

5. $^{7}P_3 = n.^{7}C_3$

$\Rightarrow\quad 5\times6\times7 = n\times5\times7$

$\Rightarrow\quad n = 6$

Alternatively : $^{n}P_r = r!(^{n}C_r)$

$\Rightarrow\quad r! = 3! = 6 = n$

6. $^{2n}C_3 : {}^{n}C_2 = 12:1$

$\Rightarrow\quad \frac{2n(2n-1)(2n-2)}{3\times2\times1} : \frac{n(n-1)}{2\times1} = 12:1$

$\Rightarrow\quad (2n-1) = 9 \Rightarrow n = 5$

Alternatively : Go through options.

7. Go through options : The least possible value of n is 8. To verify the result put the vlaue of n as given in the choices.

Let us consider choice (a) *i.e.*, $n = 6$, then

$$^{5}C_3 + {}^{5}C_4 > {}^{6}C_3$$

$10 + 5 > 20$, which is wrong.

Again consider choice (c) *i.e.*, $n = 7$, then

$$^{6}C_3 + {}^{6}C_4 > {}^{7}C_3$$

$20 + 15 > 35$, which is also wrong.

Now consider choice (b) *i.e.*, $n = 8$, then

$$^{7}C_3 + {}^{7}C_4 > {}^{8}C_3$$

$35 + 35 > 56$, which is correct.

Hence the least possible value of $n = 8$.

Now you need not to check the option (d).

8. $\frac{^{n}P_x}{^{n}C_x} = \frac{\frac{(n!)}{(n-x)!}}{\frac{(n!)}{(x!)(n-x)!}} = \frac{336}{56}$

$\Rightarrow\quad \frac{^{n}P_x}{^{n}C_x} = x! = 6$

$\Rightarrow\quad x! = 3! \Rightarrow x = 3$

Now since $^{n}P_3 = 336$

$\Rightarrow\quad n(n-1)(n-2) = 8\times7\times6$

$\Rightarrow\quad n = 8$

Alternatively : Go though options.

9. If n is odd, the greatest value of $^{n}C_r = {}^{n}C_m$

where $m = \frac{(n-1)}{2}$ or $m = \frac{(n+1)}{2}$

$\therefore\quad x = \frac{(11-1)}{2} = 5$ or $x = \frac{(11+1)}{2} = 6$

Hence (d) is the correct choice.

10. If n is even, the greatest value of $^{n}C_r = {}^{n}C_m$

where $m = \frac{n}{2}$

$x = \frac{12}{2} = 6$

11. $^{14}C_x$ is maximum when $x = \frac{14}{2} = 7$

Alternatively : Go through options.

12. Since there are total 10 persons and out of these 10 persons we have to select any 5 persons which can be done in $^{10}C_5$ ways.

$\therefore\quad ^{10}C_5 = \frac{10!}{5!\times5!} = 252$

13. Out of 10 questions, 6 questions can be selected in $^{10}C_6$ ways

$\therefore\quad ^{10}C_6 = 210$

14. Since question number 1 is compulsory so we have to choose only 5 questions from rest of the 9 questions, which can be done in $^{9}C_5$ ways

$\therefore\quad ^{9}C_5 = 126$

15. She may invite one or more friends by selecting either 1 or 2 or 3 or 4 or 5 friends out of 5 friends.

$\therefore$ 1 friend can be selected out of 5 in $^{5}C_1$ ways

2 friends can be selected out of 5 in $^{5}C_2$ ways

3 friends can be selected out of 5 in $^{5}C_3$ ways

4 friends can be selected out of 5 in $^{5}C_4$ ways

5 friends can be selected out of 5 in $^{5}C_5$ ways

Hence the required number of ways

$= {}^{5}C_1 + {}^{5}C_2 + {}^{5}C_3 + {}^{5}C_4 + {}^{5}C_5$

$= 5 + 10 + 10 + 5 + 1 = 31$

Alternatively : $^{5}C_1 + {}^{5}C_2 + {}^{5}C_3 + {}^{5}C_4 + {}^{5}C_5 = 2^5 - 1 = 31$

Since, $^{n}C_1 + {}^{n}C_2 + {}^{n}C_3 + \ldots + {}^{n}C_n = 2^n - 1$

16. 4 men can be selected out of 7 men in $^{7}C_4$ ways and 2 ladies can be selected out of 6 ladies in $^{6}C_2$ ways

Hence, the required number of ways $= {}^{7}C_4 \times {}^{6}C_2$

$= 35\times15 = 525$

17. A committee of 5 persons is to be formed from 6 gentlemen and 4 ladies by taking.

(i) 1 lady out of 4 and 4 gentlemen out of 6.

(ii) 2 ladies out of 4 and 3 gentlemen out of 6.

(iii) 3 ladies out of 4 and 2 gentlemen out of 6.

(iv) 4 ladies out of 4 and 1 gentleman out of 6.

In case I the number of ways $= {}^{4}C_1 \times {}^{6}C_4 = 4\times15 = 60$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

In case II the number of ways = ${}^4C_2 \times {}^6C_3 = 6 \times 20 = 120$

In case III the number of ways = ${}^4C_3 \times {}^6C_2 = 4 \times 15 = 60$

In case IV the number of ways = ${}^4C_4 \times {}^6C_1 = 1 \times 6 = 6$

Hence, the required number of ways

$= 60 + 120 + 60 + 6 = 246$

18. A committee of 7 persons retaining a majority of Indians can be made from 6 Americans and 7 Indians by taking

(i) 1 American out of 6 and 6 Indians out of 7.

(ii) 2 Americans out of 6 and 5 Indians out of 7.

(iii) 3 Americans out of 6 and 4 Indians out of 7.

In case I the number of ways = ${}^6C_1 \times {}^7C_6 = 6 \times 7 = 42$

In case II the number of ways = ${}^6C_2 \times {}^7C_5 = 15 \times 21 = 315$

In case III the number of ways = ${}^6C_3 \times {}^7C_4 = 20 \times 35 = 700$

Hence, the required number of ways

$= 700 + 315 + 42 = 1057$

19. Required number of ways = ${}^6C_4 \times {}^8C_3 = 15 \times 56 = 840$

20. The number of committees in which both Miss. *A* and Mr. *B* are always members. We can select rest 3 men from the remaining 5 in ${}^5C_3$ ways and rest 2 women from the remaining 7 in ${}^7C_2$ ways.

∴The required number of ways in which both Miss. *A* and Mr. *B* are always included

$= {}^5C_3 \times {}^7C_2 = 10 \times 21 = 210$

Hence the required number of ways in which Miss. *A* and Mr. *B* do not serve together = 840 – 210 = 630

21. Since 3 particular members are already selected, then we are required to select only 4 members out of the remaining 8 members.

It can be done in ${}^8C_4$ ways.

$\therefore \quad {}^8C_4 = 70$

22. Since 3 particular members are not included hence we have to select 7 members out of remaining 8 members. It can be done in ${}^8C_7$ ways

$\therefore \quad {}^8C_7 = 8$

23. 3 experts including atleast an engineer and a manager can be selected by taking.

(i) 2 managers out of 3 and 1 engineer out of 3.

(ii) 1 manager out of 3 and 2 engineer out of 3.

(iii) 2 persons out of 6 (3 managers and 3 engineers) and 1 person out of one who is both engineer and manager.

In case I, the number of ways = ${}^3C_2 \times {}^3C_1 = 9$

In case II, the number of ways = ${}^3C_1 \times {}^3C_2 = 9$

In case III, the number of ways = ${}^6C_2 \times {}^1C_1 = 15$

Hence, the required number of ways = 9 + 9 + 15 = 33.

24. A committee of 5 persons, consisting of at most two ladies, can be formed in the following ways.

(i) selecting 5 gents only out of 6.

(ii) selecting 4 gents only out of 6 and one lady out of 4.

(iii) selecting 3 gents only out of 6 and two ladies out of 4.

In case I, the number of ways = ${}^6C_5$

In case II, the number of ways = ${}^6C_4 \times {}^4C_1$

In case III, the number of ways = ${}^6C_3 \times {}^4C_2$

∴Required number of ways $= {}^6C_5 + {}^6C_4 \times {}^4C_1 + {}^6C_3 \times {}^4C_2$

$= 6 + 60 + 120 = 186$

25. A voter may cast vote for either 1 candidate or 2 candidates or 3 candidates or 4 candidates out of 7.

The voter may cast vote for 1 candidate in ${}^7C_1$ ways

The voter may cast vote for 2 candidates in ${}^7C_2$ ways

The voter may cast vote for 3 candidates in ${}^7C_3$ ways

The voter may cast vote for 4 candidate in ${}^7C_4$ ways

Hence, the required number of ways

$= {}^7C_1 + {}^7C_2 + {}^7C_3 + {}^7C_4$

$= 7 + 21 + 35 + 35 = 98$

26. Required number of ways $= {}^5C_1 + {}^5C_2 + {}^5C_3$

$= 5 + 10 + 10 = 25$

27. Required number of ways $= {}^8C_6 \times {}^8C_5$

$= 28 \times 56 = 1568$

28. Required number of ways

$= ({}^6C_4 \times {}^6C_2) + ({}^6C_3 \times {}^6C_3) + ({}^6C_2 \times {}^6C_4)$

$= (15 \times 15) + (20 \times 20) + (15 \times 15)$

$= 225 + 400 + 225 = 850$

29. Required number of ways

$= {}^7C_3 \times {}^5C_5 + {}^7C_4 \times {}^5C_4 + {}^7C_5 \times {}^5C_3$

$= 35 \times 1 + 35 \times 5 + 21 \times 10$

$= 35 + 175 + 210 = 420$

30. Required number of ways

$= ({}^5C_2 \times {}^5C_3 \times {}^4C_1) + ({}^5C_3 \times {}^5C_2 \times {}^4C_1) + ({}^5C_2 \times {}^5C_2 \times {}^4C_2)$

$= (10 \times 10 \times 4) + (10 \times 10 \times 4) + (10 \times 10 \times 6)$

$= 1400$

31. Required number of ways $= {}^{13}C_8 \times {}^5C_2$

$= 1287 \times 10 = 12870$

32. One hand shake requires two different persons.

∴ Number of hand shakes

= number of ways of selecting two persons out of 11 persons

$= {}^{11}C_2 = 55$

33. If n is even, ${}^nC_r$ is maximum when $r = \frac{n}{2}$

$\therefore \quad$ Number of invitees in a party $= \frac{24}{2} = 12$

and maximum possible number of parties $= {}^{24}C_{12}$

$= 2704156$

34. If n is odd, ${}^nC_r$ is maximum when

$$r = \frac{n-1}{2} \quad \text{or} \quad r = \frac{n+1}{2}$$

$$\therefore \quad r = \frac{11-1}{2} = 5 \quad \text{or} \quad r = \frac{11+1}{2} = 6$$

$$\therefore \quad {}^{11}C_5 = {}^{11}C_6 = 462$$

35. Possible number of members in a family is either 5 or 6.

So, the maximum possible number of family members = 6

36. Let there were n persons in the meeting, then number of handshakes = ${}^nC_2$

$\therefore \quad {}^nC_2 = 66$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmanboob.wordpress.com

$\Rightarrow \quad \frac{n!}{2!(n-2)!} = 66$

$\Rightarrow \quad \frac{n(n-1)}{2} = 66$

$\Rightarrow \quad n(n-1) = 132$

$\Rightarrow \quad n = 12$

Alternatively : Go through options.

37. Since a particular player is always chosen therefore we have to choose only 10 players out of remaining 15 players.

$\therefore$ Required number of ways $= {}^{15}C_{10} = 3003$

38. Since a player is never chosen,hence we have to choose 11 players out of remaining 15 players.

$\therefore$ Required number of ways $= {}^{15}C_{11} = 1365$

39. Total number of bowlers = 4
Total number of wicket keepers = 2
Rest (normal) players = 10
Possible Combinations :

Bowlers	Wicket keepers	Normal players
3	1	7
3	2	6
4	1	6
4	2	5

$\therefore$ Required number of ways

$= ({}^4C_3 \times {}^2C_1 \times {}^{10}C_7) + ({}^4C_3 \times {}^2C_2 \times {}^{10}C_6) + ({}^4C_4 \times {}^2C_1 \times {}^{10}C_6) + ({}^4C_4 \times {}^2C_2 \times {}^{10}C_5)$

$= (3 \times 1 \times 7) + (3 \times 2 \times 6) + (4 \times 1 \times 6) + (4 \times 2 \times 5)$

$= 960 + 840 + 420 + 252 = 2472$

40. Total number of bowles = 6
Total number of wicket keepers = 3
Total number of normal players = 11 [20 – (6 + 3)]
Possible combinations :

Bowlers	Wicket keepers	Normal players
4	2	5
5	2	4
6	2	3

$\therefore$ Required number of ways

$= ({}^6C_4 \times {}^3C_2 \times {}^{11}C_5) + ({}^6C_5 \times {}^3C_2 \times {}^{11}C_4) + ({}^6C_6 \times {}^3C_2 \times {}^{11}C_3)$

$= 20790 + 5940 + 495 = 27225$

41. Total number of wicket keepers = 2
Total number of bowlers = 8
Total number of batsman = 10
Total number of all rounders = 5
1 Wicket keeper can be selected out of 2 in ${}^2C_1$ ways
2 Bowlers can be selected out of 8 in ${}^8C_2$ ways
5 Batsmen can be selected out of 10 in ${}^{10}C_5$ ways
3 All rounders can be selected out of 5 in ${}^5C_3$ ways
$\therefore$ Required number of ways $= {}^2C_1 \times {}^8C_2 \times {}^{10}C_5 \times {}^5C_3$

$= 2 \times 28 \times 252 \times 10 = 141120$

42. 1 book on Economics can be collected out of 3 in ${}^3C_1$ ways
1 book on Corporate strategy can be collected out of 4 in ${}^4C_1$ ways 1 book on Philosophy can be collected out of 5 in ${}^5C_1$ ways
Hence the required number of collections each of which consists of exactly one book on each subject

$= {}^3C_1 \times {}^4C_1 \times {}^5C_1$

$= 3 \times 4 \times 5 = 60$

43. At least one book on Economics can be collected out of $3 = {}^3C_1 + {}^3C_2 + {}^3C_3 = 2^3 - 1 = 7$ ways.
Similarly one book on corporate strategy can be collected out of $4 = {}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4 = 2^4 - 1 = 15$ ways.
Similarly one book on Philosophy can be collected out of $5 = {}^5C_1 + {}^5C_2 + {}^5C_3 + {}^5C_4 + {}^5C_5 = 2^5 - 1 = 31$ ways.
Hence the required number of ways $= 7 \times 15 \times 31 = 3255$

44. Since all the three balls are red which can be selected from 7 red balls only.

$\therefore$ Required number of selections $= {}^7C_3 = 35$

45. Since red balls can not be taken thus we can select 3 balls out of 6 white and 4 blue balls.

$\therefore$ Required number of selections $= {}^{10}C_3 = 120$ ways

46. The required number of selections

$= {}^7C_1 \times {}^6C_1 \times {}^4C_1 = 168$

47. Total number of red balls = 5
Total number of green balls = 6
Possible combinations are :

Red	Green
2	4
3	3
4	2

$\therefore$ Required number of selections

$= ({}^5C_2 \times {}^6C_4) + ({}^5C_3 \times {}^6C_3) + ({}^5C_4 \times {}^6C_2) = 425$

48. A line requires 2 end points for its formation.
Hence required number of lines

= 2 points selected from 6 points

$= {}^6C_2 = 15$

Thus out of total 6 non-collinear points 15 lines can be drawn.

49. Total number of lines formed by 12 points $= {}^{12}C_2$
Number of lines formed by 4 points $= {}^4C_2$
$\therefore$ Required number of lines = Total lines formed by 12 points – number of lines formed by 4 collinear points + 1

$= {}^{12}C_2 - {}^4C_2 + 1 = 61$

Explanation : If no three points are collinear, a straight line can be formed by joining any two points out of given non-collinear points. But since if some points are collinear then some lines become overlapped.
Here 12 points can make ${}^{12}C_2$ lines, but since 4 points are collinear, we can find only one line by joining these 4 collinear points where as we have considered ${}^4C_2$ lines in ${}^{12}C_2$ lines.
Hence the required number of straight lines

$= {}^{12}C_2 - {}^4C_2 + 1 = 61$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

50. A decagon has 10 vertices *i.e.*, 10 non-collinear points. Hence the total number of lines formed by 10 non-collinear points $= {}^{10}C_2 = 45$

But since out of 45 lines 10 lines are the sides of the decagon and the remaining 35 lines are diagonals of a decagon.

Alternatively : Number of diagonals $= {}^{n}C_2 - n$

51. Number of diagonals in an n-sided polygon

= Number of total lines – Number of sides

$= {}^{n}C_2 - n$

$= \frac{n(n-1)}{2} - n = \frac{n(n-3)}{2}$

52. Number of diagonals $= {}^{n}C_2 - n = 54$

$\Rightarrow \quad \frac{n(n-1)}{2} - n = 54$

$\Rightarrow \quad \frac{n(n-3)}{2} = 54$

$\Rightarrow \quad n = 12$

Alternatively : Go through options.

53. A triangle can be formed by joining 3 non-collinear points. Hence number of triangles is equal to the number of ways in which 3 points can be selected out of 6 non-collinear points.

$\therefore$ Number of triangles $= {}^{6}C_3 = 20$

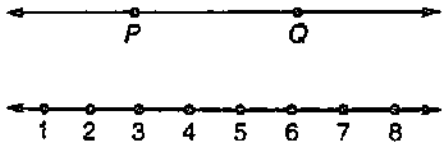
54. Number of triangles that can be formed if no three points are collinear $= {}^{12}C_3 = 220$.

But since 4 points are collinear, we cannot form any triangle from these 4 points by joining any three of them where as we have taken ${}^{4}C_3$ triangles in the above result (*i.e.*, ${}^{12}C_3$).

Hence the required number of triangles $= {}^{12}C_3 - {}^{4}C_3$

$= 220 - 4 = 216$

55.



A triangle can be formed by selecting 1 point out of P and Q and selecting 2 points out of 8 points on the other parallel line.

or a triangle can be formed by selecting 2 points out of 2 points (P and Q) and selecting 1 point out of 8 points.

$\therefore$ Required number of triangles $= ({}^{2}C_1 \times {}^{8}C_2) + ({}^{2}C_2 \times {}^{8}C_1)$

$= (2 \times 28) + (1 \times 8) = 64$

56. Number of triangles whose vertex is P = total number of triangles – number of triangles which do not contain P as a vertex.

$\therefore$ Number of triangles which do not contain P as a vertex

$= {}^{1}C_1 \times {}^{8}C_2 = 28$

$\therefore$ Required number of triangles $= 64 - 28 = 36$

57. An intersection point is formed by the intersection of two lines. Hence number of intersection points is equal to the number of ways of selecting 2 lines out of the given 20 non-parallel and non-concurrent lines.

i.e., Required number of points $= {}^{20}C_2 = 190$.

58. Number of straight lines formed by 'n' non-collinear points

$= {}^{n}C_2 = \frac{n(n-1)}{2}$.

59. Since 'p' points are in a straight line therefore they will form only one line instead of ${}^{p}C_2$ lines.

$\therefore$ Required number of lines $= {}^{n}C_2 - {}^{p}C_2 + 1$.

60. Number of triangles formed by 'n' non-collinear points is ${}^{n}C_3$.

But since m points are in a straight line therefore from these m points ${}^{m}C_3$ triangles can not be formed. Hence required number of triangles $= {}^{n}C_3 - {}^{m}C_3$.

61. A parallelogram is formed by choosing two straight lines. From the set of m parallel lines and two straight lines from the set of n parallel lines.

Two straight lines from the set of m parallel lines can be chosen in ${}^{m}C_2$ ways and two straight lines from the set of n parallel lines can be chosen in ${}^{n}C_2$ ways. Hence the number of parallelograms formed

$= {}^{m}C_2 \times {}^{n}C_2 = \frac{m(m-1)}{2} \times \frac{n(n-1)}{2}$

$= \frac{mn(m-1)(n-1)}{4}$

62. The number of points of intersection of 37 lines is ${}^{37}C_2$. But 13 straight lines out of the given 37 straight lines pass through the same point A. Therefore instead of getting ${}^{13}C_2$ points, we get only one point A. Similarly 11 straight lines out of the given 37 straight lines intersect at point B. Therefore instead of getting ${}^{11}C_2$ points, we get only one point B.

Hence the number of intersection points of the lines is

${}^{37}C_2 - {}^{13}C_2 - {}^{11}C_2 + 2 = 535$

63. Required number of words $= {}^{15}C_2 \times {}^{5}C_4$

$= 105 \times 5 = 525$

64. There are 11 letters in the given word of which 2 are A's, 2 are I's, 2 are N's and the remaining 5 letters are different. Thus we have 11 letters of 8 different kinds *viz.*, (A, A), (I, I), (N, N), E, X, M, T, O.

A group of 4 letters can be classified as follows :

(i) Two alike of one kind and two alike of another kind.

(ii) Two alike and the other two different.

(iii) All four different.

In case I, the number of ways $= {}^{3}C_2 = 3$

In case II, the number of ways $= {}^{3}C_1 \times {}^{7}C_2 = 63$

In case III, the number of ways $= {}^{8}C_4 = 70$

Hence, the required number of ways $= 3 + 63 + 70 = 136$

65. There are 11 letters in the given word where M occurs 2 times, A occurs 2 times and T occurs 2 times and rest letters occur one times each.

i.e., M, A, T, H, E, I, C, S

M, A, T

A group of 4 letters can be selected in the following manner.

(i) Two alike of one kind and two alike of other kind.

(ii) Two alike and the other two different.

(iii) All four different.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Case I. ${}^3C_2 \times \frac{4!}{2!2!} = 18$

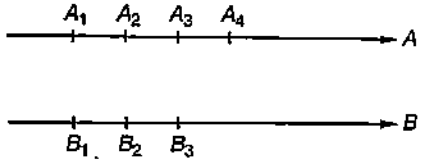
Case II. ${}^3C_1 \times {}^7C_2 \times \frac{4!}{2!} = 756$

Case III. ${}^8C_4 \times 4! = 1680$

∴ Total required words of 4 letters = 18 + 756 + 1680
= 2454

66. Let L_1, L_2, L_3 be 3 ladies and G_1, G_2, G_3 be 3 gents.
Any two of the ladies out of 3 can be selected in ${}^3C_2$ ways and can be arranged mutually in 2! ways.
Now we are left with 3 gents and 1 lady these 4 persons can be arranged at a round table in 3! ways.
Now we have only two places among 4 persons to place the two selected ladies which can be done in ${}^2P_1$ ways.
Hence the required number of arrangements
$= {}^3C_2 \times 2! \times 3! \times {}^2P_1$
$= 3 \times 2 \times 6 \times 2 = 72$

67. Since 4 guests desire to sit on one side and the other 3 guests desire to sit on the other side.



Hence we are left with 11 guests only out of which we choose 5 guests for the side A in ${}^{11}C_5$ ways and remaining 6 guests can be selected for the side B in ${}^6C_6$ ways.
Further in each side 9 guests can be arranged in 9! ways.
Hence the required number of arrangements
$= {}^{11}C_5 \times {}^6C_6 \times 9! \times 9! = 462 \times (9!)^2$

68. The number of ways in which none of the n letters goes to the correct place
$$= n!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + (-1)^n \frac{1}{n!}\right]$$

∴ Number of ways in which all the 4 letters go to the wrong place
$$= 4!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!}\right]$$
$$= 24\left[1 - \frac{1}{1} + \frac{1}{2} - \frac{1}{6} + \frac{1}{24}\right]$$
$$= 24\left[\frac{12 - 4 + 1}{24}\right] = 9$$

Now the number of ways in which all the possibilities can occur = 4! = 24.
∴ Number of ways in which atleast one letter goes to the correct place = 24 − 9 = 15.

69. Required number of ways = 9.

INTRODUCTORY EXERCISE 19.6

1. Atleast 1 green dye can be selected out of 5 green dyes in $(2^5 - 1)$ *i.e.*, 31 ways.
Similarly, atleast one blue dye can be selected out of 4 in $(2^4 - 1)$ *i.e.*, 15 ways,
and atleast 1 red or no red dye can be selected out od 3 red dyes in 2^3 *i.e.*, 8 ways.
Hence the required number of ways = 31 × 15 × 8 = 3720.

2. Out of 15 fruits, 7 are alike of one kind, 5 are alike of a second kind and 3 are alike of a third kind.
Hence, the required number of ways
$= [(7 + 1)(5 + 1)(3 + 1) - 1] = 191$

3. Out of 4 beverages he can select atleast one beverage in $2^4 - 1$ *i.e.*, 15 ways.

4. Atleast one mango can be selected in $2^4 - 1 = 15$ ways and atleast one orange can be selected in $2^5 - 1 = 31$ ways.
Hence the required number of ways = 15 × 31 = 465

5. Required number of ways = [(9 + 1)(7 + 1)(4 + 1)] − 1
= 399

6. ${}^{(2n+1)}C_1 + {}^{(2n+1)}C_2 + {}^{(2n+1)}C_2 + \ldots + {}^{(2n+1)}C_n = 63$
Now going through options we get $n = 3$.
∴ ${}^7C_1 + {}^7C_2 + {}^7C_3 = 7 + 21 + 35 = 63$
Hence $n = 3$ is the correct answer.

INTRODUCTORY EXERCISE 19.7

1. The required number of ways in which 12 students may be equally divided into three groups.
$$= \frac{12!}{3! \times (4!)^3} = 5775$$

2. The required number of ways in which 12 mangoes may be equally divided among three boys
$$= \frac{12!}{(4!)^3} = 34650$$

3. Let three boxes of different sizes be denoted by B_1, B_2, B_3. 5 balls can be placed in the three boxes so that no box is empty as follows :

Case	I	II	III	IV	V	VI
B_1	1	1	1	2	2	3
B_2	1	2	3	1	2	1
B_3	3	2	1	2	1	1

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

In case I, the number of ways $= \frac{5!}{1!\times1!\times3!} = 20$

In case II, the number of ways $= \frac{5!}{1!\times2!\times2!} = 30$

In case III, the number of ways $= \frac{5!}{1!\times3!\times1!} = 20$

In case IV, the number of ways $= \frac{5!}{2!\times1!\times2!} = 30$

In case V, the number of ways $= \frac{5!}{2!\times2!\times1!} = 30$

In case VI, the number of ways $= \frac{5!}{1!\times1!\times3!} = 20$

Hence, the required number of ways

$= 20 + 30 + 20 + 30 + 30 + 20 = 150$

4. It mean each player can receive 13 cards.

∴First player can get 13 cards in ${}^{52}C_{13}$ways.

Second player can get 13 cards in ${}^{39}C_{13}$ ways.

Third player can get 13 cards in ${}^{26}C_{13}$ ways

Fourth player can get 13 cards in ${}^{13}C_{13}$ ways.

Hence the total number of ways

$$= {}^{52}C_{13} \times {}^{39}C_{13} \times {}^{26}C_{13} \times {}^{13}C_{13} = \frac{52!}{(13!)^4}$$

Alternatively : 52 cards can be divided equally among 4 players in $\frac{52!}{(13!)^4 \times (4!)}$ ways.

But since order is important, hence the required number of ways

$$= \frac{52! \times 4!}{(13!)^4 \times (4!)} = \frac{52!}{(13!)^4}$$

5. Required number of ways $= \frac{52!}{(13!)^4 \times (4!)}$.

6. First player can get 16 cards in ${}^{52}C_{16}$ ways.

Second player can get 16 cards in ${}^{36}C_{16}$ ways.

Third player can get 16 cards in ${}^{20}C_{16}$ ways.

and fourth player can get 4 cards in ${}^{4}C_{4}$ ways.

But since first three sets can be interchanged in 3! ways.

Hence, the required number of ways

$$= {}^{52}C_{16} \times {}^{36}C_{16} \times {}^{20}C_{16} \times {}^{4}C_{4} \times \frac{1}{3!}$$

$$= \frac{52!}{(16!)^3 \times (3!)}$$

7. Since the order of the couples is not important.

So, the required number of ways $= \frac{2n!}{(2!)^n \cdot n!}$.

8. Number of ways in which 20 mangoes can be distributed among 4 persons is ${}^{20+4-1}C_{4-1} = {}^{23}C_3$.

9. Number of ways of distributing 7 identical balls among 3 boxes so that a box receives atleast one ball

$$= {}^{7-1}C_{3-1} = {}^{6}C_2 = 15$$

10. Total number of non-negative integral solutions of the given equations is same as the number of ways of distributing 44 items among 4 persons such that each person can receive any number of items.

Hence, total number of solutions $= {}^{44+4-1}C_{4-1} = {}^{47}C_3$.

11. Let $u \geq 0, v \geq 0, w \geq 0$, then

$$a \geq u+1, \quad b \geq v+2, \quad c \geq w+3$$

$$\therefore \quad a+b+c = 18$$

$$\Rightarrow \quad u+1+v+2+w+3 = 18$$

$$\Rightarrow \quad u+v+w = 12$$

∴Total number of solutions of this equations is

$${}^{12+3-1}C_{3-1} = {}^{14}C_2$$

12. Let x, y, z be the number of balls received by the three persons, then

$$x \geq 5, y \geq 5, z \geq 5 \quad \text{and} \quad x+y+z = 21$$

Let $u \geq 0, v \geq 0, w \geq 0$, then

$$\therefore \quad x+y+z = 21$$

$$\Rightarrow \quad u+5+v+5+w+5 = 21$$

$$\Rightarrow \quad u+v+w = 6$$

∴ Total number of solutions $= {}^{6+3-1}C_{3-1} = {}^{8}C_2 = 28$

LEVEL (1)

1. $9 \times 9 \times 9 \times 9 = 9^4$.

2. $9^4 - 1$ (It is considered that the lock opens in the last attempt.)

3. **Case I.** *MW MW MW MW*

Case II. *WM WM WM WM*

Let us arrange 4 men in 4! ways, then we arrange 4 women in ${}^{4}P_4$ ways at 4 places either left of the men or right of the men.

Hence required number of arrangements

$$= 4! \times {}^{4}P_4 + 4! \times {}^{4}P_4 = 2 \times 576 = 1152$$

4. First of all we arrange 4 women in 4! ways then we arrange 4 men in 5 places in ${}^{5}P_4$ ways.

Hence, the total number of arrangements $= 4! \times {}^{5}P_4 = 2880$

5. Let us arrange 4 persons in 4! ways then we place their respective wives adjacent to them in 2! ways i.e., either left or right sides of the husbands.

Hence, the required numbr of ways $= 4! \times 2! \times 2! \times 2! \times 2!$

6. Let us first of all we arrange 4 men in 3! ways then 4 women can be arranged in 4 places in ${}^{4}P_4$ ways.

Hence the required number of ways $= 3! \times {}^{4}P_4 = 144$.

7. All the 4 men can be arranged in 4! ways.

Similarly all the 4 women can be arranged in 4! ways but the two different groups of men and women can be mutually arranged in 2! ways.

Hence the required number of ways $= (4!)^2 \times 2! = 1152$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

8. $H\ L\ C\ N\ T \quad A\ U\ I\ O$
$L \quad N \quad A \quad I$

There are total 13 letters out of which 7 are consonants and 6 are vowels. Also there are $2L's$, $2N's$, $2A's$, and $2I's$.

If all the consonants are together then the number of arrangements $= \frac{7!}{2!.2!}$.

but the 7 consonants can be arranged themselves in $\frac{7!}{2!.2!}$ ways.

Hence the required number of ways $= \frac{7!}{2!\times 2!} \times \frac{7!}{2!\times 2!}$

$= (1260)^2 = 1587600$

9. There are 7 odd places and 6 vowels of which 2 vowels occur two times.

Hence number of arrangements of vowels $= \frac{{}^7P_6}{2!\times 2!}$.

Now the remaining 7 consonants of which two consonants occur two times, can be arranged in 7 places in $\frac{7!}{2!\times 2!}$ ways.

Hence, the required number of ways $= \frac{{}^7P_6}{2!\times 2!} \times \frac{7!}{2!\times 2!}$

$= 1587600$

10. $S\ U\ C\ E\ F\ L$
$S\ U\ C$
S

There are 10 letters in the word SUCCESSFUL and S occurs 3 times, U occurs 2 times and C occurs 2 times.

∴ The letters of the word SUCCESSFUL can be arranged in

$= \frac{10!}{3!\times 2!\times 2!}$ ways

$= 151200$ ways

11. Since all three $S's$ are together, hence there will be 8 letters of which C occurs 2 times and U occurs 2 times.

Thus the required number of arrangements $= \frac{8!}{2!\times 2!}$

$= 10080$

12. $(C, C), (U, U), S, S, S, E, F, L$

These 8 letters now can be arranged in $\frac{8!}{3!}$ ways.

13. Total number of 4 digit numbers without repetition of digits $= 4! = 24$.

Each of the digits in each of the 4 places (thousands, hundreds and tens and unit) place will be 3! times *i.e.*, 6 times.

Hence the sum of all the digits at each of the four places

$= 6(1 + 2 + 3 + 4) = 60$

Hence the sum of all the required 4 digit numbers

$= 60\ (1000 + 100 + 10 + 1)$

$= 60 \times 1111 = 66660$

14. Total number of 5 digit numbers (including which begins with zero) $= 5! = 120$.

Number of 5 digit numbers which begin with zero $= 4! = 24$.

Sum of all 5 digit numbers

$= (0 + 1 + 2 + 3 + 4) \times 4! \times (11111)$

$= 240 \times 11111 = 2666640$

Sum of all five digit numbers which begin with zero

$= (1 + 2 + 3 + 4) \times 3! \times (1111) = 66660$

Hence the sum of the required numbers

= sum of all 5 digit numbers including those numbers which begin with zero

− sum of all 5 digit numbers which begin with zero

$= 2666640 - 66660 = 2599980$

15. Number of polygons $= {}^nC_2$ − Number of sides (n)

$\therefore \quad {}^nC_2 - n = n$

$\Rightarrow \quad {}^nC_2 = 2n$

$\Rightarrow \quad \frac{n(n-1)}{2} = 2n$

$\Rightarrow \quad n = 5$

16. There are 10 letters in the given word PROPORTION and we have 10 letters of 6 different kinds *viz.*,

$(O, O, O), (P, P), (R, R), T, I, N$.

For a group of four letters, we have four cases :

(i) Three alike and one different.

(ii) Two alike and two other alike.

(iii) Two alike and other two different.

(iv) All four different.

In case I, the number of selections $= {}^1C_1 \times {}^5C_1 = 5$.

In case II, the number of selections $= {}^3C_2 = 3$.

[Since we can select two pairs out of 3 pairs (O, O), (P, P), (R, R)]

In case III, the number of selections $= {}^3C_1 \times {}^5C_2 = 30$.

(∵ We can select one of 3 pairs and then two from the remaining 5 letters say, P, R, T, I, N.)

In case IV, the number of selections $= {}^6C_4 = 15$.

(∵ We can select 4 different letters from 6 letters O, P, R, T, I, N.)

In case I, the number of arrangements $= 5 \times \frac{4!}{3!} = 20$.

In case II, the number of arrangements $= 3 \times \frac{4!}{2!\times 2!} = 18$.

In case III, the number of arrangements $= 30 \times \frac{4!}{2!} = 360$.

In case IV, the number of arrangements $= 15 \times 4! = 360$.

Hence the required number of arrangements

$= 20 + 18 + 360 + 360 = 758$

17. Vowels themselves can be permuted in $\frac{4!}{3!}$ ways

and consonants can be permuted in $\frac{6!}{2!\times 2!}$ ways

Therefore the required number of ways $= \frac{4!}{3!} \times \frac{6!}{2!\times 2!}$

$= 4 \times 180 = 720$

18. Total number of ways of arranging 16 people $= 15!$ ways

Number of ways in which two brothers are together $= 14! \times 2$

∴ Number of ways in which two brothers are never together

$= 15! - 14! \times 2$

$= 14!(15 - 2) = 14! \times 13$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

19. A person out of remaining 14 persons can be selected in ${}^{14}C_1 = 14$ ways.
Now consider Lehman, Mckinsey and a third (selected person) as a single person, then we can arrange all of them in 13! ways.
But the two brothers mutually can be arranged in 2! ways.
Hence the required number of ways $= 14 \times 13! \times 2 = 14! \times 2$

20. Every ball can be distributed in 4 ways.
Hence the required number of ways $= 4 \times 4 \times 4 \times 4 \times 4 \times 4$
$= 4^6 = 4096$

21. Number of ways of distribution of 'n' identical things taken r at a time $= {}^{n+r-1}C_{r-1}$
$\therefore$ Required number of ways $= {}^{6+4-1}C_{4-1} = {}^{9}C_3 = 84$

22. Number of non-negative integral solutions $= {}^{n+r-1}C_{r-1}$
$= {}^{21+3-1}C_{3-1}$
$= {}^{23}C_2 = 253$

23. Number of positive integral solutions $= {}^{n-1}C_{r-1}$
$= {}^{21-1}C_{3-1} = {}^{20}C_2$
$= 190$

24. Number of ways of selecting items from 6 different items $= 2^6$.
Similarly from second set of 6 different objects we can select items in 2^6 ways.
Therefore, total number of selections $= 2^6 \times 2^6$
but we have to select at least one item therefore we subtract one case in which none of the items is selected $= 2^6 \times 2^6 - 1$
$= 2^{12} - 1 = 4095$

25. Number of ways of selecting the items from 6 identical items $= 6 + 1 = 7$
Similarly we can select from other set of 6 identical items $= 7$
$\therefore$ Total number of ways in which at least one item can be selected from the two sets $= (7 \times 7) - 1 = 48$

Note Since we can select either 0, 1, 2, 3, 4, 5, or 6 items which can be done in 7 ways.

26. We can select atleast one item from 6 different items $= (2^6 - 1)$
Similarly we can select atleast one item from other set of 6 different items in $(2^6 - 1)$ ways.
$\therefore$ Required number of ways $= (2^6 - 1)(2^6 - 1)$
$= (2^6 - 1)^2 = 3969$

27. We can select atleast one item from 6 identical items in 6 ways.
Similarly from other set also we can select atleast one item in 6 ways.
Hence, the required number of ways $= 6 \times 6 = 36$.

28.

Single step	Double step	Number of ways
12	0	$\frac{12!}{12!} = 1$
10	1	$\frac{11!}{10!} = 11$
8	2	$\frac{10!}{(8! \times 2!)} = 45$
6	3	$\frac{9!}{(6! \times 3!)} = 84$
4	4	$\frac{8!}{(4! \times 4!)} = 70$
2	5	$\frac{7!}{(2! \times 5!)} = 21$
0	6	$\frac{6!}{6!} = 1$

$\therefore$ Total number of ways $= 1 + 11 + 45 + 84 + 70 + 21 + 1$
$= 233$

29. Using ${}^{n+r-1}C_{r-1}$, we get ${}^{8+3-1}C_{3-1} = {}^{10}C_2 = 45$

30. ${}^{100}C_{10} \times {}^{90}C_{20} \times {}^{70}C_{30} \times {}^{40}C_{40} = \frac{100!}{10! \times 20! \times 30! \times 40!}$

31. There are two sets of numbers 0, 1, 3, 5 and 0, 2, 3, 4.
Therefore number of 4 digit numbers using digits 0, 1, 3, 5
$= 3 \times 3 \times 2 \times 1 = 18$
Similarly number of 4 digit numbers using digits 0, 2, 3, 4
$= 3 \times 3 \times 2 \times 1 = 18$
Hence the total required numbers $= 18 + 18 = 36$.

32.

When 7 is at ten thousands place →	1	1	9	9	9	→ 9^3
	1	9	1	9	9	→ 9^3
	1	9	9	1	9	→ 9^3
	1	9	9	9	1	→ 9^3
When 7 is not at the ten thousands place →	8	1	1	9	9	→ 8×9^2
	8	1	9	1	9	→ 8×9^2
	8	1	9	9	1	→ 8×9^2
	8	9	1	1	9	→ 8×9^2
	8	9	1	9	1	→ 8×9^2
	8	9	9	1	1	→ 8×9^2

Hence the required number of ways $= 4 \times 9^3 + 6 \times 8 \times 9^2$
$= 9^2 (36 + 48)$
$= 81 \times 84 = 6804$

33. Required number of ways $= \frac{15!}{3! \times 4! \times 8!}$.

34. Total number of ways $= 7!$
Let A has to speak before B.
Now since there are half of the total cases in which A speaks before B (Similarly in half of the total cases B speaks before A)
$\therefore$ Required number of ways $= \frac{1}{2} \times 7! = 2520$.

35.

1	10	10	10	10	10	10	$\Rightarrow$ 10^6

Total number of 7 digit telephone numbers which begin with digit $2 = 10^6$.

36. ${}^6P_4 = 360$.
Alternatively : First son can accept any of the 6 proposals and second son can accept any of the 5 proposals and so on.
Hence required number of ways $= 6 \times 5 \times 4 \times 3 = 360$

37. ${}^{16}C_4 = 1820$

38. ${}^{16}P_4 = 43680$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

39. ${}^{16}C_4 \times {}^{12}C_4 \times {}^{8}C_4 \times {}^{4}C_4 \times \frac{1}{4!} = \frac{16!}{(4!)^5}$.

40. ${}^{n}C_2 = 105$, where n is the number of players

$$\Rightarrow \quad \frac{n(n-1)}{2} = 105 \Rightarrow n = 15$$

41. ${}^{9}P_6 = 60480$

42. $\frac{6!}{2! \times 4!} = 15$

43. $4 \times 6 = 24$

44. Total number of 5 digit numbers $= 9 \times 10^4 = 90000$
Number of 5 digit numbers without any repetition
$= 9 \times 9 \times 8 \times 7 \times 6 = 27216$
∴Required number of 5 digit numbers in which atleast one of the digit repeats $= 90000 - 27216 = 62784$

45. Total number of possible outcomes $= 6^4$.
The number of possible outcomes in which 4 does not appear on any die is 5^4.
Therefore the number of possible outcomes in which atleast one die shows digit $4 = 6^4 - 5^4 = 671$

46. Required number of signals $= {}^{5}P_1 + {}^{5}P_2 + {}^{5}P_3 + {}^{5}P_4 + {}^{5}P_5$
$= 5 + 20 + 60 + 120 + 120 = 325$

47. The number of words begin with A is 4!.
The number of words begin with E is 4!.
The number of words begin with M is 4!.
The number of words begin with R is 4!.
Number of words begin with VA is 3!
Words begin with VE are $VEAMR$
$VEARM$
$VEMAR$
$VEMRA$
$VERAM$
$VERMA$
∴ The rank of the word $VERMA = 4 \times 4! + 3! + 6$
$= 96 + 6 + 6 = 108$

48. **Case I:** 0, 1, 2, 4, 5
Case II: 1, 2, 3, 4, 5
Since we know that a number is divisible by 3, if and only if the sum of its digits is divisible by 3.
Case I, number of 5 digit numbers $= 4 \times 4 \times 3 \times 2 = 96$
Case II, number of 5 digit numbers $= 5 \times 4 \times 3 \times 2 = 120$
∴ Total required numbers $= 216$

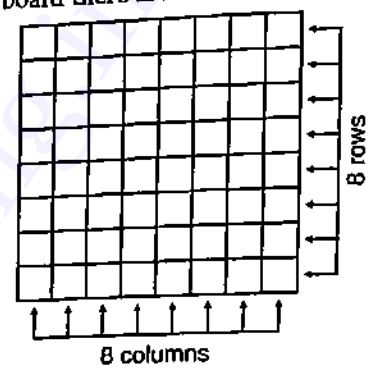
49. We can choose two men out of 8 in ${}^{8}C_2$ ways. Since no husband and wife are to play in the same game, two women out of the remaining 6 can be chosen in ${}^{6}C_2$ ways.
If M_1, M_2, W_1, W_2 are chosen, then a team may consist of M_1 and W_1 or M_1 and W_2. Thus the number of ways of arranging the game is
$= {}^{8}C_2 \times {}^{6}C_2 \times 2 = 840$

50. ${}^{n-2}C_3$.

51. Since 8 does not occur in 1000, we have to count the number of times 8 occurs when we list the integers from 1 to 999. Any number between 1 and 999 is of the form xyz, where $0 \le x, y, z \le 9$.
Let us first count the numbers in which 8 occurs exactly once.
Since 8 can occur at one place in ${}^{3}C_1$ ways. There are ${}^{3}C_1 (9 \times 9) = 3 \times 9^2$ such numbers.
Next, 8 can occur in exactly two places in ${}^{3}C_2 (9) = 3 \times 9$ such numbers. Lastly, 8 can occur in all three digits in one number only.
Hence, the number of times 8 occurs is
$1 \times (3 \times 9^2) + 2 \times (3 \times 9) + 3 \times 1 = 300$

52. **Case I** | 2 | 2 | 1 |
Number of possible arrangements $= {}^{5}C_2 \times {}^{3}C_2 \times {}^{1}C_1 \times 3 = 90$
Case II | 1 | 1 | 3 |
Number of possible arrangements $= {}^{5}C_1 \times {}^{4}C_1 \times {}^{3}C_3 \times 3 = 60$
Hence, the total number of ways is $90 + 60 = 150$.

53. In a chess board there are 8 columns and 8 rows.



Since number of columns = number of rows
Hence it is considered as a square.
∴Number of square in a square of
$n \times n = 1^2 + 2^2 + \ldots + n^2$
Hence required number of squares
$= 1^2 + 2^2 + 3^2 + \ldots + 8^2$
$= \frac{8 \times 9 \times 17}{6} = 204 \qquad \left[\because \Sigma n^2 = \frac{n(n+1)(2n+1)}{6}\right]$

54. $L \quad S_1 \quad S_2 \quad S_3 \quad S_4 \quad S_5 \quad S_6 \quad S_7 \quad M$
Since these five people enter the train during journey it means they must have enter the train after Lucknow. Hence if a person enters at S_1 he can have 7 different tickets for $S_2, S_3, \ldots S_7$ and Mumbai.
Similarly if a person enters at S_2 he can have 6 different tickets and if a person enters at S_3 he can have 5 different tickets and so on.
∴Total numer of possible tickets $= 7 + 6 + 5 + 4 + 3 + 2 + 1$
$= 28$
Now, the five different tickets must be out of these 28 tickets. Therefore, the required number of ways
$= {}^{28}C_5 = 98280$

55. The number of ways of choosing the committee when Ms. B is a member (when Mr. A refuses to serve)
$= {}^{8}C_3 \times {}^{6}C_5$
$= 56 \times 6 = 336$
The number of ways of choosing the committee when Ms. B is not a member (when Mr. A can serve)
$= {}^{8}C_4 \times {}^{7}C_5$
$= 70 \times 21 = 1470$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

Thus the required number of ways = 336 + 1470 = 1806.

Alternatively : Total number of ways of selecting 4 women and 5 men = ${}^9C_4 \times {}^7C_5 = 2646$.

Number of ways of selecting 4 women and 5 men in which Ms. B and Mr. A are not present together

$= {}^7C_3 \times {}^6C_4 = 840$

Hence the number of ways in which Ms. B and Mr. A do not serve together = 2646 − 840 = 1806

56. Required number of ways = $5 \times 5 \times 5 \times 5 \times 5 = 3125$

57. Required number of arrangements = $5 \times 5 \times 5 \times 5 = 625$

Since we fix cap C_1 in box B_1 then we arrange the remaining 4 caps in any of the 5 boxes.

Alternatively : Total number of arrangements = 5^5

Number of arrangements in which B_1 have cap $C_1 = \frac{5^5}{5} = 5^4$

58. Required number of arrangements = ${}^5P_5$

$\therefore \quad {}^5P_5 = 5 \times 4 \times 3 \times 2 \times 1 = 120$

59. 1 cap can be selected in ${}^5C_1$ ways.

Now this selected cap can be arranged in 5 boxes in ${}^5P_1$ ways.

Again 2 caps can be selected in ${}^5C_2$ ways.

Now, these selected caps can be arranged in 5 boxes in ${}^5P_2$ ways and so on.

Hence the required number of ways

$= {}^5C_1 \times {}^5P_1 + {}^5C_2 \times {}^5P_2 + {}^5C_3 \times {}^5P_3 + {}^5C_4 \times {}^5P_4 + {}^5C_5 \times {}^5P_5$

$= 25 + 200 + 600 + 600 + 120 = 1545$

60. Since all the boxes are filled with exactly one cap and all the caps are identical. Hence the required number of ways = 1.

61. Required number of ways = $\frac{5!}{2!} = 60$

(Since 2 caps are identical)

62. B_3 can be filled in 2! ways and rest can be filled in 4! ways. Hence, required number of ways = $2 \times 4! = 48$.

63. B_1 and B_5 can be filled up in $2 \times 2 = 4$ ways and B_2, B_3, B_4 can be filled up in $= 5 \times 5 \times 5 = 125$ ways.

Hence the required number of ways = $4 \times 125 = 500$

LEVEL 2

1. First and second prizes can be given in Mathematics in (30×29) ways.

First and second prizes can be given in Physics in (30×29) ways.

First prize can be given Chemistry in 30 ways.

First prize can be given in English in 30 ways.

Hence, the number of ways to give prizes in all the four subjects

$(30 \times 29) \times (30 \times 29) \times 30 \times 30 = (30)^4 \times (29)^2$

2. $1 + 1 \cdot P_1 + 2 \cdot P_2 + 3 \cdot P_3 + 4 \cdot P_4 + \ldots + n \cdot P_n$

$= 1 + 1 + 2 \cdot 2! + 3 \cdot 3! + 4 \cdot 4! + \ldots + n \cdot n!$

$= 1 + [(2! - 1!) + (3! - 2!) + (4! - 3!) + \ldots + ((n+1)! - n!)]$

$= 1 + [(n+1)! - 1!]$

$= (n+1)!$

Alternatively : Consider $n = 1, 2, 3, \ldots$ etc. and then verify the result.

3.

	Row-1	Row-2	Row-3	Row-4
Case 1.	Class 1	Class 2	Class 1	Class 2
Case 2.	Class 2	Class 1	Class 2	Class 1

Required seating arrangement can be done in two ways-case 1 and case 2.

Total numer of arrangements

= Number of arrangements in case 1 + Number of arrangements in case 2

Now, 12 students of class 1 can be seated in 12 chairs in ${}^{12}P_{12}$ ways and 12 students of class 2 can be seated in 12 chairs in ${}^{12}P_{12}$ ways.

Hence, the total number of arrangements

$= (12! \times 12!) + (12! \times 12!) \qquad (\because {}^{12}P_{12} = 12!)$

$= 2(12!)^2$

4. There are 5 even digits *viz.*, 0, 2, 4, 6 and 8. Digit 3 can not be at the tens place, since if 3 is at tens place, then 9 must be at unit place which is impossible. Hence 3 can be only at hundred's place. Now there are two cases :

(1) **When 3 is at hundred's place :** If 3 is at hundred's place, then 9 will be at ten's place. So unit place can be filled up in only 5 ways by using even digits.

So, the total number of even numbers = $1 \times 1 \times 5 = 5$.

(2) **When 3 is not at hundred's place :** In this case hundred's place can be filled up in 8 ways (0 and 3 can not be used). In tens place we can use any one of the 10 digits except 3. So tens place can be filled up in 9 ways and unit place can be filled up in 5 ways by using 0, 2, 4, 6 and 8.

So, the total number of even numbers = $8 \times 9 \times 5 = 360$

Hence the total number of required 3 digit even numbers

$= 5 + 360 = 365$

5. Number of vehicles registered upto RZ 9999

$= 18 \times 26 \times (9999)$

Number of vehicles registered between SA-0001

and SJ 9999 = $1 \times 10 \times 9999$

Therefore number of vehicles registered before SK-0123

$= 18 \times 26 \times 9999 + 1 \times 10 \times 9999 + 122$

$= 9999 \times 478 + 122 = 4779644$

6. Available digits are 1, 2, 3, 4 and 5. Now since we know that a number is divisible by 4 if and only if the number formed by last two digits is divisible by 4. So the following cases are possible.

Thousands	Hundreds	Tens	Unit
x	y	1	2
x	y	2	4
x	y	3	2
x	y	5	2

In each case thousands and hundreds places can be filled up in ${}^3P_2$ ways.

Hence the required number of ways = ${}^3P_2 \times 4 = 24$.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

7. Total number of seats = 2 (3 + 4) = 14
Total number of persons = 3 + 9 = 12
∴Total number of required arrangements = $^{14}P_{12}$.

8. Three girls can be seated together in a back row on adjacent seats in the following ways :
1, 2, 3 or 2, 3, 4 of first van and 1, 2, 3 or 2, 3 4 of second van.
In each of the above four cases all the three girls can be inter change among themselved in 3! ways.
So the total number of ways = 4 × 3!
Now, 9 boys can be seated in remaining 11 seats, which can be done in $^{11}P_9$ ways.
Hence, the required number of ways = $^{11}P_9 \times 4 \times 3!$

9. 5 '+' (plus) signs can be arranged in 1 way, then we have 6 places to arrange 5 '–' (minus) signs in $^6C_5$ ways but since all the minus (–) signs are identical, therefore minus signs can be arranged in only 1 way in 5 chosen places out of 6.
Hence the required number of arrangements = $1 \times {}^6C_5 \times 1 = 6$

10. In order that no two books on Maths be together we first arrange 11 identical books on English which can be done in 1 way.
Now we can choose 9 places out of 12 (11 + 1) places for Maths books, which can be done in $^{12}C_9$ ways.
Again since all books on Maths are identical which can be mutually arranged in only 1 way.
Hence the required number of arrangements = $1 \times {}^{12}C_9 \times 1$
$= 220$

11. $1680 = 2^4 \times 3 \times 5 \times 7$
Since if $N = a^p \cdot b^q \cdot c^r \cdot \ldots$
we have number of factors of $N = (p+1)(q+1)(r+1)\ldots$
Hence the total number of factors of the given number
$= (4+1)(1+1)(1+1)(1+1) = 40$

12. $\sum_{k=m}^{n} {}^kC_r = {}^mC_r + {}^{m+1}C_r + {}^{m+2}C_r + \ldots + {}^{(n-1)}C_r + {}^nC_r$
Now assume some values of m and n, then verify the correct result from the choices given in the question.

13. The number of students answering exactly k $(1 \le k \le n-1)$ questions wrongly is $2^{n-k} - 2^{n-k-1}$.
The number of students answering all n questions wrongly is 2^0. Thus, the total number of wrong answers is :
$1(2^{n-1} - 2^{n-2}) + 2(2^{n-2} - 2^{n-3}) + \ldots$
$\ldots + (n-1)(2^1 - 2^0) + n(2^0)$
$= 2^{n-1} + 2^{n-2} + 2^{n-3} + \ldots + 2^0 = 2^n - 1$
$\Rightarrow \quad 2^n - 1 = 2047$
$\Rightarrow \quad 2^n = 2048$
$\Rightarrow \quad n = 11$

14. Required number of triangles = Total number of triangles
– Number of triangles having one side common with the octagon
– Number of triangles having two sides common with the octagon
$= {}^8C_3 - ({}^8C_1 \times {}^4C_1) - 8 = 16$

15. The number of ways of selecting any five numbers from the given set = $^{25}C_5$.
Now, the five consecutive numbers can be chosen in the following ways
(1, 2, 3, 4, 5); (2, 3, 4, 5, 6); (3, 4, 5, 6, 7);...(21, 22, 23, 24, 25)
Thus the required number of ways = $^{25}C_5 - 21 = 53109$

16. $\because a \ge -1, b \ge -1, c \ge -1, d \ge -1$.
Let $u \ge 0, \quad v \ge 0, \quad w \ge 0, \quad x \ge 0$.
$\therefore \quad a + b + c + d = 12$
$\Rightarrow \quad (u-1) + (v-1) + (w-1) + (x-1) = 12$
$\Rightarrow \quad u + v + w + x = 16$
$\therefore$ Required number of solutions = $^{16+4-1}C_{4-1} = {}^{19}C_3$

17. Assume some value of k then find 'n' and hence $^nC_2$.
Now, verify the correct option.

18. $42 = 2 \times 3 \times 7$
Here each of a, b and c can take 3 values.
Hence the required number of solutions = $3 \times 3 \times 3 = 27$.

19. Let (r, b, g) denotes the number of red, blue and green balls respectively.
Therefore the possibilities are (0, 10, 0), (2, 7, 1), (4, 4, 2) and (6, 1, 3).
Hence, the required number of ways = 4.

20. Let x_1 be the number of stations before the first halting station, x_2 between first and second, x_3 between second and third, x_4 between third and fourth and x_5 on the right of the fourth station, then
$x_1 \ge 0, \quad x_2, x_3, x_4 \ge 1 \quad$ and $\quad x_5 \ge 0$
such that $x_1 + x_2 + x_3 + x_4 + x_5 = 6$.
Now, the total number of ways is the number of solutions of the above equation.
Let $y_2 \ge 0, \quad y_3 \ge 0, \quad y_4 \ge 0$.
$\therefore \quad x_1 + x_2 + x_3 + x_4 + x_5 = 6$
$\Rightarrow \quad x_1 + (y_2 + 1) + (y_3 + 1) + (y_4 + 1) + x_5 = 6$
$\Rightarrow \quad x_1 + y_2 + y_3 + y_4 + x_5 = 3$
∴ The number of solutions = $^{3+5-1}C_{5-1} = {}^7C_4 = 35$

21. Let x_i denote the marks assigned to the ith question. Then
$x_1 + x_2 + x_3 + \ldots + x_7 + x_8 = 30$
where $\quad i = 1, 2, 3, \ldots, 8 \quad$ and $\quad x_i \ge 2$
and $\quad y_1 + y_2 + \ldots + y_7 + y_8 = 14$
where $\quad y_i = x_i - 2, \quad i = 1, 2, 3, \ldots$
∴The total number of solutions of this equation is
$^{14+8-1}C_{8-1} = {}^{21}C_7$

22. ∴The exponent of 3 in 33! is 15.

3	33	
3	11 ←	15
3	3 ←	
	1 ←	

HINT For more information about exponents refer the first chapter of the book.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

23. There are two distinct possibilities
(i) When any of the four digits may repeat thrice.
(ii) When any two of the four digits may repeat twice.
(i) $\frac{6!}{3!} \times {}^4C_1 = 480$
(ii) $\frac{6!}{2!2!} \times {}^4C_2 = 1080$
Therefore the required number = 480 + 1080 = 1560.

24. | 9 | 10 | 10 | 10 | 10 | 10 | 5 | = 4500000
Since if we fix first digit then unit digit will determine the odd/even nature of the numbers so formed.
But unit place can assume half even and half odd. Hence unit place can assume just 5 digits which makes the sum of previous six digits even.

25. Required number of ways = ${}^6P_4$.

26. Suppose x one rupee coins, y fifty paise coins and z twenty paise coins are selected.
Then $x + y + z = 6$
∴Total number of ways
= Number of solutions of the above equation
$= {}^{6+3-1}C_{3-1} = {}^8C_2 = 28$

27. Required number of times
= Number of selections of 3 children taking at a time
$= {}^8C_3 = 56$

28. Each child will go as often as he or she can be accompanied by two others. Therefore the required number
$= {}^7C_2 = 21$

29. Total number of points = 12
Total number of triangles = ${}^{12}C_3$
But there are 3 cases which must be excluded for the required number of triangles.
Case 1. The number of triangles formed by 3 points on
$AB = {}^3C_3 = 1$
Case 2. The number of triangles formed by 4 points on
$BC = {}^4C_3 = 4$
Case 3. The number of triangles formed by 5 points on
$CA = {}^5C_3 = 10$
Hence, the required number of triangles = 220 − (10 + 4 + 1)
= 205

30. Each question can be attempted in 3 ways *i.e.*, either it can be omitted or one of the two parts can be attempted.
Hence required number of ways = $3^{10} - 1 = 59048$.

31. Required number of ways = ${}^{16}C_9$.

32. Since each bulb has two choices, either switched on or off, therefore required number = $2^{10} - 1 = 1023$.

33. Since each digit of a 10 digit number can be written as either 2 or 3, therefore required number of 10 digit number is 2^{10}.

34. Consider some value of n then verify the result.

35. Any divisor of $2^3 \cdot 3^4 \cdot 5^2$ is of the form $2^a \cdot 3^b \cdot 5^c$ where $0 \le a \le 3, 0 \le b \le 4$ and $0 \le c \le 2$. Thus, the sum of the divisors of $2^3 \cdot 3^4 \cdot 5^2$ is
$(1 + 2 + \ldots + 2^3)(1 + 3 + \ldots + 3^4)(1 + 5 + 5^2)$
$= \frac{(2^4 - 1)(3^5 - 1)(5^3 - 1)}{(2-1)(3-1)(5-1)}$
$= \frac{15 \times 242 \times 124}{1 \times 2 \times 4} = 56265$

36. Required number of ways = ${}^{10+4-1}C_{4-1} = {}^{13}C_3 = 286$

37. Since we know that out of n things arranged in a row then we can select r things out of n such that no two of them are adjacent to each other in ${}^{n+1-r}C_r$ ways.
Hence required number of ways = ${}^{10+1-3}C_3 = {}^8C_3 = 56$
Alternatively : To each selection of 3 objects we associated a binary sequence of the form 1001000100 where 1 at i^{th} place means the ith object is selected and 0 at ith place means the ith object is not selected.
There exists one to one correspondence between the set of selections of 3 objects and set of binary sequences containing 7 zeros and 3 ones.
Actually we are concerned with the binary sequences in which 2 ones are consecutive we first arrange 7 zeros.
This can be done in just one way. Now, 3 ones can be arranged at any of the 3 places marked with a cross in the following arrangement.
$\times 0 \times 0 \times 0 \times 0 \times 0 \times 0 \times 0 \times$
We can arrange 3 ones at 8 places in ${}^8C_3$ ways = 56

38. Let x_5 be such that $x_1 + x_2 + x_3 + x_4 + x_5 = n$
Now we require the non-negative integral solutions of
$x_1 + x_2 + x_3 + x_4 + x_5 = n$
The number of required solutions = ${}^{n+5-1}C_{5-1} = {}^{n+4}C_4$

39. When we arrange things one at a time, the number of possible permutations is n.
When we arrange them two at a time, the number of possible permutations is $n \times n = n^2$ and so on.
Hence, the total number of required permutations is
$n + n^2 + n^3 + \ldots + n^r = \frac{n(n^r - 1)}{n - 1}$

40. In a chess board there are 8 columns and 8 rows.
Hence the number of rectangles = $1^3 + 2^3 + 3^3 + \ldots + 8^3$
$= \left(\frac{8 \times 9}{2}\right)^2 = 1296$
Alternatively : A chessboard is a network of 9 vertical and 9 horizontal lines.
Since we know that number of quadrilaterals formed by m vertical and n horizontal lines is ${}^mC_2 \times {}^nC_2$.
Hence the required numer of rectangles = ${}^9C_2 \times {}^9C_2 = 1296$

41. Order of speaking is equivalent to the order of placing the things.
Thus A, B and C can be placed in ${}^{10}C_3$ ways and the remaining 7 persons can be placed in 7! ways.
Hence, the number of ways in which they can speak is
${}^{10}C_3 \times 7! = \frac{10!}{3!}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Alternatively : All the 10 persons can speak in 10! ways in which A, B and C can speak in 6 different ways.
But we are required to find a single case out of 6 cases in which A speaks before B and B speaks before C.

Hence the required number of ways $= \frac{10!}{3!} = \frac{10!}{6}$

42. Total one digit number = 2
Total two digit numbers = 2^2
Total three digit numbers = 2^3
Total four digit numbers = 2^4
... ...
Total eight digit numbers = 2^8
There are total 9 digit numbers = 2^9
but out of these 2^9 numbers half of the numbers begin with digit 1 and rest half numbers begin with digit 2.
The greatest possible number less than $2 \cdot 10^8$ is 122222222, which in turn is the greatest possible number of 9 digit begin with 1.
Thus the maximum possible number of 9 digit numbers

$$= \frac{2^9}{2} = 2^8$$

Hence, the required total numbers

$$= 2 + 2^2 + 2^3 + 2^4 + \ldots + 2^8 + 2^8$$

$$= \frac{2(2^8 - 1)}{2 - 1} + 2^8 = 2^9 - 2 + 2^8$$

$$= 2^8 (2 + 1) - 2 = 766$$

43. Required number of ways $= {}^{12+1-4}C_4 = {}^9C_4 = 126$.

44. Since $2k < n \Rightarrow k < \frac{n}{2}$

Hence, number of zeros is less than the number of ones.
Now, first of all we arrange $(n - k)$ ones as follows

$$\times 1 \times 1 \times 1 \times 1 \times 1 \times \ldots \times 1 \times$$

The above arrangement leaves $(n - k + 1)$ places marked with $\times$.
Now we can arrange k zeros in the $(n - k + 1)$ places in ${}^{n-k+1}C_k$ ways.

45. Part A can be attempted in 3 ways
(i) The student do not solve the question.
(ii) The student attempts the first part of the question.
(iii) The student attempts the second (*i.e.*, alternative) part of the question.

$\therefore$ Total number of choices in the first part = 3^5

Hence, the required number of ways in which a student must attempt at least one question = $3^5 - 1 = 242$
Similarly we can show that there are $2^4 - 1$
i.e., 15 choices for part B.
Hence the required number of ways = $242 \times 15 = 3630$.

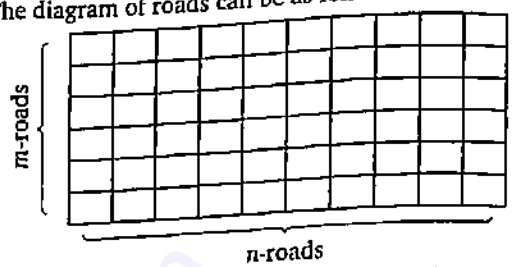
46. Required number of number plates of licences = $(26)^2 \times (999)$
= 675324

47. Required number of plates = ${}^{26}P_2 \times {}^{10}P_3 = 468000$

48. We can arrange r persons on m chairs on a particular side in ${}^mP_r$ ways and s persons on m chairs on the other side in ${}^mP_s$ ways. Now, we can arrange $(2m - r - s)$ persons on the remaining $(2m - r - s)$ chairs in ${}^{(2m-r-s)}P_{(2m-r-s)}$ ways.
Thus, the required number of ways of arranging the persons

$$= ({}^mP_r)({}^mP_s)({}^{2m-r-s}P_{2m-r-s})$$

49. The diagram of roads can be as follows :



Suppose the distance between two successive parallel roads be one unit then a person can travel $(m - 1)$ steps in north-south direction and he can travel $(n - 1)$ steps in east-west direction.
Thus he has to travel $\{(m - 1) + (n - 1)\} = (m + n - 2)$ steps to reach from one corner to diagonally opposite corner.
Thus he can arrange his steps in

$${}^{(m+n-2)}C_{(m-1)} \times {}^{(n-1)}C_{(n-1)} \text{ ways}$$

i.e., ${}^{m+n-2}C_{m-1}$ ways.

50. The required number of ways is equal to the number of ways in which they can select the same number of toys from the collection of their combined number of toys – the number of ways in which each can select her original toys.
Hence, required number of ways = $({}^{19}C_{11} \times {}^8C_8) - 1$
$= {}^{19}C_{11} - 1$

51. Go through options.

52. The number of ways of choosing 26 cards out of 52 cards is ${}^{52}C_{26}$. There are two ways in which each card can be dealt because a card can be either from a first pack or from the second pack.
Hence, the number of ways in which the card can be dealt is

$${}^{52}C_{26} \times 2^{26}$$

53. There are 4 suits and each suit contains 13 cards so we can select any one card from the first suit in 13 ways. Similarly we can select one card different from the first card out of second suit in 12 ways and so on.
Hence, the required number of ways = $13 \times 12 \times 11 \times 10$

$$= \frac{13!}{9!}$$

54. First digit (*i.e.*, ten thousands place) can be filled up in 2! ways by the digit 3, but since 3 is being repeated twice, hence the actual number of ways $= \frac{2!}{2!} = 1$
Remaining 4 places can be filled up in 4! ways but since zero is being repeated thrice hence the actual number of ways

$$= \frac{4!}{3!} = 4$$

$\therefore$ The required number of ways = $1 \times 4 = 4$.

55. Digit 5 can be placed in any one of the 4 places in 4 ways. Now the remaining 3 places can be filled up with remaining 8 digits in 8^3 ways.
Hence, the required number of ways = $4 \times 8^3 = 2048$.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

56. Required number of ways = $4 \times 8 \times 7 \times 6 = 1344$

57. Digit 2 can be arranged in two places out of 8 places in $\frac{^8P_2}{2!}$ ways.
Now, the remaining 6 places can be filled by the rest 3 digits in 3^6 ways.
Hence, the required number of ways $= \frac{^8P_2}{2!} \times 3^6$
$= {}^8C_2 \times 3^6 = 20412$

58. Required number of ways = $3^8 = 6561$.

59. Since chairs are numbered, hence these are distinguishable.
Therefore 3 boys can be arranged in 3! ways on alternate chairs and 3 girls can be arranged in remaining 3 chairs in 3! ways.
Also 3 girls can be arranged before the boys are arranged.
Hence the required number of ways = $2 \times (3!)^2 = 72$

60. First rook can be selected in 64 ways ($^{64}C_1$).
Second rook can not be selected from the same row and same column from which the first rook has been selected hence there are only 64 – (8 + 7) = 49 places for the second rook.
Hence, the second rook can be selected in $^{49}C_1$ ways.
∴ Required number of ways = $^{64}C_1 \times {}^{49}C_1 = 3136$.

61. Number of white shoes = 9 pairs
⇒ 9 left and 9 right shoes
Number of black shoes = 6 pairs
⇒ 6 left and 6 right
In such questions we always consider the worst situation.
Therefore we assume that initially we get 6 + 6 = 12 black shoes and 9 white shoes either left or right.
Thus we have taken out 12 + 9 = 21 shoes still we don't have required pair of white shoes.
Now we drawout just one white shoe which will fulfill our requirement of getting one pair of white shoes.
[One pair of white shoes means 1 left shoe and 1 right shoe.]
Thus we have to draw out 12 + 9 + 1 = 22 shoes to get correct pair of white shoes.

62. The worst possible situations are :

White	Black
9 left	6 left
9 left	6 right
9 right	6 left
9 right	6 right

Thus before getting the required correct pair of shoes we have to draw out 9 + 6 = 15 shoes.
Now when we will draw out one more shoe which will give the required combination of correct shoes either white or black.
Hence we have to draw out minimum 15 + 1 = 16 shoes.

63. We can not put only one incorrect ball since if we make a wrong choice it will simultaneously make another choice wrong. For example if we put red colour ball in either of green, blue or black box then we have to fill the red colour box with different colour ball, hence one colour combination can never be wrong.
Since atleast two combinations will be different.

64. Select any two balls out of 4 then interchange the combination of colours.
Hence the required number of ways = $^4C_2 = 6$.
Alternatively :

$\begin{Bmatrix} R & B & G & W \\ B & R & G & W' \end{Bmatrix}$ $\begin{Bmatrix} R & B & G & W \\ G & B & R & W \end{Bmatrix}$

$\begin{Bmatrix} R & B & G & W \\ W & B & G & R' \end{Bmatrix}$ $\begin{Bmatrix} R & B & G & W \\ R & G & B & W \end{Bmatrix}$

$\begin{Bmatrix} R & B & G & W \\ R & W & G & B' \end{Bmatrix}$ $\begin{Bmatrix} R & B & G & W \\ R & B & W & G \end{Bmatrix}$

65. For one digit number 1 to 9 → $1 \times 9 = 9$
For two digit numbers 10 to 99 → $2 \times 90 = 180$
For threee digit numbers 100 to 300 → $3 \times 201 = 603$
Hence, total number of required digits = 9 + 180 + 603 = 792

66. Four odd digits (1, 1, 3, 3) can be arranged in 4 odd places in $\frac{4!}{2!2!} = 6$ ways.
Now the remaining 3 even digits (2, 2, 4) can be arranged in 3 even place in $\frac{3!}{2!} = 3$ ways.
Hence the required number of ways = $6 \times 3 = 18$.

67. Each of the 7 balls can be distributed into 5 ways.
Hence the required number of ways = 5^7.

68. There can be two cases
(1) 1, 1, 1, 1, 3
(2) 1, 1, 1, 2, 2
Case 1. A box (in which 3 balls can be put) is selected in $^5C_1$ ways.
Now, 3 balls can be selected in $^7C_3$ ways and remaining 4 balls can be arranged in 4! ways.
Hence, total number of ways = $^5C_1 \times {}^7C_3 \times 4!$
Case 2. 2 boxes (in which 2–2 balls can be put) are selected in $^5C_2$ ways.
Now, any two ball can be selected in $^7C_2$ ways and for the other selected box, 2 balls can be selected from the remaining 5 balls in $^5C_2$ ways and the remaining balls can be distributed in 3! ways.
Hence, the total number of ways $= {}^5C_2 \times ({}^7C_2 \times {}^5C_2) \times 3!$
∴ The required number of ways
$= ({}^5C_1 \times {}^7C_3 \times 4!) + ({}^5C_2 + {}^7C_2 \times {}^5C_2 \times 3!) = 16800$

69. Required number of ways = $^{7+5-1}C_{5-1}$ (using $^{n+r-1}C_{r-1}$)
$= {}^{11}C_4 = 330$

70. Possible ways of distribution are as follows :
(i) (7) (ii) (6 + 1)
(iii) (5 + 2) (iv) (4 + 3)
(v) (5 + 1 + 1) (vi) (4 + 2 + 1)
(vii) (3 + 2 + 2) (viii) (3 + 3 + 1)
(ix) (4 + 1 + 1 + 1) (x) (3 + 2 + 1 + 1)
(xi) (2 + 2 + 2 + 1) (xii) (3 + 1 + 1 + 1 + 1)
(xiii) (2 + 2 + 1 + 1 + 1)
Hence total number of ways of distribution = 13

71. There is only one way, which is as follows :
1, 2, 1, 2, 1.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

72. One ball can be selected for the box 2 in ${}^7C_1$ ways and two balls can be selected out of remaining 6 balls for the box 4 in ${}^6C_2$ ways and the remaining 4 balls can be distributed in 3 boxes in 3^4 ways.
Hence, the required number of ways = ${}^7C_1 \times {}^6C_2 \times 3^4 = 8505$

73. Ball 2 can be distributed between 2 boxes (*viz.*, box 2 and box 4) in 2 ways.
Now, the remaining 6 balls can be distributed in any of the 5 boxes in 5^6 ways.
Hence, the required number of ways = 2×5^6 ways
= 31250 ways

74. Required number of distribution
= Total number of distribution
– Number of distribution in which ball 2 and ball 4 are together.
Here total number of distribution (as calculated in question number 68) = 16800.
Now, consider ball 2 and ball 4 are stuck together and this arrangement is assumed to be a single ball.
Thus we have 6 balls to be distributed into 5 boxes. Which can be done as 1, 1, 1, 1, 2 in ${}^5C_1 \times {}^6C_2 \times 4!$ ways = 1800.
${}^5C_1 \rightarrow$ Number of ways of selecting one box in which two balls are kept together.
${}^6C_2 \rightarrow$ Number of ways of selecting 2 balls out of 6 balls.
$4! \rightarrow$ Number of ways of distribution of remaining 4 balls in remaining 4 boxes.
Hence, the required number of ways = 16800 – 1800 = 15000

75. Required number of ways = 3^4.

76. Following 4 distributions are possible :
(4, 0, 0), (3, 1, 0), (2, 2, 0), (2, 1, 1)

77. Required number of ways = ${}^{4+3-1}C_{3-1}$ (using ${}^{n+r-1}C_{r-1}$)
= ${}^6C_2 = 15$

78. Since pots are identical then there will be 4 cases (4, 0, 0), (3, 1, 0), (2, 2, 0) and (2, 1, 1) but since all coins are different hence selection of coins matters.
Therefore for the first case number of selections = ${}^4C_4 = 1$
For the second case number of selections = ${}^4C_3 \times {}^1C_1 = 4$
For the third case number of selections = $\frac{{}^4C_2 \times {}^2C_2}{2!} = 3$
For the fourth case number of selections
$$= \frac{{}^4C_2 \times {}^2C_1 \times {}^1C_1}{2!} = 6$$
Hence the total number of distributions = 1 + 4 + 3 + 6 = 14.

79. Since no box is empty and all pots are identical so the possible case is (1, 1, 2).
But since all the coins are different, then the 2 balls can be selected in ${}^4C_2$ ways and rest can be put in $\frac{{}^2C_1 \times {}^1C_1}{2!}$ ways.
Hence, the required number of distribution
$$= \frac{{}^4C_2 \times {}^2C_1 \times {}^1C_1}{2!} = 6$$

80. Since no pot is empty and all coins are identical, the possible case is (1, 1, 2).
But since all three pots are different hence a pot (which contains 2 coins together) can be selected in ${}^3C_1$ ways.
Hence the required number of distribution = ${}^3C_1 \times 1 = 3$

81. The possible arrangements are as follows :
(4, 0, 0) → can be done in 2 ways
i.e., 4 balls can be put either one of the two identical pots or can be put in different pot.
(3, 1, 0) → can be done in 3 ways
either distinct pot can be filled up with 3 balls, or 1 ball or remained empty.
(2, 2, 0) → can be done in 2 ways
either distinct pot can be filled by 2 balls or remained empty.
(2, 1, 1) → can be done in 2 ways
either distinct pot can be filled up with 1 ball or 2 ball.
Hence, the required number of ways = 2 + 3 + 2 + 2 = 9.

82. The possible arrangements are as follows :
(4, 0, 0) → can be done in 3 ways
let the 4 coins be *A*, *A*, *B* and *C* then all these 4 coins as a single packet of coins can be arranged 3 different pots in 3 ways.
(3, 1, 0) → can be done in 18 ways
out of *A*, *A*, *B* and *C* we can select one coin in 3 ways *i.e.*, either *A* or *B* or *C*.
Now we can arrange this selected coin in any 3 pots in 3 ways and the remaining 3 coins as a single packet of coins can be arranged in remaining 2 pots in 2 ways.
Hence, the required number of ways = 3 × 3 × 2 = 18.
(2, 2, 0) → can be done in 12 ways
There are two possible cases :
(i) (*A*, *A*), (*B*, *C*)
(ii) (*A*, *B*), (*A*, *C*)
In each of two cases we assume that there are two packets of coins which can be arranged in 3 × 2 = 6 ways.
Since there are two cases, hence the total number of required ways = 6 × 2 = 12
(2, 1, 1) → can be done in 21 ways.
There are 4 possible cases :
(i) (*A*, *A*), (*B*), (*C*) (ii) (*A*, *B*), (*A*), (*C*)
(iii) (*A*, *C*), (*A*), (*B*) (iv) (*B*, *C*), (*A*), (*A*)
For the first 3 cases in each cases all the 3 packets of coins can be arranged in 3! ways.
Hence, the number of arrangements = 3 × 3! = 18.
Now in the fourth (iv) case two coins are identical so the third packet of coins can be arranged in 3 pots in 3 ways.
Hence, the total number of arrangements = 18 + 3 = 21.
Thus the required number of ways = 3 + 18 + 12 + 21 = 54 ways.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Final Round

1. ${}^6P_2 = 30$.
2. ${}^6C_2 = 15$.
3. 3 lines intersect each other in ${}^3C_2 = 3$ points.
   3 circles intersect each other in ${}^3P_2 = 6$ points.
   Every line cuts 3 circles into 6 points. Therefore 3 lines cuts 3 circles into 18 points.
   Therefore, the maximum number of points = 3 + 6 + 18 = 27
4. $\frac{8!}{4!4!} = 70$
5. Required number of triangles $= {}^{m+n}C_3 - {}^mC_3 - {}^nC_3$.
6. Number of even places = 4
   Number of evendigits = 5 (2, 2, 8, 8, 8)
   Number of odd places = 5
   Number of odd digits = 4 (3, 3, 5, 5)
   Odd digits can be arranged in $\frac{4!}{2!\times 2!}$ ways = 6 ways.
   Even digits can be arranged in $\frac{5!}{2!\times 3!} = 10$ ways.
   Hence the required number of ways = $6 \times 10 = 60$ ways.
7. Required number of triangles $= {}^{m+n+k}C_3 - {}^mC_3 - {}^nC_3 - {}^kC_3$
8. Let $n = 2m + 1$, for the three numbers are in AP we have the following patterns :

Common differences	Numbers	Number of ways
1	(1, 2, 3)(2, 3, 4)... $(n-2, n-1, n)$	$(n-2)$
2	(1, 3, 5), (2, 4, 6)... $(n-4, n-2, n)$	$(n-4)$
3	(1, 4, 7), (2, 5, 8)... $(n-6, n-3, n)$	$(n-6)$
—	... ...	—
—	... ...	—
m	$(1, m+1, 2m+1)$	1

   ∴ Favourable number of ways
   $= (n-2) + (n-4) + (n-6) + \ldots + 3 + 1$ (total m terms)
   $= \frac{m}{2}(n - 2 + 1)$
   $= \frac{(n-1)}{2} \cdot \frac{(n-1)}{2} = \frac{(n-1)^2}{4}$
   Alternatively : Consider some proper value of n and verify the result.
9. Required number of parts $= 1 + \sum_{r=1}^{8} r$
   $= 1 + \frac{8\times 9}{2} = 37$
10. $\frac{240}{4n+2} = k \in I$
    ∴ $k = \frac{120}{2n+1} = \frac{2^3 \times 3 \times 5}{2n+1}$
    Since probable divisors are 1, 3, 5, 7, 9, 11, 13, ..., $(2n + 1)$ but we have only 4 possible divisors 1, 3, 5, 15.
11. Total number of rectangles
    $= (1 + 2 + 3 + \ldots + 12) \times (1 + 2 + \ldots + 8)$
    $= \frac{12\times 13}{2} \times \frac{8\times 9}{2} = 2808$
    Total number of squares
    $= (12\times 8) + (11\times 7) + (10\times 6) + \ldots + (5\times 1) = 348$
    ∴ Required number of rectangles = 2808 − 348 = 2460

HINT Number of squares $= \sum_{r=1}^{8} (13 - r)(9 - r)$

12. Required number of triangles $= ({}^{2n}C_3 - {}^nC_3 - {}^nC_3) + (n \times n)$
    $= n^2 (n-1) + n^2 = n^3$
    Alternatively : Required number of triangles
    $= ({}^nC_2 \times {}^nC_1) + ({}^nC_1 \times {}^nC_2) + (n \times n) = n^3$
13. There are 12 ways as follows :
    (9, 0, 0), (8, 1, 0), (7, 2, 0), (6, 3, 0), (5, 4, 0), (7, 1, 1), (6, 2, 1), (5, 3, 1), (5, 2, 2), (4, 4, 1) (4, 3, 2), (3, 3, 3).
14. First of all we give 4 coins to each of the six persons so that each one must have at least 4 coins.
    Now we are left with = $30 - (6 \times 4) = 6$ coins.
    These remaining 6 coins can be distributed among 6 persons in such a way that any one can receive any number of coins which can be done in ${}^{6+6-1}C_{6-1} = {}^{11}C_5$ ways
    = 462 ways
15. Required number of circles $= {}^{10}C_3 - {}^7C_3 = 85$.
16. 0 can not be placed as the left most digit so we have only 9 digits to be placed.
    Also we will not consider 1 digit numbers.
    Now we can form a two digit number in ${}^9C_2$ ways.
    Now we can form a three digit number in ${}^9C_3$ ways.
    Now we can form a four digit number in ${}^9C_4$ ways.
    ... ...
    ... ...
    We can form 9 digit number in ${}^9C_9$ way.
    Hence the required numbers $= {}^9C_2 + {}^9C_3 + {}^9C_4 + \ldots + {}^9C_9$
    = 502
17. The possible ways are
    (25×4)
    $(22\times 4 + 2\times 6)$
    $(19\times 4 + 4\times 6)$
    $(16\times 4 + 6\times 6)$
    $(13\times 4 + 8\times 6)$
    $(10\times 4 + 10\times 6)$
    $(7\times 4 + 12\times 6)$
    $(4\times 4 + 14\times 6)$
    $(1\times 4 + 16\times 6)$
    Hence there are total 9 ways.
18. Total number of permutations = 8! = 40320
    Number of permutations when word LURY occurs
    = (8 − 4 + 1)! = 5! = 120

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Number of permutations when word MINA occurs = 5! = 120
Number of permutation when words LURY and MINA both occurs = 3! = 6
Required number of permutations
$= 40320 - (120 + 120) + 6 = 40086$

19. 5 students can be selected out of 10 students in ${}^{10}C_5$ ways remaining 5 students can be selected in ${}^{5}C_5$ ways. These students (in each row) can be arranged mutually in 5! × 5! ways.
Besides it the two set of papers can be arranged in 2! ways between two rows.
Hence, the required number of arrangements
$= {}^{10}C_5 \times (5!)^2 \times 2 = 7257600$

20. ${}^{n}C_2 + 2n = 65$
Now, for convenience go through options.

21. First of all deduce 3 × 10 = 30 marks to assign atleast 3 marks to each of the 10 students.
Now remaining 20 marks can be assigned to 10 students in ${}^{20+10-1}C_{10-1}$ ways $= {}^{29}C_9$ ways

22. $\left[\frac{1}{3}\right] = \left[\frac{1}{3} + \frac{1}{100}\right] = \left[\frac{1}{3} + \frac{2}{100}\right]$
$= \ldots \left[\frac{1}{3} + \frac{65}{100}\right] = \left[\frac{1}{3} + \frac{66}{100}\right] = 0$
and $\left[\frac{1}{3} + \frac{67}{100}\right] = \left[\frac{1}{3} + \frac{68}{100}\right] = \left[\frac{1}{3} + \frac{69}{100}\right]$
$= \ldots \left[\frac{1}{3} + \frac{98}{100}\right] = \left[\frac{1}{3} + \frac{99}{100}\right] = 1$
Hence, $E = \left[\frac{1}{3}\right] + \left[\frac{1}{3} + \frac{1}{100}\right] + \ldots + \left[\frac{1}{3} + \frac{99}{100}\right]$
$= 33 \times 1 = 33$

23. Total numbers $= 10^6$ $(1, 2, 3, 4, \ldots, 10^6)$
$n^2 \Rightarrow \{1, 4, 9, 16, 25, \ldots 10^6\} \rightarrow 10^3 = 1000$
$\{1, 2, 3, 4, 5, \ldots (10^3)\}$
$n^3 \Rightarrow \{1, 8, 27, 64, 125, \ldots, 10^6\} \rightarrow 10^2 = 100$
$\{1, 2, 3, 4, 5, \ldots, (10^2)\}$
$n^4 \Rightarrow \{1, 16, 81, 256, \ldots, 923521\} \rightarrow 31$
$\{1, 2, 3, 4, \ldots, 31\}$

HINT $\sqrt[4]{10^6} = 10^{6/4} = 10^{3/2} = \sqrt{1000} \Rightarrow 31$
$n^2 \cap n^3 \Rightarrow n^6 \{1, 64, 729, 4096, \ldots, 10^6\} \rightarrow 10$
$\{1, 2, 3, 4, \ldots, 10\}$
$n^3 \cap n^4 \Rightarrow n^{12} \{1^{12}, 2^{12}, 3^{12}\} \rightarrow 3$
$\{1, 2, 3\}$

HINT $\sqrt[12]{10^6} = 10^{6/12} = 10^{1/2} = \sqrt{10} \Rightarrow 3$
$n^2 \cap n^4 \Rightarrow n^4 \{1, 16, 81, 256, \ldots, (31)^4\} \rightarrow 31$
$\{1, 2, 3, 4, \ldots, 31\}$
$n^2 \cap n^3 \cap n^4 \Rightarrow n^{12} \{1^{12}, 2^{12}, 3^{12}\} \rightarrow 3$
$\{1, 2, 3\}$

Hence the number of numbers which are either perfect square or perfect cube or perfect fourth powers or all of these
$= n^2 + n^3 + n^4 - (n^2 \cap n^3 + n^3 \cap n^4 + n^2 \cap n^4)$
$= 1131 - 44 + 3 = 1090$
Hence, the required number of ways
= Total numbers − Numbers which are perfect squares or perfect cubes or perfect fourth powers
$= 10^6 - 1090 = 998910$

24. Total number of required seats = 1 + m + 2n
The Grandchildren can occupy the n seats on either side of the table in $({}^{2n}P_{2n})$ ways. Remaining seats are (1 + m).
Since grandfather can not occupy adjacent seats of the grandchildren hence the grandfather can access only m + 1 − 2 = m − 1 seats. Hence he can occupy a seat in ${}^{(m-1)}P_1$ ways.
Now the remaining seats can be occupied in ${}^{m}P_m$ ways by the 'm' sons and daughters.
Hence the required number of ways $= {}^{2n}P_{2n} \times {}^{m}P_m \times {}^{m-1}P_1$
$= (2n!)\,(m!)\,(m-1)$

25. The 4 possible cases are as follows :

C_1	C_2	C_3
2	3	1
2	2	2
1	4	1
1	3	2

$C_1 \rightarrow$ First column
$C_2 \rightarrow$ Second column
$C_3 \rightarrow$ Third column

Hence, the required number of ways
$= {}^{2}C_2 \times {}^{4}C_3 \times {}^{2}C_1 + {}^{2}C_2 \times {}^{4}C_2 \times {}^{2}C_2$
$+ {}^{2}C_1 \times {}^{4}C_4 \times {}^{2}C_1 + {}^{2}C_1 + {}^{2}C_1 \times {}^{4}C_3 \times {}^{2}C_2$
$= 1 \times 4 \times 2 + 1 \times 6 \times 1 + 2 \times 1 \times 2 + 2 \times 4 \times 1$
$= 8 + 6 + 4 + 8 = 26$

26. The total number of balls in the box = 2 + 3 + 4 = 9.
Total number of selections of 3 balls out of 9 balls $= {}^{9}C_3$
Number of selections in which no any green ball is selected $= {}^{6}C_3$
Hence the required number of selections $= {}^{9}C_3 - {}^{6}C_3 = 64$.

27. There are 4 possible cases :

$H \rightarrow$ Husband's
Relatives
$W \rightarrow$ Wife's relatives

	H	W
M	0	3
F	3	0

H	W
1	2
2	1

H	W
2	1
1	2

H	W
3	0
0	3

$M \rightarrow$ Male
$F \rightarrow$ Female

Hence, the required number of ways
$= ({}^{4}C_3 \times {}^{4}C_3) + ({}^{3}C_1 \times {}^{4}C_2)({}^{4}C_2 \times {}^{3}C_1)$
$+ ({}^{3}C_2 \times {}^{4}C_1)({}^{4}C_1 \times {}^{3}C_2) + ({}^{3}C_3 \times {}^{3}C_3)$
$= (4 \times 4) + (3 \times 6 \times 6 \times 3) + (3 \times 4 \times 4 \times 3) + (1 \times 1)$
$= 485$

28. Let the number of men participating in the tournament be n.
Since every participant played two games with every other participant.
Therefore the total number of games played among men is $2 \times {}^{n}C_2 = n\,(n-1)$.
And the number of games played with each woman = 2n.
but since there are two women, hence the total number of games men played with 2 women $= 2 \times 2n = 4n$
$\therefore \quad \{n\,(n-1)\} - 4n = 66$
$\Rightarrow \quad n^2 - 5n - 66 = 0$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

$n = 11$ ($\because n < 0$, is not possible)

$\Rightarrow$ $\therefore$ Number of participants = 11 men + 2 women = 13.

29. Number of games played by them is $2(^{13}C_2) = 156$.

30. There are two possible cases in which 12 sweets can be distributed among 10 girls.

(i) Any 9 girls get one sweets each and the remaining 1 girl gets 3 sweets.

(ii) Any 8 girls get one sweets each and the remaining 2 girls get 2 piece of sweets each.

Case 1. 3 pieces of sweets can be given to a girl in the following four way :

Burfi	3	2	1	0
Rasgulla	0	1	2	3

After giving 3 pieces of sweets to a single girl. We can distribute the remaining. 9 sweets to the 9 girls in the following ways :

$$^9C_3 \times {}^6C_6 + {}^9C_4 \times {}^5C_5 + {}^9C_5 + {}^4C_4 + {}^9C_6 \times {}^3C_3$$
$$= 2(^9C_3 + {}^9C_4)$$

One particular girl can be chosen in $^{10}C_1$ ways.

Therefore 3 sweets can be given to a single girl in

$$^{10}C_1 \times 2 \times (^9C_3 + {}^9C_4) = 4200 \text{ ways.}$$

Case II. We can give two sweets to two girls (say A and B) in the following ways :

A	Burfi	2	1	0	2	1	0	2	1	0
	Rasgulla	0	1	2	0	1	2	0	1	2
B	Burfi	2	2	2	1	1	1	0	0	0
	Rasgulla	0	0	0	1	1	1	2	2	2

Now, remaining 8 pieces of sweets can be distributed among eight girls in the following ways

$$(^8C_2 \times {}^6C_6) + (^8C_3 \times {}^5C_5) + (^8C_4 \times {}^4C_4) + (^8C_3 \times {}^5C_5)$$
$$+ (^8C_4 \times {}^4C_4) + (^8C_5 \times {}^3C_3)$$
$$+ (^8C_4 \times {}^4C_4) + (^8C_5 \times {}^5C_5) + (^8C_6 \times {}^2C_2)$$
$$= 2(^8C_2) + 4(^8C_3) + 3(^8C_4)$$

Further, two girls can be selected in $^{10}C_2$ ways.

Therefore two girls can get two sweets each in

$$(^{10}C_2)[2(^8C_2) + 4(^8C_3) + 3(^8C_4)] = 22050 \text{ way}$$

Hence, the required number of ways = 4200 + 22050

$$= 26250$$

31. Number of common children of Mr. John and Ms. Bashu

$$= 10 - (x + x + 1) = 9 - 2x$$

Let N = The number of fights between children of different parents

= (Total number of fights that can take place among all the children)

− (The number of fights among the children of the same parents)

$$= {}^{10}C_2 - [{}^xC_2 + {}^{x+1}C_2 + {}^{9-2x}C_2] \quad \ldots(i)$$

$$= 45 - \left[\frac{x(x-1)}{2} + \frac{(x+1)(x)}{2} + \frac{(9-2x)(8-2x)}{2}\right]$$

$$= 45 - \frac{1}{2}[x^2 - x + x^2 + x + 72 - 34x + 4x^2]$$

$$= \frac{397}{12} - 3\left(x - \frac{17}{6}\right)^2$$

For N to be maximum, x must be $\frac{17}{6}$. As x can not be in fraction, we take $x = 3\left(\approx \frac{17}{6}\right)$.

Thus, maximum value of $N = 33$ which is attained at $x = 3$.

Alternatively : After making the equation (i) go through options.

32. Let the form of the required numbers be $a_1, a_2 \ldots a_9$, where $0 \le a_1 \le 1$ and $0 \le a_i \le 2$ for $i = 2, 3, \ldots, 9$ and where all $a_1, a_2, \ldots, a_9$ can not be equal to zero.

Now, we can choose a_1 in two ways (0 or 1) and a_i for $i = 2, 3, \ldots 8$ in 3 ways (0, 1, 2).

After selecting $a_1, a_2, a_3 \ldots, a_8$ we find the sum $s = a_1 + a_2 + \ldots a_8$ which is of the form $3m - 2$, $3m - 1$ or $3m$.

Now we can select a_9 in just one way.

Actually a_9 can be selected out of 2, 1 or 0 depending on whether $s = 3m - 2$, $3m - 1$ or $3m$. Therefore, we can choose the numbers in $2 \times 3^7 \times 1 = 4374$ ways.

But this includes the case in which each of $a_i = 0$. Thus, the required number of numbers = 4374 − 1 = 4373

33. The digits which can be recognised as digits on the screen of a calculator when they are read inverted i.e., upside down are 0, 1, 2, 5, 6, 8 and 9.

Since a number can not begin with zero hence left most digit and right most digit can never be 0 as when an 'n' digit number read upside down it will become a number of less than n digit.

Hence,

Number of digits	Total number of numbers
1	7
2	$6 \times 6 = 6^2$
3	$6 \times 7 \times 6 = 6^2 \times 7$
4	$6 \times 7 \times 7 \times 6 = 6^2 \times 7^2$
5	$6 \times 7 \times 7 \times 7 \times 6 = 6^2 \times 7^3$
6	$6 \times 7 \times 7 \times 7 \times 7 \times 6 = 6^2 \times 7^4$

Thus, the number of required numbers

$$= 7 + 6^2 + 6^2 \cdot 7 + \ldots + 6^2 \cdot 7^4$$

$$= 7 + 6^2 \frac{(7^5 - 1)}{(7 - 1)} = 7 + 6(7^5 - 1)$$

$$= 6 \cdot 7^5 + 1 = 100843$$

34. Since rings are distinct, hence they can be named as R_1, R_2, R_3, R_4 and R_5.

The ring R_1 can be placed on any of the four fingers in 4 ways.

The ring R_2 can be placed on any of the four fingers in 5 ways since the finger in which R_1 is placed now has 2 choices, one above the R_1 and one below the ring R_1.

Similarly R_3, R_4 and R_5 can be arrange in 6, 7 and 8 ways respectively.

Hence, the required number of ways

$$= 4 \times 5 \times 6 \times 7 \times 8 = 6720$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

35. We can select first object out of n objects in ${}^nC_1$ ways.

Now, number of ways of choosing two objects such that they are always together $(n-4)$ ways.

Since we assume two objects as a single object. Further we can select three objects *viz.*, the one object which has been already selected and two objects of one either side of the first object.

Therefore the number of ways of choosing two objects such that they are not together

$$= {}^{(n-3)}C_2 - (n-4) = \frac{1}{2}(n-4)(n-5)$$

Since these two objects can be arranged in 2! ways, the number of ways of choosing three objects (in order of the first, second and third) is

$$n \times \frac{1}{2}(n-4)(n-5) \times 2 = n(n-4)(n-5)$$

But, since the order in which the objects are taken is immaterial, the number of ways of choosing the objects is $\frac{1}{6}n(n-4)(n-5)$

36.

Number of similar letters	Number of different letters	Number of selections
5	0	${}^1C_1 = 1$
4	1	${}^4C_1 \times {}^2C_1 = 8$
3	2	${}^3C_1 \times {}^4C_2 = 18$
3 of one type and 2 of another type	0	${}^3C_1 \times {}^3C_1 = 9$
2 of one type and 2 of another type	1	${}^4C_2 \times {}^3C_1 = 18$
2	3	${}^4C_1 \times {}^4C_3 = 16$
0	5	${}^5C_5 = 1$

Hence, the total number of selections

$$= 1 + 8 + 18 + 9 + 18 + 16 + 1 = 71$$

37.

	Ist paper	2nd paper	3rd paper	4th paper
Max. marks	n	n	n	$2n$

Let us consider $n = 1$, then a candidate required 3 marks out of 5 marks, which can be done in the following ways :

Ist paper	2nd paper	3rd paper	4th paper	Total
0	0	1	2	3
0	1	0	2	3
1	0	0	2	3
1	1	1	0	3
1	1	0	1	3
1	0	1	1	3
0	1	1	1	3

Hence, there are total 7 ways.

Now, go through options.

Let us consider option (b).

Putting $n = 1$, we get

$$\frac{1}{6}(1+1)(5 \times 1^2 + 10 \times 1 + 6) = 7$$

Hence the choice (b) is correct.

38. Do this problem similarly as discussed in the previous problem.

NOTE The general solution of this type of problems involves higher mathematics. But we have given in question number 37 a very simple, lucid and a novel solution for the lay students.

www.LearnEngineering.in

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

20 PROBABILITY

Generally this chapter is considered as an extension of permutation and combination, since it is as logical in nature as permutation and combination. I hope that you must have been well-versed with the permutation and combination (previous chapter) which will enable you to solve the problems of this chapter quickly and easily. This chapter is important for XAT, SNAP and other MBA entrance tests. Over the past years CAT has shown no great interest in this chapter, still you should learn it heartly due to uncertain syllabus and irregular pattern of CAT. Remember that so far CAT has asked very simple and logical problems from this chapter which could be solved by applying just commonsense.

The literal meaning of probability is the chance of occurrence of an event. For example, if a person is standing at the crossing of two roads which direct towards North, South, East and West. Thus he has total four alternatives (*i.e.* four different directions) to proceed. Now if he wish to go towards a particular direction then the probability of completing his wish is $\frac{1}{4}$ since he can choose only one direction at a time out of four directions.

Consider another example, A person has two different cars *viz.*, Scorpio and Safari, which he uses randomly,then it can be said that the probability of using Scorpio is $\frac{1}{2}$ since out of two cars he can use any one car at a time. Similarly the probability of using Safari is also $\frac{1}{2}$. Thus we can say that the probability of using any one car at a time is $\frac{1}{2}$ *i.e.*, 50%. So there are chances that in 50% cases he can use Scorpio and in other 50% cases he can use Safari.

Hence from the above illustrations we can conclude that the probability of an event

$$= \frac{\text{(No. of ways in which favourable (or desired event occurs.)}}{\text{(Total number of possible outcomes)}}$$

IMPORTANT ARTICLES/DEFINITIONS

1. Experiment
2. Random Experiment
3. Sample Space
4. Events
5. Elementary Events
6. Compound Events
7. Occurrence of Events
8. Impossible Events
9. Sure Events
10. Equally Likely Events
11. Favourable Events
12. Complementary Events
13. Algebra of Events
14. Mutually Exclusive Events
15. Mutually Exclusive and Exhaustive Events
16. Addition Theorems on Probability
17. Conditional Probability
18. Multiplication Theorems on Probability
19. Independent Events
20. The Law of Total Probability
21. Baye's Rule

Experiment : An operation which can produce some well-defined outcomes, is known as an experiment.

The various experiments, when repeated under identical conditions, results. (*i.e.*, outcomes) in each case are same *e.g.*, standard scientific experiments.

But there are some other experiment, when repeated under identical conditions, results in each case are different *e.g.*, rolling of a fair die, tossing of a coin etc.

Random Experiment : If in each trial of an experiment conducted under identical conditions, the outcome is not unique, but may be any of the possible outcomes then such an experiment is known as a random experiment. *e.g.*, rolling of an unbiased die, tossing of a fair coin, drawing of a card from a well shuffled pack of cards.

Sample Space : The set of all possible outcomes in a random experiment is called a sample space and it is generally denoted by S.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

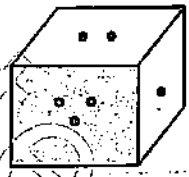
If $E_1, E_2, E_3 \ldots E_n$ are the possible outcomes of a random experiment, then $S = \{E_1, E_2, \ldots E_n\}$. Also, each of the element of sample space 'S' is called a **sample point.**

EXAMPLE 1 In tossing of a fair coin, there are two possible outcomes, *viz.*, head (H) and tail (T). So, the sample space in this random experiment is $S = \{H, T\}$

EXAMPLE 2 When two fair coins are tossed together, the possible outcomes of the experiment are HH, HT, TH and TT. So the sample space is given by $S = \{HH, HT, TH, TT\}$

EXAMPLE 3 When unbiased die is thrown, it gives 6 possible outcomes *viz.*, 1, 2, 3, 4, 5 and 6. So, the sample space $S = \{1, 2, 3, 4, 5, 6\}$

NOTE A die is a cubical solid having 6 similar faces. In each of the six faces there are unique number of holes *viz.*, 1 hole, 2 holes, 3 holes, 4 holes, 5 holes and 6 holes.
Remember that the sum of number of holes in any two opposite faces is always 7 *viz.*, 1 + 6 = 2 + 5 = 3 + 4 = 7.
- Die means one (single) die
- Dice means more than one die.
- In throwing a die, the outcome is the number of holes on the uppermost face



EXAMPLE 4 When two unbiased dice are rolled (or tossed) simultaneously then there are total $6 \times 6 = 36$ possible outcomes.

So, the sample space

$$S = \begin{Bmatrix} (1,1), (1,2), (1,3), (1,4), (1,5), (1,6) \\ (2,1), (2,2), (2,3), (2,4), (2,5), (2,6) \\ (3,1) \ldots\ldots (3,6) \\ (4,1) \ldots\ldots (4,6) \\ (5,1) \ldots\ldots (5,6) \\ (6,1) \ldots\ldots (6,6) \end{Bmatrix}$$

EXAMPLE 5 When a die and a coin are tossed simultaneously, then there are total 12 possible outcomes. So the sample space

$$S = \begin{Bmatrix} (1,H), (2,H), (3,H), (4,H), (5,H), (6,H) \\ (1,T), (2,T), (3,T), (4,T), (5,T), (6,T) \end{Bmatrix}$$

EXAMPLE 6 A coin is tossed twice. If the second throw results in a tail, then a die is thrown. So in this random experiment sample space.

$$S = \begin{Bmatrix} HH, TH, HT1, HT2, HT3, HT4, HT5, HT6, \\ TT1., TT2., TT3, TT4, TT5, TT6, \end{Bmatrix}$$

EXAMPLE 7 From a bag containing 2 black and 3 white balls we draw two balls. Let B_1, B_2 be the black balls and W_1, W_2 and W_3 be the white balls then the sample space

$$S = \begin{Bmatrix} B_1W_1, & B_1W_2, & B_1W_3 \\ B_2W_1, & B_2W_2, & B_2W_3 \\ B_1B_2, & W_1W_2, & W_2W_3 \\ W_1W_3 & & \end{Bmatrix}$$

Event : Any subset of a sample space is called an event.

EXAMPLE 1 In a single throw of a die, the event of getting an even number is given by $E = \{2, 4, 6\}$

Clearly, here the sample space $S = \{1, 2, 3, 4, 5, 6\}$

Hence $E \subset S$ *i.e.*, E is the subset of S.

EXAMPLE 2 Consider a random experiment of tossing two dice at a time. The sample space $S = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2,1), (2,2), (2,3), (2,4), (2,5) \ldots (6,1), (6,2) \ldots (6,6)\}$

Some different events associated with the above sample space are given below.

$E_1 = \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)\}$

$E_2 = \{(1,8), (2,7), (3,6), (4,5), (5,4), (6,3), (7,2), (8,1)\}$

$E_3 = \{(1,2), (1,4), (1,6), (3,2), (3,4), (3,6), (5,2), (5,4), (5,6)\}$

$E_4 = \{(2,2), (2,3), (2,5), (3,2), (3,3), (3,5), (5,2), (5,3), (5,5)\}$ etc.

Clearly $E_1 \subset S, E_2 \subset S, E_3 \subset S, E_4 \subset S$

Where E_1 is the event of getting a doublet

and E_2 is the event of getting 9 as the sum.

E_3 is the event of getting odd number on the first die and an even number on the second die

E_4 is the event of getting only prime number on each of the two dice.

EXAMPLE 3 Consider a random experiment of tossing three coins at a time. The sample space $S = \{HHH, HHT, HTH, THH, TTH, HTT, THT, TTT\}$

Following are the various events associated with the above sample space.

$E_1 = \{HHH, HHT, HTH, THH\}$

$E_2 = \{HHT, HTH, THH, TTH, HTT, THT\}$

$E_3 = \{HHH, TTT\}$

$E_4 = \{THH, TTH, THT, TTT\}$

Where E_1 is the event of getting atleast two heads and

E_2 is the event of getting atleast one head and atmost two heads.

E_3 is the event of getting all the three coins identical and E_4 is the event of getting tail on the first coin.

Elementary Events : An event containing only a single sample point is called an elementary event, or simple event.

EXAMPLE 1 In a simultaneous toss of two coins the sample space $S = \{HH, HT, TH, TT\}$

then $E_1 = \{HH\}$

$E_2 = \{TT\}$ are the elementary events.

EXAMPLE 2 Consider a random experiment in which two dice rolled simultaneously, the sample space

$S = \{(1,1), (1,2) \ldots (6,5), (6,6)\}$

then $E_1 = \{(1,1)\}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$E_2 = \{(6, 6)\}$

$E_3 = \{(5, 5)\}$ are the elementary events.

Here E_1 is the event of getting the sum of two.

E_2 is the event of getting the sum of twelve

E_3 is the event of getting the product of 25.

Compound Events : Events which are not elementary are known as compound events or the events which contains more than one element are called compound or composite events.

Example 1 Consider a random experiment in which two dice are rolled simultaneously, the sample space

$S = \{(1, 1), (1, 2) \ldots\ldots (6, 5), (6, 6)\}$.

then $E_1 = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$

$E_2 = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$

$E_3 = \{(1, 2), (2, 4), (3, 6)\}$

are the compound events.

Here E_1 is the event of getting the sum of 6

E_2 is the event of getting identical results

E_3 is the event of getting the twice number by the second die than that by first die.

Occurrence of Events : In a random experiment, let S be the sample space and E be the event such that $E \subseteq S$, Let w be an outcome of a trial such that $w \in A$, then we say that the event E has occurred. If $w \notin E$, we say that the event E has not occurred.

EXAMPLE 1 Consider the random experiment of throwing an unbiased die. Let E be the event of getting an odd number, then $E = \{1, 3, 5\}$

Now, in a trial, let the outcome be 3, Since $3 \in E$, so in this trial, the event E has occurred.

In another trial, let the outcome be 4.

Since $4 \notin E$ so in this trial, the event E has not occurred.

EXAMPLE 2 Consider the random experiment of throwing a die. Let the outcome of the trial is 6. Then we can say that the following events have occurred.

(i) Getting a number greater than or equal to 3, represented by the set $\{3, 4, 5, 6\}$

(ii) Getting an even number represented by the set $\{2, 4, 6\}$

We can also say that the following events have not occurred :

(i) Getting an odd number represented by the set $\{1, 3, 5\}$.

(ii) Getting a prime number represented by the set $\{2, 3, 5\}$

Impossible Events : Let S be a sample space associated with a random experiment,

Now, since $\phi \subseteq S$, so ϕ is an event, called an impossible event.

Sure (or certain) Event : Let S be a sample space associated with a random experiment.

Now, since $S \subseteq S$, so S is an event, called a sure or certain event.

EXAMPLE Consider the random experiment of throwing an unbiased die. The sample space $S = \{1, 2, 3, 4, 5, 6\}$.

Let E_1 be the event of getting a number less than 1.

and E_2 be the event of getting a number greater than 6.

Here E_1 and E_2 both are impossible events.

Let E_3 be the event of getting a number less than or equal to 7.

and E_4 be the event of getting a number multiple of 2 but less than 7.

Here E_3 and E_4 are the certain events.

Equally Likely Events : Events are said to be equally likely, if none of them is expected to occur in preference to the other.

EXAMPLE 1 If an unbiased die is rolled, then each outcome is equally likely to happen *i.e.*, all elementary events are equally likely to happen. If however, the die is so formed that a particular face occurs most often, then the die is biased. So in this case, the outcomes are not equally likely to happen.

Favourable Events : Let S be the sample space associated with a random experiment and let $E \subset S$. Then the elementary events belonging to E are known as the favourable events to A.

EXAMPLE 1 Consider a random experiment of throwing a die. Let E be the event of getting an even number, then $E = \{2, 4, 6\}$.

So there are three favourable events of event E *viz.* $\{2\}$, $\{4\}$ and $\{6\}$

EXAMPLE 2 Consider a random experiment of throwing a pair of dice. Let E be the event of getting the sum as 8, then

$E = \{(1, 7), (2, 6), (3, 5), (4, 4), (5, 3), (6, 2), (7, 1)\}$

So, there are 7 favourable events to event E.

Complementary Events : In a random experiment, let S be the sample space and let E be an event. Then $E \subseteq S$. Clearly, $E^c \subseteq S$. So E^c is also an event, called the complementary of E.

Sometimes E^c is denoted by $\overline{E}$ or E'.

Where $\overline{E}$ is called as "not-E"

Thus it is clear that $\overline{E}$ occurs only when E does not occur. Also in a trial one and only of E and $\overline{E}$ must occur.

Algebra of Events : In a random experiment, let the sample space be S. Let $E \subseteq S$, $F \subseteq S$ and $G \subseteq S$, then we can define the following relations in the set theory mode.

(i) $\overline{E}$ is an event that occurs only when E does not occur.

(ii) $(E \cup F)$ is an event that occurs only when E occurs or F occurs or both occurs.

(iii) $(E \cap F)$ is an event that occurs only when each one of E and F occurs.

(iv) $(E \cap \overline{F})$ is an event that occurs only when E occurs but not F.

(v) $(\overline{E} \cap \overline{F}) = \overline{(E \cup F)}$ is an event that occurs when neither E nor F occurs.

(vi) $(E \cup F \cup G)$ is an event that occurs when at least one of E, F or G occurs.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

(vii) $(E \cap F \cap G)$ is an event that occurs when all three E, F and G occurs.

(viii) $(E \cap \overline{F}) \cup (\overline{E} \cap F)$ is an event that occurs when exactly one of E and F occurs.

(ix) $(E \cap F \cap \overline{G}) \cup (E \cap \overline{F} \cap G) \cup (\overline{E} \cap F \cap G)$ is an event that occurs when exactly two of E, F and G occurs.

Mutually Exclusive Events : Let S be the sample space associated with a random experiment and let E_1 and E_2 be the two events. Then E_1 and E_2 are mutually exclusive events if $E_1 \cap E_2 = \phi$

EXAMPLE 1 Consider a random experiment of throwing a die. Let E_1, E_2 and E_3 be three events such that

$E_1 = \{1, 3, 5\}$, the event of getting an odd number

$E_2 = \{2, 4, 6\}$, the event of getting an even number

$E_3 = \{2, 3, 5\}$, the event of getting a prime number.

Clearly $E_1 \cap E_2 = \phi$ and $E_1 \cap E_3 \neq \phi$, $E_2 \cap E_3 \neq \phi$

Hence E_1 and E_2 are mutually exclusive events but E_1 and E_3, E_2 and E_3 are not mutually exclusive.

MUTUALLY EXCLUSIVE & EXHAUSTIVE SYSTEM OF EVENTS

Let S be the sample space associated with a random experiment. Let $E_1, E_2, \ldots E_n$ be the subsets of S such that

(i) $E_i \cap E_j = \phi$ for $i \neq j$ and

(ii) $E_1 \cup E_2 \cup E_3 \cup \ldots \cup E_n = S$

then the set of events $E_1, E_2, E_3 \ldots E_n$ is said to form a mutually exclusive and exhaustive system of events.

EXAMPLE 1 In a sample space all the elementary events form a mutually exclusive and exhaustive system of events.

EXAMPLE 2 Let $S = \{1, 2, 3, 4, 5, 6\}$ be the sample space when an unbiased die is rolled.

Let $E_1 = \{1, 3, 5\}$ and $E_2 = \{2, 4, 6\}$, then E_1 and E_2 form the mutually exclusive and exhaustive system of events.

Also, let $E_3 = \{1, 2\}, E_4 = \{4, 5\}$, then E_3 and E_4 do not form the mutually exclusive and exhaustive system of events since E_3 and E_4 do not include all the elementary events of S.

Again if $E_5 = \{3, 6\}$ and $E_6 = \{2, 4, 6\}$, then E_5 and E_6 do not form mutually exclusive and exhaustive system of events since $E_5 \cap E_6 \neq \phi$ and E_5, E_6 do not include all the elementary events of the sample space S.

- A pack of cards consists of 52 cards. There are four suits each containing 13 cards called as spades, clubs, hearts and diamonds.

All the spades and clubs are black faced cards while hearts and diamonds are red faced cards. The aces, kings, queens and jacks are known as face cards.

In each suit there is one ace, one king, one queen and one jack and rest 9 cards are numbered cards.

Definition of Probability : In a random experiment, let S be the sample space and let $E \subseteq S$.

where E is an event.

The probability of occurrence of the event E is defined as

$$P(E) = \frac{\text{number of favourable outcomes}}{\text{number of possible outcomes}}$$

$$= \frac{\text{number of elements in } E}{\text{number of elements in } S} = \frac{n(E)}{n(S)}$$

$$= \frac{\text{number of elementary events in } E}{\text{number of elementary events in } S}$$

From the above definitions it is clear that

(i) $0 \leq P(E) \leq 1$

(ii) $P(\phi) = 0$

(iii) $P(S) = 1$

Also, $P(\overline{E}) = \frac{\text{number of elementary events in } \overline{E}}{\text{number of elementary events in } S}$

$$= \frac{n(S) - n(E)}{n(S)} = 1 - \frac{n(E)}{n(S)} = 1 - P(E)$$

$\Rightarrow \quad P(\overline{E}) = 1 - P(E)$

$\therefore \quad P(E) + P(\overline{E}) = 1$

Odds In Favour of An Event and Odds Against An Event

In m be the number of ways in which an event occurs and n be the number of ways in which it does not occur, then

(i) odds in favour of the event $= \frac{m}{n}$ (or $m : n$)

(ii) odds against the event $= \frac{n}{m}$ (or $n : m$)

INTRODUCTORY EXERCISE-20.1

1. Find the probability of getting a head in a throw of a coin.

(a) $\frac{1}{2}$ (b) 1

(c) 2 (d) none of these

Directions for question number 2-5 : Two fair coins are tossed simultaneously. Find the probability of

2. Getting only one head.

(a) 1/2 (b) 1/3

(c) 2/3 (d) 3/4

3. Getting atleast one head.

(a) $\frac{1}{4}$ (b) $\frac{3}{4}$

(c) $\frac{1}{2}$ (d) $\frac{3}{8}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

4. Getting two heads.
(a) $\frac{2}{7}$ (b) $\frac{1}{4}$
(c) $\frac{1}{2}$ (d) $\frac{4}{5}$

5. Getting atleast two heads :
(a) $\frac{3}{4}$ (b) $\frac{1}{2}$
(c) $\frac{1}{4}$ (d) 1

Directions for question number 6-12 : Three fair coins are tossed simultaneously. Find the probability of

6. Getting one head.
(a) 0 (b) 3/4
(c) $\frac{5}{8}$ (d) $\frac{3}{8}$

7. Getting one tail.
(a) 1 (b) $\frac{1}{4}$
(c) $\frac{5}{8}$ (d) $\frac{3}{8}$

8. Getting atlest one head.
(a) $\frac{7}{8}$ (b) $\frac{1}{8}$
(c) $\frac{3}{4}$ (d) $\frac{1}{4}$

9. Getting two heads.
(a) $\frac{3}{5}$ (b) $\frac{3}{8}$
(c) $\frac{5}{8}$ (d) $\frac{2}{5}$

10. Getting atleast two heads.
(a) $\frac{3}{8}$ (b) $\frac{7}{8}$
(c) $\frac{1}{2}$ (d) $\frac{1}{4}$

11. Getting atleast one head and one tail.
(a) $\frac{2}{8}$ (b) $\frac{1}{2}$
(c) $\frac{3}{10}$ (d) $\frac{3}{4}$

12. Getting more heads than the number of tails.
(a) 2 (b) 7/8
(c) $\frac{5}{8}$ (d) $\frac{1}{2}$

Directions for question number 13-16 : An unbiased die is rolled. Find the probability of

13. Getting a number less than 7 but greater than zero.
(a) 0 (b) 3/4
(c) 1 (d) $\frac{7}{8}$

14. Getting a multiple of 3.
(a) $\frac{1}{6}$ (b) $\frac{1}{3}$
(c) $\frac{5}{6}$ (d) none of these

15. Getting a prime number.
(a) $\frac{1}{2}$ (b) $\frac{3}{5}$
(c) $\frac{5}{7}$ (d) $\frac{5}{8}$

16. Getting an even number.
(a) $\frac{1}{2}$ (b) $\frac{4}{5}$
(c) $\frac{2}{8}$ (d) $\frac{3}{4}$

Directions for question number 17 and 18 : A coin is tossed successively three times. Find the probability of

17. Getting exactly one head or two heads.
(a) $\frac{1}{4}$ (b) $\frac{3}{4}$
(c) $\frac{1}{2}$ (d) $\frac{3}{8}$

18. Getting no heads.
(a) 0 (b) $\frac{1}{7}$
(c) $\frac{1}{8}$ (d) $\frac{7}{8}$

Directions for question number 19-27 : Two dice are rolled simultaneously. Find the probability of

19. Getting a total of 9.
(a) $\frac{1}{3}$ (b) $\frac{1}{9}$
(c) $\frac{8}{9}$ (d) $\frac{9}{10}$

20. Getting a sum greater than 9.
(a) $\frac{10}{11}$ (b) $\frac{5}{6}$
(c) $\frac{1}{6}$ (d) $\frac{8}{9}$

21. Getting a total of 9 or 11.
(a) $\frac{2}{99}$ (b) $\frac{20}{99}$
(c) $\frac{1}{6}$ (d) $\frac{1}{10}$

22. Getting a doublet.
(a) 1/12 (b) 0
(c) 5/8 (d) 1/6

23. Getting a doublet of even numbers.
(a) 5/8 (b) 1/12
(c) 3/4 (d) 1/4

24. Getting a multiple of 2 on one die and a multiple of 3 on the other.
(a) $\frac{15}{36}$ (b) $\frac{25}{36}$
(c) $\frac{11}{36}$ (d) $\frac{5}{6}$

25. Getting the sum of numbers on the two faces divisible by 3 or 4.
(a) $\frac{4}{9}$ (b) $\frac{1}{7}$
(c) $\frac{5}{9}$ (d) $\frac{7}{12}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

26. Getting the sum as a prime number.
(a) $\frac{3}{5}$ (b) $\frac{5}{12}$
(c) $\frac{1}{2}$ (d) $\frac{3}{4}$

27. Getting atleast one '5'.
(a) $\frac{3}{5}$ (b) $\frac{1}{5}$
(c) $\frac{5}{36}$ (d) $\frac{11}{36}$

Directions for question number 28-35 : One card is drawn from a pack of 52 cards, each of the 52 cards being equally likely to be drawn. Find the probability that

28. The card drawn is black.
(a) $\frac{1}{2}$ (b) $\frac{1}{4}$
(c) $\frac{8}{13}$ (d) can't be determine

29. The card drawn is a queen.
(a) $\frac{1}{12}$ (b) $\frac{1}{13}$
(c) $\frac{1}{4}$ (d) $\frac{3}{4}$

30. The card drawn is black and a queen.
(a) $\frac{1}{13}$ (b) $\frac{1}{52}$
(c) $\frac{1}{26}$ (d) $\frac{5}{6}$

31. The card drawn is either black or a queen.
(a) $\frac{15}{26}$ (b) $\frac{13}{17}$
(c) $\frac{7}{13}$ (d) $\frac{15}{26}$

32. The card drawn is either king or a queen.
(a) $\frac{5}{26}$ (b) $\frac{1}{13}$
(c) $\frac{2}{13}$ (d) $\frac{12}{13}$

33. The card drawn is either a heart, a queen or a king.
(a) $\frac{17}{52}$ (b) $\frac{21}{52}$
(c) $\frac{19}{52}$ (d) $\frac{9}{26}$

34. The card drawn is neither a spade nor a king.
(a) 0 (b) $\frac{9}{13}$
(c) $\frac{1}{2}$ (d) $\frac{4}{13}$

35. The card drawn is neither an ace nor a king.
(a) $\frac{11}{13}$ (b) $\frac{1}{2}$
(c) $\frac{2}{13}$ (d) $\frac{11}{26}$

36. From a well shuffled pack of 52 cards, three cards are drawn at random. Find the probability of drawing an ace, a king and a jack.
(a) $\frac{16}{5525}$ (b) $\frac{16}{625}$
(c) $\frac{16}{3125}$ (d) none of these

37. Four cards are drawn at random from a pack of 52 cards. Find the probability of getting all the four cards of same number.
(a) $\frac{17}{1665}$ (b) $\frac{1}{20825}$
(c) $\frac{7}{25850}$ (d) none of these

38. From a well shuffled pack of 52 playing cards, four cards are accidently dropped. Find the probability that one card is missing from each suit.
(a) $\frac{17}{20825}$ (b) $\frac{2197}{20825}$
(c) $\frac{197}{1665}$ (d) none of these

39. Four cards are drawn at random from a pack of 52 cards. Find the probability of getting all the four cards of different numbers.
(a) $\frac{141}{4165}$ (b) $\frac{117}{833}$
(c) $\frac{264}{4165}$ (d) none of these

Directions for question number 40-43 : Four dice are thrown simultaneously. Find the probability that :

40. All of them show the same face.
(a) $\frac{1}{216}$ (b) $\frac{15}{16}$
(c) $\frac{15}{36}$ (d) $\frac{1}{2}$

41. All of them show the different face.
(a) $\frac{3}{28}$ (b) $\frac{5}{18}$
(c) $\frac{15}{36}$ (d) $\frac{11}{36}$

42. Two of them show the same face and remaining two show the different faces.
(a) $\frac{4}{9}$ (b) $\frac{5}{9}$
(c) $\frac{11}{18}$ (d) $\frac{7}{9}$

43. Atleast two of them show the same face.
(a) $\frac{37}{72}$ (b) $\frac{11}{36}$
(c) $\frac{47}{72}$ (d) $\frac{25}{36}$

44. What is the probability that a number selected from the numbers 1, 2, 3, ..., 20, is a prime number when each of the given numbers is equally likely to be selected?
(a) 7/10 (b) 2/15
(c) 2/5 (d) 3/5

45. Tickets are numbered from 1 to 18 are mixed up together and then 9 ticket is drawn at random. Find the probability that the ticket has a number, which is a multiple of 2 or 3.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

(a) $\frac{1}{3}$ (b) $\frac{3}{5}$
(c) $\frac{2}{3}$ (d) $\frac{5}{6}$

46. In a lottery of 100 tickets numbered 1 to 100, two tickets are drawn simultaneously. Find the probability that both the tickets drawn have prime numbers.
(a) $\frac{2}{33}$ (b) $\frac{7}{50}$
(c) $\frac{7}{20}$ (d) $\frac{5}{66}$

47. In the previous question (number 46), find the probability that none of the tickets drawn has a prime number.
(a) $\frac{29}{66}$ (b) $\frac{17}{33}$
(c) $\frac{37}{66}$ (d) $\frac{17}{50}$

48. Find the probability that a leap year selected at random will contain 53 Sundays.
(a) $\frac{5}{7}$ (b) $\frac{3}{4}$
(c) $\frac{4}{7}$ (d) $\frac{2}{7}$

Directions for question number 49-53: A bag contains 8 red and 4 green balls. Find the probability that

49. The ball drawn is red when one ball is selected at random.
(a) $\frac{2}{3}$ (b) $\frac{1}{3}$
(c) $\frac{1}{6}$ (d) $\frac{5}{6}$

50. All the 4 balls drawn are red when four balls are drawn at random.
(a) $\frac{17}{32}$ (b) $\frac{14}{99}$
(c) $\frac{7}{12}$ (d) none of these

51. All the 4 balls drawn are green when four balls are drawn at random.
(a) $\frac{1}{495}$ (b) $\frac{7}{99}$
(c) $\frac{5}{12}$ (d) $\frac{2}{3}$

52. Two balls are red and one ball is green when three balls are drawn at random.
(a) $\frac{56}{99}$ (b) $\frac{112}{495}$
(c) $\frac{78}{495}$ (d) none of these

53. Three balls are drawn and none of them is red.
(a) $\frac{68}{99}$ (b) $\frac{7}{99}$
(c) $\frac{4}{495}$ (d) none of these

54. The odds in favour of an event are 2 : 7. Find the probability of occurrence of this event.
(a) $\frac{2}{9}$ (b) $\frac{5}{12}$
(c) $\frac{7}{12}$ (d) $\frac{2}{5}$

55. The odds against of an event are 5 : 7, find the probability of occurrence of this event.
(a) $\frac{3}{8}$ (b) $\frac{7}{12}$
(c) $\frac{2}{7}$ (d) $\frac{5}{12}$

56. If there are two children in a family, find the probability that there is atleast one girl in the family.
(a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) $\frac{3}{4}$ (d) none of these

57. From a group of 3 men and 2 women, two persons are selected at random. Find the probability that atleast one woman is selected.
(a) $\frac{1}{5}$ (b) $\frac{7}{10}$
(c) $\frac{2}{5}$ (d) $\frac{5}{6}$

58. A box contains 5 defective and 15 non-defective bulbs. Two bulbs are chosen at random. Find the probability that both the bulbs are non-defective.
(a) $\frac{5}{19}$ (b) $\frac{3}{20}$
(c) $\frac{21}{38}$ (d) none of these

59. In the previous question (number 58), find the probability that atleast 3 bulbs are defective when 4 bulbs are selected at random.
(a) $\frac{31}{969}$ (b) $\frac{7}{20}$
(c) $\frac{1}{20}$ (d) none of these

IMPORTANT RESULTS

1. $P(E) \geq 0$
2. $P(\phi) = 0$
3. $P(S) = 1$

Addition Theorem 1 : If A and B are two events associated with a random experiment.

Then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\Rightarrow$ $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

Corollary: If A and B are mutually exclusive events.

then $P(A \cap B) = 0$, therefore

$P(A \cup B) = P(A) + P(B)$

Addition Theorem 2 : If A, B and C three events associated with a random experiment,

then $P(A \cup B \cup C) = P(A) + P(B) + P(C)$
$- P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Corollary : If A, B and C are the three mutually exclusive events, then

$$P(A \cap B) = P(B \cap C) = P(A \cap C) = P(A \cap B \cap C) = 0$$

$$\therefore \quad P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

Theorem 3. If A and B be two events such that $A \subseteq B$, then $P(A) \le P(B)$

Theorem 4. If E is an event associated with a random experiment, then

$$0 \le P(E) \le 1$$

Theorem 5. For any two events A and B

$$P(A - B) = P(A) - P(A \cap B)$$
$$P(B - A) = P(B) - P(A \cap B)$$
$$P(\bar{A} \cap B) = P(B) - P(A \cap B)$$
$$P(A \cap \bar{B}) = P(A) - P(A \cap B)$$

SOME IMPORTANT RESULTS

(A) If A, B and C are three events, then

(i) P [Exactly one of A, B, C occurs]

$$= P(A) + P(B) + P(C) - 2[(A \cap B) + (B \cap C) + (A \cap C)] + 3P(A \cap B \cap C)$$

(ii) P [Exactly two of A, B, C occur]

$$= P(A \cap B) + P(B \cap C) + P(A \cap C) - 3P(A \cap B \cap C)$$

(iii) P (Atleast two of A, B, C occur)

$$= P(A \cap B) + P(B \cap C) + P(A \cap C) - 2P(A \cap B \cap C)$$

(B) If A and B are two events, then

P(Exactly one of A, B occurs)

$$= P(A) + P(B) - 2P(A \cup B)$$
$$= P(A \cup B) - P(A \cap B)$$

INTRODUCTORY EXERCISE-20.2

1. The probability of occurrence of two events A and B are 1/4 and 1/2 respectively. The probability of their simultaneous occurrence is $\frac{7}{50}$. Find the probability that either A or B must occur.

(a) $\frac{61}{100}$ (b) $\frac{29}{100}$
(c) $\frac{39}{100}$ (d) $\frac{56}{99}$

2. In the previous question, find the probability that neither A nor B occurs.

(a) $\frac{25}{99}$ (b) $\frac{39}{100}$
(c) $\frac{61}{100}$ (d) $\frac{17}{100}$

3. If A and B be two events in a sample space such that $P(A) = \frac{3}{10}$ and $P(B) = \frac{1}{2}$ and $P(A \cap B) = \frac{1}{5}$. Find $P(A \cup B)$.

(a) $\frac{1}{5}$ (b) $\frac{2}{5}$
(c) $\frac{3}{5}$ (d) $\frac{4}{5}$

4. If A and B be two events in a sample space such that $P(A) = \frac{2}{5}$, $P(B) = \frac{1}{2}$ and $P(A \cup B) = \frac{3}{5}$, find $P(A \cap B)$.

(a) $\frac{3}{10}$ (b) $\frac{7}{10}$
(c) $\frac{4}{7}$ (d) $\frac{4}{15}$

Directions for question number 5-10 : If A and B be two mutually exclusive events in a sample space such that, $P(A) = \frac{2}{5}$ and $P(B) = \frac{1}{2}$, then

5. Find $P(\bar{A})$:

(a) $\frac{2}{5}$ (b) $\frac{3}{5}$
(c) $\frac{4}{5}$ (d) $\frac{6}{7}$

6. Find $P(\bar{B})$:

(a) $\frac{1}{4}$ (b) $\frac{3}{4}$
(c) $\frac{1}{2}$ (d) $\frac{4}{5}$

7. Find $P(A \cup B)$:

(a) $\frac{7}{16}$ (b) $\frac{9}{16}$
(c) $\frac{9}{10}$ (d) $\frac{1}{2}$

8. Find $P(\bar{A} \cap \bar{B})$:

(a) $\frac{4}{5}$ (b) $\frac{1}{10}$
(c) $\frac{8}{9}$ (d) $\frac{13}{20}$

9. Find $P(\bar{A} \cap B)$:

(a) $\frac{1}{2}$ (b) $\frac{3}{5}$
(c) $\frac{4}{7}$ (d) $\frac{7}{15}$

10. Find $P(A \cap \bar{B})$:

(a) $\frac{1}{5}$ (b) $\frac{2}{5}$
(c) $\frac{4}{15}$ (d) $\frac{3}{10}$

11. If $P(\bar{A}) = 0.65$, $P(A \cup B) = 0.65$, where A and B are two mutually exclusive events, then find $P(B)$.

(a) 0.60 (b) 0.30
(c) 0.70 (d) none of these

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

12. If A, B and C are three mutually exclusive and exhaustive events. Find $P(A)$, if $P(B) = \frac{3}{2}P(A)$ and $P(C) = \frac{1}{2}P(B)$.
(a) $\frac{8}{13}$ (b) $\frac{5}{13}$
(c) $\frac{4}{13}$ (d) $\frac{9}{13}$

13. Two dice are tossed once. Find the probability of getting an even number on first die, or a total of 8.
(a) $\frac{4}{9}$ (b) $\frac{2}{3}$
(c) $\frac{5}{9}$ (d) $\frac{1}{3}$

14. A die is thrown twice, what is the probability that atleast one of the two throws comes up with the number 5?
(a) $\frac{11}{36}$ (b) $\frac{5}{6}$
(c) $\frac{15}{36}$ (d) none of these

15. In a single throw of two dice, find the probability that neither a doublet nor a total of 8 will appear.
(a) $\frac{7}{15}$ (b) $\frac{5}{18}$
(c) $\frac{13}{18}$ (d) $\frac{3}{16}$

16. A die is thrown twice, what is the probability that atleast one of the two numbers is 6?
(a) $\frac{11}{12}$ (b) $\frac{11}{36}$
(c) $\frac{1}{6}$ (d) $\frac{7}{36}$

17. A card is drawn at random from a well-shuffled deck of 52 cards. Find the probability of its being a heart or a king.
(a) $\frac{4}{13}$ (b) $\frac{9}{13}$
(c) $\frac{8}{13}$ (d) $\frac{11}{26}$

18. Two cards are drawn at random from a well-shuffled pack of 52 cards. What is the probability that either both are red or both are queens?
(a) $\frac{17}{112}$ (b) $\frac{55}{221}$
(c) $\frac{55}{121}$ (d) $\frac{33}{221}$

19. A card is drawn from a deck of 52 cards. Find the probability of getting a red card or a heart or a king.
(a) $\frac{6}{13}$ (b) $\frac{7}{13}$
(c) $\frac{11}{26}$ (d) $\frac{15}{26}$

20. Four cards are drawn at a time from a pack of 52 playing cards. Find the probability of getting all the four cards of the same suit.
(a) $\frac{5}{13}$ (b) $\frac{12}{65}$
(c) $\frac{44}{4165}$ (d) $\frac{44}{169}$

21. From a well-shuffled pack of 52 cards, a card is drawn at random. Find the probability that the drawn card is a king or a queen.
(a) $\frac{2}{13}$ (b) $\frac{8}{13}$
(c) $\frac{11}{13}$ (d) none of these

22. Two cards are drawn at random from a pack of 52 cards. What is the probability that either both are black or both are jacks?
(a) $\frac{65}{121}$ (b) $\frac{55}{221}$
(c) $\frac{17}{221}$ (d) none of these

23. A natural number is chosen at random from amongst the first 300. What is the probability that the number, so chosen is divisible by 3 or 5?
(a) $\frac{48}{515}$ (b) $\frac{4}{150}$
(c) 1/2 (d) none of these

24. A natural number is chosen at random from the first 100 natural numbers. What is the probability that the number chosen is a multiple of 2 or 3 or 5?
(a) $\frac{30}{100}$ (b) $\frac{1}{33}$
(c) $\frac{74}{100}$ (d) $\frac{7}{10}$

25. A box contains 5 red balls, 8 green balls and 10 pink balls. A ball is drawn at random from the box. What is the probability that the ball drawn is either red or green?
(a) $\frac{13}{23}$ (b) $\frac{10}{23}$
(c) $\frac{11}{23}$ (d) $\frac{13}{529}$

26. A basket contains 10 apples and 20 oranges out of which 3 apples and 5 oranges are defective. If we choose two fruits at random, what is the probability that either both are oranges or both are non defective?
(a) $\frac{136}{345}$ (b) $\frac{17}{87}$
(c) $\frac{316}{435}$ (d) $\frac{158}{435}$

27. In a class 40% of the students offered Physics 20% offered Chemistry and 5% offered both. If a student is selected at random, find the probability that he has offered Physics or Chemistry only.
(a) 45% (b) 55%
(c) 36% (d) none of these

28. The probability that an MBA aspirant will join IIM is $\frac{2}{5}$ and that he will join XLRI is $\frac{1}{3}$. Find the probability that he will join IIM or XLRI.
(a) $\frac{4}{15}$ (b) $\frac{7}{15}$
(c) $\frac{11}{15}$ (d) $\frac{8}{15}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

29. In a given race, the odds in favour of horses H_1, H_2, H_3 and H_4 are 1 : 2, 1 : 3, 1 : 4, 1 : 5 respectively. Find the probability that one of them wins the race.

(a) $\frac{57}{60}$ (b) $\frac{1}{20}$ (c) $\frac{2}{7}$ (d) $\frac{7}{60}$

CONDITIONAL PROBABILITY

Let A and B be two events associated with a random experiment. Then, the probability of occurrence of A under the condition that B has already occurred and $P(B) \neq 0$, is called the conditional probability and it is denoted by $P\left(\frac{A}{B}\right)$.

Thus, $P\left(\frac{A}{B}\right)$ = Probability of occurrence of A given that B has already occurred.

Similarly, $P\left(\frac{B}{A}\right)$ = Probability of occurrence of B given that A has already occurred.

NOTE

(i) Sometimes $P\left(\frac{A}{B}\right)$ is used to denote the probability of occurrence of A when B occurs.

(ii) Similarly $P\left(\frac{B}{A}\right)$ is used to denoted the probability of occurrence of B when A occurs.

The above two cases happens due to the simultaneous occurrence of two events since the two events are the subsets of the same sample space.

EXAMPLE 1 *An urn contains 6 red and 9 blue balls. Two balls are drawn from the urn one after another without replacement. Find the probability of drawing a red ball when a blue ball has been drawn from the urn.*

SOLUTION Let A = drawing of a red ball in the second draw and B = drawing of a blue ball in the first draw

Now $P\left(\frac{A}{B}\right)$ = Probability of drawing a red ball in second draw when a blue ball has been drawn in the first draw.

Now, since there are only 14 balls after drawing a blue ball in first draw and out of these 14 balls, 6 balls are red.

Therefore $P\left(\frac{A}{B}\right) = \frac{6}{14} = \frac{3}{7}$

EXAMPLE 2 *A pair of dice is thrown simultaneously, find the probability that the sum is obtained 9 when there is an odd number on the first die.*

SOLUTION Let A be the event of getting a sum of 9 and B be the event of getting an odd number on the first die.

$\therefore$ $A = \{(3, 6) (4, 5), (5, 4), (6, 3)\}$

$B = \{(1, 1), (1, 2), (1, 3) \ldots (3, 1), (3, 2), (3, 3) \ldots (5, 1), (5, 2) \ldots (5, 6)\}$

$P(A) = \frac{4}{36} = \frac{1}{9}$ and $P(B) = \frac{18}{36} = \frac{1}{2}$

$\therefore$ $P\left(\frac{A}{B}\right)$ = Probability of occurrence of A when B Occurs.

$\Rightarrow$ $P\left(\frac{A}{B}\right)$ = Probability of getting 9 as the sum when there is an odd number on first die.

$\Rightarrow$ $P\left(\frac{A}{B}\right) = \frac{n(A \cap B)}{n(B)} = \frac{2}{18} = \frac{1}{9}$

[Here $(A \cap B) = (3, 6), (6, 3)$]

MULTIPLICATION THEOREM

Let A and B be two events associated with the same random experiment then

$$P(A \cap B) = P(A)\, P\left(\frac{B}{A}\right), \quad \text{if } P(A) \neq 0 \ldots \text{(i)}$$

or $$P(A \cap B) = P(B)\, P\left(\frac{A}{B}\right), \quad P(B) \neq 0 \ldots \text{(ii)}$$

NOTE $P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)}$ from (i)

and $P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$ from (ii)

In general, if $A_1, A_2, A_3 \ldots A_n$ are events associated with the same random experiment, then

$$P(A_1 \cap A_2 \cap A_3 \cap \ldots \cap A_n) = P(A_1)\, P\left(\frac{A_2}{A_1}\right) P\left(\frac{A_3}{A_1 \cap A_2}\right) \ldots P\left(\frac{A_n}{A_1 \cap A_2 \cap \ldots \cap A_{n-1}}\right)$$

EXAMPLE 1 *Let A and B be the two events such that $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$ and $P(A \cap B) = \frac{1}{4}$, find*

(i) $P\left(\frac{A}{B}\right)$ (ii) $P\left(\frac{B}{A}\right)$ (iii) $P(A \cup B)$ (iv) $P\left(\frac{\bar{A}}{\bar{B}}\right)$

SOLUTION (i) $P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{1/4}{1/3} = \frac{3}{4}$

(ii) $P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{1/4}{1/2} = \frac{1}{2}$

(iii) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$= \frac{1}{2} \quad \frac{1}{3} \quad \frac{1}{4} \quad \frac{7}{12}$

(iv) $P\left(\frac{\bar{A}}{\bar{B}}\right) = \frac{P(\bar{A} \cap \bar{B})}{P(\bar{B})} = \frac{P(\overline{A \cup B})}{P(\bar{B})} = \frac{1 - P(A \cup B)}{1 - P(B)}$

$= \frac{1 - \frac{7}{12}}{1 - \frac{1}{3}} = \frac{\frac{5}{12}}{\frac{2}{3}} = \frac{5}{8}$

EXAMPLE 2 *An urn contains 6 red balls and 9 green balls. Two balls are drawn in succession without replacement. What is the probability that first is red and second is green.*

SOLUTION Let A be the event of drawing a red ball in first draw and B be the event of getting a green ball in the second draw.

$\therefore \quad P(A) = \frac{{}^6C_1}{{}^{15}C_1} = \frac{6}{15} = \frac{2}{5}$

$P\left(\frac{B}{A}\right)$ = Probability of getting a green ball in second draw when a red ball has been selected in first draw $= \frac{{}^9C_1}{{}^{14}C_1} = \frac{9}{14}$

$\therefore$ Required probability $= P(A \cap B) = P(A)\,P\left(\frac{B}{A}\right)$

$= \frac{2}{5} \times \frac{9}{14} = \frac{9}{35}$

INTRODUCTORY EXERCISE-20.3

1. If A and B are two events such that $P(A) = 0 \cdot 5$, $P(B) = 0.6$ and $P(A \cup B) = 0.8$. Find $P\left(\frac{A}{B}\right)$.
   (a) $\frac{1}{3}$ (b) $\frac{1}{2}$
   (c) $\frac{1}{4}$ (d) none of these

2. If A and B are two events such that $P(A) = 0.4$, $P(B) = 0.8$ and $P\left(\frac{B}{A}\right) = 0.6$, find $P(A \cup B)$:
   (a) 0.24 (b) 0.96
   (c) 0.04 (d) none of these

3. Three fair coins are tossed. Find the probability that they are all tails, if one of the coins shows a tail.
   (a) $\frac{2}{7}$ (b) $\frac{5}{14}$
   (c) $\frac{1}{7}$ (d) none of these

4. A coin is tossed twice and the four possible outcomes are assumed to be equally likely. If E is the event, "Both head and tail have appeared and F be the event, at most one tail is observed ; find $P\left(\frac{E}{F}\right)$ and $P\left(\frac{F}{E}\right)$.
   (a) $\frac{2}{3}, 1$ (b) $\frac{1}{3}, \frac{2}{3}$
   (c) $1, \frac{2}{3}$ (d) none of these

5. A coin is tossed twice if the coin shows head it is tossed again but if it shows a tail then a die is tossed. If 8 possible outcomes are equally likely. Find the probability that the die shows a number greater than 4, if it is known that the first throw of the coin results in a tail.
   (a) $\frac{1}{3}$ (b) $\frac{2}{3}$
   (c) $\frac{2}{5}$ (d) $\frac{4}{15}$

6. A die is rolled. If the outcome is an odd number, what is the probability that it is a prime number?
   (a) $\frac{3}{8}$ (b) $\frac{7}{9}$
   (c) $\frac{2}{3}$ (d) none of these

7. A die is thrown twice and the sum of the numbers appearing is observed to be 9. What is the conditional probability that the number 4 has appeared atleast once?
   (a) $\frac{1}{2}$ (b) $\frac{2}{3}$
   (c) $\frac{3}{4}$ (d) none of these

8. Two dice are thrown. Find the probability that the sum is 8 or greater than 8, if 4 appears on the first die.
   (a) $\frac{3}{8}$ (b) $\frac{5}{8}$
   (c) $\frac{1}{2}$ (d) none of these

9. A die is rolled. If the outcome is an odd number, what is the probability that it is a number greater than 1 ?
   (a) $\frac{2}{3}$ (b) $\frac{1}{3}$
   (c) $\frac{3}{8}$ (d) $\frac{5}{6}$

10. In a class 45% students read English, 30% read French and 20% read both English and French. One student is selected at random. Find the probability that he reads English, if it is known that he reads French.
    (a) $\frac{1}{3}$ (b) $\frac{2}{3}$
    (c) $\frac{5}{6}$ (d) none of these

11. In the previous question, find the probability that he reads French, if it is known that he reads English.
    (a) $\frac{4}{9}$ (b) $\frac{5}{9}$
    (c) $\frac{2}{9}$ (d) $\frac{1}{9}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

12. A couple has two children. Find the probability that both are boys, if it is known that one of the children is a boy.
(a) $\frac{1}{9}$ (b) $\frac{1}{3}$
(c) $\frac{2}{3}$ (d) $\frac{4}{5}$

13. In the previous question find the probability that both are boys, if it is known that the older child is a boy.
(a) $\frac{3}{8}$ (b) $\frac{1}{2}$
(c) $\frac{5}{8}$ (d) $\frac{3}{4}$

Directions for question number 14-17 : A bag contains 3 red and 4 black balls and another bag has 4 red and 2 black balls. One bag is selected at random and from the selected bag a ball is drawn. Let E be the event that the first bag is selected, F be the event that the second bag is selected, G be the event that ball drawn is red.

14. Find $P(E)$.
(a) $\frac{1}{2}$ (b) $\frac{3}{4}$
(c) $\frac{1}{4}$ (d) $\frac{5}{8}$

15. Find $P(F)$.
(a) $\frac{3}{4}$ (b) $\frac{1}{2}$
(c) $\frac{1}{4}$ (d) $\frac{1}{6}$

16. Find $P\left(\frac{G}{E}\right)$.
(a) $\frac{5}{6}$ (b) $\frac{5}{14}$
(c) $\frac{3}{7}$ (d) none of these

17. Find $P\left(\frac{G}{F}\right)$.
(a) $\frac{2}{3}$ (b) $\frac{1}{9}$
(c) $\frac{5}{9}$ (d) $\frac{4}{5}$

18. Two balls are drawn from a bag containing 2 white, 3 red and 4 black balls one by one without replacement. What is the probability that atleast one ball is red?
(a) $\frac{7}{12}$ (b) $\frac{5}{12}$
(c) $\frac{3}{10}$ (d) none of these

19. A bag contains 6 red and 9 blue balls. Two successive drawing of four balls are made such that the balls are not replaced before the second draw. Find the probability that the first draw gives 4 red balls and second draw gives 4 blue balls.
(a) $\frac{3}{715}$ (b) $\frac{7}{715}$
(c) $\frac{15}{233}$ (d) none of these

20. An urn contains 4 white 6 black and 8 red balls. If 3 balls are drawn one by one without replacement, find the probability of getting all white balls.
(a) $\frac{5}{204}$ (b) $\frac{1}{204}$
(c) 13/204 (d) none of these

21. Two numbers are selected at random from the integers 1 through 9. If the sum is even, find the probability that both numbers are odd.
(a) 5/8 (b) 3/8
(c) 3/10 (d) none of these

22. A box contains 25 tickets, numbered 1, 2, 3, .. 25. A ticket is drawn and then another ticket is drawn without replacement. Find the probability that both tickets will show odd numbers.
(a) $\frac{37}{50}$ (b) $\frac{13}{50}$
(c) 13/25 (d) none of these

INDEPENDENT EVENTS

Events are said to be independent, if the occurrence of one does not depend upon the occurrence of the other.

Suppose an urn contains m red balls and n green balls. Two balls are drawn from the urn one after the other.

If the ball drawn in the first draw is not replaced back in the bag, then two events of drawing the ball are dependent because first draw of the ball determine the probability of drawing the second ball.

If the ball drawn in the first draw is replaced back in the bag, then two events are independent because first draw of a ball has no effect on the second draw.

Theorem 1. Two events A and B associated with the same sample space of a random experiment are independent if and only if

$$P(A \cap B) = P(A) \cdot P(B)$$

Theorem 2. If $A_1, A_2, A_3 \ldots A_n$ are independent events associated with a random experiment, then

$$P(A_1 \cap A_2 \cap A_3 \ldots \cap A_n) = P(A_1)\,P(A_2) \ldots P(A_n)$$

Theorem 3. If $A_1, A_2 \ldots A_n$ are n independent events associated with a random experiment, then

$$P(A_1 \cup A_2 \cup \ldots \cup A_n) = 1 - P(\bar{A}_1)\,P(\bar{A}_2) \ldots P(\bar{A}_n)$$

IMPORTANT RESULTS

If A and B are independent events then the following events are also independent.

(i) $A \cap \bar{B}$ (ii) $\bar{A} \cap B$ (iii) $\bar{A} \cap \bar{B}$

LAW OF TOTAL PROBABILITY

Let $E_1, E_2, \ldots E_n$ be n mutually exclusive and exhaustive events associated with a random experiment. If A is any event which occurs with E_1 or E_2 or ... or E_n, then

$$P(A) = P(E_1)\,P\left(\frac{A}{E_1}\right) + P(E_2)\,P\left(\frac{A}{E_2}\right) + \ldots P(E_n)\,P\left(\frac{A}{E_n}\right)$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Baye's Rule : Let $E_1, E_2, \ldots E_n$ be a mutually exclusive and exhaustive events associated with a random experiment. If A is any event which occurs with E_1 or E_2 or ... or E_n, then

$$P\left(\frac{E_i}{A}\right) = \frac{P(E_i)\,P\left(\frac{A}{E_i}\right)}{\sum_{i=1}^{n} P(E_i)\,P\left(\frac{A}{E_i}\right)}, \qquad i = 1, 2, \ldots n$$

EXAMPLE 1 *A coin is tossed twice and all 4 outcomes are equally likely. Let A be the event that first throw results in a head and B be the event that second throw results in a tail, then show that the events A and B are independent.*

SOLUTION

$S = \{HH, HT, TH, TT\}$

$A = \{HH, HT\}$

$B = \{HT, TT\}$

$A \cap B = \{HT\}$

$\therefore \quad P(A) = \frac{2}{4} = \frac{1}{2}, \; P(B) = \frac{2}{4} = \frac{1}{2}$

and $\quad P(A \cap B) = \frac{1}{4}$

Clearly $\quad P(A \cap B) = \frac{1}{4} = P(A) \cdot P(B)$

Hence A and B are independent events.

EXAMPLE 2 *A bag contains 10 red balls and 10 green balls. Two balls are drawn at random, one at a time, with replacement. Let A be the event that first ball is red, B be the event that second ball is green and C be the event that both balls are either red or green, then show that the events A, B and C are pair wise independent and A, B, C are mutually dependent.*

SOLUTION We have $P(A) = \frac{1}{2}, \quad P(B) = \frac{1}{2},$

$$P(C) = \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{1}{2}$$

$P(A \cap B)$ = Probability that the first is red and second is green

$\Rightarrow \quad P(A) \cdot P(B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

$P(B \cap C)$ = Probability that both the balls are green

$\Rightarrow \quad P(B) \cdot P(C) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

and $P(A \cap C)$ = Probability that both the balls are red

$\Rightarrow \quad P(A)\,P(C) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

Hence A, B, C are pairwise independent

Now, $P(A \cap B \cap C)$ = Probability that the first ball is red and the second ball is green and the first and second are both red or green = 0

and $\quad P(A)\,P(B)\,P(C) = \frac{1}{8}$

$\therefore \quad P(A \cap B \cap C) \neq P(A)\,P(B)\,P(C)$

Thus, A, B, C are not mutually independent.

EXAMPLE 3 *(For explanation of law of total probability) There are two bags. The first bag contains 4 white and 5 black balls and the second bag contains 5 white and 4 black balls. Two balls are drawn at random from the first bag and are put into the second bag without noticing their colours. Then two balls are drawn from the second bag. Find the probability that the balls are white and black.*

SOLUTION A white and a black ball can be drawn from the second bag in the following mutually exclusive ways :

(i) By transferring 2 white balls from the first bag to the second bag and then drawing a white and a black ball from it.

(ii) By transferring 2 black balls from the first bag to the second bag and then drawing a white and a black ball from it .

(iii) By transferring 1 white and 1 black ball from first bag to the second bag and then drawing a white and a black ball from it.

Let A, B, C and D be the events as defined below :

A = Two white balls are drawn from the first bag

B = Two black balls are drawn from the first bag.

C = One white and one black ball is drawn from the first bag.

D = Two balls drawn from the second bag are white and black.

We, have $\quad P(A) = \frac{{}^4C_2}{{}^9C_2} = \frac{6}{36} = \frac{1}{6}$

$$P(B) = \frac{{}^5C_2}{{}^9C_2} = \frac{10}{36} = \frac{5}{18}$$

$$P(C) = \frac{{}^4C_1 \times {}^5C_1}{{}^9C_2} = \frac{20}{36} = \frac{5}{9}$$

If A has already occurred, *i.e.* if two white balls have been transferred from the first bag to the second bag, then the second bag will contain 7 white and 4 black balls, therefore the probability of drawing a white and a black from the second bag is

$$P\left(\frac{D}{A}\right) = \frac{{}^7C_1 \times {}^4C_1}{{}^{11}C_2} = \frac{28}{55}$$

Similarly, $\quad P\left(\frac{D}{B}\right) = \frac{{}^5C_1 \times {}^6C_1}{{}^{11}C_2} = \frac{30}{55} = \frac{6}{11}$

and $\quad P\left(\frac{D}{C}\right) = \frac{{}^6C_1 \times {}^5C_1}{{}^{11}C_2} = \frac{30}{55} = \frac{6}{11}$

$\therefore$ By the law of total probability, we have

$$P(D) = P(A)\,P\left(\frac{D}{A}\right) + P(B)P\left(\frac{D}{B}\right) + P(C)\,P\left(\frac{D}{C}\right)$$

$$= \frac{1}{6} \times \frac{28}{55} + \frac{5}{18} \times \frac{6}{11} + \frac{5}{9} \times \frac{6}{11}$$

$$= \frac{14}{165} + \frac{5}{33} + \frac{10}{33}$$

$$= \frac{14 + 25 + 50}{165}$$

$$= \frac{89}{165}$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

EXAMPLE 4 *(For explanation of Baye's rule) Three boxes contain 6 red, 4 black, ; 5 red, 5 black and 4 red, 6 black balls respectively. One of the box is selected at random and a ball is drawn from it. If the ball drawn is red, find the probability that it is drawn from the first bag.*

SOLUTION Let A, B, C and D be the events defined as follows :

A = first box is chosen

B = second box is chosen

C = third box is chosen

D = ball drawn is red.

Since there are three boxes and one of the three boxes is chosen at random, therefore

$$P(A)=P(B)=P(C)=\frac{1}{3}$$

If A has already occurred, then first box has been chosen which contains 6 red and 4 black balls. The probability of drawing a red ball from it is $\frac{6}{10}$

So, $$P\left(\frac{D}{A}\right)=\frac{6}{10}$$

Similarly, $P\left(\frac{D}{B}\right)=\frac{5}{10}$ and $P\left(\frac{D}{C}\right)=\frac{4}{10}$

We are required to find $P\left(\frac{A}{D}\right)$ *i.e.,* given that the ball drawn is red, what is the probability that it is drawn from the first box.

By Baye's rule,

$$P\left(\frac{A}{D}\right)=\frac{P(A)\cdot P\left(\frac{D}{A}\right)}{P(A)\cdot P\left(\frac{D}{A}\right)+P(B)\cdot P\left(\frac{D}{B}\right)+P(C)\cdot P\left(\frac{D}{C}\right)}$$

$$=\frac{\frac{1}{3}\times\frac{6}{10}}{\left(\frac{1}{3}\times\frac{6}{10}\right)+\left(\frac{1}{3}\times\frac{5}{10}\right)+\left(\frac{1}{3}\times\frac{4}{10}\right)}=\frac{2}{5}$$

INTRODUCTORY EXERCISE-20.4

Directions for question number 1-5 : Let A and B be independent events such that $P(A)=0.6$ and $P(B)=0.4$

1. Find $P(A\cap B)$.
(a) 0.24 (b) 0.76
(c) 0.56 (d) none of these

2. Find $P(A\cup B)$.
(a) 0.24 (b) 0.76
(c) 0.36 (d) none of these

3. Find $P(\bar{A}\cap B)$.
(a) 0.24 (b) 0.16
(c) 0.36 (d) none of these

4. Find $P(A\cap \bar{B})$.
(a) 0.24 (b) 0.56
(c) 0.36 (d) 0.76

5. Find $P(\bar{A}\cap \bar{B})$.
(a) 0.76 (b) 0.54
(c) 0.36 (d) 0.24

6. If A and B are two independent events such that $P(\bar{A})=0.65$, $P(A\cup B)=0.65$ and $P(B)=p$, find the value of p.
(a) $\frac{7}{13}$ (b) $\frac{6}{13}$
(c) $\frac{37}{65}$ (d) none of these

7. An unbiased die is tossed twice. Find the probability of getting a 1, 2, 3 or 4 on the first toss and a 4, 5 or 6 on the second toss.
(a) $\frac{1}{3}$ (b) $\frac{2}{3}$
(c) $\frac{5}{6}$ (d) $\frac{1}{6}$

8. Two persons A and B throw a die alternatively till one of them gets a three and wins the game. Find their respective probabilities of winning.
(a) $\frac{6}{11},\frac{5}{11}$ (b) $\frac{5}{11},\frac{8}{11}$
(c) $\frac{3}{11},\frac{7}{11}$ (d) $\frac{8}{11},\frac{3}{11}$

9. Two persons A and B throw a coin alternatively till one of them gets head and wins the game. Find their respective probabilities of winning.
(a) $\frac{1}{3},\frac{5}{6}$ (b) $\frac{3}{5},\frac{4}{5}$
(c) $\frac{2}{3},\frac{1}{3}$ (d) $\frac{1}{6},\frac{5}{6}$

10. Three persons A, B, C throw a die in succession till one gets a six and wins the game. Find their respective probabilities of winning.
(a) $\frac{36}{91},\frac{30}{91},\frac{25}{91}$ (b) $\frac{10}{71},\frac{16}{91},\frac{22}{81}$
(c) $\frac{13}{61},\frac{15}{61},\frac{17}{61}$ (d) none of these

11. A and B take turn in throwing two dice; the first to throw 9 being awarded. Find the ratio of probabilities of their winning if A has the first throw.
(a) 7/8 (b) 9/8
(c) 8/7 (d) 9/10

12. From a pack of 52 cards, two are drawn one by one without replacement. Find the probabilities that both of them are kings.
(a) $\frac{11}{21}$ (b) $\frac{13}{121}$
(c) $\frac{1}{221}$ (d) $\frac{1}{121}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

13. Ashmit can solve 80% of the problem given in a book and Amisha can solve 70%. What is the probability that at least one of them will solve a problem, selected at random from the book?
(a) 0.60 (b) 0.06
(c) 0.94 (d) 0.56

14. The probability that A hits a target is $\frac{1}{3}$ and the probability that B hits it, is $\frac{2}{5}$. What is the probability that the target will be hit, if each one of A and B shoots the target?
(a) $\frac{5}{6}$ (b) $\frac{3}{5}$
(c) $\frac{11}{15}$ (d) $\frac{1}{6}$

15. A problem is given to three students whose chances of solving it are $\frac{1}{2}, \frac{1}{3}$ and $\frac{1}{4}$ respectively. What is the probability that the problem will be solved?
(a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) $\frac{3}{4}$ (d) $\frac{7}{12}$

16. The probabilities of A, B, C solving a problem are $\frac{1}{3}, \frac{2}{7}$ and $\frac{3}{8}$ respectively. If all the three try to solve the problem simultaneously, find the probability that exactly one of them will solve it.
(a) $\frac{25}{52}$ (b) $\frac{25}{56}$
(c) 13/42 (d) none of these

17. A, B and C shoot to hit a target. If A hits the target 4 times in 5 trials, B hits it 3 times in 4 trials and C hits it 2 times in 3 trials. What is the probability that the target is hit by atleast 2 persons?
(a) $\frac{5}{6}$ (b) $\frac{3}{4}$
(c) $\frac{4}{5}$ (d) $\frac{1}{9}$

18. An airgun can take a maximum of 4 shots at a balloon at some distance. The probabilities of hitting the balloon at the first, second, third and fourth shot are 0.1, 0.2, 0.3 and 0.4 respectively. What is the probability that the balloon is hit?
(a) 0.6976 (b) 0.6576
(c) 0.786 (d) none of these

19. The probability that A can solve a problem is $\frac{2}{3}$ and the probability that B can solve the same problem is $\frac{3}{5}$. Find the probability that atleast one of A and B will be able to solve the problem.
(a) $\frac{12}{15}$ (b) $\frac{13}{15}$
(c) $\frac{19}{45}$ (d) none of these

20. In the previous question (number 19), find the probability that none of the two will be able to solve the problem.
(a) $\frac{13}{15}$ (b) $\frac{4}{15}$
(c) $\frac{2}{15}$ (d) $\frac{23}{30}$

Directions for question number 21-24 : *The probabilities that a student will receive an A, B, C or D grade are 0.4, 0.3, 0.2 and 0.1 respectively. Find the probability that a student will receive*

21. Not an A grade.
(a) 0.4 (b) 0.6
(c) 0.56 (d) none of these

22. At most a C grade.
(a) 0.3 (b) 0.7
(c) 0.36 (d) none of these

23. B or C grade.
(a) 0.2 (b) 0.5
(c) 0.8 (d) 0.6

24. Atleast B grade.
(a) 0.21 (b) 0.3
(c) 0.7 (d) none of these

Directions for question number 25-28 : *Ajay and his wife Kajol appear in an interview for two vaccancies in the same post. The probability of Ajay's selection is $\frac{1}{7}$ and that of his wife Kajol's selection is $\frac{1}{5}$. What is the probability that*

25. Both of them will be selected?
(a) $\frac{1}{12}$ (b) $\frac{1}{35}$
(c) $\frac{13}{35}$ (d) $\frac{12}{35}$

26. Only one of them will be selected ?
(a) $\frac{5}{7}$ (b) $\frac{1}{5}$
(c) $\frac{2}{7}$ (d) $\frac{2}{35}$

27. None of them will be selected?
(a) $\frac{12}{35}$ (b) $\frac{6}{35}$
(c) $\frac{24}{35}$ (d) none of these

28. Atleast one of them will be selected?
(a) $\frac{11}{35}$ (b) $\frac{24}{35}$
(c) $\frac{2}{7}$ (d) $\frac{1}{35}$

29. A speaks truth in 60% and B in 80% of the cases. In what percentage of cases are they likely to contradict each other narrating the same incident?
(a) 44% (b) 36%
(c) 64% (d) 48%

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Directions for question number 30 and 31 : *The odds against a husband who is 50 years old, living till he is 70 are 7 : 5 and the odds against his wife who is now 40, living till she is 60 are 5 : 3. Find the probability that*

30. The couple will be alive 20 years hence.
(a) $\frac{21}{32}$ (b) $\frac{5}{32}$
(c) $\frac{15}{32}$ (d) $\frac{12}{35}$

31. Atleast one of them will be alive 20 years hence.
(a) $\frac{61}{96}$ (b) $\frac{31}{96}$
(c) $\frac{41}{70}$ (d) none of these

32. Three critics review a book. Odds in favour of the book are 5 : 2, 4 : 3 and 3 : 4 respectively for the critics. Find the probability that majority are in favour of the book.
(a) $\frac{108}{343}$ (b) $\frac{209}{343}$
(c) $\frac{1}{7}$ (d) $\frac{1}{243}$

33. An article manufactured by a company consists of two parts X and Y. In the process of manufacture of the part X, 9 out of 100 parts may be defective. Similarly, 5 out of 100 are likely to be defective in the manufacture of the part Y. Calculate the probability that the assembled product will not be defective.
(a) 0.6485 (b) 0.6565
(c) 0.8645 (d) none of these

34. A bag contains 4 red and 3 black balls. A second bag contains 2 red and 4 black balls. One bag is selected at random. From the selected bag, one ball is drawn. Find the probability that the ball drawn is red.
(a) $\frac{23}{42}$ (b) $\frac{19}{42}$
(c) $\frac{7}{32}$ (d) $\frac{16}{39}$

35. In a toys making factory, machine A, B and C manufacture respectively 25%, 35% and 40% of the total toys. Of their output 5%, 4% and 2% respectively are defective toys. A toy is drawn at random from the product. What is the probability that the toy drawn is defective?
(a) 0.225 (b) 0.345
(c) 0.235 (d) none of these

36. A box contains 20 bulbs. The probability that the box contains exactly 2 defective bulbs is 0.4 and the probability that the box contains exactly 3 defective bulbs is 0.6. Bulbs are drawn at random one by one without replacement and tested till the defective bulbs are found. What is the probability that the testing procedure ends at the twelfth testing?
(a) 0 (b) 1
(c) can't be determined (d) none of these

37. In a toys making factory, machines A, B and C manufacture respectively 25%, 35% and 40% of the total toys of their output 5%, 4% and 2% respectively are defective toys. A toy is drawn at random from the product. If the toy drawn is found to be defective, what is the probability that it is manufactured by the machine B?
(a) $\frac{17}{69}$ (b) $\frac{28}{69}$
(c) $\frac{35}{69}$ (d) none of these

38. An architecture company built 200 bridges 400 hospitals and 600 hotels. The probability of damage due to earthquake of a bridge, hospital and hotel is 0.01, 0.03 and 0.15 respectively. One of the construction gets damaged with earthquake. What is the probability that it is a bridge?
(a) $\frac{1}{26}$ (b) $\frac{1}{52}$
(c) $\frac{7}{52}$ (d) none of these

39. There are 3 boxes each containing 3 red and 5 green balls. Also there are 2 boxes, each containing 4 red and 2 green balls. A green ball is selected at random. Find the probability that this green ball is from a box of the first group.
(a) $\frac{54}{61}$ (b) $\frac{45}{61}$
(c) $\frac{8}{31}$ (d) none of these

40. A man speaks truth 3 out of 4 times. He throws a die and reports that it is a six. Find the probability that it is actually a six.
(a) $\frac{3}{8}$ (b) $\frac{5}{8}$
(c) $\frac{7}{8}$ (d) $\frac{1}{12}$

41. A card from a pack of 52 cards is lost. From the remaining cards of pack, two cards are drawn and are found to be diamonds. Find the probability of the missing card to be diamond.
(a) $\frac{39}{50}$ (b) $\frac{11}{50}$
(c) $\frac{23}{25}$ (d) $\frac{3}{26}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

EXERCISE

LEVEL 1

1. The digits 1, 2, 3, 4, 5, 6, 7, 8, 9 are written in random order to form a nine digit number. Find the probability that this number is divisible by 4 :
   (a) $\frac{4}{9}$ (b) $\frac{2}{9}$
   (c) $\frac{17}{81}$ (d) none of these

2. A four digit number is formed with the digits 1, 3, 4, 5 without repetition. Find the chance that the number is divisible by 5 :
   (a) $\frac{3}{4}$ (b) $\frac{1}{4}$
   (c) $\frac{9}{16}$ (d) $\frac{1}{16}$

3. Five persons entered the lift cabin on the ground floor of an 8-floor house. Suppose each of them independently and with equal probability can leave the cabin at any floor beginning with the first. Find out the probability that all persons leaving at different floors :
   (a) $\frac{365}{2401}$ (b) $\frac{360}{2401}$
   (c) $\frac{35}{2410}$ (d) none of these

4. If from each of the three boxes containing 3 white and 1 black, 2 white and 2 black, 1 white and 3 black balls, one ball is drawn at random, then the probability that 2 white and 1 black ball will be drawn is :
   (a) $\frac{13}{32}$ (b) $\frac{27}{32}$
   (c) $\frac{19}{32}$ (d) none of these

5. A fair coin is tossed repeatedly. If tail appears on first four tosses, then the probability of head appearing on fifth toss equals :
   (a) $\frac{5}{12}$ (b) $\frac{1}{2}$
   (c) $\frac{5}{6}$ (d) $\frac{1}{6}$

6. There are four calculators and it is known that exactly two of them are defective. They are tested one by one , in a random order till both the defective calculators are identified. Then the probability that only two tests are required is :
   (a) $\frac{5}{6}$ (b) $\frac{1}{2}$
   (c) $\frac{1}{6}$ (d) $\frac{1}{3}$

7. 20 girls, among whom are A and B sit down at a round table. The probability that there are 4 girls between A and B is :
   (a) $\frac{17}{19}$ (b) $\frac{2}{19}$
   (c) $\frac{13}{19}$ (d) $\frac{6}{19}$

8. Two integers x and y are chosen with replacement out of the set $\{0, 1, 2, 3, \ldots 10\}$. Then the probability that $|x - y| > 5$ is :
   (a) $\frac{7}{11}$ (b) $\frac{40}{121}$
   (c) $\frac{35}{121}$ (d) $\frac{30}{121}$

9. The probability that the birthdays of 4 different persons will fall in exactly two calendar months is :
   (a) $\frac{77}{1728}$ (b) $\frac{17}{87}$
   (c) $\frac{11}{144}$ (d) none of these

10. If 6 objects are distributed at random among 6 persons, the probability that atleast one of them will not get any thing is :
    (a) $6 \cdot (6!)$ (b) $\frac{5^6}{6!}$
    (c) $\frac{6^6 - 6!}{6^6}$ (d) none of these

11. There is 4 volume encyclopaedia among 40 books arranged on a shelf in a random order. If the volumes are not necessarily kept side by side, the probability that they occur in increasing order from left to right is :
    (a) $\frac{1}{24}$ (b) $\frac{1}{12}$
    (c) $\frac{1}{10}$ (d) none of these

12. Four numbers are multiplied together. Then the probability that the product will be divisible by 5 or 10 is :
    (a) $\frac{169}{625}$ (b) $\frac{369}{625}$
    (c) $\frac{169}{1626}$ (d) none of these

13. 8 couples (husband and wife) attend a dance show 'Nach Baliye' in a popular TV channel ; A lucky draw in which 4 persons picked up for a prize is held, then the probability that there is atleast one couple will be selected is :
    (a) $\frac{8}{39}$ (b) $\frac{15}{39}$
    (c) $\frac{12}{13}$ (d) none of these

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

14. Three persons A, B and C are to speak at a function along with 4 other persons. if they all speak in random order, the probability that A speaks before B and B speaks before C is :
(a) $\frac{5}{6}$ (b) $\frac{1}{6}$
(c) $\frac{1}{2}$ (d) $\frac{1}{3}$

15. A bag contains 16 coins of which 2 coins are counterfeit with heads on the both sides. The rest are fair coins. One is selected at random from the bag and tossed. The probability of getting a head is :
(a) $\frac{3}{16}$ (b) $\frac{13}{16}$
(c) $\frac{9}{16}$ (d) $\frac{7}{16}$

16. A committee of five persons is to be chosen from a group of 9 people. The probability that a certain married couple will either serve together or not at all is :
(a) 4/9 (b) 5/9
(c) 13/18 (d) none of these

17. A speaks truth in 60% cases and B speaks truth in 80% cases. The probability that they will say the same thing while describing a single event is :
(a) 0.36 (b) 0.56
(c) 0.48 (d) 0.20

18. Two squares are chosen at random on a chessboard, the probability that they have a side in common is :
(a) $\frac{3}{32}$ (b) $\frac{1}{32}$
(c) 1/18 (d) none of these

19. An old person forgets the last two digits of a telephone number, remembering only that these are different dialled at random. The probability that the number is dialled correctly is :
(a) 1/90 (b) 81/91
(c) 2/99 (d) none of these

20. Three squares of a chessboard are chosen at random, the probability that two are of one colour and one of another is :
(a) $\frac{67}{992}$ (b) $\frac{16}{21}$
(c) $\frac{31}{32}$ (d) none of these

21. The probability that a leap year selected at random contains either 53 Sundays or 53 Mondays, is :
(a) $\frac{17}{53}$ (b) $\frac{1}{53}$
(c) $\frac{3}{7}$ (d) none of these

22. In order to get atleast once a head with probability $P \geq 0.9$, the number of times a coin needs to be tossed is :
(a) 3 (b) 2
(c) 5 (d) 4

23. Out of 13 applicants for a job, there are 5 women and 8 men. It is desired to select 2 persons for the job. The probability that atleast one of the selected persons will be a woman is :
(a) $\frac{25}{39}$ (b) $\frac{31}{65}$
(c) $\frac{25}{69}$ (d) $\frac{5}{13}$

24. Nine squares are chosen at random on a chessboard. What is the probability that they form a square of size 3×3?
(a) $\frac{9}{{}^{64}C_9}$ (b) $\frac{36}{{}^{64}C_9}$
(c) $\frac{6}{{}^{64}C_9}$ (d) none of these

25. Seven digits from the numbers 1, 2, 3, 4, 5, 6, 7, 8 and 9 are written in random order. The probability that this seven digit number is divisible by 9 is :
(a) $\frac{7}{9}$ (b) $\frac{1}{9}$
(c) $\frac{2}{9!}$ (d) $\frac{4}{9}$

LEVEL 2

1. What is the probability that four S's come consecutively in the word MISSISSIPPI?
(a) $\frac{4}{165}$ (b) $\frac{4}{135}$
(c) $\frac{24}{165}$ (d) none of these

2. Each coefficient in the equation $ax^2 + bx + c = 0$ is determined by throwing ordinary six faced die. Find the probability that the equation will have real roots.
(a) $\frac{34}{161}$ (b) $\frac{43}{216}$
(c) $\frac{25}{36}$ (d) none of these

3. A and B throw alternately a pair of dice. A wins if he throws 6 before B throws 7 and B wins if he throws 7 before A throws 6. Find their respective chance of winning, if A begins.
(a) $\frac{13}{16}, \frac{31}{16}$ (b) $\frac{30}{61}, \frac{31}{61}$
(c) $\frac{31}{61}, \frac{41}{61}$ (d) $\frac{38}{61}, \frac{23}{61}$

4. A consignment of 15 wristwatches contains 4 defectives. The wristwatches are selected at random, one by one and examined. The ones examined are not put back. What is the probability that ninth one examined is the last defective?
(a) $\frac{11}{195}$ (b) $\frac{17}{195}$
(c) $\frac{8}{195}$ (d) $\frac{16}{195}$

5. In a test, an examinee either guesses or copies or knows the answer to a multiple choice question with four choices. The probability that he makes a guess is $\frac{1}{3}$ and the probability that he copies the answer is 1/6. The probability that his answer is correct, given that he copied it, is 1/8. Find the probability that he knew the answer to the question, given that he correctly answered it.
(a) $\frac{17}{39}$ (b) $\frac{13}{29}$
(c) $\frac{24}{29}$ (d) $\frac{24}{39}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

6. Given that the sum of two non-negative quantities is 200, the probability that their product is not less than $\frac{3}{4}$ times their greatest product value is :
(a) $\frac{99}{200}$ (b) $\frac{101}{200}$
(c) $\frac{87}{100}$ (d) none of these

7. A pack of cards consists of 9 cards numbered 1 to 9. Three cards are drawn at random with replacement. Then the probability of getting 1 even and 2 odd numbered cards is :
(a) $\frac{3}{143}$ (b) $\frac{100}{243}$
(c) $\frac{50}{343}$ (d) $\frac{7}{72}$

8. Three numbers are to be selected at random without replacement from the set of numbers $\{1, 2, \ldots n\}$. The conditional probability that the third number lies between the first two, if the first number is known to be smaller than the second is :
(a) $\frac{1}{3}$ (b) $\frac{2}{3}$
(c) $\frac{5}{6}$ (d) $\frac{7}{12}$

9. Two numbers b and c are chosen at random with replacement from the numbers 1, 2, 3, 4, 5, 6, 7, 8 and 9. The probability that $x^2 + bx + c > 0$ for all $x \in R$ is :
(a) $\frac{23}{81}$ (b) $\frac{7}{9}$
(c) $\frac{32}{81}$ (d) $\frac{65}{729}$

10. The probabilities that a student passes in Mathematics, Physics and Chemistry are m, p and c, respectively. Of these subjects the student has 75% chance of passing in atleast one, a 50% chance of passing in atleast two and a 40% chance of passing in exactly, two, which of the following relations are true.
(a) $p + m + c = \frac{27}{20}$ (b) $p + m + c = \frac{13}{20}$
(c) $pmc = \frac{1}{10}$ (d) both (a) and (c)

11. A student appears for tests A, B and C. The student is successful if he passes either in tests A and B or tests A and C. The probabilities of the student passing in tests A, B, C are p, q and 1/2 respectively. If the probability that the student is successful is $\frac{1}{2}$ then,
(a) $p = 1, q = 0$
(b) $p = 0, q = 1$
(c) $\frac{p}{q} = 1$
(d) infinitely many solutions

12. If $\frac{1+4p}{p}, \frac{1-p}{4}, \frac{1-2p}{2}$ are probabilities of three mutually exclusive events then
(a) $p = \frac{1}{2}$ (b) $p = \frac{3}{4}$
(c) $p = \frac{1}{3}$ (d) none of these

13. A letter is takenout at random from 'ASSISTANT' and another is taken out from "STATISTICS'. The probability that they are the same letters is :
(a) $\frac{35}{96}$ (b) $\frac{19}{90}$
(c) $\frac{19}{96}$ (d) none of these

14. Two numbers a and b are chosen at random from the set of first 30 natural numbers. The probability that $a^2 - b^2$ is divisible by 3 is :
(a) $\frac{37}{87}$ (b) $\frac{47}{87}$
(c) $\frac{17}{29}$ (d) none of these

15. A man takes a step forward with probability 0.4 and backward with probability 0.6. The probability that at the end of eleven steps he is one step away from the starting point is :
(a) ${}^{11}C_6(0.1)^{11}$ (b) ${}^{11}C_6(0.24)^5$
(c) ${}^{11}C_6(0.2)^{11}$ (d) none of these

16. Three of the six vertices of a regular hexagon are chosen at random. The probability that the triangle with these three vertices is equilateral equals :
(a) $\frac{1}{6}$ (b) $\frac{1}{3}$
(c) $\frac{1}{10}$ (d) none of these

17. A man can take a step forward, backward, left or right with equal probability. Find the probability that after nine steps he will be just one step away from his initial position.
(a) $\frac{3696}{7^4}$ (b) $\frac{3969}{4^7}$
(c) $\frac{4^4}{10}$ (d) none of these

18. Urn A contains six red and four black balls and urn B has four red and six black balls. One ball is drawn at random from urn A and placed in urn B. Then one ball is transfered at random from urn B to urn A. If one ball is now drawn at random from urn A, find the probability that it is red.
(a) $\frac{32}{65}$ (b) $\frac{32}{55}$
(c) $\frac{23}{55}$ (d) $\frac{56}{65}$

19. The digits 1, 2, 3, ..., 9 are written in random order to form a nine digit number. Find the probability that this number is divisible by 11.
(a) $\frac{11}{63}$ (b) $\frac{11}{81}$
(c) $\frac{11}{126}$ (d) none of these

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Answers

INTRODUCTORY EXERCISE- 20.1

1. (a)	2. (a)	3. (b)	4. (b)	5. (c)	6. (d)	7. (d)	8. (a)	9. (b)	10. (c)
11. (d)	12. (d)	13. (c)	14. (b)	15. (a)	16. (a)	17. (b)	18. (c)	19. (b)	20. (c)
21. (c)	22. (d)	23. (b)	24. (c)	25. (c)	26. (b)	27. (d)	28. (a)	29. (b)	30. (c)
31. (c)	32. (c)	33. (c)	34. (b)	35. (a)	36. (a)	37. (b)	38. (b)	39. (c)	40. (a)
41. (b)	42. (b)	43. (c)	44. (c)	45. (c)	46. (a)	47. (c)	48. (d)	49. (a)	50. (b)
51. (a)	52. (b)	53. (c)	54. (a)	55. (b)	56. (c)	57. (b)	58. (c)	59. (a)	

INTRODUCTORY EXERCISE- 20.2

1. (a)	2. (b)	3. (c)	4. (a)	5. (b)	6. (c)	7. (c)	8. (b)	9. (a)	10. (b)
11. (b)	12. (c)	13. (c)	14. (a)	15. (b)	16. (b)	17. (a)	18. (b)	19. (b)	20. (c)
21. (a)	22. (b)	23. (c)	24. (c)	25. (a)	26. (c)	27. (b)	28. (c)	29. (a)	

INTRODUCTORY EXERCISE- 20.3

1. (b)	2. (b)	3. (c)	4. (a)	5. (a)	6. (c)	7. (a)	8. (c)	9. (a)	10. (b)
11. (a)	12. (b)	13. (b)	14. (a)	15. (b)	16. (c)	17. (a)	18. (a)	19. (a)	20. (b)
21. (a)	22. (b)								

INTRODUCTORY EXERCISE- 20.4

1. (a)	2. (b)	3. (b)	4. (c)	5. (d)	6. (b)	7. (a)	8. (a)	9. (c)	10. (a)
11. (b)	12. (c)	13. (c)	14. (b)	15. (c)	16. (b)	17. (a)	18. (a)	19. (b)	20. (c)
21. (b)	22. (a)	23. (b)	24. (c)	25. (b)	26. (c)	27. (c)	28. (a)	29. (a)	30. (b)
31. (a)	32. (b)	33. (c)	34. (b)	35. (b)	36. (d)	37. (b)	38. (b)	39. (b)	40. (a)
41. (b)									

LEVEL-1

1. (b)	2. (b)	3. (b)	4. (a)	5. (b)	6. (c)	7. (b)	8. (d)	9. (a)	10. (c)
11. (a)	12. (b)	13. (b)	14. (b)	15. (c)	16. (a)	17. (b)	18. (c)	19. (a)	20. (b)
21. (c)	22. (d)	23. (a)	24. (b)	25. (b)					

LEVEL-2

1. (a)	2. (b)	3. (b)	4. (c)	5. (c)	6. (b)	7. (b)	8. (a)	9. (c)	10. (d)
11. (d)	12. (a)	13. (b)	14. (b)	15. (b)	16. (c)	17. (b)	18. (b)	19. (c)	

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

Hints & Solutions

INTRODUCTORY EXERCISE 20.1

1. Here the sample space $S = \{H, T\} \Rightarrow n(S) = 2$
Event of getting head $= \{H\} \Rightarrow n(E) = 1$
∴ Probability of getting a head is given by $P(E) = \frac{1}{2}$

Solutions for question number 2-5 :

$S = \{HH, HT, TH, TT\} \Rightarrow n(S) = 4$

2. $E = \{HT, TH\} \Rightarrow n(E) = 2$
$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{2}{4} = \frac{1}{2}$

3. $E = \{HH, HT, TH\} \Rightarrow n(E) = 3$
$\therefore P(E) = \frac{3}{4}$

4. $E = \{H, H\} \Rightarrow n(E) = 1$
$\therefore P(E) = \frac{1}{4}$

5. $E = \{H, H\} \Rightarrow n(E) = 1$
$\therefore P(E) = \frac{1}{4}$

Solutions for question number 6-12 :

$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$
$\Rightarrow n(S) = 8$

6. $E = \{HTT, THT, TTH\} \Rightarrow n(E) = 3$
$\therefore P(E) = \frac{3}{8}$

7. $E = \{HHT, HTH, THH\} \Rightarrow n(E) = 3$
$\therefore P(E) = \frac{3}{8}$

8. $E = \{HHH, HHT, HTH, HTT, THH, THT, TTH\}$
$\Rightarrow n(E) = 7$
$\therefore P(E) = \frac{7}{8}$

9. $E = \{HHT, HTH, THH\} \Rightarrow n(E) = 3$
$\therefore P(E) = \frac{3}{8}$

10. $E = \{HHH, HHT, HTH, THH\} \Rightarrow n(E) = 4$
$\therefore P(E) = \frac{4}{8} = \frac{1}{2}$

11. $E = \{HHT, HTH, HTT, THH, THT, TTH\}$
$\Rightarrow n(E) = 6$
$\therefore P(E) = \frac{6}{8} = \frac{3}{4}$

12. $E = \{HHH, HHT, HTH, THH\} \Rightarrow n(E) = 4$
$\therefore P(E) = \frac{4}{8} = \frac{1}{2}$

Solutions for question number 13-16 :

$S = \{1, 2, 3, 4, 5, 6\} \Rightarrow n(S) = 6$

13. $E = \{1, 2, 3, 4, 5, 6\} \Rightarrow n(E) = 6$
$\therefore P(E) = \frac{6}{6} = 1$

14. $E = \{3, 6\} \Rightarrow n(E) = 2$
$\therefore P(E) = \frac{2}{6} = \frac{1}{3}$

15. $E = \{2, 3, 5\} \Rightarrow n(E) = 3$
$\therefore P(E) = \frac{3}{6} = \frac{1}{2}$

16. $E = \{2, 4, 6\} \Rightarrow n(E) = 3$
$\therefore P(E) = \frac{3}{6} = \frac{1}{2}$

Solution for question number 17 and 18 :

$S = \{HHH, HHT, HTH, THH, TTH, THT, HTT, TTT\}$
$\Rightarrow n(S) = 8$

17. $E = \{HHT, HTH, THH, TTH, THT, HTT\}$
$\Rightarrow n(E) = 6 \quad \therefore P(E) = \frac{6}{8} = \frac{3}{4}$

18. $E = \{TTT\} \Rightarrow n(E) = 1$
$\therefore P(E) = \frac{1}{8}$

Solutions for question number 19-27 :

$S = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), \ldots, (6, 5), (6, 6)\}$
$\Rightarrow n(S) = 6 \times 6 = 36$

19. $E = \{(6, 3), (5, 4), (4, 5), (3, 6)\}$
$n(E) = 4$
$\therefore P(E) = \frac{4}{36} = \frac{1}{9}$

20. $E = \{(6, 4), (5, 5), (4, 6), (6, 5), (5, 6), (6, 6)\}$
$n(E) = 6$
$\therefore P(E) = \frac{6}{36} = \frac{1}{6}$

21. $E = \{(6, 3), (5, 4), (4, 5), (3, 6), (6, 5), (5, 6)\}$
$\Rightarrow n(E) = 6$
$\therefore P(E) = \frac{6}{36} = \frac{1}{6}$

22. $E = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$
$n(E) = 6$
$\therefore P(E) = \frac{6}{36} = \frac{1}{6}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

23. $E = \{(2, 2), (4, 4), (6, 6)\} \Rightarrow n(E) = 3$

$\therefore \quad P(E) = \frac{3}{36} = \frac{1}{12}$

24. $E = \{(2, 3), (2, 6), (4, 3), (4, 6), (6, 3), (6, 6) (3, 2), (6, 2), (3, 4), (6, 4), (3, 6)\}$

$\Rightarrow \quad n(E) = 11$

$\therefore \quad P(E) = \frac{11}{36}$

25. $E = \{(1, 2), (1, 5), (2, 1), (2, 4), (3, 3), (3, 6), (4, 2), (4, 5), (5, 1), (5, 4), (6, 3), (6, 6), (1, 3), (2, 2), (2, 6), (3, 1), (3, 5), (4, 4), (5, 3), (6, 2)\}$

$\Rightarrow \quad n(E) = 20$

$\therefore \quad P(E) = \frac{20}{36} = \frac{5}{9}$

26. $E = \{(1, 1), (1, 2), (1, 4), (1, 6), (2, 1), (2, 3), (2, 5), (3, 2), (3, 4), (4, 1), (4, 3), (5, 2), (5, 6), (6, 1), (6, 5)\}$

$\Rightarrow \quad n(E) = 15$

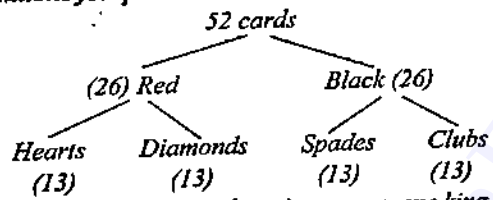
$\therefore \quad P(E) = \frac{15}{36} = \frac{5}{12}$

27. $E = \{(1, 5), (2, 5), (3, 5), (4, 5), (5, 5), (6, 5), (5, 1), (5, 2), (5, 3), (5, 4), (5, 6)\}$

$\Rightarrow \quad n(E) = 11$

$\therefore \quad P(E) = \frac{11}{36}$

✍ ***Solutions for question number 28-35 :*** $S = \{52 \text{ cards}\}$



In each of the four suits there is one ace, one king, one queen and one jack (or knave) and rest 9 cards are numbered.

$\therefore \quad n(S) = 52$

28. $n(E) = 26$

$\therefore \quad P(E) = \frac{26}{52} = \frac{1}{2}$

29. $n(E) = 4$

$\therefore \quad P(E) = \frac{4}{52} = \frac{1}{13}$

30. Since drawn card must be black so there are only two queens.

Hence $n(E) = 2$

$\therefore \quad P(E) = \frac{2}{52} = \frac{1}{26}$

31. There are 26 black cards (including two queens). Besides it there are two more queens (in red colours)

Thus $n(E) = 26 + 2 = 28$

$\therefore \quad P(E) = \frac{28}{52} = \frac{7}{13}$

32. There are 4 kings and 4 queens

$\therefore \quad E = K \cup Q$

$\therefore \quad n(E) = 4 + 4 = 8$

$\therefore \quad P(E) = \frac{8}{52} = \frac{2}{13}$

33. There are 13 hearts (including one queen and one king). Besides it there are 3 queens and 3 kings in remaining 3 suits each.

Thus $n(E) = 13 + 3 + 3 = 19$

$\therefore \quad P(E) = \frac{19}{52}$

34. There are 13 spades (including one king). Besides there are 3 more kings in remaining 3 suits).

Thus $n(E) = 13 + 3 = 16$

Hence $n(\bar{E}) = 52 - 16 = 36$

$\therefore \quad P(\bar{E}) = \frac{36}{52} = \frac{9}{13}$

35. There are 4 aces and 4 kings

$\therefore \quad n(E) = 4 + 4 = 8$

$\therefore \quad n(\bar{E}) = 52 - 8 = 44$

$\therefore \quad P(\bar{E}) = \frac{44}{52} = \frac{11}{13}$

36. There are 4 aces, 4 kings and 4 jacks and their selection can be made in following ways.

$${}^{12}C_1 \times {}^{8}C_1 \times {}^{4}C_1 = 12 \times 8 \times 4$$

$$n(E) = 12 \times 8 \times 4$$

Total selection can be made $= {}^{52}C_3 = 52 \times 51 \times 50$

$$P(E) = \frac{12 \times 8 \times 4}{52 \times 51 \times 50} = \frac{16}{5525}$$

37. $E = \{(1, 1, 1, 1), (2, 2, 2, 2) \ldots\ldots (13, 13, 13, 13)\}$

$\therefore \quad n(E) = 13$

and $n(S) = {}^{52}C_4 = 270725$

$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{13}{270725} = \frac{1}{20825}$

38. $n(E) = {}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1 = (13)^4$

$n(S) = {}^{52}C_4 = 270725$

$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{(13)^4}{270725} = \frac{2197}{20825}$

39. $n(E) = {}^{13}C_1 \times {}^{12}C_1 \times {}^{11}C_1 \times {}^{10}C_1 = 13 \times 12 \times 11 \times 10$

$n(S) = {}^{52}C_4 = 270725$

$\therefore \quad P(E) = \frac{13 \times 12 \times 11 \ 10}{270725} = \frac{264}{4165}$

✍ ***Solutions for question number 40-43 :***

$$n(S) = 6 \times 6 \times 6 \times 6 = 6^4$$

40. $n(E) = 6$

$\because \quad E = \{(1, 1, 1, 1), (2, 2, 2, 2), (3, 3, 3, 3) \ldots (6, 6, 6, 6)\}$

$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{6}{6^4} = \frac{1}{6^3} = \frac{1}{216}$

41. $n(E) = {}^{6}C_1 \times {}^{5}C_1 \times {}^{4}C_1 \times {}^{3}C_1 = 360$

$\therefore \quad P(E) = \frac{360}{6^4} = \frac{5}{18}$

42. Select a number which occurs on two dice out of six numbers (1, 2, 3, 4, 5, 6). This can be done in ${}^{6}C_1$, ways. Now select two distinct number out of remaining 5 numbers which can be done in ${}^{5}C_2$ ways. Thus these 4 numbers can be arranged in $\frac{4!}{2!}$ ways.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

So, the number of ways in which two dice show the same face and the remaining two show different faces is

$$^6C_1 \times {}^5C_2 \times \frac{4!}{2!} = 720$$

$$\therefore \quad n(E) = 720$$

$$\therefore \quad P(E) = \frac{720}{6^4} = \frac{5}{9}$$

43. There are 3 possible cases-
(i) 2 similar faces + 2 different faces
(ii) 3 similar faces + 1 different face
(iii) all 4 faces are similar
∴ Required number of ways

$$= \left({}^6C_1 \times {}^5C_2 \times \frac{4!}{2!}\right) + \left({}^6C_1 \times {}^5C_1 \times \frac{4!}{3!}\right) + \left({}^6C_1 \times \frac{4!}{4!}\right)$$

$$= (6 \times 10 \times 12) + (6 \times 5 \times 4) + 6 = 846$$

$$\therefore \quad n(E) = 846$$

and $\quad n(S) = 6^4$

$$\therefore \quad P(E) = \frac{846}{6^4} = \frac{47}{72}$$

44. $E = \{2, 3, 5, 7, 11, 13, 17, 19\}$

$$\Rightarrow \quad n(E) = 8$$

$S = \{1, 2, 3, 4, \ldots 20\}$

$$\Rightarrow \quad n(S) = 20$$

$$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{8}{20} = \frac{2}{5}$$

45. $S = \{1, 2, 3, 4, \ldots . 18\} \Rightarrow n(S) = 18$

$E_1 = \{2, 4, 6, 8, 10, 12, 14, 16, 18\}$

$$\Rightarrow \quad n(E_1) = 9$$

$E_2 = \{3, 6, 9, 12, 15, 18\} \Rightarrow n(E_2) = 6$

$(E_1 \cap E_2) = E_3 = \{6, 12, 18\} \Rightarrow n(E_3) = 3$

$$\therefore \quad E = E_1 \cup E_2 = E_1 + E_2 - E_3$$

$$n(E) = 9 + 6 - 3$$

$$n(E) = 12$$

where $E = \{2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18\}$

$$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{12}{18} = \frac{2}{3}$$

NOTE For any problem refer set theory and number system.

46. $$n(S) = {}^{100}C_2 = \frac{100 \times 99}{2} = 4950$$

$$n(E) = {}^{25}C_2 = 300$$

$$\therefore \quad P(E) = \frac{300}{4950} = \frac{2}{33}$$

HINT There are total 25 prime numbers upto 100.

47. $$n(S) = {}^{100}C_2 = 4950$$

$$n(E) = {}^{75}C_2 = 2775 \qquad (100 - 25 = 75)$$

$$\therefore \quad P(E) = \frac{2775}{4950} = \frac{37}{66}$$

48. $366 = 7 \times 52 + 2$
It means a leap year has 52 full weeks and 2 more days. These 2 days can be :
(i) Sunday and Monday
(ii) Monday and Tuesday
(iii) Tuesday and Wednesday
(iv) Wednesday and Thursday
(v) Thursday and Friday
(vi) Friday and Saturday
(vii) Saturday and Sunday
Clearly atleast there are 52 Sundays.
Now, for having 53 Sundays in the year, one of the above 2 consecutive, days must be Sunday.
Thus, out of the above 7 possibilities, 2 possibilities are in favour [(i) and (vii)] of the event that one of the two days is a Sunday.

$$\therefore \quad \text{Required probability} = \frac{2}{7}$$

49. $$n(S) = 8 + 4 = 12$$

$$n(E) = 8$$

$$\therefore \quad P(E) = \frac{8}{12} = \frac{2}{3}$$

50. $$n(S) = {}^{12}C_4 = 495$$

$$n(E) = {}^8C_4 = 70$$

$$\therefore \quad P(E) = \frac{70}{495} = \frac{14}{99}$$

51. $$n(S) = {}^{12}C_4 = 495$$

$$n(E) = {}^4C_4 = 1$$

$$\therefore \quad P(E) = \frac{1}{495}$$

52. $$n(S) = {}^{12}C_4 = 495$$

$$n(E) = {}^8C_2 \times {}^4C_1 = 112$$

$$\therefore \quad P(E) = \frac{112}{495}$$

53. $$n(S) = 495$$

$$n(E) = {}^4C_3 = 4$$

$$\therefore \quad P(E) = \frac{4}{495}$$

54. Total number of outcomes $= 2 + 7 = 9$
Favourable number of cases $= 2$

$$\therefore \quad P(E) = \frac{2}{9}$$

55. Total number of outcomes $= 5 + 7 = 12$
Number of cases against the occurrence of event $= 5$
∴ Number of cases in favour of event $= 12 - 5 = 7$

$$\therefore \quad P(E) = \frac{7}{12}$$

56. $S = \{GG, GB, BG, BB\}$

and $\quad E = \{GG, GB, BG\}$

$$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{3}{4}$$

57. $$n(S) = {}^5C_2 = 10$$

$$n(E) = ({}^2C_1 \times {}^3C_1) + ({}^2C_2) = 7$$

$$\therefore \quad P(E) = \frac{7}{10}$$

58. $$n(S) = {}^{20}C_2 = 190$$

$$n(E) = {}^{15}C_2 = 105$$

$$\therefore \quad P(E) = \frac{105}{190} = \frac{21}{38}$$

59. $$n(S) = {}^{20}C_4 = 4845$$

$$n(E) = ({}^5C_3 \times {}^{15}C_1) + ({}^5C_4) = 155$$

$$\therefore \quad P(E) = \frac{155}{4845} = \frac{31}{969}$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

INTRODUCTORY EXERCISE 20.2

1. $P(A)=\frac{1}{4}, P(B)=\frac{1}{2}$ and $P(A\cap B)=\frac{7}{50}$

$\therefore$ $P(A \text{ or } B)=P(A\cup B)$

$=P(A)+P(B)-P(A\cap B)$

$=\frac{1}{4}+\frac{1}{2}-\frac{7}{50}=\frac{61}{100}$

2. P (neither A nor B) $=P(\bar{A}$ and $\bar{B})$

$=P(\bar{A}\cap\bar{B})=P(\overline{A\cup B})=1-P(A\cup B)$

$=1-\frac{61}{100}=\frac{39}{100}$

3. $P(A\cup B)=P(A)+P(B)-P(A\cap B)$

$=\frac{3}{10}+\frac{1}{2}-\frac{1}{5}=\frac{6}{10}=\frac{3}{5}$

4. $P(A\cup B)=P(A)+P(B)-P(A\cap B)$

$\frac{3}{5}=\frac{2}{5}+\frac{1}{2}-P(A\cap B)$

$\Rightarrow$ $P(A\cap B)=\frac{3}{10}$

5. $P(\bar{A})=1-P(A)=1-\frac{2}{5}=\frac{3}{5}$

6. $P(\bar{B})=1-P(B)=1-\frac{1}{2}=\frac{1}{2}$

7. $P(A\cup B)=P(A)+P(B)-P(A\cap B)$

$=\frac{2}{5}+\frac{1}{2}-0=\frac{9}{10}$ $(\because P(A\cap B)=0)$

8. $P(\bar{A}\cap\bar{B})=P(\overline{A\cup B})=1-P(A\cup B)$

$=1-\frac{9}{10}=\frac{1}{10}$

9. $P(\bar{A}\cap B)=P(B)-P(A\cap B)$

$=P(B)=\frac{1}{2}$ $(\because P(A\cap B)=0)$

10. $P(A\cap\bar{B})=P(A)-P(A\cap B)=\frac{2}{5}$

11. $P(A\cup B)=P(A)+P(B)$

$0.65=0.35+P(B)$ $(\because P(A)=1-P(\bar{A}))$

$\Rightarrow$ $P(B)=0.30$

12. Let $P(A)=x$, then $P(B)=\frac{3}{2}x$ and $P(C)=\frac{1}{2}\times\frac{3}{2}x=\frac{3}{4}x$

$\therefore$ $P(A)+P(B)+P(C)=P(A\cup B\cup C)=P(S)=1$

$\therefore$ $x+\frac{3}{2}x+\frac{3}{4}x=1$

$\Rightarrow$ $x=\frac{4}{13}$

$\therefore$ $P(A)=\frac{4}{13}$

13. $n(S)=6\times6=36$

Let A = Event of getting an even number on the first die

$\therefore$ $n(A)=3\times6=18$

and B = Event of getting a total of 8

$n(B)=5$

$A=\begin{Bmatrix}(2,1),(2,2),(2,3)\ldots.(2,6)\\(4,1)\ldots\ldots\ldots\ldots\ldots(4,6)\\(6,1)\ldots\ldots\ldots\ldots\ldots(6,6)\end{Bmatrix}$

$B=\{(6,2),(5,3),(4,4),(3,5),(2,6)\}$

$\therefore$ $n(A)=3\times6=18$

and $n(B)=5$

$\therefore$ $(A\cap B)=\{(2,6),(6,2),(4,4)\}$

$\therefore$ $n(A\cap B)=3$

$P(A)=\frac{18}{36}=\frac{1}{2}$, $P(B)=\frac{5}{36}$

and $P(A\cap B)=\frac{3}{36}=\frac{1}{12}$

$\therefore$ P (Even number on first die or a total of 8)

$=p(A \text{ or } B)=P(A\cup B)$

$=P(A)+P(B)-P(A\cap B)$

$=\frac{1}{2}+\frac{5}{36}-\frac{1}{12}=\frac{5}{9}$

14. $A=\{(5,1),)\ (5,2),(5,3),(5,4),(5,5),(5,6)\}$

$B=\{(1,5),(2,5),(3,5),(4,5),(5,5),(6,5)\}$

$A\cap B=\{(5,5)\}$

Also $n(S)=36$

$\therefore$ $P(A)=\frac{6}{36}=\frac{1}{6}$

$P(B)=\frac{6}{36}=\frac{1}{6}$

and $P(A\cap B)=\frac{1}{36}$

$\therefore$ Required probability $=P(A)+P(B)-P(A\cap B)$

$=\frac{1}{6}+\frac{1}{6}-\frac{1}{36}=\frac{11}{36}$

15. $n(S)=36$

$A=\{(1,1),)\ (2,2),(3,3),(4,4),(5,5),(6,6)\}$

$B=\{(2,6),(3,5),(4,4),(5,3),(6,2)\}$

$(A\cap B)=\{(4,4)\}$

$n(A)=6,\ n(B)=5,\ n(A\cap B)=1$

$\therefore$ Required probability $=P(A\cup B)$

$=P(A)+P(B)-P(A\cap B)$

$=\frac{6}{36}+\frac{5}{36}-\frac{1}{36}$

$=\frac{10}{36}=\frac{5}{18}$

16. See the solution of question number 14

P (atleast one die shows 6) $=\frac{11}{36}$

17. $n(S)=52$

$A\rightarrow$ The event of getting a heart

$B\rightarrow$ The event of getting a king

then $A\cap B\rightarrow$ The event of getting a king of heart.

$\therefore$ $P(A)=\frac{13}{52}=\frac{1}{4}$, $P(B)=\frac{4}{52}=\frac{1}{13}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

and $P(A \cap B) = \frac{1}{52}$

$\therefore$ P (a heart or a king) $= P(A \text{ or } B) = P(A \cup B)$

$= P(A) + P(B) - P(A \cap B)$

$= \frac{1}{4} + \frac{1}{13} - \frac{1}{52} = \frac{4}{13}$

18. $n(S) = {}^{52}C_2 = 1326$

Let A = event of getting both red cards

and B = event of getting both queens

then $A \cap B$ = event of getting two red queens

$n(A) = {}^{26}C_2 = 325, n(B) = {}^{4}C_2 = 6$

$n(A \cap B) = {}^{2}C_2 = 1$

$\therefore$ $P(A) = \frac{325}{1326}$, $P(B) = \frac{6}{1326} = \frac{1}{221}$

$P(A \cap B) = \frac{1}{1326}$

$\therefore P$ (both red or both queens) $= P(A \cup B)$

$= P(A) + P(B) - P(A \cap B)$

$= \frac{325}{1326} + \frac{1}{221} - \frac{1}{1326} = \frac{55}{221}$

19. $n(S) = 52$

Let A, B, C be the events of getting a red card, a heart and a king respectively.

then $n(A) = 26, \quad n(B) = 13, \quad n(C) = 4$

Clearly $n(A \cap B) = 13, \quad n(B \cap C) = 1,$

$n(A \cap C) = 2, \quad n(A \cap B \cap C) = 1$

$\therefore P(A) = \frac{26}{52} = \frac{1}{2}, \quad P(B) = \frac{13}{52} = \frac{1}{4}, \quad P(C) = \frac{4}{52} = \frac{1}{13}$

$P(A \cap B) = \frac{13}{52} = \frac{1}{4}, P(B \cap C) = \frac{1}{52},$

$P(A \cap C) = \frac{2}{52} = \frac{1}{26}$

$P(A \cap B \cap C) = \frac{1}{52}$

$\therefore P$ (a red card, or a heart or a king) $= P(A \cup B \cup C)$

$= P(A) + P(B) + P(C) - (P(A \cap B) + P(B \cap C) + P(A \cap C) + P(A \cap B \cap C)$

$= \frac{1}{2} + \frac{1}{4} + \frac{1}{13} - \left(\frac{1}{4} + \frac{1}{52} + \frac{1}{26}\right) + \frac{1}{52} = \frac{7}{13}$

20. $n(S) = {}^{52}C_4$

Let E_1, E_2, E_3, E_4 be the events of getting all spades, all clubs, all hearts and all diamonds respectively.

Then $n(E_1) = {}^{13}C_4$

$n(E_2) = {}^{13}C_4$

$n(E_3) = {}^{13}C_4$

$n(E_4) = {}^{13}C_4$

$\therefore$ $P(E_1) = \frac{{}^{13}C_4}{{}^{52}C_4}, \quad P(E_2) = \frac{{}^{13}C_4}{{}^{52}C_4},$

$P(E_3) = \frac{{}^{13}C_4}{{}^{52}C_4}, \quad P(E_4) = \frac{{}^{13}C_4}{{}^{52}C_4}$

Since E_1, E_2, E_3 and E_4 are mutually exclusive events.

$\therefore P$ (getting all the 4 cards of the same suit)

$P(E_1 \text{ or } E_2 \text{ or } E_3 \text{ or } E_4) = P(E_1) + P(E_2) + P(E_3) + P(E_4)$

$= 4 \times \left(\frac{{}^{13}C_4}{{}^{52}C_4}\right) = \frac{44}{4165}$

21. $n(S) = 52$

A = The event of getting a king

B = The event of getting a queen

$(A \cap B)$ = The event of getting a king and a queen both

$\therefore$ $n(A) = 4, \quad n(B) = 4, \quad n(A \cap B) = 0$

$P(A) = \frac{4}{52}, \quad P(B) = \frac{4}{52}$

$\therefore$ $P(A \cup B) = P(A) + P(B) = \frac{4}{52} + \frac{4}{52} = \frac{2}{13}$

22. $n(S) = {}^{52}C_2 = 1326$

Let A be the event of getting two black cards

and B be the event of getting two jacks

and $(A \cap B)$ be the event of getting two black jacks.

$\therefore$ $n(A) = {}^{26}C_2, \quad n(B) = {}^{4}C_2, \quad n(A \cap B) = {}^{2}C_2$

$\therefore$ $P(A) = \frac{{}^{26}C_2}{{}^{52}C_2}, \quad P(B) = \frac{{}^{4}C_2}{{}^{52}C_2}, \quad P(A \cap B) = \frac{{}^{2}C_2}{{}^{52}C_2}$

$\therefore$ Required probability $= P(A \cup B)$

$= P(A) + P(B) - P(A \cap B)$

$= \frac{{}^{26}C_2}{{}^{52}C_2} + \frac{{}^{4}C_2}{{}^{52}C_2} - \frac{{}^{2}C_2}{{}^{52}C_2}$

$= \frac{325}{1326} + \frac{1}{221} - \frac{1}{1326} = \frac{55}{221}$

23. $n(S) = 300$

Let A be the event of getting a number divisible by 3 and

B be the event of getting a number divisible by 5 and

$(A \cap B)$ be event of getting a number divisible by 3 and 5 both

$\therefore$ $n(A) = 100, \quad n(B) = 60, \quad n(A \cap B) = 20$

$\therefore$ $P(A) = \frac{100}{300} = \frac{1}{3}, P(B) = \frac{60}{300} = \frac{1}{5}, P(A \cap B) = \frac{20}{600} = \frac{1}{30}$

$\therefore$ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= \frac{1}{3} + \frac{1}{5} - \frac{1}{30} = \frac{1}{2}$

24. $n(S) = 100$

Let A be the event of getting a number divisible by 2 and

B be the event of getting a number divisible by 3 and

C be the event of getting of number divisible by 5.

$\therefore (A \cap B)$ be the event of getting a number divisible by both 2 and 3.

$(B \cap C)$ be the event of getting a number divisible by both 3 and 5.

$(A \cap C)$ be the event of getting a number divisible by both 2 and 5.

$(A \cap B \cap C)$ be the event of getting a number divisible by A, B and C.

Now, $n(A) = 50, \quad n(B) = 33, \quad n(C) = 20, \quad n(A \cap B) = 16,$

$n(B \cap C) = 6, n(A \cap C) = 10, n(A \cap B \cap C) = 3$

$\therefore$ $P(A) = \frac{50}{100} = \frac{1}{2}, \quad P(B) = \frac{33}{100},$

$P(C) = \frac{20}{100} = \frac{1}{5}, \quad n(A \cap B) = \frac{16}{100}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmanboob.wordpress.com
For More Visit : www.LearnEngineering.in

$$n(B \cap C)=\frac{6}{100},\quad n(A \cap C)=\frac{10}{100},$$

$$n(A \cap B \cap C)=\frac{3}{100}$$

Required probability $= P(A \cup B \cup C)$

$$= P(A)+P(B)+P(C)-[P(A \cap B) + P(B \cap C)+P(A \cap C)]+P(A \cap B \cap C)$$

$$=\frac{50}{100}+\frac{33}{100}+\frac{20}{100}-\left(\frac{16}{100}+\frac{6}{100}+\frac{10}{100}\right)+\frac{3}{100}$$

$$=\frac{74}{100}$$

25. $n(S)=5+8+10=23$

$n(A)=5$

$n(B)=8$ $\quad [n(A \cap B)=0]$

$\therefore \quad P(A)=\frac{5}{23}$ and $P(B)=\frac{8}{23}$ $\quad P(A \cap B)=0$

$\therefore \quad P(A \cup B)=P(A)+P(B)=\frac{5}{23}+\frac{8}{23}=\frac{13}{23}$

26. $n(S)={}^{30}C_2$

Let A be the event of getting two oranges and
B be the event of getting two non-defective fruits.
and $(A \cap B)$ be the event of getting two non-defective oranges

$\therefore \quad P(A)=\frac{{}^{20}C_2}{{}^{30}C_2}, P(B)=\frac{{}^{22}C_2}{{}^{30}C_2}$

and $\quad P(A \cap B)=\frac{{}^{15}C_2}{{}^{30}C_2}$

$\therefore \quad P(A \cup B)=P(A)+P(B)-P(A \cap B)$

$$=\frac{{}^{20}C_2}{{}^{30}C_2}+\frac{{}^{22}C_2}{{}^{30}C_2}-\frac{{}^{15}C_2}{{}^{30}C_2}=\frac{316}{435}$$

HINT There are 20 oranges, and $30-(3+5)=22$ non-defective fruits and $20-5=15$ non-defective oranges.

27. $n(S)=100$

$n(A)=40, \quad n(B)=20, \quad n(A \cap B)=5$

$\therefore \quad P(A \cup B)=P(A)+P(B)-P(A \cap B)$

$$=\frac{40}{100}+\frac{20}{100}-\frac{5}{100}$$

$$P(A \cup B)=\frac{55}{100}=55\%$$

28. $P(A)=\frac{2}{5}, \quad P(B)=\frac{1}{3}$

$\therefore \quad P(A \cup B)=\frac{2}{5}+\frac{1}{3}=\frac{11}{15}$

($\because A$ and B are mutually exclusive events)

29. $P(H_1)=\frac{1}{3}, P(H_2)=\frac{1}{4}, P(H_3)=\frac{1}{5}, P(H_4)=\frac{1}{6}$

$\therefore P(H_1 \cup H_2 \cup H_3 \cup H_4)=P(H_1)+P(H_2)+P(H_3)+P(H_4)$

$$=\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\frac{1}{6}=\frac{57}{60}$$

HINT Here, H_1, H_2, H_3 and H_4 are the mutually exclusive events.

INTRODUCTORY EXERCISE 20.3

1. $P\left(\frac{A}{B}\right)=\frac{1}{2}$ (see the example 1)

2. $P(A \cup B)=0.96$ (see the example 1)

3. Here $S=\{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$

Let A be the event that one of the coins shows a tail

$\therefore \quad A=\{HHT, HTH, THH, HTT, THT, TTH, TTT\}$

$\therefore \quad P(A)=\frac{7}{8}$

Now, let B be the event that they are all tails

$\therefore \quad B=\{TTT\}$

$\therefore \quad P(B)=\frac{1}{8}$

$\therefore \quad (A \cap B)=\{TTT\}$

$\Rightarrow \quad P(A \cap B)=\frac{1}{8}$

$\therefore \quad P\left(\frac{B}{A}\right)=\frac{P(A \cap B)}{P(A)}=\frac{1/8}{7/8}=\frac{1}{7}$

4. Here sample space $S=\{HH, HT, TH, TT\}$

$E=\{HT, TH\}$

$F=\{HH, HT, TH\}$

$\therefore \quad E \cap F=\{HT, TH\}$

$\therefore \quad P(E)=\frac{2}{4}=\frac{1}{2}$ and $P(F)=\frac{3}{4}$

and $\quad P(E \cap F)=\frac{2}{4}=\frac{1}{2}$

$\therefore \quad P\left(\frac{E}{F}\right)=\frac{P(E \cap F)}{P(F)}=\frac{\frac{1}{2}}{\frac{3}{4}}=\frac{2}{3}$

$$P\left(\frac{F}{E}\right)=\frac{P(E \cap F)}{P(E)}=\frac{\frac{1}{2}}{\frac{1}{2}}=1$$

5. Here sample space

$S=\{HH, HT, T1, T2, T3, T4, T5, T6\}$

Let A be the event that the die shows a number greater than 4 and B be the event that the first throw of the coin results in a tail then,

$A=\{T5, T6\}$

$B=\{T1, T2, T3, T4, T5, T6\}$

$\therefore$ Required probability $= P\left(\frac{A}{B}\right)=\frac{P(A \cap B)}{P(B)}=\frac{\frac{2}{8}}{\frac{6}{8}}=\frac{1}{3}$

6. $S=\{1, 2, 3, 4, 5, 6\}$

Let A be the event of getting an odd number and
B be the event of getting a prime number

$\therefore \quad A=\{1, 3, 5\}, \quad B=\{2, 3, 5\}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

and $(A \cap B) = \{3, 5\}$

$\therefore P(A) = \frac{3}{6} = \frac{1}{2}, \quad P(B) = \frac{3}{6} = \frac{1}{2}, \quad P(A \cap B) = \frac{2}{6} = \frac{1}{3}$

$\therefore$ Required probability $= P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)}$

$= \frac{1/3}{1/2} = \frac{2}{3}$

7. Let A be the event of getting the sum 9 and B be the event of getting atleast one 4.

Then $A = \{(3, 6), (4, 5), (5, 4), (6, 3)\}$

$B = \{(1, 4), (2, 4), (3, 4), (4, 4), (5, 4), (6, 4), (4, 1), (4, 2), (4, 3), (4, 5), (4, 6)\}$

then $A \cap B = \{(4, 5) (5, 4)\}$

$\therefore$ Required probability $= P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)}$

$= \frac{n(A \cap B)}{n(A)} = \frac{2}{4} = \frac{1}{2}$

8. Let A = the event of getting 4 on the first die.

and B = the event of getting the sum 8 or greater

$\therefore A = \{(4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6)\}$

$B = \{(4, 4), (4, 5), (4, 6), (5, 4), (5, 5), (5, 6), (6, 2), (2, 6), (3, 5), (3, 6), (5, 3), (6, 3), (6, 4), (6, 5), (6, 6)\}$

$\therefore A \cap B = \{(4, 4)\}, (4, 5), (4, 6)\}$

$\therefore$ Required probability $= P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)}$

$= \frac{n(A \cap B)}{n(A)} = \frac{3}{6} = \frac{1}{2}$

9. Let A = event of getting an odd number

and B = the event of getting a number greater than 1.

$\therefore A = \{1, 3, 5\}, B = \{3, 5\}, A \cap B = \{3, 5\}$

$\therefore$ Required probability $= P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{n(A \cap B)}{n(A)} = \frac{2}{3}$

10. Let A be the event of reading English and

B be the event of reading French.

Then $P(A) = \frac{45}{100} = \frac{9}{20}, \quad P(B) = \frac{30}{100} = \frac{3}{10}$

and $P(A \cap B) = \frac{20}{100} = \frac{1}{5}$

$\therefore P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{5}}{\frac{3}{10}} = \frac{2}{3}$

11. $P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{1/5}{9/20} = \frac{4}{9}$

12. $S = (B_1B_2, B_1G_2, G_1B_2, G_1G_2)$

Where $B \rightarrow$ Boy and $G \rightarrow$ Girl

Let A be the event that both are boys

and B be the event that one of two is a boy.

Then $A = \{B_1B_2\}$ and $B = \{B_1B_2, B_1G_2, G_1B_2\}$

So $A \cap B = \{B_1B_2\}$

$\therefore$ Required probability $= P\left(\frac{A}{B}\right) = \frac{n(A \cap B)}{n(B)} = \frac{1}{3}$

13. A be the event that both children are boys

and B be the event that the other child is a boy.

then $A = \{B_1, B_2\}$ and $B = \{B_1B_2, B_1G_2\}$

So $(A \cap B) = \{B_1B_2\}$

$\therefore$ Required probability $= P\left(\frac{A}{B}\right) = \frac{n(A \cap B)}{n(B)} = \frac{1}{2}$

14. $P(E) = \frac{1}{2}$

15. $P(F) = \frac{1}{2}$

16. $P\left(\frac{G}{E}\right)$ = Probability of drawing a red ball when first bag is selected

= Probability of drawing a red ball from first bag

$= \frac{3}{7}$

17. $P\left(\frac{G}{F}\right)$ = Probability of drawing a red ball from second bag

$= \frac{4}{6} = \frac{2}{3}$

18. Let A be the event of not getting a red ball in first draw and B be the event of not getting a red ball in second draw. Then Required probability

= Probability that atleast one ball is red

= 1 – Probability that none is red

$= 1 - P(A \text{ and } B)$

$= 1 - P(A \cap B)$

$= 1 - P(A) \cdot P\left(\frac{B}{A}\right)$

$= 1 - \left(\frac{2}{3} \times \frac{5}{8}\right) = \frac{7}{12}$

Here $P(A) = \frac{6}{9} = \frac{2}{3}$

and $P\left(\frac{B}{A}\right) = \frac{5}{8}$ [There are 5 balls (excluding 3 red balls) after the selection of one non-red ball]

19. Let A be the event drawing 4 red ball in first draw and B be the event of drawing 4 blue balls in the second draw.

Then $P(A) = \frac{{}^6C_4}{{}^{15}C_4} = \frac{15}{1365} = \frac{1}{91}$

$P\left(\frac{B}{A}\right) = \frac{{}^9C_4}{{}^{11}C_4} = \frac{126}{330} = \frac{21}{55}$

Hence, the required probability $= P(A \cap B)$

$= P(A) P\left(\frac{B}{A}\right)$

$= \frac{1}{91} \times \frac{21}{55} = \frac{3}{715}$

20. Let A, B, C be the events of getting a white ball in first, second and third draw respectively, then

Required probability $= P(A \cap B \cap C)$

$= P(A) P\left(\frac{B}{A}\right) P\left(\frac{C}{A \cap B}\right)$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Now, $P(A)$ = Probability of drawing a white ball in first draw

$$= \frac{4}{18} = \frac{2}{9}$$

When a white ball is drawn in the first draw there are 17 balls left in the urn, out of which 3 are white.

$$\therefore \quad P\left(\frac{B}{A}\right) = \frac{3}{17}$$

Since the ball drawn is not replaced, therefore after drawing a white ball in the second draw there are 16 balls left in the urn, out of which 2 are white.

$$\therefore \quad P\left(\frac{C}{A \cap B}\right) = \frac{2}{16} = \frac{1}{8}$$

Hence the required probability $= \frac{2}{9} \times \frac{3}{17} \times \frac{1}{8} = \frac{1}{204}$

21. There are 4 even numbers and 5 odd numbers

Let A = the event of choosing odd numbers

B = the event of getting the sum an even number.

Then $A \cap B$ = The event of choosing odd numbers whose sum is even.

$$\therefore \quad n(B) = {}^4C_2 + {}^5C_2 = 16$$

and $\quad n(A \cap B) = {}^5C_2 = 10$

$$\therefore \quad \text{Required probability} = P\left(\frac{A}{B}\right) = \frac{n(A \cap B)}{n(B)} = \frac{10}{16} = \frac{5}{8}$$

22. Let A be the event of drawing an odd numbered ticket in the first draw and B be the event of drawing an odd numbered ticket in the second draw. Then

$$\text{Required probability} = P(A \cap B) = P(A)\,P\left(\frac{B}{A}\right)$$

$P(A) = \frac{13}{25}$, since there 13 odd numbers 1, 3, 5, ... 25.

Since the ticket drawn in the first draw is not replaced, therefore second ticket drawn is from the remaining 24 tickets, out of which 12 are odd numbered.

$$\therefore \quad P\left(\frac{B}{A}\right) = \frac{12}{24} = \frac{1}{2}$$

Hence, Required probability $= \frac{13}{25} \times \frac{1}{2} = \frac{13}{50}$

INTRODUCTORY EXERCISE 20.4

1. $P(A \cap B) = P(A) \cdot P(B) = 0.6 \times 0.4 = 0.24$

2. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= 0.6 + 0.4 - 0.24 = 0.76$

3. $P(\bar{A} \cap B) = \{1 - P(A)\}\,P(B)$

$= 0.4 \times 0.4 = 0.16$

4. $P(A \cap \bar{B}) = P(A)\{1 - P(B)\}$

$= 0.6 \times 0.6 = 0.36$

5. $P(\bar{A} \cap \bar{B}) = \{1 - P(A)\}\{1 - P(B)\}$

$= 0.4 \times 0.6 = 0.24$

6. $P(\bar{A}) = 0.65 \Rightarrow P(A) = 0.35$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\Rightarrow \quad P(A \cup B) = P(A) + P(B) - P(A) \cdot P(B)$

$\Rightarrow \quad 0.65 = 0.35 + p - 0.35p$

$\Rightarrow \quad 0.65p = 0.30$

$\Rightarrow \quad p = \frac{6}{13}$

7. $S = \{1, 2, 3, 4, 5, 6\}$, for each case

Let A = event of getting a 1,2, 3 or 4 on the first toss

and B = event of getting a 4, 5 or 6 on the second toss

Then, clearly A and B are independent events.

$$\therefore \quad P(A) = \frac{4}{6} = \frac{2}{3} \quad \text{and} \quad P(B) = \frac{3}{6} = \frac{1}{2}$$

So, required probability $= P(A \cap B)$

$$= P(A) \cdot P(B) = \frac{2}{3} \times \frac{1}{2} = \frac{1}{3}$$

8. Let E = the event that A gets a three

and F = the event that B gets a three

Then, $\quad P(E) = \frac{1}{6}, \quad P(F) = \frac{1}{6}$

$$P(\bar{E}) = \frac{5}{6}, \quad P(\bar{F}) = \frac{5}{6}$$

Suppose A wins then, he gets a three in 1st or 3rd of 5th ... throw etc.

$\therefore P$ (A wins) $= P[E$ or $(\bar{E}\,\bar{F}\,E)$ or $(\bar{E}\,\bar{F}\,\bar{E}\,\bar{F}\,E)$ or ... $\infty]$

$= P[E$ or $(\bar{E}$ and $\bar{F}$ and $E)$ or $(\bar{E}$ and $\bar{F}$ and $\bar{E}$ and $\bar{F}$ and $E)$ or ....]

$= P(E) + P(\bar{E}$ and $\bar{F}$ and $E) + P(\bar{E}$ and $\bar{F}$ and $\bar{E}$ and $\bar{F}$ and $E)$ + ... ∞

$= P(E) + P(\bar{E})\,P(\bar{F})\,P(E) + P(\bar{E})\,P(\bar{F})$ $\cdot P(\bar{E})\,P(\bar{F}) \cdot P(E)$ + ... ∞

$$= \frac{1}{6} + \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} + \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} + \ldots \infty$$

$$= \frac{1}{6} + \frac{1}{6}\left(\frac{5}{6}\right)^2 + \frac{1}{6}\left(\frac{5}{6}\right)^4 + \ldots \infty$$

$$= \frac{1}{6}\left[1 + \left(\frac{5}{6}\right)^2 + \left(\frac{5}{6}\right)^4 + \ldots \infty\right]$$

$$= \frac{1}{6} \cdot \frac{1}{\left[1 - \left(\frac{5}{6}\right)^2\right]} = \left(\frac{1}{6} \cdot \frac{36}{11}\right) = \frac{6}{11}$$

Thus, P (A wins) $= \frac{6}{11}$ and $P(B$ wins) $= \frac{5}{11}$

9. We have, $P(H) = \frac{1}{2}$ and $P(T) = \frac{1}{2}$

Now, A wins if he throws a head in 1st, or 3rd or 5th or ... draw.

$\therefore P(A$ wins) $= P[H$ or $(T\,TH)$ or $(T\,T\,T\,TH)$ or $(T\,T\,T\,T\,T\,TH)$ or ...]

$= P(H) + P(T\,TH) + P(T\,T\,T\,TH) + \ldots$

$= P(H) + P(T)\,P(T)\,P(H) + P(T)\,P(T)\,P(T)$ $P(T)\,P(H) + \cdots$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$$= \frac{1}{2} + \left(\frac{1}{2}\right)^2 \frac{1}{2} + \left(\frac{1}{2}\right)^4 \frac{1}{2} + \ldots \infty$$

$$= \frac{1}{2} + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^5 + \ldots \infty$$

$$= \frac{1}{2}\left[1 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^4 + \ldots \infty\right]$$

$$= \frac{1}{2} \cdot \frac{1}{\left[1 - \left(\frac{1}{2}\right)^2\right]} = \left(\frac{1}{2} \times \frac{4}{3}\right) = \frac{2}{3}$$

Thus $P(A \text{ wins}) = \frac{2}{3}$ and $P(B \text{ wins}) = \left(1 - \frac{2}{3}\right) = \frac{1}{3}$

10. Let E be the event of 'getting a six'. Then $\bar{E}$ is the event of 'not getting a six'.

Then $P(E) = \frac{1}{6}$ and $P(\bar{E}) = \frac{5}{6}$

Now, A wins if he throws a six in 1st, 4th or 7th or ... draw.

$\therefore P(A \text{ wins}) = P(E \text{ or } (\bar{E}\,\bar{E}\,\bar{E}\,E) \text{ or } (\bar{E}\,\bar{E}\,\bar{E}\,\bar{E}\,\bar{E}\,\bar{E}\,E) \text{ or } \ldots]$

$$= P(E) + P(\bar{E}\,\bar{E}\,\bar{E}\,E) + P(\bar{E}\,\bar{E}\,\bar{E}\,\bar{E}\,\bar{E}\,\bar{E}\,E) + \ldots$$

$$= \frac{1}{6} + \left(\frac{5}{6}\right)^3 \cdot \frac{1}{6} + \left(\frac{5}{6}\right)^6 \cdot \frac{1}{6} + \ldots \infty$$

$$= \frac{1}{6}\left[1 + \left(\frac{5}{6}\right)^3 + \left(\frac{5}{6}\right)^6 + \ldots \infty\right]$$

$$= \frac{1}{6} \cdot \frac{1}{\left[1 - \left(\frac{5}{6}\right)^3\right]} = \left(\frac{1}{6} \cdot \frac{216}{91}\right) = \frac{36}{91}$$

Now, B wins if he throws a six in 2nd or 5th or 8th or ..draw.

$\therefore P(B \text{ wins}) = P[(\bar{E}E) \text{ or } (\bar{E}\,\bar{E}\,\bar{E}\,\bar{E}\,E) \text{ or } (\bar{E}\,\bar{E}\,\bar{E}\,\bar{E}\,\bar{E}\,\bar{E}\,\bar{E}\,E) \text{ or } \ldots]$

$$= P(\bar{E}E) + P(\bar{E}\,\bar{E}\,\bar{E}\,\bar{E}\,E) + P(\bar{E}\,\bar{E}\,\bar{E}\,\bar{E}\,\bar{E}\,\bar{E}\,\bar{E}\,E) + \ldots$$

$$= \frac{5}{6} \cdot \frac{1}{6} + \left(\frac{5}{6}\right)^4 \cdot \frac{1}{6} + \left(\frac{5}{6}\right)^7 \cdot \frac{1}{6} + \ldots \infty$$

$$= \frac{5}{36}\left[1 + \left(\frac{5}{6}\right)^3 + \left(\frac{5}{6}\right)^6 + \ldots \infty\right]$$

$$= \frac{5}{36} \cdot \frac{1}{\left[1 - \left(\frac{5}{6}\right)^3\right]} = \left(\frac{5}{36} \cdot \frac{216}{91}\right) = \frac{30}{91}$$

Thus, $P(A \text{ wins}) = \frac{36}{91}$ and $P(B \text{ wins}) = \frac{30}{91}$

$\therefore P(A \text{ wins or } B \text{ wins}) = P(A \text{ wins}) + P(B \text{ wins})$

$$= \frac{36}{91} + \frac{30}{91} = \frac{66}{91}$$

and $P(C \text{ wins}) = 1 - P(A \text{ wins or } B \text{ wins})$

$$= 1 - \frac{66}{91} = \frac{25}{91}$$

11. Let E = Event of getting a sum 9 on two dice.

Then, $E = \{(3, 6), (4, 5), (5, 4), (6, 3)\}$

$\therefore PE = \frac{4}{36} = \frac{1}{9}$ and $P(\bar{E}) = \frac{8}{9}$

$\therefore P(A \text{ wins}) = P[E_1 \text{ or } \bar{E}_1 \bar{E}_2 E_3 \text{ or } \bar{E}_1 \bar{E}_2 \bar{E}_3 \bar{E}_4 E_5 \text{ or } \ldots]$

$$= \left[\frac{1}{9} + \frac{8}{9} \cdot \frac{8}{9} \cdot \frac{1}{9} + \frac{8}{9} \cdot \frac{8}{9} \cdot \frac{8}{9} \cdot \frac{8}{9} \cdot \frac{1}{9} + \ldots \infty\right]$$

$$= \frac{9}{17}$$

$\therefore$ $P(B \text{ wins}) = \left(1 - \frac{9}{17}\right) = \frac{8}{17}$

$\therefore$ Required ratio $= \frac{9}{8}$

12. Required probability $= \frac{4}{52} \times \frac{3}{51} = \frac{1}{221}$

13. Let A = the event that Ashmit solves the problem.
and B = the event that Amisha solves the problem.
Clearly, A and B are independent events.

Now, $P(A) = \frac{80}{100} = \frac{8}{10}$

and $P(B) = \frac{70}{100} = \frac{7}{10}$

$\therefore$ $P(A \cap B) = P(A) \cdot P(B) = \frac{8}{10} \times \frac{7}{10} = \frac{56}{100}$

So $P(A \text{ or } B) = P(A \cup B)$

$$= P(A) + P(B) - P(A \cap B)$$

$$= \frac{8}{10} + \frac{7}{10} - \frac{56}{100}$$

$$= \frac{94}{100} = 0.94$$

14. Let A = the event that A hits the target
and B = the event that B hits the target
As given, we have $P(A) = \frac{1}{3}$ and $P(B) = \frac{2}{5}$

Clearly, A and B are independent events

$\therefore$ $P(A \cap B) = P(A) \cdot P(B) = \frac{1}{3} \times \frac{2}{5} = \frac{2}{15}$

$\therefore$ P (target is hit) = $P(A \text{ hits or } B \text{ hits})$

$$= P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{1}{3} + \frac{2}{5} - \frac{2}{15} = \frac{3}{5}$$

15. Let A, B, C be the respective events of solving the problem and $\bar{A}, \bar{B}, \bar{C}$ be the respective events of not solving the problem. Then A, B, C are independent events
$\therefore \bar{A}, \bar{B}, \bar{C}$ are independent events

Now, $P(A) = \frac{1}{2}$ $P(B) = \frac{1}{3}$ and $P(C) = \frac{1}{4}$

$P(\bar{A}) = \frac{1}{2}$, $P(\bar{B}) = \frac{2}{3}$ and $P(\bar{C}) = \frac{3}{4}$

$\therefore P(\text{none solves the problem})$

$= P(\text{not } A) \text{ and } (\text{not } B) \text{ and } (\text{not } C)]$

$= P(\bar{A} \cap \bar{B} \cap \bar{C})$

$= P(\bar{A})\,P(\bar{B})\,P(\bar{C})$ ($\because \bar{A}, \bar{B}$ and $\bar{C}$ are independent)

$= \frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} = \frac{1}{4}$

Hence, P(the problem will be solved)

$= 1 - P$ (none solves the problem)

$= 1 - \frac{1}{4} = \frac{3}{4}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

16. Let A, B, C respectively.
Then $\bar{A}, \bar{B}, \bar{C}$ are the respective events of not solving the problem by them

Now, $P(A)=\frac{1}{3}$, $P(B)=\frac{2}{7}$ and $P(C)=\frac{3}{8}$

$\therefore$ $P(\bar{A})=\frac{2}{3}$, $P(\bar{B})=\frac{5}{7}$ and $P(\bar{C})=\frac{5}{8}$

$\therefore$ The probability that exactly one of them will solve it

$= P\{[A \cap (\text{not } B) \cap (\text{not } C)]$ or $[(\text{not } A) \cap B \cap (\text{not } C)]$ or $[(\text{not } A) \cap (\text{not } B) \cap C]\}$

$= P\{(A \cap \bar{B} \cap \bar{C}) \cup (\bar{A} \cap B \cap \bar{C}) \cup (\bar{A} \cap \bar{B} \cap C)\}$

$= P(A \cap \bar{B} \cap \bar{C}) + P(\bar{A} \cap B \cap \bar{C}) + P(\bar{A} \cap \bar{B} \cap C)$

$[\because (A \cap \bar{B} \cap \bar{C}) \cap (\bar{A} \cap B \cap \bar{C}) \cap (\bar{A} \cap \bar{B} \cap C) = \phi]$

$= P(A)P(\bar{B})P(\bar{C}) + P(\bar{A}).P(B).P(\bar{C}) + P(\bar{A}).P(\bar{B}).P(C)$

$= \left(\frac{1}{3}\times\frac{5}{7}\times\frac{5}{8}\right) + \left(\frac{2}{3}\times\frac{2}{7}\times\frac{5}{8}\right) + \left(\frac{2}{3}\times\frac{5}{7}\times\frac{3}{8}\right)$

$= \frac{25}{168} + \frac{5}{42} + \frac{5}{28} = \frac{25}{56}$

17. Let A, B, C be the events that A hits the target, B hits the target and C hits the target respectively.

Then, $P(A)=\frac{4}{5}$, $P(B)=\frac{3}{4}$, $P(C)=\frac{2}{3}$

$P(\bar{A})=\frac{1}{5}$, $P(\bar{B})=\frac{1}{4}$, $P(\bar{C})=\frac{1}{3}$

Case 1. $P(A, B,$ and $C,$ all hit the target$) = P(A \cap B \cap C)$

$= P(A)P(B)P(C) = \frac{4}{5}\times\frac{3}{4}\times\frac{2}{3} = \frac{2}{5}$

Case II. $P(A$ and B hit but not $C) = P[A \cap B \cap \bar{C}]$

$= P(A)P(B)P(\bar{C})$

$= \frac{4}{5}\times\frac{3}{4}\times\frac{1}{3} = \frac{1}{5}$

Case III. $P(A$ and C hit but not $B) = P(A \cap C \cap \bar{B})$

$= P(A)P(C)P(\bar{B})$

$= \frac{4}{5}\times\frac{2}{3}\times\frac{1}{4} = \frac{2}{15}$

Case IV. $P(B$ and C hit but not $A) = P(B \cap C \cap \bar{A})$

$= P(B)P(C)P(\bar{A})$

$= \frac{3}{4}\times\frac{2}{3}\times\frac{1}{5} = \frac{1}{10}$

All the above cases being mutually exclusive,we have the required probability

$= \frac{2}{5} + \frac{1}{5} + \frac{2}{15} + \frac{1}{10} = \frac{5}{6}$

18. Let $P_1 = 0.1, P_2 = 0.2, P_3 = 0.3, P_4 = 0.4$

$\therefore P$ (The balloon is hit) $= P$ (the balloon is hit atleast once)

$= 1 - P$(the balloon is hit in none of the shots)

$= 1 - (1-P_1)(1-P_2)(1-P_3)(1-P_4)$

$= 1 - (0.9)(0.8)(0.7)(0.6) = 0.6976$

19. $P(A)=\frac{2}{3}$, $P(B)=\frac{3}{5}$

Required probability $= P(A$ or $B) = P(A \cup B)$ be the events of solving the problem by A, B, C

$= P(A) + P(B) - P(A)P(B)$

$= \frac{2}{3} + \frac{3}{5} - \frac{2}{3}\cdot\frac{3}{5} = \frac{13}{15}$

20. $P(\text{none}) = 1 - P(\text{atleast one})$

$= 1 - P(A \cup B) = 1 - \frac{13}{15} = \frac{2}{15}$

21. $P(\text{not } A) = 1 - P(A) = 1 - 0.4 = 0.6$

22. $P(\text{at most } C) = P(D \text{ or } C) = P(D) + P(C)$

$= (0.1) + (0.2) = 0.3$

23. $P(B \text{ or } C) = P(B) + P(C)$

$= (0.3) + (0.2) = 0.5$

24. $P(\text{atleast } B) = P(B \text{ or } A) = P(B) + P(A)$

$= (0.3) + (0.4) = 0.7$

Solutions for question number 25-28 :

Let E = the event that Ajay is selected
and F = the event that Kajol is selected
Clearly, E and F are independent events

Now, $P(E)=\frac{1}{7}$ and $P(F)=\frac{1}{5}$

$\therefore$ $P(\bar{E})=\frac{6}{7}$ *and* $P(\bar{E})=\frac{4}{5}$

25. $P(\text{Both of them will be selected}) = P(E \text{ and } F) = P(E \cap F)$

$= P(E)P(F) = \frac{1}{7}\times\frac{1}{5} = \frac{1}{35}$

26. $P(\text{only one of them will be selected})$

$= P[(E \text{ and not } F) \text{ or } (F \text{ and not } E)]$

$= P[(E \cap \bar{F}) \cup (F \cap \bar{E})]$

$= P(E \cap \bar{F}) + P(F \cap \bar{E})$

$= P(E)P(\bar{F}) + P(F)P(\bar{E})$

$= \frac{1}{7}\times\frac{4}{5} + \frac{1}{5}\times\frac{6}{7} = \frac{2}{7}$

27. $P(\text{none of them will be selected})$

$= P[(\text{not } E) \text{ and } (\text{not } F)]$

$= P(\bar{E} \cap \bar{F}) = P(\bar{E}) \times P(\bar{F})$

$= \frac{6}{7}\times\frac{4}{5} = \frac{24}{35}$

28. $P(\text{atleast one of them will be selected})$

$= 1 - P(\text{none will be selected})$

$= 1 - \frac{24}{35} = \frac{11}{35}$

29. Let E = the event that A speaks the truth
and F = the event that B speaks the truth
then $\bar{E}$ = the event that A tells a lie
and $\bar{F}$ = the event that B tells a lie.
Clearly, E and F are independent events, So, E and $\bar{F}$ as well as $\bar{E}$ and F are independent.

Now, $P(E)=\frac{60}{100}=\frac{3}{5}$, $P(F)=\frac{80}{100}=\frac{4}{5}$

$\therefore$ $P(\bar{E})=\frac{2}{5}$, $P(\bar{F})=\frac{1}{5}$

$\therefore P(A$ and B contradict each other$)$

$= P(A$ speaks the truth and B tells a lie$)$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Probability

or (A tells a lie and B speaks the truth)

$$= P[(E \cap \bar{F}) \cup (\bar{E} \cup F)]$$
$$= P(E \cap \bar{F}) + P(\bar{E} \cap F)$$
$$= P(E)P(\bar{F}) + P(\bar{E})P(F)$$
$$= \frac{3}{5} \times \frac{1}{5} + \frac{2}{5} \times \frac{4}{5}$$
$$= \frac{11}{25} \Rightarrow 44\%$$

So A and B contradict each other in 44% cases.

Solutions for question number 30 and 31 : Let E = the event that the husband will be alive 20 years hence and F = the event that the wife will be alive 20 years hence.

Then $P(E) = \frac{5}{12}$ *and* $P(F) = \frac{3}{8}$

$\therefore$ $P(\bar{E}) = \frac{7}{12}$ *and* $P(\bar{F}) = \frac{5}{8}$

Clearly, E and F are independent events.

30. P(Couple will be alive 20 years hence)

$$= P(E \text{ and } F) = P(E \cap F)$$
$$= P(E).P(F) = \frac{5}{12} \times \frac{3}{8} = \frac{5}{32}$$

31. P(atleast one of them will be alive 20 years hence)

$$= 1 - P(\text{none will be alive 20 years hence})$$
$$= 1 - P(\bar{E} \cap \bar{F})$$
$$= 1 - P(\bar{E})P(\bar{F}) \qquad (\because \bar{E} \text{ and } \bar{F} \text{ are independent})$$
$$= 1 - \left(\frac{7}{12} \times \frac{5}{8}\right) = \frac{61}{96}$$

32. Let A, B, C denote the events of favouring the book by the first, the second and the third critic respectively.

Then $P(A) = \frac{5}{7}$, $P(B) = \frac{4}{7}$ and $P(C) = \frac{3}{7}$

$\therefore$ $P(\bar{A}) = \frac{2}{7}$, $P(\bar{B}) = \frac{3}{7}$ and $P(\bar{C}) = \frac{4}{7}$

$\therefore$Required probability

$= P$ (two favour the book or three favour the book)

$= P$ (two favour the book) + P (three favour the book)

$= P[\{A \text{ and } B \text{ (not } C)\} \text{ or } \{A \text{ and (not } B) \text{ and } C\}$ or $\{(\text{not } A) \text{ and } B \text{ and } C\}] + P(A \text{ and } B \text{ and } C)$

$$= P[(A \cap B \cap \bar{C}) \cup P(A \cap \bar{B} \cap C) \cup (\bar{A} \cap B \cap C)] + P(A \cap B \cap C)$$
$$= P(A)P(B)P(\bar{C}) + P(A)P(\bar{B})P(C) + P(\bar{A})P(B)P(C) + P(A)P(B)P(C)$$
$$= \left(\frac{5}{7} \times \frac{4}{7} \times \frac{4}{7}\right) + \left(\frac{5}{7} \times \frac{3}{7} \times \frac{3}{7}\right) + \left(\frac{2}{7} \times \frac{4}{7} \times \frac{3}{7}\right) + \left(\frac{5}{7} \times \frac{4}{7} \times \frac{3}{7}\right)$$
$$= \frac{209}{343}$$

33. Required probability

$$= P(X \text{ not defective and } Y \text{ not defective})$$
$$= P(\bar{X})P(\bar{Y})$$
$$= \{1 - P(X)\}\{1 - P(Y)\}$$
$$= \frac{91}{100} \times \frac{95}{100} = \frac{8645}{10000} = 0.8645$$

34. A red ball can be drawn in two mutually exclusive ways

(i) Selecting bag I and then drawing a red ball from it.

(ii) Selecting bag II and then drawing a red bal form it.

Let E_1, E_2 and A denote the events defined as follows :

E_1 = selecting bag I,

E_2 = selecting bag II

A = drawing a red ball.

Since one of the two bags is selected randomly, therefore

$$P(E_1) = \frac{1}{2} \quad \text{and} \quad P(E_2) = \frac{1}{2}.$$

Now, $P\left(\frac{A}{E_1}\right)$ = probability of drawing a red ball when the first bag has been chosen $= \frac{4}{7}$

$P\left(\frac{A}{E_2}\right)$ = probability of drawing a red ball when the second bag has been selected $= \frac{2}{6}$

Using the law of total probability, we have

$$P(\text{red ball}) = P(A) = P(E_1)P\left(\frac{A}{E_1}\right) + P(E_2)P\left(\frac{A}{E_2}\right)$$
$$= \frac{1}{2} \times \frac{4}{7} + \frac{1}{2} \times \frac{2}{6} = \frac{19}{42}$$

35. Let E_1, E_2, E_3 and A be the events defined as follows.

E_1 = the toy is manufactured by machine A

E_2 = the toy is manufactured by machine B

E_3 = the toy is manufactured by machine C

and A = the toy is defective then

$$P(E_1) = \frac{25}{100} = \frac{1}{4}, \quad P(E_2) = \frac{35}{100} = \frac{7}{20}$$
$$P(E_3) = \frac{40}{100} = \frac{2}{5}$$

and $P\left(\frac{A}{E_1}\right)$ = Probability that the toy drawn is defective given the condition that it is a manufactured by machine $A = \frac{5}{100}$

Similarly, $P\left(\frac{A}{E_2}\right) = \frac{4}{100}$ and $P\left(\frac{A}{E_3}\right) = \frac{2}{100}$

Using law of total probability, we have

$$P(A) = P(E_1)P\left(\frac{A}{E_1}\right) + P(E_2)P\left(\frac{A}{E_2}\right) + P(E_3)P\left(\frac{A}{E_3}\right)$$
$$= \frac{25}{100} \times \frac{5}{100} + \frac{35}{100} \times \frac{4}{100} + \frac{40}{100} \times \frac{2}{100}$$
$$= 0.345$$

36. The testing procedure may terminate at the twelfth testing in two mutually exclusive ways

(i) When lot contain 2 defective bulbs.

(ii) When lot contains 3 defective bulbs .

Consider the following events :

A = testing procedure ends the twelfth testing

E_1 = lot contains 2 defective bulbs.

E_2 = lot contains 3 defective bulbs.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

your-shahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Required Probability $= P(A)$

$$= P(A \cap E_1) \cap P(A \cap E_2)$$
$$= P(A \cap E_1) + P(A \cap E_2)$$
$$= P(E_1)P\left(\frac{A}{E_1}\right) + P(A_2)P\left(\frac{A}{E_2}\right)$$

Now, $P\left(\frac{A}{E_1}\right)$ = Probability that first 11 draws contain 10 non defective and one defective and 12th draw contains a defective article.

$$= \frac{{}^{18}C_{10} \times {}^{2}C_1}{{}^{20}C_{11}} \times \frac{1}{9}$$

and $P\left(\frac{A}{E_2}\right)$ = Probability that first 11 draws contain 9 non-defective and 2 defective articles and twelfth draw contains a defective article.

$$= \frac{{}^{17}C_9 \times {}^{3}C_2}{{}^{20}C_{11}} \times \frac{1}{9}$$

Hence, Required probability

$$= 0.4 \times \frac{{}^{18}C_{10} \times {}^{2}C_1}{{}^{20}C_{11}} \times \frac{1}{9} + 0.6 \times \frac{{}^{17}C_9 \times {}^{3}C_2}{{}^{20}C_{11}} \times \frac{1}{9}$$

37. Let E_1, E_2, E_3 and A be the events defined as follows :

E_1 = the toy is manufactured by machine A

E_2 = the toy is manufactured by machine B

E_3 = the toy is manufactured by machine C

A = the toy is defective

Then $P(E_1)$ = probability that the toy drawn is manufactured by machine $A = \frac{25}{100}$

Similarly, $P(E_2) = \frac{35}{100}$ and $P(E_3) = \frac{40}{100}$

$P\left(\frac{A}{E_1}\right)$ = probability that the toy drawn is defective given that it is manufactured by machine $A = \frac{5}{100}$

Similarly $P\left(\frac{A}{E_2}\right) = \frac{4}{100}$, $P\left(\frac{A}{E_3}\right) = \frac{2}{100}$

Now, required probability

$$= \frac{P(E_2)P\left(\frac{A}{E_2}\right)}{P(E_1)P\left(\frac{A}{E_1}\right) + P(E_2)P\left(\frac{A}{E_2}\right) + P(E_3)P\left(\frac{A}{E_3}\right)}$$

$$= \frac{\frac{35}{100} \times \frac{4}{100}}{\frac{25}{100} \times \frac{5}{100} + \frac{35}{100} \times \frac{4}{100} + \frac{40}{100} \times \frac{2}{100}} = \frac{28}{69}$$

38. Let E_1, E_2, E_3 and A be the events defined as follows :

E_1 = construction chosen is a bridge.

E_2 = construction chosen is a hospital

E_3 = construction chosen is a hotel.

A = construction gets damaged

Since there are 1200 constructions, therefore

$$P(E_1) = \frac{200}{1200} = \frac{1}{6}, \quad P(E_2) = \frac{400}{1200} = \frac{1}{3}$$

and $$P(E_3) = \frac{600}{1200} = \frac{1}{2}$$

It is given that $P\left(\frac{A}{E_1}\right)$ =Probability that a construction gets damaged is a bridge = 0.01

Similarly, $P\left(\frac{A}{E_2}\right) = 0.03$ and $P\left(\frac{A}{E_3}\right) = 0.15$

We are required to find $P\left(\frac{E_1}{A}\right)$ by Baye's rule.

$$P\left(\frac{E_1}{A}\right) = \frac{P(E_1)P\left(\frac{A}{E_1}\right)}{P(E_1)P\left(\frac{A}{E_1}\right) + P(E_2)P\left(\frac{A}{E_2}\right) + P(E_3)P\left(\frac{A}{E_3}\right)}$$

$$= \frac{\frac{1}{6} \times 0.01}{\frac{1}{6} \times 0.01 + \frac{1}{3} \times 0.03 + \frac{1}{2} \times 0.15}$$

$$= \frac{1}{1 + 6 + 45} = \frac{1}{52}$$

39. Let E_1, E_2 and A be the events defined as follows :

E_1 = selecting a box from the first group

E_2 = selecting a box from the second group and

A = ball drawn is green

Since there are 5 boxes out of which 3 boxes belong to first group and 2 boxes belong the second group. Therefore

$$P(E_1) = \frac{3}{5}, P(E_2) = \frac{2}{5}$$

If E_1 has already occurred, then a box from the first group is chosen. The box chosen contains 5 green balls and 3 red balls.

Therefore the probability of drawing a green ball from it is $\frac{5}{8}$.

So $$P\left(\frac{A}{E_1}\right) = \frac{5}{8}$$

Similarly, $$P\left(\frac{A}{E_2}\right) = \frac{2}{6} = \frac{1}{3}$$

Now, we have to find $P\left(\frac{E_1}{A}\right)$

By Baye's rule, we have

$$P\left(\frac{E_1}{A}\right) = \frac{P(E_1)P\left(\frac{A}{E_1}\right)}{P(E_1)P\left(\frac{A}{E_1}\right) + P(E_2)P\left(\frac{A}{E_2}\right)}$$

$$= \frac{\frac{3}{5} \times \frac{5}{8}}{\frac{3}{5} \times \frac{5}{8} + \frac{2}{5} \times \frac{1}{3}} = \frac{45}{61}$$

40. Let E_1, E_2 and A be the events defined as follows :

E_1 = six occurs, E_2 = six does not occur

and A = the man reports that it is a six.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

We have, $P(E_1)=\frac{1}{6}, P(E_2)=\frac{5}{6}$

Now $P\left(\frac{A}{E_1}\right)$ = probability that the man reports that there is a six on the die given that six has occurred on the die.

= probability that the man speaks truth $=\frac{3}{4}$

and $P\left(\frac{A}{E_2}\right)$ = probability that the man reports that there is six on the die given that six has not occurred on the die.

= probability that the man does not speak truth

$=1-\frac{3}{4}=\frac{1}{4}$

We have to find $P\left(\frac{E_1}{A}\right)$

By Baye's rule, we have

$$P\left(\frac{E_1}{A}\right)=\frac{P(E_1)P\left(\frac{A}{E_1}\right)}{P(E_1)P\left(\frac{A}{E_1}\right)+P(E_2)P\left(\frac{A}{E_2}\right)}$$

$$=\frac{\frac{1}{6}\times\frac{3}{4}}{\frac{1}{6}\times\frac{3}{4}+\frac{5}{6}\times\frac{1}{4}}=\frac{3}{8}$$

41. Let E_1, E_2, E_3, E_4 and A be the events as defined below :

E_1 = the missing card is diamond

E_2 = the missing card is heart

E_3 = the missing card is spade

E_4 = the missing card is club

A = drawing two diamonds cards from the remaining cards

Then $P(E_1)=\frac{13}{52}=\frac{1}{4}$, $P(E_2)=\frac{13}{52}=\frac{1}{4}$

$P(E_3)=\frac{13}{52}=\frac{1}{4}$

and $P(E_4)=\frac{13}{52}=\frac{1}{4}$

$P\left(\frac{A}{E_1}\right)$ = probability of drawing two diamonds cards given that one diamond card is missing $=\frac{^{12}C_2}{^{51}C_2}$

$P\left(\frac{A}{E_2}\right)$ = probability of drawing two diamond cards given that one heart card is missing.

$=\frac{^{13}C_2}{^{51}C_2}$

Similarly, $P\left(\frac{A}{E_3}\right)=\frac{^{13}C_2}{^{51}C_2}$

and $P\left(\frac{A}{E_4}\right)=\frac{^{13}C_2}{^{31}C_2}$

By, Baye's rule

Required probability $=P\left(\frac{E_1}{A}\right)$

$$=\frac{P(E_1)P\left(\frac{A}{E_1}\right)}{P(E_1)P\left(\frac{A}{E_1}\right)+P(E_2)P\left(\frac{A}{E_2}\right)+P(E_3)P\left(\frac{A}{E_3}\right)+P(E_4)P\left(\frac{A}{E_4}\right)}$$

$=\frac{11}{50}$

LEVEL 1

1. Total possible nine digit numbers = 9!
Out of these 9! numbers only those numbers are divisible by 4 which have their last digits as even natural number and the numbers formed by their last two digits are divisible by 4.
The possible numbers of last two digits are
12, 32, 52, 72, 92, 24, 64, 84, 16, 36, 56, 76, 96, 28, 48, 68.
Thus there are 16 ways of choosing the last two digits. Corresponding to each of these ways the remaining 7 digits can be arranged in 7! ways. Therefore, the total number of 9 digits numbers divisible by 4 is $16\times7!$.

Hence, required probability $=\frac{16\times7!}{9!}=\frac{2}{9}$

2. Total possible number of 4 digits = 4! = 24
The number is divisible by 5 if unit digit itself is 5. Therefore we fix 5 at unit place and then remaining 3 places can be filled up in 3! ways.

Hence, the required probability $=\frac{3!}{4!}=\frac{6}{24}=\frac{1}{4}$

3. Excluding ground floor there are 7 floors. A person can leave the cabin at any of the 7 floors. Therefore, the total number of ways in which each of the 5 persons can leave the cabin at any of the 7 floors $=7^5$.

$\therefore$ Exhaustive number of ways $=7^5$

Five persons can leave the cabin at five different floors in $^7P_5$ ways.

$\therefore$ The favourable number of cases $=^7P_5$

Hence, the required probability $=\frac{^7P_5}{7^5}=\frac{360}{2401}$

4.

Box 1	Box 2	Box 3
3W	2W	1W
1B	2B	3B

There can be three mutually exclusive cases of drawing 2 white balls and 1 black ball.

	Box 1	Box 2	Box 3
Case 1	1W	1W	1B
Case 2	1W	1B	1W
Case 3	1B	1W	1W

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$\therefore$ Required probability

$= P(W_1 \cap W_2 \cap B_3) \cup (W_1 \cap B_2 \cap W_3) \cup (B_1 \cap W_2 \cap W_3)$

$= P(W_1)P(W_2)P(B_3) + P(W_1)P(B_2)P(W_3) + P(B_1)P(W_2)P(W_3)$

$= \frac{3}{4} \times \frac{2}{4} \times \frac{3}{4} + \frac{3}{4} \times \frac{2}{4} \times \frac{1}{4} + \frac{1}{4} \times \frac{2}{4} \times \frac{1}{4} = \frac{26}{64} = \frac{13}{32}$

5. Since coin is fair *i.e.*, with equal probability a heads and a tail can be obtained. Also all the trials are independent so the probability that head appears on the fifth toss does not depend upon previous results of the tosses.

Hence, required probability $= \frac{1}{2}$

6. The total number of ways in which two calculators can be chosen out of four calculators is ${}^4C_2 = 6$. If only two tests are required to identify defective calculators, then in first two tests defective calculators are identified. This can be done in one way only.

$\therefore$ Required probability $= \frac{1}{6}$

7. 20 girls can be seated around a round table in 19! ways.

So, exhaustive number of cases = 19!

Excluding A and B, out of remaining 18 girls, 4 girls can be selected ${}^{16}C_4$ ways which can be arranged in 4! ways.

Remaining $20 - (4 - 2) = 14$ girls can be arranged in 14! ways. Also A and B mutually can be arranged in 2! ways.

$\therefore$ Required number of arrangements $= {}^{18}C_4 \times 4! \times 2! \times 14!$

$= 18! \times 2$

$\therefore$ Required probability $= \frac{18! \times 2}{19!} = \frac{2}{19}$

8. Since x and y can take values from 0 to 10. So, the total number of ways of selecting x and y is $11 \times 11 = 121$

Now, $|x - y| > 5 \Rightarrow x - y < -5$ or $x - y > 5$

There are 30 pairs of values of x and y satisfying these two inequalities, so favourable number of ways = 30

Hence, required probability $= \frac{30}{121}$

9. Since a persons's birthday can fall in any of the 12 months.

So, total number of ways $= 12^4$.

Now, any two months can be chosen in ${}^{12}C_2$ ways. The 4 persons birthday can fall in these two months in 2^4 ways. Out of these 2^4 ways there are two ways when all of the four birthdays fall in one month.

So, favourable number of ways $= \frac{{}^{12}C_2 \times (2^4 - 2)}{12^4} = \frac{77}{1728}$

10. 6 objects can be distributed among 6 persons in 6^6 ways.

$\therefore$ Total number of ways $= 6^6$

The number of ways of distribution in which each one gets only one thing is 6!. So, the number of distribution in which atleast one of them does not get any thing is $6^6 - 6!$

Hence, required probability $= \frac{6^6 - 6!}{6^6}$

11. Total number of ways of arranging 40 books on a shelf = 40!

Out of 40 places, 4 places for the four volumes can be chosen in ${}^{40}C_4$ ways. In the remaining 36 places the remaining 36 books can be arranged in 36! ways. In the 4 places four volumes of encyclopedia can be arranged in increasing order in one way only.

So, favourable number of ways $= {}^{40}C_4 \times 36!$

Hence, required probability $= \frac{{}^{40}C_4 \times 36!}{40!} = \frac{1}{24}$

12. The divisibility of the product of four numbers depends upon the value of the last digit of each number. The last digit of a number can be any of the 10 digits 0, 1, 2 ... 9 . So, the total number of ways of selecting last digits of four numbers is $10 \times 10 \times 10 \times 10 = 10^4$.

If the product of the 4 numbers is not divisible by 5 or 10. Then the number of choices for the last digit of each number is 8 (excluding 0 or 5).

So, favourable number of ways $= 8 \times 8 \times 8 \times 8 = 8^4$

$\therefore$The probability that the product is not divisible by 5 or 10

$= \frac{8^4}{10^4} = \left(\frac{8}{10}\right)^4$

Hence, Required probability $= 1 - \left(\frac{8}{10}\right)^4 = \frac{369}{625}$

13. P(selecting atleast one couple) $= 1 - P$(selecting none of the couples for the prize.)

$= 1 - \left(\frac{{}^{16}C_1 \times {}^{14}C_1 \times {}^{12}C_1 \times {}^{10}C_1}{{}^{16}C_4}\right) = \frac{15}{39}$

14. Total number of ways in which 7 $(= 4 + 3)$ persons can speak is 7!.

The number of ways in which A, B, C speak in the given order is ${}^7C_3$ ways and remaining 4 persons can be arranged in 4! ways.

$\therefore$ Favourable number of ways $= {}^7C_3 \times 4!$

$\therefore$ Required probability $= \frac{{}^7C_3 \times 4!}{7!} = \frac{1}{6}$

15. Let A be the event of selecting a counterfeit coin and B the event of getting head, then

Required probability $= P(A \cap B) \cup (\bar{A} \cap B)$

$= P(A \cap B) + P(\bar{A} \cap B)$

$= P(A)P\left(\frac{B}{A}\right) + P(\bar{A})P\left(\frac{B}{\bar{A}}\right)$

$= \frac{2}{16} \times 1 + \frac{14}{16} \times \frac{1}{2} = \frac{9}{16}$

16. Total number of ways in which 5 people can be chosen out of 9 people $= {}^9C_5 = 126$

Number of ways in which the couple serves the committee

$= {}^7C_3 \times {}^2C_2 = 35$

Number of ways in which the couple does not serve the committee $= {}^7C_5 = 21$

$\therefore$ Favourable number of cases $= 35 + 21 = 56$

Hence, the required probability $= \frac{56}{126} = \frac{4}{9}$

17. E_1 = The event in which A speaks truth

E_2 = The event in which B speaks truth

Then $P(E_1) = \frac{60}{100} = \frac{3}{5}$, $P(E_2) = \frac{80}{100} = \frac{4}{5}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

and $P(\bar{E}_1)=\frac{2}{5}, \quad P(\bar{E}_2)=\frac{1}{5}$

$\therefore$ Required probability $= P[(E_1 \cap E_2) \cup (\bar{E}_1 \cap \bar{E}_2)]$

$= P(E_1 \cap E_2) + P(\bar{E}_1 \cap \bar{E}_2)$

$= P(E_1).P(E_2) + P(\bar{E}_1).P(\bar{E}_2)$

$= \left(\frac{3}{5}\times\frac{4}{5}\right)+\left(\frac{2}{5}\times\frac{1}{5}\right)=\frac{14}{25}=0.56$

18. Two different squares can be chosen in 64×63 ways.
For each of the four corner squares, the favourable number of cases is 2.
For each of the 24 non-corner squares on all the four sides of the chessboard, the favourable number of cases is 3.
For each of the 36 remaining squares, the favourable number of cases is 4.
Thus, the total number of favourable cases

$= 4 \times 2 + 24 \times 3 + 36 \times 4 = 224$

Hence, the required probability $= \frac{224}{64 \times 63} = \frac{1}{18}$

19. The last two digits can be dialled in ${}^{10}P_2 = 90$ ways.
Out of these 90 cases only one case is favourable.

Hence, the required probability $= \frac{1}{90}$.

20. 3 squares on a chessboard can be chosen in ${}^{64}C_3$ ways.
Two squares of one colour and third square of different colour can be chosen in two mutually exclusive way.
(i) 2 white and one black (ii) 2 black and one white
Thus the favourable number of cases

$= {}^{32}C_2 \times {}^{32}C_1 + {}^{32}C_1 \times {}^{32}C_2$

$= 2({}^{32}C_2 \times {}^{32}C_1)$

Hence, the required probability $= \frac{2({}^{32}C_2 \times {}^{32}C_1)}{{}^{64}C_3} = \frac{16}{21}$

21. A leap year contains 366 days comprising of 52 full weeks and 2 extra days. Thus there can be following 7 possibilities for 2 extra days.
(i) Sunday, Monday, (ii) Monday, Tuesday
(iii) Tuesday, Wednesday (iv) Wednesday, Thursday
(v) Thursday, Friday (vi) Friday, Saturday
(vii) Saturday, Sunday
Let A be the event that the leap year contains 53 Sundays.
and B be the event that leap year contains 53 Mondays.
Then, we have

$P(A)=\frac{2}{7}, \quad P(B)=\frac{2}{7}, \quad P(A \cap B)=\frac{1}{7}$

So, required probability $= P(A \cup B)$

$= P(A) + P(B) - P(A \cap B)$

$= \frac{2}{7}+\frac{2}{7}-\frac{1}{7}=\frac{3}{7}$

22. Since, the probability of getting atleast one head in n times

$= 1-\left(\frac{1}{2}\right)^n$

Therefore, $1-\left(\frac{1}{2}\right)^n \geq 0.9$

$\Rightarrow \quad \left(\frac{1}{2}\right)^n \leq 0.1$

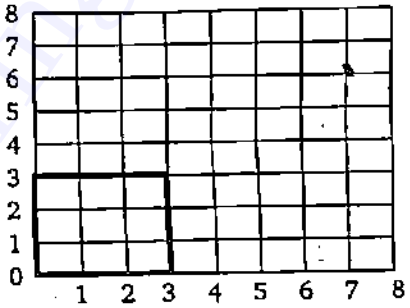
$\Rightarrow \quad 2^n \geq 10 \Rightarrow n \geq 4$

Hence, the least value of n is 4.

23. Total number of ways in which 2 persons can be selected out of 13 persons is ${}^{13}C_2$

Now, favourable number of cases $= \frac{{}^5C_1 \times {}^8C_1 + {}^5C_2}{{}^{13}C_2} = \frac{25}{39}$

24. We can choose 9 squares out of 64 squares in ${}^{64}C_9$ ways.
Hence, exhaustive number of cases $= {}^{64}C_9$
From the figure it is clear that the given square of size 3×3



can be formed by using four consecutive horizontal and 4 consecutive vertical lines, which can be done in

${}^6C_1 \times {}^6C_1 = 36$ ways

Basically you can make 6 squares of size 3×3 in vertical direction and 6 squares of the size 3×3 in horizontal direction. Hence total $6 \times 6 = 36$ squares can be chosen.

$\therefore$ The required probability $= \frac{36}{{}^{64}C_9}$

25. Total 7 digit numbers can be formed from the 9 digits $= {}^9P_7$
There are four exclusive cases of selecting 7 digits out of 9 digits which can form 7 digit numbers which are divisible by 9.
2, 3, 4, 5, 6, 7, 9 } 36removing 1 and 8
1, 3, 4, 5, 6, 8, 9 } 36 removing 2 and 7
1, 2, 4, 5, 7, 8, 9} 36 removing 3 and 6
1, 2, 3, 6, 7, 8, 9 } 36 removing 4 and 5
All the 7 numbers of each of the 4 sets can be arranged in 7! ways.
Hence the favourable number of numbers $= 4 \times 7!$

$\therefore$ Required probability $= \frac{4 \times 7!}{{}^9P_7} = \frac{1}{9}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yourshabab.wordpress.com
For More Visit : www.LearnEngineering.in

LEVEL (2)

1. Total number of words that can be formed from the letters of the word MISSISSIPPI is $\frac{11!}{4!4!2!}$

When all the S's are together then the number of words can be formed $= \frac{8!}{4!2!}$

$$\text{Required probability} = \frac{\frac{8!}{4!2!}}{\frac{11!}{4!4!2!}} = \frac{4}{165}$$

2. Since each of the coefficients a, b and c can take values from 1 to 6. Therefore the total number of equations

$$= 6 \times 6 \times 6 = 216$$

Hence the exhaustive number of cases = 216

Now, the roots of the equation $ax^2 + bx + c = 0$ will be real if $b^2 - 4ac \geq 0 \Rightarrow b^2 \geq 4ac$

Following are the number of favourable cases

a	c	ac	$4ac$	b^2 ($\geq 4ac$)	b	Number of cases
1	1	1	4	4, 9, 16, 25, 36	2, 3, 4, 5, 6	$1 \times 5 = 5$
1 2	2 1	2	8	9, 16, 25, 36	3, 4, 5, 6	$2 \times 4 = 8$
1 3	3 1	3	12	16, 25, 36	4, 5, 6	$2 \times 3 = 6$
1 2 4	4 2 2	4	16	16, 25, 36	4, 5, 6	$3 \times 3 = 9$
1 5	5 1	5	20	25, 36	5, 6	$2 \times 2 = 4$
1 2 3 6	6 3 2 1	6	24	25, 36	5, 6	$4 \times 2 = 8$
2 4	4 2	8	32	36	6	$2 \times 1 = 2$
3	3	9	36	36	6	$1 \times 1 = 1$
					Total	$= 43$

Note $\rightarrow$ $ac = 7$ is not possible

Since $b^2_{(max)} = 36$ and $4ac \leq b^2$ hence $ac = 10, 11, 12, \ldots$ etc. is not possible.

Hence, total number of favourable cases = 43

So, the required probability $= \frac{43}{216}$.

3. 6 can be thrown with a pair of dice in the following ways (1, 5), (5, 1), (2, 4), (4, 2), (3, 3)

So, probability of throwing a '6' $= \frac{5}{36}$

and probability of not throwing a '6' $= \frac{31}{36}$

And 7 can be thrown with a pair of dice in the following ways. (1, 6), (6, 1), (2, 5), (5, 2), (3, 4), (4, 3)

so probability of throwing a '7' $= \frac{6}{36} = \frac{1}{6}$

and probability of not throwing a '7' $= \frac{5}{6}$

Let E_1 be the event of the throwing a '6' in a single throw of a pair of dice and

E_2 be the event of throwing a 7 in a single throw of a pair of dice.

Then $P(E_1) = \frac{5}{36}$, $P(E_2) = \frac{1}{6}$

and $P(\bar{E}_1) = \frac{31}{36}$, $P(\bar{E}_2) = \frac{5}{6}$

A wins if he throws '6' in first or third or fifth .. throws.

Probability of A throwing a 6 in first throw $= p(E_1) = \frac{5}{36}$

and probability of A throwing a 6 in third throw

$$= P(\bar{E}_1 \cap \bar{E}_2 \cap E_1) = P(\bar{E}_1)\,P(\bar{E}_2)\,P(E_1)$$

$$= \frac{31}{36} \times \frac{5}{6} \times \frac{5}{36}$$

Similarly, probability of A throwing a '6' in fifth throw

$$= P(\bar{E}_1)\,P(\bar{E}_2)\,P(\bar{E}_1)\,P(\bar{E}_2)\,P(E_1)$$

$$= \left(\frac{31}{36}\right)^2 \times \left(\frac{5}{6}\right)^2 \times \frac{5}{36}$$

Hence, probability of winning of A

$$= P[E_1 \cup (\bar{E}_1 \cap \bar{E}_2 \cap E_1) \cup (\bar{E}_1 \cap \bar{E}_2 \cap \bar{E}_1 \cap \bar{E}_2 \cap E_1) \cup \ldots]$$

$$= P(E_1) + P(\bar{E}_1 \cap \bar{E}_2 \cap E_1) + P(\bar{E}_1 \cap \bar{E}_2 \cap \bar{E}_1 \cap \bar{E}_2 \cap \bar{E}_1) +$$

$$= \frac{5}{36} + \left(\frac{31}{36} \times \frac{5}{6}\right) \times \frac{5}{36} + \left(\frac{31}{36} \times \frac{5}{6}\right)^2 \times \frac{5}{36} + \ldots$$

$$= \frac{\frac{5}{36}}{1 - \left(\frac{31}{36}\right) \times \left(\frac{5}{6}\right)} = \frac{30}{61}$$

Thus, probability of winning of $B = 1 - \frac{30}{61} = \frac{31}{61}$

4. Let A be the event of getting exactly 3 defectives in the examination of 8 wristwatches.

And B be the event of getting nineth wristwatch defective. Then

Required probability $= P(A \cap B) = P(A)\,P\left(\frac{B}{A}\right)$

Now, $P(A) = \frac{{}^4C_3 \times {}^{11}C_5}{{}^{15}C_8}$

And $P\left(\frac{B}{A}\right)$ = Probability that the nineth examined wristwatch is defective given that there were 3 defectives in the first 8 pieces examined $= \frac{1}{7}$

Hence, required probability $= \frac{{}^4C_3 \times {}^{11}C_5}{{}^{15}C_8} \times \frac{1}{7} = \frac{8}{195}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

5. Let E_1, E_2, E_3 and A be the events defined as follows :
E_1 = the examinee guesses the answer
E_2 = the examinee copies the answer
E_3 = the examinee knows the answer and
A = the examinee answers correctly
We have $P(E_1) = \frac{1}{3}, P(E_2) = \frac{1}{6}$

Since E_1, E_2 and E_3 are mutually exclusive and exhaustive events therefore

$$P(E_1) + P(E_2) + P(E_3) = 1$$

$$\Rightarrow \quad P(E_3) = \frac{1}{2}$$

If E_1 has already occurred, then the examinee guesses. Since there are four choices out of which only one is correct, therefore the probability that he answers correctly given that he has made a guess is $\frac{1}{4}$ i.e., $P\left(\frac{A}{E_1}\right) = \frac{1}{4}$

It is a given that $P\left(\frac{A}{E_2}\right) = \frac{1}{8}$ and $P\left(\frac{A}{E_3}\right)$ is the probability that he answers correctly given that he knew the answer = 1
By Baye's rule,
Required probability $= P\left(\frac{E_3}{A}\right)$

$$= \frac{P(E_3)P\left(\frac{A}{E_3}\right)}{P(E_1)P\left(\frac{A}{E_1}\right) + P(E_2)P\left(\frac{A}{E_2}\right) + P(E_3)P\left(\frac{A}{E_3}\right)}$$

$$= \frac{\frac{1}{2} \times 1}{\frac{1}{3} \times \frac{1}{4} + \frac{1}{6} \times \frac{1}{8} + \frac{1}{2} \times 1} = \frac{24}{29}$$

6. Let x and y be the two non-negative integers since $x + y = 200$

$$\therefore \quad (xy)_{max} = 100 \times 100 = 10000 \quad (xy_{max} \text{ at } x = y)$$

Now, $xy \not< \frac{3}{4} \times 10000 \Rightarrow xy \geq \frac{3}{4} \times 10000$

$$\Rightarrow \quad xy \geq 7500 \Rightarrow x(200 - x) \geq 7500$$

$$\Rightarrow \quad x^2 - 200x + 7500 \leq 0$$

$$\Rightarrow \quad 50 \leq x \leq 150$$

So favourable number of ways = 150 − 50 + 1 = 101
Total number of ways = 200
Hence, required probability $= \frac{101}{200}$

7. Let $E_i (i = 1, 2, 3$ etc.) denote the event of drawing an even numbered card in i^{th} draw and F_i $(i = 1, 2, 3)$ denote the event of drawing an odd numbered card in i^{th} draw, then required probability

$$= P[(E_1 \cap F_2 \cap F_3) \cup (F_1 \cap E_2 \cap F_3) \cup (F_1 \cap F_2 \cap E_3)]$$
$$= P(E_1)P(F_2)P(F_3) + P(F_1)P(E_2)P(F_3) + P(F_1)P(F_2)P(E_3)$$
$$= \frac{4}{9} \times \frac{5}{9} \times \frac{5}{9} + \frac{5}{9} \times \frac{4}{9} \times \frac{5}{9} + \frac{5}{9} \times \frac{5}{9} \times \frac{4}{9}$$
$$= 3 \times \frac{4 \times (5)^2}{(9)^3} = \frac{100}{243}$$

8. Consider the following events
A = The first number is less than the second number
B = The third number lies between the first and the second.

Now, we have to find $P\left(\frac{B}{A}\right)$.

Also, we have $P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)}$

Any 3 numbers can be chosen out of n numbers in ${}^nC_3$ ways.
Let the selected numbers be x_1, x_2, x_3. Then they satisfy exactly one of the following inequalities.

$$x_1 < x_2 < x_3, \quad x_1 < x_3 < x_2, \quad x_2 < x_1 < x_3,$$
$$x_2 < x_3 < x_1, \quad x_3 < x_1 < x_2, \quad x_3 < x_2 < x_1$$

The total number of ways of selecting three numbers and then arranging them $= {}^nC_3 \times 3! = {}^nP_3$

$$\therefore \quad P(A) = \frac{{}^nC_3 \times 3}{{}^nC_3 \times 3!}$$

and $\quad P(A \cap B) = \frac{{}^nC_3}{{}^nC_3 \times 3!}$

Hence $\quad P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{1}{3}$

9. Since b and c each can assume 9 values from 1 to 9.
So, total number of ways of choosing b and c is $9 \times 9 = 81$
Now, $x^2 + bx + c > 0$ for all $x \in R$

$$\Rightarrow \quad D < 0$$
$$\Rightarrow \quad b^2 - 4ac < 0 \Rightarrow b^2 - 4c < 0 \Rightarrow b^2 < 4c$$

Alternatively : $\quad x^2 + bx + c > 0$

$$\Rightarrow \quad \left(x + \frac{b}{2}\right)^2 + \frac{4c - b^2}{4} > 0$$
$$\Rightarrow \quad 4c - b^2 > 0$$
$$\Rightarrow \quad b^2 < 4c$$

Now, the following table shows the possible values of b and c for which $b^2 < 4c$

c	b	Total
1	1,	1
2	1, 2	2
3	1, 2, 3	3
4	1, 2, 3,	3
5	1, 2, 3, 4	4
6	1, 2, 3, 4	4
7	1, 2, 3, 4, 5	5
8	1, 2, 3, 4, 5	5
9	1, 2, 3, 4, 5	5
		32

So, favourable number of cases = 32
Hence required probability $= \frac{32}{81}$

10. We have, $P(A \cup B \cup C) = \frac{3}{4}$

i.e., $\quad P(A) + P(B) + P(C) - P(A \cap B)$
$$- P(B \cap C) - P(A \cap C) + P(A \cup B \cup C) = \frac{3}{4}$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

and $P(A \cap B) + P(B \cap C) + P(A \cap C) - 2P(A \cap B \cap C) = \frac{1}{2}$

and $P(A \cap B) + P(B \cap C) + P(A \cap C) - 3P(A \cap B \cap C) = \frac{2}{5}$

Solving the above equations (last two), we get

$$P(A \cap B \cap C) = \frac{1}{2} - \frac{2}{5} = \frac{1}{10}$$

$$\Rightarrow \quad P(A)P(B)P(C) = \frac{1}{10} \Rightarrow pmc = \frac{1}{10}$$

Also, $P(A) + P(B) + P(C) - [P(A \cap B) + P(B \cap C) + P(A \cap C)] + P(A \cup B \cup C) = \frac{3}{4}$

$$p + m + c - \left(\frac{1}{2} + \frac{2}{10}\right) + \frac{1}{10} = \frac{3}{4}$$

$$\Rightarrow \quad p + m + c = \frac{27}{20}$$

11. Let E, F, G be the events that the student is successful in tests A, B and C respectively. Then the probability that the students is successful is

$$= P[(E \cap F \cap \bar{G}) \cup (E \cap \bar{F} \cap G) \cup (E \cap F \cap G)]$$
$$= P(E \cap F \cap \bar{G}) + P(A \cap \bar{F} \cap G) + P(E \cap F \cap G)$$
$$= P(E)P(F)P(\bar{G}) + P(E)P(\bar{F})P(G) + P(E)P(F)P(G)$$
$$= pq\left(1 - \frac{1}{2}\right) + p(1-q)\left(\frac{1}{2}\right) + pq\left(\frac{1}{2}\right)$$
$$= \frac{pq + p - pq + pq}{2}$$
$$= \frac{p(1+q)}{2}$$

But the probability that the student is successful $= \frac{1}{2}$

$$\therefore \quad \frac{P(1+q)}{2} = \frac{1}{2} \Rightarrow p(1+q) = 1$$

This is satisfied by $p = 1, q = 0$

Also there are other values (infinite numbers) of p, q for which the above relation is satisfied.

Hence, (d) is the correct option.

12. Since $\frac{1+4p}{p}, \frac{1-p}{4}, \frac{1-2p}{2}$ are the probabilities of 3 mutually exclusive events, therefore

$$0 \le \frac{1+4p}{p} \le 1, \quad 0 \le \frac{1-p}{4} \le 1, \quad 0 \le \frac{1-2p}{2} \le 1$$

and $0 \le \frac{1+4p}{p} + \frac{1-p}{4} + \frac{1-2p}{2} \le 1$

$$\Rightarrow \quad -\frac{1}{4} \le p \le \frac{3}{4}, \quad -1 \le p \le 1, \quad -\frac{1}{2} \le p \le \frac{1}{2}$$

and $\frac{1}{2} \le p \le \frac{5}{2}$

$$\Rightarrow \max\left\{-\frac{1}{4}, -1, -\frac{1}{2}, \frac{1}{2}\right\} \le p \le \min\left\{\frac{3}{4}, 1, \frac{1}{2}, \frac{5}{2}\right\}$$

$$\Rightarrow \quad \frac{1}{2} \le p \le \frac{1}{2} \Rightarrow p = \frac{1}{2}$$

13. ASSISTANT $\rightarrow$ *AA I N SSS TT*

STATISTICS $\rightarrow$ *A II C SSS TTT*

Here N and C are not common and same letters can be A, I, S, T. Therefore

Probability of choosing $A = \frac{{}^2C_1}{{}^9C_1} \times \frac{{}^1C_1}{{}^{10}C_1} = \frac{1}{45}$

Probability of choosing $I = \frac{1}{{}^9C_1} \times \frac{{}^2C_1}{{}^{10}C_1} = \frac{1}{45}$

Probability of choosing $S = \frac{{}^3C_1}{{}^9C_1} \times \frac{{}^3C_1}{{}^{10}C_1} = \frac{1}{10}$

Probability of choosing $T = \frac{{}^2C_1}{{}^9C_1} \times \frac{{}^3C_1}{{}^{10}C_1} = \frac{1}{15}$

Hence, required probability

$$= \frac{1}{45} + \frac{1}{45} + \frac{1}{10} + \frac{1}{15} = \frac{19}{90}$$

14. Out of 30 numbers 2 numbers can be chosen in ${}^{30}C_2$ ways.

So, exhaustive number of cases $= {}^{30}C_2 = 435$

Since $a^2 - b^2$ is divisible by 3 iff either a and b are divisible by 3 or none of a and b is divisible by 3. Thus, the favourable numbers, of cases $= {}^{10}C_2 + {}^{20}C_2 = 235$

Hence, required probability $= \frac{235}{435} = \frac{47}{87}$

15. The man will be one step away from the starting point if (A) either he is one step ahead or (B) one step behind the starting point.

Therefore, required probability $= P(A) + P(B)$

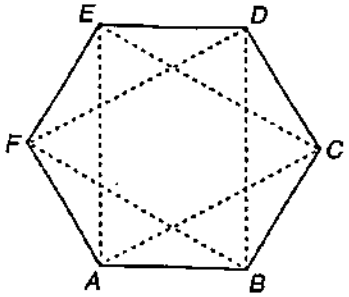
The man will be one step ahead at the end of eleven steps if he moves six steps forward and five steps backward. The probability of this event $= {}^{11}C_6(0.4)^6\,(0.6)^5$

The man will be one step behind at the end of the eleven steps if he moves six steps backward and five steps forward. The probability of this event $= {}^{11}C_6(0.6)^6\,(0.4)^5$.

Hence, the required probability

$$= {}^{11}C_6(0.4)^6(0.6)^5 + {}^{11}C_6(0.6)^6\,(0.4)^5$$
$$= {}^{11}C_6(0.4)^5\,(0.6)^5\,(0.4 + 0.6) = {}^{11}C_6(0.24)^5$$

16. There are 6 vertices in a hexagon. Using 3 vertices out of 6 vertices we can form ${}^6C_3$ triangles. But there can be only two triangles out of ${}^6C_3$ triangles which are equilateral (see the figure, (i) ΔACE, (ii) ΔBDF)



Hence, the required probability $= \frac{2}{{}^6C_3} = \frac{2}{20} = \frac{1}{10}$

17. Let F, B, L and R denote the forward, backward, left and right steps (or movements) then the following mutually exclusive ways are possible.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

F	B	L	R
0	0	4	5
1	1	3	4
2	2	2	3
3	3	1	2
4	4	0	1
0	0	5	4
1	1	4	3
2	2	3	2
3	3	2	1
4	4	1	0

In this case he cancels out his forward and backward movements by moving equal steps in forward and backward directions each and he creates a difference of 1 step by moving one step extra either in right or left direction.

F	B	L	R
4	5	0	0
3	4	1	1
2	3	2	2
1	2	3	3
0	1	4	4
5	4	0	0
4	3	1	1
3	2	2	2
2	1	3	3
1	0	4	4

In this case he cancels out his left or right movement by moving equal number of steps in left and right directions each and he creates a difference of 1 step by moving one step extra either in forward or backward directions.

The number of permutations of these five arrangements is

$$4\left[\frac{9!}{5!4!}+\frac{9!}{1!1!3!4!}+\frac{9!}{2!2!2!3!}+\frac{9}{3!3!1!2!}+\frac{9!}{4!4!1!}\right]$$

$= 4(126 + 2520 + 7560 + 5040 + 630)$

$= 4 \times 15876$

But the total number of ways of arranging nine steps $= 4^9$.

$\therefore$ The required probability $= \dfrac{4 \times 15876}{4^9} = \dfrac{3969}{4^7}$

18. Let E_{rr} denote that a red colour ball is transferred from urn A to urn B tourn then a red colour ball is transferred from urn B to urn A.

E_{rb} denote that a red colour ball is transferred from urn A to urn B then a black colour ball is transferred from urn B to urn A

E_{br} denote that a black colour ball is transferred from urn A to urn B then a red colour ball is transferred from urn B to urn A.

E_{bb} denote that a black colour ball is transferred from urn A to urn B then a black colour ball is transferred from urn B to urn A. Then

$$P(E_{rr}) = \left(\frac{6}{10}\right)\left(\frac{5}{11}\right) = \frac{3}{11},$$

$$P(E_{rb}) = \left(\frac{6}{10}\right)\left(\frac{6}{11}\right) = \frac{18}{55}$$

$$P(E_{br}) = \left(\frac{4}{10}\right)\left(\frac{4}{11}\right) = \frac{8}{55},$$

$$P(E_{bb}) = \left(\frac{4}{10}\right)\left(\frac{7}{11}\right) = \frac{14}{55}$$

Let A be the event of drawing a red colour ball after these transfers. Then

$$P\left(\frac{A}{E_{rr}}\right) = \frac{6}{10}, P\left(\frac{A}{E_{rb}}\right) = \frac{5}{10}$$

$$P\left(\frac{A}{E_{br}}\right) = \frac{7}{10}, P\left(\frac{A}{E_{bb}}\right) = \frac{6}{10}$$

Therefore, the required probability is

$$P(A) = P(E_{rr})P\left(\frac{A}{E_{rr}}\right) + P(E_{rb})P\left(\frac{A}{E_{rb}}\right) + P(E_{br})P\left(\frac{A}{E_{br}}\right) + P(E_{bb})P\left(\frac{A}{E_{bb}}\right)$$

$$= \left(\frac{3}{11}\right)\left(\frac{6}{10}\right) + \left(\frac{5}{10}\right)\left(\frac{18}{55}\right) + \left(\frac{8}{55}\right)\left(\frac{7}{10}\right) + \left(\frac{14}{55}\right)\left(\frac{6}{10}\right)$$

$$= \frac{90 + 90 + 56 + 84}{550} = \frac{32}{55}$$

19. A number is divisible by 11 only if the difference of the sum of the digits at odd places and sum of the digits at even places is divisible by 11 i.e, 0, 11, 22, 33 ....

Here the sum of all the 9 digits (1, 2, 3, ...9) is 45.

We cannot create the difference of zero

since $x + y = 45$, which is odd hence cannot be broken into two equal parts in integers.

Now, we will look for the possibilities of 11

which are as follows :

{1, 2, 6, 8}{1, 2, 5, 9}{1, 3, 6, 7}

{1, 3, 5, 8} {1, 3, 4, 9}{1, 4, 5, 7}

{2, 3, 5, 7}{2, 3, 4, 8}{2, 4, 5, 6}

and {4, 7, 8, 9}{5, 6, 8, 9}

The above set of values either gives the sum of 17 or 28.

Since if the sum of 4 digits at even places be 17 or 28 then the sum of rest of the digits (*i.e.*, digits at odd places) be 28 or 17 respectively and thus we can get the difference of 11.

Further we cannot get the difference of 22 or 33...

So there is only possible difference that can be created is 11 and there are only 11 set of values given above containing 4 digits which can be arranged in 4! ways and the remaining 5 digits can be arranged in 5! ways.

Thus the favourable number of numbers $= 11 \times 4! \times 5!$

But the total number of ways of arranging a nine digit number is ${}^9P_9 = 9!$

$\therefore$ Exclusive number of cases $= 9!$

$\therefore$ Required probability $= \dfrac{11 \times 4! \times 5!}{9!} = \dfrac{11}{126}$.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

21 CO-ORDINATE GEOMETRY

It is one of those chapters which contributes least number of problems in CAT. Perhaps this chapter is more or less algebraic instead of being logical, since most of the problems require formula to solve them.

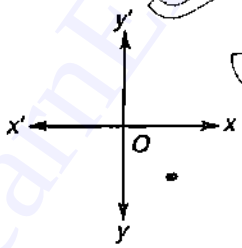
But some other MBA entrance exams ask the question from this chapter.

It is not fool proof but this chapter may be optional particularly for CAT aspirants who feel difficulty in algebraic problems. But it does not mean that this chapter is difficult or can not be learned.

1. CARTESIAN COORDINATE SYSTEM

1.1 Rectangular Co-ordinate Axes

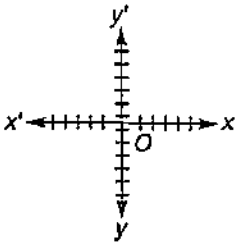
Let $X'OX$ and $Y'OY$ be the two mutually perpendicular lines through any point O in the plane of paper. The point O is called the origin.



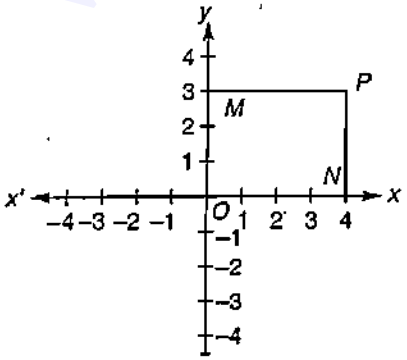
The line $X'OX$ is called the x-axis and the line $Y'OY$ is called the y-axis. These two lines taken together are called the **coordinate axes.**

1.2 Cartesian Co-ordinate of a Point

Each axes is calibrated as a scale into several points which are equidistant from each other.



The point O is assumed to be zero.

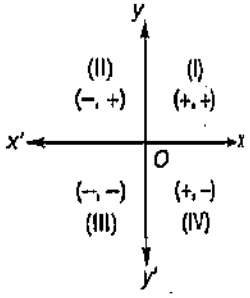


Let P be any point on the plane, then we draw a perpendicular from P on the X-axis and another on the Y-axis. Then the distance MP (which is parallel to the X-axis) is called the x-coordinate or abscissa of point P. Similarly, the length PN (which is parallel to the Y-axis) is called the y-coordinate or ordinate of point P.

Then the position of the point P in the plane with respect to the coordinate axes is represented by the ordered pair (x, y). The ordered pair (x, y) is called the coordinates of point P.

1.3 Quadrants

In the adjacent figure the two axes intersect each other perpendicular at point O.



The regions XOY, $X'OY$, $X'OY'$ and $Y'OX$ are known as first, second, third and fourth quadrant respectively.

The nature of values of x and y in different quadrants is different which is shown in figure also given below

Quadrant	Sign of (x, y)
I	$(+, +)$
II	$(-, +)$
III	$(-, -)$
IV	$(+, -)$

For example, let P be a point in second quadrant, then it is denoted as $(-1, 2)$, $(-2, 3)$, $(-3, 5)$, $(-6, 8)$, $(-7, 10)$ etc.

The coordinates of point O (i.e, origin) are taken as $(0, 0)$. The coordinates of any point on x-axis are of the form $(x, 0)$ and the coordinates of any point on y-axis are of the form $(0, y)$.

2. DISTANCE FORMULA

The distance between two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is given by

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Also, Distance of point (x_1, y_1) from origin $= \sqrt{x_1^2 + y_1^2}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

3. SECTION FORMULA

If $R(x, y)$ divides the line segment joined by the points $P(x_1, y_1)$ and $Q(x_2, y_2)$ internally in the ratio $m : n$ then

$$x = \frac{mx_2 + nx_1}{m+n}, \quad y = \frac{my_2 + ny_1}{m+n}$$

If $R(x, y)$ divides the line segment joined by the points $P(x_1, y_1)$ and $Q(x_2, y_2)$ externally in the ratio $m : n$ then

$$x = \frac{mx_2 - nx_1}{m-n}, \quad y = \frac{my_2 - ny_1}{m-n}$$

4.1 AREA OF A TRIANGLE

The area of a triangle ABC whose vertices are $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ is denoted by Δ.

$$\therefore \quad \Delta = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

$$= \frac{1}{2}[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)]$$

4.2 AREA OF POLYGON

The area of the polygon whose vertices are $(x_1, y_1), (x_2, y_2), (x_3, y_3) \ldots (x_n, y_n)$ is

$$\Delta = \frac{1}{2}[(x_1 y_2 - x_2 y_1) + (x_2 y_3 - x_3 y_2) + \ldots + (x_n y_1 - x_1 y_n)]$$

5. CONDITION OF COLLINEARITY OF THREE POINTS

The three given points are collinear *i.e.*, lie on the same straight line if

(i) Area of triangle ABC is zero.

(ii) Slope of AB = Slope of BC = Slope of AC

(iii) Distance between A and B + distance between B and C = Distance between A and C.

(iv) Find the equation of the line passing through any two points, if third point satisfies the equation of the line then three points are collinear.

6. SOME IMPORTANT RESULTS

6.1 If the vertices of a triangle have integral co-ordinates then the triangle can not be equilateral.

6.2 In order to prove that a given figure is a square, parallelogram, rectangle etc. We will prove the following points given in the table below corresponding the name of the figure.

	Name of the figure	Conditions
1.	Square	Four sides are equal and the diagonals are also equal.
2.	Rhombus	Four sides are equal
3.	Rectangle	Opposite sides are equal and diagonals are also equal
4.	Parallelogram	Opposite sides are equal
5.	Parallelogram but not a rectangle	Opposite sides are equal but the diagonals are not equal.
6.	Rhombus but not a square	All sides are equal but the diagonals are not equal

7. SOME IMPORTANT POINTS IN A TRIANGLE

7.1 Centroid

If $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3) are the vertices of a triangle, then the coordinates of its centroid are $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$

7.2 Incentre

If $A(x_1, y_1), B(x_2, y_2)$ and $C(x_3, y_3)$ are the vertices of a triangle ABC such that $BC = a$, $CA = b$ and $AB = c$, then the coordinates of its centre are $\left(\frac{ax_1 + bx_2 + cx_3}{a+b+c}, \frac{ay_1 + by_2 + cy_3}{a+b+c}\right)$

NOTE If the equations of the sides of the triangle are given then we find the bisectors of internal angles and then their point of intersection to determine the in-centre.

7.3 Circumcentre

If 'O' is the circumcentre of a triangle ABC, then $OA = OB = OC$ and OA is called the circumradius.

To find the circumcentre of ΔABC, we use the relation $OA = OB = OC$. This gives two simultaneous linear equation and their solution provides the coordinates of circumcentre.

7.4 Orthocentre

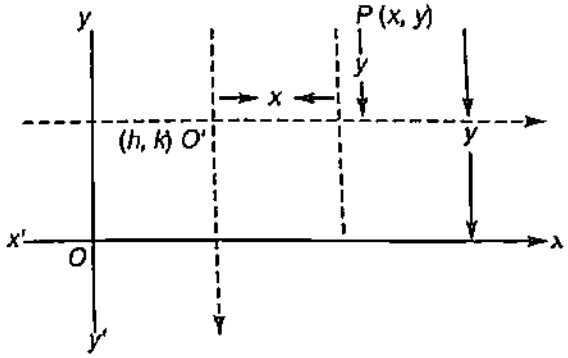
To determine the orthocentre, first we find equations of lines passing through vertices and perpendicular to the opposite sides. Solving any two of these three equations we get the coordinates of orthocentre.

NOTE The circumcentre O, centroid G and othrocentre O' of a triangle ABC are collinear such that G divides $O'O$ in the ratio 2 : 1 *i.e.*, $O'G : OG = 2 : 1$

8. SHIFTING OF ORIGIN

Let O be the origin and $X'OX$ and $Y'OY$ be the axis of x and y respectively. Let O' and P be two points in the plane having coordinates (h, k) and (x, y) respectively referred to $X'OX$ and $Y'OY$ as coordinate axes.

Let the origin be transferred to O' and let $X'O'X$ and $Y'O'Y$ be new rectangular axes. Let the co-ordinates of P referred to new axes as the co-ordinate axes be (X, Y)



Thus if (x, y) are coordinates of a point referred to old axis and (X, Y) are the coordinates of the same point referred to new axis then.

$$x = X + h \quad \text{and} \quad y = Y + k$$

If therefore the origin is shifted at a point (h, k) we must substitute $X + h$ and $Y + k$ for x and y respectively.

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

For More Visit : www.LearnEngineering.in

The transformation formula from new axes to old axes is

$$X = x - h, \quad Y = y - k$$

The coordinates of the old origin referred to the new axes are $(-h, -k)$

9. DEFINITION OF A STRAIGHT LINE

A straight line is a curve such that every point on the line segment joining any two points on it lies on it.

9.1 The general form of the equation of straight line

$$ax + by + c = 0$$

where a, b and c are real constants and x, y are two unknowns.

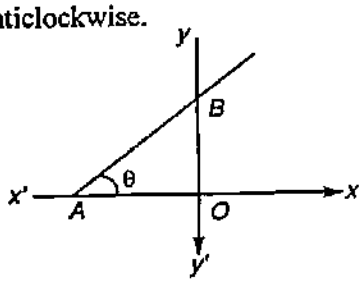
9.2 Slope (Gradient) of a Line

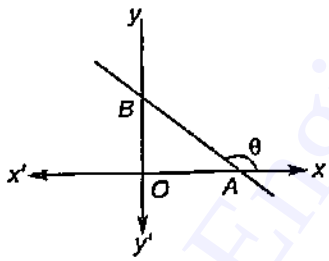
$$m = \tan\theta = -\frac{a}{b}$$

$$\{\because ax + by + c = 0 \Rightarrow by = -ax - c \Rightarrow y = -\frac{a}{b}x - \frac{c}{b}$$

$$\Rightarrow \quad y = mx + c, \quad \text{where } m = -\frac{a}{b} \text{ and } c \text{ is a constant}\}$$

Here m is called the slope or gradient of a line and c is the intercept on y-axis. The slope of a line is always measured in anticlockwise.





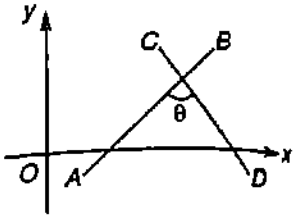
9.3 Slope of a line in terms of coordinates of any two points on it.

If (x_1, y_1) and (x_2, y_2) are coordinates of any two points on a line, then its slope is

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{Difference of ordinates}}{\text{Difference of abscissae}} = \frac{\text{rise}}{\text{run}}$$

10.1 Angle between two lines

$$\tan\theta = \pm\left(\frac{m_2 - m_1}{1 + m_1 m_2}\right)$$



10.2 Condition of Parallelism of lines

If the slopes of two lines is m_1 and m_2 and if they are parallel then,

$$m_1 = m_2$$

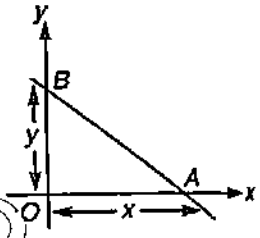
10.3 Condition of Perpendicularity of lines

If the slopes of two lines is m_1 and m_2 and if they are perpendicular to each other, then

$$m_1 \cdot m_2 = -1$$

10.4 Intercepts of a line on the axis

If a straight line cuts x-axis at A and the y-axis at B then OA and OB are known as the intercepts of the line on x-axis and y-axis respectively where O is the origin.



INTRODUCTORY EXERCISE-21.1

1. The point (– 2, 3) lies in the quadrant :
   (a) First (b) Second
   (c) Third (d) Fourth
2. The point (2, – 3) lies in the quadrant :
   (a) First (b) Second
   (c) Third (d) Fourth
3. Find the distance between the points (– 5, 3) and (3, 1):
   (a) $2\sqrt{7}$ (b) $3\sqrt{14}$
   (c) $5\sqrt{17}$ (d) $2\sqrt{17}$
4. Let the vertices of a triangle ABC be (4, 3), (7, – 1), (9, 3) then the triangle is :
   (a) Scalene (b) Isosceles
   (c) Equilateral (d) none of (a), (b),(c)
5. Let the vertices of a triangle ABC be (4, 4), (3, 5), (– 1, – 1), then the triangle is :
   (a) scalene
   (b) equilateral
   (c) right angled
   (d) none of (a), (b), (c)
6. Let the vertices of a triangle ABC be (7, 9), (3, – 7), and (- 3, 3) then the triangle is :
   (a) right angled (b) equilateral
   (c) isosceles (d) both (a) and (c)
7. A(– 3, 2) and B(5, 4) are the end points of a line segment, find the coordinates of the mid points of the line segment :
   (a) (1, 3) (b) (2, 3)
   (c) (3, 2) (d) (4, 3)
8. In the previous question (no. 7) find the co-ordinates of the point which divides AB in the ratio 2 : 3 :
   (a) $\left(\frac{1}{5}, \frac{14}{5}\right)$ (b) (2, 3)
   (c) $\left(\frac{1}{5}, 3\right)$ (d) $\left(3, \frac{1}{2}\right)$
9. Find the co-ordinates of the point which divides the join of the points (2, 4) and (6, 8) externally in the ratio 5 : 3:
   (a) (5, 6) (b) (12, 14)
   (c) (3, 8) (d) (2, 7)

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com

For More Visit : www.LearnEngineering.in

10. The coordinates of the vertices of a triangle are (3, 1), (2, 3) and (– 2, 2). Find the coordinates of the centroid of the triangle ABC :
(a) (1, 2) (b) (2, 3)
(c) (4, 5) (d) (5, 6)

11. Find the co-ordinates of the incentre of the triangle whose vertices are the points (4, – 2), (5, 5) and (– 2, 4):
(a) $\left(\frac{7}{3}, \frac{2}{3}\right)$ (b) $\left(\frac{5}{2}, \frac{5}{2}\right)$
(c) $\left(\frac{6}{5}, 5\right)$ (d) none of these

12. Find the coordinates of the circum centre of the triangle whose vertices are (8, 6), (8, – 2) and (2, – 2):
(a) (2, 3) (b) (5, 2)
(c) (5, 3) (d) (7, 2)

13. If the coordinates of the mid points of the sides of a triangle are (1, 1), (2, – 3) and (3, 4). Find its centroid :
(a) $\left(3, \frac{2}{3}\right)$ (b) $\left(2, \frac{3}{4}\right)$
(c) $\left(2, \frac{2}{3}\right)$ (d) none of these

14. If the previous problem (no. 13) find the incentre for the given triangle :
(a) $\left(8, \frac{3}{2}\right)$ (b) $\left(\frac{7}{2}, 5\right)$
(c) $(2\sqrt{3}, 5)$ (d) none of these

15. $A(-2, -1)$, $B(1, 0)$, $C(4, 3)$ and $D(1, 2)$ are the four points of a quadrilateral. The quadrilateral is a :
(a) Square (b) Rhombus
(c) Parallelogram (d) none of (a), (b),(c)

16. The three vertices of a parallelogram taken in a order are $(-1, 0)$, $(3, 1)$ and $(2, 2)$ respectively. Find the coordinates of the fourth vertex :
(a) $(-1, 2)$ (b) $(-2, 1)$
(c) (2, 3) (d) $(3, -2)$

17. If the coordinates of the mid points of the sides of a triangle are $(1, 2)$, $(0, -1)$ and $(2, -1)$. Find the coordinates of its vertices :
(a) $(1, -4)$, $(3, 2)$, $(-1, 2)$ (b) $(1, 2)$, $(2, 3)$, $(3, -4)$
(c) $(3, 4)$, $(5, 2)$, $(1, 2)$ (d) none of these

18. Points $A(4, -1)$, $B(6, 0)$, $C(7, 2)$ and $D(5, 1)$ are the vertices of the following quadrilateral which is a :
(a) Square (b) Rectangle
(c) Rhombus (d) none of (a), (b), (c)

19. Find the area of quadrilateral formed by joining the points (– 4, 2), (1, – 1), (4, 1) and (2, 5):
(a) 25.4 (b) 20.5
(c) 24.5 (d) none of these

20. Find (x, y) if $(3, 2)$, $(6, 3)$, (x, y) and $(6, 5)$ are the vertices of a parallelogram :
(a) (5, 6) (b) (6, 5)
(c) (9, 6) (d) (9, 5)

21. Determine the ratio in which $y - x + 2 = 0$ divides the line joining (3, – 1) and (8, 9):
(a) 3 : 5 (b) 4 : 3
(c) 2 : 3 (d) none of these

22. If the area of the quadrilateral whose angular points taken in order are (1, 2), (– 5, 6), (7, – 4) and $(h - 2)$ be zero, find the value of h :
(a) 2 (b) 3
(c) 5 (d) 6

23. The co-ordinates of A, B, C are (6, 3), (– 3, 5), (4, – 2) respectively and P is a point (x, y), then find the value of $\frac{\Delta PBC}{\Delta ABC}$:
(a) $\frac{x + y - 2}{7}$ (b) $\frac{x - y}{4}$
(c) $\frac{x + y - 3}{5}$ (d) none of these

24. Find the slope of the line joining the points (7, 5) and (9, 7):
(a) 1 (b) 2
(c) $\frac{1}{2}$ (d) 3

25. If $A(-2, 1)$, $B(2, 3)$ and $C(-2, -4)$ are three points, find the angle between BA and BC :
(a) $\tan^{-1}\left(\frac{3}{5}\right)$ (b) $\tan^{-1}\left(\frac{2}{3}\right)$
(c) $\tan^{-1}\left(\frac{3}{2}\right)$ (d) none of these

26. What is the slope of the line perpendicular to the line passing through the points (3, 5) and (– 4, 2)?
(a) $-\frac{3}{5}$ (b) $-\frac{2}{3}$
(c) $-\frac{7}{3}$ (d) none of these

27. A line passes through the points $A(2, -3)$ and $B(6, 3)$. Find the slope of the line which is parallel to AB :
(a) $\frac{2}{3}$ (b) $\frac{3}{2}$
(c) $\frac{1}{2}$ (d) $\frac{3}{4}$

28. In the previous question (no. 27), find the slope of the line which is perpendicular to AB :
(a) $-\frac{4}{3}$ (b) $\frac{5}{6}$
(c) $-\frac{2}{3}$ (d) none of these

29. Points $A(6, 6)$, $B(2, 3)$ and $C(4, 7)$ are the vertices of a triangle which is :
(a) right angled (b) acute angled
(c) obtuse angled (d) none of these

30. Three points $A(1, -2)$, $B(3, 4)$ and $C(4, 7)$ form :
(a) a straight line
(b) an equilateral triangle
(c) a right angled triangle
(d) none of (a), (b), (c)

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

11. EQUATIONS OF LINES PARALLEL TO THE COORDINATE AXES

11.1 Equation of a line parallel to x-axis and at a distance k from it is $y = k$

11.2 Equation of a line parallel to the y-axis and at a distance h from it is $x = h$.

11.3 Equation of the x-axis is $y = 0$

11.4 Equation of the y-axis is $x = 0$

12. DIFFERENT STANDARD FORMS OF THE EQUATIONS OF A STRAIGHT LINE

12.1 Slope Intercept Form

(a) The equation of a line with slope m and making an intercept on y-axis is $y = mx + c$.

(b) The equation of a line with slope m and passing through the origin (here $c = 0$) is $y = mx$.

12.2 Point Slope Form

The equation of a line passes through the point (x_1, y_1) and has the slope 'm' is $y - y_1 = m(x - x_1)$

12.3 Two Point Form

The equation of a line passing through two points (x_1, y_1) and (x_2, y_2) is $y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)(x - x_1)$

12.4 Intercept Form

The equation of a line which cuts off intercepts a and b respectively from the x-axis and y-axis is $\frac{x}{a} + \frac{y}{b} = 1$.

12.5 Normal (or Perpendicular) Form

The equation of the straight line on which the length of the perpendicular from the origin (0, 0) is p and this perpendicular makes an angle α with x-axis is $x\cos\alpha + y\sin\alpha = p$.

12.6 Distance (or Symmetrical) Form

The equation of the straight line passing through (x_1, y_1) and making an angle θ with positive direction of x-axis is $\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r$

Where r is the distance of the point (x, y) on the line from the point (x_1, y_1)

- Since different values of r determine different points on the line, therefore the above form of the line is also called parametric form or symmetric form.

13. REDUCTION (OR TRANSFORMATION) OF THE GENERAL EQUATION

The general equation of a straight line is $Ax + By + C = 0$ which can be transformed into various standard forms.

13.1 General Equation ($Ax + By + C = 0$) → Slope Intercept form ($y = mx + c$)

$$Ax + By + C = 0 \Rightarrow By = -Ax - C$$

$$\Rightarrow y = \left(-\frac{A}{B}\right)x + \left(-\frac{C}{B}\right)$$

$$\Rightarrow y = mx + c$$

where $m = -\frac{A}{B} = -\frac{\text{Coefficient of } x}{\text{coefficient of } y}$

and intercept on y-axis $= -\frac{C}{B} = -\frac{\text{Constant trem}}{\text{coefficient of } y}$

13.2 General Equation ($Ax + By + C = 0$) → Intercept form $\left(\frac{x}{a} + \frac{y}{b} = 1\right)$

$$Ax + By + C = 0 \Rightarrow \frac{Ax}{-C} + \frac{By}{-C} = 1$$

$$\Rightarrow \frac{x}{\left(-\frac{C}{A}\right)} + \frac{y}{\left(-\frac{C}{B}\right)} = 1$$

$$\Rightarrow \frac{x}{a} + \frac{y}{b} = 1$$

$\Rightarrow$ Where, intercept on x-axis $= -\frac{C}{A} = -\frac{\text{Constant term}}{\text{Coefficient of } x}$

Intercept on y-axis $= -\frac{C}{B} = -\frac{\text{Constant term}}{\text{Coefficient of } y}$

13.3 General Equation ($Ax + By + C = 0$) → Normal Form ($x\cos\alpha + y\sin\alpha = p$)

$$Ax + By + C = 0 \quad \ldots(1)$$

$$x\cos\alpha + y\sin\alpha - p = 0 \quad \ldots(2)$$

Comparing (1) and (2)

$$\frac{A}{\cos\alpha} = \frac{B}{\sin\alpha} = \frac{C}{-p}$$

$$\Rightarrow \cos\alpha = -\frac{Ap}{C} \text{ and } \sin\alpha = -\frac{Bp}{C}$$

$$\Rightarrow 1 = \cos^2\alpha + \sin^2\alpha = \frac{p^2}{C^2}(A^2 + B^2)$$

$$\text{or } p^2 = \frac{C^2}{A^2 + B^2} \Rightarrow p = \frac{C}{\sqrt{A^2 + B^2}}$$

$$\therefore \cos\alpha = \frac{-A}{\sqrt{A^2 + B^2}}, \sin\alpha = \frac{B}{\sqrt{A^2 + B^2}}$$

$\therefore$ The required equation is

$$-\frac{A}{\sqrt{A^2 + B^2}}x - \frac{B}{\sqrt{A^2 + B^2}}y = \frac{C}{\sqrt{A^2 + B^2}}$$

14. POINT OF INTERSECTION OF TWO LINES

Let the equations of two lines be

$$a_1x + b_1y + c_1 = 0 \quad \ldots(1)$$

and $$a_2x + b_2y + c_2 = 0 \quad \ldots(2)$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Let (x_1, y_1) be the coordinates of the point of intersection of the two lines. Then

$$a_1x_1 + b_1y_1 + c_1 = 0 \quad \ldots(3)$$

and $$a_2x_1 + b_2y_1 + c_2 = 0 \quad \ldots(4)$$

$\therefore$ By the method of cross multiplication

$$\frac{x_1}{b_1c_2 - b_2c_1} = \frac{y_1}{c_1a_2 - c_2a_1} = \frac{1}{a_1b_2 - a_2b_1}$$

$$\therefore \quad x_1 = \frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \; y_1 = \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1},$$

where $a_1b_2 - a_2b_1 \neq 0$

Thus the coordinates of the point of intersection of the two lines are

$$\frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}$$

Remark : To find the coordinates of the point of intersection of two non parallel lines, we solve the given equations simultaneously and the values of x and y so obtained determine the coordinates of the point of intersection.

15. CONDITION OF CONCURRENCY OF THREE LINES

Three lines are said to be concurrent if they pass through a common point *i.e.*, they meet at a single common point.

Thus if three lines are concurrent the point of intersection of any two lines lies on the third line.

Let $$a_1x + b_1y + c_1 = 0 \quad \ldots(1)$$

$$a_2x + b_2y + c_2 = 0 \quad \ldots(2)$$

$$a_3x + b_3y + c_3 = 0 \quad \ldots(3)$$

be three concurrent lines then

$$a_3(b_1c_2 - b_2c_1) + b_3(c_1a_2 - c_2a_1) + c_3(a_1b_2 - a_2b_1) = 0$$

$$\Rightarrow \quad \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

Which is the required condition for concurrency of three lines.

- Find the point of intersection of any two lines by solving them simultaneously. If this point satisfies the third equation also, then the given lines are concurrent.

16. ANGLE BETWEEN TWO STRAIGHT LINES

Let the two straight lines be

$$y = m_1x + c_1 \quad \ldots(1)$$

and $$y = m_2x + c_2 \quad \ldots(2)$$

and θ be the angle between them, then

$$\theta = \tan^{-1}\left|\frac{a_2b_1 - a_1b_2}{a_1a_2 + b_1b_2}\right|$$

17. CONDITIONS FOR PARALLELISM AND PERPENDICULARITY OF TWO LINES

17.1 If the two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are parallel, then

$$m_1 = m_2 \Rightarrow -\frac{a_1}{b_1} = \frac{-a_2}{b_2} \Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2}$$

17.2 If the two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are perpendicular, then

$$m_1.m_2 = -1 \Rightarrow -\frac{a_1}{b_1} \times -\frac{a_2}{b_2} = -1$$

$$\Rightarrow \quad a_1a_2 + b_1b_2 = 0$$

Thus two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are

(i) Coincident, if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

(ii) Parallel, if $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

(iii) Intersecting, if $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$

(iv) Perpendicular, if $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$

18. DISTANCE OF A POINT FROM A GIVEN LINE OR LENGTH OF PERPENDICULAR FROM THE POINT (x_1, y_1) TO THE STRAIGHT LINE $ax + by + c = 0$

Let the length of perpendicular be p, then

$$p = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

19. DISTANCE BETWEEN TWO PARALLEL LINES

If two lines are parallel, then they have the same distance between these throughout. Therefore to find the distance between two parallel lines find the coordinates of any point on one of the given lines, preferably putting $x = 0$ or $y = 0$. Then the perpendicular distance of this point from the other line is the required distance between the lines.

Alternatively : Let the two parallel lines be $ax + by + c_1 = 0$ and $ax + by + c_2 = 0$, then the perpendicular distance between the lines is $\left|\frac{c_2 - c_1}{\sqrt{a^2 + b^2}}\right|$

20. EQUATIONS OF STRAIGHT LINES PASSING THROUGH (x_1, y_1) MAKING ANGLE θ WITH THE GIVEN LINE $y = mx + c$ ARE

$$y - y_1 = \frac{m \pm \tan\theta}{1 \mp m\tan\theta}(x - x_1)$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

21. EQUATION OF ANY LINE PASSING THROUGH THE POINT OF INTERSECTION OF TWO GIVEN LINES

Any line passing through the point of intersection of the lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ can be represented by the equation

$$(a_1x + b_1y + c_1) + \lambda (a_2x + b_2y + c_2) = 0$$

where λ is an arbitrary constant.

22. POSITION OF A POINT WITH RESPECT TO A LINE

Any line $ax + by + c = 0$ $(c \neq 0)$ divides the whole plane into two parts (i) One part containing the origin called the origin side, (ii) the other part not containing the origin called non-origin side of the line.

The length of the perpendicular from the origin on the line $ax + by + c = 0$ is $\frac{c}{\sqrt{a^2 + b^2}}$; if $c > 0$.

If $c > 0$, we say that the origin is on the positive side of the line and if $c < 0$, the origin is said to be on the negative side of the line.

A point $P(x_1, y_1)$ is on the origin side or non-origin side of the line $ax + by + c = 0$ according as $ax_1 + by_1 + c$ and c are the same or opposite sign.

If $c > 0$, then $P(x_1, y_1)$ is on the origin side or non-origin side of line $ax + by + c = 0$ according as $ax_1 + by_1 + c$ is positive or negative.

23. EQUATIONS OF BISECTORS OF THE ANGLES BETWEEN TWO GIVEN LINES

Let $a_1x + b_1y + c = 0$ and $a_2x + b_2y + c_2 = 0$ be the equations of two given lines AB and AC respectively then the equations of the two bisectors are

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

INTRODUCTORY EXERCISE-21.2

1. Find the equation of the line with slope $\frac{2}{3}$ and intercept on the y-axis is 5 :
   (a) $y = \frac{2}{3}x + 5$ (b) $y = \frac{3}{2}x + 5$
   (c) $y = 3x + 6$ (d) none of these
2. Find the slope and the intercept on the y-axis of the line $\sqrt{3}x + 3y = 6$:
   (a) $\frac{1}{\sqrt{2}}$ (b) $\frac{1}{5}$
   (c) $\frac{2}{\sqrt{3}}$ (d) $-\frac{1}{\sqrt{3}}$
3. Find the equation of the line passing through the point $(2, -3)$ and having its slope $\frac{5}{4}$:
   (a) $4x - 5y = 20$ (b) $3x - 2y = 5$
   (c) $5x - 4y = 22$ (d) none of these
4. Find the equation of the line which cuts off intercepts 2 and 3 from the axes :
   (a) $9x - 7y = 6$ (b) $3x + 2y = 6$
   (c) $4x + 3y = 7$ (d) none of these
5. Find the intercepts made by the line $3x + 4y - 12 = 0$ on the axes :
   (a) 2 and 3 (b) 4 and 3
   (c) 3 and 5 (d) none of these
6. Find the equation of the line through the points $(-1, -2)$ and $(-5, 2)$:
   (a) $2x + y = 3$ (b) $3x - 2y + 7 = 0$
   (c) $x + y + 3 = 0$ (d) none of these
7. Find the equation of the straight line passing through the point $(-1, 4)$ and having a gradient of 2.5 :
   (a) $2x - 5y + 13 = 0$ (b) $5x - 9y = 13$
   (c) $13x - 15y + 17 = 0$ (d) $5x - 2y + 13 = 0$
8. Find the equation of the straight line which makes equal intercepts on the axes passes through the point $(3, -5)$:
   (a) $x - y = 2$ (b) $x + y + 2 = 0$
   (c) $y - x + 2 = 0$ (d) none of these
9. Find the equation of the straight line making intercepts on the axes equal in magnitude but opposite in sign and passing through the point $(-5, -8)$:
   (a) $x - y = 7$ (b) $2x + y = 3$
   (c) $x - y = 3$ (d) none of these
10. Find the equation of the line passing through the point $(-4, -5)$ and perpendicular to the line joining the points $(1, 2)$ and $(5, 6)$:
    (a) $x + y + 17 = 0$ (b) $3x + 2y + 11 = 0$
    (c) $x + y + 9 = 0$ (d) $x - y + 20 = 0$
11. A straight line intersects the x-axis at A and the y-axis at B. AB is divided internally at $C(8, 10)$ in the ratio 5 : 4. Find the equation of AB :
    (a) $x + y = 18$ (b) $x + y + 2 = 0$
    (c) $x + y - 2 = 0$ (d) none of these
12. Find the equation of the straight line which passes through the point $(3, 4)$ and has intercepts on the axes such that their sum is 14 :
    (a) $4x + 3y = 24$ (b) $x + y = 7$
    (c) $3x + 7y = 43$ (d) both (a) and (b)
13. A straight line passes through the points $(a, 0)$ and $(0, b)$. The length of the line segment contained between the axis is 13 and the product of the intercepts on the axes is 60. Find the equation of the straight line :
    (a) $5x + 12y = 60$ (b) $7x - 12y = 50$
    (c) $5x + 12y + 60 = 0$ (d) both (a) and (c)
14. A firm produces 50 units of a good for Rs. 320 and 80 units for Rs. 380. Supposing that the cost curve is a straight line, estimate the cost of producing 110 units :

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

(a) Rs. 330 (b) Rs. 1665
(c) Rs. 440 (d) Rs. 365

15. Find the equation of the line on which length of the perpendicular from the origin is 5 and the angle which this perpendicular makes with the x-axis is 60° :
(a) $x\sqrt{3} + 2y + 8 = 0$ (b) $x + \sqrt{2}y - 7 = 0$
(c) $x + \sqrt{3}y = 10$ (d) none of (a),(b), (c)

16. Find the equation of the line which passes through the point $(3, -4)$ and makes an angle of 60° with the positive direction of x-axis :
(a) $x\sqrt{2} + y\sqrt{3} = 0$ (b) $\sqrt{3}x - y = 4 + 3\sqrt{3}$
(c) $x\sqrt{3} + y = 3\sqrt{2} + 5$ (d) none of (a), (b),(c)

17. Find the equation of the line joining the points of intersection of $2x + y = 4$ with $x - y + 1 = 0$ and $2x - y - 1 = 0$ with $x + y - 8 = 0$:
(a) $2x + 3y + 6 = 0$ (b) $3x + 2y + 12 = 0$
(c) $3x - 2y + 1 = 0$ (d) none of (a), (b),(c)

18. Find the equation of one of the two lines which pass through the point $(4, 5)$ which make an acute angle 45° with the line $2x - y + 7 = 0$:
(a) $x - 2y = 0$ (b) $7x + 5y - 3 = 0$
(c) $3x + y + 8 = 0$ (d) $x - 3y + 11 = 0$

19. Find the equation of the straight line which passes through the point $(5, -6)$ which is parallel to the line $8x + 7y + 5 = 0$:
(a) $3x - 5y + 8 = 0$ (b) $7x + 8y + 5 = 0$
(c) $7x - 8y + 2 = 0$ (d) $8x + 7y + 2 = 0$

20. Find the equation of the straight line which passes through the point of intersection of the straight lines $x + y = 8$ and $3x - 2y + 1 = 0$ and is parallel to the straight line joining the points $(3, 4)$ and $(5, 6)$:
(a) $x - y + 2 = 0$ (b) $x + y - 2 = 0$
(c) $3x - 4y + 8 = 0$ (d) none of these

21. Find the length of the perpendicular from the point $(3, -2)$ to the straight line $12x - 5y + 6 = 0$:
(a) 5 (b) 4
(c) 6 (d) 8

22. Find the distance between two parallel lines $5x + 12y - 30 = 0$ and $5x + 12y - 4 = 0$
(a) 3 (b) 7
(c) $\frac{5}{2}$ (d) 2

23. Find the equation of the line through the point of intersection of $2x - 3y + 1 = 0$ and $x + y - 2 = 0$ which is parallel to the y-axis.
(a) $x = 1$ (b) $8x = 7$
(c) $x + 3 = 0$ (d) $x = 6$

24. Find the equation of the line which passes through the point of intersection of the lines $x + 2y - 3 = 0$ and $4x - y + 7 = 0$ and is parallel to the line $y - x + 10 = 0$.
(a) $2x + 2y + 5 = 0$ (b) $3x - 3y = 10$
(c) $3x + 2y - 8 = 0$ (d) none of these

25. Find the equation of the line which passes through the point of intersection of the lines $2x - y + 5 = 0$ and $5x + 3y - 4 = 0$ and is perpendicular to the line $x - 3y + 21 = 0$
(a) $2x + y + 8 = 0$ (b) $3x + 4y - 7 = 0$
(c) $3x + y = 0$ (d) none of these

26. Find the equation of the line through the intersection of the lines $3x + 4y = 7$ and $x - y + 2 = 0$ having slope 3.
(a) $4x - 3y + 7 = 0$ (b) $21x - 7y + 16 = 0$
(c) $8x + y + 8 = 0$ (d) none of these

27. Find the equation of the straight line which passes through the point of intersection of the straight lines $3x - 4y + 1 = 0$ and $5x + y - 1 = 0$ and cuts off equal intercepts from the axis.
(a) $32x + 32y + 11 = 0$ (b) $23x + 23y = 11$
(c) $9x + 18y + 5 = 0$ (d) none of these

28. A straight line $\frac{x}{a} - \frac{y}{b} = 1$ passes through the point $(8, 6)$ and cuts off a triangle of area 12 units from the axes of coordinates. Find the equations of the straight line.
(a) $3x - 2y = 12$ (b) $4x - 3y = 12$
(c) $3x - 8y + 24 = 0$ (d) both (a) and (c)

29. Find the equations of the bisectors of the angle between the straight line $3x + 4y + 2 = 0$ and $5x - 12y - 6 = 0$.
(a) $8x + y + 7 = 0$ (b) $16x - 12y - 1 = 0$
(c) $x + 8y + 4 = 0$ (d) both (b) and (c)

30. Find the area of the triangle formed by the lines whose equations are $2y - x = 5$, $y + 2x = 7$ and $y - x = 1$.
(a) $\frac{3}{10}$ (b) 10
(c) 6 (d) $\frac{2}{5}$

31. Find the coordinates of the orthocentre of the triangle whose vertices are $(1, 2)$, $(2, 3)$ and $(4, 3)$.
(a) (2, 5) (b) (3, 4)
(c) (1, 6) (d) none of these

32. Two vertices of a triangle ABC are $B(5, -1)$ and $C(-2, 3)$. If the orthocentre of the triangle is the origin, find the third vertex.
(a) $\left(\frac{7}{2}, \frac{13}{2}\right)$ (b) $\left(\frac{3}{2}, \frac{11}{2}\right)$
(c) $(-4, -7)$ (d) none of these

33. The area of a triangle is 5. Two of its vertices are $(2, 1)$ and $(3, -2)$. The third vertex lies on $y = x + 3$. Find the third vertex :
(a) $\left(\frac{2}{7}, \frac{13}{5}\right)$ (b) $\left(\frac{7}{2}, \frac{13}{2}\right)$
(c) $\left(\frac{9}{2}, \frac{13}{2}\right)$ (d) $\left(\frac{7}{2}, \frac{13}{2}\right)$ or $\left(-\frac{3}{2}, \frac{3}{2}\right)$

34. A straight line L is perpendicular to the line $5x - y = 1$. The area of the triangle formed by the line L and coordinate axes is 5. Find the equation of the line :
(a) $x + 5y = \pm 5\sqrt{2}$ (b) $x - 3y = 0$
(c) $2x + y = 0$ (d) $x + 4y = 5\sqrt{2}$

35. $(1, 2)$ and $(3, 8)$ are a pair of a opposite vertices of square. Find the diagonals of the square passing through $(1, 2)$:
(a) $x - 2y = 1$ (b) $2x + 7y = 0$
(c) $3x + 2y + 7 = 0$ (d) $3x - y = 1$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Answers

INTRODUCTORY EXERCISE- 21.1

1. (b)	2. (d)	3. (d)	4. (b)	5. (c)	6. (d)	7. (a)	8. (a)	9. (b)	10. (a)
11. (b)	12. (b)	13. (c)	14. (d)	15. (c)	16. (b)	17. (a)	18. (c)	19. (c)	20. (c)
21. (c)	22. (b)	23. (a)	24. (a)	25. (b)	26. (c)	27. (b)	28. (c)	29. (a)	30. (d)

INTRODUCTORY EXERCISE- 21.1

1. (a)	2. (d)	3. (c)	4. (b)	5. (b)	6. (c)	7. (d)	8. (b)	9. (c)	10. (c)
11. (a)	12. (d)	13. (d)	14. (c)	15. (c)	16. (b)	17. (c)	18. (d)	19. (d)	20. (a)
21. (b)	22. (d)	23. (a)	24. (b)	25. (c)	26. (b)	27. (b)	28. (d)	29. (d)	30. (a)
31. (a)	32. (c)	33. (d)	34. (d)	35. (d)					

www.LearnEngineering.in

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

Hints & Solutions

INTRODUCTORY EXERCISE 21.1

1. The point $(-2, 3)$ lies in the second quadrant.

2. The point $(2, -3)$ lies in the fourth quadrant

3. Distance between two points $= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Let $(x_1, y_1) \equiv (-5, 3)$ and $(x_2, y_2) \equiv (3, 1)$

$\therefore$ Required distance $= \sqrt{(3+5)^2 + (1-3)^2}$

$= \sqrt{64+4} = \sqrt{68} = 2\sqrt{17}$ unit

4. Let $A \equiv (4, 3), B \equiv (7, -1), C \equiv (9, 3)$

$AB = \sqrt{(7-4)^2 + (-1-3)^2} = \sqrt{25} = 5$

$BC = \sqrt{(9-7)^2 + (3+1)^2} = \sqrt{20} = 2\sqrt{5}$

$CA = \sqrt{(4-9)^2 + (3-3)^2} = \sqrt{25} = 5$

$\therefore \quad AB = CA = 5$

Hence ABC is an isosceles triangle.

5. Let $A \equiv (4, 4), B \equiv (3, 5), C \equiv (-1, -1)$

Then $AB = \sqrt{(3-4)^2 + (5-4)^2} = \sqrt{2}$

$BC = \sqrt{(-1-3)^2 + (-1-5)^2} = \sqrt{52}$

$AC = \sqrt{(-1-4)^2 + (-1-4)^2} = \sqrt{50}$

$\therefore \quad AB^2 + AC^2 = BC^2$

Hence by the Pythagorus Theorem, ABC is a right angled triangle.

6. Let $A \equiv (7, 9), B \equiv (3, -7)$ and $C \equiv (-3, 3)$

Then $AB = \sqrt{(3-7)^2 + (-7-9)^2} = \sqrt{272}$

$BC = \sqrt{(-3-3)^2 + (3+7)^2} = \sqrt{136}$

$AC = \sqrt{(7+3)^2 + (9-3)^2} = \sqrt{136}$

$\therefore \quad AB^2 = BC^2 + AC^2$

Hence by Pythagorus Theorem, triangle ABC is a right angled triangle.

Also, since $BC = CA$, hence ABC is an isosceles triangle.

Thus ABC is a right angled isoceles triangle.

7. The coordinates of the mid points of AB

$= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) = \left(\frac{-3+5}{2}, \frac{2+4}{2}\right) = (1, 3)$

8. Here $(x_1, y_1) = (-3, 2)$ and $(x_2, y_2) = (5, 4)$

The coordinates of the point which divides AB in the ratio $2:3$

$= \left(\frac{2\times 5 + 3\times -3}{2+3}, \frac{2\times 4 + 3\times 2}{2+3}\right) = \left(\frac{1}{5}, \frac{14}{5}\right)$

9. The required coordinates of the point which divides the join of $(2, 4)$ and $(6, 8)$ externally in the ratio $5:3$ are $\left(\frac{mx_2 - nx_1}{m-n}, \frac{my_2 - ny_1}{m-n}\right)$

Here, $m : n = 5 : 3, \quad (x_1, y_1) = (2, 4), \quad (x_2, y_2) = (6, 8)$

Hence the required co-ordinates $= (12, 14)$.

10. Let the co-ordinate of the centroid of ΔABC be (x, y) then

$(x, y) = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$

$= \left(\frac{3+2-2}{3}, \frac{1+3+2}{3}\right) = (1, 2)$

11. Let $A \equiv (4, -2), B \equiv (5, 5)$ and $C \equiv (-2, 4)$

Then $a = BC = \sqrt{(-2-5)^2 + (4-5)^2} = 5\sqrt{2}$

$b = AC = \sqrt{(4+2)^2 + (-2-4)^2} = 6\sqrt{2}$

$c = AB = \sqrt{(5-4)^2 + (5+2)^2} = 5\sqrt{2}$

and $(x_1, y_1) \equiv (4, -2), (x_2, y_2) \equiv (5, 5),$

$(x_3, y_3) \equiv (-2, 4)$

$\therefore$ The coordinates of the incentre of the ΔABC are

$\left(\frac{ax_1 + bx_2 + cx_3}{a+b+c}, \frac{ay_1 + by_2 + cy_3}{a+b+c}\right)$

$= \left(\frac{5}{2}, \frac{5}{2}\right)$ {Substitute the values of (a, b, c)

x_1, x_2, x_3, y_1, y_2 and y_3}

12. Let $S(x, y)$ be the circumcentre, then $AS = BS = CS = R$, where R is the circumradius

Now $(AS)^2 = (BS)^2$ where $S \equiv (x, y), A \equiv (8, 6), B \equiv (8, -2)$

$\therefore \quad (x-8)^2 + (y-6)^2 = (x-8)^2 + (y+2)^2$

$\Rightarrow \quad 36 - 12y = 4 + 4y$

$\Rightarrow \quad y = 2$

Again $(BS)^2 = (CS)^2$, where $S \equiv (x, y) B \equiv (8, -2), C \equiv (2, -2)$

$\therefore \quad (x-8)^2 + (y+2)^2 = (x-2)^2 + (y+2)^2$

$\Rightarrow \quad x^2 + 64 - 16x + y^2 + 4 + 4y$

$= x^2 + 4 - 4x + y^2 + 4 + 4y$

$\Rightarrow \quad -12x + 60 = 0$

$\Rightarrow \quad x = 5$

$\therefore$ The circumcentre $S \equiv (x, y) = (5, 2)$

13. Let $P(1, 1), Q(2, -3), R(3, 4)$ be the midpoints of sides AB, BC and CA respectively triangle ABC.

Let $A(x_1, y_1), B(x_2, y_2)$ and $C(x_3, y_3)$ be the vertices of triangle ABC.

Then, P is the mid point of AB

$\Rightarrow \quad \frac{x_1 + x_2}{2} = 1, \frac{y_1 + y_2}{2} = 1$

$\Rightarrow \quad x_1 + x_2 = 2$ and $y_1 + y_2 = 2$ ...(1)

Q is the mid point of BC

$\Rightarrow \quad \frac{x_2 + x_3}{2} = 2, \frac{y_2 + y_3}{2} = -3$

$\Rightarrow \quad x_2 + x_3 = 4$ and $y_2 + y_3 = -6$

R is the mid point of AC

$\Rightarrow \quad x_1 + x_3 = 6$ and $y_1 + y_3 = 8$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

From (1), (2) and (3), we get

$(x_1, y_1) \equiv (2, 8)$, $(x_2, y_2) \equiv (0, -6)$

and $(x_3, y_3) \equiv (4, 0)$

Then the coordinates of the centroid

$$= \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$$

$$= \left(\frac{2+0+4}{3}, \frac{8-6+0}{3}\right) = \left(2, \frac{2}{3}\right)$$

14. Since $(x_1, y_1) \equiv A(2, 8)$, $(x_2\ y_2) \equiv B(0, -6)$

and $(x_3, y_3) \equiv C(4, 0)$

Now, $a = BC = 2\sqrt{3}$, $b = AC = 2\sqrt{17}$, $c = AB = 10\sqrt{2}$

The coordinates of the in-centre of the triangle ABC are $\left(\frac{ax_1 + bx_2 + cx_3}{a+b+c}, \frac{ay_1 + by_2 + cy_3}{a+b+c}\right)$

$\therefore$ Required coordinates of incentre are

$$\left(\frac{2\sqrt{13} + 20\sqrt{2}}{\sqrt{13} + \sqrt{17} + 5\sqrt{2}}, \frac{2\sqrt{13} - 6\sqrt{17}}{\sqrt{13} + \sqrt{17} + 5\sqrt{2}}\right)$$

15. Let A, B, C and D be the vertices of the quadrilateral whose coordinates are $(-2, -1)$ $(1, 0)$, $(4, 3)$ and $(1, 2)$ respectively.

Now, $AB = \sqrt{10}$, $BC = \sqrt{18}$, $DC = \sqrt{10}$, $AD = \sqrt{18}$

$\therefore$ $AB = CD$ and $BC = AD$ *i.e.*, the opposite sides are equal.

Hence $ABCD$ is a parallelogram.

16. Let $A(-1, 0)$, $B(3, 1)$, $C(2, 2)$ and $D(x, y)$ be the vertices of a parallelogram $ABCD$ taken in order. Since the diagonals of a parallelogram bisect each other.

$\therefore$ Coordinates of the mid point of AC = Coordinates of the mid point of BD

$$\Rightarrow \quad \left(\frac{-1+2}{2}, \frac{0+2}{2}\right) = \left(\frac{3+x}{2}, \frac{1+y}{2}\right)$$

$$\Rightarrow \quad \left(\frac{1}{2}, 1\right) = \left(\frac{3+x}{2}, \frac{y+1}{2}\right)$$

$$\Rightarrow \quad \frac{3+x}{2} = \frac{1}{2} \text{ and } \frac{y+1}{2} = 1$$

$$\Rightarrow \quad x = -2 \text{ and } y = 1$$

Hence the fourth, vertex of the parallelogram is $(-2, 1)$

17. Solve as the question no. 13 has been solved. The required vertices of the triangle are $(1, -4)$, $(3, 2)$, $(-1, 2)$.

18. Let the vertices be $A(4, -1)$, $B(6, 0)$, $C(7, 2)$, $D(5, 1)$ then the coordinates of the mid point of AC are

$$\left(\frac{4+7}{2}, \frac{-1+2}{2}\right) = \left(\frac{11}{2}, \frac{1}{2}\right)$$

Coordinates of the mid point of BD are

$$\left(\frac{6+5}{2}, \frac{0+1}{2}\right) = \left(\frac{11}{2}, \frac{1}{2}\right)$$

Thus AC and BD have the same mid point. Hence $ABCD$ is a parallelogram.

Now, $AB = \sqrt{5}$, $BC = \sqrt{5}$ $\therefore AB = BC$

So, $ABCD$ is a parallelogram whose adjacent sides are equal. Hence, $ABCD$ is a rhombus.

Also, we have $AC = 3\sqrt{2}$ and $BD = \sqrt{2}$

Clearly, $AC \neq BD$. So $ABCD$ is not a square.

19. Let A, B, C, D be the points $(-4, 2)$, $(1, -1)$, $(4, 1)$ and $(2, 5)$ respectively. Then the area of the quadrilateral $ABCD$

$$= \frac{1}{2}\{-4 \times -1 - 2 \times 1 + 1 \times 1 - 4 \times -1 + 4 \times 5 - 2 \times 1 + 2 \times 2 \times -5 \times -4\}$$

$$= \frac{1}{2}(4 - 2 + 1 + 4 + 20 - 2 + 4 + 20)$$

$$= \frac{49}{2} = 24 \cdot 5 \text{ square units.}$$

HINT $-4 \quad 1 \quad 4 \quad 2 \quad -4$
$\quad 2 \quad -1 \quad 1 \quad 5 \quad 2$

NOTE Area of quadrilateral $= \frac{1}{2}\{(x_1y_2 - x_2y_1) + (x_2y_3 - x_2y_3) + (x_3y_4 - x_4y_3) + (x_4y_1 - x_1y_4)\}$

20. Let A, B, C, D be the points $(3, 2)$, $(6, 3)$, $(x\ y)$ and $(6, 5)$ respectively.

Since $ABCD$ is a parallelogram, the diagonals AC and BD must bisect each other *i.e.*, the mid points of AC and the mid point of BD must coincide and hence the coordinates of the two mid points are the same

$$\therefore \quad \frac{3+x}{2} = \frac{6+6}{2} \text{ and } \frac{2+y}{2} = \frac{3+5}{2}$$

or $\quad 3 + x = 12$ and $2 + y = 8$

or $\quad x = 9$ and $y = 6$

Hence $\quad (x, y) = (9, 6)$

21. Let the line $y - x + 2 = 0$ divide the join of $(3, -1)$ and $(8, 9)$ at the point P in the ratio $k : 1$. Then the coordinates of P are

$$\left(\frac{k \cdot 8 + 1 \cdot 3}{k+1}, \frac{k \cdot 9 + 1(-1)}{k+1}\right) = \left(\frac{8k+3}{k+1}, \frac{9k-1}{k+1}\right)$$

Since, this point lies on the line $y - x + 2 = 0$, we have

$$\frac{9k-1}{k+1} - \frac{8k+3}{k+1} + 2 = 0$$

or $\quad k = \frac{2}{3}$

Hence the required ratio is $k : 1 = \frac{2}{3} : 1 = 2 : 3$

22. Let the points be $A(1, 2)$, $B(-5, 6)$, $C(7, -4)$, $D(h, -2)$

Given, area of the quadrilateral $ABCD = 0$

or $\quad \frac{1}{2}\{(6 + 10) + (20 - 42) + (-14 + 4h) + (2h + 2)\} = 0$

or $\quad h = 3$

23. Area of $\Delta PBC = \frac{1}{2}\{(5x + 3y) + (6 - 20) + (4y + 2x)\}$

$$= \frac{1}{2}(7x + 7y - 14)$$

$$= \frac{7}{2}(x + y - 2)$$

and area of $\Delta ABC = \frac{1}{2}\{(30 + 9) + (6 - 20) + (12 + 12)\}$

$$= \frac{49}{2}$$

Hence $\quad \frac{\Delta PBC}{\Delta ABC} = \frac{x + y - 2}{7}$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

24. Slope of the line $= \frac{y_2 - y_1}{x_2 - x_1} = \frac{7-5}{9-7} = 1$

Here $(x_1, y_1) \equiv (7, 5)$ and $(x_2, y_2) \equiv (9, 7)$

25. Let m_1 and m_2 be the slopes of BA and BC respectively. Then

$$m_1 = \frac{3-1}{2-(-2)} = \frac{1}{2} \quad \text{and} \quad m_2 = \frac{-4-3}{-2-2} = \frac{7}{4}$$

Let θ be the angle between BA and BC. Then

$$\tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right| = \left|\frac{\frac{7}{4} - \frac{1}{2}}{1 + \frac{7}{4} \times \frac{1}{2}}\right| = \pm\frac{2}{3}$$

$$\Rightarrow \quad \theta = \tan^{-1}\left(\frac{2}{3}\right)$$

26. If m_1 be the slope of the line passing through the points $A(3, 5)$ and $B(-4, 2)$, then

$$m_1 = \frac{2-5}{-4-3} = \frac{3}{7}$$

If m_2 be the slope of the perpendicular line CD then

$m_1 \cdot m_2 = -1$ or $\frac{3}{7} m_2 = -1$

or $$m_2 = -\frac{7}{3}$$

27. Let m_1 be the slope of line AB

$\therefore$ $$m_1 = \frac{3-(-3)}{6-2} = \frac{3}{2}$$

If m_2 be the slope of a line parallel to AB, then

$$m_2 = m_1 = \frac{3}{2}$$

28. Let m_3 be the slope of line perpendicular to AB, then

$$m_1 \cdot m_3 = -1$$

$$\Rightarrow \quad \frac{3}{2} \cdot m_3 = -1$$

$$\Rightarrow \quad m_3 = -\frac{2}{3}$$

29. $$m_1 = \text{Slope of } AB = \frac{3-6}{2-6} = \frac{3}{4}$$

$$m_2 = \text{Slope of } BC = \frac{7-3}{4-2} = 2$$

and $$m_3 = \text{Slope of } AC = \frac{7-6}{4-6} = -\frac{1}{2}$$

$\therefore$ $m_2 \cdot m_3 = 2 \times -\frac{1}{2} = -1$

This shows that BC is perpendicular to AC.
Hence, ABC is a right-angled triangle.

30. $$m_1 = \text{Slope of } AB = \frac{4-(-2)}{3-1} = 3$$

and $$m_2 = \text{Slope of } BC = \frac{7-4}{4-3} = 3$$

$\therefore$ $m_1 = m_2$

$\therefore AB$ is parallel to BC and B is common to both AB and BC.
Hence, the point $A(1, -2)$, $B(3, 4)$ and $C(4, 7)$ are collinear.

INTRODUCTORY EXERCISE 21.2

1. The equation of the line with slope 2/3 and intercept on the y-axis 5 is $y = \frac{2}{3}x + 5 (\because y = mx + c)$

2. We have $\sqrt{3}x + 3y = 6$

or $$3y = -\sqrt{3}x + 6$$

or $$y = -\frac{1}{\sqrt{3}}x + 2$$

Comparing the above equation with $y = mx + c$

we get $$m = -\frac{1}{\sqrt{3}} \quad \text{and} \quad c = 2$$

Hence slope is $\left(-\frac{1}{\sqrt{3}}\right)$ and intercept on the y-axis is 2.

3. We have $m = \frac{5}{4}$ and $(x_1, y_1) = (2, -3)$

$\therefore$ The equation of the line as point slope form is

$$y - y_1 = m(x - x_1)$$

or $$y - (-3) = \frac{5}{4}(x - 2)$$

or $$y + 3 = \frac{5}{4}(x - 2)$$

or $$5x - 4y = 22$$

4. Here $a = 2$ and $b = 3$

$\therefore$ The required equation of the line is $\frac{x}{2} + \frac{y}{3} = 1$

$\Rightarrow$ $$3x + 2y = 6$$

5. We have $3x + 4y - 12 = 0$

$\Rightarrow$ $$3x + 4y = 12$$

$\Rightarrow$ $$\frac{3x}{12} + \frac{4y}{12} = 1 \Rightarrow \frac{x}{4} + \frac{y}{3} = 1$$

Which is of the form $\frac{x}{a} + \frac{y}{b} = 1$

Thus the required intercepts on the axes are 4 and 3.

6. The equation of the line through the points $(-1, -2)$ and $(-5, 2)$ is $(y - y_1) = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

where $(x_1, y_1) \equiv (-1, -2)$

and $(x_2, y_2) \equiv (-5, 2)$

$\therefore$ Required equation is

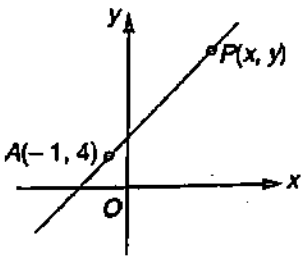
$$y - (-2) = \frac{2-(-2)}{-5-(-1)}[x - (-1)]$$

or $$y + 2 = \frac{4}{-4}(x + 1)$$

or $$x + y + 3 = 0$$

7. Let $(-1, 4)$ be the point as shown in figure and let $P(x, y)$ be any point on the line. Then the gradient (or slope) of the line is

$$\frac{y-4}{x-(-1)} = 2.5$$



For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$\Rightarrow \quad \frac{y-4}{x+1}=\frac{5}{2}$

$\Rightarrow \quad 5x-2y+13=0$

8. Let the equation of the straight line in the intercept form be

$$\frac{x}{a}+\frac{y}{b}=1 \qquad \ldots(1)$$

Since the intercepts are equal, therefore $a=b$

$\therefore$ From equation (1)

$$x+y=a \qquad \ldots(2)$$

Since this line passes through the point $(3,-5)$

$\therefore \quad 3+(-5)=a$

$\Rightarrow \quad a=-2$

$\therefore$ From equation (2), the required equation of the straight line is $x+y=-2$ or $x+y+2=0$

9. Let the equation of the straight line be

$$\frac{x}{a}+\frac{y}{b}=1 \qquad \ldots(1)$$

Since intercepts a, b are equal in magnitude but opposite in sign.

$\therefore \quad b=-a$

$\therefore$ From eq. (1) $\frac{x}{a}+\frac{y}{(-a)}=1$

or $\quad x-y=a \qquad \ldots(2)$

Since this line passes through the point $(-5,-8)$

$\therefore \quad -5-(-8)=a$

$\Rightarrow \quad a=3$

Hence, from (2) the required equation of the line is $x-y=3$

10. Let m_1 = slope of the line 'joining' $(1, 2)$ and $(5, 6)$

$\therefore \quad m_1=\frac{6-2}{5-1}=\frac{4}{4}=1$

If m_2 be the slope of the perpendicular line, then $m_1 m_2=-1$

$\Rightarrow m_2=-1 \ (\because m_1=1)$

$\therefore$ The required line has slope $m_2=-1$ and passes through the point $(-4,-5)$

Hence, the required equation of the line in the point slope form is

$$(y-y_1)=m_2(x_2-x_1)$$

or $\quad y-(-5)=-1\{x-(-4)\}$

or $\quad x+y+9=0$

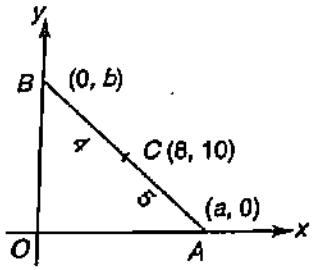
11. Let the equation of the line AB be $\frac{x}{a}+\frac{y}{b}=1 \qquad \ldots(1)$

Then the coordinates of A and B are respectively $(a, 0)$ and $(0, b)$.

Since $C(8, 10)$ divides AB in the ratio 5 : 4, we have

$$\frac{5\times 0+4\times a}{5+4}=8$$

and $\quad \frac{5\times b+4\times 0}{5+4}=10$



or $\quad a=18 \quad$ and $\quad b=18$

Hence from (1), the required equation of the line AB is

$$\frac{x}{18}+\frac{y}{18}=1 \quad \text{or} \quad x+y=18$$

12. Let the equation of the line in the intercept form be $\frac{x}{a}+\frac{y}{b}=1$,

where a and b are intercepts on the axes.

Then $\quad a+b=14 \quad$ or $\quad b=14-a$

Since the line $\frac{x}{a}+\frac{y}{b}=1$ passes through the point $(3, 4)$;

$\therefore \quad \frac{3}{a}+\frac{4}{b}=1 \quad$ or $\quad \frac{3}{a}+\frac{4}{14-a}=1$

or $\quad a^2-13a+42=0$

or $\quad (a-6)(a-7)=0$

$\therefore \quad a=6, 7$

If $a=6$ then $b=8$

If $a=7$ then $b=7$

Hence the required equation of the line are

$$\frac{x}{6}+\frac{y}{8}=1 \quad \text{and} \quad \frac{x}{7}+\frac{y}{7}=1$$

or $\quad 4x+3y=24 \quad$ and $\quad x+y=7$

13. Since the line passes through $A(a, 0)$ and $B(0, b)$, it makes intercepts a and b on the axes of x and y. Let the equation of the line be

$$\frac{x}{a}+\frac{y}{b}=1 \qquad \ldots(1)$$

By the given conditions,

$$AB=13 \ \Rightarrow \ a.b=60 \qquad \ldots(2)$$

From (1) $\sqrt{a^2+b^2}=13$

$\Rightarrow \quad a^2+b^2=169$

$\Rightarrow \quad a+b=\pm 17$

Again $(a-b)^2=(a+b)^2-4ab=289-240=49$

$\therefore$ we get $a=12, b=5$ and $a=-12, b=-5$

$\therefore$ The required equations of the straight line are

$$\frac{x}{12}+\frac{y}{5}=1 \quad \text{and} \quad \frac{x}{-12}+\frac{y}{-5}=1$$

i.e., $\quad 5x+12y=60$ and $5x+12y+60=0$

14. Let the equation of the cost curve as a straight line be

$$y=mx+c \qquad \ldots(1)$$

where x = number of units of a good produced

and y = cost of x units in rupees.

Given, when $x=50$, $y=320$ and when $x=80$, $y=380$

from (1) $\quad 320=50m+c \qquad \ldots(2)$

and $\quad 380=80m+c \qquad \ldots(3)$

Subtracting (2) from (3), we get $m=2$

Substituting $m=2$ in equation (2), we get $c=220$

$\therefore$ From (1) $y=2x+220$

when $x=110$, $y=2\times 110+220=440$

Hence, the required cost of producing 110 units is Rs. 440.

15. Here $p=5$ and $\alpha=60°$

$\therefore$ The required equation of the line is

$$x\cos\alpha+y\sin\alpha=p$$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

or $x\cos 60° + y\sin 60° = 5$

$\Rightarrow$ $x + \sqrt{3}y = 10$ $\left(\because \sin 60° = \frac{\sqrt{3}}{2} \text{ and } \cos 60° = \frac{1}{2}\right)$

16. Here $(x_1, y_1) \equiv (3, -4)$ and $\theta = 60°$

The required equation of the line in the symmetric form is

$\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta}$

$\Rightarrow$ $\frac{x-3}{\cos 60°} = \frac{y-(-4)}{\sin 60°}$

$\Rightarrow$ $\sqrt{3}x - y = 4 + 3\sqrt{3}$

17. We have $2x + y = 4$ ...(1)

and $x - y + 1 = 0$ ...(2)

Solving equations (1) and (2), we get

$x = 1$ and $y = 2$

$\therefore$The point of intersection of (1) and (2) is (1, 2).

Again $2x - y - 1 = 0$ ...(3)

$x + y - 8 = 0$ ...(4)

Solving equations (3) and (4), we get $(x, y) \equiv (3, 5)$

$\therefore$The point of intersection of (3) and (4) is (3, 5)

$\therefore$ The required equation of the straight line joining the points of intersection (1, 2), (3, 5) is

$y - 2 = \frac{(5-2)}{(3-1)} \cdot (x - 1)$

$\Rightarrow$ $3x - 2y + 1 = 0$

18. The equation of the line through the point (4, 5) is

$y - 5 = m(x - 4)$ ...(1)

where m is the slope of the line.

Now the given line is $2x - y + 7 = 0$

$\Rightarrow$ $y = 2x + 7$ ...(2)

If m_1 be the slope of the line (2), then $m_1 = 2$

If equation (1) makes an angle of 45° with equation (2), then we have

$\tan 45° = \frac{m_1 - m}{1 + m \cdot m_1} = \frac{2-m}{1+2m}$ *i.e.*, $\frac{2-m}{1+2m}$

$\therefore$either $1 = \frac{m-2}{1+2m}$ or $1 = \frac{2-m}{1+2m}$

If $\frac{m-2}{1+2m} = 1$ then $m = -3$

If $\frac{2-m}{1+2m} = 1$ then $m = \frac{1}{3}$

Hence, from (1) the required equation of the two lines is

$y - 5 = -3(x - 4)$ and $y - 5 = \frac{1}{3}(x - 4)$

$\Rightarrow$ $3x - y - 17 = 0$ and $x - 3y + 11 = 0$

19. The equation of any straight line parallel to the line $8x + 7y + 5 = 0$ is $8x + 7y + c = 0$ ...(1)

where c is an arbitrary constant.

If the line (1) passes through the point (5, – 6), then

$8 \times 5 + 7 \times (-6) + c = 0 \Rightarrow c = 2$

Hence from (1), the required equation of the straight line is $8x + 7y + 2 = 0$

20. Solving $x + y = 8$ and $3x - 2y + 1 = 0$, we get the point of intersection.

$\therefore$The point of intersection is (3, 5).

Now, the equation of the line joining the points (3, 4) and (5, 6) is $(y - 4) = \frac{(6-4)}{(5-3)}(x - 3)$

$\Rightarrow$ $x - y + 1 = 0$ ...(1)

$\therefore$The equation of the line parallel to the line $x - y + 1 = 0$ is

$x - y + c = 0$ ...(2)

Where c is an arbitrary constant. If the line passes through the point (3, 5) then

$3 - 5 + c = 0$ or $c = 2$

Hence from (2), the required equation of the line is $x - y + 2 = 0$.

21. Length of the perpendicular from the point (3, – 2) to the straight line $12x - 5y + 6 = 0$ is

$\frac{12\times 3 - 5\times -2 + 6}{\sqrt{(12)^2 + (-5)^2}} = \frac{36 + 10 + 6}{\sqrt{169}} = 4$ units.

22. Putting $y = 0$ in $5x + 12y - 30 = 0$, we get

$5x - 30 = 0$ or $x = 6$

$\therefore$ (6, 0) is a point on the first line $5x + 12y - 30 = 0$

Required distance between the parallel lines = Perpendicular distance of the point (6, 0) from the second line $5x + 12y - 4 = 0$.

$= \frac{5.6 + 12.0 - 4}{\sqrt{5^2 + 12^2}} = \frac{30 - 4}{13} = 2$ units

23. The equation of the line through the point of intersection of $2x - 3y + 1 = 0$ and $x + y - 2 = 0$, is

$(2x - 3y + 1) + k(x + y - 2) = 0$

i.e., $(2 + k)x + (k - 3)y + (1 - 2k) = 0$ ...(1)

If this line is parallel to the y-axis, then its equation must be of the form $x = h$, *i.e*, the coefficient of y in (1) must be zero.

$\therefore$ $k - 3 = 0$ or $k = 3$

Hence, from (1) the required equation of the line is $(2 + 3)x + 0 \cdot y + (1 - 2 \times 3) = 0$ [Putting $k = 3$]

$\Rightarrow$ $x = 1$

24. The equation of any line passing through the point of intersection of the lines $x + 2y - 3 = 0$ and $4x - y + 7 = 0$ is

$x + 2y - 3 + k(4x - y + 7) = 0$ ...(1)

or $(1 + 4k)x + (2 - k)y + (7k - 3) = 0$ ...(2)

m_1 = Slope of the line (2) $= \frac{4k+1}{k-2}$

and m_2 =(Slope of the line $y - x + 10 = 0$) = 1

If the line (1) be parallel to the line $y - x + 10 = 0$

then $\frac{4k+1}{k-2} = 1 \Rightarrow k = -1$

Hence from (1), the required equation of the line is

$(x + 2y - 3) - 1 \cdot (4x - y + 7) = 0$

$\Rightarrow$ $3x - 3y + 10 = 0$

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

25. Solving $2x - y + 5 = 0$ and $5x + 3y - 4 = 0$, we get $x = -1$ and $y = 3$ i.e., the point of intersection of the given lines is $(-1, 3)$.

∴ The equation of any line perpendicular to the line

$$x - 3y + 21 = 0 \text{ is } 3x + y + k = 0 \quad \ldots(1)$$

If this line (1) passes through the point $(-1, 3)$ then

$$3 \times -1 + 3 + k = 0 \Rightarrow k = 0$$

∴ From (1), the required equation of the line is $3x + y = 0$.

26. The equation of any line passing through the intersection of the lines $3x + 4y - 7 = 0$ and $x - y + 2 = 0$ is

$$(3x + 4y - 7) + k(x - y + 2) = 0 \quad \ldots(1)$$

Slope of the line $= -\dfrac{3+k}{4-k} = 3$

$$\Rightarrow \quad k = \frac{15}{2}$$

Hence, from (1) the required equation of the line is

$$(3x + 4y - 7) + \frac{15}{2}(x - y + 2) = 0$$

$$\Rightarrow \quad 21x - 7y + 16 = 0$$

27. The equation of any line passing through the point of intersection of the lines $3x - 4y + 1 = 0$ and $5x + y - 1 = 0$ is

$$(3x - 4y + 1) + k(5x + y - 1) = 0 \quad \ldots(1)$$

for intercept of this line with the x-axis, $y = 0$

$$\therefore \quad 3x + 1 + k(5x - 1) = 0$$

$$\Rightarrow \quad x = \frac{k-1}{5k+3}$$

For intercept of the line (1) on the y-axis, $x = 0$

$$\therefore \quad -4y + 1 + k(y - 1) = 0$$

$$\Rightarrow \quad y = \frac{k-1}{k-4}$$

Since the intercepts on the axes are equal.

$$\therefore \quad \frac{k-1}{5k+3} = \frac{k-1}{k-4} \Rightarrow k = 1, \text{ or } x = -\frac{7}{4}$$

But $k \neq 1$, because if $k = 1$, the line (1) becomes $8x - 3y = 0$ which passes through the origin and therefore cannot make non-zero intercepts on the axis.

$$\therefore \quad k = -\frac{7}{4} \text{ and from (1), we get}$$

$$3x - 4y + 1 - \frac{7}{4}(5x + y - 1) = 0$$

$\Rightarrow 23x + 23y = 11$, which is the required equation of the line.

28. We have $\dfrac{x}{a} - \dfrac{y}{b} = 1 \quad \ldots(1)$

Since (1) passes through the point $(8, 6)$

$$\therefore \quad \frac{8}{a} - \frac{6}{b} = 1 \quad \ldots(2)$$

The line (1) meets the x-axis at the point given by $y = 0$ and from (1) $x = a$ i.e, the line (1) meets the x-axis at the point $A(a, 0)$.

Similarly, the line meets the y-axis $(x = 0)$ at the point $B(0, -b)$.

By the given condition, area of $\Delta = 12$

$$\Rightarrow \quad \frac{1}{2}ab = 12$$

$$\Rightarrow \quad ab = 24 \Rightarrow b = \frac{24}{a}$$

Substituting $b = \dfrac{24}{a}$ in (2), we get

$$\frac{8}{a} - \frac{6}{\frac{24}{a}} = 1 \Rightarrow a = 4 \text{ or } -8 \text{ and } b = 6 \text{ or } -3$$

Hence, from (1) the equation of the straight line are

$$\frac{x}{4} - \frac{y}{6} = 1 \text{ and } \frac{x}{-8} - \frac{-y}{-3} = 1$$

$$\Rightarrow \quad 3x - 2y = 12 \text{ and } 3x - 8y + 24 = 0$$

29. The equation of the lines may be written as $3x + 4y + 2 = 0$ and $-5x + 12y + 6 = 0$ in which the constant terms 2 and 6 are both positive.

The equation of the bisector of the angle in which the origin lies is $\dfrac{3x + 4y + 2}{\sqrt{3^3 + 4^2}} = \dfrac{-5x + 12y + 6}{\sqrt{(-5)^2 + (12)^2}}$

$$\Rightarrow \quad 16x - 12y - 1 = 0$$

The equation of the other bisector is

$$\frac{3x + 4y + 2}{\sqrt{3^2 + 4^2}} = \frac{-(-5x + 12y + 6)}{\sqrt{(-5)^2 + (12)^2}}$$

$$\Rightarrow \quad x + 8y + 4 = 0$$

30. Let the equations of the sides BC, CA and AB of the triangle ABC be represented by

$$2y - x = 5 \quad \ldots(1)$$

$$y + 2x = 7 \quad \ldots(2)$$

$$y - x = 1 \quad \ldots(3)$$

Solving the above 3 equations (1), (2) and (3), we get $A(2, 3)$, $B(3, 4)$ and $C\left(\dfrac{9}{5}, \dfrac{17}{5}\right)$

∴ The area of ΔABC

$$= \frac{1}{2}\left\{2 \times 4 - 3 \times 3 + 3 \times \frac{17}{5} - 4 \times \frac{9}{5} + \frac{9}{5} \times 3 - \frac{17}{5} \times 2\right\}$$

$$= \frac{1}{2}\left(8 - 9 + \frac{51}{5} - \frac{36}{5} + \frac{27}{5} - \frac{34}{5}\right) = \frac{3}{10} \text{ units.}$$

31. Let $A(1, 2)$, $B(2, 3)$ and $C(4, 3)$ be the vertices of ΔABC.

$m_1 =$ Slope of $BC = \dfrac{3-3}{4-2} = 0$ i.e., BC is parallel to the x-axis.

∴ The perpendicular from $A(1, 2)$ to BC is parallel to the y-axis and its equation is $x = h$, which passes through $A(1, 2)$

∴ $h = 1$ i.e, the equation of the perpendicular from $A(1, 2)$ on BC is $x = 1 \quad \ldots(1)$

$$m_2 = \text{Slope of } AC = \frac{3-2}{4-1} = \frac{1}{3}$$

If m_2' be the slope of a line perpendicular to AC then $m_2 m_2' = -1$ or $\dfrac{1}{3} \cdot m_2' = -1$ or $m_2' = -3$

∴ The equation of the perpendicular from $B(2, 3)$ on AC whose slope is -3 is

For More Visit : www.LearnEngineering.in
Scanned by CamScanner

yoursmahboob.wordpress.com
For More Visit : www.LearnEngineering.in

$$y - 3 = -3(x - 2)$$

$$\Rightarrow \quad 3x + y = 9 \qquad \ldots(2)$$

The orthocentre is the point of intersection of two lines (1) and (2)

$\therefore$ From (1) and (3), we get $3 \times 1 + y = 9$

$$\Rightarrow \quad y = 6$$

$\therefore$ The required coordinates of the orthocentre are (1, 6)

32. Let $A(x_1, y_1)$ be the third vertex. Let AD, BE, CF be the perpendiculars from the vertices on the opposite side BC, CA, AB respectively. Then the orthocentre $O(0, 0)$ is the point of intersection of AD, BE and CF.

Since AD i.e. OA is perpendicular to BC.

$\therefore$ Slope of $BA \times$ slope of $BC = -1$

or $$\frac{y_1 - 0}{x_1 - 0} \times \frac{3 - (-1)}{-2 - 5} = -1$$

$$\Rightarrow \quad y_1 = \frac{7x_1}{4} \qquad \ldots(1)$$

Again, since OB is perpendicular to CA

$$\therefore \quad \frac{-1 - 0}{5 - 0} \times \frac{y_1 - 3}{x_1 + 2} = -1$$

$$\Rightarrow \quad 5x_1 + 10 = y_1 - 3$$

$$\Rightarrow \quad x_1 = -4 \qquad \left(\because y_1 = \frac{7x_1}{4}\right)$$

From (1), $y_1 = \frac{7x_1}{4} = \frac{7 \times (-4)}{4} = -7$

Hence, the required coordinates of the third vertex A are $(x_1, y_1) = (-4, -7)$

33. Let (x_1, y_1) be the third vertex then $y_1 = x_1 + 3 \ldots(1)$

The area of the triangle formed by the points (2, 1), (3, – 2) and (x_1, y_1)

$$= \frac{1}{2}\{-4 - 3 + 3y_1 + 2x_1 + x_1 - 2y_1\}$$

$$= \frac{1}{2}(3x_1 + y_1 - 7)$$

By the given condition,

$$\pm \frac{1}{2}(3x_1 + y_1 - 7) = 5$$

$$\Rightarrow \quad 3x_1 + y_1 - 7 = \pm 10$$

$$\Rightarrow \quad 3x_1 + y_1 = 17 \qquad \ldots(2)$$

and $$3x_1 + y_1 = -3 \qquad \ldots(3)$$

Solving (1) and (2) we get $x_1 = \frac{7}{2}, y_1 = \frac{13}{2}$

Solving (1) and (3) we get $x_1 = -\frac{3}{2}$ and $y_1 = \frac{3}{2}$

Hence, the coordinates of the third vertex is either $\left(\frac{7}{2}, \frac{13}{2}\right)$ or $\left(-\frac{3}{2}, \frac{3}{2}\right)$

34. Equation of any line L perpendicular to

$$5x - y = 1 \text{ is } x + 5y = k \qquad \ldots(1)$$

Where k is an arbitrary constant.

If this line cuts x-axis at A and y-axis at B, then for A, $y = 0$ and from (1) $x = k$ i.e., A is the point $(k, 0)$ for B, $x = 0$ and from (1) $y = \frac{k}{5}$ i.e., B is the point $\left(0, \frac{k}{5}\right)$

$\therefore$ Area of the given $\Delta OAB = \frac{1}{2}(x_1 y_2 - x_2, y_1)$

$$= \frac{1}{2}\left(\frac{k^2}{5} - 0\right) = \frac{k^2}{10}$$

But the given condition $\frac{k^2}{10} = 5$

or $$k^2 = 50 \quad \therefore k = \pm 5\sqrt{2}$$

Hence, from (1), the required equation of the line is

$$x + 5y = 5\sqrt{2} \quad \text{or} \quad x + 5y = -5\sqrt{2}$$

35. Let $ABCD$ be the square and let (1, 2) and (3, 8) be the coordinates of opposite vertices A and C respectively.

The equation of the diagonal AC is $y - 2 = \frac{8 - 2}{3 - 1}(x - 1)$

$$\Rightarrow \quad 3x - y = 1$$



For More Visit : www.LearnEngineering.in
Scanned by CamScanner